INITIO INTELLIGENCE

Four-dimensional Yang–Mills fields from the Wilson measure

Conditional construction and collected proofs

For a fixed group SU(N), the continuum theorem concerns the 21 gauge-invariant curvature bilinears on one sequence of Wilson lattices. It requires a common physical normalization and uniform estimates for the transported fields, their histories and their Schwinger functions.

The theorem is conditional. In particular the all-history C-05 boundary estimate requires control of a composed reconstruction while earlier innovations stay fixed. The finite-cutoff statements and conditional implications in the annexes retain their individual hypotheses.

The 65 mathematical annexes are followed by detailed estimates for the coupling, boundary laws and insertions. The closing catalogue gives the statements and dependencies of the open problems. The Wilson-source volumes retain their own equation numbering and pagination.

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A conditional construction of four-dimensional Yang–Mills fields from the Wilson measure

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Abstract

We formulate a conditional existence theorem for the gauge-invariant curvature bilinears of four-dimensional $SU(N)$ Wilson lattice theory. The lattice fields have 21 components and share one physical normalization. Exact conditional transport of their sources gives a representation of their joint moments. Uniform bounds on these moments and on the rotation defects would yield a common Euclidean limit. Ultraviolet and energy estimates would identify a vacuum phase of the reconstructed fields. The unresolved analytic and representation hypotheses are stated explicitly below.

1 The continuum statement

Fix $G = SU(N)$, $N \geq 2$, and give its Hermitian generators the normalization $\mathrm{Tr}(T^a T^b) = \delta^{ab}/2$. Let

$$E = \mathrm{Sym}^2(\Lambda^2 \mathbb{R}^4), \qquad D(R)A = (\Lambda^2 R) A (\Lambda^2 R)^t.$$

The fields considered below take values in this real, 21-dimensional space. Its scalar, pseudoscalar, traceless Ricci and Weyl summands under orthogonal coordinate changes have dimensions $1, 1, 9, 10$.

Write $X = (x_1, \ldots, x_m)$. Let $\mathcal{S}_0((\mathbb{R}^4)^m)$ be the Schwartz functions all of whose derivatives vanish on every partial diagonal $x_i = x_j$, $i \neq j$. For an $E^{\otimes m}$-valued test set

$$\nu_s(F) = \max_{|\alpha| \leq s} \sup_X (1 + |X|^2)^{s/2} |\partial^\alpha F(X)|.$$

A fixed component norm is understood. At order one there is no diagonal condition.

Conditional existence theorem. Assume GC, CL, the analytic estimates and the reconstruction and phase hypotheses stated below, excluding the final identification A-03 from the hypotheses, with their stated uniformity for the same Wilson action, physical normalization and cofinal sequence. There is a subsequence along which all full moments of the normalized curvature bilinears converge on $\mathcal{S}_0$. For constants C, p, q independent of m, the limits satisfy

$$|S_m(F)| \leq C^{m+1} (m!)^p \, \nu_{qm}(F).$$

They satisfy Euclidean covariance with tensor action D, permutation symmetry and reflection positivity. Corrected Osterwalder–Schrader reconstruction yields relativistic local fields with the spectrum condition. There is a pure vacuum phase with the stipulated connected and disconnected symmetries, a unique vacuum up to scalars, clustering in the stated vacuum-phase sense, and polynomial energy bounds for the 21 fields. In this phase the scalar has the separated two-point asymptotic and nonzero connected three-point asymptotic specified below. The theorem concerns the Wilson continuum hierarchy and a phase of its reconstruction. A-01 and A-02 specify the common subsequence and the reconstruction on its test space. A-03 is the final identification to be verified by these implications. No positive spectral gap is asserted.

2 Wilson measure and physical scale

On the torus $\Lambda_{a,T} = (a\mathbb{Z}/T\mathbb{Z})^4$, give each positive link ℓ a matrix $U_\ell \in G$ and set $U_{-\ell} = U_\ell^\dagger$ for the reversed link. Let U_p be the ordered product around p. Counting each positive plaquette once, put

$$s(U) = N - \operatorname{Re}\operatorname{Tr} U, \qquad d\mu_{a,T} = \frac{1}{Z_{a,T}} \exp\left(-\beta_a \sum_p s(U_p)\right) \prod_\ell dU_\ell.$$

Here dU_ℓ is probability Haar measure. For the bare parameter $x_{0,a}$ the convention is $\beta_a = 2Nx_{0,a}$. The inverse coupling in the source-free renormalization is $Y_0 = N\beta_a = 2N^2 x_{0,a}$. Use an admitted odd blocking factor L, fixed reference length ℓ_0, and cubic tori with

$$a_n = L^{-K_n}\ell_0, \qquad T_n = 2L^{M_n}\ell_0, \qquad K_n, M_n \longrightarrow \infty.$$

At physical intermediate scale $h = a_n L^j$, the period and each retained block grid must be compatible.

On the admitted source-free domains of Bałaban [1, 2], let Y_j be the inverse coupling determined by the marginal curvature projection of the homogeneous Wilson action. The Gaussian coefficient must be calibrated in this same marginal projection. Let C_n be the curvature projection of the composite constrained Gaussian log density after n steps, and let d_j^G be the projected Gaussian increment at step j. Both use the original coarea, gauge and block average, relative to coarse-link Haar measure.

Open proposition GC. *Uniform Gaussian calibration.* For these objects the projection commutes with the finite compositions and the required infinite-volume limit, and

$$C_n = \sum_{j<n} d_j^G, \qquad \sup_n \left| C_n - \frac{11N^2 \log L}{12\pi^2} n \right| < \infty.$$

The equalities and uniform bound hold with the background derivatives and spatial moments defining the marginal coordinate. In the Schur decomposition of the composite Gaussian, the transverse blocked-covariance remainder is uniformly bounded in depth.

A logarithmic asymptotic with an $o(n)$ remainder would identify the slope but would not supply the bounded correction used here. We retain the full estimate in GC as a hypothesis.

Conditional on GC and the common native-chart and reset-comparison hypotheses, the weak-step estimate through the weak stopping depth [8, Apps. F and G] has the form

$$Y_j - Y_{j+1} = d_j^G + e_j, \qquad |e_j| \le A/Y_j, \qquad \sum_{i<j} d_i^G = dj + q_j, \quad |q_j| \le Q, \quad d = \frac{11N^2}{12\pi^2} \log L.$$

To see where the error bound enters, multiply the complete mature reset insertions by a common complex parameter while keeping the numerical couplings fixed. At each earlier homogeneous integration subtract the curvature projection of that same integration. On the admitted source charts the common parameter disc has radius of order g_j^{-D}, where $g_j = Y_j^{-1/2}$ and $D > 4$ is fixed. One Cauchy estimate for the terminal projection, including the native two-derivative cost, bounds the change between parameters zero and one by $O(g_j^{D-2})$ up to the retained logarithmic factors. At zero the even fluctuation-amplitude expansion on the fixed physical chart differs from the calibrated Gaussian projection by $O(g_j^2)$. At one, installing the outgoing coupling after

the raw integration reproduces the Wilson recursion. This comparison controls the effect of mature resets on all intervening homogeneous terms [8, BP.17–34, DE.1–15].

The bounded correction $Z_j = Y_j + q_j$ gives $Z_j - Z_{j+1} = d + e_j$. In the ultraviolet direction $u_s = 1/Z_{K-s}$ therefore satisfies $u_{s+1} = u_s - du_s^2 + O(u_s^3)$. In particular the sign comes from reversing the blocking direction. As long as Y_j is sufficiently large, Z_j has strictly positive decrement. Stopping at a weak threshold then gives trajectories with positive subsequential terminal coupling. A specified terminal normalization at ℓ_0 imposes a further equation on the bare data.

A useful common choice of depths can be made before fixing those data. Let $G_j = \sum_{i<j} d_i^G$, take the weak floor F with $\varepsilon = A/F < d$, and fix $0 < b < d - \varepsilon$. Choose K to be a running maximum of $G_j - (b+\varepsilon)j$. Since the latter equals $(d - b - \varepsilon)j + q_j$, these depths have bounded gaps. On any admitted trajectory reaching such a depth, summation gives

$$Y_j - Y_K = G_K - G_j + \sum_{i=j}^{K-1} e_i \;\geq\; b(K-j).$$

Thus the lower comparison holds at the same depths for different bare parameters. Obtaining $Y_K = Y_*$ also requires that the terminal readout attain its prescribed value across changes of the incoming block grid.

Open proposition B-05. *Matching at the physical scale.* For the prescribed weak inverse coupling Y_* at ℓ_0, there is a single cofinal family of bare Wilson coefficients β_n and depths K_n such that $Y_{K_n} = Y_*$. Every intermediate step satisfies the source-free chart, cutoff and scale-comparison bounds. The readouts at different cutoffs admit the same physical prescription and compatible limiting data.

3 Local fields and their normalization

Index the two-forms by $A \in \{01, 02, 03, 12, 13, 23\}$. For the plaquette $P_{u,v}(x)$ traversed first in direction u and then v, let $\pi_0 H = H - \operatorname{Tr}(H)I/N$, and define

$$C_{\mu\nu}(x) = \frac{1}{8i} \sum_{\sigma,\tau=\pm1} \sigma\tau\, \pi_0\Big(P_{\sigma e_\mu, \tau e_\nu}(x) - P_{\sigma e_\mu, \tau e_\nu}(x)^\dagger\Big), \qquad q_{AB} = a^{-4} \operatorname{Tr}(C_A C_B).$$

The vertex average w_a fixes the scalar component by

$$w_a(x) = \frac{1}{4a^4} \sum_{\mu<\nu} \sum_{\sigma,\tau=\pm1} s(P_{\sigma e_\mu, \tau e_\nu}(x)), \qquad \widetilde{q} = q + \frac{I_6}{6}(2w_a - \operatorname{Tr} q).$$

Gauge conjugations at the rooted vertex disappear under the trace. Each unoriented elementary plaquette occurs at its four vertices with the same value of s. Hence $\frac{1}{2}\operatorname{Tr}\widetilde{q} = w_a$ and $a^4 \sum_x w_a(x) = \sum_p s(U_p)$.

Let P_h denote the orthogonal projectors for the signed permutations on E. In the diagonal plane entries there are a scalar, a traceless Ricci diagonal and a Weyl diagonal, of dimensions $1, 3, 2$. The entries with a shared index split into symmetric and antisymmetric combinations of the two complementary choices (dimensions $6, 6$). The entries with disjoint indices split into a pseudoscalar and its orthogonal complement (dimensions $1, 2$). In these decompositions pair antisymmetry supplies the orientation signs. Write $e_S = I_6/\sqrt{6}$. Use

$$\Phi_a = Z_a \widetilde{q} - c_a e_S, \qquad Z_a = \sum_h z_{h,a} P_h.$$

The coefficients depend on the mesh, torus and fixed physical scale μ.

For time reflection θ, write $r_0 = \mathrm{diag}(-1, 1, 1, 1)$. Take the radial bump $\eta(x) \propto \exp[-1/(1 - |x|^2)]$ for $|x| < 1$, zero outside, with unit integral. With $\epsilon = 1/64$, use $f_\mu(x) = (\mu/\epsilon)^4 \eta((\mu x - e_0)/\epsilon)$. In each channel let n_h be the mean square in the Wilson reflection inner product of the smeared centered components of $P_h \widetilde{q}$, averaged over an orthonormal basis of that channel. The reflected tensor carries $D(r_0)$. Let λ_h be the same pairing for the unit-coupling free Wick bilinears. Their separated covariance operator on E is

$$\frac{N^2 - 1}{2\pi^4 |x|^8} D(I - 2\widehat{x}\,\widehat{x}^{\,t}).$$

In the reflected integral the direction differs from the time axis by $O(1/64)$. The tensor factor differs from the identity after multiplication by $D(r_0)$ by at most $8/63$ in operator norm. This proves $\lambda_h > 0$. Reflection positivity gives $n_h \geq 0$. On the domain where all seven $n_h > 0$, the positive normalization and the scalar mean subtraction are uniquely

$$z_{h,a} = \sqrt{\lambda_h / n_h}, \qquad c_a = z_{S,a}\, \mathbb{E}_{\mu_{a,T}} \langle e_S, \widetilde{q} \rangle.$$

To see the role of the first mixing coefficients, write the seven orthogonal channel projectors as P_h, and let $\Gamma = \sum_h \gamma_h P_h$. For a weak step with $M_j = I + u_j \Gamma + O(u_j^2)$ in this commuting algebra, write $M_j = \sum_h m_{h,j} P_h$. On a sufficiently weak tail,

$$\log m_{h,j} = u_j \gamma_h + O(u_j^2), \qquad \prod_{j=k}^{l} m_{h,j} = \exp\left(\gamma_h \sum_{j=k}^{l} u_j + O\left(\sum_{j=k}^{l} u_j^2 \right) \right).$$

The square error has finite sum along the ultraviolet tail. After removal of the displayed leading exponential, each product converges to a finite nonzero factor. The same calculation controls its inverse. The actual matrix estimate and the data at the finite normalization scale enter separately in F-03–F-05.

Open proposition F-03. *First-order mixing.* In the seven hypercubic channels of the fixed clover basis, projection of the exact regular one-field transport gives a matrix M_j with first-order part $I + u_j \Gamma$. Each of the seven entries of Γ is identified by an explicit coefficient or invariant integral with its trace, field and coupling normalization evaluated.

Open proposition F-04. *Uniform mixing remainder.* On the regular histories under consideration, $\|M_j - I - u_j \Gamma\| \leq C u_j^2$, uniformly in cutoff, carrier, and retained background, or the remainder obeys a uniform summable bound sufficient for the same running product. The bound includes all derivatives contributing to the exact matrix.

Open proposition F-05. *Running field and identity coefficients.* Along the matched cofinal sequence, the finite normalization equations at μ have solutions with nondegenerate normalization determinant. The field matrices, their inverses and the scalar identity coefficient obey the logarithmic bounds obtained from the channel products of M_j, in the same seven measurements and identity convention.

Open proposition F-06. *Compatible local products.* The collision coefficients of the exact source jets define real, reflection-compatible tensor distributions carried by point jets or by stencils of vanishing physical diameter. Restriction to lower source degree agrees with the

lower-order construction. Their distribution orders grow at most linearly with the number of insertions and their norms obey the uniform hierarchy bounds.

Open proposition F-07. *Weighted additive terms.* Write T_{j+1} for the running linear product through the indicated step and R_j for its additive response, including identity and collision terms. The sum $\sum_j \|T_{j+1}^{-1} R_j\|$ is uniformly finite in the joint observable norm, with source-order growth bounded by a fixed exponential and factorial power sufficient for S-06.

4 Conditional measure and derivatives of its action

Use a single positive augmented Wilson law throughout the conditional construction. Its retained sigma-field contains the variables of the source normal form which have not been integrated. Normal-form history labels are retained until their stipulated integration. A primitive label already summed in forming the normal form is integrated at that operation. For a second-class component, its internal coordinates are retained until the containing successful operation or the terminal integration, unless the entire normalized component kernel is integrated at its birth with all its outer factors retained. A projection to a coarser sigma-field is then a separate conditional expectation.

Open proposition CL. *Compatibility of the conditional representation.* Choose either the literal source construction or a specified localized component construction, including its block tests, coordinate charts, gauge frames and cutoffs. Its finite integral has the original Wilson marginal. The complete conditional densities, their current domains and the complex reconstruction maps satisfy the uniform source identities and bounds used below on that same retained sigma-field. All completed component masses, Haar and coarea factors and changes of gauge frame are assigned to their actual conditional or outer densities. Any local replacement of a test satisfies the support, reflection covariance, admission and decay estimates for its added terms. Exact transport of the marked numerator and the spatial reset estimates hold for this chosen representation, including marks depending on earlier preparation-stage variables.

The local component variant additionally requires the multilinear response estimates on the walk tiling, the necessary buffer layers for its box problems, and the enlarged construction constants to be compatible with one common coupling ceiling. These requirements enter CL. Equality of unmarked partition functions alone does not imply them.

Assuming CL, at any finite cutoff retain the gauge data, inactive exterior variables and discrete history variables needed to continue the source-free construction. Disintegrate the positive joint law over these variables. For independent interior coordinates z of an admitted regular block the conditional law has density proportional to its positive weight on the actual product of free-coordinate balls. Its precision is the constrained principal block K_{II}. The term $-z^t K_{IO} y$ in the exponent retains the exterior coordinate y. The rest contains the constrained classical remainder, Haar and coarea terms, regular activities, and boundary terms including responses of the older Gaussian and Schur coefficients. Completed-component integrations carry their normalizing masses in the outer weight.

For example, write V for the full negative log density on such a block and P for the current real cutoff. The unperturbed precision has an admitted positive spectral lower bound λ_0. A sufficient boundary input includes a bound C/P on its score at the origin and a Hessian with row norm C/P^2 in the exponentially weighted block distance, uniformly on the requisite chart. An entrywise majorant at a strictly smaller exponential rate follows by summing four-dimensional distance shells. Together with the separate regular and local bounds this retains a positive

lower Hessian bound λ. A linear tilt leaves the Hessian unchanged. For a linear functional $t \cdot z$, integrating the tilted variance bound gives

$$\mathbb{E} \exp\{t \cdot (z - \mathbb{E}z)\} \leq \exp\{|t|^2/(2\lambda)\}.$$

The bound yields centered L^p estimates of order $\sqrt{p}$. The mean requires the exterior force as well as the score. The complete boundary estimate must hold uniformly through repeated components.

For an old local term, denote its birth scale by b, its evaluation layer by n, and the integration stage by j, with $b \leq n < j$. Set $a = n - b$ and $c = j - n$. Suppose the same Wilson-subtracted difference on both stopped reconstructions has a Ward bound proportional to $L^{-\lambda a}$ and a response bound proportional to $e^{-\Gamma c}$, up to fixed polynomial factors. For $0 < \eta < 1$, the geometric mean of the two bounds gives a finite root sum whenever

$$\eta\lambda > 4, \qquad (1 - \eta)\Gamma > 4 \log L.$$

Indeed, the four-dimensional count contributes $L^{4(a+c)}$, and the two residual geometric series absorb fixed powers of $1 + a + c$. This calculation isolates the required response. It is the derivative of the composed stopped reconstruction at fixed earlier innovations. Varying the determining data of a single minimizing field also moves those innovations under a gauge change. Equivariance cancels that extra term in a complete gauge integral under simultaneous transport; the partially integrated density retains a fixed-exterior response. The required uniform estimate for that response on the joint chart remains an input.

Open proposition C-05. *Boundary activity derivatives.* On the common physical current support and smaller native chart, the complete boundary action, including collars, constraints and responses induced by earlier component operations, satisfies the local native and current derivative bounds of the bulk activity estimate. The weighted current Hessian is small relative to the principal coercivity gap. The first current derivatives have the uniform score bound needed for the conditional means. Constants are independent of volume, depth and distances inside the activity supports.

Open proposition C-06. *Coercivity in heterogeneous blocks.* For every admitted retained background and layer geometry on the common chart, the full regular potential has a uniformly positive Hessian on independent interior coordinates and a uniformly exponentially decaying inverse in the current spatial metric. The local derivative estimates hold with uniform weighted norms and domain margins, retaining the actual exterior force.

Open proposition C-08. *Regular conditional moments.* For the exact regular conditional law, the field and its covariant finite gradients satisfy the current-scale mean bounds including the exterior force, centered moment estimates at every finite moment order, and spatially decaying cumulant estimates with a fixed factorial power. The constants are uniform in the admitted geometry and volume.

On a fixed product of convex cutoff balls, write $A = K_{II}$, $B = K_{IO}$ and let W be the smooth inner perturbation. Suppose $A \succeq \lambda_0 I$ and $\|W_{zz}\|, \|W_{zy}\| \leq \varepsilon < \lambda_0/2$. For fixed exterior y, the complete inner score is $S = -B^t z + W_y$. The convex-domain variance inequality applied to $u^t S$ gives

$$\mathrm{Cov}_{\nu_y}(S, S) \preceq B^t A^{-1} B + \left(\frac{\|B\|^2 \varepsilon}{\lambda_0(\lambda_0 - \varepsilon)} + \frac{2\|B\|\varepsilon + \varepsilon^2}{\lambda_0 - \varepsilon} \right) I.$$

To obtain this estimate, bound the variance by the inverse Hessian form with gradient $(-B + W_{zy})u$, then expand the two mixed terms and use the resolvent identity. Only the upper comparison is required. The outer density retains its own potential and its support; these must be controlled to deduce concentration for the full law.

5 Transport of insertions and averaging of histories

On the source-free joint probability space let Q_j take conditional expectation from the complete state at one cut to the next retained state. Attach distinct commuting variables s_i with $s_i^2 = 0$ to finitely many microscopic insertions Φ_i. With $Q_{0:j}$ the composed conditional kernel, put

$$\mathcal{Z}_j(s) = Q_{0:j} \prod_i (1 + s_i \Phi_i), \qquad W_j(s) = \log \mathcal{Z}_j(s).$$

Both are finite polynomials, since $\mathcal{Z}_j(0) = 1$. The tower identity gives $\mathcal{Z}_{j+1} = Q_j \mathcal{Z}_j$ and $W_{j+1} = \log Q_j e^{W_j}$. In particular the coefficient of $\prod_{i \in I} s_i$ in $\mathcal{Z}_j$ is the conditional moment of the original product at its original source positions. For $I = \{1, 2\}$,

$$W_{j+1,12} = Q_j W_{j,12} + \mathrm{Cov}_{Q_j}(W_{j,1}, W_{j,2}).$$

For general I, sum over set partitions of I the corresponding conditional cumulants of the lower coefficients. Thus a change of history which erases a variable creates genuine connected sources of higher order.

For real retained parameter v, a smooth density $p_v = w_v / \int w_v$ on fixed support obeys

$$\partial_v Q_v f = Q_v(\partial_v f) + \mathrm{Cov}_{Q_v}(f, \partial_v \log w_v).$$

Changes of support require the signed variation of the kernel on a common raw space. This distinction enters the localization norm for a marked coefficient. At scale h, order m, its physical prefactor is h^{4m}. A carrier containing the source anchors is charged its size and the length of its connecting tree. Its value, its direct retained-coordinate differences and the differences generated by the kernel are controlled separately, in the moment and weighted supremum norms. In particular, L^p contraction alone leaves $(h/a)^{4m}$ if it is applied only to a microscopic bound of order a^{-4m}.

For precision, let X be a carrier in the block graph at mesh h, enlarged to contain the insertion anchors and all blocks on which its coordinates depend. Let $t_h(X)$ be the minimum number of block edges in a connected subgraph containing X. The regulator on this carrier is

$$R_X = 1 + \sum_{p:\, \mathrm{foot}(p) \subset X} \beta_p s(V_p) + \sum_{i:\, \mathrm{foot}(i) \subset X} |z_i|^2.$$

Only retained plaquettes and free coordinates occur in these sums. For a group coordinate use left multiplication by $\exp(tT/\sqrt{1+\beta_e})$, with T an orthonormal Lie algebra direction and $0 < |t| \le 1$. Use translation by t in a free coordinate and replacement in a discrete label. Continuous differences are divided by t. Write $A_X^\sharp$ for the supremum of the absolute value over the identity and these one-coordinate perturbations, and put $\widehat{R}_X = R_X + R_X^\sharp$. Decompose the difference as $\delta A_X = B_X + J_X$, with B_X the direct insertion variation and J_X the signed response of the conditional kernel. For either response, the superscript $\natural$ denotes the sum over coordinate directions of the suprema over the allowed perturbation sizes. At degree $d > 0$ set

$$D_{d,X}(G) = \sup_{p \ge 1} (dp)^{-d} \|G\|_{L^p(\mu_h)} + d^{-d} \sup_x \frac{|G(x)|}{\widehat{R}_X(x)^d}.$$

The norm for a localized connected mark with m insertions is the sum over its carriers of

$$e^{\kappa(|X|+t_h(X))}h^{4m}\big[D_{m,X}(A^{\sharp}_X)+D_{m+1/2,X}(B^{\natural}_X)+D_{m+1/2,X}(J^{\natural}_X)\big].$$

Here μ_h is the normalized outer law, and the suprema use the fixed ambient versions of the retained coordinates. Bounds are uniform in the choice of the first anchor. Disconnected pieces carry the product of the weights of their connected components.

At three fixed regular blocking layers the microscopic scalar admits a more detailed decomposition. In a fixed physical tube the local Ward projection removes the marginal Wilson completion from its smooth quadratic jet. Simultaneous dilation gives the leading coefficient $\alpha(g_3/g_0)^2$; the separate current estimates retain the responses of the innovation kernels and of the insertion itself. These statements concern the specified three-layer input. Iteration requires uniform transfer of the whole quadratic function in the physical carrier norm, as well as the contributions of complementary histories.

Open proposition M-04. *Scalar transport in the incoming basis.* For inputs satisfying the one-step current-scale budget, the conditional expectation of the incoming scalar decomposes exactly into the current scalar basis element, an identity coefficient and a spatially localized remainder bounded uniformly in the current-scale mark norm. The coefficients include the constrained background, bulk, boundary, remainder and coarea responses.

Open proposition M-05. *Tensor transport.* The corresponding one-step identity holds for each of the 21 curvature bilinears in the fixed reference basis, including both Weyl sectors under the connected rotation group. The retained-field responses obey the same current-scale norm and the coefficients intertwine the appropriate hypercubic tensor actions.

Open proposition M-06. *Local source jets.* For every source degree m, conditional integration of the incoming jet has an exact expansion into localized output vertices. It includes all collisions, resets and remainders. The sum over carriers obeys uniform bounds with fixed factorial power and derivative order linear in m. The expansions agree upon truncation in source degree.

Open proposition M-07. *Uniform iterated matching.* The successive one-step expansions along the full normalized history law represent each transported jet by current-scale local fields, their response kernels and a terminal remainder. The irrelevant contributions are summable in scale, including the outer retained-data responses, with constants independent of volume and total depth.

The same law gives the outer covariance identity, with ξ denoting a retained state,

$$\operatorname{Cov}(F,G)=\mathbb{E}\,\operatorname{Cov}(F,G\mid\xi)+\operatorname{Cov}(\mathbb{E}[F\mid\xi],\mathbb{E}[G\mid\xi]).$$

For the conditional reset of successful components, spatial cumulant decay is measured between complete component footprints, including the random geometry inside the conditional average. This estimate controls that conditional law. To iterate the marked identity one must also integrate its bounds against the true probabilities of the low-margin and unreplaced sectors and estimate the second term above. The corresponding requirements at all orders are the following.

For retained bad-sector events an integrated estimate may be stated without taking a supremum over field configurations conditioned to lie inside the event. For example, if $E_{Y,b}$ specifies a component Y at scale j with oldest birth b, a useful joint hypothesis is

$$\mathbb{P}\Big(\bigcap_i E_{Y_i,b_i}\Big)\le\prod_i a(Y_i,b_i),\qquad \sup_x\sum_{b\le j}\sum_{Y\ni x}L^{4(j-b)}e^{\sigma|Y|_\natural}a(Y,b)\le\delta<1$$

for the specified disjoint components, in the actual normalized joint law. Such a bound must be combined with the physical insertion and multi-anchor costs. To use it in place of a pointwise carrier estimate, one must prove the resulting integrated marked bounds, their stability under reset, and their attribution to the exact physical shells in S-04–S-05. A rare-event bound gives no bound on a supremum taken after conditioning on that event.

Open proposition H-02. *Inherited low-margin and unreplaced events.* The joint law of inherited low-margin and unreplaced events, or its directly marked version, obeys bounds conditional on the specified retained data that include all overlaps. Its decay pays the geometric size and the actual insertion losses of each affected carrier in the current-scale norm.

Open proposition H-04. *One step in the full history law.* For incoming jets satisfying the one-step hypotheses of M-06, summing regular, repaired, residual and inherited sectors in the normalized law preserves all output responses and gives a uniform current-scale estimate of fixed factorial order. The bound includes the full conditional normalizers and is uniform in volume, scale and retained history.

Open proposition H-05. *Cumulants of the retained means.* The retained-data variations of the conditional means and their higher joint cumulants obey the required spatial estimates at intermediate scales and the terminal distribution bounds at ℓ_0. In particular these estimates apply to the outer covariance in the law of total covariance.

6 From a connected kernel to a Schwinger distribution

Let $C_{m,n}$ be the connected distribution and $S_{m,n}$ the full distribution of the prescribed fields. In this section enumerate the shells from the reference scale toward short distances, so that $h_j = \ell_0 L^{-j}$. On compactly supported tests in a central torus chart, the needed connected identity consists of a terminal distribution, a sum of physical kernels over the shells, and the prescribed contact contributions. A contact supported strictly on a partial diagonal and having finite differential order pairs to zero with $F \in \mathcal{S}_0$. An extended lattice stencil needs its own estimate as the mesh tends to zero.

Open proposition S-03. *Terminal and one-point distributions.* At ℓ_0, the one-point functions have the prescribed identity subtraction and the normalized terminal terms of all orders act on joint tests with bounds $C^{m+1}(m!)^p \nu_{qm}$, uniformly along the expanding tori, for the same fields and coefficients as in F-05.

Open proposition S-04. *Exact physical shells.* At every finite cutoff the connected Wilson correlations equal the sum of the localized physical shell kernels and the terminal distribution on joint tests, with the explicitly identified contact and approximation terms. The identity includes outer means, source collisions, history resets and the chosen normalization.

Open proposition S-05. *Full-law tree estimate.* Let h_j denote shell mesh and τ the minimal tree length of the insertion points in physical units. The absolute shell kernels are bounded by

$$A^{m+1}(m!)^{p_0} h_j^{-4m} w_j \exp[-c\tau/((m+d_0)h_j)] \,.$$

There is a cutoff-independent sequence $\bar{w}_j \geq w_j \geq 0$, indexed from the reference scale towards the ultraviolet, with $\sum_{j\geq 0} h_j \bar{w}_j < \infty$. All constants apply to the complete normalized law at

every cutoff and volume. A replacement kernel estimate suffices if its shell sum is proved with the same uniform fixed-factorial growth and linear test order.

Open proposition S-06. *Contact and approximation error.* Stencil, normalization and contact errors, including those on partial diagonals, admit one vanishing cofinal error estimate or a summable error with a controlled terminal contribution, of fixed factorial power and test order linear in m. It is compatible with the positive-time algebra. Conversion between full and connected errors preserves these budgets by partition inversion.

Here is the estimate which turns the shell hypotheses into an all-order bound. Suppose the order-m kernel on a shell h, $m \geq 2$, is bounded by

$$A^{m+1}(m!)^{p_0} h^{-4m} w_j \exp[-c\,\tau(X)/((m+d_0)h)], \qquad \sum_{j\geq 0} h_j w_j < \infty,$$

where τ is the minimum spanning-tree length of the insertion positions. Taylor expand a flat test from $(x_1, \dots, x_1)$ to X through order four. Its magnitude is bounded by $C\nu_{6m}(F)(1 + |x_1|^2)^{-3}(\sum_i |x_i - x_1|_1)^5$. Every displacement is bounded by a fixed multiple of τ. Half the exponential pays the fifth moment and contributes h^5 times a polynomial in m. Bound the remaining exponential by the sum over the m^{m-2} labeled trees. Each successive leaf integration contributes at most $C(m + d_0)^4 h^4$. The identical estimate holds for lattice sums with cell measure a_n^4 and $a_n \leq h$. The integral in x_1 is bounded by the displayed integrable weight. Since $h^{-4m} h^5 h^{4(m-1)} = h$, Stirling's inequality gives a shell bound

$$C_1^{m+1}(m!)^{p_0+5}\, \nu_{6m}(F)\, h_j w_j.$$

Its ultraviolet tail tends to zero uniformly at every fixed order.

For the isolated regular conditional block one can use sixth-order flatness at a single pair of coincident points. Under the spatial conditional cumulant bound and the native field-size bound $P(h)$, the resulting joint test estimate is

$$|\mathcal{K}_{m,h,v}(F)| \leq C^{m+1}(m!)^7 h^2 P(h)^m\, \nu_{6m}(F), \qquad m \geq 2.$$

The sixth-order zero pays h^6, the root sum costs h^{-4}, and summation over relative locations costs at most $C^m(m!)^5$. The conditional moment factors supply the other two factorial powers. The normalized microscopic hierarchy also contains outer cumulants of conditional expectations and residual terms from other histories. Those terms are among the full-law hypotheses above.

Include the terminal distribution and the one-point normalization. The partition identity is

$$S_{m,n} = \sum_{\pi \in \mathcal{P}([m])} \bigotimes_{D\in\pi} C_{|D|,n}$$

with arguments restored to their original order. One can apply the block distributions successively to disjoint groups of variables of a joint test. Flatness in the other variables is preserved. A change of extension on any internal diagonal acts by zero. The allocated derivative orders add to at most qm, while the product of block Schwartz weights is bounded by the joint weight of that order. For $p \geq 1$, $\sum_\pi \prod_{D\in\pi}(|D|!)^p \leq 2^{m-1}(m!)^p$: order the blocks, group by compositions of m, and use the multinomial coefficient to count placements. Powers of the fixed constants incurred at each block cost at most C^m. This gives the full-moment bound of the theorem on joint tests.

7 Limits, reflection and rotations

For each fixed order and label tuple, take a countable dense set of flat smooth compact tests in the norm controlling its moments. Compact cutoffs preserve flatness and approximate Schwartz tests after increasing the weight order by one. The moment bounds give a convergent diagonal subsequence on the union of these sets over all orders. Linearity, estimated on the same subsequence, extends the limits to their test spaces. Use growing central cutoffs on expanding tori, so that the coincidence diagonals on every fixed compact test are the physical ones.

Take positive-time polynomials whose joint tests stay a fixed distance from the reflection plane and are flat at coincidences within each term. Their reflected products are then flat across both halves. The real local Wilson insertion and its tensor reflection give $\mu_{a,T}[(\Theta P)P] \geq 0$ whenever all the stencils fit on their respective sides. Terms concentrated on true coincidences give zero on these tests. All finite-width errors must tend to zero by the contact bound. Convergence of the finitely many moments in this quadratic form passes its nonnegativity to the limit.

Open proposition P-01. *Positive-time splitting.* The actual field normalization and local product subtractions are real, local and compatible with time reflection including tensor Wick phases. On products with the two time supports separated, they split into reflected and positive-time substitutions, preserving the Wilson Gram form.

Open proposition P-02. *Translations and permutations.* The full moments are symmetric under simultaneous permutations of insertions and labels. Fine-lattice translations preserve them exactly or have a vanishing error in a common joint-test seminorm, including any dependence introduced by the block grids.

Translation by a fixed vector is approximated by lattice translations at distance $O(a_n)$. The mean-value theorem and the common Schwartz bound make the resulting test error tend to zero. Permutations pass directly on the same hierarchy. To prove covariance under general rotations one needs a bound on the full defect. The signed-permutation representation supplies the finite lattice subgroup. For $R \in SO(4)$ let $R \cdot F$ denote simultaneous rotation of arguments with tensor action $D(R)^{\otimes m}$. Integrating the infinitesimal defect along a smooth path from 1 to R, under the next uniform estimates, gives $S_m(R \cdot F) = S_m(F)$. The reflected component transformation and the action of a fixed coordinate reflection extend this relation to $O(4)$.

Open proposition R-02. *Angular variations of the action.* For each infinitesimal rotation, the angular defect of the normalized action is a sum of local insertions and a controlled remainder. Their contribution in the complete law tends to zero on each fixed physical test, including the contributions from spatial tails and inherited histories.

Open proposition R-03. *Angular variations of the fields.* For the same infinitesimal generators the combined defects of the matched tensor fields, running matrices and contact terms tend to zero on joint tests, for all 21 labels and all source degrees in the physical normalization.

Open proposition R-04. *Uniformity along rotation paths.* For each fixed rotation and joint test the combined action and observable defects are bounded along its generating paths by an integrable error tending to zero with the cutoff. The bound is uniform over the intermediate rotations required for each such path.

Open proposition R-05. *Orthogonal covariance.* Integration of the angular estimates for the limiting hierarchy, followed by the exact lattice reflection with its tensor action, yields full

$O(4)$ covariance of those same limiting distributions.

8 The separated ultraviolet coefficients

Let $d_G = N^2 - 1$, $b_0 = 11/(48\pi^2)$, and use the canonical scalar

$$Q = \tfrac{1}{2} \sum_{\mu<\nu,a} (F^a_{\mu\nu})^2, \qquad g^2 = N g^2_{\mathrm{YM}}.$$

For the free curvature, each color has the same two-form covariance

$$K(x) = \frac{\Lambda^2 R_x}{\pi^2 |x|^4}, \qquad R_x = I - 2\widehat{x}\widehat{x}^t.$$

Wick pairing at distinct points gives $\langle Q(x)Q(0)\rangle = 3d_G/(\pi^4|x|^8)$. At three points the connected Wick term is a trace of a product of three such two-form matrices. The Hodge star anticommutes with $\Lambda^2 R_x$, since R_x reverses orientation. Conjugating this product by the Hodge star changes its sign, so its trace vanishes. The coefficient to be identified in the interacting scalar three-point function is the term of order g^2.

In the stated convention the separated perturbative coefficient for $Q = F^2/4$ is $3d_G b_0/\pi^6$, using the scalar two-point normalization and the F^2 Wilson coefficient in the separated three-point function [3, 4]. Its identification in the Wilson limit is an independent estimate. Put

$$c_2 = \frac{3d_G}{\pi^4}, \qquad c_3 = \frac{3d_G b_0}{\pi^6} = \frac{11 d_G}{16\pi^8},$$
$$K_2(u,v) = |u-v|^{-8}, \quad K_3(u,v,w) = (|u-v|\,|v-w|\,|w-u|)^{-4}.$$

For separated unit-scale tests let $I_m = \int K_m \prod_i f_i$ and $f_{i,\rho}(x) = \rho^{-4} f_i((x - x_*)/\rho)$. The physically normalized scalar at scale μ has matching factor z_ρ relative to Q at scale $1/\rho$, with coupling g_ρ on the same trajectory. The ultraviolet assertions below must identify this factor from the finite normalization and yield

$$\rho^8 z_\rho^{-2} S^c_2(f_{1,\rho}, f_{2,\rho}) = c_2 I_2 + o(1),$$
$$\rho^{12} z_\rho^{-3} g_\rho^{-2} S^c_3(f_{1,\rho}, f_{2,\rho}, f_{3,\rho}) = c_3 I_3 + o(1).$$

At each fixed $\rho > 0$ take $a_n/\rho \to 0$ and $\rho/T_n \to 0$ before sending $\rho \downarrow 0$. Cutoff and volume errors must tend to zero after the displayed rescalings. For a scalar readout with leading factor $g_\rho^2 Q$ the three-point function has coupling power eight, including the g_ρ^2 in the canonical three-point coefficient.

An explicit witness uses a smooth unit-mass bump η. Set $\eta_\epsilon(u) = \epsilon^{-4}\eta(u/\epsilon)$ and $f_i(u) = -\partial_0 \eta_\epsilon(u - t_i e_0)$ for $(t_1, t_2, t_3) = (0, 1, 3)$. For sufficiently small ϵ the supports are disjoint. Three integrations by parts show

$$\lim_{\epsilon \downarrow 0} I_3 = \partial_{t_1}\partial_{t_2}\partial_{t_3} \prod_{i<j} (t_i - t_j)^{-4} \Big|_{(0,1,3)} = \frac{25}{729}.$$

Thus the specified profile has nonzero coefficient. Zero integrals of its individual tests remove translation-invariant one-point terms.

Open proposition U-01. *Two-point coefficient.* For the prescribed separated profiles and scalar or tensor projections, the first ultraviolet two-point distribution of the exactly matched fields has the prescribed coefficient, including logarithmic running and the fixed physical normalization.

Open proposition U-02. *Connected three-point coefficient.* There exists an explicit admissible separated profile and field contraction for which the prescribed leading connected three-point coefficient, evaluated in the same normalization as U-01, has an explicitly identified nonzero value.

Open proposition U-03. *Interacting ultraviolet remainder.* The two- and three-point asymptotics have remainders uniform under cutoff removal and increasing volume before the short-distance limit. Exact matching, outer means and all history terms are included. In this order of limits the coefficient in U-02 retains its nonzero value.

Open proposition U-04. *Reflected block variances.* For each of the two positive-time ultraviolet block observables of degrees two and three, the vacuum norm of the block minus its prescribed scalar coefficient has the summable logarithmic-scale error required for phase transfer. The reflected products have degrees four and six and obey the same test-space control.

Open proposition U-05. *Scale and profile regularity.* The logarithmic scale derivative of each centered ultraviolet block obeys the integrable norm estimate required for continuous-scale phase transfer. The block estimates extend uniformly to the stipulated bounded profile sets with the required modulus of continuity.

9 Reconstruction, energy and a vacuum phase

The corrected Osterwalder–Schrader reconstruction theorem uses regularity on coincidence-flat tests, Euclidean covariance, symmetry, reflection positivity and the linear-order growth condition [7]. The fixed factorial power and linear derivative budget give the required growth. Tensor reflection and the indexed-field extension are taken in the conventions of [6, §6]. Applying reconstruction without a clustering assumption initially allows a translation-invariant vacuum subspace of dimension greater than one. All energy and phase assertions concern the fields of this same reconstruction.

For a closed smeared field $\phi(f)$, suppose $e^{-tH}\mathcal{H} \subset D(\phi(f))$ and $\|\phi(f)e^{-tH}\| \le C\nu_s(f)t^{-r}$ for $0 < t \le 1$, where $H \ge 0$ is the reconstructed Hamiltonian. The large-time bound follows by contractivity. For $R > r$, the identity

$$(1+H)^{-R} = \frac{1}{\Gamma(R)} \int_0^\infty t^{R-1} e^{-t} e^{-tH}\, dt$$

converges also after applying $\phi(f)$, by closedness and absolute integrability in the graph norm. It gives $\|\phi(f)(1+H)^{-R}\| \le C_R\nu_s(f)$, an arbitrary-vector bound. At finite periodic cutoff a sufficient insertion estimate has the following form. Let P be a fixed positive-time polynomial, τ_t a positive time shift, $p_n(P)^2 = S_n((\Theta P)P)$, g a spatial Schwartz test, and χ_ϵ a nonnegative even time mollifier with $\epsilon < t/8$. For each physically normalized seed Q_n, require

$$|S_n((\Theta\tau_t P)Q_n(\chi_\epsilon \otimes g)(\tau_t P))| \le Ct^{-\kappa}\nu_s(g)\, p_n(P)^2 + o_n(1), \qquad 0 < t \le 1.$$

The error tends to zero at fixed tests, t and ϵ. The constant on the right is independent of the degree of P and the mollifier width. Removing the cutoff, identifying the time-zero field and

extending from the common core are part of E-05–E-06. Covariance and the finite Lorentz orbit of the scalar, pseudoscalar and off-diagonal magnetic seeds propagate these estimates to the full tensor family.

Open proposition E-02. *Scalar energy bound.* The physically normalized scalar insertion, including its identity term, satisfies a cutoff-uniform arbitrary-vector heat or graph estimate on the common field domain, through its transfer response or its periodic reflection-positive insertion.

Open proposition E-03. *Pseudoscalar energy bound.* The physically normalized pseudoscalar, with its electric and magnetic components and time-reflection convention, satisfies the corresponding operator estimate on the same domain with a uniform energy weight and Schwartz seminorm.

Open proposition E-04. *Magnetic tensor energy bound.* The renormalized off-diagonal magnetic seed B_1B_2, with its physical scaling, mixing factors and temporal admixtures, obeys the required cutoff-uniform arbitrary-vector estimate.

There is a finite-volume reduction of one sufficient state criterion. Let Q be the positive physical Wilson transfer operator on a spatial torus with Lm sites per direction, $\lambda_0 > 0$ its top eigenvalue, and $H = -a^{-1}\log(Q/\lambda_0)$. For $f(t) = \log \operatorname{Tr} e^{-tH}$ and $T = Lma$, set

$$\Gamma = f(T/2) - f(T), \qquad \Phi = L^4 \log Z_{\mathrm{cube}}(m) - \log Z_{\mathrm{cube}}(Lm).$$

Whenever the indicated lattice times are integral,

$$0 \le \Gamma \le f(T/L) - f(T) \le \Phi/L, \qquad \Phi \ge 0.$$

Indeed $H \ge 0$ makes f decreasing. The top-eigenvalue bound $\lambda_0^{Lm} \le \operatorname{Tr} Q^{Lm}$ gives $f(T/L) - f(T) \le \log \operatorname{Tr} Q^m - L^{-1} \log \operatorname{Tr} Q^{Lm}$. The physical trace is the Wilson integral on the corresponding rectangular torus. Successively in each of the other three directions, $\operatorname{Tr} A^{Lr} \le (\operatorname{Tr} A^r)^L$ for $A \ge 0$ bounds its partition function by $Z_{\mathrm{cube}}(m)^{L^3}$. The same trace inequality in all four directions gives $\Phi \ge 0$. Any one-link scalar normalization of Q cancels in these differences. For application one still needs the quantitative bound on this same matched family, with the required insertion norms and operator domains.

Open proposition E-05. *Periodic state comparison.* Along the matched cofinal cubic sequence, local periodic state comparison, the thermal trace-increment criterion, or a direct periodic reflection-positive insertion estimate holds for the physically normalized hierarchy at every fixed polynomial degree, with the rates required to pass the seed bounds.

Open proposition E-06. *Common field domain.* The limits of the three seed estimates act on the reconstructed fields on a common graph core. Their Lorentz transforms span the 21 components and yield polynomial energy bounds for the actual fields with the stated smearing norms on the common domain.

Apply field decomposition on the separable observable algebra when its spectral, localization, domain and measurability hypotheses hold [5]. For a positive-time block $A_{m,\rho}(F)$ of degree $m = 2, 3$, with the appropriate centering and ultraviolet rescaling, the vacuum norm

$$\big\|(A_{m,\rho}(F) - c_m(F))\Omega\big\|^2$$

is a reflected correlation of order at most $2m$. In a direct-integral decomposition with normalized vacuum vectors Ω_λ, Cauchy–Schwarz gives

$$\int |\langle \Omega_\lambda, A_{m,\rho}(F)\Omega_\lambda\rangle - c_m(F)|^2 \, d\pi(\lambda) \le \big\|(A_{m,\rho}(F) - c_m(F))\Omega\big\|^2.$$

If the right-hand side is summable on a sequence of logarithmic scales, Tonelli's theorem gives convergence for almost every phase on that sequence. For continuous scales, let $t = -\log \rho$ and let $v_t(F)$ be the centered block vector. On each compact off-diagonal profile set the required estimates take the form

$$\|v_t(F)\| \le e_0(t)\|F\|_{C^r}, \qquad \|\partial_t v_t(F)\| \le e_1(t)\|F\|_{C^r}, \qquad \int_{t_0}^{\infty} (e_0^2 + e_1^2)\, dt < \infty.$$

Expanding compact profiles in a sufficiently high Sobolev basis gives a square-summable bound on the corresponding dual Sobolev norm and its time derivative. In almost every fiber this yields an H^1 function of logarithmic scale with values in that dual space. Its norm tends to zero at infinity by the one-dimensional Sobolev inequality on tails. One first intersects the full-measure sets for a countable dense set of profiles and the finite field family. This argument transfers a coefficient to the same phases as the fields and energy bounds.

For a required disconnected symmetry γ and bounded algebra element A, the translation-invariant projection P_0 satisfies, in a translation-ergodic decomposition of the state,

$$\|P_0(\alpha_\gamma(A) - A)\Omega\|^2 = \int |\omega_\lambda(\alpha_\gamma(A)) - \omega_\lambda(A)|^2 \, d\pi(\lambda)$$

for the bounded symmetry action in the stated convention. Anti-linear actions require the corresponding conjugation of the scalar expectation. Vanishing of the central defects on a countable determining algebra gives the required symmetry almost everywhere. Intersect this set with the full-measure sets for the field domain, energy bound and ultraviolet limits to select a single component.

Open proposition V-02. *Decomposition into phases.* The reconstructed fields and symmetry representation satisfy the field phase-decomposition hypotheses, including the requisite generalized energy bounds or intrinsic locality. They have a measurable field decomposition with common phasewise energy domains, proper-orthochronous covariance, a unique translation-invariant vacuum in each selected fiber, and clustering.

Open proposition V-03. *Phasewise ultraviolet limits.* The reflected block estimates and continuous-scale control transfer both prescribed ultraviolet coefficients to one common full-measure set of phases, simultaneously for the required profiles in the fixed normalization.

Open proposition V-04. *Disconnected symmetries in a phase.* The central defects of each required disconnected symmetry vanish on a common full-measure set, or a phase-selection theorem supplies the corresponding implementation for the selected phase and the given fields.

Open proposition V-05. *Selection of one phase.* One phase satisfies simultaneously the field energy bounds on their common domain, proper-orthochronous covariance, the stipulated disconnected symmetries, uniqueness of the translation-invariant vacuum, clustering and both normalized ultraviolet coefficients.

10 Identification of the continuum theory

The final identifications concern a single subsequence and its selected phase. In particular, a weak limit of moments of the original microscopic insertions and a separately reconstructed hierarchy can be identified only when the coupling, seven normalization measurements, tensor signs and local product prescription agree.

Open proposition A-01. *Common continuum subsequence.* All orders and field tuples converge on one cofinal subsequence on the coincidence-flat Schwartz spaces. The joint-test bounds extend convergence from countable dense families and preserve the finite normalization, reflection positivity and the symmetries established at this stage.

Open proposition A-02. *Reconstruction.* The full mixed moments satisfy the corrected Osterwalder–Schrader reconstruction hypotheses, including linear growth and the required tensor covariance. The reconstructed relativistic fields have the identified common domains. Initial reconstruction uses the mixed hierarchy before phase selection.

Open proposition A-03. *Identification of the constructed theory.* The Wilson subsequence, limiting moments, reconstructed fields and selected phase appearing in the preceding assertions obey the conclusions of the main theorem with the same cutoff sequence, field normalization, energy domains and ultraviolet coefficients. Every identification between these objects holds on the specified test spaces and domains.

Under the stated analytic hypotheses the shell bounds give the common full hierarchy. The symmetry and positivity estimates identify its Euclidean transformations and reflection form. The reconstruction and domain identifications transfer the three seed estimates to local fields. The phase selection above retains their ultraviolet coefficients and symmetries on one component. This proves the stated implication, subject to the open estimates and the common-sequence identifications. □

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Annex A — Compact Wilson energy and arbitrary-event estimates

Annex A. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Scope

This note proves two improvements on the actual compact Wilson probability law. It does not prove the inherited low-margin estimate in the complete Balaban history measure, the matching of transported marks to current-scale local fields, or four-dimensional Yang–Mills existence.

1. An exact maximal-tree gauge fixing and an open-box plaquette-frame comparison remove the polynomial factor in the bare inverse coupling from the local energy reserve along the proposed cofinal lattice sequence. The finite-volume correction is explicit.
2. Once an event has probability q, polynomial moment bounds give a marked estimate with q itself, multiplied by a power of $m + \log(1/q)$. This applies to inherited events as well as fresh ones. It avoids the stronger, generally false requirement that every original-input-measurable weight inherit the same probability factor.

The first result is an unconditional, interacting, large-fields-retained estimate. The second is a general probability estimate applied to the first result and the exact positive transports. Neither identifies a coarse effective energy with the transported raw Wilson energy.

The source-free inputs are Wilson reflection positivity, the chessboard and conditional-cumulant estimates, and the vanishing raw bad-bridge sector. The fresh-reference argument uses a predictable supermartingale and treats the inherited low-margin remainder separately.

1. Conventions and geometry

Let $N \geq 2$ and

$$G = SU(N), \qquad D = N^2 - 1, \qquad d \geq 2, \qquad \Lambda_n = (\mathbb{Z}/n\mathbb{Z})^d, \qquad V = n^d,$$

with even $n \geq 4$. Links and elementary unoriented plaquettes are counted once in positive coordinate orientations. Haar measures are normalized. Define

$$s(U) = N - \operatorname{Re}\operatorname{Tr} U = \tfrac{1}{2}\|U - I\|_{\mathrm{HS}}^2, \quad S = \sum_p s(U_p), \quad Z_n^P(\beta) = \int e^{-\beta S}\, dH, \quad e_p = \beta s(U_p). \tag{1.1}$$

Here P means periodic boundary conditions. The coupling convention is:

$$\beta = 2Nx_0, \qquad X_B = N\beta = 2N^2 x_0. \tag{1.2}$$

The results hold for every $\beta \geq 0$; the asymptotically free trajectory is needed only when applying the cofinal corollary to the proposed Yang–Mills sequence.

We use the open box with the same n^d vertices and only links whose two endpoints are in that box. Write Z_n^F for its free-boundary partition function. The counts of independent links after spanning-tree gauge fixing are

$$q_P = dV - (V - 1) = (d - 1)V + 1,$$

$$q_F = d(n - 1)n^{d-1} - (V - 1) = (d - 1)V - dn^{d-1} + 1. \tag{1.3}$$

Thus

$$q_P - q_F = dn^{d-1}. \tag{1.4}$$

The difference is a surface term. Keeping this difference, rather than replacing both counts by unspecified multiples of V, is essential.

2. Exact gauge-fixed partition-function comparison

Theorem 2.1 — matched ultraviolet powers There are constants $0 < c_\ell \le 1 \le C_u$, depending only on N, d, such that

$$\boxed{c_\ell^{q_P}(1+\beta)^{-Dq_P/2} \;\le\; Z_n^P(\beta) \;\le\; Z_n^F(\beta) \;\le\; z(\beta)^{q_F} \;\le\; C_u^{q_F}(1+\beta)^{-Dq_F/2},} \tag{2.1}$$

where

$$z(\beta) = \int_{SU(N)} e^{-\beta s(U)}\, dU. \tag{2.2}$$

All constants are independent of the number of sites and of β. In particular, no factor of the form β^{cV} is hidden in a constant.

Proof: exact tree gauge fixing For a connected finite graph choose a root and a spanning tree. Successively change variables by vertex gauge transformations, fixing every tree link to I, while retaining the $V-1$ tree transport variables as independent group variables. Multiplication and inversion preserve normalized Haar measure. A gauge-invariant integrand does not depend on those tree transport variables, which integrate to one. The resulting integral is exactly the integral over the $E - V + 1$ non-tree links with tree links set to I. No Faddeev–Popov determinant or residual gauge-volume factor occurs in this finite compact change of variables. The root gauge is fixed throughout.

For the periodic graph there are q_P free links. Choose constants $r_0 > 0$ and $c_G > 0$ such that

$$\mathrm{Haar}\{U : \|U - I\|_{\mathrm{HS}} \le r\} \ge c_G r^D, \qquad 0 < r \le r_0. \tag{2.3}$$

This follows from nonsingular Haar density in a coordinate chart at the identity. Restrict each free link to radius $r = r_0(1+\beta)^{-1/2}$. A product of four unitaries telescopes, and inverse links have the same distance to the identity. Consequently

$$s(U_p) \le 2 \sum_{l \in \partial p,\ l \text{ free}} \|U_l - I\|_{\mathrm{HS}}^2. \tag{2.4}$$

Each free link belongs to $2(d-1)$ elementary plaquettes, so

$$S \le 4(d-1) \sum_{l \text{ free}} \|U_l - I\|_{\mathrm{HS}}^2. \tag{2.5}$$

On the restricted event,

$$\beta S \le 4(d-1) r_0^2 q_P.$$

Its Haar measure is at least

$$(c_G r_0^D)^{q_P}(1+\beta)^{-Dq_P/2}.$$

After decreasing a fixed positive constant to at most one this proves the first inequality in (2.1).

Proof: deleting wrap plaquettes Delete all plaquettes which cross a periodic seam. Their action terms are nonnegative. The remaining action depends only on the open-box links, and all wrap links integrate to one. Therefore

$$Z_n^P(\beta) \le Z_n^F(\beta). \tag{2.6}$$

This comparison is between the unnormalized partition functions. It makes no claim about monotonicity of normalized plaquette observables.

Proof: an exactly integrable open-box plaquette frame The required subset of plaquettes is constructed by induction on dimension. For $d = 1$ the graph is a tree and the gauge-fixed integral is one, so $q_F = 0$.

In dimension d, label one coordinate by $t = 0, \dots, n-1$. Fix to the identity all links in the t direction. These links form a forest of open chains. Add a spanning tree in the $t = 0$ slice; together these links form a spanning tree of the full box.

Keep all plaquettes in the (t, i) planes. Also keep the recursively constructed $(d-1)$-dimensional plaquette frame in the $t = 0$ slice. Delete every other plaquette action. In this gauge the kept temporal plaquette through a spatial link e has action

$$s\big(U_e(t+1)U_e(t)^{-1}\big). \tag{2.7}$$

For each spatial edge, integrate $U_e(n-1)$, then $U_e(n-2)$, and so on down to $U_e(1)$. At each integration the upper member of that chain occurs only in one remaining plaquette factor; Haar invariance gives exactly $z(\beta)$, independently of the next link below it. The final integral in the $t = 0$ slice is the lower-dimensional frame integral.

There are

$$E_{d-1} = (d-1)(n-1)n^{d-2}$$

spatial edges per slice. Thus the number of factors satisfies

$$q_d = (n-1)E_{d-1} + q_{d-1}, \qquad q_1 = 0. \tag{2.8}$$

An elementary induction gives

$$q_d = (d-1)n^d - dn^{d-1} + 1 = q_F. \tag{2.9}$$

Deleting nonnegative action terms increases the integral. The retained frame integral equals $z(\beta)^{q_F}$ exactly. This proves the third inequality in (2.1).

Proof: the one-plaquette Laplace integral The function $s(U)$ is nonnegative and vanishes only at I. In a sufficiently small exponential coordinate chart at I, its Hessian is strictly positive and Haar density is bounded above. Hence that part of the integral is at most a constant times $(1+\beta)^{-D/2}$. On the compact complement, s has a strictly positive minimum δ; the remaining integral is at most $\exp(-\delta\beta)$, which obeys the same polynomial upper bound after increasing the constant. This proves the last inequality in (2.1). All constants depend only on the compact group and on d. ■

Corollary 2.2 — local pressure ratio with an explicit surface correction For every β' in $[\beta/2, \beta]$,

$$\boxed{\left(\frac{Z_n^P(\beta')}{Z_n^P(\beta)}\right)^{1/V} \le C_{N,d}(1+\beta)^{Dd/(2n)}.} \tag{2.10}$$

Indeed, (2.1) gives

$$\frac{Z_n^P(\beta')}{Z_n^P(\beta)} \le C_u^{q_F} c_\ell^{-q_P} 2^{Dq_F/2} (1+\beta)^{D(q_P-q_F)/2}. \tag{2.11}$$

Now use $q_P/V \le d$, $q_F/V \le d-1$, and (1.4). For example, one may take

$$C_{N,d} = c_\ell^{-d} C_u^{d-1} 2^{D(d-1)/2}. \tag{2.12}$$

The surface correction has not been suppressed in (2.10).

3. Constant local reserve on the cofinal Wilson sequence

For a closed unit d-cube C, let E_C be the sum of e_p over all its plaquette faces. Every plaquette belongs to

$$M_d = 2^{d-2}$$

such cubes. For a plaquette set X, let $b(X)$ be the minimum number of closed unit cubes covering it, and let $E_X = sum_{p\in X} e_p$.

The Wilson site-reflection positivity and closed-block chessboard inequality (Section 8) give, for a set A of distinct cubes,

$$\mathbb{E}_{\beta,n} \exp\left(\eta \sum_{C\in A} E_C\right) \le \left[\frac{Z_n^P((1-\eta M_d)\beta)}{Z_n^P(\beta)}\right]^{|A|/V}. \tag{3.1}$$

The abstract chessboard input uses even block counts; it does not require a power of two. Shared boundary links are allowed. This is the standard Wilson site-reflection geometry, not a new reflection-positivity assertion about an RG history measure.

Theorem 3.1 — cofinal constant reserve For $0 \le \eta \le 1/(2M_d)$ and every plaquette set X,

$$\boxed{\mathbb{E}_{\beta,n} e^{\eta E_X} \le \left[C_{N,d}(1+\beta)^{Dd/(2n)}\right]^{b(X)}.} \tag{3.2}$$

If a sequence (β_ν, n_ν) obeys

$$B_* := \sup_\nu \frac{\log(1+\beta_\nu)}{n_\nu} < \infty, \tag{3.3}$$

then, with $K = C_{N,d} \exp(DdB_*/2)$,

$$\boxed{\mathbb{E}_{\beta_\nu,n_\nu} e^{\eta E_X} \le K^{b(X)}} \tag{3.4}$$

uniformly in the cutoff, volume, plaquette locations and shape of X.

Proof. In (3.1), $(1-\eta M_d)\beta \ge \beta/2$. Apply Corollary 2.2. For a cover of X, positivity gives $E_X \le \sum_{C\in A} E_C$; minimize the cover. Equation (3.3) gives (3.4). ■

For $d = 4$, the allowed exponent is any $\eta \le 1/8$. The particular cofinal geometry

$$a_\nu = L^{-K_\nu}\ell_0, \qquad T_\nu = 2L^{M_\nu}\ell_0, \qquad n_\nu = 2L^{K_\nu+M_\nu}$$

satisfies (3.3) if the independently verified AF trajectory gives

$$\beta_\nu \le C(1+K_\nu). \tag{3.5}$$

Equation (3.5) is an application hypothesis here, not a newly proved coupling-flow theorem. The reserve theorem itself needs only (3.3).

Corollary 3.2 — uniform moments of the raw finite field family For a stencil with $b(X) \le \upsilon_0$ and $r \ge 1$, exponential-tail integration gives

$$\|E_X\|_r \le \eta^{-1}\big(\upsilon_0 \log K + 2r\big) \le C_{\upsilon_0} r. \tag{3.6}$$

For integer r one may replace $2r$ by r. The factor two avoids any need for a sharp real-order Gamma-function estimate.

Let C_A be any member of a fixed finite plaquette-curvature stencil, formed from Hermitian sine parts of plaquette holonomies, finite parallel transports and fixed linear combinations. Since

$$\left\| \frac{U - U^*}{2i} \right\|_{\mathrm{HS}}^2 \le \|U - I\|_{\mathrm{HS}}^2 = 2s(U),$$

unitary parallel transports preserve the norm, and trace Cauchy–Schwarz bounds a bilinear by a constant times the sum of the corresponding plaquette actions. Therefore every fixed component of an energy-normalized bilinear satisfies

$$|\beta\ \mathrm{Tr}(C_A C_B)| \le C E_X, \qquad \|\beta\ \mathrm{Tr}(C_A C_B)\|_r \le C r. \tag{3.7}$$

In four dimensions the bare physical local field

$$O_{\alpha,a}(x) = a^{-4} \beta\ \mathrm{Tr}(C_A C_B)$$

thus obeys

$$\|O_{\alpha,a}(x)\|_r \le C a^{-4} r. \tag{3.8}$$

The same estimate holds after subtracting its expectation, at the cost of a factor two. Additional running mixing coefficients and product subtractions are not silently included in C: they need their own bounds. Most importantly, (3.8) still has the physical cutoff power a^{-4}.

Corollary 3.3 — exact all-history average For every composite frozen positive transport Q from the Wilson input to an output space containing its actual positive histories,

$$\int Q e^{\eta E_X}\, d\mu_{\mathrm{out}} = \mathbb{E}_{\beta,n} e^{\eta E_X} \le K^{b(X)}. \tag{3.9}$$

Conditional Jensen gives the corresponding L^r contraction for transported original marks. This uses the actual positive history lift before signed polymer expansion. It does not require a pointwise reserve on each conditioned history, and it does not grant reflection positivity to the history variables.

4. Rare inherited histories: a probability factor with a logarithmic cost

Lemma 4.1 — optimal-tail form for an arbitrary event Let $Y \ge 0$ satisfy

$$\mathbb{E} e^{\eta Y} \le K_0, \qquad K_0 \ge 1,$$

and let B be any event of probability q. For $q > 0$ and $r \ge 1$,

$$\boxed{\mathbb{E}[1_B Y^r] \le q\, \eta^{-r} \big[\log(K_0/q) + 2r\big]^r.} \tag{4.1}$$

For integer r, $2r$ may again be replaced by r. When $q = 0$, the left side is zero.

Proof. Markov's inequality and the event probability give

$$\mathbb{P}(B \cap \{Y > t\}) \le \min(q, K_0 e^{-\eta t}). \tag{4.2}$$

Conditional on B, this tail is bounded by that of

$$T + Z, \qquad T = \eta^{-1} \log(K_0/q),$$

where Z is exponential with rate η. Tail integration and Minkowski give

$$\left(\mathbb{E}[Y^r \mid B]\right)^{1/r} \le T + \eta^{-1}\Gamma(r+1)^{1/r}. \tag{4.3}$$

For real $r \ge 1$, comparison with the integer order $\lceil r \rceil$ bounds the last Gamma root by $2r$. For integer r, $(r!)^{1/r} \le r$. Multiply by q. ■

There is no independence, fresh-noise assumption, or conditioning-margin hypothesis in this lemma. In the Wilson application take $Y = E_X$ and $K_0 = K^{(}b(X))$. The event may depend on the entire original input, on later auxiliary randomness, or on both, because passing to the joint history coupling leaves the marginal exponential bound on Y unchanged.

If B is output-measurable and Q is the conditional expectation of an original mark given that output, conditional Jensen also gives

$$\mathbb{E}[1_B |QF|^r] \le \mathbb{E}[1_B |F|^r]. \tag{4.4}$$

This can be combined directly with (4.1) when F is bounded by a power of the original energy.

Theorem 4.2 — arbitrary-event bounds from a polynomial moment budget Suppose marks $A_1, \dots, A_m$ satisfy, for all real $r \ge 1$,

$$\|A_i\|_r \le b_i C r^s \tag{4.5}$$

with fixed $s \ge 0$. For every event B of probability $q > 0$, put $\lambda = \log(1/q)$. Then

$$\boxed{\mathbb{E}\left[1_B \prod_{i=1}^{m} |A_i|\right] \le q(eC)^m (m+\lambda)^{sm} \prod_i b_i.} \tag{4.6}$$

Let Q be a positive normalized conditional-expectation transport and let

$$J_m = \kappa_Q(A_1, \dots, A_m)$$

be the corresponding conditional cumulant, equivalently the mixed derivative of $\log Q \exp(\sum t_i A_i)$ at zero. Then

$$\boxed{\mathbb{E}[1_B |J_m|] \le q(2eC)^m m! \, (m+\lambda)^{sm} \prod_i b_i.} \tag{4.7}$$

The same bound as (4.6) holds for the full transported coefficient $Q(prod_i A_i)$. For that conclusion B is an event on the output probability space, or any joint probability space extending it. Formula (4.7) has the same interpretation. All these conclusions are zero when $q = 0$.

Proof. If $q = 1$, use Hölder with exponent m for each mark. Otherwise set

$$p = 1 + \lambda/m > 1.$$

Hölder first on the event and then among the m marks gives

$$\mathbb{E}\left[1_B \prod_i |A_i|\right] \le q^{1-1/p} \prod_i \|A_i\|_{mp}. \tag{4.8}$$

Here $mp = m + \lambda$ and

$$q^{1-1/p} = qe^{\lambda/p} \le qe^m.$$

This proves (4.6). Conditional contraction proves the same product-moment bound for $Q(prod_i A_i)$.

For conditional cumulants the previously proved partition estimate is

$$\|J_m\|_p \le m! 2^{m-1} \prod_i \|A_i\|_{pm}. \tag{4.9}$$

For completeness, each partition term is a product of conditional moments. Hölder across its blocks, conditional contraction and Hölder inside each block bound its L^p norm by the displayed product of L^{pm} norms. The sum of absolute cumulant coefficients is at most $m!2^{m-1}$, by encoding ordered partitions by a permutation and cuts. Apply the same event Hölder inequality with the chosen p. ■

For the fields in (3.8), $s = 1$ and $b_i = a^{-4}$. Thus an arbitrary inherited event of probability q has the explicit original-mark estimate

$$\mathbb{E}[1_B |J_m|] \le qC^m m! \, (m + \log(1/q))^m a^{-4m}. \tag{4.10}$$

The probability factor has no fixed Hölder loss. The cost is an explicit logarithmic factor. This is stronger than choosing one fixed $p > 1$ in the earlier rare-history estimate; it is not a spatial-decay theorem or a renormalization theorem.

5. Why the fresh weighted factor cannot hold for arbitrary inherited events

Proposition 5.1 — exact obstruction Let $\mathcal{F}_0$ be the sigma-field containing the complete original input. For an event B, the following are equivalent:

$$\mathbb{E}[W 1_B] \le q\mathbb{E}W \quad \text{for every nonnegative integrable } F_0\text{-measurable } W; \tag{5.1}$$

$$\mathbb{P}(B \mid F_0) \le q \quad \text{almost surely.} \tag{5.2}$$

In particular, if B is $\mathcal{F}_0$-measurable and has positive probability, then (5.1) requires $q \ge 1$.

Proof. The conditional-expectation identity proves (5.2) ⇒ (5.1). Conversely take W to be the indicator of each $\mathcal{F}_0$-measurable set on which the conditional probability exceeds q; a positive-measure such set contradicts (5.1). If B itself is $\mathcal{F}_0$-measurable, take $W = 1_B$. ■

Consequently the fresh-reference theorem should retain its exact predictable scope. It proves a factor for weights known before the specified genuinely fresh draws because the conditional probability is bounded there. For original-input-determined badness, the same factor for all original weights is generally impossible. An inherited event involving intermediate randomness need not be $\mathcal{F}_0$-measurable; nevertheless (5.2), rather than its unconditional probability, remains the exact requirement for a universal original-weight bound.

The appropriate replacement is the actual averaged estimate (4.6) or (4.7), with a specified polynomial moment class. The original energy reserve supplies such a class. A current-scale matching theorem would have to supply it for the current local renormalized marks.

6. What an inherited low-margin theorem still has to prove

The decomposition has the form

$$\mathbb{E}[W1_{\{\text{many failures}\}}] \le \mathbb{E}[W1_{\{\text{many low margins}\}}] + w\mathbb{E}W. \tag{6.1}$$

For the first term there are now two distinct required inputs, rather than a demand for a universal conditional factor:

1. Actual unweighted geometric probability. Bound the probability of the specified inherited low-margin event, with its preparatory failures and unreplaced components, in the normalized joint measure of the complete positive Wilson histories. For a multi-component estimate, the bound must retain the dependence on the selected component set or connecting geometry.
2. Current-scale moment budget. Match the transported source jets, including responses and remainders, to renormalized current-scale fields or vertices with a bound of the form $\|A_{\alpha,k}\|_r \le Cb_{\alpha,k}r^s$. The desired physical prefactor is governed by the current scale h_k, not the original cutoff a, after the required local subtractions.

Given those two inputs, Theorem 4.2 supplies their marked combination with no pointwise all-history estimate. For example, if an event indexed by D has probability q_D, the product estimate is

$$\mathbb{E}\left[1_{E_D}\prod_{i=1}^{m}|A_{\alpha_i,k_i}|\right] \le q_D C^m[m+\log(1/q_D)]^{sm}\prod_i b_{\alpha_i,k_i}. \tag{6.2}$$

For conditional-cumulant jets add the explicit $m!$ factor from (4.7). The estimate is directly an averaged bound under the actual normalized law. It does not require the false inequality (5.1) for every possible weight.

Neither required input has been established for the complete low-margin sector here. The constant Wilson reserve makes the original moment budget cleaner, but it does not replace current-scale matching.

Rate conditions: why isolated rarity is still insufficient Write $t = \log(a^{-1})$. On the indicated weak-coupling trajectory, a representative available isolated-failure probability is

$$q_a = \exp\{-c(\log t)^\gamma\} \tag{6.3}$$

for fixed $\gamma > 0$. For every fixed positive m, s,

$$a^{-4m}q_a[m+\log(1/q_a)]^{sm} \longrightarrow \infty. \tag{6.4}$$

Indeed the logarithm is $4mt - c(\log t)^\gamma + O(\log\log t)$, and the linear term in t dominates. Thus even the improved q factor does not make an estimate with raw physical fields summable. This is failure of this proposed bound to close the argument, not a proof that the actual correlations diverge.

The previously proved raw bad-bridge sector has the additional probability $\exp(-cr/a)$ when a bad cluster connects separated supports. That factor can absorb the raw powers and gives coincidence-flat vanishing. A general inherited event is not automatically such a raw connecting cluster: many fine plaquettes below a raw threshold can collectively influence a coarse field.

7. Conditions of possible closure mechanisms

Mechanism	What is established	Additional statement required
Raw Wilson pressure and chessboard	The constant reserve (3.4), with all original large fields retained	Identification of a later effective energy or margin with a quantity controlled by this reserve

Mechanism	What is established	Additional statement required
Fresh reference randomness	the conditional reserve and predictable rare-failure factor	Probability of low-margin reference centers, preparatory failures and unreplaced components in the outer law
Arbitrary inherited event	Marked bound (4.7) once its probability and mark moments are known	Actual geometric event probabilities and a current-scale moment budget
Compact conditional energy	A conditional reserve for excess over a conditional minimum	A retained flux can already be bad at that minimum; a bound on excess does not pay that retained flux
A larger core for surgery	Geometrically available in parts of the RG construction	An injective or controlled-multiplicity change of the full history, with action gain, boundary compatibility, measure/Jacobian control and the complete reference/counterterm ratio
Coarse reflection/chessboard	Original Wilson reflection positivity	A reflection-local pullback of the actual coarse selector and a bound on its fully disseminated constrained partition ratio
Frozen observable transport	Exact averages and original-mark L^r contraction	Localization and cancellation producing current-scale fields plus controlled remainder vertices

Why the coarse chessboard idea is not yet a proof A deterministic block observable can be reflection positive under the original law if its pullback respects the reflection and the positive-half-space algebra. A positive Markov map alone does not imply this property. The localized R construction additionally has history selectors, backgrounds, retained data and reference draws. Before applying a chessboard theorem to the actual low-margin flag, one must prove that these data yield the requisite local reflection-compatible observable or enlarged probability law.

Even after that is proved, the fully disseminated event inserts constraints into a positive density. Source-free global upper and lower stability bounds, with uncanceled factors proportional to the entire volume, do not by themselves bound the probability of one local event uniformly in the expanding volume. An analogue of the matched partition-ratio calculation in Sections 2–3 is needed for the *constrained coarse effective density*. No such analogue has been proved here.

Why retaining the conditional minimum does not pay the outer flux The existing two-plaquette $SU(2)$ example fixes a retained product for which a filling under the old wider cutoff exists while no filling under the new sharper cutoff exists. Its conditional fluctuation minimum is regular, and its excess-energy law has good moments, but the sharper failure has probability one at that fixed retained datum. This does not claim that every such datum is an admitted Balaban history. It does show why an estimate only on fluctuations around a fixed conditional minimum cannot replace a probability bound on the retained datum in the outer law.

The published thresholds retain this distinction: LFR I's extension construction uses a wider cutoff; the CMP 119 new cutoff is narrower; and the old allowed last-layer field size used after the R operation is wider still. The implication checked in LFR II is from success of the new cutoff to success of preparatory selectors, not a uniform margin in the reverse direction. Those facts cannot be promoted to a fresh-failure estimate for every inherited center.

8. Related literature

Related primary context for the partition powers and multiple reflections is:

Paulo A. Faria da Veiga and Michael O'Carroll, On Yang–Mills Stability Bounds and Plaquette Field Generating Function, *Reports on Mathematical Physics* 95 (2025), 303–380, DOI 10.1016/S0034-4877(25)00035-7. Preprint: arXiv:2205.07376v2, 54 pages.

Its main generating field in equations (30) and (32), printed page 11, is a scaled trace-sine plaquette field, not the full dimension-four bilinear family. For $SU(2)$ in the fundamental representation the trace of every group element is real, so this particular trace-sine field is identically zero. Remark 16, printed page 30, also mentions an action-based alternative; that remark should not be omitted when describing the paper's scope.

Theorem 4, equation (83), printed page 31, retains different periodic lower- and upper-bound link counts; their surface difference matters in a simultaneous cofinal limit. The present note proves its own explicit correction in (2.10) and checks the sufficient cofinal condition (3.3). No continuum physical-field identification, tensor-family theorem or inherited RG-selector estimate is imported from the paper.

Other published inputs:

- T. Balaban, *Large Field Renormalization. I*, *Communications in Mathematical Physics* 122 (1989), 175–202: the localized R replacement and integral preservation, equations (1.100) and (1.102), and the distinct reference/extension cutoff information.
- T. Balaban, *Large Field Renormalization. II*, *Communications in Mathematical Physics* 122 (1989), 355–392: source-free stability, the normalized reference estimates, and the retained large-field components. Its component activities are not being reinterpreted here as normalized outer event probabilities.
- T. Balaban, *Convergent expansions in lattice gauge theories*, *Communications in Mathematical Physics* 119 (1988), 243–285: the source-free induction and new small-field selector, with the precise domains and cutoffs retained.
- M. Biskup, *Reflection positivity and phase transitions in lattice spin models*, Theorem 5.8, equation (5.34): the abstract even-block chessboard inequality used with the explicitly verified Wilson site-reflection geometry. This import is only the abstract reflection theorem, not an RG-history assertion.

9. Remaining theorem statements after the argument

The two improvements remove the bare-coupling loss in the raw reserve on the chosen geometry and the unnecessary fixed Hölder loss in marked rare-event estimates. They do not remove the following missing results:

1. A fully verified J0 trajectory and source-free construction with the stated normalization and admissible cofinal geometry.
2. Actual normalized geometric probabilities for all inherited low-margin, preparatory-failure and unreplaced sectors of the complete positive history law, or a different proved mechanism that controls their marked averages directly.
3. Full transported-jet matching and localization at the current physical scale, retaining every response and remainder generated by the finite field family, with contact subtractions compatible at all orders.
4. Uniform distribution-space and order-growth estimates for that full hierarchy, beyond the raw magnitude bounds proved here.
5. Rotation restoration, uniform short-distance identification/nontriviality, and the separately required operator/phase-transfer estimates.

The gap identified in item 2 is formulated precisely as follows: it is not a need for an impossible universal weighted probability bound at fixed original input. It is a need for a true averaged probability or direct marked estimate under the full normalized law, together with the current-scale information in item 3.

Annex B — Regular-action Hessians and homogeneous coercivity

Annex B. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Scope

Uniform real weighted coercivity of the full conditional kernel requires bounds on the complete boundary part of CMP119 (3.27) throughout its real cutoff support.

Scope. The fixed-domain CMP109/CMP116 inherited regular-action estimate is valid, including its old-scale resummation. Its extension with a fixed radius and prefactor to all complete CMP119 interior histories is not justified. CMP119 uses running multiscale analytic domains. Both the admissible free-coordinate radius and the constants in the activity estimate must be tracked through that transfer. Section 3 separates the fixed-domain theorem from this still-conditional extension; Annex G gives the source analysis and explicit running-radius formula.

A sharper vanishing Hessian estimate is proved for the explicitly cubic localized classical summands in CMP116 (1.38)–(1.39). It is not promoted to all of the classical fluctuation action: total cubicity does not, by itself, prove cubicity of every localized carrier. Section 4 explains the distinction.

The direct calculation in Section 9 closes the complete homogeneous classical-action Hessian without localizing its Taylor remainder. It uses the scale normalization in CMP102 (190), converts fine-link variations to physical curvature variations, and retains the compensating η^4 volume factor. Combined with the regular E estimate and the new coarea bound, this proves full conditional coercivity for the explicitly specified homogeneous no-R density (9.9) on its stated regular window. It is not identified with the complete RG/R homogeneous sector, because inherited R activities can survive there.

Finally, recentering the exact native constraint chart gives complex tubes about every point of its real regular cutoff. Thus a comparison of two radii measured from zero is not a necessary obstruction. The unfinished issue is admission of the full inherited multiscale pullback, including the current and localized activity bounds, into these tubes. Section 5 proves the native-chart statement and identifies this remaining step precisely.

Labels: P means proved here, with stated source inputs; L means the precisely scoped statement imported from a primary paper; O means not proved in this argument.

1. The exact measure being estimated

Use the source-free chart and all the retained history/exterior data. On a regular current interior the positive conditional density has the form

$$d\nu(z) = Z^{-1}\chi(z)\exp\left[-\tfrac{1}{2}\langle z, H_0 z\rangle + W_k(z)\right]dz.$$

Here H_0 is the actual constrained classical quadratic form. In particular the quadratic contribution generated by the nonlinear averaging constraint belongs to H_0; replacing it by an unconstrained curl Hessian would change the measure. The complete W_k retains the new coarea logarithm once, all inherited E_k, R_k, B_k, exterior dependence, nonconstant coupling, and the coordinate-localization corrections. The selector is fixed at zero source. A Hessian statement below concerns the interior of its regular product-cutoff support, not a derivative of its discontinuous boundary.

The desired estimate is

$$\eta_\gamma(W_k) := \sup_z \sup_i \sum_j e^{\gamma d(i,j)} |\partial_i \partial_j W_k(z)| < \lambda_0, \qquad H_0 \geq \lambda_0 I. \tag{1.1}$$

The available covariance/precision input is the source's constrained one. CMP99, Section E and Theorem 3.15, supplies uniform exponential bounds for the current fluctuation covariance, with the positivity statement for $C^*\Delta_k C$ on printed p.428. This is not a substitute for estimating W_k.

2. Published source estimates

2.1 Exact geometry See CMP119, pp. 254–257, 261, 265, 268–270 and 276–277.

$$\Omega_1 \supset \Lambda_1 \supset \Omega_2 \supset \Lambda_2 \supset \cdots \supset \Omega_k \supset \Lambda_k, \qquad Z_j = \Lambda_j^c. \tag{2.1}$$

These are CMP119 (2.1) and (2.3), not inferred nesting conventions. Printed p.256 requires at least $2MR_j$ separation between relevant boundaries for domains created by the RG transformation rather than an R operation. In the next step, printed p.265 first surrounds the previous large-field region by several layers of current LMR_{k+1}-cubes. Printed pp.269–270 then define the regular integration region by removing the enlarged bad-field/exterior region. Old boundary carriers consequently have genuine collars to cross; this is relevant to the missing uniform activity estimate, but does not by itself estimate its derivatives.

2.2 The cutoff margin that is actually proved Set

$$\delta_k = g_k A_1 (\log g_k^{-2})^{p_0}, \qquad P_k = \delta_k / g_k. \tag{2.2}$$

CMP119 p.267 makes the scaling $A'_k = g_k A_k$. Formula (3.16) on p.268 literally prints the good condition $\sup |A_k| < \delta_k$; it does not print the corresponding factor g_k^{-1}. Either the literal displayed condition, for $g_k \le 1$, or the intended physical condition $\sup |g_k A_k| < \delta_k$, implies the conservative inclusion

$$\|A_k\|_\infty \le P_k. \tag{2.3}$$

For every fixed admissible $\epsilon_1 > 0$,

$$\frac{P_k}{\epsilon_1 / g_k} = \frac{\delta_k}{\epsilon_1} \longrightarrow 0. \tag{2.4}$$

This provides a genuine complex margin on the fixed-domain CMP116 free-field analytic ball $\|B\|_\infty < \epsilon_1 / g_k$. It supplies no margin on a running radius $\epsilon_{1,k}/g_k$ without a separate comparison. The complete CMP119 transfer does not establish that $\epsilon_{1,k}$, or its activity prefactor, stays fixed.

2.3 Complete-action transfer, with its scope retained CMP119 p.276 describes generalized random-walk localization on the full sequence of domains, applies CMP99 Theorems 3.7–3.10, and resums into current-scale carriers $Y \in D_k$. It then explicitly states that the summands whose carriers satisfy

$$Y \subset \Lambda_{k+1} \tag{2.5}$$

satisfy exactly the small-field lemmas from [I]. Those lemmas are CMP116 Lemmas 1 and 2 in the split publication. This identifies the already-renormalized full inherited E_k fluctuation action on these carriers. The domains and parameter dependence of those lemmas still have to be retained. In particular, their fixed-domain constants cannot be imported unchanged into running CMP119 domains. The identification is stronger than a bare-action statement, but it is not yet the required uniform derivative bound.

For the remaining carriers the displayed analytic space is instead

$$\widetilde{U}^c_{k+1}(Y, (1+\beta)\widetilde{\alpha}_0, (1+\beta)\widetilde{\alpha}_1) \times \{A : \operatorname{supp} A \subset Y \cap \Lambda^{(k)*}_{k+1},\ |A| < C_1 p_1(g_k)\}. \tag{2.6}$$

See CMP119 (3.43). The printed radius is $C_1 p_1(g_k)$. No checked definition in that paper currently identifies it with P_k, with p_0, or with the later LFR-I sharper selector also named p_1. No such identification is made here.

The text then treats R_k by the same analysis as E_k, except that the coupling subtraction is not needed; it treats B_k using the localized versions of CMP109 (3.6)–(3.7). On p.277 it states a boundary-activity bound of the form (2.42), with an extra $O(\epsilon_k)$ factor. That page also explicitly treats the nonconstant-coupling difference, supported in Z_k, and the coordinate-localization correction. Large coefficients in the coupling difference are controlled there by decay of the minimizing response. Thus none of these terms may be dropped from the requested full theorem.

2.4 The precise unresolved source-reading issue CMP116 Lemma 1 supplies an analytic activity $V'_{k,Y}$ on (1.34), with

$$|V'_{k,Y}(B)| \le E_0\epsilon_1 C_1 M^q e^{C_2\kappa_1} e^{-(1-2\vartheta)\kappa d_k(Y)}. \tag{2.7}$$

Here ϑ denotes the paper's fixed exponent-loss parameter, not the running cutoff δ_k. The constants in front are fixed when the CMP109/CMP116 analytic parameters are fixed. On a changed multiscale domain their dependence must be retained. Formula (1.36) is readable in the article.

CMP116 Lemma 2 writes

$$V_{k,Y}(B) = \langle B, A_{k,Y}(B)B\rangle + V'_{k,Y}(B). \tag{2.8}$$

The estimates below use the explicit cubic terms. Extending them to the regrouped quadratic term requires the matrix-element estimate (1.43) of CMP116.

3. Fixed-domain regular-action Hessian and the conditional running-domain extension

Theorem 3.1 (P in the fixed CMP109/CMP116 analytic domain) Let $W_k^{E,\mathrm{bulk}}$ be the inherited renormalized regular-action fluctuation sum with the fixed analytic domain and the fixed activity bound (2.7), in the source's actual free coordinates. Suppose its real product cutoff obeys $P_k \le \epsilon_1/(2g_k)$, with $\epsilon_1 > 0$ fixed. For sufficiently small admitted shell coupling and sufficiently small fixed $\gamma > 0$,

$$\eta_\gamma(W_k^{E,\mathrm{bulk}}) \le C_{\mathrm{constr},\gamma} E_0 \frac{g_k^2}{\epsilon_1}. \tag{3.1}$$

The bound is uniform in volume and shell index within that fixed-domain construction. It does not assert the same constants for arbitrary CMP119 histories or their changed analytic domains, and it does not estimate the complete $R_k + B_k$ residual action.

Counterterm bookkeeping. The exact source identity being estimated is CMP116 (1.33),

$$E_k(U_k(e^{iB'}V^{(k)})) - E_k(U_k(V^{(k)})) = \sum_Y V'_{k,Y}(B), \qquad B' = g_k CB - hD(g_k CB).$$

The E_k here is the already-renormalized inherited regular action. CMP119 (2.25), p. 259, defines it as

$$E_k(U) = \sum_{j=1}^{k}\big[E^{(j)}(\Lambda_j, g_{j-1}, U) - E^{(j)}(\Lambda_j, g_{j-1}, 1) - \beta_j(g_{j-1})A(\phi_j, U)\big].$$

Thus its past coupling subtractions, their nonzero fluctuation derivatives, and old Jacobian contributions are included in the identity. The subtracted base value $E_k(U_k(V^{(k)}))$ is constant in the current free coordinates. The classical Hessian H_0 uses the already-flowed coupling g_k; no extra marginal quadratic is removed from V'_Y, and no past β_j-term is added to H_0 a second time. The new β_{k+1} is generated only after integrating the current shell.

Proof. By (2.4), reduce the coupling ceiling so that $P_k \le \epsilon_1/(2g_k)$. Every real point z of the actual cutoff has a complex coordinate polydisc of radius $c_G\epsilon_1/g_k$ contained in the activity domain, where $c_G > 0$ is a fixed norm-equivalence factor for the finite-dimensional Lie algebra. Mixed and diagonal coordinate Cauchy estimates applied to (2.7) give

$$|\partial_i\partial_j V'_{k,Y}(z)| \le C_{\mathrm{constr}} E_0 \frac{g_k^2}{\epsilon_1} e^{-\kappa_* d_k(Y)}, \qquad \kappa_* = (1-2\vartheta)\kappa. \tag{3.2}$$

The derivative is zero unless both free coordinates belong to the carrier's permitted bond set, enlarged by the fixed localization convention if required.

Coordinate identification. In this theorem z means the actual surviving free-bond coordinates after eliminating the exceptional bonds in the linearized constraint. These are named B in CMP109/CMP116 and A in CMP119 (3.15)/(3.25). They are not the ambient physical link field B', nor its division by g. The source already evaluates its analytic activity at the complete argument $gCB - hD(gCB)$, or its multiscale counterpart. Thus the nonlinear chart

and minimizing-field pullback are inside the holomorphic activity to which (3.2) is applied. They are not a further unestimated substitution after Cauchy. Conditioning exterior coordinates gives the affine embedding $B = \zeta^0 + \iota_I z$; on a carrier contained in the interior only the indicated current free coordinates occur. This embedding has weighted row/column norm one and zero second derivative. Fixed finite-range reindexings or retained gauge-frame changes cost only a fixed weighted norm constant.

For any genuinely different nonlinear free coordinates $B = \Psi(\tilde{z})$, the Hessian would instead be

$$D^2_{\tilde{z}}(V' \circ \Psi) = (D\Psi)^T (D^2_B V')(D\Psi) + \sum_a (\partial_{B_a} V') D^2 \Psi_a.$$

That extra first-derivative term must be estimated. The theorem does not make that substitution; it uses the source's actual free coordinates.

The carrier sum must be performed on the current M-cube graph, not on unit-lattice animals. A carrier with n current M-cubes contains at most $c_G M^4 n$ free coordinates. Its current-lattice diameter is at most cMn, and the geometric estimates used in CMP116 (1.26)–(1.28) bound n by a fixed multiple of $1 + d_k(Y)$. The number of connected carriers of size n containing a fixed cube is at most c_d^n. Choosing $\gamma > 0$ sufficiently small compared with the source's remaining decay exponent, and using the same entropy reserve as the source localization sum, yields

$$\sup_i \sum_{Y \ni i} \sum_{j \in Y} e^{\gamma d(i,j)} e^{-\kappa_* d_k(Y)} \le C_{M,\kappa_*,\gamma,G} < \infty. \tag{3.3}$$

Singleton carriers and any fixed enlargement only change this constant. Summing (3.2) and applying (3.3) proves (3.1). No additional sum over old shell indices is introduced: (2.7) applies to the regular effective-action fluctuation expansion after its old-scale renormalization and resummation. In particular the proof does not replace this by the generally divergent surrogate $\sum_{j \le k} g_j^2$. □

What this removes. The inherited regular action is controlled in the specified fixed-domain homogeneous model. This is the estimate used in Section 9.4 and in Annex F. It is not an established complete-history E-bulk estimate.

Conditional running-domain version — C If the actual fully pulled-back carrier has the bound

$$|V^E_{k,Y}(B)| \le E_0 \epsilon_{1,k} K^E_k e^{-\kappa_* d_k(Y)} \quad \text{for } \|B\|_\infty < \epsilon_{1,k}/g_k,$$

and $\delta_k = g_k P_k < \epsilon_{1,k}$, the same Cauchy/carrier proof gives

$$\boxed{\eta_\gamma(W^{E,\mathrm{bulk}}_k) \le C E_0 K^E_k \frac{\epsilon_{1,k} g_k^2}{(\epsilon_{1,k} - \delta_k)^2}.} \tag{3.4}$$

Here a uniform carrier entropy reserve is also required. The hypothesis concerns the complete localized activity after the nonlinear constrained substitution, including its currents. Neither a native chart tube nor a real activity bound establishes it. Uniform admission, the size of K^E_k, and a small right side in (3.4) remain open for complete histories. Annex G checks the source inequalities that force this qualification.

4. Sharper estimate for the explicitly cubic classical carriers

CMP116 (1.38) localizes the classical remainder $V(H_1 B')$, where

$$B' = g_k C B - hD(g_k C B).$$

Its bound (1.39) is explicitly cubic in B'. With the fixed construction parameters and the source's carrier resummation absorbed into constants, this gives, for each such localized classical summand,

$$|P^{\mathrm{cub}}_{k,Y}(B)| \le C g_k^{-2} \rho^3 e^{-\kappa_* d_k(Y)} \quad \text{if } \|g_k B\|_\infty \le \rho \le \rho_0, \tag{4.1}$$

where $\rho_0 > 0$ is a fixed admitted physical radius in this fixed-domain construction. The small correction hD and the relevant localization operators have been retained. For these particular summands, vanishing through degree two follows already from the cubic bound; it is not inferred from cancellation in the total action. Its extension to running complete-history domains requires the corresponding radius and constant analysis.

Proposition 4.1 (P for these explicitly cubic localized summands) On the actual regular cutoff (2.3), for $2\delta_k \le \rho_0$,

$$\eta_\gamma\left(\sum_{Y\subset\Lambda_{k+1}} P^{\mathrm{cub}}_{k,Y}\right) \le C_{\mathrm{constr},\gamma}\delta_k = C_{\mathrm{constr},\gamma} g_k A_1 (\log g_k^{-2})^{p_0}. \tag{4.2}$$

Proof. Apply (4.1) at physical radius $2\delta_k$. Every real point of the cutoff then has complex coordinate margin $c_G\delta_k/g_k$. Cauchy gives

$$|\partial_i\partial_j P^{\mathrm{cub}}_{k,Y}(z)| \le C\frac{g_k^{-2}\delta_k^3}{(\delta_k/g_k)^2} e^{-\kappa_* d_k(Y)} = C\delta_k e^{-\kappa_* d_k(Y)}.$$

Use (3.3). □

The same radius scaling can be proved from a bound at one fixed radius by applying the one-variable Schwarz lemma to $t \mapsto P^{\mathrm{cub}}_{k,Y}(tB)$, since the first three Taylor coefficients vanish. This requires vanishing for that carrier.

Why this is not yet the complete classical bulk theorem The total exact constrained classical remainder is cubic after the true constrained quadratic form is extracted. This alone does not imply that every term in an arbitrary localized decomposition is cubic. For example,

$$z^3 = (z^2 + z^3) + (-z^2)$$

is a decomposition into analytic localized functions with nonzero individual quadratic jets. Even though the cancellation is exact, a bound obtained by absolute summation of individual Hessians must account for those jets.

CMP116 states after (1.40) that the other terms of $P^{(k)}$ are treated by the same decompositions and bounds, but the exact complete regrouped quadratic coefficient is precisely (1.43), which is not used below. Therefore (4.2) is claimed only for the explicitly cubic construction (1.38)–(1.39), and for any further summand whose cubic integral-remainder representation is separately verified. For a general localized complete-action extension, one can inspect and use the actual (1.43), or verify carrier by carrier that localization preserves the required jets. That carrier verification remains open. Section 9 below gives a separate direct proof for the complete homogeneous classical action, without making this carrier-level assertion.

5. Recentered complex tubes for the exact native chart

Proposition 5.1 (P, exact regular chart) Use the chart on a regular current core,

$$a(\zeta) = gC\zeta - hD(gC\zeta), \qquad D(q) = N(q - hD(q)), \quad N(0) = DN(0) = 0. \tag{5.1}$$

Suppose ζ_0 is any real point of the cutoff, including its fixed exterior coordinates, such that the same frozen chart has a fixed physical margin. There are fixed $c, C, \gamma > 0$, independent of the number of copies of its regular local template, such that for

$$\|\eta\|_\infty < r, \qquad gr < c,$$

the exact relative link field

$$B_{\zeta_0,e}(\eta) = \frac{1}{i}\log\left(e^{ia_e(\zeta_0+\iota_I\eta)}e^{-ia_e(\zeta_0)}\right) \tag{5.2}$$

is holomorphic on that tube and satisfies

$$|B_{\zeta_0,e}(\eta)| \le Cg \sum_{i\in I} e^{-\gamma d(e,i)} |\eta_i|, \tag{5.3}$$

with weighted first- and second-derivative bounds of order g and g^2, respectively. The background, integration map, coordinate inverse h, and retained variables have not been changed.

Proof. In a fixed physical complex neighborhood, the Neumann inverse $(I+DN(a)h)^{-1}$ appearing in the exact implicit derivatives is uniformly bounded in the weighted matrix algebra. The derivative formulas consequently hold at every $\zeta_0 + \iota_I \eta$ in the tube, not just at 0. Integrating the first derivative along $s \mapsto \zeta_0 + s\iota_I\eta$ gives the exponentially localized difference bound for $a(\zeta_0+\iota_I\eta)-a(\zeta_0)$. Matrix multiplication, exponentiation and the local logarithm are holomorphic on a common fixed neighborhood of the identity. Their first two derivatives are uniformly bounded there. Applying their chain rule to (5.2) yields (5.3) and its weighted derivative versions. In particular, no false identity $e^{ia}e^{-ia_0} = e^{i(a-a_0)}$ is used. □

Consequence for the radius issue. If a local activity has a uniformly bounded holomorphic extension on relative complex gauge fields of physical size α_1 about every admitted real base, (5.3) allows a coordinate tube of width $c\alpha_1/g$ about every admitted real cutoff point. One need not have $\alpha_1/g > P_k$, because both radii are not measured from zero. This corrects an unnecessarily restrictive origin-ball criterion.

What the primary multiscale minimizer theorem already gives CMP102, Proposition 9, printed pp.305–309, proves precisely a recentered analytic extension

$$V = V'V_0 \longmapsto U_k(V'V_0)$$

about each admitted real regular base V_0. Its (182)–(190) derive the first-derivative kernel and give the same regularity/decay pattern as the minimizing propagator H. This source result is real support for the moving-tube strategy. It prevents dismissing recentering merely because the original activity proof was written around one fixed source-free background.

What is not yet proved for the actual complete pullback The original analytic activity spaces in CMP119 (2.34)–(2.39) require more than a small link variation. They simultaneously constrain multiscale real regularity, reconstructed fields, currents $J, J_{p,X}$, local gauges, and relative complex fields with their spatial derivatives. In particular:

1. The actual real point must have a quantitative margin in the current constraints, not merely belong to an open regularity set. CMP119 gives $\alpha_{0,j} = g_j C_0 (\log g_j^{-2})^{q_0}$, $\alpha_{1,j} = g_j C_1 (\log g_j^{-2})^{q_1}$, with $q_0, q_1 > 1$ and construction restrictions. No checked explicit relation $q_0 \ge p_0$ is available, and the constants cannot be enlarged freely after the source construction is fixed.
2. The pullback U_k, its currents and their required spatial derivatives must map the complete coordinate tube into every old activity domain. Proposition 9 supplies the first derivative pattern; translating it into these simultaneous quantitative domain margins, including second spatial derivatives of currents and all old layers, has not been performed here.
3. The bound must hold for the localized complex activities with decoupled propagators, not just for the unlocalized action evaluated at a real field. Recentered analyticity of the unlocalized minimizing configuration alone does not prove a bound for every localized carrier.
4. The carrier estimates must be summable uniformly over old scales and histories. The source's E renormalization and the R small powers are essential. Replacing them by an accumulation of $O(g_j^2)$ bounds would not be valid.

Consequently Proposition 5.1 is a genuine exact-chart improvement but not a proof of the complete boundary Hessian budget.

6. Why the current origin-ball information is insufficient by itself

This elementary counterexample pinpoints the missing inference; it is not a Yang–Mills counterexample.
Let $P \geq 6R > 0$, $b = P/2$, $\varepsilon > 0$, and

$$W_s(z) = -\varepsilon \exp[-(z-b)^2/s^2]. \tag{6.1}$$

It is entire, real on the real axis, and $|W_s(x)| \leq \varepsilon$ for every real x. If $|z| \leq R$, then

$$\mathrm{Re}(z-b)^2 \geq (b-R)^2 - R^2 = b^2 - 2bR > 0,$$

so $|W_s(z)| \leq \varepsilon$ on the entire origin-centered complex disc. Nevertheless,

$$W_s''(b) = 2\varepsilon/s^2. \tag{6.2}$$

Thus the strictly positive cutoff density

$$1_{[-P,P]}(x) e^{-x^2/2 + W_s(x)} dx$$

has potential Hessian $1 - 2\varepsilon/s^2$ at the interior point b, which becomes arbitrarily negative as $s \downarrow 0$. An arbitrarily good real action bound, plus a uniformly bounded holomorphic extension on a smaller ball about zero, does not control the Hessian on a larger real cutoff.

The moving-tube premise would exclude this example. It is therefore an appropriate strengthening to investigate; the unavailable comparison of p_1 and p_0 is not by itself a mathematical impossibility result.

7. Remaining budget, item by item

Part of the actual action	What is established here	Remaining issue
Actual constrained Gaussian precision	Primary constrained positivity/covariance input retained	Match the precise regular-core coordinate/gauge convention when assembling (1.1)
New exact chart and coarea	the native weighted derivatives retained; Proposition 5.1 supplies recentered exact-chart tubes	Arbitrary multiscale interfaces are not covered by the repeated strict-core template alone
Inherited regular E action in the fixed CMP109/CMP116 domain	Theorem 3.1: $\eta_\gamma \leq CE_0 g_k^2/\epsilon_1$, with the old-scale resummation retained	Complete CMP119 histories require the running-radius/prefactor premises of (3.4); this extension is O
Explicitly cubic localized classical terms	Proposition 4.1: $\eta_\gamma \leq C\delta_k$ on the entire conservative cutoff	Do not identify this subset with all of the localized classical action without the remaining carrier-jet analysis
Other bulk terms packaged in CMP116 Lemma 2	Actual CMP119 bulk transfer is explicit	Read exact (1.43), or independently bound every packaged classical/Jacobian term without dropping quadratic jets
Inherited R action	CMP119 (2.31) gives a complex multiscale activity bound; (2.46) has the six-power old-scale loss	Prove current-coordinate tube admission and the complex marginal-transfer/carrier estimate of Annex G; the full Hessian remains O
Inherited B action	Source has genuine multiscale analytic domains and boundary carrier bounds, with an extra fluctuation factor on p.277	Prove tube admission and complex activity norms on the whole real cutoff, including current margins

Part of the actual action	What is established here	Remaining issue
Exterior, nonconstant coupling, coordinate-localization corrections	All explicitly present in CMP119 pp.276–277 and assigned to boundary terms	Same complete boundary derivative estimate, with coefficient and carrier sums checked
Full selected conditional measure	Positive exact density retained	Sum the above budgets below the actual precision gap; verify its selected support is the required convex product core before applying Brascamp–Lieb
All histories and observable transport	No history or remainder vertex is discarded	Even full native conditional coercivity would still not identify transported microscopic marks with current native fields or close their UV OS hierarchy

8. References

1. T. Balaban, Renormalization group approach to lattice gauge field theories. I, CMP109 (1987), 249–301. DOI: https://doi.org/10.1007/BF01215223. Used: (2.12), (3.1)–(3.2), regular action scopes.
2. T. Balaban, Renormalization group approach to lattice gauge field theories. II. Cluster expansions, CMP116 (1988), 1–22. DOI: https://doi.org/10.1007/BF01239022. Used: (1.19)–(1.20), (1.26), (1.28), Lemma 1 (1.33)–(1.36), the explicit cubic construction (1.38)–(1.40), Lemma 2 representation (1.41)–(1.42).
3. T. Balaban, Convergent renormalization expansions for lattice gauge theories, CMP119 (1988), 243–285, (3.43). DOI: https://doi.org/10.1007/BF01217741.
4. T. Balaban, The variational problem and background fields in renormalization group method for lattice gauge theories, CMP102 (1985), 277–309. Proposition 9 and formulas (172)–(190) provide the recentered minimizing-field statement; they do not themselves contain the complete marked/boundary Hessian theorem sought here.
5. T. Balaban, Propagators for lattice gauge theories in a background field, CMP99 (1985), 389–434. The multi-scale regularity and exponential-decay statements, and the current constrained covariance, are retained in their stated scopes.

This note makes no claim that either the full Yang–Mills existence theorem or its complete native conditional coercivity input has been proved.

9. Direct complete classical estimate in the homogeneous regular geometry

This argument avoids both (1.43) and the per-carrier cancellation issue for the complete classical action in the homogeneous geometry. It does not discard R terms from the complete RG/R measure; Section 9.5 explains why those terms remain a separate issue even when the current large-field set is empty.

9.1 The normalized source input is visible in the published text of CMP102 printed pp.308–309 were given there. In (190), for input on layer j' and output on layer j, the first two entries have the scale factors

$$\left|\frac{\delta \mathcal{H}_\mu(B,x)}{\delta B_\nu(y')}\right| \le C(L^j\eta)^{-1}(L^{j'}\eta)^{-4}e^{-cd(y,y')},$$

$$\left|\nabla_x \frac{\delta \mathcal{H}_\mu(B,x)}{\delta B_\nu(y')}\right| \le C(L^j\eta)^{-2}(L^{j'}\eta)^{-4}e^{-cd(y,y')}. \tag{9.1}$$

The functional-derivative convention includes the corresponding input lattice measure. Proposition 9 explicitly states complex analyticity and that the derivative satisfies (190). In the homogeneous case, rescale the current lattice to unit

spacing: $j = j' = k$, $L^k\eta = 1$. Both factors in (9.1) are then one. The resulting first derivative and its spatial gradient are bounded uniformly, with exponential decay in current-scale distance, throughout an admitted fixed physical complex neighborhood. This is a curvature/scaled-difference statement, not merely a bound on a fine link.

CMP109 pp.265–267 identifies the map used in its exact conditional integral with this minimizing map in Landau gauge; its (2.6)–(2.8) includes the full nonlinear minimizing correction. Its (2.11) identifies the true constrained quadratic form, including the contribution from the quadratic constraint correction.

9.2 Curvature scaling and higher derivatives Let q be the physical free coordinates in the exact current constraint chart, and let $U(q)$ be the complete fine minimizing configuration. The current map from q into the coarse input of Proposition 9 is the exact analytic map, including hD. Its uniformly localized derivatives were proved. Composition with (9.1) therefore gives, in the homogeneous geometry, the same first-derivative and gradient estimate with input q_i.

Work in the Landau representative supplied by Proposition 9. This is permitted because each Wilson plaquette trace is gauge invariant; no large derivative from a subsequent change to axial gauge need be estimated in the action. Define the physical curvature coordinate

$$\mathfrak{F}_p(q) = \frac{1}{i\eta^2} \log U(q)_p. \tag{9.2}$$

The principal local logarithm is admissible throughout a smaller fixed complex neighborhood by the regularity estimates. Exact variation of a plaquette is a covariant curl of its tangent link field. In fine lattice notation the link tangent has a factor η, and its covariant difference supplies the second factor η. Equation (9.1) consequently gives

$$|\partial_{q_i} U(q)_p| \le C\eta^2 e^{-cd(x_p,i)}, \qquad |\partial_{q_i} \mathfrak{F}_p(q)| \le C e^{-cd(x_p,i)}. \tag{9.3}$$

The uniform physical curvature bound used here follows from explicit source regularity, not from fine-link boundedness. The real base has $|U_{0,p} - 1| \le C\epsilon_0\eta^2$ in this homogeneous specialization. Proposition 9 supplies (19)–(21) on the same complex neighborhood, with $\epsilon_2 = B_5\epsilon_1$, hence $\|\mathcal{H}\| + \|\nabla_{U_0}\mathcal{H}\| \le C\epsilon_1$ in current physical units. Expanding the four relative link factors, the plaquette change is bounded by

$$|U_p(q) - U_{0,p}| \le C\eta^2 \left(\|\nabla_{U_0}\mathcal{H}\| + \|\mathcal{H}\|^2 + \epsilon_0 \|\mathcal{H}\| \right).$$

Thus $\|\mathfrak{F}_p(q)\| \le C(\epsilon_0 + \epsilon_1 + \epsilon_1^2)$ uniformly on that complex neighborhood, after choosing the fixed source parameters small enough for the logarithm. This justifies the common bounded set used for the derivatives of $\mathcal{L}_\eta$ in (9.5).

For clarity, if $U_e(q) = e^{i\eta\mathcal{H}_e(q)} U_{0,e}$, its tangent is

$$T_{i,e}(q) = \int_0^1 \mathrm{Ad}(e^{it\eta\mathcal{H}_e(q)}) \partial_{q_i}\mathcal{H}_e(q)\, dt.$$

The covariant derivative of T_i is controlled by $\nabla\partial_{q_i}\mathcal{H}$, together with bounded products of $\mathcal{H}, \nabla\mathcal{H}, \partial_{q_i}\mathcal{H}$. Exact telescoping around the four plaquette edges gives (9.3); it does not estimate four link variations separately and thereby lose one power of η.

Uniform complex Cauchy estimates now supply the higher derivatives. On a fixed smaller complex neighborhood,

$$|\partial_{q_{i_1}} \cdots \partial_{q_{i_r}} \mathfrak{F}_p(q)| \le C_r e^{-c_r \tau(x_p, i_1, \ldots, i_r)}, \qquad r = 1, 2, 3. \tag{9.4}$$

Here τ is a shortest connecting-tree length in current-scale distance. Indeed, apply coordinate Cauchy to (9.3) $r - 1$ times. This first yields the exponential involving any chosen one of the input indices. Symmetry of the mixed derivatives permits the minimum over all such indices. Since

$$\tau(x_p, i_1, \ldots, i_r) \le \sum_{a=1}^{r} d(x_p, i_a) \le r \max_a d(x_p, i_a),$$

the minimum of the exponentials is bounded by the right side of (9.4), with a fixed loss in decay. The constants depend on the fixed physical analytic radius, not on g, η, k, or the torus volume.

9.3 The complete classical third derivative, with its volume factor retained In four dimensions write the Wilson action exactly as

$$A^{\eta}(U(q)) = \sum_{p} \eta^4 \mathcal{L}_{\eta}(\mathfrak{F}_p(q)),$$

$$\mathcal{L}_{\eta}(F) = \eta^{-4}\left[1 - \frac{1}{2N}\operatorname{Tr}(e^{i\eta^2 F} + e^{-i\eta^2 F})\right]. \tag{9.5}$$

All normalizing group constants can be replaced by the fixed convention used here. The derivatives of $\mathcal{L}_{\eta}$ through order three are uniformly bounded on the admitted physical curvature neighborhood, for $0 < \eta \le 1$. This follows directly from the even power series in $\eta^2 F$; its lowest nonconstant term has degree two and carries η^4.

The third chain rule has the three kinds of terms

$$D^3\mathcal{L}_{\eta}(\mathfrak{F})[D\mathfrak{F}]^3, \qquad D^2\mathcal{L}_{\eta}(\mathfrak{F})[D^2\mathfrak{F}, D\mathfrak{F}], \qquad D\mathcal{L}_{\eta}(\mathfrak{F})[D^3\mathfrak{F}]. \tag{9.6}$$

The last term is essential: this calculation does not discard the analogue of $A'a'''$, or the other derivatives of the nonlinear chart/minimizer. By (9.4), every term has tree decay joining the plaquette position and the three input coordinates. The factor η^4 in (9.5) survives in every term. Therefore

$$\sup_{i} \sum_{j,l} e^{\gamma d(i,j)} \left|\partial_{q_i}\partial_{q_j}\partial_{q_l} A^{\eta}(U(q))\right| \le C. \tag{9.7}$$

To see the ultraviolet cancellation explicitly, the plaquette sum is bounded by $\sum_p \eta^4 e^{-c' d(x_p,i)} \le C$, uniformly in η. The two remaining current-coordinate sums are exponentially summable. Estimating the fine link derivatives without the curvature curl in (9.3) would instead produce an inadmissible η^{-4} factor.

For the hard axial gauge used in the exact constrained-coordinate formulation one sets $G = 0$. The optional soft-gauge version has a concrete source check as well: CMP109 (2.5), printed p.266, writes its actual $G(B')$ as a quadratic block term plus an analytic $G_3(B')$ beginning at degree three and localized in current blocks. Its uniformly bounded physical third derivatives follow from finite block size and that fixed physical analytic neighborhood. This particular source term therefore also satisfies (9.7), without a fine-lattice volume sum. No assertion about a generic unspecified gauge-fixing functional is needed.

Proposition 9.1 (P using the homogeneous specialization of Proposition 9) In the homogeneous regular geometry, let

$$S_g(z) = g^{-2}\big[A^{\eta}(U(gz)) + G(gz)\big]$$

be the complete exact classical/gauge-fixing action in the physical chart, and let $H_0 = D_z^2 S_g(0)$ be its true constrained Hessian. If $\|z\|_\infty \le P(g)$, $gP(g)$ is sufficiently small, and the line segment to zero lies in the same regular chart, then

$$\sup_{z}\sup_{i}\sum_{j} e^{\gamma d(i,j)} \left|\partial_i\partial_j \left(S_g(z) - S_g(0) - DS_g(0)z - \tfrac{1}{2}\langle z, H_0 z\rangle\right)\right| \le CgP(g). \tag{9.8}$$

At the source-free constrained minimum $DS_g(0) = 0$.

Proof. Scaling (9.7) by $q = gz$ and the action prefactor g^{-2} gives a weighted third-derivative row sum bounded by Cg. Integrate the third derivative along tz, $0 \le t \le 1$, and use $\sup_l |z_l| \le P(g)$. □

This is a complete classical-action statement in this homogeneous scope. It uses no artificial assignment of cubicity to each carrier and no numerical coefficient from (1.43).

9.4 A rigorously specified no-R conditional density Consider the explicitly specified homogeneous density

$$d\nu_{\mathrm{sf}}(z) \propto 1_{\{\|z\|_\infty \le P(g)\}} J(gz) \exp\{-S_g(z) + E_k(U(gz))\}\, dz, \tag{9.9}$$

where E_k is the complete inherited regular action in the source's small-field construction. Older Jacobian terms are included in this inherited action; J is the new exact coarea factor only. This is the actual form of the CMP109 small-field conditional integrand, with the indicated further regular-window restriction if its chosen cutoff is larger. It is not asserted to equal the homogeneous sector of the complete RG/R construction.

Combining (9.8), the corrected fixed-domain Theorem 3.1 on the whole torus in this geometry, and the $O(g^2)$ coarea Hessian gives

$$\eta_\gamma\big(S_g - \tfrac{1}{2} z H_0 z - \log J - E_k\big) \le C\left[gP(g) + g^2 + E_0 g^2/\epsilon_1\right]. \tag{9.10}$$

Choose the fixed physical window $gP(g)$ sufficiently small and then g sufficiently small. The right side is below $\lambda_0/2$, and the complete real potential in (9.9) has Hessian at least $\lambda_0 I/2$. The cutoff is a product of convex finite-dimensional norm balls. Consequently the weighted Brascamp–Lieb/covariance argument applies to this density, and its native marked cut estimates become unconditional for this specified density and regular window.

In particular, for the actual finite native curvature-bilinear family let B_* bound the reference physical curvature divided by g, and put $Q = P + B_*$. The exact-chart value/gradient estimates and this coercivity give, for $m \ge 2$,

$$|\kappa_{\nu_{\mathrm{sf}}}(O_{\alpha_1,x_1}, \dots, O_{\alpha_m,x_m})| \le C^{m+1} m! Q^{2m-2} e^{-\gamma\tau(x_1,\dots,x_m)/(m-1)}. \tag{9.12}$$

The one-point and full-moment bounds are respectively CQ^2 and $C^m Q^{2m}$. If the retained reference has the actual logarithmic-window bound $B_* \le CP$, replace Q by a constant times P. That background condition is stated explicitly; a fixed ϵ_0 curvature bound alone does not imply it. Restoring physical dimension-four normalizations multiplies the order-m bound by the appropriate physical factor h^{-4m}; nothing here bounds that ultraviolet factor or identifies transported microscopic insertions with these native fields.

There is no source-dependent change of variables in this conclusion. The normalized conditional measure, all regular old effective terms, and the exact new coarea are retained. This result establishes neither comparison with the full Wilson measure nor a negligible-error removal of the further cutoff if one was imposed.

9.5 Why R cannot be silently removed from the complete homogeneous sector In CMP119, the support property of a B activity requires $X \cap Z_j \neq \emptyset$; hence B activities vanish if every relevant old Z_j is empty. The exterior/nonconstant-coupling/coordinate-localization corrections also vanish in the genuinely homogeneous geometry. In contrast, R activities in (2.30)–(2.31) are supported inside Λ_j and can remain after earlier R replacements. An empty current large-field set is not an assertion that $R_k = 0$.

CMP119 p.263 also fixes the old-scale loss in its available real value estimate:

$$|R_k(U_k)| \le R_1 \sum_{n=1}^{k} |\Gamma_n| \sum_{j=1}^{n} g_j^{\kappa_0} < R_1 \sum_{n=1}^{k} |\Gamma_n| g_n^{\kappa_0 - 6}, \qquad \kappa_0 \ge 7. \tag{9.11}$$

This is (2.46); the source prints the exponent $\kappa_0 - 6$. It must not be replaced by a guessed $\kappa_0 - 2$ in an attribution to this theorem. Formula (2.44) has the unsummed $g_j^{\kappa_0}$. Those real marginal statements alone do not give the current-coordinate R-bulk Cauchy estimate. The source additionally gives the genuinely complex activity bound in (2.31); the localized current-coordinate pullback and its simultaneous U, J domain margins are still required.

To identify (9.9) with a positive no-large-field-history branch of the full construction, one would have to show in the original positive history representation that every R map on that branch is the identity, and verify the identification before signed activity regrouping/exponentiation. This has not been done here. Alternatively one can estimate the complete inherited R bulk Hessian. Annex G records the original complex activity norm and a conditional old-scale derivative sum. Its current-coordinate domain admission and complex marginal-transfer premise remain open; the text saying that R parallels E does not discharge them.

In heterogeneous histories, (9.1) retains its factors $(L^j\eta)^{-1}$ and $(L^j\eta)^{-2}$. The simple homogeneous η^4-sum argument does not remove them. The collar, layer, and activity sums must be performed with those factors. Thus Proposition 9.1 and (9.10) do not close the complete-history conditional theorem.

Annex C — Exact scalar transfer response and the field domain

Annex C. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Status

The reduced interacting periodic OS quadratic inequality is not proved here. The available source-free Wilson stability, bare transfer estimates, finite native jets and numerical coupling trajectory do not control its arbitrary-vector norm after physical renormalization.

This annex supplies two exact further results and a diagnostic:

1. A scalar Wilson coupling insertion is related to the logarithmic response of the normalized transfer operator by an explicit completely positive real-time averaging map. The inverse has a controlled imaginary-time strip. Thus the physical scalar slab target can be stated as a local coupling-response bound, with exact identity subtraction and no assumed differentiability of an unbounded logarithm.
2. The full Schwartz Borchers domain is an energy graph core. Once the bounds and canonical identification are available, covariance, locality and the minimal closed field realization extend to the whole smooth-energy domain. This supplies the domain argument; it is not an interacting estimate.
3. A central random rescaling can preserve the entire two- and three-point functions, including their quantitative short-distance laws, while violating every polynomial energy bound in the mixed theory. A related additive construction preserves all connected functions of order at least three. These examples are not Wilson limits; they test the proposed inference from vacuum information to operator estimates.

The source geometry check still does not license independent temporal enlargement. CMP109 specifies common side parameter $L_\mu = L^m$; CMP119 explicitly inherits that geometry. A rectangular extension may be possible, but is not imported as an already proved result.

Status notation: P means a proof from the displayed hypotheses is supplied; L means a scoped primary-source statement; O means an unproved Yang–Mills estimate. No novelty claim is made.

1. Exact normalized scalar source in the full compact transfer kernel

Fix a finite spatial lattice, spacing $a > 0$, and the physical gauge-invariant Hilbert space. Use the full compact Wilson transfer operator T_θ, with no small-field restriction. A real time-independent spatial profile f changes the scalar plaquette couplings from β to $\beta + \theta f_p$, with θ small enough that all temporal couplings remain positive. The spatial action is distributed equally between the two ends of a time step. The temporal-link integrations give the usual gauge projection. One-link scalar normalization factors may be held fixed during the source variation. A different scalar convention multiplies T_θ by an overall positive scalar, which cancels in $P_\theta = T_\theta/\lambda_0(\theta)$; it only shifts the unnormalized raw insertion by a known identity term.

At fixed cutoff T_θ is a positive selfadjoint trace-class operator, is injective on the physical space, and has a simple positive largest eigenvalue $\lambda_0(\theta)$. Its kernel and its source derivative are bounded on the compact configuration space. Write

$$P_\theta = \frac{T_\theta}{\lambda_0(\theta)}, \qquad P = P_0 = e^{-aH}, \qquad H \geq 0, \qquad \dot{P} = \left.\frac{dP_\theta}{d\theta}\right|_0 . \tag{1.1}$$

The derivative $\dot{P}$ is a bounded selfadjoint operator. No global differentiability statement for the unbounded family $-a^{-1}\log P_\theta$ is assumed.

Let $\mathcal{D}_{\mathrm{fin}}$ be the finite span of spectral vectors of H. At fixed spatial cutoff every bounded-energy spectral projection has finite rank. Define the centered one-step midpoint insertion as a form on this domain by

$$\mathcal{O} = -P^{-1/2}\dot{P}P^{-1/2}. \tag{1.2}$$

If $T_f = -\dot{T}$ denotes the transfer kernel with the full one-step scalar action mark, then

$$\mathcal{O} = P^{-1/2}\frac{T_f}{\lambda_0}P^{-1/2} + \frac{\dot{\lambda}_0}{\lambda_0}I. \tag{1.3}$$

Thus the identity term is fixed by normalization of the actual transfer operator. In particular, with $P\Omega = \Omega$ and $\|\Omega\| = 1$,

$$\langle \Omega, \mathcal{O}\Omega \rangle = 0. \tag{1.4}$$

To see this without differentiating the vacuum vector, use the simple-eigenvalue formula $\dot{\lambda}_0 = \langle \Omega, \dot{T}\Omega \rangle$. The two terms in (1.3) cancel in the vacuum expectation.

For $t \geq a/2$, its heat sandwich is the bounded exact marked transfer slab

$$e^{-tH}\mathcal{O}e^{-tH} = -P^{t/a-1/2}\dot{P}P^{t/a-1/2}. \tag{1.5}$$

This formula includes all spatial interactions, gauge projection, large fields and the identity subtraction. It establishes boundedness at each fixed cutoff, not a bound uniform after physical normalization.

1.1 Spectral logarithmic response Choose an orthonormal spectral basis with

$$He_i = E_i e_i, \qquad p_i = e^{-aE_i} > 0.$$

Define a Hermitian form $\mathfrak{h}$ on $\mathcal{D}_{\text{fin}}$ by

$$\mathfrak{h}_{ij} = -\frac{1}{a}\,\frac{\log p_i - \log p_j}{p_i - p_j}\,\dot{P}_{ij}, \tag{1.6}$$

where the diagonal divided difference is $1/p_i$. The definition is basis independent within degenerate energy eigenspaces. It is the divided-difference logarithmic response of the normalized transfer operator. Whenever an unbounded logarithmic derivative is separately justified, it agrees with that derivative on these matrix elements. Equation (1.6) itself requires no such unproved differentiability theorem.

Put

$$k_a(\omega) = \frac{a\omega/2}{\sinh(a\omega/2)}, \qquad \rho_a(s) = \frac{\pi}{2a}\operatorname{sech}^2(\pi s/a). \tag{1.7}$$

Then $\rho_a \geq 0$, $\int \rho_a(s)\,ds = 1$, and

$$\int_{\mathbb{R}} \rho_a(s)e^{is\omega}\,ds = k_a(\omega). \tag{1.8}$$

For example, after $x = \pi s/a$ and $u = e^{2x}$ the integral becomes $\int_0^\infty u^{ia\omega/(2\pi)}(1+u)^{-2}du$; the beta-integral evaluation gives (1.8).

Proposition 1.1 (P: exact response transforms). On every finite-energy spectral band,

$$\boxed{\mathfrak{h} = \frac{1}{a}\int_{\mathbb{R}} \rho_a(s)e^{isH}\mathcal{O}e^{-isH}\,ds,} \tag{1.9}$$

and

$$\boxed{\frac{\mathcal{O}}{a} = \int_{-1/2}^{1/2} e^{asH}\mathfrak{h}e^{-asH}\,ds.} \tag{1.10}$$

Proof. Writing $\Delta = E_i - E_j$ and using (1.2),

$$\mathfrak{h}_{ij} = \frac{1}{a}\frac{a\Delta/2}{\sinh(a\Delta/2)}\mathcal{O}_{ij}. \tag{1.11}$$

Equation (1.9) follows from (1.8). The coefficient of $\mathfrak{h}_{ij}$ on the right of (1.10) is $\int_{-1/2}^{1/2} e^{as\Delta}ds = \sinh(a\Delta/2)/(a\Delta/2)$, proving the inverse formula. All expressions on a bounded-energy band are finite dimensional and the zero-gap values are obtained continuously. ■

The map in (1.9), apart from its factor a^{-1}, is unital and completely positive on bounded operators, since it is an average of unitary conjugations. In particular it is contractive in operator norm. This property is stronger than an entrywise divided-difference bound, which would not by itself control matrix operator norms.

1.2 Uniform slab comparison with no loss of a spatial volume factor Let $Q_E = 1_{[0,E]}(H)$. On a finite-energy band,

$$\|Q_E\mathfrak{h}Q_E\| \le a^{-1}\|Q_E\mathcal{O}Q_E\|, \qquad \|Q_E\mathcal{O}Q_E/a\| \le e^{aE/2}\|Q_E\mathfrak{h}Q_E\|. \tag{1.12}$$

The second estimate uses the spectral range $[0, E]$ in (1.10). For a more quantitative low-energy comparison,

$$\|Q_E(\mathcal{O}/a - \mathfrak{h})Q_E\| \le \frac{(aE)^2}{24}e^{aE/2}\|Q_E\mathfrak{h}Q_E\|. \tag{1.13}$$

Indeed the norm of $\mathrm{ad}_{H|_{Q_E}}$ is at most E, after subtracting $E/2$ from H, and the odd term in the symmetric integral vanishes. Taylor's remainder bounds the rest by the displayed constant.

For heat sandwiches, the completely positive transform gives

$$\boxed{\|e^{-tH}\mathfrak{h}e^{-tH}\| \le a^{-1}\|e^{-tH}\mathcal{O}e^{-tH}\|.} \tag{1.14}$$

This is first proved by spectral truncation and then by the bounded unitary average of the right-hand sandwich. It is valid whenever that sandwich is bounded, in particular at finite cutoff for $t \ge a/2$ by (1.5).

Conversely, suppose a compatible family of bounded form sandwiches satisfies

$$\|e^{-uH}\mathfrak{h}e^{-uH}\| \le Cu^{-\kappa}\rho_s(f) \quad (0 < u \le 1). \tag{1.15}$$

For $0 < t - a/2 \le 1$, insert (1.10) and write every integrand as

$$e^{-(t-as)H}\mathfrak{h}e^{-(t+as)H}.$$

Both heat times are at least $t - a/2$. Factor out the symmetric sandwich at that smaller time; the remaining semigroups are contractions. This gives

$$\boxed{\|e^{-tH}(\mathcal{O}/a)e^{-tH}\| \le C(t - a/2)^{-\kappa}\rho_s(f). \quad (0 < t - a/2 \le 1)} \tag{1.16}$$

For arbitrary $t > a/2$, the same argument uses the sandwich at $u = \min(t - a/2, 1)$ and gives the bound $C[\min(t - a/2, 1)]^{-\kappa}\rho_s(f)$. Therefore, at fixed physical time, a uniform bound for the centered scalar slab and one for its logarithmic coupling response are equivalent up to a harmless time shortening. In the small-time range $a < t \le 1$, the loss relative to $Ct^{-\kappa}$ is at most 2^{κ}.

1.3 The physical normalization and what remains unknown The plaquette coefficient convention is the one retained throughout: $s_p = N - \mathrm{Re}\,\mathrm{Tr}\,U_p$ has coefficient $\beta = 2Nx_0$, and Balaban's normalized-trace coefficient is $X_B = N\beta = 2N^2x_0$.

For a spatial profile f, the dimension-four physical normalization of a one-step coupling insertion has spatial smearing

$$a^3 \sum_{\mathrm{x}} f(\mathrm{x})\big(a^{-4}Z_a \text{ ground-centered plaquette insertion}\big) = \frac{Z_a}{a}\mathcal{O}_a(f), \tag{1.17}$$

with the finite orientation and half-step weights absorbed in the definition of the source. Ground centering is not silently identified with the chosen finite normalization prescription at μ. Let

$$\mathcal{A}_a(f) = P^{-1/2}(T_f/\lambda_0)P^{-1/2}, \qquad m_{a,V}(f) = \langle \Omega, \mathcal{A}_a(f)\Omega\rangle = -\dot{\lambda}_0/\lambda_0. \tag{1.18}$$

If the prescribed scalar insertion is $\mathcal{Q}_a^R(f) = Z_a\mathcal{A}_a(f)/a - c_{a,\mu}(f)I$, then exactly

$$\mathcal{Q}_a^R(f) = Z_a\mathcal{O}_a(f)/a + d_{a,V,\mu}(f)I, \qquad d_{a,V,\mu}(f) = Z_a m_{a,V}(f)/a - c_{a,\mu}(f). \tag{1.19}$$

The source prescription must control this remaining identity coefficient. The formulas do not establish its volume independence or equality with a periodic-state subtraction. Define $\mathfrak{h}_a^R(f) = Z_a\mathfrak{h}_{a,f} + d_{a,V,\mu}(f)I$. Let $\Theta_a(X) = \int \rho_a(s)e^{isH_a}Xe^{-isH_a}ds$ and, on finite-energy bands, $\Psi_a(X) = \int_{-1/2}^{1/2} e^{asH_a}Xe^{-asH_a}ds$. Both maps preserve the identity, so exactly $\mathfrak{h}_a^R = \Theta_a(\mathcal{Q}_a^R)$ and $\mathcal{Q}_a^R = \Psi_a(\mathfrak{h}_a^R)$. In particular the forward comparison is $\|e^{-tH_a}\mathfrak{h}_a^R e^{-tH_a}\| \le \|e^{-tH_a}\mathcal{Q}_a^R e^{-tH_a}\|$, with no additional factor a^{-1}. The inverse comparison has the time shortening in (1.16). The exact scalar response target is therefore

$$\boxed{\|e^{-tH_a}\mathfrak{h}_a^R(f)e^{-tH_a}\| \le Ct^{-\varkappa}\rho_s(f), \quad 0 < t \le 1, \quad \text{uniformly in cutoff and expanding volume.}} \tag{R}$$

This is still O. The transform proves that (R) is an alternative formulation of the scalar physical slab estimate; it does not supply the norm on its left. The microscopic power cancellation is now contained in the logarithmic response and the explicitly retained identity term, rather than proved by rewriting them. If the prescribed scheme is separately proved to agree with this ground-state centering, then $d_{a,V,\mu} = 0$; that agreement is not assumed here.

The numerical AF trajectory controls a scalar coupling recursion. It does not control arbitrary-vector matrix elements of $\mathfrak{h}_{a,f}$, nor derivatives of the transfer eigenvectors. Bounds on eigenvalues or partition functions alone would not control those off-diagonal responses. A local source theorem in an operator or boundary-kernel norm is additional information.

The definition in (1.17) covers the scalar coupling-derivative channel. It does not identify the other tensor sources with a variation through positive transfer operators, or establish their running mixing. An actual tensor estimate must retain their Wick/adjoint conventions and their marked half-step placements.

Finally, (R) is a Hamiltonian norm estimate. The periodic-state transfer issue remains separate: a ground-state norm bound cannot simply be substituted into the periodic OS quadratic inequality without controlling the exact thermal factors or proving an admissible temporal diagonal-limit argument.

2. The Schwartz Borchers domain is an energy graph core

This section fills a domain detail in the identification. It assumes that the reconstruction and the energy graph bounds have already been obtained. It does not help prove the interacting hypothesis used to obtain them.

Let $\mathcal{D}_0$ be the canonical Wightman domain generated by the full Schwartz Borchers algebra: vectors are finite sums $\Psi_n(F_n)$ with joint $F_n \in \mathcal{S}(\mathbb{R}^{4n})$, together with the vacuum. It is important to use this full domain, not merely algebraic finite sums of tensor-product tests. Put $R = 1 + H$.

Lemma 2.1 (P).

1. $\mathcal{D}_0 \subset \mathcal{D}_\infty = \bigcap_{r\ge 0} D(R^r)$.

2. If $\rho \in C_c^\infty(\mathbb{R})$, then $\rho(H)\mathcal{D}_0 \subset \mathcal{D}_0$.
3. $\mathcal{D}_0$ is a graph core for R^r for every fixed integer $r \geq 0$.

Proof. Common time translation acts on F_n by translating all its time variables. The Schwartz-valued map is C^∞. The Hilbert-valued Wightman map is continuous, because its squared norm is a tempered $(2n)$-point distribution. Thus $t \mapsto U(t)\Psi_n(F_n)$ is norm C^∞, proving the first assertion by the generator criterion.

For the second, write $\rho(H)$ as the common-time Fourier integral of the Schwartz function $\hat{\rho}$. The resulting test is

$$F_{n,\rho}(x_1, \dots, x_n) = c \int_{\mathbb{R}} \hat{\rho}(t) F_n(x_1 - te_0, \dots, x_n - te_0)\, dt, \tag{2.1}$$

which is a joint Schwartz test. The integral converges in Schwartz topology and hence in Hilbert norm. Therefore $\rho(H)\Psi_n(F_n) = \Psi_n(F_{n,\rho})$.

Choose smooth energy cutoffs ρ_M equal to one on $[0, M]$ and supported in $[-1, 2M]$. For $\psi \in D(R^r)$, spectral calculus gives $\rho_M(H)\psi \to \psi$ in the R^r graph norm. For fixed M, choose $v_j \in \mathcal{D}_0$ tending to ψ in Hilbert norm. Then $\rho_M(H)v_j \in \mathcal{D}_0$ and

$$\|R^r \rho_M(H)(v_j - \psi)\| \leq \|R^r \rho_M(H)\|\, \|v_j - \psi\| \longrightarrow 0.$$

A diagonal choice proves graph-core density. ■

Corollary 2.2 (P: domain completion). Suppose the realization agrees with the canonical fields on $\mathcal{D}_0$, and has, for each integer $r \geq 0$,

$$\|R^r \phi_\alpha(f) R^{-(r+b)}\| \leq c_{r,\alpha}\, \mathcal{N}_r(f) \tag{2.2}$$

with one finite b (the construction has $b = 2\ell$). Then:

- The closure of the field restricted to $\mathcal{D}_0$ equals the closure of its $\mathcal{D}_\infty$ realization.
- Spacelike commutativity holds on all of $\mathcal{D}_\infty$.
- Poincare covariance holds on all of $\mathcal{D}_\infty$, with the stipulated finite tensor transformation matrices.

Proof. For $\psi \in \mathcal{D}_\infty$, approximate in the graph norm of R^b by $v_j \in \mathcal{D}_0$. The $r = 0$ bound gives convergence of the field values, so $\mathcal{D}_0$ is a core for the field on $\mathcal{D}_\infty$. The reverse closure inclusion is immediate.

For a product $\phi_\alpha(f)\phi_\beta(g)$, approximate in the graph norm of R^{2b}; apply (2.2) with $r = b$ to the inner field and $r = 0$ to the outer field. Both orderings converge in Hilbert norm. Their difference vanishes on $\mathcal{D}_0$ for spacelike separated supports, so it vanishes on $\mathcal{D}_\infty$.

The forward-cone spectrum implies $U(\Lambda)^* H U(\Lambda) \leq c_\Lambda H$ in joint spectral calculus, and the inverse Lorentz transformation gives the reverse domain comparison. Thus $U(\Lambda)$ is continuous between the corresponding energy graph norms and preserves $\mathcal{D}_\infty$. Approximate there by $\mathcal{D}_0$ and pass the canonical covariance identity using (2.2). Translations commute with H and are simpler. ■

This strengthens the conclusion to the actual minimal closed canonical fields on the common smooth-energy domain. It still does not identify arbitrary other closed extensions of those operators.

3. Exact two- and three-point data do not imply the operator bound

The old constant-Gaussian example showed that excellent vacuum regularity alone need not give an H-bound. The following example shows that even preserving prescribed two- and three-point functions exactly is insufficient before controlling the center or phase transfer.

3.1 Random rescaling Let ψ_α be a nontrivial centered Wightman field family with unique vacuum Ω_0, positive Hamiltonian H_0, corrected OS bounds and a polynomial energy estimate. Assume some real smear f of a Hermitian component has $\psi(f)\Omega_0 \neq 0$ and nonzero even vacuum moments. A concrete scalar dimension-four choice is the Wick fourth power of a massive free scalar field; curvature Wick bilinears of free Maxwell fields provide another dimension-four example. No Yang–Mills claim is made for either.

Let G be an independent standard Gaussian and define

$$A = m + \sigma G, \qquad m = \frac{\sqrt{3}-1}{2}, \qquad \sigma^2 = \frac{\sqrt{3}}{2}. \tag{3.1}$$

Direct calculation gives

$$\mathbb{E}A^2 = m^2 + \sigma^2 = 1, \qquad \mathbb{E}A^3 = m^3 + 3m\sigma^2 = 1. \tag{3.2}$$

On $L^2(\mathbb{P}_A) \otimes \mathcal{H}_0$ put

$$\Phi_\alpha(f) = M_A \otimes \psi_\alpha(f), \qquad U = I \otimes U_0, \qquad H = I \otimes H_0, \qquad \Omega = 1 \otimes \Omega_0, \tag{3.3}$$

and restrict to the cyclic Wightman subspace generated by these fields.

Proposition 3.1 (P). This hierarchy has the same full two- and three-point functions as ψ, retains Euclidean covariance, symmetry and reflection positivity, and retains a fixed factorial OS growth bound. It nevertheless has no polynomial energy estimate for any component smear f with $\psi(f)\Omega_0 \neq 0$.

Proof of the hierarchy statements. Its full Euclidean moments are

$$S_m^\Phi = \mathbb{E}A^m \, S_m^\psi. \tag{3.4}$$

Centering and (3.2) give exact equality at orders two and three, including connected functions. Tensor covariance and permutation symmetry are unchanged. Reflection positivity follows by integrating the positive quadratic form for the scaled field $a\psi$ over real a. Gaussian moments satisfy

$$\mathbb{E}|A|^m \leq C^m (m!)^{1/2}, \tag{3.5}$$

so an OS bound with exponent p becomes one with exponent $p+1/2$, with the same linear derivative-order parameter. The bound can instead be rounded to any larger integer exponent.

Proof of failure of the energy bound. Let $P_0 = I \otimes |\Omega_0\rangle\langle\Omega_0|$. For each n, a field word with nonzero even vacuum moment gives

$$P_0 \Phi(f)^{2n} \Omega = c_{2n} A^{2n} \otimes \Omega_0, \qquad c_{2n} \neq 0. \tag{3.6}$$

The cyclic space is invariant under time translations and their spectral projections, so these central vectors belong to it. They also belong to the domain of the minimal closed field, rather than merely the tensor-product extension. To verify this, apply common-time heat smearing to the field word: $e^{-TH_0}\psi(f)^{2n}\Omega_0$ tends to $c_{2n}\Omega_0$ in every fixed energy graph norm. For each $T > 0$, the multiplier e^{-TE} on $E \geq 0$ has a Schwartz extension to $E \in \mathbb{R}$, so the common-time smear lies in the full Borchers domain. The base polynomial energy estimate shows convergence after applying $\psi(f)$ as well. The factor A^{2n+1} is square integrable. Thus (3.6)'s central vector is obtained in the graph closure of $\Phi(f)$.

Normalize $v_n = A^{2n} \otimes \Omega_0 / \|A^{2n}\|_2$. Then $Hv_n = 0$, while

$$\|\overline{\Phi(f)} v_n\| = \|\psi(f)\Omega_0\| \left(\frac{\mathbb{E}|A|^{4n+2}}{\mathbb{E}|A|^{4n}} \right)^{1/2} \longrightarrow \infty. \tag{3.7}$$

For an unbounded random variable with all moments, these moment ratios diverge: log-convexity makes the successive ratios nondecreasing, and a bounded limit would force the support of A^2 to be bounded. Since $(1+H)^{-r} v_n = v_n$, no finite r gives a bounded regularized field. ■

The base's two- and three-point short-distance asymptotics are preserved exactly, whatever their detailed form. For any hypothetical base with the required Yang–Mills ultraviolet coefficients, this operation would preserve those two functions while destroying the mixed-theory energy estimate. For the concrete free composite example the ultraviolet dimension is four and the three-point coefficient is nonzero, but its law is not the Yang–Mills AF law.

In the displayed conditional decomposition at fixed $A = a$, the two- and three-point coefficients become a^2 and a^3 times the base values. For both prescribed nonzero coefficients to remain unchanged with the fixed field normalization requires $a = 1$, a set of zero Gaussian measure. Thus scalar averaged coefficients do not imply an almost-everywhere or positive-measure phase transfer. This does not exclude a conceivable selected zero-weight phase at $a = 1$, and does not exclude a new phase-dependent finite normalization. Those are different claims requiring their own specification.

3.2 Additive central noise preserves all higher connected functions For a scalar field one can instead put

$$\widetilde{\Phi}(f) = I \otimes \psi(f) + M_G \otimes I \int f, \tag{3.8}$$

with centered independent Gaussian G. Independence gives

$$S^{\widetilde{\Phi}}_{2,c}(f,g) = S^{\psi}_{2,c}(f,g) + \int f \int g, \qquad S^{\widetilde{\Phi}}_{m,c} = S^{\psi}_{m,c} \quad (m \geq 3). \tag{3.9}$$

The added two-point term is smooth, so all leading singular ultraviolet coefficients are unchanged. Reflection positivity, locality, covariance and a fixed factorial OS bound are retained. The same heat-projection argument now gives all Gaussian polynomial vacuum vectors in the cyclic space, and multiplication by G defeats every global energy bound when $\int f \neq 0$.

This second example separates a stronger global energy requirement from energy bounds inside a chosen phase: each fixed-G phase has only an identity shift of the base field. A phase-first construction might avoid this particular mixed-center obstruction. It would still need an actual phase-selection theorem and a proof of the required field bounds in the selected phase; it is not supplied by the Wilson vacuum hierarchy alone.

4. Geometry conditions of the proposed time-first limit

L. CMP109, p.251, defines the torus with common side parameter $L_\mu = L^m$, with odd $L > 11$. Its symmetry discussion on p.269 explicitly restricts to Euclidean transformations preserving the chosen torus. CMP119, pp.243–244, says its geometric definitions are those in the introduction of CMP109 and the preceding papers. The available source text therefore does not provide a stated theorem with an arbitrarily long temporal side at each fixed spatial lattice.

At each fixed spatial lattice, the trace-to-vacuum comparison is valid and allows an arbitrarily long time circle. The missing point is uniform local RG and observable control on that enlarged geometry. A proof that every relevant local construction and global polymer count extends to the rectangular sequence would remove this particular geometric condition. It must also account for the lost global time-space permutation symmetries at finite rectangular volume and prove their restoration locally. Neither the mere locality of individual maps nor the cubic stability theorem by itself performs that check.

No anisotropic lattice-spacing or Hamiltonian-limit construction is introduced here. Changing the temporal lattice spacing independently would be an additional universality problem beyond changing only the torus aspect ratio.

5. Remaining operator input after the argument

The decisive estimate is still one of the following, with the actual field normalization and complete history law:

1. Directly prove the reduced periodic OS form bound, uniformly in polynomial vector degree and time-mollifier width.
2. Prove a full interacting physical-time slab norm for the renormalized tensor fields, and also justify its passage from the transfer state to the selected periodic hierarchy.

3. In the scalar coupling channel, prove (R), then use the exact response transforms (1.14)–(1.16); the periodic-state passage and tensor extension still remain.

The old local transfer comparison has a local deletion energy increment. It removes an extensive-volume loss, but after physical normalization leaves an ultraviolet power. Centering only cancels the vacuum expectation; it does not bound the logarithmic response on all low-energy vectors. The source-free pressure theorem and the numerical running coupling do not supply that bound.

A uniform source theorem useful here would have to control the normalized boundary operator or kernel, with the vacuum identity subtraction inside the derivative and a sufficient boundary norm. A scalar real pressure estimate, even combined with reflection positivity, does not do so. Nonlinear source counterterms also cannot be discarded from a moment argument. These limitations are already demonstrated by the earlier finite probability examples and are strengthened by Section 3.

The domain/core part is complete, conditional on the insertion estimate. The interacting renormalized operator inequality, its full tensor normalization and any needed phase-first alternative remain open. No part of this note proves four-dimensional Yang–Mills existence.

Primary sources and exact scope

- T. Balaban, CMP 109 (1987), 249–301, *Renormalization Group Approach to Lattice Gauge Field Theories I*: DOI. Printed p.251 gives the common-side geometry; p.269 specifies the torus-preserving symmetries.
- T. Balaban, CMP 119 (1988), 243–285, *Convergent Renormalization Expansions for Lattice Gauge Theories*: pp.243–244, for the explicit inheritance of geometric definitions. No rectangular extension is asserted.
- The positive Wilson transfer representation and bare operator conventions are those of Lüscher and Sint. The new response formulas above are proved from that representation; they are not quoted as a published interacting renormalization theorem.
- The graph-core argument is an elementary consequence of the canonical Schwartz Borchers domain, translation covariance and the stated energy graph bounds. The analytic identification it completes retains the precise OS II imports stated in the main paper.

Annex D — Finite tensor mixing and rotation constraints

Annex D. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Status: (P), finite-dimensional algebra. This classifies the one-field, dimension-four curvature-bilinear mixing problem. It does not prove rotation restoration of any Schwinger function, determine an RG coefficient, or control contact/derivative remainders.

Let $V = \mathbb{R}^4$, $E = \mathrm{Sym}^2(\Lambda^2 V)$, and $H_4 = (\mathbb{Z}_2)^4 \rtimes S_4$, the full signed-permutation group including reflections. Use formal bilinears $Q_{ij,kl} = \mathrm{Tr}(F_{ij}F_{kl})$; they are antisymmetric within each pair and symmetric between the pairs. The invariant color quadratic form is fixed. There are 21 independent tensor labels.

1. Decomposition under coordinate sign changes

For $d = \mathrm{diag}(\epsilon_1, \dots, \epsilon_4)$,

$$dQ_{ij,kl} = \epsilon_i\epsilon_j\epsilon_k\epsilon_l Q_{ij,kl}.$$

The sign characters split E into three invariant types:

1. The six squares $Q_{ij,ij}$, with trivial sign character.
2. The twelve shared-index bilinears. For each unordered pair $\{i, j\}$, there are two terms $Q_{ik,jk}$, where $k \notin \{i, j\}$, both with character $\epsilon_i\epsilon_j$.
3. The three disjoint-index terms, all with character $\epsilon_1\epsilon_2\epsilon_3\epsilon_4$.

Normality of the sign subgroup forces any equivariant endomorphism to preserve these isotypic pieces.

2. The seven irreducible summands

The six square terms: dimensions 1, 3 and 2 Their permutation representation is the action of S_4 on the edges of a tetrahedron. It decomposes as follows:

- Scalar: $S = \sum_{i<j} Q_{ij,ij}$, dimension one.
- Diagonal traceless Ricci: coefficients $q_{ij} = r_i + r_j$, with $\sum_i r_i = 0$, dimension three.
- Diagonal Weyl: $\sum_{j\neq i} q_{ij} = 0$ for every i, dimension two.

The last space consists of

$$q_{12} = q_{34} = x, \quad q_{13} = q_{24} = y, \quad q_{14} = q_{23} = z, \quad x + y + z = 0.$$

It is the standard two-dimensional representation obtained by permuting the three perfect matchings. The three-dimensional piece is the standard zero-sum permutation representation of S_4. Both are absolutely irreducible: subtracting a permutation of any nonconstant vector produces coordinate differences, whose permutation orbit spans the respective zero-sum space. The scalar is inequivalent to either.

The shared-index terms: dimensions 6 and 6 For each $\{i, j\}$, let k, l be the two complementary indices. The two lines in that sign-character space are

$$T_{ij} = Q_{ik,jk} + Q_{il,jl}, \qquad W_{ij} = Q_{ik,jk} - Q_{il,jl}.$$

The sums form the six off-diagonal Ricci components. The differences span a six-dimensional Weyl subspace. Their sign-character lines are permuted transitively by S_4. Any invariant subspace is a sum of its sign-character projections,

so a nonzero invariant subspace in either six-dimensional piece contains every line. This proves irreducibility over $\mathbb{C}$ as well as $\mathbb{R}$.

The two representations are inequivalent: in the sign-character space indexed by $\{i, j\}$, the stabilizer transposition $k \leftrightarrow l$ acts by $+1$ on T_{ij} and by -1 on W_{ij}. An intertwiner must preserve this stabilizer action.

The disjoint-index terms: dimensions 1 and 2 Set

$$p_1 = Q_{12,34}, \qquad p_2 = -Q_{13,24}, \qquad p_3 = Q_{14,23}.$$

A signed permutation acts on these three variables by its determinant times the permutation of the three perfect matchings. Hence the line $p_1 + p_2 + p_3$ is the pseudoscalar representation. The plane of zero-sum coefficients is a second two-dimensional Weyl representation. It is inequivalent to the earlier two-dimensional representation because their sign-subgroup characters differ.

Thus the seven mutually inequivalent absolutely irreducible summands have dimensions

$$1,\ 3,\ 2,\ 6,\ 6,\ 1,\ 2.$$

By Schur's lemma, every H_4-equivariant real linear mixing map on E is a scalar on each summand. Its commutant has dimension seven.

3. Full rotations impose exactly three additional equalities

Group the summands as follows:

Rotational tensor type	Hypercubic pieces	Dimension
Scalar	Scalar	1
Pseudoscalar	Pseudoscalar	1
Traceless Ricci/stress type	Diagonal 3 + off-diagonal 6	9
Weyl type	Diagonal 2 + shared-index 6 + disjoint-index 2	10

These four grouped spaces are invariant under $O(4)$. This follows directly from the coordinate-invariant trace, complete alternation, and Ricci contraction of a pair-antisymmetric, pair-symmetric tensor. On the kernel of complete alternation, remove the scalar and traceless Ricci parts; the remaining ten-dimensional kernel is the Weyl part.

The three equalities needed to commute with continuous rotations are necessary by direct differentiation. Write

$$a = F_{12},\ b = F_{13},\ c = F_{14},\ d = F_{23},\ e = F_{24},\ f = F_{34}$$

inside the symmetric invariant bilinear notation. The generator J_{12} acts by

$$J_{12}b = d, \quad J_{12}c = e, \quad J_{12}d = -b, \quad J_{12}e = -c, \quad J_{12}a = J_{12}f = 0.$$

It therefore gives nonzero links

$$J_{12}(b^2 + c^2 - d^2 - e^2) = 4(bd + ce),$$

$$J_{12}(a^2 + f^2 - b^2 - e^2) = -2(bd - ce),$$

$$J_{12}(af + be) = de - bc.$$

These connect the two stress pieces and connect both two-dimensional Weyl pieces to the shared-index Weyl piece. Consequently an equivariant mixing map with coefficients $z_\bullet$ commutes with all of $O(4)$ if and only if

$$z_{T,3} = z_{T,6}, \qquad z_{W,2d} = z_{W,6} = z_{W,2p}.$$

Necessity follows from the displayed nonzero links. Sufficiency follows because the map is then scalar on each of the four coordinate-invariant grouped spaces. The $O(4)$ commutant has dimension four.

The independent identity subtraction lies only in the scalar channel under full hypercubic covariance. It does not change the three tensor equalities above.

4. Consequences and limits for the YM route

1. In the local gauge-invariant polynomial class, the only parity-even hypercubic scalar of engineering dimension four is $\mathrm{Tr}\, F_{\mu\nu}F_{\mu\nu}$, up to normalization. The pseudoscalar changes sign under reflections. Thus there is no additional parity-even marginal bulk action anisotropy of this elementary type.
2. The one-field tensor mixing matrix still has seven hypercubic channels. Lattice symmetry does not impose the three equalities required of a rotational intertwiner.
3. The equalities classify a map between two copies of the canonical tensor representation. They do not state that a particular bare-to-renormalized lattice mixing matrix must itself commute with $O(4)$: finite counterterms may compensate anisotropic cutoff representatives. The actual requirement remains rotational covariance of the normalized limiting correlation hierarchy.
4. Even an exactly $O(4)$-equivariant marginal mixing matrix does not bound nonlocal or irrelevant rotation defects. The angular-defect estimates of the route still have to be proved.
5. The classification concerns the fixed finite family at one-field marginal order. It does not close higher source contacts, derivative vertices, or the full marked RG induction.

5. Dimensions and character orthogonality

The signed-permutation group has order 384. The seven invariant subspaces have dimensions 1, 3, 2, 6, 6, 1 and 2. In this order their character inner-product matrix is the identity. The six infinitesimal coordinate rotations impose the three compatibility relations above.

Annex E — Explicit Wilson tensor hierarchy, positivity and contact locality

Annex E. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Scope

This note completes a concrete cutoff positivity and translation architecture for the full 21-component curvature-bilinear family. It specifies the lattice fields, including an exact Wilson coupling-derivative scalar; gives a source-jet prescription compatible at every order; proves its partial-diagonal support and reflection positivity; and proves a quantitative removal of microscopic overlapping stencils on coincidence-flat tests.

There are two different conclusions.

1. P: structural P0 theorem. The explicitly defined Wilson hierarchy below has permutation symmetry, exact fine-lattice translations, the full signed-permutation tensor action, and the required reflection positivity on every fixed positive-time test algebra once its support margin fits the stencil. These statements hold for arbitrary finite running real equivariant mixing coefficients and real scalar identity subtractions. Higher point-local source contacts are compatible at all orders, preserve positivity, and vanish exactly on the coincidence-flat test space.
2. P with an explicit normalization budget: microscopic contact theorem. A coherent finite-stencil exclusion of overlapping insertion centers preserves the same positivity and symmetries. Its difference from the ordinary moments is at most
$$a\, C^{m+1} m!\, \nu_{qm}(F)$$
for fixed q, C, if the one-field normalization has a fixed polynomial cutoff bound. The theorem applies unconditionally to the specified energy-normalized, vacuum-centered cutoff family. It also applies to any eventual logarithmically running normalization for which that bound is established.

These results do not prove that the ultraviolet matching coefficients required by F0 exist in this architecture, that the prescribed normalization at μ is achieved, or that its off-diagonal correlations satisfy S0. Neither full $O(4)$ restoration nor phase/operator control is claimed. In particular, the renormalization/matching obligation is not renamed as a positivity result.

The proofs use the actual compact Wilson measure. They do not use a reflection-positive law for auxiliary RG histories and do not require source-dependent integration maps.

1. The actual finite field family

1.1 Links and the four-quadrant clover Let
$$\Lambda_{a,T} = a\mathbb{Z}^4/T\mathbb{Z}^4, \qquad T = na,$$
with even $n \geq 4$. Time is coordinate 0. The gauge group is $SU(N)$, $N \geq 2$, with normalized Haar measure and one variable on each positively oriented link. For the reverse link set $U_{yx} = U_{xy}^{-1}$. The probability law is
$$d\mu_{a,T,\beta}(U) = Z^{-1} \exp\left[-\beta \sum_p \big(N - \operatorname{Re}\operatorname{Tr} U_p\big)\right] \prod_\ell dU_\ell, \qquad \beta \geq 0. \tag{1.1}$$

The normalization dictionary remains $\beta = 2Nx_0$, and the coefficient of Balaban's normalized-trace action is $X_B = N\beta = 2N^2 x_0$. Thus the total scalar in (1.6) is the β-derivative of the specified P2 action, while its x_0-derivative is $2N$ times that scalar. No convention change is hidden in the tensor definition.

For two orthogonal signed unit vectors u, v, define the plaquette based at x by
$$P_{u,v}(x) = U_{x,x+au}\, U_{x+au,x+a(u+v)}\, U_{x+a(u+v),x+av}\, U_{x+av,x}. \tag{1.2}$$

In particular $P_{v,u}(x) = P_{u,v}(x)^{-1}$.

Let $\pi_0 H = H - \mathrm{Tr}(H)I/N$. For $0 \le \mu < \nu \le 3$, put

$$C_{\mu\nu,a}(x) = \frac{1}{8i} \sum_{\sigma,\tau=\pm1} \sigma\tau\, \pi_0\Big(P_{\sigma e_\mu, \tau e_\nu}(x) - P_{\sigma e_\mu, \tau e_\nu}(x)^\dagger\Big), \qquad C_{\nu\mu,a} = -C_{\mu\nu,a}. \tag{1.3}$$

Each $C_{\mu\nu,a}$ is traceless Hermitian. All plaquette loops are based at the same vertex; no comparison of Lie algebra variables in different gauge frames is implicit.

Let $\mathcal{I} = \{(\mu,\nu) : 0 \le \mu < \nu \le 3\}$, so $|\mathcal{I}| = 6$. The real symmetric matrix

$$q_{AB,a}(x) = a^{-4}\,\mathrm{Tr}\big(C_{A,a}(x) C_{B,a}(x)\big), \qquad A, B \in \mathcal{I}, \tag{1.4}$$

has 21 independent components. Its trace will be denoted

$$t_a(x) = \sum_{A\in\mathcal{I}} q_{AA,a}(x).$$

The matrix entries are real: for Hermitian C_A, C_B, $\overline{\mathrm{Tr}(C_A C_B)} = \mathrm{Tr}(C_B C_A) = \mathrm{Tr}(C_A C_B)$. The matrix $q(x)$ is positive semidefinite as a Gram matrix, although positivity of individual field values is not needed for OS positivity.

1.2 Retain the exact Wilson scalar channel Define the vertex action density

$$w_a(x) = \frac{1}{4a^4} \sum_{\mu<\nu} \sum_{\sigma,\tau=\pm1} \Big[N - \mathrm{Re}\,\mathrm{Tr}\, P_{\sigma e_\mu, \tau e_\nu}(x)\Big]. \tag{1.5}$$

Every unoriented elementary plaquette occurs once at each of its four vertices. Its trace action is unchanged by orientation reversal and by moving the root around the plaquette. Hence, exactly,

$$\boxed{a^4 \sum_{x\in\Lambda_{a,T}} w_a(x) = \sum_p s(U_p).} \tag{1.6}$$

More generally, for any vertex profile f,

$$\boxed{a^4 \sum_x f(x) w_a(x) = \sum_p s(U_p)\, \frac{1}{4} \sum_{v\in \mathrm{Vert}(p)} f(v).} \tag{1.7}$$

Thus the scalar source has the explicit four-corner plaquette coefficient. For $f = 1$, it is the exact total coupling insertion $\partial_\beta(\beta S)$. A source factor $e^{\lambda a^4 \sum f w_a}$ changes the Wilson plaquette coefficient to $\beta - \lambda \frac{1}{4} \sum_{v\in p} f(v)$; this fixes the sign convention.

To keep this scalar together with the other twenty tensor directions, set

$$\boxed{\tilde q_{AB,a}(x) = q_{AB,a}(x) + \frac{\delta_{AB}}{6}\,[\,2w_a(x) - t_a(x)\,].} \tag{1.8}$$

Then

$$\frac{1}{2} \sum_A \tilde q_{AA,a}(x) = w_a(x). \tag{1.9}$$

Only the scalar component is changed. The traceless Ricci/stress, pseudoscalar and Weyl components remain those of the clover bilinear.

For a fixed smooth classical connection, plaquette expansion gives

$$C_{\mu\nu,a}(x) = a^2 F_{\mu\nu}(x) + O(a^3),$$

$$w_a(x) = \frac{1}{2}\sum_{\mu<\nu} \mathrm{Tr}\big(F_{\mu\nu}(x)^2\big) + O(a), \qquad \tilde{q}_{AB,a}(x) = \mathrm{Tr}(F_A F_B)(x) + O(a). \tag{1.10}$$

The conservative errors shown here suffice to identify the engineering tensor type; the symmetric stencil allows sharper classical errors. This is a classical consistency calculation, not a theorem of quantum universality.

There is also a useful exact bound. For a unitary matrix P, let $H(P) = (P - P^\dagger)/(2i)$. Spectral calculus gives

$$\|H(P)\|_{\mathrm{HS}}^2 = \sum_j \sin^2\theta_j \le 2\sum_j (1-\cos\theta_j) = 2s(P).$$

Since (1.3) is the average of four signed $\pi_0 H(P)$'s,

$$\|C_{A,a}(x)\|_{\mathrm{HS}}^2 \le \frac{1}{2}\sum_{\sigma,\tau} s(P_{\sigma A_1,\tau A_2}(x)).$$

Summing over A proves

$$0 \le t_a(x) \le 2w_a(x). \tag{1.11}$$

Consequently $\tilde{q}(x)$ is also positive semidefinite and has trace $2w_a(x)$. In particular,

$$|\tilde{q}_{AB,a}(x)| \le 2w_a(x). \tag{1.12}$$

1.3 Exact gauge, translation and hypercubic transformations A gauge transformation sends every plaquette in (1.2) to $\Omega_x P_{u,v}(x)\Omega_x^{-1}$. Therefore $C_A(x)$ transforms by conjugation at x, and every component of $\tilde{q}(x)$ is gauge invariant.

Let $g \in H_4 = (\mathbb{Z}_2)^4 \rtimes S_4$, and write

$$ge_\mu = \varepsilon_\mu e_{\pi(\mu)}.$$

Use the pullback convention $(gU)_{xy} = U_{gx,gy}$. Relabeling the four quadrants in (1.3) gives exactly

$$C_{\mu\nu,a}(x; gU) = \varepsilon_\mu\varepsilon_\nu C_{\pi(\mu)\pi(\nu),a}(gx; U), \tag{1.13}$$

with the displayed antisymmetry used to reorder a pair. The action density obeys $w_a(x; gU) = w_a(gx; U)$. Thus (1.8) transforms in the canonical representation

$$E = \mathrm{Sym}^2(\Lambda^2\mathbb{R}^4).$$

Denote its matrix by $D(g)$. In an orthonormal component basis,

$$\tilde{q}_a(x; gU) = D(g)\tilde{q}_a(gx; U). \tag{1.14}$$

Translations by $b \in a\mathbb{Z}^4$ obey the analogous identity with no tensor matrix. All these identities are exact on the torus. The Wilson measure is invariant under the corresponding change of link variables, since it preserves Haar measure and permutes unoriented plaquettes.

Each field depends only on plaquettes incident on x, hence on links with endpoints in $x + [-a, a]^4$. The same support statement holds after (1.8). This is the actual support margin used below.

1.4 Running one-field mixing Choose, at each cutoff, a finite real matrix Z_a commuting with every $D(g)$, and a finite real vector $c_a \in E^{H_4}$. Define

$$\Phi_{\alpha,a}(x) = \sum_\beta (Z_a)_{\alpha\beta}\tilde{q}_{\beta,a}(x) - (c_a)_\alpha. \tag{1.15}$$

No uniform bound on Z_a, Z_a^{-1} or c_a is assumed for the positivity theorem. An inverse is not needed. The seven real hypercubic channel coefficients from the classification give an explicit parametrization of Z_a. The invariant vector c_a is the scalar identity-mixing direction.

The coefficients are independent of x. Their normalization at μ, cutoff dependence and permitted volume dependence are questions for F0; the positivity theorem does not select their values. Its conclusions hold in particular for any prescribed $Z(a,\mu)$ and $c(a,\mu)$ that meet the displayed reality and symmetry conditions.

Equations (1.14) and the exact lattice translation identity continue to hold for Φ.

2. Reflection positivity proved directly on the Wilson integral

2.1 Site-reflection factorization Let $\theta(x_0, \mathrm{x}) = (-x_0, \mathrm{x})$ modulo T. The two fixed planes are $x_0 = 0$ and $x_0 = T/2$. Decompose the link variables into:

- B: spatial links lying in the two fixed planes;
- U_+: all other links in the closed positive half-torus;
- U_-: their reflected partners in the negative half.

Every elementary plaquette lies entirely in one closed half or entirely in a fixed plane. The action therefore splits as

$$S(U) = S_+(U_+, B) + S_-(U_-, B) + S_0(B),$$

and reflection identifies the two first terms. Haar measure has the corresponding product decomposition. For every bounded complex function $A(U_+, B)$,

$$\begin{aligned} \mathbb{E}_\mu[\overline{A(\theta U)}A(U)] = Z^{-1} \int dB\, e^{-\beta S_0(B)} \\ \times \left| \int dU_+\, e^{-\beta S_+(U_+,B)} A(U_+, B) \right|^2 \geq 0. \end{aligned} \tag{2.1}$$

This proof uses the original unconstrained compact link integral. It introduces no gauge fixing or signed activity law. It is the site-reflection property needed here; no assertion about an independently defined auxiliary-history reflection is made.

2.2 Tensor reflection and Wick phases Let $r = \mathrm{diag}(-1, 1, 1, 1)$. In the real tensor-component basis,

$$D(r)_{\alpha\alpha} = (-1)^{n_0(\alpha)},$$

where $n_0(\alpha) \in \{0, 1, 2\}$ counts the occurrences of the time index in the two antisymmetric index pairs. Thus

$$\overline{\Phi_\alpha(x; \theta U)} = \sum_\beta D(r)_{\alpha\beta} \Phi_\beta(\theta x; U). \tag{2.2}$$

For a component test kernel F_m, define ΘF_m by reflecting all arguments, complex conjugating its coefficients, applying $D(r)$ to each component index, and reversing their order. The reversal makes it the usual Borchers involution; the cutoff random variables themselves commute.

If a Hermitian Wick component convention is desired, put

$$\widehat{\Phi}_\alpha = i^{n_0(\alpha)} \Phi_\alpha. \tag{2.3}$$

The antilinearity in (2.2) cancels the reflection sign:

$$\overline{i^{n_0(\alpha)}} (-1)^{n_0(\alpha)} = i^{n_0(\alpha)}.$$

In that basis the antilinear reflection has no remaining component sign. General Euclidean tensor transformations are conjugated by this fixed Wick matrix. One must use a single convention throughout the hierarchy; a real Euclidean tensor component with an odd number of time indices is not assigned the wrong scalar reflection rule.

2.3 The full cutoff hierarchy For a finite joint test vector $F = (F_0, F_1, \dots, F_M)$, let

$$\mathcal{A}_a(F; U) = \sum_{m=0}^{M} a^{4m} \sum_{x_1,\dots,x_m} \sum_{\alpha_1,\dots,\alpha_m} F_{m;\alpha}(x_1, \dots, x_m) \prod_{j=1}^{m} \Phi_{\alpha_j, a}(x_j). \tag{2.4}$$

At finite cutoff the sum is finite. For continuum tests use their interior torus embedding, specified in Section 6.

Suppose the center supports of all nonzero F_m lie in

$$\delta < x_{j,0} < T/2 - \delta$$

for a fixed $\delta > 0$, and $a < \delta$. The actual link support of every term in (2.4) then lies in the positive half. Equation (2.1) proves

$$\boxed{\sum_{r,s} S_{r+s,a,T}(\Theta F_r \otimes F_s) = \mathbb{E}_\mu[\overline{\mathcal{A}_a(F;\theta U)}\mathcal{A}_a(F;U)] \geq 0.} \tag{2.5}$$

This is a statement about the full moments, for arbitrary polynomial degree and arbitrary joint test kernels. It is not an estimate only on connected derivatives or on factorized tests. Permutation symmetry, exact lattice translations and hypercubic covariance follow directly from their proved field/measure identities.

3. A concrete all-order point-contact prescription

3.1 Formal source maps Use smooth compactly supported sources $j : \mathbb{R}^4 \to E$, sampled together with their derivatives at lattice vertices. This is a permitted cutoff source prescription: source jets are evaluated at a point even though the marked field has a neighboring plaquette support.

Put

$$\mathcal{L}_a(j;U) = a^4 \sum_x \langle j(x), \Phi_a(x;U)\rangle.$$

Choose formal coefficient sequences, compatible under truncation in source degree,

$$R_a[j](x) = j(x) + \sum_{r\geq 2} \frac{1}{r!} T_{a,r}\big(j(x), \dots, j(x)\big), \tag{3.1}$$

and

$$C_a[j] = a^4 \sum_x \sum_{r\geq 2} \frac{1}{r!} \sum_{\sum_i |\gamma_i| \leq 4} c_{a,r}^{\gamma_1,\dots,\gamma_r}\big(\partial^{\gamma_1} j(x), \dots, \partial^{\gamma_r} j(x)\big). \tag{3.2}$$

Here $T_{a,r} : \mathrm{Sym}^r E \to E$ is real and H_4-equivariant. The real coefficients of C_a are invariant for the simultaneous tensor and source-derivative action, and are symmetrized in the pairs of source labels and derivative indices. All coefficients are independent of x.

The derivative order four in (3.2) is the vacuum contact degree in the dimension-four power-counting class. Dimension-four operator-valued outputs in (3.1) carry no source derivatives. The formulas do not assert that these are the only outputs of the interacting induction: irrelevant vertices remain in that induction. They specify the local relevant/contact prescription used for the external hierarchy.

Define, as a formal source jet,

$$G_a[j;U] = \exp\{\mathcal{L}_a(R_a[j];U) - C_a[j]\}, \qquad \mathcal{M}_a^{R,C}[j] = \mathbb{E}_\mu G_a[j;U]. \tag{3.3}$$

At any finite source order M, truncate all series above M. Only finitely many coefficients and finite cutoff moments occur. The logarithm/exponential composition is algebraic; no cutoff-independent complex source disc or convergence of the infinite source series is assumed. The constant coefficient is one.

The one-field coefficient is exactly the field (1.15), since $R_a[j] = j + O(j^2)$ and $C_a[j] = O(j^2)$. Its identity subtraction is not confused with a higher-source contact subtraction.

3.2 Explicit positivity and triangularity For two sources supported in disjoint time half-spaces,

$$R_a[j_- + j_+] = R_a[j_-] + R_a[j_+], \qquad C_a[j_- + j_+] = C_a[j_-] + C_a[j_+]. \tag{3.4}$$

All derivatives of the absent source vanish at every root in the other support. Consequently

$$G_a[j_- + j_+;U] = G_a[j_-;U]G_a[j_+;U] \tag{3.5}$$

coefficient by coefficient. Reality and equivariance give

$$G_a[\Theta j; U] = \overline{G_a[j; \theta U]}. \tag{3.6}$$

For a positive-time polynomial test vector F, extract its finite coefficients from G_a to obtain a bounded positive-half-link function $\mathcal{J}_a F$. This is a triangular map: the degree-m coefficient has leading term the ordinary product of m fields (1.15), while every nonlinear R or C vertex lowers the number of field factors. The finite source degree ensures that the map is defined even if the full source series diverges.

Equations (3.5)–(3.6) and the Wilson square formula (2.1) prove

$$\boxed{\sum_{r,s} S^{R,C}_{r+s,a,T}(\Theta F_r \otimes F_s) = \mathbb{E}_\mu[\overline{(\mathcal{J}_a F)(\theta U)}(\mathcal{J}_a F)(U)] \geq 0.} \tag{3.7}$$

The same proof works for joint test kernels. At finite cutoff, coefficient extraction produces finite sums of values and derivatives of those kernels on partial diagonals. Thus the polynomial map extends directly; an unproved nuclearity passage is not being used for this finite-cutoff statement.

3.3 Composition closes in the same class Let $\mathfrak{T}_{R,C}$ act on a normalized formal functional by

$$(\mathfrak{T}_{R,C}\mathcal{M})[j] = e^{-C[j]}\mathcal{M}[R[j]].$$

Then

$$\boxed{\mathfrak{T}_{R_2,C_2}\mathfrak{T}_{R_1,C_1} = \mathfrak{T}_{R_1\circ R_2,\, C_2 + C_1\circ R_2}.} \tag{3.8}$$

Since R_2 has no source derivatives, differentiating $R_2[j]$ in a monomial of total derivative order at most four leaves the total derivative order at most four. Thus $C_1 \circ R_2$ is again of the form (3.2). The composed R is again point-local, equivariant and real, with identity linear term. The formal inverse also exists:

$$(R, C)^{-1} = (R^{-1}, -C \circ R^{-1}).$$

No estimate for a uniformly bounded inverse mixing matrix is involved.

This proves all-order compatibility for repeated local renormalization at the same external source point. It does not identify a finite-size carrier or a coarse source stencil with that point without a remainder estimate; such an identification remains part of F0/M0.

3.4 Symmetry can be imposed at the coefficient level The finite-group Reynolds map gives an explicit projection onto the permitted source coefficients:

$$T^{\mathrm{sym}}_{a,r}(v_1, \dots, v_r) = \frac{1}{|H_4|} \sum_{g \in H_4} D(g)^{-1} \widetilde{T}_{a,r}(D(g)v_1, \dots, D(g)v_r). \tag{3.9}$$

Apply the analogous invariant projection to the order-four source-jet tensors of C_a, and take real coefficients. The group acts isometrically on the finite component/derivative spaces, so these projections do not increase the corresponding multilinear operator norm. They commute with truncation in source degree.

This is a coefficient-level construction. Averaging an arbitrary nonsymmetric renormalized hierarchy is not claimed to preserve its ultraviolet normalization. The one-field coefficients in (1.15) are chosen translation independent and equivariant from the outset.

4. Exact contact support on the required test space

Let

$$\Delta_m = \bigcup_{i<j} \{x_i = x_j\}, \qquad \mathcal{S}_0(\mathbb{R}^{4m}) = \{F \in \mathcal{S} : \partial^\alpha F|_{\Delta_m} = 0 \text{ for every } \alpha\}. \tag{4.1}$$

Theorem 4.1 — exact partial-diagonal support At fixed cutoff and fixed source order m, the difference between the hierarchy (3.3) and the ordinary full moments of (1.15) is a finite-order distribution supported on partial coincidence diagonals. Its spatial derivative order is at most $2m$. For compact tests embedded without torus seam identifications,

$$\boxed{S^{R,C}_{m,a,T}(F) = S^{\Phi}_{m,a,T}(F) \quad \text{for every } F \in \mathcal{S}_0.} \tag{4.2}$$

Proof. In a term containing a higher R vertex of source degree $r \geq 2$, all r differentiated source slots are evaluated at one lattice root x. Hence their external source coordinates coincide. The same is true for a higher C vertex, possibly with source derivatives. The remaining field/source factors can be at arbitrary other roots.

Thus every non-baseline term contains a block B with $|B| \geq 2$, supported on $\{x_i = x_j : i, j \in B\}$. Different contact vertices use disjoint source-label blocks in the coefficient extraction. A C vertex differentiates its sources to total order at most four, and there are at most $\lfloor m/2 \rfloor$ such vertices. Higher R vertices add no spatial derivative. The total distribution order is therefore at most $4\lfloor m/2 \rfloor \leq 2m$.

All these derivatives annihilate $F \in \mathcal{S}_0$ on the indicated partial diagonal. Finite cutoff sums and finite source order justify this termwise, without any coefficient bound. ■

The statement is exact even if the contact coefficients diverge with the cutoff or grow rapidly with source order: on the required flat test space their contribution is identically zero at each order. Such growth could matter for an extension to non-flat Euclidean product tests, but no such extension is silently included in this theorem.

The one-field identity term c_a, and the one-field matrix Z_a, are excluded from this cancellation: they affect separated moments and remain inside the ultraviolet/operator estimates.

Point-local source derivatives and field support are different notions here. The field at root x uses neighboring links, but all external source slots of a higher local vertex are evaluated at x. Its source contact is therefore exact, despite its larger gauge-variable carrier.

5. A coherent microscopic exclusion with a quantitative contact error

This section deals with a shrinking finite stencil, rather than an exact point contact. It supplies an additional way to remove microscopic overlapping insertions while retaining positivity.

5.1 The finite source polynomial Fix $R > 0$. Put an edge between distinct vertices x, y when their periodic Euclidean distance satisfies $d_T(x, y) \leq Ra$. Let $\mathfrak{F}_{a,T,R}$ be the independent subsets of this finite graph. Define

$$\mathcal{G}_{a,R}[j; U] = \sum_{I \in \mathfrak{F}_{a,T,R}} \prod_{x \in I} \left[a^4 \langle j(x), \Phi_a(x; U) \rangle \right]. \tag{5.1}$$

This is one finite polynomial, with constant coefficient one, so all its source orders are compatible. Its m-th full source coefficient is

$$S^{\mathrm{sep}}_{m,a,T}(F) = a^{4m} \sum_X 1_{\{d_T(x_i,x_j) > Ra \ \forall i<j\}} \mathbb{E}_\mu \left[\prod_i \Phi_{\alpha_i,a}(x_i) \right] F_\alpha(X), \tag{5.2}$$

summed over component labels as usual. Repeated vertices are excluded. For example, any $R > 2\sqrt{2}$ excludes overlapping clover link supports.

For positive/negative source supports with a fixed time margin δ, their periodic separation is at least 2δ, including the second reflection plane. Once $Ra < 2\delta$, no graph edge joins these supports. Every independent set therefore splits uniquely into its positive and negative parts:

$$\mathcal{G}_{a,R}[j_- + j_+; U] = \mathcal{G}_{a,R}[j_-; U] \mathcal{G}_{a,R}[j_+; U]. \tag{5.3}$$

The graph is reflection and translation invariant, and the coefficients obey the tensor reflection rule. Equations (2.1) and (5.3) prove reflection positivity of the entire full hierarchy (5.2), on every fixed positive-time polynomial algebra for all sufficiently fine cutoffs. This is not an assertion that (5.1) is a probability Laplace transform.

For joint kernels the same statement is especially transparent: the mask for the combined reflected and positive test factors into the separate internal masks, because every cross-half pair has distance greater than Ra. The OS form is the Wilson Gram form of the masked polynomial test vector.

Permutation symmetry, fine-lattice translations and the full hypercubic tensor covariance are exact. One can also apply the point-contact transformation (3.3) to this source polynomial; the same splitting proof and exact flat-test identity remain valid.

5.2 The actual raw moment budget Here $\beta > 0$, as on the admitted AF sequence.

Let X_x be the fixed set of plaquettes incident on x, and let

$$E_{X_x} = \beta \sum_{p \in X_x} s(U_p).$$

The Wilson reserve gives

$$\|E_{X_x}\|_{L^p(\mu)} \le Cp, \qquad p \ge 1, \tag{5.4}$$

uniformly along the admitted cofinal sequence. The number of covering cubes for X_x is a fixed geometric constant.

Equation (1.12) now gives

$$\|\tilde{q}_{\alpha,a}(x)\|_p \le Ca^{-4}\beta^{-1}p. \tag{5.5}$$

Suppose the prescribed one-field coefficients satisfy, for some fixed $A, \zeta \ge 0$,

$$\boxed{\beta^{-1} \max_\alpha \sum_\gamma |(Z_a)_{\alpha\gamma}| + a^4 \max_\alpha |(c_a)_\alpha| \le Aa^{-\zeta}.} \tag{5.6}$$

Then, with $b = 4 + \zeta$,

$$\|\Phi_{\alpha,a}(x)\|_p \le CA\, p\, a^{-b}. \tag{5.7}$$

The hypothesis includes fixed powers of logarithmic running after choosing any fixed positive ζ and enlarging the constant. It is not inferred for an as-yet-unconstructed F0 normalization.

A completely specified unconditional instance is $Z_a = \beta I$ and $c_a = \mathbb{E}_\mu[\beta \tilde{q}_a(x)]$. Translation and hypercubic invariance make this vector independent of x and scalar. Equations (5.4)–(5.5) prove (5.6) with $\zeta = 0$ for that finite-cutoff family. This choice is not declared to be the final μ-normalization or to have the prescribed Yang–Mills short-distance coefficients.

Theorem 5.1 — all-order vanishing of the microscopic collision strip Assume (5.7), with fixed b, A. Choose an integer

$$q \ge \max\{6, \lceil b \rceil + 2\}.$$

Use the interior torus embedding in Section 6. There is a constant C, independent of m, a, T along the sequence, such that for every coincidence-flat Schwartz test,

$$\boxed{\left|S^{\Phi}_{m,a,T}(F) - S^{\mathrm{sep}}_{m,a,T}(F)\right| \le a\, C^{m+1} m!\, \nu_{qm}(F).} \tag{5.8}$$

For $m = 0, 1$ the difference is zero. The statement holds for every fixed R, with C depending on R. Point-contact transformations of Section 3 can be applied on either side without changing (5.8) on the flat test space.

Proof. Hölder with exponent m and (5.7) give

$$\left|\mathbb{E} \prod_{i=1}^m \Phi_{\alpha_i,a}(x_i)\right| \le (CA)^m m^m a^{-bm} \le (eCA)^m m! a^{-bm}. \tag{5.9}$$

The difference in (5.8) is supported where at least one pair has periodic separation at most Ra. Within the central embedding this is an ordinary Euclidean separation. Choose such a nearest pair, and let Y be the point obtained by replacing that pair by its midpoint. Then $Y \in \Delta_m$, and $V = X - Y$ satisfies

$$\|V\|_1 \le 2Ra, \qquad |V|^2 \le R^2 a^2/2, \qquad Y \perp V.$$

Take $r_m = \lfloor bm \rfloor + 1$. Since $m \geq 2$, the chosen q gives $r_m \leq qm$. Flatness and the directional Taylor remainder give

$$|F(X)| \leq \frac{(2Ra)^{r_m}}{r_m!} C_R^{qm/2} \nu_{qm}(F)(1 + |X|^2)^{-qm/2}. \tag{5.10}$$

Here the weight comparison along $Y + tV$ follows from orthogonality:

$$\frac{1 + |X|^2}{1 + |Y + tV|^2} \leq 1 + R^2 a^2/2 \leq 1 + R^2/2$$

for $a \leq 1$. The power a^{-bm} in (5.9) leaves at least one power of a after multiplication by (5.10).

Finally,

$$(1 + \textstyle\sum_i |x_i|^2)^{-qm/2} \leq \prod_i (1 + |x_i|^2)^{-q/2},$$

and for $q > 4$,

$$a^4 \sum_{x \in a\mathbb{Z}^4} (1 + |x|^2)^{-q/2} \leq C_q.$$

Thus the full lattice sum contributes at most C_q^m. The 21^m component sum is exponential and is absorbed into C^{m+1}. The powers of R and the weight-comparison constant are likewise exponential in m; the Taylor factorial need not even be used to improve the result.

The interior cutoff factors are kept bounded by one throughout this estimate; Taylor expansion is applied to F, not to those factors. No derivative of the torus-embedding cutoff is taken. This proves (5.8). ■

The theorem removes the entire microscopic collision strip with one fixed linear test order and one fixed factorial exponent. It is stronger than proving vanishing separately at each source order with arbitrary constants. It still says nothing about the separated correlation kernel outside that strip; the full observable/matching estimate remains necessary there.

6. Torus embeddings and continuum inheritance

6.1 Avoiding the seam Choose smooth cutoffs $0 \leq \chi_T \leq 1$, invariant under H_4, equal to one on a growing central box and supported in

$$(-T/4, T/4)^4.$$

For a joint test F_m, use

$$F_{m,T}(X) = \prod_{i=1}^{m} \chi_T(x_i)\, F_m(X) \tag{6.1}$$

in the lattice sums, with the ordinary central representatives of the torus sites. Every fixed compact test is unchanged for sufficiently large T. On this support, each coordinate difference has absolute value less than $T/2$, so periodic distance and ordinary Euclidean distance agree.

Consequently a periodic-near pair in Theorem 5.1 is genuinely near a Euclidean coincidence diagonal. Without this embedding, two points near opposite torus seams would not be controlled by coincidence-flat Taylor expansion alone. No extra relation between T and a^{-1} is used.

The products in (6.1) preserve flatness. Exact point contacts therefore vanish as in Theorem 4.1, including those with source derivatives. For (5.8), the proof uses only $0 \leq \chi_T \leq 1$, so it incurs no derivative-order or factorial cost from the embedding.

6.2 Translation covariance of any controlled limit Let a subsequence of one of these hierarchies converge on $\mathcal{S}_0$, with the common Schwartz seminorm bounds required by S0. For a compact test F and a fixed translation $b \in \mathbb{R}^4$, choose $b_a \in a\mathbb{Z}^4$ with $b_a \to b$. Once the original and translated supports lie where $\chi_T = 1$, exact lattice translation covariance gives

$$S_{m,a,T}(\tau_{b_a} F) = S_{m,a,T}(F). \tag{6.2}$$

Continuity of translations in the common Schwartz seminorm gives

$$S_{m,a,T}(\tau_b F - \tau_{b_a} F) \longrightarrow 0.$$

Pass to the limit and then approximate a general flat Schwartz test by compact flat tests. The limit is translation covariant. Thus no separate physical-cell-grid translation defect remains in this specified hierarchy.

The common seminorm bound is part of the still-open S0 estimate. This section proves the symmetry implication once that estimate is available; it does not derive S0 from positivity.

6.3 Positivity and the finite symmetry group in the limit First take a finite OS vector of compact coincidence-flat kernels supported strictly inside positive time. There is then a common positive support margin. Its combined reflected test kernels are still coincidence-flat: internal collisions are killed by flatness, and cross-half collisions are excluded by that margin. For all sufficiently fine lattices and large tori, the stencil and, if used, the exclusion graph fit the margin. Equations (2.5), (3.7) or (5.3) give nonnegative cutoff OS forms. Their finite sums of convergent Schwinger evaluations have a nonnegative limit.

A general positive-time Schwartz kernel need not have a strictly positive support margin. Approximate it by multiplying by a compact cutoff and by smooth cutoffs that vanish near every time boundary. A smooth function supported in the closed positive-time region is flat at those boundaries; its weighted derivatives therefore make this approximation converge in each required Schwartz seminorm. Multiplication by these cutoffs also preserves coincidence-flatness. The assumed common S0 bound makes the finite OS quadratic form continuous in this approximation. Its nonnegativity extends to the full positive-time Schwartz algebra. This step uses the separately required S0 estimate; it does not assert uniform cutoff positivity for tests whose stencils cross the reflection plane.

Permutation symmetry and the full signed-permutation tensor covariance pass directly. The correct Wick phases are fixed once, before taking the limit. Full continuous rotations remain a separate defect theorem; no rotation average is performed.

The exact source-free positive transports may be used to calculate these cutoff moments. Since they preserve every marked integral, their fixed cell grids and selectors do not change the symmetry identities of the original hierarchy. They need not themselves form a reflection-positive random process.

7. Completion conditions for P0 and F0

Statement	Status
Explicit actual 21-component Wilson fields	P: vertex clovers with the scalar trace replaced by the exact Wilson action density
Exact one-field gauge covariance, translations and hypercubic tensor transformations	P
Reflection positivity of the full finite-cutoff moment hierarchy	P on every fixed positive-time test algebra once its support margin fits the stencil
Running equivariant one-field mixing and scalar identity subtraction preserve these properties	P for arbitrary finite coefficients at each cutoff
Compatible local source jets at every order	P: one formal functional and its composition law
Exact partial-diagonal support and vanishing on the corrected OS test space	P; no contact-coefficient norm is needed for exact point contacts
Microscopic finite-stencil collision removal with fixed factorial/linear-order bounds	P under (5.6), and instantiated unconditionally for the stated energy-normalized vacuum-centered cutoff family
Positivity, translations and the finite tensor symmetry group of a controlled continuum limit	P implication using the separately required convergence/OS seminorm bound
The ultraviolet renormalization coefficients of the actual RG induction belong to this architecture and meet the prescribed μ-conditions	O: this is F0/M0, not proved by the algebra
Current-scale transported-field bounds, full history control and the off-diagonal hierarchy	O
Full $O(4)$ restoration, quantitative AF identification and phase/operator control	O

P0 is realized by one concrete definition on the actual Wilson measure. It is unnecessary to leave reflection positivity as an unspecified property of a future global source normal form. The remaining F0 problem is to prove that the normalization and local extractions required by the observable induction can be implemented in this explicit class, with their errors controlled. Point-contact positivity itself then requires no additional estimate.

Source and proof dependencies

The compact Wilson action and the reduced finite tensor target are those of the main paper. The exact source-free transport interpretation and the coincidence-flat OS test space are proved or scoped in the other annexes. The uniform raw energy reserve used only in Section 5.2 is the theorem of Annex A, with its cofinal geometric condition retained.

The site-reflection square formula, explicit field covariance, Wilson scalar identity, formal source composition, partial-diagonal statement and finite-stencil positivity/error theorem are proved in this note. No unproved observable theorem, phase decomposition or unpublished Yang–Mills construction is imported.

Annex F — Conditional means, native moments and Gaussian comparison

Annex F. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Conditional moment estimate

The full transported-observable matching theorem M0 remains open. This note closes a quantitative premise that had been left explicit in the native moment theorems: the local mean of the nonlinear conditional fluctuation field. It also proves a volume-uniform, all-order comparison of the native marked cumulants with their constrained truncated-Gaussian counterparts.

Both conclusions apply to the specified homogeneous, fixed-domain, no-R regular integral with the complete inherited regular E action and the new coarea factor retained. They do not apply to an arbitrary CMP119 multiscale history merely because its current large-field set is empty. In particular, the fixed-domain CMP109/CMP116 activity norm used below must not be transplanted to a changed CMP119 domain with a different activity prefactor.

The labels are P for a result proved from the displayed source-specific inputs, L for those scoped primary inputs, and O for an estimate not proved for the full Wilson/RG law. No novelty or independent formal-validation claim is made.

The new mean estimate is stronger than substituting the cutoff into the field bounds:

$$\boxed{\sup_i |\mathbb{E}_\nu z_i| \le Cg, \qquad \sup_i \|z_i\|_{L^p(\nu)} \le C\sqrt{p}, \quad p \ge 1.} \tag{0.1}$$

It holds uniformly in volume and in every admissible common symmetric product cutoff P with $gP \le \delta$, where $\delta > 0$ is fixed and sufficiently small. No auxiliary condition $gP^2 \to 0$ is needed. The actual logarithmic window is included.

For a native field at physical scale h, this yields

$$\boxed{\|O_{\alpha,x}\|_p \le Ch^{-4}(B_x + \sqrt{p})^2,} \tag{0.2}$$

$$\boxed{\|O_{\alpha,x} - \mathbb{E}O_{\alpha,x}\|_p \le Ch^{-4}\big[(1+B_x)\sqrt{p} + p\big], \qquad \|\|\nabla O_{\alpha,x}\|_{\gamma,x}\|_p \le Ch^{-4}(B_x + \sqrt{p}).} \tag{0.3}$$

Here B_x is the actual reference-curvature bound divided by g, with the fixed local enlargement required by the stencil. It has not been bounded in the outer history law. If the admitted background has $B_x \le CP(g)$, the resulting conditional bounds have the corresponding explicit logarithmic background factor. A fixed physical regularity bound alone permits B_x to grow like g^{-1}.

1. The actual conditional density and its source inputs

Let the free-coordinate block set $\mathcal{I}$ be a finite homogeneous current lattice or torus. Each block has dimension bounded by a fixed number, and the current metric satisfies

$$\sup_i \sum_j e^{-cd(i,j)} < \infty \quad (c > 0) \tag{1.1}$$

uniformly in volume. The current lattice has dimension four. Group and admitted block-size constants are fixed.

The domain is a common symmetric convex product

$$\mathcal{D}_P = \prod_{i\in\mathcal{I}} \{z_i \,:\, |z_i| \le P\}. \tag{1.2}$$

Scalar coordinate cubes or the source's finite-dimensional norm balls are allowed. Every norm used on an individual block is fixed, so their equivalence constants are independent of volume. The domain and selector are independent of the interpolation parameter introduced below.

The reference field, exact averaging chart and gauge are frozen. Let

$$d\nu(z) = Z^{-1} 1_{\mathcal{D}_P}(z) J(gz) \exp\{-S_g(z) + E_k(U(gz))\}\, dz. \tag{1.3}$$

The classical action is the exact source-free constrained action, including the particular gauge-fixing term if that version is used. Its stationary point is at $z = 0$. Write

$$H_0 = D^2 S_g(0), \qquad R_{\mathrm{cl}}(z) = S_g(z) - S_g(0) - \tfrac{1}{2}\langle z, H_0 z\rangle. \tag{1.4}$$

Thus $DR_{\mathrm{cl}}(0) = D^2 R_{\mathrm{cl}}(0) = 0$. The term E_k is the full inherited regular action after its coupling subtractions, so the flowed classical coefficient in H_0 is not counted a second time. The new coarea factor J occurs once.

Define

$$W(z) = -R_{\mathrm{cl}}(z) + E_k(U(gz)) + \log J(gz). \tag{1.5}$$

Additive constants in W do not matter. No R activity, boundary B activity, nonconstant old coupling, moving selector or arbitrary conditioned exterior is hidden in this notation.

The classical argument and the fixed-domain regular activity lemma supply the following actual estimates, for $gP \le \delta$ inside a sufficiently small physical analytic neighborhood.

1.1 Classical and Gaussian input The true constrained precision has a uniform positive gap and the weighted locality needed for the previous real covariance theorem. The complete homogeneous classical calculation gives a nonnegative tree majorant K such that

$$|D_i D_j D_l R_{\mathrm{cl}}(z)| \le Cg\, K(i, j, l), \tag{1.6}$$

$$\sup_i \sum_{j,l} e^{\gamma_0 \tau(i,j,l)} K(i, j, l) \le C. \tag{1.7}$$

Replace an initially obtained majorant by the sum of its six permutations in the indices i, j, l. This makes K symmetric, preserves (1.6), and changes the uniform sum in (1.7) by only a fixed factor. One may equivalently take a fixed exponential tree majorant with a smaller decay rate. The source of (1.6)–(1.7) is the homogeneous specialization of CMP102 Proposition 9 and (190), together with the exact curvature cancellation and η^4-weighted fine plaquette sum proved. The complete third chain rule is retained.

Since R_{cl} has zero first and second jets,

$$|D_i R_{\mathrm{cl}}(z)| \le Cg \sum_{j,l} K(i, j, l)|z_j||z_l|, \tag{1.8}$$

$$|D_i D_j R_{\mathrm{cl}}(z)| \le Cg \sum_{l} K(i, j, l)|z_l|. \tag{1.9}$$

These follow by Taylor's integral formula along sz, $0 \le s \le 1$. They are bounds for the actual remainder, not only its cubic Taylor polynomial.

1.2 The inherited E first derivative is controlled as well as its Hessian On the fixed small-field analytic domain, CMP116 Lemma 1 bounds the complete regular fluctuation activities by

$$|V'_{k,Y}(B)| \le CE_0\epsilon_1 e^{-\kappa_* d_k(Y)} \quad \text{for } \|B\|_\infty < \epsilon_1/g. \tag{1.10}$$

The exact free-coordinate argument already contains the nonlinear chart. The base value subtracted in the source identity is constant in B.

Choose $\delta < \epsilon_1/2$. Coordinate Cauchy estimates at every real point of $\mathcal{D}_P$ then give

$$|D_i V'_{k,Y}(z)| \le CE_0 g\, e^{-\kappa_* d_k(Y)}, \qquad |D_i D_j V'_{k,Y}(z)| \le CE_0 g^2 \epsilon_1^{-1} e^{-\kappa_* d_k(Y)}. \tag{1.11}$$

The current M-cell carrier summation has the same entropy reserve as in the source lemma. Therefore

$$\sup_i |D_i E_k(U(gz))| \le CE_0 g, \qquad \sup_i \sum_j e^{\gamma_0 d(i,j)} |D_i D_j E_k(U(gz))| \le CE_0 g^2/\epsilon_1. \tag{1.12}$$

The old-scale sum is the source's already-renormalized sum. No new unweighted sum of g_j^2 is introduced.

The constant $CE_0 g$ in the first estimate may depend on the fixed admitted source parameters. It does not depend on P, the shell depth or the number of current blocks within this fixed-domain scope. A value bound on a different domain does not imply (1.12).

1.3 The exact coarea first derivative In the actual chart,

$$a(z) = gCz - hD(gCz), \qquad J(gz) = \text{constant}\, \frac{J_{\text{Haar}}(a(z))}{\det[I + DN(a(z))h]}. \tag{1.13}$$

The physical chart and its weighted derivatives are uniform in the homogeneous regular template. Put $A(z) = DN(a(z))h$. Its weighted row and column norm is below 1/2 after reducing the physical radius.

For the Haar logarithm, differentiate its local analytic summands and use $D_i a = O(g)$ with exponential localization. For the determinant,

$$D_i \log\det(I + A) = \mathrm{Tr}[(I + A)^{-1} D_i A]. \tag{1.14}$$

The inverse is a convergent weighted Neumann series. Each term in $D_i A$ has a factor g and a tree connection from i to its two matrix indices. The closed-walk trace sum is therefore bounded by Cg; there is no volume factor. Differentiating once more gives either one second derivative or two first derivatives, each with total order g^2. The same anchored sums give

$$\sup_i |D_i \log J(gz)| \le Cg, \qquad \sup_i \sum_j e^{\gamma_0 d(i,j)} |D_i D_j \log J(gz)| \le Cg^2. \tag{1.15}$$

This extends the previously proved coarea Hessian estimate by the first derivative needed here. The determinant is the new exact constraint determinant; older measure terms remain in E.

1.4 One uniform covariance estimate for the full interpolation For $0 \le t \le 1$, define

$$d\nu_t(z) = Z_t^{-1} 1_{\mathcal{D}_P}(z) \exp[-\tfrac{1}{2}\langle z, H_0 z\rangle + tW(z)]\, dz. \tag{1.16}$$

The Hessian estimate gives

$$\eta_{\gamma_0}(W) \le C\delta + Cg^2(1 + E_0/\epsilon_1). \tag{1.17}$$

Choose a fixed small δ, then a sufficiently small g, so this budget fits inside the true constrained gap. Choose $0 < \gamma < \gamma_0$ so that the weighted conjugation error of H_0 is also inside that gap. The previous vector-resolvent proof then supplies constants $\lambda, c > 0$, independent of t, P and volume, with

$$D^2\Big(\tfrac{1}{2} z H_0 z - tW(z)\Big) \ge \lambda I, \tag{1.18}$$

$$|\operatorname{Cov}_{\nu_t}(F, G)| \le C \sum_{i,j} e^{-\gamma d(i,j)} \|D_i F\|_{L^2(\nu_t)} \|D_j G\|_{L^2(\nu_t)}. \tag{1.19}$$

The product convex boundary is part of this statement. An arbitrary selected interface does not inherit it. The fixed source-dependent cutoff has not been differentiated.

2. A volume-uniform mean theorem

Theorem 2.1 — nonlinear conditional means and all-p fluctuations, P For the exact density (1.3) in the scope above,

$$\sup_{0\le t\le 1}\sup_{i}|\mathbb{E}_{\nu_t} z_i| \le Cg, \tag{2.1}$$

$$\sup_{0\le t\le 1}\sup_{i}\|z_i\|_{L^p(\nu_t)} \le C\sqrt{p}, \qquad p\ge 1. \tag{2.2}$$

The constants are independent of volume and of the admissible symmetric cutoff P. The reference density ν_0 is the actual constrained Gaussian restricted to the same product domain, not an unconditioned replacement on a different domain.

Proof. A linear real tilt leaves the Hessian lower bound (1.18) unchanged. For any scalar component $X = v\cdot z_i$, the unweighted covariance inequality in the tilted measure gives

$$\frac{d^2}{ds^2}\log \mathbb{E}_{\nu_t} e^{sX} = \mathrm{Var}_{\nu_{t,s}}(X) \le \lambda^{-1}|v|^2. \tag{2.3}$$

Integrating twice proves

$$\mathbb{E}_{\nu_t} e^{s(X-\mathbb{E}X)} \le e^{s^2|v|^2/(2\lambda)}, \qquad \|z_i - \mathbb{E}_{\nu_t} z_i\|_p \le C\sqrt{p}. \tag{2.4}$$

The block dimension is fixed. These estimates do not yet bound the mean.

Put

$$m_i(t) = \mathbb{E}_{\nu_t} z_i, \qquad M(t) = \max_i |m_i(t)|. \tag{2.5}$$

At finite volume, every function is integrable on the bounded domain, and

$$m_i'(t) = \mathrm{Cov}_{\nu_t}(z_i, W). \tag{2.6}$$

Using (2.4) at order four,

$$\|z_j z_l\|_2 \le \|z_j\|_4\|z_l\|_4 \le C[1+M(t)^2]. \tag{2.7}$$

Equations (1.8), (1.12), (1.15) and the summability of K therefore imply

$$\sup_j \|D_j W\|_{L^2(\nu_t)} \le Cg[1+M(t)^2]. \tag{2.8}$$

Insert (2.8) into (1.19), with $F = z_i$, and sum the exponentially decaying kernel:

$$|m_i'(t)| \le Cg[1+M(t)^2]. \tag{2.9}$$

The domain is symmetric and the quadratic reference density is even, so $m_i(0) = 0$. Consequently

$$M(t) \le Cg\int_0^t [1+M(s)^2]\,ds. \tag{2.10}$$

Let the right side be $y(t)$. Then $M \le y$, $y(0) = 0$, and

$$y'(t) \le Cg(1 + y(t)^2).$$

Thus $\arctan y(t) \le Cgt$, and

$$M(t) \le \tan(Cgt) \le 2Cg, \qquad 0 \le t \le 1, \tag{2.11}$$

after fixing a small coupling ceiling. Combining (2.11) with (2.4) gives (2.2). All constants used before taking the maximum are uniform in the finite block set, so no factor of its size occurs. This proves the theorem. □

The Riccati comparison is useful here because the crude cutoff estimate would give gP^2, which need not be small on the largest fixed physical window. The exact quadratic drift bound and the centered concentration estimate remove that artificial restriction.

3. Verification of the native moment budget

Let $O_{\alpha,x}$ be any of the 21 native curvature-bilinear stencils in the exact nonlinear chart, with physical factor h^{-4} retained. The scalar Wilson-action mark is also included. The deterministic theorem supplies

$$|O_{\alpha,x}(z)| \le Ch^{-4}[B_x + Z_x(z)]^2, \qquad Z_x(z) = \sum_j e^{-2\gamma d(x,j)}|z_j|, \tag{3.1}$$

$$\|\nabla O_{\alpha,x}(z)\|_{\gamma,x} := \left(\sum_j e^{2\gamma d(x,j)}|D_j O_{\alpha,x}(z)|^2\right)^{1/2} \le Ch^{-4}[B_x + Z_x(z)]. \tag{3.2}$$

The first two native derivative estimates and finite local template also give the corresponding mean-value inequality along a segment in the convex product domain.

Theorem 3.1 — all-p native field and gradient norms, P Uniformly for ν_t, $0 \le t \le 1$, and $p \ge 1$,

$$\|O_{\alpha,x}\|_p \le Ch^{-4}(B_x + \sqrt{p})^2, \tag{3.3}$$

$$\|O_{\alpha,x} - \mathbb{E}_{\nu_t} O_{\alpha,x}\|_p \le Ch^{-4}\big[(1 + B_x)\sqrt{p} + p\big], \tag{3.4}$$

$$\|\|\nabla O_{\alpha,x}\|_{\gamma,x}\|_p \le Ch^{-4}(B_x + \sqrt{p}). \tag{3.5}$$

In particular one can take the quadratic-growth cumulant budget

$$B_{i,m} \le A_i m, \qquad D_{i,m} \le A_i\sqrt{m}, \qquad A_i = Ch^{-4}(1 + B_{x_i}) \tag{3.6}$$

for centered marks. The uncentered full-product budget has $A_i^{\text{full}} = Ch^{-4}(1 + B_{x_i})^2$.

Proof. Minkowski and (2.2) give $\|Z_x(z)\|_p \le C\sqrt{p}$. This proves (3.3) and (3.5).

For the sharper centered bound, write $m = \mathbb{E}_{\nu_t} z$, $\xi = z - m$, and

$$X_x(\xi) = \sum_j e^{-\gamma d(x,j)}|\xi_j|.$$

Then $\|X_x\|_p \le C\sqrt{p}$, and $\sum_j e^{-2\gamma d(x,j)}|m_j| \le Cg$. The segment from m to z remains in $\mathcal{D}_P$, because a mean of a measure supported in a convex set lies in that set.

Apply the fundamental theorem of calculus to $O(z) - O(m)$. Cauchy–Schwarz with the reciprocal gradient weight, followed by (3.2), gives

$$|O_{\alpha,x}(z) - O_{\alpha,x}(m)| \le Ch^{-4}\big[(B_x + Cg)X_x(\xi) + X_x(\xi)^2\big]. \tag{3.7}$$

Taking L^p norms, using the $2p$ norm for the square, and then subtracting the expectation costs at most a factor two. This proves (3.4). Substitution of $p = 2m$ gives (3.6). □

This is an actual conditional norm theorem, with the complete inherited regular E and exact new coarea retained. It is no longer a hypothesis about an unspecified conditional mean. The reference-curvature factor and the restriction to the selected homogeneous regular law remain explicit.

Corollary 3.2 — native cumulants and full products, P For $m \ge 2$, the previous real cut theorem now gives

$$\boxed{|\kappa_\nu(O_{\alpha_1,x_1}, \dots, O_{\alpha_m,x_m})| \le C^{m+1}(m!)^2 h^{-4m} \prod_{i=1}^{m}(1 + B_{x_i}) \exp\Big[-\frac{\gamma\,\tau(x_1, \dots, x_m)}{m-1}\Big]}. \tag{3.8}$$

The full moments satisfy

$$\Big|\mathbb{E}_\nu \prod_{i=1}^{m} O_{\alpha_i,x_i}\Big| \le C^m m!\, h^{-4m} \prod_i (1 + B_{x_i})^2. \tag{3.9}$$

For (3.8), center the marks and insert (3.6) into the already proved real cut estimate. The two differentiated factors contribute $\sqrt{m}$ each, the other $m-2$ factors contribute m each, and the partition coefficient is bounded by $(m-1)!2^{m-2}$. Deleting the longest edge of a minimum spanning tree supplies the stated range; $m^m \le e^m m!$ yields the fixed factorial exponent two. Formula (3.9) is Hölder with exponent m, (3.3), and the same elementary factorial inequality.

These are point-mark conditional estimates. They have not been identified with a joint OS test-space bound on the complete Wilson hierarchy. The current h^{-4m}, the outer background distribution and the change from native to exactly transported marks still have to be treated.

4. An all-order, volume-uniform comparison with the actual Gaussian reference

The same estimates give a quantitative comparison theorem for the real conditional cumulants, without requiring a common exponential-source disc. It retains the same product cutoff in the reference Gaussian.

4.1 A local decomposition of the complete perturbation For each coordinate block j put

$$R_j(z) = \int_0^1 \langle z_j, D_j W(sz)\rangle\, ds. \tag{4.1}$$

Then, at every finite volume,

$$W(z) - W(0) = \sum_{j \in \mathcal{J}} R_j(z). \tag{4.2}$$

This identity is the radial fundamental theorem of calculus. It does not replace W by its Taylor polynomial or assume a source-dependent minimizing map.

For $p \ge 1$, (1.8) and the all-p coordinate moments give

$$\|D_j W\|_{L^p(\nu_t)} \le Cgp. \tag{4.3}$$

Differentiate (4.1):

First decrease the working γ, if necessary, so that 4γ is strictly smaller than every tree and Hessian decay rate used in (1.7), (1.12), (1.15), and the native derivative bound. All earlier covariance estimates remain valid at this smaller weight. The unused exponential decay pays the anchor weights and subsequent convolutions in this section.

$$D_i R_j(z) = \delta_{ij} \int_0^1 D_j W(sz)\, ds + \int_0^1 s\, D_i D_j W(sz) z_j\, ds, \tag{4.4}$$

with the evident block contraction. Using (1.9), the E/coarea Hessian estimates and the tree weights yields

$$\boxed{\|R_j - \mathbb{E}_{\nu_t} R_j\|_p \le C g p^{3/2}, \qquad \|\|\nabla R_j\|_{\gamma,j}\|_p \le C g p.} \tag{4.5}$$

For the first bound, $z_j D_j W$ contains at most cubic products of coordinates in its majorant. For the second, the non-trivial term is $z_j D_i D_j R_{\mathrm{cl}}$, a quadratic product times g. The exponential tree majorant makes the weighted sum over i finite. The E/coarea terms have first derivative $O(g)$ and Hessian $O(g^2)$; they fit the same estimates. The evaluation at sz does not change the coordinate moment majorants, since $0 \le s \le 1$.

The summability in (4.5) is the needed locality of the perturbation current. Merely knowing that W has zero mean, or is small as an extensive scalar, would not prove it.

Theorem 4.1 — native cumulant comparison, P For the same native marks and the same fixed reference Gaussian ν_0,

$$\boxed{\begin{aligned} &\big|\kappa_{\nu_1}(O_{\alpha_1,x_1}, \dots, O_{\alpha_m,x_m}) - \kappa_{\nu_0}(O_{\alpha_1,x_1}, \dots, O_{\alpha_m,x_m})\big| \\ &\quad \le g C^{m+1} (m!)^2 h^{-4m} \prod_{i=1}^m (1 + B_{x_i}) \exp\Big[-\frac{\gamma\, \tau(x_1, \dots, x_m)}{4m}\Big], \qquad m \ge 1. \end{aligned}} \tag{4.6}$$

Set $\tau(x_1) = 0$ for $m = 1$. Constants are independent of the volume, cutoff P and source order m.

Proof. For fixed m and finite volume, differentiate the finite moments defining the cumulant:

$$\frac{d}{dt} \kappa_{\nu_t}(O_1, \dots, O_m) = \kappa_{\nu_t}(O_1, \dots, O_m, W) = \sum_j \kappa_{\nu_t}(O_1, \dots, O_m, R_j). \tag{4.7}$$

The marks do not depend on t. An additive constant in W contributes zero.

Apply the real cut theorem to m quadratic-growth marks and the one cubic-growth mark R_j. At order $2(m+1)$, (3.6) and (4.5) give

$$B_i \le C A_i (m+1), \quad D_i \le C A_i \sqrt{m+1}, \qquad B_R \le C g (m+1)^{3/2}, \quad D_R \le C g (m+1).$$

Whether the differentiated pair consists of two O marks or one O mark and R, the polynomial factor before the pair count is bounded by

$$C^{m+1} g (m+1)^{m+1/2} \prod_i A_i.$$

The partition coefficient is at most $m! 2^{m-1}$, and there are at most $(m+1)^2$ crossing pairs. The inequality $m^m \le e^m m!$ absorbs this into $g C^{m+1} (m!)^2 \prod_i A_i$.

Delete a longest edge of a minimum spanning tree on $x_1, \dots, x_m, j$. The resulting cut gives

$$|\kappa_{\nu_t}(O_1, \dots, O_m, R_j)| \le g C^{m+1} (m!)^2 e^{-\gamma \tau(x_1, \dots, x_m, j)/m} \prod_i A_i. \tag{4.8}$$

The tree length with the additional point obeys

$$\tau(x_1, \dots, x_m, j) \ge \tfrac{1}{2} \tau(x_1, \dots, x_m), \qquad \tau(x_1, \dots, x_m, j) \ge d(j, x_1). \tag{4.9}$$

The first inequality follows by doubling the enlarged tree, traversing it through the marked vertices, and shortcutting the walk; the second follows from its path from j to x_1. Consequently

$$e^{-\gamma\tau(x_1,\dots,x_m,j)/m} \le e^{-\gamma\tau(x_1,\dots,x_m)/(4m)} e^{-\gamma d(j,x_1)/(2m)}.$$

The sum over j is at most $C(m+1)^4$ on a four-dimensional current lattice. This polynomial is absorbed into another fixed exponential in m. Integrate (4.7) from zero to one. This proves (4.6). □

The same interpolation and the product covariance formula give

$$\left| \mathbb{E}_{\nu_1} \prod_i O_i - \mathbb{E}_{\nu_0} \prod_i O_i \right| \le g C^{m+1} m! h^{-4m} \prod_i (1 + B_{x_i})^2. \tag{4.10}$$

For the first moment, the sharper direct estimate is

$$|\mathbb{E}_{\nu_1} O_{\alpha,x} - \mathbb{E}_{\nu_0} O_{\alpha,x}| \le C g h^{-4} (1 + B_x). \tag{4.11}$$

To prove (4.10), differentiate the product expectation and use the vector covariance representation. In each of its m terms there is one native gradient, $m-1$ native values and a first gradient of W. Apply Hölder at order $2m$ to the native factors, use (4.3) at order two for W, and sum the exponentially localized covariance kernel. The resulting $g\, m^{m+1/2}$ factor is bounded by $gC^m m!$.

Scope of this comparison The Gaussian reference is constrained and truncated on the same domain. The theorem does not remove that cutoff, expand the native marks into a polynomial, or identify the Gaussian comparison with the exact microscopic transport through prior RG steps. Such additional identifications require their own estimates.

In particular, (4.6) is not a theorem that the full Yang–Mills hierarchy differs from a free Gaussian hierarchy by O(g). It is a theorem about one specified current conditional law and its actual native marks. The explicit physical and reference-curvature factors remain.

The O(g) factor is a one-step perturbation size. It is not a summable multiscale remainder on a typical AF trajectory: $\sum_j g_j$ need not converge. Extracting the marginal mixing, proving its actual running normalization and assigning irrelevant scale gains to the remainder are still required. No unproved $O(g^2)$ marginal improvement or coupling derivative has been inferred from this comparison.

5. Why the scalar coupling derivative still does not close full matching

The full Wilson total action has the exact relation

$$\mathbb{E}_\beta e^{s\beta S} = Z(\beta(1-s))/Z(\beta). \tag{5.1}$$

For a differentiable family of positive integral-preserving maps P_β, the earlier exact identity is

$$Q_\beta S = -\partial_\beta \log(P_\beta e^{-\beta S}) + \frac{(\partial_\beta P_\beta) e^{-\beta S}}{P_\beta e^{-\beta S}}. \tag{5.2}$$

Freezing the map eliminates its derivative but does not supply an effective-action formula at nearby coupling for that frozen family. Moving with the source-free construction gives the known source-free representation but restores the current in the second term. The normalization derivative of each reference kernel, derivative of any moving coordinates, and selector changes belong in that current.

For an individual positive replacement with normalized reference kernel $K_\beta(u'|v)$, the current is exactly $\partial_\beta \log K_\beta(u'|v)$. The original mark after that replacement is a conditional expectation at fixed v and does not depend on the fresh variable u'. The current cancels the artificial reference-score contribution in the adaptive output density. A zero average of that current does not bound its product with another retained mark, and does not make it contact-local.

The fixed-domain estimates above are derivatives in physical free-field coordinates. They are not estimates of $\partial_\beta E_k$, the coupling derivative of the renormalized trajectory, or the coupling derivative of the complete normalized history density. Those derivative estimates have not been established here. A small field-analytic remainder may have large parameter derivatives unless a corresponding parameter regularity theorem is proved.

Therefore the scalar total-coupling identity does not currently yield a current-scale bound for $Q_{0:k}F_0$. Its special fixed-box cancellation is retained, but no new claim of protected scalar transport through the complete construction is made.

6. What the argument changes in the decomposition

The following local regular obligations are now P in the displayed fixed-domain, homogeneous, no-R scope:

1. The complete E/coarea perturbation has the first-gradient budget needed by the real conditional mean theorem.
2. The actual conditional fluctuation mean is $O(g)$, uniformly in volume and in all admissible symmetric product windows $gP \le \delta$.
3. The native all-p moment and weighted gradient bounds have the quadratic growth exponents needed by the real cumulant theorem, with fixed factorial growth at every source order.
4. Native marked cumulants admit a volume-uniform localized $O(g)$ comparison with the constrained Gaussian on the same domain.

The following remain O:

- The required estimate for the actual full CMP119 boundary and inherited R law.
- A bound on the reference background B_x after averaging all retained data and histories.
- Matching the exactly transported microscopic marks to current native fields and generated remainders.
- The full-law product-error or joint-kernel estimate, its contact assembly and its OS growth conclusion.
- The interacting operator estimate, tensor rotation restoration, quantitative ultraviolet identification and phase transfer.

The new conditional moment theorem is useful because it removes an explicit premise from an already established covariance-to-cumulant argument. It does not complete the full M0 or H0 estimate.

Source scope

- CMP109 (1987), 249–301: exact fixed-domain homogeneous small-field conditional integral, true constrained quadratic form and free coordinates.
- CMP116 (1988), 1–22, Lemma 1 and its fixed analytic ball: complete inherited regular E fluctuation activity norm used for (1.10)–(1.12).
- CMP102 (1985), 277–309, Proposition 9 and (190): homogeneous minimizing-field derivative and spatial-gradient input; the curvature-normalized full classical derivative proof is the result.
- The weighted covariance and real cut-cumulant estimates require the product domain, nonlinear chart and coarea density used above.
- Annex B, Section 9: the explicitly specified homogeneous no-R coercivity theorem. A transfer of the same fixed E prefactor to an arbitrary changed CMP119 domain is not used.

No unpublished coupling-response theorem, complete all-source action normal form or unproved tensor-product-to-joint-kernel passage is imported.

Annex G — Complete-action analytic domains and the remaining R-bulk theorem

Annex G. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

1. Hypotheses for the R Hessian

A real value bound gives no uniform R Hessian estimate. The R activities have a complex norm on specified multiscale spaces. Their pullback to the current free coordinates requires a quantitative domain and localization argument, as in Annexes AE and AH for the regular native law under CL.

In Annex B, Theorem 3.1, a fixed physical radius and prefactor apply to the fixed-domain small-field construction. The complete CMP119 construction has running constants. The transferred small-field lemma must retain both the actual radius and the activity majorant until their uniformity has been established.

The complete homogeneous classical-action estimate, native-chart bounds, and fixed-parameter no-R comparison have their separate hypotheses.

The results below are:

1. A source-specific identification of the actual complex R norm, the distinction between its original and current localization scales, and the source's six-power loss in the old-scale sum.
2. A proved Cauchy theorem that keeps actual analytic margins and activity majorants explicit. It gives the precise statement for E and the correct form of a future R theorem.
3. A proved old-scale summation calculation, conditional on an explicitly stated complex marginal-transfer estimate. It keeps the two analytic derivative powers separate from the six old-scale powers and from the current-coordinate normalization.
4. An exact statement of the remaining domain and localization premise. Neither selecting convenient logarithmic exponents nor recentering the native chart alone supplies that premise.

Here P means a mathematical implication proved in this note, L a precisely scoped source result, C a conditional application, and O an unproved application to the complete construction.

2. The actual R activity is complex analytic

2.1 Original activity space and norm — L CMP119, printed pp.259–260, separates the inherited effective action into E, R, and B. R has only its vacuum value subtracted:

$$R_k(U_k) = \sum_{j=1}^{k} \left[R^{(j)}(\Lambda_j, U_k) - R^{(j)}(\Lambda_j, 1) \right]. \tag{2.1}$$

Each $R^{(j)}$ has a localized representation over its specified $X \in \mathcal{D}_j$. Its terms have the same three properties as the preceding E activities: localized dependence; extension to a complex analytic function of (U, J); and gauge invariance. The analytic space is the specified $\mathcal{U}_j(X, \alpha_{0,j}, \alpha_{1,j})$, not a ball of arbitrary link values. Formula (2.28) is

$$\alpha_{0,j} = g_j C_0 (\log g_j^{-2})^{q_0}, \qquad \alpha_{1,j} = g_j C_1 (\log g_j^{-2})^{q_1}. \tag{2.2}$$

The integers $q_0, q_1 > 1$ and the constants are subject to the construction's restrictions. Formula (2.31), is

$$\left| R^{(j)}(X, (U, J)) \right| \le g_j^{\varkappa_0} e^{-\varkappa d_j(X)}. \tag{2.3}$$

The construction can arrange an arbitrarily large $\varkappa_0$ by fixing its parameters appropriately. This is a complex norm on the stated domain. It is neither a norm on every complex gauge field nor already a norm on the current free fluctuation coordinates.

The distinction matters because admission to these activity spaces involves several conditions simultaneously: plaquette regularity, currents, reconstructed configurations, gauges on the indicated local cubes, and relative complex fields with their required spatial regularity. The multiscale definition on CMP119 p.261, (2.34)–(2.39), makes this explicit for the complete construction. A bound for one relative link coordinate is not the whole admission test.

LFR II, printed p.390, (1.98)–(1.100), verifies the generation of new R activities. The particularly strong new-activity estimate is

$$|R'^{(k)}(X,(U,J))| \le e^{-p_0(g_k)} e^{-\kappa d_k(X)} \tag{2.4}$$

on its stated complex space. This explains the freely large power in (2.3). It also confirms that R can remain after a previous replacement has removed a large-field region. An empty current large-field set does not imply $R_k = 0$.

2.2 What the real marginal theorem proves — L CMP119 Theorem 2, (2.44), controls the vacuum-subtracted R sum for regular real backgrounds. Its proof on printed p.283 applies the same gauge identities as for E, but stops at (3.49), without subtracting a coupling term. The resulting small-carrier factor is

$$O(1)(L^{j-n})^4 g_j^{\kappa_0} e^{-\kappa d_j(X)}. \tag{2.5}$$

The factor $(L^{j-n})^4$ compensates the number of old-scale cells per current physical volume. Large carriers are handled by the stronger exponential in (3.48). The result is a marginal contribution, whose summability comes from $g_j^{\kappa_0}$, rather than from an extra irrelevant scale factor.

The old-scale estimate printed in (2.46) is

$$|R_k(U_k)| \le R_1 \sum_n |\Gamma_n| \sum_{j \le n} g_j^{\kappa_0} \le R_1 \sum_n |\Gamma_n| g_n^{\kappa_0 - 6}. \tag{2.6}$$

The source loses six powers here. A two-power loss is not what this source states. Formula (2.6) concerns regular real fields. Differentiating it is not a valid way to obtain a Hessian estimate.

2.3 The current localization transfer — L, with scope CMP119 printed p.276, first performs the generalized random-walk localization and resums onto current carriers $Y \in \mathcal{D}_k$. For carriers inside Λ_{k+1}, it transfers the small-field lemmas. The subsequent paragraph explicitly treats R by the corresponding analysis with the coupling-subtraction step omitted. This is genuine evidence for a complex R fluctuation theorem; R is not being replaced by E.

There are two points that the transfer sentence does not make numerically explicit for the present application:

- the full dependence of the fluctuation majorant on the running analytic radii (2.2);
- a complex tube, with quantitative margin, about every point of the actual real cutoff, admitted by every localized old activity and all its auxiliary current variables.

For the other carriers, CMP119 gives in (3.43) the radius $C_1 p_1(g_k)$. It does not print a fixed radius divided by g_k. No identification of that symbol with the later LFR I sharper selector is used here.

3. Fixed radii in the original theorem and running radii in the complete construction

3.1 The source dependence is real CMP109, printed p.263, (1.18), assumes fixed α_0, α_1, independent of the old index. CMP116 (1.19)–(1.22) admits the free fluctuation field on $g_k \|B\| < \epsilon_1$, subject to

$$4B_0 C_1 e^{16\kappa_1} \epsilon_1 \le \frac{1}{2}\alpha_2. \tag{3.1}$$

The proof's interpolation derivative costs $O(1)\epsilon_1/\alpha_2$. CMP109 p.280, immediately following (3.54), states this dependence explicitly. Its Lemma 4 also imposes restrictions including

$$B_3\alpha_3 < \frac{1}{2}\alpha_1, \qquad 2B_3\alpha_3 \le \beta L^{-2}\alpha_0. \tag{3.2}$$

CMP116 p.8 describes the final prefactor as obtained from these parameter restrictions. With fixed parameters this is a fixed constant. Replacing their role by the sequence (2.2) requires another quantitative argument.

Thus the proof step

$$\delta_k/\epsilon_1 \longrightarrow 0 \quad \text{with one fixed admissible } \epsilon_1 \tag{3.3}$$

has not been established for the complete-history inherited E action. Furthermore, merely changing ϵ_1 to $\epsilon_{1,k}$ while continuing to declare the prefactor fixed is not enough. Some radius ratios in the proof can cancel, but that cancellation has to be shown.

This is a correction to our previous application. It is not a counterexample to the transferred lemma or to Balaban's construction.

3.2 A precise Cauchy replacement — P Let I_k be the current free coordinates, K_k a real set, and V_Y a localized analytic activity. Suppose for every $z \in K_k$ the complex polydisc

$$\mathbb{D}_I(z, r_k) = \{z + w : \|w\|_\infty < r_k\} \tag{3.4}$$

is in its actual pullback domain, and on this disc

$$|V_Y(z + w)| \le A_{k,Y}. \tag{3.5}$$

The activity depends only on the permitted coordinates of Y. Define

$$\mathcal{A}_{k,\gamma} := \sup_i \sum_{Y \ni i} \sum_{l \in Y} e^{\gamma d(i,l)} A_{k,Y}. \tag{3.6}$$

Then

$$\sup_{z \in K_k} \sup_i \sum_l e^{\gamma d(i,l)} \left| \partial_i \partial_l \sum_Y V_Y(z) \right| \le \frac{8\mathcal{A}_{k,\gamma}}{r_k^2}. \tag{3.7}$$

Proof. Use coordinate circles of radius $r_k/2$. For different coordinates, the two-variable Cauchy formula bounds the mixed derivative by $4A_{k,Y}/r_k^2$. For the same coordinate, the one-variable second-derivative estimate gives $8A_{k,Y}/r_k^2$. A derivative vanishes when a coordinate is absent from the activity. Sum these bounds using (3.6). Finite-volume sums are immediate; uniformly absolutely convergent activity sums follow by the same majorant. This proves (3.7). No continuation or nonvanishing statement for a logarithm is used. □

The actual radius can depend on a real point or on a carrier. Replacing the right side of (3.6) by the corresponding sum of $A_{k,Y}/r_{k,Y}(z)^2$, and then taking the supremum, gives the identical proof. A common radius is merely a convenient sufficient version.

3.3 Correct E statement — C Suppose the transferred current E activities have the proved majorant

$$|V^E_{k,Y}(B)| \le E_0 \epsilon_{1,k} K_k^E e^{-\kappa_* d_k(Y)}, \qquad \|B\| < \epsilon_{1,k}/g_k, \tag{3.8}$$

where K_k^E includes all radius dependence. On a real window $\|B\| \le P_k$, set $\delta_k = g_k P_k$. If $\delta_k < \epsilon_{1,k}$, the actual Cauchy margin gives

$$\eta_\gamma(W_k^{E,\mathrm{bulk}}) \le C E_0 K_k^E \frac{\epsilon_{1,k} g_k^2}{(\epsilon_{1,k} - \delta_k)^2}. \tag{3.9}$$

The carrier entropy constant is the usual current-scale constant, provided $\varkappa_*$ retains its entropy reserve. There is no new old-scale sum: (3.8) is already the resummed E estimate. Formula (3.9) is the valid replacement for the fixed-radius inference.

For example, if a separate proof establishes $\delta_k \le \theta\epsilon_{1,k}$, $\theta < 1$, and K_k^E bounded, then the right side is $O(g_k^2/\epsilon_{1,k})$. If the established majorant is instead uniformly $O(E_0)$, and the free-coordinate margin is of order $(\log g_k^{-2})^q$, (3.7) gives $O(E_0(\log g_k^{-2})^{-2q})$. Both are sufficient smallness mechanisms. Neither set of premises has been proved here for the complete history construction.

The literal cutoff printed in CMP119 (3.16) deserves separate bookkeeping: if it indeed bounds the already normalized free variable by δ_k, it is smaller than the conservative window $P_k = \delta_k/g_k$ used. It could improve the margin. This note does not silently resolve the source's variable convention in order to assert a larger theorem.

4. R derivative costs and the old-scale sum

4.1 What follows directly from the primitive complex norm — P with L input Fix an old activity $R_X^{(j)}$ and an analytic coordinate chart whose polydisc of radius a lies inside its actual (U, J) domain. Cauchy's formula applied to (2.3) gives, at its center,

$$|\partial_u \partial_v R_X^{(j)}| \le 2a^{-2} g_j^{\varkappa_0} e^{-\varkappa d_j(X)} \tag{4.1}$$

for unit coordinate directions, using the full radius by a limiting argument. This is an actual complex second-derivative estimate for the original R activity. It does not concern the current coordinate Hessian until the pullback, localization, and admissible radius have been established. In particular, the current derivative of a composite function includes

$$D^2(R \circ \Phi) = (D\Phi)^T (D^2 R)(D\Phi) + (DR) D^2\Phi. \tag{4.2}$$

Ignoring the second term away from the zero configuration is invalid. Applying (3.7) to the full pulled-back activity would automatically include it.

4.2 A radius-explicit marginal-transfer premise — O The following is one sufficient quantitative form of what still has to be proved. It is stated as a premise, not as an extraction already completed from the transfer sentence.

Let $V_{j,k,Y}^R$ be the current localized vacuum-subtracted activity originating at old scale j. For every real cutoff point suppose its admitted complex current tube has common radius r_k, and suppose

$$\sup_i \sum_{Y \ni i} \sum_{l \in Y} e^{\gamma d(i,l)} \sup_{\mathbb{D}(z,r_k)} |V_{j,k,Y}^R| \le C \ell_j^A g_j^{\varkappa_0} \frac{Q_k^2}{a_j^2}, \quad \ell_j = \log g_j^{-2}. \tag{4.3}$$

Here a_j is an actual admitted old-field radius, Q_k bounds the physical field/curvature amplitude needed by the marginal estimate on the tube, and A is fixed. The four powers of the scale ratio have already compensated the old-cell count in (4.3). This is exactly where the marginal, gauge-invariant localization enters. It must also include the large-carrier tail, the localized decoupling parameters, and both terms of (4.2).

The original analytic norm (2.3), by itself, does not prove (4.3). Without the marginal factor one can retain an old-cell multiplicity growing as $L^{4(k-j)}$, which no logarithmic asymptotic-freedom bound can absorb.

4.3 Summation theorem with all coupling costs visible — P as an implication Assume (4.3), $a_j \ge c g_j$, and

$$Q_k \le \Theta_k g_k r_k. \tag{4.4}$$

Assume the numerical trajectory satisfies the old-scale bound used in CMP119 (2.46), in the form

$$\sum_{j \le k} g_j^s \le C_s g_k^{s-6}, \qquad s > 6. \tag{4.5}$$

Then for every fixed $\varepsilon > 0$ and $\kappa_0 > 8 + \varepsilon$,

$$\eta_\gamma(W_k^{R,\mathrm{bulk}}) \le C_\varepsilon \Theta_k^2 g_k^{\kappa_0 - 6 - \varepsilon}. \tag{4.6}$$

If the logarithmic factor in (4.3) is absent, one can take $\varepsilon = 0$.

Proof. Apply (3.7) to each old scale, use (4.3), and sum. Equations (4.4) and $a_j \ge c g_j$ give

$$\eta_\gamma(W_k^{R,\mathrm{bulk}}) \le C \Theta_k^2 g_k^2 \sum_{j \le k} \ell_j^A g_j^{\kappa_0 - 2}. \tag{4.7}$$

For fixed A and $\varepsilon > 0$, elementary calculus gives $\ell^A \le C_{A,\varepsilon} g^{-\varepsilon}$ on a fixed sufficiently small coupling interval. Apply (4.5) with $s = \kappa_0 - 2 - \varepsilon$:

$$g_k^2 \sum_{j \le k} g_j^{\kappa_0 - 2 - \varepsilon} \le C g_k^2 g_k^{\kappa_0 - 8 - \varepsilon} = C g_k^{\kappa_0 - 6 - \varepsilon}. \tag{4.8}$$

This proves (4.6). □

The bookkeeping is deliberate: two powers can be lost to an old radius proportional to g_j; six more are lost in the available old-scale bound; and two powers return because current free-coordinate derivatives carry g_k^2. A claim of $g_k^{\kappa_0 - 6}$ without identifying these three mechanisms would not be a proof. If a different admission or localization argument loses additional fixed powers, those powers must be displayed. The ability to choose κ_0 large can absorb any proved fixed loss; it cannot replace a proof that the loss is fixed.

The statement also shows why a common tube and its amplitude must be recorded together. A vanishing tube radius is not necessarily fatal when Q_k / r_k has the physical normalization in (4.4). Conversely, small values on a domain disjoint from parts of the real cutoff give no Hessian control there.

5. Can logarithmic exponents or recentering finish the admission step?

5.1 Choosing exponents before the construction For a prospective origin-centered tube, suppose a checked choice gave

$$\epsilon_{1,k} = c g_k \ell_k^q, \qquad P_k = A_1 \ell_k^{p_0}. \tag{5.1}$$

Then the ratio of the real window to the analytic radius is exactly

$$\frac{g_k P_k}{\epsilon_{1,k}} = \frac{A_1}{c} \ell_k^{p_0 - q}. \tag{5.2}$$

Thus $q > p_0$ would provide an eventual strict margin; $q = p_0$ would require an explicit constant margin; $q < p_0$ would not work for this particular origin-ball argument.

CMP119 Theorem 1 quantifies the construction parameters existentially and requires all the inherited restrictions. Its (2.28) gives only $q_0, q_1 > 1$. A further hypothesis is needed to prescribe $q_0, q_1 > p_0$ while maintaining every restriction in CMP109, CMP116, CMP119, and LFR I–II. Several checked LFR estimates contain a factor g_k times a fixed polynomial in logarithms, which would tolerate arbitrarily large fixed logarithmic powers. This is consistent with the requirement but is not a check of all restrictions. No exponent choice is asserted here as an already verified construction.

In particular, LFR I's later auxiliary exponent $p_1 < p_0$ and the symbol in CMP119 (3.43) must not be identified simply because they have the same name.

5.2 Recentered tubes The native-chart lemma and CMP102 Proposition 9 support recentering at an admitted real point. Recentered tubes can avoid requiring an origin radius larger than the entire real support. They must still preserve the actual analytic constraints.

A sufficient complete admission theorem would show, uniformly in all relevant histories and real cutoff points, that for a specified $r_k > 0$, every localized pullback on $\mathbb{D}(z, r_k)$:

1. satisfies the original and reconstructed (U, J) conditions at every old layer with a fixed fractional margin;
2. satisfies the local gauge and spatial-derivative bounds, including the required current bounds;
3. remains admissible when all localization/decoupling parameters run over the complex contours used by the source;
4. obeys the current activity bound (4.3), including the old-scale sum and large-carrier tail.

Old-layer gains such as L^{j-k} in relative fields and $L^{2(j-k)}$ in curvature are relevant. They do not prove the simultaneous assertions above without tracking the radii and the current norms. In particular, (2.3) is a bound only after this admission has been verified.

6. Scope for assembly

Item	Status
Primitive inherited R complex activity norm	L: exact (2.31) verified
Generation of superpolynomially small new R activities	L: LFR II (1.100) verified
R real marginal factor and old-scale sum	L: CMP119 (2.44), (2.46), proof on p.283; six-power loss
Primitive R second derivatives inside an admitted chart	P with L input: (4.1)
Current localized R Hessian on every real cutoff point	O: (4.3) and full tube admission remain
Current R summation once that premise holds	P implication: (4.6)–(4.8)
Complete-history E-bulk bound with fixed ϵ_1 and fixed prefactor	The fixed-radius, fixed-prefactor import is not justified; use (3.9) until its premises are checked
Homogeneous classical-action estimate and exact coarea bound	Unaffected by this radius analysis
Fixed-parameter pure CMP109 no-R conditional model	Unaffected within its explicitly stated scope
Complete homogeneous RG/R coercivity	Not closed: inherited R remains, and the complete-history E domain import is not justified
General-history coercivity and observable hierarchy	Not closed; boundary terms and marked transport remain in addition

For the regular domains, Annexes AD, AE and AH track the running radii in the current contour and its derivative majorants. Application to the complete conditional representation requires CL, including the heterogeneous boundary terms.

7. References

- T. Balaban, Convergent renormalization expansions for lattice gauge theories, CMP119 (1988), 243–285, (2.31) and p. 276. The R portion of the proof of Theorem 2 appears on p. 283.
- T. Balaban, Large field renormalization. I, II, CMP122 (1989), 175–202 and 355–392. R generation and complex norm at II (1.98)–(1.100). Auxiliary exponent restrictions at I p.183 and II p.383.
- T. Balaban, Renormalization group approach to lattice gauge field theories. I, CMP109 (1987), 249–301. DOI. Fixed-domain quantifiers at p.263; Lemma 4 and radius dependence at p.280.
- T. Balaban, Renormalization group approach to lattice gauge field theories. II. Cluster expansions, CMP116 (1988), 1–22. DOI. The ϵ_1/α_2 dependence is in (1.21)–(1.24); parameter dependence in the prefactor is discussed after (1.29). Formula (1.43) is not used.

- T. Balaban, The variational problem and background fields in renormalization group method for lattice gauge theories, CMP102 (1985), 277–309, Proposition 9. Its recentered analytic minimizer theorem remains an input to the proposed tube argument, not a substitute for admission of every localized activity.

This note makes no claim that four-dimensional Yang–Mills existence or the complete conditional-coercivity obligation has been proved.

Annex H — Current domains, preparation budgets and E/R arithmetic

Annex H. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

1. Result and scope

The complete-history C0 estimate is not closed in this argument. In particular, the fixed-radius extension to all complete histories (Annex B) is not used here; it is not justified.

Several narrower questions about the domains are settled here:

1. CMP102 (190) controls the covariant second-order outputs of the unlocalized minimizing-field derivative. The current is not an additional unknown derivative in that theorem.
2. The loss caused by the geometrical overlap depth N_0 in one LFR preparation is explicitly logarithmic: $N_0 = O(\log\log g_k^{-2})$, and every fixed power of $L_0^{N_0}$ has an explicit finite logarithmic bound.
3. The dyadic coefficients in the LFR complex domains have a uniform capacity if the admitted parameter satisfies $\beta < 1/4$. At the endpoint $\beta = 1/4$, one of the capacities need not have a uniform positive lower bound. This is separate from the distance of an actual real cutoff configuration to the boundary.
4. Global semisimple gauge invariance kills the first holomorphic jet of each original R activity at the identity. An admitted radial complex chart therefore gives a quantitative quadratic, or marginal, R bound. The required chart and current-carrier summation have not been proved for every complete-history decoupling contour.
5. A near/far old-scale split absorbs any fixed logarithmic prefactor in an E estimate, provided both a marginal analytic estimate and an irrelevant estimate are available on the same admitted current tube. Thus proving that every intermediate prefactor is bounded is stronger than necessary.

Here P denotes a proof in this note, L a primary result with its scope retained, and O an application still to be established. None of the arithmetic or algebraic results below completes a hypothesis of the Yang–Mills existence theorem.

2. What CMP102 actually controls

2.1 The five outputs in (190) — L Write $h_j = L^j\eta$, where j is an output layer of the determining set. Proposition 9 of CMP102 concerns the analytic minimizing-field map $\mathcal{H}(B;U)$, with a fixed admissible external background U, and includes the kernel estimate (190). CMP102, p. 308, has five outputs:

$$\frac{\delta\mathcal{H}}{\delta B},\qquad \nabla^\eta\frac{\delta\mathcal{H}}{\delta B},\qquad \left\|\zeta\nabla^\eta\frac{\delta\mathcal{H}}{\delta B}\right\|_\beta,\qquad D_U^*D_U\frac{\delta\mathcal{H}}{\delta B},\qquad \Delta_U\frac{\delta\mathcal{H}}{\delta B}. \tag{2.1}$$

Their scale factors are, respectively,

$$C\left[h_j^{-1},\quad h_j^{-2},\quad (\|\zeta\|_\beta+|\zeta|)h_j^{-2-\beta},\quad h_j^{-3},\quad h_j^{-3}\right]h_{j'}^{-d}e^{-\delta_0 d(y,y')/8}. \tag{2.2}$$

Here j' is the input layer. The input-cell volume in the source's functional-derivative convention compensates $h_{j'}^{-d}$ when the kernel is integrated against a coarse field. The background-dependent operators in (2.1) are fixed while differentiating B.

Proposition 9, printed p.309, explicitly includes analytic dependence on the external regular field. This is an unlocalized minimizing-field theorem on its stated regularity spaces. It is not a theorem that every independently decoupled random-walk term satisfies the same Landau-gauge identity.

2.2 Finite differences within one admitted chart — P with L input Suppose the line segment $B_t = B_0 + t(B_1 - B_0)$, $0 \le t \le 1$, stays in one chart to which (2.2) applies uniformly. For any of the linear, fixed-background operators in (2.1),

$$\mathcal{D}\{\mathcal{H}(B_1) - \mathcal{H}(B_0)\} = \int_0^1 \mathcal{D}\, D_B\mathcal{H}(B_t)[B_1 - B_0]\, dt. \tag{2.3}$$

Consequently, with the source's input-cell measure convention,

$$h_j^s|\mathcal{D}_s\{\mathcal{H}(B_1) - \mathcal{H}(B_0)\}(x)| \le C \sum_{j'} \sum_{y' \in \Gamma_{j'}} e^{-cd(y,y')}\, |B_1(y') - B_0(y')|, \tag{2.4}$$

where $s = 1, 2, 3$ for the field, first derivative, and either second-order output. The Hölder version follows with its displayed cutoff factor. In particular, perturbing one or two input coordinates gives a volume-independent estimate immediately, without first proving a row-sum estimate for an arbitrary extensive perturbation.

Proof. Apply the fundamental theorem of calculus in the finite-dimensional complex chart, move the fixed linear spatial operator under the integral, apply (2.2), and cancel the input-cell volume against $h_{j'}^{-d}$. No extension beyond the admitted chart is involved. □

This settles an input needed for a possible *pair-coordinate* Cauchy argument. A full polydisc in all free coordinates is a convenient sufficient condition for a Hessian bound, but a verified bidisc for each pair of differentiated coordinates would suffice.

2.3 The unlocalized current and the localized current are different issues CMP109 (3.11), printed p.272, writes the current of an inserted field in the form

$$J(\exp(i\xi A)U) = J(U) + D_U^* D_U A + F(U, A), \tag{2.5}$$

where F is local and depends on $U, \partial U, A, \nabla_U A$. Differentiating (2.5) with $A = \mathcal{H}(B)$ gives

$$D_B J(\exp(i\xi\mathcal{H}(B))U)[v] = D_U^* D_U D_B\mathcal{H}(B)[v] + D_A F(U, \mathcal{H}(B))[D_B\mathcal{H}(B)[v]]. \tag{2.6}$$

Thus (2.1) supplies the second-order output needed for the unlocalized current variation; the remaining term uses the first-order outputs and the displayed local function. There is no need to postulate an additional third spatial derivative of $\mathcal{H}$ to obtain (2.6).

However, the same primary page explicitly explains why one must not apply this argument to each localized summand. The localized expansion has good covariant-Laplacian bounds, but does not retain the corresponding D^*D bounds term by term. The source uses the Landau identity before localization and replaces

$$D^*D \;\longmapsto\; D^*D + DRD^* = \Delta_U - P + \text{local lower-order terms}. \tag{2.7}$$

Its analytic extension (3.13) has an independent current variable. The admissible insertion norm (3.14) controls

$$|A|, \quad |P_U A|, \quad |\nabla_U A|, \quad |\Delta_U A|. \tag{2.8}$$

This is the correct current mechanism for the localized activities. It is not legitimate to substitute the physical current of each localized field and invoke (190). Conversely, the failure of that substitution does not mean that the source lacks a complex-current mechanism: (2.7)–(2.8) are precisely that mechanism.

The remaining application is to bound the correctly extended current, and every reconstructed-field constraint, on a quantitative current free-coordinate tube centered at each actual real cutoff point. Neither (2.4) nor the existence of the extension alone specifies that tube.

3. Exact dyadic domain budgets

In this section, β is the source's domain-budget parameter, distinct from the Wilson inverse coupling. The notation is retained to permit checking the cited formulas.

3.1 The CMP119 coefficient The complete-history space on CMP119 p.261 uses

$$s_{j,n} = 1 - \beta(1 - 2^{-(j-n)}). \tag{3.1}$$

Hence

$$s_{j,n} \geq 1 - \beta, \qquad s_{j,n} - s_{j+1,n} = \beta\, 2^{-(j+1-n)}. \tag{3.2}$$

The second quantity is the *one-step capacity consumed at layer n*. A lower bound on the first quantity is not a lower bound on the margin of a particular field satisfying the strict domain inequality.

3.2 The two-sum coefficient in LFR I — P LFR I (1.64), printed p.190, contains

$$\Theta_{m;h,j,k} = 1 - \beta \sum_{i=h+1}^{j} 2^{-|m-i|} - \beta \sum_{q=m+1}^{k} 2^{-(q-m)}. \tag{3.3}$$

For all permitted indices,

$$\Theta_{m;h,j,k} \geq 1 - 4\beta. \tag{3.4}$$

The corresponding coefficients in (1.66) and (1.68) satisfy the stronger lower bounds $1 - 3\beta$ and $1 - 2\beta$, respectively.

Proof. The sum over all integer i of $2^{-|m-i|}$ is 3, while the one-sided strictly positive sum is 1. This proves (3.4). In (1.66) the first sum is one-sided with its endpoint included, so is at most 2; the second is at most 1. In (1.68), $m > j$, so the first sum is at most 1, as is the second. □

Consequently, for any fixed admitted $0 < \beta < 1/4$, these *capacities* stay uniformly positive, independently of the number of preparatory integrations. No assumption on the logarithmic exponents is needed for this calculation.

The endpoint $\beta = 1/4$ requires care. If $h < m \leq j$, the exact formula is

$$\Theta_{m;h,j,k} = \frac{1}{4}\left[2^{-(m-h-1)} + 2^{-(j-m)} + 2^{-(k-m)}\right]. \tag{3.5}$$

Taking m well inside a long interval $[h+1, j]$, with $k - m$ large, makes this coefficient arbitrarily small. Thus the convenient example $\beta = 1/4$ in the source cannot be treated as giving a fixed positive capacity in every preparatory layer. One may use a smaller admitted β, or keep the depth-dependent loss explicitly.

This does not show a defect in the LFR construction. The source requires descending domains and controlled successive changes, and its inequalities are compatible with small positive capacities. It does identify a loss that must not be suppressed in an additional Cauchy-radius argument.

3.3 The actual R-preparation age decay LFR I p.186, (1.47), proves for the preparatory geometry

$$\exp[-\delta\, d(y, \Omega_{j+1}^{c} \setminus Z''_{j+1})] \leq \exp[-\delta(M/M_1)(j-i)] \quad (y \in \Gamma_i'',\ i < j). \tag{3.6}$$

This is a proved geometrical inequality for that preparation. It should not be replaced by the unsupported assertion that every old large-field layer survives every later R map.

The arithmetic of paying a dyadic reserve is exact: if a layer change is bounded by

$$A(1+t)^b e^{-\lambda t}\alpha_i, \qquad t = j - i, \tag{3.7}$$

then, for $\lambda > \log 2$,

$$(1+t)^b e^{-\lambda t} \leq K_{b,\lambda}\, 2^{-t}, \qquad K_{b,\lambda} := \sup_{t \geq 0} (1+t)^b e^{-(\lambda - \log 2)t} < \infty. \tag{3.8}$$

Choosing the permitted small numerical factor A so that $AK_{b,\lambda} \le \beta/2$ pays a reserve of size $\beta 2^{-(t+1)}\alpha_i$. This proves the age-budget calculation. LFR I (1.46)–(1.48) applies this mechanism to its actual preparatory fields, using the coupling comparisons and the exponential (3.6).

What is not supplied by (3.8) is the analogous bound for *every complex decoupling parameter and every reconstructed constraint* of the current E/R activity. It is that map estimate, not the summability of 2^{-t}, that is still needed.

4. An explicit bound for the R overlap depth — P with L inputs

Put

$$\ell_k = \log g_k^{-2} \ge 1.$$

CMP119 (2.5) chooses R_k to be the smallest power of L at least ℓ_k^r, so

$$\ell_k^r \le R_k < L\ell_k^r. \tag{4.1}$$

LFR I p.192 gives

$$L^{N_0-1} = R_{k-N_0+1} \le (L+1)(N_0+1)^{\beta_0} R_k, \qquad L_0^2 \le L/3. \tag{4.2}$$

These inequalities imply the sharper bounds

$$N_0 \le C + \frac{2r}{\log L}\log \ell_k, \tag{4.3}$$

and, writing

$$\sigma = \frac{2\log L_0}{\log L} \in [0,1) \quad \text{for } 1 \le L_0,$$

$$L_0^{2N_0} \le C\,\ell_k^{r\sigma}(1+\log \ell_k)^{\beta_0\sigma}. \tag{4.4}$$

Here and below constants depend only on the fixed construction parameters.

Proof. Let $a = \log L > 0$, $b = \beta_0 \ge 0$, and

$$M_{a,b} := \sup_{x\ge 0}\{b\log(x+1) - ax/2\} < \infty.$$

Taking logarithms of (4.2) yields

$$aN_0 \le \log[L(L+1)R_k] + b\log(N_0+1).$$

By the definition of $M_{a,b}$,

$$N_0 \le \frac{2}{a}\{\log[L(L+1)R_k] + M_{a,b}\}.$$

Use (4.1) to obtain (4.3). Next raise

$$L^{N_0} \le L(L+1)(N_0+1)^{\beta_0}R_k$$

to the power σ and use (4.1), (4.3). This proves (4.4). The endpoint $L_0 = 1$, if used, has $\sigma = 0$ and is immediate. □

Every fixed power of $L_0^{N_0}$, and every fixed polynomial in N_0, therefore costs only a finite power of ℓ_k, with at most a power of $\log \ell_k$.

Two limits of this result. First, N_0 is the *overlap depth*, not the full preparatory depth N. The source explicitly takes $N > N_0$ and eventually requires N substantially larger. Second, this calculation concerns one preparation; it does not permit multiplying losses from an unbounded number of histories and declaring the product logarithmic. A complete-history induction must reset the activity norm rather than accumulate an uncontrolled product of these constants.

5. The first R jet vanishes, and what that buys

5.1 Semisimple gauge invariance — P Let G be a compact connected semisimple group, and let f be holomorphic near the origin of a finite product of copies of $\mathfrak{g}_{\mathbb{C}}$. Suppose f is invariant under simultaneous global adjoint action. Then

$$Df(0) = 0. \tag{5.1}$$

Proof. Write $Df(0)[X] = \sum_b \lambda_b(X_b)$. Set all components except one equal to zero. Adjoint invariance gives $\lambda_b(\mathrm{Ad}_g X) = \lambda_b(X)$. Differentiating at $g = \exp(tY)$ gives $\lambda_b([Y, X]) = 0$. Since $[\mathfrak{g}_{\mathbb{C}}, \mathfrak{g}_{\mathbb{C}}] = \mathfrak{g}_{\mathbb{C}}$, each $\lambda_b = 0$. □

For the original analytic R activity of CMP119 (2.31), global gauge transformations fix the identity configuration and zero current and act adjointly on the link logarithms and current variables. Consequently, its first joint holomorphic jet at that point vanishes. This fact does not require a perturbative expansion, hypercubic covariance, or a coupling subtraction.

5.2 A quantitative radial bound — P Suppose a gauge-compatible chart has norm q, and the radial path of the chart through (U, J) is admitted for complex parameter $|t| < a/q$. Suppose the primitive activity has bound

$$|R_X| \le M_X, \qquad M_X = g_j^{\kappa_0} e^{-\kappa d_j(X)} \tag{5.2}$$

throughout that radial disc and at the target configuration. Then

$$|R_X(U, J) - R_X(1, 0)| \le 2M_X \min\{1, q^2/a^2\}. \tag{5.3}$$

Proof. If $a/q > 1$, compose the activity with the radial path. The resulting holomorphic function minus its value at zero has a double zero by (5.1) and has modulus at most $2M_X$. The maximum-modulus principle applied after division by t^2, first on smaller circles and then in the limit, gives $2M_X(q/a)^2$ at $t = 1$. If $a/q \le 1$, the difference of the two admitted values is bounded by $2M_X$. □

In a scale-compatible local gauge, the desired marginal chart estimate has the form

$$q \le C\, P(|X|)(L^{j-k})^2 Q_k, \tag{5.4}$$

where P is a fixed polynomial in carrier size and Q_k bounds the current physical curvature/current amplitude. Inserting (5.4) into (5.3) yields

$$|R_X - R_X(1, 0)| \le C\, P(|X|)^2 (L^{j-k})^4 g_j^{\kappa_0} \frac{Q_k^2}{a^2} e^{-\kappa d_j(X)}. \tag{5.5}$$

The factor $(L^{j-k})^4$ is exactly the one that compensates the old-cell count in four dimensions.

Status of (5.4). Equation (5.5) is a proved implication; (5.4) together with admission of its radial path has not been proved here for the independently extended currents on every decoupling contour. The real version is what CMP119 uses in its proof of the marginal R estimate on p.283. A complex version cannot be obtained by analytically continuing the inequality. It must use the actual extension (2.7), the corresponding local gauge and reconstructed fields, and a retained analytic margin.

This identifies the algebraic cancellation needed for R quite sharply. The missing R theorem is not the cancellation of the linear term. It is the uniform, localized complex realization of the chart estimate and its subsequent current-carrier summation.

6. E does not require every intermediate prefactor to be bounded

The Cauchy estimate retains a radius $\epsilon_{1,k}/g_k$ and a prefactor K_k^E. Combining a marginal estimate with an irrelevant estimate before summing over old scales avoids requiring both to be uniform.

Proposition: near/far summation — P Let $\ell \geq e$, $q > 0$, $\lambda > 0$, and $D, b \geq 0$ be fixed. Suppose nonnegative old-age contributions obey

$$H_a \leq A(1+a)^b \min\{\ell^{-2q}, e^{-\lambda a}\ell^D\}, \qquad a = 0, 1, 2, \dots . \tag{6.1}$$

Then

$$\sum_{a\geq 0} H_a \leq CA\,\ell^{-2q}(1+\log\ell)^{b+1} \longrightarrow 0 \quad (\ell \longrightarrow \infty). \tag{6.2}$$

Proof. Set

$$n = \left\lceil \frac{D+2q+1}{\lambda} \log\ell \right\rceil.$$

For $a \leq n$, use the first bound and sum $(1+a)^b$, obtaining

$$CA\ell^{-2q}(n+1)^{b+1}.$$

For $a > n$, the exponential tail gives

$$\sum_{a>n} (1+a)^b e^{-\lambda a} \leq C_{b,\lambda}(n+1)^b e^{-\lambda n}.$$

Its contribution is therefore at most

$$CA\ell^{-2q-1}(n+1)^b.$$

Combining the two estimates proves (6.2). □

The role in the E problem is concrete. A genuine marginal estimate on a current tube of free radius comparable to ℓ_k^q would give the first branch of (6.1) after two Cauchy derivatives. The renormalized fifth-order/localization gain would give the second branch, with whatever *fixed finite* logarithmic loss the domain calculation actually produces. The coupling and radius comparisons in CMP119 (2.6)–(2.9) can contribute polynomial age factors, which are allowed in $(1+a)^b$.

Thus a finite logarithmic loss, including one of the form proved in (4.4), is not by itself fatal. Nor does the arithmetic require the unsupported choice $q_1 > p_0$. It requires $q > 0$ for an actually proved free-coordinate margin, and both bounds on the same localized activities and contours.

This proposition is not an E-bulk theorem. The complex marginal branch, the irrelevant branch with all radius costs, and their common tube still have to be obtained from the complete construction. It gives a less demanding numerical closing criterion than insisting on a bounded K_k^E.

7. What remains for the actual complete-history C0 application

The unresolved step can now be stated without conflating algebra, geometry, or analyticity.

D1. Admission of the actual current-coordinate contours — O For every positive history, every actual real free-field cutoff point z, every old E/R activity that contributes, and every localization/decoupling parameter used in its expansion, prove an admitted complex contour for the differentiated current coordinates. A pair-coordinate bidisc is sufficient; a full polydisc is stronger.

The verification must include all of CMP119 (2.34)–(2.39): the field and real-reference plaquettes, independently extended current, every specified reconstructed field and current, local gauge, and relative complex field with its spatial norm. During R preparation it must include the modified LFR I (1.64)–(1.69) conditions.

The radius must be quantitative. Membership in an open set, uniform positivity of the coefficients in Section 3, or a small physical cutoff alone does not give a lower bound for the distance to its boundary. Recentring the real reference field may avoid imposing $q_1 > p_0$, but that procedure has not been verified simultaneously for all the reconstructed fields and decoupling maps.

D2. Radius-aware complex bulk majorants — O On those admitted contours, prove the current-carrier norm estimates. For R, Section 5 supplies the first-jet cancellation; its scale-compatible complex chart, radial path, carrier entropy reserve and full localization still need proof. For E, either the resummed majorant with a controlled radius/prefactor ratio, or the two bounds in Section 6, would suffice.

The primitive bounds

$$|E_X^{(j)}| \le E_0 e^{-\kappa d_j(X)}, \qquad |R_X^{(j)}| \le g_j^{\kappa_0} e^{-\kappa d_j(X)}$$

are not yet current-carrier bounds. Without the marginal scale factor, an old-cell multiplicity $L^{4(k-j)}$ remains. Logarithmic asymptotic freedom cannot absorb that exponential multiplicity.

D3. Completing C0 requires the whole conditional law — O Even after E/R bulk closure, the existing separate accounting of boundary terms, coordinate/Jacobian terms, real cutoff support and positive history conditioning must be joined with the classical-action estimate on the same law. The fixed-domain no-R theorem remains valid in its stated model. It must not be called the complete conditional law merely because the current large-field region is empty.

The inherited-R old-scale loss remains the six-power estimate printed in CMP119 (2.46). No result here changes that to two powers. The new logarithmic-overlap calculation also does not establish the independent AF trajectory or its Wilson normalization.

8. Primary-source estimates

Source and printed location	Use and limit
CMP102, pp.307–309, (182)–(190), Proposition 9	Five derivative outputs, including D^*D and Δ; unlocalized admitted minimizing-field chart
CMP109, pp.272–273, (3.10)–(3.17)	Physical-current formula, reason it fails termwise after localization, independent-current repair and insertion norm
CMP109, pp.262–264, (1.11)–(1.18)	Analytic activity domain includes reconstructed fields; fixed-parameter primitive E norm
CMP119, p.255, (2.4)–(2.7)	Exact ϵ_j, R_j, coupling and logarithmic comparisons
CMP119, pp.259–261, (2.28), (2.31), (2.34)–(2.39)	Running radii, primitive complex R norm, all multiscale domain constraints
CMP119, pp.276–277, (3.43) and surrounding localization argument	Full-history localization transfer and dyadic domain reserve; no radius/prefactor uniformity inferred from the transfer sentence
CMP119, p.283 and (2.46)	Real marginal R factor and six-power old-scale loss
LFR I, pp.184–186, (1.31)–(1.48)	Actual preparatory current geometry; in particular the age-distance estimate (1.47)
LFR I, pp.190–192, (1.64)–(1.70) and following paragraph	Two-sum domain coefficients, preparation-map inclusion, overlap-depth and L_0 inequalities
LFR II, pp.358, 367–368, 380, 390	Complex reference denominator, examples of polynomial logarithmic costs, new-R complex bound (1.100); not a complete current-coordinate Hessian theorem

CMP109: DOI 10.1007/BF01215223.

9. Final conditions of the argument

Closed: the precise unlocalized current derivative input; the dyadic capacity arithmetic including its endpoint loss; the explicit finite logarithmic overlap loss; semisimple vanishing of the first R jet; and the near/far numerical criterion for absorbing logarithmic E prefactors.

Not closed: the simultaneous complete-history complex contour admission and the corresponding radius-aware, spatially weighted E/R bulk majorants. These are the remaining mathematical estimates, not bibliographic formalities. The current-coordinate C0 theorem, and consequently the full Yang–Mills existence target, cannot be asserted on the basis of this argument.

Annex I — Exact history resampling and normalized source scores

Annex I. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Scope

The complete transported-observable theorem M0 is not proved in this note. The native conditional moment theorem concerns a specified homogeneous regular law and its newly evaluated current fields. It does not identify those fields with the conditional expectations of renormalized microscopic Wilson insertions. The new work below does not change that distinction.

There is an exact improvement to the way the remaining theorem can be formulated. Auxiliary history variables can be averaged at every step, even when the next source-free map needs such variables, provided they are resampled from their baseline conditional law before reuse. The whole marked source jet is transported through this extra positive operation. The unmarked state and every final Schwinger coefficient are preserved. This removes an unnecessary prohibition on averaging live auxiliary labels.

The price is explicit: history averaging creates conditional covariance and higher cumulant responses. Its conditional kernel can be nonlocal. Neither its positivity nor its norm contraction supplies their spatial decay or current-scale normalization. Thus this is a useful exact representation theorem, not a missing ultraviolet estimate hidden in a change of notation.

The second part identifies exactly when a reference-kernel score disappears under such an averaging. The entire normalized reference likelihood jet cancels when all of its reference variables are averaged and the other marked coefficient depends only on the retained data. First-score mean zero does not justify deleting individual higher logarithmic derivatives or correlations with marks that still depend on those reference variables.

The source analysis also verifies the coupling-regularity statement in CMP109 and its qualification: the notation for a beta function suppresses its dependence on all preceding couplings. The checked one-parameter smoothness is not a proved local-source derivative bound or a bound on every total derivative along the complete history construction.

Here P denotes a proof from the stated hypotheses, L a scoped primary input, and O an unproved application to the actual complete Wilson observable induction.

1. The positive augmented state

At a finite cutoff, work before the introduction of signed polymer activities. Let the source-free augmented state be

$$X = (V, \omega) \in \mathcal{V} \times \Omega, \qquad \mu(dV, d\omega) = \nu(dV)\, \pi(d\omega \mid V). \tag{1.1}$$

Both spaces are standard Borel. In the finite lattice construction this includes compact link variables, finitely many discrete selectors/history labels, and the finite-dimensional chart variables used on a branch. An admissible-state subset can be incorporated by declaring that $\pi(\cdot \mid V)$ is supported on its fiber.

The field V is the coordinate or collection of retained data chosen for the next matching norm. The symbol ω denotes everything one proposes to average. Some components may be needed by the next source-free operation; they will be resampled rather than erased from the state space. The augmented state must contain every input used by that next operation. If the construction has further path-dependent inputs, include them in X first; no Markov property after an unauthorized omission is assumed.

The disintegration in (1.1) exists. At points with zero ν-mass its value is irrelevant and can be fixed arbitrarily. Where a density or a finite history sum is available, it is simply the ratio of the known source-free joint density to its marginal. No source-dependent denominator or new sourced minimizing field is used.

Define

$$(\Pi F)(V) = \int F(V, \omega)\, \pi(d\omega \mid V). \tag{1.2}$$

2. Exact averaging and resampling

Theorem 2.1 — history resampling with unchanged baseline law, P Define the source-free Markov kernel

$$\mathcal{R}\big((V,\omega), dV'\, d\omega'\big) = \delta_V(dV')\,\pi(d\omega' \mid V). \tag{2.1}$$

It has the following properties.

1. $\mu\mathcal{R} = \mu$.
2. On functions it is the conditional-expectation projection Π, lifted back to the augmented state:

$$(\mathcal{R}F)(V,\omega) = \Pi F(V). \tag{2.2}$$

3. If the old state is sampled from μ and then $\mathcal{R}$ is applied, the marked numerator transport is the same projection:

$$Q_{\mathcal{R}}F = \mathbb{E}[F(X) \mid X'] = \Pi F(V'). \tag{2.3}$$

4. $\Pi^2 = \Pi$, $\Pi 1 = 1$, and

$$\int \Pi F\, d\mu = \int F\, d\mu, \qquad \|\Pi F\|_{L^p(\mu)} \le \|F\|_{L^p(\mu)} \quad (1 \le p \le \infty). \tag{2.4}$$

5. On $L^2(\mu)$, the projection is self-adjoint. In particular,

$$\|F - \Pi F\|_2^2 = \|F\|_2^2 - \|\Pi F\|_2^2. \tag{2.5}$$

Proof. Under the joint old/new law, conditional on V, the two labels ω, ω' are independent with the same distribution $\pi(\cdot \mid V)$. This proves (2.2)–(2.3). Integrating the new conditional law gives (1), and the conditional-expectation identities and Jensen inequality give (2.4). For $F, G \in L^2$, $\int \overline{F}\,\Pi G\, d\mu = \int \overline{\Pi F}\,\Pi G\, d\nu = \int \overline{\Pi F}\, G\, d\mu$. This proves self-adjointness, idempotence, and the orthogonal decomposition (2.5). □

Corollary 2.2 — future maps may still use the labels, P Let $\mathcal{K}_k$ be the next source-free Markov operation, carrying μ_k to μ_{k+1}. Its selectors, integration coordinates and reference factors may depend on ω_k. Insert $\mathcal{R}_k$ before it. Then

$$\mu_k\mathcal{R}_k\mathcal{K}_k = \mu_k\mathcal{K}_k = \mu_{k+1}. \tag{2.6}$$

The marked transport for the inserted two-step operation is

$$Q_{\mathcal{R}_k\mathcal{K}_k}F = Q_k\Pi_k F. \tag{2.7}$$

The label resampled by $\mathcal{R}_k$ is admissible for its current field because it is sampled from the original conditional fiber. Thus the next operation remains well-defined almost surely. One has changed the joint coupling between the incoming mark and future history choices, but not any unmarked state law or integrated marked expectation.

Proof. Equation (2.6) uses stationarity from Theorem 2.1. For an output test H, the defining joint-law identity for conditional expectation gives

$$\int H\, Q_{\mathcal{R}_k\mathcal{K}_k}F\, d\mu_{k+1} = \int (\mathcal{K}_k H)\,\Pi_k F\, d\mu_k = \int H\, Q_k\Pi_k F\, d\mu_{k+1}.$$

This proves (2.7). □

An arbitrary finite sequence can therefore have such a reset at every stage. Every unmarked marginal of the original source-free construction remains the same. Future histories are still sampled from their actual baseline laws at each stage; what changes is their pathwise correlation with the original marked variables.

This corrects the overly restrictive formulation that a live label cannot be averaged because a later step uses it. It can be averaged and resampled. Simply deleting it while keeping a next map that requires it would not define an operation.

Corollary 2.3 — the complete source jet is preserved, P Let $G_k(t)$ be a marked polynomial or formal series truncated above source degree M, with $G_k(0) = 1$. Apply Π_k coefficient by coefficient, then apply the next original positive transport. At every finite source order,

$$\int \Pi_k G_k(t)\, d\mu_k = \int G_k(t)\, d\mu_k. \tag{2.8}$$

After any number of inserted resets and original operations, every integrated coefficient equals its original cutoff Wilson coefficient.

The proof is linearity and (2.4), applied to finitely many coefficients. No positivity of the coefficients or convergence of an infinite source series is needed. The baseline Markov law is positive; its signed marked coefficients need not be.

It is essential to reset G_k itself. Replacing it by $\exp(\sum_i t_i \Pi_k F_{i,k})$ generally changes every order above one.

3. The responses generated by a reset

3.1 Exact source-jet rule Put $\Psi = \log G$ and $A_I = \partial_I \Psi(0)$ for nonempty sets of distinct source labels. The reset logarithm is

$$\Psi^+ = \log \Pi e^{\Psi}.$$

The finite cumulant expansion gives

$$\boxed{A_I^+ = \sum_{\mathcal{P} \in \Pi(I)} \kappa_\pi(A_B : B \in \mathcal{P}).} \tag{3.1}$$

Here κ_π is a conditional cumulant in the label law $\pi(\cdot \mid V)$. In particular,

$$A_i^+ = \Pi A_i, \qquad A_{ij}^+ = \Pi A_{ij} + \mathrm{Cov}_\pi(A_i, A_j). \tag{3.2}$$

The same formula includes all responses already created by earlier integrations. For instance, A_{ij} need not vanish before the reset.

Proof. In the truncated polynomial algebra,

$$\log \Pi e^{\Psi} = \sum_{r=1}^{M} \frac{1}{r!} \kappa_\pi(\Psi, \dots, \Psi).$$

Extracting the coefficient with label set I groups its labels into a partition; all sums are finite. This yields (3.1). □

3.2 Direct product errors and their norms For single marks $F_1, \dots, F_m$, the extra product error is

$$D_m^\pi = \Pi\Big(\prod_{i=1}^{m} F_i\Big) - \prod_{i=1}^{m} \Pi F_i. \tag{3.3}$$

Let $\eta_i = F_i - \Pi F_i$. Conditional centering gives the exact expansion

$$D_m^\pi = \sum_{\substack{I \subset \{1,\dots,m\} \\ |I| \geq 2}} \Pi\Big(\prod_{i \in I} \eta_i\Big) \prod_{j \notin I} \Pi F_j. \tag{3.4}$$

For $1 \leq p < \infty$,

$$\|D_m^\pi\|_p \leq 2 \prod_i \|F_i\|_{pm}, \tag{3.5}$$

and also

$$\|D_m^\pi\|_p \le \sum_{|I|\ge 2} \prod_{i\in I} \|\eta_i\|_{pm} \prod_{j\notin I} \|F_j\|_{pm}. \tag{3.6}$$

The first inequality is conditional Jensen and Hölder for the two terms of (3.3); the second applies the same inequalities to (3.4). The corresponding $p = \infty$ statements use the sup norms.

For conditional cumulants of arbitrary marks,

$$\boxed{\|\kappa_\pi(F_1, \dots, F_m)\|_p \le m!\, 2^{m-1} \prod_i \|F_i\|_{pm}.} \tag{3.7}$$

For a partition with blocks B, conditional Jensen and Hölder bound $\prod_B \Pi(\prod_{i\in B} F_i)$ in L^p by the same product of L^{pm} norms. The absolute partition coefficient sum is $\sum_{\mathcal{P}}(|\mathcal{P}|-1)!$, bounded by $m!2^{m-1}$ using ordered partitions encoded by a permutation and cuts. This proves (3.7).

These bounds are denominator-free after averaging over the physical marginal. They can control response magnitudes when the input marks have useful norms. They supply no separation factor on their own.

3.3 Variance removed from a mark is retained in the hierarchy For real $F, G \in L^2(\mu)$,

$$\mathrm{Cov}_\mu(F, G) = \mathbb{E}_\nu[\mathrm{Cov}_\pi(F, G)] + \mathrm{Cov}_\nu(\Pi F, \Pi G). \tag{3.8}$$

In particular, the norm decrease (2.5) reappears as the integrated second-order response. It is not erased.

A simple finite example makes the spatial limitation explicit. Take a single auxiliary sign $\omega = \pm 1$ with equal probabilities, no physical retained variable, and two marks assigned to arbitrarily separated labels,

$$F_n(x) = n\omega, \qquad F_n(y) = n\omega.$$

The reset makes both single marks zero, but

$$D_2^\pi(x, y) = n^2. \tag{3.9}$$

Thus arbitrarily good post-reset single-mark norms do not imply a bounded or decaying full two-point function. This is a counterexample to that inference, not a model of Yang–Mills.

The conditional law $\pi(d\omega \mid V)$ can itself correlate remote regions. Applying Theorem 2.1 therefore does not preserve a previously available spatial polymer norm without an additional theorem. Its exactness is compatible with a difficult new response estimate.

4. Exactly when a reference score cancels

4.1 A normalized reference family Let $K_\theta(da \mid \upsilon)$ be a differentiable family of probability kernels on a common support, with K_0 positive there. Assume the required finite derivatives are integrable so that differentiation under the integral is justified. Define its likelihood jet and logarithmic score jets by

$$L_\theta(a, \upsilon) = \frac{dK_\theta}{dK_0}(a \mid \upsilon), \qquad c_r(a, \upsilon) = \partial_\theta^r \log L_\theta(a, \upsilon)\big|_{\theta=0}. \tag{4.1}$$

Let Π_K average the entire reference variable a, retaining only υ. Normalization gives the exact formal identity

$$\boxed{\Pi_K L_\theta = 1.} \tag{4.2}$$

If a marked jet $G(t, \upsilon)$ depends only on the retained data, then

$$\boxed{\Pi_K[G(t, \upsilon) L_\theta(a, \upsilon)] = G(t, \upsilon).} \tag{4.3}$$

Every derivative in both t and θ follows coefficient by coefficient. This is the precise all-order cancellation of an independent normalized reference likelihood.

At first and second orders, normalization says

$$\Pi_K c_1 = 0, \qquad \Pi_K(c_2 + c_1^2) = 0. \tag{4.4}$$

Equivalently,

$$\Pi_K c_2 + \mathrm{Var}_{K_0}(c_1) = 0. \tag{4.5}$$

The second logarithmic derivative therefore does not separately average to zero. Its cancellation uses the covariance response in the reset logarithm.

For example, take $K_\theta = N(\theta, 1)$. Then $c_1(a) = a$, $c_2(a) = -1$. Their separate expectations are $0, -1$, while the full second jet has expectation $-1 + \mathbb{E}a^2 = 0$.

4.2 Application to a single positive R replacement For a source-independent integration of the old variable u, followed by a normalized reference draw u', write

$$\rho'_\beta(u', v) = K_\beta(u' \mid v)\, M_\beta(v), \qquad M_\beta(v) = \int \rho_\beta(u, v)\, du. \tag{4.6}$$

The exact transport of an old observable is

$$Q_\beta F(u', v) = \frac{\int \rho_\beta(u, v) F(u, v)\, du}{M_\beta(v)}. \tag{4.7}$$

It depends only on v. The same is true for the entire transported old source jet. Consequently (4.3) applies to a variation of the normalized reference factor at this operation, if all of u' is averaged and the retained-data law M_β is treated separately.

For the bare scalar score $\rho_\beta = e^{-\beta S}$,

$$-\partial_\beta \log \rho'_\beta = Q_\beta S - \partial_\beta \log K_\beta. \tag{4.8}$$

Averaging in u' removes the first reference score and leaves $-\partial_\beta \log M_\beta = Q_\beta S$. The full likelihood-jet identity, rather than a separate deletion of higher logarithmic scores, gives its all-order version.

This sharpens the earlier generic warning about mean-zero currents. A reference score is indeed orthogonal to every transported old mark that has become a function only of v at this single replacement. It is not automatically orthogonal to a mark that depends on the fresh reference field, or to a later quantity whose construction has already used that field.

4.3 The cancellation has precise limits If $G = G(a, v)$ still depends on a, generally

$$\Pi_K(Gc_1) \neq 0. \tag{4.9}$$

For $K_\theta = N(\theta, 1)$ and $G(a) = a$, the left side is one.

If only some reference variables are averaged and a is retained in the matching state, the score need not disappear. If a subsequent coarse map mixes a into its new retained variable, the projected score can again be nonzero. A later global mean-zero identity does not supply spatial locality.

Finally, the theorem assumes a common-support differentiable reference family. Moving hard selectors, changes of charts or deterministic maps, and singular source-dependent supports require their own derivatives or interface terms. Equation (4.3) does not make those terms vanish.

The representation (4.6) separates the normalized reference factor from the old integrated density. It does not bound the derivative of $M_\beta(v)$, which is precisely the scalar matching quantity that remains to be estimated.

5. Scalar scores after arbitrary frozen positive channels

This section records a precise target for a possible coupling-response proof. It does not derive the needed uniform bound.

Let μ_θ be the finite-cutoff Wilson measure with real local parameters,

$$d\mu_\theta(U) = \frac{e^{-\sum_i \theta_i A_i(U)}}{\mathbb{E}_{\mu_0} e^{-\sum_i \theta_i A_i}} d\mu_0(U). \tag{5.1}$$

The finitely many A_i may be plaquette energies or fixed smeared linear combinations. At a finite compact cutoff they are bounded. Let $\mathcal{K}$ be any fixed source-free positive channel, including any finite sequence of the resets above, and let V be its output. Write $\nu_\theta = \mu_\theta \mathcal{K}$.

The normalized likelihood ratio of the output is exactly

$$\frac{d\nu_\theta}{d\nu_0}(V) = \frac{\mathbb{E}_{\mu_0,\mathcal{K}}[e^{-\sum_i \theta_i A_i} \mid V]}{\mathbb{E}_{\mu_0} e^{-\sum_i \theta_i A_i}}. \tag{5.2}$$

Its first two logarithmic derivatives are

$$\partial_i \log \frac{d\nu_\theta}{d\nu_0}\Big|_0 = -\mathbb{E}[A_i \mid V] + \mathbb{E}A_i, \tag{5.3}$$

$$\partial_i \partial_j \log \frac{d\nu_\theta}{d\nu_0}\Big|_0 = \mathrm{Cov}(A_i, A_j \mid V) - \mathrm{Cov}(A_i, A_j). \tag{5.4}$$

At every order, the derivative is $(-1)^m$ times the corresponding conditional cumulant minus the unconditional cumulant. These formulas follow by differentiating (5.2), justified by boundedness at the finite cutoff.

For the first score vector, conditional Jensen gives the Fisher-information matrix inequality

$$\mathrm{Cov}(\mathbb{E}[A \mid V]) \preceq \mathrm{Cov}(A). \tag{5.5}$$

The difference is the positive matrix $\mathbb{E}\ \mathrm{Cov}(A \mid V)$. This is exactly the second-order response discarded by single-mark projection.

For the global scalar coupling, $A = S$, so the current scalar insertion is a derivative of the frozen-channel physical marginal. This avoids fictitious derivatives of a source-dependent reference draw, but its uniform local derivative norm remains unknown. The right side of (5.5) has the microscopic ultraviolet size. The inequality alone does not replace that size by the current physical scale.

An effective-action formula proved for a different, coupling-dependent channel cannot be substituted into (5.3) without the channel-current terms. Conversely, keeping the channel frozen makes (5.3) exact but requires the actual frozen-channel derivative estimates.

6. What the primary coupling regularity does and does not supply

6.1 Checked source statements — L The article of CMP109, printed p.263, states that a localized $E^{(j)}$ is analytic in its admitted gauge-field/current domain, is C^∞ in the displayed coupling parameter on a fixed interval (or analytic), and obeys the field activity bound (1.18).

Printed p.264, following (1.22), states that the beta function is smooth on its admitted interval and uniformly bounded there together with all derivatives. Theorem 3 asserts the displayed small-field inductive assumptions when the couplings and construction parameters satisfy its restrictions.

Printed p.298, in the final paragraph before the references, explicitly explains that β_j also depends on all the preceding couplings, although the notation displays only g_{j-1}.

These are meaningful parameter-regularity inputs in the stated small-field construction. They are not a theorem about the complete local source field $j(x)$, nor a checked bound on every mixed derivative in all earlier coupling variables and moving history data.

6.2 The missing total derivative is identifiable If one writes the actual dependence as

$$b_j = b_j(g_0, \dots, g_{j-1}),$$

then along a bare-parameter path $\lambda \mapsto (g_0(\lambda), \dots, g_{j-1}(\lambda))$,

$$\frac{d}{d\lambda} b_j = \sum_{r=0}^{j-1} \partial_{g_r} b_j \, \frac{dg_r}{d\lambda}. \tag{6.1}$$

Higher derivatives contain the corresponding mixed history derivatives. Uniform bounds on derivatives in the last displayed argument alone do not bound (6.1).

This failure is elementary even for analytic bounded functions. On a fixed real coupling cube, take

$$b_j(g_0, \dots, g_{j-1}) = b_* + \varepsilon \sin(N_j g_0), \qquad j \geq 2. \tag{6.2}$$

The values are uniformly bounded and every positive derivative in g_{j-1} is zero. Nevertheless $\partial_{g_0} b_j = \varepsilon N_j \cos(N_j g_0)$ can be arbitrarily large. This is a counterexample to the proposed inference from displayed-argument bounds; it is not asserted to be a Balaban beta function.

Likewise, a uniform activity bound on a complex field domain does not imply a uniform derivative in a different real coupling parameter. Multiplication by $\sin(Ng)$ preserves the field-domain norm and smoothness while making the coupling derivative large. A uniform complex coupling neighborhood or a proved differentiated activity induction would rule out this example; neither can be inferred merely from field analyticity.

Even a total derivative for a homogeneous coupling would still be only the integral of the local insertion. A local source theorem has to distinguish source positions, control the corresponding carrier norms, and retain the generated partial-collision responses.

7. The remaining matching theorem after the exact reset reduction

The new representation permits choosing the single-mark state after each reset to be a function of V_k alone. Write

$$\widehat{F}_{i,k} = \Pi_k F_{i,k}.$$

This removes artificial large norms caused solely by an unnecessarily retained auxiliary label. A sufficient next theorem must nevertheless prove both:

$$\|\widehat{F}_{i,k}\|_{L^p(\nu_k)} \leq A p^s h_k^{-4} \ell(h_k)^b, \tag{7.1}$$

and a current-scale, spatially decaying bound for every product error from the original integration operations and the new resets. In particular, it must include

$$\Pi_k\Big(\prod_i F_{i,k}\Big) - \prod_i \Pi_k F_{i,k} \tag{7.2}$$

with the same type of shell decay required in the original S0 interface. The exact full-product telescope remains valid when these additional positive operations are inserted as extra steps.

The conditional moments can be applied to (7.1) only after $\widehat{F}_{i,k}$ has been identified with the controlled native fields plus all remainders. A constant or background-dependent coefficient does not become a deterministic vacuum subtraction merely because a history was averaged.

For the scalar route, one possible concrete proof would establish a localized derivative expansion for the frozen-channel marginal in (5.2), with all vacuum terms separated and the actual current coupling dictionary retained. The missing estimate is its field-dependent local derivative norm, not the elementary score identity.

The complete construction still needs normalized inherited-history bounds, retained-data covariance, rotation restoration, ultraviolet identification, and the operator/phase theorem. None of those statements follows from the reset representation.

8. Completion conditions and references

Statement	Status
Averaging and resampling auxiliary labels while preserving the complete unmarked construction	P: Theorem 2.1 and Corollary 2.2
Preservation of every finite full source jet through those resets	P: Corollary 2.3
Exact covariance/cumulant responses and integrated L^p bounds for a reset	P: Section 3
All-order cancellation of a fully averaged normalized reference likelihood	P under the stated common-support differentiability; Section 4
Cancellation of arbitrary moving-selector or later coupled reference currents	Not asserted
Scalar local-source score identity for an arbitrary frozen positive channel	P at finite cutoff; Section 5
Uniform current-scale norm or spatial decay for the reset responses	O
Local-source derivatives of the complete effective marginal	O
Exact matching of transported microscopic Wilson fields to native current fields	O
Full M0/F0 or Yang–Mills existence	Not proved

Source for the coupling statements:

T. Balaban, *Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions*, Communications in Mathematical Physics 109 (1987), 249–301, DOI 10.1007/BF01215223. The statements above concern printed pp.263–264 and p.298. The field-domain, coupling and source-variable scopes are kept distinct.

The positive-history and localized R framework is the one specified in the main paper and in the source-free construction of the earlier annexes. The exact conditional-expectation, reset, likelihood and Fisher-information identities in this note are proved directly; no additional unpublished theorem is imported.

Annex J — The even native expansion and its fourth-order remainder

Annex J. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Result and scope

This note strengthens the native Gaussian comparison in one precise case. At the identity retained background, freeze the inherited regular action at its actual shell value, and use the same symmetric product cutoff throughout a signed computational coupling interpolation. Then every integrated native cumulant is even in that interpolation parameter. Its second coefficient has an explicit constrained truncated-Gaussian formula, and the remainder after that coefficient is $O(g^4)$, uniformly in volume, cutoff size, and source order with the same factorial exponent two.

The theorem is P using the fixed-domain regular inputs already specified in Annex F. It does not differentiate the inherited action along the RG trajectory, remove the Gaussian cutoff, cover a nonzero retained background, or identify native fields with $Q_{0:k}F_0$. The complete-history M0/C0/H0 obligations remain open.

The improvement is an actual second-order expansion of the nonlinear conditional integral, including its exact classical action, inherited E, nonlinear constraint chart, and new coarea. It does not come from replacing the full integral by its quadratic approximation.

1. Freeze the shell before varying the computational parameter

Fix one admitted homogeneous, no-R regular integral, with the retained background equal to the identity. All old couplings, the physical nonlinear chart, the minimizing map, and the inherited regular action E_* are frozen at their actual values in that shell. Let $\mathcal{J}$ be its finite four-dimensional current coordinate lattice. Its metric satisfies

$$\sup_x \sum_{y\in\mathcal{J}} e^{-ad(x,y)} \le C(1+a^{-4}), \qquad 0<a\le 1, \tag{1.1}$$

uniformly over the admitted tori. Coordinate-block dimension is fixed.

Write $U(q)$ for the exact physical chart, including the constraint correction and minimizing-field pullback, with $U(0)=1$. Write

$$\mathcal{A}(q) = A^{\eta}(U(q)) + G(q), \qquad \mathcal{A}(0) = D\mathcal{A}(0) = 0, \qquad H = D^2\mathcal{A}(0) > 0. \tag{1.2}$$

Here $G = 0$ in the hard-gauge version; the specified analytic soft-gauge term of the previous proof may instead be included. The true constrained precision H, including the constraint contribution, has the uniform gap and weighted decay retained in the earlier covariance theorem.

Let $\mathcal{E}(q) = E_*(U(q))$, and let $J(q)$ be the new exact coarea factor. Older measure terms occur in $\mathcal{E}$, so they are not included a second time in J. Define

$$\mathcal{K}(q) = \mathcal{E}(q) - \mathcal{E}(0) + \log J(q) - \log J(0). \tag{1.3}$$

The logarithm is the real branch at the positive local density, continued on the admitted complex chart. Its nonvanishing follows from the Haar chart and the same small Neumann determinant used in the coarea proof.

Choose a common symmetric convex product

$$\mathcal{D}_P = \prod_{i\in\mathcal{J}} \{z_i \,:\, |z_i| \le P\}. \tag{1.4}$$

The signed parameter g varies over $[-g_*, g_*]$, where g_* is the actual positive shell coupling, g_* is below a fixed small ceiling, and $g_*P \le \delta$. The value of P is fixed during this interpolation. Set

$$W_g(z) = -\Big[g^{-2}\mathcal{A}(gz) - \tfrac{1}{2}zHz\Big] + \mathcal{K}(gz), \tag{1.5}$$

$$d\nu_g(z) = Z_g^{-1} 1_{\mathcal{D}_P}(z) \exp\{-\tfrac{1}{2}zHz + W_g(z)\}\, dz. \tag{1.6}$$

The removable value at $g = 0$ is $W_0 = 0$. Thus ν_0 is exactly the constrained Gaussian restricted to this same product cutoff. Overall coordinate Jacobian powers of $|g|$ cancel from normalized expectations and are not part of W_g.

For each native curvature-bilinear stencil or scalar Wilson-action stencil at x, let $\mathcal{P}_{\alpha,x}(q)$ be its physical numerator. At the identity,

$$\mathcal{P}_{\alpha,x}(0) = D\mathcal{P}_{\alpha,x}(0) = 0.$$

Its normalized native field at physical scale h is

$$O_{\alpha,x}(g, z) = h^{-4} g^{-2} \mathcal{P}_{\alpha,x}(gz). \tag{1.7}$$

Again, its value at zero is defined by the removable Taylor continuation. Every finite component in the specified native family is included.

This is a computational interpolation of one frozen shell. In particular, $\partial_g \mathcal{E}(gz) = D\mathcal{E}(gz)[z]$, without a term $\partial_g E_*$. The latter would belong to a different, presently unproved coupling-flow derivative theorem.

2. The physical derivative bounds needed through order six

The following finite-order extension of the previous physical-chart estimates is useful because it preserves the polynomial degree of each coupling derivative.

Lemma 2.1 — fixed-order physical locality, P with the earlier scoped inputs On a fixed smaller physical complex neighborhood, there are $C_n, c_n > 0$, for $1 \le n \le 6$, such that the native numerators obey

$$\big|\partial_{i_1} \cdots \partial_{i_n} \mathcal{P}_{\alpha,x}(q)\big| \le C_n e^{-c_n \tau(x, i_1, \dots, i_n)}. \tag{2.1}$$

For the extensive functions $\mathcal{A}, \mathcal{K}$, their differentiated tensors, with at least one fixed coordinate index, obey the corresponding volume-independent tree majorants

$$\big|\partial_{i_1} \cdots \partial_{i_n} \mathcal{F}(q)\big| \le C_n e^{-c_n \tau(i_1, \dots, i_n)}, \qquad \mathcal{F} = \mathcal{A}, \mathcal{K}. \tag{2.2}$$

At order one, the right side means a uniform bound for each fixed derivative. Constants may depend on the fixed group, block size, physical analytic radius and regular-action parameters. They do not depend on P, volume, fine spacing η, or shell depth within the fixed-domain scope.

Proof. The native exact chart has an analytic first derivative with exponential localization on a fixed physical complex neighborhood. Coordinate Cauchy estimates for the other $n - 1$ derivatives preserve the exponential involving any selected one of the input indices. Symmetry of mixed derivatives permits taking the minimum of these bounds. Since

$$\tau(x, i_1, \dots, i_n) \le \sum_a d(x, i_a) \le n \max_a d(x, i_a),$$

one obtains tree decay with a fixed loss in exponent. The same argument applies after composing the finite plaquette and transport stencil. Only the finite orders through six are used here.

For the complete classical action, use precisely the curvature normalization retained in the proof:

$$\mathcal{A}(q) = \sum_p \eta^4 \mathcal{L}_\eta(\mathfrak{F}_p(q)) + G(q), \qquad \mathfrak{F}_p(q) = \frac{\log U(q)_p}{i\eta^2}. \tag{2.3}$$

The homogeneous specialization of CMP102 Proposition 9 and (190) supplies the physical first-variation and spatial-gradient bounds. They give exponential localization for $D\mathfrak{F}_p$, with no inverse power of η. Repeated coordinate Cauchy, as above, extends the already proved orders one through three to the finite orders through six.

Every derivative through order six of

$$\mathcal{L}_\eta(F) = \eta^{-4}\left[1 - \frac{1}{2N}\operatorname{Tr}(e^{i\eta^2 F} + e^{-i\eta^2 F})\right]$$

is uniformly bounded on the smaller physical curvature ball. This follows either from its convergent even power series or from its analytic bound on a slightly larger fixed ball. The full chain rule retains the factor η^4 in every term. After fixing one current coordinate, the fine-plaquette sum satisfies $\sum_p \eta^4 e^{-cd(x_p,i_1)} \le C$. The remaining current-coordinate sums are exponential. This proves (2.2) for $\mathcal{A}$, retaining all nonlinear chain-rule terms. The specified block-local analytic G has the same finite-order bounds.

For $\mathcal{E}$, apply coordinate Cauchy to the fixed-domain CMP116 regular activity bound used. Differentiation adds at most a fixed power of the activity-carrier size. Its exponential carrier entropy reserve absorbs each such power through order six. The inherited old-scale sum is the one already performed in that regular bound.

For $\log J$, the differentiated Haar terms are local analytic terms. The constraint determinant has its convergent trace/Neumann expansion. Each first derivative anchors the trace; each second derivative has the weighted decay already proved. These estimates hold on a smaller complex neighborhood. Applying Cauchy to a second derivative preserves decay between any selected pair of indices. Taking the largest pair distance gives tree decay, since the tree on n points has length at most $(n-1)$ times their diameter. This proves the remaining finite-order bounds without a volume factor. □

The restriction to the fixed homogeneous analytic domain is essential. This lemma does not assert the missing complex tube for arbitrary CMP119 histories.

3. Coupling derivatives are controlled polynomial marks

Write

$$O_i^{(a)}(g,z) = \partial_g^a O_i(g,z), \qquad W_r(g,z) = \partial_g^r W_g(z). \tag{3.1}$$

For $0 \le a \le 4$, Taylor's integral identity gives

$$O_i^{(a)}(g,z) = h^{-4}\int_0^1 (1-t)t^a D^{a+2}\mathcal{P}_i(tgz)[z^{a+2}]\,dt, \tag{3.2}$$

and the particularly useful first-gradient identity is

$$D_{z_y} O_i^{(a)}(g,z) = h^{-4}\int_0^1 t^a D^{a+2}\mathcal{P}_i(tgz)[e_y, z^{a+1}]\,dt. \tag{3.3}$$

To obtain (3.3), first write $D_{z_y}(g^{-2}\mathcal{P}_i(gz)) = g^{-1}D_y\mathcal{P}_i(gz)$, use $D\mathcal{P}_i(0) = 0$, and only then differentiate in g. This avoids an unnecessary apparent extra polynomial degree.

Likewise, for $1 \le r \le 4$,

$$D_{z_y}\partial_g^r[g^{-2}\mathcal{A}(gz)] = \int_0^1 t^r D^{r+2}\mathcal{A}(tgz)[e_y, z^{r+1}]\,dt. \tag{3.4}$$

For the second coordinate derivative there is a simpler exact identity:

$$D_{z_i}D_{z_y}\partial_g^r[g^{-2}\mathcal{A}(gz)] = D^{r+2}\mathcal{A}(gz)[e_i, e_y, z^r]. \tag{3.4a}$$

Indeed, taking the two coordinate derivatives first cancels g^{-2}, leaving $D^2\mathcal{A}(gz)[e_i, e_y]$; then differentiate r times in g. This gives a degree-r majorant directly and needs no seventh physical derivative. The identities extend to $g = 0$ by their removable continuations.

For the analytic remainder,

$$\partial_g^r \mathcal{K}(gz) = D^r \mathcal{K}(gz)[z^r].$$

Its first two coordinate derivatives have lower polynomial degree than the bounds just described; factors gz_j are treated in the same way.

Choose one sufficiently small decay rate $\gamma > 0$ below all the finite-order rates in Lemma 2.1 and the covariance rate. Put

$$Z_x(z) = \sum_j e^{-2\gamma d(x,j)} |z_j|.$$

The mean and concentration proof applies uniformly to all signed g in the present interval: the classical Hessian perturbation is $O(|g|P)$, the E/coarea perturbation is $O(g^2)$, and the linear-to-full-density interpolation has the same gap. Its first-gradient argument uses $|g|$ and yields

$$\sup_{g,i} \| z_i \|_{L^p(\nu_g)} \le C\sqrt{p}, \qquad p \ge 1. \tag{3.5}$$

Consequently (3.2)–(3.3), Lemma 2.1 and Hölder/Minkowski give

$$\boxed{\|O_i^{(a)}\|_p \le C_a h^{-4} p^{1+a/2}, \qquad \left\| \|\nabla O_i^{(a)}\|_{\gamma,x_i} \right\|_p \le C_a h^{-4} p^{(1+a)/2}.} \tag{3.6}$$

The weighted gradient norm is the one used in the real cut theorem. Centering the first quantity costs at most a factor two.

3.1 Exact anchored decomposition of every W_r Since $W_r(g,0) = 0$, define

$$R_{r,y}(g,z) = \int_0^1 \langle z_y, D_y W_r(g, sz) \rangle \, ds. \tag{3.7}$$

The radial fundamental theorem of calculus gives the exact identity

$$W_r(g,z) = \sum_{y \in \mathcal{J}} R_{r,y}(g,z). \tag{3.8}$$

The degree bounds following (3.4), and exponential summation of their fixed-order tensors, imply

$$\begin{aligned} |R_{r,y}(g,z)| &\le C_r [1 + Z_y(z)]^{r+2}, \\ \|\nabla R_{r,y}(g,z)\|_{\gamma,y} &\le C_r [1 + Z_y(z)]^{r+1}. \end{aligned} \tag{3.9}$$

For the gradient, differentiate (3.7). The two terms are $D_y W_r$ and $z_y D_i D_y W_r$; their degrees are at most $r+1$. The unused exponential decay pays the gradient weight. The estimates for $\mathcal{K}$ are of lower degree and fit (3.9).

Thus

$$\boxed{\|R_{r,y} - \mathbb{E}_{\nu_g} R_{r,y}\|_p \le C_r p^{1+r/2}, \qquad \left\| \|\nabla R_{r,y}\|_{\gamma,y} \right\|_p \le C_r p^{(1+r)/2}.} \tag{3.10}$$

All these estimates are uniform in P, volume, and the computational parameter. They concern the actual nonlinear functions, not just their Taylor polynomials at zero.

4. Four parameter derivatives retain factorial exponent two

Let

$$K_m(g) = \kappa_{\nu_g}\big(O_1(g), \dots, O_m(g)\big), \qquad m \geq 1, \tag{4.1}$$

with anchors $x_1, \dots, x_m$. For $m = 1$, the cumulant means expectation and $\tau(x_1) = 0$.

Proposition 4.1 — localized derivative bound, P For $0 \leq k \leq 4$,

$$\boxed{|K_m^{(k)}(g)| \leq C^{m+1}(m!)^2 h^{-4m} \exp\left[-\frac{\gamma\, \tau(x_1, \dots, x_m)}{4(m+4)}\right].} \tag{4.2}$$

The constant can be chosen simultaneously for those five derivative orders, for all source orders m, and for all admitted finite volumes and common cutoffs.

Proof. At finite volume the domain is fixed and bounded. The removable continuations are smooth, and ordinary differentiation of finite moments is legitimate. If a $= (a_1, \dots, a_m)$ and b_r is the multiplicity of the inserted response W_r, the exact formula is

$$\begin{aligned} K_m^{(k)}(g) = \sum_{\substack{a_i, b_r \geq 0 \\ \sum_i a_i + \sum_{r=1}^k r b_r = k}} & \frac{k!}{\prod_i a_i! \prod_{r=1}^k b_r! (r!)^{b_r}} \\ & \cdot \kappa_{\nu_g}\Big(O_1^{(a_1)}, \dots, O_m^{(a_m)}, W_1^{[b_1]}, \dots, W_k^{[b_k]}\Big). \end{aligned} \tag{4.3}$$

The notation $[b]$ means b repeated cumulant arguments, not a product inserted as one argument. Formula (4.3) follows by expanding, at the current g,

$$\log \mathbb{E}_{\nu_g} \exp\left[W_{g+v} - W_g + \sum_i t_i O_i(g+v)\right]$$

and extracting the coefficient $v^k \prod_i t_i$. The normalization term with no t_i has no effect for $m \geq 1$.

Let $n = \sum_r b_r \leq k$, $N = m + n$, and replace every W_r by (3.8). At fixed inserted anchors $y_1, \dots, y_n$, there are N polynomial-growth cumulant arguments. Their total growth degree is

$$\sum_i (2 + a_i) + \sum_r b_r (2 + r) = 2N + k. \tag{4.4}$$

Apply the real cut theorem at moment order $2N$. Its two gradient factors reduce the total power of N by one. The product of the value/gradient budgets is therefore at most

$$C^{N+1} h^{-4m} N^{N+k/2-1}. \tag{4.5}$$

The cut partition coefficient is at most $(N-1)!2^{N-2}$, and there are at most N^2 crossing pairs. Since $N^N \leq e^N N!$, $k \leq 4$, and $N \leq m + 4$, these factors are bounded by

$$C^{m+1}(m!)^2 h^{-4m}$$

times a fixed polynomial in $m+4$. Such a polynomial is absorbed by increasing the fixed exponential constant C^{m+1}.

Delete a longest edge of a minimum spanning tree on all N anchors. The cut theorem supplies the spatial factor

$$\exp[-\gamma\tau(x_1, \dots, x_m, y_1, \dots, y_n)/(N-1)]. \tag{4.6}$$

When $n \geq 1$, write the enlarged tree length as T. Doubling and shortcutting its walk through the external anchors shows

$$T \geq \tfrac{1}{2}\tau(x_1, \dots, x_m).$$

Its paths also give $T \geq d(y_l, x_1)$ for each l, hence

$$T \geq \frac{1}{n}\sum_{l=1}^{n} d(y_l, x_1).$$

Splitting (4.6) between these two bounds gives

$$e^{-\gamma T/(N-1)} \leq e^{-\gamma\tau(x_1,\dots,x_m)/(4(N-1))} \prod_{l=1}^{n} e^{-\gamma d(y_l,x_1)/(2n(N-1))}. \tag{4.7}$$

By (1.1), summing the inserted anchors costs at most

$$C_n[1 + N^{4n}], \qquad n \leq 4, \tag{4.8}$$

another fixed polynomial. For $n = 0$, no anchor sum is needed. The case $N = 1$ is the direct first-moment bound (3.6). Finally, the number and coefficients of terms in (4.3), for $k \leq 4$, are bounded by a fixed polynomial in $m + 4$. Combining these bounds proves (4.2), with its slightly weakened common range. □

This proof does not apply a general degree-six cumulant estimate to every mark. Only at most four coupling derivatives occur. The total degree increment is k, so its cost is polynomial in the source order, preserving the exponent $(m!)^2$.

5. Exact parity and the Gaussian second coefficient

Theorem 5.1 — even native expansion, P In the stated fixed zero-background regular law,

$$K_m(-g) = K_m(g) \tag{5.1}$$

for every source order and field-label tuple. Moreover,

$$\boxed{K_m(g) = K_m(0) + \frac{g^2}{2}\,\mathcal{C}_{2,m} + \mathcal{R}_{4,m}(g),} \tag{5.2}$$

where $\mathcal{C}_{2,m} = K_m''(0)$ and

$$\boxed{|\mathcal{R}_{4,m}(g)| \leq |g|^4 C^{m+1}(m!)^2 h^{-4m} \exp\!\left[-\frac{\gamma\,\tau(x_1, \dots, x_m)}{4(m+4)}\right].} \tag{5.3}$$

The coefficient $\mathcal{C}_{2,m}$ obeys the same bound without the factor $|g|^4$.

Proof. Under $(g, z) \mapsto (-g, -z)$, the physical argument gz is unchanged, g^{-2} is unchanged, and zHz is unchanged. Thus

$$W_{-g}(-z) = W_g(z), \qquad O_i(-g, -z) = O_i(g, z).$$

The domain and Lebesgue measure are invariant under $z \mapsto -z$. Every joint moment, and therefore every cumulant, satisfies (5.1). In particular $K_m'(0) = K_m'''(0) = 0$.

Taylor's formula with integral remainder gives

$$\mathcal{R}_{4,m}(g) = \frac{g^4}{6}\int_0^1 (1-s)^3 K_m^{(4)}(sg)\,ds. \tag{5.4}$$

Apply Proposition 4.1. Its second-derivative bound also controls the displayed coefficient. □

5.1 Explicit coefficient Use the following polynomials, with all derivatives evaluated at the identity physical field:

$$O_{i,r} := O_i^{(r)}(0,z) = \frac{h^{-4}}{(r+1)(r+2)} D^{r+2}\mathcal{P}_i(0)[z^{r+2}], \qquad r = 0,1,2, \tag{5.5}$$

$$W_r := W_r(0,z) = -\frac{1}{(r+1)(r+2)} D^{r+2}\mathcal{A}(0)[z^{r+2}] + D^r\mathcal{K}(0)[z^r], \qquad r = 1,2. \tag{5.6}$$

Here $O_{i,0}$ is quadratic, $O_{i,1}$ is cubic, and $O_{i,2}$ is quartic. W_1 is a sum of cubic and linear terms; W_2 is a sum of quartic and quadratic terms. No first-jet vanishing of E or the coarea is needed.

Write κ_0 for cumulants in the actual constrained truncated Gaussian ν_0. Formula (4.3) at $k = 2$ gives

$$\begin{aligned} \mathcal{C}_{2,m} = &\sum_i \kappa_0(O_{1,0}, \dots, O_{i,2}, \dots, O_{m,0}) \\ &+ 2\sum_{i<j} \kappa_0(O_{1,0}, \dots, O_{i,1}, \dots, O_{j,1}, \dots, O_{m,0}) \\ &+ 2\sum_i \kappa_0(O_{1,0}, \dots, O_{i,1}, \dots, O_{m,0}, W_1) \\ &+ \kappa_0(O_{1,0}, \dots, O_{m,0}, W_2) \\ &+ \kappa_0(O_{1,0}, \dots, O_{m,0}, W_1, W_1). \end{aligned} \tag{5.7}$$

All normalization responses are included through cumulants. For example, for one external field the last term is the third cumulant, not merely the expectation of $O_{1,0}W_1^2$. The anchored decomposition in Section 3 gives an absolutely summed, volume-uniform meaning to the extensive W_r insertions after at least one external mark is present.

The leading term is

$$K_m(0) = \kappa_0(O_{1,0}, \dots, O_{m,0}). \tag{5.8}$$

These are exact formulas with the common hard cutoff retained. Ordinary untruncated Wick formulas must not be substituted for κ_0 without a separate cutoff-removal estimate.

6. What this does and does not settle

The $O(g)$ native comparison of Annex F is sharpened, at identity retained background and on this fixed law, to an $O(g^2)$ comparison with the quadratic native Gaussian fields. More precisely, its g^2 coefficient is (5.7), and its remaining error is the localized $O(g^4)$ bound (5.3), uniformly at every source order.

The new result includes four important restrictions.

- The inherited action is held fixed while computing the parameter derivatives. Its derivatives along the actual coupling trajectory and its dependence on earlier couplings are not bounded here.
- The retained background is the identity. At a nonzero background, $g^{-2}\mathcal{P}(gz)$ need not have a removable zero-coupling limit; this proof does not cover that case.
- The cutoff is common and symmetric. If it is changed with g, its boundary derivatives must be restored, and the displayed differentiation formula is no longer the whole response.
- The law is the specified homogeneous regular integral. It includes no arbitrary inherited R/boundary activity, and its native marks have not been identified with exact microscopic transported marks.

In particular, this theorem supplies neither the full coupling-flow derivative theorem nor a multiscale matching theorem. Its second coefficient can depend on the frozen shell action E_*. Calling it the universal beta-function coefficient would require a separate identification.

Source scope and proof dependencies

The fixed-law density, uniform mean and all-p coordinate theorem, and weighted covariance budget are those proved in Annex F, Sections 1–4. The native physical chart and coarea hypotheses are those of Annex B, Section 5. The complete classical curvature normalization and homogeneous use of CMP102 are in Annex B, Section 9. The real cut bound is the real-cumulant theorem.

The primary scope remains CMP102 Proposition 9 and (190) for admitted analytic minimizing fields; CMP109/CMP116 for the fixed-domain complete regular activities and their carrier summation; and the exact source-free constraint chart. No complete-history complex domain or coupling derivative is imported from those field-analytic statements.

The extension to six fixed physical derivatives, polynomial coupling-derivative budgets, finite-anchor summation, parity identity, and second-order expansion are proved in this note.

Annex K — Gaussian product-cutoff removal and low-order Wick comparison

Annex K. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

1. Main result and its scope

The Gaussian cutoff-removal lemma below is proved. It compares a centered, uniformly massive finite-dimensional Gaussian with its conditioning on a product of symmetric Euclidean balls. The error for every fixed polynomial moment and cumulant is bounded by a polynomial in the cutoff radius times $\exp(-cP^2)$, uniformly in the number of blocks.

The proof uses the explicit full Gaussian Ornstein–Uhlenbeck semigroup and a divergence identity on the product domain. It does not assume existence, stationarity or regularity of a reflected stochastic process. No estimate of the probability that *all* Gaussian coordinates satisfy the cutoff is used.

The result also covers polynomials whose coefficient kernels have a uniform absolute-sum norm, including exponentially localized polynomial observables with noncompact spatial support. A separate, precisely stated tree-bound input permits summation over a fixed number of extensive anchors. The pointwise error by itself is not summed over the whole volume.

This theorem removes a cutoff from a Gaussian reference law. It does not remove the regular-window cutoff from the interacting native law or prove the complete-history C0 estimate.

2. Assumptions and notation

Let I be any finite block set and

$$\mathcal{X}_I = \bigoplus_{i\in I} \mathbb{R}^{d_i}, \qquad 1 \le d_i \le d_*. \tag{2.1}$$

Let H be a real symmetric matrix on this space. Assume

$$H \ge \lambda I, \qquad H_{ii} \le \Lambda I_{d_i}, \qquad \lambda > 0, \quad \Lambda < \infty, \tag{2.2}$$

with constants independent of the volume. Write

$$A_H := \sup_{i\in I} \sum_{j\in I} \int_0^\infty \|(e^{-tH})_{ij}\|_{\mathrm{op}}\, dt \le A < \infty. \tag{2.3}$$

Condition (2.3) is the exact semigroup locality assumption needed below. A sufficient stronger condition is

$$\sup_i \sum_j e^{\mu d(i,j)} \|(e^{-tH})_{ij}\|_{\mathrm{op}} \le Ce^{-\lambda_1 t}, \tag{2.4}$$

which gives $A \le C/\lambda_1$. The spectral gap and the diagonal-block bound alone are not silently substituted for (2.3).

Define

$$\nu_\infty(dz) = Z_\infty^{-1} e^{-\frac{1}{2}z^T Hz}\, dz, \qquad K_P = \prod_{i\in I} \overline{B}_{d_i}(0,P), \qquad \nu_P = \nu_\infty(\,\cdot \mid K_P), \quad P \ge 1. \tag{2.5}$$

Thus $Z_P = \int_{K_P} \exp(-z^T Hz/2)\, dz$ is the unnormalized constrained integral. Matrix inequalities in (2.2) use the quadratic-form order.

The boundary of each factor is a sphere, including the two endpoints when $d_i = 1$. The products of two or more factor boundaries have zero surface measure in the full boundary of K_P. The usual divergence theorem applies face by face, or directly to this bounded convex Lipschitz domain.

For a bounded Lipschitz function f, set

$$\mathrm{Lip}_i(f) = \operatorname*{ess\,sup}_z |\nabla_i f(z)|. \tag{2.6}$$

All constants below depend only on the displayed structural parameters and the indicated fixed polynomial degrees or cumulant orders.

3. A one-block marginal has a uniform boundary density

Lemma 3.1 — effective-potential sandwich Fix i, write $z = (x, w)$ with $x = z_i$, and integrate over the other constrained blocks:

$$e^{-v_i(x)} := \int_{\prod_{j\neq i} B_{d_j}(0,P)} \exp\left[-\frac{1}{2}(x,w)^T H(x,w)\right] dw. \tag{3.1}$$

Then v_i is a smooth even function on $\mathbb{R}^{d_i}$, and

$$\lambda I \le v_i''(x) \le H_{ii} \le \Lambda I. \tag{3.2}$$

Consequently,

$$\frac{\lambda}{2}|x|^2 \le v_i(x) - v_i(0) \le \frac{\Lambda}{2}|x|^2. \tag{3.3}$$

Proof. The integration domain in (3.1) is compact and independent of x, so differentiation is justified to every finite order. Evenness follows by replacing w by $-w$. Differentiating twice gives

$$v_i''(x) = H_{ii} - H_{i,-i}\,\mathrm{Cov}_{x,P}(w) H_{-i,i} \le H_{ii}. \tag{3.4}$$

For the lower bound, the function

$$\frac{1}{2}(x,w)^T H(x,w) - \frac{\lambda}{2}|x|^2 \tag{3.5}$$

is convex: its Hessian is $H - \lambda\,\mathrm{diag}(I_{d_i}, 0) \ge 0$. Its exponential with negative sign is log-concave. Integrating over the convex set of the other blocks preserves log-concavity by Prékopa's marginal theorem. Therefore $v_i(x) - \lambda|x|^2/2$ is convex, proving the lower bound in (3.2). Evenness gives $v_i'(0) = 0$, and integration along the segment from 0 to x gives (3.3). □

The sole non-elementary convexity import here is Prékopa's marginal theorem: Theorem 6, printed p.342, in *On logarithmic concave measures and functions*, Acta Sci. Math. 34 (1973), 335–343. A copy of the paper is available at `https://rutcor.rutgers.edu/Prekopa/pdf/SCIENT2.pdf`.

Corollary 3.2 — boundary flux and block moments The i-block marginal density of ν_P, with its continuous boundary value, satisfies

$$p_{i,P}(x) \le C_{\lambda,\Lambda,d_*} e^{-\lambda|x|^2/2} \quad (|x| \le P). \tag{3.6}$$

In particular the normalized boundary flux of face i obeys

$$b_i(P) := \frac{1}{Z_P} \int_{\substack{|z_i|=P \\ z_{-i}\in K_{P,-i}}} e^{-\frac{1}{2}z^T H z}\, d\sigma_i(z_i)\, dz_{-i} \le C P^{d_*-1} e^{-\lambda P^2/2}. \tag{3.7}$$

Every fixed block moment under ν_P is bounded uniformly in the volume and $P \ge 1$.

Proof. The denominator of the marginal is at least

$$e^{-v_i(0)} \int_{|x|\le 1} e^{-\Lambda|x|^2/2}\, dx. \tag{3.8}$$

The minimum of the last positive integral over $1 \le d_i \le d_*$ depends only on Λ, d_*. Apply the lower bound in (3.3) to the numerator. Integrating (3.6) on the sphere gives (3.7), and integrating it against $|x|^r$ gives the moment assertion. □

This is where the potentially tiny global normalization Z_P/Z_∞ is handled. It is absorbed in the single-block marginal; it never becomes a volume-dependent inverse probability in the final bound.

4. Local Lipschitz comparison through the full Gaussian semigroup

Theorem 4.1 — volume-uniform Lipschitz estimate Under (2.1)–(2.3), for every bounded Lipschitz f,

$$|\nu_P(f) - \nu_\infty(f)| \le C A P^{d_*-1} e^{-\lambda P^2/2} \sum_{i\in I} \mathrm{Lip}_i(f). \tag{4.1}$$

The estimate is useful when the sum of block Lipschitz constants is bounded independently of the volume, as it is for fixed local observables and the clipped polynomials considered below.

Proof. First take smooth f with bounded derivatives. The full Gaussian generator and semigroup are

$$\mathcal{L} = \Delta - (Hz)\cdot\nabla, \qquad \mathsf{P}_t f(z) = \mathbb{E} f(e^{-tH}z + \xi_t), \tag{4.2}$$

where ξ_t is centered Gaussian with covariance $H^{-1}(I - e^{-2tH})$. For $T > 0$, let

$$u_T(z) = \int_0^T [\mathsf{P}_t f(z) - \nu_\infty(f)]\, dt. \tag{4.3}$$

Then

$$\mathcal{L} u_T = \mathsf{P}_T f - f. \tag{4.4}$$

Writing $V(z) = z^T H z/2$, the divergence identity

$$e^{-V}\mathcal{L} u_T = \mathrm{div}(e^{-V}\nabla u_T)$$

and the divergence theorem on K_P give

$$\nu_P(\mathsf{P}_T f - f) = \frac{1}{Z_P} \sum_j \int_{\partial_j K_P} \partial_{n_j} u_T\, e^{-V}\, d\sigma. \tag{4.5}$$

No boundary condition for u_T is imposed. This is an identity for the full-space OU solution restricted to the product domain.

Differentiating the explicit semigroup yields

$$|\nabla_j \mathsf{P}_t f(z)| \le \sum_i \|(e^{-tH})_{ij}\|_{\mathrm{op}}\, \mathrm{Lip}_i(f). \tag{4.6}$$

Hence the absolute value of the right side of (4.5) is at most

$$\sum_i \mathrm{Lip}_i(f) \sum_j b_j(P) \int_0^T \|(e^{-tH})_{ij}\|_{\mathrm{op}} dt \le CAP^{d_*-1} e^{-\lambda P^2/2} \sum_i \mathrm{Lip}_i(f). \tag{4.7}$$

For each fixed finite volume, the gap implies $\mathsf{P}_T f \to \nu_\infty(f)$ uniformly on the compact set K_P. Let $T \to \infty$ in (4.5). The bound in (4.7) is already independent of the volume and T, giving (4.1).

For bounded Lipschitz f, convolve with smooth approximate identities. The block Lipschitz bounds do not increase, and the functions converge uniformly. Pass to the limit in (4.1). □

The proof justifies all boundary contributions directly. Intersections of faces contribute no surface term; no boundary local-time law or invariant coupling is being assumed.

5. Polynomial moments, including absolutely summable kernels

Fix an orthonormal basis in every block and let $a = (i, \alpha)$ denote a scalar coordinate. Write a polynomial of degree at most D as

$$Q(z) = \sum_{d=0}^{D} \sum_{a_1,\dots,a_d} c^{(d)}_{a_1\cdots a_d} z_{a_1} \cdots z_{a_d}, \qquad \|Q\|_1 := \sum_{d=0}^{D} \sum_{a_1,\dots,a_d} |c^{(d)}_{a_1\cdots a_d}|. \tag{5.1}$$

The given coefficient representation is used; no minimization over representations is required. Products satisfy $\|Q_1 Q_2\|_1 \le \|Q_1\|_1 \|Q_2\|_1$. An exponentially localized coefficient kernel with a uniform absolute-sum bound is admissible even when its support grows with the volume.

Theorem 5.1 — polynomial cutoff removal For each fixed D, there is C_D, independent of the volume and $P \ge 1$, such that

$$|\nu_P(Q) - \nu_\infty(Q)| \le C_D \|Q\|_1 (1+P)^{D+d_*} e^{-\lambda P^2/8}. \tag{5.2}$$

Proof. Let π_P be blockwise radial projection onto the radius-P ball and put $Q_P = Q \circ \pi_P$. Projection is 1-Lipschitz on each block, Q_P is bounded Lipschitz, and $Q_P = Q$ on K_P. At differentiability points,

$$\sum_i \mathrm{Lip}_i(Q_P) \le D\|Q\|_1 (1+P)^{D-1}. \tag{5.3}$$

To verify (5.3), differentiate a monomial. Each occurrence of a coordinate contributes at most $(1+P)^{d-1}$, and summing its multiplicities over the blocks gives $d \le D$. Apply Theorem 4.1:

$$|\nu_P(Q) - \nu_\infty(Q_P)| \le CAD\|Q\|_1 (1+P)^{D+d_*-2} e^{-\lambda P^2/2}. \tag{5.4}$$

It remains to compare Q and Q_P under the full Gaussian. For a monomial of degree d, telescope its product after projection. Each term is bounded by

$$\prod_{\ell=1}^{d} |z_{a_\ell}|\, 1_{\{|z_{i_r}|>P\}} \tag{5.5}$$

for one of its d coordinate occurrences, since radial projection decreases the absolute value of each scalar coordinate.

The covariance of every Gaussian block is bounded by $\lambda^{-1} I$, so

$$\mathbb{P}_{\nu_\infty}(|z_i| > P) \le 2^{d_*/2} e^{-\lambda P^2/4}. \tag{5.6}$$

This follows by applying Markov's inequality to $\exp(\lambda|z_i|^2/4)$. By Hölder, the L^2 norm of the product in (5.5) is bounded by a constant depending only on d, λ, including when indices coincide. Cauchy–Schwarz and (5.6) therefore give

$$\nu_\infty|Q - Q_P| \le C_D\|Q\|_1 e^{-\lambda P^2/8}. \tag{5.7}$$

Combine (5.4) and (5.7), enlarging the harmless polynomial prefactor to obtain (5.2). The constant polynomial case is exact. □

The estimate retains the actual coefficient norm. For an observable carrying a physical factor h^{-4}, that factor is not discarded; it must be included in $\|Q\|_1$ or removed by the chosen physical normalization.

5.1 Other fixed symmetric norm balls The same proof applies to products $K_P = \prod_i PC_i$ of centrally symmetric convex block bodies when, uniformly in the block index,

$$r_-B_{d_i} \subset C_i \subset r_+B_{d_i}, \qquad \mathcal{H}^{d_i-1}(\partial C_i) \le S_*, \quad r_- > 0. \tag{5.8}$$

These are explicit fixed-template assumptions; no nonsymmetric or shifting cutoff is included. The effective potential remains even. Its denominator is bounded using the ball of radius r_-, every boundary point has norm at least r_-P, and the boundary area scales as P^{d_i-1}. Use Euclidean nearest-point projection onto PC_i, which is 1-Lipschitz and satisfies $|\pi_{PC_i}z_i| \le |z_i|$. In the polynomial tail estimate, replace scalar-coordinate products by the corresponding block-norm products, and use $z_i \notin PC_i \Rightarrow |z_i| > r_-P$. All conclusions remain valid with changed constants and $c = \lambda r_-^2/8$ in place of $\lambda/8$.

This version covers a specified finite family of symmetric convex norm balls, provided the template constants in (5.8) are included in the application. It is not necessary to change the cutoff during the coupling interpolation.

6. Joint moments and cumulants of fixed order

Corollary 6.1 Let $Q_1, \dots, Q_m$ have degrees at most d_0, and let $A_i = \|Q_i\|_1$. For every fixed m, in particular $m \le 6$,

$$\left|\nu_P\left(\prod_{i=1}^m Q_i\right) - \nu_\infty\left(\prod_{i=1}^m Q_i\right)\right| \le C_{m,d_0}\left(\prod_i A_i\right)(1+P)^{md_0+d_*}e^{-\lambda P^2/8}, \tag{6.1}$$

and

$$|\kappa_{\nu_P}(Q_1, \dots, Q_m) - \kappa_{\nu_\infty}(Q_1, \dots, Q_m)| \le C_{m,d_0}\left(\prod_i A_i\right)(1+P)^{md_0+d_*}e^{-\lambda P^2/8}. \tag{6.2}$$

Proof. Apply Theorem 5.1 to the product polynomial. For cumulants use

$$\kappa(Q_1, \dots, Q_m) = \sum_{\pi \in \Pi_m}(|\pi|-1)!(-1)^{|\pi|-1}\prod_{B\in\pi}\nu\left(\prod_{i\in B}Q_i\right). \tag{6.3}$$

Every fixed joint moment under both laws is bounded by a constant times the product of the coefficient norms. Under ν_P, this follows from the block moment bound after (3.7) and Hölder; under ν_∞, it follows from its covariance bound. Telescope the product over blocks of each partition in (6.3), apply (6.1) to the changed factor, and sum the finitely many partitions. □

The constants are controlled at every fixed order. This corollary is not an all-orders OS-growth theorem. Both measures are even, so moments of odd total polynomial parity vanish exactly; no error bound is needed for those expressions.

7. Extensive anchors: interpolation with tree decay

The small bound in (6.2) is pointwise in the anchor positions. Summing it directly over all possible positions would introduce powers of the volume. The following separate implication is the correct way to use it.

Proposition 7.1 — anchored summation, with a tree-bound premise Let $Q_u^{(a)}$ be polynomial marks anchored at sites u, with degrees and coefficient norms bounded uniformly in u and the volume. Let m be fixed. Suppose the cumulants of both laws obey

$$\begin{aligned} &|\kappa_{\nu_P}(Q_{u_1}^{(1)}, \dots, Q_{u_m}^{(m)})| + |\kappa_{\nu_\infty}(Q_{u_1}^{(1)}, \dots, Q_{u_m}^{(m)})| \\ &\qquad \le B_m(1+P)^{b_m} \sum_{T \in \mathcal{T}_m} \prod_{\{a,b\} \in T} e^{-\mu d(u_a, u_b)}, \end{aligned} \tag{7.1}$$

with volume-uniform constants. Also suppose the anchor geometry satisfies

$$S_\gamma := \sup_u \sum_v e^{-\gamma d(u,v)} < \infty \tag{7.2}$$

uniformly in volume at the positive exponents used below.

Then, for any $r \le m-1$, summing the cutoff error over r anchor positions while retaining at least one fixed anchor gives

$$\sum_{u_{m-r+1}, \dots, u_m} |\Delta\kappa_m(u_1, \dots, u_m)| \le C_m(1+P)^{c_m} e^{-\lambda P^2/16}. \tag{7.3}$$

If the unsummed anchors are $u_1, \dots, u_{m-r}$, the bound can retain

$$C_m(1+P)^{c_m} e^{-\lambda P^2/16} e^{-\mu\tau(u_1, \dots, u_{m-r})/8}, \tag{7.4}$$

where τ is minimum spanning-tree length in the metric d.

Proof. Equation (6.2) gives

$$|\Delta\kappa_m| \le \varepsilon_m(P), \qquad \varepsilon_m(P) \le C_m(1+P)^{md_0 + d_*} e^{-\lambda P^2/8}.$$

There are at most m^{m-2} labeled trees for $m \ge 2$, and every tree has length at least $\tau(u_1, \dots, u_m)$. Thus (7.1) also gives

$$|\Delta\kappa_m| \le C_m(1+P)^{b_m} e^{-\mu\tau(u_1, \dots, u_m)}.$$

Use $\min(a,b) \le \sqrt{ab}$ to obtain

$$|\Delta\kappa_m| \le C_m(1+P)^{c_m} e^{-\lambda P^2/16} e^{-\mu\tau(u_1, \dots, u_m)/2}. \tag{7.5}$$

For any positive γ,

$$e^{-\gamma\tau(u_1, \dots, u_m)} \le \sum_{T \in \mathcal{T}_m} \prod_{\{a,b\} \in T} e^{-\gamma d(u_a, u_b)}.$$

Root a tree at a fixed anchor and sum leaves successively. Summing all its other vertices costs at most S_γ^{m-1}; summing fewer is bounded by the same full sum. This proves (7.3).

For (7.4), a doubled traversal of a tree on all anchors, followed by metric shortcuts, connects the unsummed anchors with length at most twice the original tree length. Hence

$$\tau(u_1, \dots, u_m) \ge \tfrac{1}{2}\tau(u_1, \dots, u_{m-r}).$$

Split the last exponential in (7.5) into two equal factors. Use the preceding inequality on one factor and the leaf summation at exponent $\mu/4$ on the other. □

The proposition applies at every fixed total order, whenever the corresponding two tree bounds are available. In the native second-coefficient application with at most six external fields, there are at most two extensive anchors and at most eight cumulant arguments. Four extensive anchors are also allowed at any fixed total order with at least one unsummed anchor. Bounded coefficients multiplying the anchor sums merely change C_m.

Scope of the premise. Theorem 4.1 and Corollary 6.1 prove the cutoff error under (2.1)–(2.3). This note does not derive all higher truncated-Gaussian tree bounds merely from the full Gaussian semigroup estimate. To use (7.3) for extensive anchors, apply the separately established tree theorem for the specified product-cutoff Gaussian law and check that it supplies (7.1), with the same normalization. If it does not, (7.1) remains an explicit additional input.

8. Logarithmic cutoffs and small coupling

Let

$$P(g) = a[\log(g^{-1})]^p, \qquad a > 0, \quad p > \tfrac{1}{2}. \tag{8.1}$$

For every fixed $B, c, R > 0$,

$$P(g)^B e^{-cP(g)^2} = o(g^R) \quad (g \downarrow 0). \tag{8.2}$$

Proof. Put $t = \log(g^{-1})$. The logarithm of the ratio to g^R is

$$B \log a + Bp \log t - ca^2 t^{2p} + Rt.$$

Since $2p > 1$, this tends to $-\infty$. □

Thus the fixed-degree Gaussian moments, cumulants, and the anchor sums admitted by Section 7 can be replaced by their untruncated Gaussian counterparts with an error smaller than every fixed power of g. If additional explicit fixed powers of g^{-1} occur in an observable's normalization, they can be included in this comparison. Uniformity is not claimed for an unbounded number of factors or for degrees growing with the cutoff.

9. Application conditions

Question	Result
Is the small probability of the full product-cutoff event divided out naively?	No. Lemma 3.1 controls normalized one-block boundary densities directly.
Is reflection at intersecting ball boundaries assumed to be well behaved?	No reflected process is used; the proof applies a finite-dimensional divergence theorem on a convex Lipschitz product domain.
Is a stationary synchronous coupling required?	No. The explicit full Gaussian semigroup supplies the comparison identity.
Are noncompact but exponentially localized polynomial marks covered?	Yes, when their displayed coefficient absolute-sum norms are uniform.
Can a pointwise cumulant error be summed over four extensive anchors?	Only after applying Section 7 with the two volume-uniform tree bounds; direct summation is not valid.
Does this replace an identity-background truncated Gaussian coefficient by the ordinary Gaussian coefficient?	Yes, for a fixed finite combination of the covered polynomial cumulants, with the anchor proviso above.
Does it remove the cutoff from the interacting YM conditional law?	No; the comparison here is precisely ν_P versus its own full Gaussian reference ν_∞.

Final status. The volume-uniform Gaussian product-cutoff removal theorem for fixed-order polynomial moments and cumulants is proved. Extensive-anchor removal is a proved implication with the explicit tree-bound premise (7.1), rather than an unqualified consequence of pointwise cutoff errors.

Supplement to Annex K: low-order native Wick comparison

Result and scope

The parity theorem and the Gaussian product-cutoff lemma combine into a volume-uniform expansion whose leading and second coefficients can be evaluated by ordinary finite-dimensional Gaussian Wick contraction. The result is proved for each source order $1 \le m \le 6$, including all extra extensive vertices in the second coefficient.

The inherited action E_* remains frozen at its actual shell value. The Gaussian precision is the actual finite-volume constrained precision H. This note makes no infinite-volume, continuum, universal-coefficient, or AF-identification claim. It removes the cutoff from the two Gaussian coefficients; the interacting native integral still has its specified regular-window cutoff.

The underlying results are Annex J and Annex K, Gaussian cutoff-removal theorem. The latter's boundary-flux proof has been independently checked: its effective-potential sandwich, full-space OU divergence identity, clipping estimate and fixed-order moment/cumulant telescoping are valid under its displayed hypotheses. Sections 1–3 below verify those hypotheses and the required anchored tree bounds for the present native Gaussian.

1. Gaussian semigroup locality follows from the actual precision bounds

Keep the identity-background homogeneous fixed-domain setup of the parity theorem. For this corollary, take the common cutoff to be the product of Euclidean coordinate-block balls

$$\mathcal{D}_P = \prod_{i\in\mathcal{J}} \overline{B}_{d_i}(0,P), \qquad P \ge 1, \quad d_i \le d_*. \tag{W.1}$$

Scalar coordinate cubes are included by treating their scalar coordinates as one-dimensional blocks. The displayed constants below use these Euclidean factors. The fixed centrally symmetric convex block templates covered by Section 5.1 of the Gaussian cutoff theorem give the same conclusions, with constants depending on their stated inradius, outradius and surface-area bounds and with $\lambda P^2/16$ replaced by $\lambda r_-^2 P^2/16$ in the Gaussian error. This extension keeps the actual chosen cutoff unchanged throughout the interpolation.

The two Gaussian laws are

$$d\nu_P = Z_P^{-1} 1_{\mathcal{D}_P} e^{-zHz/2}\, dz, \qquad d\nu_\infty = Z_\infty^{-1} e^{-zHz/2}\, dz. \tag{W.2}$$

The actual precision has a volume-uniform gap and physical locality:

$$H \ge \lambda I, \qquad \|H_{ij}\|_{\mathrm{op}} \le C e^{-\mu_0 d(i,j)}. \tag{W.3}$$

The gap is the retained constrained Gaussian input. The second estimate follows from the homogeneous physical second-derivative estimate for $\mathcal{A}$, evaluated at zero. In particular $H_{ii} \le \Lambda I$ for a fixed Λ.

To verify the additional semigroup hypothesis in the cutoff lemma, define the block-diagonal weight

$$(W_x v)_i = e^{ad(i,x)} v_i.$$

Metric triangle inequalities and the exponential row/column sums in (W.3) imply

$$\sup_x \|W_x H W_x^{-1} - H\|_{2\to 2} \le \rho(a), \qquad \rho(a) \longrightarrow 0 \quad (a \downarrow 0), \tag{W.4}$$

uniformly in volume. For example, the Schur estimate is bounded by a fixed multiple of $\sup_i \sum_j (e^{ad(i,j)} - 1) e^{-\mu_0 d(i,j)}$. The four-dimensional current-lattice volume bound makes this expression tend to zero uniformly.

Choose $a > 0$ with $\rho(a) \le \lambda/2$. The real part of $W_x H W_x^{-1}$ is then at least $\lambda/2$, so its matrix semigroup obeys

$$\|W_x e^{-tH} W_x^{-1}\|_{2\to 2} \le e^{-\lambda t/2}. \tag{W.5}$$

Taking $x = i$, bounding the block from i to j, and using symmetry of e^{-tH} gives

$$\|(e^{-tH})_{ij}\|_{\mathrm{op}} \le e^{-ad(i,j)} e^{-\lambda t/2}. \tag{W.6}$$

Consequently

$$\sup_i \sum_j \int_0^\infty \|(e^{-tH})_{ij}\|_{\mathrm{op}}\, dt \le \frac{2}{\lambda} \sup_i \sum_j e^{-ad(i,j)} < \infty, \tag{W.7}$$

uniformly in volume. Thus the cutoff lemma's semigroup assumption is supplied by the existing homogeneous precision estimate; it is not inferred from the gap alone.

2. A weighted polynomial coefficient norm

Fix orthonormal coordinates in every block, and write a scalar coordinate as $b = (j, \alpha)$, with spatial block $j(b) = j$. For a polynomial anchored at x, choose a coefficient representation

$$Q_x(z) = \sum_{d=0}^{D} \sum_{b_1,\dots,b_d} c^{(d)}_{b_1\cdots b_d} z_{b_1} \cdots z_{b_d}. \tag{W.8}$$

Define its weighted absolute-sum norm by

$$\|Q_x\|_{\theta,1} = \sum_{d=0}^{D} \sum_{b_1,\dots,b_d} |c^{(d)}_{b_1\cdots b_d}| e^{\theta\tau(x, j(b_1),\dots,j(b_d))}. \tag{W.9}$$

The degree-zero term has weight one. This is a norm on the chosen coefficient representation; no minimization over representations is needed. In particular it dominates the coefficient norm used in the Gaussian cutoff lemma.

Let

$$O_{i,r} = \partial_g^r O_i(g, z)|_{g=0}, \quad r = 0, 1, 2,$$

and let

$$W_r = \partial_g^r W_g(z)|_{g=0}, \quad r = 1, 2. \tag{W.10}$$

These are exactly the polynomials in the parity theorem. Its finite physical derivative bounds give, for one fixed $\theta > 0$,

$$\deg O_{i,r} \le r + 2, \qquad \|O_{i,r}\|_{\theta,1} \le C h^{-4}. \tag{W.11}$$

The constants are uniform over the finite native field family, volume, and admitted shell data. To verify the absolute summation, the physical derivative tensor is bounded by $C_r e^{-c_r \tau}$. Choose $\theta < c_r$ with a fixed margin. The sum of the remaining exponential over its at most four coordinate positions is finite uniformly in volume: bound it by a sum over finitely many trees rooted at the anchor and sum successive leaves.

2.1 The extensive responses retain their own anchors Use the exact radial decomposition, evaluated at zero computational coupling:

$$W_r = \sum_y R_{r,y}, \qquad R_{r,y}(z) = \int_0^1 \langle z_y, D_y W_r(sz) \rangle\, ds. \tag{W.12}$$

It gives

$$\deg R_{1,y} \le 3, \qquad \deg R_{2,y} \le 4, \qquad \|R_{r,y}\|_{\theta,1} \le C. \tag{W.13}$$

Indeed W_r is the sum of its classical homogeneous term of degree $r+2$ and its E/coarea homogeneous term of degree r. Radial decomposition assigns one coordinate slot to y; the remaining physical derivative tensor has the same tree majorant anchored there. The fixed derivative locality therefore proves (W.13) directly. No volume-sized norm is assigned to the entire W_r.

Only derivatives through order four are needed for the coefficient polynomials in this note. The parity remainder uses its separately established physical derivatives through order six.

3. Both Gaussian laws have the required cut-tree bounds

The same weighted conjugation budget (W.4) gives the real covariance estimate for ν_∞ and for the Gaussian restricted to the convex product (W.1). For the restricted law, the product-boundary contribution is nonnegative and the weight is scalar on each coordinate block, as required by the previous homogeneous covariance proof.

Both laws are even under $z \mapsto -z$, so all coordinate means vanish. Linear tilting leaves their potential Hessian equal to H. The established concentration estimate consequently gives

$$\sup_{i,P} \|z_i\|_{L^p(\nu_P)} + \sup_i \|z_i\|_{L^p(\nu_\infty)} \le C\sqrt{p}. \tag{W.14}$$

Here $P \ge 1$, and constants are independent of volume. For a polynomial of fixed degree D, (W.9), Hölder, and the fact that $d(x,j) \le \tau(x, j(b_1), \dots, j(b_d))$ for every participating coordinate imply

$$\begin{aligned} \|Q_x\|_p &\le C_D \|Q_x\|_{\theta,1}\, p^{D/2}, \\ \Big\| \|\nabla Q_x\|_{\gamma,x} \Big\|_p &\le C_D \|Q_x\|_{\theta,1}\, p^{(D-1)/2}, \quad 0 < \gamma < \theta. \end{aligned} \tag{W.15}$$

These estimates hold under either Gaussian law. Centering costs at most a factor two. Thus the real cut theorem supplies, for every fixed $N \ge 2$,

$$\begin{aligned} &|\kappa_{\nu_P}(Q^{(1)}_{u_1}, \dots, Q^{(N)}_{u_N})| + |\kappa_{\nu_\infty}(Q^{(1)}_{u_1}, \dots, Q^{(N)}_{u_N})| \\ &\qquad \le C_{N,D} \prod_{j=1}^{N} \|Q^{(j)}_{u_j}\|_{\theta,1} \exp\Big[-\frac{\gamma \tau(u_1, \dots, u_N)}{N-1} \Big]. \end{aligned} \tag{W.16}$$

The constants are independent of P. This verifies the two-law spatial premise of the Gaussian cutoff lemma for the present polynomials. A sum over labeled trees is an equivalent sufficient majorant here, since the exponential of the minimum tree length is one term bounded by that tree sum.

3.1 Cutoff error with spatial decay For fixed polynomial degrees $d_1, \dots, d_N$, the cutoff lemma and its moment-partition proof give

$$|\Delta\kappa_N| \le C \left(\prod_j \|Q^{(j)}_{u_j}\|_{\theta,1} \right) (1+P)^{d_1 + \cdots + d_N + d_*} e^{-\lambda P^2/8}. \tag{W.17}$$

Combining this bound with (W.16), using $\min(A,B) \le \sqrt{AB}$, gives

$$|\Delta\kappa_N| \le C \left(\prod_j \|Q^{(j)}_{u_j}\|_{\theta,1} \right) (1+P)^{D} e^{-\lambda P^2/16} e^{-\gamma \tau(u_1, \dots, u_N)/(2(N-1))}, \tag{W.18}$$

where one can take the harmless enlarged power $D = d_1 + \cdots + d_N + d_*$. The one-argument case uses (W.17) directly and has no spatial factor.

For $N \le 8$, use the common rate $\mu = \gamma/7$. If any $n \le N-1$ anchors are summed and at least one remains fixed, split the tree exponential in (W.18) into two factors. A doubled traversal gives the lower bound of one half of the unsummed minimum tree length; the second factor is summed using exponential kernels. This yields

$$\sum_{\text{summed anchors}} |\Delta\kappa_N| \le C\left(\prod_j A_j\right)(1+P)^D e^{-\lambda P^2/16} e^{-\gamma\tau(\text{unsummed anchors})/56}, \tag{W.19}$$

when $\|Q_{u_j}^{(j)}\|_{\theta,1} \le A_j$ uniformly in any summed position. Constants depend on the fixed N, D, not on volume. This is the required use of the cutoff estimate for extensive vertices; direct summation of (W.17) would not suffice.

4. Replacement of both Gaussian coefficients

For $1 \le m \le 6$, write

$$L_m^P = \kappa_{\nu_P}(O_{1,0}, \dots, O_{m,0}), \qquad L_m^\infty = \kappa_{\nu_\infty}(O_{1,0}, \dots, O_{m,0}). \tag{W.20}$$

Let $C_{2,m}^P$ be the second coefficient in Annex J (5.7), the parity theorem, and define $C_{2,m}^\infty$ by the identical formula with every κ_{ν_P} replaced by κ_{ν_∞}. Explicitly,

$$\begin{aligned} C_{2,m}^\infty = &\sum_i \kappa_\infty(O_{1,0}, \dots, O_{i,2}, \dots, O_{m,0}) \\ &+ 2\sum_{i<j} \kappa_\infty(O_{1,0}, \dots, O_{i,1}, \dots, O_{j,1}, \dots, O_{m,0}) \\ &+ 2\sum_i \kappa_\infty(O_{1,0}, \dots, O_{i,1}, \dots, O_{m,0}, W_1) \\ &+ \kappa_\infty(O_{1,0}, \dots, O_{m,0}, W_2) \\ &+ \kappa_\infty(O_{1,0}, \dots, O_{m,0}, W_1, W_1). \end{aligned} \tag{W.21}$$

Here $\kappa_\infty = \kappa_{\nu_\infty}$. The two W_1 factors in the last line remain two cumulant arguments.

Corollary 4.1 — uniform Gaussian coefficient replacement, P For $1 \le m \le 6$,

$$\boxed{|L_m^P - L_m^\infty| + |C_{2,m}^P - C_{2,m}^\infty| \le C_m h^{-4m}(1+P)^{D_m} e^{-\lambda P^2/16} e^{-\gamma\tau(x_1,\dots,x_m)/56},} \tag{W.22}$$

where one may take $D_m = 2m + 6 + d_*$. All constants are uniform in the finite volume and admissible frozen-shell data.

Proof. The largest number of Gaussian cumulant arguments in (W.21) is $N = m + 2 \le 8$. There are at most two additional anchors, from the last line. The maximum total polynomial degree is $2m + 6$: each native $O_{i,0}$ contributes two, and each $R_{1,y}$ contributes at most three. The remaining lines have total degrees at most $2m + 2$ or $2m + 4$.

Use (W.12) in every occurrence of W_1, W_2. At fixed added anchors, (W.11) and (W.13) bound the coefficient-norm product by $C_m h^{-4m}$. Apply (W.19) to sum the zero, one or two extra anchors in the cutoff difference. For L_m there are no extra anchors. The finitely many sums over external derivative placements in (W.21) change only C_m. This proves (W.22). □

All sums of the extensive Gaussian coefficient insertions are absolutely controlled by the same tree argument. Their volume-uniform meaning is therefore established before comparing their cutoff and untruncated values.

5. The combined native expansion

Let $K_m^P(g)$ be the interacting native cumulant from the parity theorem, with the common cutoff fixed at P during the signed interpolation. Combining that theorem with (W.22) gives, for $1 \le m \le 6$,

$$\boxed{\begin{aligned} K_m^P(g) &= L_m^\infty + \frac{g^2}{2} C_{2,m}^\infty + \mathcal{E}_m(g,P), \\ |\mathcal{E}_m(g,P)| &\le C_m h^{-4m} \left[|g|^4 + (1+P)^{D_m} e^{-\lambda P^2/16} \right] e^{-\gamma \tau(x_1,\dots,x_m)/56}. \end{aligned}} \tag{W.23}$$

The working decay rate can be decreased once to be common to the parity and Gaussian cut estimates. The parity remainder has the stronger range $4(m+4) \le 40$ at these orders, so the displayed denominator 56 is a valid common weakening. The small coupling ceiling is fixed; the harmless factor $1+g^2$ multiplying (W.22) has been absorbed into C_m.

Equation (W.23) has ordinary untruncated finite-volume Gaussian coefficients. Wick's rule now applies directly to the displayed polynomials with covariance H^{-1}. It does not identify that covariance with a continuum propagator, or compute a universal Yang–Mills ultraviolet coefficient.

5.1 Logarithmic windows at the physical endpoint If the family of admitted endpoints uses

$$P(g_*) = a[\log(g_*^{-1})]^p, \qquad p > \frac{1}{2}, \tag{W.24}$$

then the cutoff term in (W.23) is smaller than every fixed power of g_*. Hence, at the physical endpoint $g = g_*$, it can be absorbed into the $O(g_*^4)$ remainder. For each member of this family, $P(g_*)$ is held constant throughout the computation of the signed g-derivatives. No derivative of a moving cutoff has been suppressed.

The fixed-order constants are uniform within the admitted shell bounds, while H, E_*, and the coefficient $C_{2,m}^\infty$ may still vary from shell to shell. In particular, the result is not a derivative estimate for that variation.

What remains

This corollary settles the Gaussian-cutoff issue for the second native coefficient at the low source orders used here, including its extensive insertion sums. The parity theorem separately retains its all-source-order factorial bound with the cutoff present.

The step from these native regular integrals to the actual transported Wilson hierarchy, the complete-history domain and geometry estimates, the identification of the running operator normalization, rotation restoration, the interacting phasewise UV estimate, and the operator/phase hypotheses remain separate obligations. No part of (W.23) supplies them by itself.

Annex L — Phasewise ultraviolet coefficients from vacuum block estimates

Annex L. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Scope

This note proves a phase-transfer theorem that uses only vacuum norms of local positive-time blocks and their scale derivatives, once a valid field/phase disintegration exists. For two- and three-field ultraviolet coefficients, its inputs are mixed-vacuum four- and six-field distributions. It does not require an operator-norm OPE bound to transfer those coefficients.

The implication is P. Its application to the interacting Wilson limit remains C: neither the required four/six-field ultraviolet block estimates nor the decomposition hypotheses have been established for that limit here. In particular, this result does not prove the requested field energy bound or construct the continuum hierarchy.

The distinction from a two- or three-point calculation is exact. Knowing the mixed-state mean of a candidate coefficient identifies its average. The additional reflected block norm controls its dispersion across components. That second piece is what rules out different ultraviolet coefficients in different phases.

1. The actual positive-time block and its vacuum norm

Assume an OS hierarchy has already been constructed and reconstructed. Let $\mathcal{A}_+$ be its polynomial positive-time test algebra. The coefficient tests are joint coincidence-flat kernels, with the component reflection/adjoint convention fixed once.

For $F \in \mathcal{A}_+$, denote the corresponding reconstructed vacuum vector by

$$\Psi(F) = [F], \qquad \langle \Psi(F), \Psi(G)\rangle = S(\Theta F \otimes G). \tag{1.1}$$

No operator closure or energy estimate is needed to define these vectors. A degree-q joint kernel is a single element of the algebra; it need not be a tensor product.

Fix $q = 2$ or $q = 3$, a compact set K of relative q-point configurations away from all coincidence diagonals, and a physical center x_* strictly in positive time. For sufficiently small $\rho > 0$, the configurations $x_* + \rho K$ remain strictly in positive time. Let $A_\rho(F) \in \mathcal{A}_+$, for $F \in C_c^\infty(K)$, be the specified rescaled q-field block, including the normalization required by the desired AF coefficient. The source order is q, although lower-order terms can be included if prescribed centering requires them.

Let $c_q(F)$ be the proposed limiting coefficient, a continuous linear functional of the relative test. Set

$$B_\rho(F) = A_\rho(F) - c_q(F)1. \tag{1.2}$$

The exact norm identity is

$$\begin{aligned}\|\Psi(B_\rho(F))\|^2 = {} & S(\Theta A_\rho(F) \otimes A_\rho(F)) \\ & - 2\,\mathrm{Re}\Big[\overline{c_q(F)} S(A_\rho(F))\Big] + |c_q(F)|^2.\end{aligned} \tag{1.3}$$

For a homogeneous block of degree q, this uses S_q and S_{2q}. The reflection keeps the two shrinking clusters a fixed physical distance apart. Internal points within either cluster remain separated by order ρ. All tests in (1.3) remain in the corrected coincidence-flat test space.

Consequently a uniform cutoff estimate for the right side, before the continuum limit and with the same field normalization, passes to the reconstructed norm at each fixed ρ. It must be an estimate for the full hierarchy. A connected $2q$-point function alone is not the right-hand side of (1.3).

2. A precise phase-disintegration input

Assume a valid decomposition of the reconstructed field system:

$$\mathcal{H} = \int^{\oplus} \mathcal{H}_\omega \, d\nu(\omega), \qquad \Omega = \int^{\oplus} \Omega_\omega \, d\nu(\omega), \qquad \|\Omega_\omega\| = 1 \quad \text{a.e.} \tag{2.1}$$

Assume the polynomial vacuum vectors disintegrate consistently:

$$\Psi(F) = \int^{\oplus} \Psi_\omega(F) \, d\nu(\omega), \quad S_\omega(F) = \langle \Omega_\omega, \Psi_\omega(F) \rangle. \tag{2.2}$$

The phase fields and their vacuum distributions must have the continuity and common-core properties supplied by the chosen valid decomposition theorem. These are hypotheses, not consequences of merely writing a direct integral.

The elementary estimate is

$$\boxed{\int |S_\omega(A_\rho(F)) - c_q(F)|^2 \, d\nu(\omega) \le \|\Psi(B_\rho(F))\|^2.} \tag{2.3}$$

Proof. In each fiber, apply Cauchy–Schwarz to $\langle \Omega_\omega, \Psi_\omega(B_\rho(F)) \rangle$, using the normalization in (2.1). Integrate the squared inequality and use the direct-integral norm identity. □

This is a variance estimate across phases. It uses the full reflected block norm and retains the one-field identity terms in the definition of the block.

3. Phase transfer on a countable ultraviolet sequence

Theorem 3.1 — phase coefficient from a summable vacuum error, P For one fixed block profile F, suppose $\rho_n \downarrow 0$ and

$$\sum_n \|\Psi(B_{\rho_n}(F))\|^2 < \infty. \tag{3.1}$$

Then

$$S_\omega(A_{\rho_n}(F)) \longrightarrow c_q(F) \quad \text{for a.e. } \omega. \tag{3.2}$$

The conclusion holds simultaneously for any countable family of profiles, field labels and coefficients for which (3.1) holds.

Proof. Sum (2.3) and apply Tonelli:

$$\int \sum_n |S_\omega(A_{\rho_n}(F)) - c_q(F)|^2 \, d\nu(\omega) < \infty.$$

The nonnegative series is therefore finite for almost every ω, and its terms tend to zero. Intersect the resulting countably many full-measure sets. □

An error $C\rho^\alpha[1 + |\log \rho|]^b$, with $\alpha > 0$, is summable on every fixed geometric sequence. This conclusion is stronger than finding one phase with a nonzero coefficient: the prescribed value transfers to almost every phase in the decomposition.

Without quantitative summability, (2.3) gives convergence in $L^2(d\nu)$, and a subsequence gives almost-everywhere convergence. It does not automatically give pointwise convergence along every prescribed sequence.

4. The continuous short-distance limit for all compact profiles

The countable result can be strengthened without assuming a phasewise ultraviolet bound in advance.

Put $t = -\log \rho$ and $B_t = B_{e^{-t}}$. Fix one compact relative configuration set K, with an open neighborhood disjoint from the coincidence diagonals. Let $\|\cdot\|_r$ denote the maximum of derivatives through order r on a fixed surrounding box. Tensor labels range over a fixed finite set.

Assume $t \mapsto A_t(F)$ is continuously differentiable in the positive-time test-algebra topology for every profile, and the phase distributions in (2.2) are continuous on that algebra. Suppose, for some fixed r,

$$\|\Psi(B_t(F))\| \le e_0(t)\|F\|_r, \qquad \|\partial_t \Psi(B_t(F))\| \le e_1(t)\|F\|_r, \tag{4.1}$$

where

$$\int_{t_0}^{\infty} \big[e_0(t)^2 + e_1(t)^2\big] dt < \infty. \tag{4.2}$$

Scale-dependent AF powers and logarithms are part of A_t, so their derivatives must be retained in the second inequality.

Theorem 4.1 — distributional phase transfer for every short-distance scale, P Under (2.1)–(2.2) and (4.1)–(4.2), there is a Sobolev order s, depending only on r and the finite profile dimension, such that for almost every phase

$$\boxed{\sup_{\substack{F \in C_c^\infty(K) \\ \|F\|_{H^s} \le 1}} |S_\omega(A_\rho(F)) - c_q(F)| \longrightarrow 0 \quad (\rho \downarrow 0).} \tag{4.3}$$

Thus the limiting coefficient holds as a distribution on the off-diagonal relative configuration space. One common full-measure set works for $q = 2, 3$, the finite field family and a countable compact exhaustion.

Proof. The profile dimension is $d = 4q$. Embed the surrounding box into a fixed flat torus and extend the compact profiles by zero. Sobolev embedding supplies an integer $\ell > r + d/2$ and a constant C_K with

$$\|F\|_r \le C_K \|F\|_{H^\ell}. \tag{4.4}$$

Write $E_v = \overline{C_c^\infty(K)}^{H^v}$ inside this torus. For an integer $u > d/2$, the inclusion

$$J : E_{\ell+u} \longrightarrow E_\ell \tag{4.5}$$

is Hilbert–Schmidt: the ambient torus inclusion has squared Hilbert–Schmidt norm a fixed multiple of $\sum_{n \in \mathbb{Z}^d} (1 + |n|^2)^{-u} < \infty$. Restricting its domain and codomain to these supported-profile closures does not increase this norm. Take $s = \ell + u$.

Define

$$(T_t F)(\omega) = S_\omega(A_t(F)) - c_q(F).$$

By (2.3), (4.1) and (4.4), T_t and its weak t-derivative extend from the profile space to bounded maps $E_\ell \to L^2(d\nu)$, with norms at most $C_K e_0(t)$ and $C_K e_1(t)$. Composing with (4.5) gives Hilbert–Schmidt maps.

Choose an orthonormal basis (f_j) of E_s. Their coefficient functions define a random dual vector $b_\omega(t) \in E_s^*$, with

$$\int \|b_\omega(t)\|_{E_s^*}^2 \, d\nu(\omega) \le C_K e_0(t)^2,$$

$$\int \|\partial_t b_\omega(t)\|_{E_s^*}^2 \, d\nu(\omega) \le C_K e_1(t)^2. \tag{4.6}$$

These statements follow by summing the squared coefficient norms and using the Hilbert–Schmidt bound. They also construct consistent measurable versions. On each bounded time interval, integrating the derivative estimate shows that $T_t J$ has an H^1 representative with values in the Hilbert–Schmidt maps $E_s \to L^2(d\nu)$. Its continuous representative equals the pointwise-defined $T_t J$ at every t; this follows first on smooth profiles from test-algebra continuity and then by the common norm bound. Evaluation at a fixed time consequently agrees in $L^2(d\nu)$ with the Fubini H^1 representative.

To identify the phasewise continuous representatives with the original phase evaluations, first use rational t and a countable set dense in the C_K^∞ topology, discarding one null set. The assumed continuity of $t \mapsto A_t(F)$, the phase distributions and the H^1 representatives gives equality for every t on this set. Approximation in C_K^∞, which also implies convergence in E_s, then gives equality for every smooth profile. The construction therefore controls the actual phase distributions, rather than only unrelated almost-everywhere versions at each separate scale.

For a Hilbert-space-valued H^1 function on a unit interval $I_n = [n, n+1]$,

$$\sup_{t\in I_n} \|b(t)\|^2 \le C \int_{I_n} \big(\|b(t)\|^2 + \|b'(t)\|^2\big) dt. \tag{4.7}$$

For example, choose a point whose norm is no larger than the interval L^2 norm, apply the fundamental theorem of calculus, and then Cauchy–Schwarz. Fubini and (4.6) show that the phase curves have the required H^1 representatives almost surely.

Integrating (4.7), summing over n, and using (4.2) gives

$$\int \sum_n \sup_{t\in I_n} \|b_\omega(t)\|^2 \, d\nu(\omega) < \infty.$$

The suprema therefore tend to zero for almost every ω. This is (4.3). Countable intersections over compact sets and the finite labels prove the final assertion. □

The test-space step is explicit. It does not identify a bound only on tensor-product tests with a bound on arbitrary joint kernels. The norm premises (4.1) are themselves for all joint profiles.

5. Connected two- and three-point coefficients

For a translation-invariant phase, the one-point distribution of a scalar field is a constant. Restrict first to block profiles satisfying $\int F(u_1, \dots, u_q)\, du_i = 0$ as a function of all remaining variables, for each i. Vanishing of the total integral alone is not this condition. Then every one-point contribution vanishes in every phase.

For $q = 2, 3$, the full block expectation on these profiles equals its connected version. Theorems 3.1 and 4.1 therefore transfer the prescribed connected coefficients, including a specified nonzero three-point value, once their vacuum block premises hold. More general profiles can be treated only after the relevant lower-order phase terms and normalization have been controlled.

This avoids the incorrect inference that an averaged nonzero three-point coefficient by itself gives the prescribed coefficient in a selected phase. The reflected four/six-field estimates supply the missing phase dispersion control.

If the decomposition theorem and the energy/covariance argument already provide their required properties on a full-measure set, intersect that set with the one supplied by Theorem 4.1. Every phase in the intersection retains both the prescribed coefficients and those properties. The present theorem does not create that energy/covariance set.

6. A Gaussian benchmark with the correct two-block geometry

The vacuum norm premise is natural in the frozen Gaussian test, but it remains an interacting obligation.

Consider a reflection-positive, stationary Gaussian curvature field whose off-diagonal covariance $C_{\alpha\beta}(x - y)$ is homogeneous of degree -4, with

$$|C_{\alpha\beta}(x - y)| \le C|x - y|^{-4}. \tag{6.1}$$

Take quadratic Wick fields O of dimension four. Their Schwinger functions on separated supports are defined by Wick contraction, with the stated tensor reflection convention. One need not claim that every Euclidean composite extends as an ordinary random distribution across its self-coincidence diagonals.

For $q = 2$ or $q = 3$, define

$$A_\rho(F) = \rho^{4q} \int F(u_1, \dots, u_q) O(x_* + \rho u_1) \cdots O(x_* + \rho u_q)\, du. \tag{6.2}$$

The profiles are supported in the fixed compact K away from internal coincidences. Gaussian homogeneity makes the vacuum coefficient $c_q(F) = S(A_\rho(F))$ independent of ρ.

Proposition 6.1 — Gaussian block-norm remainder, P For sufficiently small ρ,

$$\|\Psi(A_\rho(F) - c_q(F))\|^2 \le C_{K,q,x_*}\rho^8\|F\|_0^2. \tag{6.3}$$

Its logarithmic scale derivative has the same power, with a fixed first-derivative profile norm.

Proof. In the reflected $2q$-field moment there are $4q$ Gaussian legs and $2q$ contraction edges. The two blocks are separated by a fixed physical time gap. The diagrams with no edge between blocks give exactly $|c_q(F)|^2$, and cancel in (1.3).

Every remaining diagram has an even number $r \ge 2$ of cross-block edges: each block has $2q$ legs, and internal contractions consume an even number of them. Cross-block kernels are uniformly bounded by the fixed gap. Each of the other $2q - r$ edges costs at most $C\rho^{-4}$. The two block normalizations contribute ρ^{8q}, leaving

$$\rho^{8q}\rho^{-4(2q-r)} = \rho^{4r} \le C\rho^8 \quad (0 < \rho \le 1).$$

There are finitely many diagrams for $q = 2, 3$, and the compact profile integrals are bounded by a fixed multiple of $\|F\|_0^2$.

For the scale derivative, the smeared block in (6.2) is equivalently the degree-q field evaluated on $F((x_i - x_*)/\rho)$. With $t = -\log\rho$, differentiation replaces F by $\mathcal{D}F = \sum_i u_i \cdot \nabla_{u_i}F$. Homogeneity gives $c_q(\mathcal{D}F) = 0$. The same contraction count, with the fixed derivative profile norm, proves the derivative claim. □

Thus $e_0(t), e_1(t)$ can be taken proportional to e^{-4t} in this Gaussian benchmark. Fixed logarithmic losses remain integrable in (4.2). The calculation does not determine the interacting scalar three-point coefficient: in channels where the Gaussian coefficient vanishes, the required higher perturbative coefficient still has to be established with its uniform remainder.

7. Why this is weaker than an operator-norm OPE estimate

Let a central real Gaussian variable A act by multiplication on L^2 of its law, with vacuum 1. The block $B_\rho = \rho A$ obeys

$$\|B_\rho\Omega\| = \rho, \qquad \|B_\rho\| = \infty \quad (\rho > 0).$$

Disintegration over $A = a$ gives $B_{\rho,a} = \rho a \to 0$ in every finite-a component.

This example shows that a vacuum block estimate can transfer a coefficient even when a global operator-norm bound is false. It is not a Yang–Mills model. It also explains why the present improvement cannot supply P1's separate field energy requirement.

8. Exact remaining Yang–Mills input

The sufficient UV-transfer estimate still to be proved is the following: on the actual normalized cutoff hierarchy and uniformly before taking the cutoff/volume limit, the two-block full-moment bounds corresponding to (4.1), for $q = 2, 3$, including the logarithmic scale derivative and the actual field/coupling normalization.

The means alone are J4's usual two/three-point coefficient calculations. The norm remainders add reflected four/six-point information with a fixed macroscopic separation between the shrinking blocks. This is an observable statement; it is not an arbitrary-vector operator theorem.

No complete Wilson estimate of this strength is proved in this note. The Gaussian benchmark verifies the mechanism in its stated model, and the phase theorem shows exactly how the interacting estimate would transfer the coefficients. The independent decomposition, rotation/covariance and energy requirements remain explicit.

Source scope

The OS positive-time vector and reconstruction setup is the one retained from Osterwalder–Schrader II, CMP 42 (1975), 281–305, with the corrected flat test space. The disintegration is conditional on the hypotheses of the chosen field decomposition theorem, such as Driessler–Summers, Helv. Phys. Acta 59 (1986), 331–348. No extra decomposition theorem is imported from vacuum moments.

The phase L^2 inequality, countable and continuous-profile transfer theorems, Sobolev/Hilbert–Schmidt argument, Gaussian block calculation and central example are proved above. They are not quotations of an interacting Yang–Mills result.

Annex M — Compact SU(2) operators and the periodic-state criterion

Annex M. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Scope

The interacting physical operator obligation E0 remains open. The new results are:

1. P — an actual compact-Wilson estimate. For SU(2), a representation-uniform kinetic multiplier inequality improves the bare temporal plaquette bound from the previous square-root weak-coupling estimate to $O((1+\log\beta)/\beta)$, with an actual closed-field domain. Spatial interactions and every compact field configuration are retained.
2. C — a proved bridge on the existing cubic sequence. Uniform renormalized ground-Hamiltonian slabs imply the required periodic OS insertion inequality if a specified thermal trace increment is sublinear in the physical time side and each fixed polynomial has a controlled cutoff norm. The proof does not assume a rectangular extension of Balaban's construction, does not require a uniform mass gap, and does not replace a periodic state by a ground state without an estimate.
3. P — a sharp obstruction to omitting state control. Strictly positive finite transfer matrices can satisfy uniform ground-Hamiltonian slabs, Gaussian full-hierarchy and complex-source bounds, and an extensive pressure bound, while their periodic limits have an unbounded constant field and zero Hamiltonian. Thus those hypotheses alone do not prove the periodic-limit operator estimate.

Here P means the displayed assertion is proved, C a proved implication with unverified Yang–Mills premises, L a scoped primary-source input, and O an unproved Yang–Mills assertion. No novelty claim is made. The separate vacuum-block criterion in Annex L can transfer a prescribed ultraviolet coefficient without first deriving this operator estimate. The operator work remains necessary for the requested energy bounds and for any phase-decomposition route that uses them.

1. Normalization and the retained compact transfer comparison

Use normalized Haar measure and

$$s(g) = N - \operatorname{Re}\operatorname{Tr} g, \qquad \beta = 2Nx_0, \qquad X_B = N\beta = 2N^2x_0. \tag{1.1}$$

At a finite spatial cutoff, on the full link Hilbert space, put

$$\begin{aligned} Z_1(\beta) &= \int_G e^{-\beta s(g)}\,dg, \\ (C_\beta f)(u) &= Z_1(\beta)^{-1}\int_G e^{-\beta s(uv^{-1})} f(v)\,dv, \\ K &= \bigotimes_l C_\beta, \qquad V_s = \beta \sum_{p\ \text{spatial}} s(U_p), \\ A &= e^{-V_s/2}, \qquad T_0 = AKA. \end{aligned} \tag{1.2}$$

Gauge projection commutes with these operators. The physical restriction T has the same simple positive largest eigenvalue λ_0 as T_0. Set $H = -a^{-1}\log(T/\lambda_0) \geq 0$. The compact positive kernel and positive Fourier coefficients are used here, rather than a small-field approximation. The source normalization dictionary in (1.1) is unchanged.

The following already-proved local comparison is the starting point. For a finite link set Γ, delete exactly the spatial plaquettes incident on Γ, denote the top eigenvalue of the remaining transfer factor by λ_r, and set

$$\Delta_\Gamma = \log(\lambda_r/\lambda_0) \geq 0. \tag{1.3}$$

If $K^{1/2}A = WT_0^{1/2}$ is the polar decomposition, then W is unitary, commutes with gauge projection, and

$$\boxed{W^* L_\Gamma W \le aH + \Delta_\Gamma I, \qquad L_\Gamma = -\log K_\Gamma} \tag{1.4}$$

as closed quadratic forms. The direct bound is $\Delta_\Gamma \le 2N\beta|\mathcal{P}_\Gamma|$. On the compatible spatial geometry used there, with all incident plaquettes fitting in a fixed B-site cube,

$$0 \le \Delta_\Gamma \le B^3\{c_N + 2(N^2-1)\log(1+\beta)\}. \tag{1.5}$$

Equation (1.5) was proved by a compact-Wilson cylinder reserve and a trace limit at each fixed spatial cutoff. That elementary auxiliary Wilson integral does not import a rectangular source-free RG theorem. The unitary W is not asserted to be spatially local.

One also retains the proved closed-domain consequence. If a central multiplier on Γ satisfies

$$|F_\lambda| \le A_F(b + \ell_\Gamma(\lambda))^r, \qquad \ell_\Gamma = \sum_{l\in\Gamma} -\log c_{\lambda_l}, \quad r \ge 1, \tag{1.6}$$

then its exact midpoint realization $O_F = W^* F W$ is closed, its Hermitian version is selfadjoint, and

$$\begin{aligned} &D((1+H)^{r+1}) \subset D(O_F), \\ &\|O_F(1+H)^{-(r+1)}\| \le 2^{r+1}A_F(eb + 2r + \Delta_\Gamma + a)^r. \end{aligned} \tag{1.7}$$

This statement comes from range factorization of fractional transfer powers, followed by a dyadic spectral sum; no form inequality is squared.

2. A representation-uniform SU(2) kinetic inequality

This section specializes to $G = SU(2)$. Write $n = 1, 2, \ldots$ for the dimension of an irreducible representation, and let I_ν denote the modified Bessel function.

2.1 Exact one-link Fourier coefficients For class angle θ, normalized Haar class measure is $2\pi^{-1}\sin^2\theta\, d\theta$, $s = 2 - 2\cos\theta$, and $\chi_n(\theta) = \sin(n\theta)/\sin\theta$. Consequently

$$\begin{aligned} a_n(\beta) &= \frac{1}{n}\int_{SU(2)} e^{-\beta s(g)}\chi_n(g)\, dg \\ &= \frac{e^{-2\beta}}{n}\{I_{n-1}(2\beta) - I_{n+1}(2\beta)\} = e^{-2\beta}\frac{I_n(2\beta)}{\beta}. \end{aligned} \tag{2.1}$$

The last equality follows either by termwise differentiation of the power series or by integration by parts. Therefore

$$c_n(\beta) = \frac{a_n(\beta)}{a_1(\beta)} = \frac{I_n(2\beta)}{I_1(2\beta)}, \qquad \ell_n(\beta) = -\log c_n(\beta). \tag{2.2}$$

Only standard defining Bessel formulas are used below. They are recorded in the official DLMF at 10.25.2, 10.32.2, and 10.32.3. The inequality that follows is proved here rather than imported from that reference.

2.2 Ratio lemma Lemma 2.1 (P). For $\nu \ge 1$, $x > 0$, and $r_\nu(x) = I_{\nu+1}(x)/I_\nu(x)$,

$$0 < r_\nu(x) < 1, \qquad 0 \le x\frac{r_\nu'(x)}{r_\nu(x)} \le \frac{-\log r_\nu(x)}{\log 2}. \tag{2.3}$$

Proof. The integral representation

$$I_\nu(x) = \frac{(x/2)^\nu}{\sqrt{\pi}\,\Gamma(\nu+1/2)} \int_{-1}^{1} e^{xt}(1-t^2)^{\nu-1/2}\,dt \tag{2.4}$$

and $I_\nu' = I_{\nu+1} + \nu I_\nu/x$ identify r_ν with the mean of t under the probability density proportional to the integrand. Thus $0 < r_\nu < 1$ and $r_\nu' = \mathrm{Var}(t) > 0$.

There are two elementary upper bounds. First, write

$$I_\nu(x) = \frac{(x/2)^\nu}{\Gamma(\nu+1)} A_\nu(x^2/4), \qquad A_\nu(z) = \sum_{k\ge 0} \frac{z^k}{k!(\nu+1)_k}. \tag{2.5}$$

Here $(b)_k = b(b+1)\cdots(b+k-1)$, with $(b)_0 = 1$. Then $A_{\nu+1}(z)/A_\nu(z)$ is the mean of $(\nu+1)/(\nu+1+k)$ under the probability weights proportional to the summands of A_ν. Its derivative with respect to $\log z$ is the covariance of this decreasing function with k, and is nonpositive. Since

$$\frac{r_\nu(x)}{x} = \frac{1}{2(\nu+1)} \frac{A_{\nu+1}(x^2/4)}{A_\nu(x^2/4)},$$

we obtain

$$x r_\nu'(x) \le r_\nu(x). \tag{2.6}$$

For the second bound set $u = 1-t$ and $A_\nu^0 = \nu - \frac{1}{2} > 0$. The density of u is proportional to $e^{-xu}[u(2-u)]^{A_\nu^0}$, $0 < u < 2$. Integration of the derivative of u times that density gives

$$x\,\mathbb{E}u = A_\nu^0 + 1 - A_\nu^0 \mathbb{E}\frac{u}{2-u}. \tag{2.7}$$

The endpoint terms vanish; the displayed expectations and their x-derivatives are integrable for $\nu \ge 1$. Differentiation yields

$$x\,\mathrm{Var}(u) = \mathbb{E}u - A_\nu^0\,\mathrm{Cov}\Big(u, \frac{u}{2-u}\Big) \le \mathbb{E}u. \tag{2.8}$$

The covariance is nonnegative because both functions increase. Therefore

$$x r_\nu'(x) \le 1 - r_\nu(x). \tag{2.9}$$

Combining (2.6) and (2.9),

$$x(\log r_\nu)' \le \min\{1, (1-r_\nu)/r_\nu\} \le (-\log r_\nu)/\log 2.$$

For $r_\nu \le 1/2$, the last assertion follows from $-\log r_\nu \ge \log 2$. For $1/2 \le r_\nu < 1$, concavity of $-r\log r$ above the chord joining its values at $1/2$ and 1 gives $-r\log r \ge (1-r)\log 2$. This proves (2.3). ■

Theorem 2.2 (P: all representations and all couplings). For every $n \ge 1$ and $\beta > 0$,

$$\boxed{0 \le \beta\,\partial_\beta \log c_n(\beta) \le \frac{\ell_n(\beta)}{\log 2}.} \tag{2.10}$$

Proof. For $n \ge 2$, $c_n(\beta) = \prod_{\nu=1}^{n-1} r_\nu(2\beta)$. Sum (2.3), using $\beta\partial_\beta = x\partial_x$ at $x = 2\beta$. For $n = 1$, both sides vanish. ■

Let

$$\mu_\beta = \frac{\int s(g) e^{-\beta s(g)}\,dg}{Z_1(\beta)} = -\partial_\beta \log Z_1(\beta). \tag{2.11}$$

Equations (2.1) and (2.4) give

$$\mu_\beta = 2(1 - r_1(2\beta)), \qquad 0 < \mu_\beta \le \min\{2, 3/(2\beta)\}. \tag{2.12}$$

The $3/(2\beta)$ bound follows directly from (2.7), dropping its last nonnegative term at $\nu = 1$.

The exact multiplier of the unnormalized temporal mark s is $F_n = -a_n'/a_n$. Therefore

$$\boxed{0 \le \mu_\beta - F_n = \partial_\beta \log c_n \le \frac{\ell_n}{\beta \log 2}.} \tag{2.13}$$

This inequality includes high representations. It is stronger than a fixed-representation weak-coupling expansion.

3. Consequences for the full interacting compact-Wilson operator

Take one temporal plaquette based on a spatial link l, and let Γ contain that link. Its exact selfadjoint midpoint field is $O_s = W^* F_l W$. All spatial action factors remain in W and H.

Theorem 3.1 (P). Define $D_l = \mu_\beta I - O_s$. Then

$$\boxed{0 \le D_l \le \frac{aH + \Delta_\Gamma I}{\beta \log 2}} \tag{3.1}$$

as closed forms, and

$$\begin{aligned} &D((1+H)^2) \subset D(D_l), \\ &\|D_l (1+H)^{-2}\| \le \frac{4(2 + \Delta_\Gamma + a)}{\beta \log 2}. \end{aligned} \tag{3.2}$$

Let Ω be the normalized physical ground vector and set

$$m_l = \langle \Omega, O_s \Omega \rangle, \qquad Q_l = O_s - m_l I. \tag{3.3}$$

Then, for every $t > 0$,

$$\boxed{\|e^{-tH} Q_l e^{-tH}\| \le \frac{\Delta_\Gamma + a/(2et)}{\beta \log 2}} \tag{3.4}$$

and

$$\boxed{\|Q_l (1+H)^{-2}\| \le \frac{8 + 5\Delta_\Gamma + 4a}{\beta \log 2}, \qquad D((1+H)^2) \subset D(Q_l).} \tag{3.5}$$

Proof. Combine (2.13) and (1.4) for (3.1). Apply (1.7) to the nonnegative multiplier $\mu_\beta - F_l$, with $r = 1$, $b = 0$, and $A_F = (\beta \log 2)^{-1}$, to obtain (3.2). This also places Ω in the actual operator domain.

Let $c_l = \mu_\beta - m_l = \langle \Omega, D_l \Omega \rangle$. Then

$$0 \le c_l \le \Delta_\Gamma / (\beta \log 2). \tag{3.6}$$

The heat-domain result underlying (1.7) makes $B_D = e^{-tH} D_l e^{-tH}$ an actual bounded positive operator. The form bound gives

$$0 \le B_D, \qquad \|B_D\| \le \frac{\Delta_\Gamma + a \sup_{h \ge 0} h e^{-2th}}{\beta \log 2} = \frac{\Delta_\Gamma + a/(2et)}{\beta \log 2}. \tag{3.7}$$

Now $e^{-tH} Q_l e^{-tH} = c_l e^{-2tH} - B_D$. For positive bounded operators B, C, the selfadjoint difference satisfies $\|B - C\| \le \max(\|B\|, \|C\|)$; this follows from $-\|C\| I \le B - C \le \|B\| I$. Use this with (3.6)–(3.7), giving (3.4) without a factor of two. Finally $Q_l = c_l I - D_l$, so (3.2) and (3.6) give (3.5). ■

On the compatible geometry of (1.5), both constants are $O((1 + \log \beta)/\beta)$ at fixed $t > 0$ and $0 < a \le 1$, uniformly in total spatial volume. This improves the earlier square-root logarithmic estimate for this SU(2) channel.

The physical ultraviolet loss has not disappeared. If the prescribed dimension-four field is

$$Q_{l,a}^{R} = z_a a^{-4} Q_l + d_{a,V,\mu} I, \tag{3.8}$$

then the theorem proves only

$$\|e^{-tH} Q_{l,a}^{R} e^{-tH}\| \le |z_a| a^{-4} \frac{\Delta_\Gamma + a/(2et)}{\beta \log 2} + |d_{a,V,\mu}|. \tag{3.9}$$

Spatial smearing costs the lattice L^1 norm of the test and a finite orientation factor, rather than the full spatial volume. The factor a^{-4}, the correct subtraction $d_{a,V,\mu}$, mixing, and the remaining tensor channels still require renormalized estimates. This theorem does not extend (2.10) to arbitrary $SU(N)$.

For a temporal-only coupling variation, ground normalization makes Q_l the exact centered transfer response. The completely positive response transform therefore gives the corresponding logarithmic-Hamiltonian response bound with its explicit factor a^{-1}. That identity supplies no additional ultraviolet cancellation.

4. A same-cubic-geometry bridge from Hamiltonian slabs to periodic OS forms

4.1 Setup and precise state hypothesis Let $a_n \downarrow 0$ and $T_n \uparrow \infty$ be the existing cofinal cubic sequence, with T_n/a_n even. At the finite spatial cutoff of side T_n, let

$$P_n = e^{-a_n H_n}, \qquad H_n \ge 0, \qquad \mathcal{Z}_n(u) = \operatorname{Tr} e^{-uH_n}. \tag{4.1}$$

The normalization is by the largest eigenvalue of the same Wilson transfer operator as above.

Define the spectral trace increment

$$\Gamma_n = \log \frac{\mathcal{Z}_n(T_n/2)}{\mathcal{Z}_n(T_n)} \ge 0. \tag{4.2}$$

This is a trace of an already defined finite-spatial-volume operator. No source-free RG theorem on a rectangular torus of temporal side $T_n/2$ is assumed. Trace class at these long times follows at fixed cutoff from the trace-class transfer operator.

For each fixed positive-time polynomial test F, with all its actual lattice stencils contained in a fixed compact time interval, let $A_{F,n}$ be its positive-half-circle transfer kernel. Write

$$A_{F,n} = e^{-(T_n/2 - r_{F,n})H_n} X_{F,n}, \qquad 0 \le r_{F,n} \le R_F < \infty, \tag{4.3}$$

where $X_{F,n}$ is a bounded marked slab. The last support time and stencil margin can be included in $r_{F,n}$. Exact reflection gives

$$p_n(F)^2 = \frac{\|A_{F,n}\|_2^2}{\mathcal{Z}_n(T_n)}. \tag{4.4}$$

Assume the selected full hierarchy converges, so $p_n(F)^2 \to p(F)^2 < \infty$.

The sufficient state-and-mark condition is

$$\boxed{\frac{\Gamma_n + 2\log^+ \|X_{F,n}\|}{T_n} \longrightarrow 0 \quad \text{for every fixed } F.} \tag{4.5}$$

Here $\log^+ x = \max(0, \log x)$, with $\log^+ 0 = 0$. A convenient stronger formulation is

$$\boxed{\Gamma_n/T_n \to 0, \qquad \log(1/a_n)/T_n \to 0, \qquad \|X_{F,n}\| \le C_F a_n^{-d_F}.} \tag{4.6}$$

The exponent d_F and constant C_F may depend on the fixed polynomial, its degree, and its component tests. They are not required to be uniform over all polynomials. The final slab constant will be uniform over them.

The geometric middle condition is compatible with the proposed sequence $a_n = L^{-K_n}\ell_0$, $T_n = 2L^{M_n}\ell_0$; for example $K_n = M_n = n$ satisfies it. This geometric choice does not prove the first condition.

4.2 Hilbert–Schmidt interpolation, including OS-null limits Lemma 4.1 (P). Under (4.3)–(4.5), for every bounded sequence $t_n \geq 0$ converging to a fixed t,

$$\boxed{\frac{\|e^{t_n H_n} A_{F,n}\|_2^2}{\mathcal{Z}_n(T_n)} - p_n(F)^2 \longrightarrow 0.} \tag{4.7}$$

All terms are defined once n is large enough. In particular this includes limits with $p(F) = 0$.

Proof. Put

$$S_{F,n} = T_n/4 - r_{F,n} > 0, \qquad q_{F,n}(u) = \frac{\|e^{uH_n} A_{F,n}\|_2^2}{\mathcal{Z}_n(T_n)}, \quad 0 \leq u \leq S_{F,n}. \tag{4.8}$$

At the upper endpoint,

$$\begin{aligned} q_{F,n}(S_{F,n}) &= \frac{\mathrm{Tr}[X_{F,n}^* e^{-(T_n/2)H_n} X_{F,n}]}{\mathcal{Z}_n(T_n)} \\ &\leq \|X_{F,n}\|^2 e^{\Gamma_n}. \end{aligned} \tag{4.9}$$

Thus $\log^+ q_{F,n}(S_{F,n}) = o(S_{F,n})$.

Using the spectral resolution of H_n, $q_{F,n}(u)$ is the Laplace transform of a positive measure with variable $2u$. It is nondecreasing, since $H_n \geq 0$, and is log-convex by Hölder. Therefore, for $0 \leq t_n \leq S_{F,n}$,

$$q_{F,n}(0) \leq q_{F,n}(t_n) \leq q_{F,n}(0)^{1-t_n/S_{F,n}} q_{F,n}(S_{F,n})^{t_n/S_{F,n}}. \tag{4.10}$$

If $q_{F,n}(0) = 0$, then $A_{F,n} = 0$ and every term is zero. Otherwise use $q_{F,n}(0) = p_n(F)^2$. When $p(F)^2 > 0$, (4.10) and (4.9) squeeze the limit to $p(F)^2$. When $p(F) = 0$, eventually $q_{F,n}(0) \leq 1$ and $t_n/S_{F,n} \leq 1/2$, so

$$q_{F,n}(t_n) \leq q_{F,n}(0)^{1/2} \exp\left[\frac{t_n}{S_{F,n}} \log^+ q_{F,n}(S_{F,n})\right] \longrightarrow 0. \tag{4.11}$$

This proves (4.7), including null limits. ■

4.3 The field norm that this interpolation can transfer Assume, in addition, an actual Hermitian renormalized one-field insertion $q_{\alpha,n}(g)$ whose midpoint placements agree with the chosen hierarchy. Its compatible form sandwiches are assumed to satisfy

$$\boxed{\|e^{-uH_n} q_{\alpha,n}(g) e^{-uH_n}\| \leq C u^{-\kappa} \rho_s(g), \qquad 0 < u \leq 1,} \tag{4.12}$$

uniformly in n, with fixed C, κ, s, where $\rho_s(g) = \sum_{|\beta| \leq s} \|\partial^\beta g\|_{L^1(\mathbb{R}^3)}$. Reality and adjoint labels include the stipulated Wick phases. This is the renormalized Hamiltonian slab premise; the bare estimate in Section 3 does not prove it.

For lattice shifts $t_n \to t > 0$ and a central time δ, the exact thermal insertion identity is

$$\begin{aligned} N_{n,t_n,\delta}(F,G) &= \frac{1}{\mathcal{Z}_n(T_n)} \mathrm{Tr}\big[A_{F,n}^* e^{2t_n H_n} A_{G,n} B_{n,t_n,\delta}\big], \\ B_{n,t_n,\delta} &= e^{-(t_n-\delta)H_n} q_{\alpha,n}(g) e^{-(t_n+\delta)H_n}. \end{aligned} \tag{4.13}$$

The formula follows from $A_{\tau_t F} = e^{tH_n} A_F e^{-tH_n}$, then cyclicity of the trace. Exact half-step midpoint locations are retained; a finite shrinking stencil is accommodated by its actual support buffer.

For $|\delta| < t/8$, and eventually t_n sufficiently close to t, both heat times, after allowing this stencil buffer, exceed $t/2$. Factor out the symmetric sandwich at $t/2$. Equation (4.12) gives

$$\|B_{n,t_n,\delta}\| \leq 2^\kappa C t^{-\kappa} \rho_s(g). \tag{4.14}$$

Apply the Hilbert–Schmidt Cauchy–Schwarz inequality in (4.13):

$$|N_{n,t_n,\delta}(F,G)| \leq 2^\kappa C t^{-\kappa} \rho_s(g) \sqrt{q_{F,n}(t_n) q_{G,n}(t_n)}. \tag{4.15}$$

This bound is uniform in the allowed central location. It is not obtained by declaring periodic translations to be contractions.

Theorem 4.2 (C: same-cubic transfer to the reduced periodic OS hypothesis). Assume exact cutoff transfer realizations (4.3)–(4.4), convergence of the chosen hierarchy, the condition (4.5), and the renormalized bound (4.12). For every fixed positive-time complex polynomial F, every fixed real spatial test g, $0 < t \le 1$, and every fixed nonnegative even mollifier $\chi_\epsilon(u) = \epsilon^{-1}\chi(u/\epsilon)$ with $\epsilon < t/8$, the actual periodic hierarchy satisfies

$$\begin{aligned} &\left|S_n\big((\Theta\tau_t F)Q_{\alpha,n}(\chi_\epsilon \otimes g)(\tau_t F)\big)\right| \\ &\qquad \le C' t^{-\kappa}\rho_s(g)\, p_n(F)^2 + \varepsilon_n(F,t,\epsilon,g,\alpha), \qquad \varepsilon_n \longrightarrow 0, \end{aligned} \tag{4.16}$$

where C' is independent of F, its polynomial degree, and ϵ. Consequently the periodic OS slab-to-domain theorem applies, subject to its J1–J3 premises.

Proof. Integrate (4.15), with $G = F$, against the nonnegative lattice sampling of χ_ϵ. At each fixed ϵ, its lattice L^1 sum tends to one and is eventually at most two, including a fixed half-step sampling shift. Thus $C' = 2^{\kappa+1}C$ is sufficient. Lemma 4.1 replaces $q_{F,n}(t_n)$ by $p_n(F)^2$, with an error tending to zero for this fixed F, t.

Fine-lattice shifts $t_n \to t$ and the actual stencil placements converge in the chosen hierarchy by its uniform distribution continuity. That convergence adds another test-dependent vanishing error. Products are evaluated on the coincidence-flat positive-time test algebra, so separated support groups do not introduce an uncontrolled contact distribution. The finite-cutoff scheme must actually have these transfer realizations; they are not inferred merely from symmetry of a counterterm.

The exponents d_F in (4.6), if used, occur only in the error through $\log \|X_{F,n}\|/T_n$. They do not enter C', so the estimate retains the degree-independent constant required in the previous operator theorem. ■

The order of limits is essential and unchanged: first $n \to \infty$ at every fixed $\epsilon > 0$; then $\epsilon \downarrow 0$ in the already constructed continuum bounded forms. The previous semigroup regularity and convolution argument gives operator-norm unsmearing and the common spacetime-field domain. No estimate for $\epsilon = \epsilon(a_n)$ is asserted.

4.4 Where the polynomial mark budget comes from There is a simple sufficient cutoff bound for $X_{F,n}$. Suppose the explicitly chosen compact one-field representatives, including their mixing and identity coefficients, have pointwise bound

$$|\Phi^R_{\alpha,n}| \le C_\alpha a_n^{-b_\alpha}. \tag{4.17}$$

This is a finite polynomial normalization budget, not a uniform renormalized estimate. Logarithmically running coefficients satisfy such a budget after increasing the exponents. Required finite product-subtraction coefficients must obey corresponding finite budgets, or annihilate the tests by their shrinking coincidence support.

For a fixed polynomial F, compact smearing and bounded lattice L^1 sums yield a Euclidean path-mark bound

$$\|F_n\|_\infty \le C_F a_n^{-d_F}. \tag{4.18}$$

The exponent can increase with the degree of F. No assertion of a single exponent for all degrees is needed.

Let $K_{F,n}(u,v)$ be the unnormalized marked path kernel through its finite time interval and $K_{r,n}$ the unmarked kernel. Positivity of every compact Wilson path density gives

$$|K_{F,n}(u,v)| \le \|F_n\|_\infty K_{r,n}(u,v). \tag{4.19}$$

For $f \in L^2$, this implies pointwise $|K_{F,n}f| \le \|F_n\|_\infty K_{r,n}|f|$. Consequently, after division by the appropriate power of λ_0,

$$\|X_{F,n}\| \le \|F_n\|_\infty \|e^{-rH_n}\| \le C_F a_n^{-d_F}. \tag{4.20}$$

Gauge projection is contractive and preserves the same estimate. This argument treats all time-overlapping compact stencils together as a path mark, so it does not incorrectly replace an unbounded midpoint operator by its Euclidean sup norm. It incurs no total-volume factor.

4.5 What the thermal condition means and what has not been proved Whenever the derivative is taken at these trace-class times,

$$\Gamma_n = \int_{T_n/2}^{T_n} \frac{\operatorname{Tr}(H_n e^{-uH_n})}{\operatorname{Tr} e^{-uH_n}}\, du. \tag{4.21}$$

Thus $\Gamma_n = o(T_n)$ asks for a vanishing average *total* thermal excitation energy in this low-temperature interval. It does not require the thermal ground-state probability to tend to one, and it does not assume a positive mass gap. A cubic sequence still changes the spatial transfer operator when time grows.

The source-free pressure estimate of order physical four-volume, by itself, does not give (4.5) or (4.6). Nor does finite-cutoff uniqueness of the ground state. No available Balaban theorem is claimed here to establish this trace increment. The next section shows why an additional assertion of this type, or a direct periodic inequality, is needed for the proposed Hamiltonian bridge.

5. A positive-transfer counterexample to an automatic periodic energy bound

This example is finite dimensional at every cutoff and has a strictly positive transfer kernel. It is not a Wilson gauge model and is not claimed to satisfy Yang–Mills field identification. Its purpose is to refute a general inference from ground-Hamiltonian slab estimates and source/pressure bounds to an operator bound in an expanding periodic limit.

5.1 Cofinal parameters and a strictly positive transfer matrix Fix an odd $L \geq 3$, set the physical reference length to one, and use

$$a_n = L^{-n}, \qquad T_n = 2L^n, \qquad N_n = T_n/a_n = 2L^{2n}, \qquad E_n = T_n^2. \tag{5.1}$$

Thus the time step count is even and these are the same cubic cofinal scales. Let

$$D_n = 2\left\lceil e^{2T_nE_n}/2 \right\rceil, \qquad \mathcal{H}_n = \mathbb{C}^{D_n}, \qquad \Omega_n = D_n^{-1/2}(1, \dots, 1). \tag{5.2}$$

Write $\Pi_n = |\Omega_n\rangle\langle\Omega_n|$, $R_n = I - \Pi_n$, and define

$$H_n = E_nR_n, \qquad P_n = e^{-a_nH_n} = \eta_nI + (1-\eta_n)\Pi_n, \quad \eta_n = e^{-a_nE_n}. \tag{5.3}$$

Every matrix entry of P_n is strictly positive. It is symmetric, positive definite, stochastic, and has unique normalized ground vector Ω_n. Its nonzero excitation energy is E_n, which tends to infinity.

5.2 A bounded cutoff field with uniform Gaussian marginals Let Φ be the standard Gaussian distribution function and define

$$q_{j,n} = D_n \int_{(j-1)/D_n}^{j/D_n} \Phi^{-1}(u)\, du, \qquad Q_n = \operatorname{diag}(q_{1,n}, \dots, q_{D_n,n}). \tag{5.4}$$

The even number of bins and Gaussian symmetry give

$$\langle \Omega_n, Q_n\Omega_n \rangle = 0. \tag{5.5}$$

Conditional Jensen gives, for every $p \geq 1$,

$$\frac{1}{D_n}\sum_j |q_{j,n}|^p \leq \mathbb{E}|G|^p, \qquad G \sim N(0,1). \tag{5.6}$$

In particular $\|Q_n\Omega_n\| \leq 1$. For any bin of probability D_n^{-1} and real z,

$$e^{zq_{j,n}} \leq D_n\, \mathbb{E}e^{zG} = D_n e^{z^2/2}.$$

Optimizing z with either sign gives

$$\|Q_n\| \leq \sqrt{2\log D_n} \leq 3E_n. \tag{5.7}$$

The last inequality holds for the present $T_n \geq 1$, since $\log D_n \leq 2T_n^3 + \log 3$.

The quantile step functions in (5.4) converge pointwise on $0 < u < 1$ to $\Phi^{-1}(u)$. Their L^{2p} norms are uniformly bounded by Jensen. Hence they converge in L^p for each finite p, and

$$\mu_{m,n} := D_n^{-1}\operatorname{Tr} Q_n^m \longrightarrow \mathbb{E}G^m. \tag{5.8}$$

5.3 Uniform ground-Hamiltonian slab estimates Since $\Pi_n Q_n \Pi_n = 0$,

$$e^{-tH_n} Q_n e^{-tH_n} = e^{-tE_n}(\Pi_n Q_n R_n + R_n Q_n \Pi_n) + e^{-2tE_n} R_n Q_n R_n. \tag{5.9}$$

Equations (5.6)–(5.7) imply, for all $t > 0$,

$$\begin{aligned} \|e^{-tH_n} Q_n e^{-tH_n}\| &\le 2e^{-tE_n} + 3E_n e^{-2tE_n} \\ &\le \frac{7}{2e}\, t^{-1}. \end{aligned} \tag{5.10}$$

The constant is independent of n. Declaring the field spatially constant makes its spatially smeared bound cost only the lattice L^1 norm of the spatial test. Thus this is precisely a uniform ground-Hamiltonian slab bound of the relevant kind, with $\kappa = 1$.

5.4 The actual periodic hierarchy has a constant Gaussian limit Use the positive periodic Markov-chain law of length N_n with weight $\prod_{k=0}^{N_n-1} (P_n)_{x_k x_{k+1}}$, $x_{N_n} = x_0$. Its normalization is

$$\mathcal{Z}_n(T_n) = \operatorname{Tr} P_n^{N_n} = 1 + (D_n - 1)e^{-T_n E_n}. \tag{5.11}$$

Every one-time marginal is exactly uniform, since every diagonal entry of $P_n^{N_n}$ is equal. Hölder and (5.6) therefore give, at arbitrary lattice time arguments,

$$\left| \mathbb{E}_{\mathrm{per},n} \prod_{j=1}^{m} Q_n(t_j) \right| \le M_m, \qquad M_m = \mathbb{E}|G|^m. \tag{5.12}$$

For a sharper convergence statement, arrange the m insertions cyclically, with integer gaps $r_i \ge 0$ summing to N_n. In the marked trace, expand each

$$P_n^{r_i} = \eta_n^{r_i} I + (1 - \eta_n^{r_i})\Pi_n.$$

The term with no Π_n is $D_n e^{-T_n E_n} \mu_{m,n}$. Every other term is a product of moments $\mu_{k_1,n} \cdots \mu_{k_j,n}$, with $\sum k_i = m$, multiplied by a coefficient of magnitude at most one. Hölder bounds each product by M_m. Since

$$1 - \frac{D_n e^{-T_n E_n}}{\mathcal{Z}_n(T_n)} = \frac{1 - e^{-T_n E_n}}{\mathcal{Z}_n(T_n)},$$

we obtain the uniform-in-position estimate

$$\boxed{\left| S_{m,n}(t_1, \dots, t_m) - \mu_{m,n} \right| \le \frac{2^m M_m}{\mathcal{Z}_n(T_n)} \longrightarrow 0.} \tag{5.13}$$

Thus the full hierarchy converges to a single standard Gaussian variable G, constant at every spacetime point.

Reflection positivity at each cutoff follows directly by reflecting the positive half-chain: its quadratic form is the Hilbert–Schmidt norm of the half-chain marked matrix, divided by (5.11). The limiting hierarchy is symmetric, translation invariant, fully Euclidean invariant, and reflection positive.

It also has the required type of factorial Schwartz bound. For $q > 4$, the lattice L^1 sum of a Schwartz test in $4m$ variables is bounded by $C_q^m \nu_{qm}(F)$, uniformly for $a_n \le 1$. One may use

$$(1 + \textstyle\sum_{i=1}^m |x_i|^2)^{-qm/2} \le \prod_{i=1}^m (1 + |x_i|^2)^{-q/2}$$

and factor the lattice sums. Since $M_m \le C^m (m!)^{1/2}$, (5.12) gives an OS-type linear-order factorial bound for the entire cutoff hierarchy, not only separated two-point functions.

There are even uniform complex exponential-source bounds. For complex coefficients z_i, put $S = \sum_i |z_i|$. Generalized Hölder, the exactly uniform marginals, and conditional Gaussian Jensen give

$$\left| \mathbb{E}_{\mathrm{per},n} e^{\sum_i z_i Q_n(t_i)} \right| \le \exp(S^2/2). \tag{5.14}$$

The statement for $S = 0$ is immediate; otherwise use Hölder weights $|z_i|/S$, taking real parts in the modulus. The same bound applies to smeared lattice sources with their lattice L^1 norm.

5.5 Failure of the limiting energy bound and of the thermal criterion OS reconstruction of the limiting constant field gives

$$\mathcal{H} = L^2(\mathbb{R}, d\gamma), \qquad \Omega = 1, \qquad H = 0, \qquad \phi(f) = M_G \int_{\mathbb{R}^4} f(x)\, dx, \tag{5.15}$$

where M_G is multiplication by the Gaussian coordinate. Polynomials are dense, so this is the reconstructed cyclic Hilbert space. For $\int f \neq 0$, multiplication by G is unbounded. Hence

$$\boxed{\|\phi(f)(1+H)^{-r}\| = \infty \quad \text{for every } r \geq 0.} \tag{5.16}$$

Thus the uniform finite-cutoff ground slabs in (5.10) do not imply the desired operator estimate in this periodic limit.

Even a strong extensive pressure statement does not repair the inference. Here

$$\log \mathcal{Z}_n(T_n) = T_n E_n + o(1) = T_n^3 + o(1), \qquad T_n^{-4} \log \mathcal{Z}_n(T_n) \longrightarrow 0. \tag{5.17}$$

In contrast,

$$\Gamma_n = \log \frac{1 + (D_n - 1)e^{-T_n E_n/2}}{1 + (D_n - 1)e^{-T_n E_n}} = \tfrac{1}{2} T_n E_n + o(1), \quad \Gamma_n / T_n \longrightarrow \infty. \tag{5.18}$$

The state criterion in Section 4 correctly fails.

This is a global finite-state model, not a local compact gauge action. It proves a logical obstruction to an abstract bridge, not a counterexample to Yang–Mills existence or to a Wilson-specific theorem. A phase decomposition of the constant Gaussian example yields point masses, so it also does not disprove a separately proved phase-first strategy. It shows why that strategy must be stated and verified, rather than inferred from mixed-state vacuum bounds.

6. Remaining operator and phase premises

Obligation	What this argument proves	What is still required
Bare temporal operator, SU(2)	All-representation inequality (2.10), sign-controlled centering (2.13), interacting compact bounds and domains (3.1)–(3.5)	Physical a^{-4} cancellation, all tensor components, prescribed finite normalization
Ground-Hamiltonian scalar response	Exact earlier completely positive transform combines with the improved bare bound	Uniform renormalized response in the same field convention; control of the identity shift
Periodic state versus ground Hamiltonian	Theorem 4.2 on the existing cubic geometry	A proof of (4.5), or its stronger version (4.6), together with actual renormalized slabs (4.12); alternatively the direct periodic OS inequality
Fixed-polynomial kernel norm	Positive path-kernel domination (4.19)–(4.20)	A verified finite cutoff-power budget for the chosen operator normalization and any surviving product redefinitions
Limiting operator domains	The prior periodic slab-to-domain theorem applies after (4.16)	Its interacting periodic premise remains unproved; excellent vacuum moments do not supply it
Phase decomposition	No new decomposition theorem is imported	Check the actual hypotheses of the chosen decomposition theorem, or supply a different proved phase construction
Ultraviolet coefficient in a phase	The separate vacuum-block criterion of Annex L is available	Prove its actual interacting block estimates; an operator-bound assumption is not automatically necessary for this transfer

The main unresolved operator estimate is still (4.12), or directly (4.16), for the full renormalized finite tensor family in the prescribed Wilson hierarchy. Neither the representation estimate nor the thermal interpolation supplies the nonlinear marked ultraviolet analysis.

7. Source formulas

Source	Exact location	Scope used here
M. Lüscher, *Construction of a selfadjoint, strictly positive transfer matrix for Euclidean lattice gauge theories*, DESY 76/54; published CMP 54 (1977), 283–292	DESY preprint, printed preprint pp.7–9: (17), (20)–(23), Proposition 1; pp.10–11: (25)–(29), Proposition 2.	Compact Wilson transfer factors, strict positivity and gauge invariance, periodic trace interpretation. No continuum bound, uniform thermal estimate, or pure-phase theorem is imported.
NIST Digital Library of Mathematical Functions	10.25.2, 10.32.2, 10.32.3	Official reference for defining series and integral identities only. The representation-uniform derivative inequality and its consequences are derived in this annex.
Local transfer-energy comparison and volume-uniform midpoint bounds	Local deletion form and logarithmic increment	Range factorization and the closed-domain multiplier theorem on their stated geometry.
Periodic OS slabs and the reconstructed operator domain	Periodic trace identity and reduced even-mollifier hypothesis	Conditional passage to the common field domain.
Scalar transfer response	Annex C, Sections 1–2	Exact scalar transfer/logarithmic-response relation and the distinction between ground centering and the prescribed finite normalization.

The same-side geometry of the source-free Balaban construction is cubic. In particular, $\mathcal{Z}_n(T_n/2)$ in this annex is not presented as the output of an independently admitted rectangular RG construction. No unpublished announcement or proof is used.

Annex N — All-order Gaussian coefficient removal and joint-test budgets

Annex N. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Result and exact scope

This note removes the restriction $m \le 6$ from the native Wick comparison. The new bound has a fixed factorial exponent two, with constants uniform in the source order, volume and admitted Gaussian cutoff. Its cutoff error is e^{-cP^2}, without a factor P^{2m+6}. That factor is absorbed into the Gaussian tail with its entire order cost recorded in $(m!)^2$.

The main statement is

$$\boxed{\begin{aligned} &K_m^P(g) = L_m^\infty + \frac{g^2}{2} C_{2,m}^\infty + \mathcal{E}_m(g,P), \\ &|\mathcal{E}_m(g,P)| \le C^{m+1}(m!)^2 h^{-4m}\big(|g|^4 + e^{-cP^2}\big) \exp\!\left[-\frac{\gamma\tau(x_1,\dots,x_m)}{8(m+4)}\right], \qquad m \ge 1. \end{aligned}} \tag{T.1}$$

The law is exactly the fixed homogeneous no-R, identity-background native integral of the parity theorem. Its inherited regular action, constrained physical chart and cutoff are frozen during the signed computational coupling interpolation. Its leading and second coefficients are the ordinary finite-volume Gaussian Wick coefficients of its actual constrained precision. No microscopic-transport identification or complete-history estimate is added by (T.1).

The note also proves:

1. the corresponding full-moment expansion, with the same fixed factorial exponent;
2. a genuine joint Schwartz-test estimate, of order $6m$, at each fixed physical shell scale h;
3. a full-diagonal ultraviolet summation lemma with explicit $(q, p) = (6, 7)$, whenever the stated physical shell-kernel majorant is available.

The factor h^{-4m} in the unsummed fixed-shell estimate is retained. The third statement handles localized connected shell kernels on coincidence-flat tests; it does not identify the full Wilson connected hierarchy with a sum of those native kernels. Exact matching, retained-data cumulants, generated collision vertices, history control, rotations, operator bounds and phase hypotheses remain separate.

Status: P means the displayed assertion is proved from the stated inputs. The native application uses the scoped regular-law results already proved.

1. Inputs retained

Let $\mathcal{J}$ be the finite current coordinate lattice, with a metric satisfying

$$\sup_x \sum_{y\in\mathcal{J}} e^{-ad(x,y)} \le C(1 + a^{-4}), \qquad 0 < a \le 1, \tag{T.2}$$

uniformly over the admitted volumes. Coordinate-block dimensions are bounded by a fixed d_*.

For the centered Gaussian precision H, retain

$$H \ge \lambda I, \qquad H_{ii} \le \Lambda I, \qquad A_H = \sup_i \sum_j \int_0^\infty \|(e^{-tH})_{ij}\|_{\mathrm{op}}\, dt \le A. \tag{T.3}$$

The two Gaussian laws are

$$\nu_\infty(dz) = Z_\infty^{-1} e^{-zHz/2}\, dz, \qquad \nu_P = \nu_\infty\Big(\cdot \,\Big|\, \prod_i \{|z_i| \le P\}\Big), \qquad P \ge 1. \tag{T.4}$$

For the native precision, the Wick supplement derives the semigroup bound in (T.3) from the uniform gap and exponential matrix locality. It also verifies the weighted real covariance estimate under both laws. These are retained inputs, not conclusions of a gap alone.

The Gaussian cutoff theorem gives

$$|\nu_P(f) - \nu_\infty(f)| \le C A_H P^{d_*-1} e^{-\lambda P^2/2} \sum_i \mathrm{Lip}_i(f) \tag{T.5}$$

for bounded blockwise Lipschitz f. It proves a uniform one-block Gaussian density bound under ν_P, so that

$$\sup_{i,P} \|z_i\|_{L^p(\nu_P)} + \sup_i \|z_i\|_{L^p(\nu_\infty)} \le C\sqrt{p}, \qquad p \ge 1. \tag{T.6}$$

The proof of (T.5) is the normalized boundary-flux/OU-divergence proof in the Gaussian cutoff theorem. In particular, no division by the probability of the entire cutoff event is used here.

For a polynomial Q in scalar coordinates, use the displayed coefficient representation

$$Q(z) = \sum_{d=0}^{D} \sum_{a_1,\dots,a_d} c^{(d)}_{a_1\cdots a_d} z_{a_1} \cdots z_{a_d}, \qquad \|Q\|_1 = \sum_{d,a_1,\dots,a_d} |c^{(d)}_{a_1\cdots a_d}|. \tag{T.7}$$

For one anchored at x, also use

$$\|Q_x\|_{\theta,1} = \sum_{d,a_1,\dots,a_d} |c^{(d)}_{a_1\cdots a_d}| e^{\theta\tau(x,j(a_1),\dots,j(a_d))}. \tag{T.8}$$

Here $j(a)$ is the coordinate block. The degree-zero term has weight one. These norms refer to the chosen representation, and are multiplicative upper bounds under products. No minimization over polynomial representations is needed.

2. Quantitative polynomial cutoff removal at arbitrary degree

Proposition 2.1 — the degree cost is explicit, P There are constants $C \ge 1$ and $\kappa = d_*/2 + 1$, depending only on the parameters in (T.3), such that for every polynomial of degree at most $D \ge 1$,

$$|\nu_\#(Q)| \le \|Q\|_1 C^{D+1} (1+D)^{D/2}, \qquad \# \in \{P, \infty\}, \tag{T.9}$$

and

$$\boxed{|\nu_P(Q) - \nu_\infty(Q)| \le \|Q\|_1 C^{D+1} (1+D)^{D/2+\kappa} e^{-\lambda P^2/8}.} \tag{T.10}$$

Both constants are independent of the volume, D, and P. A constant polynomial has zero cutoff error.

Proof. For a monomial of degree d, Hölder with exponent d, followed by (T.6), gives

$$\nu_\# |z_{a_1} \cdots z_{a_d}| \le (C\sqrt{\max(1,d)})^d.$$

Summing coefficients, and enlarging C, proves (T.9).

Let π_P be blockwise radial projection onto the radius-P balls and set $Q_P = Q \circ \pi_P$. It agrees with Q on the constrained domain and satisfies

$$\sum_i \mathrm{Lip}_i(Q_P) \le D\|Q\|_1(1+P)^{D-1}.$$

Therefore (T.5) bounds the clipped comparison by

$$|\nu_P(Q) - \nu_\infty(Q_P)| \le CA_H D\|Q\|_1(1+P)^{D+d_*-2}e^{-\lambda P^2/2}. \tag{T.11}$$

The crucial all-order step is to absorb the polynomial in P now, before summing partitions. For every $s \ge 0$,

$$\sup_{P\ge 1}(1+P)^s e^{-3\lambda P^2/8} \le C_\lambda^{s+1}(1+s)^{s/2}. \tag{T.12}$$

For example, use $1+P \le 2P$ and maximize $P^s e^{-3\lambda P^2/8}$; the $s=0$ case is immediate. Replacing the harmless power in (T.11) by $D+d_*$, and retaining $e^{-\lambda P^2/8}$, gives

$$|\nu_P(Q) - \nu_\infty(Q_P)| \le \|Q\|_1 C^{D+1}(1+D)^{D/2+d_*/2+1}e^{-\lambda P^2/8}. \tag{T.13}$$

For the remaining full-Gaussian projection tail, telescope a degree-d monomial across its d coordinate occurrences. Radial projection decreases the absolute value of each scalar coordinate, so each term is bounded by

$$\prod_{\ell=1}^{d} |z_{a_\ell}|\, 1_{\{|z_{i_r}|>P\}}$$

for one occurrence r. The Gaussian block tail is

$$\mathbb{P}_{\nu_\infty}(|z_i| > P) \le 2^{d_*/2}e^{-\lambda P^2/4}.$$

Cauchy–Schwarz and Hölder at order $2d$ consequently give

$$\nu_\infty|Q - Q_P| \le \|Q\|_1 C^{D+1}(1+D)^{D/2+1}e^{-\lambda P^2/8}. \tag{T.14}$$

Combine (T.13) and (T.14). □

The point of (T.10) is that it contains no unrecorded C_D. It remains useful when the total polynomial degree grows linearly with the source order.

For the fixed centrally symmetric convex block templates covered by the Gaussian theorem, the same proof works with $c > 0$ in place of $\lambda/8$. The constant depends on the specified inradius, outradius and surface-area bounds; scalar-coordinate products in the tail proof are bounded by block-norm products. No moving or nonsymmetric cutoff is included.

3. All-order cumulant comparison with a fixed factorial exponent

Proposition 3.1 — arbitrary total degree, P Let $Q_1, \dots, Q_N$ have degrees $d_1, \dots, d_N$, put $D = \sum_i d_i \ge 1$, and let $A_i = \|Q_i\|_1$. Then

$$\boxed{|\kappa_{\nu_P}(Q_1, \dots, Q_N) - \kappa_{\nu_\infty}(Q_1, \dots, Q_N)| \le C^{D+N+1}N!(1+D)^{D/2+\kappa}e^{-\lambda P^2/8}\prod_i A_i.} \tag{T.15}$$

The formula includes $N = 1$. If all degrees vanish, the difference is zero.

Proof. Write the cumulant as

$$\kappa_\nu(Q_1, \dots, Q_N) = \sum_{\pi\in\Pi_N}(-1)^{|\pi|-1}(|\pi|-1)!\prod_{B\in\pi}\nu\left(\prod_{i\in B}Q_i\right). \tag{T.16}$$

For a block B, its product polynomial has degree $D_B = \sum_{i\in B} d_i$ and coefficient norm at most $\prod_{i\in B} A_i$. Telescope the product of block moments between the two laws. Each term has exactly one moment difference, controlled by (T.10), and ordinary moments controlled by (T.9).

The moment-degree factors obey

$$\prod_{B\in\pi}(1+D_B)^{D_B/2} \le (1+D)^{D/2}.$$

The extra factor from the changed block is at most $(1+D)^{\kappa}$. Constants in all blocks together cost at most C^{D+N+1}, uniformly in the partition. The number of telescoping terms is $|\pi|$, so the entire coefficient sum is

$$\sum_{\pi\in\Pi_N} |\pi|! \le N!2^{N-1}. \tag{T.17}$$

To see (T.17), an ordered partition can be encoded by a permutation of the labels and a choice of cuts between consecutive labels; allowing all such encodings only overcounts. Absorb 2^{N-1} into C^{D+N+1}. □

Corollary 3.2 — the almost-quadratic degree budget, P Fix $b \ge 0$. If

$$\sum_i d_i \le 2N + b, \tag{T.18}$$

then

$$\boxed{|\Delta\kappa_N| \le C_b^{N+1}(N!)^2 e^{-\lambda P^2/8}\prod_i A_i.} \tag{T.19}$$

The exponent two is independent of N.

Proof. Under (T.18),

$$(1+D)^{D/2+\kappa} \le C_b^{N+1} N^{N+b/2+\kappa}.$$

Use $N^N \le e^N N!$. The remaining power $N^{b/2+\kappa}$ has a fixed exponent and is absorbed into C_b^{N+1}. Insert this in (T.15). □

This is sharper than treating every argument as a degree-four polynomial. That treatment would lose the almost-quadratic total degree and produce an unnecessarily larger factorial exponent.

4. Spatial decay and the at-most-two extensive anchors

For this section, retain the weighted real covariance estimate under both Gaussian laws, as verified in the Wick supplement. Fix one $\theta > 0$, and choose $0 < \gamma < \theta$ within that covariance budget.

Suppose Q_i is anchored at u_i, has degree $d_i \le 4$, and $\|Q_i\|_{\theta,1} \le A_i$. Equation (T.6), coefficient summation and the weighted first derivative give, for $p \ge 1$,

$$\|Q_i - \nu_{\#}Q_i\|_p \le CA_i p^{d_i/2}, \qquad \Big\|\|\nabla Q_i\|_{\gamma,u_i}\Big\|_p \le CA_i p^{(d_i-1)/2}. \tag{T.20}$$

Constants are uniform because the individual degrees are at most four. A degree-zero argument contributes no cumulant of order at least two and may be omitted from that case.

Lemma 4.1 — uniform two-law tree bound, P If $\sum_i d_i \le 2N+2$, then for $N \ge 2$,

$$|\kappa_{\nu_p}(Q_1,\dots,Q_N)| + |\kappa_{\nu_\infty}(Q_1,\dots,Q_N)| \le C^{N+1}(N!)^2 \exp\left[-\frac{\gamma\tau(u_1,\dots,u_N)}{N-1}\right]\prod_i A_i. \tag{T.21}$$

Proof. Use the real cut theorem at moment order $2N$. Its partition coefficient is at most $(N-1)!2^{N-2}$, and there are at most N^2 crossing gradient pairs. The two gradient factors in (T.20) reduce the total moment power by one. Thus the moment part is at most

$$C^N\left(\prod_i A_i\right)N^{(\sum_i d_i)/2-1} \le C^N\left(\prod_i A_i\right)N^N.$$

Combining the factors and using $N^N \le e^N N!$ leaves a fixed polynomial in N times $C^N(N!)^2$. Absorb that polynomial into C^{N+1}.

Delete a longest edge of a minimum spanning tree on the anchors. Every crossing edge of the resulting cut has length at least $\tau/(N-1)$; the real cut theorem therefore supplies the displayed decay. This is precisely the two-gradient cut argument, with the total degree tracked rather than replaced by four for every mark. □

Combining (T.19), with $b = 2$, and (T.21) by $\min(a,b) \le \sqrt{ab}$, gives

$$\boxed{|\Delta\kappa_N| \le C^{N+1}(N!)^2 e^{-\lambda P^2/16}\exp\left[-\frac{\gamma\tau(u_1,\dots,u_N)}{2(N-1)}\right]\prod_i A_i.} \tag{T.22}$$

For $N = 1$, use (T.19) directly and omit the spatial factor.

Lemma 4.2 — fixed-number anchor sums do not change the factorial exponent, P Retain $m \ge 1$ external anchors $x_1,\dots,x_m$ and sum $r \in \{0,1,2\}$ additional anchors $y_1,\dots,y_r$, with $N = m + r$. Suppose all coefficient norms of the additional marks are bounded uniformly in their anchors. Then the bound in (T.22), summed over those r anchors, is at most

$$C^{m+1}(m!)^2 e^{-\lambda P^2/16}\exp\left[-\frac{\gamma\tau(x_1,\dots,x_m)}{8(m+4)}\right]\prod_{i=1}^m A_i, \tag{T.23}$$

with the uniform norms of the at-most-two added marks absorbed into C.

Proof. For $r = 0$, the assertion is a weakening of (T.22); the one-argument case has no spatial factor. For $r \ge 1$, put $T = \tau(x_1,\dots,x_m,y_1,\dots,y_r)$. Doubling a tree walk and shortcutting through the external anchors gives

$$T \ge \frac{1}{2}\tau(x_1,\dots,x_m).$$

Also $T \ge d(y_l,x_1)$ for every l, and consequently $T \ge r^{-1}\sum_l d(y_l,x_1)$. Split the exponential in (T.22) into two equal factors to obtain

$$e^{-\gamma T/[2(N-1)]} \le e^{-\gamma\tau(x_1,\dots,x_m)/[8(N-1)]}\prod_{l=1}^r e^{-\gamma d(y_l,x_1)/[4r(N-1)]}. \tag{T.24}$$

Equation (T.2) bounds the sum of the last product by $C_r(1+N^{4r})$. This is a fixed polynomial because $r \le 2$. Furthermore,

$$(N!)^2 \le (m!)^2(m+2)^{2r}.$$

All polynomial costs are absorbed into C^{m+1}, and the displayed spatial denominator is weakened to $8(m+4)$. □

Summing a labeled-tree bound over all $N-1$ vertices would introduce an unnecessary additional factorial cost. Here only two extensive vertices are summed, and (T.24) records that fact.

5. The all-order native Wick theorem

Keep the fixed shell, frozen inherited action E_*, fixed nonlinear chart and common cutoff of the parity theorem. Its Gaussian coefficient polynomials are

$$O_{i,r} = \frac{h^{-4}}{(r+1)(r+2)} D^{r+2} \mathcal{P}_i(0)[z^{r+2}], \qquad r = 0, 1, 2, \tag{T.25}$$

$$W_r = -\frac{1}{(r+1)(r+2)} D^{r+2} \mathcal{A}(0)[z^{r+2}] + D^r \mathcal{K}(0)[z^r], \qquad r = 1, 2. \tag{T.26}$$

The proven physical derivative estimates give

$$\deg O_{i,r} \le r + 2, \qquad \|O_{i,r}\|_{\theta,1} \le C h^{-4}. \tag{T.27}$$

Use the exact radial decomposition

$$W_r = \sum_y R_{r,y}, \qquad R_{r,y}(z) = \int_0^1 \langle z_y, D_y W_r(sz) \rangle \, ds. \tag{T.28}$$

Its coefficient budgets are

$$\deg R_{1,y} \le 3, \quad \deg R_{2,y} \le 4, \quad \|R_{r,y}\|_{\theta,1} \le C. \tag{T.29}$$

These are fixed-degree tensors; taking an arbitrary number of external marks requires no physical derivatives beyond those already used in the coefficient and remainder proofs.

Define

$$L_m^{\#} = \kappa_{\nu_\#}(O_{1,0}, \dots, O_{m,0}), \qquad \# \in \{P, \infty\}. \tag{T.30}$$

Define $C_{2,m}^{\#}$ by the exact five-term second-response formula:

$$\begin{aligned} C_{2,m}^{\#} = & \sum_i \kappa_\#(O_{1,0}, \dots, O_{i,2}, \dots, O_{m,0}) \\ & + 2 \sum_{i<j} \kappa_\#(O_{1,0}, \dots, O_{i,1}, \dots, O_{j,1}, \dots, O_{m,0}) \\ & + 2 \sum_i \kappa_\#(O_{1,0}, \dots, O_{i,1}, \dots, O_{m,0}, W_1) \\ & + \kappa_\#(O_{1,0}, \dots, O_{m,0}, W_2) \\ & + \kappa_\#(O_{1,0}, \dots, O_{m,0}, W_1, W_1). \end{aligned} \tag{T.31}$$

Every occurrence of W_r means the anchored sum (T.28). The last line has two separate cumulant arguments. At least one external mark is always present.

Theorem 5.1 — all-order Gaussian coefficient replacement, P For every $m \ge 1$,

$$\boxed{|L_m^P - L_m^\infty| + |C_{2,m}^P - C_{2,m}^\infty| \le C^{m+1} (m!)^2 h^{-4m} e^{-\lambda P^2/16} \exp\left[-\frac{\gamma \tau(x_1, \dots, x_m)}{8(m+4)}\right].} \tag{T.32}$$

Both coefficients separately have the same factorial and spatial bound without the Gaussian error factor. All anchor sums are absolutely controlled, uniformly in volume.

Proof. Every term in (T.31), after (T.28), has $N = m + r$ arguments with $r \le 2$. Individual degrees are at most four. Its total degree is at most

$$D \le 2N + 2. \tag{T.33}$$

For clarity, the five lines have respective maximum degrees

$$2m+2, \quad 2m+2, \quad 2m+4, \quad 2m+4, \quad 2m+6,$$

while their argument counts are

$$m, \quad m, \quad m+1, \quad m+1, \quad m+2.$$

Thus (T.33) holds in every case. The leading coefficient has $D = 2m$.

The coefficient-norm product is bounded by $C^{m+2}h^{-4m}$. Apply (T.22) and Lemma 4.2. There are at most $C(m+1)^2$ external derivative placements in (T.31), another fixed polynomial absorbed by C^{m+1}. This proves (T.32).

For the separate coefficient bounds, use (T.21) and the same anchor summation without the cutoff-comparison factor. The $N = 1$ moment is controlled by (T.9). □

Theorem 5.2 — all-order native expansion with ordinary Wick coefficients, P Under the stated native-law assumptions, for all $m \ge 1$,

$$\boxed{\begin{aligned} &K_m^P(g) = L_m^\infty + \frac{g^2}{2} C_{2,m}^\infty + \mathcal{E}_m(g,P), \\ &|\mathcal{E}_m(g,P)| \le C^{m+1}(m!)^2 h^{-4m}\big(|g|^4 + e^{-cP^2}\big) \exp\!\left[-\frac{\gamma\tau(x_1,\dots,x_m)}{8(m+4)}\right]. \end{aligned}} \tag{T.34}$$

For the Euclidean block balls one may take $c = \lambda/16$, after choosing a common smaller spatial rate γ. Constants are uniform in the admitted finite volume, P, g, and m.

Proof. The parity theorem already gives

$$K_m^P(g) = L_m^P + \frac{g^2}{2} C_{2,m}^P + R_{4,m}^P(g),$$

$$|R_{4,m}^P(g)| \le |g|^4 C^{m+1}(m!)^2 h^{-4m} e^{-\gamma\tau/[4(m+4)]}.$$

Insert (T.32), absorb the fixed bound on $1 + g^2$, and weaken the common spatial rate as displayed. □

The laws defining L_m^∞ and $C_{2,m}^\infty$ are ordinary finite-dimensional centered Gaussians with covariance H^{-1}. Wick contraction is therefore legitimate for every coefficient in (T.31). This does not identify that covariance with a continuum propagator, and does not declare the second coefficient universal.

Corollary 5.3 — logarithmic windows uniformly in the source order, P Suppose the admitted physical endpoints have

$$P(g_*) = a_0[\log(g_*^{-1})]^p, \qquad p > \frac{1}{2}. \tag{T.35}$$

There exists a single $g_0 > 0$, independent of m, such that $e^{-cP(g_*)^2} \le g_*^4$ for $0 < g_* \le g_0$. Consequently (T.34) at $g = g_*$ has an $O(g_*^4)$ remainder with the same fixed factorial exponent for all source orders simultaneously.

Proof. With $t = \log(g_*^{-1})$, the required inequality is $ca_0^2 t^{2p} \ge 4t$, which holds for all sufficiently large t. Its threshold has no source-order dependence because the P-power was already paid in the factorial bound. □

The cutoff is held at the endpoint value $P(g_*)$ throughout each signed computational interpolation. No derivative of a moving cutoff is omitted. This is a fourth-order parameter remainder at arbitrary source order, not an all-orders perturbative expansion in g.

6. Full moments: the factorial exponent stays two

Let

$$M_m^P(g) = \nu_g\left(\prod_{i=1}^m O_i(g)\right).$$

For each nonempty label subset B, let L_B^∞ and $C_{2,B}^\infty$ denote the corresponding connected coefficients. Set

$$\mathfrak{M}_m^\infty = \sum_{\pi\in\Pi_m}\prod_{B\in\pi} L_B^\infty, \tag{T.36}$$

$$\mathfrak{C}_{2,m}^\infty = \sum_{\pi\in\Pi_m}\sum_{B\in\pi} C_{2,B}^\infty \prod_{A\in\pi,\ A\neq B} L_A^\infty. \tag{T.37}$$

The first quantity is exactly the full Gaussian moment of the quadratic native fields. The second is the compatible normalized full-moment response assembled from (T.31).

Lemma 6.1 — partition budget, P For every $m \geq 1$,

$$\sum_{\pi\in\Pi_m}\prod_{B\in\pi}(|B|!)^2 \leq 2^{m-1}(m!)^2. \tag{T.38}$$

Proof. Order the k blocks and write their sizes as a composition $n_1 + \cdots + n_k = m$. Dividing the ordered count by $k!$ gives

$$\sum_{\pi\in\Pi_m}\prod_{B\in\pi}(|B|!)^2 = m!\sum_{k=1}^m \frac{1}{k!}\sum_{\substack{n_1,\dots,n_k\geq 1\\ n_1+\cdots+n_k=m}}\prod_{j=1}^k n_j!.$$

Since $\prod_j n_j! \leq m!$, the right side is at most $(m!)^2\sum_k \binom{m-1}{k-1}/k! \leq 2^{m-1}(m!)^2$. □

Theorem 6.2 — full-moment expansion, P For every source order,

$$\boxed{M_m^P(g) = \mathfrak{M}_m^\infty + \frac{g^2}{2}\mathfrak{C}_{2,m}^\infty + \mathfrak{E}_m(g,P),} \tag{T.39}$$

$$\boxed{|\mathfrak{E}_m(g,P)| \leq C^{m+1}(m!)^2 h^{-4m}\big(|g|^4 + e^{-cP^2}\big).} \tag{T.40}$$

The full coefficients themselves obey the same factorial bound without the small error factor.

Proof. The full moment is the partition polynomial in the connected functions. The parity theorem bounds every connected derivative through order four by $C^{|B|+1}(|B|!)^2 h^{-4|B|}$, after dropping its spatial factor. Differentiating the partition polynomial at most four times adds a factor bounded by a fixed power of the number of blocks, hence by m^4. Equation (T.38) bounds the sum of products of the connected budgets. The factors $C^{m+|\pi|}$ and m^4 are absorbed into one C^{m+1}.

The full moment is even in g by the same exact $(g,z) \mapsto (-g,-z)$ change of variables. Taylor's theorem therefore gives its fourth-order remainder with the bound in (T.40).

To replace its zeroth and second cutoff-Gaussian coefficients, use the partition formulas (T.36)–(T.37) with P in place of ∞, telescope each product, and insert (T.32). At most two marked block choices occur: one for the second response and one for the changed coefficient. Their cost is at most m^2, absorbed in C^{m+1}. Every product carries exactly h^{-4m}, and (T.38) retains factorial exponent two. □

No connected spatial decay is asserted for a full moment; its disconnected products need not decay when their clusters are separated.

7. Direct joint-test continuity at a fixed physical scale

This section makes the distribution-space statement explicit. It does not infer a joint-kernel estimate from a bound only on tensor-product tests.

Write the current coordinate sites as integer vectors x, with physical positions hx in the grid $h\mathbb{Z}^4$, with $0 < h \le 1$, and use any finite subset or chosen torus fundamental domain. For a full-moment kernel define the lattice distribution

$$\mathcal{M}^P_{m,h}(g;F) = h^{4m} \sum_{x_1,\dots,x_m} M^P_m(g;x_1,\dots,x_m)F(hx_1,\dots,hx_m), \quad F \in \mathcal{S}(\mathbb{R}^{4m}). \tag{T.41}$$

Define the coefficient and error distributions by the identical summation. The maximum over the fixed finite field-label family changes only C^{m+1}.

Proposition 7.1 — fixed linear test order, P For

$$\nu_s(F) = \max_{|\alpha|\le s} \sup_X (1+|X|^2)^{s/2}|\partial^\alpha F(X)|,$$

the full-moment error obeys

$$\boxed{|\mathfrak{C}_{m,h}(g,P;F)| \le C^{m+1}(m!)^2 h^{-4m}\big(|g|^4 + e^{-cP^2}\big)\nu_{6m}(F).} \tag{T.42}$$

The full moments and their coefficients have the corresponding bound without the small error factor.

Proof. The elementary lattice estimate

$$\sup_{0<h\le 1} h^4 \sum_{x\in\mathbb{Z}^4} (1+|hx|^2)^{-3} < \infty \tag{T.43}$$

follows by comparison with the integrable weight on neighboring cubes. Moreover

$$(1+\textstyle\sum_i |X_i|^2)^{-3m} \le \prod_i (1+|X_i|^2)^{-3}.$$

Thus

$$h^{4m} \sum_{x_1,\dots,x_m} |F(hx_1,\dots,hx_m)| \le C^m \nu_{6m}(F).$$

Insert (T.40). The argument applies to arbitrary joint F, using its actual kernel and a direct absolutely convergent sum. □

Equation (T.42) is uniform in volume and Gaussian cutoff, but not in the ultraviolet shell scale h. Its h^{-4m} is the physical dimension-four normalization, and has not been removed by changing the field definition. In particular, (T.42) alone is not J2 or S0 for the Wilson continuum hierarchy.

The estimate applies to coincidence-flat tests as a subspace, but does not itself exploit their flatness. That exploitation is the next lemma.

8. What coincidence flatness buys for a localized connected shell

The following statement records the physical summation cost. It is useful for the new all-order error kernels and for any eventual exact transported kernel with the same majorant.

Proposition 8.1 — full-diagonal shell summation, P Let $m \geq 2$, $0 < a \leq h \leq 1$, and let a connected kernel on the Euclidean physical grid $a\mathbb{Z}^4$ satisfy

$$|T_{m,h}(X_1, \dots, X_m)| \leq C_0^{m+1}(m!)^2 h^{-4m} \varepsilon_h \exp\left[-\frac{\gamma\tau(X_1, \dots, X_m)}{8(m+4)h}\right]. \tag{T.44}$$

Here τ is the minimum-tree length for the ordinary physical metric, and $\varepsilon_h \geq 0$. Let F be a Schwartz function whose normal derivatives through order four, including its value at order zero, vanish on the full diagonal. Coincidence-flat Schwartz functions satisfy this condition. Then

$$\boxed{\left|a^{4m} \sum_{X_1,\dots,X_m} T_{m,h}(X_1, \dots, X_m) F(X_1, \dots, X_m)\right| \leq C^{m+1}(m!)^7 h\, \varepsilon_h\, \nu_{6m}(F).} \tag{T.45}$$

The same assertion holds with integrals in place of the grid sums.

Proof. Put $r_i = X_i - X_1$, $i \geq 2$. Taylor's integral remainder in the relative coordinates, through degree four at $r = 0$, gives

$$|F(X_1, \dots, X_m)| \leq C\Big(\sum_{i=2}^{m} |r_i|_1\Big)^5 (1 + |X_1|^2)^{-3} \nu_{6m}(F). \tag{T.46}$$

All intermediate points in that Taylor formula retain the first coordinate X_1, so the Schwartz weight has the stated lower bound. The multi-index coefficient sum is included exactly by the fifth power of the ℓ^1 relative-coordinate norm.

Every r_i is a path displacement in a spanning tree; hence $\sum_{i\geq 2} |r_i|_1 \leq C(m-1)\tau$. Absorb τ^5 into half the exponential in (T.44):

$$\tau^5 e^{-\gamma\tau/[8(m+4)h]} \leq C_\gamma (m+4)^5 h^5 e^{-\gamma\tau/[16(m+4)h]}. \tag{T.47}$$

Bound the last minimum-tree exponential by the sum over all labeled trees. There are m^{m-2} such trees. Successively summing leaves, using $a \leq h$, bounds each tree's relative-coordinate integral by

$$\left[C_\gamma (m+4)^4 h^4\right]^{m-1}. \tag{T.48}$$

The remaining X_1 sum of the weight in (T.46) is uniformly bounded by (T.43), with a in place of h. The powers of h are

$$h^{-4m} h^5 h^{4(m-1)} = h.$$

The tree count and range factors obey

$$m^{m-2}(m+4)^{4(m-1)} \leq C^m (m!)^5.$$

The extra fixed powers from (T.46)–(T.47) are absorbed into C^{m+1}. Multiplying by the $(m!)^2$ in (T.44) gives (T.45). □

For a geometric shell sequence $h_j = L^{-j} h_0$, Proposition 8.1 gives

$$\left|\sum_j T_{m,h_j}(F)\right| \leq C^{m+1}(m!)^7 \nu_{6m}(F) \sum_j h_j \varepsilon_{h_j}, \tag{T.49}$$

whenever the last sum is finite. In particular a uniformly bounded ε_{h_j} suffices. The new native Gaussian-cutoff error has $\varepsilon_{h_j} = e^{-cP_j^2}$; its fourth-order comparison error has $\varepsilon_{h_j} = |g_j|^4 + e^{-cP_j^2}$.

Geometry restriction. The estimate uses physical Euclidean distances. On finite tori it applies locally when the compact test support lies in a nonwrapping patch so that the required tree-distance majorant is the same there. It is not an

unproved global replacement of periodic distance by Euclidean distance. The separate fixed-shell joint bound (T.42) has no such spatial premise.

Contact restriction. This is a full-diagonal Taylor subtraction statement with normal order four. It does not prove the complete hierarchy of partial-diagonal counterterms generated by collisions inside transported subclusters. The one-point term, for which there is no relative coordinate, still requires its separate identity subtraction.

Matching restriction. The actual Wilson connected hierarchy has not been shown to equal a sum of the native connected kernels in (T.44). Conditional cumulants of retained means and all generated response vertices remain necessary. Formula (T.49) proves summability after the physical kernel representation is supplied; it does not supply that representation.

9. What is closed, and what remains

Item	Status
Gaussian cutoff removal only at fixed source order	Closed for the almost-quadratic degree family: (T.19) is uniform in all source orders with factorial exponent two.
Wick coefficient comparison restricted to $m \le 6$	Closed in the fixed native regular law: Theorems 5.1–5.2 hold for every m.
Extensive insertion sums in the second coefficient	Closed in that scope: there are at most two additional anchors, and their order cost is polynomial.
Cutoff error $P^{2m+6}e^{-cP^2}$ obstructing uniformity in m	Closed by the degree-explicit estimate; the error is $C^{m+1}(m!)^2e^{-cP^2}$.
Compatible full moments	Closed in that scope through the partition formulas and Lemma 6.1.
Joint-test continuity rather than tensor-product-only estimates	Proved at each fixed h, with explicit $q = 6$, $p = 2$, and the retained physical factor h^{-4m}.
Ultraviolet summation of a localized connected shell majorant	Proved on the stated flat-test/physical-distance domain, with $q = 6$, $p = 7$, and shell gain h.
Exact microscopic matching and complete normalized history law	Still open. Native/reference theorems do not establish M0, C0, H0 or the actual full-law S0 estimate.
Running normalization, rotations, physical UV coefficients and operator/phase estimates	Still open in their previously stated scopes.

This annex removes an all-order limitation in the regular Gaussian comparison and records the actual factorial and distribution-space budgets. It does not complete four-dimensional Yang–Mills existence.

Proof dependencies

- The parity theorem (Annex J) supplies the actual fixed native law, physical derivatives through order six, and its all-order fourth-parameter-derivative bound.
- The Gaussian cutoff theorem (Annex K) supplies normalized boundary flux, the full-space OU comparison, and uniform block moments. Its sole convexity import is the precisely scoped Prékopa marginal theorem already cited there.
- The Wick comparison (Annex K, Section 5) supplies the actual homogeneous precision/semigroup identification, weighted polynomial coefficient norms and the two-law covariance scope.
- The real cut theorem supplies the two-gradient cumulant estimate; its partition coefficient is explicitly tracked in Lemma 4.1.

The degree-explicit cutoff bound, all-order replacement, fixed-factorial full-moment assembly, and explicit joint-test estimates above are derived here.

Annex O — Normalized microscopic defects, source jets and exact matching

Annex O. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Result and scope

The complete-history obligations C0/H0 and the exact transported-field matching obligation M0 remain open. This note proves a new microscopic, fully interacting statement that addresses one of their genuine local obstructions: cancellation of the staples in a conditional link law.

For the full compact SU(2) Wilson measure:

1. The conditional minimum of the incident plaquette energy is an explicit nonnegative function of the surrounding links. Joint low-staple events, for arbitrarily dependent link neighborhoods, have a normalized exponential bound controlled by that minimum energy.
2. Whenever the resultant staple has a fixed positive lower bound, the centered one-link source logarithm has a common complex source ball and a bound linear in the conditional minimum energy. Consequently every conditional scalar cumulant has a factorial bound with that same linear dependence. No small plaquette condition on the boundary, Gaussian replacement, or removal of exceptional interior configurations is used.
3. Averaging the remaining low-staple sector against the actual Wilson law controls every conditional scalar response, uniformly in volume and coupling. A family of links with no plaquette containing two of them gives an exact, local, all-order elimination layer.

These are initial-lattice statements. The exact coarse constrained fibers in the RG/R construction are not one-link Wilson laws. The result therefore does not identify coarse inherited low-margin events with these microscopic defects, and does not remove the physical factor a^{-4} in a dimension-four field.

Here P denotes a proof below, L a scoped standard special-function input, and O an application still missing for the Yang–Mills construction.

1. The actual Wilson law and the already-proved reserve

Let $d \geq 2$, $M = 2(d-1)$, and let the periodic lattice have an even side $n \geq 4$. Use normalized Haar measure and the exact compact SU(2) Wilson law

$$d\mu_{\beta,n}(U) = Z_{\beta,n}^{-1} \exp\left[-\sum_p e_p(U)\right] dH(U), \qquad e_p = \beta\left[2 - \operatorname{Re}\operatorname{Tr} U_p\right]. \tag{1.1}$$

The coefficient dictionary remains $\beta = 2Nx_0 = 4x_0$ for $N = 2$, and $X_B = N\beta = 2\beta$.

We use the constant local exponential reserve proved in Annex A:

$$\mathbb{E}_{\mu_{\beta,n}} \exp\left(\eta \sum_{p \in X} e_p\right) \leq K^{b(X)}. \tag{1.2}$$

Here $\eta > 0$ and K are fixed along the cofinal family, $b(X)$ is the unit-cube cover count, and the sufficient geometric condition is $\sup \log(1+\beta)/n < \infty$. In dimension four one may take $0 < \eta \leq 1/8$. At a single finite lattice the same proof uses $K_{\beta,n} = C_{2,d}(1+\beta)^{3d/(2n)}$; all statements below remain true with that explicit value.

Equation (1.2) is a bound for the normalized full Wilson probability law. It retains every compact field configuration. In particular, for any fixed plaquette stencil X,

$$\left\| \sum_{p \in X} e_p \right\|_{L^r(\mu_{\beta,n})} \leq C_X r, \qquad r \geq 1. \tag{1.3}$$

The only earlier constructive input in this note is (1.2). Its proof uses the actual Wilson reflection/chessboard geometry and matched partition-function powers, as in Annex A. No reflection property of an auxiliary RG history law is assumed here.

2. An exact conditional minimum at one link

Fix an oriented link l, and condition on every other link. Identifying SU(2) with the unit quaternions S^3, write its remaining quaternion as u. Every incident plaquette contribution has the form

$$e_{l,p}(u) = 2\beta(1 - v_{l,p} \cdot u), \qquad v_{l,p} \in S^3, \qquad p \ni l. \tag{2.1}$$

Orientation or an inverse occurrence of U_l is absorbed into the unit vector $v_{l,p}$. The real part of the fundamental trace is twice the quaternion scalar product. Define

$$v_l = \sum_{p \ni l} v_{l,p}, \qquad \rho_l = |v_l| \in [0, M], \qquad B_l = 2\beta(M - \rho_l). \tag{2.2}$$

These quantities depend on the links other than l. The norm ρ_l, and hence B_l, is gauge invariant: a gauge transformation rotates all the relevant quaternion vectors by the same orthogonal left/right multiplication at the endpoints of l.

The exact conditional law is

$$Q_l(du \mid U_{\neq l}) = \frac{\exp(2\beta v_l \cdot u)}{\mathcal{Z}(2\beta\rho_l)}\, d\sigma_{S^3}(u), \qquad \mathcal{Z}(r) = \frac{2I_1(r)}{r}, \qquad \mathcal{Z}(0) = 1. \tag{2.3}$$

The normalization identity follows by integrating the normalized S^3 marginal $(2/\pi)(1 - t^2)^{1/2}\, dt$; it is also the $\nu = 1$ case of the integral representation in NIST DLMF 10.32.2.

Lemma 2.1 — conditional minimum, P

$$\boxed{B_l = \min_{u \in S^3} \sum_{p \ni l} e_{l,p}(u) \le E_l(U) := \sum_{p \ni l} e_p(U).} \tag{2.4}$$

If $\rho_l > 0$, the minimizing direction is $u_l^* = v_l/\rho_l$. Put $a_{l,p} = v_{l,p} \cdot u_l^*$ and $v_{l,p}^\perp = v_{l,p} - a_{l,p} u_l^*$. Then

$$\boxed{\sum_{p \ni l} |v_{l,p}^\perp|^2 \le 2(M - \rho_l) = B_l/\beta.} \tag{2.5}$$

The last expression is used only for $\beta > 0$; for $\beta = 0$ every marked energy is zero.

Proof. The incident energy equals $2\beta(M - v_l \cdot u)$. Its minimum is achieved by maximizing the scalar product, which gives (2.4). Also $\sum_p a_{l,p} = \rho_l$ and $1 - a_{l,p}^2 \le 2(1 - a_{l,p})$ for $-1 \le a_{l,p} \le 1$. Sum this inequality. □

The boundary may be very far from a regular plaquette configuration. In particular, the canceled-staple example $\rho_l = 0$ is included. It has $B_l = 2\beta M$, so its arbitrary-boundary conditional fluctuations can be large while its occurrence in the normalized outer law can be estimated.

3. Joint normalized probabilities of low-staple events

Let Γ be any finite set of distinct links. Define

$$X_\Gamma = \{p : \partial p \cap \Gamma \neq \varnothing\}, \qquad m_\Gamma = \max_p |\partial p \cap \Gamma| \le 4. \tag{3.1}$$

For a nonempty Γ, $1 \le m_\Gamma \le 4$. No disjoint-neighborhood or independence assumption is imposed.

Theorem 3.1 — full-law low-staple suppression, P For prescribed $0 < \delta_l \le M$,

$$\boxed{\mathbb{P}_{\mu_{\beta,n}}\left(\rho_l \le M - \delta_l \text{ for every } l \in \Gamma\right) \le K^{b(X_\Gamma)} \exp\left[-\frac{2\eta\beta}{m_\Gamma}\sum_{l\in\Gamma}\delta_l\right].} \tag{3.2}$$

One always has $b(X_\Gamma) \le M|\Gamma|$. In particular, for a fixed $\rho_* \in (0, M)$, with $\delta = M - \rho_*$,

$$\mathbb{P}(\rho_l < \rho_* \text{ for all } l \in \Gamma) \le \exp\left[-|\Gamma|\left(\frac{\eta\beta\delta}{2} - M\log K\right)\right]. \tag{3.3}$$

Proof. Pointwise on the full configuration space,

$$\sum_{l\in\Gamma} B_l \le \sum_{l\in\Gamma} E_l \le m_\Gamma \sum_{p\in X_\Gamma} e_p. \tag{3.4}$$

On the event in (3.2), its left side is at least $2\beta\sum_l \delta_l$. Apply exponential Markov with (1.2). A link has M incident plaquettes, and covering each such plaquette separately proves $b(X_\Gamma) \le M|\Gamma|$. Use $m_\Gamma \le 4$ for (3.3). □

This is a normalized, volume-uniform estimate. It applies to the original configurations on the joint space of any later positive history coupling because that coupling preserves their Wilson marginal. It is not a claim that a coarse current-selector failure implies $\rho_l < \rho_*$ at some original link.

Corollary 3.2 — marked version with the full rarity factor, P Suppose original marks obey $\|A_i\|_r \le C b_i r$ for all $r \ge 1$. Let $\mathcal{Q}$ be any composite frozen positive transport on an augmented output that retains the event in (3.3) as a history label. Thus $\mathcal{Q}$ denotes conditional transport to that augmented output; it is not an unchanged conditional kernel subsequently conditioned on an omitted event. Put

$$\sigma_\beta = \frac{\eta\beta\delta}{2} - M\log K, \qquad r_\Gamma = |\Gamma|.$$

When $\sigma_\beta \ge 0$,

$$\boxed{\mathbb{E}\left[1_{\{\rho_l<\rho_*:\ l\in\Gamma\}}\ |\kappa_{\mathcal{Q}}(A_1,\dots,A_m)|\right] \le e^{-\sigma_\beta r_\Gamma} C^{m+1} m!\,(m+\sigma_\beta r_\Gamma)^m \prod_i b_i.} \tag{3.5}$$

For a full transported product coefficient, omit the extra $m!$.

Proof. Use the arbitrary-event estimate with the true event probability q. The function

$$e^{-\lambda}(m+\lambda)^m$$

is nonincreasing for $\lambda \ge 0$, since its logarithmic derivative is $-1 + m/(m+\lambda) \le 0$. Hence the upper bound $q \le e^{-\sigma_\beta r_\Gamma}$ may replace q in that estimate. Conditional contraction applies to the composite positive transport. □

For dimension-four bare fields b_i still contains a^{-4}. The theorem does not assert that an isolated bad-link event absorbs this factor for every source order. It supplies the actual normalized probability that the earlier generic rare-event lemma required for this specific microscopic event.

4. A common complex source ball with linear background-energy cost

The next estimate uses the exact compact conditional integral, rather than a Gaussian approximation. Fix

$$0 < \rho_* < M.$$

The constants below may depend on d, ρ_*, but not on β, n, boundary data, or the number of source labels.

Theorem 4.1 — regular-resultant conditional log jets, P with a scoped L input Suppose $\rho_l \geq \rho_*$. Let $p_1, \dots, p_m$ be any incident plaquettes, with repetitions permitted, and set

$$\tilde{e}_i = e_{l,p_i} - Q_l e_{l,p_i}.$$

There is $r_* > 0$, independent of m, such that

$$\Phi_l(t) := \log Q_l \exp\left(\sum_{i=1}^{m} t_i \tilde{e}_i\right) \tag{4.1}$$

has its branch $\Phi_l(0) = 0$ throughout the complex ℓ^1 ball $\sum_i |t_i| < r_*$, and

$$\boxed{|\Phi_l(t)| \leq C(1 + B_l)\left(\sum_i |t_i|\right)^2.} \tag{4.2}$$

Consequently, for every $m \geq 2$,

$$\boxed{|\kappa_{Q_l}(e_{l,p_1}, \dots, e_{l,p_m})| \leq C^{m+1} m!\,(1 + B_l).} \tag{4.3}$$

This is not a uniform source logarithm for arbitrary coarse Wilson actions. It is a theorem about the explicit one-link law (2.3).

Proof: the exact complex reduction Suppress the link index, put $v = \rho u_*$, and write

$$w(t) = \sum_i t_i v_{p_i}, \qquad w_\parallel = u_* \cdot w, \qquad w_\perp = w - w_\parallel u_*.$$

The dot product in the following expression is the complex bilinear extension of the real Euclidean scalar product. For sufficiently small $s = \sum_i |t_i|$, define

$$q(t) = \sqrt{(\rho - w_\parallel)^2 + w_\perp \cdot w_\perp}, \qquad q(0) = \rho. \tag{4.4}$$

Choose a fixed $s_0 > 0$, depending only on ρ_*, small enough that

$$|q - \rho| \leq Cs, \qquad \operatorname{Re} q \geq \rho/2, \qquad |\arg q| \leq \pi/4, \qquad |q + \rho - w_\parallel| \geq c\rho \tag{4.5}$$

whenever $s < s_0$. These follow by the analytic square root around ρ^2 and $|w| \leq s$.

The source integral is exactly

$$Q_l \exp\left(\sum_i t_i e_{l,p_i}\right) = \exp\left(2\beta \sum_i t_i\right) \frac{\mathcal{Z}(2\beta q(t))}{\mathcal{Z}(2\beta\rho)}. \tag{4.6}$$

For real source vectors this follows by rotation of the quaternion integral. Both sides are holomorphic near zero, and the power-series rotational identity extends it to the stated complex neighborhood. Equivalently, $\mathcal{Z}(\sqrt{z \cdot z})$ is entire in the four complex components of z, because $\mathcal{Z}$ has an even power series.

Proof: the large-argument branch and its estimate The only additional literature input is the fixed-order modified-Bessel asymptotic in a closed sector:

$$I_1(z) = \frac{e^z}{\sqrt{2\pi z}}\,[1 + O(|z|^{-1})], \qquad |\arg z| \leq \pi/4, \quad |z| \longrightarrow \infty, \tag{4.7}$$

with a uniform remainder there. This is NIST DLMF 10.40.1; the error-bound discussion is in 10.40(iii). It implies that, on this sector above a fixed radius,

$$\log \mathcal{Z}(z) = z - \frac{3}{2}\log z + c_0 + R(z), \qquad |R(z)| \le C/|z|, \tag{4.8}$$

with a holomorphic bounded remainder and no zeros. The zero-free statement here follows directly from the relative error $1 + O(1/z)$ after increasing that fixed radius; a zero theorem outside the sector is not used.

If $2\beta\rho$ is above a sufficiently large fixed threshold, (4.5) puts every $2\beta q(t)$, $s < s_0$, in this same domain. Therefore (4.6) has a well-defined logarithm on the full source ball. Up to a linear term, the logarithm is the sum of

$$2\beta q(t), \qquad -\frac{3}{2}\log(q(t)/\rho), \qquad R(2\beta q(t)) - R(2\beta\rho). \tag{4.9}$$

The first term has an exact quadratic remainder:

$$q - \rho + w_{\parallel} = \frac{w_{\perp}\cdot w_{\perp}}{q + \rho - w_{\parallel}}. \tag{4.10}$$

By (2.5),

$$|w_{\perp}|^2 \le s^2 \sum_{p \ni l} |v_p^{\perp}|^2 \le s^2 B_l/\beta. \tag{4.11}$$

Together with (4.5), this bounds the centered contribution of the first term in (4.9) by $CB_l s^2$.

The other two terms in (4.9) are uniformly bounded holomorphic functions on the larger source ball $s < s_0$. Subtracting their constant and linear terms leaves a bound Cs^2 on a fixed smaller ball. To see the dimension independence explicitly, fix a source direction t with $\|t\|_1 = s$ and apply the one-variable Taylor estimate to $z \mapsto f(zt/s)$, on a disc of radius s_0. The bound depends only on the ℓ^1 radius, not on the number of coordinates.

Subtracting the full linear term in the logarithm of (4.6) is exactly the centering in (4.1). This proves (4.2) in the large-argument regime.

Proof: bounded arguments and cumulants If $2\beta\rho$ stays below the fixed threshold, then $\beta \le C/\rho_*$. All incident energies and their centered versions are uniformly bounded. On a sufficiently small common ℓ^1 source ball,

$$\left|Q_l e^{\sum_i t_i \tilde{e}_i} - 1\right| < 1/2.$$

The logarithm is then holomorphic and uniformly bounded. Its constant and linear terms vanish, so the same one-variable Taylor estimate proves (4.2), with $1 + B_l \ge 1$.

Finally apply multivariable Cauchy to (4.2) on coordinate radii $r_*/(2m)$. The mixed derivative is bounded by

$$C(1 + B_l)(2m/r_*)^m.$$

Use $m^m \le e^m m!$ and enlarge C. For $m \ge 2$, centering does not change the cumulant. This proves (4.3). □

Corollary 4.2 — actual conditional centered moments, P For $\rho_l \ge \rho_*$, every incident plaquette satisfies

$$\boxed{\|e_{l,p} - Q_l e_{l,p}\|_{L^r(Q_l)} \le C\big[r + \sqrt{(1 + B_l)r}\big], \qquad r \ge 1.} \tag{4.12}$$

Proof. The real-source case of (4.2) gives $\log Q_l e^{t\tilde{e}} \le C(1 + B_l)t^2$ for $|t| \le r_*/2$. Chernoff optimization with

$$|t| = \min\{r_*/2,\ u/[2C(1 + B_l)]\}$$

gives a two-sided tail

$$Q_l(|\tilde{e}| \ge u) \le 2\exp\{-c\min(u^2/(1 + B_l), u)\}.$$

Tail integration yields (4.12). □

The dependence on B_l cannot simply be discarded: a large retained boundary mismatch can generate large individual-plaquette fluctuations even though the conditional total excess has good moments. The result identifies that dependence and supplies its actual outer-law control.

5. Averaging every conditional response over the full Wilson law

Theorem 5.1 — averaged conditional scalar hierarchy, P For every one-link conditional kernel and every tuple of its incident plaquettes,

$$J_{m,l} = \kappa_{Q_l}(e_{l,p_1}, \dots, e_{l,p_m}), \qquad m \geq 2,$$

the full Wilson law obeys

$$\boxed{\mathbb{E}|J_{m,l}| \leq C^{m+1}(m!)^2.} \tag{5.1}$$

More precisely,

$$\mathbb{E}[1_{\{\rho_l \geq \rho_*\}}|J_{m,l}|] \leq C^{m+1} m!, \tag{5.2}$$

while, for every fixed m,

$$\boxed{\mathbb{E}[1_{\{\rho_l < \rho_*\}}|J_{m,l}|] \leq C^{m+1} m!\, \beta^m e^{-c\beta}.} \tag{5.3}$$

Constants are independent of cutoff and volume along the cofinal reserve family. The last estimate shows that the degenerate-resultant contribution is smaller than every inverse power of β at every fixed source order.

Proof. On $\rho_l \geq \rho_*$, use (4.3), $B_l \leq E_l$, and the fixed-stencil reserve. This gives (5.2).

For arbitrary boundary data, $0 \leq e_{l,p} \leq 4\beta$. A centered bounded family has a uniformly bounded logarithmic generating function on an ℓ^1 source ball of radius c/β, so Cauchy gives

$$|J_{m,l}| \leq C^{m+1} m!\, \beta^m. \tag{5.4}$$

At $\beta = 0$ this is immediate. The one-link case of (3.2) gives $\mathbb{P}(\rho_l < \rho_*) \leq Ce^{-c\beta}$, proving (5.3). Finally,

$$\sup_{\beta \geq 0} \beta^m e^{-c\beta} = \Big(\frac{m}{ce}\Big)^m \leq c^{-m} m!, \tag{5.5}$$

where the last inequality follows from $\log m! \geq \int_1^m \log x\, dx = m \log m - m + 1$. Combining (5.2)–(5.5) proves (5.1). □

The general conditional-cumulant contraction bound already gave a magnitude estimate of the form (5.1). The new content here is the explicit linear-B_l bound before outer averaging, its sharper $m!$ regular-resultant part, and a normalized exponentially small estimate for the degenerate-resultant part. These statements do not assume a uniform good boundary.

6. An exact local elimination layer

Let Γ now satisfy the concrete geometry condition

$$\text{no plaquette contains two distinct links of } \Gamma. \tag{6.1}$$

Condition on all links outside Γ. Every Wilson plaquette factor then depends on at most one remaining link. Thus

$$\mu(dU_\Gamma \mid U_{\Gamma^c}) = \prod_{l \in \Gamma} Q_l(dU_l \mid U_{\Gamma^c}) \tag{6.2}$$

exactly, including all compact configurations. In particular, the staples $v_{l,p}$, and the functions ρ_l, B_l, depend only on the retained links.

For sources on scalar plaquette energies, partition the marked plaquettes according to the unique eliminated link they meet, leaving the plaquettes meeting no eliminated link as retained-data functions. The normalized source functional factors:

$$Q_\Gamma e^{\sum_p t_p e_p} = e^{\sum_{p:\,\partial p\cap\Gamma=\varnothing} t_p e_p} \prod_{l\in\Gamma} Q_l e^{\sum_{p\ni l} t_p e_p}. \tag{6.3}$$

The identity holds for actual finite sources and hence coefficientwise for every finite source jet.

Corollary 6.1 — local all-order responses, P Every logarithmic response of order at least two created by this elimination vanishes unless all its marked plaquettes meet the same eliminated link. Each surviving response depends only on the retained links in that link's plaquette star. On $\rho_l \geq \rho_*$ it satisfies (4.3), and its full-law average satisfies (5.1)–(5.3).

Proof. Take the logarithm of (6.3) and differentiate. Each summand has only the corresponding one-link source arguments. The scalar plaquette data entering (2.1) use the other three links of the incident plaquette, so the response support is precisely local in the stated retained variables. □

Such layers exist on every sufficiently large periodic cubic lattice: greedily color the finite link-conflict graph in which two links are adjacent if they lie in one plaquette. Its degree is bounded only by dimension. Each color class obeys (6.1). This observation proves existence of a uniformly finite number of admissible *first* layers; it does not make successive conditional laws after eliminating different colors remain Wilson one-link laws.

This is a genuine local source-jet calculation for the complete microscopic measure. It supplies no long-distance decay for the covariance of the resulting retained-data means. After the retained variables are integrated further, their law has new interactions; restarting (2.3) without accounting for those interactions would be incorrect.

6.2 Full tensor products: the one-layer error is a controlled contact strip The conditional product structure has a consequence for the actual finite field family, beyond the scalar logarithmic jets. It is an application of the earlier flat-test estimate to a newly identified exact response kernel; the Taylor estimate itself is not being claimed as a new mechanism.

Let $\{O_{\alpha,a}(x)\}$ be a fixed finite family of microscopic gauge-invariant fields. Assume:

1. Each $O_{\alpha,a}(x)$ depends only on links in a stencil contained in a physical ball of radius Ra, with R independent of the cutoff.
2. On the actual Wilson law,
$$\|O_{\alpha,a}(x)\|_{L^r(\mu_{\beta,n})} \leq Ca^{-4}r, \qquad r \geq 1, \tag{6.4}$$
uniformly in labels and positions.

The explicit instance is $O_{\alpha,a} = \beta\tilde{q}_{\alpha,a} - \mathbb{E}[\beta\tilde{q}_{\alpha,a}]$ for the 21-field family of Annex E (scalar trace replaced by the Wilson action density). It satisfies these two hypotheses: unitary parallel transports preserve curvature norms, $|\tilde{q}_{AB,a}| \leq 2w_a$, and (1.2) gives (6.4). Vacuum centering preserves the bound with a fixed constant. A different running normalization requires its own coefficient budget and is not included merely by calling the field renormalized.

For a set Γ obeying (6.1), define the projected fields by the actual conditional expectation

$$\widehat{O}_{\alpha,a}(x) = Q_\Gamma O_{\alpha,a}(x) = \mathbb{E}[O_{\alpha,a}(x) \mid U_{\Gamma^c}], \tag{6.5}$$

and their full-moment difference kernel by

$$D_{m,a}(x) = \mathbb{E}\left[\prod_{i=1}^m O_{\alpha_i,a}(x_i)\right] - \mathbb{E}\left[\prod_{i=1}^m \widehat{O}_{\alpha_i,a}(x_i)\right]. \tag{6.6}$$

The field in (6.5) is a function of retained links in a stencil enlarged by only a fixed number of lattice spacings: only the conditional kernels of eliminated links actually appearing in the original mark need to be integrated, and each such kernel depends on its plaquette star.

Theorem 6.2 — exact one-layer matching on coincidence-flat tests, P For $m \geq 2$,

$$\boxed{D_{m,a}(x) = 0 \quad \text{if the original marked stencils share no eliminated link,}} \tag{6.7}$$

and

$$\boxed{|D_{m,a}(x)| \leq C^{m+1} m!\, a^{-4m} 1_{\{\min_{i<j} d_T(x_i,x_j) \leq 2Ra\}}.} \tag{6.8}$$

Here d_T is the torus distance. The equality $D_{1,a} = 0$ is exact.

For a joint Schwartz test F that is coincidence-flat, use an interior torus embedding: choose fixed weights $0 \leq \zeta_T \leq 1$ supported in a central cube of side at most $T/2$, and assume $T \geq T_0(R)$, $0 < a \leq 1$, so that two anchors in that cube which share one microscopic stencil link are also within $2Ra$ in the ordinary physical metric. Then

$$\boxed{\left| a^{4m} \sum_x D_{m,a}(x) F(x_1, \dots, x_m) \prod_i \zeta_T(x_i) \right| \leq a\, C^{m+1} m!\, \nu_{6m}(F),} \tag{6.9}$$

where

$$\nu_s(F) = \max_{|\alpha| \leq s} \sup_X (1 + |X|^2)^{s/2} |\partial^\alpha F(X)|. \tag{6.10}$$

The same assertion holds for compact tests supported in such a nonwrapping patch, without the weights. For comparison along expanding tori one may take $\zeta_T = 1$ on expanding compact subsets of the interior. All constants are independent of the number of eliminated links, volume, cutoff and source order. Every field-label tuple in the finite family is included. An arbitrary choice of Γ is not asserted to preserve finite-cutoff translations or reflection positivity of the projected-field hierarchy.

Proof of the exact support statement. Let $I_i \subset \Gamma$ be the set of eliminated links on which the i-th mark actually depends. Given the retained variables, (6.2) makes the eliminated variables independent. If the sets I_i are pairwise disjoint, the conditional expectation of the product is the product of the conditional expectations. This proves (6.7). If two marked stencils share an eliminated link, their anchors have torus distance at most $2Ra$, proving the support inclusion in (6.8).

Proof of the magnitude estimate. Conditional contraction and Hölder give

$$\begin{aligned} |D_{m,a}(x)| &\leq \mathbb{E} \prod_i |O_{\alpha_i,a}(x_i)| + \mathbb{E} \prod_i |\widehat{O}_{\alpha_i,a}(x_i)| \\ &\leq 2 \prod_i \|O_{\alpha_i,a}(x_i)\|_m \leq 2C^m m^m a^{-4m} \leq C^{m+1} m! a^{-4m}. \end{aligned} \tag{6.11}$$

This bound is for the full products. It already includes every response generated by the one-layer conditional integration; no product of single means has been substituted without estimating its difference.

Proof of the joint flat-test estimate. At every point contributing to (6.9), choose one colliding pair with ordinary separation at most $2Ra$. Taylor-expand F along the displacement taking one of its coordinates to the other, to order

$$k_m = 4m + 1 \leq 6m.$$

Coincidence flatness annihilates every lower term. The displacement is $O(a)$, so the integral remainder gives

$$|F(X)| \leq C^{m+1} a^{4m+1} (1 + |X|^2)^{-3m} \nu_{6m}(F) \tag{6.12}$$

on this collision strip. The intermediate points differ from X in one coordinate by $O(a)$; since $a \leq 1$, their Schwartz weights differ by a factor at most C^{m+1}. The directional multi-index sum is bounded by the k_m-th power of the displacement's ℓ^1 norm. The Taylor factorial could be retained and only improves the displayed bound.

The elementary weighted lattice estimate is

$$a^{4m}\sum_{x}(1+\sum_{i}|x_i|^2)^{-3m} \le \left[a^4\sum_{x\in a\mathbb{Z}^4}(1+|x|^2)^{-3}\right]^m \le C^m. \tag{6.13}$$

Multiply (6.8) by (6.12), use (6.13), and keep the weights ζ_T bounded by one. No derivative of these weights is needed in this estimate. This proves (6.9). □

Thus this one microscopic elimination layer preserves the full off-diagonal Schwinger hierarchy up to an error that vanishes in a fixed OS-type seminorm, provided the field budget (6.4) is the one being used. It proves an exact matching statement modulo a controlled contact strip for that layer.

Group scope of Theorem 6.2. Its proof uses only the factorization of a Wilson plaquette action under (6.1), conditional contraction, and (6.4). It does not use quaternions, Bessel functions or the scalar-source theorem. Consequently Theorem 6.2 applies unchanged to every fixed $SU(N)$ covered by the existing Wilson reserve and 21-field moment budget, with $e_p = \beta[N - \mathrm{Re}\,\mathrm{Tr}\,U_p]$. The low-staple and Bessel-source theorems in Sections 2–5 remain SU(2) statements.

The theorem does not show that a second colored elimination has the product conditional law (6.2), nor that an unbounded number of such projections retains a shrinking physical stencil. After a fixed positive physical scale is reached, a local stencil need no longer have radius $O(a)$. The all-scale matching theorem M0 therefore remains open. This result supplies a verified microscopic starting step, not a replacement for the missing induction.

7. Why this does not close the coarse inherited-history theorem

The microscopic estimate resolves the canceled-staple obstruction in an averaged form: arbitrary-boundary conditional bounds can fail, yet the bad boundary geometry has a quantified cost under the actual normalized Wilson law. It also shows that a source logarithm can have a useful common complex radius in a restricted conditional system, with its background dependence kept visible.

Three missing identifications prevent promotion to H0 or M0:

1. Coarse margins are different objects. A coarse selector depends on nonlinear constrained minimizing fields, currents, preparatory geometry and old history. No theorem identifies its failure with the events $\rho_l < \rho_*$ or bounds its retained minimum by the microscopic B_l's with a sufficient current-scale factor.
2. Exact transported marks are different functions. The one-link marks in (4.1) are original plaquette energies. After RG integration and history resets, the full source jet has generated responses and remainder vertices. They are not automatically native plaquette sources in a new one-link Wilson kernel.
3. Outer covariance remains. Even when every conditional fluctuation is controlled, the covariance of the retained conditional means remains in the exact law of total covariance. The local response identity (6.3) does not estimate it.

The actual all-history estimate still needs a current-scale normalization h_k^{-4}, spatial localization for every generated response, and a bound for the complete outer law. The power a^{-4} on an original physical insertion has not disappeared in this note.

8. Assessment of a history-fugacity route

An analytic signed expansion may legitimately compute a generating function of a positive history law. It need not itself be a probability decomposition. That observation suggests inserting local history fugacities

$$Z(z) = \sum_{\omega}\int \rho_\omega(V)\exp\left(\sum_{B} z_B\,\xi_B(V,\omega)\right)dV, \tag{8.1}$$

where $0 \le \xi_B \le 1$ are specified local selector flags.

At every finite cutoff, (8.1) exists and is entire in a finite collection of z_B's. For its normalized functional $M(z) = Z(z)/Z(0)$, boundedness and positivity give the precise uniform estimate

$$|M(z) - 1| \le e^{\sum_B |z_B|} - 1. \tag{8.2}$$

Consequently there is a zero-free complex ℓ^1 ball, with a bounded logarithm, whose radius is independent of the number of blocks. This useful elementary ball is not a common per-coordinate ℓ^∞ ball: a fixed amplitude at each of many blocks has a growing ℓ^1 norm. Nor does (8.2) give a carrier-localized logarithm or spatial decay of its mixed derivatives. The missing history-fugacity theorem concerns those stronger spatial and extensive-source norms, not existence of the elementary ℓ^1 ball.

The source-free signed expansion can be used for this purpose only after proving that the fugacity-weighted same positive-history kernels obey the required absolute activity norms, with the selectors, current domains and background dependence retained. Multiplication by a real bounded flag preserves a real mass bound, but a flag may be a hard selector depending on a moving background. It does not automatically preserve the complex-background analyticity used in the effective-action cluster norm. Nor does an estimate at $z = 0$ control derivatives in z.

The earlier LFR residual factor supports a localized source theorem for bounded marks at fixed retained data once the cross-interaction norm is verified; the positive-interpolation theorem already captures that restricted use. It does not control fugacities that change the outer selected geometry or the covariance of their retained-data means. None of the published sources used here supplies the missing uniform fugacity induction. The idea remains a possible proof route, not an imported theorem.

9. References and completion conditions

The one-link formulas and estimates use:

- the proof of the compact Wilson reserve in Annex A. This is the only prior lattice-probability input to (3.2).
- NIST Digital Library of Mathematical Functions, 10.32.2, the modified-Bessel integral representation, and 10.40.1, the fixed-order sectorial large-argument asymptotic. The complex error discussion is 10.40(iii). These standard special-function inputs are used only to establish the one-link source ball in Section 4. The normalized probability and locality arguments are proved directly.
- The frozen positive transport and history-resampling identities proved in Annexes A and I. They are used for the explicit marked corollary, not to infer spatial mixing.

Statement	Status
Joint normalized probabilities for arbitrary sets of microscopic low-resultant-staple links	P, Theorem 3.1
Their transported marked contribution with a full rarity factor	P, Corollary 3.2
Common complex source ball for one-link scalar jets, with linear conditional-minimum dependence	P with the stated special-function input, Theorem 4.1
Full-law averaged conditional responses, including canceled-staple sectors	P, Theorem 5.1
Exact spatial locality of an admissible microscopic elimination layer	P, Corollary 6.1
Full 21-field one-layer matching modulo a contact strip with $aC^{m+1}m!\nu_{6m}$ error	P for the stated actual field budget, Theorem 6.2
Coarse inherited-selector probabilities and full-history spatial responses	O
Complete current-coordinate E/R contour and majorant theorem	O
Exact transported Wilson-to-native-field matching	O
Four-dimensional Yang–Mills existence	Not proved

Annex P — Seven-channel running normalization and weighted remainders

Annex P. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue. Identifying the calibrated Gaussian marginal with the Wilson trajectory also requires GC, including the bounded error uniform in depth.

Purpose

P: finite-dimensional product theorem. C: its application to the actual Yang–Mills observable induction. The tensor decomposition of Annex D has a useful consequence for the running product: on the 21 curvature-bilinear components, the full hypercubic commutant is commutative. A controlled order-g^4 error in an equivariant *mixing matrix* can therefore be summed without a noncommuting conjugation loss. This does not establish that the exact transported fields have the required mixing matrices, and does not sum arbitrary additive observable remainders.

All assertions here are finite-dimensional derivations. No anomalous dimension or beta-function coefficient is imported or computed.

1. Fixed projectors, including reflections

Let

$$E = \mathrm{Sym}^2(\Lambda^2\mathbb{R}^4), \qquad \dim E = 21,$$

with its invariant Euclidean inner product, and let $H_4 = (\mathbb{Z}_2)^4 \rtimes S_4$ act by the tensor representation. The decomposition proved in Annex D consists of seven inequivalent, absolutely irreducible real summands, with dimensions

$$1,\ 3,\ 2,\ 6,\ 6,\ 1,\ 2. \tag{R.1}$$

Let P_ρ, $1 \le \rho \le 7$, be their orthogonal projectors. They are fixed once the tensor convention is fixed, satisfy

$$P_\rho P_\sigma = \delta_{\rho\sigma} P_\rho, \qquad \sum_\rho P_\rho = I,$$

and every real H_4-equivariant endomorphism has the form

$$M = \sum_{\rho=1}^{7} m_\rho P_\rho, \qquad m_\rho \in \mathbb{R}. \tag{R.2}$$

Absolute irreducibility is relevant: there are no complex or quaternionic endomorphism algebras hidden inside the individual summands. Consequently every two such endomorphisms commute.

The identity field is not included in E. Its additive mixing is handled in Section 4. The full group includes coordinate reflections; restricting to a smaller symmetry group can change the commutant.

2. Product theorem with a varying leading coefficient

Let $u_j \ge 0$, $j \ge 0$, satisfy

$$\sum_{j=0}^{\infty} u_j^2 < \infty. \tag{R.3}$$

Suppose that the real equivariant matrices M_j, Γ_j, R_j obey

$$M_j = I + u_j\Gamma_j + R_j, \qquad \|\Gamma_j\| \le G, \qquad \|R_j\| \le B u_j^2, \tag{R.4}$$

and

$$0 \le u_j \le 1, \qquad G u_j + B u_j^2 \le \frac{1}{2}. \tag{R.5}$$

A finite initial segment that does not meet (R.5) can instead be retained as a fixed positive definite equivariant prefactor, provided that property is separately known. Its constants are then included in the bounds. It is not discarded. An arbitrary invertible prefactor preserves convergence and invertibility but need not preserve the positivity conclusions below.

Define

$$T_0 = I, \qquad T_n = M_{n-1} \cdots M_0, \qquad A_n = \sum_{j=0}^{n-1} u_j \Gamma_j, \qquad V = \sum_{j=0}^{\infty} u_j^2. \tag{R.6}$$

Theorem 2.1 Every M_j and T_n is positive definite as a matrix on E. There is a positive definite, equivariant, invertible matrix R_∞ such that

$$\boxed{e^{-A_n} T_n = \mathsf{R}_n \longrightarrow \mathsf{R}_\infty.} \tag{R.7}$$

For constants C_1, C_2 depending only on G, B and the smallness ceiling in (R.5),

$$\|\mathsf{R}_n\| + \|\mathsf{R}_n^{-1}\| \quad + \|\mathsf{R}_\infty\| + \|\mathsf{R}_\infty^{-1}\| \le 4e^{C_1 V}, \tag{R.8}$$

$$\|\mathsf{R}_\infty - \mathsf{R}_n\| \le C_2 e^{C_1 V} \sum_{j=n}^{\infty} u_j^2, \tag{R.9}$$

and

$$\boxed{\|T_n\| + \|T_n^{-1}\| \le 2 \exp\left(G \sum_{j<n} u_j + C_1 V \right).} \tag{R.10}$$

No convergence of Γ_j is required.

Proof. Write

$$M_j = \sum_{\rho} (1 + u_j \gamma_{\rho,j} + r_{\rho,j}) P_\rho.$$

All eigenvalues lie in $[1/2, 3/2]$. The real matrix logarithm is therefore well defined in the same commutative algebra. For $|x| \le 1/2$,

$$|\log(1+x) - x| \le \sum_{k \ge 2} |x|^k / k \le |x|^2.$$

Since $u_j \le 1$, the elementary estimate, with (R.4), gives

$$\Delta_j := \log M_j - u_j \Gamma_j, \qquad \|\Delta_j\| \le C_1 u_j^2. \tag{R.11}$$

The explicit choice $C_1 = B + (G+B)^2$ suffices. Any finite terms outside the ceiling are covered by the prefactor convention.

Every Δ_j commutes with every Γ_k. Thus the product identity is exact:

$$T_n = \exp\left(A_n + \sum_{j<n} \Delta_j \right), \qquad \mathsf{R}_n = \exp\left(\sum_{j<n} \Delta_j \right). \tag{R.12}$$

The sums of Δ_j converge absolutely. Taking their exponential gives (R.7), invertibility and (R.8). For selfadjoint matrices in this fixed commuting algebra, the scalar mean-value estimate for the exponential gives (R.9), after increasing the constant if needed. Finally $\|A_n\| \le G \sum_{j<n} u_j$, which proves (R.10). □

The proof explicitly separates the generally divergent first-order sum $\sum u_j \Gamma_j$ from the convergent second-order correction. Symmetry alone would not bound the former.

3. Logarithmic running on a quantified AF sequence

Assume, in addition, that for fixed $b > 0$, $\kappa \geq 1$,

$$u_j = \frac{1}{b(j+\kappa)} + \varepsilon_j, \qquad |\varepsilon_j| \leq C \frac{\log(j+\kappa+1)}{(j+\kappa)^2}, \tag{R.13}$$

and that one equivariant matrix Γ satisfies

$$\Gamma_j = \Gamma + \Xi_j, \qquad \|\Xi_j\| \leq C u_j. \tag{R.14}$$

The bounded finite-dimensional eigenvalues of Γ are denoted γ_ρ. These numbers must be obtained from the actual observable matching theorem before this corollary describes its renormalization.

Corollary 3.1 There is an equivariant positive definite matrix L_∞ such that

$$\boxed{\left(\frac{n+\kappa}{\kappa}\right)^{-\Gamma/b} T_n \longrightarrow L_\infty,} \tag{R.15}$$

and the difference from its limit is

$$O\left(\frac{\log(n+\kappa+1)}{n+\kappa}\right) \tag{R.16}$$

in matrix norm. Here $x^{\Gamma/b}$ means $\sum_\rho x^{\gamma_\rho/b} P_\rho$.

Proof. The term $u_j \Xi_j$ is absolutely summable by (R.3). The series of ε_j converges absolutely, with tail $O(\log(n+\kappa+1)/(n+\kappa))$. Also

$$\sum_{j<n} \frac{1}{j+\kappa} - \log \frac{n+\kappa}{\kappa}$$

has a finite limit with tail $O((n+\kappa)^{-1})$, as follows by comparing each reciprocal with the integral over its unit interval. Substitute these statements and (R.11) in (R.12). All exponents commute, and the remaining exponent has an absolutely convergent series with the stated tail. Exponentiation proves the result. □

For a geometric ultraviolet sequence $a_n = \ell_0 L^{-n}$, the factor $n^{\gamma_\rho/b}$ is a power of $\log(\ell_0/a_n)$. A running normalization can therefore be defined by the exact inverse T_n^{-1}, while its explicit channel asymptotics are given by (R.15). Neither T_n nor its inverse must remain bounded in the original bare basis.

This corollary does not verify (R.13) for the AF input of the construction, nor (R.14) for its observable matrices. In particular the native cumulant coefficient is not silently identified with Γ.

3.1 What this says about rotations If two hypercubic pieces belong to one rotational tensor type, the ratio of their product eigenvalues has the asymptotic form

$$\frac{t_{\rho,n}}{t_{\sigma,n}} = C_{\rho\sigma} \left(\frac{n+\kappa}{\kappa}\right)^{(\gamma_\rho - \gamma_\sigma)/b} (1 + o(1)), \qquad C_{\rho\sigma} > 0. \tag{R.17}$$

Equality of the leading exponents removes this particular relative power. It does not force the finite constants to agree, nor estimate any nonlocal or irrelevant rotation defect. Conversely, the running bare-to-renormalized matrix need not itself be a rotational intertwiner when its normalization compensates anisotropic cutoff representatives. The required statement is covariance of the normalized limiting hierarchy. No rotation-restoration theorem is claimed from (R.17).

4. Identity mixing and positivity of a realized field change

At a finite cutoff, an equivariant affine marginal recursion has the form

$$X_{j+1} = M_j X_j + c_j 1. \tag{R.18}$$

The vector $c_j \in E$ lies in the scalar invariant line, because 1 is fixed by the full hypercubic group and that line is the unique trivial summand in E.

For any chosen initial centering v_0 in that line, set

$$v_{j+1} = M_j v_j + c_j, \qquad Y_j = T_j^{-1}(X_j - v_j 1). \tag{R.19}$$

Then $Y_{j+1} = Y_j$ exactly in this finite-dimensional recursion. No bound on the bare numbers c_j, v_j is used.

If such a real field mixing and identity subtraction are implemented on the actual finite-cutoff positive-time polynomial algebra, with the reflection action respected, reflection positivity is preserved. Indeed every new polynomial is an old polynomial obtained by the same real linear-affine substitution; its OS quadratic form is the old nonnegative form on that substituted polynomial. At finite source order this is the corresponding finite triangular substitution. This is the positivity mechanism already proved for the explicit field/contact architecture, not a proof that a proposed running normalization meets the required continuum conditions.

5. Why summable matrix errors do not finish marked matching

An exact transported observable generally has an additional field-valued remainder:

$$X_{j+1} = M_j X_j + c_j 1 + \mathcal{R}_j. \tag{R.20}$$

After (R.19), the exact identity becomes

$$\boxed{Y_{j+1} - Y_j = T_{j+1}^{-1}\mathcal{R}_j.} \tag{R.21}$$

Therefore the relevant sufficient condition in any fixed normed space of observable distributions is

$$\sum_j \|T_{j+1}^{-1}\mathcal{R}_j\| < \infty. \tag{R.22}$$

The matrix theorem estimates the weights in (R.22). It does not supply the spatial norms, history averages or decay of $\mathcal{R}_j$.

Proposition 5.1: raw order-u_j^2 summability can fail Fix $A \geq 1$ and $\kappa > 2A$. In one real channel, put

$$u_j = (j+\kappa)^{-1}, \qquad M_j = 1 - Au_j > 0, \qquad \mathcal{R}_j = u_j^2. \tag{R.23}$$

There is no matrix error in (R.4), and $\sum_j |\mathcal{R}_j| < \infty$. Nevertheless, the normalized additive increments in (R.21) have a divergent sum.

Proof. Summing $\log(1 - A/(j+\kappa)) = -A/(j+\kappa) + O((j+\kappa)^{-2})$ gives $T_j \asymp (j+\kappa)^{-A}$. Hence

$$T_{j+1}^{-1}\mathcal{R}_j \asymp (j+\kappa)^{A-2} > 0.$$

The series diverges for $A \geq 1$, logarithmically at $A = 1$. □

An actual irrelevant gain, for example a bound $\|\mathcal{R}_j\| \leq C(j+\kappa)^d L^{-\epsilon j}$ in the same norm with $\epsilon > 0$, survives every fixed polynomial weight furnished by (R.10) under the AF hypothesis (R.13), and is summable. Even (R.3) alone gives $\sum_{i<j} u_i \leq \sqrt{jV}$; the resulting $\exp(G\sqrt{jV})$ weight is still beaten by $L^{-\epsilon j}$. Such a gain must be proved for the actual transported remainder; it is not inferred from its engineering name or from an $O(g^4)$ number in a one-shell cumulant expansion.

6. Consequence for the running-normalization problem

The finite-dimensional noncommuting-product concern is removed for the specified fully hypercubic equivariant 21-component marginal matrices, once the quantitative one-step hypotheses are supplied. This is an improvement in the running-normalization subproblem F0. It is not a claim that arbitrary history-dependent insertions have already been projected onto that commutative algebra.

The missing steps remain the identification of the actual matrices, their coefficient estimates and prescribed finite normalization; the compatible contact terms; the weighted remainder estimate (R.22) with spatial and source-order control; and the complete history law. M0, F0 and R0 remain open in the full Yang–Mills construction.

Annex Q — Compact magnetic operator bounds for every SU(N)

Annex Q. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Scope

The full renormalized energy obligation E0 is still open. This note proves a new estimate on the actual compact Wilson transfer operator for every $SU(N)$, rather than only the SU(2) temporal scalar of Annex M.

The result controls every spatial-slice multiplication observable q satisfying a pointwise bound

$$|q| \le BS_\Gamma,$$

where S_Γ is the sum of the spatial Wilson plaquette actions incident on a fixed finite link set. It gives a volume-uniform heat-sandwich estimate and a one-sided graph estimate of order $(1+\log(1+\beta))/\beta$ for the ground-centered bare field. The proof retains all spatial interactions and all compact configurations, and it does not square a quadratic-form inequality.

All six raw magnetic clover bilinears satisfy the hypothesis. There is a necessary qualification for the exact field architecture chosen: its scalar repair adds a temporal contribution to the diagonal entries. Consequently the theorem directly covers the five trace-free magnetic combinations in that exact architecture; the raw magnetic trace is an additional auxiliary representative. This distinction is kept explicit below.

The physical a^{-4} normalization remains. The same-cubic periodic-state premise has not been proved, and a choice of a cofinal sequence alone does not establish it. Section 6 gives the exact quantifier obstruction.

The standard compact transfer representation is the pure-gauge specialization of Lüscher, *Construction of a selfadjoint, strictly positive transfer matrix for Euclidean lattice gauge theories*, CMP 54 (1977), 283–292, preprint. All local comparisons used below are proved here or are the already-proved local deletion reserve of Annex M; no new complete-history RG estimate is imported.

1. The actual local deletion identity

Fix a finite periodic spatial lattice, spacing $a > 0$, gauge group $SU(N)$, and the physical gauge-invariant subspace of

$$\mathcal{H}_0 = L^2(SU(N)^{\mathcal{L}_s}, dH).$$

Use the unchanged convention

$$s(U) = N - \operatorname{Re}\operatorname{Tr} U, \qquad 0 \le s(U) \le 2N, \qquad \beta = 2Nx_0, \qquad X_B = N\beta. \tag{1.1}$$

Let $K = \bigotimes_l C_\beta$ be the normalized electric convolution, let

$$V_s = \beta \sum_{p \text{ spatial}} s(U_p), \qquad A = e^{-V_s/2}, \qquad T_0 = AKA, \tag{1.2}$$

and let $\lambda_0 = \|T_0\|$. Positivity improvement and gauge invariance put the unique positive top vector in the physical subspace, so the physical transfer operator T has the same norm. Normalize

$$P = T/\lambda_0 = e^{-aH}, \qquad H \ge 0. \tag{1.3}$$

For a fixed finite spatial link set Γ, let $\mathcal{P}_\Gamma$ be all spatial plaquettes incident on Γ. Set

$$S_\Gamma = \sum_{p \in \mathcal{P}_\Gamma} s(U_p), \qquad A_\Gamma = e^{-\beta S_\Gamma/2}, \qquad A = A_\Gamma A_r. \tag{1.4}$$

The multiplication operator A_r is independent of all links in Γ. Therefore, on the full link Hilbert space,

$$T_0 = A_\Gamma T_{\text{del}} A_\Gamma, \qquad T_{\text{del}} = K_\Gamma \otimes (A_r K_r A_r). \tag{1.5}$$

This is an identity, with every undeleted action factor retained. Write

$$\lambda_{\mathrm{del}} = \|T_{\mathrm{del}}\|, \qquad \Delta_\Gamma = \log(\lambda_{\mathrm{del}}/\lambda_0) \geq 0. \tag{1.6}$$

Both operators commute with gauge projection. The strictly positive top-vector argument applies to T_{del} as well, so its full and physical norms agree.

Since $0 \leq T_{\mathrm{del}} \leq \lambda_{\mathrm{del}} I$, (1.5) gives

$$\boxed{T \leq \lambda_{\mathrm{del}} e^{-\beta S_\Gamma}, \qquad P \leq e^{\Delta_\Gamma} e^{-\beta S_\Gamma}.} \tag{1.7}$$

The multiplier S_Γ is bounded at each fixed cutoff, nonnegative, and gauge invariant.

Two previously proved bounds remain available:

$$0 \leq \Delta_\Gamma \leq 2N\beta|\mathcal{P}_\Gamma|, \tag{1.8}$$

and, when the containing fixed B-site spatial cube fits the compatible even-block geometry,

$$\boxed{\Delta_\Gamma \leq B^3\{c_N + 2(N^2-1)\log(1+\beta)\}.} \tag{1.9}$$

The latter is the cylinder reserve and fixed-spatial-lattice trace limit. The value of B depends only on the finite local stencil. It is independent of the total spatial volume. No temporal aspect-ratio extension of the source-free RG is used in that elementary comparison.

2. Local spatial energy as a quadratic form

Theorem 2.1 — local magnetic form comparison, P On the physical Hilbert space,

$$\boxed{\beta S_\Gamma \leq aH + \Delta_\Gamma I} \tag{2.1}$$

as closed quadratic forms.

Proof. Apply operator monotonicity of the logarithm to (1.7), with a small positive regularization if necessary. Explicitly, for bounded strictly positive $X \leq Y$,

$$\log Y - \log X = \int_0^\infty \left[(X+t)^{-1} - (Y+t)^{-1}\right] dt$$

is nonnegative in the quadratic-form sense. To cover P, which has no bounded inverse, replace both sides of (1.7) by their sums with ϵI, use the resolvent integral, and let $\epsilon \downarrow 0$. The spectral theorem gives the closed form of $\log P = -aH$. The right comparison operator is boundedly invertible at this fixed cutoff, and its logarithm is $\Delta_\Gamma - \beta S_\Gamma$. Hence

$$-aH \leq \Delta_\Gamma - \beta S_\Gamma,$$

which is (2.1). □

In contrast to the earlier electric comparison, no polar unitary occurs in (2.1): S_Γ is the actual time-zero multiplication operator.

For any real gauge-invariant multiplication observable q with

$$|q(U)| \leq B S_\Gamma(U), \tag{2.2}$$

one immediately has

$$|\langle\psi, q\psi\rangle| \leq \frac{B}{\beta}\left(a\langle\psi, H\psi\rangle + \Delta_\Gamma\|\psi\|^2\right). \tag{2.3}$$

If Ω is the normalized ground vector, then

$$|\langle\Omega, q\Omega\rangle| \leq B\Delta_\Gamma/\beta. \tag{2.4}$$

Corollary 2.2 — heat-sandwich bounds, P For $t > 0$,

$$\boxed{\|e^{-tH} q e^{-tH}\| \le \frac{B}{\beta}\left(\Delta_\Gamma + \frac{a}{2et}\right).} \tag{2.5}$$

For $q^\circ = q - \langle \Omega, q\Omega\rangle I$,

$$\boxed{\|e^{-tH} q^\circ e^{-tH}\| \le \frac{B}{\beta}\left(2\Delta_\Gamma + \frac{a}{2et}\right).} \tag{2.6}$$

When $q \ge 0$, the coefficient $2\Delta_\Gamma$ in (2.6) can be replaced by Δ_Γ.

Proof. Substitute $e^{-tH}\psi$ in (2.3) and use

$$\|He^{-2tH}\| = \sup_{x\ge 0} xe^{-2tx} = (2et)^{-1}.$$

The centered bound follows from (2.4) and $\|e^{-2tH}\| \le 1$. In the positive case, both $e^{-tH} q e^{-tH}$ and $\langle q\rangle_\Omega e^{-2tH}$ are positive. The norm of their selfadjoint difference is bounded by the larger of their norms, giving the improvement. □

These are arbitrary-vector operator bounds for the actual cutoff ground Hamiltonian. They are not merely estimates of vacuum matrix elements.

3. One-sided graph bounds without squaring forms

The form bound (2.1) must not be squared to infer a one-sided operator estimate. The stronger exponential comparison (1.7) supplies the missing argument.

Lemma 3.1 — local exponential range estimate, P For $0 < s \le 1/2$,

$$\boxed{\|e^{s\beta S_\Gamma} e^{-saH}\| \le e^{s\Delta_\Gamma}.} \tag{3.1}$$

Proof. The function $x \mapsto x^{2s}$ is operator monotone for $0 < 2s \le 1$. For completeness, if $0 < r < 1$, its representation

$$x^r = \frac{\sin \pi r}{\pi} \int_0^\infty t^{r-1} x (t + x)^{-1}\, dt$$

proves this from resolvent monotonicity; the case $r = 1$ is immediate. Applying it to (1.7) gives

$$P^{2s} \le e^{2s\Delta_\Gamma} e^{-2s\beta S_\Gamma}. \tag{3.2}$$

Conjugation by $e^{s\beta S_\Gamma}$, which is bounded at fixed cutoff, gives

$$\|P^s e^{s\beta S_\Gamma}\|^2 \le e^{2s\Delta_\Gamma}.$$

Taking the adjoint and using $P^s = e^{-saH}$ proves (3.1). This is the relevant range-factorization step. □

Let $\Pi_E = 1_{[0,E]}(H)$. From (3.1),

$$\|e^{s\beta S_\Gamma}\Pi_E\| \le e^{s(\Delta_\Gamma + aE)}. \tag{3.3}$$

For q satisfying (2.2),

$$\|qe^{-s\beta S_\Gamma}\| \le B \sup_{u\ge 0} ue^{-s\beta u} = \frac{B}{es\beta}. \tag{3.4}$$

Choose $s = (2 + \Delta_\Gamma + aE)^{-1}$. This is at most $1/2$, and $s(\Delta_\Gamma + aE) \le 1$, so

$$\boxed{\|q\Pi_E\| \le \frac{B}{\beta}(2 + \Delta_\Gamma + aE).} \tag{3.5}$$

Theorem 3.2 — local spatial field graph bound, P For every real multiplication q satisfying (2.2),

$$\boxed{\|q(1+H)^{-2}\| \le \frac{4B}{\beta}(2+\Delta_\Gamma+a).} \tag{3.6}$$

For its ground-centered version,

$$\boxed{\|q^\circ(1+H)^{-2}\| \le \frac{B}{\beta}(8+5\Delta_\Gamma+4a).} \tag{3.7}$$

Proof. Decompose the spectrum with

$$R_j = 1_{\{2^j \le 1+H<2^{j+1}\}}, \qquad j \ge 0.$$

Using (3.5) with $E = 2^{j+1}$ and the triangle inequality for the operator-norm convergent sum,

$$\begin{aligned}\|q(1+H)^{-2}\| &\le \frac{B}{\beta}\sum_{j\ge 0}(2+\Delta_\Gamma+a2^{j+1})\,2^{-2j}\\ &= \frac{B}{\beta}\left\{\frac{4}{3}(2+\Delta_\Gamma)+4a\right\}\\ &\le \frac{4B}{\beta}(2+\Delta_\Gamma+a).\end{aligned}$$

Adding the identity subtraction bound (2.4) proves (3.7). □

At each fixed cutoff q is a bounded multiplication operator, so its domain is the full physical Hilbert space. The important content of (3.6)–(3.7) is the uniform cutoff and spatial-volume coefficient in the specified normalization. They make no claim that an unconstructed renormalized continuum operator has already been obtained.

A more general polynomial local version follows from the same proof. If $|q| \le BS_\Gamma^r$, with $r \ge 1$, choose $s = r/(2r+\Delta_\Gamma+aE)$. Then

$$\|q\Pi_E\| \le B\beta^{-r}(2r+\Delta_\Gamma+aE)^r, \tag{3.8}$$

and

$$\boxed{\|q(1+H)^{-(r+1)}\| \le 2B\beta^{-r}(2r+\Delta_\Gamma+2a)^r.} \tag{3.9}$$

Indeed, on the j-th spectral band the numerator in (3.8) is at most $2^{jr}(2r+\Delta_\Gamma+2a)^r$, leaving the summable factor 2^{-j}. This supplies a controlled finite-product extension on any fixed local stencil.

4. Exact application to the magnetic clover tensor

Use the same four-quadrant clover as the architecture. For $A=(i,j)$, $1 \le i < j \le 3$, let

$$C_A(x) = \frac{1}{8i}\sum_{\sigma,\tau=\pm1}\sigma\tau\,\pi_0\big(P_{\sigma e_i,\tau e_j}(x) - P_{\sigma e_i,\tau e_j}(x)^\dagger\big), \tag{4.1}$$

where all loops are based at x, and π_0 removes the scalar matrix part. All links in these stencils lie in the same time slice.

Write $\mathcal{I}_B = \{(1,2),(1,3),(2,3)\}$. For a real symmetric 3-by-3 matrix M, define the dimensionless raw magnetic field

$$\mathfrak{q}_M(x) = \sum_{A,B\in\mathcal{I}_B} M_{AB}\,\mathrm{Tr}(C_A(x)C_B(x)). \tag{4.2}$$

The six independent choices of M are the six raw spatial curvature-bilinear components.

For a unitary matrix U,

$$\left\|\frac{U-U^\dagger}{2i}\right\|_{\mathrm{HS}}^2 = \sum_j \sin^2\theta_j \le 2\sum_j(1-\cos\theta_j) = 2s(U). \tag{4.3}$$

Orthogonal projection π_0 contracts the Hilbert–Schmidt norm. Convexity of that norm squared therefore gives

$$\|C_A(x)\|_{\mathrm{HS}}^2 \le \frac{1}{2}\sum_{\sigma,\tau} s(P_{\sigma e_i,\tau e_j}(x)). \tag{4.4}$$

Let Γ contain the finitely many links needed to include these magnetic plaquettes in $\mathcal{P}_\Gamma$. The Gram matrix

$$G_{AB}(x) = \mathrm{Tr}(C_A(x)C_B(x))$$

is positive semidefinite. Hence

$$\begin{aligned} |\mathfrak{q}_M(x)| &\le \|M\|_{\mathrm{op}} \,\mathrm{Tr}\, G(x) \\ &\le \frac{\|M\|_{\mathrm{op}}}{2} \sum_{A\in\mathcal{J}_B} \sum_{\sigma,\tau} s(P_{\sigma A_1,\tau A_2}(x)) \\ &\le \frac{\|M\|_{\mathrm{op}}}{2} S_\Gamma. \end{aligned} \tag{4.5}$$

Each plaquette on the second line is one of the finitely many geometric plaquettes in the stencil. On the nondegenerate tori used in the construction, different quadrants and magnetic orientations give distinct such plaquettes. A fixed multiplicity factor can be inserted on smaller exceptional geometries; those are excluded by the cofinal minimum-size conditions.

Corollary 4.1 — all-$SU(N)$ magnetic clover bound, P For every fixed real symmetric M, define

$$\mathfrak{q}^\circ_M(x) = \mathfrak{q}_M(x) - \langle\Omega, \mathfrak{q}_M(x)\Omega\rangle I.$$

Then

$$\boxed{\|\mathfrak{q}^\circ_M(x)(1+H)^{-2}\| \le \frac{\|M\|_{\mathrm{op}}}{2\beta}(8 + 5\Delta_\Gamma + 4a).} \tag{4.6}$$

The heat estimate is

$$\boxed{\|e^{-tH}\mathfrak{q}^\circ_M(x)e^{-tH}\| \le \frac{\|M\|_{\mathrm{op}}}{2\beta}\left(2\Delta_\Gamma + \frac{a}{2et}\right).} \tag{4.7}$$

If $M \ge 0$, the $2\Delta_\Gamma$ in (4.7) can be replaced by Δ_Γ.

With (1.9), these estimates are

$$O\left(\frac{1+\log(1+\beta)}{\beta}\right)$$

at fixed $t > 0$, uniformly in compatible spatial volume and $0 < a \le 1$. They apply to arbitrary vectors in the physical Hilbert space and retain the entire compact Wilson action.

4.1 The scalar-repair caveat is necessary The exact hierarchy uses

$$\tilde{q}_{AB,a} = a^{-4}\,\mathrm{Tr}(C_A C_B) + \frac{\delta_{AB}}{6}(2w_a - t_a), \tag{4.8}$$

where w_a is the full four-dimensional Wilson action density and t_a is the trace over all six curvature planes. This ensures that the scalar trace is the exact Wilson coupling derivative.

For magnetic M,

$$\sum_{A,B\in\mathcal{J}_B} M_{AB}\tilde{q}_{AB,a} = a^{-4}\mathfrak{q}_M + \frac{\mathrm{Tr}\,M}{6}(2w_a - t_a). \tag{4.9}$$

Thus if $\mathrm{Tr}\,M = 0$, the correction cancels exactly. The five-dimensional trace-free magnetic subspace consists of three off-diagonal entries and two independent diagonal differences. It is directly covered by (4.6)–(4.7), with the factor a^{-4} restored.

If $\mathrm{Tr}\,M \ne 0$, the repaired field contains temporal plaquettes and is not a time-zero multiplication observable. Equations (4.6)–(4.7) control its raw magnetic part only. The raw magnetic trace is a legitimate auxiliary cutoff field, but it is not silently substituted for the repaired scalar channel.

Likewise, changing which coordinate is called time gives a related estimate in a different reflection reconstruction. That does not, by itself, give all 21 fields a bound with respect to one common Hamiltonian.

5. Physical smearing and the ultraviolet loss

Let g be a fixed compact spatial test, and let

$$\mathrm{Q}_{M,a}(g) = a^3 \sum_{\mathrm{x}} g(\mathrm{x})\, \mathrm{q}^{\circ}_M(\mathrm{x}). \tag{5.1}$$

Translation invariance and the uniform local deletion constant imply

$$\|\mathrm{Q}_{M,a}(g)(1+H)^{-2}\| \le \frac{\|M\|_{\mathrm{op}}}{2\beta}(8 + 5\Delta_* + 4a)\left(a^3 \sum_{\mathrm{x}} |g(\mathrm{x})|\right), \tag{5.2}$$

with Δ_* bounded by the same fixed-stencil version of (1.9). The same argument applies to the heat sandwiches. There is no factor from the total spatial volume.

For a physical dimension-four normalization

$$Q^R_{M,a}(g) = z_a a^{-4} \mathrm{Q}_{M,a}(g) + d_a(g) I, \tag{5.3}$$

the theorem yields only

$$\boxed{\|Q^R_{M,a}(g)(1+H)^{-2}\| \le |z_a| a^{-4} \frac{\|M\|_{\mathrm{op}}}{2\beta}(8 + 5\Delta_* + 4a)\|g\|_{\ell^1_a} + |d_a(g)|.} \tag{5.4}$$

The factor a^{-4} remains. Along an AF trajectory, a logarithmic gain in β does not cancel it. The theorem therefore supplies a sharper actual cutoff operator estimate, not the renormalized E0 conclusion.

There are two further distinctions. First, the center in this proof is the finite-volume ground-state mean, while the continuum prescription must use the normalization and identity coefficient fixed in F0. Second, the periodic Wilson hierarchy need not be replaced by ground-state matrix elements without the state-control step. The periodic bridge remains conditional.

6. Why the same-cubic thermal premise is not a diagonal-choice theorem

For the spatial lattice of side T, denote its normalized transfer Hamiltonian by $H_{a,T}$. The required the quantity is

$$\Gamma(a,T) = \log \frac{\operatorname{Tr} e^{-(T/2)H_{a,T}}}{\operatorname{Tr} e^{-TH_{a,T}}}. \tag{6.1}$$

At every *fixed* a, T, trace class and a unique ground state imply

$$\lim_{u\to\infty} \log \frac{\operatorname{Tr} e^{-(u/2)H_{a,T}}}{\operatorname{Tr} e^{-uH_{a,T}}} = 0. \tag{6.2}$$

This follows by spectral expansion and dominated convergence at a fixed trace-class reference time. It is a statement about one spatial operator while the temporal length u grows independently.

The target instead sets $u = T$ and changes $H_{a,T}$ when T changes. Formula (6.2) does not establish

$$\Gamma(a_n, T_n)/T_n \longrightarrow 0 \tag{6.3}$$

on a cubic sequence.

A genuine diagonal selection would be justified, for example, if one first proved that, for each sufficiently small admitted a,

$$\liminf_{\substack{T\to\infty \\ T \text{ admitted}}} \Gamma(a,T)/T = 0. \tag{6.4}$$

Then one could select admitted T_n with $\Gamma(a_n, T_n)/T_n < 1/n$, $T_n \to \infty$, and $\log(1/a_n)/T_n \to 0$. But (6.4) is an additional spatially uniform spectral assertion. Neither finite-cutoff ground-state uniqueness nor the known local deletion reserve proves it.

The positive-transfer counterexample demonstrates the logical gap even in finite-dimensional transfer families with good ground-state insertion estimates. Locality of the Wilson kernel may provide further information, but that information must enter through a proved estimate. No such thermal estimate is established in this argument.

The new local magnetic bounds also do not imply (6.3). Their Hamiltonian is the ground-normalized one, and their local energy cost Δ_Γ controls a fixed insertion region. The thermal trace increment sums excitations throughout the whole growing spatial torus. A uniform local form bound is not a bound on that total spectral multiplicity.

7. Operator conditions

Closed in this note: local spatial action form control; a local exponential range estimate; one-sided polynomial graph bounds for dominated spatial multiplication marks; and their exact all-$SU(N)$ magnetic clover application, including the five trace-free magnetic combinations in the chosen scalar-repaired field architecture.

Still open for E0/V0: renormalized physical cancellation of the a^{-4} loss; correct running mixing and identity normalization; temporal and mixed tensor representatives on one common reconstructed domain; the same-cubic periodic state estimate or another rigorously verified state passage; and the interacting block/disintegration hypotheses required by the phase-transfer theorem.

The new results concern the complete compact Wilson cutoff theory in their stated channels. They do not close the renormalized continuum operator obligation, and they do not complete four-dimensional Yang–Mills existence.

Annex R — Three continuum seed estimates for the full tensor energy bound

Annex R. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Scope

P: a finite tensor-orbit theorem and its energy-bound implication. C: application to the requested continuum theory. Once a Poincare-covariant 21-component curvature-bilinear field family has been reconstructed, its spacetime-field energy estimates can be reduced to three specified seed fields: the scalar, the pseudoscalar, and one off-diagonal magnetic component. This is not a proof of any of those three renormalized estimates, or of rotation restoration.

The reduction uses finitely many fixed Lorentz transformations. It does not infer operator bounds by differentiating unbounded boost generators, or by choosing a different time direction at the cutoff.

1. A real Lorentz orbit has dimension nineteen

Use the Minkowski field labels

$$E = (F_{01}, F_{02}, F_{03}), \qquad B = (F_{23}, F_{31}, F_{12}). \tag{L.1}$$

In this section a quadratic monomial means its symmetric color bilinear, for example $B_1B_2 = \mathrm{Tr}(B_1B_2)$. The cyclic trace makes the two factors symmetric. There are 21 such labels, the real vector space

$$\mathcal{E} = \mathrm{Sym}^2(\mathbb{R}^6).$$

This is the Wick-continued version of the Euclidean tensor family; the appropriate fixed phases of components containing time indices are included in its Minkowski field convention.

Choose the infinitesimal rotations and boosts to act by

$$\begin{aligned} J_aE &= e_a \times E, \qquad & J_aB &= e_a \times B, \\ K_aE &= e_a \times B, \qquad & K_aB &= -e_a \times E. \end{aligned} \tag{L.2}$$

They act on the quadratic labels by the Leibniz rule. Reversing all boost conventions gives the same invariant spans.

The scalar and pseudoscalar labels are

$$s = |B|^2 - |E|^2, \qquad p = E \cdot B. \tag{L.3}$$

They are invariant under the connected proper Lorentz group. They remain separate fields because their transformation under the disconnected parity operation is different.

Theorem 1.1 Let $v = B_1B_2$. Its real linear orbit under the connected proper Lorentz group spans a nineteen-dimensional subspace $\mathcal{W}$. Together with s and p, it spans all of $\mathcal{E}$:

$$\boxed{\mathcal{E} = \mathbb{R}s \oplus \mathbb{R}p \oplus \mathcal{W}.} \tag{L.4}$$

Consequently there are nineteen fixed finite Lorentz transformations $\Lambda_1, \dots, \Lambda_{19}$ such that

$$s,\ p,\ D(\Lambda_1)v, \dots, D(\Lambda_{19})v$$

are a basis. No limit of these transformations is needed in the operator argument below.

Proof. Work first with the invariant span of the generator actions on v.

Spatial rotations of B_1B_2 generate every off-diagonal B_iB_j. A rotation by $\pi/4$ in a coordinate plane generates a difference of diagonal squares. Therefore the span contains the five-dimensional space

$$\mathcal{B}_0 = \{B^{\mathsf{T}}MB : M = M^{\mathsf{T}},\ \mathrm{Tr}\, M = 0\}. \tag{L.5}$$

The exact boost identity $K_1(B_1B_2) = E_3B_1$, followed by spatial rotations, gives all off-diagonal E_iB_j, $i \neq j$. Applying a $\pi/4$ coordinate rotation to one such label gives a diagonal difference $E_iB_i - E_jB_j$, after subtracting the off-diagonal terms already in the span. Thus it contains the eight-dimensional space

$$\mathcal{C}_0 = \{E^{\mathsf{T}}MB : \operatorname{Tr} M = 0\}. \tag{L.6}$$

No symmetry of M is imposed in (L.6).

For example the boost identities give

$$K_1(E_3B_2) = B_2^2 + E_3^2, \qquad K_3(E_2B_1) = B_1^2 + E_2^2. \tag{L.7}$$

Subtract these and use the magnetic diagonal differences in $\mathcal{B}_0$. The result includes $E_3^2 - E_2^2$. Its spatial rotations generate

$$\mathcal{E}_0 = \{E^{\mathsf{T}}ME : M = M^{\mathsf{T}},\ \operatorname{Tr} M = 0\}, \tag{L.8}$$

another five-dimensional space. Finally subtract $B_2^2 - |B|^2/3$ and $E_3^2 - |E|^2/3$ from the first expression in (L.7). This supplies $(|B|^2 + |E|^2)/3$.

The displayed spaces and that last line are linearly independent. Their dimensions sum to $5 + 8 + 5 + 1 = 19$. Their span is exactly the simultaneous kernel of the two linear coefficient functionals

$$\ell_s(Q) = \sum_i [B_i^2]Q - \sum_i [E_i^2]Q, \qquad \ell_p(Q) = \sum_i [E_iB_i]Q, \tag{L.9}$$

where brackets denote the coefficient in the monomial basis. Direct substitution in (L.2) gives $\ell_s(J_aQ) = \ell_s(K_aQ) = \ell_p(J_aQ) = \ell_p(K_aQ) = 0$. This codimension-two kernel is invariant and contains v; the orbit span has no additional directions. Also $\ell_s(s) = 6$, $\ell_p(p) = 3$, and the cross evaluations vanish. This proves (L.4).

To pass from generators to finite transformations, the linear span of a group orbit is a subspace of a finite-dimensional space and is closed. It contains the derivative at zero of every curve $D(e^{tX})w$ with w already in that span. Hence it contains all the generator directions just exhibited. Conversely the generator-invariant finite-dimensional span is invariant under their exponentials and under the connected group they generate. The two spans agree. A basis can therefore be selected from nineteen actual finite orbit elements. □

As a separate exact-arithmetic check, integer generator matrices in the ordered monomial basis $(E_1^2, E_1E_2, \dots, B_3^2)$ give nineteen independent generator words on B_1B_2. Adding s, p gives determinant -36. The proof above does not depend on a numerical rank tolerance or on treating a generator derivative as an energy-bounded operator.

2. Energy bounds pass through a fixed Lorentz transformation

Assume a reconstructed covariant field family $v \mapsto \Phi_v(f)$ with a common invariant field core, a unitary Poincare representation U, and joint translation spectrum in the closed forward cone. Require this core also to be graph dense for the energy powers used below, as in the established common smooth-energy core construction. This makes the passage from core equalities to the closed-field domains explicit. Write $H = P^0 \geq 0$. Use the convention

$$\Phi_{D(\Lambda)v}(f) = U(\Lambda)\Phi_v(f_\Lambda)U(\Lambda)^{-1} \tag{L.10}$$

on that core. The transformation $f \mapsto f_\Lambda$ is a fixed linear change of spacetime variables and is continuous on Schwartz space.

Suppose a seed field has an actual one-sided bound, with $r \geq 0$,

$$\|\Phi_v(f)(1 + H)^{-r}\| \leq C_v\rho_s(f), \tag{L.11}$$

including a justified bounded extension and operator-domain interpretation on the common core. More precisely, the closed seed field has $D((1 + H)^r) \subset D(\overline{\Phi_v(f)})$, and its restriction to the specified common core is the reconstructed field. A formal vacuum estimate is not the hypothesis (L.11).

Lemma 2.1 For each fixed proper orthochronous Lorentz transformation Λ, there is $c_\Lambda \geq 1$ such that

$$c_\Lambda^{-1} H \leq U(\Lambda) H U(\Lambda)^{-1} \leq c_\Lambda H \tag{L.12}$$

in the joint translation spectral calculus. Consequently

$$\boxed{\|(1+H)^r U(\Lambda)^{-1}(1+H)^{-r}\| \leq c_\Lambda^r.} \tag{L.13}$$

The transformed field in (L.10) therefore has the same energy exponent r, with a different finite constant and Schwartz seminorm.

Proof. If p lies in the closed forward cone, then $|\mathrm{p}| \leq p^0$. The time component of a fixed Lorentz transform lies between positive fixed multiples of p^0; for a boost of rapidity χ, $e^{-|\chi|}$ and $e^{|\chi|}$ suffice. The translation spectral theorem proves (L.12). Both linear combinations of the P^μ in that inequality commute. The identity

$$(1+H)^r U(\Lambda)^{-1} = U(\Lambda)^{-1}(1 + U(\Lambda) H U(\Lambda)^{-1})^r$$

reduces (L.13) to the pointwise ratio of their positive spectral powers. It also preserves the required energy domains. Finally factor, on the common core,

$$\begin{aligned} \Phi_{D(\Lambda)v}(f)(1+H)^{-r} = U(\Lambda)\Phi_v(f_\Lambda)(1+H)^{-r} \\ \cdot(1+H)^r U(\Lambda)^{-1}(1+H)^{-r}. \end{aligned} \tag{L.14}$$

The two bounded factors give the extension and its norm bound. Covariance on the core identifies that extension with the specified field. □

3. Reduction to three renormalized seed fields

Theorem 3.1 Under the covariance, spectrum and domain hypotheses above, assume (L.11) for the three seeds

$$s = |B|^2 - |E|^2, \qquad p = E \cdot B, \qquad v = B_1 B_2. \tag{L.15}$$

The exponents and seminorms may differ between seeds. Then there are a common finite energy exponent $r = \max(r_s, r_p, r_v)$, a Schwartz seminorm ρ, and a constant C such that every one of the 21 tensor components satisfies

$$\boxed{\|\Phi_\alpha(f)(1+H)^{-r}\| \leq C\rho(f).} \tag{L.16}$$

The same conclusion holds for a finite direct sum of fixed group or color-bilinear sectors when the corresponding seeds are supplied.

Proof. Increasing a seed exponent to r only composes its bounded extension with a contraction. Use the nineteen finite transformations in Theorem 1.1 and Lemma 2.1. Every field label is a finite linear combination of those nineteen transformed labels and s, p. The finite coefficient matrix and finitely many pullback seminorms can be dominated by one fixed Schwartz seminorm with one constant. Summing the bounds gives (L.16). The equality is first an equality on the common covariant field core, so the bounded extensions have the required field identification. □

This is a spacetime-smeared field theorem. It neither constructs a sharp-time continuum field nor infers an operator norm from a vacuum moment. The common invariant smooth-energy domain can then be obtained through the already proved field/domain assembly once its remaining hypotheses are in place.

4. How this changes the problem, and what it cannot remove

The full tensor energy conclusion need not require 21 independent analytic estimates after covariance has been proved. Three canonically identified, renormalized seed estimates suffice. The scalar and pseudoscalar cannot be dropped: the magnetic off-diagonal orbit misses precisely those two invariant directions.

The compact spatial theorem of Annex Q applies directly to the bare off-diagonal magnetic clover seed, which is one of the five trace-free magnetic combinations left unchanged by the scalar repair. Its physical renormalized estimate is still missing. In particular the running mixing matrix can mix bare magnetic and temporal components within the

stress and Weyl sectors. The continuum seed Φ_{υ} need not be represented by a pure magnetic cutoff multiplication operator after that mixing. The exact F0 prescription must be retained in using (L.11).

Nor may (L.10) be used before full covariance R0 has been established. A finite hypercubic cutoff symmetry does not provide the boosts, the spectrum comparison, or the nineteen-dimensional continuous orbit. The cutoff magnetic estimate and this continuum reduction are distinct statements with distinct hypotheses.

The result reduces the field-count part of E0 once J0–J3 and the seed estimates hold. It does not supply their ultraviolet cancellation, the periodic-state passage, the phase decomposition, or the required interacting ultraviolet coefficients. E0, R0 and the full existence theorem remain open.

Annex S — Numerical AF stability and logarithmic factorial budgets

Annex S. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue. Identifying the calibrated Gaussian marginal with the Wilson trajectory also requires GC, including the bounded error uniform in depth.

Mathematical consequences

This note proves statements B-04 and F-02. Both are numerical bridge theorems with complete elementary proofs. Neither identifies an actual Yang–Mills coupling recursion, mixing coefficient, or contact coefficient. Those applications remain separate in B-03/B-05 and F-03–F-07.

The first theorem supplies the precise asymptotically free numerical asymptotic required by the commuting-product theorem in Annex P, once its actual recurrence has been established. The second theorem shows that a power of a running logarithm per insertion can be absorbed into a fixed factorial exponent, while retaining a positive cutoff rate. It applies in particular to the new mesoscopic Wilson contact error.

1. A perturbed asymptotically free recurrence

Fix $b > 0$ and $A \geq 0$. Choose $u_* > 0$ such that

$$u_* \leq 1, \qquad Au_* \leq b/2, \qquad \frac{3b}{2}u_* \leq \frac{1}{2}. \tag{B.1}$$

For $A = 0$, the middle restriction is empty. Let $0 < u_0 \leq u_*$ and suppose a real sequence is generated by

$$u_{j+1} = u_j - bu_j^2 + r_j, \qquad |r_j| \leq Au_j^3. \tag{B.2}$$

The remainder may depend on the complete earlier sequence. No autonomous beta function, differentiability, or fixed sign of r_j is assumed. The bound must hold at each step that is generated.

Write

$$v_j = u_j^{-1}, \qquad \kappa = (bu_0)^{-1} \geq 3. \tag{B.3}$$

Theorem 1.1 — stability, asymptotics and summable tails Every generated u_j is positive and at most u_0. Moreover,

$$\boxed{v_0 + \frac{b}{2}j \leq v_j \leq v_0 + 3bj.} \tag{B.4}$$

There is a constant C, depending only on b, A, u_*, such that

$$\boxed{\left|u_j - \frac{1}{b(j+\kappa)}\right| \leq C\,\frac{\log(1 + j/\kappa)}{(j+\kappa)^2}.} \tag{B.5}$$

The right side is zero at $j = 0$, as it should be. In addition,

$$\sum_{j=n}^{\infty} u_j^2 \leq \frac{8}{b^2(n+\kappa)}. \tag{B.6}$$

Finally the sequence

$$\sum_{j=0}^{n-1} u_j - \frac{1}{b}\log\frac{n+\kappa}{\kappa} \tag{B.7}$$

has a finite limit. Its tail is at most

$$C'\,\frac{1 + \log(1 + n/\kappa)}{n+\kappa}, \tag{B.8}$$

with a constant that can be chosen uniformly for $u_0 \leq u_*$.

All conclusions are uniform over the admissible remainders in (B.2).

Proof. Suppose inductively that $0 < u_j \le u_*$, and set

$$d_j = bu_j - r_j/u_j.$$

The assumptions imply

$$\frac{b}{2}u_j \le d_j \le \frac{3b}{2}u_j \le \frac{1}{2}.$$

Consequently $u_{j+1} = u_j(1-d_j)$ lies in $[u_j/2, u_j)$, proving positivity and invariance of the small interval by induction.

Taking reciprocals gives the exact formula

$$v_{j+1} - v_j = \frac{b - r_j/u_j^2}{1 - bu_j + r_j/u_j}. \tag{B.9}$$

The numerator lies between $b/2$ and $3b/2$, and the denominator between $1/2$ and 1. This proves (B.4).

Subtract b in (B.9). Directly,

$$\begin{aligned} v_{j+1} - v_j - b &= \frac{b^2u_j - br_j/u_j - r_j/u_j^2}{1 - bu_j + r_j/u_j}, \\ |v_{j+1} - v_j - b| &\le C_0 u_j, \qquad C_0 = 2(b^2 + A + bAu_*). \end{aligned} \tag{B.10}$$

From (B.4),

$$u_j \le \frac{1}{b(\kappa + j/2)} \le \frac{2}{b(\kappa + j)}. \tag{B.11}$$

A comparison with the integral of $(\kappa + x/2)^{-1}$ gives, for $n \ge 1$,

$$\sum_{j<n} u_j \le \frac{3}{b}\log(1 + n/\kappa). \tag{B.12}$$

For an explicit justification, on each interval $[j, j+1]$ the ratio of the value at j to the minimum value is at most $1 + (2\kappa)^{-1} \le 7/6$. Thus the sum is bounded by $(7/3b)\log(1 + n/(2\kappa))$, which is no larger than the right side of (B.12).

Summing (B.10) now gives

$$|v_n - b(n+\kappa)| \le \frac{3C_0}{b}\log(1 + n/\kappa). \tag{B.13}$$

Also (B.4) implies $v_n \ge \frac{b}{2}(n+\kappa)$. Divide (B.13) by $v_n\, b(n+\kappa)$ to obtain (B.5), for example with $C = 6C_0/b^3$. At $n = 0$ the equality is exact.

Squaring (B.11) and comparing the decreasing reciprocal-square sum with its integral proves

$$\sum_{j=n}^{\infty} u_j^2 \le \frac{4}{b^2}\left[\frac{1}{(n+\kappa)^2} + \frac{1}{n+\kappa}\right] \le \frac{8}{b^2(n+\kappa)},$$

which is (B.6).

For the last assertion, split (B.7) as

$$\sum_{j<n}\left(u_j - \frac{1}{b(j+\kappa)}\right) + \frac{1}{b}\left[\sum_{j<n}\frac{1}{j+\kappa} - \log\frac{n+\kappa}{\kappa}\right]. \tag{B.14}$$

The first series converges absolutely by (B.5). For clarity, if $g(x) = \log(1 + x/\kappa)/(x+\kappa)^2$, then for $x \in [j, j+1]$,

$$g(j) \le (1 + 1/\kappa)^2 g(x) \le (16/9)g(x).$$

Its discrete tail is therefore at most 16/9 times the exact integral

$$\int_n^{\infty} g(x)\,dx = \frac{1 + \log(1 + n/\kappa)}{n+\kappa}.$$

This comparison does not incorrectly assume that g is decreasing near zero. For the second bracket, each difference between $(j+\kappa)^{-1}$ and its integral over $[j, j+1]$ is nonnegative and at most $(j+\kappa)^{-2}$. That series also converges, with tail at most $C/(n+\kappa)$. This proves (B.7)–(B.8). □

1.1 What this closes, and what it does not The theorem verifies the numerical hypothesis used in Annex P from the one-step recurrence (B.2). Its $O(\log j/j^2)$ error and square-summable tail are sufficient for the logarithmic product asymptotics stated there.

It does not prove (B.2) for the Wilson RG map. In particular, it does not infer b from a convention, identify u with a specific multiple of the inverse Wilson coefficient, solve a physical normalization/shooting problem, or establish the admissibility of all RG histories. Those are B-03 and B-05. A coherent family of finite trajectories must be checked there; the present theorem cannot replace that check by unrelated numerical runs.

2. Logarithmic running and factorial growth

Use a dimensionless cutoff $0 < \rho = a/\ell_0 \le 1$, and set

$$\ell(\rho) = 1 + \log(1/\rho). \tag{F.1}$$

Let $m \ge 1$. A norm below may be any nonnegative quantity, including the dual of a joint Schwartz seminorm or a local response norm. Let $\delta > 0$, $p, \gamma, \eta \ge 0$, and $A \ge 1$ be fixed.

Theorem 2.1 — logarithmic loss absorption with a vanishing rate Suppose

$$\|E_{m,\rho}\|_{qm,*} \le A^{m+1}(m!)^p \rho^\delta \ell(\rho)^{\gamma m+\eta}. \tag{F.2}$$

Then for every fixed $0 < \theta < \delta$, a constant A_θ independent of m, ρ satisfies

$$\boxed{\|E_{m,\rho}\|_{qm,*} \le A_\theta^{m+1}(m!)^{p+\gamma}\rho^\theta.} \tag{F.3}$$

There is no increase in the derivative-order parameter q. The factorial exponent $p + \gamma$ is fixed; if an integer exponent is desired, it may be enlarged to its ceiling.

Proof. Put $t = \log(1/\rho) \ge 0$ and $\epsilon = (\delta - \theta)/2 > 0$. The leftover logarithmic factor is

$$e^{-(\delta-\theta)t}(1+t)^{\gamma m+\eta} = \left[e^{-\epsilon t}(1+t)^\eta\right]\left[e^{-\epsilon t}(1+t)^{\gamma m}\right]. \tag{F.4}$$

The first bracket has a finite supremum $K_{\epsilon,\eta}$ independent of m. If $\gamma = 0$, the second bracket is at most one. If $\gamma > 0$, maximizing its logarithm on $t \ge 0$ gives

$$e^{-\epsilon t}(1+t)^{\gamma m} \le e^\epsilon \max\left\{1, \left(\frac{\gamma m}{e\epsilon}\right)^{\gamma m}\right\}. \tag{F.5}$$

The elementary integral estimate

$$\log(m!) = \sum_{j=1}^{m} \log j \ge \int_1^m \log x\, dx \ge m\log m - m$$

implies $(m/e)^m \le m!$. Hence (F.5) is at most

$$e^\epsilon \max\{1, (\gamma/\epsilon)^\gamma\}^m (m!)^\gamma. \tag{F.6}$$

Multiply (F.6) by $K_{\epsilon,\eta}$, insert into (F.2), and absorb fixed constants into A_θ^{m+1}. This proves (F.3). □

The loss in the factorial exponent is governed by the logarithm power per insertion, γ. A fixed extra logarithm power η changes only the exponential constant. Replacing γ by $\gamma/(\delta - \theta)$ in the factorial exponent would be an unnecessary loss; the parameter $\delta - \theta$ belongs in the constant in (F.6).

Corollary 2.2 — logarithmically many controlled errors Use a norm or seminorm satisfying the triangle inequality. Suppose $E_{m,\rho} = \sum_{\ell=1}^{N(\rho)} E_{m,\rho}^{(\ell)}$, each term has (F.2) with the same constants, and

$$N(\rho) \le B\,\ell(\rho)^\sigma \tag{F.7}$$

for fixed $B \ge 1, \sigma \ge 0$. Then (F.3) still holds with exponent $p + \gamma$.

Proof. Sum (F.2), replace η by $\eta + \sigma$, and apply Theorem 2.1. □

The common-constant premise is essential. A factor of the form $C^{mN(\rho)}$ accumulated by an uncontrolled iteration is not bounded by a fixed power of $\ell(\rho)$ per insertion, and this corollary does not remove it.

Corollary 2.3 — geometric ultraviolet tails Let $\rho_j = L^{-j}$, $L > 1$, and let $E_{m,j}$ satisfy (F.2) at ρ_j, uniformly in j. For $J \geq 0$,

$$\boxed{\left\| \sum_{j \geq J} E_{m,j} \right\|_{qm,*} \leq C_\theta^{m+1} (m!)^{p+\gamma} L^{-\theta J}.} \tag{F.8}$$

The series converges in the stated norm.

Proof. Apply (F.3) term by term and use $\sum_{j \geq J} L^{-\theta j} = L^{-\theta J}/(1 - L^{-\theta})$. □

This is an abstract sufficient bound. A marginal remainder with no positive scale gain δ does not satisfy its hypotheses. In particular it does not repair the inverse-running-product counterexample in Annex P.

3. Exact relation to mesoscopic Wilson matching

The mesoscopic matching theorem in the proof of Annex T starts with the original cutoff-field moment estimate

$$\|O_{\alpha,a}(x)\|_{L^r} \leq C a^{-4} r, \qquad r \geq 1, \tag{F.9}$$

on the actual Wilson law. It projects onto retained variables by an actual conditional expectation Q, not a formal coarse action. For mesoscopic blocks of physical size at most $C\sqrt{a}$, its full-product matching error on flat joint tests obeys

$$|D_{m,a}(F)| \leq a^{1/2} C^{m+1} m!\, \nu_{10m}(F) \tag{F.10}$$

in units $\ell_0 = 1$. The proof retains the original a^{-4} budget; it does not claim a new current-scale moment estimate.

Suppose, as an additional application premise, that actual one-field normalization coefficients satisfy

$$\|Z_a\| \leq C \ell(a)^\gamma, \qquad |c_{\alpha,a}| \leq C a^{-4} \ell(a)^\gamma. \tag{F.11}$$

For the finite family

$$O^R_{\alpha,a} = \sum_\beta (Z_a)_{\alpha\beta} O_{\beta,a} - c_{\alpha,a},$$

finite dimensionality and (F.9) give

$$\|O^R_{\alpha,a}\|_{L^r} \leq C' a^{-4} \ell(a)^\gamma r. \tag{F.12}$$

Conditional expectation commutes with the deterministic linear combination and preserves constants. The same mesoscopic proof therefore changes (F.10) only by $\ell(a)^{\gamma m}$ and a new exponential constant. Theorem 2.1 yields, for instance,

$$\boxed{|D^R_{m,a}(F)| \leq a^{1/4} C_\gamma^{m+1} (m!)^{1+\gamma} \nu_{10m}(F).} \tag{F.13}$$

Thus logarithmic running does not itself destroy the all-order shrinking-contact theorem. The estimate is valid uniformly over all source orders with one fixed factorial power.

The proof of (F.11) for the actual desired normalization remains F-05. Contact coefficients of products are not all individual field shifts and remain F-06/F-07. Most importantly, neither (F.10) nor (F.13) bounds separated correlations: away from the shrinking contact set the exact projected fields have the same unknown correlations as the original fields.

4. Application conditions

B-04: complete. The theorem has a specified invariant interval, explicit reciprocal bounds, a uniform logarithmic asymptotic and square-summable tails. Its input is a numerical recurrence; B-03 and B-05 remain open.

F-02: complete. The theorem has fixed all-order constants and exponent, gives a positive cutoff rate, and handles a fixed logarithm power, a power per insertion, finite tensor mixing and logarithmically many common-budget errors. Its actual-field application is conditional on (F.11).

Annex T — Positive-chain tails and exact mesoscopic Wilson matching

Annex T. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Status, scope, and overlap conditions

The following results concern M-01 and M-02. M-01 is an abstract positive-kernel theorem. M-02 is an exact matching theorem on the compact Wilson measure, including all compact configurations. Neither theorem supplies current-scale moment bounds for the conditional means, complete-history decay, or the physical renormalization. The full Yang–Mills existence theorem and the estimates M0/H0 remain open.

The finite-step covariance telescope has a conditional-cumulant interpretation; a shell estimate of order $h^{-8}e^{-cr/h}$ controls its separated contribution. The history reset and its finite source-jet rule are already proved in Annex I. Annex O proves one-link-color matching.

The additions are:

1. a reverse-tail realization for arbitrary positive RG/R chains, with orthogonal innovations, absolute L^2 summability, and the exact infinite-depth terminal term;
2. a matching theorem valid for a growing number of exact eliminations with their true inherited factors, and for direct mesoscopic block elimination. The discrepancy is controlled on joint coincidence-flat tests with constants uniform in the elimination depth, cutoff, volume, and source order. A reflection-compatible block choice preserves the positive-time observable algebra for each fixed support margin.

All length estimates below use the fixed physical reference length as the unit. Thus $0 < a \le 1$. Restoring that length only changes the displayed fixed constants and the corresponding powers of the reference length.

1. M-01: the tail term of a positive RG/R chain

Statement and requirements Let $X_0, X_1, \dots$ be a Markov chain on standard Borel spaces, with source-independent kernels P_k, initial law μ_0, and marginals μ_k. Its state at each step includes every input used by the next operation, including any retained history variables. The operation may integrate variables, add labels, change coordinates, or replace variables by fresh reference variables. It need not be deterministic.

The target is to realize the exact marked transports on a decreasing path-space filtration, prove their L^2 tail limit, and identify the terminal covariance without assuming nested state spaces or discarding replacement noise. The accepted result must retain the tail covariance and state exactly when it is zero.

1.1 The path-space construction The same construction starts directly from the observable interface. Suppose $Q_k(y, dx)$ is a probability kernel satisfying

$$\int \mu_{k+1}(dy) Q_k(y, A) = \mu_k(A).$$

Disintegrate the joint probability $\mu_{k+1}(dy)Q_k(y, dx)$ in the opposite direction as $\mu_k(dx)P_k(x, dy)$. Standard Borel state spaces ensure that this forward kernel exists. Thus every such integral-preserving positive observable kernel, including the exact frozen RG/R kernels, has a forward Markov realization with the required one-step joint law.

Equip the product path space with the iterated-kernel probability law

$$\mathbb{P}(dx_0\, dx_1 \cdots) = \mu_0(dx_0)P_0(x_0, dx_1)P_1(x_1, dx_2)\cdots. \tag{1.1}$$

For a finite chain, extend it by identity kernels after its last state. This does not change any finite-cutoff observable or the terminal term.

The reverse kernel is the conditional expectation

$$Q_k f(x_{k+1}) = \mathbb{E}[f(X_k) \mid X_{k+1} = x_{k+1}]. \tag{1.2}$$

The definition is up to μ_{k+1}-null sets, which do not affect any identity. Put $Q_{0:k} = Q_{k-1} \cdots Q_0$, and $Q_{0:0} = I$.

For $F \in L^2(\mu_0)$, define

$$F_k = Q_{0:k} F, \qquad M_k^F = F_k(X_k). \tag{1.3}$$

Let

$$\mathcal{G}_k = \sigma(X_k, X_{k+1}, \ldots), \qquad \mathcal{G}_\infty = \bigcap_{k \geq 0} \mathcal{G}_k, \tag{1.4}$$

where all sigma fields are completed in the path law. The filtration $\mathcal{G}_k$ decreases. It is the future path filtration, not the generally nonnested family $\sigma(X_k)$.

Theorem 1.1 – reverse-tail realization and orthogonal innovations For every $F \in L^2(\mu_0)$:

1. The transported field is the path-space conditional expectation

$$M_k^F = \mathbb{E}[F(X_0) \mid \mathcal{G}_k]. \tag{1.5}$$

2. In $L^2(\mathbb{P})$,

$$M_k^F \longrightarrow M_\infty^F := \mathbb{E}[F(X_0) \mid \mathcal{G}_\infty]. \tag{1.6}$$

3. Define the innovation

$$\eta_k^F = M_k^F - M_{k+1}^F. \tag{1.7}$$

Then $\mathbb{E}[\eta_k^F \mid \mathcal{G}_{k+1}] = 0$. For $j \neq k$,

$$\mathbb{E}[\overline{\eta_j^F} \eta_k^G] = 0 \tag{1.8}$$

whenever $F, G \in L^2(\mu_0)$. Each innovation is also orthogonal to every $\mathcal{G}_\infty$-measurable L^2 variable.

4. The exact L^2 expansion is

$$F(X_0) = M_\infty^F + \sum_{k \geq 0} \eta_k^F, \tag{1.9}$$

and

$$\sum_{k \geq 0} \|\eta_k^F\|_2^2 = \|F\|_{L^2(\mu_0)}^2 - \|M_\infty^F\|_2^2. \tag{1.10}$$

Proof. First, the reverse-kernel composition equals conditional expectation of $F(X_0)$ given X_k. This follows inductively from the Markov property: conditioning on X_k, the future transition is independent of the past. For a bounded cylinder function $H(X_k, \ldots, X_N)$, its conditional expectation given $X_0, \ldots, X_k$ is a function only of X_k. Hence

$$\mathbb{E}[F(X_0)H] = \mathbb{E}[F_k(X_k)H].$$

A monotone-class argument extends the equality to every bounded $\mathcal{G}_k$-measurable H, proving (1.5).

For completeness, the L^2 convergence can be seen directly through orthogonal projections. Let $V_k = L^2(\mathcal{G}_k)$; these are decreasing closed subspaces. If $l \geq k$, the nesting and the projection identity give

$$\|M_k^F - M_l^F\|_2^2 = \|M_k^F\|_2^2 - \|M_l^F\|_2^2. \tag{1.11}$$

The squared norms decrease to a nonnegative limit, so M_k^F is Cauchy. Its limit M belongs to every V_k, since a tail of the sequence lies in V_k and V_k is closed. One may also choose an almost surely convergent subsequence of M_k^F. Its pointwise limit, defined by the limsup of the real and imaginary parts and set to zero on the null exceptional set, is measurable with respect to every $\mathcal{G}_k$: discarding the finitely many earlier subsequence terms leaves only $\mathcal{G}_k$-measurable variables. Thus M has a $\mathcal{G}_\infty$-measurable version. For every tail-measurable L^2 variable, the projection identity at finite k passes to the limit. It follows that M is the conditional expectation in (1.6).

Conditioning (1.7) on $\mathcal{G}_{k+1}$ gives zero by the nested projection identity. If $j < k$, then η_k^G is $\mathcal{G}_{j+1}$-measurable, so conditioning η_j^F on that sigma field proves (1.8). The same argument handles a tail-measurable variable. The finite telescope and (1.6) prove (1.9); orthogonality proves (1.10). □

The completed tail sigma field in this theorem is the decreasing intersection in the path probability space. No assertion that it equals an uncompleted coordinate tail sigma field is needed.

Corollary 1.2 – absolutely summable covariance and the exact tail term Use the sesquilinear convention

$$\mathrm{Cov}(F,G) = \mathbb{E}[\overline{F - \mathbb{E}F}(G - \mathbb{E}G)].$$

Then

$$\boxed{\mathrm{Cov}_{\mu_0}(F,G) = \sum_{k\geq 0} \mathbb{E}[\overline{\eta_k^F}\eta_k^G] + \mathrm{Cov}(M_\infty^F, M_\infty^G).} \tag{1.12}$$

The series converges absolutely, with

$$\begin{aligned}\sum_{k\geq 0} \left|\mathbb{E}[\overline{\eta_k^F}\eta_k^G]\right| &\leq \left(\sum_k \|\eta_k^F\|_2^2\right)^{1/2} \left(\sum_k \|\eta_k^G\|_2^2\right)^{1/2} \\ &\leq \sqrt{\mathrm{Var}_{\mu_0}(F)\,\mathrm{Var}_{\mu_0}(G)}.\end{aligned} \tag{1.13}$$

The individual term is exactly the integrated one-step conditional covariance:

$$\mathbb{E}[\overline{\eta_k^F}\eta_k^G] = \int \left[Q_k(\overline{F_k}G_k) - \overline{Q_k F_k}\, Q_k G_k\right] d\mu_{k+1}. \tag{1.14}$$

For a finite list of fields the matrix of these terms is positive semidefinite at every scale.

Proof. Apply the orthogonal expansion to the centered fields. Cauchy–Schwarz first in L^2, then in the scale index, proves (1.13). Expanding (1.7), conditioning on X_{k+1}, and using (1.2) proves (1.14). A finite linear combination of innovations has nonnegative squared norm, giving the matrix assertion. □

There is also the stronger unweighted L^1 estimate. Write

$$D_k(F,G) = Q_k(\overline{F}_k G_k) - \overline{Q_k F_k} Q_k G_k.$$

Then

$$\boxed{\begin{aligned}\sum_{k\geq 0} \int |D_k(F,G)|\, d\mu_{k+1} \leq \left[\mathrm{Var}_{\mu_0}(F) - \mathrm{Var}(M_\infty^F)\right]^{1/2} \\ \cdot \left[\mathrm{Var}_{\mu_0}(G) - \mathrm{Var}(M_\infty^G)\right]^{1/2}.\end{aligned}} \tag{1.14a}$$

To prove it, use conditional Cauchy–Schwarz:

$$|D_k(F,G)| \leq \sqrt{D_k(F,F) D_k(G,G)}.$$

Both diagonal terms are nonnegative and their integrals are $\|\eta_k^F\|_2^2$, $\|\eta_k^G\|_2^2$, respectively. Cauchy–Schwarz first in the output measure and then in the scale index, followed by (1.10), proves (1.14a).

This is genuine absolute L^1 summability at each fixed cutoff. It contains no spatial decay weight and its right side is not automatically uniform in the ultraviolet cutoff.

Corollary 1.3 – the precise condition for discarding the terminal term For a specified class $\mathcal{A} \subset L^2(\mu_0)$, the terminal covariance in (1.12) is zero for every pair $F, G \in \mathcal{A}$ if and only if

$$\mathbb{E}[F(X_0) \mid \mathcal{G}_\infty] = \mathbb{E}F \quad \text{for every } F \in \mathcal{A}. \tag{1.15}$$

Triviality of $\mathcal{G}_\infty$ is sufficient, but is stronger than needed: it is enough that this tail carries no information about the chosen initial observable class.

Proof. The condition immediately makes every terminal covariance zero. Conversely take $G = F$. Zero terminal variance says exactly that the conditional expectation is almost surely constant; its constant is its preserved mean. □

For the identity chain $X_k = X_0$, every innovation is zero and the entire original covariance is the tail term. Therefore arbitrarily many positive steps do not, by themselves, eliminate the terminal contribution.

1.2 Consequence and remaining spatial input For finitely many scales, (1.12) reduces to the already-proved finite telescope. Its additional content is the tail realization and the absolute unweighted L^1 summability of the conditional covariance kernels, valid with replacement noise and arbitrary positive history resets included.

Suppose the exact renormalized transported fields additionally satisfy the localized shell and terminal bounds M2a–M2b of the earlier reduction:

$$\int |D_{k,n}(x,y)|\, d\mu_{k+1,n} \le B h_k^{-8}(1+\log(\ell_0/h_k))^b e^{-c|x-y|/h_k}, \tag{1.16}$$

and a terminal covariance bound of order $B\ell_0^{-8}$. The previously proved geometric scale sum then gives the separated $|x-y|^{-8}$ bound with the corresponding logarithm.

The new L^2 identity does not prove (1.16). At a microscopic cutoff the available bound on a dimension-four insertion still carries a^{-4}, so its squared L^2 norm can carry a^{-8}. Absolute summability at each fixed cutoff does not make its total uniform as $a \to 0$, and no spatial distance appears in (1.13). In the stopped physical-scale construction the identity extension retains exactly the coarse covariance that must be bounded.

2. M-02: exact elimination with the inherited factors retained

Statement and requirements Start from the actual compact $SU(N)$ Wilson law and the explicit dimension-four field family with the already-proved microscopic moment budget. Eliminate local variables through their actual conditional laws. The target is an all-order matching estimate for repeated eliminations, with an explicit locality radius and fixed q, p in the joint test norm.

The statement requires: exact formulas for the inherited marginal factors; no assumption that they remain Wilson plaquette factors; a support proof for the full-product discrepancy; constants independent of the number of layers in the stated range; and a concrete admissible geometry. The renormalized coefficient budget is not to be silently supplied by the word "field."

2.1 Finite positive factor laws Consider a finite collection $\mathcal{V}_r$ of remaining compact link variables. Suppose its actual marginal has density

$$\mu_r(dU) = Z_r^{-1} \prod_{A\in\mathcal{F}_r} \psi_A(U_A) \prod_{l\in\mathcal{V}_r} dU_l, \qquad \psi_A > 0, \tag{2.1}$$

where the ψ_A are continuous on their compact domains. Multiplicity of factors with the same scope is allowed. This class includes the original Wilson law and all marginals constructed below.

Choose $\Gamma_r \subset \mathcal{V}_r$ such that

$$|A \cap \Gamma_r| \le 1 \quad \text{for every } A \in \mathcal{F}_r. \tag{2.2}$$

For $l \in \Gamma_r$, set

$$N_r(l) = \bigcup_{\substack{A\in\mathcal{F}_r \\ l\in A}} (A \setminus \{l\}), \qquad \Psi_l(U_{N_r(l)}) = \int \prod_{A\ni l} \psi_A(U_A)\, dU_l. \tag{2.3}$$

An isolated eliminated variable contributes the constant factor one.

Lemma 2.1 – exact inherited factors and locality recurrence The true marginal on $\mathcal{V}_{r+1} = \mathcal{V}_r \setminus \Gamma_r$ has exactly the factors

$$\{\psi_A : A \cap \Gamma_r = \varnothing\} \quad \text{and} \quad \{\Psi_l : l \in \Gamma_r\}. \tag{2.4}$$

Its conditional elimination kernel is a product over $l \in \Gamma_r$, with the exact normalized local factors in (2.3).

If every old factor scope has torus diameter at most $d_r a$, then every new factor scope has diameter at most

$$d_{r+1}a, \qquad d_{r+1} \le 2d_r. \tag{2.5}$$

If a mark is supported within radius $R_r a$ of its original anchor, its conditional expectation is supported within radius $R_{r+1}a$, where

$$R_{r+1} \le R_r + d_r. \tag{2.6}$$

Consequently one may take

$$d_r = 2^r d_0, \qquad R_r = R_0 + d_0(2^r - 1). \tag{2.7}$$

Proof. Because of (2.2), every affected factor is assigned to a unique eliminated variable. The conditional density is therefore the product of its one-variable factors and the integrals factor. This gives (2.3)–(2.4) exactly. Every new factor is finite, strictly positive and continuous by compactness and integration.

Two variables in $N_r(l)$ lie in old scopes that both contain l, so their distance is at most $2d_r a$, proving (2.5). A mark only uses the elimination kernels of the variables in its own current support. Each such kernel depends on retained variables at distance at most $d_r a$ from the eliminated variable. This proves (2.6) and then (2.7). □

There is no uniform lower bound on the local denominator in this proof, and none is needed for the later L^p estimate. Conditional contraction is with respect to the actual joint law.

Admissible nonempty layers always exist. Form the graph whose vertices are remaining links and whose edges join links in a common factor scope. Its degree is at most the number of lattice links in a radius-$d_r a$ ball, minus one, hence at most $C_d(1 + d_r)^4$, independently of the torus volume. A greedy proper coloring has at most one more than this many colors. Each color class satisfies (2.2); the largest has at least the reciprocal fraction of all remaining links. This proves existence of a layer at each step, not a fixed positive elimination fraction uniformly in the number of steps.

2.2 The full-moment discrepancy Let the original finite family satisfy

$$\|O_{\alpha,a}(x)\|_{L^p(\mu_0)} \le C_0 a^{-4} p, \qquad p \ge 1, \tag{2.8}$$

and have support radius $R_0 a$. The explicit field budget is that of Annex O, Theorem 6.2:

$$O_{\alpha,a} = \beta \tilde{q}_{\alpha,a} - \mathbb{E}[\beta \tilde{q}_{\alpha,a}] \tag{2.9}$$

for the scalar-repaired 21-component family. A different field normalization must satisfy its own version of (2.8).

Let Q_r be the true conditional expectation eliminating Γ_r, and put

$$O^{(r)}_{\alpha,a} = Q_{r-1} \cdots Q_0 O_{\alpha,a}. \tag{2.10}$$

The marginal sigma fields are nested for these deterministic eliminations, so (2.10) is also the direct conditional expectation of the original mark given all variables remaining after r steps. In particular,

$$\|O^{(r)}_{\alpha,a}(x)\|_{L^p(\mu_r)} \le C_0 a^{-4} p \tag{2.11}$$

with no loss depending on r.

Define

$$D^{[r]}_{m,a}(x) = \mathbb{E}_{\mu_0} \prod_{i=1}^{m} O_{\alpha_i,a}(x_i) - \mathbb{E}_{\mu_r} \prod_{i=1}^{m} O^{(r)}_{\alpha_i,a}(x_i). \tag{2.12}$$

Theorem 2.2 – repeated exact matching in a shrinking strip For every finite r, every $m \ge 2$, and every field-label tuple,

$$\boxed{|D^{[r]}_{m,a}(x)| \le C^{m+1} m!\, a^{-4m} 1_{\{\min_{i<j} d_T(x_i,x_j) \le 2R_r a\}}.} \tag{2.13}$$

The constant is independent of r, the volume, a, and m. Moreover $D^{[r]}_{1,a} = 0$.

Suppose

$$R_{r(a)} a \le R_* a^{\alpha}, \qquad 0 < \alpha \le 1, \tag{2.14}$$

with R_* fixed. For any fixed $\delta > 0$, set

$$q = \max\left\{5, \left\lceil \frac{4+\delta}{\alpha} \right\rceil + 1\right\}. \tag{2.15}$$

Use the joint Schwartz seminorm

$$\nu_s(F) = \max_{|\gamma| \le s} \sup_{X \in \mathbb{R}^{4m}} (1 + |X|^2)^{s/2} |\partial^\gamma F(X)|.$$

Coincidence flatness means that every derivative vanishes on each pair diagonal $x_i = x_j$. Then on coincidence-flat joint Schwartz tests, embedded in a central nonwrapping torus patch,

$$\boxed{\left| a^{4m} \sum_x D^{[r(a)]}_{m,a}(x) F(x) \prod_i \zeta_T(x_i) \right| \le a^{\delta} C^{m+1}_{\alpha,\delta} m!\, \nu_{qm}(F).} \tag{2.16}$$

Here $0 \le \zeta_T \le 1$ is supported in a central cube of side at most $T/2$, and may equal one on expanding compact subsets. The result holds without ζ_T for compact tests in the same nonwrapping patch. The constants can depend on R_* and on the fixed original stencil. The torus side is assumed larger than a fixed multiple of these radii.

Proof of support. At one step, if the current supports of the marks share no eliminated variable, their conditional expectations factor because the elimination kernel is a product. Hence the full-moment discrepancy at that step vanishes unless two current marked stencils overlap on an eliminated variable. Their anchors must then be at distance at most $2R_k a$. Telescoping the full-moment expectations for $0 \le k < r$, and using $R_k \le R_r$, proves the support assertion. This use of the telescope keeps every generated product response.

Proof of magnitude without a depth loss. Estimate (2.12) directly at its two endpoints, rather than summing the separate magnitude bounds:

$$\begin{aligned} |D^{[r]}_{m,a}| &\le \prod_i \|O_{\alpha_i,a}\|_m + \prod_i \|O^{(r)}_{\alpha_i,a}\|_m \\ &\le 2C_0^m m^m a^{-4m} \le C^{m+1} m! a^{-4m}. \end{aligned} \tag{2.17}$$

Thus no factor r appears, even if $r = r(a) \to \infty$. The one-point identity is preservation of each mean.

Proof on joint tests. At a contributing configuration choose one pair with separation at most $2R_* a^{\alpha}$. Move only the four-vector coordinate of one member of that pair to the other. Put

$$N_m = \left\lceil \frac{4m + \delta}{\alpha} \right\rceil. \tag{2.18}$$

Every Taylor coefficient of lower order is zero by coincidence flatness. The integral Taylor remainder and $N_m \le qm$ give

$$|F(X)| \le C^{m+1}_{\alpha,\delta} a^{4m+\delta} (1 + |X|^2)^{-qm/2} \nu_{qm}(F) \tag{2.19}$$

on the strip. The displacement is in a single four-dimensional coordinate; its directional multi-index sum costs a constant to the power N_m, rather than an additional m^{N_m}. The factorial in the Taylor remainder only improves the bound. The intermediate points are at a fixed bounded distance from X, so replacing their Schwartz weight by the one at X costs C^{qm}.

Since $q > 4$,

$$a^{4m} \sum_x (1 + \textstyle\sum_i |x_i|^2)^{-qm/2} \le \left[a^4 \sum_{x \in a\mathbb{Z}^4} (1 + |x|^2)^{-q/2} \right]^m \le C_q^m. \tag{2.20}$$

Combining (2.13), (2.19), and (2.20) proves (2.16). No derivatives of ζ_T enter. □

Corollary 2.3 – logarithmically many admissible layers With the exact inherited factor recurrence (2.7), any choice

$$r(a) \le (1 - \alpha) \log_2(a^{-1}) \tag{2.21}$$

satisfies (2.14), after increasing the fixed radius constant. In particular, for

$$r(a) \le \tfrac{1}{2} \log_2(a^{-1})$$

one obtains

$$\boxed{|D^{[r(a)]}_{m,a}(F)| \le a^{1/2} C^{m+1} m!\, \nu_{10m}(F).} \tag{2.22}$$

This is a growing-depth result for exact microscopic eliminations. It does not assert that the sequence performs a complete block RG integration or removes a fixed fraction of all links at every stage. In particular, (2.21) keeps the total support radius shrinking to zero, rather than reaching the fixed physical scale ℓ_0.

3. A concrete mesoscopic block elimination on the Wilson law

The following geometry gives an explicit substantial elimination rather than only existence of single-link color classes.

Let the periodic fine lattice have n sites on every axis. Choose an integer $b \geq 2$ dividing n. Retain every link with at least one endpoint on one of the coordinate planes

$$x_\mu = kb, \qquad k \in \mathbb{Z}, \quad \mu = 1, \dots, 4, \tag{3.1}$$

in lattice units. Eliminate all other links. They split into the interiors of the b-side coordinate blocks.

Lemma 3.1 – exact block conditional independence Conditional on the retained links, the eliminated variables in different blocks are independent under the actual compact Wilson law. The factor inside a block is its complete compact conditional integral with the specified retained boundary data.

Proof. An elementary plaquette has coordinate width at most one fine lattice unit in each direction. It cannot contain an eliminated interior link from each of two different blocks: two strictly interior coordinate levels on opposite sides of a boundary plane are separated by at least two lattice units. Consequently every Wilson plaquette factor meets eliminated links in at most one block. Group the plaquette factors according to that block and integrate product Haar measure. The full conditional law factors exactly. □

An initial radius-$R_0 a$ mark may meet several blocks. Its projected conditional expectation depends only on the retained links of those blocks and the original retained stencil. Its support radius is at most

$$R_0 a + 2ba + d_0 a, \tag{3.2}$$

for a fixed plaquette constant d_0.

Theorem 3.2 – full-Wilson mesoscopic matching Let Q_b be the exact conditional expectation over all eliminated block interiors and $\widehat{O}_{\alpha,a} = Q_b O_{\alpha,a}$. If

$$ba \leq C_* a^\alpha, \qquad 0 < \alpha \leq 1, \tag{3.3}$$

then the full-moment difference between the original fields and the $\widehat{O}_{\alpha,a}$ obeys (2.16), with the same q in (2.15) and the same factorial exponent 1.

For $\alpha = \delta = \frac{1}{2}$,

$$\boxed{|S_{m,a}(F) - \hat{S}_{m,a}(F)| \leq a^{1/2} C^{m+1} m!\, \nu_{10m}(F).} \tag{3.4}$$

All 21 field components and every fixed $SU(N)$ in the Wilson reserve are included.

Proof. If the sets of eliminated blocks met by distinct original marks are pairwise disjoint, block conditional independence gives equality of the conditional product and the product of conditional means. Otherwise two marks meet the same eliminated block. Their anchors are at distance at most

$$2R_0 a + 2ba.$$

Thus the difference is supported in a strip of radius Ca^α. Conditional contraction proves its endpoint magnitude bound (2.17) without a factor depending on the number of eliminated blocks or on the number of variables in a block. The proof of (2.16) now applies unchanged. □

On the cofinal family $a = L^{-K}$, $n = 2L^{K+M}$, choose

$$b = L^{\lfloor K/2 \rfloor}. \tag{3.5}$$

It divides n, and $ba \leq \sqrt{a}$, so (3.4) applies.

The retained set is sparse in the following exact quantitative sense. There are at most

$$n^4[1 - (1 - 1/b)^4] \leq 4n^4/b \tag{3.6}$$

vertices on the boundary planes. At most eight positively oriented or incoming links touch each vertex, so the number of retained links is at most $32n^4/b$. Since the full lattice has $4n^4$ links, the retained fraction is at most $8/b$. Thus for (3.5) the fraction tends to zero. This count does not assert any weak-coupling or locality estimate on the strength of the induced boundary factors.

4. Reflection compatibility of the block projection

Assume the original Wilson torus has even side length in fine lattice sites and carries the standard site-reflection-positive Wilson measure. The cofinal geometry $n = 2L^{K+M}$ has this property.

Choose the plane family (3.1) with the site-reflection plane at time zero among its boundary planes. The retained set is invariant under that reflection. Let Θ include the original Wilson reflection and the specified action on the field labels.

Corollary 4.1 – positive-time algebra for a fixed margin The block conditional expectation commutes with reflection:

$$Q_b(\Theta F) = \Theta(Q_b F). \tag{4.1}$$

For sources whose original anchors have a fixed positive-time margin $\epsilon > 0$ from both reflection planes, the projected fields remain functions of positive-time links when the support bound (3.2) is smaller than ϵ. Their polynomial Gram forms are therefore reflection positive under the original Wilson measure, and hence under the retained marginal.

Proof. Reflection preserves the original law and the retained sigma field. The defining integral identity for conditional expectation, tested against retained measurable functions and then reflected, proves (4.1). The exact support estimate (3.2) places the projected fields in the same positive-time observable algebra for small enough a. Every finite polynomial in them is then an allowed positive-time observable for the original Wilson reflection inequality. Its expectation under the retained marginal equals its expectation under the original measure. This proves the claim. □

The conclusion is for each fixed positive support margin, as needed in passage to the continuum OS algebra. The selected block grid does not preserve every fine-lattice translation. A translation-covariant continuum limit still requires the common distribution bounds and the identification with the original hierarchy. Nor is reflection positivity asserted for arbitrary elimination sequences in Section 2.

5. What these two theorems do and do not establish

M-01 identifies exactly where a terminal covariance survives an unlimited sequence of positive replacements. It removes an ambiguity about filtrations and gives a useful cutoff-level L^2 accounting identity. It does not remove the ultraviolet loss or prove spatial decay.

M-02 establishes that the original full hierarchy on coincidence-flat tests may be replaced, with a quantitative vanishing error, by the hierarchy of conditional means on an explicitly sparse mesoscopic skeleton. The effective boundary measure is the exact Wilson marginal, and all its induced factors are retained. This is an actual-model matching theorem, beyond the one-link-color layer of Annex O.

The following statements remain false as deductions from M-02 alone:

- that the conditional means now obey an h^{-4} budget with $h = ba$; their proved budget is still a^{-4};
- that their retained marginal is a Wilson measure at a new coupling;
- that the resulting effective action obeys the complete Balaban activity norms, with all current-coordinate domains admitted;
- that arbitrary fine-lattice running factors preserve (3.4) without their own coefficient budget;
- that a uniform Schwinger bound follows from the vanishing matching error when neither of the two compared hierarchies has yet been bounded uniformly.

These are the remaining analytic problems, not consequences of positivity or exact conditioning.

Annex U — Joint-distribution hierarchy and connected-shell assembly

Annex U. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Scope

This note closes two interface statements, not the actual interacting Schwinger-hierarchy estimate.

- S-01 — proved abstract theorem: connected distributions with a fixed linear Schwartz order and a fixed factorial exponent give full distributions on arbitrary joint coincidence-flat tests with the same order and exponent. The construction is independent of extensions across coincidence diagonals. The converse Möbius transformation, a quantitative stability estimate, and weak sequential continuity have the same property.
- S-02 — proved conditional assembly theorem: an exact connected-shell representation, a summable physical tree-kernel majorant, a terminal estimate, and a separately normalized one-point function give the full OS growth bound and a uniform ultraviolet-tail estimate. They also give convergence of the full hierarchy once the finitely resolved connected shells and the terminal terms converge.

The S-02 shell majorant and exact representation have not been proved for the complete normalized Wilson/RG histories. Thus S-02 is a completed implication whose principal Yang–Mills premises remain open. It is not a completed actual-model S0 estimate.

The derivations below use standard Hahn–Banach extension and finite regular measure representation of bounded functionals on C_0. Their use is made explicit, including the norm budgets; no unquantified nuclearity argument or assertion about tensor-product tests is inserted.

1. Test spaces and the statement boundary

For $m \geq 1$, put

$$X_m = (\mathbb{R}^4)^m, \qquad \Delta_m = \bigcup_{1\leq i<j\leq m} \{x_i = x_j\}. \tag{S.1}$$

Let $\mathcal{S}_0(X_m)$ be the closed subspace of Schwartz functions whose derivatives of every order vanish on Δ_m. Set $\mathcal{S}_0(X_1) = \mathcal{S}(\mathbb{R}^4)$. The required seminorm is

$$\nu_r(F) = \max_{|\alpha|\leq r} \sup_{X\in X_m} (1 + |X|^2)^{r/2} |\partial^\alpha F(X)|, \qquad r \in \mathbb{N}_0. \tag{S.2}$$

All weights use the ordinary Euclidean norm on all $4m$ coordinates. The space dimension is allowed to grow with m; the estimates below record that growth rather than holding dimension artificially fixed.

A component-label tuple from a fixed finite field family is understood in every distribution. The estimates are componentwise and uniform in those labels. Summing the labels changes only an exponential-in-m constant.

For a functional T on $\mathcal{S}_0(X_m)$, write

$$\|T\|_{r,*} = \sup_{F\in\mathcal{S}_0(X_m), F\neq 0} \frac{|T(F)|}{\nu_r(F)}. \tag{S.3}$$

A bound in this norm is stronger than merely having an unspecified Schwartz-continuity bound at each fixed order.

If π is a partition of $\{1, \dots, m\}$, denote its coordinate blocks by B. A joint function flat on every diagonal internal to each B belongs to a larger space $\mathcal{S}_\pi$. In particular,

$$\mathcal{S}_0(X_m) \subset \mathcal{S}_\pi. \tag{S.4}$$

Cross-block flatness will not be needed to define a block tensor product.

2. An explicit tensor-product construction with no order loss

Lemma 2.1 — a bounded measure representation Let T be a functional on $\mathcal{S}_0(\mathbb{R}^d)$ obeying

$$|T(f)| \le B\,\nu_r(f) \tag{S.5}$$

for an integer $r \ge 0$. Here the notation $\mathcal{S}_0$ can mean any fixed closed subspace of Schwartz space, including the internal coincidence-flat space for one coordinate block. There exist finite complex measures μ_α, $|\alpha| \le r$, such that

$$\widetilde{T}(f) = \sum_{|\alpha|\le r} \int_{\mathbb{R}^d} (1+|x|^2)^{r/2} \partial^\alpha f(x)\, d\mu_\alpha(x), \tag{S.6}$$

defines a tempered-distribution extension of T, and

$$\sum_{|\alpha|\le r} \|\mu_\alpha\|_{\mathrm{TV}} \le B. \tag{S.7}$$

In particular, no factor equal to the number of derivatives occurs.

Proof. Let $I_r = \{\alpha : |\alpha| \le r\}$, and map f into

$$J_r f(x,\alpha) = (1+|x|^2)^{r/2} \partial^\alpha f(x)$$

in $C_0(\mathbb{R}^d \times I_r)$, with its supremum norm. Schwartz decay makes every component vanish at infinity, and $\|J_r f\|_\infty = \nu_r(f)$. The $\alpha = 0$ component makes this map injective. Thus $J_r f \mapsto T(f)$ is a bounded functional of norm at most B on the image of the stated test subspace.

Hahn–Banach extends it to $C_0(\mathbb{R}^d \times I_r)$ with the same norm. Its finite complex measure has total variation at most B. Restricting that measure to the finitely many copies indexed by α gives (S.6)–(S.7). For an arbitrary Schwartz function the right side is absolutely convergent and bounded by $B\nu_r(f)$, so it is a tempered distribution extending T. □

The choice of extension need not be unique. The next lemma proves that its relevant tensor products on flat tests are unique.

Lemma 2.2 — joint-test tensor estimate and extension independence For the blocks $B \in \pi$, let T_B be defined on the internal coincidence-flat Schwartz space of its variables, and suppose

$$|T_B(f)| \le b_B \nu_{r_B}(f), \qquad r_B \in \mathbb{N}_0. \tag{S.8}$$

For every $F \in \mathcal{S}_\pi$, the block product

$$\left(\bigotimes_{B\in\pi} T_B\right)(F) \tag{S.9}$$

is well-defined, independent of the extensions in Lemma 2.1, and satisfies

$$\boxed{\left|\left(\bigotimes_{B\in\pi} T_B\right)(F)\right| \le \left(\prod_{B\in\pi} b_B\right) \nu_{\sum_B r_B}(F).} \tag{S.10}$$

It is associative, covariant under permutations of coordinate blocks, and multilinear in the T_B's.

Proof. Choose the measure representations (S.6) for every block. Define (S.9) by the finite sum of the integrals

$$\int \left[\prod_{B\in\pi} (1+|x_B|^2)^{r_B/2}\right] \left[\prod_{B\in\pi} \partial_{x_B}^{\alpha_B}\right] F(X) \prod_{B\in\pi} d\mu_{B,\alpha_B}(x_B). \tag{S.11}$$

The order of the derivative is at most $\sum_B r_B$, while

$$\prod_{B\in\pi} (1+|x_B|^2)^{r_B/2} \le (1+|X|^2)^{(\sum_B r_B)/2}. \tag{S.12}$$

Consequently the integrand in (S.11) is bounded by $\nu_{\sum_B r_B}(F)$. Summing the products of measure variations and using (S.7) gives (S.10). This also proves the absolute integrability required to change the integration order.

Now change the extension of one block B_0. Hold the other block coordinates fixed, as well as their derivatives in (S.11). The resulting function of x_{B_0} is Schwartz and flat on all its internal coincidence diagonals: derivatives in other coordinates preserve that property. The difference of the two extensions is zero on that function. Integrating first over x_{B_0}, and then over the other variables, gives zero. Repeating this for each block proves independence. Absolute convergence and ordinary product measures give associativity, permutation covariance and multilinearity. □

This is a construction for an arbitrary joint F. It is not an estimate first stated only for a simple tensor and then promoted without proof to the joint test space. It also works when the connected distributions are supplied only on $\mathcal{S}_0$; there is no requirement that a chosen extension across diagonals be physically preferred.

3. S-01: connected and full hierarchies have the same OS budget

Theorem 3.1 — quantitative partition and Möbius passage Let $q \in \mathbb{N}_0$, $p \geq 1$, and $A \geq 1$. For every $k \geq 1$, suppose the connected distribution T_k is defined on $\mathcal{S}_0(X_k)$ and

$$|T_k(f)| \leq A^{k+1}(k!)^p \nu_{qk}(f). \tag{S.13}$$

Define $S_0 = 1$, and on $\mathcal{S}_0(X_m)$ set

$$S_m = \sum_{\pi \in \Pi_m} \bigotimes_{B \in \pi} T_{|B|}. \tag{S.14}$$

The coordinate and field labels in each block are the inherited ones. Then (S.14) is well-defined, independent of every diagonal extension, and

$$\boxed{|S_m(F)| \leq C^{m+1}(m!)^p \nu_{qm}(F).} \tag{S.15}$$

One can take $C = 2A^2$. There is no increase in q or p.

Conversely, if full distributions S_k have a bound of the form (S.13), then

$$T_m = \sum_{\pi \in \Pi_m} (-1)^{|\pi|-1}(|\pi|-1)! \bigotimes_{B \in \pi} S_{|B|} \tag{S.16}$$

has the same type of bound, with the same q, p and a changed exponential constant. The two operations are inverses on the flat test spaces.

Proof. Lemma 2.2 applies to each partition with $r_B = q|B|$, so its total order is exactly qm. Its contribution is bounded by

$$A^{m+|\pi|} \prod_{B \in \pi} (|B|!)^p \, \nu_{qm}(F). \tag{S.17}$$

Since $A \geq 1$ and $|\pi| \leq m$, the power of A is at most A^{2m}.

For $p \geq 1$,

$$\prod_{B \in \pi} (|B|!)^p \leq (m!)^{p-1} \prod_{B \in \pi} |B|!. \tag{S.18}$$

Ordering r blocks and summing over their positive sizes gives

$$\sum_{\substack{\pi \in \Pi_m \\ |\pi| = r}} \prod_{B \in \pi} |B|! = \frac{m!}{r!} \binom{m-1}{r-1}. \tag{S.19}$$

Thus

$$\sum_{\pi \in \Pi_m} \prod_{B \in \pi} (|B|!)^p \leq 2^{m-1}(m!)^p. \tag{S.20}$$

Equations (S.17)–(S.20) prove (S.15).

For the inverse formula, the extra Möbius coefficient gives

$$\begin{aligned}\sum_{\pi\in\Pi_m}(|\pi|-1)!\prod_{B\in\pi}(|B|!)^p &\le (m!)^p\sum_{r=1}^{m}\frac{1}{r}\binom{m-1}{r-1}\\ &= (m!)^p\,\frac{2^m-1}{m}\le 2^m(m!)^p.\end{aligned}\tag{S.21}$$

Lemma 2.2 gives the same test order. Finally, multilinearity and associativity make the usual finite partition cancellation proving the moment–cumulant inversion valid here. Each computation involves only finitely many partitions at a fixed m; no exponential-source series is assumed to converge. □

If a starting estimate has $0 \le p < 1$, it may first be enlarged to $p = 1$. No assertion of preserving an exponent less than one is needed for the OS application.

Corollary 3.2 — quantitative stability and contact invariance Suppose $T_k, \widehat{T}_k$ satisfy (S.13) with the same A, q, p, and, for a number $\varepsilon \ge 0$,

$$\|(T_k-\widehat{T}_k)\|_{qk,*}\le \varepsilon A^{k+1}(k!)^p,\qquad k\ge 1.\tag{S.22}$$

Their full hierarchies satisfy

$$\boxed{|(S_m-\hat{S}_m)(F)|\le \varepsilon C^{m+1}(m!)^p\nu_{qm}(F),}\tag{S.23}$$

with C independent of m, ε. The analogous statement holds for the inverse Möbius operation.

Proof. Telescope a product over the blocks of one partition. There are $|\pi| \le m$ changed-factor terms, each with the same product budget as (S.17) and the extra factor ε. Multiply (S.20) by m and absorb $m \le 2^m$ into C^{m+1}. For the inverse operation the product-telescope factor changes $(|\pi|-1)!$ to $|\pi|!$; (S.19) then gives the bound $(m!)^p 2^{m-1}$ directly. □

If two connected extensions differ only by distributions annihilating the flat test spaces, their restricted T_k's coincide. Their full restrictions therefore coincide as well. In particular, finite-order partial-diagonal contact changes do not create a hidden separated ambiguity through the partition operation. This does not choose or bound a contact extension on non-flat tests.

Proposition 3.3 — weak sequential continuity on the flat spaces Let $T_{k,n}$ satisfy a common bound (S.13), and suppose

$$T_{k,n}(f)\longrightarrow T_k(f)\quad\text{for every } k,\ f\in\mathcal{S}_0(X_k).\tag{S.24}$$

Then their full hierarchies satisfy

$$S_{m,n}(F)\longrightarrow S_m(F)\quad\text{for every } m,\ F\in\mathcal{S}_0(X_m),\tag{S.25}$$

where S is formed from the limits by (S.14). The same assertion holds for the inverse Möbius transformation.

Proof. Fix m, F, and any subsequence of n. For the finitely many orders and component tuples needed by the partitions of $\{1, \dots, m\}$, choose the measure extensions in Lemma 2.1. Their variations have uniform bounds.

For completeness, their needed compactness follows by diagonal extraction of integrals against a countable dense subset of each $C_0(\mathbb{R}^{4k}\times I_{qk})$. The common norm bound extends the limit to all of C_0; measure representation gives a finite measure limit with no larger variation. A further subsequence therefore has simultaneous weak-star measure limits for these finitely many blocks. Each limit represents an extension of T_k, by (S.24).

For fixed multi-indices, the weighted derivative of F in (S.11) belongs to C_0 on the product coordinate space. Products of the bounded measures converge against that function: on a large compact box approximate a continuous function uniformly by finite sums of products of one-block C_0 functions, and then use its vanishing at infinity and the common variation bounds to control the rest. Such finite-sum approximation can be made directly with a finite rectangular partition of unity.

It follows that along this further subsequence every term of (S.14) converges to the product of the limiting extensions. Lemma 2.2 makes that answer independent of which extensions or further subsequence were selected. Every subsequence thus has a further subsequence with the same numerical limit $S_m(F)$, proving (S.25). The finite sum (S.16) is handled identically. □

The proposition supplies the continuity that cannot be obtained by merely invoking pointwise convergence on simple tensor tests.

4. S-02: connected shells and a terminal law assemble the hierarchy

4.1 Precise premises Set the fixed physical stopping scale to one. Let

$$h_j = L^{-j}, \qquad L > 1, \qquad K_n \longrightarrow \infty, \qquad 0 < a_n \le h_{K_n}. \tag{S.26}$$

All torus tests use a nonwrapping interior embedding as in Annex E. Equivalently, the assumptions below may be made on the ordinary Euclidean grid $a_n\mathbb{Z}^4$, with compactly supported embedding weights bounded by one. The metric in the tree bound below is the ordinary physical Euclidean metric on the relevant supports, not an unjustified replacement of periodic distance.

Assume an actual compatible cutoff hierarchy has connected distributions $C_{m,n}$. For $m \ge 2$, require the exact identity on $\mathcal{S}_0(X_m)$

$$C_{m,n} = C^{\mathrm{ter}}_{m,n} + \sum_{j=0}^{K_n} T_{m,n,j}. \tag{S.27}$$

Identity (S.27) must be established for the same initial renormalized fields and normalized full law. It is not licensed by having a useful native kernel at each scale.

The following are quantitative inputs, uniform in m, n, j and the field-label tuple:

1. One-point normalization. For fixed A_1, q_1,

$$|C_{1,n}(f)| \le A_1^2 \nu_{q_1}(f). \tag{S.28}$$

   This can include an exact zero normalization, but no subtraction is supplied by this theorem.
2. Terminal connected bound. For fixed $A_T \ge 1$, $p_T \ge 0$, $q_T \in \mathbb{N}_0$,

$$|C^{\mathrm{ter}}_{m,n}(F)| \le A_T^{m+1} (m!)^{p_T} \nu_{q_T m}(F). \tag{S.29}$$

3. Physical connected-shell majorant. The shell distribution is the actual sum or integral of a kernel satisfying

$$|t_{m,n,j}(X)| \le A_0^{m+1} (m!)^{p_0} h_j^{-4m} w_j \exp\left[-\frac{c\,\tau(X)}{(m+d_0)h_j}\right], \tag{S.30}$$

   where $p_0 \ge 0$, $c > 0$, $d_0 \ge 0$ are fixed, and $\tau(X)$ is the minimum spanning-tree length. The fixed nonnegative sequence w_j satisfies

$$W = \sum_{j \ge 0} h_j w_j < \infty. \tag{S.31}$$

   On the fine grid, the distribution includes the measure factor a_n^{4m}; the kernel is not a distributional delta function with its measure factor omitted.

Exact point contacts may be included in (S.27), since they vanish on the stated test space. Nonzero-width contact remainders require their own quantitative norm bound; they cannot be dropped by naming them local.

Lemma 4.1 — the shell norm Premise (S.30) implies

$$\boxed{\|T_{m,n,j}\|_{6m,*} \le A_2^{m+1} (m!)^{p_0+5} h_j w_j, \qquad m \ge 2.} \tag{S.32}$$

The constant is independent of a_n, h_j, m, n.

Proof and prior-result attribution. This is the physical shell lemma already proved in Annex N, Proposition 8.1, with its initial factorial exponent 2 replaced by p_0. Its proof gives precisely an additional factorial power five.

For $r_i = X_i - X_1$, flatness at the full diagonal and fifth-order Taylor remainder give

$$|F(X)| \le C \left(\sum_{i=2}^{m} |r_i|_1\right)^5 (1 + |X_1|^2)^{-3} \nu_{6m}(F). \tag{S.33}$$

The intermediate points retain X_1, so the stated weight holds. Every r_i is a path displacement in a minimum tree; their total norm is at most $C(m-1)\tau(X)$. Absorb the fifth power of τ into half the exponential in (S.30). This costs $C(m+d_0)^5 h_j^5$, besides the preceding factor $C(m-1)^5$.

The remaining minimum-tree exponential is bounded by the sum over the m^{m-2} labeled trees. For each tree, successive leaf sums or integrals cost

$$\left[C(m+d_0)^4 h_j^4\right]^{m-1};$$

the lattice estimate is uniform because $a_n \le h_j$. The remaining weighted root sum or integral is bounded by a fixed constant. The physical powers multiply to

$$h_j^{-4m} h_j^5 h_j^{4(m-1)} = h_j. \tag{S.34}$$

The tree count and the range factors are bounded by $C^{m+1}(m!)^5$; the extra fixed powers of m are absorbed in the exponential constant. This proves (S.32). No new analytic shell estimate beyond the cited prior lemma is claimed here. □

Theorem 4.2 — full OS budget and a uniform shell tail Under (S.26)–(S.31), set

$$q = \max\{6, q_1, q_T\}, \qquad p = \max\{1, p_T, p_0 + 5\}. \tag{S.35}$$

Then the actual full hierarchy obeys

$$\boxed{|S_{m,n}(F)| \le C^{m+1}(m!)^p \nu_{qm}(F)} \tag{S.36}$$

uniformly in n, m. In particular, for $p_0 = 2$, $q_T, q_1 \le 6$, and $p_T \le 7$, the output is

$$(q, p) = (6, 7). \tag{S.37}$$

For $J \ge 0$, form an auxiliary connected hierarchy by keeping the same one-point and terminal terms and only the shells with $j \le \min(J, K_n)$. Let $S_{m,n}^{\le J}$ be its full partition hierarchy. Then

$$\boxed{|(S_{m,n} - S_{m,n}^{\le J})(F)| \le W(J)\, C^{m+1}(m!)^p \nu_{qm}(F), \quad W(J) = \sum_{j>J} h_j w_j \longrightarrow 0.} \tag{S.38}$$

No positivity assertion is made for the auxiliary truncated hierarchy.

Proof. Summing (S.32) in (S.27), adding (S.29), and using (S.28) for order one gives

$$|C_{m,n}(F)| \le A^{m+1}(m!)^p \nu_{qm}(F)$$

for one fixed A. Monotonicity of the seminorm in its integer index justifies using the common q. The factor W is absorbed in A^{m+1}. Theorem 3.1 gives (S.36) for joint tests.

Both the full and truncated connected hierarchies have the same uniform budget. Their difference is bounded by (S.32) summed over $j > J$, with the factor $W(J)$. Apply Corollary 3.2 to get (S.38). Compatibility with the given cutoff full moments follows from the exact connected identity and the ordinary finite moment–cumulant identity at that cutoff; Lemma 2.2 supplies its joint-test meaning on $\mathcal{S}_0$. □

Theorem 4.3 — dominated cofinal passage for the full hierarchy In addition, suppose for every fixed m, j and every flat joint test the shell distributions $T_{m,n,j}$ converge as $n \to \infty$ once $K_n \ge j$. Suppose the terminal distributions and the one-point distributions also converge on their stated test spaces. Then

$$C_m = C_m^{\mathrm{ter}} + \sum_{j \ge 0} T_{m,j}, \qquad m \ge 2, \tag{S.39}$$

is a well-defined distributional sum with the same connected budget, and

$$S_{m,n}(F) \longrightarrow \sum_{\pi \in \Pi_m} \left(\bigotimes_{B \in \pi} C_{|B|} \right)(F). \tag{S.40}$$

The output obeys (S.36).

Proof. For a fixed finite J, use the assumed convergence of its finitely many shells and the terminal term. The tail of the connected sum is bounded uniformly by $W(J)$ times the fixed order's seminorm budget, so it tends to zero independently of n. This proves connected convergence to (S.39). Proposition 3.3 gives full convergence. Alternatively, (S.38) first removes the full-hierarchy tail uniformly and Proposition 3.3 applies to the finitely resolved connected sum. □

If those resolved-shell limits have not been shown, the common bound still supports the previously proved cofinal compactness argument. No uniqueness of the resulting subsequence follows from the present theorem.

5. How these two results differ from the earlier annexes

Earlier result	What was already proved	What is added here
Annex N, Lemma 6.1 and Theorem 6.2	The partition factorial budget and a full-moment expansion for the specified native point kernels	S-01 starts with general connected distributions on the flat test spaces, constructs their products on arbitrary joint tests, proves extension independence, preserves the exact q, p, and proves stability and weak continuity
Annex N, Proposition 7.1	Direct joint-test summation for an actual fixed-shell kernel, retaining its ultraviolet factor	S-01 no longer requires a pointwise kernel or uses a tensor-test bound as a substitute
Annex N, Proposition 8.1	The physical full-diagonal shell estimate and its summable h_j gain	S-02 uses that lemma as an explicit input; it supplies the terminal-plus-shell-to-full distribution construction, the full-hierarchy tail estimate, and dominated cofinal passage
Annex E and Annex O	Exact contact algebra and a one-layer microscopic contact error	No all-scale matching or new contact-support result is claimed in S-01/S-02
Earlier compactness theorem	A uniformly bounded full hierarchy has a subsequential continuum limit	S-02 supplies a sufficient distribution-level input interface with explicit q, p, without claiming its Wilson premises

The shell lemma and partition counting of Annex N are not new here. The new content is the general joint distribution passage and its complete conditional assembly interface.

Annex V — Local periodic-state comparison and phase-symmetry defects

Annex V. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Scope

This annex proves two precisely scoped intermediate statements:

- E-01: an exact local periodic-to-vacuum comparison identity and a quantitative theorem transferring renormalized vacuum slab estimates to the same periodic hierarchy.
- V-01: a criterion for almost-everywhere preservation of a time-preserving disconnected symmetry in a valid phase decomposition, expressed as a large-distance positive-time two-block estimate.

Neither statement proves its actual Yang–Mills analytic premise. In particular, E-05, the required local state-comparison rate on the changing cubic Wilson family, and V-04, the vanishing of the actual phase-reflection defects, remain open.

Annex R reduces the covariant tensor energy estimate to three seed inequalities.

The periodic bridge uses the sufficient global trace condition

$$\frac{1}{T}\log\frac{\operatorname{Tr} e^{-(T/2)H_{a,T}}}{\operatorname{Tr} e^{-TH_{a,T}}}\longrightarrow 0. \tag{E.1}$$

The new comparison route does not assume (E.1). It concerns the restriction of the actual path laws to the support of each fixed observable. It is an alternative sufficient state input, not a proof that local vacuum equivalence holds in Yang–Mills.

1. An exact endpoint density for the compact Wilson transfer law

Fix one finite cutoff. Let X be the compact space of spatial-link configurations with its Haar reference measure dx. Gauge projection can be included in the transfer kernel. Let P be the nonnegative, selfadjoint, ground-normalized transfer operator, represented by a nonnegative kernel $p(x,y)$, and suppose

$$P\Omega=\Omega,\qquad \Omega(x)>0,\qquad \int_X \Omega(x)^2dx=1. \tag{E.2}$$

These are the finite-cutoff transfer objects retained in Annex Q. The trace below is the physical trace; representing the gauge-projected operator on the full link space gives the same trace, since its orthogonal complement is annihilated. Put

$$\mathcal{Z}_N=\operatorname{Tr} P^N,\qquad P=e^{-aH}\text{ on the physical space.} \tag{E.3}$$

Take integers $N>m\geq 1$. The temporal circumference is $T=Na$. The m-step segment is chosen to contain every link and auxiliary transition variable used by the marks under discussion.

The stationary vacuum path law on $x_0,\dots,x_m$ has density

$$d\nu_m=\Omega(x_0)\left[\prod_{j=1}^{m}p(x_{j-1},x_j)\right]\Omega(x_m)\prod_{j=0}^{m}dx_j. \tag{E.4}$$

It is a probability measure because $\langle\Omega,P^m\Omega\rangle=1$. Equivalently it is the stationary law of the Doob transition

$$Q(x,dy)=p(x,y)\frac{\Omega(y)}{\Omega(x)}dy,\qquad \pi(dx)=\Omega(x)^2dx. \tag{E.5}$$

The marginal of the periodic N-step path law on the same segment is

$$d\mu_{N,m}=\frac{1}{\mathcal{Z}_N}\left[\prod_{j=1}^{m}p(x_{j-1},x_j)\right]p^{(N-m)}(x_m,x_0)\prod_{j=0}^{m}dx_j, \tag{E.6}$$

where $p^{(k)}$ is the kernel of P^k.

Theorem E.1 — exact local comparison identity The Radon–Nikodym density of (E.6) relative to (E.4) is

$$\boxed{R_{N,m}(x_0, x_m) = \frac{p^{(N-m)}(x_m, x_0)}{\mathcal{Z}_N \Omega(x_0)\Omega(x_m)}.} \tag{E.7}$$

In particular $\nu_m(R_{N,m}) = 1$.

Let $\mathcal{C}$ be any sub-sigma-algebra describing a smaller spacetime cylinder inside this segment. If $\mu_{\mathcal{C}}, \nu_{\mathcal{C}}$ are the restricted laws, then

$$\boxed{\frac{d\mu_{\mathcal{C}}}{d\nu_{\mathcal{C}}} = r_{\mathcal{C}} := \mathbb{E}_{\nu_m}[R_{N,m} \mid \mathcal{C}].} \tag{E.8}$$

With total variation normalized as $\delta(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|$,

$$\boxed{\delta_{\mathcal{C}} = \frac{1}{2}\mathbb{E}_{\nu_m}|r_{\mathcal{C}} - 1|, \qquad |\mu(W) - \nu(W)| \le 2\delta_{\mathcal{C}}\|W\|_\infty} \tag{E.9}$$

for every bounded complex $\mathcal{C}$-measurable mark W.

Proof. Integrating out the complement of the segment in the periodic path gives the closing kernel $p^{(N-m)}$, proving (E.6). Dividing the two displayed nonnegative densities gives (E.7), up to irrelevant null sets. The conditional-expectation identity

$$\mathbb{E}_{\nu_m}[R_{N,m}W] = \mathbb{E}_{\nu_m}[r_{\mathcal{C}}W]$$

for $\mathcal{C}$-measurable W proves (E.8). The signed measure $(r_{\mathcal{C}} - 1)d\nu_{\mathcal{C}}$ has total mass zero, so the integral of its positive part and of the absolute value of its negative part both equal $\delta_{\mathcal{C}}$. This proves (E.9), also for complex marks by the total-variation norm inequality. □

The same proof retains auxiliary transition variables. If a Wilson transfer kernel is obtained by integrating a nonnegative density $p(x, z, y)$ over temporal links or other transition variables z, use that density in every segment factor in both laws. Their ratio is still (E.7). Consequently temporal plaquettes and finite temporal clovers are covered. The statement is not restricted to time-zero multiplication observables.

The relevant target for E-05 is the conditional density (E.8), on the support of the observable. A bound on the full endpoint ratio is sufficient but can be unnecessarily strong. Conditioning can discard a large spectral contribution carried by degrees of freedom outside that cylinder.

Entropy version, with its proof An alternative local input is

$$D_{\mathcal{C}} = \int r_{\mathcal{C}} \log r_{\mathcal{C}}\, d\nu_{\mathcal{C}}. \tag{E.10}$$

Then

$$\delta_{\mathcal{C}} \le \sqrt{D_{\mathcal{C}}/2}, \qquad |\mu(W) - \nu(W)| \le \sqrt{2D_{\mathcal{C}}}\,\|W\|_\infty. \tag{E.11}$$

To verify the constant, put $A = \{r_{\mathcal{C}} \ge 1\}$, $p = \nu_{\mathcal{C}}(A)$, and $q = \mu_{\mathcal{C}}(A)$, so $q - p = \delta_{\mathcal{C}}$. Jensen on A and its complement gives

$$D_{\mathcal{C}} \ge q \log(q/p) + (1 - q) \log((1 - q)/(1 - p)).$$

As a function of q, the right side and its first derivative vanish at $q = p$, while its second derivative is $q^{-1} + (1-q)^{-1} \ge 4$. It is therefore at least $2(q - p)^2$. The limiting cases $p = 0, 1$ follow by continuity or give infinite relative entropy. This proves (E.11). No entropy estimate for the actual Wilson family is asserted here.

2. Quantitative transfer with the physical cutoff factors retained

Let $a_n \downarrow 0$, $T_n \uparrow \infty$ be an admitted cubic sequence. At each n, μ_n is its actual periodic Wilson law and ν_n is the vacuum path law constructed from the same finite-spatial-volume transfer operator. Thus the spatial side of both laws is T_n. This definition does not change the temporal side while silently keeping a different spatial Hamiltonian fixed.

Choose nested physical cylinders $\mathcal{C}_R$ exhausting compact spacetime sets. Give each test enough margin for every lattice stencil in its representative. Denote the total variation of the two restrictions by $\delta_n(R)$. The exact formula (E.8) defines it even when its decay is unknown.

The coefficient-budget hypothesis is the following. For every fixed compact polynomial or joint-test path mark W_n used in the hierarchy, including its running field coefficients, identity subtractions and finite product redefinitions, there are $R_W, d_W, C_W < \infty$ such that

$$W_n \text{ is } \mathcal{C}_{R_W}\text{-measurable}, \qquad \|W_n\|_\infty \le C_W a_n^{-d_W}. \tag{E.12}$$

The constants and exponent may depend on the fixed degree and tests. They are uniform in n and in the selected volume. They must be verified for the actual field prescription.

For a degree-m product of dimension-four representatives, the bound can contain a_n^{-4m}, multiplied by running normalization factors. It is not legitimate to treat its supremum as cutoff independent. A fixed logarithmic factor can be absorbed into an additional fixed power of a_n^{-1}; an unrestricted subtraction coefficient cannot.

Theorem E.2 — local state comparison transfers the hierarchy Suppose (E.12) holds and

$$\boxed{a_n^{-k}\delta_n(R) \longrightarrow 0 \quad \text{for every fixed } R < \infty \text{ and integer } k \ge 0.} \tag{E.13}$$

Then, for every such fixed marked hierarchy evaluation,

$$\mu_n(W_n) - \nu_n(W_n) \longrightarrow 0. \tag{E.14}$$

Thus whenever either of these two normalized hierarchies converges on these tests, the other has the same limit. If the common uniform Schwartz bounds needed for reconstruction have already been established, the equality extends by continuity from compact coincidence-flat tests to the full test spaces.

Proof. Equation (E.9) gives

$$|\mu_n(W_n) - \nu_n(W_n)| \le 2C_W a_n^{-d_W}\delta_n(R_W).$$

Choose an integer $k \ge d_W$, eventually $a_n \le 1$, and apply (E.13). The same common limit and the claimed continuity extension then follow directly. □

For a particular finite set of evaluations, (E.13) can be weakened to precisely the finitely many rates

$$a_n^{-d_W}\delta_n(R_W) \longrightarrow 0. \tag{E.15}$$

The superpolynomial formulation is a convenient single sufficient condition for the whole hierarchy. It is not declared necessary.

Plain local total-variation convergence $\delta_n(R) \to 0$ is insufficient with no rate. For example, a bound $\delta_n = a_n$ leaves the estimate for an a_n^{-8}-bounded two-field mark divergent. This is a defect of that proposed inference, not a claim that the actual moment difference must diverge.

Theorem E.3 — local comparison transfers the periodic OS insertion inequality Assume the exact cutoff transfer realization of the chosen Hermitian renormalized insertion $q_{\alpha,n}(g)$, with the correct identity coefficient and midpoint placement. Assume

$$\|e^{-uH_n} q_{\alpha,n}(g) e^{-uH_n}\| \le C u^{-\kappa}\rho_s(g), \qquad 0 < u \le 1, \tag{E.16}$$

uniformly in n. Here $\rho_s(g) = \sum_{|\beta|\le s} \|\partial^\beta g\|_{L^1(\mathbb{R}^3)}$.

Assume (E.12)–(E.13) for the actual cutoff polynomial products below. Fix a positive-time complex polynomial F, $0 < t \le 1$, real spatial g, and a nonnegative even mollifier χ_ϵ of integral one with $0 < \epsilon < t/8$. Then

$$\begin{aligned} &\left|\mu_n\big[(\Theta\tau_t F)\, Q_{\alpha,n}(\chi_\epsilon \otimes g)\,(\tau_t F)\big]\right| \\ &\qquad \le C' t^{-\kappa}\rho_s(g)\,\mu_n[(\Theta F)F] + \varepsilon_n(F,t,\epsilon,g,\alpha), \qquad \varepsilon_n \longrightarrow 0, \end{aligned} \tag{E.17}$$

with C' independent of F, its degree, and the fixed ϵ. In this cutoff formula, time translations mean the fixed lattice realizations τ_{t_n}, with $t_n \to t$, and all marks use one specified lattice sampling. The order of limits remains: first $n \to \infty$ at every fixed $\epsilon > 0$, then unsmear in the reconstructed continuum theory.

Proof. In the vacuum transfer representation, write $\Psi_n(F)$ for the corresponding positive-time vector. Translation by t gives the heat factor e^{-tH_n}. At a central insertion time δ in the mollifier support, the relevant matrix element is

$$\left\langle \Psi_n(F), e^{-(t_n-\delta)H_n} q_{\alpha,n}(g) e^{-(t_n+\delta)H_n} \Psi_n(F) \right\rangle, \tag{E.18}$$

with the actual lattice shifts $t_n \to t$ and the fixed stencil buffer included. Eventually both heat intervals exceed $t/2$. Factoring out the sandwich in (E.16) at $t/2$ bounds the absolute value by

$$2^{\kappa} C t^{-\kappa} \rho_s(g) \|\Psi_n(F)\|^2.$$

At each fixed ϵ, the lattice L^1 sum of the sampled mollifier is eventually at most two. Hence the vacuum version of (E.17) holds with $C' = 2^{\kappa+1} C$.

Set

$$W_n = (\Theta \tau_t F) Q_{\alpha,n}(\chi_\epsilon \otimes g)(\tau_t F), \qquad V_n = (\Theta F) F.$$

Apply (E.9) separately to W_n and V_n. The error can be taken to be

$$\begin{aligned} \varepsilon_n = {} & 2\delta_n(R_W) \|W_n\|_\infty \\ & + 2C' t^{-\kappa} \rho_s(g) \delta_n(R_V) \|V_n\|_\infty, \end{aligned} \tag{E.19}$$

for these fixed sampled marks; no additional sampling error occurs in this identity. The two terms tend to zero by (E.12)–(E.13). Reflection positivity makes the quadratic norm terms nonnegative in their respective laws. The constants in (E.12) affect the error, not C'. This proves (E.17). Replacing the specified sampling or rounded shifts by another approximation of the continuum tests requires the common S0 seminorm bounds and continuity; only under that additional premise does the resulting sampling difference tend to zero. □

If F has degree d, the central mark W_n can have degree $2d+1$, and its physical coefficient budget can contain $a_n^{-4(2d+1)}$. This dependence is included in (E.19). The theorem does not infer (E.16) from the dimensionless magnetic estimate in Annex Q.

The equality of the insertion prescriptions matters. A ground-centered observable and the prescribed F0-centered observable differ by an identity term, and that term must be bounded in (E.16) or explicitly included in the normalization. Local comparison does not erase it.

3. The exact diagonal-selection premise

The following is a useful way to formulate the open E-05 estimate without requiring a rate uniform in both a and T from the start.

For each fixed admitted a, let $\delta(a, T; R)$ be the local comparison defect between the periodic cube of side T and the vacuum path law of its finite-spatial-volume transfer operator $H_{a,T}$. Assume

$$\boxed{\liminf_{\substack{T\to\infty \\ T \text{ admitted at } a}} \delta(a, T; R) = 0 \quad \text{for every fixed } a, R.} \tag{E.20}$$

The term $H_{a,T}$ changes with T. This is an additional cubic-family assertion, not the fixed-spatial-operator zero-temperature limit.

Proposition E.4 — one cofinal sequence suffices Assume (E.20) on an admitted family of arbitrarily large cubes, and assume the source-free and marked hypotheses being used remain valid on the selected family. Let $a_n \downarrow 0$ be any admitted cutoff sequence. One can choose admitted $T_n \to \infty$ such that (E.13) holds, while also imposing any prescribed finite lower bound on T_n, for example

$$T_n \geq n(1 + \log(1/a_n)). \tag{E.21}$$

Proof. Choose nested cylinders $\mathcal{C}_n$ containing every fixed compact cylinder eventually. At the fixed value $a = a_n$, (E.20) with $R = n$ permits an arbitrarily large admitted T_n with

$$\delta(a_n, T_n; n) \leq a_n^n / n. \tag{E.22}$$

Choose it also large enough for (E.21), the cylinder margin, the source-free minimum size, and $T_n > T_{n-1}$. Restricting measures to a smaller sigma-algebra cannot increase total variation. Thus, for fixed R, k and all sufficiently large n,

$$a_n^{-k}\delta_n(R) \le a_n^{n-k}/n \longrightarrow 0.$$

This is (E.13). □

This diagonal argument is valid only after (E.20) is proved. Finite-cutoff spectral convergence at fixed spatial volume does not prove (E.20).

4. What the elementary spectral estimate does and does not give

At one fixed cutoff, suppose the physical normalized transfer operator has a unique ground state and gap $\gamma > 0$. Define

$$\mathcal{K} = \sup_{x\in X}\left(\frac{p^{(2)}(x,x)}{\Omega(x)^2} - 1\right). \tag{E.23}$$

For the continuous strictly positive compact kernel at a fixed finite cutoff, $\mathcal{K} < \infty$. This fact is not a bound uniform in cutoff or spatial volume.

For every integer $k \ge 2$,

$$\left|\frac{p^{(k)}(x,y)}{\Omega(x)\Omega(y)} - 1\right| \le \mathcal{K}e^{-(k-2)a\gamma}. \tag{E.24}$$

Indeed, set $u_x(z) = p(x,z) - \Omega(x)\Omega(z)$. It is orthogonal to Ω, and

$$\|u_x\|^2 = p^{(2)}(x,x) - \Omega(x)^2.$$

The kernel difference in (E.24) is $\langle u_x, P^{k-2}u_y\rangle$, up to the harmless real-kernel ordering. On $\Omega^\perp$, the operator norm is at most $e^{-(k-2)a\gamma}$. Cauchy–Schwarz proves (E.24).

Integrating the diagonal gives

$$0 \le \mathcal{Z}_N - 1 \le \mathcal{K}e^{-(N-2)a\gamma}.$$

If $N - m \ge 2$, (E.7) consequently yields

$$\|R_{N,m} - 1\|_\infty \le 2\mathcal{K}e^{-(T-ma-2a)\gamma}, \qquad \delta_{\mathcal{C}} \le \mathcal{K}e^{-(T-ma-2a)\gamma}. \tag{E.25}$$

This is a valid finite-cutoff bound for the actual kernel. In the cubic family it contains

$$\mathcal{K}_{a,T} \quad \text{and} \quad \gamma_{a,T}.$$

Neither is controlled by the local deletion cost in Annex Q. Therefore (E.25) cannot be used to select T_n by pretending that these quantities stay fixed as the spatial side grows. A useful actual-model estimate may instead exploit the conditional expectation in (E.8), which can be much smaller than the full endpoint supremum.

The two sufficient criteria are not comparable in general The next finite-dimensional positive-transfer example shows that the local comparison criterion can hold when the old trace criterion fails. It is not a Yang–Mills or translation-invariant lattice model, and it does not by itself prove an implication between the two criteria.

Let the observable subsystem be two-dimensional, with unique strictly positive ground vector and Hamiltonian H_{vis} having spectrum $0, 1$. Choose it so its heat kernel is strictly positive in the two-state basis. Tensor it with a spectator Hamiltonian $H_{\mathrm{sp},n}$ having one ground level and d_n levels at energy one, with

$$d_n = \lceil e^{cT_n}\rceil, \qquad \frac{1}{2} < c < 1.$$

A strictly positive kernel is obtained by taking the ground projection onto the uniform vector and $H_{\mathrm{sp},n} = I - \Pi_{\mathrm{ground}}$. All specified local marks act only on the observable subsystem.

The periodic and vacuum path restrictions of those marks differ by

$$O_R(e^{-(T_n-R)}).$$

For example, this follows directly from (E.7) for the fixed two-dimensional observable subsystem. Taking $a_n = L^{-n}$ and $T_n = 2L^n$ makes the local error smaller than every power of a_n.

In contrast, the spectator partition function is

$$\mathcal{Z}_{\mathrm{sp},n}(u) = 1 + d_n e^{-u},$$

so

$$\frac{1}{T_n} \log \frac{\mathcal{Z}_{\mathrm{sp},n}(T_n/2)}{\mathcal{Z}_{\mathrm{sp},n}(T_n)} \longrightarrow c - \tfrac{1}{2} > 0. \tag{E.26}$$

Thus the local comparison route can succeed when (E.1) fails. The spectator may be viewed as a degree of freedom outside all specified local cylinders. This example proves no estimate for actual Wilson local marginals; it only prevents treating the global trace condition as a necessary state criterion.

The converse implication also fails. In a two-state space choose $\Omega = (\sqrt{p}, \sqrt{1-p})$, with $0 < p < 1$ and $p \neq 1/2$, and put

$$H_n = \epsilon_n (I - \Pi_\Omega), \qquad \epsilon_n = T_n^{-2}. \tag{E.27}$$

Every positive-time kernel is strictly positive, and the ground state is unique. Its partition function is $\mathcal{Z}_n(u) = 1 + e^{-u\epsilon_n}$. Hence

$$\frac{1}{T_n} \log \frac{\mathcal{Z}_n(T_n/2)}{\mathcal{Z}_n(T_n)} \longrightarrow 0.$$

Nevertheless, the periodic time-zero probability of the first state is

$$\mu_n(\{1\}) = \frac{e^{-T_n\epsilon_n} + (1 - e^{-T_n\epsilon_n})p}{1 + e^{-T_n\epsilon_n}} \longrightarrow \frac{1}{2},$$

whereas its vacuum probability is p. The local total variation therefore stays bounded away from zero. The old trace criterion can transfer a slab inequality without asserting equality of these two local laws.

Thus E-01 supplies an alternative sufficient route. It is not a uniformly weaker replacement. Its local norm comparison can be stronger than necessary for the energy conclusion, and the original trace route and the direct periodic OS route remain available.

5. A phase-reflection criterion in the actual positive-time test algebra

This section uses the corrected OS test spaces and their actual tensor reflection conventions. No statement about ordinary probability positivity after contact renormalization is assumed.

Assume that a continuum hierarchy and a valid phase-field disintegration have already been constructed:

$$\mathcal{H} = \int^{\oplus} \mathcal{H}_\omega \, d\nu(\omega), \quad H = \int^{\oplus} H_\omega \, d\nu(\omega), \quad \Omega = \int^{\oplus} \Omega_\omega \, d\nu(\omega), \quad \|\Omega_\omega\| = 1. \tag{V.1}$$

Assume the positive-time polynomial vectors disintegrate consistently,

$$\Psi(F) = \int^{\oplus} \Psi_\omega(F) \, d\nu(\omega), \qquad S_\omega(F) = \langle \Omega_\omega, \Psi_\omega(F) \rangle. \tag{V.2}$$

Assume the fibers are translation-ergodic phases in the following precise sense:

$$\ker H_\omega = \mathbb{C}\Omega_\omega \quad \text{for almost every } \omega. \tag{V.3}$$

The forward-cone spectrum makes $\ker H$ equal the space invariant under all translations: on the joint spectrum, $p^0 = 0$ implies $\mathrm{p} = 0$. The same applies in each fiber. Thus (V.3) is the unique translation-invariant-vacuum condition, rather than an unsupported assertion about arbitrary factor states.

Let G be a finite group of time-preserving spatial orthogonal transformations, including any desired spatial reflection. Use its stipulated action on field labels. Such a transformation preserves the positive-time test algebra and commutes with the time translations used below. Write

$$D_g F = gF - F, \qquad P_0 = 1_{\{0\}}(H). \tag{V.4}$$

Require that the phase distributions are continuous on the relevant coincidence-flat test spaces and translation invariant. These properties must come from the valid phase construction, for example the previously proved energy-to-phase-regularity implication once its energy input is available.

Choose a countable complex rational-linear test core $\mathcal{D}_+$, dense at each degree and field-label tuple on every compact positive-time support with positive margin. Such cores are obtained by a countable compact exhaustion and separability of the supported smooth test spaces; the coincidence-flat subspaces inherit a countable dense set. Enlarge the core by its finitely many G-images. No norm separability of a local von Neumann algebra is assumed.

Theorem V.1 — exact equivalence for phase reflection covariance Under (V.1)–(V.4), the following are equivalent:

1. On one full-measure set of phases,
$$S_\omega(gF) = S_\omega(F) \tag{V.5}$$
for all $g \in G$ and all tests in the full continuum hierarchy.

2. For every $g \in G$ and every $F \in \mathcal{D}_+$,
$$\boxed{\|P_0 \Psi(D_g F)\| = 0.} \tag{V.6}$$

3. For every $g \in G$ and every $F \in \mathcal{D}_+$,
$$\boxed{\lim_{R\to\infty} S\big((\Theta\tau_R F_g)(\tau_R F_g)\big) = 0, \qquad F_g = D_g F.} \tag{V.7}$$
Here products mean the full OS quadratic-form evaluation, including joint polynomial kernels; the expression is nonnegative.

The precise identity behind these equivalences is

$$\boxed{\lim_{R\to\infty} S\big((\Theta\tau_R D_g F)(\tau_R D_g F)\big) = \|P_0\Psi(D_g F)\|^2 = \int |S_\omega(D_g F)|^2 d\nu(\omega).} \tag{V.8}$$

Proof. Spectral calculus and (V.3) give
$$P_0 = \int^{\oplus} |\Omega_\omega\rangle\langle\Omega_\omega|\, d\nu(\omega).$$
Using (V.2), the fiber of $P_0\Psi(D_g F)$ is
$$S_\omega(D_g F)\Omega_\omega.$$
Taking the direct-integral norm gives the second equality in (V.8).

OS time translation gives
$$\Psi(\tau_R F_g) = e^{-RH}\Psi(F_g).$$
The spectral theorem implies $e^{-RH} \to P_0$ strongly as $R \to \infty$. Consequently
$$S((\Theta\tau_R F_g)(\tau_R F_g)) = \|e^{-RH}\Psi(F_g)\|^2 \longrightarrow \|P_0\Psi(F_g)\|^2.$$
This proves the first equality. No gap or quantitative clustering rate is used.

Condition (V.5) immediately implies (V.6) by (V.8), and (V.6) and (V.7) are equivalent by the same identity. If (V.6) holds, then $S_\omega(D_g F) = 0$ for almost every ω for each pair g, F in the countable core. Intersect these countably many full-measure sets. Continuity of each phase distribution extends the equality to all compact positive-time tests, at every degree and label tuple.

Every compact joint test can be shifted by one common time translation far enough to lie strictly in positive time. Since g preserves time and the phase distributions are translation invariant, the equality for the translated test implies the equality for the original test. Finally, compact cutoffs preserve coincidence flatness and converge in the Schwartz topology. Phase distribution continuity extends the identity to all the full hierarchy tests. Thus (V.5) holds on one full-measure set. □

Bounded-algebra and macroscopic-average form For the stated theorem, the same result has a bounded-algebra formulation. Let $\mathfrak{A}_{\mathrm{sep}}$ be a specified norm-separable unital C^*-algebra of bounded decomposable observables, invariant under the finite symmetry automorphisms α_g. Suppose its represented phase states are

$$\omega(A) = \langle \Omega_\omega, A_\omega \Omega_\omega \rangle.$$

Do not identify $\mathfrak{A}_{\mathrm{sep}}$ with an arbitrary local von Neumann algebra, which need not be norm separable. If invariance of this algebra is to determine all field distributions, that determining property is a separate application hypothesis.

Choose a countable norm-dense set A_j, including its finitely many symmetry images and rational linear combinations. Then the phase states are G-invariant on $\mathfrak{A}_{\mathrm{sep}}$ on one full-measure set if and only if

$$\|P_0(\alpha_g(A_j) - A_j)\Omega\| = 0 \quad \text{for all } g, j. \tag{V.12}$$

Indeed, direct integration gives the exact identity

$$\|P_0(\alpha_g(A) - A)\Omega\|^2 = \int |\omega(\alpha_g(A) - A)|^2 d\nu(\omega). \tag{V.13}$$

After the countable intersection, norm continuity of every state extends the equality from A_j to all A. Conversely invariant phase states make the right side zero.

For a macroscopic translation-average version, choose nonnegative $\chi \in C_c^\infty(\mathbb{R}^4)$ of integral one and put $\chi_R(x) = R^{-4}\chi(x/R)$. Let $U(x)$ be the strongly continuous translation representation and set

$$v_R = \int_{\mathbb{R}^4} \chi_R(x) U(x)(\alpha_g(A) - A)\Omega \, dx. \tag{V.14}$$

This is a norm-convergent vector integral; no operator-norm continuity of translations on a local von Neumann algebra is required. With the usual Fourier convention,

$$v_R = \widehat{\chi}(RP)(\alpha_g(A) - A)\Omega.$$

Here $\widehat{\chi}(0) = 1$, $|\widehat{\chi}| \le 1$, and $\widehat{\chi}(Rp) \to 0$ for $p \neq 0$, by integration by parts for the smooth compactly supported function. Dominated convergence in the joint spectral measure and the forward-cone identification of P_0 give

$$v_R \longrightarrow P_0(\alpha_g(A) - A)\Omega \quad \text{in norm.} \tag{V.15}$$

Therefore the limit of the squared macroscopic-average norm is exactly (V.13). Equivalently it is the double integral of the corresponding translated two-block vacuum correlation. This proves the bounded-algebra statement of V-01 and its simultaneous-null-set requirement. The positive-time formulation (V.8) is preferable for the present observable route because its test algebra and full Schwinger functions are already specified.

What this criterion buys The open V-04 target can now be stated as a concrete estimate on the mixed hierarchy:

$$S((\Theta\tau_R D_g F)(\tau_R D_g F)) \longrightarrow 0 \quad (R \to \infty) \tag{V.9}$$

for a countable determining family of positive-time tests and the required spatial reflection.

This is a two-block estimate at large Euclidean time separation. It is weaker than imposing clustering for every pair of observables: only the symmetry-defect vectors are required to have zero projection onto the vacuum subspace. It also differs from the ultraviolet block estimates of Annex L, which shrink the internal geometry of blocks while keeping their centers a fixed distance apart.

If the phase construction separately guarantees connected Euclidean covariance, one spatial reflection together with $SO(4)$ generates $O(4)$. Then (V.5) for that reflection gives full $O(4)$ covariance of almost every phase hierarchy. This inference does not supply connected covariance or prove that a symmetry which permutes phases acts within each phase.

The conclusion is an almost-everywhere theorem. Existence of just one symmetric phase is weaker and could in principle be proved by another argument even if (V.9) fails. The criterion is offered as a clean sufficient route with an exact necessity statement for its stronger almost-everywhere conclusion.

Mixed symmetry only fixes the mean If the mixed hierarchy is G-invariant, one knows

$$S(D_g F) = \int S_\omega(D_g F)\, d\nu(\omega) = 0. \tag{V.10}$$

Equation (V.8) requires the squared modulus of this phase defect to have mean zero. Equation (V.10) does not imply that assertion.

A minimal Hilbert-space example makes the distinction exact. Take

$$\mathcal{H} = \mathbb{C}^2, \quad H = 0, \quad \Omega = 2^{-1/2}(1,1), \quad A = \mathrm{diag}(v, -v),$$

and let a $\mathbb{Z}_2$ unitary swap the two coordinates. The mixed state is invariant, each one-dimensional component has a unique vacuum, and the symmetry sends A to $-A$. Hence

$$\langle \Omega, (\alpha_g(A) - A)\Omega \rangle = 0, \qquad \|P_0(\alpha_g(A) - A)\Omega\|^2 = 4v^2. \tag{V.11}$$

The two pure components are exchanged. This is an internal-symmetry finite-dimensional example, not a spacetime field construction or a Yang–Mills model. It illustrates precisely the mean-versus-variance obstruction; for spacetime reflection, (V.8) must be evaluated with the actual tensor and Wick conventions.

The same distinction explains why common ultraviolet coefficients alone cannot establish reflection invariance of phases. A fixed central order parameter can be invisible in a rescaled short-distance leading coefficient and still produce a nonzero defect in (V.8). The ultraviolet and large-distance tests have different limits and neither is substituted for the other.

6. Simultaneous phase selection once the separate premises are proved

The following is an application specification, not a third new theorem.

Suppose the chosen valid phase decomposition supplies energy/domain control and connected covariance on a full-measure set $\mathcal{G}_{\text{field}}$. Suppose the actual interacting four/six-block estimates satisfy the hypotheses of Annex L, giving the specified ultraviolet coefficients on a full-measure set $\mathcal{G}_{\text{UV}}$. Suppose (V.9) is proved, giving $\mathcal{G}_{\text{refl}}$.

Then

$$\mathcal{G}_{\text{field}} \cap \mathcal{G}_{\text{UV}} \cap \mathcal{G}_{\text{refl}}$$

has full measure. Every phase in that set has all three properties. The countable-profile and continuous-scale arguments needed for $\mathcal{G}_{\text{UV}}$ remain those of Annex L; they are not proved a second time here. The target field family and normalization must be the same in each input.

Annex W — Source-free identities, complete conditional density and normalized histories

Annex W. The conclusions below hold on the stated source domains. Applying them to the complete Wilson law requires compatibility of the conditional representation (CL), including the retained variables, densities, and marked transport. See the theorem catalogue. Identifying the calibrated Gaussian marginal with the Wilson trajectory also requires GC, including the bounded error uniform in depth.

0. Result and scope

This chapter gives the source formulas and a finite-cutoff construction for B-01, C-01, and H-01, subject to the published source-free construction's stated admissibility hypotheses. It is not a proof that the actual Wilson coupling trajectory satisfies those hypotheses. B-02 remains open as a whole: its geometric part can be made explicit, but its requested coupling-ceiling verification is not supplied by geometric arithmetic.

The distinction between an exposed source input and a new analytic estimate is essential. The localized reference density below is the actual one in Balaban's LFR I (1.100), not the schematic full density of its introduction. All history probabilities come from nonnegative primitive integrals. Individual cluster activities and their absolute values are never treated as probabilities.

The conditional-density identity is written directly from the physical exponent in CMP119 (2.23). This avoids an unresolved sign comparison with the decomposed expression (3.15). All variables and Jacobians are retained; no estimate from a homogeneous conditional law is imported for the complete law by this rewriting.

The finite conditional construction applies to the Wilson measure under CL. Its uniform coupling ceiling is a separate hypothesis.

1. Conventions and primary scope

Fix $SU(N)$, $N \geq 2$, with normalized Haar measure on each positively oriented link. For a periodic four-dimensional lattice, write

$$s_p(U) = N - \operatorname{Re}\operatorname{Tr} U_p, \qquad A(U) = \sum_p \Big(1 - \frac{1}{N}\operatorname{Re}\operatorname{Tr} U_p\Big). \tag{1.1}$$

The actual Wilson law and the source's initial density are

$$d\mu_W(U) = Z_W^{-1} e^{-\beta_W \sum_p s_p(U)} dH(U), \qquad \rho_0(U) = e^{-g_0^{-2}A(U) - \mathcal{E}_0}. \tag{1.2}$$

Here

$$\boxed{\beta_W = 2Nx_0, \qquad g_0^{-2} = X_{B,0} = N\beta_W = 2N^2 x_0.} \tag{1.3}$$

The symbol β_W is deliberately distinct from the small domain parameter called β in the Balaban papers, and from the functions β_j in the coupling recursion. In particular,

$$\mathcal{Z} := \int \rho_0 \, dH = e^{-\mathcal{E}_0} Z_W. \tag{1.4}$$

The initial scalar $\mathcal{E}_0$ cancels from normalized expectations. Later history-dependent vacuum constants do not separately cancel from history probabilities.

The following published formulas fix the conventions.

- CMP119: *Convergent Renormalization Expansions for Lattice Gauge Theories*, CMP 119 (1988), 243–285, pp. 251–252, 257–263 and 267–270.

- LFR I: *Large Field Renormalization. I. The Basic Step of the R Operation*, CMP 122 (1989), 175–202. The localized quotient and its normalization appear on p. 201.
- LFR II: *Large Field Renormalization. II. Localization, Exponentiation, and Bounds for the R Operation*, CMP 122 (1989), 355–392, Theorem 1 on p.355 retains a small-effective-coupling hypothesis. The reference-denominator calculation is (1.2)–(1.11), pp.356–358.
- CMP109: *Renormalization Group Approach to Lattice Gauge Field Theories. I*, CMP 109 (1987), 249–301, DOI `https://doi.org/10.1007/BF01215223`. Its p.251 gives the torus geometry, p.252 the normalized trace, and pp.267–268 the constraint-linearization and elimination conventions.

LFR II Theorem 1 assumes $0 < g_j \le \gamma$ for sufficiently small γ, along the source-defined effective trajectory. Its discussion expressly does not provide the second-order calculation proposed to remove that assumption. Neither this note nor the numerical recurrence theorem B-04 supplies that missing application. The underlying averaging, variational, propagator, and cluster-expansion theorems remain scoped source inputs, rather than newly reproved results.

2. B-01: the actual positive integral construction

2.1 Blocking and gauge normalization For the source averaging map $\mathcal{M}_j$, the complete blocking operator is

$$(T_j\rho)(V) = \int dH(U)\, \delta_G(\mathcal{M}_j(U)V^{-1})\rho(U). \tag{2.1}$$

This is CMP119 (0.1), with the index and averaging map displayed. The group delta is normalized against Haar measure, so

$$\int dH(V)(T_j\rho)(V) = \int dH(U)\rho(U). \tag{2.2}$$

Equivalently, the forward blocking kernel is the Dirac probability at $\mathcal{M}_j(U)$. The delta notation is not an extra signed density. All formulas containing gauge deltas may be read as finite positive measures on the constrained variables. Absolute continuity with respect to unrestricted link Haar measure is not assumed for an intermediate gauge-fixed output.

The smoothed axial gauge factor of CMP119 (1.5)–(1.7) uses the one-variable normalization

$$z_j = \int_G 1_{\{|W-1|<\varepsilon_j\}} \exp[-g_j^{-2}(1 - \operatorname{Re}\operatorname{tr} W)]\, dH(W). \tag{2.3}$$

Each noncentral site of a block contributes the corresponding normalized factor. The contour variable transforms by the endpoint gauge transformations, so Haar invariance makes its gauge integral exactly z_j. The source's factor $z_j^{-(L^4-1)|\Gamma|}$ is therefore required. It is retained before passing to conditional or history probabilities.

The tree-gauge alternative uses normalized Haar deltas on the chosen tree. It is an exact gauge disintegration, not a replacement of an integral by its value at a gauge-fixed field. If the gauge orientation is averaged, use the probability average over signed coordinate permutations; in four dimensions each of the $2^4 4! = 384$ choices has weight $1/384$. Section 6 records the source notation issue which makes it preferable to write this probability explicitly.

2.2 Selector and history inventory At finite cutoff the primitive history records the labels introduced by the following decompositions, with all complement branches kept.

Operation	Source	Actual label and retained restriction
Initial plaquette split	CMP119 (1.1)	P_0, from the good/bad plaquette test on the LM_2R_0 cubes.

Operation	Source	Actual label and retained restriction
Coarse-field split	CMP119 (1.4); general step (3.2)	P_{j+1}, testing the local minimizing background determined by the coarse field.
Approximate-fluctuation split	CMP119 (1.8); general step (3.3)	Q_{j+1}, before the normalized free fluctuation variable has been introduced.
Current free-field split	CMP119 (1.18); general step (3.16)	R_{j+1}, with both good and bad free-field restrictions. The literal current-coordinate cutoff is retained; see Section 6.
Enlarged geometric domains	CMP119 (1.10), (1.20), (2.1)–(2.3), (3.5), (3.20)	Ω_j, Λ_j, S_j, obtained from the preceding labels and their specified cube layers. Small eligible complements are replaced by their specified rectangular hulls.
Preparatory R tests	LFR I (1.24)–(1.29), (1.64)–(1.73)	The success/failure of each preliminary integration test and the remaining selected components. A failed component is retained in the nonreplaced sector.
Exterior extension test	LFR I (1.75)	$\chi_{k,\Lambda}$, a condition on the exterior data admitting a regular extension; it is not a condition on a newly sampled reference field.
Preparatory fluctuation and final boundary tests	LFR I (1.82), (1.88)–(1.89)	χ', boundary half-threshold labels, and the corresponding excluded components.
Gauge choices	CMP119 (1.5)–(1.7); LFR I (1.100)	Axial/tree graphs and their explicitly normalized finite averages.
Replacement components	LFR I (1.99)–(1.100)	Disjoint selected components X_i, their cores Λ_i, and their local determining sets.

Each elementary indicator insertion has the form $1 = \chi + (1-\chi)$, or the corresponding finite product expansion. Thus a primitive branch has a nonnegative weight. Resummation over labels which produce the same geometry produces sums of such weights. It does not turn a polymer activity into a branch probability.

The geometric labels determine the determining set of CMP119 (2.2), the retained fields of (2.10), and the minimizing background $U_k = U_{\mathcal{B}_k}(V)$ of (2.12). All minima, selectors and coordinate maps remain fixed at zero external source for marked calculations.

2.3 The source inductive integral and its terms Use bold symbols for the three activity classes and $\mathcal{E}_k$ for the scalar vacuum term. CMP119 (2.18)–(2.23) reads

$$\rho_k(V_k) = \sum_{\{\Omega_j\},\{\Lambda_j\}} \chi_k(\Omega_k)\, \mathcal{T}_k(\{\Omega_j\},\{\Lambda_j\}) \exp \mathcal{A}_k(w_k, U_k), \tag{2.4}$$

$$\mathcal{A}_k(w_k, U_k) = -A(w_k, U_k) + \mathrm{E}_k(U_k) + \mathrm{R}_k(U_k) + \mathrm{B}_k(U_k, \mathrm{a}) - \mathcal{E}_k. \tag{2.5}$$

Here $A(w,U) = \sum_p w(x(p))(1 - \mathrm{Re\,tr}\, U_p)$, $w_k(x) = g_k(x)^{-2}$, and a denotes the retained earlier fluctuation fields. The sum over S_j is inside $\mathcal{T}_k$, as specified in the source.

For disjoint large-field components X_i, the operations factor and commute; within one component their scale order is fixed:

$$\mathcal{T}_k\Big(\bigcup_i X_i\Big) = \prod_i \mathcal{T}_k(X_i), \qquad \mathcal{T}_k(X) = \mathcal{T}^{(k-1)}(Z_k \cap X) \cdots \mathcal{T}^{(0)}(Z_1 \cap X). \tag{2.6}$$

Before an R modification, the elementary retained operation of CMP119 (2.21) has the following explicit content:

$$\begin{aligned} \mathcal{T}^{(j)}(Z_{j+1} \cap X) = & \int dV_j\big|_{\Omega^c_{j+1}\cap X} \delta_G(\overline{V}_j V_{j+1}^{-1}) \zeta(\Omega^c_{j+1}) \\ & \cdot \int da_j\big|_{Z_{j+1}\cap\Omega_{j+1}\cap X} \chi(Z_{j+1} \cap \Omega_{j+1} \cap X) \\ & \cdot \exp\Big[-\frac{1}{2}\langle a_j, K_j a_j\rangle + \frac{1}{2}\langle a_j, K_j C^{(j)}(\Lambda_{j+1}) K_j a_j\rangle\Big]. \end{aligned} \tag{2.7}$$

The quadratic form is $K_j = C_j^* \Delta^{(j)} C_j$, with the component restrictions in the source. Cross-component quadratic terms are assigned to B; they are not included a second time in (2.7). The function ζ is the resummed product of the specified outer characteristic and normalized gauge factors. The R operation changes the selector/domain content and adds the localized reference integrals below. The recursive primitive description gives every resulting case; it does not require pretending that all histories have the unchanged form of one homogeneous Gaussian fiber.

The three activity classes in (2.5) are distinct:

$$\begin{aligned} \mathrm{E}_k(U) = \sum_{j=1}^{k} \big[& \mathrm{E}^{(j)}(\Lambda_j, g_{j-1}, U) - \mathrm{E}^{(j)}(\Lambda_j, g_{j-1}, 1) \\ & - \beta_j(g_{j-1}) A(\phi_j, U)\big], \end{aligned} \tag{2.8}$$

$$\mathrm{R}_k(U) = \sum_{j=1}^{k} \big[\mathrm{R}^{(j)}(\Lambda_j, U) - \mathrm{R}^{(j)}(\Lambda_j, 1)\big], \tag{2.9}$$

$$\mathrm{B}_k(U, \mathrm{a}) = \sum_{j=1}^{k} \sum_{X} \mathrm{B}^{(j)}(X, U, \mathrm{a}, \{S_i \cap X\}). \tag{2.10}$$

These are respectively CMP119 (2.25), (2.30), and (2.40)–(2.41). The E activities also have their point/carrier expansion (2.26)–(2.27). The B carriers meet the current old large-field boundary as specified in (2.41); they include boundary cross terms and changes of variables. The E term has coupling and vacuum subtraction, the R term has vacuum subtraction, and B has neither additional subtraction. No new counterterm is inserted by merely rewriting (2.8)–(2.10).

The source's local coupling identity is

$$w_{j-1}(x) = w_j(x) + \beta_j(g_{j-1})\phi_j(x). \tag{2.11}$$

The scalar $\mathcal{E}_k$ depends on the geometry history. The statement on CMP119 p.262 is why it must stay inside (2.4) and inside the normalized history weights below.

2.4 The localized R kernel, rather than its introductory surrogate Fix a selected component X, its core Λ, its retained exterior data υ, and a selected axial graph G. Let $V_\Lambda(\upsilon)$ be the source's constrained minimizing filling, and $U_{k,X}$ its localized minimizing-background map. Let ζ_X be exactly the localization function in LFR I (1.100). Define the positive reference measure

$$\begin{aligned} h_{k,X,G}(dV'; \upsilon) = & \, dH(V')\big|_\Lambda \, \delta_G(V') 1_{\{|(1/i)\log V'(b)| < M_0 \varepsilon_k \,:\, b \in \Lambda\}} \\ & \cdot \exp\big[-g_k^{-2} A\big(\zeta_X, U_{k,X}(V' V_\Lambda(\upsilon))\big)\big], \\ H_{k,X,G}(\upsilon) = & \int h_{k,X,G}(dV'; \upsilon), \qquad K_{k,X,G}(dV' \mid \upsilon) = \frac{h_{k,X,G}(dV'; \upsilon)}{H_{k,X,G}(\upsilon)}. \end{aligned} \tag{2.12}$$

This is the numerator and denominator of the localized quotient of LFR I (1.100), with its cutoff (1.101). It is not the complete effective action evaluated at a fresh field. The exterior selector $\chi_{k,\Lambda}(v)$, the preparatory selectors, earlier integrations, and the numerator action $\mathcal{A}''_k$ remain in the old branch weight. They do not enter a different reference denominator.

On the active exterior set, LFR II (1.2)–(1.11) gives a finite, positive reference normalization. The kernel may be extended arbitrarily as a probability on unused null or inactive fibers; the inactive branch itself uses the identity operation.

For a branch whose old variables inside the replacement are u, the operation is exactly

$$(\mathcal{R}_{k,X,G}f)(dV', dv) = K_{k,X,G}(dV' \mid v) \int f(du, dv). \tag{2.13}$$

For the disjoint selected components in (1.100), take the product of (2.12), retain the old-component numerator integrations appearing there, and then sum over the admissible component histories and normalized graph choices. Equation (2.13) is the quotient cancellation performed at each such component, with the actual localized reference specified by (2.12). It proves integral preservation by Tonelli:

$$\int \mathcal{R}_{k,X,G}f = \int f, \qquad \int (R'_k\rho_k) = \int \rho_k. \tag{2.14}$$

The latter is LFR I (1.102). The localization and exponentiation in LFR II reorganize these same integrals. Their signed or complex analytic activities are a representation of the integral and are not an additional stochastic branching rule.

2.5 Exact equality with the normalized Wilson law At every finite cutoff in the admitted source-free domain, expose all primitive selectors, gauge choices, old integration variables, and fresh reference variables. Blocking is (2.1); gauge and selector insertion are normalized identities; coordinate changes have the Jacobians of Section 3; each replacement is (2.13). Consequently their composed positive integral map $\mathcal{P}_{0:k}$ obeys

$$\rho_k = \mathcal{P}_{0:k}\rho_0, \qquad \int \rho_k = \mathcal{Z} = e^{-\mathcal{E}_0} Z_W. \tag{2.15}$$

For any integrable microscopic mark F, its transported numerator is $\mathcal{P}_{0:k}(\rho_0 F)$. Hence

$$\boxed{\frac{\int \mathcal{P}_{0:k}(\rho_0 F)}{\int \rho_k} = \int F \, d\mu_W.} \tag{2.16}$$

The same formula holds coefficientwise for a finite exponential source jet. It does not mean that the original mark may simply be multiplied into the final unmarked density.

3. C-01: one complete current-coordinate density

3.1 Constraint elimination and its determinant At a regular step, choose the same free link coordinates as the source. Let $Q = (Q_E, Q_R)$ be the linearized averaging constraint, including the source's fixed scale factor when present, where the square block Q_E acts on the eliminated coordinates and is invertible in the admitted chart. Put

$$Cz = \begin{pmatrix} -Q_E^{-1}Q_R z \\ z \end{pmatrix}, \qquad QC = 0. \tag{3.1}$$

The exact Euclidean delta identity is

$$\int dB\, \delta(QB) f(B) = \frac{1}{|\det Q_E|} \int dz\, f(Cz). \tag{3.2}$$

It follows by integrating the eliminated variables. It is consistent with the coarea formula, since

$$\frac{\sqrt{\det(C^*C)}}{\sqrt{\det(QQ^*)}} = \frac{1}{|\det Q_E|}. \tag{3.3}$$

To verify (3.3), write $C^*C = I + Q_R^*(Q_E Q_E^*)^{-1}Q_R$ and use the finite-dimensional determinant identity $\det(I+AB) = \det(I+BA)$. Thus no additional factor $\sqrt{\det C^*C}$ may be inserted after (3.2); that would count part of coarea twice. The source factor denoted $z^{(k)}$ includes this linear constraint normalization and its stated convention-dependent constants.

Let D be the source's analytic linearizing function, with $D(0) = D'(0) = 0$, and let h be its right-inverse insertion operator. In the source's normalized free variable,

$$\theta_{k,v}(z) = g_k C z - hD(g_k C z), \qquad V_k(z) = e^{i\theta_{k,v}(z)} V^{(k)}. \tag{3.4}$$

Here all operators and backgrounds depend on retained data v. They are fixed when differentiating in the current free variable z. The full configuration entering the old effective action is

$$\Phi_{k,v}(z) = U_k(V_k(z)), \qquad \Phi_{k,v}(0) = U_{k+1}(v). \tag{3.5}$$

Before the delta constraints are eliminated, the change $B \mapsto B - hD(B)$ contributes the determinant $\det(I - hD'(B))$. Write $\sigma_{\mathrm{in},k}$ for the total Haar density in the chosen real coordinates on all current integration links. Thus it is already a product over those links, not a one-coordinate constant. After fixed normalization factors are separated, the remaining coordinate density is

$$J_{k,v}(z) = \frac{\sigma_{\mathrm{in},k}(\theta_{k,v}(z))}{\sigma_{\mathrm{in},k}(0)} \left|\det(I - hD'(g_k C z))\right|. \tag{3.6}$$

On an oriented real analytic chart the determinant has its continuously chosen positive sign; the absolute value in (3.6) states the real change-of-variable formula independently of that convention.

The fixed factor can be stated without guessing a power of a Haar normalization. Let m be the real dimension before linear-constraint elimination, let r be the real rank of Q, and let $\sigma_{\mathrm{con},k}(0)$ be the total Haar density at the identity in the real coordinates used for the group-valued constraints. Normalized group delta becomes $\sigma_{\mathrm{con},k}(0)^{-1}\delta_{\mathbb{R}^r}$ in those coordinates. The scaling $B = g_k b$ contributes g_k^m, while its linear delta contributes g_k^{-r}. Equation (3.2) then gives the fixed prefactor

$$\mathfrak{c}_{k,v} = g_k^{m-r}\, \frac{\sigma_{\mathrm{in},k}(0)}{\sigma_{\mathrm{con},k}(0)} \frac{1}{|\det Q_E(v)|}. \tag{3.6a}$$

The normalized gauge factor from (2.3), and any separately imposed graph-constraint normalization, remain separate factors with their actual source conventions. Formula (3.6a) concerns precisely the current Haar measure, group averaging constraints, scaling, and linear elimination just described. In particular there is no factor $\sigma_{\mathrm{in},k}(0)^{\dim z}$: that would raise an already multiplied Haar density to an unjustified power. All these fixed factors cancel only when conditioning on the same retained datum. They remain in its marginal and in the history normalizers.

3.2 The physical action identity fixes every sign Let $G_k(z;v)$ be the actual axial gauge action generated by (2.3), evaluated on (3.4). It has $G_k(0;v) = 0$. Define the complete homogeneous classical difference

$$\mathcal{S}_{k,v}(z) = g_k^{-2}\left[A(\Phi_{k,v}(z)) - A(U_{k+1}(v)) + G_k(z;v)\right]. \tag{3.7}$$

The stationary constrained background gives $D\mathcal{S}_{k,v}(0) = 0$. Define its actual constrained Hessian and exact nonlinear remainder by

$$K_{k,v} = D^2\mathcal{S}_{k,v}(0), \qquad \mathcal{V}_{k,v}(z) = \mathcal{S}_{k,v}(z) - \tfrac{1}{2}\langle z, K_{k,v} z\rangle. \tag{3.8}$$

In the source coordinates $K_{k,v} = C^* \Delta^{(k)} C$, with the full background-current contribution already inside $\Delta^{(k)}$. It is not obtained by discarding the nonlinear constraint before differentiating.

For any function F of the old background and retained old fields, write

$$\Delta_z F = F(\Phi_{k,v}(z), \mathrm{a}) - F(U_{k+1}(v), \mathrm{a}). \tag{3.9}$$

For the physical coefficient $w_k(x) = g_k(x)^{-2}$, the elementary identity

$$-\Delta_z A(w_k, \cdot) = -g_k^{-2} \Delta_z A(1, \cdot) - \Delta_z A(w_k - g_k^{-2}, \cdot) \tag{3.10}$$

determines the heterogeneous Wilson term. In particular the last sign is fixed by the original negative action, independently of any shorthand for a remainder.

After dividing by the value of the background exponential at $z = 0$, the complete unshifted current-coordinate exponent is therefore

$$\boxed{\begin{aligned} \mathcal{L}_{k,v}(z) = & -\tfrac{1}{2}\langle z, K_{k,v} z\rangle - \mathcal{V}_{k,v}(z) + \log J_{k,v}(z) \\ & + \Delta_z \mathrm{E}_k + \Delta_z \mathrm{R}_k + \Delta_z \mathrm{B}_k - \Delta_z A(w_k - g_k^{-2}, \cdot). \end{aligned}} \tag{3.11}$$

Equation (3.11) is an exact identity obtained from (2.5) and the change of variables. It is not a claim that its seven displayed contributions are small. The E differences include the already chosen local coupling counterterms of (2.8). The R differences include their actual local carriers. The B differences include all old boundary activities and their dependence on retained old fluctuations. None is discarded because it is absent from a homogeneous approximation.

Older Gaussian factors. On the C-01 coordinates used here, fixing an older fluctuation or selector can leave its Gaussian density dependent on the current field. For any such unintegrated factor, (3.11) includes the logarithm of its ratio at the current field to its value at zero. In a completed component the source integrates the entire boundary exponential. Conditional density and older Gaussian factors

The factors of the older operation $\mathcal{T}_k$ retained outside the current integral in CMP119 (3.15) and (3.25) stay in the outer law. This separation is the one provided by the source's supported operations, not a claim of independence for an arbitrary choice of a block. If a different coordinate selection makes one of those factors depend on the free variable, its exact logarithmic ratio must be retained in (3.11); there is no permission to freeze it merely by notation.

3.3 Boundary relocation, including its full Jacobian The boundary localization of CMP119 (1.22) and (3.22) uses a map supported on the specified set S_{k+1}. With a cutoff ϑ_b taking values in $[0, 1]$, the linkwise formula is

$$\mathcal{F}_{H,b}(t_b) = \frac{1}{i g_k} \log\left(e^{i g_k t_b} e^{-i \vartheta_b(t_b) H_b}\right). \tag{3.12}$$

The cutoff equals the source's smooth interpolant on the appropriate bad-fluctuation boundary links and is identically one on the other shifted links. The background difference H is exactly (3.22), obtained from the two source minimizing fields with the specified determining sets. On unshifted coordinates $\mathcal{F}_H$ is the identity. All these choices are frozen at zero source.

The exact new boundary contribution can be written without losing a Taylor remainder:

$$\mathcal{B}_{\partial,k,v}(t) = \mathcal{L}_{k,v}(\mathcal{F}_H(t)) - \mathcal{L}_{k,v}(t) + \log|\det D\mathcal{F}_H(t)|. \tag{3.13}$$

This is a complete definition of the source's boundary-change contribution before its subsequent localized expansion. It includes the change of every old term and the full Jacobian. In particular one may either use the pulled-back $\mathcal{L}$ and the Jacobian, or use $\mathcal{L} + \mathcal{B}_\partial$; using both would count the change twice.

Let $\mathcal{D}_{k,v}$ be the intersection of the transformed real selector domains specified by the chosen primitive history. It includes the current good-field tests, the retained large-field tests, the exterior extension test where relevant, and every surviving preparatory test. The exact normalized law is

$$\boxed{d\nu_{k,v}(t) = \frac{1_{\mathcal{D}_{k,v}}(t)\exp[\mathcal{L}_{k,v}(t) + \mathcal{B}_{\partial,k,v}(t)]\,dt}{\displaystyle\int_{\mathcal{D}_{k,v}} \exp[\mathcal{L}_{k,v}(s) + \mathcal{B}_{\partial,k,v}(s)]\,ds}.} \tag{3.14}$$

Only active positive-mass fibers are used. An R-reference fiber uses (2.12) instead; an arbitrary inherited coarse fiber is not substituted for that reference law.

3.4 Conditioning on the interior: the Schur complement is not a second action To identify the quadratic terms in CMP119 (3.23)–(3.25), write $t = (A, y)$, where A is free in Λ_{k+1} and y is the retained current fluctuation outside it. Put $C_\Lambda = K_{\Lambda\Lambda}^{-1}$, extended by zero outside Λ. Then

$$\begin{aligned} -\tfrac{1}{2}\langle A + y, K(A + y)\rangle = &\left[-\tfrac{1}{2}\langle y, Ky\rangle + \tfrac{1}{2}\langle y, KC_\Lambda Ky\rangle\right] \\ &+ \left[-\tfrac{1}{2}\langle A, KA\rangle - \langle A, Ky\rangle - \tfrac{1}{2}\langle y, KC_\Lambda Ky\rangle\right]. \end{aligned} \tag{3.15}$$

The first bracket is the retained Gaussian factor in the new $\mathcal{J}^{(k)}$. The second is the inner Gaussian exponent, equivalently the completed square around $-C_\Lambda Ky$. Their sum is the original quadratic form. This proves the bookkeeping identity and shows exactly where the retained mean occurs. Dropping the cross term would change the conditional law.

The normalized conditional law of A follows by fixing y in (3.14). All nonlinear terms in (3.11) and (3.13) are evaluated at the same pair (A, y). Constants independent of A cancel in this conditional normalization, but they remain in the marginal law of y and in history weights.

3.5 Terms and domains

Contribution	Exact place in the current law	Allocation
Full constrained homogeneous Hessian	$-\langle z, Kz\rangle/2$	Retain the source background-current contribution and constraint curvature.
All higher classical/gauge terms	$-\mathcal{V}$	Defined as the exact difference in (3.8); no nonlinear term omitted.
Heterogeneous Wilson coefficient	$-\Delta A(w_k - g_k^{-2}, \cdot)$	Sign follows from (2.5), not from a shorthand P.
Haar density	$\log[\sigma_{\mathrm{in},k}(\theta)/\sigma_{\mathrm{in},k}(0)]$ inside $\log J$	Insert once; the total input/constraint constants occur in (3.6a).
Nonlinear constraint-linearization Jacobian	$\log\lvert\det(I - hD')\rvert$ inside $\log J$	Insert once before elimination, as in the source change of variables.
Linear constraint coarea	$\lvert\det Q_E\rvert^{-1}$, together with fixed scale factors	Outer factor for fixed v; keep in marginal/history normalizers.
Inherited E activities and their local subtraction	$\Delta\mathrm{E}_k$, expanded by (2.8)	Do not add the same coupling subtraction again.
Inherited R activities	$\Delta\mathrm{R}_k$, expanded by (2.9)	R analytic sizes are not normalized probabilities.
Inherited boundary activities	$\Delta\mathrm{B}_k$, expanded by (2.10)	Keep old fluctuation and boundary-data dependence.
Boundary-coordinate change	$\mathcal{B}_\partial$	Either use (3.13), or pull back all terms and include its Jacobian.
Current and historical selectors	$1_{\mathcal{D}_{k,v}}$	Keep actual real support; differentiating a moving support is a separate estimate.

Contribution	Exact place in the current law	Allocation
Retained Gaussian/mean terms	The two brackets of (3.15)	Preserve cross terms and outer Schur complement.
History-dependent vacuum scalar	Outer $e^{-\mathcal{E}_k(h)}$	Cancels within a fixed conditional fiber, not between histories.
Fresh replacement normalization	$H_{k,X,G}(v)$ in (2.12)	Use the localized reference action and its own graph/cutoff.

This closes an identification problem. It does not give the current-coordinate complex tubes, derivative-carrier bounds, selector boundary estimates, or full conditional moment and decay estimate required by C-02 through C-08.

4. H-01: exact normalized history and reset laws

4.1 Primitive path construction For a fixed finite construction, start with $U_0 \sim \mu_W$. At each chronological operation:

1. expose the next finite selector label according to its actual partition of unity;
2. expose any gauge choice using its normalized gauge or finite graph probability;
3. apply the deterministic blocking or coordinate map, retaining old variables as latent variables;
4. for each active R replacement, sample from the actual kernel (2.12).

The resulting finite path law is the product of μ_W and these probability kernels. All compact field spaces, their admitted finite-dimensional coordinate charts, and the finite label spaces are standard measurable spaces; regular conditional laws can be used on their completed probability spaces. The original Wilson field remains a coordinate of this construction, with exactly its original marginal.

Let h denote the exposed history through scale k, including primitive selector and graph labels before any resummation. Let $r_{k,h}(dV_k)$ denote its unnormalized output measure. Expanding only primitive positive integrals in (2.4) defines these measures, and

$$r_{k,h} \geq 0, \qquad \rho_k(dV_k) = \sum_h r_{k,h}(dV_k), \qquad Z_{k,h} = \int r_{k,h}(dV_k), \qquad \sum_h Z_{k,h} = \mathcal{Z}. \tag{4.1}$$

If some retained labels are continuous, the same formulas use the corresponding disintegration rather than a finite sum.

The exact normalized laws are

$$\boxed{\mathbb{P}(H_k = h, dV_k) = \frac{r_{k,h}(dV_k)}{\mathcal{Z}}, \quad \mathbb{P}(H_k = h) = \frac{Z_{k,h}}{\mathcal{Z}}, \quad \mathbb{P}(H_k = h \mid V_k) = \frac{dr_{k,h}}{d\rho_k}(V_k).} \tag{4.2}$$

The Radon–Nikodym derivative exists since $r_{k,h} \leq \rho_k$. When both measures are represented against a common reference measure it is the familiar quotient $r_{k,h}(V_k)/\rho_k(V_k)$. This formulation also covers singular gauge-fixed outputs without a false Haar-density assumption. All formulas are on nonzero-mass sets; arbitrary versions on null sets make no difference. They give the law of a retained history as an exact integral of the actual normalized Wilson construction. The denominator is not replaced by the mass of a convenient reference Gaussian.

Vacuum-normalization warning. The common factor $e^{-\mathcal{E}_0}$ cancels from (4.2). The factors $e^{-\mathcal{E}_k(h)}$ generally depend on the history and are part of $r_{k,h}$. Removing them from individual branch weights would usually change both probabilities in (4.2).

4.2 Old and fresh conditional laws at a replacement For an old normalized branch density $\rho(du, dv)/\mathcal{Z}$, the exact old–fresh coupling is

$$d\Pi(u, u', v) = \frac{\rho(du, dv)}{\mathcal{Z}} \frac{h_{k,X,G}(du'; v)}{H_{k,X,G}(v)}. \tag{4.3}$$

Its new marginal is the R quotient (2.13). Its old conditional law, given the augmented output (u', υ), is

$$\Pi(du \mid u', \upsilon) = \frac{\rho(du, \upsilon)}{\int \rho(d\tilde{u}, \upsilon)}. \tag{4.4}$$

The fresh reference factor cancels from this conditional law. Thus the transported observable is

$$(QF)(u', \upsilon) = \frac{\int F(u, \upsilon)\rho(du, \upsilon)}{\int \rho(du, \upsilon)}. \tag{4.5}$$

The retained data υ must include the history, graph and exterior data deciding which kernel was applied. Conditioning later on an omitted rare event would give a different kernel; (4.5) is not permission to ignore that event's normalizer.

The fresh denominator $H_{k,X,G}$, the old conditional denominator $\int \rho(du, \upsilon)$, and the history normalizer $\rho_k(V_k)$ in (4.2) are different objects. Keeping all three separate prevents the incorrect transfer of a fresh-reference estimate to an inherited conditional history.

4.3 Resetting an auxiliary history label Let the actual joint law at a reset be

$$d\mu(x, h) = d\bar{\mu}(x)\, \pi(dh \mid x), \tag{4.6}$$

where π is obtained from (4.2) or its corresponding conditional version. The exact reset joint law is

$$d\Pi_{\text{reset}}(x, h, h') = d\bar{\mu}(x)\, \pi(dh \mid x)\, \pi(dh' \mid x). \tag{4.7}$$

Both old and new marginals equal (4.6). If the mark is $F(x, h)$, the transported new mark is

$$\widetilde{F}(x, h') = \int F(x, h)\pi(dh \mid x). \tag{4.8}$$

For a full finite source jet $G(x, h; t)$, apply (4.8) to the entire truncated polynomial:

$$\widetilde{G}(x, h'; t) = \int G(x, h; t)\pi(dh \mid x). \tag{4.9}$$

Every integrated coefficient is preserved. If $G = e^{\Psi}$, taking the finite formal logarithm after averaging generates all conditional covariance and higher cumulant responses. Averaging only each one-field coefficient and then multiplying would omit those responses.

Equations (4.1)–(4.9), with the actual primitive selectors and reference kernel (2.12), are the normalized-history construction required for marked expectations. They establish no smallness of a bad-history probability. H-02 through H-05 retain that analytic task.

5. B-02: explicit geometric progress and the remaining obstruction

5.1 Compatible cubic tori CMP109 p.251 takes an odd block integer with a lower restriction at 11, torus half-side a power of that integer, and nested center lattices with spacing a negative power. We use an admitted odd $L \geq 13$, so the endpoint ambiguity in the printed text is immaterial. These are cubic tori. No rectangular extension is required for the following construction.

Fix an admitted odd L and fixed block constants supplied by the source hypotheses. The elementary numerical choices $L = 13$, $L_0 = 2$, and domain parameter $\beta_{\text{dom}} = 1/8$ satisfy the displayed restrictions $L \geq 11$, $2 \leq L_0 < L/2$, $L_0^2 \leq L/3$, and $0 < \beta_{\text{dom}} < 1/4$. This checks these numerical restrictions only; it is not a claim that every other source parameter or coupling condition is thereby verified.

Write the source radii at a finite cutoff as

$$R_{j,n} = L^{r_{j,n}}, \qquad R_{j,n} \geq [\log g_{j,n}^{-2}]^r, \qquad 0 \leq j \leq K_n, \tag{5.1}$$

with the source's least-power choice and fixed $r \geq 2$. Choose an integer m_* large enough for all the fixed block exponents and finite halo multipliers being used. An explicit adaptive sequence is

$$K_n = n, \qquad M_n = n + m_* + 2 + \max_{0 \leq j \leq n} r_{j,n}, \qquad a_n = L^{-n}\ell_0, \quad T_n = 2L^{M_n}\ell_0. \tag{5.2}$$

Every source grid whose size is a fixed power of L times $R_{j,n}$ at one of the first n scales divides the period when m_* includes its fixed exponent. Its finite surrounding layers fit inside the period for sufficiently large m_*. The extra factor L^n gives an arbitrarily large separation from any fixed support and from these fixed multiples. Any further primary geometric condition involving a different scale-dependent family must be inserted into the maximum before claiming the complete B-02 test.

The number of microscopic sites per direction is

$$N_n = T_n/a_n = 2L^{M_n+n}, \tag{5.3}$$

which is even. In the source's original coordinates, reflection of the center lattice through time zero sends $a_n(j+1/2)$ to $a_n(-j-1+1/2)$. This proves lattice invariance, but the original coordinate plane is a bond plane. It does not by itself verify the site-reflection convention used by the chosen field architecture.

The conventions agree after one fixed translation: move the coordinate origin to the coarsest center $\ell_0/2$ in each coordinate. At stopped scale j, put $h = L^{j-K_n}\ell_0$. Since L is odd,

$$h(\mathbb{Z} + \tfrac{1}{2}) - \tfrac{\ell_0}{2} = h\left(\mathbb{Z} + \frac{1 - L^{K_n - j}}{2}\right) = h\mathbb{Z}. \tag{5.3a}$$

Moreover $T_n/(2h) = L^{M_n+K_n-j}$ is an integer. Thus the planes with translated time coordinates 0 and $T_n/2$ are site planes at every stopped scale. This exact coordinate dictionary permits the site-reflection convention of the field construction to be used on the same cubic sequence; it is stronger than mere reflection invariance of the unshifted center grid. Since $T_n \to \infty$, every fixed compact support eventually embeds with a fixed positive boundary margin.

The choice also pays the elementary geometric condition used in the compact reserve. Indeed $\beta_{W,0,n} = g_{0,n}^{-2}/N$, while N_n dominates $R_{0,n}$. Thus, for weak couplings bounded above by a fixed small number,

$$\sup_n \frac{\log(1 + \beta_{W,0,n})}{N_n} < \infty. \tag{5.4}$$

For large $\log g_{0,n}^{-2}$, its numerator grows linearly in that logarithm and $R_{0,n}$ grows at least as its power $r \geq 2$; the remaining finite range is bounded. This does not use asymptotic freedom to prove a coupling recursion.

5.2 Why the complete statement remains open The B-02 statement asks for every actual minimum-size, coarse-grid, reflection-plane, and coupling-ceiling condition. Equations (5.2)–(5.4) settle a useful geometric subproblem. They do not establish

$$0 < g_{j,n} \leq \gamma \quad \text{for every } 0 \leq j \leq K_n \tag{5.5}$$

for the actual Wilson-normalized trajectory required by the source-free theorem. Nor does naming an admitted L replace a final exhaustive constants ledger. The actual trajectory and its source identification belong to B-03 and B-05. The stated application requires the trajectory and its uniform coupling ceiling.

6. Literal-source comparisons and their handling

6.1 Graph-average normalization On LFR I p.201 the display (1.100) places $1/N_i$ before a graph sum, while the following prose writes $N_i = 1/(2^d d!)$ for the signed-coordinate averaging choice. Read literally, those two uses of N_i conflict. This note uses the unambiguous normalized averaging measure: each signed-coordinate choice has mass $1/(2^d d!)$. This agrees with the word "average" and with the integral normalization (1.102). The notation clash is not interpreted as a failure of the R mechanism.

6.2 The heterogeneous action sign CMP119 (2.23) has $-A(w_k, U)$. The displayed curly bracket in (3.15), on p.268, has

$$+A(w_k - g_k^{-2}, \Phi(z)) - A(w_k - g_k^{-2}, U_{k+1}). \tag{6.1}$$

The direct physical-action decomposition (3.10) has the opposite sign. CMP109's prose after (2.12) identifies $P^{(k)}$ with the homogeneous action and measure contribution. This is therefore recorded as an unresolved cross-paper sign/convention comparison, not as a proved misprint.

The current-density identity (3.11) does not rely on resolving that comparison: it is derived directly from the physical exponent (2.5). Before importing a bound stated for the source's shorthand $P^{(k)}$ into this exact density, the correspondence must be checked. This is part of the remaining conditional-law estimates, not an extra unstated premise of (3.11).

6.3 The printed free-field cutoff CMP119 p.267 states the scaling $A'_k = g_k A_k$; its (3.16) on p.268 prints a good-field test $|A_k| < \delta_k$. The initial-step expression (1.18) instead displays $|A| < g_0^{-1}\delta_0$. This note does not silently identify the two radii. Any literal cutoff gives an exact partition of unity when its complement is kept, so the normalized identities above remain valid. Which current normalized radius is intended by a subsequently quoted analytic bound must be checked before that bound is used. The real domain $\mathcal{D}_{k,v}$ in (3.14) is the actual selected domain, not an enlarged Gaussian domain chosen for convenience.

6.4 No hidden source imports The published source-free construction assumes small effective couplings. A Wilson application must verify this hypothesis on its normalized trajectory.

Every remaining application condition is assigned to an existing open statement:

Remaining condition	Catalogue statement retaining it
Derive the actual effective coupling recurrence and its quantitative remainder.	B-03
Identify that trajectory with the physical Wilson coefficient and verify all source-free coupling hypotheses.	B-05
Exhaust all source parameter/minimum-size conditions and verify the ceilings along one complete cofinal sequence.	B-02
Admit the complete E contribution, with the correct current coordinates, classical term and heterogeneous correction, on a common contour.	C-02
Admit the actual R activities and their current-coordinate marginal transfer.	C-03
Bound the resulting actual E/R derivative carriers and the constants needed to compare any imported shorthand $P^{(k)}$ with (3.11).	C-04
Bound old boundary activities and the full boundary-coordinate contribution (3.13).	C-05
Control heterogeneous determining sets, backgrounds and nonconstant coefficients in (3.11).	C-06

Remaining condition	Catalogue statement retaining it
Fix the actual current cutoff convention for the estimates, retain its real support, and treat selectors and their boundaries.	C-07
Prove the complete conditional moment and spatial decay theorem for (3.14), rather than a homogeneous surrogate.	C-08
Identify and bound inherited or unreplaced bad events under (4.2).	H-02
Localize all responses generated by (4.9).	H-03
Sum marked bad-history contributions with their actual normalizers.	H-04
Control the outer retained-mean covariance and higher cumulants.	H-05

Annex X — A separated normalization prescription and rotational compatibility equations

Annex X. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

Finite normalization and continuum hypotheses

The finite construction gives seven separated physical-scale normalization conditions and explicit rotational compatibility equations. Applying them to the limiting hierarchy requires control of the generated contact terms and a common hierarchy estimate.

The seven-channel representation classification, clover fields, Wilson scalar repair, and positive-half-space square identity are retained results from Annex D and Annex E. The new finite construction gives explicit orthogonal projectors, seven separated reflection-positive normalization functionals at a fixed physical scale, a positive free reference value for each, the resulting normalization equations and determinant, and the three rotational compatibility defects expressed in those actual finite-cutoff data.

No numerical coefficient from a Gaussian benchmark is identified with an interacting RG coefficient. The actual nondegeneracy and logarithmic bounds along the cofinal sequence are still F-05. The vanishing of the rotational defects is not proved here.

1. Conventions and the twenty-one representatives

Work on the even-sided Wilson torus with gauge group $SU(N)$, normalized Haar measure and action

$$d\mu_{a,T}(U) = Z_{a,T}^{-1} \exp\left[-\beta_a \sum_p s(U_p)\right] \prod_\ell dU_\ell, \qquad s(U) = N - \operatorname{Re}\operatorname{Tr} U. \tag{F.1}$$

The normalization dictionary is fixed:

$$\beta_a = 2N x_{0,a}, \qquad X_{B,a} = N\beta_a = 2N^2 x_{0,a}. \tag{F.2}$$

Coordinates are 0, 1, 2, 3, with time coordinate 0. Let

$$\mathcal{J} = (01, 02, 03, 12, 13, 23), \qquad E = \operatorname{Sym}^2(\mathbb{R}^{\mathcal{J}}), \qquad \langle Q, Q'\rangle_E = \operatorname{Tr}(QQ'). \tag{F.3}$$

Thus E has dimension 21. Its orthonormal component basis consists of the six diagonal matrices and $(E_{AB}+E_{BA})/\sqrt{2}$ for $A<B$. A symmetric matrix Q therefore has off-diagonal orthonormal coordinates $\sqrt{2}Q_{AB}$, rather than Q_{AB}. This convention removes an otherwise easy factor-of-two ambiguity in finite Gram systems.

For signed orthogonal unit vectors u, v, write $P_{u,v}(x)$ for the oriented elementary plaquette rooted at x. Put

$$C_{\mu\nu,a}(x) = \frac{1}{8i} \sum_{\sigma,\tau=\pm1} \sigma\tau\, \pi_0\big(P_{\sigma e_\mu, \tau e_\nu}(x) - P_{\sigma e_\mu,\tau e_\nu}(x)^\dagger\big), \qquad C_{\nu\mu,a} = -C_{\mu\nu,a}, \tag{F.4}$$

where $\pi_0 H = H - \operatorname{Tr}(H) I/N$. Define

$$q_{AB,a}(x) = a^{-4} \operatorname{Tr}(C_{A,a}(x) C_{B,a}(x)), \qquad t_a = \sum_A q_{AA,a}, \tag{F.5}$$

and retain the exact Wilson scalar through

$$w_a(x) = \frac{1}{4a^4} \sum_{\mu<\nu} \sum_{\sigma,\tau=\pm1} s(P_{\sigma e_\mu, \tau e_\nu}(x)), \qquad \tilde{q}_a = q_a + \frac{I_6}{6}(2w_a - t_a). \tag{F.6}$$

These are the retained scalar-repaired clover representatives. In particular,

$$\frac{1}{2}\operatorname{Tr}\tilde{q}_a = w_a, \qquad a^4 \sum_x w_a(x) = \sum_p s(U_p). \tag{F.7}$$

The 21 orthonormal components of $\tilde{q}$ are the field representatives in this prescription. Every component is real, gauge invariant, and supported on links with endpoints in $x + [-a, a]^4$. Their exact signed-permutation action is

$$\tilde{q}_a(x; gU) = D(g)\tilde{q}_a(gx; U), \qquad D(g)Q = (\Lambda^2 g)Q(\Lambda^2 g)^t. \tag{F.8}$$

Fine-lattice translations have no tensor factor. The scalar repair changes no nonscalar channel.

2. Explicit orthogonal projectors

Extend $Q_{ij,kl}$ antisymmetrically in each index pair and symmetrically between pairs. All formulas below are linear on E. Entries not specified in a projector are zero.

2.1 The diagonal sector

Let

$$d_{ij} = Q_{ij,ij} = d_{ji}, \qquad s = \sum_{i<j} d_{ij}, \qquad r_i = \sum_{j\neq i} d_{ij} - \frac{s}{2}. \tag{F.9}$$

Then $\sum_i r_i = 0$. Define

$$(P_S Q)_{ij,ij} = \frac{s}{6}, \qquad (P_{T_d} Q)_{ij,ij} = \frac{r_i + r_j}{2}, \tag{F.10}$$

$$(P_{W_d} Q)_{ij,ij} = d_{ij} - \frac{s}{6} - \frac{r_i + r_j}{2}. \tag{F.11}$$

The images have dimensions 1, 3 and 2. The factor 1/2 in P_{T_d} is essential: an input $d_{ij} = u_i + u_j$, $\sum_i u_i = 0$, has $r_i = 2u_i$.

2.2 The shared-index sector

For $i < j$, let $k < l$ be the complementary indices and put

$$x = Q_{ik,jk}, \qquad y = Q_{il,jl}.$$

Use the already specified pair antisymmetries to convert these entries back to the ordered-pair matrix. Set

$$P_{T_o} : (x, y) \longmapsto \left(\frac{x+y}{2}, \frac{x+y}{2}\right),$$

$$P_{W_s} : (x, y) \longmapsto \left(\frac{x-y}{2}, \frac{y-x}{2}\right). \tag{F.12}$$

Their image dimensions are 6 and 6. Every shared-index matrix entry belongs to exactly one such two-coordinate block, with the stipulated orientation signs.

2.3 The disjoint-index sector

Let

$$p_1 = Q_{01,23}, \qquad p_2 = -Q_{02,13}, \qquad p_3 = Q_{03,12}, \qquad \bar{p} = \frac{p_1 + p_2 + p_3}{3}. \tag{F.13}$$

Define

$$P_P : (p_1, p_2, p_3) \longmapsto (\bar{p}, \bar{p}, \bar{p}),$$

$$P_{W_p} : (p_1, p_2, p_3) \longmapsto (p_1 - \bar{p}, p_2 - \bar{p}, p_3 - \bar{p}). \tag{F.14}$$

Their dimensions are 1 and 2. P_P is the pseudoscalar projector; the minus sign in p_2 is part of the convention.

These seven maps obey

$$P_h^2 = P_h = P_h^*, \qquad P_h P_k = 0\ (h \neq k), \qquad \sum_h P_h = I_E, \tag{F.15}$$

with

$$h = (S, T_d, W_d, T_o, W_s, P, W_p), \quad d_h = (1, 3, 2, 6, 6, 1, 2). \tag{F.16}$$

Indeed, the three diagonal spaces are mutually orthogonal: the stress coefficients have zero sum and a diagonal Weyl coefficient has zero incident sum at each vertex. The latter property also proves stress/Weyl orthogonality. The other two decompositions are orthogonal sum/difference or mean/zero-sum decompositions in equal-weight coordinates. Their sum is the identity separately on the diagonal, shared-index and disjoint-index sectors. This proves (F.15) directly. Their equivariance and their correspondence to the seven irreducible signed-hypercubic channels are the retained classification.

The unit scalar vector is

$$e_S = I_6/\sqrt{6}. \tag{F.17}$$

It spans the only identity-mixing direction. The invariant expectation is

$$\mathbb{E}\tilde{q}_a(x) = m_{a,T} e_S, \qquad m_{a,T} = \mathbb{E}\langle e_S, \tilde{q}_a(0)\rangle_E. \tag{F.18}$$

Here the origin can be replaced by any lattice vertex. The equality follows from exact translation and hypercubic symmetry of the actual Wilson measure; no limiting theorem is used.

3. A separated normalization measurement at the physical scale

The prescription must not normalize using an ordinary same-support variance that may be dominated by contact divergences. The following measurements are genuine separated positive-half-space pairings.

Fix $\mu > 0$, $\varepsilon = 1/64$, and the explicit radial bump

$$\eta(z) = c_\eta \begin{cases} \exp[-1/(1-|z|^2)], & |z| < 1, \\ 0, & |z| \geq 1, \end{cases} \qquad \int_{\mathbb{R}^4} \eta(z)\, dz = 1. \tag{F.19}$$

Set

$$f_\mu(x) = \mu^4 \varepsilon^{-4} \eta\Big(\frac{\mu x - e_0}{\varepsilon}\Big). \tag{F.20}$$

Its support lies in a ball of radius ε/μ centered at e_0/μ. Write $r_0 = \mathrm{diag}(-1,1,1,1)$ and $\theta x = r_0 x$. The two supports $\mathrm{supp}\, f_\mu$ and $\theta\, \mathrm{supp}\, f_\mu$ have distance at least $2(1-\varepsilon)/\mu$.

Use tori with $T > 8/\mu$, and $a < (1-\varepsilon)/(2\mu)$, so the positive field stencils fit strictly inside the positive half and neither lobe approaches a torus seam. Every admissible cofinal sequence eventually meets these inequalities. This uses only its eventual geometric embedding; it does not verify the separate RG geometry statement B-02.

For any real equivariant field A_a, define its centered smeared vector

$$B_a(A) = a^4 \sum_x f_\mu(x)\big(A_a(x) - \mathbb{E}A_a(x)\big) \in E. \tag{F.21}$$

For an orthonormal basis $e_{h,1}, \dots, e_{h,d_h}$ of $P_h E$, put

$$\mathcal{N}_{h,a,T}(A) = \frac{1}{d_h} \sum_{i=1}^{d_h} \mathbb{E}\Big[\overline{\langle e_{h,i}, B_a(A;\theta U)\rangle_E} \langle e_{h,i}, B_a(A;U)\rangle_E\Big]. \tag{F.22}$$

This is independent of the chosen orthonormal basis. In terms of the connected two-point distribution it is

$$\mathcal{N}_{h,a,T}(A) = \frac{a^8}{d_h} \sum_{x,y} f_\mu(x) f_\mu(y) \mathbb{E}\left[\langle D(r_0) P_h A_a^c(\theta x), P_h A_a^c(y)\rangle_E\right]. \tag{F.23}$$

It is a finite normalization functional on the two-point distribution, with the fixed tensor contraction shown. Its two external source arguments stay a fixed physical distance apart. Point-local contact changes do not alter it.

Proposition F.1 — the actual normalization data are nonnegative For the compact Wilson law and the stated cutoff margins,

$$0 \leq n_{h,a,T} := \mathcal{N}_{h,a,T}(\tilde{q}) < \infty. \tag{F.24}$$

Proof. Each summand of (F.22) is the original Wilson site-reflection Gram form of a bounded positive-half-link function. The compactness of the finite link space gives finiteness. The exact site-reflection square formula proves

nonnegativity. Summation and division by d_h preserve it. This uses the actual compact measure, not an auxiliary RG law. ■

Strict positivity is not asserted merely from compactness or from reflection positivity. For example, at $\beta = 0$, a centered field supported strictly away from the boundary has zero conditional mean on the reflecting-plane links, so its site-reflection norm vanishes. The asymptotically free sequence has positive large β, but excluding small or zero values of (F.24) on that sequence is still a real nondegeneracy task.

4. Explicit positive reference values

Fix generators T^a with $\mathrm{Tr}(T^a T^b) = \delta^{ab}/2$. As a normalization benchmark only, take independent unit-coupling free Euclidean potentials with covariance $\delta^{ab}\delta_{\mu\nu}/(4\pi^2|x-y|^2)$. Their curvature covariance at distinct points is

$$\mathbb{E}F_A^a(x)F_B^b(y) = \frac{\delta^{ab}}{\pi^2|x-y|^4}(\Lambda^2 R_{x-y})_{AB}, \qquad R_r = I_4 - 2\hat{r}\hat{r}^t. \tag{F.25}$$

This follows by differentiating the displayed potential covariance. The quartic term in the wedge of $I - 2\hat{r}\hat{r}^t$ cancels, giving the familiar curl-curl expression directly.

Let $Q_{AB}^* =: \mathrm{Tr}(F_A F_B) :$ in the benchmark. Wick contraction gives, for $u, v \in E$,

$$\mathbb{E}\langle u, Q^*(x)\rangle_E \langle v, Q^*(y)\rangle_E = c_N|x-y|^{-8}\langle u, D(R_{x-y})v\rangle_E, \quad c_N = \frac{N^2-1}{2\pi^4}. \tag{F.26}$$

Only distinct-point pairings are used, so no extension of this benchmark through coincidence sets is needed here.

Define the seven fixed reference numbers by the finite integrals

$$\lambda_h(\mu) = \frac{c_N}{d_h}\int f_\mu(x) f_\mu(y)|\theta x - y|^{-8}\,\mathrm{Tr}_E\left[P_h D(r_0) P_h D(R_{\theta x-y}) P_h\right] dx\, dy. \tag{F.27}$$

These are explicit real numbers determined by (F.10)–(F.20) and the fixed trace convention. In particular $\lambda_h(\mu) = \mu^8 \lambda_h(1)$. They are fully specified finite reference integrals, rather than unspecified matching coefficients; no numerical quadrature is required by this prescription.

Proposition F.2 — every reference value is strictly positive For every channel,

$$\frac{c_N(1 - 8\varepsilon/(1-\varepsilon))}{[2(1+\varepsilon)]^8}\,\mu^8 \le \lambda_h(\mu) \le \frac{c_N}{[2(1-\varepsilon)]^8}\,\mu^8. \tag{F.28}$$

Proof. On the two supports,

$$2(1-\varepsilon)/\mu \le |\theta x - y| \le 2(1+\varepsilon)/\mu.$$

The sine of the angle of $\theta x - y$ to the time axis is at most $\varepsilon/(1-\varepsilon)$. Hence

$$\|R_{\theta x-y} - r_0\| \le 2\varepsilon/(1-\varepsilon).$$

The tensor action D is a restriction of the fourth tensor power, so telescoping the four tensor factors gives

$$\|D(R_{\theta x-y}) - D(r_0)\| \le 8\varepsilon/(1-\varepsilon).$$

The projector P_h commutes with $D(r_0)$, and the latter is an orthogonal involution. At $R = r_0$, the trace in (F.27) equals d_h; its difference from d_h is at most d_h times the last norm bound. Its absolute value is at most d_h, because it is the trace of a product of contractions on $P_h E$. Integrating against the two probability densities proves (F.28). With $\varepsilon = 1/64$, the lower prefactor is positive. ■

The free value fixes a convention for the finite normalization. It does not prove that the interacting data $n_{h,a,T}$ have this asymptotic, nor that the induced fields have the desired interacting short-distance coefficients. Those are F-05 and U-01–U-03.

5. The finite system, identity coefficient and determinant

The proposed physical-scale conditions are

$$\mathbb{E}\Phi_a(x) = 0, \qquad \mathcal{N}_{h,a,T}(\Phi) = \lambda_h(\mu) \quad (h = S, T_d, W_d, T_o, W_s, P, W_p), \tag{F.29}$$

with positive channel orientation. Write

$$Z_{a,T,\mu} = \sum_h z_{h,a,T,\mu} P_h, \qquad \Phi_a = Z_{a,T,\mu}\tilde{q}_a - c_{a,T,\mu} e_S. \tag{F.30}$$

Because $P_h Z = z_h P_h$ and the connected normalization functional removes constants,

$$\mathcal{N}_{h,a,T}(\Phi) = z_h^2 n_{h,a,T}, \qquad c = z_S m_{a,T}. \tag{F.31}$$

Consequently, on the explicitly checkable domain

$$\boxed{n_{h,a,T} > 0 \quad \text{for all seven channels,}} \tag{F.32}$$

the unique solution with $z_h > 0$ is

$$\boxed{z_{h,a,T,\mu} = \sqrt{\lambda_h(\mu)/n_{h,a,T}}, \qquad c_{a,T,\mu} = z_{S,a,T,\mu} m_{a,T}.} \tag{F.33}$$

The seven measurements therefore separate the seven channel coefficients, and the mean separates the identity coefficient. The determinant of the eight-variable normalization Jacobian, ordering the seven two-point equations before the scalar mean equation, is

$$\boxed{-2^7 \prod_h z_h n_{h,a,T} \neq 0.} \tag{F.34}$$

Changing the order of equations changes only its sign. This is the promised explicit finite nondegeneracy test.

This completes F-01 as stated: it asks for a concrete prescription and checkable nondegeneracy, leaving actual invertibility and its cutoff control to F-05. The explicit tests require only eventual embedding along B-02, not a new geometric hypothesis.

A total finite-cutoff definition, without hiding nondegeneracy If a coefficient must be defined at every cutoff obeying the geometric margins of Section 3 before (F.32) is established, one may use the following explicitly marked totalization:

$$z_h^{\mathrm{tot}} = \begin{cases} \sqrt{\lambda_h/n_h}, & n_h > 0, \\ 1, & n_h = 0, \end{cases} \qquad c^{\mathrm{tot}} = z_S^{\mathrm{tot}} m. \tag{F.35}$$

Every coefficient in (F.35) is a finite real number determined by the actual compact law, so this defines an actual cutoff field rather than an arbitrary-coefficient placeholder. If $n_h = 0$, it deliberately does not claim to satisfy that channel's nonzero condition in (F.29). F-05 must show that the totalization agrees with (F.33) on the required tail of the cofinal sequence and has the prescribed logarithmic budgets. The fallback is bookkeeping for the cutoff construction, not a solution of a singular normalization system.

The coefficients in either prescription are constant in spacetime. Translating the normalization lobe and its reflecting plane together by a fine-lattice translation leaves the data unchanged by the change of link variables. Applying a signed permutation to the lobe, reflection and channel basis together likewise leaves the data unchanged. Thus using a particular time axis and origin in (F.20) does not insert an uncontrolled spatially varying coefficient.

6. Wick and reflection conventions for this prescription

In the real Euclidean component basis, a component with n_0 occurrences of the time index obeys the reflection sign $(-1)^{n_0}$, where $n_0 \in \{0, 1, 2\}$. All projectors commute with the time reflection, and all coefficients in (F.33) or (F.35) are real. Therefore the normalized fields satisfy exactly

$$\overline{\Phi_a(x; \theta U)} = D(r_0)\Phi_a(\theta x; U). \tag{F.36}$$

For the Hermitian Wick component convention set

$$W_{\alpha\alpha} = i^{n_0(\alpha)}, \qquad \widehat{\Phi}_a = W\Phi_a. \tag{F.37}$$

Then $\overline{W}D(r_0) = W$, so the antilinear reflection on $\widehat{\Phi}$ has no remaining component sign. Other tensor transformations are $\widehat{D}(g) = WD(g)W^{-1}$. The identity vector also changes to We_S; it must not be silently kept as the real Euclidean scalar vector after the Wick basis change.

Use either this Wick convention everywhere or the real Euclidean convention with $D(r_0)$ everywhere. They are equivalent descriptions of the same cutoff Gram forms.

7. R-01: the three conditions in the actual normalization data

The retained classification groups the seven projectors into four rotational types:

$$P_S, \quad P_P, \quad P_T = P_{T_d} + P_{T_o}, \quad P_W = P_{W_d} + P_{W_s} + P_{W_p}. \tag{R.1}$$

For the explicitly constructed normalization map (F.30), its commutation with the canonical $O(4)$ tensor action is equivalent to the three scalar relations

$$z_{T_d} = z_{T_o}, \qquad z_{W_d} = z_{W_s}, \qquad z_{W_p} = z_{W_s}. \tag{R.2}$$

The representation classification proving this equivalence is reused. The new concrete normalization conditions are obtained by substituting (F.33):

$$\boxed{\frac{n_{T_d,a,T}}{\lambda_{T_d}(\mu)} = \frac{n_{T_o,a,T}}{\lambda_{T_o}(\mu)},} \tag{R.3}$$

$$\boxed{\frac{n_{W_d,a,T}}{\lambda_{W_d}(\mu)} = \frac{n_{W_s,a,T}}{\lambda_{W_s}(\mu)}, \qquad \frac{n_{W_p,a,T}}{\lambda_{W_p}(\mu)} = \frac{n_{W_s,a,T}}{\lambda_{W_s}(\mu)}.} \tag{R.4}$$

All quantities are now specified compact Wilson integrals or the fixed positive integrals (F.27). The equivalent logarithmic defects are

$$d_T = \frac{1}{2}\log\frac{n_{T_o}/\lambda_{T_o}}{n_{T_d}/\lambda_{T_d}},$$

$$d_{W_d} = \frac{1}{2}\log\frac{n_{W_s}/\lambda_{W_s}}{n_{W_d}/\lambda_{W_d}}, \qquad d_{W_p} = \frac{1}{2}\log\frac{n_{W_s}/\lambda_{W_s}}{n_{W_p}/\lambda_{W_p}}. \tag{R.5}$$

They are $\log(z_{T_d}/z_{T_o})$, $\log(z_{W_d}/z_{W_s})$, and $\log(z_{W_p}/z_{W_s})$, respectively.

There is also an explicit finite metric for this defect. The Hilbert–Schmidt orthogonal projection of Z onto the four-channel rotational commutant is

$$\overline{Z} = z_S P_S + z_P P_P + \bar{z}_T P_T + \bar{z}_W P_W,$$

$$\bar{z}_T = \frac{3z_{T_d} + 6z_{T_o}}{9}, \qquad \bar{z}_W = \frac{2z_{W_d} + 6z_{W_s} + 2z_{W_p}}{10}. \tag{R.6}$$

The exact squared distance is

$$\boxed{\begin{aligned}\|Z - \overline{Z}\|^2_{\mathrm{HS}} = 2(z_{T_d} - z_{T_o})^2 + \frac{6}{5}(z_{W_d} - z_{W_s})^2 \\ + \frac{2}{5}(z_{W_d} - z_{W_p})^2 + \frac{6}{5}(z_{W_s} - z_{W_p})^2.\end{aligned}} \tag{R.7}$$

This follows from the weighted-variance identity $\sum_i d_i(z_i - \bar{z})^2 = (\sum_i d_i)^{-1}\sum_{i<j} d_i d_j (z_i - z_j)^2$.

If an actual one-step matrix is later proved to be

$$M_j = I + u_j \sum_h \gamma_h P_h + \mathcal{E}_j,$$

its first-order rotational compatibility relations are correspondingly

$$\gamma_{T_d} = \gamma_{T_o}, \quad \gamma_{W_d} = \gamma_{W_s}, \quad \gamma_{W_p} = \gamma_{W_s}. \tag{R.8}$$

This writes the conditions without assigning values to the uncomputed actual coefficients in F-03.

Equations (R.3)–(R.7) complete the finite construction required by R-01 for the prescription in this note. They do not assert that the data satisfy them. Nor is exact rotational commutation of a finite-cutoff bare-to-normalized map a necessary substitute for covariance of the limiting hierarchy: anisotropic cutoff representatives can require compensating finite coefficients. R-02 through R-05 still concern the actual hierarchy's action, observable and contact defects. No rotation-restoration estimate is inferred from this finite construction.

8. The actual cutoff positivity and translation verification obtained here

The totalization (F.35), with the scalar identity coefficient shown there, is an actual finite-cutoff function of the Wilson measure. Define its ordinary full moment hierarchy by

$$S^{\mathrm{tot}}_{m,a,T}(F) = a^{4m} \sum_{X,\alpha} F_\alpha(X) \mathbb{E} \prod_{i=1}^{m} \Phi^{\mathrm{tot}}_{\alpha_i,a}(x_i). \tag{P.1}$$

These finite sums are always defined. The same statements hold for the exactly normalized prescription on (F.32).

Proposition P.1 — concrete cutoff instance The hierarchy (P.1) has permutation symmetry, exact fine-lattice translation symmetry, hypercubic tensor covariance, and reflection positivity on every fixed positive-time test algebra once its stencil margin fits. The Wick convention is exactly (F.37).

Proof. The coefficients are real and independent of position; their mixing matrix is the displayed sum of the equivariant projectors; their identity direction is e_S. Thus (F.8) and (F.36) hold with these actual coefficients. For a finite positive-time polynomial test vector, the corresponding polynomial in Φ^{tot} is a bounded positive-half-link function at each cutoff. The Wilson square formula gives its nonnegative Gram form. Commutativity gives permutation symmetry, and changing all link variables by a fine-lattice translation proves exact translation covariance of the finite sums. No bound on Z is needed for these finite-cutoff statements. ■

The proposition also verifies any explicitly chosen point-local real equivariant source substitution that has been constructed for this hierarchy and splits across separated half-spaces. In particular the concrete choice $R[j] = j, C[j] = 0$ defines a compatible all-order external source jet and needs no contact coefficients. This observation must be used with its exact scope: it does not identify all contact terms generated by the marked RG induction with that zero-contact representative, or bound their finite-stencil and normalization remainders.

Exact positive transports may be used to evaluate (P.1). If the whole product is transported, its integral is unchanged, and an auxiliary skeleton cannot break the original fine-translation identity. If the product of the separately transported marks is used instead, the matching discrepancy must be retained. Its vanishing requires the actual matching and coefficient budgets; the unnormalized mesoscopic theorem alone does not establish them for (F.35).

9. Why P-01 and P-02 remain open under their stated mathematical requirements

P-01: the missing actual contact realization The finite one-field coefficients in this note are actual real coefficients, so no arbitrary-coefficient placeholder is needed for this cutoff prescription. Nevertheless P-01 asks for the constructed field/source substitutions of the target hierarchy, including its required contacts, and explicitly depends on F-05, F-06 and S-06.

The following have not been proved here:

1. That (F.32) holds on a cofinal tail and that (F.33) is the normalization produced and controlled by the actual marked induction.
2. That every local extraction generated by that induction has the required point-jet or shrinking-stencil realization, compatible in source degree and with the stipulated spatial-order bounds.

3. That the complete contact/normalization approximation error vanishes in the common joint-test norm for the same hierarchy.

Proposition P.1 therefore completes a useful concrete cutoff verification, not the P-01 statement. Calling the external contact coefficients zero does not prove the missing equality between that chosen representative and the exact transported marked hierarchy.

P-02: the missing common continuity and matching budget Exact fine-lattice translation symmetry and permutation symmetry are proved for (P.1). For a continuum shift b, however, the passage using fine shifts $b_a \to b$ still needs a uniform estimate of the form

$$|S_{m,a,T}(\tau_b F - \tau_{b_a} F)| \le C^{m+1}(m!)^p \nu_{qm}(\tau_b F - \tau_{b_a} F), \tag{P.2}$$

for the actual normalized hierarchy, with the common parameters supplied by S-05/S-06. No such estimate is proved by cutoff symmetry. If an approximate skeleton representation is used, its corresponding translation defect also requires the matching budget just described.

Passing from (P.2) to continuum translation covariance requires the stated common seminorm estimate.

Annex Y — A localized value, gradient and current norm for transported marks

Annex Y. The conclusions below hold on the stated source domains. Applying them to the complete Wilson law requires compatibility of the conditional representation (CL), including the retained variables, densities, and marked transport. See the theorem catalogue.

Finite norm and uniform transport

This note completes the finite specification in M-03. It gives a concrete carrier norm for value, retained-coordinate difference and normalized-kernel current records; proves initial bounds for the scalar-repaired 21-field family used by F-01; and gives exact product, source-jet and conditional-expectation rules. The one-step operator costs are explicit integrals of the actual normalized kernels. They are not asserted to be bounded uniformly, or even finite in the strong current norm, on every selected history.

M-06 remains open. Its required all-order carrier-bound sum for the actual RG/R step has not been proved. The algebraic identities below are not its actual-model estimate. In particular, the new norm does not turn the existing bare a^{-4} budget into a current-scale h^{-4} budget.

This distinction is part of the result: M-03 asks for a finite norm construction and its initial bounds; it does not assert the boundedness of the actual RG/R operator in that norm. A discontinuous selected kernel can have an infinite current cost in the construction below. The norm detects that failure instead of assuming a smooth score that does not exist.

1. Coordinates, carriers and reference points

At a fixed cutoff and fixed stage, let h be the current physical block size and let $\mathcal{B}_h$ be the finite periodic block graph. Use the augmented source-free state, including every retained continuous coordinate and every history variable needed by a later map. Inactive variables in a history sector are padded by fixed values. The resulting ambient space is a finite product

$$\mathsf{X}_h = \prod_{b\in\mathcal{B}_h} \mathsf{X}_{h,b}. \tag{M.1}$$

Its probability law μ_h is the actual normalized law, supported on its admissible subset. No independence of these blocks is assumed. A finite tuple of group, Lie-algebra and discrete coordinates may be assigned to a block. A coordinate whose definition uses several blocks is assigned its full dependency footprint, even if it is stored at one block. A carrier must contain that footprint. The storage block of a variable and its full dependency footprint are distinguished. A global history flag therefore has a global footprint unless an actual locality identity reduces it. Write $\mathrm{cl}(Y)$ for the deterministic footprint closure of a block set: start with Y, add the full footprints of all coordinates stored in its blocks, and repeat until stable. This finite construction may yield the full torus.

Fix the ambient reference configuration x_*: identity group variables, zero Euclidean coordinates, and the first discrete value in the already specified enumeration of each history coordinate. A marked expression that has a natural global compact-link formula retains that formula, independently of any auxiliary history labels. This applies in particular to the microscopic F-01 fields, including after a positive history lift. Only a primitive expression that is defined solely in a restricted coordinate chart is extended by zero where its chart formula has no meaning. Fix these primitive versions once. A normalized conditional kernel on an inactive or zero-normalizer retained datum is extended by the point mass at its reference input; a coordinate map outside its admitted chart is extended by its old-state reference value. Every generated output is thereafter defined recursively by these chosen kernel and coordinate formulas on the whole ambient space. It is not independently re-zeroed on inactive sectors. This convention keeps the constant jet equal to one and makes the exact identities valid also at the reset reference points used below. These are conventions for a norm and a localization formula, not changes to the probability law on its support. An actual proof that uses a better extension must record that choice explicitly.

For a finite block set X, let $\mathrm{t}_h(X)$ be the number of edges in a shortest connected subgraph of the block graph containing X. Set $\mathrm{t}_h(\varnothing) = 0$, and use

$$w_h(X) = \exp\{\kappa[|X| + \mathrm{t}_h(X)]\}, \qquad \kappa > 0. \tag{M.2}$$

In every displayed carrier array the field-insertion anchors and all dependency footprints are contained in its carrier. A constant identity subtraction can be placed on its insertion anchor. Carrier size and distance are measured in current units h, not in the original number of fine links.

The norm below is a norm of a specified carrier representation. Cancellation between two different activities is not used to reduce its size. This is the usual purpose of keeping the marked activities and generated responses as separate vertices.

2. Allowed differences and native regulators

For a group coordinate with source-free inverse-coupling label $\beta_e \geq 0$, choose a fixed Hilbert–Schmidt orthonormal basis of its Lie algebra and use the left-invariant perturbations

$$U_e \longmapsto \exp\left(\frac{tT}{\sqrt{1+\beta_e}}\right) U_e, \qquad |t| \leq 1. \tag{M.3}$$

Each elementary perturbation acts only in its storage block. Its basis, scale label and β_e in (M.3) are fixed external labels of that padded coordinate, independent of the values of all other coordinates and history tags. If a chart uses moving conventions, their changes belong to the separate coordinate pullback, not to these elementary perturbations. This ensures the commuting reset identities in Section 7. For a Euclidean coordinate already in the source-free dimensionless free-field convention, use $z_i \mapsto z_i + te_j$. A different physical chart is first pulled back to that specified convention; its Jacobian or finite difference is not replaced by one. Discrete history changes are replacements of one coordinate by another value in its finite enumeration, with denominator one. Their whole dependency footprint is retained.

Write τ for one such elementary perturbation. For a continuous perturbation with parameter $t \neq 0$, set

$$\delta_\tau A(x) = \frac{A(\tau x) - A(x)}{t}; \tag{M.4}$$

for a discrete change omit the division by t. The supremum over t below is a genuine supremum of finite differences; it is not an assumption of differentiability. Supremum envelopes can be interpreted by upper measurable envelopes if necessary. All input Wilson functions are continuous, so this qualification has no effect on their initial estimates.

Use a fixed native energy regulator

$$R_{h,X}(x) = 1 + \sum_{p\colon \mathrm{foot}(p)\subset X} \beta_p s(V_p(x)) + \sum_{i\colon \mathrm{foot}(i)\subset X} |z_i(x)|^2, \qquad s(V) = N - \mathrm{Re}\,\mathrm{Tr}\,V. \tag{M.5}$$

The plaquette and free-coordinate list is the actual retained/native list in the source-free representation, with its stated scale labels. It does not contain an unobservable eliminated microscopic energy. If no such coordinate is present, its sum is empty. At the microscopic input it is simply one plus the local Wilson plaquette energy. The list is finite at each cutoff; an extensive inherited list is not silently replaced by a bounded one.

Let $\mathcal{T}_X$ contain the identity and all elementary perturbations whose footprints lie in X. Define

$$A_X^\sharp(x) = \sup_{\tau\in\mathcal{T}_X} |A_X(\tau x)|, \qquad \widehat{R}_{h,X}(x) = R_{h,X}(x) + \sup_{\tau\in\mathcal{T}_X} R_{h,X}(\tau x). \tag{M.6}$$

For a difference record $B_{X,\tau}$, put

$$B_X^\natural(x) = \sum_{d\colon \mathrm{foot}(d)\subset X} \sup_{0<|t|\leq 1} |B_{X,d,t}(x)|, \tag{M.7}$$

with the analogous finite sum over discrete replacements. Each Lie-algebra basis direction is counted separately. This keeps dependence on all retained directions, including directions on which a selected background depends.

3. The graded value/gradient/current norm

For $d > 0$, define the averaged moment norm

$$\mathfrak{m}_d(F;\mu) = \sup_{p\geq 1} (dp)^{-d} \|F\|_{L^p(\mu)}. \tag{M.8}$$

Degree zero uses the essential supremum. A source coefficient with $m \geq 1$ labels is assigned value grade m and difference/current grade $m + \frac{1}{2}$. The half grade permits the normalized-kernel gradient to carry a natural square-root energy factor; it is not an assertion that every actual kernel has that bound.

A decorated coefficient is a triple

$$\mathrm{A}_{I,X} = (A_{I,X}, B_{I,X}, J_{I,X}), \qquad |I| = m, \tag{M.9}$$

with the exact compatibility identity for each tagged activity

$$\delta_\tau A_{I,X} = B_{I,X,\tau} + J_{I,X,\tau}. \tag{M.10}$$

It follows that the summed coefficient has the same identity after summing its records. A coordinate difference can enlarge the declared carrier, in which case the whole compatible triple is assigned that enlarged carrier. Multiple activities with the same carrier retain distinct activity tags and are summed in the norm before cancellation. Thus X in a carrier sum can occur more than once with different tags. The compatible localization rule in Section 7 is chosen specifically to preserve (M.10); naive localization of all three components by the same reset operator would not do so. Here B records direct differences of the insertion and its explicit pullbacks; J records currents generated when a normalized conditional law, selector, reference law or support changes. They are retained separately even when their sum simplifies. Initially $B = \delta A$ and $J = 0$.

For a fixed triple, put

$$\begin{aligned}
\upsilon_{h,m,X}(A) &= h^{4m}\left[\mathfrak{m}_m(A_X^\sharp;\mu_h) + \frac{1}{m^m}\sup_x \frac{A_X^\sharp(x)}{\widehat{R}_{h,X}(x)^m}\right],\\
g_{h,m,X}(B) &= h^{4m}\left[\mathfrak{m}_{m+1/2}(B_X^\natural;\mu_h) + \frac{1}{(m+1/2)^{m+1/2}}\sup_x \frac{B_X^\natural(x)}{\widehat{R}_{h,X}(x)^{m+1/2}}\right],\\
j_{h,m,X}(J) &= h^{4m}\left[\mathfrak{m}_{m+1/2}(J_X^\natural;\mu_h) + \frac{1}{(m+1/2)^{m+1/2}}\sup_x \frac{J_X^\natural(x)}{\widehat{R}_{h,X}(x)^{m+1/2}}\right].
\end{aligned} \tag{M.11}$$

The supremum is over the specified ambient versions. Each term is a norm on the finite-valued records for which it is finite. On arbitrary measurable records it is an extended norm and may be infinite. Define the carrier norm, for a fixed label/anchor tuple, by

$$\boxed{\|\mathrm{A}_I\|_{h,\varkappa} = \sum_{X \supset \mathrm{anchors}(I)} w_h(X)[\upsilon_{h,m,X}(A) + g_{h,m,X}(B) + j_{h,m,X}(J)].} \tag{M.12}$$

For a uniformly anchored family take the supremum over its first anchor and translate the other fixed relative positions with it. A full exponential jet may retain a forest of disconnected carriers; the weight of such a forest is the product of the weights of its connected components. Joining components into one connected response requires an actual kernel localization estimate and pays the connector lengths in (M.2).

This is not a renamed bare L^p norm. It contains a native-scale polynomial regulator, all retained finite differences, separately measured kernel currents, actual history variables, and a spatial weight. The physical factor is h^{4m}. Conditional contraction of a microscopic bound alone leaves the explicit loss $(h/a)^{4m}$ in this norm.

4. The microscopic input, including the F-01 coefficients

Use exactly the scalar-repaired clover matrix $\tilde{q}_a$ of F-01, in its orthonormal 21-component basis. Put

$$E_X = \beta_a \sum_{p \in \mathcal{P}(X)} s(U_p), \qquad O_{\alpha,a} = \beta_a \tilde{q}_{\alpha,a} - \mathbb{E}[\beta_a \tilde{q}_{\alpha,a}]. \tag{M.13}$$

Here $\mathcal{P}(X)$ is a fixed enlarged plaquette stencil containing the clover support and every plaquette affected by the allowed perturbations of its links. Its size depends only on the fixed stencil and N. The retained compact Wilson reserve gives

$$\|E_X\|_p \leq C p, \qquad p \geq 1, \tag{M.14}$$

on its stated cofinal/AF-bounded geometry. At a single arbitrary finite cutoff the same statements are finite, with the compact bound replacing the uniform reserve constant if its hypotheses have not been checked.

Proposition M.1 – an initial retained-link difference estimate For the scalar-repaired 21 fields and the perturbations (M.3), uniformly in $|t| \le 1$,

$$|O_{\alpha,a}(\tau U)| \le C a^{-4}(1 + E_X(U)), \tag{M.15}$$

$$|\delta_\tau O_{\alpha,a}(U)| \le C a^{-4}[\sqrt{E_X(U)} + 1], \tag{M.16}$$

and hence

$$a^4 \|O^\sharp_{\alpha,a}\|_p \le C(p+1), \qquad a^4 \|(\delta O_{\alpha,a})^\natural\|_p \le C(\sqrt{p}+1). \tag{M.17}$$

The initial current record is zero. Consequently the initial one-field carrier norm is bounded by a constant depending on N, the fixed stencil and κ, not on the cutoff or torus, under (M.14).

Proof. For a unitary plaquette,

$$\|U_p - I\|^2_{\mathrm{HS}} = 2s(U_p). \tag{M.18}$$

Changing one link by (M.3) changes any affected plaquette by a left or right multiplication, or a conjugate multiplication, with tangent norm at most $(1+\beta_a)^{-1/2}$. The first derivative of $s(P_t)$ is bounded by $C(1+\beta_a)^{-1/2}\sqrt{s(P_t)}$; its second derivative is bounded by $C(1+\beta_a)^{-1}$. Taylor's formula at zero therefore gives

$$\beta_a \left| \frac{s(P_t) - s(P_0)}{t} \right| \le C\sqrt{\beta_a s(P_0)} + C|t|, \tag{M.19}$$

because $\beta_a/(1+\beta_a) \le 1$. It also gives $\beta_a s(P_t) \le C[\beta_a s(P_0) + t^2]$.

For a clover curvature, the unitary product rule gives $\|C_A(t) - C_A(0)\|_{\mathrm{HS}} \le C|t|/\sqrt{1+\beta_a}$. The exact compact inequality used in the earlier field construction gives $\beta_a \|C_A(0)\|^2_{\mathrm{HS}} \le C E_X$. Expanding the two factors of $\beta_a \operatorname{Tr}(C_A(t) C_B(t))$ yields the same right side as (M.19). The scalar repair is a fixed linear combination of these bilinears and the action density, so it obeys the same estimate. Its orthonormal off-diagonal factors $\sqrt{2}$ change only the fixed constant. The centering term is independent of the link variables; its magnitude is bounded by (M.14).

There are only a fixed number of directions in the stencil. Taking their maximum or sum preserves the estimates with a fixed constant. Taking moments in (M.15)–(M.16), and using (M.14), proves (M.17). The regulator (M.5) and its shifted version dominate the pointwise right sides; the initial carrier has bounded diameter in units $h = a$. This proves the norm bound. □

Now use the actual proposed F-01 prescription, wherever its finite nondegeneracy test holds:

$$\Phi_a = Z_{a,T,\mu} \tilde{q}_a - c_{a,T,\mu} e_S, \qquad Z = \sum_h z_h P_h. \tag{M.20}$$

The seven z_h and the scalar c are the real finite coefficients specified there. The same norm statement also applies to the total finite-cutoff definition F.35, with its explicitly marked fallback on a zero normalization measurement. That fallback is not an inverse or a solution of the failed normalization equation. For $\beta_a > 0$, define the exposed coefficient budget

$$b_a = \frac{\|Z_{a,T,\mu}\|}{\beta_a} + a^4 |c_{a,T,\mu}|. \tag{M.21}$$

Then the same proof gives

$$\|\mathbf{\Phi}_{\alpha,a}\|_{a,\kappa} \le C b_a, \qquad a^4 \|(\delta \Phi_{\alpha,a})^\natural\|_p \le C \frac{\|Z\|}{\beta_a}(\sqrt{p}+1). \tag{M.22}$$

For the F-01 centering equation $c = z_S m_{a,T}$, the reserve also gives $a^4|c| \le C\|Z\|/\beta_a$. Thus (M.22) is finite for every nondegenerate finite normalization system, with its coefficient dependence shown. No uniform or logarithmic bound on b_a is asserted. That remains F-05. The construction does not define an inverse when the F-01 normalization determinant vanishes. The diagonal Wick phases of F.37 have modulus one, so the estimates also hold in that convention, with its transformed identity vector retained. If $\beta_a = 0$, the simple compact bound gives a finite input estimate proportional to $\|Z\| + a^4|c|$; the asymptotically free application uses the positive-coupling formula (M.21).

5. Product and finite source-jet compatibility

Lemma M.2 – graded Holder multiplication For $d, e > 0$,

$$\mathfrak{m}_{d+e}(FG;\mu) \le \mathfrak{m}_d(F;\mu)\mathfrak{m}_e(G;\mu). \tag{M.23}$$

Proof. Holder with exponents $(d+e)/d$ and $(d+e)/e$ gives

$$\|FG\|_p \le \|F\|_{p(d+e)/d}\|G\|_{p(d+e)/e} \le [(d+e)p]^{d+e}\mathfrak{m}_d(F)\mathfrak{m}_e(G).$$

Divide and take the supremum. □

For compatible records $\delta A = B_A + J_A$ and $\delta C = B_C + J_C$, the exact finite-difference product rule is

$$\begin{aligned} B_{AC,\tau} &= B_{A,\tau}\, C\circ\tau + A\, B_{C,\tau}, \\ J_{AC,\tau} &= J_{A,\tau}\, C\circ\tau + A\, J_{C,\tau}. \end{aligned} \tag{M.24}$$

Their sum equals $\delta_\tau(AC)$. The value envelopes multiply, and the difference/current envelopes are bounded by the corresponding two products of an envelope of grade $r + 1/2$ and one of grade s, or vice versa. Lemma M.2 bounds their averaged parts. For the regulator parts use $\widehat{R}_X, \widehat{R}_Y \le \widehat{R}_{X\cup Y}$, and $d^d e^e \le (d+e)^{d+e}$. Thus the same source grades are preserved by multiplication, with the two displayed Leibniz terms retained.

If $X \cap Y \neq \emptyset$, then

$$w_h(X\cup Y) \le w_h(X)w_h(Y). \tag{M.25}$$

If they are disjoint, the same inequality has the additional explicit factor $\exp[\varkappa\, \mathrm{dist}_h(X,Y)]$, obtained by joining shortest carrier trees by a shortest connecting path. One may instead keep the two carriers as a forest. These formulas expose the exact geometric cost of multiplying or joining vertices; they do not bound an actual sum over connecting kernels.

For the initial exponential jet

$$G_0(t) = \exp\left(\sum_i t_i\Phi_i\right) \tag{M.26}$$

the coefficient with m distinct labels is $\prod_i \Phi_i$. Lemma M.2 and (M.22), or direct Holder, give

$$\left\|\prod_{i=1}^m \Phi_i\right\|_p \le C^m b_a^m a^{-4m}(mp)^m \le (Ce)^m b_a^m a^{-4m} m! p^m. \tag{M.27}$$

The direct-difference record is the sum of the m terms in (M.24); it has the corresponding fixed grade $m + 1/2$ and the same factorial type. Initially every current record is zero. These statements hold coefficientwise for every finite truncation, and truncating a higher-order jet gives exactly the lower-order jet. They do not assume convergence of an infinite exponential series.

The logarithm and exponential in the algebra of source jets are finite polynomial operations once a maximal source order is fixed. Consequently the decorated product rule preserves compatibility under both operations. The conditional cumulant partition formulas retained from the earlier passes then give the exact response coefficients. No finite bound on the number of one-field labels truncates the number or valence of the generated vertices.

6. Conditional expectation with the exact kernel current

Write one actual source-free integration, replacement or reset in common coordinates as

$$(QA)(y) = \int A(u,y)\, q_y(du), \qquad q_y(1) = 1. \tag{M.28}$$

All discrete labels and all constraints belong either to u, to y, or to the kernel. The common input space for u can be a disjoint union of padded sectors. Domains, coarea factors, selectors and all normalization constants are part of q_y. A fixed coordinate pullback $x = \chi(u,y)$ is performed before (M.28); it is an explicit substitution, whose finite differences are retained in the direct record. No derivative of a coordinate map is discarded. The norm of $A\circ\chi$ is not

automatically controlled by the previous elementary-coordinate norm. The additional data are the regulator pullback $\widehat{R}_{h,X}(\chi(u,y))$, the dependency footprint of χ, and its normalized coordinate increments along $y \mapsto \tau y$. One sufficient bound is obtained by joining $\chi(u,y)$ to $\chi(u,\tau y)$ through the explicit sequence of old coordinate changes: sum their normalized lengths times the old regulator to the difference grade along that sequence, and divide by the new step size. The old pointwise difference bound then controls the pullback by telescoping. If no such controlled path is available, its cost is recorded as infinite. The exact change of rank is treated by the padded sectors and discrete steps, not by an assumed common-rank differential. These path/regulator costs must be bounded in the actual matching theorem; (M.38) below does not pay them on its own.

For an explicit path version of this cost, choose a finite sequence $z^0 = \chi(u,y), z^1, \dots, z^J = \chi(u,\tau_t y)$ of old elementary coordinate changes, with normalized step sizes $|s_j| \le 1$; a discrete replacement has size one. With $d = m + \frac{1}{2}$, define

$$\mathcal{L}^{(d)}_{\chi,X,\tau}(u,y) = \frac{1}{|t|}\sum_{j=1}^{J} |s_j| \widehat{R}_{h,X}(z^{j-1})^d. \tag{M.28a}$$

If the new change is discrete, its denominator is one. A source-order- m input activity whose total pointwise difference record has coefficient c_X in (M.11) obeys, by telescoping,

$$\left|\frac{A(\chi(u,\tau_t y)) - A(\chi(u,y))}{t}\right| \le c_X h^{-4m} d^d \mathcal{L}^{(d)}_{\chi,X,\tau}(u,y). \tag{M.28b}$$

The two types of record remain separate under this pullback. If the j-th old step is $z^j = \tau_{d_j,s_j} z^{j-1}$, use

$$B^{\chi}_{\tau_t}(u,y) = \frac{1}{t}\sum_{j=1}^{J} s_j B_{d_j,s_j}(z^{j-1}), \qquad J^{\chi}_{\tau_t}(u,y) = \frac{1}{t}\sum_{j=1}^{J} s_j J_{d_j,s_j}(z^{j-1}). \tag{M.28c}$$

For a discrete old step take $s_j = 1$; for a discrete new change omit the overall division by t. The sum is exactly $\delta_{\tau_t}(A \circ \chi)$ by telescoping, and (M.28a) bounds both records with their respective input coefficients. No previous kernel current is folded into a putative bare gradient.

The regulator pullback and (M.28a), including their actual carriers, are explicit extra data to be bounded, not unit-cost coordinate changes. If a continuous curve is used instead, the right side is the corresponding weighted path-length integral, justified by finite-difference bounds along its elementary subdivisions.

There is also an exact representation requiring no controlled chart path: push q_y to the full old-coordinate state, $\tilde{q}_y = \chi(\cdot,y)_* q_y$, and apply (M.29) to $\tilde{q}_y$ with the old mark independent of y. All chart motion is then in the normalized-kernel current. Its total- variation upper bound can be infinite even for a smooth deterministic coordinate change, because distinct point masses are mutually singular. That is a valid diagnostic representation, not a finite bound; a useful application must supply the path/gradient control or another rigorous estimate. Neither choice assumes a constant rank or a unit Jacobian.

For a continuous retained perturbation $y_t = \tau_t y$, define the signed normalized-kernel current

$$\mathcal{K}_{y,\tau} = \frac{q_{y_t} - q_y}{t}, \qquad \mathcal{K}_{y,\tau}(1) = 0. \tag{M.29}$$

For a discrete change use the difference without division. This signed measure always exists for fixed nonzero t, even for hard selectors, changing supports or mutually singular conditional laws. Its total variation is at most $2/|t|$; this elementary bound is not a useful uniform gradient estimate.

Theorem M.3 – exact decorated conditional transport Assume the displayed integrals are absolutely defined under both the unshifted and shifted kernels. This holds for the bounded marked coefficients at every fixed compact cutoff and fixed nonzero difference step. If $\delta_\tau A = B_\tau + J_\tau$, set

$$\begin{aligned} A'(y) &= Q_y A, \\ B'_\tau(y) &= \int B_\tau(u,y)\, q_{y_t}(du), \\ J'_\tau(y) &= \int J_\tau(u,y)\, q_{y_t}(du) + \int A(u,y)\, \mathcal{K}_{y,\tau}(du). \end{aligned} \tag{M.30}$$

Then

$$\delta_\tau A' = B'_\tau + J'_\tau. \tag{M.31}$$

Furthermore the newly generated term can be written in the centered form

$$\int [A(u,y) - Q_y A]\, \mathcal{K}_{y,\tau}(du). \tag{M.32}$$

The unshifted value obeys conditional-expectation contraction under the actual incoming and outgoing marginal laws. The shifted values, direct differences and currents obey (M.30), with their actual shifted kernels; contraction does not replace those kernels by q_y.

Proof. Subtract $Q_y A$ from $Q_{y_t} A(\cdot, y_t)$, add and subtract $Q_{y_t} A(\cdot, y)$, and divide by t. This gives

$$\delta_\tau(QA) = Q_{y_t}(\delta_\tau A) + \mathcal{K}_{y,\tau}(A).$$

Substitute the compatible record. Normalization gives (M.32). The proof for a discrete change is the same subtraction without division. Conditional Jensen proves the value contraction. □

On a common smooth support, with a positive unnormalized density $w(u,y)$, the limit of (M.29), when justified in the required norm, is

$$Dq_y = q_y\,[D\log w - Q_y D\log w],$$

so the new current reduces to

$$\mathrm{Cov}_{q_y}(A, D\log w). \tag{M.33}$$

The finite-difference theorem is stronger in scope: it does not infer this smooth formula at a moving selector boundary. Boundary flux and changes in the conditional normalizer are already present in (M.29).

For the frozen positive R quotient, the conditional mark is

$$(Q_R F)(u', v) = \frac{\int \rho(u,v) F(u,v)\, du}{\int \rho(u,v)\, du}. \tag{M.34}$$

The normalized replacement factor cancels. Thus the mark is independent of the freshly sampled u', and its finite difference in that variable is exactly zero. Its retained-v current remains (M.29). This cancellation does not set the retained current or the higher source responses to zero.

7. Explicit carrier costs for the actual kernel

There is a finite, fully specified carrier decomposition even when locality is poor. For a function of retained blocks, let

$$(E_b f)(y) = f(y \text{ with block } b \text{ reset to } y_{*,b}),$$

and set

$$L_Y = \prod_{b\in Y} (I - E_b) \prod_{b\notin Y} E_b. \tag{M.35}$$

The operators commute, and

$$f = \sum_{Y\subset\mathcal{B}_{h'}} L_Y f. \tag{M.36}$$

For a total compatible output record (A', B', J'), localize its value by $A^{[Y]} = L_Y A'$. The operator L_Y need not commute with δ_τ. The precise compatible rule uses the storage block b of the varied ambient coordinate. If $b \notin Y$, set both localized difference and current records to zero. If $b \in Y$, define

$$L^{(b)}_{Y\setminus\{b\}} = \prod_{c\in Y\setminus\{b\}} (I - E_c) \prod_{c\notin Y} E_c, \tag{M.36a}$$

which leaves block b free, and put

$$B^{[Y]}_{b,t} = L^{(b)}_{Y\setminus\{b\}} B'_{b,t}, \qquad J^{[Y]}_{b,t} = L^{(b)}_{Y\setminus\{b\}} J'_{b,t}. \tag{M.36b}$$

Then, exactly,

$$\delta_{b,t} L_Y A' = \begin{cases} 0, & b \notin Y, \\ L^{(b)}_{Y\setminus\{b\}} \delta_{b,t} A', & b \in Y, \end{cases} \tag{M.36c}$$

so $\delta_{b,t} A^{[Y]} = B^{[Y]}_{b,t} + J^{[Y]}_{b,t}$. To prove (M.36c), observe that a finite difference in block b annihilates $E_b f$ and commutes with every E_c, $c \neq b$. In particular $L^{(b)}_{Y\setminus\{b\}} = L_Y + L_{Y\setminus\{b\}}$; this formula accounts for the localization commutator without losing its contribution. It holds for continuous and discrete coordinate perturbations alike.

The value index Y need not contain the insertion anchors. Declare the actual carrier to be

$$\overline{Y}_I = \mathrm{cl}(Y \cup \mathrm{anchors}(I)). \tag{M.36d}$$

It contains the full footprint of each varied coordinate stored in Y, as well as every insertion anchor. Regroup the compatible triples on these declared carriers, retaining distinct activity tags before taking absolute norms. Use $w_{h'}(\overline{Y}_I)$, not $w_{h'}(Y)$, in (M.12). This deterministic enlargement is part of the cost. It is used for this output localization and can be overinclusive or global. It is not imposed on an already localized microscopic activity independent of other, globally supported flags that happen to be stored at the same block. In particular, the empty index $Y = \varnothing$ is retained and assigned to $\mathrm{cl}(\mathrm{anchors}(I))$; constant components are not lost from the anchored norm.

If QA_X is independent of a block, all its value coefficients containing that block vanish by (M.35). If the actual conditional law has global dependence, the formula produces global carriers instead of claiming local ones.

For clarity, the cost is given by actual signed kernels, not an unnamed operator constant. Write y^Z for the configuration with the blocks in Z retained and all others reset to the reference point. On the common (u, v) input chart define

$$K^V_{Y\leftarrow X}(y; du, dv) = \sum_{Z\subset Y} (-1)^{|Y|-|Z|} q_{y^Z}(du)\, \delta_{y^Z}(dv), \tag{M.37}$$

restricted to the actual coordinates needed by the pulled-back input carrier. For a difference/current direction stored in block b, both record kernels are zero when $b \notin Y$. When $b \in Y$, the compatible direct kernel is

$$K^B_{Y\leftarrow X,b,t}(y; du, dv) = \sum_{Z\subset Y\setminus\{b\}} (-1)^{|Y|-1-|Z|} q_{\tau_{b,t} y^{Z\cup\{b\}}}(du)\, \delta_{y^{Z\cup\{b\}}}(dv), \tag{M.38}$$

and the new kernel-current part is

$$K^J_{Y\leftarrow X,b,t}(y; du, dv) = \sum_{Z\subset Y\setminus\{b\}} (-1)^{|Y|-1-|Z|} \mathcal{K}_{y^{Z\cup\{b\}},b,t}(du)\, \delta_{y^{Z\cup\{b\}}}(dv). \tag{M.39}$$

These are the inclusion-exclusion formulas for (M.36b); retaining block b in $Z\cup\{b\}$ is essential. Their declared carriers are enlarged by (M.36d). Thus, if an input value coefficient obeys $|A_X| \le a_X h^{-4m} m^m \widehat{R}^m_{h,X}$, its output value is bounded by

$$|L_Y Q A_X| \le a_X h^{-4m} m^m \int \widehat{R}^m_{h,X}\, |K^V_{Y\leftarrow X}|. \tag{M.40}$$

The direct record is bounded using $\int \widehat{R}^{m+1/2}_{h,X} |K^B_{Y\leftarrow X,\tau}|$. The current record has both this term applied to the previous current and $\int \widehat{R}^m_{h,X} |K^J_{Y\leftarrow X,\tau}|$ applied to the previous value. The changed inclusion-exclusion operator in (M.38)–(M.39) already accounts for the noncommuting localization reset. Multiplication by the precise output weights in (M.11)–(M.12), taking the displayed supremum envelopes and the actual $\mu_{h'}$ moment norms, gives a nonnegative matrix of transport costs depending only on the actual source-free kernels, charts, regulator and geometry. Every factor $(h'/h)^{4m}$ is retained.

These formulas can be evaluated before choosing a marked input. No spatial bound on their sum has been inserted. In particular, the current entries can be infinite after the supremum over $t \to 0$. Proving useful bounds on these data is precisely the content of M-04–M-07, C-02–C-08 and H-02–H-05. An effective locality proof may replace the reference-point expansion by a sharper explicit carrier expansion, but must control the same exact output.

8. A nonvacuity and failure test

The norm is nonvacuous on the actual microscopic field family by Proposition M.1 and (M.22). It also detects a failure that the bare moment norm misses.

Let y be uniform on $(-1,1)$ and let the conditional variable $u \in \{-1,1\}$ have law

$$q_y = \begin{cases} \delta_{-1}, & y < 0, \\ \delta_1, & y \geq 0. \end{cases}$$

The input mark $A(u,y) = u$ is bounded, independent of y, and has zero direct difference and zero initial current. But

$$QA(y) = \operatorname{sgn}(y),$$

and for $t > 0$, on $-t < y < 0$,

$$\mathcal{K}_{y,t}(A) = 2/t. \tag{M.41}$$

For any fixed $p > 1$, its L^p norm grows like $t^{-1+1/p}$, and its pointwise norm grows like t^{-1}. The strong current norm is infinite. Formula (M.30) is still exact. This is why a hard-selector current cannot be replaced by a formal smooth log score without a support theorem.

The example is not asserted to model a Yang–Mills selector. It proves that the finite construction does not smuggle in a false universal contraction statement.

9. Exact residual obligation for M-06

For every finite source order, exact transport remains

$$G' = QG, \qquad \Psi' = \log(Qe^{\Psi}),$$

and hence, writing $A_I = \partial_I \Psi(0)$,

$$A_I' = \sum_{\pi \in \Pi(I)} \kappa_Q(A_B : B \in \pi). \tag{M.42}$$

The existing moment estimate gives magnitudes of conditional cumulants after outer averaging, and Sections 5–7 give exact carrier/current bookkeeping. These facts alone do not establish the M-06 hypotheses.

The missing statement is an all-order estimate of the actual carrier-cost sums arising from (M.37)–(M.42), with constants independent of current scale, UV depth, volume and admitted history, after the stated current-field matching and local subtractions. It must include:

- every higher vertex already present in the incoming jet;
- the conditional cumulants joining input carriers;
- all selector, coordinate, coarea and normalized-reference currents;
- the reset responses and their possibly global conditional dependence;
- the physical current-scale budget and a fixed factorial exponent in m.

Neither the initial bound (M.27), a formal partition count, nor an infinite diagnostic current cost is that estimate. The statement therefore remains open.

10. Application conditions

Requirement	Result in this annex
Explicit current-scale value norm	(M.5), (M.8), (M.11); scale factor h^{4m}, native regulator and actual full-law moments.
Retained-variable gradients	Exact dimensionless finite differences (M.3)–(M.7), including discrete history variations.
Currents and hard support changes	Normalized signed kernel differences (M.29)–(M.32); no common-support score assumption.
Initial finite-cutoff fields	Proposition M.1 for all 21 scalar-repaired fields; F-01 coefficients exposed in (M.20)–(M.22).

Requirement	Result in this annex
Carrier geometry	Dependency footprints, current block tree weight, forest convention and exact localization (M.35)–(M.39).
Conditional-expectation compatibility	Exact transport (M.30); actual kernel/regulator integrals (M.40) and their direct/current analogues.
Full finite-jet algebra	Graded Holder/product rules (M.23)–(M.27), exact finite cumulant rule (M.42), compatibility under truncation.
Actual bounded transport in this norm	Not proved; not part of the finite M-03 statement.
M-06 uniform actual carrier-bound sum	Not proved; statement remains open.

The only imported estimates in the initial bound are the compact Wilson energy reserve of Annex A and the already specified scalar-repaired clover inequalities. All other identities and norm estimates in this note are proved directly. No new primary-source theorem about marked Balaban histories is claimed.

Annex Z — Current-coordinate contours, nonlinear marginal factors and order-uniform scale budgets

Annex Z. All estimates have the hypotheses stated below. Applications to the complete Wilson conditional law use the common assumptions in the theorem catalogue.

0. Results and hypotheses

The contour criteria below apply to regular and heterogeneous histories when the requisite joint domains and carrier bounds hold. Annexes AD–AH treat the regular and boundary contributions separately.

The new work has three concrete uses.

1. It gives an anisotropic contour-to-Hessian test that retains the radius of each differentiated coordinate and each carrier. This avoids imposing the minimum radius of an unrelated distant part of the history on every term.
2. It proves a marginal derivative theorem using complex tubes centered along a real radial segment. A complex disc centered at the identity need not reach the real background. The theorem retains the nonlinear second-coordinate-map contribution, and extends to every derivative order with a fixed factorial bound.
3. It proves an all-order old-age summation theorem and a sharp limitation of a different proposed majorant. Exponential decay in old age can pay order-dependent polynomial age losses within one factorial power. Suppression of the form `exp[-c(log(2+age))^r]`, by itself, cannot do so for every source order with one fixed factorial power.

The last distinction is especially relevant to the full Schwinger hierarchy. Summability at every fixed order is weaker than the fixed growth condition required in J2.

All new mathematical results below are direct finite-dimensional proofs or elementary series estimates. Their actual-model premises are listed explicitly in Section 7.

1. Source conditions: what is available and what the proposed derivative must preserve

The hypotheses below use CMP119, pp. 244, 255, 259–263, 267–270, 276–277 and 283.

- The inherited regular action is already renormalized. CMP119 (2.25) contains its old vacuum and coupling subtractions; differentiating it must retain their field dependence.
- Each primitive E carrier has a complex extension with its stated decay norm, on the regular primitive space with radii `alpha_{0,j}=g_j C_0(log g_j^{-2})^{q_0}` and `alpha_{1,j}=g_j C_1(log g_j^{-2})^{q_1}`. The source introduces fixed integers `q_0,q_1>1`; this note does not infer an additional inequality between them and the real cutoff exponent.
- Source scope: CMP119 (2.34)-(2.39) defines the multiscale boundary domain, including independently extended currents, reconstructed fields and currents, a real-reference representative, local gauge bounds, and a complex relative field. It is not the definition of every primitive E domain. The appropriate domain at each actual preparation stage must be preserved by its contour; Annex AD makes this distinction explicit.
- CMP119 p.276 transfers the small-field localization construction to the complete geometry. Its boundary extension is (3.43). On p.277 the source explains the consumption of the dyadic analytic reserve and the extra fluctuation factor. These statements must be used together with their radii and preparation losses.
- CMP119 Theorem 2, p.263, is a regular-configuration estimate. Its proof obtains a four-power marginal factor for R and an irrelevant gain after coupling subtraction for E. An inequality proved on the real regular configurations is not, merely by analytic continuation, a modulus estimate on a proposed complex contour.

The source uses an independent current precisely because an individual localized field no longer satisfies the full unlocalized Landau-gauge identity. CMP109 (3.10)-(3.17), pp.272-273, makes this distinction. Its insertion norm controls the field, its first derivative, the appropriate projection and covariant Laplacian. The unlocalized minimizing-field derivative in CMP102 Proposition 9 and (190), pp.308-309, also controls the covariant second-order outputs.

These are useful map inputs; they do not identify every localized independent-current pullback with the physical current of that localized field.

The exact conditional density is the one derived in Annex W. The results below do not discard its R terms, B terms, heterogeneous coefficient, coordinate factors, retained mean, or selected support. The support theorem of this argument is a separate result.

2. An anisotropic contour-to-Hessian certificate

Let `I` be the finite free-coordinate index set at a cutoff, with a metric `d(i,j)`. A coordinate can mean one real component of a Lie-algebra block; alternatively one may work with block operator norms throughout. Let `Y` run over a finite family of localized carriers, and let `I_Y` be the coordinates on which its pulled-back activity `W_Y` depends.

For each real configuration `z` in the actual support, suppose a product contour centered at `z` is admitted for the relevant coordinates, with radii

$$r_{Y,i}(z) > 0, \qquad i \in I_Y, \tag{EC.1}$$

and with

$$|W_Y(z+\zeta)| \le M_Y(z) \quad \text{on that contour.} \tag{EC.2}$$

For a Hessian bound it is enough to have the corresponding one- and two-coordinate product contours with the same bounds. A collection of pair contours alone does not prove the arbitrary-order estimate stated below.

Let `omega_i>0` be the weights of the intended physical quadratic norm, and put

$$\mathcal{E}_\gamma = 2 \sup_z \sup_i \sum_{Y\,:\,i\in I_Y} \frac{M_Y(z)}{\omega_i r_{Y,i}(z)} \sum_{j\in I_Y} \frac{e^{\gamma d(i,j)}}{\omega_j r_{Y,j}(z)}. \tag{EC.3}$$

Proposition EC.1 – local radii give the exact weighted budget Under (EC.1)-(EC.2),

$$\sup_z \sup_i \sum_j \frac{e^{\gamma d(i,j)}}{\omega_i \omega_j} \left| \partial_i \partial_j \sum_Y W_Y(z) \right| \le \mathcal{E}_\gamma. \tag{EC.4}$$

For any multi-index `alpha` supported in a verified full product contour,

$$|\partial^\alpha W_Y(z)| \le \alpha! M_Y(z) \prod_{i\in I_Y} r_{Y,i}(z)^{-\alpha_i}. \tag{EC.5}$$

Proof. Cauchy's formula gives `M_Y/(r_{Y,i}r_{Y,j})` for distinct coordinate indices and `2M_Y/r_{Y,i}^2` for a repeated index. Derivatives outside `I_Y` vanish. Sum those estimates, take absolute values before summation, and use the common factor two to get (EC.4). Applying the multi-variable Cauchy formula gives (EC.5). The proof does not differentiate the integration-domain indicator. □

The same proposition applies to joint current and retained-background variables: include both types in `I` and use their actual radii and coordinate weights. It is invalid to hold a background constant in a derivative for which that background is an output of the differentiated map.

The activity in (EC.2) must already be the complete pullback used by the construction. If instead `W_Y=f_Y circ Psi_Y`, the nonlinear map derivatives remain to be estimated; Proposition EC.3 below gives one way to do so.

Corollary EC.2 – heterogeneous matrix coercivity and inverse bounds Let `D_omega=diag(omega_i)`. Assume each `W_Y` is real-valued on the real support, so the perturbation Hessian is real symmetric. Suppose, on that support,

$$V(z) = \tfrac{1}{2}\langle z, H_0 z\rangle - \sum_Y W_Y(z), \qquad H_0 \geq \lambda D_\omega^2, \quad \mathcal{E}_0 < \lambda. \tag{EC.6}$$

Then

$$\nabla^2 V(z) \geq (\lambda - \mathcal{E}_0) D_\omega^2. \tag{EC.7}$$

Define the weighted matrix norm

$$\|A\|_\gamma = \sup_i \sum_j e^{\gamma d(i,j)} |A_{ij}|. \tag{EC.8}$$

If `C_0=H_0^{-1}` and

$$\|D_\omega C_0 D_\omega\|_\gamma \leq K_\gamma, \qquad K_\gamma \mathcal{E}_\gamma < 1, \tag{EC.9}$$

then the inverse of the actual Hessian obeys

$$\left\| D_\omega (\nabla^2 V(z))^{-1} D_\omega \right\|_\gamma \leq \frac{K_\gamma}{1 - K_\gamma \mathcal{E}_\gamma}. \tag{EC.10}$$

Proof. The balanced perturbation `P=D_omega^{-1}(sum_Y Hess W_Y)D_omega^{-1}` is real symmetric and has row-sum norm at most `mathcal E_0`. The symmetric Schur estimate bounds its Euclidean operator norm by the same number, proving (EC.7). The triangle inequality for `d` makes (EC.8) submultiplicative. Apply its convergent Neumann series to

$$[D_\omega^{-1} H_0 D_\omega^{-1} - P]^{-1} = \sum_{n \geq 0} [(D_\omega C_0 D_\omega) P]^n D_\omega C_0 D_\omega.$$

This proves (EC.10). □

Equation (EC.10) is a matrix inverse estimate, not by itself a covariance theorem on a selected integration domain. The latter also uses the support and integration-by-parts theorem and the relevant field domains.

These formulas expose a concrete sufficient target for C-06. They do not infer (EC.3) from primitive activity norms. In particular, the physically relevant radius is the dimensionless product `omega_i r_{Y,i}`, and it can vary with layer, carrier and retained data.

3. Recentered tubes along real backgrounds preserve marginal derivatives

A previous sufficient radial estimate required a complex disc centered at the identity whose radius reached the background. That condition can be stronger than necessary. The following theorem uses a chain of complex neighborhoods centered along a real segment instead.

Let `E` be a finite-dimensional complex normed space, let `b` belong to its real form, and suppose `f` is holomorphic and bounded by `M` on

$$\mathcal{T}(b, \rho) = \bigcup_{0 \leq t \leq 1} \{tb + w : \|w\| < \rho\}, \qquad 0 < \rho \leq 1. \tag{EC.11}$$

The norm can be the joint norm of link logarithms and independent-current variables. Its precise definition must be fixed in an application.

Lemma EC.3 – the first derivative at a real background If `Df(0)=0`, then

$$\|D^r f(tb)\| \le r! e^r M \rho^{-r} \quad (r \ge 1), \tag{EC.12}$$

$$\|Df(b)\| \le 4M\rho^{-2}\|b\|, \qquad |f(b) - f(0)| \le 2M\rho^{-2}\|b\|^2. \tag{EC.13}$$

The right side of the value bound can also be replaced by its minimum with `2M`.

Proof. For unit directions `v_1,...,v_r`, apply Cauchy's formula to `f(tb+sum_l t_l v_l)` on the polydisc of radius `rho/r` in each scalar parameter. This gives `r^r M rho^{-r}`, and `r^r<=e^r r!`. For the Hessian one can use two radii `rho/2`, giving the sharper bound `4M rho^{-2}`. Now integrate

$$Df(b) - Df(0) = \int_0^1 D^2 f(tb)[b, \cdot]\, dt$$

and then integrate once more for the value difference, using `Df(0)=0`. □

The condition `Df(0)=0` is an exact consequence of semisimple global gauge invariance for the original activity expressed jointly in its Lie-algebra and current variables. Indeed, write a linear functional at the origin as a sum over its variables. Invariance under simultaneous adjoint action, tested with all other variables zero, makes each summand annihilate every commutator. Since the complexified Lie algebra of `SU(N)` equals its commutator algebra, every summand is zero.

This conclusion must not be applied to an arbitrary already-centered function with a nontrivial frozen background: its origin need not be the gauge-fixed flat configuration. The flat origin in this lemma belongs to the original activity variables, before the nonlinear current-coordinate pullback.

Proposition EC.4 – all-order composite marginal bound Let `Psi` be a holomorphic map from the current coordinates to `E`, with `Psi(z_0)=b`. Let `0<theta<=1`, `Q>=0`, `K>=1`, and `g>0`. Suppose

$$\|b\| \le Q\theta, \qquad \|D^s \Psi(z_0)\| \le s! K^s g^s \theta \quad (s \ge 1). \tag{EC.14}$$

Suppose the tube (EC.11) is admitted for `f`, with the same `rho` along its whole real segment, and `Df(0)=0`. For every integer `m>=2`,

$$\boxed{\|D^m(f \circ \Psi)(z_0)\| \le M(1+4Q)m! \left[\frac{(1+e)Kg}{\rho}\right]^m \theta^2.} \tag{EC.15}$$

For the first derivative,

$$\|D(f \circ \Psi)(z_0)\| \le 4MQKg\rho^{-2}\theta^2. \tag{EC.16}$$

The bounds use the derivatives of the actual map `Psi`; no linearization of it is substituted for the exact pullback.

Proof. Faà di Bruno's partition formula gives

$$D^m(f \circ \Psi)[v_1, \dots, v_m] = \sum_{\pi \in \Pi_m} D^{|\pi|} f(b)[D^{|B|}\Psi[v_i : i \in B] : B \in \pi]. \tag{EC.17}$$

The one-block term is bounded, by (EC.13)-(EC.14), by

$$4MQ\, m!(Kg)^m \rho^{-2}\theta^2. \tag{EC.18}$$

For `r>=2` blocks, (EC.12) bounds the sum by

$$Mm!(Kg)^m e^r \binom{m-1}{r-1} (\theta/\rho)^r. \tag{EC.19}$$

To obtain its combinatorial coefficient, order the `r` blocks. For prescribed positive sizes summing to `m`, the number of ordered labeled partitions is `m!/(d_1!...d_r!)`. Multiplication by the native derivative factors `prod d_i!`, division by `r!`, and multiplication by the `r!` in (EC.12) leave `m!` times the number of ordered compositions, namely `binom(m-1,r-1)`.

Because `theta<=1`, `rho<=1`, and `2<=r<=m`,

$$(\theta/\rho)^r \le \theta^2 \rho^{-m}.$$

The sum of $e^r\binom{m-1}{r-1}$ is at most $(1+e)^m$. The one-block term also has $\rho^{-2} \le \rho^{-m}$. Combining these estimates gives (EC.15). Equation (EC.16) follows from the ordinary chain rule and (EC.13). □

3.1 Why the nonlinear term matters At second order the identity is

$$D^2(f \circ \Psi) = D^2 f(\Psi)[D\Psi, D\Psi] + Df(\Psi) D^2\Psi. \tag{EC.20}$$

The second term in (EC.20) is generally nonzero. Lemma EC.3 gives its extra small factor from the background `b`, so both terms retain `theta^2`. A bound on `D^2 f` alone would not have justified this conclusion.

In the intended four-dimensional application, `theta=L^{-2a}` for an old age `a=k-j`. Then `theta^2=L^{-4a}` compensates the four-dimensional old-cell count. The coordinate map includes the local gauge, the constrained minimizing field and the correct independent-current extension. Establishing (EC.11) and (EC.14) for every such map is still an actual-source obligation.

There is no need for `||b||<rho` in this theorem. The real background may be larger than the imaginary width of its analytic tube. What is required is a uniformly admitted tube along the real segment, and that is a quantitative condition stronger than knowing the endpoint lies in an open analytic domain.

3.2 A native analytic map can supply all its derivative orders at once Suppose, in its own native variables, `Psi` is holomorphic on the norm ball of radius `R/g` about `z_0`, with

$$\sup_{\|\zeta\|<R/g} \|\Psi(z_0+\zeta)\| \le B\theta. \tag{EC.21}$$

The same mixed Cauchy argument gives

$$\|D^s\Psi(z_0)\| \le s! \left[\frac{e\max(1,B)}{R}\right]^s g^s \theta. \tag{EC.22}$$

Thus (EC.14) follows without separately estimating every derivative. The large native ball in (EC.21) is used only to estimate derivatives of `Psi`; the activity `f` is not assumed to be admissible on its entire image. The chain rule uses the separate tube (EC.11). Confusing these two domains would recreate the missing contour step.

4. Carrier polynomial losses have a fixed factorial price

Lemma EC.5 – exponential carrier entropy absorbs polynomial derivative losses Let `b,c>=0` and `delta>0` be fixed. There is a constant `C`, independent of `m`, such that

$$\sum_{n\ge 1} n^{bm+c} e^{-\delta n} \le C^{m+1}(m!)^b, \qquad m \ge 1. \tag{EC.23}$$

Proof. Put t=bm+c. Split exp(-delta n) into two equal factors. The first factor times n^t is at most (2t/(e delta))^t for t>0; the other factor is summable. For fixed b,c, this upper bound is at most C^{m+1}m^{bm}m^c. The polynomial m^c is absorbed by another fixed exponential, and m^{bm}<=(e^m m!)^b. The case t=0 is immediate. □

Suppose the number of carriers of size n containing a prescribed current block is at most mu^n, their activity decay is exp(-kappa n), and the geometric weight costs at most exp(gamma D_0 n). If

$$\delta = \kappa - \log\mu - \gamma D_0 > 0, \tag{EC.24}$$

then (EC.23) pays any fixed polynomial carrier factor raised to source order m. The factorial exponent increases by the degree of that polynomial. This is compatible with J2 when that degree is fixed; it is not compatible if the degree itself grows uncontrollably during the induction.

The actual source's carrier geometry and decay reserve must be checked before identifying its parameters with (EC.24). A unit-lattice animal count cannot silently replace the current-block carrier count.

5. An all-order near/far old-age summation theorem

The fixed-order near/far calculation of Annex H can be strengthened to permit polynomial age losses whose exponent grows linearly with the derivative order.

Theorem EC.6 – one fixed factorial budget for all derivative orders Fix A>=1, p>=0, q>0, lambda>0, and b,sigma,D,tau>=0. Suppose nonnegative numbers H_{a,m}(ell) satisfy, for integers a>=0, m>=2, and ell>=e,

$$H_{a,m}(\ell) \le A^{m+1}(m!)^p(1+a)^{b+\sigma m}\min\{\ell^{-qm}, e^{-\lambda a}\ell^{D+\tau m}\}. \tag{EC.25}$$

For every 0<q'<q, there is a constant C, depending only on the displayed fixed parameters and on q', such that

$$\boxed{\sum_{a\ge 0} H_{a,m}(\ell) \le C^{m+1}(m!)^{p+\sigma}\ell^{-q'm}.} \tag{EC.26}$$

In particular the order-two sum tends to zero as ell tends to infinity, while all orders share the same factorial exponent p+sigma.

Proof. Write beta_m=b+sigma m and choose

$$n = \left\lceil \frac{D+(\tau+q)m}{\lambda}\log\ell + \frac{2\beta_m}{\lambda} + 1 \right\rceil. \tag{EC.27}$$

This choice has both

$$\ell^{D+\tau m}e^{-\lambda n} \le \ell^{-qm}, \qquad \frac{\beta_m}{1+n} \le \lambda/2, \tag{EC.28}$$

and n+1<=C_0(m+1)(1+log ell).

For a<=n, use the first branch of (EC.25) and sum at most n+1 terms. For a=n+t>n,

$$\frac{(1+n+t)^{\beta_m}}{(1+n)^{\beta_m}} \le \exp\!\left(\frac{\beta_m t}{1+n}\right) \le e^{\lambda t/2}.$$

Use the second branch, (EC.28), and the remaining geometric series. The two portions are therefore bounded by

$$C_1A^{m+1}(m!)^p\ell^{-qm}[C_0(m+1)(1+\log\ell)]^{b+\sigma m+1}. \tag{EC.29}$$

The factor (m+1)^{sigma m+b+1} is at most C_2^{m+1}(m!)^sigma. Since m>=2,

$$(1+\log \ell)^{b+\sigma m+1}\ell^{-(q-q')m} \le \left[(1+\log \ell)^{\sigma+(b+1)/2}\ell^{-(q-q')}\right]^m .$$

The quantity in square brackets has a finite supremum for `ell>=e`. Combining the bounds proves (EC.26). □

The term `2 beta_m/lambda` in (EC.27) matters. Without it, a tail estimate with a constant depending on `beta_m` could hide a new order-dependent growth loss.

5.1 How the source coupling comparison enters For an admitted source coupling array, CMP119 (2.6)-(2.7) gives a polynomial comparison between `g_k` and `g_j` and a fixed comparison of the logarithms. If an old analytic radius has a proved lower bound

$$\rho_{j,k} \ge c\, g_j \ell_j^q (1+k-j)^{-s_0}, \qquad \ell_j = \log g_j^{-2}, \tag{EC.30}$$

then, with `a=k-j`,

$$g_k/\rho_{j,k} \le C(1+a)^{s_0+1/2}\ell_k^{-q}. \tag{EC.31}$$

The case `a=0` is included by `1+a`. This is arithmetic using an admitted coupling comparison; it is not a derivation of the actual trajectory.

Combining (EC.15), a native polynomial age loss, (EC.31), the old-cell count and a carrier sum of the form (EC.23) yields the first branch of (EC.25), provided the displayed radius and native-map estimates have actually been established. An independently proved irrelevant branch on the same objects then invokes Theorem EC.6.

Thus a finite polynomial loss in age per derivative is not automatically fatal. Its factorial price is explicit. What cannot be done is to assume (EC.30), or to use the marginal and irrelevant branches on different unverified contours.

6. A sharp failure of a weaker all-order majorant

The previous theorem uses decay exponential in the old age `a`. A large-field small factor expressed in the logarithm of that age can be much weaker for purposes of uniform source-order growth.

Proposition EC.7 – logarithmic age suppression alone is not a factorial budget Fix `c>0`, `sigma>0`, and `r>1`, and define

$$S_m = \sum_{a=0}^{\infty} (1+a)^{\sigma m}\exp[-c(\log(2+a))^r], \qquad m \ge 1. \tag{EC.32}$$

Every `S_m` is finite. Nevertheless there are positive constants `c_1`,`c_2` and `m_0`, depending only on the fixed parameters, such that

$$c_1 m^{r/(r-1)} \le \log S_m \le c_2 (m+1)^{r/(r-1)}, \qquad m \ge m_0. \tag{EC.33}$$

Consequently there are no fixed constants `B`,`P<infinity` for which

$$S_m \le B^{m+1}(m!)^P \quad \text{for every } m. \tag{EC.34}$$

Proof. For the upper bound, divide the integers `a` into exponential bins `e^n<=a<e^{n+1}`. Each bin contains at most `e^{n+1}` integers. Up to a factor exponential in `m`, its contribution is bounded by

$$\exp[(\sigma m+1)n - cn^r].$$

Young's elementary maximization inequality gives

$$bn - cn^r \le C_{c,r} b^{r/(r-1)} - (c/2)n^r, \qquad b \ge 0.$$

The remaining sum over n is finite. This proves finiteness and the upper bound in (EC.33).

For the lower bound, let

$$t_m = \Big(\frac{\sigma m}{cr}\Big)^{1/(r-1)}$$

and take an integer a_m within one of exp(t_m). For large m,

$$\log(1 + a_m) \geq t_m - 1, \qquad \log(2 + a_m) \leq t_m + 1.$$

A single summand then has logarithm at least

$$\sigma m(t_m - 1) - c(t_m + 1)^r = \frac{r-1}{r}\sigma m t_m - O(m) - O(1).$$

The positive leading term has order m^{r/(r-1)}, proving the lower bound. Finally log(m!)<=m log m, while m^{r/(r-1)}/(m log m) tends to infinity. This excludes (EC.34). □

6.1 What this diagnosis says about the Yang-Mills route Under a reciprocal-linear ultraviolet coupling comparison, an old-scale small factor exp[-c(log g_j^{-2})^r] can behave like exp[-c'(log(2+a))^r] as a function of the number of older shells. If a proposed absolute majorant simultaneously pays an age factor (1+a)^{sigma m} at derivative order m, the model sum (EC.32) shows that fixed-order convergence of that majorant is not enough for a fixed OS factorial exponent.

This does not prove that actual Yang-Mills correlations have this growth. It proves that this particular class of upper-bound arguments is too weak without another input. Extra exponential spatial/age decay, a better derivative map estimate, resummation before absolute values, or another norm can avoid the example. A bound by a large majorant is not a lower bound on the quantity being estimated.

The argument also does not contradict the completed logarithmic-running budget F-02. There the variable is the absolute ultraviolet logarithm and there is a genuine positive power of the lattice spacing. A factor exp(-delta t) pays powers t^{gamma m} at a factorial cost. In (EC.32), after setting t=log(2+a), the dangerous derivative factor is exp(sigma m t), and the available damping is exp(-c t^r). These are different optimization problems.

For E, the explicit irrelevant branch in Theorem EC.6 is a way to avoid this loss. For R, a fixed polynomial small-coupling bound, or even the stronger displayed logarithmic large-field factor, cannot alone be declared sufficient at every order without checking the age dependence of the derivative and carrier costs.

7. Analytic conditions and the exact remaining certificates

C-02 — hypotheses for simultaneous E contours (Annex AD) Proposition EC.1 identifies the quantitative output required of a contour theorem, and Proposition EC.4 shows how one can use recentered real-segment tubes instead of an unnecessarily large identity-centered disc.

The following actual estimates are still missing:

1. For every active real current configuration, every contributing E carrier and every localization/preparation parameter, an admitted joint tube for all the original variables entering that carrier.
2. Verification of every reconstructed-field and independently extended-current condition, not just the physical plaquette inequality.
3. Quantitative lower bounds on the coordinate radii after composition with the actual constrained, local-gauge and preparation maps, uniform enough to feed (EC.3) and (EC.30).
4. A complex activity majorant on those same tubes, with all carrier and preparation prefactors retained.

A point lying in the source's open analytic set has some positive radius at that finite cutoff. That observation does not prove any of these uniform statements. No missing map estimate is replaced by a choice of names for its radius.

C-04 — hypotheses for the regular E/R derivative-carrier sum (Annex AH) The new composite theorem proves the marginal four-power mechanism, including the nonlinear map contribution, if its tube and map estimates are available. The all-order summation theorem gives a fixed factorial budget for an explicit pair of majorants. The logarithmic-age example shows why order-by-order summability is insufficient.

To finish the application one must prove those majorants for the actual E and R pullbacks and sum them with the actual radii, coupling sequence, carrier geometry and preparation losses. In particular the small weighted-Hessian number in (EC.3) has not been obtained for the complete potential. The primitive source norms are not substituted for that sum.

C-06 – heterogeneous retained backgrounds: OPEN Corollary EC.2 supplies an exact weighted matrix formulation suited to different physical coordinate weights. It prevents a homogeneous estimate from being extended simply by renaming the metric.

The actual source geometry must still supply the weighted principal lower bound and covariance norm in that same coordinate system, together with the complete E/R/B perturbation budget and quantitative domain margins. The unlocalized background theorem is a relevant source input, but it does not by itself verify every localized current and nonlinear preparation output. The complete boundary estimates remain in C-05.

No claim is made that the C-07 support result or an arbitrary-vector moment comparison fills these missing analytic certificates.

8. References

Primary references

- T. Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, Communications in Mathematical Physics 119 (1988), 243-285. The relevant equations are listed in Section 1.
- T. Balaban, *Variational Problem and Background Fields in Renormalization Group Method for Lattice Gauge Theories*, Communications in Mathematical Physics 102 (1985), 277-309, Proposition 9 and (190), pp.308-309.

Additional reference:

- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. I*, Communications in Mathematical Physics 109 (1987), 249-301, especially (3.10)-(3.17), pp.272-273. DOI.

Annex AA — Actual regular product support and the remaining R and boundary estimates

Annex AA. The conclusions below hold on the stated source domains. Applying them to the complete Wilson law requires compatibility of the conditional representation (CL), including the retained variables, densities, and marked transport. See the theorem catalogue.

0. Result and precise scope

This annex distinguishes two different conditional laws that the abstract notation `D_{k,v}` does not separate.

1. A fiber conditioned on every primitive archived selector label can have a complicated selected domain. It must not be declared a convex product merely because the effective potential is regular.
2. The source-normal-form regular marginal, obtained after the exact integrations and resummations prescribed by the source, has a product of fixed free-bond norm balls as its inner integration domain. This remains true after the complete R operation has restored the inductive form. The full marked integral can be written on this marginal by conditional expectation of the entire marked jet. That replacement is exact; it is not a pointwise identification of marks on the two different fibers.

The main result below proves the support assertion for the second law, supplies the exact marked disintegration, and checks the hard-boundary argument needed by the weighted covariance theorem. This is the natural regular law to use for C-08. All E, R, B, Jacobian, heterogeneous-coupling and retained-mean terms remain in its density.

The proof does not make the complete positive history law convex. The responses generated by averaging archived labels and by crossing live-history activation boundaries are retained explicitly. They are inputs to H-03/H-04 and M-04/M-06, rather than silently deleted from the marked induction. Likewise, H1 regularity and the size of derivatives of transported marks still require their own estimates.

Under CL, the support and integration-by-parts statement for the regular conditional law applies to the chosen Wilson representation. The R and B derivative estimates have separate hypotheses.

No complete C-08 estimate or Yang-Mills existence theorem is asserted.

1. Published support and coordinate formulas

The table gives the published formulas used in the support argument.

Source	Exact input used
CMP119, p.249, between (1.14) and (1.15); CMP99, p.428, (3.157)-(3.158) and preceding paragraph	The remaining integration coordinates are individual non-exceptional bond values. The map `C` solves for the exceptional values and is the identity on the remaining bonds. The cutoff on the free variables is not a constraint on every component of `C A`.
CMP119, pp.257-258, (2.17)-(2.23)	The effective density is the positive source construction expressed by outer operations supported in the large-field region, multiplied by the effective action. The old operations factor by connected large-field components. Their characteristic functions can change under R, but their supported-operation form is retained.
CMP119, p.267, (3.15) and preceding text	Nonlinear averaging constraints are linearized, exceptional bond variables are eliminated, and the remaining current variables are free fluctuation variables.

Source	Exact input used
CMP119, p.268, (3.16)-(3.19)	A new decomposition restricts the individual free-bond values. On the good cubes the earlier approximate-fluctuation selector is identically one, by the quantitative comparison (3.19).
CMP119, p.269, (3.20)-(3.22)	On the regular interior, including the stated collar, only the good free-field selectors remain. Bad selectors and the residual old selector occur in the exterior. The coordinate relocation acts on the specified exterior set and leaves the other characteristic functions unchanged.
CMP119, p.270, (3.23)-(3.25)	The exterior variables are included in the new outer operation. The inner integral is explicitly over free variables on the regular interior, with the product cutoff and the complete effective exponent.
LFR I, p.201, (1.100)-(1.102)	A fresh R reference field has a fixed cutoff in its relative link logarithms; its normalized quotient preserves the integral.
LFR II, p.378, the decomposition after (1.70)	On newly good replacement components the old exterior-extension and fresh-reference cutoffs are exactly one and are removed. Surviving nontrivial selectors belong to the remaining large components. Gauge elimination is described as integral equivalence, so the marked construction must carry the corresponding normalized gauge map.
LFR II, pp.390-391, (1.101)-(1.102) and the sentence following them	The complete R operation returns the density to CMP119 (2.18), with all the stated inductive hypotheses.
LFR II, pp.391-392, (1.103)-(1.104)	In the alternative exact form, the remaining large-component factors depend only on new field variables restricted to those components and do not participate in the next regular integration.

The source's regularization and parameter hypotheses remain in force. In particular this note does not verify the actual asymptotically free trajectory or the admitted small-coupling ceilings. It applies at every finite step for which those source hypotheses hold.

The literal threshold in CMP119 (3.16) is denoted below by `r_k`. Its printed symbol is `delta_k`. The relationship between that printed threshold and the preceding normalization of the free variable is discussed in Annex W, Section 6.3. Nothing below chooses a different radius: the product-domain argument is valid for either consistently specified convention, with the actual threshold retained throughout.

2. C-07: exact regular product support after the complete R operation

2.1 Notation and the conditional law Fix an admitted finite scale and an output of the complete source-free construction in the inductive form of CMP119 (2.18). Fix its live geometry, gauge graphs, determining sets and outer variables. Call these retained data `v`; in particular the source coupling labels are fixed, rather than varied with the integration variable.

Let `Lambda=Lambda_{k+1}` be the regular interior of CMP119 (3.20). Let `I_Lambda` be the remaining free bond coordinates on that interior after the constraint elimination in (3.15). Each free coordinate takes values in a copy `E_b` of the finite-dimensional real Lie algebra. Denote the norm used by the literal source cutoff on that bond by `||.||_b`.

This identification of free coordinates is substantive. CMP119 p.249 explicitly defines the starred bond set by deleting one exceptional bond `b_0(c)` for each eliminated constraint, names the remaining bond values `A`, and writes the constrained field as `A'=C A`. CMP99 p.428 says that `C` is the identity except at the solved exceptional bonds. The general step on CMP119 p.267 invokes exactly this elimination; its exponent contains `C A_k`, whereas (3.16) tests `A_k`

on the starred free-bond set. The cutoff is therefore a product in independent coordinates. Had it instead tested all components of `C A_k`, this product conclusion would not follow merely from convexity of the resulting intersection.

Define

$$K_{k,\Lambda} = \prod_{b \in I_\Lambda} \{A_b \in E_b \,:\, \|A_b\|_b < r_k\}. \tag{C7.1}$$

Repeated good-cube tests in (3.16) do not produce a new constraint: a product of repeated indicators `1_{||A_b||<r_k}` is that same indicator. Some tests may contain already fixed exterior coordinates. On an active exterior fiber those factors equal one. Thus the restriction on the remaining inner variables is exactly (C7.1).

Let `L_reg(A;v)` be the complete logarithmic density furnished by the source identity (3.25), including the full terms identified in the C-01 density identity. Constants independent of `A` can be kept in the outer weight. The conditional law is

$$q_{k,v}(dA) = Z_{k,v}^{-1} 1_{K_{k,\Lambda}}(A) e^{\mathcal{L}_{\mathrm{reg}}(A;v)}\, dA, \qquad Z_{k,v} = \int_{K_{k,\Lambda}} e^{\mathcal{L}_{\mathrm{reg}}(A;v)}\, dA. \tag{C7.2}$$

The retained Gaussian cross term belongs to `L_reg`. It is a linear term in `A` and changes the conditional mean. Its presence does not change the support in (C7.1).

Theorem C7.1 – the support of the actual regular marginal For the exact source-normal-form regular marginal just specified, on every active nonzero-mass retained fiber:

1. the normalized inner law is (C7.2);
2. `K_{k,Lambda}` is a convex product domain in the free coordinates;
3. this domain is independent of the values of the retained continuous variables within the fixed live geometry and fixed coupling chart;
4. no inherited R, E or B term has been discarded to obtain this assertion;
5. this assertion applies after every admitted complete R step, not only before the first replacement or in the homogeneous no-R model.

Proof. Begin with the exact normal form (2.18), not an arbitrary conditional law obtained by fixing extra latent labels. The source applies the next blocking operation, inserts the new coarse and approximate-fluctuation partitions, and performs the exact constraint linearization and exceptional-coordinate elimination. Formula (3.15) is the result in free coordinates. Consequently the old group and nonlinear averaging constraints are not additional delta surfaces inside the `A` integral. Their coordinate factors belong to the density already identified by C-01.

The remaining approximate-fluctuation selector in (3.15) depends on the coarse field through the constrained background. The new decomposition (3.16) tests the supremum of the individual free-bond norms on its specified cubes. On a good cube, the source comparison (3.19) makes the earlier selector exactly one. This is an exact identity on the stated cutoff, not an estimate which replaces a nonzero boundary contribution by zero.

The geometry in (3.20) chooses the regular interior with the indicated collars. The text immediately after (3.20) states that only the good free-field tests remain there, in fact on the indicated enlargement of that interior. Their product is `chi^(k)(Lambda)`. All complementary bad tests and the surviving approximate-fluctuation test occur in the exterior factor (3.21).

The coordinate relocation (3.22), referring to (1.22), acts on the exterior set `S_{k+1}`. The source states that the other characteristic functions are unchanged. The complete change of variables, including its Jacobian and its effects on every old action term, generates the boundary correction `V^(k)` in (3.25). It does not shift the inner ball or remove that correction from the exponent.

Formula (3.23) puts the exterior integration, its selectors and its retained quadratic factor into the new outer operation. Formula (3.25) then has exactly the inner factor

$$\int dA|_\Lambda\, \chi^{(k)}(\Lambda) e^{\mathcal{L}_{\mathrm{reg}}(A;v)}. \tag{C7.3}$$

Conditioning the outer variables in this equality gives (C7.2). The cutoff in (C7.3) is a product of norm inequalities, hence a convex product as in (C7.1). It contains neither `v` nor the old selectors as an additional constraint on `A`. The dependence on `v` remains in the full potential and in the outer weight.

For inherited operations the argument uses their actual source-supported form: CMP119 (2.19)-(2.22), rather than an assertion that every background-dependent expression is local. These old operators are supported in the large-field region and remain outside the inner integral in (3.25). Their smooth dependence, if present in the particular expression for the regular exponent, is retained there. The exact formula is the justification for separating the support; no additional operation is commuted across a mark or a selector.

Finally LFR II (1.101)-(1.102) reconstructs the new action including the new R and B terms and states that the resulting density has precisely the form (2.18), with all inductive hypotheses. Induction therefore permits the same argument at the next regular step. The stronger representation (1.104) gives another explicit check: the remaining large-component factors depend only on variables restricted to those components, and the source states that they do not enter the next regular renormalization integration. This does not assert that those factors are small. It locates their support.

All integrations and domain assertions above are identities of the admitted source construction. This proves the theorem. □

2.2 Fresh R-reference support There is a second product support that must not be confused with the old selected numerator. At one localized R quotient, write the new relative field as

$$V'(b) = \exp(ig_k B'(b)), \qquad B'(b) = 0 \quad \text{on the chosen axial graph.} \tag{C7.4}$$

LFR I (1.101) gives the fixed cutoff

$$\prod_{b \notin G} 1_{\{\|B'(b)\| < M_0 \varepsilon_k / g_k\}}. \tag{C7.5}$$

The independent non-graph coordinates therefore also have fixed convex product support. Its radius is exactly the one in the relative-log chart. The reference density includes its full Haar factors and localized action. The exterior-extension selector only decides whether this component is active; it is not an additional inequality on `B'` once the retained data are fixed.

The old numerator conditional law in the R quotient need not have this support. The positive quotient transports an old mark by its old conditional expectation. It does not identify the old law with the fresh reference law. This distinction is essential for every subsequent use of (C7.5).

3. Exact marked passage to the regular marginal

Theorem C7.1 concerns an unmarked marginal. To use it in the proposed route, one must check that no observable is lost when archived histories are integrated out.

Let `Xi` denote the variables retained in (C7.2), including `A,v`, and let `omega` denote archived primitive histories and any latent old variables. On the positive finite source construction there is a joint probability measure `P(d omega,d Xi)` whose `Xi` marginal is the source-normal-form measure. This is the positive lift of B-01/H-01; no signed polymer activity is used as a probability. When the source removes a gauge delta by integral equivalence rather than pointwise equality, as on LFR II p.378, use its normalized gauge disintegration/averaging as a Markov step on this enlarged space. Carry the mark through that same step. The requested field family is gauge invariant at the original Wilson level. This procedure preserves its marked integrals without multiplying a merely gauge-equivalent final density by an untransported mark.

For a finite source jet `G(omega,Xi;t)` with integrable coefficients, define coefficientwise

$$\overline{G}(\Xi; t) = \mathbb{E}[G(\omega, \Xi; t) \mid \Xi]. \tag{C7.6}$$

Proposition C7.2 – full-jet disintegration without support identification For every retained test function `f(Xi)` and every coefficient of the finite jet,

$$\mathbb{E}[f(\Xi)G(\omega,\Xi;t)] = \mathbb{E}[f(\Xi)\overline{G}(\Xi;t)]. \tag{C7.7}$$

Truncation in source degree commutes with this operation. If the original constant coefficient is one, the new constant coefficient is one. If `Psi=log G`, the logarithm of `Gbar` includes the conditional covariance, higher cumulants and all pre-existing higher source responses. In particular,

$$\overline{\Psi}_i = \mathbb{E}[\Psi_i \mid \Xi], \qquad \overline{\Psi}_{ij} = \mathbb{E}[\Psi_{ij} \mid \Xi] + \mathrm{Cov}(\Psi_i, \Psi_j \mid \Xi). \tag{C7.8}$$

Proof. Equation (C7.7) is the defining identity for conditional expectation. At finite cutoff there are finitely many jet coefficients, so it applies to all of them and commutes with truncation. Differentiating the finite algebraic identity `Gbar=exp(Psibar)` proves (C7.8) and the corresponding partition formula at higher orders. □

Thus the regular covariance theorem may be applied to `Gbar` on the domain of Theorem C7.1 once its gradient and current budgets have been proved. It may not instead be applied to the original mark on a finer selector-conditioned fiber, nor may `Gbar` be replaced by a freshly inserted coarse plaquette.

This exact change of conditioning is allowed by the frozen-source-free route: it uses the same positive source construction and conditional expectation of marks. It requires no source-dependent background. It also shows exactly where archived-selector responses go. They become derivatives and cumulants of (C7.6), which remain part of the obligations M-04/M-06 and H-03/H-04.

4. The hard boundary in the actual regular covariance argument

Let

$$V(A;v) = -\mathcal{L}_{\mathrm{reg}}(A;v), \qquad q_v(dA) = Z_v^{-1} e^{-V(A;v)} 1_K(A)\, dA, \tag{C7.9}$$

where `K=K_{k,Lambda}` is the domain just proved. Suppose the complete potential satisfies the derivative and coercivity hypotheses being sought in C-04-C-06. This subsection checks the support and boundary part of the ensuing estimate, rather than assuming those potential estimates.

Write

$$V = \tfrac{1}{2}\langle A, K_0 A\rangle + \langle \ell, A\rangle + W(A), \qquad K_0 \geq \lambda_0 I, \tag{C7.10}$$

and, for the spatial metric on free bonds, set

$$a_0(\gamma) = \sup_i \sum_j \|(K_0)_{ij}\|(e^{\gamma d(i,j)} - 1), \quad \eta_\gamma = \sup_i \sum_j e^{\gamma d(i,j)} \sup_{A\in K} \|\nabla_i \nabla_j W(A)\|. \tag{C7.11}$$

If `c_0=lambda_0-a_0(gamma)-eta_gamma>0`, the existing weighted covariance argument is valid on the actual domain and gives

$$|\mathrm{Cov}_{q_v}(F_x, G_y)| \leq c_0^{-1} e^{-\gamma d(x,y)} \left(\sum_i e^{2\gamma d(i,x)} q_v |\nabla_i F_x|^2\right)^{1/2} \left(\sum_i e^{2\gamma d(i,y)} q_v |\nabla_i G_y|^2\right)^{1/2}. \tag{C7.12}$$

Here the marks must belong to the indicated weighted Sobolev spaces; (C7.12) does not prove that transported marks have those norms.

For completeness the boundary check is repeated explicitly. On a smooth convex product approximation to `K`, solve the weighted Neumann equation

$$Lu = F - q_v F, \qquad L = -\Delta + \nabla V \cdot \nabla, \qquad \partial_n u = 0. \tag{C7.13}$$

Set `w=grad u`, and let `M_i=exp(gamma d(i,x))`, scalar on each Lie-algebra block. Differentiating (C7.13) gives the interior equation

$$(L \otimes I + \nabla^2 V)w = \nabla F. \tag{C7.14}$$

After multiplication by `M^2w`, the boundary term on a face belonging to block `i` is

$$M_i^2 \int_{\partial K_i \times \prod_{j \neq i} K_j} \mathrm{II}_i(w_{i,\tan}, w_{i,\tan})\, Z_v^{-1} e^{-V}\, d\sigma_i\, dA_{\neq i} \geq 0. \tag{C7.15}$$

All other block directions have zero second fundamental form on that face. The scalar-by-block weighting preserves the tangent/normal condition. Consequently the boundary has a favorable sign and may be retained or discarded in the lower bound; it is not set to zero by pretending that the sharp cutoff has no derivative.

The conjugation cost of `K_0` is at most `a_0(gamma)`, by the symmetric Schur majorant. The conjugated perturbation Hessian costs at most `eta_gamma`. This yields

$$c_0 \|Mw\|_2^2 \leq \langle Mw, M\nabla F \rangle, \qquad \mathrm{Cov}_{q_v}(F, G) = q_v[\nabla u \cdot \nabla G], \tag{C7.16}$$

from which (C7.12) follows by Cauchy-Schwarz and the triangle inequality for `d`.

Norm balls need not have smooth boundaries. At each finite dimension one approximates each convex ball from inside by smooth convex bodies, uses the same Hessian bounds on their product, and passes to the limit for smooth marks by dominated convergence. Their weighted gradients have the same dominating integrable functions on the compact product. Density in the weighted Sobolev space gives the general case. Corners of product domains produce no additional negative boundary term. Thus the argument applies to the literal bond norm in the source, without replacing that norm by an unrelated spherical cutoff.

The lower bound `c_0>0` is a remaining actual-potential premise; the theorem above supplies the complete support justification for using it on the source regular marginal.

5. Retained derivatives and history interfaces are different from inner IBP

5.1 Within one active live-history chart Since `K` in (C7.9) is independent of the retained field values, a derivative within one active fixed-geometry chart gives, when the differentiated integrands are integrable,

$$\partial_{v_a} q_v F = q_v(\partial_{v_a} F) - \mathrm{Cov}_{q_v}(F, \partial_{v_a} V). \tag{C7.17}$$

All coordinate-map, Haar, coarea, inherited-activity and boundary-relocation derivatives belong to `partial_v V` and `partial_v F`. The product cutoff itself has zero `v` derivative. This is now a source-specific support fact, not an assumption about every primitive history kernel.

If the source parameters, cutoff radius or coupling label are varied instead of held fixed, (C7.17) is no longer the full formula. A radius derivative produces the ordinary surface term. This statement concerns fixed source-free parameters, as required by the proposed frozen-map route.

5.2 Exact interface formula for live activation changes Suppose in a common retained chart the live-history branch density is

$$w_h(v) = 1_{D_h}(v) z_h(v), \qquad m_h(v) = q_{h,v} F_h, \quad w(v) = \sum_h w_h(v), \quad m(v) = \frac{\sum_h w_h(v) m_h(v)}{w(v)}. \tag{C7.18}$$

For piecewise smooth activation domains, the distributional derivative of the unnormalized numerator in a retained direction `a` is exactly

$$\partial_a \sum_h 1_{D_h} z_h m_h = \sum_h 1_{D_h} \partial_a (z_h m_h) - \sum_h n_{h,a} z_h m_h \, \sigma_{\partial D_h}. \tag{C7.19}$$

The minus sign uses the outward normal and the convention `partial_a 1_D=-n_a sigma_boundary`. The denominator has the same formula with `m_h` replaced by one. A singular quotient derivative must be computed from the two actual traces, rather than by dividing a surface measure by an unspecified value of a discontinuous function.

At a regular interface `Sigma`, let `n` point from its minus side to its plus side, write `N=sum_h w_h m_h`, and suppose both traces `w^-,w^+` are positive. Then the exact singular derivative is

$$(D_a m)_\Sigma = \left(\frac{N^+}{w^+} - \frac{N^-}{w^-}\right) n_a \, \sigma_\Sigma = \frac{[N] - \overline{m}\,[w]}{\overline{w}} n_a \, \sigma_\Sigma, \qquad \overline{m} = \frac{m^+ + m^-}{2}, \quad \overline{w} = \frac{w^+ + w^-}{2}. \tag{C7.20}$$

Here `[f]=f^+-f^-`. The second equality is the elementary product identity `[wm]=wbar [m]+mbar [w]`; it specifies the arithmetic trace convention completely. If a side has zero marginal weight, the normalized conditional mean on that side has no canonical value, and no quotient distribution across it is asserted. The exact finite-difference kernel below remains the correct formulation on nonzero fibers.

For a nonsmooth selector one uses the exact finite signed-kernel difference of Annex Y rather than assuming a classical surface density exists. Equations (C7.19)-(C7.20) apply whenever the indicated boundary measures and traces are defined. No regularity of every source selector is assumed merely to display them.

Across an interface separating two primitive cells, their boundary terms cancel only if the two inward limits of the unnormalized marked density agree. Equality of the sum of the unmarked weights is not sufficient for this marked cancellation. In particular, the conditional means on two branches need not agree. This is why the live-history current survives in H-03/M-06.

Equation (C7.20) is outside the inner `A` integration by parts of Section 4. It has not been bounded here. The distinction is mathematical: it differentiates the marginal history/retained-data law, while (C7.15) is the reflecting boundary form of the conditional regular field law. The statement support result does not identify these two derivatives or erase either one.

5.3 Centered finite differences and integrated denominator cancellation The same separation is available without differentiability or surface geometry. Let `r_v` be the finite unnormalized conditional measure on a common padded input space, `Z_v=r_v(1)>0`, and `q_v=r_v/Z_v`. Then, for a finite increment `v->v'`,

$$q_{v'} F - q_v F = \frac{(r_{v'} - r_v)(F - q_v F)}{Z_{v'}}. \tag{C7.21}$$

If the mark itself changes, add `q_{v'}(F_{v'}-F_v)` to the right-hand side. This follows by subtracting the two fractions and uses `(r_{v'}-r_v)(1)=Z_{v'}-Z_v`.

On multiplying by the true marginal factor `Z_{v'}`, the displayed fiber denominator cancels exactly. Integration against the corresponding output marginal uses its actual underlying measure and any coordinate-change Jacobian. This is useful for an integrated-current estimate, but does not establish its spatial localization or a uniform pointwise retained-data norm. It also does not permit cancellation against a different reference normalizer.

6. C-03: R activity norms and the required quantitative pullback (Annex AE)

6.1 A nonvanishing reference normalization is already part of the source LFR II (1.10)-(1.11), printed p.358, represents the localized reference integral after removal of its large background Wilson factor as the exponential of an analytic function `I(U_0)` plus the stated constant normalization terms. On its source-defined complex domain,

$$|I(U_0)| \le C \log(M) \, |\Lambda|. \tag{C3.1}$$

In particular the normalization is nonzero on that specified domain. An additional generic argument trying to derive zero-freeness from a real lower bound is unnecessary there and would be invalid in general. What remains necessary is admission of the actual differentiated current-coordinate contour to that domain.

For example, if a particular one- or two-coordinate pullback disc of radius `r` is verified to lie in that source domain, Cauchy's formula gives

$$|\partial_a I| \le C \log(M)|\Lambda|/r, \qquad |\partial_a \partial_b I| \le 2C \log(M)|\Lambda|/r^2. \tag{C3.2}$$

The full logarithmic derivative of the original reference denominator also contains the derivative of the extracted background action and of any retained normalization factors. It is not equal to the derivative of `I` alone. At the frozen quotient, the reference factor cancels from the old conditional mark, but it still contributes to the output reference density and to later transformations.

6.2 Primitive R norm versus current marginal norm CMP119 (2.31), p.260, gives the primitive complex bound

$$|R_X^{(j)}(U,J)| \le g_j^{\kappa_0} e^{-\kappa d_j(X)} \tag{C3.3}$$

on the specified old multiscale space. LFR II (1.99)-(1.100), p.390, proves the new activity bound on its own complex space, with the stronger small factor `exp[-p_0(g_k)]` away from the surviving large components.

LFR II (1.65), pp.374-375, supplies an analytic old-field preparation map from its new `(U,J)` domain into the enlarged old domain of LFR I (1.64)-(1.69). This is a real source theorem. It cannot be replaced by the unsupported claim that arbitrary fluctuations around an admitted real point remain in its domain with one fixed current-coordinate radius.

The exact missing C-03 estimate can be expressed without an unnamed constant. For every actual current real point `A` and every contributing old carrier, one needs a quantitative admitted current bidisc, including independently extended currents and all decoupling parameters, and a localized marginal transfer of the form

$$\sup_i \sum_{Y \ni i} \sum_{l \in Y} e^{\gamma d(i,l)} \sup_{\zeta \in \mathbb{D}_{i,l}(A, r_k)} |V^R_{j,k,Y}(\zeta)| \le C \ell_j^D g_j^{\kappa_0} \frac{Q_k^2}{a_j^2}. \tag{C3.4}$$

The notation and the necessity of the marginal scale gain are as in Annex G: `a_j` is an admitted old radius, `Q_k` controls the physical amplitude on the same tube, and `ell_j=log(g_j^-2)`. The four marginal scale powers must compensate the old-cell count before the old-scale sum. The two current Cauchy derivatives then cost `r_k^-2`.

The existing semisimple first-jet cancellation and old-scale summation do not prove the contour or (C3.4). The fresh denominator source theorem (C3.1) removes a possible extra zero-free subproblem on its admitted domain, but does not establish (C3.4). C-03 remains open.

7. C-05: boundary activities have complex source norms, but no complete current estimate yet

CMP119 (2.40)-(2.42), pp.261-262, gives

$$\mathrm{B}_k = \sum_{j=1}^{k} \sum_X B_X^{(j)}(U, J, \mathrm{A}, \{S_i \cap X\}), \qquad |B_X^{(j)}| \le B_0 e^{-\kappa d_j(X)}, \tag{C5.1}$$

with localization meeting the appropriate large-field region, and analytic extension on the specified multiscale spaces. Its old fields and history labels remain arguments of the activity. The source's component refinement on p.262 prevents dependence on disconnected exterior pieces that do not connect to the regular region; it does not make the whole boundary activity zero in the regular interior.

On pp.276-277 the source localizes the actual pulled-back B differences and states the additional small factor from the fluctuation. It also explains that the decreasing dyadic domain reserve pays for the dependence on the minimizing-field map. The geometric and analytic changes of variables producing `V^(k)` are included. Thus a nontrivial boundary derivative theory exists in the source; C-05 is not a request to invent a boundary expansion from nothing.

For the observable application, however, the complete quantitative estimate is still missing. A sufficient target, after current localization, is

$$\sup_{A\in K_{k,\Lambda}} \sup_{i} \sum_{j,X} \sum_{l} e^{\gamma d(i,l)} \left\| \partial_i \partial_l [B_X^{(j)} \circ \Phi_{j,k}](A) \right\| \le \eta^B_{k,\gamma}, \tag{C5.2}$$

with a budget compatible with the complete coercivity margin, together with the relevant retained-variable/source derivative estimates. The pullback `Phi_{j,k}` includes the boundary relocation and all old fluctuation arguments; its second derivative cannot be omitted.

The value bound (C5.1) and the extensive real bound (2.48) do not imply (C5.2). Two Cauchy radii, all old-layer gains, the collar geometry, and the radius costs of the boundary relocation must be retained. In particular a boundary component whose distance grows with the volume cannot be charged by an unrecorded constant. Neither the product support result nor the normalized interface identity supplies that missing current-carrier budget. C-05 remains open.

8. Conditional support and derivative estimates

Statement	Result of this argument	Exact remaining qualification
C-07	The actual source-normal-form regular marginal has fixed convex product support after the complete R operation; exact full-jet disintegration preserves the original marked integral; the Neumann boundary form and retained/chart/interface derivatives are distinguished and explicitly accounted for.	Complete at its regular covariance/IBP support scope, under the admitted source construction. Potential coercivity remains C-04-C-06; regularity of transported marks remains M-04/M-06; history interface estimates remain H-03/H-04.
C-03	The primitive R norm and the reference logarithm are controlled on their stated domains.	Annex AE derives the regular native contour under its hypotheses.
C-05	The B arguments, support, complex domain and boundary-relocation terms enter (C5.2).	The Wilson application of the boundary induction requires CL and its uniform collar and derivative budgets.

The useful change is that the current regular support need no longer be carried as an unspecified nonconvex `D_{k,v}` when the exact source-normal-form marginal is used. This removes one obstruction to applying a proved weighted covariance theorem. It does not supply the complete potential budget, all-order marked estimates, ultraviolet matching, rotation restoration, operator bounds or phase selection.

9. Source references

- T. Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, Communications in Mathematical Physics 119 (1988), 243-285: the printed locations listed in Section 1.
- T. Balaban, *Propagators for Lattice Gauge Theories in a Background Field*, Communications in Mathematical Physics 99 (1985), 389-434. p.428, the free-coordinate parametrization preceding (3.157).
- T. Balaban, *Large Field Renormalization. I. The Basic Step of the R Operation*, Communications in Mathematical Physics 122 (1989), 175-202. (1.100)-(1.102), p.201.
- T. Balaban, *Large Field Renormalization. II. Localization, Exponentiation, and Bounds for the R Operation*, Communications in Mathematical Physics 122 (1989), 355-392. (1.10)-(1.11), p.358; (1.65), pp.374-375; (1.99)-(1.104), pp.390-392.

Annex AB — Complete conditional-law criteria and exact scalar integration by parts

Annex AB. The conclusions below hold on the stated source domains. Applying them to the complete Wilson law requires compatibility of the conditional representation (CL), including the retained variables, densities, and marked transport. See the theorem catalogue.

0. Disposition and new results

C-08 and M-04 remain open. The complete current-coordinate activity and background-derivative budgets have not been established for every admitted history. A new calculation of the homogeneous no-R law, or an abstract criterion which assumes those budgets, would not pass their stated mathematical requirements.

This annex gives three concrete results. None of them is a completed statement for the actual model.

1. It gives a local, volume-independent conditional-mean criterion which retains the exact exterior Schur force and the first derivative of the complete interaction. This specifies the mean input which is absent from a coercivity inequality alone.
2. It proves an exact scalar quadratic readout for the actual incoming mark under the full regular conditional law. The identity isolates a constrained trace term and retains the exterior mean, all inherited E/R/B and measure gradients, and the incoming Taylor remainder. A compactly supported integration-by-parts field replaces the hard-boundary flux by an explicit ordinary expectation on coordinate collars. No coupling derivative or source-dependent minimizer is used.
3. It gives a local rare-collar bound and the exact first and second retained-parameter response formulas. These formulas expose the additional costs needed for the M-03 current norm and for the scalar marginal coefficient.

The new exact identities apply at finite cutoff whenever their displayed integrals exist. Their quantitative consequences retain explicit hypotheses. They do not turn an unspecified transported mark into a freshly inserted native coarse plaquette, or prove a summable multiscale matching error.

1. The actual regular law and the actual incoming mark

1.1 Source support and the change of conditioning The primary formulas used here are CMP119, printed pp.267-270, (3.15)-(3.25). The source first eliminates the exceptional variables of the linearized averaging constraint; the constrained quadratic form subsequently appears as $C^*\Delta^{(k)}C$. It then inserts the free-bond partition (3.16). On the regular interior the old approximate-fluctuation selector is exactly one. Formula (3.25) displays only the product good-field cutoff in the inner integral. The boundary relocation is retained in the exponent.

The support calculation in Annex AA checks the complete R return to that inductive form. Its result concerns the source-normal-form regular marginal, not a finer conditional law obtained by fixing every archived selector label. Let υ contain the live geometry and all exterior variables retained by that normal form. Write its free variables as $z = (z_i)_{i\in I}$, with bounded-dimensional real Lie-algebra blocks. The inner domain is

$$D = \prod_{i\in I} D_i, \qquad D_i = P_i D_i^0, \tag{CM.1}$$

where each D_i^0 is the fixed symmetric convex bond-norm unit ball. The radii P_i are the actual source radii in these coordinates. They are fixed while the retained field values vary within one admitted live-geometry chart. No alternative radius convention is substituted.

Let F^{aug} be the incoming scalar mark on the positive augmented construction, and let ω denote the variables already integrated out in obtaining this marginal. The mark to be inserted in the regular integral is

$$F(z;\upsilon) = \mathbb{E}[F^{\mathrm{aug}} \mid z, \upsilon]. \tag{CM.2}$$

For a full finite source jet, (CM.2) is applied to every coefficient of the whole jet. Its logarithm retains the resulting conditional cumulants. The identity

$$\mathbb{E}[F^{\mathrm{aug}} f(z,\upsilon)] = \mathbb{E}[F(z;\upsilon) f(z,\upsilon)] \tag{CM.3}$$

holds for every bounded retained test f. This justifies using the regular marginal, but gives no pointwise or Sobolev identification of F with a native field. In particular, a smooth marginal density need not make every projected mark smooth. Regularity and localization of (CM.2) remain part of the marked induction.

1.2 Complete potential and exterior force Split the true constrained quadratic form into interior and retained current coordinates. In the notation of C-01, let y be the retained exterior current field and put

$$K = K_{\Lambda\Lambda}, \qquad b = -K_{\Lambda\Lambda^c} y, \qquad C = K^{-1}. \tag{CM.4}$$

At fixed retained data, the complete regular conditional law is

$$d\nu_\upsilon(z) = Z_\upsilon^{-1} 1_D(z) \exp\left[-\frac{1}{2} z^T K z + b^T z + W(z;\upsilon)\right] dz. \tag{CM.5}$$

Here W, up to a retained-data-only constant, contains precisely

$$\begin{aligned} W = & -\mathcal{V} + \log J + \Delta\mathrm{E} + \Delta\mathrm{R} + \Delta\mathrm{B} \\ & - \Delta A(w_k - g_k^{-2}, \cdot) + \mathcal{B}_\partial, \end{aligned} \tag{CM.6}$$

evaluated on the actual interior/exterior coordinate map. Any older operation factor which depends on this free variable in the chosen exact representation also belongs in W. The quadratic exterior constant belongs in the outer weight; its linear cross term is $b^T z$ and remains in (CM.5). The same convention is used for the fixed Haar/constraint prefactor: it cancels from this conditional ratio but remains in the outer marginal and history law.

The estimates below never assert that W or its derivatives are small merely because it has been written in the form (CM.6).

2. A local mean criterion with the exterior force retained

2.1 Quantitative hypotheses Let the block metric obey

$$S_a := \sup_i \sum_j e^{-a d(i,j)} < \infty \quad (a > 0), \tag{CM.7}$$

with the desired volume-independent constants. For $0 \le s \le 1$, let

$$d\nu_s = Z_s^{-1} 1_D e^{-\frac{1}{2} z^T K z + b^T z + s W(z)} dz. \tag{CM.8}$$

Assume the complete potentials on this same product domain, including their real linear tilts, have the uniform coercivity and weighted covariance estimate

$$|\operatorname{Cov}_{\nu_s}(z_i, f)| \le \sum_j G_{ij} \|\nabla_j f\|_{L^2(\nu_s)}, \qquad G_{ij} = c_* e^{-\gamma d(i,j)}. \tag{CM.9}$$

The norm on the left is the Euclidean norm of the block covariance. For the arbitrary-mark cumulant and retained-current consequences below, retain explicitly the stronger bilinear estimate

$$|\operatorname{Cov}_{\nu_s}(f, g)| \le \sum_{i,j} G_{ij} \|\nabla_i f\|_{L^2(\nu_s)} \|\nabla_j g\|_{L^2(\nu_s)}. \tag{CM.9a}$$

It is required for the relevant weighted Sobolev functions and the same real linear tilts. The mean proof uses (CM.9); the coordinate estimate alone is not asserted to imply (CM.9a).

Assume also that every real linear tilt of these measures has Hessian at least λI, with the same convex support. These conclusions follow from the existing weighted Neumann/convex-domain argument when the actual constrained precision has its admitted gap and weighted conjugation bound and the complete W Hessian fits inside the corresponding numerical margin. C-04-C-06 must supply that application; (CM.9)-(CM.9a) are not asserted without it.

Define the nonnegative matrix and local zero-field force

$$A_{ij} = \sup_{z\in D} \|\nabla_i \nabla_j W(z)\|, \qquad d_i = |\nabla_i W(0)|. \tag{CM.10}$$

Use the weighted positive-kernel row norm

$$\|T\|_a = \sup_i \sum_j e^{ad(i,j)} T_{ij}. \tag{CM.11}$$

For one fixed $0 < a < \gamma$, require

$$\|GA\|_a < 1. \tag{CM.12}$$

This is an explicit additional smallness condition, not a hidden use of an unweighted value bound. A sufficient condition is $c_* S_{\gamma-a} \|A\|_a < 1$.

Theorem CM.1 – force-resolvent mean and moment bounds Under (CM.7)-(CM.12), let

$$H = (I - GA)^{-1} G, \qquad \sigma = \sqrt{d_{\max}/\lambda}, \tag{CM.13}$$

where $d_{\max}$ is the largest real block dimension. Then

$$\sup_{0\le s\le 1} |\nu_s z_i| \le \sum_j H_{ij} \left(|b_j| + d_j + \sigma \sum_l A_{jl} \right). \tag{CM.14}$$

Moreover, for every $p \ge 1$,

$$\|z_i - \nu_s z_i\|_{L^p(\nu_s)} \le C\sqrt{p/\lambda}, \tag{CM.15}$$

with C depending only on the bounded block dimension. Finally,

$$\|H\|_a \le \frac{c_* S_{\gamma-a}}{1 - \|GA\|_a}. \tag{CM.16}$$

Thus the local mean is controlled by a spatially decaying convolution of the actual exterior force and actual interaction gradient, not by setting the retained Gaussian mean to zero.

Proof. A linear tilt leaves the Hessian unchanged. Applying the convex-domain variance bound to that tilted measure gives, for a unit vector u in one block,

$$\frac{d^2}{dt^2} \log \nu_s e^{tu\cdot z_i} = \mathrm{Var}_{\nu_{s,t}}(u \cdot z_i) \le \lambda^{-1}. \tag{CM.17}$$

Integrate twice and subtract the mean. The usual exponential Markov inequality gives a sub-Gaussian tail for each real block component. Integrating that tail and summing over at most $d_{\max}$ components proves (CM.15). In particular its L^2 block fluctuation is at most σ.

At $s = 0, b = 0$, the symmetric product domain and even quadratic density give zero mean. Interpolate the linear force from zero to b, and apply (CM.9), which has the same constants under this linear tilt. The resulting Gaussian reference mean satisfies

$$|\nu_0 z_i| \le \sum_j G_{ij}|b_j|. \tag{CM.18}$$

The line segment from zero to z lies in D. Taylor's integral formula therefore yields

$$|\nabla_j W(z)| \le d_j + \sum_l A_{jl}|z_l|. \tag{CM.19}$$

Let $M_i = \sup_{0\le s\le 1} |\nu_s z_i|$. Differentiation in s, the exact normalized covariance identity, (CM.9), (CM.15), and (CM.19) give

$$M \le G|b| + Gd + GA(M + \sigma 1). \tag{CM.20}$$

All objects at finite cutoff have finitely many blocks. By (CM.12) the positive Neumann series for $(I-GA)^{-1}$ converges, proving (CM.14). The weighted row norm is submultiplicative by the triangle inequality for the metric. Since $\|G\|_a \le c_* S_{\gamma-a}$, the same series gives (CM.16). No determinant ratio or total-volume factor appears. □

2.2 Consequence for local quadratic field budgets Let x be an insertion anchor. Suppose a native mark, or an actual transported mark for which this has been proved, obeys

$$|O_x(z)| \le Ch^{-4}\left(B_x + \sum_i e^{-\gamma_1 d(x,i)}|z_i|\right)^2, \tag{CM.21}$$

$$|\nabla_j O_x(z)| \le Ch^{-4}e^{-\gamma_1 d(x,j)}\left(B_x + \sum_i e^{-\gamma_1 d(x,i)}|z_i|\right). \tag{CM.22}$$

Write M_i for the right side of (CM.14), and put

$$\mathcal{B}_x = 1 + B_x + \sum_i e^{-\gamma_1 d(x,i)} M_i. \tag{CM.23}$$

Minkowski's inequality and Theorem CM.1 give

$$\|O_x\|_p \le Ch^{-4}(\mathcal{B}_x + \sqrt{p})^2, \qquad \|\nabla_j O_x\|_p \le Ch^{-4}e^{-\gamma_1 d(x,j)}(\mathcal{B}_x + \sqrt{p}). \tag{CM.24}$$

Inserting these bounds into the existing real cut-cumulant theorem gives a fixed factorial bound of the form

$$|\kappa_m(O_{x_1}, \dots, O_{x_m})| \le C^m (m!)^2 h^{-4m} \prod_{r=1}^m \mathcal{B}_{x_r}^2 \exp\left[-\frac{c\tau(x_1, \dots, x_m)}{m-1}\right], \qquad m \ge 2, \tag{CM.25}$$

with the same conditional scope. The constants in (CM.24)-(CM.25) depend on the fixed lower bound λ, the fixed covariance constants c_*, γ, the localization rate γ_1, the geometric sums (CM.7), the fixed block dimensions, and the constants in (CM.21)-(CM.22). They are uniform in volume, source order and the cutoff only when those inputs and the force-resolvent margin are uniform; no uniformity as λ tends to zero is claimed.

One can check the factorial growth directly: apply the covariance bound to the two products across a longest-edge cut of a minimal spanning tree; use Holder with moment order at most $2m$; and sum the cumulant partition coefficients. Each undifferentiated quadratic mark costs at most $Cm\mathcal{B}_x^2$, each differentiated mark at most $C\sqrt{m}\mathcal{B}_x$. The fixed polynomial factors in m, the partition sum, and $m^m \le e^m m!$ fit in $C^m(m!)^2$. The longest-edge cut provides the

displayed order-dependent range. This is the same cumulant mechanism already proved in the collection; the new point is the explicit nonzero-force budget (CM.14).

Neither (CM.24) nor (CM.25) bounds $\mathcal{B}_x$ in the outer retained/history law. That is H-05 and the relevant part of H-04. Nor does the native-field bound (CM.21) automatically hold for the projected mark (CM.2).

3. Exact scalar readout without a coupling derivative

3.1 A quadratic readout of the actual incoming function At a fixed retained datum, choose a quadratic readout of the actual function F in (CM.2):

$$F(z) = a + \ell^T z + \frac{1}{2} z^T V z + T(z), \qquad V = V^T. \tag{CM.26}$$

When a common differentiable chart has been proved, one may use its Taylor coefficients at zero. Otherwise (CM.26) simply defines the exact residual T for the chosen quadratic readout. Its mere existence proves no smallness. In particular, an arbitrary measurable transported mark is not declared analytic by this notation.

Set

$$B = CV, \qquad \Psi(z) = \frac{1}{2} z^T K z - b^T z - W(z). \tag{CM.27}$$

For each block choose a fixed smooth cutoff $0 \le \chi_i \le 1$, equal to one on $\frac{1}{2}D_i$, with support in $\frac{3}{4}D_i$, and

$$|\nabla \chi_i| \le c_\chi / P_i. \tag{CM.28}$$

Fixed-dimensional norm equivalence and mollification supply these cutoffs for the actual bond-norm balls. They do not change the probability law. Their sole purpose is to make the integration-by-parts vector field vanish at its coordinate boundary.

Define the explicit collar expectation

$$\begin{aligned} \mathcal{E}_\chi(B) = \nu_v \sum_i \big[& (\chi_i - 1) \operatorname{tr} B_{ii} + \nabla \chi_i \cdot (Bz)_i \\ & + (1 - \chi_i)(Bz)_i \cdot \nabla_i \Psi(z) \big]. \end{aligned} \tag{CM.29}$$

Every summand is supported in the coordinate collar $E_i = \{z_i \notin \frac{1}{2}D_i\}$. Unlike a surface trace, (CM.29) is an expectation in the actual normalized law.

Theorem CM.2 – exact selected-law virial readout Assume the integrals in the following formula are absolutely finite. Then

$$\boxed{\begin{aligned} \nu_v F = a + \ell^T \nu_v z + \frac{1}{2} \operatorname{Tr}(CV) \\ + \frac{1}{2} \nu_v [(CVz) \cdot (b + \nabla W)] + \frac{1}{2} \mathcal{E}_\chi(CV) + \nu_v T. \end{aligned}} \tag{CM.30}$$

At every finite source regular fiber this applies to the actual incoming mark and chosen readout whenever the displayed derivative and integrability conditions hold. It uses the full W in (CM.6), not a Gaussian replacement of the density.

Proof. The vector field with block components $X_i(z) = \chi_i(z_i)(Bz)_i$ vanishes near the boundary of its own integration block. Integration by parts in that block, followed by Fubini, gives

$$0 = \nu_v \sum_i \left[\chi_i \operatorname{tr} B_{ii} + \nabla \chi_i \cdot (Bz)_i - \chi_i (Bz)_i \cdot \nabla_i \Psi \right]. \tag{CM.31}$$

No assertion that the sharp cutoff has zero derivative is used. Rearrange (CM.31), add the complementary $1-\chi_i$ terms, and use $\nabla\Psi = Kz - b - \nabla W$. Because K and V are symmetric,

$$(CVz)\cdot Kz = z^T Vz. \tag{CM.32}$$

The result is

$$\nu_v(z^T Vz) = \mathrm{Tr}(CV) + \nu_v[(CVz)\cdot(b+\nabla W)] + \mathcal{E}_\chi(CV). \tag{CM.33}$$

Insert (CM.26). This proves (CM.30). □

3.2 Checks and interpretation For an untruncated Gaussian with $W = 0$, the collar term is absent and $m = Cb$. Formula (CM.30) becomes

$$\nu F = a + \ell^T m + \tfrac{1}{2}\,\mathrm{Tr}(CV) + \tfrac{1}{2} m^T V m + \nu T, \tag{CM.34}$$

the exact Gaussian expectation. Thus the retained mean is not hidden in the determinant term.

For a standard one-dimensional Gaussian truncated to $(-P, P)$, $b = W = 0$ and $V = K = 1$, (CM.33) gives

$$\mathcal{E}_\chi(1) = \nu(z^2) - 1 = -\frac{2P\varphi(P)}{2\Phi(P) - 1}, \tag{CM.35}$$

where φ, Φ are the standard Gaussian density and distribution. In particular the exact collar expression has the nonzero boundary correction with the correct sign, independently of which admissible cutoff χ was used.

For the scalar YM mark, (CM.30) isolates the constrained trace term which appears in the Gaussian diagnostic, but its complete correction is now explicit. It contains the inner exterior-force response, the gradients of inherited E/R/B and coarea terms, the heterogeneous action correction, the boundary-relocation contribution, the actual incoming remainder, and a coordinate-collar expectation. No parameter derivative of the running coupling or of an adaptive R reference kernel has been used.

The readout $a(v) + \frac{1}{2}\,\mathrm{Tr}(C(v)V(v))$ is still a function of retained data. It has not yet been identified with a scalar coarse field plus a scalar constant in the F-01 prescription. That projection, its exact normalization, and its localized error remain part of M-04/F-03.

4. Local quantitative control of the collar term

This section proves that the ordinary expectation in (CM.29) can be bounded without a global partition-function comparison or a surface-density estimate.

Suppose, at the fixed retained datum, that for every $p \geq 1$,

$$\|z_j\|_p \leq M_j + c_z\sqrt{p}, \qquad \|\nabla_i\Psi\|_p \leq U_i + c_\Psi k_i\sqrt{p}. \tag{CM.36}$$

Theorem CM.1 and (CM.19) give one sufficient instance:

$$\begin{aligned} U_i &= |b_i| + d_i + \sum_j (\|K_{ij}\| + A_{ij})M_j, \\ k_i &= \sum_j (\|K_{ij}\| + A_{ij}), \end{aligned} \tag{CM.37}$$

with fixed c_z, c_Ψ. The following result only needs (CM.36).

Proposition CM.3 – rare-collar estimate with anchored summation Let $q_i = \nu_v(E_i)$, and put $L_i = 1 + \log(1/q_i)$ when $q_i > 0$. Then

$$|\mathcal{E}_\chi(B)| \le C \sum_{i:\, q_i>0} q_i \Big[|\operatorname{tr} B_{ii}| + \sum_j \|B_{ij}\|(M_j + c_z\sqrt{L_i})\left(P_i^{-1} + U_i + c_\Psi k_i\sqrt{L_i}\right) \Big]. \tag{CM.38}$$

The constant depends only on the fixed cutoff template and bounded block dimension. A term with $q_i = 0$ is zero. No total-volume factor is implicit in (CM.38).

Proof. For an event of probability $q > 0$, choose $r = \max\{2, \log(1/q)\}$. Holder gives

$$\nu(1_E|Y|) \le q^{1-1/r}\|Y\|_r \le e\, q\|Y\|_r. \tag{CM.39}$$

Here the last inequality also holds when $r = 2$, since then $q \ge e^{-2}$. The trace term in (CM.29) is bounded directly by $q_i|\operatorname{tr} B_{ii}|$. For the derivative-cutoff term use (CM.28), (CM.39), and the first estimate in (CM.36). For the last term apply Holder once more to the product of z_j and $\nabla_i\Psi$, at moment order $2r$, and use both estimates in (CM.36). Since $r \le 2L_i$, all factors fit in (CM.38) after a fixed change of constant. Sum the explicit matrix entries. □

For an insertion anchor x, a useful input is an actual weighted matrix bound

$$\sum_{i,j} e^{\varkappa(d(x,i)+d(x,j))}\|B_{ij}\| \le V_x. \tag{CM.40}$$

When the other local quantities in (CM.38) have a controlled envelope, (CM.40) makes the collar sum local and volume independent. It is not enough to bound the trace alone: the full matrix entries occur in (CM.38).

If, in addition, D_i contains a Euclidean ball of radius $c_D P_i$ and the mean obeys $|\nu z_i| \le c_D P_i/4$, the concentration proof of (CM.15) gives

$$q_i \le C\exp[-c\lambda P_i^2]. \tag{CM.41}$$

Indeed, outside $\frac{1}{2}D_i$, the block is at Euclidean distance at least $c_D P_i/2$ from zero, hence at least $c_D P_i/4$ from its mean. A finite union over real components and (CM.17) gives (CM.41). The required mean margin is a numerical condition on the actual right side of (CM.14). The fact that the support is a product does not establish that margin.

For a homogeneous radius P, uniform force envelopes and (CM.40), (CM.38)-(CM.41) yield an exponentially small collar error with an explicit polynomial prefactor in P. To insert a numerical upper bound on q_i in (CM.38), one can either use the same Holder proof with the numerical rate in place of the optimization or use the monotonicity of $q(1 + \log(1/q))^r$ once q is sufficiently small. No monotonicity outside that range is assumed.

This is an actual normalized-measure estimate whenever its moment and probability inputs have been verified. It is not yet a proof of those inputs on every interacting RG history.

5. Retained-variable derivatives and the scalar marginal coefficient

5.1 Exact current within one fixed regular chart Let $v(s)$ be a differentiable retained-data path which stays within one fixed live geometry and coupling chart. The product support (CM.1) is then fixed. Define the unnormalized exponent

$$L(z; v) = -\frac{1}{2}z^T K(v)z + b(v)^T z + W(z; v). \tag{CM.42}$$

Whenever differentiation is justified by the actual local bounds,

$$\frac{d}{ds}\nu_{\upsilon(s)}F_{\upsilon(s)} = \nu_{\upsilon(s)}\dot{F} + \mathrm{Cov}_{\nu_{\upsilon(s)}}(F, \dot{L}). \tag{CM.43}$$

In particular,

$$\dot{L} = -\frac{1}{2}z^T\dot{K}z + \dot{b}^T z + \dot{W}. \tag{CM.44}$$

Every background, chart, E/R/B, coarea and boundary-relocation derivative is retained in (CM.44). A retained-only additive constant in L disappears from the covariance, as it must for the same conditional law; its derivative remains in the outer weight. This is a conditional normalization cancellation, not permission to discard relative history weights.

Under the weighted covariance estimate, (CM.43) yields the local current bound

$$\left|\frac{d}{ds}\nu F - \nu\dot{F}\right| \le C\sum_{i,j} e^{-\gamma d(i,j)}\|\nabla_i F\|_2\|\nabla_j \dot{L}\|_2. \tag{CM.45}$$

It is the gradient of the complete retained score which must be localized and estimated. A real bound on the conditional partition function does not imply this estimate.

Integrating (CM.43) along an admitted path gives the finite-difference current needed by M-03. If the path crosses a live-selector interface, its interface current must additionally be included. The fixed regular support theorem does not remove those currents, and the formula is not asserted across a change of geometry without them.

5.2 The exact second retained derivative For retained coordinates υ_a, υ_b inside such a chart, put $S_a = \partial_a L$. Direct differentiation of the normalized integrals gives

$$\begin{aligned}\partial_a\partial_b(\nu F) = \nu F_{ab} + \mathrm{Cov}(F_a, S_b) + \mathrm{Cov}(F_b, S_a)\\ + \mathrm{Cov}(F, \partial_a\partial_b L) + \kappa_3(F, S_a, S_b).\end{aligned} \tag{CM.46}$$

Proof. Differentiate (CM.43) in the second coordinate. Differentiating the two factors in the covariance produces the two covariance terms involving derivatives of its arguments. The change of the normalized law produces the third joint cumulant. Expanding those expectations proves (CM.46); retained-only constants in S_a, S_b cancel from every covariance and cumulant. □

This gives a direct formula for the quadratic retained-background coefficient of the exact transported scalar mark, once the appropriate scalar projection and physical normalization have been fixed. It includes the actual first and second retained derivatives of the complete density and of the projected incoming mark. Using only the derivative of the native classical action would omit four classes of terms in (CM.46).

No uniform bound on those derivatives or a numerical evaluation of the interacting scalar coefficient is supplied by this identity. Thus it does not complete F-03 or M-04.

6. The precise remaining inequalities

C-08 The fixed product support, granted the source estimates of Annex AA, removes the unspecified inner-support premise for the source-normal-form regular marginal. The remaining quantitative theorem must verify, on the actual complete family (CM.5):

- the constrained precision gap and weighted localization on the heterogeneous admitted class;
- the full E/R/B/measure/heterogeneous Hessian bound strong enough for the numerical covariance margin and (CM.12);
- the local zero-field gradient and exterior-force envelopes in (CM.14), including any collar-margin requirement used in (CM.41);
- the actual field and gradient estimates for the projected marks, with the fixed factorial growth needed at every source order;

- the uniformity in scale and volume, with retained-background factors exposed before outer-history averaging.

Neither a small value bound for the action nor the covariance inequality for the earlier homogeneous no-R model establishes these statements. C-08 therefore remains open.

M-04 The exact incoming scalar transport now admits the readout (CM.30), and the finite retained coefficient can be expressed by (CM.46). To meet the stated requirements one must still prove, for that same incoming mark and full law:

1. a localized representation and quantitative readout bound for (CM.2), rather than assuming that it is a native field;
2. the scalar projection of $a(\upsilon) + \frac{1}{2}\operatorname{Tr}(C(\upsilon)V(\upsilon))$ into the prescribed current scalar basis and identity coefficient;
3. a uniform bound in M-03 for the exterior-mean, complete-interaction-gradient, incoming-remainder and collar terms in (CM.30);
4. localized retained-score and direct-mark derivatives for (CM.45), including interfaces encountered by the actual finite-difference norm;
5. the scale gain or coefficient budget that makes the subsequent all-scale induction valid. A one-step $O(g)$ comparison alone is not summable on the expected asymptotically free trajectory.

7. References

- T. Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, CMP 119 (1988), 243-285: (3.15)-(3.25), printed pp.267-270. The actual current cutoff and exponent are retained.
- Annex W: complete current-coordinate law, coarea and Haar normalization, boundary relocation, and Schur-complement bookkeeping.
- Annex AA: source-normal-form product support after complete R, exact projection of marked jets, and the weighted Neumann boundary check.
- The conditional-mean and real cut-cumulant proofs of the earlier annexes: used only for their stated weighted covariance and cumulant mechanism, not as estimates for a different interacting law.

All new elementary identities are proved above. The proposed force and moment criteria retain their unverified actual-model inputs explicitly.

Annex AC — Latent rare histories, full reset jets and a spatial mixing certificate

Annex AC. The conclusions below hold on the stated source domains. Applying them to the complete Wilson law requires compatibility of the conditional representation (CL), including the retained variables, densities, and marked transport. See the theorem catalogue.

0. Results and hypotheses

Neither H-02 nor H-03 is completed by this note. The actual inherited low-margin probabilities and the spatial mixing of the actual normalized history posterior have not been bounded. The results below make two further parts of those problems exact.

First, a rare history may be latent inside the conditional law being averaged. There is still a bound with its full unconditional probability factor for the change of conditional cumulants caused by modifying marks only on that history. One does not have to retain the rare event as an output coordinate, normalize on its complement, or assume a lower bound on its conditional probability. The result extends to a complete finite logarithmic source jet without adding a factorial exponent to its existing combinatorial budget.

Second, an averaged coupling bound for a cut of the actual history posterior implies an all-order spatial cumulant estimate. A longest-edge cut of a minimum spanning tree gives the required tree-length range, with a fixed factorial exponent. This is a concrete sufficient mixing certificate; the certificate itself remains unproved for the full Balaban history law.

All probability laws used below are normalized positive laws. Signed polymer activities are never used as history probabilities. Every construction is finite at a cutoff, and every statement about arbitrary source order is an identity or estimate for finite jets of every order, not a claim about convergence of an infinite source series.

1. Conditional replicas retain the correct probability

Let a finite-cutoff positive history space have law

$$d\mu(y,h) = d\nu(y)\,\pi_y(dh). \tag{H.1}$$

The retained datum is y. The variable h includes every variable to be averaged at the specified operation. It need not consist only of discrete labels. The standard Borel spaces and actual disintegration are the ones constructed in Annex W. Write

$$(QF)(y) = \int F(y,h)\,\pi_y(dh). \tag{H.2}$$

For a partition $\mathcal{P}$ of $\{1,\dots,k\}$, introduce one independent conditional replica h_B for each block:

$$d\mu_{\mathcal{P}} = d\nu(y) \prod_{B\in\mathcal{P}} \pi_y(dh_B). \tag{H.3}$$

Independence is conditional on y, and is asserted only for the new replicas in (H.3). The original histories need not have any independence property. Every single pair (y,h_B) has marginal μ.

Let E be an event on the input space, with $q = \mu(E)$. It need not be measurable with respect to y. If B_* is one block of a partition, then

$$\mu_{\mathcal{P}}\big((y,h_{B_*}) \in E\big) = q. \tag{H.4}$$

This identity is what permits an event factor after products of conditional moments have appeared. It does not replace $\pi_y(E)$ by q pointwise.

Lemma H.1 — a replica event estimate Let positive integers $d_1, \dots, d_k$ have sum m. Suppose functions Z_i obey

$$\|Z_i\|_{L^p(\mu)} \le b_i p^{s d_i} \quad (p \ge 1), \qquad s \ge 0. \tag{H.5}$$

Choose any partition $\mathcal{P}$, write $B(i)$ for the block containing i, and choose a distinguished block B_*. For $0 < q \le 1$, put $\lambda = \log(1/q)$. Then

$$\int 1_E(y, h_{B_*}) \prod_{i=1}^{k} |Z_i(y, h_{B(i)})| \, d\mu_{\mathcal{P}} \le q e^m (m+\lambda)^{sm} \prod_i b_i. \tag{H.6}$$

The bound is zero when $q = 0$.

Proof. For $0 < q < 1$, set $r = 1 + \lambda/m$. Apply Hölder to the event with exponent $r/(r-1)$, and to the i-th function with exponent rm/d_i. Their reciprocal exponents sum to one. Since each replica has the correct marginal, (H.5) yields

$$q^{1-1/r} \prod_i b_i (rm/d_i)^{s d_i} \le q e^{\lambda/r} (rm)^{sm} \prod_i b_i.$$

Here $\lambda/r \le m$ and $rm = m + \lambda$. This proves (H.6). For $q = 1$, omit the event and use Hölder with exponents m/d_i. For $q = 0$, (H.4) proves the assertion. No conditional normalizer was estimated. □

2. Changing marks on a latent bad history

For functions $A_1, \dots, A_k$, their conditional cumulant is the finite expression

$$\kappa_Q(A_1, \dots, A_k) = \sum_{\mathcal{P} \in \Pi_k} (-1)^{|\mathcal{P}|-1} (|\mathcal{P}| - 1)! \prod_{B \in \mathcal{P}} Q\left(\prod_{i \in B} A_i \right). \tag{H.7}$$

All the products in (H.7) are integrable under the moment assumptions below. Complex marks are allowed; the cumulant is multilinear, with any desired conjugations included in the specified marks themselves.

Theorem H.2 — latent rare-edit stability Let A_i, B_i, $1 \le i \le k$, satisfy $A_i = B_i$ outside the same event E, and suppose

$$\max\{\|A_i\|_p, \|B_i\|_p\} \le b_i p^{s d_i}, \qquad p \ge 1, \qquad \sum_i d_i = m. \tag{H.8}$$

Then

$$\int |\kappa_Q(A_1, \dots, A_k) - \kappa_Q(B_1, \dots, B_k)| \, d\nu \le q \, 2^k k k! e^m (m+\lambda)^{sm} \prod_i b_i. \tag{H.9}$$

For $k = 1$, the left side is the difference of conditional means and the formula remains valid. The right side is interpreted as zero for $q = 0$.

Proof. Multilinearity gives the exact telescope

$$\kappa_Q(A_1, \dots, A_k) - \kappa_Q(B_1, \dots, B_k) = \sum_{j=1}^{k} \kappa_Q(B_1, \dots, B_{j-1}, A_j - B_j, A_{j+1}, \dots, A_k). \tag{H.10}$$

For each term in its partition formula, designate the block containing the difference. Its absolute value is bounded by the product of conditional absolute moments, with 1_E in the designated block. Integrating represents that product on (H.3). Apply Lemma H.1, using the bound $2b_j p^{s d_j}$ for the difference and the original bounds for the other marks.

The sum of absolute cumulant coefficients is at most $k!2^{k-1}$. Indeed, ordered set partitions can be encoded by a permutation of the k labels and a choice of cuts in its $k-1$ gaps. Their count bounds the coefficient sum. Multiplying by the two from the difference and by the k telescoping terms proves (H.9). □

The event in this theorem is inside a conditional replica. This differs from a bound of the form $\int 1_E|\kappa_Q|$, which places the event outside the conditional cumulant. The latter cannot be used when an input event has been averaged away without explicitly enlarging the output. Theorem H.2 avoids that enlargement for the stated comparison of cumulants.

For full product coefficients, telescoping directly before conditioning yields

$$\int \left| Q\left(\prod_i A_i\right) - Q\left(\prod_i B_i\right)\right| d\nu \le 2kqe^m(m+\lambda)^{sm}\prod_i b_i. \tag{H.11}$$

There is no cumulant partition factor in (H.11).

Application to the actual compact Wilson input On the cofinal reserve class of Annex A, the energy-normalized 21-field representatives obey

$$\|O_{\alpha,a}(x)\|_p \le Ca^{-4}p, \qquad p \ge 1. \tag{H.12}$$

The microscopic Wilson marginal is unchanged by the complete positive history lift. Therefore (H.12) holds on that lift for every actual inherited event E, with its actual normalized probability q. Taking $B_i = 1_{E^c}A_i$, and using conditional Jensen when A_i is an exactly transported original mark, gives

$$\int |\kappa_Q(A_1,\dots,A_m) - \kappa_Q(1_{E^c}A_1,\dots,1_{E^c}A_m)|d\nu \le qC^{m+1}m!(m+\log(1/q))^m a^{-4m}. \tag{H.13}$$

The same coefficient dictionary and source-free positive law as Annex W are used. Additional running mixing coefficients must be included explicitly in the input budgets; (H.13) does not absorb them into C.

This is an actual finite-Wilson comparison for arbitrary inherited events. It does not give the missing upper bound on q, and its physical field budget is still a^{-4}, not the desired current-scale h^{-4}. The clipping in (H.13) is an estimation device, not a proposed replacement of the physical field or a claim that clipping preserves locality or OS positivity.

3. Stability of the whole reset jet

Fix a finite set of distinct source labels. For every nonempty subset I, let A_I and B_I be the coefficients of two incoming logarithmic jets. Repeated field types are allowed: the auxiliary source labels remain distinct. Set

$$A_I^+ = \sum_{\mathcal{P}\in\Pi(I)} \kappa_Q(A_J : J\in\mathcal{P}), \qquad B_I^+ = \sum_{\mathcal{P}\in\Pi(I)} \kappa_Q(B_J : J\in\mathcal{P}). \tag{H.14}$$

These are exactly the coefficients of $\log Qe^{\Psi_A}$ and $\log Qe^{\Psi_B}$. Thus every already generated multi-field vertex is retained.

Theorem H.3 — complete rare-edit jet bound Assume $p_0 \ge 1$, $s \ge 0$, $C \ge 1$, $b > 0$, and

$$A_J = B_J \quad \text{outside } E,$$

$$\max\{\|A_J\|_p, \|B_J\|_p\} \le C^{d+1}(d!)^{p_0}b^d p^{sd}, \qquad d = |J|, \quad p \ge 1. \tag{H.15}$$

There is C_*, depending only on C, such that for every I of size $m \ge 1$,

$$\boxed{\int |A_I^+ - B_I^+|\, d\nu \le q C_*^{m+1} (m!)^{p_0} (m + \log(1/q))^{sm} b^m.} \tag{H.16}$$

This construction is compatible under truncation at every source order.

Proof. Apply Theorem H.2 to the k block marks in each partition of I, with block degrees $d_1, \dots, d_k$. Their total degree is m. The product of moment constants contributes at most

$$C^{m+k} b^m \prod_{j=1}^{k} (d_j!)^{p_0}.$$

It remains to bound a purely finite partition sum. For fixed k,

$$\sum_{\substack{\mathcal{P} \in \Pi_m \\ |\mathcal{P}|=k}} k! \prod_{J \in \mathcal{P}} (|J|!)^{p_0} = m! \sum_{\substack{d_1 + \cdots + d_k = m \\ d_j \ge 1}} \prod_{j=1}^{k} (d_j!)^{p_0 - 1}. \tag{H.17}$$

This follows by first ordering the blocks and then choosing their labels. Since $p_0 \ge 1$, each product on the right is at most $(m!)^{p_0 - 1}$. There are $\binom{m-1}{k-1}$ ordered compositions. The remaining factors $k 2^k C^k$, summed over k, are bounded by a constant to the power $m + 1$, because $k \le m \le 2^m$. Equation (H.16) follows. The formula for a coefficient of order m uses no coefficient of higher order, proving truncation compatibility. □

The estimate does not add a factorial exponent to p_0 through the reset partition sum. The degree-dependent moment factor remains explicit. An actual iteration still needs a useful current-scale input budget, spatial localization, and a summable probability/geometry budget. None follows just from (H.16).

Corollary H.3a — an explicit carrier-sum threshold Suppose corrections are indexed by finite carriers D containing a fixed anchor, with integer size $r(D) \ge 1$. Assume that at most A^r such carriers have size r, with $A \ge 1$, that each correction is a comparison of the form (H.16) with the same constants, and that its actual event probability satisfies, for $\sigma > 0$,

$$q_D \le e^{-\sigma r(D)}. \tag{H.16a}$$

For a carrier weight $e^{\kappa r(D)}$, the strict inequality

$$\sigma > \kappa + \log A \tag{H.16b}$$

is sufficient to give

$$\sum_{D \ni x} e^{\kappa r(D)} \int |A_{I,D}^+ - B_{I,D}^+|\, d\nu \le C_{C,\sigma,\kappa,A,s}^{m+1} (m!)^{p_0 + s} b^m. \tag{H.16c}$$

No independence or disjointness among the carrier events is required.

Proof. Choose $\varepsilon \in (0,1)$ such that $(1-\varepsilon)\sigma > \kappa + \log A$. For $\lambda \ge 0$, maximizing $e^{-\varepsilon\lambda}(m+\lambda)^{sm}$ gives

$$e^{-\lambda}(m+\lambda)^{sm} \le C_{s,\varepsilon}^{m+1} (m!)^s e^{-(1-\varepsilon)\lambda}.$$

This includes $s = 0$. Apply it to $\lambda_D = \log(1/q_D)$, use (H.16a), and sum the resulting geometric series over $r \ge 1$. Its ratio is $\exp[\kappa + \log A - (1-\varepsilon)\sigma] < 1$. This proves (H.16c). □

This identifies an explicit competition between probability suppression and carrier count. It does not prove (H.16a) for an inherited sector, or a current-scale bound for the input jets. If the actual carriers have extra factors growing with their size, those factors must be included in the sum rather than hidden in C.

4. A concrete spatial certificate for the actual posterior

Let $A_i(y,h)$, $i=1,\dots,m$, be marks assigned to spatial anchors x_i, and suppose

$$\|A_i\|_p \le C b_i p^s, \qquad p \ge 1. \tag{H.18}$$

For a nontrivial cut $I \sqcup J = \{1,\dots,m\}$, let σ_y be the joint conditional law of the two vectors A_I, A_J, and let $\sigma_y^I \otimes \sigma_y^J$ be the product of its two marginals. Define the actual conditional cut defect

$$q_{I|J} = \int \|\sigma_y - \sigma_y^I \otimes \sigma_y^J\|_{\mathrm{TV}}\, d\nu(y), \tag{H.19}$$

where total variation is $\sup_E |P(E)-Q(E)| \in [0,1]$. The definition is in terms of the normalized history posterior, not the fresh R reference law.

At a finite lattice, for finite history labels, a measurable maximal coupling is explicit. On a common finite value set its common part is the pointwise minimum of the two probability masses; couple that part identically and couple their disjoint residual parts. More generally the same construction uses the two densities with respect to their sum on a standard Borel space. The conditional densities and resulting kernels may be chosen measurably; alternatively the following theorem can take a specified measurable coupling with failure probability q as its input, without invoking optimality.

Theorem H.4 — averaged cut mixing implies a cumulant bound For every nontrivial cut and $q = q_{I|J} > 0$,

$$\int |\kappa_{\pi_y}(A_1,\dots,A_m)|\, d\nu(y) \le q C_1^{m+1} m!(m+\log(1/q))^{sm} \prod_i b_i. \tag{H.20}$$

It is zero if $q = 0$. The same statement holds using any proved upper-bound coupling failure probability, with the monotonic replacement described below.

Proof. Conditional on y, couple the original vector A to a vector B with law $\sigma_y^I \otimes \sigma_y^J$. The two vectors agree outside a coupling failure event of unconditional probability q. Their individual coordinates have the same marginal laws, so (H.18) holds for both. The conditional cumulant of all B_i is zero: its formal logarithmic generating jet is a sum of the jets for the two independent subvectors, hence has no mixed coefficient involving both. Apply Theorem H.2 on this enlarged conditional probability space, with all degrees equal to one, and absorb $m2^m$ into C_1^{m+1}. □

For $s > 1$, the function $q(m+\log(1/q))^{sm}$ need not be monotone on all $(0,1]$. A safe consequence of (H.20), valid for every $s \ge 0$, is

$$\boxed{\int |\kappa_{\pi_y}(A_1,\dots,A_m)|\, d\nu(y) \le C_s^{m+1}(m!)^{s+1} q^{1/2} \prod_i b_i.} \tag{H.21}$$

To prove it, write $q = e^{-\lambda}$, $\lambda \ge 0$, and maximize $e^{-\lambda/2}(m+\lambda)^{sm}$. If $s = 0$ the expression is at most one. If $s > 0$, the maximum occurs at $\lambda = \max(0,(2s-1)m)$, and is at most $C_s^m m^{sm} \le C_s'^m (m!)^s$. Substitution in (H.20) proves (H.21). This bound is monotone in the proved upper bound on q.

Corollary H.5 — a tree-length estimate with a fixed factorial exponent Let $\tau(x_1,\dots,x_m)$ be the length of a minimum spanning tree in the chosen physical metric, and let $h > 0$ be the current scale. Suppose, for the actual marks and posterior, every nontrivial cut satisfies the quantitative certificate

$$q_{I|J} \le \min\{1, D^m \exp[-c\, d(x_I,x_J)/h]\}, \qquad D \ge 1, \quad c > 0. \tag{H.22}$$

Then, for $m \ge 2$,

$$\boxed{\int |\kappa_{\pi_y}(A_1, \dots, A_m)|\, d\nu(y) \le C_{s,C,D}^{m+1} (m!)^{s+1} \exp\Big[-\frac{c\,\tau(x_1, \dots, x_m)}{2(m-1)h}\Big] \prod_i b_i.} \tag{H.23}$$

Proof. A longest edge of a minimum spanning tree has length at least $\tau/(m-1)$. Removing it gives a cut. Every edge crossing that cut has length at least the removed edge: a shorter one would produce a spanning tree of smaller total length. Consequently the cut distance in (H.22) is at least $\tau/(m-1)$. Apply (H.21), and absorb $D^{m/2}$ into the constant. □

For marks supported on spatial carriers, use their actual supports in the cut distance. Anchor distance can replace support distance only after subtracting the corresponding carrier radii or proving a localization estimate. A nominal insertion label does not make a globally history-dependent mark local.

The conclusion has the range dependence needed by the existing shell-summation architecture. If the input budget were $b_i \le Ch^{-4}$, it would also have the current-scale physical prefactor. This conditional statement supplies neither that budget nor (H.22) for the interacting history posterior.

The theorem bounds the conditional cumulant averaged over y. An additional external history parameter can be held fixed throughout the proof, but uniformity in that parameter requires the input moment and cut-coupling bounds with the same constants for every admitted parameter. It cannot be obtained by averaging over that parameter first. Retained-coordinate derivatives require the corresponding score/current marks and their own moment and coupling budgets. The outer cumulants of conditional means remain the separate H-05 obligation.

5. What the new estimates do and do not remove

5.1 No hidden normalization on a good sector Equations (H.9), (H.13) and (H.16) do not form $\pi_y(\cdot \mid E^c)$. They remain valid on retained data where $\pi_y(E) = 1$, and they have no factor $1/\pi_y(E^c)$. The reference or clipping values are compared inside the unchanged normalized law. This removes one avoidable denominator from estimates of proposed rare-history corrections.

The probability q remains the actual full-law probability. A small unnormalized branch mass or a fresh-reference failure probability cannot replace it.

5.2 A mixture can still have nonlocal reset responses Let $h = \pm 1$ have equal probability, and place $A_1 = h, A_2 = h$ at arbitrarily distant anchors. There is no retained datum. Each mark has bounded moments, while $\kappa(A_1, A_2) = 1$. The joint sign law has total variation distance $1/2$ from the product of its marginals, independent of the anchor distance. Thus (H.22) fails, exactly where the proof of spatial decay requires it.

This is a check of the certificate's necessity as an added input to the displayed sufficient argument. It is not a Yang–Mills counterexample and does not assert that cut total variation is the only possible route to H-03. Weighted couplings or direct cumulant estimates may be more efficient for continuous or strongly constrained histories.

5.3 Full jet comparison is not locality of its coefficients The factorial estimate (H.16) is integrated over retained data and has no spatial factor. Even if an event is rare, a correction supported on it may depend on a large carrier. Its carrier norm must be paid using the actual geometry. At the same time, a cutoff-wide clipping event can destroy the locality of a mark. The theorem licenses the stated bound, not an unexamined field redefinition.

6. Application conditions

Statement	New proved work	Still required by the original hypotheses
H-02	Rare-edit estimates for latent history events under the actual normalized law; all-order extension to complete reset jets; no good-sector conditional denominator.	A current-scale bound for the actual inherited low-margin and unreplaced events, with overlap and geometric costs sufficient to pay the insertion/carrier losses. Neither the needed probability nor the current-scale field budget follows from the new theorem.
H-03	An averaged conditional cut-coupling criterion implying all-order tree-length decay with fixed factorial growth; actual normalized posterior specified; full jets retained.	Prove the cut certificate or a sufficient direct spatial response estimate for the actual history posterior, uniformly in the remaining history and at the required current scale.

Sources

The probability identities and estimates in this note are proved directly. Their finite Wilson application uses the already proved compact Wilson reserve and field bounds retained in Annex A, the positive-history construction in Annex W, and the exact reset-jet identity retained in Annex I. No new external mathematical theorem about Yang–Mills is imported. The maximal-coupling formulation can be replaced throughout by a supplied finite measurable coupling, as stated above.

Annex AD — Actual E contours and uniformly summable derivatives

Annex AD. The conclusions below hold on the stated source domains. Applying them to the complete Wilson law requires compatibility of the conditional representation (CL), including the retained variables, densities, and marked transport. See the theorem catalogue.

1. Result and a correction to the preceding domain conditions

Under the stated source-free construction hypotheses, Theorem E14.3 supplies a common native complex contour for the regular E contribution and summable old-age majorants before Cauchy differentiation. The contour includes the native background, the independently extended current and the preparation parameters. Boundary activities B, complete conditional coercivity and matching after a change of retained coordinates require separate estimates. Identification with the Wilson conditional law requires CL.

There is a useful correction to the source estimates. CMP119 p.259 places a primitive E activity on the regular space $U_j^c(X, \alpha_{0,j}, \alpha_{1,j})$ of CMP109 (1.11)-(1.16). The multiscale space defined in CMP119 (2.34)-(2.39), pp.260-261, is introduced for the boundary activities B. A primitive E activity should not itself be assigned all the B-domain conditions. The latter re-enter when an E expression is localized across a multiscale boundary. This distinguishes the interior and boundary admission problems; it does not discard the boundary contribution.

The primary source already resums the old E scales before stating its current-scale analytic activity bound. Differentiating that resummed identity on a verified common complex domain is a viable alternative to charging every primitive old-scale Cauchy cost separately. Sections 5.1–5.6 control its running-domain margin and retain the coefficient-order dependence of the old-scale constants.

No Gaussian substitution, source-dependent background, or deletion of an inherited counterterm is made below.

2. The precise primary inputs

See CMP119, pp. 247, 250, 255, 259–263, 265, 267–268 and 275–284, CMP102, p. 285, (44), Proposition 9 and (172), (179)–(190), and CMP99, pp. 391–392 and 404–406.

The source chain used here is:

1. Primitive regular domain. CMP109 (1.11)-(1.16) distinguishes a real representative, its relative complex field, the independent current, and reconstructed fields. The paragraph on p.263 after (1.16) gives a sufficient smaller basic domain: the basic conditions (i)-(iii), with appropriately reduced constants, imply the reconstructed conditions. Its displayed estimates use the constant $2B_3$ from the minimizing-field theorem. One therefore need not independently rebuild every reconstructed constraint after proving admission to that smaller basic domain.
2. Independent-current insertion. CMP109 (3.11)-(3.14) replaces the second-order operator in the localized current by the covariant Laplacian, the projection term and the displayed local corrections. The controlled insertion norms include $A, P_U A, \nabla_U A, \Delta_U A$. The physical-current identity for an unlocalized Landau-gauge minimizer is not imposed separately on each localized summand.
3. A native additive margin. CMP109 Lemma 4, (3.53), admits an extra coarse determining-field input B', uniformly along the real parameter interval of the expansion. The proof displays restrictions of the form

$$B_3\alpha_3 < \alpha_1/2, \qquad C_{51}\alpha_3 \le \beta L^{-2}\alpha_0. \tag{E14.1}$$

   Here C_{51} denotes the fixed coefficient in the derivative estimate (3.51); no ambiguous printed superscript is used as a numerical value. Thus there is a source-native additive margin with a linear dependence on the displayed radii, after the other hypotheses of that lemma have been met. This is more information than membership in an open analytic set. It is not yet a margin on the full current-coordinate support.
4. Preparation and old-scale summation. CMP116 (1.11)-(1.22) control the actual localized minimizing map with complex decoupling parameters. Lemma 1, (1.33)-(1.36), then states the jointly analytic, localized E fluctuation identity after the old scales have been summed. CMP119 p.276 transfers the interior construction to the complete geometry and gives a different multiscale domain for the remaining boundary terms.

These are source-scoped inputs, with their published parameter and regularity hypotheses retained.

3. Quantifying the actual native preparation

3.1 The maps and the source losses Write z for the free current fluctuation field called B in CMP116. This temporary notation avoids confusing it with the coarse determining-field input in CMP109 Lemma 4. The exact input map in CMP116 (1.19) is

$$b'(z) = g_k C z - hD(g_k C z). \tag{E14.2}$$

Here C, h, D are the actual constraint-elimination and nonlinear linearization maps of that construction. They are not replaced by their linear parts.

For a localized decoupling family s, the minimizing response is

$$\mathcal{H}_s(b') = H_0(s)b' + A_0(s, H_0(s)b') - H(s)D(H(s), H_0(s)b' + A_0(s, H_0(s)b'))\,. \tag{E14.3}$$

This is CMP116 (1.17). Its operators depend on the same external field and independent current as the original construction.

Put

$$K = B_0 e^{16\kappa_1}. \tag{E14.4}$$

The source controls the localized propagators by K, in all their admitted regularity norms, simultaneously for

$$|s(\Delta)| \le e^{\kappa_1}. \tag{E14.5}$$

With C_B denoting the source constant in (1.20), the ensuing bounds are

$$\|b'(z)\|_\infty \le C_B g_k \|z\|_\infty, \qquad \|\mathcal{H}_s(b'(z))\|_{\mathrm{adm}} \le 4KC_B g_k \|z\|_\infty. \tag{E14.6}$$

The norm $\|\cdot\|_{\mathrm{adm}}$ means the collection of regularity norms actually used in this source step, including the relevant second-order output. Uniformity in the allowed external variables is part of the stated native domain. It is not asserted on an arbitrary enlarged domain.

The proof in CMP116 uses two auxiliary radii. The displayed sufficient inequalities include

$$4C_2 K\varepsilon_2 \le 1, \qquad 4C_4 K^2 \varepsilon_3 \le 1, \qquad 2K\varepsilon_3 \le \varepsilon_2. \tag{E14.7}$$

These inequalities expose the preparation cost. In particular, it is not legitimate to call the localized map a contraction with its original unlocalized constant and omit $e^{16\kappa_1}$.

Proposition E14.1 — an explicit native reserve Retain all the source's external-field and propagator hypotheses. Let $\varepsilon_{\mathrm{abs}} > 0$ be a fixed admitted ceiling for its local analytic maps and the remaining fixed smallness conditions. Let $C_\zeta \ge 1$ bound multiplication by the cutoffs used in the insertion norm. Define

$$\begin{aligned} e_2 &= \min\{\varepsilon_{\mathrm{abs}}, (16C_2K)^{-1}\},\\ e_3 &= \min\{\varepsilon_{\mathrm{abs}}, e_2/(4K), (16C_4K^2)^{-1}\},\\ e_1 &= \min\{\varepsilon_{\mathrm{abs}}, e_3/(2C_B), \alpha_2/(32C_\zeta KC_B)\}. \end{aligned} \tag{E14.8}$$

The actual map (E14.3) is analytic for the source's external variables, the simultaneous preparation parameters (E14.5), and

$$g_k\|z\|_\infty < e_1. \tag{E14.9}$$

All the inequalities (E14.7) hold with a strict numerical reserve. Moreover, its insertion output has

$$C_\zeta \|\mathcal{H}_s(b'(z))\|_{\mathrm{adm}} \le \alpha_2/8. \tag{E14.10}$$

The extra insertion parameter $t_\square$ can be admitted simultaneously on

$$|t_\square| \le T_*, \qquad T_* = \frac{\alpha_2}{16 C_\zeta K C_B e_1} \ge 2, \tag{E14.11}$$

along with the real parameter $0 \le t \le 1$. In particular the sum $(t\zeta + t_\square \zeta_\square)\mathcal{H}_s(b'(z))$ has insertion norm at most $3\alpha_2/8$.

Proof. The first two minima give $4C_2 K e_2 \le 1/4$, $4C_4 K^2 e_3 \le 1/4$, and $2Ke_3 \le e_2/2$. The next minimum gives $\|b'(z)\| < C_B e_1 \le e_3/2$. The analytic fixed-point construction in CMP116 (1.12)-(1.18) therefore applies on these smaller domains, with its displayed margins, proving (E14.9) and (E14.6). The last term in the minimum gives (E14.10). The second summand with $t_\square$ costs at most

$$C_\zeta 4 K C_B e_1 T_* = \alpha_2/4.$$

Adding the bound $\alpha_2/8$ for the first summand gives the claimed strict insertion reserve. □

This is a quantitative statement about the actual native nonlinear maps and simultaneous preparation variables. It does not assume a bound for a substituted Gaussian map. The source constant α_2 and the external-field domain have not been declared fixed under the subsequent running-domain transfer.

3.2 The independently extended current has a genuine native direction Within the regular domain, the independent variable J occurs in the basic current norm constraint. The reconstructed currents in condition (iv) are constructed from the field U; they do not turn the independent J into a function of U.

Consequently, with U fixed in an admitted smaller basic domain, an independent-current center J_0 with $\|J_0\|_\infty < \gamma_*$ has the explicit native affine current margin

$$\rho_J = \gamma_* - \|J_0\|_\infty. \tag{E14.12}$$

For $\|\delta J\|_\infty < \rho_J$, all the current constraints on this independent variable remain satisfied. This is an immediate check of the source definition, and it remains simultaneous with (E14.9) and (E14.5). The actual real-center margin and the running prepared-current substitutions are quantified separately in Sections 5.2, 5.5 and 5.6.

Corollary E14.2 — all native current and preparation derivatives Let z_0 be a real center with $\|z_0\|_\infty \le P$ and

$$P < e_1/g_k, \qquad \rho_z = e_1/g_k - P > 0. \tag{E14.13}$$

Take the external background in the smaller native domain just specified, J_0 as in (E14.12), and a real preparation center $s_0(\Delta) \in [0,1]$. After including a fixed Lie-algebra coordinate norm factor $c_G > 0$, the radii

$$r_z = c_G \rho_z, \qquad r_J = c_G \rho_J, \qquad r_s = (e^{\varkappa_1} - 1)/2 \tag{E14.14}$$

define a simultaneous product contour in all these variables. For multi-indices a, b, c, the complete minimizing output obeys

$$\|\partial_z^a \partial_J^b \partial_s^c \mathcal{H}_s(b'(z))\|_{\mathrm{adm}} \le a!\, b!\, c!\, 4KC_B e_1\, r_z^{-|a|} r_J^{-|b|} r_s^{-|c|}. \tag{E14.15}$$

Proof. The current polydisc stays in (E14.9), the independent-current polydisc stays in its basic constraint, and $|s_0 + \delta s| < e^{\varkappa_1}$. The same native bound (E14.6) holds uniformly on their product, since the native source construction is jointly analytic there. Multivariable Cauchy, applied to each of the admitted output norms, proves (E14.15). The finite Lie-algebra dimension determines c_G; it does not introduce a factor equal to the number of lattice sites. □

In particular, this calculation retains the nonlinear constraint map, independent current and complex preparation parameters simultaneously. It is stronger than estimating only the physical current of an unlocalized minimizing field. Its scope is the verified native domain, and its radius is displayed rather than presumed uniform.

3.3 A linear reserve in the nearby-carrier preparation The nearby-carrier proof supplies a further quantitative fact. In CMP109 (3.49), the extra coarse field $Q'(A)$ generated by the insertion satisfies

$$\|Q'(A)\|_\infty \le C_Q L^j \eta \, \|A\|_{\mathrm{reg}}, \qquad L^j \eta \le 1, \tag{E14.15a}$$

where C_Q is the fixed source constant and the regular norm is the one in (3.31). The argument (3.50)-(3.53) admits an independent additive coarse input of radius α_3, subject to (E14.1). Fix an admitted absolute ceiling ϵ_*, retaining the source's other background hypotheses, and choose

$$r_3 = \min\left\{\epsilon_*, \frac{\alpha_1}{4B_3}, \frac{\beta L^{-2}\alpha_0}{4C_{51}}\right\}, \qquad a_{\mathrm{near}} = \min\left\{\epsilon_*, \frac{r_3}{4C_Q}\right\}. \tag{E14.15b}$$

For $\|A\|_{\mathrm{reg}} < a_{\mathrm{near}}$, $0 \le \tau \le 1$, and an independent additive perturbation $\|B''\|_\infty < r_3/4$, one has

$$\|\tau Q'(A) + B''\|_\infty < r_3/2. \tag{E14.15c}$$

The source's simultaneous native inclusion therefore applies with a strict reserve. The proof is direct: (E14.15a) bounds the first summand by $r_3/4$, the second has the same bound, and (E14.15b) makes both inequalities (E14.1) strict. This retains the actual nonlinear Q', rather than replacing it by its first derivative.

With L, M, β fixed and α_1/α_0 fixed in the range required by the source, this yields

$$a_{\mathrm{near}} \ge c_{\mathrm{near}}\alpha_0 \tag{E14.15d}$$

once α_0 is below the fixed absolute ceilings. Thus the additional coarse input in the nearby-carrier argument does not itself force a superlinear loss of the running analytic radius. Its background-ratio hypothesis is real: the proof preceding (3.37) requires α_1 large compared with a fixed multiple of $M\alpha_0$. Equal logarithmic exponents with an appropriate fixed ratio of their coefficients meet this displayed restriction. This calculation does not by itself prove the corresponding far-carrier current bound or admission of all running multiscale centers.

4. The resummed E identity and what its bound already accomplishes

CMP116 Lemma 1 identifies

$$\begin{aligned} E_k\big(U_k(e^{ib'(z)}V^{(k)})\big) - E_k\big(U_k(V^{(k)})\big) \\ = \sum_Y V'_{k,Y}(U_{k+1}, J_{k+1}, z). \end{aligned} \tag{E14.16}$$

Here E_k is the renormalized inherited action; its old coupling subtractions have already been included. The source proves a product analytic domain and a bound of the form

$$|V'_{k,Y}| \le \mathcal{K}_E e_1 e^{-(1-2\vartheta)\varkappa d_k(Y)}, \qquad \mathcal{K}_E = E_0 C_1 M^q e^{C_2\varkappa_1}. \tag{E14.17}$$

The notation ϑ distinguishes the fixed exponent loss from the cutoff radius. The constants in $\mathcal{K}_E$ are those fixed by the native parameter choice. Their dependence must be retained if that choice changes.

The source summation preceding this lemma is substantive. In the nearby-carrier case its fifth-order factor pays the four-dimensional old-cell count, leaving a summable old-scale factor. In the distant-carrier case the source uses the spatial exponential and its scale-transfer inequality. Thus (E14.17) is a bound after old-scale summation, on one common native domain. It is not a primitive activity bound with an uncounted old-cell multiplicity.

Combining this source result with (E14.13)-(E14.14), for example, gives

$$|\partial_z^a V'_{k,Y}(z_0)| \le a!\, \mathcal{K}_E e_1\, r_z^{-|a|} e^{-(1-2\vartheta)\varkappa d_k(Y)}. \tag{E14.18}$$

At $P \le e_1/(2g_k)$, the current derivative cost is explicitly

$$a!\,\mathcal{K}_E e_1 \left(\frac{2g_k}{c_G e_1}\right)^{|a|}. \tag{E14.19}$$

Preparation differentiations commute with this estimate on a smaller common contour: the source's convergent expansion has a uniform majorant there. This step does not require a zero-free generating functional.

Equations (E14.15) and (E14.18) are source-domain consequences for actual maps and actual E activities. The running-background substitutions are verified in Sections 5.2–5.6; the native formulas alone would not justify an unverified substitution.

5. Testing an explicit running parameter choice

The conservative normalized real cutoff in the complete construction is

$$P_k = A_1 \ell_k^{p_0}, \qquad \delta_k = g_k P_k, \qquad \ell_k = \log g_k^{-2}. \tag{E14.20}$$

This uses the normalized bound, regardless of the ambiguity in the printed later cutoff formula. Choose

$$e_{1,k} = 2\delta_k \tag{E14.21}$$

gives $\mathcal{R}_{z,k} = e_{1,k}/g_k = 2P_k$ and a margin $\mathcal{R}_{z,k} - P_k = P_k$. This analytic radius is distinct from the source's geometric collar range R_k. Substitution into the *actual* preparation budget (E14.8) reduces its principal insertion condition to

$$64 C_\zeta K C_B\, \delta_k \le \alpha_{2,k}. \tag{E14.22}$$

The other fixed native smallness conditions are then eventually satisfied because $g_k \ell_k^{p_0}$ tends to zero. The right side of (E14.22) is supplied by the native local admission calculation and the whole-map argument in Sections 5.2 and 5.5.

Those calculations give

$$\alpha_{2,k} \ge c\, g_k \ell_k^q, \qquad q > p_0, \tag{E14.23}$$

with the displayed preparation constants fixed; (E14.22) therefore holds whenever

$$\ell_k^{q-p_0} \ge 64 C_\zeta K C_B A_1 / c. \tag{E14.24}$$

Any finitely many additional inequalities $C g_k \ell_k^b \le c_0$ have an explicit sufficiently small coupling ceiling. Indeed, for $\ell \ge 16 b^2$, the bound $\log \ell \le \sqrt{\ell}$ gives $e^{-\ell/2}\ell^b \le e^{-\ell/4}$; it remains to require $\ell \ge 4\log(C/c_0)$.

Section 3.3 proves the needed linear behavior for the extra coarse input in the nearby-carrier preparation. Sections 5.2–5.6 supply the far insertion, actual real-center and common native background/current margins. CMP119 (2.28) states $q_0, q_1 > 1$ and sufficiently large constants. Equal exponents $q_0 = q_1 = q > p_0$, with a suitable fixed ratio of coefficients, satisfy the displayed nearby-carrier requirements. The remaining fixed native map restrictions are imposed through the displayed minima and the common coupling ceiling. The real-center and old-scale comparison proofs use the complete construction on its actual admitted regular geometry; the numerical choice of exponents alone would not imply those inclusions.

Nor should one use the minimum of all unrescaled old $\alpha_{2,j}$ as a common current radius: on an expanding ultraviolet trajectory it can tend to zero. The scale improvement of the actual old-coordinate map must enter before taking that minimum, or one must use the already-resummed common-domain theorem with its running constants.

5.1 The actual coupling comparison pays the nearby scale change The coupling comparison in CMP119 (2.6)-(2.7), pp.255-256, gives a quantitative improvement. Set $a = k - j$, and take one fixed exponent q in $\alpha_{0,j} = C_0 g_j \ell_j^q$. For the admitted source trajectory,

$$\frac{\alpha_{0,k}}{\alpha_{0,j}} \le C_{\mathrm{AF}} \sqrt{1+a}, \qquad C_{\mathrm{AF}} = (1+\beta_0)^2. \tag{E14.24c}$$

For $a \geq 1$, multiply the source's ratio bound for g_k/g_j by its bound for ℓ_k^q/ℓ_j^q; for $a = 0$, the ratio equals one. This uses the actual age comparison, not a bound by the absolute ultraviolet cutoff logarithm.

At a native background for which the source's nearby preparation is admitted, (E14.6) and (E14.15a) give its complete current-dependent extra coarse field the bound

$$\|Q_j'(\mathcal{H}_s(b'(z)))\|_\infty \leq D g_k \|z\|_\infty L^{-a}, \quad D = 4C_Q C_{\hat{\varsigma}} K C_B, \tag{E14.24d}$$

Here $\eta = L^{-k}$, so the factor $L^j\eta$ in the displayed source estimate is exactly L^{-a}. Suppose the absolute ceilings are imposed so that the actual nearby coarse radius from (E14.15b) satisfies $r_{3,j} \geq c_3\alpha_{0,j}$. On the full product ball $\|z\|_\infty < 2P_k$, equations (E14.20), (E14.24c)-(E14.24d) imply

$$\frac{\|Q_j'(\mathcal{H}_s(b'(z)))\|_\infty}{r_{3,j}} \leq \frac{2DA_1C_{\mathrm{AF}}}{c_3C_0}\ell_k^{p_0-q}L^{-a}\sqrt{1+a}. \tag{E14.24e}$$

For $L \geq \sqrt{2}$, one has $L^{-a}\sqrt{1+a} \leq 1$ for every integer $a \geq 0$: use $1 + a \leq 2^a$ for $a \geq 1$. Therefore the explicit condition

$$\ell_k^{q-p_0} \geq \frac{8DA_1C_{\mathrm{AF}}}{c_3C_0} \tag{E14.24f}$$

places this current-dependent input inside one quarter of its native coarse margin simultaneously at every old nearby scale. This calculation removes the putative obstruction from taking the minimum of unrescaled old radii. The real-center reserve is supplied in Section 5.2; the joint external-variable reserve is supplied in Section 5.6.

The fifth-order nearby remainder has a related summation. The source's coarse determining field has size at most a fixed multiple of $\alpha_{0,k}L^{-a}$ on the admitted contour. Here the remainder is the integral Taylor remainder along the segment τB, $0 \leq \tau \leq 1$. Lemma 4 supplies the same additive $r_{3,j}$-tube at every point of that segment. Apply fifth-order Cauchy there and integrate the Taylor weight. No assertion that the Taylor series at zero converges all the way to B is needed. The resulting age factor is

$$C\left(\frac{\alpha_{0,k}}{\alpha_{0,j}}\right)^5 L^{-5a} \leq CC_{\mathrm{AF}}^5(1+a)^{5/2}L^{-5a}. \tag{E14.24g}$$

Multiplying by the four-dimensional old-cell count $O(L^{4a})$ leaves

$$\sum_{a\geq 0}(1+a)^{5/2}L^{-a} < \infty. \tag{E14.24h}$$

The convergence is uniform in total depth. Once all the activity arguments are admitted on the same current ball, further current differentiation can be performed after this majorized sum and costs $m!P_k^{-m}$, with the fixed Lie-coordinate norm factor retained. This closes the running-age bookkeeping for this nearby fifth-order remainder. It does not replace the source's treatment of the lower-order subtraction terms, the distant carriers, or the complete running-background admission test.

5.2 Actual real centers and the local relative-field comparison The actual real centers are not arbitrary elements of the largest regular domain. The complete construction supplies stronger bounds. CMP119 p.275 identifies the minimizing field on each regular layer with a minimizing field at that layer's scale and transfers its regularity bounds; p.276 transfers the interior arguments of CMP109 Sections 3–4 with that scale. In CMP102 Theorem 1, membership in the regular space, defined in (2), includes both the plaquette and the physical-current bound. Its local gauges satisfy (9)–(10), including the derivative and second-order bounds. Thus the transfer is not a curvature-only assertion.

In a layer of scale $n \geq j$, the resulting bounds in old j-scale units are

$$b_j(U_0) \leq C_{\mathrm{reg}}\epsilon_n L^{-2(n-j)}, \qquad \|J_j^{\mathrm{phys}}(U_0)\|_\infty \leq C_{\mathrm{reg}}\epsilon_n L^{-3(n-j)}. \tag{E14.26}$$

The regular-gauge field and its first derivative have the corresponding factors $L^{-(n-j)}$ and $L^{-2(n-j)}$. Here C_{reg} is a fixed maximum of the constants in those source bounds, at fixed L, M. This statement uses the actual local gauges specified in the source; it does not assert the existence of one globally small gauge potential across all history interfaces.

The actual coupling comparison gives $\epsilon_n/\epsilon_j \le C_{\mathrm{AF}}\sqrt{1+n-j}$. Consequently each of these native regularity bounds is at most a fixed multiple $C'_{\mathrm{reg}}\epsilon_j$. Since

$$\frac{C'_{\mathrm{reg}}\epsilon_j}{\alpha_{0,j}} = \frac{C'_{\mathrm{reg}}A_0}{C_0}\ell_j^{p_0-q}, \tag{E14.27}$$

any fixed requested fractional reserve is obtained at every old scale by the explicit lower bound on $\ell_j^{q-p_0}$ displayed in (E14.27). For example, the bound by $\alpha_{0,j}/(64B_3)$ follows if $\ell_j^{q-p_0} \ge 64B_3C'_{\mathrm{reg}}A_0/C_0$. At a real center the relative complex field can be taken equal to zero. These deductions establish the smaller *real-center* reserve for the actual regular layers, under the source-scoped complete inductive representation. They do not by themselves establish a simultaneous complex tube through every subsequent substitution.

There is also a direct local comparison needed by that tube. On one of the admitted local gauges write $U_0 = e^{i\xi a_0}$, and take $\|a_0\|_\infty$ below a fixed ceiling. Since

$$\xi^{-1}\|\mathrm{Ad}_{e^{i\xi a_0}} - I\| \le 2e^{2\xi\|a_0\|}\|a_0\|, \tag{E14.28}$$

the ordinary and covariant first differences of any field B obey

$$\|\nabla B - \nabla_{U_0}B\|_\infty \le C_G\|a_0\|_\infty\|B\|_\infty. \tag{E14.29}$$

This is a scale-independent estimate; the link displacement supplies the power ξ needed to cancel the difference quotient. The same observation controls composition of relative fields. The analytic map

$$\mathcal{B}_\xi(A,B) = \frac{1}{i\xi}\log(e^{i\xi A}e^{i\xi B}) \tag{E14.30}$$

has bounded first derivatives on a fixed small matrix chart, uniformly for $0 < \xi \le 1$. The convergent BCH series starts with $A+B$; every nonlinear term has at least one positive power of ξ. Applying the discrete product rule, and (E14.29) to change the reference connection, gives

$$\|\mathcal{B}_\xi(A,B)\|_{H^1_\infty} \le C_{\mathrm{cmp}}(\|A\|_{H^1_\infty} + \|B\|_{H^1_\infty}), \tag{E14.31}$$

with a fixed constant on that chart. The H^1_∞ notation denotes the supremum and first-difference norms used here, not an L^2 Sobolev norm. A fixed Hölder first-difference norm can be included by the same product-rule estimate when it is required by the local regularity chart.

Together with the actual-current calculation in the supplement, (E14.31) removes the local coefficient obstruction to a linear insertion radius. More precisely, choose the real center as in (E14.27), an independent-current center satisfying the same smaller current bound, and relative-field and insertion perturbations whose required norms are all less than

$$u_j = c_{\mathrm{loc}}\min\{\alpha_{0,j}, \alpha_{1,j}\}, \tag{E14.32}$$

where $c_{\mathrm{loc}} > 0$ is smaller than the reciprocals of the fixed constants in (E14.31), (LC.7), (LC.24), and the factor $16B_3$ in the smaller-domain reconstruction test. The plaquette, relative-field and independent-current clauses then lie strictly inside the smaller basic domain. CMP109 p.263 supplies its reconstructed-field clauses, with its displayed factor $2B_3$. This is a quantitative native local admission calculation, including reconstruction. Its restrictions are on the actual native norms; a bound on an input in different coordinates must still be transported to these norms by the actual source maps.

For a local coordinate description, the native norm ball is quantitative even on the fine lattice. The first-difference constraint is satisfied by a simultaneous scalar-coordinate polydisc of radii $c_G\xi u_j$; the unscaled independent-current

coordinates have radii $c_G u_j$. This follows from the bounded degree of the lattice stencil and (E14.29), with no factor equal to the total number of sites. The much larger current-coordinate radius in (E14.24f) is obtained only after using the smoothing preparation map; it is not inferred from these raw-link coordinate radii.

5.3 Far-carrier summation with the actual derivative bound For a distant old E carrier the native source gives more than a generic analytic function bound. CMP109 (3.8)–(3.9) applies the derivative of the actual old minimizing map to the prepared current input. It bounds the resulting field and each of the first- and second-order operators required in (3.14), with the factor $L^j\eta$ and the separation exponential. Equation (3.17) performs the Cauchy insertion using that same collection of norms. The paragraph following it explicitly retains these bounds under the subsequent localization. For a large carrier the primitive carrier decay supplies the corresponding small exponential instead.

Write $a = k - j$, so $L^j\eta = L^{-a}$. On an admitted joint native contour, (E14.6) and that actual source bound give

$$|\mathcal{E}^{\mathrm{far}}_{j,X,\square}| \le CE_0 e^{-\kappa' d_j(X)} \frac{g_k \|z\|_\infty}{\alpha_{2,j}} L^{-a} e^{-cML^a}. \tag{E14.33}$$

All fixed constants in (E14.33) include the preparation factor K, the native norm-comparison constants and the fixed loss of carrier decay. For a large carrier, a fraction of its decay supplies e^{-cML^a} and the remaining fraction is the displayed carrier weight. This is the source's actual far-carrier derivative estimate; it is not an estimate obtained by replacing the nonlinear minimizing map with a Gaussian map or by dilating a projection from a different determining geometry.

In this common notation choose $c \le \delta\kappa/M$ for the large-carrier branch, whose direct source factor is $e^{-\delta\kappa L^a}$, as well as the separation constant for the other branch. The fixed M-dependence is retained; an M-independent value of c is not claimed.

If its insertion arguments satisfy the native norm reserve (E14.32), then $\alpha_{2,j}$ may be chosen equal to $c_2\alpha_{0,j}$, with fixed $c_2 > 0$. On $\|z\|_\infty < 2P_k$, (E14.24c) yields the explicit age majorant

$$|\mathcal{E}^{\mathrm{far}}_{j,X,\square}| \le CE_0 e^{-\kappa' d_j(X)} \varrho_k^{p_0-q} \sqrt{1+a}\, L^{-a} e^{-cML^a}. \tag{E14.34}$$

After the four-dimensional old-cell count, the remaining sum is bounded by

$$\sum_{a\ge 0} \sqrt{1+a}\, L^{3a} e^{-cML^a} < \infty. \tag{E14.35}$$

This convergence is uniform in the total depth. For example, $\sqrt{1+a} \le C_L L^a$, and $x^4 e^{-cMx} \le C_{cM} e^{-cMx/2}$; the remaining lacunary exponential series is finite. On one verified common current ball, the sum is holomorphic by this majorant and subsequent current Cauchy derivatives cost $m!(c_G P_k)^{-m}$, independent of the old age. Thus the far *summation* is no longer an independent obstacle once its arguments meet the displayed native norm reserve.

The precise domain proviso matters. The source needs both the background part of the old insertion and the localized variation to satisfy (3.14), with the displayed spare fractions. The derivative estimate (3.9) supplies the latter. For the former, integrate the actual derivative along its defining coarse-input segment and use its value at zero, retaining the nonlinear reconstructed background. A change of determining geometry is handled by CMP102 (190), as in the source proof; it is not handled by declaring the old projection to be a dilation of the new one. Section 5.5 verifies this entire background segment on a fixed internal-map domain before testing the smaller E-activity radius. Equations (E14.33)–(E14.35) by themselves would not suppress that prerequisite.

5.4 The Ward and coupling terms: a degree-preserving radius bounds The fifth-order tail is not the complete inherited E action. The remaining terms can nevertheless be given a running-radius majorant without demanding that the final constant equal the strongest small native constant in (E14.17). This subsection concerns the actual nearby E decomposition of CMP109 Sections 4–5, with the matching source Wilson subtraction retained, on the common native contour used by that decomposition.

The source operations have the following finite decomposition. Let $Q_k \le c_Q\alpha_{0,k}$ bound the regular coarse-background and prepared-insertion norms on this contour. Put $r_j = c_r\alpha_{0,j}$, where $c_r > 0$ is a fixed smaller native analytic radius. Fix $0 < \theta < 1$ in the source's irrelevant-term estimates.

Actual source operation	Coefficient order and radius cost	Field factors retained	Geometric factor after subtraction
Integral tail in CMP109 (3.34), Lemma 4	fifth derivative, $CE_0 r_j^{-5}$	five factors, each of size at most CQ_k before its scale weight	L^{-5a}
Fourth-order terms, (4.21)–(4.29)	derivatives of orders $r = 2, 3, 4$, each costing $CE_0 r_j^{-r}$	total field degree four; replacing a field by a difference does not lower that degree	$L^{-(5-\theta)a}$ for the discarded terms
Third-order terms, (4.30)–(4.34)	orders $r = 2, 3$	total field degree three	$L^{-(5-\theta)a}$ for the discarded terms
Vacuum-polarization remainder, Section 5	order two, $CE_0 r_j^{-2}$	degree two, or higher commutator terms with the same coefficient	$L^{-(5-\theta)a}$
Matching Wilson subtraction, (4.38)–(4.45)	$\lvert\beta_j\rvert \le CE_0 r_j^{-2}$	degree at least two; the marginal degree-two expression cancels before summing old scales	$L^{-(5-\theta)a}$ for its remaining terms

The constants absorb fixed spatial moments of the source's exponentially decaying kernels and a fixed loss of carrier decay. They are independent of the old age. If polynomial factors in the length of an auxiliary localization box are retained instead of absorbed, write a fixed factor $(1 + a)^{b_*}$ throughout; no derivative-order-dependent value of b_* is introduced.

Here is why the radius bookkeeping in this table is valid. The coefficients in (4.3) are obtained by a finite chain rule. A term involving r derivatives of the primitive E activity has r Cauchy denominators, while the derivatives of the minimizing map have the fixed regularity and tree-decay bounds given there. Its total degree n in the displayed fields is at least r. The Ward identities (4.24)–(4.31) are linear identities in these coefficients. They replace a higher coefficient by lower coefficients multiplied by commutators or coordinate differences. They do not multiply two activity coefficients or introduce an extra Cauchy denominator. For example, (4.22) displays the fourth-order cost as the fourth power of the field-radius/analytic-radius ratio. The passage from (4.34) to the polarization tensor uses only linear kernel summation, its fixed spatial moments, and the stated symmetries. The coefficient defining β_j is a second-order coefficient with a second spatial moment, so it has the bound in the table.

The spatial gains in the last column are precisely the gains of the source's irrelevant remainders. CMP119 (3.49)–(3.67) gives the same accounting after the layer transfer, with gain $5 - \theta$. In particular the marginal F^2 term must be cancelled against the *same* β_j Wilson term before applying the table. Bounding those two terms separately would leave an unsummable marginal expression.

Since Q_k is below a fixed ceiling less than one, a degree-n term with coefficient order $r \le n$, $2 \le r \le 5$, is bounded by

$$CE_0 r_j^{-r} Q_k^n \le CE_0 \left(\frac{\alpha_{0,k}}{\alpha_{0,j}}\right)^r, \qquad 2 \le r \le 5. \tag{E14.36}$$

One must retain the value of r in this inequality. Replacing every denominator by $\alpha_{0,j}^{-5}$ while leaving a quadratic numerator would not prove it. The Wilson remainder is handled with its coefficient order two and its actual field degree; its higher-degree factors only improve (E14.36).

For the Wilson analytic tail, degrees above five are bounded by their convergent fixed-chart majorant, retaining the degree-five factor and its coefficient order two. The argument does not truncate that analytic tail or declare all its terms to have degree at most five.

The AF comparison therefore bounds the complete nearby irrelevant majorant by

$$CE_0(1 + a)^{b_*} \left[\sum_{r=2}^{5} C_{\mathrm{AF}}^r (1 + a)^{r/2}\right] L^{-(5-\theta)a} e^{-\kappa' d_j(X)}. \tag{E14.37}$$

After the four-dimensional old-cell count, the explicit remaining sum is

$$\sum_{a\geq 0}(1+a)^{b_*+5/2}L^{-(1-\theta)a}<\infty. \tag{E14.38}$$

For fixed θ, L, b_*, this is a numerical constant uniform in the total RG depth. This proves the running-radius *majorant* for all terms in the nearby E decomposition, provided the common native domain, the regularity size Q_k, and the source's exact matching subtraction are retained. It supplies no weighted Hessian smallness by itself.

The role of the independent current in this argument is specific. CMP109 (3.36) defines the nearby pair by the minimizing pullback of the coarse determining field B. The Taylor formula (3.34) has the arbitrary regular field A and the external background U; the external independent current enters when A is subsequently replaced by the actual $\mathcal{H}_s(U,J,z)$. Section 4 then evaluates its Taylor coefficients at the primitive pair $U=1, J=0$. CMP116 begins its localization from the already-subtracted local analytic functions of (U,J,A) and performs this actual preparation substitution. Thus the cancellation is an identity in the coarse determining field followed by analytic substitution. It is not an assertion that a holomorphic identity on the physical-current graph determines an arbitrary independently extended function.

5.5 Admitting the whole far insertion before testing the E radius The coarse average in the far formula need not be identified with the modified nearby Q_j'. An actual unmodified-average estimate is available in the published papers. CMP102 p.285, (44), writes

$$Q_j(\eta A)=L^j\eta\, Q_jA+C_j(L^j\eta A),\qquad |C_j(L^j\eta A)|\leq C_2(L^j\eta)^2\|A\|_\infty^2. \tag{E14.39}$$

The left-hand Q_j is the nonlinear relative average and the Q_j in its first term is its linearization. These are not the different nearby map Q_j'.

CMP99 p.406, immediately after (3.78), gives a direct uniform estimate for the nonlinear relative averages. On $\|A\|<\alpha_1/(L^j\eta)$, their logarithms at the intermediate steps satisfy $\|B_n\|\leq C\alpha_1L^n\xi$, with $\xi=L^{-j}$ and $0\leq n\leq j-1$. The remaining one-block operation has a fixed number of inputs once L, d are fixed. It is analytic on a fixed small chart and vanishes at zero relative input. Equation (3.79) controls its linearized averaging operator; converting this derivative to the fixed logarithmic coordinates introduces only a fixed chart factor. Thus its derivative is uniformly bounded on that smaller chart. It therefore sends $\|B_{j-1}\|\leq C\alpha_1/L$ to a logarithm of norm at most $C_L\alpha_1$. Choosing $\alpha_1=2L^j\eta\|A\|$, below that fixed chart ceiling, proves

$$\|Q_j(\eta A)\|_\infty\leq C_{\mathrm{av}}L^j\eta\|A\|_\infty. \tag{E14.40}$$

The case $j=0$ is the direct relative link input and has the same bound. The constants are uniform in the old index and the number of fine bonds; the last step is a fixed L-block operation, not a chain of uncontrolled Lipschitz constants. The same argument holds on the source's admitted complex-background domain. The independent current is not an argument of this relative averaging operation.

We now separate two different domains. Choose a fixed absolute small regular domain for the internal averaging and minimizing maps of CMP102 Proposition 9 and CMP116 (1.11)–(1.18). Call its admitted coarse-input radius $\rho_{\mathrm{map}}>0$. This is chosen before, and independently of, the shrinking primitive E-activity radii $\alpha_{0,j}$. The real regular centers from (E14.26) and sufficiently small native background/current perturbations lie in this fixed domain. The geometric and localized-propagator hypotheses of those primary results are retained.

In CMP109 (3.6) the relative coarse input is formed from tA, with A the prepared field, and it vanishes at $A=0$. Equations (179)–(180) of CMP102 define the corresponding relative minimizing map with value zero at zero input. The localized formula (1.17) of CMP116 preserves that zero: all its terms contain the input, and its nonlinear correction is the unique small solution with zero value there. These facts remain true when the source current is independently extended.

Let C_H be the fixed operator bound, in the four output norms required by CMP109 (3.14), for the first derivative of that relative minimizing map on the smaller absolute map domain. This bound is the integrated kernel estimate of CMP102 (190). Its scale measure cancels the kernel's input-cell volume factor; the remaining exponential has a uniformly bounded row sum under the source geometry. The localized versions obey the same estimate with their

fixed preparation loss, as stated after CMP109 (3.17) and quantified by CMP116's propagator bound. Thus, for coarse inputs B in $\|B\| < \rho_{\mathrm{map}}/2$, integrating the actual derivative on the segment υB, $0 \le \upsilon \le 1$, gives

$$\|\mathcal{H}_{j,s}(B)\|_{\mathrm{adm},j} \le C_H \|B\|_\infty. \tag{E14.41}$$

When using the current-scale partition to sum localized derivatives, apply the bound at a fixed output point first. For $\lambda = L^{-a} \le 1$, the sum over partition cubes is bounded by $C_d \sum_{r\ge 0}(1+r)^{d-1}e^{-cMr}$, independently of a. Taking the output supremum afterwards avoids a factor equal to the carrier's volume. The same pointwise argument applies to the source's first-difference, Hölder and second-order output bounds. This is the row-sum interpretation used in C_H.

This segment lies in the fixed map domain before any E-activity condition is imposed. There is consequently no circular use of the desired smaller E-domain admission. Equation (E14.41) also retains the projection term: it uses all the source's required second-order output norms, not a pure-dilation claim for projections defined on different determining geometries.

Combining (E14.6), (E14.40) and (E14.41) for the actual current input yields

$$\begin{aligned} \|B_{j,s}(t,z)\|_\infty &\le D_{\mathrm{av}} L^{-a} g_k \|z\|_\infty, \\ \|\mathcal{H}_{j,s}(B_{j,s}(t,z))\|_{\mathrm{adm},j} &\le D_{\mathrm{far}} L^{-a} g_k \|z\|_\infty, \end{aligned} \qquad 0 \le t \le 1, \tag{E14.42}$$

where one may take $D_{\mathrm{av}} = 4C_{\mathrm{av}}C_\zeta K C_B$ and $D_{\mathrm{far}} = C_H D_{\mathrm{av}}$, enlarged by the fixed localization constants if the source puts them outside C_H. Both inequalities hold jointly with the native external variables and the complex preparation parameters. First impose $2D_{\mathrm{av}}\delta_k \le \rho_{\mathrm{map}}/2$; this admits all coarse inputs to the fixed map domain. Then, with $\alpha_{2,j} = c_2\alpha_{0,j}$, impose

$$\ell_k^{q-p_0} \ge \frac{24 D_{\mathrm{far}} A_1 C_{\mathrm{AF}}}{c_2 C_0}. \tag{E14.43}$$

For $\|z\| < 2P_k$, the AF comparison and $L^{-a}\sqrt{1+a} \le 1$ give

$$\|\mathcal{H}_{j,s}(B_{j,s}(t,z))\|_{\mathrm{adm},j} < \alpha_{2,j}/12 \quad \text{at every old scale and every } 0 \le t \le 1. \tag{E14.44}$$

The spare factor admits the localized insertion-parameter variation as in (E14.11). It is the whole background insertion in the far formula, not merely its infinitesimal localized response, that is now controlled. The independently varied background and current still have to be kept in the common native external domain; their actual real-center and native local reserves were proved in Section 5.2. This calculation resolves the far coarse-input mismatch without replacing Q_j by Q_j'.

5.6 A common native background/current contour and its summed prefactor The scale transfer of the independent current is fixed before introducing it as an independent variable. If h and ξ are two choices of lattice length unit, the physical current definition $J_h = D_h^* h^{-2}\pi \operatorname{Im} dU$ gives

$$J_\xi = (h/\xi)^3 J_h. \tag{E14.45}$$

For the current-scale and old-scale units in CMP109 (3.12), this gives the conservative coefficient bound $|\tau_j| \le C_J L^{j-k}$, with a fixed one-step unit-convention factor C_J. Either $h = \eta$ or $h = L^{-1}\eta$, with $\xi = L^{-j}$, gives the bound with $C_J = 1$; retaining $C_J = L^3$ also covers a fixed adjacent-step convention. This uses the physical definition in the substitution being extended and the known relative scale, not a reading of an unclear printed superscript or an assumed exact convention. The subsequent independent replacement in (3.13) changes the variable, not that coefficient. No identity between the independently varied J and the physical current of the independently varied U is imposed afterwards.

For clarity, use the native external background coordinate b, the relative regular field in the source local charts, and its source norm. This is the coordinate of the analytic source theorem; it is not an arbitrary reparameterization of the entire retained configuration. The old reconstructed relative field caused by a change b is bounded by

$$\|b_j\|_{\mathrm{native},j} \le C_{\mathrm{ext}} L^{-a} \|b\|_{\mathrm{native},k}. \tag{E14.46}$$

Indeed, the relative average has the actual bound (E14.40), the relative minimizing map has (E14.41), and the relative-gauge compositions have (E14.31). Their fixed constants may be multiplied into C_{ext}. Where the field is merely written in old length units, the bound is its direct scaling; where its determining geometry changes, the proof uses (E14.41) on that actual geometry. This argument does not identify different projections under dilation. The same local gauges and reconstruction domains are used as in the native source extension. The analytic gauge conjugations of the independent current in CMP109 (3.26)–(3.28) have uniformly bounded norm on the smaller fixed chart. Their changes are bounded by a fixed constant times the relative-background change; multiplying the small current center and retaining the scale coefficient gives the same bound, with that fixed constant included in C_{ext}.

Choose $c_{\mathrm{ext}} > 0$ so that

$$c_{\mathrm{ext}} C_{\mathrm{AF}} \max\{C_{\mathrm{ext}}, C_J, 1\} \le \frac{1}{128} \min\{c_{\mathrm{loc}}, c_2, c_3, c_{\mathrm{near}}\}. \tag{E14.47}$$

On the common background and independent-current balls

$$\|b\|_{\mathrm{native},k} < c_{\mathrm{ext}} \alpha_{0,k}, \qquad \|J - J_{0,k}\|_\infty < c_{\mathrm{ext}} \alpha_{0,k}, \tag{E14.48}$$

(E14.45)–(E14.47) and the AF comparison put every old native background/current perturbation inside its reserved fractional radius. The actual real centers already lie in the smaller domain by (E14.26)–(E14.27). The local-current and relative-field proof supplies the basic-domain inclusion; the primary smaller-domain theorem supplies reconstruction. The same inequalities apply along the coarse-field Taylor segments used by the nearby analysis. In particular Q_k in Section 5.4 may be taken to be a fixed small multiple of $\alpha_{0,k}$, after imposing the already displayed normalized-current cutoff inequalities.

Take one fixed $q > p_0$, and take the uniform coupling ceiling small enough to meet (E14.24), (E14.24f), (E14.27), (E14.43), the fixed map-domain ceiling in Section 5.5, and the absolute smallness conditions of Section 3. Each inequality has the form already quantified there: either a lower bound on $\ell_k^{q-p_0}$, or $C g_k \ell_k^b \le c_0$. No new choice of q is made for a higher observable derivative.

The result is the simultaneous product contour

$$\begin{aligned} &\|z\|_\infty < 2P_k, \qquad \|b\|_{\mathrm{native},k} < c_{\mathrm{ext}} \alpha_{0,k}, \\ &\|J - J_{0,k}\|_\infty < c_{\mathrm{ext}} \alpha_{0,k}, \qquad |s(\Delta)| < e^{\varkappa_1}. \end{aligned} \tag{E14.49}$$

All internal activity arguments are admitted jointly on this contour, with the insertion-parameter reserves above. Boundary-crossing contributions assigned to B by the actual source localization remain in C-05; this statement applies to the E carriers of the C-01 decomposition. It does not discard those boundary contributions from the complete law.

To display the prefactor, fix $0 < \theta < 1$, and put

$$\begin{aligned} S_{\mathrm{near}} &= \sum_{a \ge 0} (1+a)^{b_* + 5/2} L^{-(1-\theta)a}, \\ S_{\mathrm{far}} &= \sum_{a \ge 0} \sqrt{1+a}\, L^{3a} e^{-cML^a}, \\ \mathfrak{K}_E &= C_{\mathrm{src}} E_0 (1 + S_{\mathrm{near}} + S_{\mathrm{far}}). \end{aligned} \tag{E14.50}$$

Here C_{src} is a fixed maximum/product of the displayed preparation, local-comparison and source carrier-resummation constants. The finite factors $C_{\mathrm{AF}}^r c_r^{-r}$, $2 \le r \le 5$, are included; the constants are independent of the cutoff depth, torus volume, history, and the order of subsequent differentiation. Let $\varkappa_* > 0$ be the decay rate left after the fixed carrier losses in CMP116 Lemma 1. Its geometric localization argument is unchanged: only the old-age scalar majorant has been replaced by (E14.37) and (E14.34). The source localization and the convergent sums give

$$|\mathcal{E}_{k,Y}(z,b,J,s)| \le \mathfrak{K}_E e^{-\kappa_* d_k(Y)} \quad \text{on (E14.49).} \tag{E14.51}$$

For a real current center $\|z_0\| \le P_k$, the coordinate radii may therefore be taken as

$$r_z = c_G P_k, \quad r_b = c_G c_{\mathrm{ext}} \alpha_{0,k}/2, \quad r_J = c_G c_{\mathrm{ext}} \alpha_{0,k}/2, \quad r_s = (e^{\kappa_1} - 1)/2. \tag{E14.52}$$

For b, this is the radius in the stated native norm. For the supremum and first-difference norm, the simultaneous fine-link scalar-coordinate polydisc has radius ξr_b, as in Section 5.2. If a fixed first-difference Hölder seminorm of exponent β is included, use instead $\xi^{1+\beta} r_b$: the distance between distinct lattice points is at least ξ, so the first-difference Hölder norm of a scalar-coordinate perturbation is bounded by a fixed constant times its amplitude divided by $\xi^{1+\beta}$. No second-order or nonlocal projection bound for an arbitrary raw coordinate direction is inferred from its first-difference norm.

For current, native-background, independent-current and preparation multi-indices u, v, w, t, multivariable Cauchy gives

$$|\partial_z^u D_b^v \partial_J^w \partial_s^t \mathcal{E}_{k,Y}| \le u!w!t!\, e^{|v|}|v|!\, \mathfrak{K}_E e^{-\kappa_* d_k(Y)} r_z^{-|u|} r_b^{-|v|} r_J^{-|w|} r_s^{-|t|}. \tag{E14.53}$$

Native-background derivatives in (E14.53) mean directions of norm at most one in the stated native norm. For $m = |v|$ such directions, use scalar radii r_b/m; their derivative cost is $m^m r_b^{-m} \le e^m m! r_b^{-m}$. For raw coordinate derivatives the simultaneous polydisc instead gives $v!(\xi r_b)^{-|v|}$, or $v!(\xi^{1+\beta} r_b)^{-|v|}$ when the stated Hölder norm is used. This distinction avoids interpreting a norm ball as an unrestricted product of arbitrary unit directions. The primitive old-age sums are absolutely convergent before differentiation on this common contour, so their Cauchy bounds have no uncounted inverse old-scale radius. This is the quantitative contour and summed-prefactor statement requested by C-02. It supplies neither the C-04 weighted-Hessian smallness nor the C-05 boundary estimates, and it does not perform the further arbitrary-mark coordinate transport assigned to M-04/M-06.

The direct native-domain shortcut and its polynomial ambiguity The product domain printed in CMP119 (3.43) would directly give a current margin

$$\rho_k = C_1 p_1(g_k) - P_k \tag{E14.24a}$$

if this number were known to be positive on the actual support. This approach legitimately uses the source's completed localization and preparation maps. It need not reconstruct every internal substitution again. The enlarged background domain in the same formula is also a genuine native domain that can be used directly after quantifying its margin at the actual background and independent-current center.

A notation check prevents using this shortcut without another step. The paragraph after CMP119 (3.4), p.265, defines the unprimed δ_k using $p_0(g_k)$. The normalized cutoff $g_0^{-1}\delta_0$ appears explicitly already in (1.18), p.250. The smaller polynomial p_1 introduced later in LFR I belongs instead to the primed preparation threshold δ_j', with $p_1 < p_0$; it is not the threshold in LFR I (1.9). The $p_1(g_k)$ printed in CMP119 (3.43) has not been identified with either polynomial by a checked definition in that paper.

In particular, identifying it with the sharper LFR polynomial would give

$$\frac{C_1 p_1(g_k)}{P_k} = \frac{C_1}{A_1} \ell_k^{p_1 - p_0} \longrightarrow 0, \tag{E14.24b}$$

so no choice of a fixed larger coefficient would repair the reserve. This disproves that proposed identification as a way to justify (E14.24a); it does not disprove the construction. The underlying-map proof in Sections 5.2–5.6 resolves C-02 without using that unidentified polynomial: it directly admits the conservative full support with a strict margin. The printed polynomial ambiguity remains an observation, not a premise of Theorem E14.3.

6. The completed contour theorem and its scope

Theorem E14.3 — the actual regular E contour Retain the source-free complete RG representation, its admitted geometry, regular and large-field history decomposition, and consistently normalized trajectory as in J0/C-01. Use the actual E/B assignment of that representation. Fix the source group, one admissible block size, the localization parameters, and one integer $q > p_0$. Choose $\alpha_{0,k} = C_0 g_k \ell_k^q$, $\alpha_{1,k} = C_1 g_k \ell_k^q$, with the fixed ratio required by the nearby-source inclusion. Impose the common small-coupling ceiling specified in Section 5.6.

Then every actual localized regular E activity has the simultaneous native contour (E14.49), containing the entire real normalized-current support $\|z\| \le P_k = \delta_k / g_k$, and the uniform carrier bound (E14.51) with the explicit summable prefactor (E14.50). Its current, native-background, independently extended current and preparation derivatives satisfy (E14.53), with radii (E14.52). The constants are uniform in the cutoff depth, torus volume and remaining admissible history. In particular the old-scale sums are controlled before differentiation on a common contour; differentiating does not introduce an uncounted inverse old-scale radius.

Proof. The actual real centers have the strict smaller-domain reserve (E14.26)–(E14.27). The local relative-field comparison (E14.28)–(E14.31), the Wilson-current calculation in the supplement, and the primary smaller-basic-domain inclusion give the native local reserve (E14.32), including reconstruction. The complete nonlinear preparation is admitted by the explicit minima (E14.8). The independent additive nearby input has radius (E14.15b), and its running current input is admitted at every old scale by (E14.24f).

For far insertions, the unmodified nonlinear averaging estimate (E14.40) first admits the whole input path to a fixed absolute internal-map domain. The old relative minimizing map vanishes at zero and satisfies the actual four-norm derivative estimate there; integrating it proves (E14.41). Equations (E14.42)–(E14.44) then put the whole old insertion path strictly inside the running E radius. This order of the argument avoids using the desired E-domain conclusion to justify the derivative estimate used to prove it.

The native background/current transfer (E14.45)–(E14.48) admits the external variables on the same contour at every old scale. The near fifth-order tail, all lower-order Ward and matching-coupling remainders, and far terms have respectively the bounds (E14.24g), (E14.37) and (E14.34). The marginal quadratic coefficient is cancelled by its prescribed Wilson subtraction before summation. The explicit age sums (E14.38) and (E14.35), with the source's geometric carrier resummation, give (E14.50)–(E14.51). Normal convergence gives joint holomorphy. Cauchy on the product contour gives (E14.53), with the stated directional-versus-coordinate convention for the background norm. □

The complex margin and the old-scale summable derivative prefactor include the independently extended current and preparation factors. The uniform Wilson trajectory and the compatible conditional representation remain hypotheses.

The following obligations remain distinct:

- C-03 concerns the actual R marginal and its additional transfers and denominators.
- C-04 asks for the small weighted derivative/Hessian budget; the finite contour prefactor here is deliberately not called that bound.
- C-05 includes the E-origin expressions which the source actually assigns to B at a multiscale boundary.
- C-06 and M-04/M-06 include further conditional and marked-coordinate estimates beyond the native contour proved here.

No term is omitted from the complete law by these distinctions. There is no inference of the full Schwinger hierarchy or Yang–Mills existence from this one completed statement.

7. Primary references

- T. Balaban, *Propagators for Lattice Gauge Theories in a Background Field*, CMP 99 (1985), 389-434, (3.3)-(3.12), (3.69)-(3.81), especially pp.391-392 and 404-406.
- T. Balaban, *The Variational Problem and Background Fields in Renormalization Group Method for Lattice Gauge Theories*, CMP 102 (1985), 277-309, especially Proposition 9 and (172), (179)-(190).
- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. I*, CMP 109 (1987), 249-301, especially (1.11)-(1.18), (3.11)-(3.17), and Lemma 4/(3.53). DOI.

- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. II. Cluster Expansions*, CMP 116 (1988), 1-22, especially (1.11)-(1.22) and Lemma 1/(1.33)-(1.36). DOI.
- T. Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, CMP 119 (1988), 243-285, (2.25)-(2.28), (2.34)-(2.39), (3.43), and the transfer discussion on pp.276-277.

The quantitative contour theorem is proved above at the stated source scope. No primary theorem is used as an unqualified proof of the full four-dimensional observable hierarchy.

8. Supplement: quantitative local-current proof

The following calculation uses CMP99, pp. 391–392 and 404–405.

1. Result and scope This part resolves a local coefficient question in the contour analysis. The nonlinear current remainder in CMP109 (3.11) can be bounded with a constant independent of the fine lattice spacing. Its control does not require an unspecified compactness argument, or the first variational paper.

The proof uses the explicit current, Hessian and covariant-difference formulas in CMP99, pp.391–392 and 404–405. It gives a source-specific bound for the local current extension:

$$\|F(U,A)\|_\infty \le C_G\left(b\|A\|_\infty + \|A\|_\infty\|\nabla_U A\|_\infty + \|A\|_\infty^3\right), \qquad b = \sup_p \xi^{-2}\|U_p - I\|. \tag{LC.1}$$

All norms are taken on the fixed enlargement of the output bond star needed by the displayed local formulas. The constant depends on the fixed dimension, group and regular-chart bounds, not on ξ, carrier size, torus volume or old-scale index.

Together with the exact discrete Weitzenböck identity, this yields a current-clause insertion radius $c_G\alpha_0$ on a quantitatively smaller input current domain. It identifies and proves the needed local coefficient bound. It does not by itself prove all reconstructed-field and preparation-domain conditions of C-02/C-03.

2. Coordinates and actual primary formulas Work in a regular representative on a lattice of spacing $0 < \xi \le 1$. Assume that one-link parallel transports and their inverses have norm at most a fixed K_0, and that the plaquette bound $b \le b_*$ holds. In the source regular complex chart, the real compact-group factor has norm one and the relative complex factor is small; K_0 can therefore be fixed uniformly.

For Lie-algebra fields define the covariant translation and difference

$$(T_\mu f)(x) = \mathrm{Ad}_{U_\mu(x)}\, f(x+\xi e_\mu), \qquad D_\mu = \xi^{-1}(T_\mu - I), \qquad D_\mu^* = \xi^{-1}(T_\mu^{-1} - I). \tag{LC.2}$$

The exterior covariant curl is

$$(D_U A)_{\mu\nu} = D_\mu A_\nu - D_\nu A_\mu. \tag{LC.3}$$

These are CMP99 (3.3)–(3.9). The adjoint here is the algebraic transpose for the invariant bilinear form when the fields are complex; it is not an antilinear operation on the holomorphic variables.

Let $\mathcal{A}(U)$ be the source Wilson action with its lattice scaling and normalized trace. CMP99 (3.11)–(3.12) identify its left-coordinate gradient and Hessian:

$$\mathcal{A}(e^{i\xi B}U) = \mathcal{A}(U) + \langle B, J(U)\rangle + \frac{1}{2}\langle B, \Delta(U)B\rangle + \cdots, \qquad J(U) = D_U^*\xi^{-2}\pi\,\mathrm{Im}\,dU, \tag{LC.4}$$

where π is the fixed Lie-algebra projection. Formula (3.10) gives

$$\Delta(U) = D_U^* D_U + \Delta'(U). \tag{LC.5}$$

Its two error terms have coefficients $\xi^{-2}(\operatorname{Re} U_p - I)$ multiplying unscaled covariant curls, and $\xi^{-2} \operatorname{Im} U_p$ multiplying products of transported bond fields. This is the actual source Hessian, including the curvature term.

For $U_t = e^{it\xi A}U$, CMP99 (3.71)–(3.73) give the explicit local operator comparison

$$\begin{aligned}\|(D^*_{U_t}D_{U_t} - D^*_U D_U)B\|_\infty \le C_G\big(|t|a\|\nabla_U B\|_\infty + |t|a_1\|B\|_\infty + t^2a^2\|B\|_\infty\big),\\ a = \|A\|_\infty, \qquad a_1 = \|\nabla_U A\|_\infty.\end{aligned} \tag{LC.6}$$

The formula is obtained by expanding the transport factors $e^{it\xi \operatorname{ad} A}$. Each first difference supplies its compensating ξ^{-1}; the remainder is bounded by $C\xi^2a^2$, so the two differences leave Ca^2, not Ca^2/ξ. Equation (LC.6) is also read directly from the source's displayed formula. For a complex initial regular representative the same finite-product proof applies, with a changed fixed K_0-dependent constant.

3. The three scale-independent estimates 3.1 Curvature along the insertion path

For a, a_1, b in a fixed bounded small range and $0 \le t \le 1$,

$$b_t := \sup_p \xi^{-2}\|(U_t)_p - I\| \le C_G(b + a_1 + a^2). \tag{LC.7}$$

Here is the cancellation in the matrix product. For a positively oriented $\mu\nu$ plaquette at x, put $P = U_{\mu\nu}(x)$. Moving the inserted exponentials to the initial vertex writes the new plaquette as

$$(U_t)_p = e^{it\xi A_\mu(x)}e^{it\xi T_\mu A_\nu(x)}e^{-it\xi \operatorname{Ad}_P T_\nu A_\mu(x)}e^{-it\xi \operatorname{Ad}_P A_\nu(x)}P. \tag{LC.8}$$

The sum of the four linear exponents is

$$\xi(D_\mu A_\nu - D_\nu A_\mu) - (\operatorname{Ad}_P - I)(T_\nu A_\mu + A_\nu). \tag{LC.9}$$

The first term, after multiplication by $t\xi$, is bounded by $C\xi^2 a_1$. The second is bounded by $C\xi^3 ba$. The terms at least quadratic in the exponents are bounded by $C\xi^2a^2$, using the convergent exponential series and the fixed transport bound. Adding $P - I$ proves (LC.7). There is no division by an uncontrolled first-order plaquette variation.

3.2 Current and Hessian-error estimates

The local formula for D^*_U gives the crude but sufficient bound

$$\xi\, \|J(U_t)\|_\infty \le C_G b_t. \tag{LC.10}$$

Only this bound is needed below; a uniform bound for the physical current itself is not inferred from curvature.

The identity

$$\operatorname{Re} P - I = \tfrac{1}{2}(P - I)^2 P^{-1} \tag{LC.11}$$

and the complex definition $\operatorname{Im} P = (P - P^{-1})/(2i)$ imply

$$\|\operatorname{Re} P - I\| \le C\xi^4 b_t^2, \qquad \|\operatorname{Im} P\| \le C\xi^2 b_t. \tag{LC.12}$$

Applying the finite-stencil Hessian formula (3.10), or the estimate preceding (3.69), now gives

$$\|\Delta'(U_t)B\|_\infty \le C_G(b_t + \xi^2 b_t^2)\|B\|_\infty \le C'_G b_t \|B\|_\infty. \tag{LC.13}$$

The first term in the source quadratic form is written with unscaled curls; their operator norm on a fixed bond star is bounded. Equivalently, if scaled curls are used, the two factors ξ^{-1} consume two of the four powers in the first estimate in (LC.12). Either convention gives the explicit $\xi^2 b_t^2$ remainder in (LC.13).

3.3 Exact moving-gradient identity

The current gradient cannot be differentiated as though the left-coordinate chart were fixed. Its exact identity is

$$\frac{d}{dt}J(U_t) = \Delta(U_t)A + \frac{i\xi}{2}[A, J(U_t)]. \tag{LC.14}$$

To prove it, fix a test direction B. The mixed coefficient in $\mathcal{A}(e^{is\xi B}e^{it\xi A}U)$ can be computed from

$$\frac{1}{i\xi}\log(e^{is\xi B}e^{it\xi A}) = sB + tA + \frac{i\xi}{2}st[B, A] + O(s^2t + st^2). \tag{LC.15}$$

Using (LC.4) and trace invariance gives $\langle B, J_0'\rangle = \langle B, \Delta(U)A + \frac{i\xi}{2}[A, J(U)]\rangle$. The same argument at U_t proves (LC.14). This is a finite-dimensional exact identity. It does not require a convergent source generating functional.

4. Proof of the local nonlinear-current bound Integrate (LC.14), insert (LC.5), and subtract the linear term $D_U^* D_U A$. This gives the exact remainder

$$F(U, A) = \int_0^1 \left[(D_{U_t}^* D_{U_t} - D_U^* D_U)A + \Delta'(U_t)A + \tfrac{i\xi}{2}[A, J(U_t)]\right] dt. \tag{LC.16}$$

The first term is bounded by $C(aa_1 + a^3)$ by (LC.6). The second is bounded by $Cb_t a$ by (LC.13). The last is bounded by Cab_t by (LC.10): its explicit factor ξ cancels the only crude ξ^{-1} current estimate. Using (LC.7) and integrating proves

$$\|F(U, A)\|_\infty \le C_G(ba + aa_1 + a^3). \tag{LC.17}$$

This proves (LC.1) for the actual Wilson current. Every expression is holomorphic in the regular complex links and A. The proof also displays why a bound obtained by estimating each unpaired fine-lattice difference separately would introduce a spurious inverse-spacing loss.

5. The local part of the independent-current extension Define the componentwise covariant Laplacian $\Delta_{\mathrm{cov}} = \sum_\nu D_\nu^* D_\nu$. The exact identity on one-forms is

$$D_U^* D_U + D_U D_U^* = \Delta_{\mathrm{cov}} + \mathcal{L}_U, \qquad (\mathcal{L}_U A)_\mu = \sum_\nu [D_\mu, D_\nu^*]A_\nu. \tag{LC.18}$$

Each commutator is a difference of two two-link transports divided by ξ^2. Their quotient is a plaquette transport. Thus

$$\|\mathcal{L}_U A\|_\infty \le C_G b\, a. \tag{LC.19}$$

This derivation uses only the source covariant differences; no second derivative of the background enters.

Write the source projection as $R = I - P_0$, and put $\mathcal{P}_U = D_U P_0 D_U^*$. Then

$$D_U^* D_U + D_U R D_U^* = \Delta_{\mathrm{cov}} - \mathcal{P}_U + \mathcal{L}_U. \tag{LC.20}$$

For the Landau-gauge minimizing field the added term vanishes. After localization, (LC.20) supplies the source's independent extension rather than reimposing that gauge identity on each localized summand.

Let τJ denote the retained-current term after its exact source rescaling. Suppose $|\tau| \le 1$ and use a smaller input domain

$$\|J\|_\infty \le \alpha_0/4, \quad b \le b_*, \quad \|A\|_\infty, \|\nabla_U A\|_\infty, \|\Delta_{\mathrm{cov}} A\|_\infty, \|\mathcal{P}_U A\|_\infty \le u. \tag{LC.21}$$

The extended current

$$J_{\mathrm{ext}} = \tau J + (\Delta_{\mathrm{cov}} - \mathcal{P}_U + \mathcal{L}_U)A + F(U, A) \tag{LC.22}$$

therefore satisfies, for $u \le 1$,

$$\|J_{\mathrm{ext}} - \tau J\|_\infty \le 2u + C_G(bu + u^2 + u^3) \le K_* u, \tag{LC.23}$$

where $K_* = 2 + C_G(b_* + 2)$ is fixed. Choosing

$$u \le \frac{\alpha_0}{4K_*} \tag{LC.24}$$

gives $\|J_{\mathrm{ext}}\|_\infty \le \alpha_0/2$, with a fixed further reserve.

This proves a linear current-clause radius. It avoids relying on an ambiguous off-by-one reading of the rescaling factor in CMP109 (3.12). If the larger input current ball is used instead, its exact scale factor must leave a quantitative reserve before (LC.24) is substituted.

6. What remains to obtain the full α_2 admission statement The proved bound removes the possible hidden inverse-lattice-spacing cost from the local current formula. In a smaller regular basic input domain, a choice

$$\alpha_2 = c_* \min(\alpha_0, \alpha_1) \tag{LC.25}$$

is sufficient for the current clause by (LC.24), and for the direct plaquette clause by (LC.7), after reducing c_* and reserving enough of the initial α_0 bound.

Two identifications must still be supplied when applying (LC.25):

1. the relative complex-field and first-difference part of the actual source domain must use the same covariant/local regular norm, or be compared to it with uniform constants;
2. the input must lie in a sufficiently smaller basic domain so that the reconstructed-field clause follows from the precise CMP109 p.263 subdomain statement, or the reconstructed-field change must be estimated directly.

These are separate domain conditions. The local remainder (LC.17) does not imply them. In particular, a curvature bound alone does not bound the physical current without the factor ξ in (LC.10).

If the actual background and current are proved to occupy the smaller basic domain, and the actual prepared insertion satisfies the four norms in (LC.21), the local-current step causes only a fixed multiplicative loss. It then permits the proposed running choice $\alpha_{0,k}, \alpha_{1,k} \asymp g_k \ell_k^q$ with an exponent large enough to contain the actual cutoff and verified fixed preparation losses. This does not complete the statement.

7. Sources CMP99 gives the formulas on pp.392, 404 and 405 used in this argument. It supplies the exact action gradient and Hessian, the curvature error formula, and the explicit covariant-operator perturbation estimate. The difference convention is the one on p.391.

CMP109 (3.11)–(3.14) specifies the independent-current construction.

Annex AE — Actual normalized R references and complex marginal transfer

Annex AE. The conclusions below hold on the stated source domains. Applying them to the complete Wilson law requires compatibility of the conditional representation (CL), including the retained variables, densities, and marked transport. See the theorem catalogue.

1. Target and result

This chapter concerns the normalized R contour. The statement asks for a simultaneous current-coordinate contour for every actual localized R pullback, including its marginal transfer, the independently extended current, and the denominators in the source construction.

C-03 is proved at the stated source J0/C-01 scope. The completed theorem, incorporated in full in Section 10, gives the actual regular R activities a simultaneous native contour containing their full real regular current support. On that contour the transferred, vacuum-subtracted R activity has the bound $C_R g_k^{\varkappa_0 - 11} e^{-\varkappa_* d_k(Y)}$, with explicit current, native-background, independently extended-current and preparation derivative radii. The five-power primitive-radius loss and the source's six-power old-scale summation loss are both retained.

The shared quantitative map-domain theorem is proved in Theorem E14.3. Section 10 checks its outputs against every domain input of the R proof, then uses the actual R source prescription and its Ward identities. No analytic continuation from the physical graph $J = J(U)$ is asserted. The true denominators are already included in the completed source R norm; a separate proof below bounds the actual fresh reference in complex total variation and explains its cancellation mechanism.

The auxiliary complex axial-gauge lemma and the optional repeated-reference norm bound retain their explicit premises. They are not needed to claim an uncontrolled bound along an entire marked history. The final theorem does not establish boundary estimates, action/Hessian smallness, arbitrary transported-mark regularity, or history-interface bounds. Status: L is a primary source result in its stated scope; P is proved here; C is an implication whose extra hypotheses remain explicit.

2. Primary domains and the actual reference

2.1 Primitive R and prepared R are different domain statements — L CMP119, pp.259–260, (2.27)–(2.31), gives the primitive R activities the same localized analytic and gauge-invariance properties as the primitive E activities, with

$$|R_X^{(j)}(U,J)| \le g_j^{\varkappa_0} e^{-\varkappa d_j(X)}, \qquad \alpha_{0,j} = C_0 g_j \ell_j^{q_0}, \quad \alpha_{1,j} = C_1 g_j \ell_j^{q_1}, \quad \ell_j = \log g_j^{-2}. \tag{R14.1}$$

The primitive domain is the regular space from CMP109, (1.10)–(1.16). It is not, by definition, the complete multiscale boundary domain introduced later in CMP119, (2.34)–(2.39). CMP109 p.262 defines the primitive domain as a union of complex gauge orbits. Its conditions include a real regular representative, a small complex relative field with a first-difference bound, a current bound, and regularity of reconstructed configurations.

CMP109 p.263 supplies a useful sufficient subdomain: conditions (i)–(iii) with sufficiently smaller constants imply condition (iv), by its displayed application of Proposition 9 of CMP102. In particular the reconstructed curvature bound is at most $2B_3\alpha_0'$ in the source's scale normalization. Thus choosing $\alpha_0' \le \alpha_0/(4B_3)$, together with the other stated smallness restrictions, leaves a quantitative reconstruction reserve. This fact can simplify an admission proof; it is not itself an admission proof for the current free variables.

After actual R preparation, LFR II p.375, (1.65), gives an analytic map from the new expanded multiscale domain to the old expanded domain defined in LFR I (1.64)–(1.69). The target is contained in the domains of the old E, R and B terms. LFR II p.390, (1.100), then gives the newly generated bulk R activity the complex bound

$$|R'^{(k)}(X,(U,J))| \le e^{-p_0(g_k)} e^{-\varkappa d_k(X)}. \tag{R14.2}$$

These are strong source statements. To use their norms for current-coordinate differentiation one must still map a quantitative contour about every actual cutoff configuration into the stated input domain. Openness alone does not give the needed uniform radius.

2.2 The reference is a finite-core Wilson reference — L Fix one replacement component and one axial graph in LFR I p.201, (1.100). Its fresh reference has the form

$$h_v(dV') = 1_{D_g}(V')\,\delta_{G_0}(V')\exp\{-g^{-2}A(\zeta_0, U_{k,Z}(V'v))\}\,dV', \qquad H(v) = \int h_v(dV'). \tag{R14.3}$$

Here v denotes the retained data in the source-admitted complex extension. The core Λ is a rectangular parallelepiped contained in a cube of side $100M$ in current lattice units. The cutoff (1.101) restricts the fresh coordinates only:

$$V' = \exp(igB), \qquad |B(b)| < M_0\varepsilon_k/g, \qquad b \in \Lambda \setminus G_0. \tag{R14.4}$$

In particular this integration domain does not move with v. Let n_Λ be the number of free bonds and $D = \dim \mathfrak{g}\, n_\Lambda$. Then $D \le C_G(100M)^4$. This bound concerns the replacement core; the outer component Z need not have bounded diameter.

Writing $U_0 = U_{k,Z}(v)$ for the reference minimizing field, LFR II p.358, (1.10)–(1.11), proves on its admitted complex domain

$$H(v)^{-1} = \exp\{g^{-2}A(\zeta_0, U_0) + I(v) - E_k(\Lambda)\}, \qquad |I(v)| \le C\log M\,|\Lambda|. \tag{R14.5}$$

This is a holomorphic reference logarithm, hence a nonvanishing statement on that domain. It is stronger than a real lower bound. The extensive retained Wilson term in (R14.5) must be cancelled before estimating the normalized kernel.

3. Uniform complex norm of the actual fresh R reference

3.1 Explicit source estimates entering the norm — L with their complex scope The expansion in LFR II pp.356–358, (1.2)–(1.11), after the change (R14.4), is

$$g^{-2}\{A(\zeta_0, U_{k,Z}(V'v)) - A(\zeta_0, U_0)\} = \tfrac{1}{2}B^{\mathsf{T}}K(v)B + L(v,B) + N(v,B). \tag{R14.6}$$

The notation uses a complex symmetric bilinear quadratic form; B is still integrated over the real Lie-algebra coordinates. Put

$$a_k = \alpha_{0,k} + \alpha_{1,k}, \qquad \lambda_M = \frac{\gamma_0}{2d(100M)^5}, \qquad d = 4. \tag{R14.7}$$

The comparison of the quadratic form with the zero-background form, followed by the axial Poincaré bound (1.8), gives

$$\mathrm{Re}(B^{\mathsf{T}}K(v)B) \ge \lambda_M|B|^2 \tag{R14.8}$$

provided the source's smallness restriction

$$C\{M^6R_ka_k + e^{-R_k}\} \le \tfrac{1}{2}\frac{\gamma_0}{d(100M)^5} \tag{R14.9}$$

holds. For real backgrounds this is (1.7)–(1.9). The paragraph introducing the complex reference on p.358 explicitly retains the preceding bounds with ε_k replaced by $O(a_k)$; applying the absolute-value bound for the quadratic-form error gives (R14.8). A complex quadratic form is not ordered without taking its real part.

The linear term is exponentially small because the full reference minimizer is stationary on the core. The source moves the factor ζ_0 to $1-\zeta_0$ in (1.5), and (1.6) gives the real estimate. The same complex estimate has the explicit form

$$|L(v,B)| \le b_{L,k} := C_M p_0(g_k)\frac{a_k}{g_k}e^{-R_k}. \tag{R14.10}$$

The harmless constant C_M includes the displayed M_0, the fixed source constants, and the bound $|\Lambda| \le (100M)^4$. The cubic-and-higher remainder obeys the source's (1.2) estimate

$$|N(\upsilon, B)| \le b_{N,k} := C g_k^{1-\vartheta} |\Lambda|, \qquad 0 < \vartheta < 1, \tag{R14.11}$$

after the fixed logarithmic cutoff has been bounded by the allowed small negative power of g_k. We do not infer this estimate from Gaussian positivity; it is the actual nonlinear reference estimate on the source-admitted domain.

The Haar coordinate Jacobian has the form $e^{E_k(\Lambda)} J_g(B) dB$, with the source constant normalization absorbed into E_k, $J_g(0) = 1$, and

$$|J_g(B)| \le e^{b_{J,k}}, \qquad b_{J,k} \le CD(M_0 \varepsilon_k)^2. \tag{R14.12}$$

This follows directly from the analytic Haar density in exponential coordinates on the fixed compact Lie group. Its logarithm has no linear term and is bounded quadratically on a fixed chart. The g_k^D and constant Haar-density factors are exactly the factors represented by $E_k(\Lambda)$.

For fixed admitted construction parameters, (R14.9) holds and $b_{L,k} + b_{N,k} + b_{J,k}$ is uniformly bounded for sufficiently small g_k. The factors $g_k \ell_k^q$, and e^{-R_k} against fixed logarithmic powers, are retained explicitly. Section 8 checks their compatibility with the fixed choice $q_0 = q_1 = q > p_0$ used in the actual native contour theorem.

3.2 Actual normalized-reference theorem — P using §3.1 Define the complex measure $\nu_\upsilon = H(\upsilon)^{-1} h_\upsilon$. On the admitted source domain,

$$\nu_\upsilon(dB) = e^{I(\upsilon)} 1_{D_g}(B) J_g(B) e^{-\frac{1}{2} B^\top K(\upsilon) B - L(\upsilon, B) - N(\upsilon, B)} \, dB. \tag{R14.13}$$

Its total mass is one, and

$$\begin{aligned} \|\nu_\upsilon\|_{\mathrm{TV}} &\le \exp\{C \log M \, |\Lambda| + b_{L,k} + b_{N,k} + b_{J,k}\} \left(\frac{2\pi}{\lambda_M}\right)^{D/2} \\ &\le K_M, \qquad K_M \le \exp\{C_G M^4 \log(2M)\} \le \exp\{C'_G M^5\}. \end{aligned} \tag{R14.14}$$

For the last two bounds, reduce the admitted coupling ceiling so that $b_{L,k} + b_{N,k} + b_{J,k} \le C_0 D$, with fixed C_0. The first bound holds without that additional simplification. Constants may depend on the fixed source parameters and gauge group, but not on the cutoff, the outer volume, or the diameter of Z. Graph averaging is a finite convex average and obeys the same bound.

Proof. Multiply (R14.3), expressed in B, by (R14.5). The factors $e^{-g^{-2} A(\zeta_0, U_0)}$ and $e^{E_k(\Lambda)}$ cancel exactly. This gives (R14.13). Take the absolute value, use (R14.8)–(R14.12), enlarge the real integration domain to all of $\mathbb{R}^D$, and evaluate the Gaussian integral. The core dimension and the explicit M^{-5} coercivity give the final bounds. The defining integral and (R14.5) give mass one. Uniform Gaussian domination on compact subsets of the admitted domain proves holomorphy as an $L^1(dB)$-valued function. □

Consequently, for a jointly analytic mark $F(\upsilon, B)$ with a uniform norm on this domain,

$$\left| \int F(\upsilon, B) \, \nu_\upsilon(dB) \right| \le K_M \sup_B |F(\upsilon, B)|. \tag{R14.15}$$

This is an actual estimate for the full fixed-core fresh reference on the complex reference domain used in (1.10). It is not a termwise estimate for independently localized random-walk expressions. The final R theorem in Section 10 uses the completed source R norm and its verified native contour for those expressions. This reference theorem is also not an estimate for the posterior of an archived history label, and it does not identify the signed polymer expansion with a probability distribution.

4. Contour shrink makes the normalization loss arbitrarily small

4.1 Local normalized kernel estimate — P Suppose an actual source chart contains the complex ball

$$v_0 + \{w : \|w\|_* < r\}, \tag{R14.16}$$

where v_0 is real and ν_{v_0} is the positive reference probability measure. Assume (R14.14) throughout this ball. For $s = \|w\|_*/r < 1$,

$$\|\nu_{v_0+w} - \nu_{v_0}\|_{\mathrm{TV}} \le \frac{K_M s}{1-s}, \qquad \|\nu_{v_0+w}\|_{\mathrm{TV}} \le 1 + \frac{K_M s}{1-s}. \tag{R14.17}$$

In particular the smaller radius

$$r_\delta = r\frac{\delta}{K_M + \delta} \tag{R14.18}$$

gives the bound $1 + \delta$, for any fixed $\delta > 0$.

Proof. Restrict the L^1-valued analytic function to the complex line in the direction of w. Cauchy's formula bounds every positive Taylor coefficient by K_M; the geometric sum gives the first inequality. Positivity and normalization give $\|\nu_{v_0}\|_{\mathrm{TV}} = 1$. The second inequality and (R14.18) follow. This proof uses one simultaneous direction, so it introduces no factor proportional to the number of retained coordinates. □

The same argument for an order-m derivative in one unit direction gives

$$\|D^m \nu_{v_0}[w, \dots, w]\|_{\mathrm{TV}} \le m! K_M r^{-m} \|w\|_*^m. \tag{R14.19}$$

Mixed derivatives can instead use an explicitly admitted polydisc. No derivative bound is obtained when (R14.16) has not been verified for the coordinates being differentiated.

4.2 Exact local carrier cost — P as an implication For n_X fresh reference integrations contributing to one carrier, if the same admitted shrunken contour is stable under their actual substitutions, their norm cost is at most

$$(1+\delta)^{n_X} = \exp\{n_X \log(1+\delta)\}. \tag{R14.20}$$

If the geometric counting available for that carrier is $n_X \le a_M d(X) + b_M$, then

$$e^{-\kappa d(X)}(1+\delta)^{n_X} \le (1+\delta)^{b_M} e^{-[\kappa - a_M \log(1+\delta)]d(X)}. \tag{R14.21}$$

Thus any fixed positive carrier-decay reserve can pay this normalization cost by choosing δ sufficiently small. The radius loss is the fixed factor in (R14.18).

For one generation of disjoint replacement cores, the number of cores is bounded by the number of current M-cubes in a connected carrier. This gives the required linear counting bound in that case. For an arbitrarily long prepared history the relevant n_X, the stable common contour, and the counting bound must still be supplied; (R14.21) does not assert them.

5. A complex axial-gauge estimate with all path costs

5.1 Finite rectangular lattice lemma — P Let $\mathcal{Q} = \{0, \dots, r\}^d$, with complex matrix-group links U_e satisfying

$$\|U_e\|, \|U_e^{-1}\| \le e^{\delta}, \qquad \|U_p - I\| \le \epsilon \tag{R14.22}$$

for all oriented links and plaquettes needed below. For each x, let P_x be the path from the origin to x that traverses coordinates in the order $1, \dots, d$, and set $g_x = U(P_x)$. We use the convention $U^g_{xy} = g_x U_{xy} g_y^{-1}$. Then

$$\|g_x\|, \|g_x^{-1}\| \le e^{dr\delta}, \qquad \|U_e^g - I\| \le dr\,\epsilon\, e^{(2dr+1)\delta}. \tag{R14.23}$$

For a current transforming at the initial endpoint,

$$\|J_e^g\| \le e^{2dr\delta}\|J_e\|. \tag{R14.24}$$

If $b = dr\epsilon e^{(2dr+1)\delta} \le 1/2$, the principal logarithm of U_e^g is holomorphic and

$$\|\log U_e^g\| \le 2b. \tag{R14.25}$$

Proof. For $e = (x, x+e_i)$, compare P_x followed by e with P_{x+e_i}. Move the last e_i-step to its ordered position by at most dr adjacent interchanges. Each interchange crosses one plaquette. The difference of the two path transports is bounded by $\epsilon e^{(dr+1)\delta}$, because the total path length is at most $dr+1$. Telescoping all interchanges and multiplying by $U(P_{x+e_i})^{-1}$ proves (R14.23). Conjugation proves (R14.24). The absolutely convergent logarithm series proves (R14.25). Every map used is holomorphic in the links and their inverses. □

At a real compact-group configuration the path transports have norm one. If the complex field is a multiplicative relative perturbation, only the relative complex size contributes to δ; an arbitrary real pure-gauge part does not create an exponential transport cost.

5.2 What this gives for a marginal R chart — C Let $t = k - j$, $\theta_t = L^{-2t}$, and suppose a particular actual coarse pair has, on a box containing its localized carrier,

$$\|U_p - I\| \le Q_k\theta_t, \quad \|J\| \le Q_k\theta_t, \quad \|U_e^{\pm 1}\| \le e^{Q_k L^{-t}}. \tag{R14.26}$$

If the box radius is r_X and

$$dr_X Q_k\theta_t \exp\{(2dr_X+1)Q_k L^{-t}\} \le \tfrac{1}{2}, \tag{R14.26a}$$

then (R14.23)–(R14.25) give

$$\|\log U^g\| + \|J^g\| \le C(1+r_X)Q_k\theta_t \exp\{Cr_X Q_k L^{-t}\}. \tag{R14.27}$$

Suppose additionally that the joint coarse pullback, including its independent current, has an admitted ball of radius a_j in these variables and the primitive bound (R14.1) there. Its first derivative at the zero pair vanishes: global semisimple gauge invariance makes each linear coefficient an invariant linear functional on the complex Lie algebra, hence zero. Applying the second-order Schwarz lemma along a radial line gives, whenever the point and the needed radial chart are admitted,

$$|R_X^{(j)}(U,J) - R_X^{(j)}(1,0)| \le 2g_j^{\varkappa_0} e^{-\varkappa d_j(X)} \min\left\{1, \left[\frac{C(1+r_X)Q_k}{a_j} e^{Cr_X Q_k L^{-t}}\right]^2 L^{-4t}\right\}. \tag{R14.28}$$

This displays the four-power marginal factor without treating complex gauge transformations as norm-preserving. If (R14.26a) fails, only the trivial bound $2g_j^{\varkappa_0} e^{-\varkappa d_j(X)}$ follows from this argument; the large-carrier estimate must then do the required work. A fixed polynomial in the carrier radius can be paid from a verified carrier-decay reserve. If r_X grows with age or with a history component, that dependence must remain in the bound until its sum is estimated.

The premise about the coarse pullback is essential. The original activity may depend on a fine reconstructed field. Axial-gauging an averaged coarse field does not, by itself, identify that fine field with the required minimizing-field substitution. Nor does the lemma identify a localized independently extended current with the physical current of a localized gauge field. Those are actual map questions, not gauge identities.

6. Exact normalization and the marked quotient

For the frozen replacement $\rho'(u',v) = h_v(u')H(v)^{-1}\int \rho(u,v)\,du$, the transported mark is

$$QF(u',v) = \frac{\int \rho(u,v)F(u,v)\,du}{\int \rho(u,v)\,du}. \tag{R14.29}$$

The fresh reference cancels from this ratio exactly. Equations (R14.13)–(R14.21) control the fresh reference when it subsequently occurs in a complex integration or coordinate transfer. They do not remove the old conditional denominator in (R14.29), or establish holomorphy of an old posterior that contains hard history selectors. Those responses belong to the marked/history estimates and must retain their interface currents.

There are therefore three distinct statements: exact real marked integral preservation; bounded analytic fresh-reference integration on an admitted domain; and spatial/analytic estimates for the old projected mark. The first is already proved by the frozen-R mechanism, the second is proved above in its source scope, and the third is not supplied by either one.

7. The quantitative targets and their resolution

Write $\Phi_{j,k,X}$ for the actual localized current-coordinate substitution, with all preparation variables and the independent current retained. Let $\mathcal{K}_k$ be the full actual regular free-coordinate cutoff from C-01/C-07. The required statements are recorded here to make the final verification explicit.

Admission. For every $z_0 \in \mathcal{K}_k$, the simultaneous contour $\mathfrak{D}_{k,X}(z_0)$ must be stable under all decoupling parameters used by the localized expression and satisfy

$$\Phi_{j,k,X}(\mathfrak{D}_{k,X}(z_0)) \subset \mathcal{U}_{j,X}^{\mathrm{actual}}, \tag{R14.30}$$

with an explicit lower bound for its current-coordinate radii. The right-hand side is the regular primitive domain or the actual prepared domain appropriate to that stage. Theorem E14.3 now supplies the actual contour (E14.49), using the full unprimed cutoff $P_k = \delta_k/g_k$, the actual relative averaging and minimizing maps, and all nonlinear preparation variables. Section 10.6 maps each required R-domain input to that proof and to the exact return of the source R operation to its regular normal form.

Complex marginal transfer. A sufficient target is a carrier norm of the form

$$\sup_i \sum_{Y\ni i} e^{\gamma\,\mathrm{diam}\,Y} \sup_{z\in\mathfrak{D}_{k,Y}(z_0)} |V^R_{j,k,Y}(z)| \le C\, g_j^{\varkappa_0}\ell_j^A \frac{Q_k^2}{a_j^2}. \tag{R14.31}$$

Here the $L^{-4(k-j)}$ factor has already paid the old-cell multiplicity. The completed proof instead retains the more conservative fixed five-power inverse-radius loss in (RM.6). After the same carrier localization and old-cell cancellation it gives (RM.9)–(RM.10), and hence the explicit uniform holomorphic carrier bound (RM.13). The primitive R exponent can be chosen arbitrarily large, so this verified fixed loss is sufficient for C-03 as stated. The independently extended current, the large-carrier tail, every nonlinear preparation term and the fixed localization losses are included in that proof. Formula (R14.28) remains a separate optional mechanism; it is not used as an unsupported replacement for the actual R marginal calculation.

Old-scale summation. The source estimate in CMP119 (2.46) loses six coupling powers. Equations (RM.6)–(RM.10) keep this loss after paying the five primitive-radius powers. The admitted current and native external radii are then used after the normally convergent old-scale sum, giving (RM.15). No unproved contour radius or uncancelled $L^{4(k-j)}$ factor is paid by an appeal to a large coupling exponent.

The source-normal-form support theorem C-07 and the holomorphic reference logarithm alone do not prove these targets. Their resolution uses the additional map-domain proof and the explicit marginal calculation incorporated in Section 10. This distinction matters for the remaining history and marked-coordinate obligations.

8. R-side compatibility of larger running analytic exponents

The following inequalities concern the R-side compatibility of the larger running analytic domains. The remaining construction parameters must satisfy their stated restrictions.

LFR I p.198, (1.94), explicitly permits a preparation depth satisfying

$$N \le C\ell_k^{\upsilon}, \qquad \tfrac{1}{2}\upsilon \le p_0 - p_1 - 1. \tag{R14.32}$$

LFR II p.384, (1.80), takes the required waiting depth to be $N = R_j$, with $R_j \asymp \ell_j^r$. Thus it is sufficient here that

$$\tfrac{1}{2}r < p_0 - p_1 - 1. \tag{R14.33}$$

The rare-history budget on LFR II p.385 has the explicit sufficient inequality

$$c\, p_0(g_j)^2 \ge C_M R_j^{d+2}; \tag{R14.34}$$

the exponent condition $2p_0 > r(d+2)$ makes this true for a sufficiently small coupling ceiling. These two requirements are consistent by choosing the fixed integer p_0 sufficiently large after p_1, r are fixed.

The complex Wilson cancellation on LFR II p.368, (1.47), costs

$$C_M p_0(g_k) q_1(g_k) R_k^7 L^{-N}. \tag{R14.35}$$

For $N \asymp \ell_k^r$, this tends to zero for every fixed finite q_1. The component variation on p.379, immediately before (1.73), costs

$$C_M g_k R_k^{d+2} p_1(g_k)^2 q_0(g_k), \tag{R14.36}$$

which also tends to zero for every fixed finite q_0. The reference coercivity condition (R14.9) has the same property. Here $q_i(g)$ denotes the source's fixed logarithmic power, with its constant; it is distinct from the integer exponent q_i.

Choosing equal analytic exponents $q_0 = q_1 = q$ is a convenient safe choice for these inequalities. The displayed bound on p.368 packages $\alpha_0 + \alpha_1$ using $C_1 q_1$, whereas the bound on p.379 packages it using $C_0 q_0$; equal powers allow both with fixed constants. Any necessary fixed ratio between the analytic amplitudes can be imposed through C_0, C_1, without claiming an incompatible asymptotic ordering of the two powers.

Consequently these R inequalities impose no upper bound on a finite common q. If a complete current-map proof produces only a verified fixed logarithmic preparation loss ℓ_k^A, choosing $q > p_0 + A$ makes

$$\frac{g_k \ell_k^q}{g_k \ell_k^{p_0+A}} = \ell_k^{q-p_0-A} \longrightarrow \infty, \tag{R14.37}$$

while preserving (R14.9), (R14.35), and (R14.36). The completed native-map proof retains its actual preparation constants and requires one fixed $q > p_0$. The more general observation (R14.37) is not used to assume that an unexamined map loss is a logarithmic power, or to prove the boundary-domain conditions remaining in C-05.

9. References and hypotheses

Primary references are LFR I, pp. 198 and 201, LFR II, pp. 357–360, 368, 374–375, 379, 385 and 390, and CMP119, pp. 255, 257, 259–260 and 276–277.

CMP109, pp. 262–263, states the smaller-domain inclusion and defines the complex gauge orbit.

The reference-kernel proof concerns the actual finite-core Wilson reference, with its actual cutoff, graph, minimizing field, and normalization. Its validity does not depend on replacing the archived history posterior by the fresh reference law. The completed C-03 proof is given below; the supporting lemmas are auxiliary to it.

10. Complete R marginal and derivative proof

The following argument gives the R marginal transfer and its derivative bounds on the stated domain.

10.1. Result and scope The finite Taylor/Ward part of the R transfer loses at most five powers of a verified old primitive analytic radius. Together with the proved shared actual preparation-domain inclusion, this gives an explicit conservative old-scale budget. The source's six-power coupling summation loss is retained. No identity theorem on the graph $J = J(U)$ is used.

Theorem E14.3 in Annex AD now proves that shared native interior contour, using the actual relative averaging and minimizing maps. Section 10.6 below checks each required output against its equations. The R argument uses those map-domain conclusions; it does not assume an E value bound to obtain an R value bound.

Only the source-defined regular R carriers are considered here. CMP119, p.276, treats inherited R as E without the coupling subtraction. Its subsequent localization assigns carriers crossing the multiscale boundary to B. LFR II, p.390, (1.99)–(1.100), makes the same regular-R/boundary separation after the large-field operation. Those B carriers remain C-05; they have not been deleted or relabelled by this supplement.

10.2. The actual joint composition, not a uniqueness assertion CMP109, pp.275–277, gives the needed source prescription:

1. Before (3.24), replace the external J_{k+1} independently by J, and replace the background in the prepared operators by the specified local reconstructed field $U_{k+1}(\square_0, M'(U))$.
2. Immediately after (3.24), interpret the notation $E(X, U_j)$ as the jointly extended expression with J_j, constructed as in (3.13).
3. In (3.28), transform the external J and the coarse fluctuation together with the background. The explicit transformations occur inside H_k. They are analytic local functions of the external field.
4. In (3.33)–(3.36), form the actual coarse determining field $B = B(U, J, z)$, and evaluate the physical minimizing pullback
$$\Phi_{j,X}(B) = R^{(j)}\left(X, U_j(\square_0, e^{iB}), J_j(\square_0, e^{iB})\right). \tag{RM.1}$$
The independent external J remains in $B(U, J, z)$. It is not identified with the physical current of U.

The actual inherited R expression in CMP119 (2.30) is vacuum-subtracted. Accordingly, the expression expanded below is $\Phi_{j,X}(B) - \Phi_{j,X}(0)$. Its constant coefficient belongs to the separately constructed source-free vacuum scalar. It is neither assigned a false fourth-order scale gain nor silently dropped from the exact action.

The finite Taylor and Ward identities apply to $\Phi_{j,X}$ as identities in its determining variable B. Substitution of the actual jointly analytic $B(U, J, z)$ preserves such an identity. CMP109, p.276, explicitly allows the prepared field to be replaced by an arbitrary regular A satisfying (3.31), because final localization does make such replacements. CMP116, §1, implements that replacement with the actual localized $H_k(s)$, including (1.17).

This is a prescription for a particular joint analytic extension of the physical transferred expression. A different holomorphic extension agreeing only on $J = J(U)$ need not agree with it away from that graph. No such equality is asserted here.

10.3. Shared domain inputs with the needed regularity outputs Fix the source parameters L, M, κ_1 and one admitted trajectory. For each current regular carrier Y, let $\mathcal{C}_{k,Y}$ be the simultaneous contour (E14.49) in the current fluctuation, native background, independent current and source preparation variables, containing the complete real regular support. The R proof uses the following source-native properties on that contour. Their quantitative verification is recorded in Section 10.6.

- Every actual far and nearby primitive argument is admitted in its regular source domain.
- The primitive insertion radius and the nearby determining-field perturbation radius admit a common lower radius
$$a_j \geq c_* g_j, \qquad 0 < a_j \leq 1, \tag{RM.2}$$

with $c_* > 0$ independent of old depth, cutoff, volume and history. For example, a proved $a_j \geq c\alpha_{0,j}$, with $\alpha_{0,j} = C_0 g_j \ell_j^q$, gives (RM.2) after reducing the coupling ceiling.

- The actual prepared current-scale field satisfies the source's regularity bounds (3.31)–(3.32) with a uniform ceiling $Q_k \leq 1$, simultaneously for the complex decoupling parameters. Consequently its old determining field satisfies, with $\lambda = L^{j-k}$,

$$\|B_j\| \leq CQ_k\lambda, \qquad \|\partial B_j\| \leq CQ_k\lambda^2, \qquad \|\partial^2 B_j\| \leq CQ_k\lambda^{2+\beta}, \tag{RM.3}$$

  for one fixed $0 < \beta < 1$. These are precisely the scaling bounds proved in CMP109 (4.16)–(4.18), including the actual averaging operation. The same bounds hold for the source's insertion variations when those are used.
- The source's far-carrier and localization bounds remain valid on the same contour, with their actual preparation factor $K = B_0 e^{16\kappa_1}$, fixed cutoff-multiplier constants, and the exponent reserve used in CMP116. In particular, neither an independent-current substitution nor a decoupling variable is omitted from this inclusion.

These are domain and map-regularity assertions. They do not assume the desired R marginal estimate.

10.4. Finite Taylor/Ward radius budget The primitive source input is

$$\sup_{\mathcal{D}_{j,X}} |R^{(j)}(X,U,J)| \leq g_j^{\kappa_0} e^{-\kappa d_j(X)}. \tag{RM.4}$$

CMP119, p.260, (2.31), gives (RM.4) and explicitly permits κ_0 to be arbitrarily large after fixing the other parameters properly. LFR II (1.100) supplies an even smaller newly generated regular R bound. Thus choosing a fixed larger κ_0 below is within the source's stated scope.

The coarse Taylor expansion uses orders at most four and the fifth-order integral remainder. On a contour satisfying (RM.2), its order-m coefficient has a Cauchy cost at most $m! a_j^{-m}$, for $m \leq 5$. The coefficient kernel moment estimates used in the Ward calculation have the same maximum inverse-radius power. This is visible in CMP109 (4.3)–(4.5), where the derivatives of the minimizing map at the vacuum have fixed regular kernel bounds, and in (4.22), whose displayed bound contains $(\alpha_1/\alpha_2)^4$. The fifth remainder is (3.54), with its fifth inverse coarse-radius power.

The remaining constants in these finite-order estimates are fixed spatial kernel moments. They depend on L, M, κ and the admitted propagator constants, but not on the number of fine sites. The exponential kernel sums in the proof of (4.22) are essential for this statement; a pointwise Cauchy coefficient bound followed by an unweighted sum over fine coordinates would not suffice.

The source subsequently differentiates the localized expression in its insertion parameter $t_\square$, as stated immediately before CMP119 (3.67). This does not require a sixth inverse old activity radius. The nearby and far estimates above are uniform on the reserved complex insertion tube. For $0 \leq t_\square \leq 1$, the disc $|h| < 1$ has $|t_\square + h| \leq 2$. The preparation reserve (E14.11) and the whole-path bound (E14.44) admit this disc, with their spare factors. Apply the fixed unit-radius Cauchy formula to the already bounded expression. Thus the $t_\square$ derivative costs a fixed factor, independent of a_j, old age and cutoff. The localization parameters are kept simultaneous as in CMP116; they are not differentiated by a separate unsupported real argument.

The Ward identities (4.14)–(4.15) are complex algebraic identities for the gauge-invariant function (RM.1). The first derivative vanishes. The quadratic, cubic and quartic terms acquire, respectively, two, one and zero extra old-scale difference factors. By (RM.3), each surviving term has a factor λ^4. The explicitly retained Hölder remainders improve this to $\lambda^{4+\beta}$, and the fifth Taylor remainder has λ^5. Therefore, before summing old cells, the native nearby bounds have the conservative form

$$\begin{aligned} |\mathcal{R}^{\mathrm{marg}}_{j,X;k}| &\leq C_{\mathrm{W}}\, g_j^{\kappa_0} \sum_{m=2}^{4} a_j^{-m} Q_k^m\, \lambda^4 e^{-\kappa' d_j(X)}, \\ |\mathcal{R}^{\mathrm{irr}}_{j,X;k}| &\leq C_{\mathrm{W}}\, g_j^{\kappa_0} \left(\sum_{m=2}^{4} a_j^{-m} Q_k^m\, \lambda^{4+\beta} + a_j^{-5} Q_k^5\, \lambda^5 \right) e^{-\kappa' d_j(X)}. \end{aligned} \tag{RM.5}$$

Here $\kappa' < \kappa$ is the fixed source decay reserve after the finite coefficient-moment sums. CMP119, p.283, says to stop the R analysis at its (3.49); there is no coupling subtraction to perform. That is exactly the marginal term retained in (RM.5).

Using (RM.2), $Q_k \le 1$, $g_j \le 1$, and absorbing c_*^{-m} into a fixed constant gives

$$|\mathcal{R}^{\mathrm{marg}}_{j,X;k}| + |\mathcal{R}^{\mathrm{irr}}_{j,X;k}| \le C_* g_j^{\kappa_0-5} \lambda^4 e^{-\kappa' d_j(X)}. \tag{RM.6}$$

This deliberately weaker bound keeps a single five-power loss. It does not assert the sharper source coefficient is unchanged when the primitive radii run.

Far carriers have instead the source factor $e^{-c/\lambda}$, or the corresponding large-carrier exponential. Their one primitive Cauchy cost is a_j^{-1}, as in CMP109 (3.15)–(3.17). For any fixed N,

$$e^{-c/\lambda} \le C_{c,N} \lambda^N, \tag{RM.7}$$

so $N \ge 4$ gives the same conservative marginal budget (RM.6), retaining the remaining carrier decay.

10.5. Localization and the six-power sum The preparations and localization operations are the source's actual ones: CMP116 §1, then CMP119 pp.275–277 for the complete geometry. The current interior terms satisfy the original regular-source conditions; the boundary-crossing terms are precisely the B terms identified there. The decoupling contour is simultaneous, $|s(\Delta)| \le e^{\kappa_1}$, and its preparation factor is retained in the hypothesis of §10.3.

Applying the source's localization/resummation estimates to (RM.6), at a fixed current carrier Y, cancels the old-cell count $O(\lambda^{-4})$ against the factor λ^4. The localization prefactor is the fixed one from that source construction, of the form

$$C_{\mathrm{loc}} = C_{\mathrm{W}} C_1 M^{q_{\mathrm{loc}}} e^{C_2 \kappa_1}, \tag{RM.8}$$

with its stated fixed-parameter dependence. A fixed reduction to an admitted $\kappa_* > 0$ accounts for the coefficient-moment and preparation decay losses. No cutoff-dependent change of M, κ_1, or unproved extra logarithmic loss is hidden in C_{loc}.

Thus the remaining old-scale majorant is

$$C_{\mathrm{loc}} e^{-\kappa_* d_k(Y)} \sum_{j \le k} g_j^{\kappa_0-5}. \tag{RM.9}$$

Use the six-power estimate actually displayed in CMP119 (2.46), under its source trajectory hypotheses. Taking $\kappa_0 - 5 \ge 7$, and imposing the corresponding coupling ceiling, gives

$$\boxed{\sup_{\mathcal{C}_{k,Y}} |\mathcal{R}^{\mathrm{transferred}}_{k,Y}| \le C_{\mathrm{loc}}\, g_k^{\kappa_0-11}\, e^{-\kappa_* d_k(Y)}.} \tag{RM.10}$$

The loss is five inverse-radius powers plus the source's six-power summation loss. No age upper bound is inferred from the wrong direction of CMP119 (2.6). In particular, this proof does not require a new lower inverse-coupling increment in order to absorb an axial-gauge age polynomial.

For a simultaneous scalar-coordinate polydisc with radii $r_i > 0$ contained in the same verified contour, Cauchy gives

$$|\partial^a \mathcal{R}^{\mathrm{transferred}}_{k,Y}| \le a!\, C_{\mathrm{loc}}\, g_k^{\kappa_0-11} e^{-\kappa_* d_k(Y)} \prod_i r_i^{-a_i}. \tag{RM.11}$$

When the admitted full current ball is $\|z\|_\infty < 2P_k$, every real center with $\|z\|_\infty \le P_k$ has simultaneous scalar Lie-coordinate radii $P_k/(2\sqrt{\dim G})$. Native-background balls require the directional convention stated in Theorem R14.3 below; they cannot be interpreted as unrestricted fine-coordinate polydiscs.

10.6. Verification of the domain inputs and the true denominators The common contour has the explicit form

$$\begin{aligned} &\|z\|_\infty < 2P_k, \qquad \|b\|_{\mathrm{native},k} < c_{\mathrm{ext}}\alpha_{0,k}, \\ &\|J - J_{0,k}\|_\infty < c_{\mathrm{ext}}\alpha_{0,k}, \qquad |s(\Delta)| < e^{\varkappa_1}. \end{aligned} \tag{RM.12}$$

Here $P_k = \delta_k/g_k = A_1\ell_k^{p_0}$ is the full unprimed source cutoff, $\alpha_{i,k} = C_i g_k \ell_k^q$ with one fixed $q > p_0$, and c_{ext} is fixed by (E14.47). The ratio C_1/C_0, the source parameters, and the common small-coupling ceiling are chosen exactly as in Theorem E14.3. In particular they do not depend on later derivative order.

Required R input	Quantitative verification on (RM.12)
Actual real centers and primitive reconstructed-field domains	(E14.26)–(E14.32) establish a strict smaller-basic-domain reserve at the actual centers, using layerwise minimizing-field estimates, the local Wilson-current formula, and the primary reconstructed-domain inclusion. CMP119 (2.30)–(2.31) gives primitive R exactly the same regular analytic and gauge-invariance properties as E.
Actual nonlinear preparation, independent current and decoupling variables	The minima (E14.8), with $K = B_0 e^{16\varkappa_1}$, admit the actual CMP116 constraint/minimizer maps. Equations (E14.45)–(E14.48) transfer the native external background and independently varied current to every old scale with conservative coefficient $C_J L^{j-k}$, including gauge-conjugation costs.
Whole far insertion path	(E14.39)–(E14.44) first admit the actual unmodified Q_j input to the fixed absolute map domain, integrate the actual relative-minimizer derivative from zero there, and then compare the whole output with the running primitive radius. No $Q_j = Q_j'$ identification or assumption of the desired activity-domain conclusion is used.
Nearby determining-field radius and the common lower bound (RM.2)	(E14.15b), (E14.24f) and (E14.32) give $r_{3,j} \geq c_3\alpha_{0,j}$ and $\alpha_{2,j} = c_2\alpha_{0,j}$, with uniform reserves along the actual Taylor segments. Taking a_j to be a fixed fraction of their minimum and of the remaining native radius gives $a_j \geq c_* g_j$ and $a_j \leq 1$ after reducing the coupling ceiling.
Prepared-field regularity (RM.3)	The actual preparation bounds in Sections 3 and 5.6 of the E note give $Q_k \leq c_Q\alpha_{0,k} \leq 1$. CMP109 (4.16)–(4.18) then give the displayed determining-field and difference bounds on this jointly admitted domain. The independent current enters through the prescribed actual prepared field as in (RM.1).
Fixed insertion differentiation and carrier localization	(E14.11) and (E14.44) give the unit insertion-parameter disc used in Section 10.4 above. The same simultaneous preparation contour is used in CMP116 Lemma 1, so its fixed localization losses apply. The E proof's map admission is used here; its matching coupling subtraction is not imported into R.

This table checks map-domain assertions rather than appealing solely to the completion label of C-02. The construction hypotheses retained in J0, including the admitted history geometry and source propagator bounds, remain hypotheses of both contour theorems.

The primitive bound (RM.4) concerns the actual completed R activities, including the denominators and preliminary integrations used to create them. LFR II, pp.374–375, (1.63)–(1.65), constructs their joint analytic pullbacks and

(1.99)–(1.100) gives the final convergent activity bound. The fresh finite-core normalized reference itself has the separate complex total-variation bound proved in Annex AE, §§2–4. That reference law is not confused with the archived-history posterior.

More precisely, LFR II (1.98)–(1.100) first completes the R operation and then assigns its boundary-intersecting outputs to B. Its remaining regular R activities satisfy the regular primitive domain and (RM.4), after choosing the coupling ceiling so that $e^{-p_0(g_j)} \le g_j^{\kappa_0}$. This uses the source's logarithmic polynomial p_0 of degree greater than one. Subsequent regular current-coordinate pullback is therefore into the very primitive domain verified in the table. No arbitrary multiscale boundary tuple is asserted to lie in the regular domain. Its estimates remain in C-05.

The true source denominators are thus already included in the completed primitive holomorphic norm being pulled back. The separate normalized-reference estimate verifies their cancellation mechanism and gives a useful mark-integration norm, but the proof does not multiply an uncontrolled number of reference norms along an entire history. It also does not infer an analytic old-history posterior from a fresh reference denominator.

10.7. Completed R contour theorem

Theorem R14.3 — actual regular R contour and marginal transfer Retain the source-free complete RG representation, its admitted geometry, regular and large-field history decomposition, and consistently normalized trajectory as in J0/C-01. Use its actual E/R/B classification. Choose the fixed source parameters and the common running analytic domains as in Theorem E14.3. Choose one fixed $\kappa_0 \ge 12$, permissible by CMP119 p.260 after (2.31), and reduce the common coupling ceiling to meet (RM.2), the reference restrictions when that reference is used, and the source summation estimate.

Then every actual vacuum-subtracted localized regular R pullback, including the inherited marginal/coordinate transfer and the new completed R activities, is jointly holomorphic on (RM.12). There are fixed $C_R < \infty$ and $\kappa_* > 0$, independent of cutoff depth, torus volume, remaining admitted history and subsequent derivative order, such that

$$\boxed{\sup_{\mathcal{C}_{k,Y}} |\mathcal{R}_{k,Y}| \le C_R g_k^{\kappa_0 - 11} e^{-\kappa_* d_k(Y)}.} \tag{RM.13}$$

Here $\mathcal{R}_{k,Y}$ is the actual source-normal-form R contribution; its vacuum scalar is separated by the exact source prescription (2.30). The old-cell count is paid by the four marginal scale powers before the old-scale sum. The denominators and preparation factors are those of the actual construction.

For a real current center $\|z_0\|_\infty \le P_k$, take

$$r_z = c_G P_k, \qquad r_b = r_J = c_G c_{\mathrm{ext}} \alpha_{0,k}/2, \qquad r_s = (e^{\kappa_1} - 1)/2, \tag{RM.14}$$

with the fixed Lie-coordinate factor c_G from (E14.52). For scalar-coordinate multi-indices u, w, t, and m native-background derivative directions of norm at most one, one has

$$\begin{aligned} &\left|\partial_z^u D_{b_1} \cdots D_{b_m} \partial_J^w \partial_s^t \mathcal{R}_{k,Y}\right| \\ &\quad \le u!w!t!\, e^m m!\, C_R g_k^{\kappa_0 - 11} e^{-\kappa_* d_k(Y)} r_z^{-|u|} r_b^{-m} r_J^{-|w|} r_s^{-|t|}. \end{aligned} \tag{RM.15}$$

For raw scalar link-coordinate derivatives of multi-index υ, the native first-difference norm instead admits the simultaneous coordinate radius ξr_b, giving $\upsilon!(\xi r_b)^{-|\upsilon|}$ in place of $e^m m! r_b^{-m}$. If the chosen native norm includes a first-difference Hölder seminorm of exponent β, use $\xi^{1+\beta} r_b$. No second-order or nonlocal projection estimate for arbitrary raw coordinate directions is asserted from a first-difference norm.

Proof. Section 10.6 verifies every shared-domain input in Section 10.3 on the same product contour, retaining actual independent-current and nonlinear preparation maps. The exact prescribed joint composition in Section 10.2 permits the finite Taylor/Ward calculation there, with its vacuum subtraction. The kernel-moment bounds and reserved insertion disc give (RM.5)–(RM.7) with at most five inverse primitive-radius powers. The source's carrier localization and old-cell counting give (RM.9); its actual six-power coupling sum gives (RM.10). The newly generated completed R activities obey the stronger primitive bound and fit the same majorant. Enlarge the fixed constant to C_R. Normal

convergence gives joint holomorphy and (RM.13). Cauchy on the scalar-coordinate factors and on discs of radius r_b/m for the m background directions gives (RM.15), using $m^m \le e^m m!$. The stated raw-coordinate alternatives follow from the explicit native-norm embedding, exactly as in (E14.52)–(E14.53). □

Under the stated construction assumptions, the regular R contour has a uniform holomorphic norm with carrier decay and the indicated preparation losses. Its application to the Wilson representation requires CL. The exponent in the preceding estimate is a sufficient bound.

This theorem does not prove the full source-free construction or its coupling trajectory independently. It does not prove C-04 action/Hessian smallness, C-05 boundary-domain estimates, C-06 conditional coercivity, arbitrary later marked-coordinate transport, or spatial control of archived-history posteriors and selector currents. Those remain distinct obligations, and every such term remains in the complete law. No Yang–Mills existence claim follows from C-03 alone.

Annex AF — Boundary normalization, current Hessian rows and collar sums

Annex AF. The conclusions below hold on the stated source domains. Applying them to the complete Wilson law requires compatibility of the conditional representation (CL), including the retained variables, densities, and marked transport. See the theorem catalogue.

0. Results and hypotheses

This annex proves three analytic reductions and locates a more useful actual source domain for the boundary fluctuation expansion. It does not complete C-05. The missing application is a uniform estimate for the complete actual boundary pullback, including every inherited fluctuation, relocation, independent-current and retained-variable argument, in the current coordinates used by the marked induction.

The new reductions are:

1. a retained-data-dependent boundary constant cancels exactly from a normalized regular conditional law, including all its normalized retained derivatives; the factor remains in the outer marginal;
2. a simultaneous complex polydisc gives a weighted Hessian row bound without the unnecessary number-of-coordinates factor obtained by summing separate pairwise Cauchy estimates;
3. an actual collar factor of the form e^{-cRL^a} pays dyadic radius losses 2^{am}, polynomial carrier multiplicity, and source-order-dependent polynomial age losses within one fixed factorial exponent.

The third statement is deliberately different from the insufficient logarithmic-age suppression diagnosed in Annex Z. Its stronger decay must be established for the actual differentiated terms before using the theorem.

1. Primary source conditions

See CMP119, pp. 254, 259, 261–262 and 276–277.

- Formula (2.1) interlaces the domains as $\Omega_1 \supset \Lambda_1 \supset \Omega_2 \supset \Lambda_2 \supset \cdots$. Formula (2.3) names $Z_j = \Lambda_j^c$.
- The primitive E activities on p.259 use the regular space from the earlier small-field construction. The separate multiscale conditions (2.34)-(2.39), p.261, are introduced for boundary terms. A statement that every primitive E activity itself has the full B-domain definition would conflate these two inputs.
- The B expansion (2.40)-(2.42) retains old fields and history data and has primitive bound $B_0 e^{-\kappa d_j(X)}$. The source selects this full-tree bound. The weaker component-resummed alternative with $d_j(X \setminus Z_j)$, discussed on p.262, must not silently replace it in a spatial estimate.
- The localized boundary fluctuation expansion on p.276 is analytic on the explicit product space (3.43), whose current fluctuation factor is $\sup |A| < C_1 p_1(g_k)$ on the displayed free-bond set. Its background factor is the stated enlarged multiscale complex space. This is stronger information than bare membership in an unspecified open set.
- Page 277 assigns an additional $O(\varepsilon_k)$ to localized B fluctuation terms and explains how minimizing-field decay pays the shrinking dyadic analytic reserve. It also retains the heterogeneous-action and boundary-relocation terms. Extracting a single complete, current-coordinate derivative majorant from all these statements still requires the original indices, resummations, maps and radius costs.

The precise representation in (3.43) concerns the localized action/fluctuation expressions at this stage of the construction. It does not justify assigning its native current radius to every independently transported history observable. It also does not identify the native background coordinates with arbitrary retained variables without a pullback estimate.

The earlier effective-action paper, CMP109, equations (3.9), (3.14)-(3.17), supplies the far-field decay and the native differentiated-insertion contour used by the source argument.

2. Remove only the conditional boundary constant

Let υ denote retained data, let A range over the fixed regular product domain D of C-07, and let

$$r_\upsilon(dA) = 1_D(A)\exp\{L(A,\upsilon) + B(A,\upsilon)\}\,dA, \qquad Z(\upsilon) = r_\upsilon(1) > 0. \tag{BD.1}$$

All other action terms remain in L; B can be the complete boundary contribution. Fix a reference $A_*(\upsilon) \in D$ for which the displayed quantities exist, and define

$$b_*(\upsilon) = B(A_*(\upsilon),\upsilon), \qquad \widetilde{B}(A,\upsilon) = B(A,\upsilon) - b_*(\upsilon), \qquad \tilde{r}_\upsilon = e^{-b_*(\upsilon)} r_\upsilon. \tag{BD.2}$$

Proposition BD.1 – exact conditional cancellation The normalized law $q_\upsilon = r_\upsilon/Z(\upsilon)$ equals $\tilde{r}_\upsilon/\tilde{r}_\upsilon(1)$. For every integrable marked numerator $F(A,\upsilon)$, its conditional expectation is unchanged. Whenever retained derivatives of these equal expressions exist, all orders of their normalized retained derivatives agree as well.

For an outer measure $\eta(d\upsilon)$, the unnormalized retained marginal is

$$Z(\upsilon)\eta(d\upsilon) = e^{b_*(\upsilon)}\tilde{r}_\upsilon(1)\eta(d\upsilon). \tag{BD.3}$$

Thus this operation does not delete b_* from the full law.

Proof. Both numerator and denominator of every conditional expectation acquire the same nonzero scalar factor $e^{-b_*(\upsilon)}$, so it cancels. Differentiation preserves the resulting exact equality whenever the chosen derivatives are justified. Equivalently, in a finite source-jet algebra, an invertible scalar series cancels algebraically, without an analytic convergence assumption. Formula (BD.3) follows by integrating (BD.2). □

In a smooth fixed chart the normalized score is

$$\partial_a(L+B) - q_\upsilon\partial_a(L+B) = \partial_a(L+\widetilde{B}) - q_\upsilon\partial_a(L+\widetilde{B}). \tag{BD.4}$$

This observation allows conditional estimates to be phrased in terms of the actual boundary fluctuation difference. Retained-only factors remain in the outer history law, and their cumulants are still part of H-03/H-05. If the live geometry changes across an interface, this smooth-chart identity does not remove the two-sided interface current from Annex AA. No global positivity or phase property is inferred from this cancellation.

3. A simultaneous contour gives a sharper Hessian budget

Let f be holomorphic and bounded in modulus by M on the polydisc

$$\mathcal{P}(z,r) = \{z+\zeta : |\zeta_j| < r_j \text{ for every } j \in I\}, \qquad r_j > 0. \tag{BD.5}$$

The coordinates here are scalar complex coordinates. A fixed finite Lie-algebra block dimension changes only the constants, as explained below.

Theorem BD.2 – one anchored row without a carrier-volume loss For every $i \in I$,

$$\sum_{j\in I} r_j|\partial_i\partial_j f(z)| \le \frac{4M}{r_i}. \tag{BD.6}$$

Consequently, for positive coordinate weights ω_i and any nonnegative spatial weights w_{ij},

$$\sum_j \frac{w_{ij}}{\omega_i\omega_j}|\partial_i\partial_j f(z)| \le \frac{4M}{\omega_i r_i}\max_j \frac{w_{ij}}{\omega_j r_j}. \tag{BD.7}$$

Proof. Choose phases u_j with $|u_j| = 1$ so that every nonzero number $u_j \partial_i \partial_j f(z)$ is real and nonnegative. For $|t|, |s| < 1$, the function

$$F(t,s) = f\left(z + \tfrac{1}{2} r_i t e_i + \tfrac{1}{2} s \sum_j r_j u_j e_j \right) \tag{BD.8}$$

lies in (BD.5), including when the two variations affect coordinate i. The two-variable Cauchy estimate gives $|\partial_t \partial_s F(0,0)| \le M$. Its derivative is

$$\partial_t \partial_s F(0,0) = \tfrac{1}{4} r_i \sum_j r_j u_j \partial_i \partial_j f(z).$$

The choice of phases proves (BD.6); multiplying by the maximum displayed in (BD.7) proves the second claim. Boundary radii are obtained as limits from smaller discs. □

This is stronger than summing the separate estimate $2M/(r_i r_j)$ over j. It uses a simultaneous contour: a collection of two-coordinate discs does not suffice for (BD.8).

For Lie-algebra blocks choose a fixed scalar basis and its norm-equivalence constants. Apply (BD.6) to each output component and each input component, reducing each scalar radius by the fixed amount needed to remain inside the block ball. Summing over the fixed number of components yields

$$\sum_j \frac{w_{ij}}{\omega_i \omega_j} \| D_i D_j f(z) \|_{\mathrm{op}} \le \frac{C_G M}{\omega_i r_i} \max_j \frac{w_{ij}}{\omega_j r_j}. \tag{BD.9}$$

Here C_G depends on the fixed gauge group and chosen block norms, not on the number of bonds, carrier size or torus volume.

Application to a native boundary vertex For a localized vertex $b_{k,Y}(A; U, J, \mathrm{A}_{\mathrm{old}})$ on the source domain (3.43), keep its native background arguments in that domain. Suppose its source bound on that domain is $M_{k,Y}$. At a real current field with

$$\sup_{i \in I_Y} |A_i| \le P_k < R_k^A, \quad R_k^A = C_1 p_1(g_k), \quad \rho_k = R_k^A - P_k > 0, \tag{BD.10}$$

the actual translated current polydisc has radius ρ_k. Formula (BD.9), with $w_{il} = e^{\gamma d(i,l)}$, gives

$$\sum_l e^{\gamma d(i,l)} \| D_i D_l b_{k,Y}(A) \|_{\mathrm{op}} \le C_G M_{k,Y} \rho_k^{-2} e^{\gamma \operatorname{diam}(I_Y)}. \tag{BD.11}$$

The same source domain gives all finite current-direction derivatives by multivariable Cauchy, with the usual $m^m \le e^m m!$ factor. This does not imply a dimension-free absolute sum of every order-m tensor coefficient. The row argument works because all but one index form a linear functional after fixing the first index. General higher-order absolute coefficient sums require their own localization or tensor norm.

The still-required normalization check in (BD.10) is explicit. The literal cutoff symbol in CMP119 (3.16) and the preceding rescaling must be reconciled before declaring a lower bound for ρ_k on the entire actual support. Using $P_k = \delta_k$ in one formula and $P_k = \delta_k / g_k$ in another would be invalid.

The source comparison makes this issue more specific. CMP119 after (3.4), p.265, defines the unprimed cutoff using $p_0(g_k)$. The smaller polynomial introduced for a primed preparation selector in LFR I is a different quantity. The printed $p_1(g_k)$ in CMP119 (3.43) has not been identified here with either quantity by a verified dictionary. One must not import the sharper primed-selector radius for the full original cutoff. If one were to identify the native radius with $C\ell^{p_1}$ and the normalized original cutoff with $A\ell^{p_0}$, where $p_1 < p_0$ and $\ell = \log g^{-2}$, then

$$\frac{C\ell^{p_1}}{A\ell^{p_0}} = \frac{C}{A} \ell^{-(p_0 - p_1)} \longrightarrow 0. \tag{BD.10a}$$

No fixed enlargement of C would produce a uniform margin in that identification. This is a test of the proposed identification, not a claim that the source construction has been disproved. The alternatives are a correct polynomial dictionary or a new quantitative admission argument using the underlying fluctuation theorem. Annex AD records the sharper remaining comparison.

4. Carrier sums and the complete current Hessian

Suppose a current-scale expansion has actual vertices $b_{k,Y}$, all with the current radius ρ_k, such that

$$M_{k,Y} \le K_k e^{-\kappa n(Y)}, \qquad \operatorname{diam}(I_Y) \le c_d n(Y), \tag{BD.12}$$

and, for every fixed current bond i, at most $C_D A_D^n$ of the carriers containing i have integer size $n \ge 1$. If

$$\delta = \kappa - c_d\gamma - \log A_D > 0, \tag{BD.13}$$

then (BD.11) proves

$$\sup_i \sum_l e^{\gamma d(i,l)} \left\| D_i D_l \sum_Y b_{k,Y} \right\|_{\mathrm{op}} \le \frac{C_G C_D K_k}{\rho_k^2} \frac{e^{-\delta}}{1-e^{-\delta}}. \tag{BD.14}$$

Proof. Sum (BD.11), first over carriers at fixed size and then over n. The scalar series is geometric by (BD.13). All constants are independent of the total number of bonds. □

For nearest-neighbor connected cell carriers a crude exponential counting constant follows by recording a depth-first traversal of a spanning tree, with a number of possible steps bounded by the fixed lattice degree. More general source domains must be counted in their actual cell convention. One cannot assign this counting constant to an arbitrary collection of resummed or component-contracted carriers without checking that convention.

For the B-origin fluctuation terms the source's additional $O(\varepsilon_k)$ is potentially the factor K_k in (BD.12). This identification must include the complete sum over old terms and the actual localization constants. The heterogeneous-action and relocation contributions require the same check; the displayed inequality does not assert that their constants already equal those of a single primitive B vertex.

For a pullback not already included in the native expansion, one still has

$$D^2(b \circ \Phi) = D^2 b(\Phi)[D\Phi, D\Phi] + Db(\Phi) D^2\Phi. \tag{BD.15}$$

The second term remains. A direct native estimate (BD.11) is useful precisely when the source has already performed that pullback and supplied its joint norm; it cannot be transported to a different coordinate system by changing its name.

5. What a genuine collar gain pays at every source order

Lemma BD.3 – elementary scaled collar geometry Let the physical cell sizes be $h_j = L^j a$, with $L > 1$. Suppose a current support point x lies at physical distance at least bRh_k from Z_k, and $Z_j \subset Z_k$ for $j \le k$. Every connected tree meeting both x and Z_j has length in j-scale units at least

$$bR\frac{h_k}{h_j} = bRL^{k-j}. \tag{BD.16}$$

Proof. A connected tree joining the two sets has physical length at least their distance. Nested bad regions give $\operatorname{dist}(x, Z_j) \ge \operatorname{dist}(x, Z_k)$. Divide by h_j. The same proof applies with the torus metric whenever the stated collar exists. □

The interlacing of source domains supplies the nesting. The current collar and the tree connecting the *actual differentiated influence* to the old boundary must still be identified in the localized expression. A native vertex's nominal

label is not such a tree when an unlocalized minimizing-field map remains attached to it. Derivatives of a localization or preparation factor must retain their own connecting paths.

Theorem BD.4 – dyadic radius losses with a collar factor Fix $L > 1$, $c > 0$, $d, \sigma, \eta \geq 0$, $A \geq 1$, and $p \geq 0$. Put $\beta = \log 2/\log L$. Suppose nonnegative terms, for integers $a \geq 0, m \geq 1$ and $R \geq 1$, satisfy

$$T_{a,m}(R) \leq A^{m+1}(m!)^p \rho^{-m} 2^{am} L^{da}(1+a)^{\sigma m+\eta} e^{-cRL^a}, \qquad \rho > 0. \tag{BD.17}$$

For every $\epsilon > 0$ there is a constant C_ϵ, depending only on the fixed displayed parameters and ϵ, such that

$$\sum_{a\geq 0} T_{a,m}(R) \leq C_\epsilon^{m+1}(m!)^{p+\beta+\epsilon}\rho^{-m}e^{-cR/2}. \tag{BD.18}$$

If $\sigma = 0$, one may take $\epsilon = 0$.

Proof. If $\sigma > 0$, the elementary exponential-versus-polynomial bound gives

$$(1+a)^\sigma \leq K_\epsilon L^{\epsilon a} \quad (a \geq 0). \tag{BD.19}$$

If $\sigma = 0$, use $K_0 = 1$ and no additional power. Similarly $(1+a)^\eta \leq KL^a$. With $b = \beta + \epsilon$, or $b = \beta$ when $\sigma = 0$, it suffices to bound

$$\sum_{a\geq 0} (L^a)^s e^{-cRL^a}, \qquad s = d + 1 + bm. \tag{BD.20}$$

Split the exponential into factors of relative weights $1/2, 1/4, 1/4$. Since $L^a \geq 1$, the first factor is at most $e^{-cR/2}$. For the second factor,

$$\sup_{x\geq 0} x^s e^{-cRx/4} = \left(\frac{4s}{ecR}\right)^s \leq \left(\frac{4s}{ec}\right)^s. \tag{BD.21}$$

The sum of the remaining factors is bounded by

$$\sum_{a\geq 0} e^{-cL^a/4} \leq \frac{e^{-c/4}}{1-e^{-c(L-1)/4}}, \tag{BD.22}$$

because $L^a \geq 1 + a(L-1)$. Finally, for fixed $d, b > 0$,

$$s^s \leq C^{m+1}(m!)^b. \tag{BD.23}$$

To verify this last step, write $s = bm + D$, with fixed $D = d + 1$. The factor $(1 + D/(bm))^{bm+D}$ is uniformly bounded up to a fixed constant; the remaining factor is a constant to the power m, times m^{bm+D}. Use $m! \geq (m/e)^m$ and absorb the fixed power m^D into another constant to the power m. Combining these bounds proves (BD.18). □

Thus a verified dyadic radius loss does not by itself destroy the fixed OS factorial budget when it is paired with this stronger physical collar decay. The theorem retains the common current radius ρ^{-m}; it has not proved that this radius has the required physical scaling. It also does not supply (BD.17) for every boundary response. Source-order growth of the preparation maps and carrier multiplicity must be included in the parameters before applying it.

6. Application conditions

The current proof obligation for C-05 can now be stated more economically, without discarding any term:

1. identify the complete native boundary fluctuation expansion in the actual free current variable, resolving its normalization and giving a positive uniform reserve in (BD.10);

2. prove the fully resummed anchored majorant (BD.12), or a directly sufficient weighted sum, for B-origin, heterogeneous-action and relocation terms together;
3. verify the pullbacks of all independently extended current and retained-data derivatives, including both terms of (BD.15);
4. retain outer boundary factors and live-history interface currents in the full law, where Proposition BD.1 does not eliminate them.

The first source domain is now explicit, and the analytic and summation implications needed after these checks are proved. Their actual joint hypotheses have not all been verified. C-05 therefore remains open under its unchanged hypotheses.

7. References

- T. Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, Communications in Mathematical Physics 119 (1988), 243-285. pp.254, 259, 261-262, 276-277; equations (2.1), (2.28), (2.34)-(2.42), (3.43).
- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. I. Generation of Effective Actions in a Small Field Approximation and a Coupling Constant Renormalization in Four Dimensions*, Communications in Mathematical Physics 109 (1987), 249-301, equations (3.9), (3.14)-(3.17). DOI: https://doi.org/10.1007/BF01215223.

All new propositions above are proved in this note. Source-specific estimates are imported only in the scopes described in Section 1. No inaccessible announced Yang-Mills proof is used.

Supplement AF.1: Weighted layer geometry and the actual local averaging response

1. The metric is specified in the primary source The weighted distance used in CMP99 and CMP102 is the distance defined in CMP96, p.231, (2.46). The reference to (2.36) in CMP99's discussion after (3.41) is an indexing error: (2.36) in CMP96 is not the metric definition. See CMP96, pp. 224 and 231–234.

Write $h_r = L^r\eta$, let $\Omega_0 \supset \cdots \supset \Omega_K$ be the admitted determining geometry, and put $A_r = \Omega_r \setminus \Omega_{r+1}$, with $\Omega_{K+1} = \varnothing$. In CMP96 (2.2) the separation condition is

$$\operatorname{dist}(\Omega_r^c, \Omega_{r+1}) > RMh_r. \tag{BM.1}$$

The fixed RM here is the separation parameter of the admitted propagator geometry. It is not an old-scale Euclidean tree distance and need not be identified with a running R_r.

For determining-block centers y, y', the actual metric is

$$d_\Omega(y, y') = \inf_\Gamma \sum_{r=0}^{K} h_r^{-1} |\Gamma \cap A_r|, \tag{BM.2}$$

with the source's admissible paths and block-center convention. The conclusions below allow the resulting fixed endpoint offsets. The actual complete construction invokes the propagator/minimizer results only on their admitted determining geometries; (BM.1) is retained as that primary hypothesis. This does not assert that every arbitrary interlaced family satisfies it.

2. Layer separation and the row bound

Lemma BM.1 — weighted age separation If y belongs to layer n and y' belongs to layer k, then

$$d_\Omega(y, y') \geq RM(|k - n| - 1)_+. \tag{BM.3}$$

Proof. Decompose an admissible path at the interfaces $\Sigma_r = \partial\Omega_r$, exactly as in CMP96 (2.47)–(2.48). A passage between successive distinct interfaces Σ_r and Σ_{r+1} lies in layer r. Its length is at least the separation in (BM.1); dividing by h_r gives at least RM, as displayed in source (2.57). A path between layers n and k has at least $(|k-n|-1)_+$ such complete passages. Repeated exits, re-entries and detours add nonnegative length. Taking the infimum proves (BM.3). □

For any fixed kernel rate $\delta > 0$, split an endpoint exponential into two halves and use Lemma 2.1 of CMP96 for the remaining row sum. With RM satisfying that lemma's fixed-parameter requirement,

$$\sup_{y\in A_n} \sum_{y'\in\Gamma_k} e^{-\delta d_\Omega(y,y')} \le C_{\mathrm{row}}(\delta/2) e^{-\frac{1}{2}\delta RM(|k-n|-1)_+}. \tag{BM.4}$$

The sum is over determining-block centers, in the source convention. Formula (2.58) proves the uniform row sum by counting interface passages and spatial lattice steps; (2.59) pays that counting. In the source normalization it suffices to require, for the appropriate fixed $0 < a < 1$,

$$\tfrac{1}{4} a\delta_0 RM > 2d \log c_0(a/2) + \gamma + 1. \tag{BM.5}$$

The proof then retains a prescribed fixed age rate γ. Alternatively, (BM.4) retains the rate $\delta RM/2$ once its row condition is met. There is no intrinsic limit at the rate one displayed in the source's convenient final choice. One can fix $\gamma > 4\log L + \log 2 + 2$ if a subsequent activity sum needs the extra four-dimensional count. For the normalized single-map row alone, $\gamma > \log 2$ is sufficient.

The source permits the fixed geometric parameters to be chosen sufficiently large. The propagator rate can be fixed positively before this enlargement; CMP99's random-walk construction retains a fixed rate below that of its local propagators. The small-coupling ceiling can subsequently be reduced to meet the remaining fixed-parameter requirements. No choice here depends on observable order or total depth.

3. Normalized minimizing-map outputs CMP102 (190) bounds the derivative of the actual relative minimizing map in five output norms. On output layer n, their scale factors are

$$h_n^{-1}, \quad h_n^{-2}, \quad h_n^{-2-\beta}, \quad h_n^{-3}, \quad h_n^{-3}, \tag{BM.6}$$

and on an input layer k the kernel contains the factor h_k^{-d}. The input integration/summation measure is h_k^d; these factors cancel. After multiplying by the output normalization in (BM.6), the remaining kernel is bounded by a fixed multiple of $e^{-\delta d_\Omega}$. Consequently, for input B supported on layer k,

$$\|D\mathcal{H}(B_0)[B]\|_{\mathrm{norm},n} \le C_H \|B\|_\infty e^{-\gamma(|k-n|-1)_+}. \tag{BM.7}$$

Here B_0 is any actual admitted background input in the smaller absolute map domain, and $\|\cdot\|_{\mathrm{norm},n}$ denotes the physically normalized output norms supplied by (190). The corresponding covariant Laplacian/projection combinations used in CMP109 (3.14) are controlled by the stated propagator identities and their kernel bounds; they are not obtained by dilating a projection defined on another determining geometry. The bound is uniform in the admitted remaining history. It does not require $B_0 = 0$.

The normalized row estimate has no factor $L^{d(k-n)}$. The cancellation of input cell volume and the source's weighted row sum are the reasons. If a later sum over activity carriers contributes such a factor, it must be charged separately; (BM.5) allows a fixed larger rate for that purpose.

Preparation parameters The same argument can retain a fixed positive spatial rate after the actual complex preparation. CMP116 (1.7) contains both a walk-localization exponential and an endpoint exponential. The estimate (1.11) uses the former to pay the complex decouplings. Taking M so that the displayed $\delta_1 M$ condition has a strict factor-of-two reserve leaves an endpoint rate $\delta_* > 0$, with the fixed norm loss $e^{16\kappa_1}$. All source output norms are included in (1.7).

For completeness, the nonlinear step can be made in the weighted norm

$$\|A\|_{w,S} = \sup_y e^{\delta_* d_\Omega(y,S)} |A(y)|_{\text{normalized}}, \tag{BM.8}$$

and the corresponding derivative/Hölder norms. The triangle inequality makes each prepared propagator bounded in this norm after reserving part of its endpoint exponential for the row sum. The actual nonlinear terms in CMP116 (1.3)–(1.4) and (1.17) are local analytic operations and the displayed propagator compositions. In a difference of two nonlinear expressions, place the weight on the one differing factor and bound the other factors in the ordinary small norm. The local operations have bounded weighted diameter on their proper-scale blocks, giving only a fixed factor $e^{\delta_* D_{\text{loc}}}$. Their weighted Lipschitz constant is therefore bounded by

$$C_{\text{nl}} e^{\delta_* D_{\text{loc}}} K_w^r \epsilon, \tag{BM.9}$$

for a fixed finite r, where K_w is the prepared weighted propagator norm and ϵ is the source's ordinary map-input ceiling. The analytic nonlinear remainders entering the contraction start at degree at least two. The explicitly linear source terms, including the $G\Delta^{(2)}H_0B$ forcing in CMP102 (180), are retained separately and obey the same weighted propagator estimate with their source coefficient bounds; they are not falsely declared quadratic. Reduce that fixed ceiling until (BM.9) is at most 1/2. The actual contraction proof then holds in the weighted norm, and its resolvent costs at most two. This proves the analogue of (BM.7) for the actual prepared nonlinear map, with a fixed changed constant and rate. It does not presume that a bound in an unweighted norm automatically implies spatial decay.

The local constraint map $g_k Cz - hD(g_k Cz)$ enlarges the support of a current variation by only its source proper-scale stencil. If that stencil stays in the same deepest determining layer, (BM.7) applies directly. A fixed adjacent-layer enlargement changes $(k-n-1)_+$ to $(k-n-c_0)_+$, with a fixed integer c_0. The actual current collar admits such an enlargement; this support check cannot be replaced by merely naming a current carrier.

For a current variation on its actual deep support, the resulting bound is

$$\|\delta\mathcal{H}_k\|_{\text{norm},n} \le C_{\text{prep}} g_k \|\delta z\|_\infty e^{-\gamma(k-n-c_0)_+}. \tag{BM.10}$$

Integrating the derivative along an admitted current segment gives the same bound for the finite difference from a reference current field. In particular, it is appropriate for the exact boundary fluctuation difference of BD.1; no source-free boundary constant is removed from the outer marginal.

4. The basic current and the dyadic reserve Let u_n be the right side of (BM.10). The exact local-current estimate LC.17 gives, in physically normalized layer units,

$$\|\delta J\|_{\text{norm},n} \le C\big[u_n + \alpha_{0,n} u_n + u_n^2 + u_n^3\big]. \tag{BM.11}$$

The terms ΔA and PA use the normalized second-order output bounds. For example $h_n^3 b_n |A| \le C\alpha_{0,n} u_n$, and $h_n^3 |A||\nabla A| \le u_n^2$. Thus no power of the fine spacing is hidden in (BM.11). The plaquette and relative-first-difference clauses have the same linear bound with higher powers of u_n absorbed at a fixed small ceiling. This covers the basic field/current clauses on the original geometry, including the independent-current extension used in the local source formula.

For an old boundary activity of birth scale $j \le k$, the coefficient gap between its domain and the current $k+1$ domain in CMP119 (2.34)–(2.39) is

$$\beta\big(2^{-(j-n)} - 2^{-(k+1-n)}\big)\alpha_{i,n} \ge \frac{\beta}{2}\, 2^{-(j-n)} \alpha_{i,n}, \qquad n \le j. \tag{BM.12}$$

Using $\alpha_{0,k}/\alpha_{0,n} \le C_{\text{AF}}\sqrt{1+k-n}$, the ratio of (BM.10) at $\|\delta z\| \le CP_k$ to the right side of (BM.12) is at most

$$C\ell_k^{p_0-q}\, 2^{j-n}\sqrt{1+k-n}\, e^{-\gamma(k-n-c_0)_+}. \tag{BM.13}$$

For fixed $\gamma > \log 2$, its age supremum is finite. One fixed small-coupling ceiling therefore gives a strict common current contour for these original-geometry output clauses, before any source differentiation. This is the valid use

of exponential age suppression. An order-m derivative is not first charged $2^{m(k-n)}$ and then incorrectly paid by one fixed exponential rate.

5. A separate current collar, when its anchor is present The current construction in CMP119 (3.2)–(3.5) separates the current forcing support from Z_k by a physical collar of width at least $c_{\mathrm{col}} M R_{k+1} h_k$, after the fixed stencil enlargement. Let an old boundary carrier X of scale $j \le k$ meet $Z_j \subset Z_k$, and let $x \in X$ be the point at which its argument responds to the current forcing S. Because every weight in (BM.2) is at least h_k^{-1}, while source $d_j(X)$ is the old tree length divided by the block side M, and $h_j \le h_k$,

$$d_\Omega(x, S) + C_{\mathrm{tree}} M d_j(X) \ge c_{\mathrm{col}} M R_{k+1} - C_{\mathrm{end}} M. \tag{BM.14}$$

Proof. Choose a boundary anchor $y \in X \cap Z_j$. The first distance bounds $\mathrm{dist}_{\mathrm{phys}}(x, S)/h_k$, up to the fixed endpoint convention. The carrier tree bounds $\mathrm{dist}_{\mathrm{phys}}(x, y)/h_k$ by $C_{\mathrm{tree}} M L^{j-k}(d_j(X) + 1)$, hence by $C_{\mathrm{tree}} M(d_j(X) + 1)$. This factor M follows from the tree-distance normalization in CMP109, immediately before (0.24). The physical triangle inequality bounds their sum below by the stated separation of S and Z_k, divided by h_k. □

Splitting the endpoint and carrier exponentials into fixed fractions therefore retains an additional factor $e^{-c_{\mathrm{eff}} R_{k+1}}$, with c_{eff} proportional to $\min\{\delta M, \varkappa\}$ up to fixed geometric constants, while leaving a positive age rate and a positive carrier decay rate. Equivalently one may write $e^{-cMR_{k+1}}$ only with c allowed to depend on M^{-1}. Fixed RM can be enlarged after this splitting so that the remaining age rate still satisfies the desired inequality in (BM.5). This does not substitute an ordinary old-scale tree distance for d_Ω, and it does not claim a collar factor for an arbitrary output point with no boundary anchor.

6. The exact part not proved here The full B domain also contains the reconstructed pair

$$U_{p,X}(M^\bullet(U)) = U\big(B_p(X) \cup \{\Gamma_i\}_{i<p}, M^\bullet(U)\big), \qquad J_{p,X}(M^\bullet(U)). \tag{BM.15}$$

The current restriction in CMP119 (2.37) has the exact weight

$$\alpha_{0,n} L^{n-\min\{p,n\}} (L^n L^{-p})^{-3}. \tag{BM.16}$$

For $n > p$, this is proportional to $\alpha_{0,n} L^{-2(n-p)}$. A generic relative-potential input estimate gives only one improved scale power and is not enough. The reconstruction must preserve the smoother layer's curvature/difference input, giving the two-power bound, on the changed determining geometry and with the independent-current extension retained.

Two further checks are therefore not consequences of (BM.3) alone:

1. the compatible determining-set union in (BM.15) must preserve or dominate the original weighted geometry in the kernel compositions actually used, or its geometry must be estimated directly;
2. the reconstructed pair must obey (BM.16), and the corresponding curvature bound (2.35), with the same layer-wise exponential response and preparation reserve.

The source's localized determining set in (2.14) and the reconstruction union in (2.35) are not silently identified. A purely formal mesh-cap argument which discards overlapping fine constraints is not used. The normalized row theorem proves the original minimizing-map response and resolves the previous metric ambiguity; it does not by itself complete this reconstructed tuple or the separate relocation/value-majorant estimates of C-05.

7. Local two-power averaging gain The local nonlinear averaging part of the $n > p$ difficulty can be settled without assuming a property of the changed minimizing geometry.

Lemma BM.2 — an actual local averaging response Let $h_p \le h_n$, and let a cube of physical side $C_b h_p$ contain all bonds used in one fixed family of the actual p-scale averages. Write η for the underlying fine spacing. Suppose on this cube that U_0 is a compact-group field multiplied by a complex relative field of size at most Ca/h_n, its plaquette curvature is bounded by

$$\|dU_0 - I\| \le Ca\eta^2 h_n^{-2}, \tag{BM.17}$$

and $U_1 = e^{i\eta A} U_0$ with

$$\|A\| \le u h_n^{-1}, \qquad \|\nabla_{U_0} A\| \le u h_n^{-2}, \qquad 0 < u \le c_a a. \tag{BM.18}$$

Here a is below one fixed small ceiling and $c_a > 0$ is a sufficiently small fixed number. Let g_t be the ordered axial gauge on this cube, anchored at a fixed point, for $U_t = e^{i\eta t A} U_0$. In the resulting gauged determining coordinates, the actual nonlinear average has logarithm $B_p^g(t)$, and

$$\|B_p^g(1) - B_p^g(0)\| \le C_{\mathrm{av}} u (h_p/h_n)^2. \tag{BM.19}$$

All constants depend on the fixed block geometry and group but not on the number of fine bonds, p, n, or total depth.

Proof. Take $R = c\,a/u$, choosing c_a so that $R \ge 4$. On the complex disc $|t| < R$, the elementary plaquette expansion gives

$$\|dU_t - I\| \le Ca\eta^2 h_n^{-2}.$$

Indeed its terms are the original curvature, the covariant curl of tA, and products bounded by $|tA|^2$, with higher analytic terms paid by the fixed small ceiling. The link and inverse-link norms are bounded by $\exp(C\eta a/h_n)$; a compact-group part costs no norm.

The axial paths have at most Ch_p/η links. The exact path-interchange proof of (R14.23) therefore yields

$$\|U_t^{g_t} - I\| \le Ca\eta h_p h_n^{-2} \exp(Cah_p/h_n).$$

The exponential is bounded by a fixed constant. The principal logarithm is consequently defined, and $U_t^{g_t} = e^{i\eta A_t^g}$ with

$$\|A_t^g\| \le Cah_p h_n^{-2}. \tag{BM.20}$$

These gauges and logarithms are holomorphic throughout the disc.

Use the actual relative-average bound proved from CMP99, p.406 after (3.78), and the fixed final averaging step, as recorded in (E14.39)–(E14.41). It gives

$$\|B_p^g(t)\| \le Ch_p \|A_t^g\| \le Ca(h_p/h_n)^2 \tag{BM.21}$$

on $|t| < R$. This is the unmodified nonlinear average, not an identification with its linearization or with a different Q_p'. Cauchy on a disc of radius at least $R/2$ around each $t \in [0,1]$, followed by integration over that segment, gives

$$\|B_p^g(1) - B_p^g(0)\| \le CR^{-1} a (h_p/h_n)^2 \le Cu(h_p/h_n)^2.$$

This proves (BM.19). If $u = 0$, the conclusion is immediate. □

The corresponding value gain is already visible in CMP109, p.263, immediately after (1.16): the actual averaged plaquette bound contains the square of the averaging length. Lemma BM.2 makes its response version explicit and retains complex parallel-transport costs. An axial gauge anchored at an output point equals the identity there, so it does not change a current evaluated at that point; it still changes the surrounding determining data and must be retained in any subsequent reconstruction.

In a basic n-regular layer chart one may take $a = C \max\{\alpha_{0,n}, \alpha_{1,n}\}$ and u from (BM.10), provided the required averaging cube and its fixed margin are in that chart. The reserve (BM.13) makes u/a small. Equation (BM.19) then supplies precisely the two scale powers needed for the local determining input when $n > p$.

It remains necessary to control the full p-reconstructed output, whose determining data outside this local cube also contribute. That requires the actual joined determining geometry in (BM.15), its normalized kernel composition, and the independently extended current. Lemma BM.2 does not replace those checks by an unproved minimizer-consistency assertion under refinement or under a change of underlying lattice.

8. The single remaining reconstruction-transfer statement Here is a quantitative formulation of the surviving reconstruction issue, rather than a further generic sufficient contour theorem. Denote the actual prepared current connection in CMP109 (3.6)–(3.13), with the full multiscale construction of CMP119 pp.275–277, by $\mathcal{V}_{k,s}(z;U,J)$. Its comparison value at $z=0$ is the prescribed unperturbed connection. Keep the independently varied external J and the complex preparations s in that actual formula. Form the prescribed reconstructed pair

$$\mathcal{T}_{p,X;k,s}(z;U,J)=\big(U_{p,X}(M^{\bullet}(\mathcal{V}_{k,s}(z;U,J))),J_{p,X}(M^{\bullet}(\mathcal{V}_{k,s}(z;U,J)))\big) \tag{BM.22}$$

using exactly the determining data and gauge conventions in CMP119 (2.35)–(2.37). Define its two normalization weights by

$$w_F(n,p)=L^{-2p}(L^nL^{-p})^{-2},\qquad w_J(n,p)=L^{n-\min\{p,n\}}(L^nL^{-p})^{-3}. \tag{BM.23}$$

Remaining statement. On the actual current contour $\|z\|_\infty < 2P_k$, $|s(\Delta)| < e^{\varkappa_1}$, and the prescribed joint background/independent-current domain, prove that every pair (BM.22) is holomorphic on a common fixed-fraction reserve and that its finite response obeys

$$\sup_{x\in X\cap A_n}\max\left\{\frac{\|\delta[dU_{p,X}](x)\|}{w_F(n,p)},\frac{\|\delta J_{p,X}(x)\|}{w_J(n,p)}\right\}\le C_{\mathrm{rec}}g_k\|z\|_\infty e^{-\Gamma(k-n-c_0)_+},\quad p\le j\le k,\quad n\le j. \tag{BM.24}$$

The deltas compare z with zero using the source's fixed comparison gauges; $C_{\mathrm{rec}},c_0,\Gamma$ must be independent of p,j,k,X, volume and remaining admitted history. A fixed $\Gamma > \log 2$ suffices for the domain gap; a larger fixed reserve such as the one in (BM.5) also pays the subsequent carrier sums. The argument must first define the maps on their absolute domains and then prove this smaller-domain inclusion, so it cannot assume the desired B contour to apply Cauchy.

The weighted-age proof and Lemma BM.2 supply, respectively, the original-map response and the local smoother-input gain needed by (BM.24). They do not yet prove that estimate for the full joined reconstruction.

The exact source-geometry issue is as follows. CMP119 (2.14) defines a localized joining operation by retaining the original determining set away from a proper-scale boundary collar and using the localized determining set in that collar. LFR II p.373 explicitly confirms this same-lattice localization property, including the at-most $2M_1$ boundary width. Its notation uses the local determining set first, so argument order is not a reliable way to read the operation. A possible proof of (BM.24) would identify (BM.15) with the actual localization of the original determining geometry truncated at level p, with the same averaged boundary data. That would permit a pointwise mesh comparison and normalized kernel composition.

This identification has not been verified here. In particular, the boundary data inserted in (BM.15) are $M^{\bullet}(U)$; localizing a separately constructed global p-minimizer would insert its own $M^{\bullet}(U_p)$. These data are not identified by a general minimality principle. The change of the underlying reconstructed lattice and the fine boundary collars also have to respect the current weight $w_J(n,p)$. An ordinary union of overlapping coarse and fine constraints, a formal intersection of layer sets, or a same-lattice identity from LFR II does not by itself prove the required statement.

Thus C-05 remains open at the precise full reconstruction-transfer estimate (BM.24), followed by its application to all of the actual B contributions. The preceding metric ambiguity and the local two-power averaging loss have been resolved; they are not listed again as missing inputs.

9. Source references

- Balaban, CMP96 (1984), 223–250, (2.2), (2.46)–(2.59), Lemma 2.1. pp. 224 and 231–234.
- Balaban, CMP99 (1985), 389–434, (3.41), Theorems 3.1 and 3.10 and the random-walk kernel bounds.
- Balaban, CMP102 (1985), 277–309, Proposition 9 and (190).
- Balaban, CMP116 (1988), 1–22, (1.3)–(1.7), (1.11)–(1.21). article.
- Balaban, CMP119 (1988), 243–285, (2.34)–(2.39), (3.2)–(3.5), (3.20), pp.276–277.

The direct metric, normalized-row, weighted-preparation, anchored-collar and local averaging-response deductions are proved above with their exact scopes. The full joined reconstruction-transfer estimate (BM.24) remains explicit.

Supplement AF.2: Complete boundary terms and value sums

1. Exact scope This argument concerns the current-dependent boundary perturbation in the actual conditional integral of CMP119 (3.25), with all older fluctuation coordinates and history geometry retained as parameters. The present conditional Gaussian quadratic form, its cross force and its Schur-complement scalar remain in the exact Gaussian split of C-01. A boundary perturbation value at the reference current field is separated only by the exact conditional cancellation of Proposition BD.1. That scalar remains in the outer marginal.

The actual common multiscale B contour and the required complete map-output kernel bounds are being proved separately. The value-summation deduction below states their precise uses. Merely knowing that a contour exists does not sum an unweighted collection of old B bounds.

The primary primitive boundary bound is the full-tree bound

$$|B^{(j)}(X, U, J, \mathrm{A}, \{S\})| \leq B_0 e^{-\varkappa d_j(X)}. \tag{BV.1}$$

This is CMP119 (2.40)–(2.42), pp.261–262. The source explicitly chooses it instead of the optional component-resummed bound with $d_j(X \setminus Z_j)$. The latter would not justify the connector argument below.

2. The actual sectors are exhaustive CMP119, pp. 267–277, separates the following contributions.

Sector in the conditional action	Exact source location and operation
Classical nonlinear fluctuation, constraint substitution, Haar and coarea terms	The full formula (3.15), pp.267–268, uses the actual constrained field $g_k CA - hD(g_k CA)$. Page 276 applies the actual localized small-field construction, including $P^{(k)}$, with its multiscale operators. The relevant classical quadratic form remains in the source Gaussian precision.
Inherited E and R contributions assigned to B	Page 276 repeats CMP109 Sections 3–4 on a layer of scale n, replacing the old unit parameter by L^{-n}, and then localizes with the full multiscale random-walk expansions. Terms whose localization crosses the current regular boundary are assigned to B. R is treated in the same way except that its coupling subtraction is absent.
Inherited B fluctuation differences	The end of p.276 applies CMP109 (3.6)–(3.7), then localizes. These carriers intersect the current regular integration region. Page 277 states the primitive boundary decay bound with an additional $O(\varepsilon_k)$, and explains that minimizing-field decay pays the shrinking dyadic domains.
Heterogeneous Wilson coefficient	The last term in the braces of (3.15)/(3.27) is localized by the multiscale partition of unity in the plaquette sum. Page 277 puts it in B because the coefficient difference is supported in Z_k, and explicitly suppresses its large coefficient using decay of the minimizing map. Its coefficient is the localized recursion (2.24), not simply one homogeneous g_j^{-2} throughout a transition collar.
Relocation and its Jacobian	Pages 251–252, (1.21)–(1.24), give the first-step operation. Pages 269–270 repeat it with $H^{(k)}$, and define $V^{(k)}(S_{k+1}, A_k, H^{(k)})$. Every previous action term contributes, along with the exponentiated Jacobian. Page 277 retains these terms and their additional localization in S_{k+1}.

Sector in the conditional action	Exact source location and operation
Gaussian cross-component terms already placed in B	Page 258 after (2.21) keeps only within-component quadratic terms in the operation $T^{(j)}$; the other quadratic terms are included in B. They are therefore among the inherited B terms in (BV.1), not omitted Gaussian terms.

The last row concerns older terms already assigned to B by the source. It is distinct from splitting the present Gaussian form between the currently integrated field z and retained field v. The latter has a cross term $z^{\mathsf{T}} H_{IO} v$. Its current Hessian is zero, but its force $H_{IO} v$ need not be small. It remains in the exact conditional Gaussian mean and in C-08; it is not included in the small perturbation prefactor below. A calculation involving only the nonlinear Wilson remainder and a determinant would not cover the complete action.

The relocation cutoff in (1.22), denoted $g(A')$ in the source, is C^∞, not claimed analytic in the older real coordinates A'. These coordinates remain parameters in the native B theorem. Dependence on the small relocation field H, and on the native complex U, J arguments, is analytic. Arbitrary derivatives in old coordinates, changing selectors or history interfaces are not supplied by a native current contour.

3. The localized heterogeneous coefficient has a safe age bound Write $w_j = g_j^{-2} > 0$, and let $w_j(x)$ be the coefficient in CMP119 (2.24). The exact homogeneous and localized recursions give

$$\beta_l = w_{l-1} - w_l, \qquad w_k(x) - w_k = \sum_{l=1}^{k} \beta_l (1 - \phi_l(x)). \tag{BV.2}$$

Let n be such that $\phi_l(x) = 1$ for $l \le n$. This may be the earliest active cutoff index at the point; identifying it with a geometric layer requires the source partition and cutoff support, not a change of notation. Assume the fixed cutoff bound $|1 - \phi_l| \le C_\phi$.

The checked source AF comparison yields

$$w_l \le C'_{\mathrm{AF}} (1 + k - l) w_k. \tag{BV.3}$$

It follows, without assuming the sign of β_l, that

$$\begin{aligned} |w_k(x) - w_k| &\le C_\phi \sum_{l=n+1}^{k} (w_{l-1} + w_l) \\ &\le C_\phi C''_{\mathrm{AF}} (1 + k - n)^2 g_k^{-2}. \end{aligned} \tag{BV.4}$$

Thus transition collars cost at most a quadratic age factor and the explicit g_k^{-2}. The constant-stratum estimate can be sharper, but is not needed for a summation with an exponential age reserve. The small physical current collar must pay the running g_k^{-2}; a fixed number $e^{-cMR_{\min}}$ cannot do so.

The geometric cutoff radius is $R_k \ge \ell_k^r$, $\ell_k = \log g_k^{-2}$, by CMP119 (2.5), p.255. For a fixed admitted choice $r > 1$ and every fixed s, N,

$$g_k^{-s} \ell_k^N e^{-cR_k} \longrightarrow 0. \tag{BV.5}$$

This follows directly by taking logarithms: $s\ell_k/2 + N \log \ell_k - c\ell_k^r \to -\infty$. The source's description of $H^{(k)} = O(\varepsilon_k e^{-R_k})$ on p.269 as smaller than every positive coupling power uses exactly this type of reserve.

4. One connector covers both possible locations of an old B carrier Let i be a current input anchor, and let y be an attachment point of a localized minimizing-map variation to a primitive carrier X, with $X \cap Z_j \neq \varnothing$. Let d_Ω be the actual weighted distance of CMP96 (2.46), not the Euclidean distance in old units. Its weight on a layer of physical mesh h_n is h_n^{-1}.

There are two cases.

1. If $y \notin \Lambda_j$, every path from the current region to y already exits Λ_j. Thus the minimizing-map path supplies the needed connection by itself.
2. If $y \in \Lambda_j$, choose a path in a full tree for X, from y toward $X \cap Z_j$, and stop at its first exit from Λ_j. Before this exit it lies in $\Lambda_j \subset \Omega_j$. Every distance weight there is at most h_j^{-1}. Convert the segment to an admissible block-center path by the fixed source block rounding: replacing each visited block segment by its prescribed center connections costs at most a fixed multiple of its weighted length plus a fixed cost per visited block. All visited blocks have scale at least h_j, so their number is bounded by a fixed multiple of the old j-grid length plus an endpoint constant. The resulting weighted length is at most $C_M(d_j(X) + 1)$, with $C_M = CM$ in the source's M-cell convention. There is no factor growing with old age.

In both cases,

$$d_\Omega(i, \partial_{\text{in}}\Lambda_j) \leq d_\Omega(i, y) + C_M d_j(X) + C_{\text{edge}}. \tag{BV.6}$$

The fixed edge term accommodates the first-exit convention. It has no age or volume dependence. Outputs outside the entire active domain have no current variation. Attachments in an earlier active layer $n < j$ need not be incorrectly declared zero: the minimizing path to them carries the stronger age $k - n$.

Suppose an actual localized kernel/tree contribution has the available decay

$$e^{-c_H d_\Omega(i,y) - \kappa d_j(X)}. \tag{BV.7}$$

Split each exponent into four fixed shares. With

$$\mu = \frac{1}{4}\min\{c_H, \kappa/C_M\} > 0, \tag{BV.8}$$

the first share gives the connector bound $Ce^{-\mu d_\Omega(i,\partial_{\text{in}}\Lambda_j)}$, used to pay the current collar. The second share pays the age: for the innermost restriction $n = j$, use the same connector and the layer-crossing lower bound; for an earlier-layer restriction $n < j$, use the endpoint y in that layer and the stronger $k - n$ lower bound on the H path itself. The third shares are reserved for the normalized kernel/carrier row sums, and the fourth shares for final current-carrier localization. In particular the age $k - n$ is not inferred from a path that merely reaches $\partial\Lambda_j$. No exponent is spent twice.

The separate geometry/kernel proof must supply these actual lower bounds in the weighted metric, with its fixed adjacent-step conventions. The distinct shares just described give the simultaneous factor

$$e^{-c_{\text{col}} M R_k} e^{-\Gamma(k-j)} \tag{BV.9}$$

at reduced fixed constants. For earlier-layer restrictions use $k-n$, by the separate H-endpoint argument. Equivalently one may first use that the maximum of collar and age lower bounds is at least half their sum, with correspondingly smaller constants. The effective rate must be chosen after the reduction (BV.8); increasing the fixed admissible $R_{\min}$ can pay the entropy and scaling losses in CMP96 (2.58)–(2.59). An ordinary old-scale tree length is not substituted into d_Ω.

5. The resulting scalar age sum This section gives a quantitative implication once the complete map/domain proof supplies (BV.7)–(BV.9), all relevant normalized kernel row sums, and the source localization on the common contour. It exposes the costs required of that proof.

For an inherited B activity of index j and one of its output-layer restrictions $n \le j$, the dyadic source radius costs $2^{j-n}/\alpha_{*,n}$. The checked running comparison gives

$$\frac{g_k P_k}{\alpha_{*,n}} \le C\sqrt{1+k-n}\,\ell_k^{p_0-q}. \tag{BV.10}$$

Retaining the conservative old-cell factor $L^{4(k-n)}$, the sum over $n \le j \le k$ is bounded by

$$\begin{aligned} &CB_0\ell_k^{p_0-q}e^{-c_{\mathrm{col}}MR_k} \\ &\quad\times\sum_{a\ge 0}(1+a)^{3/2}\exp[-(\Gamma-4\log L-\log 2)a]. \end{aligned} \tag{BV.11}$$

The factor $1+a$ includes the number of admissible j at fixed $a = k-n$. If the normalized source row/column estimate removes an old-cell factor, the bound improves; (BV.11) does not require that improvement. The integration measures and scale-normalized outputs must be those of the actual kernel theorem. An unnormalized sum over every fine bond is not used as that row sum.

Choose

$$\Gamma > 4\log L + \log 2 + \delta, \qquad \delta > 0. \tag{BV.12}$$

Then (BV.11) is uniform in cutoff depth. Any additional *fixed* polynomial age factors from the heterogeneous coefficient (BV.4), finite kernel moments or fixed-order classical expansion can be included by replacing 3/2 by one fixed exponent b. The numerical sum

$$S_{b,\delta} = \sum_{a\ge 0}(1+a)^b e^{-\delta a} < \infty \tag{BV.13}$$

remains uniform.

For E- and R-origin boundary terms, p.276 first performs the same regular calculation on the appropriate layer n. The inner $j \le n$ sum is therefore controlled by the already established E and R majorants on their admitted layer domains. Carriers crossing the newest boundary of Λ_{k+1} may touch the current support. No old-Z_k collar factor is available or claimed for them: the already finite regular E/R value majorant is used directly, on the common B background contour supplied by the admission proof. For a genuinely older layer $n < k$, the remaining remote current-to-layer dependence uses the connector just described. These activities remain B at the final scale under the actual source classification.

The heterogeneous term retains the extra coefficient (BV.4); its g_k^{-2} is paid only with the actual current-collar factor (BV.5). The relocation terms retain the analytic $H^{(k)}$ and their source bound $O(\varepsilon_k e^{-R_k})$, including all previous-action and Jacobian contributions. Their stated arbitrarily high positive power of g_k remains uniform only on the native contour and with the older cutoff arguments fixed.

Finally reserve enough of the source's *full current-carrier* decay to perform the anchored M-cell sum. The carrier count and spatial weight must use the source's face-connected M-cube convention. If

$$\sup_{\mathcal{C}_k}|\widetilde{B}_{k,Y}| \le K_{B,k}e^{-\kappa_B d_k(Y)} \tag{BV.14}$$

is the fully localized majorant obtained above, then for a sufficiently small fixed spatial $\gamma > 0$, and with the required entropy reserve,

$$\sup_i \sum_{Y:\, i\in I_Y} e^{\gamma\,\mathrm{diam}(I_Y)} \sup_{\mathcal{C}_k}|\widetilde{B}_{k,Y}| \le C_{\mathrm{car}}K_{B,k}. \tag{BV.15}$$

The fixed localization prefactors, kernel moments, Lie-algebra dimension and powers of M belong in C_{car}. They may not contain total volume, hidden distance to the current region, or an unsummed old-scale index.

On a common current-coordinate polydisc of radius $c_G P_k$, BD.2 then gives the complete conditional boundary Hessian row bound

$$\sup_i \sum_l e^{\gamma d(i,l)}\|D_iD_l\widetilde{B}_k\|_{\mathrm{op}} \le C_G C_{\mathrm{car}}K_{B,k}P_k^{-2}. \tag{BV.16}$$

Higher current derivatives are taken after the normally convergent age/carrier sum. Native external derivative radii must be those proved by the complete B-domain admission theorem; this deduction does not assign them unverified regular-E radii.

For a concrete sufficient prefactor, put $\delta_k = g_k P_k$, $\theta_k = \ell_k^{p_0-q} \le 1$, and retain the standard cubic classical bound and bounded Haar/coarea remainder from CMP116 Lemmas 1–2, transferred exactly as specified on CMP119 p.276. Once the complete map and connector estimates just stated hold, a conservative sector majorant is

$$\begin{aligned} K_{B,k} \le C_{\mathrm{src}}(1+S_{b,\delta})\big[&\mathfrak{K}_E + C_R g_k^{\kappa_0-11} + B_0\theta_k \\ &+ g_k^{-2}\delta_k^3 + \delta_k + g_k^{-2}\delta_k e^{-c_{\mathrm{col}}MR_k} + g_k^{\nu}\big], \end{aligned} \tag{BV.17}$$

where $\nu > 0$ is any one fixed relocation power allowed by the source, and b contains the fixed age-polynomial costs, including (BV.4). The first two terms use the complete layerwise E/R majorants and claim no extra collar factor for newly boundary-crossing carriers. For their remote older-layer part, the additional current-fluctuation ratio is at most one after the chosen coupling ceiling. The third is (BV.11), with its helpful collar suppressed from the bound. The fourth and fifth are the current classical cubic and measure remainders. The sixth retains the heterogeneous coefficient until its actual current collar pays it. The last is the source relocation/Jacobian bound, with its full fixed-parameter constant. The present exact Gaussian cross force is not part of $K_{B,k}$. All fixed preparation and localization constants belong in C_{src}.

This displayed prefactor is an application consequence of the specified actual sector estimates. It is not obtained from a bound on one representative Wilson term alone. In particular the heterogeneous and relocation rows of the source formulas remain necessary to use it.

Equations (BV.5) and (BV.17) yield

$$K_{B,k}P_k^{-2} \longrightarrow 0 \quad \text{as } g_k \downarrow 0, \tag{BV.18}$$

uniformly in depth and volume after the fixed source choices. For example the classical contribution is $g_k^{-2}\delta_k^3 P_k^{-2} = g_k P_k \to 0$; the unenhanced E term is $\mathfrak{K}_E P_k^{-2} \to 0$. Thus a source $O(\varepsilon_k)$ improvement for every individual B row is stronger than needed for the small conditional Hessian conclusion. The current collar is nevertheless indispensable for the explicit running g_k^{-2} coefficient in the heterogeneous row.

6. Exact remaining application checks There is no extra algebraic sector missing from the ledger in Section 2. The actual open checks in applying (BV.11)–(BV.16) are quantitative:

- all required output restrictions of the actual B pullback must obey the common-contour layerwise kernel estimate, including the reconstructed current, independent current and relocation;
- the current collar and age rate must be retained after the exponent split (BV.8), with the normalized row sums and fixed entropy costs, not just in one pointwise kernel entry;
- the heterogeneous coefficient must use its actual localized recursion and its earliest active cutoff layer, including transition collars;
- the actual localization must preserve enough full-tree/current-carrier decay after the anchor connector is paid;
- old real cutoff arguments and outer retained-only factors must not be mistaken for analytic native variables already controlled by this theorem.

Once those checks are supplied, the displayed sums are finite with no remaining unexplained old-age or volume factor. This note does not itself declare C-05 complete before they are verified.

7. The exact reconstructed-data issue still preventing C-05 completion The published construction distinguishes two operations which must not be equated without checking their data.

CMP119 (2.14), p.257, defines a localized determining set by

$$\mathrm{B} \cup \mathrm{B}_j(\Omega) = (\mathrm{B} \cap \Omega^{-2}) \cup \big(\mathrm{B}_j(\Omega) \cap (\Omega \setminus \Omega^{-2})\big). \tag{BV.19}$$

The original determining set is preserved in the interior. The specifically constructed local determining set supplies the boundary collar. The notation denotes this prescribed construction, not an unqualified union of arbitrary overlapping fine and coarse constraints.

LFR II, p.373, (1.161)/(1.61), explicitly uses the identity

$$U_k^0 = U\big(\mathrm{B}_k''(Y_1), M^\bullet(U_k^0)\big), \qquad \mathrm{B}_k''(Y_1) = \mathrm{B}(Y_1) \cup \mathrm{B}_k''. \tag{BV.20}$$

Its neighboring (1.62) uses the corresponding identity for U_1. The fields and gauge transformations in these formulas are retained explicitly. This is localization of an already constructed minimizing field on the same underlying lattice with the specified determining data.

The B-domain reconstruction in CMP119 p.261 is instead defined by

$$U_{p,X}(M^\bullet(U)) = U\big(\mathrm{B}_p(X) \cup \{\Gamma_i\}_{i<p}, M^\bullet(U)\big), \tag{BV.21}$$

with the additional curvature and current bounds (2.35), (2.37) in their p, n scale convention. In particular the $n > p$ current bound requires the two-power improvement $\alpha_{0,n} L^{-2(n-p)}$.

Equation (BV.20) does not identify (BV.21) with a simple average of the original finer minimizing field. Nor does localizing a separately constructed global p-minimizer automatically identify its new boundary data $M^\bullet(U_p)$ with the required $M^\bullet(U)$. Such a claim would change either the determining data or the underlying minimization problem.

The newly proved weighted-age row bound on the original admitted determining geometry, and the local two-power averaging lemma in Supplement AF.1, therefore do not alone settle the full joined p-reconstruction. One still needs a direct theorem for its actual compatible determining geometry and data, or an exact source identity whose hypotheses and data match (BV.21), followed by the normalized kernel composition and independent-current output estimate. This is the concrete remaining C-05 domain step. No counterexample to the source construction is asserted.

Primary record

- CMP119 (1988), 243–285: (1.21)–(1.24), (2.21)–(2.24), (2.40)–(2.42), (3.15), (3.23)–(3.29), and pp.269–270, 275–277.
- CMP96 (1984), 223–250: weighted distance (2.46), transition estimate (2.57), entropy estimate (2.58)–(2.59), and Lemma 2.1. The actual kernel admission proof must retain its source norm and integration conventions.
- CMP109 (1987), 249–301, and CMP116 (1988), 1–22: the localization and layer-transfer uses stated explicitly in CMP119 pp.276–277. The fixed source localization constants remain scoped inputs, as in the completed regular E/R contour proofs.

Supplement AF.3: Final boundary application conditions

Theorem E14.3 removes the regular E contour obstruction. The subsequent Supplement AF.1 proof resolves the weighted-age geometry, normalized prepared response on the original determining geometry, and the local two-power averaging response. The connector and scalar summations in Supplement AF.2 also pass their stated application checks. These results do not yet discharge the complete boundary statement C-05. The surviving step is the actual joined p-reconstruction and its global outputs on the full source tuple, followed by verification that the complete localized boundary action meets the displayed carrier/application bounds.

1. What the new result supplies E14.39–E14.44 use the actual nonlinear relative average, a fixed absolute domain for the internal maps, the zero insertion, and the full source derivative output norms. They admit the entire relative insertion segment before testing the smaller running primitive E radius. E14.49–E14.53 then provide a simultaneous native current/background/independent-current/preparation contour and a summable regular E prefactor. The near- and far-age summations can therefore be differentiated on that contour without an old-age-dependent Cauchy radius.

The theorem retains the actual source E/B assignment. It expressly does not assert the corresponding multiscale B-domain inclusion. Native background directions in it also do not stand for arbitrary raw retained or selector variations.

2. The actual boundary margins have an additional layer index CMP119 gives the formulas on pp.254–256, 261–262 and 276–277; the construction of the current collar is on pp.264–269. In CMP119 (2.34)–(2.39), the restrictions on layer

$$\Omega_n \setminus \Omega_{n+1}, \qquad n < j,$$

and on the innermost region Ω_j, include the coefficient

$$\vartheta_{j,n} = 1 - \beta\big(1 - 2^{-(j-n)}\big). \tag{BA.1}$$

It multiplies the relevant running radius and the displayed physical scale weights. The restrictions include the original plaquette and independently extended current, the reconstructed-field and reconstructed-current quantities, and the complex relative field and its first derivative. These are not merely the regular E restrictions at index j.

In particular, the published text of p.261 gives the reconstructed curvature and current weights, respectively, as

$$L^{-2p}(L^n L^{-p})^{-2}, \qquad L^{n-\min\{p,n\}}(L^n L^{-p})^{-3}. \tag{BA.1a}$$

The additional reconstruction index p belongs to the tuple and cannot be dropped. The second weight equals $L^{3(p-n)}$ for $n \le p$, and $L^{-2(n-p)}$ for $n > p$. The latter improvement involves the curvature/difference regularity of the actual coarse determining field; a bound on its undifferentiated relative-potential supremum is not that estimate.

When a center meets the corresponding later $(k+1)$-domain restriction, the algebraic reserve in an old j-domain, with $n \le j \le k$, is

$$\vartheta_{j,n} - \vartheta_{k+1,n} = \beta\left(2^{-(j-n)} - 2^{-(k+1-n)}\right) \ge (\beta/2)\, 2^{-(j-n)}. \tag{BA.2}$$

This is a reserve statement for those exact source restrictions, not a proof that an arbitrary center or pullback meets them. It quantifies the additional application required by the source explanation on p.277. For each restriction, the reserve must be multiplied by that restriction's own $\alpha_{0,n}$ or $\alpha_{1,n}$ and its physical normalization. A single unspecified old radius cannot replace them all.

The product domain (3.43), p.276, is genuinely useful: it already applies to the localized boundary fluctuation expressions after their source substitutions. Its displayed current radius is $C_1 p_1(g_k)$. The identification needed to compare that printed polynomial with the full normalized real support $P_k = \delta_k/g_k$ remains unverified. E14.3 bypassed that identification for the E component through its underlying maps; the corresponding B-domain argument must also be supplied before making that bypass for B.

3. An ordinary old-scale collar and a multiscale kernel have different metrics The geometric collar is real. CMP119 pp.264–265 surrounds the old bad region Z_k by several layers of current-size cubes before defining Ω_{k+1}; p.269 takes a further interior set for the regular integration. This is stronger information than interlacing alone.

Consequently BD.3 applies to an ordinary connected tree, measured in old j-scale units, joining the current region to $Z_j \subset Z_k$: its length is at least a fixed collar constant times $R_k L^{k-j}$, with the source's fixed adjacent-step conventions absorbed into that constant. The full primitive B bound (2.42), chosen by the source rather than its weaker component-resummed alternative, can pay this length when that connection lies in the primitive carrier itself.

There is a second way a B vertex can depend on a current variable. A small old carrier close to Z_j can receive a remote current dependence through the minimizing-field pullback. For that dependence the relevant exponential in CMP102 (190) is

$$\exp\{-c\, d_\Omega(y, y')\}, \tag{BA.3}$$

where d_Ω is the source's weighted multiscale distance. CMP99 p.397 explicitly identifies this weighted distance and refers to its earlier definition. It is not automatically the Euclidean distance divided by the old mesh. A route through a coarser layer has a different distance weight. Thus BD.3 does not by itself justify substituting $R_k L^{k-j}$ into (BA.3).

This distinction matters in precisely the application at issue. The primitive-tree weight controls a carrier which reaches the current region; the minimizing kernel controls the additional connection for a carrier which does not. A complete anchored estimate must combine the two with their actual metrics and their actual localization/preparation factors. It may prove more decay than a separate estimate, but that combined bound is not the ordinary-tree statement BD.3.

3.1 The weighted-age geometry CMP96 gives the formulas on printed pp.224, 231 and 233. Formula (2.2) requires separation of successive nested regions by more than RM in the local scale. Formula (2.46) defines the actual path metric by adding the length spent in each determining layer, weighted by the inverse local mesh. The surface decomposition (2.47)–(2.48), followed by (2.57), charges more than RM for every complete layer transition. A path from level n to level k therefore has at least $(|k-n|-1)_+$ such transitions.

The source row-sum proof (2.58) also retains this information after summing its spatial and path indices. To display it without changing the source metric, let $0 < a < 1$, let $\delta_0 > 0$ be the kernel decay rate, and use the source's finite number $c_0(a/2)$. The row restricted to an input layer differing by s is bounded by

$$c_0(a/2)^d \sum_{m \geq (s-1)_+} \exp\left[-m\left(\frac{a\delta_0 RM}{2} - 2d \log c_0(a/2)\right)\right]. \tag{BA.3a}$$

Thus, for any fixed $\gamma > 0$, strengthening the source parameter condition to

$$\frac{a\delta_0 RM}{4} > 2d \log c_0(a/2) + \gamma + 1 \tag{BA.3b}$$

gives a bound $C_\gamma e^{-\gamma s}$ for that row, with a fixed C_γ. This follows directly by summing the geometric series in (BA.3a), with its missing first transition absorbed in the fixed constant. The source's normalized integration measures and physical output weights must still accompany an application to a particular operator. The strengthened parameter choice must also be checked jointly with the other retained source choices; it is not an unconditional certificate for J0.

This settles the metric and base-row-summation part. Sections 3–4 of Supplement AF.1 further retain it under the actual original-geometry nonlinear preparation, with weighted contraction and the source current-output formula. None of these statements replaces d_Ω by old-scale Euclidean distance or proves a factor $e^{-cR_kL^{k-j}}$ for every multiscale kernel. Applying them to the changed p-reconstructed B outputs, with their actual p, n weights in (BA.1a), is the remaining operator step.

The source's assertion on p.277 that minimizing decay pays the shrinking domains is retained as a scoped source input. The verified age estimate and prepared original-geometry response now quantify that payment for the basic field/current clauses. Turning it into the whole reconstructed tuple still requires the changed-geometry application. The source regular contour theorem does not replace it.

3.2 The local two-power input response is also proved Lemma BM.2 in Supplement AF.1 settles the local determining-input issue for $n > p$, with an explicit local cube margin. For $U_t = e^{i\eta tA}U_0$, whose physically normalized variation has size u in the field and first-difference norms, take a complex parameter disc of radius proportional to a/u, where a is the admitted local background ceiling. The actual plaquette expansion stays within curvature size $Ca\eta^2 h_n^{-2}$ on that disc. The ordered axial gauge gives a potential of size $Cah_p h_n^{-2}$, with bounded complex parallel-transport factors. The actual nonlinear average then has size $Ca(h_p/h_n)^2$. Cauchy and integration over the real parameter segment yield

$$\|B_p^g(1) - B_p^g(0)\| \leq Cu(h_p/h_n)^2. \tag{BA.3c}$$

The local-input estimate does not imply that the global reconstructed minimum depends only on that cube. Its other determining variables, its joined geometry, and the reconstructed independent-current output remain to be controlled. No exact consistency of minimizers under a change of lattice has been assumed.

4. A precise sufficient remaining kernel test The stronger superexponential old-age bound in BD.4 is one sufficient route, not the only possible one. There is another route that avoids paying 2^{am} separately at derivative order m.

For each source output restriction on layer n, measure its increment in the physical normalization appearing in (2.34)–(2.39). Let $\mathcal{K}_{n,k}$ denote a common bound for the corresponding normalized row sums of the actual complete current-to-output derivative, including the prepared minimizing map, independently extended current outputs, and the required boundary relocation. The necessary row sum includes its source integration measure; it is not an unnormalized sum over all fine bonds. Suppose, on a previously admitted fixed absolute map domain, that these bounds imply

$$\|\Delta\Phi_{n,k}\|_n \le C g_k P_k \mathcal{K}_{n,k} \quad \text{for } \|z\|_\infty < 2P_k, \tag{BA.4}$$

jointly with the proposed external native variables and preparation parameters. A sufficient B-domain admission test, for each of its output restrictions, is

$$\frac{C g_k P_k \mathcal{K}_{n,k}}{\alpha_{*,n}} \le \frac{\beta}{8} 2^{-(j-n)}, \qquad n \le j \le k, \tag{BA.5}$$

where $\alpha_{*,n}$ is that restriction's own radius. Equations (BA.2) and (BA.5) then leave a strict reserve. The whole source segment and all output restrictions must satisfy the test; bounding only the first plaquette variation is insufficient.

The source geometry and base-kernel calculation above now provide the candidate age rate. The needed full-output conclusion is

$$\mathcal{K}_{n,k} \le C_H e^{-\gamma(k-n)}. \tag{BA.6}$$

Its application to the entire reconstructed tuple and preparation map is not proved here. A stronger factor involving $\sum_r R_r$ is unnecessary for the following implication. With one fixed $q > p_0$, equal logarithmic exponents for the two running radii, their fixed source-admissible ratio, and the common admitted coupling ceiling for that fixed exponent as in E14.3, the already checked AF comparison gives

$$\frac{g_k P_k}{\alpha_{*,n}} \le C_* \sqrt{1+k-n}\, \varrho_k^{p_0-q}. \tag{BA.7}$$

For fixed $\gamma > \log 2$, the right-hand side of (BA.7) multiplied by (BA.6) and 2^{j-n} is uniformly bounded in n, j, k; the fixed small-coupling ceiling could then make it meet (BA.5). This implication uses $j - n \le k - n$, and a strict exponential in $k-n$ dominates its square-root factor. It does not silently extend the verified base-kernel row bound to an unchecked reconstructed output. A larger fixed reserve, for example $\gamma > 4\log L + \log 2 + 1$, also pays a conservative four-dimensional old-cell count in a subsequent value estimate.

Once such a single common contour and a summable complete boundary value/carrier majorant have been established, arbitrary current derivatives can be taken after the old-age sum, as in E14.3. The Cauchy factor would then be $m!(cP_k)^{-m}$, with no additional 2^{am} from differentiation. This avoids requiring a stronger metric estimate solely to repair an artifact of differentiating too early.

5. What remains for C-05 After the contour, the remaining steps are:

1. Prove (BA.5), or an equivalent direct simultaneous B-domain inclusion, for the complete actual pullback. The weighted-age geometry, original-geometry preparation, and local two-power determining input are verified. The exact joined p-reconstruction geometry and the global reconstructed curvature/current outputs, with every weight in (BA.1a), remain to be established.
2. Verify the complete anchored current-carrier majorant for that admitted pullback, retaining old B terms, E/R terms assigned to B, the heterogeneous-action difference supported in Z_k, and relocation terms supported in S_{k+1}. Supplement AF.2 supplies the exhaustive source formulas, the combined primitive-tree/minimizer connector, four separate exponent reserves, the double old-index summation, and a sufficient sector prefactor. The full-map and localization hypotheses are required here. BD.2 then gives the required weighted Hessian row without a spurious carrier-volume factor. Newest-boundary E/R terms do not require an unavailable old-Z_k collar; the current Gaussian exterior force stays in its exact Gaussian/Schur split.

3. Check that the resulting budget fits the complete coercivity margin and retain the stated native external derivatives. The additional arbitrary retained/selector pullbacks and history interface currents must not be declared controlled by a native-domain theorem.

The conditional subtraction of BD.1 remains exact and useful, but its discarded-in-the-conditional-law scalar still belongs to the outer marginal. It cannot supply the missing spatial or current estimate.

Boundary-domain hypothesis. Application of the boundary induction to the Wilson law requires CL. See the boundary induction. The weighted distance itself is known and checked, the relevant original-geometry row and preparation bounds are proved, the local two-power response is proved, and the final scalar/carrier deductions are explicit. The surviving actual-model application is the joined p-reconstruction, its global outputs, and the complete localized boundary bound on that admitted contour. Arbitrary retained-coordinate and selector-current estimates have not been folded into a native-domain completion claim.

References

- T. Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, CMP 119 (1988), 243–285: especially (2.34)–(2.42), pp.261–262, and (3.43) with the discussion on pp.276–277. Geometric construction at pp.254–256 and pp.264–269.
- T. Balaban, *The Variational Problem and Background Fields in Renormalization Group Method for Lattice Gauge Theories*, CMP 102 (1985), 277–309: Proposition 9 and (182)–(190). The estimate uses its actual derivative kernel and retains its weighted distance.
- T. Balaban, *Propagators for Lattice Gauge Theories in a Background Field*, CMP 99 (1985), 389–434: definition/reference to the weighted distance preceding the propagator theorems.
- T. Balaban, *Propagators and Renormalization Transformations for Lattice Gauge Theories II*, CMP 96 (1984), 223–250: pp.224, 231 and 233, especially (2.2), (2.46)–(2.48), and (2.57)–(2.59). This supplies the verified age geometry and row sum of Section 3.1.

Annex AG — Actual component history influence, spatial tails and retained currents

Annex AG. The conclusions below hold on the stated source domains. Applying them to the complete Wilson law requires compatibility of the conditional representation (CL), including the retained variables, densities, and marked transport. See the theorem catalogue.

Scope

H-03 remains open. This note proves new estimates for an actual part of the positive source construction: averaging the variables of one fixed selected component in Balaban's LFR II (1.71)–(1.72). The estimates concern its actual normalized posterior, including its old histories and its interaction with the other components. They are not estimates for a newly sampled independent reference field.

The results are:

1. An exact positive component-posterior representation, with the large self-component action retained in its base measure.
2. A quantitative influence bound for changing another component's history, obtained directly from the published interaction bound (1.68). It includes the conditional denominator.
3. A complete source-jet comparison and a local truncation of that actual posterior. The bounds retain a fixed factorial exponent.
4. Native-background-coordinate current bounds for the same component reset, including a current bound for its localization error. The admitted radius, rather than only the supremum of the action, is displayed.
5. A geometric summation of the actual relative-tree carrier bound for a reset restricted to eligible components. It gives a radius of order $MR_k h$, followed by exponential tails, uniformly over the remaining fixed eligible geometry. The reset need not average a giant component.

The eligible-component spatial bound is uniform over the admitted remaining geometries in the source coordinate description. The native analytic bounds are uniform over the values admitted by their stated charts. They do not prove the full retained-coordinate carrier/current bound of the H-03 statement. The exact outstanding inequality is stated in Section 8.

The primary source input is source-free; its small-coupling and geometric hypotheses are retained. Identifying the Wilson trajectory with those hypotheses is not discharged here.

1. Primary source formulas and the particular reset

The following published formulas specify the support and the component operations.

Source	Exact input and its scope
CMP 109 (1987), p.257, geometric definitions before (0.24), article	A localization domain is a finite face-connected union of M-cubes. The neighborhood operation is the fixed cube enlargement.
LFR I, CMP 122 (1989), pp.177 and 201, conditions (i)–(ii), (1.100)–(1.102)	The selected components fit in cubes of side $100MR_k$; the actual local replacement uses its localized reference density and preserves the integral. Components outside the selected union are not replaced.
LFR II, CMP 122 (1989), pp.376–377, (1.67)–(1.69)	The cross-component functions $V(Y)$ have a uniform analytic bound in a specified relative tree size. A function whose carrier is a whole component has a different, generally nonsmall, bound.

Source	Exact input and its scope
LFR II, pp.378–379, (1.71)–(1.72)	The component integrations and old-history sums factor over fixed components. Each component operation is localized in that component. The cross action depends on the old histories of components contained in its carrier, and the old-history sum acts on this action too.
LFR II, pp.379–380, (1.73)–(1.75)	The real component operation is positive. Its analytic deformation is a small exponential tilt. Its normalized analytic operation obeys the stated supremum bound even when its real unnormalized mass is very small.
LFR II, pp.388–389, (1.92)–(1.97)	The unmarked expansion pays a component-volume cost with the small unnormalized component mass. This payment is not available after conditioning on the component.
LFR II, pp.391–392, (1.103)–(1.104)	The normalized component operations can be integrated out. The remaining domains are fixed in that expansion; the source explicitly does not require their own tree-decay factors for that unmarked convergence statement.

The source notation for the component operation is denoted here by $\mathsf{T}_i = \mathsf{T}'_k(X_i)$. The index i fixes a component, not an individual microscopic plaquette.

Fix a term of the positive source representation with its outer component geometry, gauge choices that remain external, and retained field data υ. Only sectors with positive real normalizers are used. Let η_i comprise all variables still integrated or summed by T_i: its old local histories, continuous integration variables, and local graph choices. These variables can be placed in the common padded disjoint union of sectors already specified in H-01. The operation defines a finite positive measure

$$\mathsf{T}_i F = \int F(\eta_i)\, M_i(d\eta_i; \upsilon), \qquad m_i(\upsilon) = M_i(1;\upsilon) > 0, \qquad \nu_i(d\eta_i;\upsilon) = m_i(\upsilon)^{-1} M_i(d\eta_i;\upsilon). \tag{HL.1}$$

The self-component expression $V(X_i^{\sim})$, whose bound is (1.69), is inside M_i. It has not been omitted or estimated as a small perturbation. The same is true of the old characteristic functions, local reference factors, integration Jacobians and graph averages present in (1.71).

Let $\mathcal{Y}$ be the finite family of the remaining carriers in (1.72). Write $J(Y)$ for the components whose variables occur in V_Y. A carrier may involve one component and retained fields outside it; it need not involve two distinct components. The source locality statement gives

$$V_Y = V_Y\big(\upsilon|_Y, (\eta_j)_{j\in J(Y)}\big). \tag{HL.2}$$

Here locality is in the source variables appearing in that formula. In particular, replacing a minimizing background by a storage label is not a proof that its footprint in the actual retained-link coordinates equals Y.

Theorem HL.1 – the fixed-geometry posterior On this fixed positive sector, the conditional law of the variables $\eta = (\eta_i)$, given υ, is

$$\pi_\upsilon(d\eta) = \frac{\exp\{\sum_{Y\in\mathcal{Y}} V_Y(\upsilon,\eta)\} \prod_i \nu_i(d\eta_i;\upsilon)}{\displaystyle\int \exp\{\sum_{Y\in\mathcal{Y}} V_Y(\upsilon,\eta)\} \prod_i \nu_i(d\eta_i;\upsilon)}. \tag{HL.3}$$

In particular, conditioning on all components other than i gives

$$\pi_i(d\eta_i \mid \zeta) = \frac{e^{H_i(\eta_i;\zeta)}\nu_i(d\eta_i;\upsilon)}{\nu_i(e^{H_i(\cdot;\zeta)})}, \qquad H_i = \sum_{Y\,:\, i\in J(Y)} V_Y, \tag{HL.4}$$

where ζ denotes the retained data and the other component variables.

Proof. On a fixed term of (1.72), the operations (1.71) are integrations against the product of the positive measures M_i. The only remaining dependence on their integration variables is the last exponential in (1.72); the source expressly says that the local history sums act on it. Thus the joint density before normalization is precisely the numerator of (HL.3) with M_i in place of ν_i. The factor $\prod_i m_i(\upsilon)$ cancels between its numerator and denominator. The terms of its exponent independent of η_i cancel on forming the conditional law, giving (HL.4). No cancellation has been made in a sum over different outer geometries. □

The reset considered below averages the complete incoming source jet using (HL.4). If a new component label is retained, it can subsequently be sampled from the chosen source-independent replacement law; the marked quotient is still the conditional expectation under the old posterior (HL.4). In particular, this old posterior is not replaced by that new sampling law.

All incoming marks in this note are the exact marks already transported to the positive representation (1.72). If earlier gauge averaging or an exact change of variables has occurred, its projected mark is used. Reinserting an original microscopic observable into this representation without its transport would not be Theorem HL.1.

The product-base representation is a scoped primary input from (1.71)–(1.72), not a consequence of the small cross-interaction bound alone. We apply it only to geometries admitted by that source factorization. If a different proposed geometry has a nonsmall self interaction coupling several claimed component blocks, that interaction must be included in their common base, or that cluster must be left live. It cannot be discarded or charged to (1.68). The abstract carrier count in Section 6 also covers merging collars, but does not extend the source's posterior factorization to such a different geometry.

2. A direct influence bound from the actual interaction

Set

$$b_Y = \alpha \exp\{-\bar\kappa\, d_{k,Z}(Y)\}, \qquad \bar\kappa = (1+2\beta)\kappa, \tag{HL.5}$$

for the non-self carriers of (1.68). The published estimate is $\sup|V_Y| \le b_Y$, on its stated domain and uniformly in the variables still integrated by the component operations. The real slice suffices for this section.

The relative size in (1.67) is the length, divided by M, of a shortest tree contained in Y meeting the specified M-cubes of $Y \setminus Z$. The required terminals do not include arbitrary insertion points inside components of Z. This omission, rather than an assumed identification with an ordinary physical metric, is used in the argument below.

We use $\|P-Q\|_{\mathrm{TV}} = \sup_E |P(E)-Q(E)|$ for two probability measures. For a finite complex measure, the distinct notation $\|\mu\|_{\mathrm{var}} = |\mu|(\Omega) = \sup_{\|F\|_\infty \le 1} |\mu(F)|$ is its variation norm. A positive probability has variation norm one, and $\|P-Q\|_{\mathrm{var}} = 2\|P-Q\|_{\mathrm{TV}}$.

Lemma HL.2 – a normalized exponential tilt Let P be a probability measure and $Q = e^D P/P(e^D)$, with D real and of finite essential oscillation $\omega = \operatorname{ess\,sup} D - \operatorname{ess\,inf} D$. Then

$$\|P-Q\|_{\mathrm{TV}} \le \tanh(\omega/4). \tag{HL.6}$$

Proof. After subtracting the essential infimum, put $T = e^D \in [1,R]$, $R = e^\omega$, and $a = P(T)$. If $R = 1$ the result is immediate. The chord bound for the convex function $(T-a)_+$ gives

$$\|P-Q\|_{\mathrm{TV}} = a^{-1}P[(T-a)_+] \le \frac{(R-a)(a-1)}{a(R-1)}. \tag{HL.7}$$

The right side is largest at $a = \sqrt{R}$, where it is $(\sqrt{R}-1)/(\sqrt{R}+1) = \tanh(\omega/4)$. This includes the conditional normalizer exactly. □

Theorem HL.3 – one-component posterior influence Suppose two external data sets ζ, ζ' keep the geometry and the component base ν_i fixed. Let $\mathcal{A}_i(\zeta,\zeta')$ be the carriers in H_i that can change under this replacement. Then

$$\|\pi_i(\cdot \mid \zeta') - \pi_i(\cdot \mid \zeta)\|_{\mathrm{TV}} \le \tanh\left(\sum_{Y \in \mathcal{A}_i(\zeta,\zeta')} b_Y\right). \tag{HL.8}$$

For replacement of only another fixed component history η_j, with native background data held fixed, one obtains the actual influence coefficient bound

$$c_{ij} \le \tanh\left[\alpha \sum_{Y:\, i,j\in J(Y)} e^{-\bar{\kappa} d_{k,Z}(Y)}\right], \qquad i \ne j. \tag{HL.9}$$

The supremum defining c_{ij} is over all admissible values in those fixed sectors; the right side does not contain the conditional masses $m_i(\upsilon)$.

Proof. Relative to the law at ζ, the law at ζ' is the normalized exponential tilt with

$$D(\eta_i) = H_i(\eta_i;\zeta') - H_i(\eta_i;\zeta). \tag{HL.10}$$

All unaffected terms cancel. Each affected summand has supremum at most $2b_Y$, and hence oscillation at most $4b_Y$. Apply Lemma HL.2. For a change of η_j, (HL.2) limits the affected set to the one in (HL.9). □

This is an estimate for the actual conditional posterior. A hypothetical posterior mixing certificate is not a premise of (HL.8). However, its numerical carrier sum is still geometry dependent. It is not, by itself, an estimate uniform in physical separation for every allowed remaining history.

3. Entire finite jets, rather than only single marks

For a finite label set I, let

$$\Psi(t) = \sum_{\emptyset\ne J\subseteq I} t_J A_J \tag{HL.11}$$

be its square-free logarithmic source jet; A_J is the derivative coefficient, so $t_J = \prod_{j\in J} t_j$ has no additional factorial. For repetitions one first distinguishes the labels and then identifies the arguments. All formulas take place in the finite truncated source algebra.

Under a probability measure P, the reset coefficient is

$$A_I^P = \partial_I \log P(e^{\Psi})\big|_{t=0} = \sum_{\mathcal{P}\in\Pi(I)} \kappa_P(A_B : B\in\mathcal{P}). \tag{HL.12}$$

Every incoming higher vertex is present in (HL.12).

Lemma HL.4 – bounded full jets under a change of posterior Suppose P, Q are two probability measures on a common input space and $\delta = \|P-Q\|_{\mathrm{TV}}$. Suppose, for every nonempty $J\subseteq I$,

$$\|A_J\|_\infty \le C^{d+1}(d!)^{p_0} b^d, \qquad d=|J|, \quad C\ge 1, \quad p_0\ge 1. \tag{HL.13}$$

Then, for $m=|I|$,

$$|A_I^P - A_I^Q| \le \delta\,[C(1+2C)]^m (m!)^{p_0} b^m. \tag{HL.14}$$

Proof. For bounded $F_1,\dots,F_k$, the moment of a product changes by at most $2\delta\prod_\ell \|F_\ell\|_\infty$. Telescoping a product of moments in the cumulant partition formula therefore gives

$$|\kappa_P(F_1,\dots,F_k) - \kappa_Q(F_1,\dots,F_k)| \le 2\delta \sum_{\pi\in\Pi_k} |\pi|! \prod_\ell \|F_\ell\|_\infty \le 2^k \delta\, k! \prod_\ell \|F_\ell\|_\infty. \tag{HL.15}$$

The last count uses the injection of ordered set partitions into permutations with cuts.

For partitions of the m source labels into k blocks, ordering the blocks and then their sizes gives

$$k! \sum_{\substack{\mathcal{P}\in\Pi(I)\\ |\mathcal{P}|=k}} \prod_{B\in\mathcal{P}} (|B|!)^{p_0} = m! \sum_{\substack{d_1+\cdots+d_k=m\\ d_\ell\ge 1}} \prod_{\ell=1}^{k} (d_\ell!)^{p_0-1} \le (m!)^{p_0} \binom{m-1}{k-1}. \tag{HL.16}$$

The remaining sum is

$$C^m \sum_{k=1}^{m} (2C)^k \binom{m-1}{k-1} = 2C^{m+1}(1+2C)^{m-1} \le [C(1+2C)]^m. \tag{HL.17}$$

Combining (HL.12), (HL.15), and (HL.16) proves the result. □

Thus the actual coefficient change from a remote component replacement, with the incoming coefficients unchanged by that replacement, is bounded by (HL.14) with the right side of (HL.8) for δ. If the incoming coefficients themselves change, their direct and previous-current records must also be transported. This theorem does not discard those records.

The bounded hypothesis is automatic at a single finite compact cutoff for the exact bounded incoming expressions that admit such a representation, but the constants can depend badly on that cutoff. It does not establish the desired current-scale h^{-4m} budget. The more general actual-law moment bounds remain useful when a uniform supremum budget is unavailable; replacing the supremum hypothesis by such bounds requires the corresponding coupling/Holder estimate.

Corollary HL.5 – actual local truncation of one reset Fix a source-coordinate region E containing the source dependence of the component base (including its stated collar) and the external dependency footprints of the incoming coefficients under consideration. Form an auxiliary conditional law by retaining only the terms of H_i whose carriers lie in E:

$$H_i^E = \sum_{\substack{Y:\, i\in J(Y)\\ Y\subseteq E}} V_Y, \qquad \pi_i^E = \frac{e^{H_i^E}\nu_i}{\nu_i(e^{H_i^E})}, \qquad \lambda_i(E) = \sum_{\substack{Y:\, i\in J(Y)\\ Y\not\subseteq E}} b_Y. \tag{HL.18}$$

Then the exact reset jet obeys

$$|A_I^{\pi_i} - A_I^{\pi_i^E}| \le [C(1+2C)]^m (m!)^{p_0} b^m \tanh\big(\lambda_i(E)/2\big). \tag{HL.19}$$

The approximating coefficient depends only on the source variables in E, with their actual dependency footprints retained.

Proof. The difference of the exponents in (HL.18) has supremum at most $\lambda_i(E)$, hence oscillation at most $2\lambda_i(E)$. Apply Lemmas HL.2 and HL.4. Locality follows from (HL.1)–(HL.2), with the stated assumption about the incoming coefficients. □

The auxiliary law in (HL.18) is used only to estimate the unchanged exact reset. It is not a proposed replacement of the Wilson measure. Nested regions give an exact finite telescoping carrier decomposition of the reset coefficient. A useful norm estimate still requires summing its actual carrier costs; replacing those costs by one would lose H-03.

4. The normalized analytic component operation is a source input

The source provides more than real positivity for a fixed component. LFR II (1.73)–(1.75), printed p.380, explicitly gives

$$\big|[\mathsf{T}_i(U,J)1]^{-1}\mathsf{T}_i(U,J)F\big| \le e^{3\sup|\sigma_i|}\sup|F|. \tag{HL.20}$$

Here $\mathsf{T}_i(U,J)$ is represented by the real positive component integral at $(U,0)$, multiplied by e^{σ_i}. The small complex variation σ_i is described on p.379, including the variation of the large self-component term $V(X_i^{\sim})$.

Equation (HL.20) controls the normalized old component operation. Its normalizer may be very small; its reciprocal is not separately bounded. This is a different input from the earlier estimate for the freshly sampled reference density in LFR I (1.100).

The domain in (HL.20) is the domain actually admitted by the source. It does not assert an entire function of arbitrary retained-link coordinates, or of variations that change the outer selectors and component geometry.

The following argument includes the cross action H_i, which need not have been part of the isolated normalized operation (HL.20).

Theorem HL.6 – a quantified analytic tube for the actual posterior Fix a real source datum and a one-complex-parameter native chart $|z| < \rho$ admitted by the source with the geometry and the other component histories fixed. Suppose its integral representation is the one used in (HL.20):

$$M_i(z; d\eta_i) = e^{\sigma_i(z,\eta_i)} M_i(0; d\eta_i), \quad \sigma_i(0,\eta_i) = 0, \quad \sup_{|z|<\rho,\eta_i} |\sigma_i| \le s_i. \tag{HL.21}$$

The source supplies this representation on a common fixed support in its admitted native chart. At a real datum with a nonzero current coordinate, one recenters the source exponent: if its unrecentered form is $\sigma_i^{\mathrm{src}}(z)$, the exponent in (HL.21) is $\sigma_i^{\mathrm{src}}(z) - \sigma_i^{\mathrm{src}}(0)$. Thus s_i can be taken as twice the source supremum bound on that chart. The real measure at its center is positive. Neither recentering nor a common support is assumed across a selector or geometry change. Define the explicit cross-action majorant

$$\lambda_i = \sum_{Y\,:\, i\in J(Y)} b_Y, \qquad r_* = \log(3/2), \qquad \rho_i = \frac{\rho\, r_*}{s_i + \lambda_i + r_*}. \tag{HL.22}$$

For every $0 \le t \le 1$, let

$$H_{i,t} = H_i^E + t(H_i - H_i^E) \tag{HL.23}$$

and let $Q_{i,t,z}$ be the normalized component integral with this cross action and the base (HL.21). Then throughout $|z| < \rho_i$ its normalizer is nonzero, and its complex measure has total variation norm

$$\|Q_{i,t,z}\|_{\mathrm{var}} \le 3. \tag{HL.24}$$

The constants are uniform over t, the integrated component variables, and the remaining variable values admitted by the fixed chart.

Proof. At $z = 0$, $Q_{i,t,0}$ is a real probability measure. Relative to this measure the complex operation is exactly

$$Q_{i,t,z}F = \frac{Q_{i,t,0}(e^{\theta_{i,t}(z)}F)}{Q_{i,t,0}(e^{\theta_{i,t}(z)})}, \quad \theta_{i,t}(z) = \sigma_i(z) + H_{i,t}(z) - H_{i,t}(0). \tag{HL.25}$$

The analytic function $\sigma_i + H_{i,t}$ has supremum at most $s_i + \lambda_i$ on the disc of radius ρ. Its power series and Cauchy's coefficient bound give

$$\sup_{\eta_i} |\theta_{i,t}(z)| \le (s_i + \lambda_i)\frac{|z|}{\rho - |z|} \le r_*, \qquad |z| \le \rho_i. \tag{HL.26}$$

For any probability measure μ and $|\theta| \le r_*$,

$$|\mu(e^\theta)| \ge 1 - \mu|e^\theta - 1| \ge 2 - e^{r_*} = \frac{1}{2}, \qquad \mu|e^\theta| \le e^{r_*} = \frac{3}{2}. \tag{HL.27}$$

Their ratio proves (HL.24). Every small real component mass has already cancelled in (HL.25). □

The radius ρ_i is an output of the estimate, not a fixed universal radius. If the actual carrier sum λ_i grows, the derivative cost grows. If no admitted native chart is available for a proposed retained-coordinate perturbation, Theorem HL.6 does not apply to it.

5. Source-coordinate currents and a current bound for the tail

Assume the incoming coefficients have analytic versions in the chart of Theorem HL.6 and obey (HL.13) there, with the supremum taken over that chart and the integrated variables. The argument also applies with the coefficients frozen at any fixed real datum; that version isolates the newly generated kernel current from direct input derivatives.

For a complex measure Q of mass one and variation norm at most $K \ge 1$, the cumulant partition formula gives

$$|\kappa_Q(F_1, \dots, F_k)| \le k! 2^{k-1} K^k \prod_{\ell=1}^{k} \|F_\ell\|_\infty. \tag{HL.28}$$

Indeed, each product of moments has at most k factors, each costing at most K. The same ordered-composition calculation as (HL.16) implies for $K = 3$

$$|A_I^{Q_{i,t,z}}| \le B_C^m (m!)^{p_0} b^m, \qquad B_C = C(1 + 6C), \qquad |z| < \rho_i. \tag{HL.29}$$

Consequently Cauchy's formula yields, for $q \geq 0$,

$$\left| \frac{d^q}{dz^q} A_I^{Q_{i,t,z}} \Big|_{z=0} \right| \leq q! \rho_i^{-q} B_C^m (m!)^{p_0} b^m. \tag{HL.30}$$

For a real segment in the concentric half-disc, the corresponding first finite-difference bound has $2/\rho_i$ in place of ρ_i^{-1}. These bounds include every source order m with the same factorial exponent p_0.

Theorem HL.7 – localization of the native kernel currents Let $\lambda_i(E)$ be the explicit discarded-carrier majorant in (HL.18). With the same analytic input budget and the radius (HL.22), there is D_C, depending only on C, such that

$$\sup_{|z|<\rho_i} |A_I^{Q_{i,1,z}} - A_I^{Q_{i,0,z}}| \leq D_C^{m+1} (m!)^{p_0} b^m \, \lambda_i(E), \tag{HL.31}$$

and, for every $q \geq 0$,

$$\left| \frac{d^q}{dz^q} (A_I^{Q_{i,1,z}} - A_I^{Q_{i,0,z}}) \Big|_{z=0} \right| \leq q! \rho_i^{-q} D_C^{m+1} (m!)^{p_0} b^m \, \lambda_i(E). \tag{HL.32}$$

Proof. Put $R_i = H_i - H_i^E$, with $\|R_i\|_\infty \leq \lambda_i(E)$ on the admitted chart. Differentiating the normalized finite integral in t, or its bounded holomorphic integral version, gives the exact centered formula

$$\partial_t Q_{i,t,z} F = Q_{i,t,z}(FR_i) - Q_{i,t,z} F \, Q_{i,t,z} R_i. \tag{HL.33}$$

For a bounded F independent of t, (HL.24) therefore implies

$$|(Q_{i,1,z} - Q_{i,0,z})F| \leq 12 \lambda_i(E) \|F\|_\infty. \tag{HL.34}$$

When a product of r moments in a cumulant is telescoped, the difference costs at most $12\lambda_i(E)\, r\, 3^{r-1} \prod \|F_\ell\|_\infty$. Summing its cumulant coefficients bounds it by

$$12\lambda_i(E)\, k! 6^{k-1} \prod_{\ell=1}^{k} \|F_\ell\|_\infty. \tag{HL.35}$$

Use (HL.16) once more in the complete reset formula (HL.12). For example, $D_C = 12C(1+6C)$ is more than sufficient for (HL.31). Analyticity and Cauchy's formula prove (HL.32). □

For a retained variation outside the incoming source carriers, the input coefficients are fixed and (HL.32) directly bounds the kernel part of the response. For variations inside those carriers, one must separately carry the direct and previous-current records of the inputs. Freezing their values while differentiating the kernel gives the new current in isolation; the remaining derivative is handled by the exact decorated product and conditional-transport rules of M-03. A bound on the total derivative alone is not used to bound two separately measured records through cancellation.

In particular, in one source direction the single-mark current is

$$(DQ_i)F = Q_i(F\, D \log w_i) - Q_i F \, Q_i(D \log w_i) \tag{HL.36}$$

on the common source chart. The normalizer's derivative is present. When a current-direction change alters a hard selector, the component geometry, or the underlying padded support outside this analytic chart, the finite signed-kernel difference of M-03 must be used instead. Equations (HL.30)–(HL.36) make no assertion about such a change.

6. An eligible-only reset and a geometric tail bound

The source permits a precise reset design: average the old component variables only for the eligible components selected by LFR I (i)–(ii). Retain the other histories for later steps. This is an exact choice of conditional projection and does not require averaging an arbitrarily large component at this step. Consequently the shared-sign test below does not obstruct a bounded-radius statement for this particular design.

This section proves that the relative-carrier sum in (HL.18) can be controlled uniformly in the source coordinates for that design. No independence hypothesis, contractive influence matrix, or small conditional component mass is used.

Let h be the physical lattice unit at the step, so that the source M-cube has physical side Mh. Distances in the following counting argument are measured in units of M-cubes. Write $R = R_k \geq 1$. Every selected component X_j has

$$|X_j|_M \leq (100R)^d, \qquad \operatorname{diam}_M(X_j) \leq 100\sqrt{d}\,R, \tag{HL.G1}$$

where $|X_j|_M$ counts its M-cubes. The constants can be enlarged by a dimension-only factor for closed-cube boundary conventions.

The superscript in LFR II p.377 is $Z^{\sim}$, a neighborhood, not a negative exponent denoting a shrunken set. Its stated carrier rule implies that a carrier Y contains a whole selected component whenever it meets that component. It may contain its larger neighborhood as well. In particular

$$Y = (Y \setminus Z) \cup \bigcup_{j:\, X_j \subseteq Y} X_j. \tag{HL.G2}$$

Every $Y \in \mathcal{Y}$ is a connected localization domain, and a non-self carrier touching X_i has $Y \setminus Z \neq \emptyset$. These are source geometric properties, not probabilistic assumptions. No bound on the diameter of a component of $Z^{\sim}$ is used below. Even if several enlarged components join, all their new collar and connector cubes outside Z belong to $S = Y \setminus Z$ and are terminals that the relative tree must meet. The remaining pieces are the original eligible components X_j, each satisfying (HL.G1). This observation concerns the counting lemma. Its posterior application still retains the source-factorization guard stated after Theorem HL.1.

Lemma HL.8 – counting relative carriers around an eligible component There are dimension-only constants $A_d > 1$, $B_d < \infty$, and D_d such that, for every fixed admissible selected geometry, component X_i, and integer $n \geq 0$,

$$\#\{Y \in \mathcal{Y} \,:\, i \in J(Y),\ n \leq d_{k,Z}(Y) < n+1\} \leq B_d(1+R)^d A_d^{n+1}. \tag{HL.G3}$$

Every such carrier satisfies

$$Y \subseteq N^M_{n+1+D_d R}(X_i), \tag{HL.G4}$$

where N^M_r is the physical rMh-neighborhood in the periodic metric on the finite torus. A nonwrapping Euclidean lift gives the same estimate whenever it contains the carriers in question. No distance across a chosen fundamental-box seam is treated as a large Euclidean separation. Both estimates are independent of the volume, the number and positions of the other components, and their internal history values.

Proof. Put $S = Y \setminus Z$. Choose a tree of length less than $n+1$, in M-units, meeting every closed cube of S, as in (1.67). An arbitrarily small extra length can be allowed if the infimum is not initially attained.

A finite tree of length at most $n+1$ admits a traversal of length at most $2(n+1)$. Sampling that traversal at intervals of $1/4$ shows that all cubes it meets lie in a connected set W of at most $K_d(n+2)$ cubes, for a dimension-only constant K_d. One can take adjacency by intersecting closures for this counting upper bound; that graph has bounded degree. Enlarging the sampled cubes by one fixed layer ensures connectedness and covers cubes met only on a boundary. In particular $S \subseteq W$.

Because Y is connected, contains X_i, and is not contained in that component, S includes a cube adjacent to X_i. There are at most $C_d|X_i|_M \leq C_d(100R)^d$ choices for this root cube. The number of connected cube sets of size at most $K_d(n+2)$ containing a fixed root is at most a constant to that size: choose a spanning tree and encode its depth-first traversal in the bounded-degree graph. Choosing S within such a W adds at most another factor $2^{|W|}$.

Each selected component included in Y is adjacent to S. Indeed, distinct components of Z are not face-connected; a path in Y leaving any one of them must enter S. There are at most $2d|S|$ components adjacent by a face to S. The choices of which are present add at most $2^{2d|S|}$. Once S and these components are chosen, (HL.G2) determines Y. Multiplying these counts proves (HL.G3), after changing the dimension-only constants. This is an upper bound and may count a carrier more than once. For the shell $n = 0$, the same construction uses a dimension-only number of cubes: a length-zero tree can meet only the bounded family of closed cubes incident to its point. Thus no positive minimum tree length is assumed. For example, the sampling and traversal argument permits the deliberately generous choices

$$K_d = 10 \cdot 7^d, \qquad A_d = \left[2^{2d+4}(3^d+1)^2\right]^{K_d}, \tag{HL.G3a}$$

with a larger dimension-only B_d absorbing a single extra factor A_d and the root count. Optimizing this entropy constant is not needed here.

For (HL.G4), fix a cube of S adjacent to X_i and a point of the tree in that cube. Every point of any cube of S lies within the tree's total length plus a dimension-only distance from X_i. A point of $Y \setminus S$ is in a selected component touching one of those cubes, and (HL.G1) bounds its extra distance by $D_d R$. This proves (HL.G4). □

The proof only used the length of the tree that actually appears in (1.67). It did not replace that length by a quotient metric in which travel through a component is free.

Theorem HL.9 – an actual eligible-component source-coordinate tail Choose the source decay constant with the explicit lattice-entropy margin

$$\omega = \bar{\kappa} - \log A_d > 0. \tag{HL.G5}$$

This is a displayed additional parameter condition to be checked jointly with all other source parameter constraints. The note does not assert that the already retained J0 parameter tuple has passed that extra check. There is $C_{d,\bar{\kappa}}$, independent of volume and the remaining history, such that

$$\lambda_i \le C_{d,\bar{\kappa}} \alpha (1 + R_k)^d, \tag{HL.G6}$$

and

$$\sum_{\substack{Y\,:\, i \in J(Y) \\ Y \not\subset N_r^M(X_i)}} b_Y \le C_{d,\bar{\kappa}} \alpha (1 + R_k)^d e^{-\omega [\, r - D_d R_k - 2 \,]_+}. \tag{HL.G7}$$

Proof. In the n-th shell, (HL.G3) and (HL.5) bound the sum by

$$B_d (1 + R_k)^d A_d^{n+1} \alpha e^{-\bar{\kappa} n}.$$

Its geometric sum proves (HL.G6). By (HL.G4), a carrier exiting $N_r^M(X_i)$ can occur only in a shell with $n+1+D_d R_k \ge r$. Starting the same geometric sum at that shell proves (HL.G7), with the harmless integer-rounding factor included in $C_{d,\bar{\kappa}}$. □

Combining (HL.G7) with (HL.19) proves a spatial response estimate for the actual full finite reset jet in source coordinates. If an incoming jet has source carrier S_I, use the approximation region $E = S_I \cup X_i^{\sim} \cup N_r^M(X_i)$, retaining any stated base-kernel collar. Its error has the bound (HL.G7) multiplied by the explicit jet factor in (HL.19). The output carrier is $S_I \cup X_i^{\sim} \cup N_r^M(X_i)$, not a reassigned point anchor.

Combining the same tail with (HL.32) gives the native current estimate, including all source orders, with an admitted radius at least

$$\frac{\rho\, r_*}{s_i + C_{d,\bar{\kappa}} \alpha (1 + R_k)^d + r_*}. \tag{HL.G8}$$

The cutoff dependence displayed in (HL.G8) is retained. This proof does not need $\alpha(1+R_k)^d < 1$: positivity, the exact tilt formula, and a possibly smaller analytic radius suffice for a single-component reset.

This result answers the geometric part of the eligible-only design. The reset can leave giant histories live for H-04/H-05. The remaining gap for the H-03 statement is the transport of these estimates to the actual retained coordinates and their separate current records, including the boundaries between the locally selected sectors. Even if a source expression is localized in E, its true dependency footprint after such a pullback must still be bounded.

7. What the new estimates do not justify

7.1 The self-component law has not become independent All internal old histories in X_i are still integrated jointly in ν_i. It can be a highly correlated law. The non-small bound (1.69) is specifically why its action was put in the base rather than discarded.

As a logical test of the scope, take one component with a single shared random sign H and assign $F_x = H$ to every site in that component. Take all cross interactions to vanish. Then every cross bound (HL.5) holds with $\alpha = 0$, the normalized base operation is entire in unused external parameters, and

$$\kappa(F_x, F_y) = 1 \tag{HL.37}$$

for every pair of internal sites. Thus the stated positive-product and cross-interaction hypotheses alone imply no decay between arbitrary interior insertion points. This is not a Yang–Mills counterexample. It shows exactly which stronger conclusion cannot be extracted from these source facts alone.

The component size restriction in LFR I bounds selected components by $100MR_k$, not by a scale-independent number of current blocks. Theorem HL.9 retains precisely that radius. Unselected components are left live under the design in Section 6. Thus internal giant-component mixing is not required for that reset; it would be needed, or replaced by an appropriate full-law reserve, for a later reset that actually averages a giant component.

7.2 The published unmarked expansion uses an unavailable conditional payment LFR II p.389 combines the component cost

$$\exp\{2\kappa(100R_k)^d\} \tag{HL.38}$$

with the unnormalized bound for $\mathsf{T}'_k(X)1$. The source chooses parameters so that, as displayed immediately before (1.95),

$$-2(1+\beta_0)^{-1}p_0(g_k)+1+2\kappa(100R_k)^d \le -\frac{3}{2}p_0(g_k). \tag{HL.39}$$

This is a concrete place where probability mass pays for geometric entropy in the source-free expansion. In the fixed-component posterior (HL.3)–(HL.4), that mass cancels. It cannot then be used again as a small conditional reset probability.

An unconditional marked proof can instead try to retain the actual probability of such a sector and pay (HL.38) before conditioning. It must prove that probability estimate with the exact marked law. The rare-edit and counted-carrier results specify a sufficient payment mechanism, but do not give its missing inherited probabilities.

7.3 Erasing a geometry label is a different posterior To see the distinction without notation from the action, suppose an old sector h has unnormalized density $\rho_h(u,v)$, and the frozen replacement draws u' from a normalized reference kernel $k_h(u' \mid v)$. Let

$$N_h(v)=\int \rho_h(u,v)\,du. \tag{HL.40}$$

After forgetting h, its posterior is

$$\Pr(h \mid u', v)=\frac{N_h(v)k_h(u' \mid v)}{\sum_{\tilde{h}} N_{\tilde{h}}(v)k_{\tilde{h}}(u' \mid v)}. \tag{HL.41}$$

The reference factor cancels in the old-field conditional expectation at a fixed h; it does not cancel across the sum in (HL.41). If geometries change with h, both the base measures and their dependency footprints change. Theorem HL.3, whose comparison keeps the base fixed, is not an estimate for (HL.41).

7.4 Source background locality is not yet current-link locality The source activity V_Y is local in the admitted background variables (U,J). The current norm M-03 assigns a coordinate its complete dependency footprint, not merely its storage block. A minimizing background field can depend on retained links outside the displayed source carrier. Its pullback, its derivative bounds, the coarea normalization and all selection changes must be carried.

Theorem HL.6 makes this distinction quantitative: its derivative bound contains the admitted source radius ρ and the actual interaction sum λ_i. It does not set either to a favorable constant. Furthermore the analytic input bound (HL.13) has not been derived for all actual renormalized incoming jets in the required current-scale regulator norm.

The source's normalized expansion (1.103) is compatible with these observations. Its fixed residual domains are not summed there, so the source does not need their own tree factors for that unmarked convergence proof. A bound with insertion anchors and current directions must pay the additional geometry it actually uses.

8. Exact residual inequality

The remaining estimate concerns a spatial estimate for the complete reset jet, uniform in the remaining history, including derivative/current marks. It is not just an estimate on the normalized value of a single fixed component operation.

One sufficient form, in the already fixed M-03 norm, is the following. For every admissible incoming decorated jet, the exact reset must admit compatible carrier records such that

$$\sum_{X\supset \mathrm{anchors}(I)} w_h(X)\left[\upsilon_{h,m,X}(A_I^{\mathrm{reset}})+g_{h,m,X}(B_I^{\mathrm{reset}})+j_{h,m,X}(J_I^{\mathrm{reset}})\right] \le C_*^{m+1}(m!)^{p_*}\mathcal{B}_I. \tag{HL.42}$$

Here C_*, p_* are independent of the cutoff depth, volume, current scale and admitted remaining history, and $\mathcal{B}_I$ is the specified incoming current-scale carrier budget with its actual running mixing coefficients. The current-scale normalization and the native regulators are those already fixed in M-03. All higher input vertices are included.

To obtain (HL.42) from the actual one-component estimates above still requires the following concrete bounds:

- A source-to-current-coordinate carrier and radius bound for the normalized component law, including every direct and current pullback in the strong norm.
- A carrier-norm summation after the actual coordinate pullback, with the insertion anchors included. Theorem HL.9 already supplies a spatial source-coordinate tail for every eligible component. It does not replace a proof for the actual retained-coordinate footprints, regulators and separate difference/current records.
- If a later reset averages the remaining giant components, either a local/internal marked estimate for those laws or an actual full-law probability reserve sufficient to pay their carrier size and marked losses. The eligible-only reset may instead leave these histories to H-04/H-05.
- A treatment of any geometry-changing or geometry-erasing reset, including the applicable finite-difference currents, and, for (HL.41), its old masses, reference likelihoods and support currents.
- The stipulated current-scale input budget for the actual complete renormalized jet. A compact bare bound at scale a does not turn into a budget at scale h just because a conditional expectation was taken.

An alternative route would prove a sufficiently weighted influence or cut-coupling theorem for the full physical posterior and combine it with the cumulant theorem. The actual one-component influences (HL.9) are inputs to such a route. A contractive row sum uniform in the physical geometry, or an appropriate replacement for it, has not been established here.

9. Application conditions

H-03 requirement	Result of this argument
The actual posterior, rather than the fresh reference distribution	Identified in (HL.3)–(HL.4), directly from the positive component operations.
Conditional covariance and all higher reset responses	Full formula (HL.12), bounds (HL.14), (HL.19), (HL.29)–(HL.32); all incoming vertices retained.
A genuine spatial input from the source	Actual remote-component influence (HL.9); Theorem HL.9 sums the discarded-carrier bound uniformly for eligible-only resets, with radius $O(MR_k h)$ and exponential tails.
Normalizer and self-action control	Exact normalization throughout; the self action remains in M_i; source (1.75) used only on its admitted native domain.
Current response control	Proved in admitted native charts, with the explicit radius (HL.22); physical pullbacks and geometry-changing currents remain unbounded.
Uniformity in the remaining fixed eligible geometry	Proved in source coordinates by (HL.G3)–(HL.G8), with the source entropy condition (HL.G5) displayed.
Actual retained-coordinate carrier/current norm, including sector changes	Not proved. The missing inequality is (HL.42), with the listed pullback, current and sector costs.

References

1. T. Balaban, *Large Field Renormalization. I. The Basic Step of the R Operation*, Communications in Mathematical Physics 122 (1989), 175–202.

2. T. Balaban, *Large Field Renormalization. II. Localization, Exponentiation, and Bounds for the R Operation*, Communications in Mathematical Physics 122 (1989), 355–392, The scope of every imported statement is specified in Section 1.

3. T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. I*, Communications in Mathematical Physics 109 (1987), 249–301, p.257, DOI https://doi.org/10.1007/BF01215223 . Only its connected-cube and tree-size definitions are used as an additional input in Section 6; the counting proof is given here.

Supplement AG.1: Application to the actual old-component posterior

The completed regular E and R theorems remove two useful native-contour obligations. Their scope does not yet identify an admitted current-coordinate contour for the archived-component posterior of HL.4.

The operation that still needs a current pullback The reset of Annex AG averages an arbitrary incoming full jet under the old component operation $\mathsf{T}'_k(X_i)$ in LFR II (1.71), tilted by the cross action in (1.72). The component's nonsmall self action from (1.69), old selectors, graph choices, local integrations and old-history sums remain inside that operation. Its joint analytic domain is inherited from the source's prepared multiscale map, including (1.65); its normalized native analytic estimate is (1.75).

The concrete application would be:

1. Compose the actual current/native external coordinate map, at the full real support and with fixed admitted outer geometry, into the input of LFR II (1.65).
2. Verify a simultaneous quantitative image reserve for every argument of the component operation (1.71) and its cross action (1.72), including the independent current and the complete self-component exponent. The estimate must hold for every old component history still integrated by that operation.
3. Use the resulting actual radius in HL.22. HL.6 then provides the normalized complex posterior bound and HL.7 its derivative/current-tail bound; HL.9 already supplies the eligible-component source-coordinate spatial tail.
4. Localize that coordinate pullback in the fixed M-03 norm, retaining the direct input derivative, the new conditional-kernel current, the previous currents, and any finite differences caused by sector changes.

These are actual applications to the source operation, not a request for another abstract cumulant theorem.

Why the completed regular R bound is not already this application Theorem R14.3 concerns the regular R activities after the source has completed the component integrations and returned to its E/R/B normal form. The source assigns its boundary-intersecting outputs to B. The proof therefore uses the returned regular primitive domain for those R activities. It does not identify that regular domain with the prepared multiscale domain of the unintegrated $\mathsf{T}'_k(X_i)$.

Consequently its holomorphic norm controls the already integrated action. A derivative bound on that action cannot simply be reused as a bound for the same component integral with an arbitrary incoming marked numerator. In particular it does not justify changing the current radius in HL.22 to the regular R radius without checking the input map above.

The finite-core reference estimate in Annex AE is also valuable but concerns a different probability kernel. In the exact frozen R quotient, the newly sampled reference factor cancels from the old conditional expectation. Its complex total-variation estimate cannot replace the normalized old-component estimate (1.75), whose base includes the old histories and self action. Both kernels are retained, with their distinct roles.

Exact remaining scope The first two steps of the concrete application share the multiscale-domain transfer issue isolated in Supplement AF.3. The remaining spatial pullback, regulator and sector-current estimates in HL.42 are additional requirements; neither current contour theorem declares them proved.

The source-factorization guard in Annex AG remains unchanged. The product-base posterior is used only on geometries admitted by LFR II (1.71)–(1.72). A hypothetical nonsmall interaction joining different proposed component blocks must be placed in their common base or that cluster must be left live. The geometric carrier count does not authorize a new factorization.

The use of (1.65)–(1.75) requires admission on the actual multiscale domain and the strong retained-coordinate bound (HL.42).

See LFR II, CMP 122 (1989), pp. 374–380 and 390–392, Annex AD, Theorem E14.3, and Annex AE, Theorem R14.3 and Section 10.6.

Annex AH — Complete regular E/R derivative-carrier sums

Annex AH. The conclusions below hold on the stated source domains. Applying them to the complete Wilson law requires compatibility of the conditional representation (CL), including the retained variables, densities, and marked transport. See the theorem catalogue.

1. Exact target and inputs

C-04 asks for the actual current-coordinate value, gradient and Hessian majorants, summed over old scales and carriers with the running radii and preparation losses retained. It requires a small weighted Hessian budget and local derivative budgets. The theorem below supplies these for the source's regular E/R contribution. B, the heterogeneous classical action, coordinate Jacobians and posterior/interface currents have their own original statements; they are not omitted from the complete law by completing C-04.

Use the actual current free coordinates of C-01/C-07, with their unit-scale Euclidean quadratic norm. A bond has a fixed orthonormal Lie-algebra basis of dimension $D = \dim \mathfrak{g}$. Archived history data remain fixed parameters in this conditional statement. Let $\rho_k(i, j)$ be periodic lattice distance between current-coordinate bonds in current lattice units. Fix any admitted torus large enough for the source's localization partition.

The completed Theorems E14.3 and R14.3 supply actual localized functions

$$V_{k,Y} = \mathcal{E}_{k,Y} + \mathcal{R}_{k,Y}, \qquad W_k^{ER} = \sum_Y V_{k,Y}, \tag{DS.1}$$

on one common contour containing $\|z\|_\infty \le P_k$, where

$$P_k = A_1 \ell_k^{p_0}, \quad \ell_k = \log g_k^{-2}, \quad |V_{k,Y}| \le K_k e^{-\kappa_* d_k(Y)}, \qquad K_k = \mathfrak{K}_E + C_R g_k^{\kappa_0 - 11}, \quad \kappa_0 \ge 12. \tag{DS.2}$$

The current scalar-coordinate radii are $r_z = c_G P_k$, with one fixed $c_G > 0$. The native background, independent current and preparation radii r_b, r_J, r_s are those of (E14.52)/(RM.14). These radii are independent of the later derivative order. All old-scale summations, the matching E coupling subtraction, the R vacuum subtraction and the true completed R denominators are already included in (DS.2). This proof does not differentiate a source-free real value bound.

The source's localization is exact in these variables: a term with carrier Y depends only on the admitted fields restricted to Y. A derivative in a current coordinate outside that carrier vanishes. Source-local field or current extensions are part of that actual function, not additional substitutions suppressed here.

The two source geometry facts used below are explicit: $d_k(Y)$ is shortest connecting-tree length divided by the localization cube size M, with an equivalent tree made of cube edges; the localized sums have exact carrier support and allow a sufficiently large fixed decay parameter. See CMP109 p.257 before (0.24), and CMP116 (1.26), (1.33)–(1.36).

2. An explicit anchored carrier count

Work in the rescaled M-cube lattice. Include the finite zero-length cases. Let $n(Y)$ be the integer length of the source's equivalent cube-edge tree. To avoid assuming that two equivalent length conventions are exactly equal, fix their geometric comparison constant $C_{\text{tree}} \ge 1$ so that

$$n(Y) \le C_{\text{tree}}\big(d_k(Y) + 1\big), \qquad \hat{\kappa} = \kappa_*/C_{\text{tree}}, \qquad \widehat{K}_k = e^{\kappa_*} K_k. \tag{DS.2a}$$

The comparison is a fixed cube-geometry fact: replace a connecting tree by its cube-edge representative. The source states this equivalent definition; allowing C_{tree} and an endpoint cube makes the argument valid whether or not its convention gives equality. Neither constant depends on cutoff, volume or history. In particular (DS.2) implies

$$|V_{k,Y}| \le \widehat{K}_k e^{-\hat{\kappa} n(Y)}. \tag{DS.2b}$$

The metric-equivalence loss is charged to the decay rate, not merely to a prefactor.

For a current bond i belonging to Y, choose an incident cube and one of its vertices on a shortest edge tree. There are at most 2^{2d} choices, where $d = 4$. A rooted tree with n edges is encoded by its canonical depth-first walk, of length $2n$, with at most $2d$ choices at each step. At most $2^d(n+1)$ cubes can meet its vertices. A carrier associated to that tree is a subset of these cubes, so the number of possible subsets is at most $2^{2^d(n+1)}$. Conditions requiring a domain to be connected, regular or contained in an admitted region only reduce this bound.

Consequently the number of actual distinct carriers at length n, after the source has combined terms having the same carrier, is bounded by

$$N_i(n) \le B_d A_d^n, \qquad B_d = 2^{2d+2^d}, \qquad A_d = (2d)^2 2^{2^d}. \tag{DS.3}$$

This deliberately large bound is independent of torus volume. The same encoding works on a periodic cube graph: its degree is at most $2d$, and a tree traversal determines its vertices on that graph. Winding does not introduce an uncounted root or translate.

If i, j are coordinates in Y, connect each to a point of this tree in its cube, and connect those points along the tree. Thus, with a fixed endpoint constant $c_d = 2d + 4$,

$$\rho_k(i,j) \le M\big(n(Y) + c_d\big). \tag{DS.4}$$

Distances on a torus are periodic distances, so this remains an upper bound there. Fixed carrier enlargements in an equivalent source convention merely enlarge B_d, c_d by fixed factors. The present proof uses the actual final carrier, not a smaller pre-localization support.

Choose the source decay and preparation parameters once, before the running coupling ceiling, so that

$$\hat\kappa > \log A_d + 1, \qquad 0 < \gamma M < \hat\kappa - \log A_d. \tag{DS.5}$$

This is a finite strengthening of the source's sufficiently-large decay choice. CMP116 allows the localization loss to leave a fixed fraction of the primitive decay, with its fixed preparation constants retained; first choose that primitive decay and the required decoupling parameter, then a sufficiently large fixed M as in its (1.11). The theorems E14.3/R14.3 are used with this common choice. No parameter is increased with derivative order or cutoff. The independent complete construction/trajectory certification remains part of J0.

Put

$$\mathcal{S}_\gamma = \frac{B_d \exp(\gamma M c_d)}{1 - A_d \exp[-(\hat\kappa - \gamma M)]} < \infty. \tag{DS.6}$$

Equations (DS.3)–(DS.4) imply the uniform anchored sum

$$\sup_i \sum_{Y : i \in I_Y} e^{\gamma \operatorname{diam}(I_Y)} e^{-\hat\kappa n(Y)} \le \mathcal{S}_\gamma. \tag{DS.7}$$

All multiplicities from old scales and different source preparations are already included in the per-carrier majorant (DS.2). They are not counted again as new carriers in (DS.3).

3. Value, gradient and weighted Hessian bounds

Theorem DS.1 — the actual regular E/R Hessian is uniformly small On every admitted real regular support, the actual contributions (DS.1) satisfy

$$\sup_i \sum_{Y : i \in I_Y} |V_{k,Y}| \le \widehat{K}_k \mathcal{S}_0, \tag{DS.8}$$

$$\sup_i \sum_{Y : i \in I_Y} e^{\gamma \operatorname{diam}(I_Y)} \|D_i V_{k,Y}\| \le \frac{C_{G,1} \widehat{K}_k \mathcal{S}_\gamma}{P_k}, \tag{DS.9}$$

and

$$\boxed{\sup_i \sum_j e^{\gamma\rho_k(i,j)} \|D_i D_j W_k^{ER}\|_{\mathrm{op}} \le \varepsilon_{ER}(g_k) := \frac{4D\,\widehat{K}_k \mathcal{S}_\gamma}{c_G^2 P_k^2}.} \tag{DS.10}$$

Here $C_{G,1}$ is a fixed basis constant, for example $\sqrt{D}/c_G$. Given any prescribed $\eta > 0$, one common small-coupling ceiling makes $\varepsilon_{ER}(g_k) \le \eta$ for every scale in the admitted trajectory and every remaining admitted history.

Proof. The value bound follows immediately from (DS.2b) and (DS.7). Scalar-coordinate Cauchy bounds each first derivative by the enlarged carrier majorant (DS.2b) divided by r_z. The Euclidean norm of the fixed-dimensional block gradient costs at most $\sqrt{D}$, proving (DS.9).

For the Hessian, apply Theorem BD.2 to the actual simultaneous current polydisc, keeping the native external variables fixed in their admitted contour. For each scalar output component a of a bond i,

$$\sum_{j,b} |\partial_{ia}\partial_{jb} V_{k,Y}| \le \frac{4\widehat{K}_k e^{-\widehat{\kappa} n(Y)}}{r_z^2}. \tag{DS.11}$$

Each block operator norm is bounded by the sum of the absolute values of its matrix entries. Summing the D output components, and using $e^{\gamma\rho_k(i,j)} \le e^{\gamma\,\mathrm{diam}(I_Y)}$, gives the factor $4D$ in (DS.10), followed by (DS.7). Derivatives of a carrier not containing i vanish. Absolute convergence of these anchored sums justifies the differentiated sum, uniformly in the volume. There is no spurious factor equal to the number of fine coordinates inside a carrier.

For the last assertion use $g_k \le 1$, $\kappa_0 - 11 \ge 1$, and set $K_* = e^{\varkappa_*}(\mathfrak{K}_E + C_R)$. All constants in K_* and $\mathcal{S}_\gamma$ were fixed before reducing the coupling ceiling. It suffices to impose

$$\ell_k \ge \left(\frac{4DK_*\mathcal{S}_\gamma}{c_G^2 A_1^2 \eta}\right)^{1/(2p_0)}. \tag{DS.12}$$

This is compatible with the fixed logarithmic and small-coupling conditions of the two contour theorems. It uses their source-order-independent $q > p_0$, without changing q. □

For scalar physical weights $\omega_i \ge \omega_* > 0$ in these same coordinates, the weighted version of (DS.10) is at most $\omega_*^{-2}\varepsilon_{ER}(g_k)$. The unit current coordinates here take $\omega_i = 1$. A new heterogeneous norm with no verified lower bound is not covered by renaming these weights; its precision, coordinate transfer and coercivity requirements remain C-06. This theorem does not assert the complete action is convex.

4. Fixed all-order local jets, including native external variables

Here a local analytic source direction means an explicitly given perturbation of the admitted slots (z, b, J, s). It does not mean a new inhomogeneous Wilson-action source whose induced E/R vertices have not yet been constructed. The latter marked matching remains M-04/M-06.

For directions $h_l = (h_l^z, h_l^b, h_l^J, h_l^s)$, define the actual finite-parameter pullback

$$\mathcal{V}_{k,Y}(t_1, \dots, t_m) = V_{k,Y}\left(z + \sum_l t_l h_l^z, b + \sum_l t_l h_l^b, J + \sum_l t_l h_l^J, s + \sum_l t_l h_l^s\right). \tag{DS.13}$$

Use the verified native norms, and write

$$q_l = \max\left\{\frac{\|h_l^z\|_\infty}{r_z}, \frac{\|h_l^b\|_{\mathrm{native}}}{r_b}, \frac{\|h_l^J\|_\infty}{r_J}, \frac{\|h_l^s\|_\infty}{r_s}\right\}. \tag{DS.14}$$

If $q_l = 0$, the corresponding derivative vanishes. Otherwise the polydisc $|t_l| < (mq_l)^{-1}$ lies in the same admitted contour, by the triangle inequality in each slot. Cauchy therefore gives

$$\left|\partial_{t_1}\cdots\partial_{t_m}\mathcal{V}_{k,Y}(0)\right| \le e^m m!\widehat{K}_k e^{-\hat{\kappa}n(Y)}\prod_{l=1}^{m} q_l. \tag{DS.15}$$

For local directions, a term can contribute only if its carrier meets the support of every differentiated direction. Anchoring on the first direction's support S_1, let $N(S_1)$ be the number of localization cubes meeting that support, with the fixed endpoint convention used in (DS.3). Then

$$\sum_{Y} e^{\gamma\,\mathrm{diam}(I_Y)}\left|\partial_{t_1}\cdots\partial_{t_m}\mathcal{V}_{k,Y}(0)\right| \le N(S_1)\mathcal{S}_\gamma e^m m!\widehat{K}_k\prod_{l=1}^{m} q_l. \tag{DS.16}$$

One may anchor on any chosen direction. The support factor is local; it does not become the torus volume for a fixed local test. Pure current directions of supremum norm one have $q_l = (c_G P_k)^{-1}$. Native external directions retain their actual $\alpha_{0,k}$ radii. Raw fine-link directions retain the extra ξ, or $\xi^{1+\beta}$, radius from E14.3/R14.3. Thus no uniform bare external-coordinate bound is asserted where the admitted radius runs to zero.

The exponent of $m!$ in this directional jet estimate is fixed at one. The proof bounds directional multilinear forms and anchored Hessian rows; it does not claim a dimension-free sum of the absolute coefficients of every high-order derivative tensor. A nonlinear parameter-to-slot map must be substituted into (DS.13) with its actual higher jets and chain-rule terms; the theorem supplies the slot derivative budgets but does not assume that additional map is linear.

5. Application and remaining applications

Equations (DS.8)–(DS.12) give the required complete regular E/R value, gradient and arbitrarily small weighted Hessian budgets. Equations (DS.13)–(DS.16) give explicit local joint jets with fixed factorial growth and the actual running radii. All old-scale and preparation losses are already accounted for by the completed native contour theorems, and the actual carrier geometry is summed in (DS.3)–(DS.7).

For the regular E/R part, the preceding estimates bound the current-coordinate value, gradient and Hessian. Their application to the complete Wilson density requires CL and the remaining boundary, classical, Jacobian and posterior contributions.

Sources

- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. I*, CMP 109 (1987), 249–301, p.257, definition of localization domains and their tree length; DOI.
- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. II. Cluster Expansions*, CMP 116 (1988), 1–22, (1.11), (1.26), and Lemma 1, (1.33)–(1.36); DOI.
- The complete actual-domain and old-scale proofs used here are Theorems E14.3 and R14.3 in Annexes AD and AE. The simultaneous-row argument is proved in Annex AF, Theorem BD.2.

Annex AI — Heterogeneous constrained coercivity, conditional on the boundary estimate

Annex AI. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

1. Conditional statement and inputs

The unchanged C-06 statement is:

> Extend the admitted regular derivative/coercivity estimates to the heterogeneous retained backgrounds and layer geometry actually encountered, with explicit weighted norms and domain margins.

The goal is a uniform estimate on the stated heterogeneous class, with its explicit boundary and regular-action budgets.

This note proves the C-06 implication with those predecessor conclusions as inputs. C-04 has been proved. C-05 has not been proved. Accordingly the conclusion for the complete interacting conditional law is conditional on C-05; neither that conclusion nor C-05's boundary budget is asserted unconditionally. The new actual-model checks in this note are the identification and admission of the heterogeneous principal form, its uniform gap and weighted locality in the exact free coordinates, the bulk classical and measure budget on that class, and the common-domain assembly. The elementary matrix perturbation argument by itself would not discharge C-06.

We retain the source-free construction and coupling assumptions of J0/C-01. An independent proof of that input, current-scale means and all-order interacting cumulants, arbitrary retained-coordinate changes, and selector/interface currents are separate obligations.

2. Which coordinates are integrated

Fix an admitted cutoff, scale, positive history sector and retained datum υ. Use the regular inner expression in CMP119 (3.23)–(3.25), identified directly in C-01. This pointwise Hessian argument does not use C-07 or its support/Neumann theorem; C-07 is a downstream application, so there is no circular dependency. Its variables are the Lie-algebra values of current-unit free bonds in $\Lambda = \Lambda_{k+1}$. Old-layer fields and the exterior current field y belong to υ. They are parameters, not additional coordinates in this integral.

Let I denote these free bonds and $D = \dim \mathfrak{g}$. On the real vector space $(\mathfrak{g})^I$ use

$$\|z\|_2^2 = \sum_{i\in I} |z_i|^2, \qquad \|z\|_\infty = \max_{i\in I} |z_i|. \tag{HC.1}$$

The block dimension is fixed. Distances $\rho_k(i,j)$ are measured on the current-unit lattice, using the shortest periodic lattice distance when appropriate. The free-bond set has at most d positively oriented bonds per site. Thus there is an explicitly finite lattice-counting majorant

$$\mathcal{N}_d(t) = d\left(1 + \frac{2}{e^t - 1}\right)^d, \qquad \sup_i \sum_j e^{-t\rho_k(i,j)} \le C_d \mathcal{N}_d(t), \quad t > 0. \tag{HC.2}$$

The harmless fixed C_d allows the chosen bond-origin convention. Counting shortest representatives of periodic displacement classes proves the same inequality on every torus, including small ones. Restrictions to I can only reduce the sum.

For block matrices set

$$\|A\|_\gamma = \max\left\{\sup_i \sum_j e^{\gamma\rho_k(i,j)}\|A_{ij}\|_{\mathrm{op}}, \sup_j \sum_i e^{\gamma\rho_k(i,j)}\|A_{ij}\|_{\mathrm{op}}\right\}. \tag{HC.3}$$

This norm is submultiplicative by the triangle inequality for ρ_k. The two sums agree for a real symmetric matrix. The norm does not insert a fine-lattice coordinate weight into a current-unit coordinate. Heterogeneity remains in the background, determining geometry, and precision matrix.

There is also a genuine layer-weighted norm, used for background admission rather than for the coordinates in (HC.1). With $h_n = L^n\eta$, use the actual local gauges of CMP99 (3.35)–(3.38) and the quantities

$$\|A\|_{\mathrm{lay},2} = \sup_n \max\left\{h_n\|A\|_n, h_n^2\|\nabla A\|_n, h_n^3\|d^*dA\|_n\right\}, \qquad \|J\|_{\mathrm{lay,J}} = \sup_n h_n^3\|J\|_n. \tag{HC.4}$$

If a source chart requires a Hölder norm, append its specified term $h_n^{2+\beta}\|\nabla A\|_{\beta,n}$. Current extensions are evaluated in their actual independent J slot. Equation (HC.4) does not assert that a first-derivative bound controls a second derivative; the latter is checked from the variational theorem below. For the reconstructed-output restrictions of B, use C-05's full domain, including its additional p, n weights. This note does not replace those restrictions by (HC.4).

3. Admission of the actual heterogeneous source class

Lemma HC.1 — geometry and real background admission The regular inner domain and real minimizing backgrounds of the retained construction satisfy the hypotheses of the unit-lattice propagator result, uniformly in the remaining admitted history, after one fixed reduction of the coupling ceiling.

Proof. Section E of CMP99, pp.427–428, requires $\Lambda \subset A_k = \Omega_k^{(k)}$, a union of the admitted blocks, separated from A_k^c by more than the fixed propagator distance RM. CMP119 (3.2)–(3.5), pp.264–265, constructs the next regular region by removing the old bad region and several layers of current-size cubes. The interior used in the Gaussian integral is smaller still. The rectangularization of the bad complement in CMP119 p.251, repeated on p.269, enlarges that complement; it does not enlarge Λ towards A_k^c. Consequently its distance is at least a fixed positive multiple of the running collar width, up to a fixed endpoint offset. Choose the fixed lower admissible running radius so that

$$c_{\mathrm{col}}LMR_{k+1} - C_{\mathrm{edge}}M > RM. \tag{HC.5}$$

The source running radii tend to infinity as the common coupling ceiling is reduced, so this is a uniform fixed-parameter condition. The determining geometries retain the inter-layer separation conditions required by CMP99; they are the admitted geometries in J0, not arbitrary nested sets.

For the background, CMP102 Theorem 1, pp.278–279, gives both its gauge-invariant field/current bounds and, on the proper-scale local gauge cubes, its first- and second-order bounds (9)–(10). In particular it supplies the d^*dA condition (3.36) of CMP99. A plaquette bound alone would not supply that condition.

Write ε_* for a uniform upper bound on the admitted determining-data regularity, after the fixed selection constants are included. For the running thresholds used in the construction a sufficient choice is

$$\varepsilon_* = C_{\mathrm{sel}}A_0 g_* \ell_*^{p_0}, \qquad \ell_* = \log g_*^{-2} > 2p_0. \tag{HC.6}$$

Indeed $g(\log g^{-2})^{p_0}$ is increasing in g on that range. Hence every admitted $g_n \le g_*$ obeys the same bound. Choose g_* so small that, with the constants in CMP102 Theorem 1 and CMP99,

$$\varepsilon_* < a_1/2, \quad B_3\varepsilon_* < a_0/2, \quad M\varepsilon_* < R_1M_1a_1/2, \quad MC_{\mathrm{reg}}B_3\varepsilon_* < a_{99}/8. \tag{HC.7}$$

The third condition puts the required local cubes within the range $M < M(\varepsilon_*)$ of CMP102. The fourth is the smallness required for the background propagators. The fixed constant C_{reg} accounts for the specified overlap of proper-scale cubes and gauge conventions. All choices are independent of the number of layers, volume, and derivative order. The

source places no regularity restriction on the gauge field outside its active background domain; none has been added here. This proves admission of the actual, possibly nonconstant and heterogeneous, real background class. □

This lemma retains the source bounds on determining data as part of the admitted source-free construction. It does not independently establish J0, its trajectory, or its history decomposition.

4. The principal form is the actual constrained form

The source's free-coordinate precision is

$$K_{k,\upsilon} = C_{k,\upsilon}^{*}\Delta_{\upsilon}^{(k)}C_{k,\upsilon}, \qquad K_{\Lambda} = (K_{k,\upsilon})_{II}. \tag{HC.8}$$

Here C solves the linearized averaging constraint on its exceptional bonds. The quadratic form $\Delta^{(k)}$ includes the term involving $H_1 D^{(2)}$ and the background current in CMP99 (3.156). Removing that term would change the constrained Hessian.

The precise primary inputs are CMP99 (3.155)–(3.158), the paragraph after (3.158), and Theorem 3.15. The paragraph after (3.158) states the uniform positive lower bound for the free-coordinate form on arbitrary backgrounds satisfying (3.35)–(3.36), and its uniform exponential locality. The covariance in Theorem 3.15 has exponential decay, and (3.158) identifies its restriction with the inverse free-coordinate precision. Those inputs concern the actual heterogeneous geometry admitted in Lemma HC.1.

Consequently there are fixed $\lambda_0, K_0, \delta_K > 0$, independent of scale, volume, retained datum and admitted history, such that

$$K_{\Lambda} \geq \lambda_0 I, \qquad \|(K_{\Lambda})_{ij}\|_{\mathrm{op}} \leq K_0 e^{-\delta_K \rho_k(i,j)}. \tag{HC.9}$$

Principal restriction preserves the lower bound and the entrywise estimate. The source estimate is first applied on the full admitted current fluctuation region with its already fixed map C; only then is the free-coordinate matrix restricted to I. The constraints are not re-solved inside Λ after conditioning. This order also covers a free bond whose C-image contains a neighboring exceptional bond. No translation invariance of K is assumed.

The map C introduces no depth-dependent loss. It is local to finitely many current blocks and equals the identity on free bonds. In local axial gauge its exceptional block $Q_E(U)$ is a finite-dimensional perturbation of $Q_E(1)$. If $\sigma_E > 0$ is the latter's least singular value and c_E bounds the derivative of that fixed block, the explicit condition

$$c_E \|U - 1\|_{\mathrm{block}} \leq \sigma_E/4 \tag{HC.10}$$

gives $\|Q_E(U)^{-1}\| \leq 2/\sigma_E$ and a common complex neighborhood of invertibility. Compact gauge changes act unitarily and do not change this singular-value bound. Finite block size and (HC.7) give (HC.10) uniformly. This is a local inverse estimate, not a bound on an extensive determinant. The coarea factor depending only on υ stays in the outer law.

Lemma HC.2 — explicit weighted principal bounds Let $0 < \gamma \leq \delta_K/2$, and define

$$A_K = K_0 \sup_i \sum_j \rho_k(i,j) e^{-\delta_K \rho_k(i,j)/2} < \infty. \tag{HC.11}$$

The right side can be replaced by a fixed lattice sum using (HC.2). For every real 1-Lipschitz function f on the current lattice and $D_f = \mathrm{diag}(e^{\gamma f_i})$,

$$\|D_f K_{\Lambda} D_f^{-1} - K_{\Lambda}\|_{2\to 2} \leq \gamma A_K. \tag{HC.12}$$

Choose

$$0 < \gamma \leq \min\{\delta_K/2, \gamma_{ER}, \gamma_B, \lambda_0/(4A_K)\}, \tag{HC.13}$$

omitting the last bound if $A_K = 0$. Then K_Λ^{-1} obeys

$$\|(K_\Lambda^{-1})_{ij}\|_{\mathrm{op}} \le \frac{4}{3\lambda_0} e^{-\gamma\rho_k(i,j)}, \quad \|K_\Lambda^{-1}\|_{\gamma/2} \le \frac{4C_d\mathcal{N}_d(\gamma/2)}{3\lambda_0}. \tag{HC.14}$$

Proof. The inequality $|e^t - 1| \le |t|e^{|t|}$ and (HC.9) bound both Schur sums of the difference by γA_K. The block Schur test proves (HC.12). For the inverse bound choose $f_i = \rho_k(i, j)$, truncated if desired; it is 1-Lipschitz. The real part of the conjugated quadratic form is at least $3\lambda_0/4$. Its inverse has operator norm at most $4/(3\lambda_0)$, and the ij matrix element gives the first estimate. Summation using (HC.2) gives the second. All arguments apply to periodic distance and arbitrary admitted restrictions. □

5. Complete perturbation accounting

For fixed v, the actual regular inner potential, up to a constant independent of z, is

$$V_{k,v}(z) = \tfrac{1}{2}\langle z, K_\Lambda z\rangle + \langle z, K_{I,I^c} y\rangle - W_{ER}(z; v) - W_B(z; v) - W_{\mathrm{loc}}(z; v). \tag{HC.15}$$

This is the C-01 identity, not an approximation. W_{loc} consists of the bulk nonlinear classical/gauge remainder and the current Haar/constraint density. W_B includes the inherited B activities, E/R activities assigned to B, the heterogeneous coefficient difference and the complete boundary relocation with its Jacobian. Components are assigned once, according to the source classification. If a C-05 bookkeeping convention already includes a local classical term, omit that term from W_{loc}; the estimates below then only improve.

In particular, the nonconstant coefficient is retained by the identity

$$-\Delta A(g_k(\cdot)^{-2}, U) = -g_k^{-2}\Delta A(1, U) - \Delta A(g_k(\cdot)^{-2} - g_k^{-2}, U). \tag{HC.16}$$

The last term is bounded by C-05. It is not used to replace the principal form with a fictitious homogeneous full action. The exterior force in (HC.15) is exact and has zero current Hessian; its possibly large effect on the mean remains a C-08 task. Its accompanying Schur complement stays in the outer marginal.

5.1 What is taken from the declared predecessors C-04 supplies the actual regular E/R bound

$$\sup_z \|D_z^2 W_{ER}(z; v)\|_\gamma \le \varepsilon_{ER}(g_k), \qquad \sup_{0<g\le g_*} \varepsilon_{ER}(g) \longrightarrow 0 \quad (g_* \downarrow 0). \tag{HC.17}$$

C-05, when completed, must supply the corresponding bound on its actual admitted multiscale domain,

$$\sup_z \|D_z^2 W_B(z; v)\|_\gamma \le \varepsilon_B(g_k), \qquad \sup_{0<g\le g_*} \varepsilon_B(g) \longrightarrow 0, \tag{HC.18}$$

as well as its local native derivative bounds and quantitative domain reserve. The current boundary proof records precisely such an output in (BV.14)–(BV.18), conditionally on the remaining joined reconstruction and localization checks. Equation (HC.18) is an input here, not a new proof of those checks. No coercivity assumption is included in (HC.18); the argument below establishes coercivity from it.

5.2 Bulk classical and measure terms on the actual background class Take a fixed $\epsilon > 0$, below the absolute preparation radius in CMP116 (1.15)–(1.21), and work first on the larger current disc

$$\|z\|_\infty < \epsilon/g_k. \tag{HC.19}$$

This denotes the absolute small-map domain for the full current input (the exterior part will be held fixed as specified below). It is used only for the bulk classical and measure terms. It does not assert that old B activities are admitted on (HC.19). Their domain remains C-05's domain.

The prepared nonlinear constraint obeys $B' = g_k Cz - hD(g_k Cz)$; the source bounds its map and all required output norms by a fixed multiple of $g_k\|z\|_\infty$. The cubic classical remainder is after removal of the *complete* constrained quadratic term in (HC.8). CMP116 (1.37)–(1.40) localizes this remainder; its proof, before Lemma 2, treats the other classical $P^{(k)}$ terms by the same expansion. Lemma 1 treats the separate old-E contribution. Therefore using the classical part here does not count the E contribution twice.

CMP119 pp.275–277 transfers the classical calculation to the proper-scale cube cover and the full multiscale random walks. Lemma HC.1 admits the background hypotheses for those maps. The bulk terms on the current layer have fixed-parameter carrier bounds of the form

$$\sup_{\|z\|_\infty < \epsilon/g_k} |W_{\mathrm{cl},Y}| \le C_{\mathrm{cl}}\, g_k^{-2} \epsilon^3 e^{-\kappa_{\mathrm{cl}} d_k(Y)}. \tag{HC.20}$$

The supremum is on the disc in (HC.19). A fixed carrier-volume factor from the plaquette partition has been paid with part of the available decay. Remote proper-scale and boundary-classified terms are in W_B, not in an unproved homogeneous extension of (HC.20). The nonlinearity is cubic or higher after the exact quadratic subtraction, including the quadratic constraint-current contribution; this is essential for the factor ϵ^3.

For the measure terms, the local Haar logarithm is analytic on the same small-map domain. The nonlinear constraint determinant has the form $\det(I - T)$, where the local analytic T has weighted norm $O(\epsilon)$. Reducing ϵ until that norm is at most $1/2$, the convergent expansion

$$\log\det(I - T) = -\sum_{r \ge 1} \frac{1}{r} \operatorname{Tr}(T^r) \tag{HC.21}$$

is a sum over local loops, not one volume-dependent determinant bound. The local factors and the source localization yield

$$\sup_{\|z\|_\infty < \epsilon/g_k} |W_{\mathrm{meas},Y}| \le C_{\mathrm{meas}} \epsilon e^{-\kappa_{\mathrm{meas}} d_k(Y)}. \tag{HC.22}$$

The exceptional linear-constraint coefficient is controlled by (HC.10); its determinant independent of current coordinates is not inserted into (HC.22). Boundary-change Jacobians remain in C-05. All fixed preparation, block, group and localization constants are retained in $C_{\mathrm{cl}}, C_{\mathrm{meas}}$.

The actual real support is $\|z_i\| \le P_k = A_1 \rho_k^{p_0}$. There is a separate amplitude check for the retained exterior current y. The classical action is evaluated at the full current field $t = z + y$, before the inner/exterior split. CMP119 (3.14) bounds the physical current fluctuation on its whole new active domain before that split; after the fixed local coordinate constants this gives

$$g_k \|y\|_\infty \le C_{\mathrm{ret}} \varepsilon_k. \tag{HC.23a}$$

Old layers outside that current domain remain source retained variables; their boundary-classified contributions are in C-05. Reduce g_* so that $C_{\mathrm{ret}} \varepsilon_k \le \epsilon/4$, and also so that

$$g_k P_k \le \epsilon/4 \quad \text{for every admitted scale.} \tag{HC.23}$$

Let $c_{\mathfrak{g}} \ge 1$ be the fixed conversion constant from scalar-coordinate sup norm to the chosen Lie-algebra norm. A simultaneous scalar-coordinate polydisc of radius $\epsilon/(2c_{\mathfrak{g}} g_k)$ about each such real point, with y held fixed, keeps the full current field $t = z + y$ in the absolute map domain (HC.19). The supports of z and y are disjoint. Thus this is uniform on the actual admitted exterior-current support, not for arbitrary unbounded exterior amplitudes. Apply the simultaneous Hessian-row estimate BD.2 to (HC.20) and (HC.22), then sum the actual carriers as in DS.3–DS.7. Choose γ smaller if necessary to retain that carrier entropy reserve. This gives

$$\sup_z \|D_z^2 W_{\mathrm{loc}}(z; \upsilon)\|_\gamma \le C'_{\mathrm{cl}} \epsilon + C'_{\mathrm{meas}} g_k^2/\epsilon. \tag{HC.24}$$

No total number of current coordinates occurs. This proof uses the larger disc only for terms whose actual absolute domains were admitted, and never differentiates a characteristic function. The same Cauchy argument gives each finite local current derivative and joint native derivative with the actual radii retained. For example the order-m current derivative of one classical carrier is bounded by

$$m!\, C_{\mathrm{cl}} g_k^{-2} \epsilon^3 (2c_{\mathfrak{g}} g_k/\epsilon)^m e^{-\kappa' d_k(Y)} \tag{HC.25}$$

for coordinate derivatives, and by an additional fixed e^m factor for multilinear unit directions in a norm ball. The resulting factorial exponent is fixed. No source-order-dependent change of ϵ is made.

6. Common domains and a single parameter choice

The real admission inequalities are supplemented by an actual complex propagator reserve. CMP99 Theorem 3.4 and (3.37)–(3.38) give a positive absolute analytic ceiling a_{ana} for relative fields in the full layer norm (HC.4). Let $a_J > 0$ be a fixed ceiling for the independent current in the already specified source extension; it can be reduced so that the perturbation estimate for the constraint-current term in CMP99 (3.134)–(3.138) has Neumann norm at most 1/4. The original real center has the strict slack in (HC.7). One sufficient quantitative choice is

$$r_{\mathrm{prop}} = \min\{a_{\mathrm{ana}}/4, a_{99}/(16MC_{\mathrm{reg}})\}, \quad r_{\mathrm{prop,J}} = a_J/4, \quad r_C = \sigma_E/(4c_E C_{\mathrm{block}}), \tag{HC.26a}$$

where C_{block} converts the strong native norm to the actual fixed unit-block chart. These radii are fixed before the final coupling ceiling. The strong complex field norm includes the second-order term; this assertion is not made for a first-derivative-only or arbitrary raw-coordinate direction. The explicit independent-current slot retains its own radius. Later changes to weaker coordinates must carry their actual norm conversion, which belongs to their matching estimate.

Keep the common current contour and native external domains supplied by C-04 and by the declared C-05 conclusion. Intersect them with the absolute local-map domain in (HC.19), the propagator domain from (HC.7), and the exceptional-coordinate neighborhood (HC.10), taking one half of each verified reserve. This is a quantitative construction: in each native direction use

$$r_{\mathrm{joint}} = \frac{1}{2}\min\{r_{ER}, r_B, r_{\mathrm{prop}}, r_{\mathrm{prop,J}}, r_C, r_{\mathrm{loc}}\}, \tag{HC.26}$$

in that direction's own normalization. For the current coordinate the first two radii are fixed multiples of P_k, while (HC.23) supplies a fixed larger absolute-map reserve. In this current direction the propagator and exceptional-coordinate reserves are first pulled back through the actual g_k-scaled current map: their radii are fixed absolute constants divided by g_k, not the unconverted field-slot constants in (HC.26a). For the layer-weighted background/current slots, r_B includes all C-05 reconstructed-output restrictions; it is not replaced by an unrestricted raw-coordinate ball. Thus the intersection adds only fixed losses to Cauchy estimates, and no new factor depending on the number of layers. If C-05 does not supply those uniform reserves, its hypothesis here has not been met.

Fix parameters in the following order:

1. The group, an admitted L, source geometry and localization parameters, with the required carrier and collar reserves.
2. The constants λ_0, K_0, δ_K of (HC.9) and a positive γ satisfying (HC.13) and the local-carrier reserve.
3. One absolute ϵ satisfying the local map conditions and $C'_{\mathrm{cl}}\epsilon \le \lambda_0/16$.
4. One coupling ceiling g_* satisfying (HC.5)–(HC.7), (HC.10), (HC.23)–(HC.23a), the predecessor domain conditions, and

$$\sup_{0<g\le g_*} [\varepsilon_{ER}(g) + \varepsilon_B(g) + C'_{\mathrm{meas}} g^2/\epsilon] \le 3\lambda_0/16. \tag{HC.27}$$

This last choice is available from the predecessor conclusions and the explicit local remainder. It is not asserted to have been verified while C-05 is open. Once it is available, the complete perturbation satisfies

$$\eta_\gamma := \sup_{z,\upsilon} \|D_z^2(W_{ER} + W_B + W_{\mathrm{loc}})\|_\gamma \le \lambda_0/4. \tag{HC.28}$$

The source field radii $\alpha_{i,n} = C_i g_n \ell_n^q$, with the fixed $q > p_0$ of the contour theorems, can simultaneously be made smaller than all absolute background ceilings by imposing $\ell_* > 2q$ and reducing g_*. The polynomial logarithmic factors do not obstruct any fixed absolute smallness condition. No parameter is adjusted with derivative order, volume or cutoff depth.

7. The heterogeneous extension theorem

Theorem HC.3 — uniform complete regular coercivity on the admitted heterogeneous class Retain the J0/C-01 construction hypotheses. Assume the original C-04 and C-05 conclusions, with their actual complete native domains and uniform bounds as specified in Section 5.1. Then the actual inner potential (HC.15), on its actual regular support and for every admitted heterogeneous retained background/history, satisfies

$$D_z^2 V_{k,\upsilon}(z) \ge \tfrac{3}{4}\lambda_0 I. \tag{HC.29}$$

For every real 1-Lipschitz f, its weighted conjugate has

$$\mathrm{Re}\langle u, D_f D_z^2 V_{k,\upsilon}(z) D_f^{-1} u\rangle \ge \tfrac{1}{2}\lambda_0 \|u\|_2^2. \tag{HC.30}$$

Consequently its pointwise matrix inverse obeys

$$\|[(D_z^2 V_{k,\upsilon}(z))^{-1}]_{ij}\|_{\mathrm{op}} \le \frac{2}{\lambda_0} e^{-\gamma\rho_k(i,j)}, \qquad \|(D_z^2 V_{k,\upsilon}(z))^{-1}\|_{\gamma/2} \le \frac{2C_d\mathcal{N}_d(\gamma/2)}{\lambda_0}. \tag{HC.31}$$

The current and native local derivative estimates hold on the explicit common domain (HC.26), with their original running radii and one fixed factorial exponent. All constants are uniform in cutoff depth, expanding volume, and the admitted remaining history. Real linear tilts of this conditional law have the same Hessian estimates.

Proof. Equations (HC.8)–(HC.10) identify and admit the actual principal form without setting its background to the identity. The source-normal-form identity (HC.15) includes every current-dependent contribution once. The exterior force and current-independent constants have zero Hessian; they have not been removed from the corresponding probability or outer law.

The symmetric block Schur estimate bounds the perturbation operator norm by $\eta_0 \le \eta_\gamma$. Combining (HC.9) and (HC.28) gives (HC.29). For (HC.30), (HC.12) loses at most $\lambda_0/4$, and conjugation of the perturbation has both Schur sums at most $\eta_\gamma \le \lambda_0/4$. The remaining lower bound is $\lambda_0/2$. The same distance-weight argument as in Lemma HC.2 gives (HC.31). The domain and derivative statements follow from the predecessor bounds, (HC.25), and the half-reserve intersection (HC.26). The proof differentiates analytic coefficient functions within a fixed history/chart, not moving selectors. Adding a real linear function changes neither the Hessian nor the real support. □

This is a precision/inverse/coercivity theorem. It is not an identity between the inverse Hessian and the interacting covariance. The latter requires the support and Neumann argument already proved in C-07, and its complete moment/decay application belongs to C-08. The theorem does not bound the conditional mean merely from coercivity. It does not assume the missing C-05 estimate and then claim that estimate has been established.

8. Application and remaining application

The new contribution is the actual heterogeneous application: the correct constrained precision and nonlinear constraint-current term; the source eligibility of the regular region; the second-order local background bounds; the current-unit coordinate identification; the bulk local remainder; and the fixed domain/parameter choices. It uses no identity-background extrapolation, no unverified balancing weights, no discarded heterogeneous coefficient, and no extensive determinant estimate.

The heterogeneous coercivity theorem is conditional on the complete regular and boundary derivative estimates for the same conditional representation.

The original C-05 reconstruction and spatial-localization problem remains open, as do H-03 and all later applications not separately established. No four-dimensional Yang–Mills existence theorem is claimed.

Primary references

- Balaban, *Propagators for Lattice Gauge Theories in a Background Field*, CMP 99 (1985), 389–434: (3.35)–(3.38), Section E, (3.155)–(3.158), the lower-bound/locality paragraph on p.428, and Theorem 3.15.
- Balaban, *The Variational Problem and Background Fields in Renormalization Group Method for Lattice Gauge Theories*, CMP 102 (1985), 277–309: Theorem 1, (2), (9)–(10), and the analytic relative-map construction.
- Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories II*, CMP 116 (1988), 1–22: (1.15)–(1.21), (1.37)–(1.40), and the separate treatment of the classical and old-E terms before Lemma 2.
- Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, CMP 119 (1988), 243–285: p.251; (3.2)–(3.5); (3.15)–(3.25); and pp.275–277.
- Existing estimate proofs: C-01/Annex W; C-07/Annex AA; C-04/Annex AH; C-05's conditional boundary output (BV.14)–(BV.18) in Annex AF, whose actual reconstruction premise is expressly still open.

Supplement AI.1. Admission of the heterogeneous principal form

1. Scope and decision The source inputs support a uniform principal-operator theorem on the actual admitted heterogeneous determining geometries and nonconstant backgrounds. This is stronger than extrapolating an identity-background estimate.

A full C-06 theorem may use the declared predecessor conclusions C-04 and C-05 as hypotheses for the regular E/R and complete boundary perturbation sums. It must keep C-05 visibly open and must prove its own principal-operator and remaining classical/measure estimates. The following checks show that this is a noncircular mathematical step: C-05's required derivative estimates do not assume coercivity of the full conditional action, whereas C-06 obtains that coercivity from the source principal operator and the predecessor bounds.

2. The exact Gaussian operator and free coordinates See CMP99, p.428. Formula (3.156) is

$$\langle B, (QG_1Q^*)^{-1}B\rangle - a\langle B, B\rangle - 2\langle H_1\widetilde{D}^{(2)}(B), J\rangle = \langle B, \Delta_k B\rangle. \tag{HA15.1}$$

The background-current contribution from the quadratic nonlinear constraint is part of Δ_k. It must not be dropped or reinserted as an unrelated perturbation.

The constraints are the axial conditions and the one-step linear averaging condition $Q_1B = 0$, together with the Dirichlet condition outside the current region Λ. The source chooses the actual nonexceptional bond coordinates z and writes

$$B = Cz, \qquad K = C^*\Delta_k C. \tag{HA15.2}$$

The map C is the identity on nonexceptional bonds. Its other entries solve the local linear averaging equations and depend only on a fixed number of surrounding current unit blocks. In particular, this is not a change to unrelated curl variables.

The exact identity (3.158) identifies the inverse of K with the restriction of the source covariance to the free bonds. The paragraph immediately after (3.158) states a positive lower bound for K, independent of k and of the admitted background U, under the full conditions (3.35)–(3.36) and sufficiently small $M\alpha_0$. It also explicitly retains uniform exponential decay of K. This passage first mentions the identity-field proof and then states the nonconstant-background extension using the local analysis of its Section B. Reading only the identity-field sentence would omit the source's actual scope.

Theorem 3.15, (3.187), p.432, gives exponential decay of the unit-lattice covariance with constants depending on the fixed dimension and block size. Its underlying operators depend on the entire determining sequence. They are not operators defined only for a homogeneous history.

3. The current domain has the required collar CMP99 Section E, starting on p.427, assumes a current unit-lattice domain $\Lambda \subset A_k = \Omega_k^{(k)}$, consisting of admitted big blocks, with

$$\operatorname{dist}(\Lambda, A_k^c) > R_{\text{prop}} M_{\text{prop}}. \tag{HA15.3}$$

The actual CMP119 construction has this property after a fixed choice of its separation parameters. Equations (3.2)–(3.5), pp.264–265, first surround the old bad region Z_k by several layers of current-size cubes. The new region Ω_{k+1} excludes those layers. The final regular integration region in (3.20), p.269, is a further interior subset of Ω_{k+1}. Since $\Omega_k^c \subset Z_k$, this supplies a collar to the old determining-region complement as well.

The smallest constant needed in (HA15.3) is fixed. The construction's running cube radius is bounded below by a freely chosen fixed minimum, so one chooses that minimum to make the construction's collar larger than the right side of (HA15.3), with a fixed reserve for local stencils and block conventions. This is one parameter choice; it does not depend on cutoff depth or on the individual history.

The source's rectangularization does not enlarge the regular region across a bad set. On p.251 it explicitly takes small connected components of Λ_1^c and replaces them by enclosing rectangles. Page 269 repeats the same operation on components of Λ_{k+1}^c. Thus it enlarges the bad complement and shrinks the regular region. The collar in (HA15.3) cannot deteriorate under this operation. Both pages were in these hypotheses.

The source's periodic block and locality conventions are retained. On an admissible torus the distances are periodic; if the active determining region is the whole torus, the complement condition is void. No rectangular extension of a theorem for arbitrary nonadmitted geometry is assumed.

4. Actual minimizing backgrounds satisfy the full source class CMP99 (3.35) requires a local gauge with physically normalized potential and first-difference bounds. Its additional (3.36) is a second-order bound. A curvature bound alone is not a justification for the latter.

The correct source input is CMP102 Theorem 1, pp.278–279. Its regular space (2) includes the plaquette and physical-current bounds. Its local gauges satisfy (9), and (10) gives both the $\partial^{\eta *}\partial^{\eta} A$ and the component Laplacian bounds. These are exactly the kinds of norms used to admit CMP99's operator construction.

The background in the exact C-01/C-07 current law is the source's constrained Wilson minimum with the actual determining data. The localized coefficient $g_k(x)^{-2}$ is not silently substituted into the defining variational problem: the exact action split separates its difference from the homogeneous current coefficient as a boundary perturbation. Thus the source minimizing-background theorem applies to the principal operator under discussion.

For clarity, the common smallness choice can be made using one uniform ceiling, rather than trying to prove a new running-weight estimate for every operator entry. Let

$$\epsilon_j = A_0 g_j (\log g_j^{-2})^{p_0}, \qquad g_j \le \bar{g}, \qquad \bar{\ell} = \log \bar{g}^{-2} > 2p_0. \tag{HA15.4}$$

The function $g \mapsto g(\log g^{-2})^{p_0}$ is increasing on that interval. Hence

$$\epsilon_j \le \epsilon_* := A_0 \bar{g} \bar{\ell}^{p_0}. \tag{HA15.5}$$

Include the fixed source comparison and local averaging constants in C_{sel}. The actual selected determining data obey the regular input bound with $\epsilon_1 = C_{\text{sel}}\epsilon_*$. The local source gauge constants then give an admissible uniform parameter $\alpha_* = C_{\text{reg}}\epsilon_*$. Choose $\bar{g}$ so that

$$C_{\text{sel}}\epsilon_* \le a_1, \qquad M_{\text{prop}}\alpha_* \le a_{99}, \qquad M_{\text{gauge}} \le R_1 M_1 \frac{a_1}{C_{\text{sel}}\epsilon_*}. \tag{HA15.6}$$

The last inequality is the explicit local-gauge cube-size condition $M \le M(\epsilon_1)$ in CMP102. The fixed M_{gauge} accommodates the cubes used by CMP99. All these inequalities can be met after fixing the geometry and localization constants, because $\epsilon_* \to 0$. They are uniform in the number of layers.

This verification includes the nonconstant physical current and the second-order local gauge information. It is not an extrapolation from $U = 1$. As elsewhere, it is a source-scoped assertion on actual admitted source histories; an independent verification of J0 and the coupling-trajectory construction remain open.

5. The correct norm is a current-scale norm The inner conditional integration is over free bonds of one current lattice. Old determining layers, old fluctuation fields and the retained exterior current are parameters. Consequently the natural quadratic Hilbert norm for this theorem is

$$\|z\|_{\text{cur}}^2 = \sum_{b \in I_\Lambda} \|z_b\|_{\mathfrak{g}}^2. \tag{HA15.7}$$

The coordinate weights on the integrated variables are $\omega_b = 1$. The source's full layer dependence resides in $K(U, \{\Omega_j\})$, the admitted native external norms, and the nonlinear perturbation. Assigning different physical-layer weights to these current bonds would describe a different coordinate space.

For the raw dimensionless fluctuation $q_b = g_k z_b$, the same statement has weights g_k^{-1}. For a physical potential with $q_b = h_k A_b$, the weights are h_k/g_k. These are exact scalar changes of coordinates at the current scale. They do not prove a coercivity theorem on a simultaneous integration over every archived scale.

Native retained directions still use their true layerwise radii. In particular the background Laplace/current restrictions and the boundary reconstructed-data restrictions must not be replaced by a raw unweighted link supremum. Arbitrary selector or history-interface currents remain separate obligations.

6. Remaining classical and measure contributions can be bounded separately CMP116 Section 1 separates the inherited E fluctuation, treated by Lemma 1, from the first, classical part of the fluctuation action. The exact classical algebra after eliminating the constraints is recorded in Annex AI, Section 1. Its quadratic constraint contribution belongs to (HA15.1); every residual classical term begins at degree at least three.

The localized cubic bound in CMP116 (1.37)–(1.39), together with its stated extension to the other classical terms before Lemma 2, gives an analytic activity norm on a fixed absolute free-current disc of radius $\epsilon_{\text{abs}}/g_k$. Its value budget is proportional to $g_k^{-2}\epsilon_{\text{abs}}^3$, with the source's fixed carrier-polynomial factors and exponential decay. If

$$g_k P_k \le \epsilon_{\text{abs}}/4, \tag{HA15.8}$$

there is a translated simultaneous disc of radius at least $\epsilon_{\text{abs}}/(2g_k)$ around every point of the actual support. The BD.2 row estimate, followed by the actual carrier count, therefore yields a classical Hessian budget of order $C_{\text{src}}\epsilon_{\text{abs}}$.

This argument does not need to reconstruct the unread detailed coefficient display (1.43). It uses the source's complete classical decomposition, the proved representative analytic bound and its explicitly stated extension to the remaining classical terms. It also does not double-count the inherited E fluctuation from Lemma 1.

The Haar and exceptional-constraint density can be treated directly in the actual finite unit-block chart. After dividing out the background-only linear coarea factor, the variable determinant is $\det(I + A(q))^{-1}$, with $A(0) = 0$, fixed interaction range and absolute row/column norm at most $C\epsilon_{\text{abs}}$. Uniform invertibility of the exceptional linear coefficient follows from its nonzero finite-block coefficient at the identity and a fixed small-neighborhood estimate, using the admitted local gauge. The logarithmic determinant is its convergent trace-loop series. The anchored loop norm is $O(\epsilon_{\text{abs}})$, hence Cauchy gives a current Hessian budget $O(g_k^2/\epsilon_{\text{abs}})$. The Haar logarithms obey the same conservative bound.

The principal gap is fixed before choosing ϵ_{abs}; take that absolute radius sufficiently small and then take $\bar{g}$ sufficiently small for (HA15.8) and the measure budget. This is compatible with the source's absolute minimizing-map domain and a wider fixed native regularity class. The classical sector has no old-activity dyadic-domain requirement of its own. Boundary-classified outputs, heterogeneous coefficient differences and relocation are assigned to C-05 once, as required by the exact density ledger.

7. Scope of the predecessor-conditional conclusion With the actual principal estimates above, a proof of C-06 can combine:

- C-04's already established regular E/R weighted derivative budget;
- C-05's full boundary derivative conclusion, retained as an unproved predecessor;
- the classical and measure estimates in Section 6;
- the actual precision/covariance bounds in Sections 2–4.

The retained cross term has the form $\langle z, K_{IO} y\rangle$. It changes the conditional mean but has zero current Hessian, so it does not alter coercivity. It must not be declared small or removed from subsequent moment bounds. The shifted Gaussian normalization stays in the outer marginal.

This is a genuine actual-model uniform step conditional on its stated predecessor: its new content is the checked applicability of the source principal operator and complete remaining perturbation estimates on the actual heterogeneous class. It is not an unconditional full-potential theorem before C-05 is supplied, a statement about convexity of the complete positive history space, or a mean/decay theorem for marked observables.

Sources

- CMP99, pp.396–399 and 427–432: source class, uniformity in the determining sequence, exact constrained quadratic form, free-coordinate covariance, nonconstant-background gap and exponential bounds.
- CMP102, pp.278–279, (2), Theorem 1 and (9)–(10): actual minimizing-field regularity including second-order local gauges.
- CMP119, pp.251, 264–270: current geometry, actual support, complement rectangularization, relocation and Gaussian split.
- CMP116, Section 1, (1.19)–(1.21), (1.33)–(1.43): primary equations. Annex B gives the classical and measure bounds within their stated scope.

Supplement AI.2. Covariance, perturbations and domain margins

Decision and exact scope The C-06 implication assumes the regular and boundary derivative estimates of C-04 and C-05. Uniformity on all required Wilson histories remains conditional.

The complete perturbation budget includes the boundary action B.

Checks performed

1. Actual variables and principal form The integrated variables are the current-unit free bonds in the regular inner expression of CMP119 (3.23)–(3.25). Archived layers and the exterior current are fixed parameters. Consequently the Hilbert norm in (HC.1) and the current-lattice weighted matrix norm in (HC.3) are the correct norms for this conditional Hessian. Arbitrary weighted products over all historical layers would describe a different integration space.

The exact precision includes the background-current quadratic constraint contribution in CMP99 (3.156). The source on p.428 explicitly identifies the free precision, its covariance inverse, its nonconstant-background lower bound under (3.35)–(3.36), and its exponential locality. This is an actual heterogeneous source result, not an extrapolation of its preceding identity-background example.

The argument fixes the full current map C before conditioning and then takes the principal matrix restriction. This matters for free bonds whose images include neighboring exceptional bonds; re-solving a different constraint problem inside the inner region would not give the same conditional precision. The stated order preserves both the lower bound and the entrywise decay.

2. Source geometry and background admission CMP99 Section E requires an admitted current domain with a fixed collar to the previous active-region complement. CMP119 (3.2)–(3.5) removes layers around the old bad set, and its final inner region is smaller. The source rectangularization enlarges small components of the bad complement. Thus it cannot destroy that collar. The inequality (HC.5) is a fixed parameter choice, with source-admitted geometries retained.

CMP119 p.256 defines the determining field by its layer values and requires regularity on the determining set before invoking the constrained minimum. CMP102 Theorem 1 supplies the first- and second-order local-gauge bounds needed by CMP99; a plaquette bound is not substituted for the second-order condition. The common ceiling in (HC.6)–(HC.7) is legitimate because $g(\log g^{-2})^{p_0}$ is increasing for $\log g^{-2} > 2p_0$ and tends to zero at the origin. These checks retain the admitted source-free construction and trajectory as inputs.

3. Weighted matrix proof For a 1-Lipschitz distance weight, the entrywise conjugation error has both Schur sums bounded by γA_K. The block Schur test therefore proves (HC.12), with no translation-invariance assumption. For $\gamma A_K \le \lambda_0/4$, the conjugated principal matrix has real part at least $3\lambda_0/4$. Applying its inverse to a vector supported at one bond gives the entrywise exponential bound; the current lattice count gives the row and column sum at half the exponent.

The same argument for the full Hessian uses the weighted perturbation norm and loses another $\lambda_0/4$, leaving $\lambda_0/2$. The resulting inverse matrix estimate in (HC.31) is correct. It is not identified with the interacting covariance. The theorem neither assumes nor proves a bound for the conditional mean.

The counting estimate is uniform on an admitted torus by choosing shortest periodic representatives. A principal restriction only decreases the relevant sums.

4. Exhaustive action accounting The conditional potential retains the exact exterior cross force. This term has zero current Hessian but can affect the mean, so its omission from a coercivity budget does not justify its omission from future moment estimates.

The homogeneous current coefficient supplies the principal quadratic form. The heterogeneous coefficient difference is an exact residual term assigned to C-05. It has not been set to zero. The old E/R activities, inherited B activities, boundary relocation and its Jacobians, bulk classical remainder, and current Haar/constraint density are assigned once.

C-05 is assumed only through its original required complete derivative and domain conclusions. No coercivity assumption is included in this premise. In particular the new theorem does not assume its own principal-plus-perturbation conclusion.

5. Classical and measure bounds on the real support The exact constrained quadratic term must be removed before using cubic vanishing. The algebra retained from CMP109/CMP116 does this, including the quadratic nonlinear-constraint current. The proof of the localized classical bound in CMP116 explicitly extends its treatment to the remaining classical terms. CMP119 pp.275–277 uses the actual proper-scale cube cover and full determining-sequence propagators: interior terms retain the stated regular bounds, and remote or boundary-classified terms are assigned to B.

The absolute current disc is used for the local classical and measure sector only. It is not silently imposed on the narrower old-B domain. The retained exterior current satisfies the smallness bound from CMP119 (3.14), on the whole new active domain before the split, so the full current field remains inside the local map disc when the inner coordinates move in the displayed simultaneous polydisc.

The local Haar logarithms and finite-range nonlinear constraint determinant have convergent analytic logarithms on that disc. The determinant is expanded as anchored loops with a uniform row/column norm; no extensive determinant ratio is used. With the local value estimates, simultaneous Cauchy and the admitted carrier count yield

$$\|D^2 W_{\mathrm{loc}}\|_\gamma \le C'_{\mathrm{cl}}\epsilon + C'_{\mathrm{meas}}\mathrm{g}_k^2/\epsilon.$$

The order of choices, first ϵ and then the coupling ceiling, makes both terms small and does not depend on volume or derivative order.

6. Native domains and absence of a circular premise The complex propagator domain includes the actual second-order background norm and independent-current radius. Neither a first-derivative norm nor an arbitrary retained-coordinate direction is substituted. The actual current-coordinate reserve is pulled back through its g_k-scaled map; one does not incorrectly intersect an $O(P_k)$ current disc with an unconverted $O(1)$ physical-field radius.

The half-reserve intersection in (HC.26) uses C-05's complete reconstructed-output restrictions. If those uniform reserves are not proved, the hypothesis is not available. This retains the precise open reconstruction issue.

The derivation uses the C-01 source normal form directly. C-07 is a downstream support/IBP result and is not a premise. Thus its catalogue dependency on C-06 causes no cycle.

Remaining limits The stated theorem does not establish:

- C-05's actual joined reconstruction, current response, carrier localization or complete boundary budget;
- selector and retained-history currents outside a fixed admitted native chart;
- current-scale conditional means or interacting cumulant decay;
- a full-law history estimate, the local renormalized Schwinger hierarchy, rotation restoration, operator bounds, or phase transfer.

Sources

- CMP99: source class (3.35)–(3.38) and Section E, especially p.428 and (3.155)–(3.158).
- CMP102, Theorem 1, pp. 278–279, (2), (9)–(10).
- CMP119, pp. 251, 256–257, 261, 269 and 275–277, (3.10)–(3.14).
- CMP116 and CMP109 supply the sector decomposition, local map bounds and classical and measure estimates on their stated domains.

No new theorem of the source-free construction, coupling flow or four-dimensional existence is imported through these references.

Annex AJ — Joined reconstruction, fine-collar currents and the remaining boundary application

Annex AJ. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

The full reconstructed-current estimate (BM.24) requires a source-domain pullback. The following metric comparison concerns the localized operator on its actual geometry.

1. Source facts used

See CMP119, pp.255–257 and 261. In particular, (2.14) splices the original determining set in the interior to a newly constructed local determining set in the boundary collar. The reconstruction in (2.35) is written using the determining tuple

$$\mathfrak{B}_p(X) \cup \{\Gamma_i\}_{i<p} \tag{RT.1}$$

and the data are the actual averages $M^\bullet(U)$. The output restrictions (2.35)–(2.37) are imposed on the displayed intersections with X; no interior restriction is printed there. By contrast, the simpler regular reconstruction condition CMP109 (1.16) explicitly uses X^{-2}. The source-localization statement in LFR II, printed p.373, concerns a fixed underlying variational problem and its boundary data.

These facts have different roles. The fixed-geometry localization statement does not identify the data of a separately reconstructed minimizer, and the interior restriction from the regular case is not inserted into the multiscale definition without a proof.

2. A metric lemma for the actual boundary splice

Let an admitted determining geometry have local mesh function $h(x)$, constant on the interiors of its determining blocks. Let X be an admitted source localization domain. On a patch of ∂X lying in original layer r, apply the source construction (2.14): keep the original determining set in the interior and use the local determining set of depth r in its proper-scale collar. Let $\widehat{h}(x)$ be the resulting local mesh.

Lemma RT.1 — localization refines the boundary mesh On such a patch, away from the conventions that assign points lying on block boundaries,

$$\widehat{h}(x) \le h(x). \tag{RT.2}$$

Consequently the path distance defined by integrating inverse local mesh satisfies

$$\widehat{d}(x,y) \ge d_\Omega(x,y) \tag{RT.3}$$

when both sides use the same physical endpoint convention and the left-hand paths are restricted to the localization domain. Passing to the source's determining-block centers changes only fixed endpoint constants. The analogous statement holds for the finitely many boundary-layer patches in the generalized source construction.

Proof. In the interior the determining sets coincide, so the mesh functions are equal. In the collar, the source hypothesis puts the original boundary in layer r, at the stated distance from every original layer boundary lying inside X.

The original local mesh there is h_r. The new minimal local determining set has levels only through r, so all of its local meshes are at most h_r. This proves (RT.2). Every common physical path therefore has at least as large an integral of inverse mesh for the new geometry. The infimum over paths restricted to X is no smaller than the unrestricted old infimum, proving (RT.3). In the multiple-patch version the same pointwise argument is applied on each source proper-scale patch. Endpoint assignment changes distances over only fixed proper-scale collars. □

This argument uses the actual boundary splice. It does not replace (2.14) by an intersection of arbitrary nested sets. Such an intersection generally changes the interior, whereas the source operation leaves the original determining set there.

The extra identification for (RT.1) For comparison, truncating the original determining sequence at p produces

$$\mathfrak{B}^{\langle p\rangle} = \bigcup_{i<p} \Gamma_i \cup \Omega_p^{(p)}. \tag{RT.4}$$

Its mesh is $\min\{h(x), h_p\}$, with the usual block-center conventions. If the tuple (RT.1), with its prescribed data and collars, is identified with the source localization of (RT.4), Lemma RT.1 gives

$$h_{p,X}(x) \le \min\{h(x), h_p\}. \tag{RT.5}$$

The conditional statement is useful, but (RT.5) is not asserted for (RT.1) solely from its notation. The joining symbol is used in different argument orders in the sources, and a union with the incomplete family $\{\Gamma_i\}_{i<p}$ must specify how the remaining coarse determining data and boundary blocks are supplied. The data remain $M^\bullet(U)$, regardless of how the geometry is described.

3. Fixed-lattice dilation and genuine reconstruction

For a fixed bond graph with the same link matrices, a change of units $x' = \lambda x$, $\eta' = \lambda\eta$ gives

$$F_{\eta'} = \lambda^{-2} F_\eta, \qquad J_{\eta'} = \lambda^{-3} J_\eta. \tag{RT.6}$$

The raw plaquette variable dU is unchanged. These are direct substitutions in $F_\eta = \eta^{-2}\pi \operatorname{Im} dU$ and $J_\eta = D^*_{\eta,U} F_\eta$. They do not compare two different constrained projectors or variational problems.

In a reconstruction indexed by p, the exact weights from CMP119 are

$$w_F(n,p) = L^{-2p}(L^n L^{-p})^{-2} = L^{-2n}, \tag{RT.7}$$

$$w_J(n,p) = L^{n-\min(p,n)}(L^n L^{-p})^{-3} = \begin{cases} L^{3(p-n)}, & n \le p, \\ L^{-2(n-p)}, & n > p. \end{cases} \tag{RT.8}$$

Thus the smoother case $n > p$ asks for a two-power gain in the reconstructed current. It is not obtained by treating all output currents as having the same unweighted supremum norm. Nor is the change in units (RT.6) an identity between the original and reconstructed minimizers.

4. What the local two-power gain proves, and where it loses a power

Take a local determining block of mesh $h_s \le h_n$ inside an original smooth layer n. Suppose the prepared input variation satisfies the already proved original-geometry bounds

$$|\delta A| \le u_n h_n^{-1}, \qquad |\nabla \delta A| \le u_n h_n^{-2}. \tag{RT.9}$$

Lemma BM.2 gives the response of its actual gauged determining average:

$$|\delta B_s| \le C u_n (h_s/h_n)^2. \tag{RT.10}$$

For a generic application of the minimizing-map derivative kernel, the normalized curvature and current output rows have scales h_s^{-2} and h_s^{-3}, respectively. Thus (RT.10) gives the curvature estimate

$$h_s^{-2}|\delta B_s| \le C u_n h_n^{-2}, \tag{RT.11}$$

with no loss when the determining collar is finer than the original layer. For the current it gives only

$$h_s^{-3}|\delta B_s| \le C u_n h_s^{-1} h_n^{-2}. \tag{RT.12}$$

The desired current scale in unrescaled units is

$$u_n h_{\min(n,p)}^{-1} h_n^{-2}. \tag{RT.13}$$

Their ratio is $h_{\min(n,p)}/h_s$. It can be arbitrarily large at a fine collar. Since the input and output can lie in the same determining block, there need not be any positive endpoint distance that could absorb this ratio. A scalar exponential in $k-n$ does not repair it either: the ratio depends on the collar refinement even with $k-n$ fixed.

This is a failure of this particular norm estimate, not a counterexample to the actual reconstruction theorem. It identifies the cancellation or regularity statement that a successful proof must preserve. The refined metric and the local two-power averaging gain are insufficient by themselves.

5. A kernel-composition statement with explicit scope

The following elementary statement makes precise what can be done after the geometric and output-norm issues are settled. It is not one of the catalogue estimates.

Let $K_1(x,y)$, $K_2(y,z)$ be nonnegative normalized kernel majorants on a common measured discrete space. Assume

$$\sup_x \sum_y \mu(y) e^{ad(x,y)} K_1(x,y) \le C_1, \qquad \sup_y \sum_z \mu(z) e^{ad(y,z)} K_2(y,z) \le C_2. \tag{RT.14}$$

Then, for their composition with the same normalized measure,

$$\sup_x \sum_z \mu(z) e^{ad(x,z)} (K_1 K_2)(x,z) \le C_1 C_2. \tag{RT.15}$$

Proof. Use $d(x,z) \le d(x,y) + d(y,z)$, exchange the nonnegative sums, apply the second row bound, and then the first. □

If changing from one output normalization to the other inserts a factor bounded by $Ce^{b|r-s|}$, and every complete layer crossing costs at least RM in d, reserve a rate $b/(RM)$ plus a fixed margin from the original exponential before applying (RT.15). The fixed missing endpoint crossing is included in C. A positive endpoint rate and an age rate can then be retained simultaneously.

This composition principle pays genuine mesh ratios between separated layers. It does not pay the zero-distance current loss (RT.12). It also does not create a localized walk expansion: the prepared source kernel expansions must supply that expansion, with the same normalized measures and with sufficient exponential reserve for the carrier count.

6. Exact cancellation for a structured residual

This section uses the actual operators of CMP99, Section D, printed pp.417–426. The argument remains on one admitted determining geometry throughout. It does not identify two minimizers on different geometries or change the source's averaging map.

6.1 Definitions and conventions Use the source's physical inner products: fine-lattice counting measure is multiplied by η^d, and the measure of a determining variable in a block of side h_b is multiplied by h_b^d. Let Q be the dimensionful linearized averaging operator, so that the dimensionless determining variation in (3.110) is $h_b(QA)(b)$. Let D be the covariant derivative, R the source scalar projection, and G' the scalar propagator. Define

$$K = DG'RD^*, \qquad \Pi = I - K. \tag{RT.16}$$

The source gauge projection identities give

$$K^2 = K, \qquad QK = 0, \qquad RD^*\Pi = 0. \tag{RT.17}$$

In particular, Π takes a field into the chosen gauge slice without changing its determining average. These are the identities used in CMP99 (3.115), (3.119), and the derivations following them.

Write Δ for the Wilson quadratic Hessian and D_2 for the symmetric operator defined by the quadratic form in (3.134):

$$\langle A, D_2 A\rangle = 2\langle HC^{(2)}(A), J\rangle. \tag{RT.18}$$

Here H is the earlier minimizing operator in (3.126); it is distinguished from H_1 below. The normalization of $C^{(2)}$ is matched to this dimensionful H, as specified in the paragraph following source (3.127); the alternative convention that moves a factor of h between H and $C^{(2)}$ is not mixed with it. To fix the sign directly from the defining quadratic form (3.128), set

$$T = \Delta - D_2, \qquad L = \Pi^* T \Pi, \qquad G_1^{-1} = L + DRD^* + Q^* a Q. \tag{RT.19}$$

Thus L denotes the complete gauge-invariant Hessian appearing in (3.128); the argument does not depend on the signs used in later shorthand names for its two summands. All inverses below are taken on the source's specified finite spaces, with the common null modes removed as in the construction.

The exact minimizing operator, projection, and constrained covariance are

$$H_1 = G_1 Q^* (QG_1 Q^*)^{-1}, \qquad \psi = I - H_1 Q - G_1 DRD^*, \qquad \mathcal{G} = \psi G_1 = G_1 \psi^*. \tag{RT.20}$$

These are CMP99 (3.129), (3.147), and (3.153). In particular, the last expression is $G_1\psi^*$, with the adjoint; it is not generally $G_1\psi$. The source also proves

$$G_1 DR = DG'R, \qquad RD^* G_1 = RG'D^*. \tag{RT.21}$$

The stars in this subsection refer to the real physical inner products. Their holomorphic continuations mean the corresponding bilinear transposes, as required when continuing the algebraic identities to complex coordinates. No conjugate-dependent operation is inserted into a holomorphic source map.

Proposition RT.2 — exact structured-residual identity For these operators on one fixed admitted geometry,

$$\mathcal{G}L = \psi, \qquad L\mathcal{G} = \psi^*, \qquad H_1 QA = A - KA - \mathcal{G}LA. \tag{RT.22}$$

If a forcing has the structured form $f = LA$, then

$$RD^* G_1 f = 0, \qquad L\mathcal{G}f = f - Q^* H_1^* f. \tag{RT.23}$$

Proof. The range of ψ, and hence of $\mathcal{G}$, is the simultaneous kernel of Q and RD^*. Self-adjointness of the real covariance therefore gives

$$Q\mathcal{G} = 0, \quad RD^*\mathcal{G} = 0, \quad \mathcal{G}Q^* = 0, \quad \mathcal{G}DR = 0.$$

Multiplying the identity for G_1^{-1} in (RT.19) on the left by $\mathcal{G}$, and using (RT.20), gives $\mathcal{G}L = \psi$; taking the adjoint gives $L\mathcal{G} = \psi^*$. Equations (RT.20)–(RT.21) then imply the last identity in (RT.22).

Because $\Pi DG'R = 0$, the definition of L gives $LDG'R = 0$. Transposing and using (RT.21) yields $RD^*G_1L = 0$. For $f = LA$, the last term in

$$\psi^* f = f - Q^* H_1^* f - DRD^* G_1 f$$

therefore vanishes. This proves (RT.23). □

The printed derivation (3.145) first writes its cancellation with the covariance constrained only by Q, denoted $\overline{G}_1$. Its difference from G_1 is

$$G_1 Q^* (QG_1 Q^*)^{-1} QG_1.$$

The source identity $RD^*G_1Q^* = 0$ kills this difference. Thus the full- G_1 cancellation in (RT.23) agrees with the source calculation; no distinction between the two covariances has been suppressed.

The proof is algebraic in T. In particular, it remains valid if the coefficient J of the quadratic form (RT.18) is an independently specified small current, provided all displayed operators are constructed together from that same quadratic form and the same gauge projection, on a domain where their inverses exist. It is not an inference from the lower-dimensional relation $J = J(U)$. Conversely, it does not prove that an arbitrary prepared family obtained by separately altering kernels continues to satisfy those identities. That matching must be checked for the actual preparation parameters.

Proposition RT.3 — weighted adjoint-minimizer bound Here a genuine part of the current normalization can be verified from the source estimates. Work in dimension $d = 4$, with one admitted determining geometry satisfying CMP96 Lemma 2.1. Let $\delta > 0$ be an exponential rate in the value estimate (3.133) for H or H_1. Reserve a positive rate κ such that

$$\delta - \kappa - \frac{\log L}{RM} > 0, \tag{RT.24}$$

and retain a further positive margin for the summation condition of that lemma. This is an explicit condition on the chosen geometric constants; it is not asserted for an arbitrary smaller choice of R, M.

For any source set A_0 of determining blocks, define

$$\|f\|_{-3,\kappa;A_0} = \sup_a h_a^3 e^{\kappa d(a,A_0)} \sup_{x\in\Delta(a)} |f(x)|. \tag{RT.25}$$

Then

$$\sup_b h_b^3 e^{\kappa d(b,A_0)} |(H_1^* f)(b)| \le C\|f\|_{-3,\kappa;A_0}, \tag{RT.26}$$

with a uniform constant on the source geometries and regular backgrounds covered by the cited estimate. The same assertion holds with H in place of H_1.

Proof. The dimensionful convention in (3.126), (3.129), and (3.133) gives

$$|H_1(x,b)| \le C h_b^{-d} e^{-\delta d(a,b)}, \qquad x \in \Delta(a). \tag{RT.27}$$

Taking the adjoint with the physical fine and coarse measures and grouping the fine variables into their determining blocks gives

$$|(H_1^* f)(b)| \le C \sum_a h_a^d h_b^{-d} e^{-\delta d(a,b)} \sup_{\Delta(a)} |f|. \tag{RT.28}$$

Fixed bond-direction multiplicities and endpoint conventions are included in C. By the triangle inequality for the source distance,

$$h_b^3 e^{\varkappa d(b,A_0)} |(H_1^* f)(b)| \le C \|f\|_{-3,\varkappa;A_0} \sum_a \left(\frac{h_a}{h_b}\right)^{d-3} e^{-(\delta-\varkappa)d(a,b)}. \tag{RT.29}$$

In dimension four the displayed mesh ratio has exponent one. If a, b have layer indices r, s, CMP96 (2.60) at printed p.234, gives

$$d(a,b) \ge RM \max\{|r-s|-1, 0\}, \qquad \frac{h_a}{h_b} \le L \exp\left(\frac{\log L}{RM} d(a,b)\right). \tag{RT.30}$$

The reserved rate in (RT.24) and the exponential row sum (2.61) now bound (RT.29). This proves (RT.26). □

This is a source-weighted normalized kernel estimate, not merely a qualitative boundedness assertion. It retains an endpoint exponential from an arbitrary fixed source set. Its unweighted specialization agrees with the bound on H^*J stated immediately after CMP99 (3.136).

Using the dimensionless determining operator $Q_B = hQ$ would replace H_1 by $H_{1,B} = H_1 h^{-1}$. The combination occurring in (RT.23) is unchanged:

$$Q_B^* H_{1,B}^* = Q^* H_1^*. \tag{RT.31}$$

Therefore a change between those conventions must not create or erase a power of mesh in the multiplier term.

6.2 What remains to convert the cancellation into a physical current bound Equation (RT.23) controls the complete constrained Hessian. The current required in BM.24 is obtained from the actual Wilson field. It is not legitimate to identify these operators without retaining the remaining terms.

Set $a = \mathcal{G} f$. Since $\Pi a = a$, the definitions give the exact identity

$$\Delta a = La + D_2 a + K^*(\Delta - D_2)a. \tag{RT.32}$$

For a structured residual this becomes

$$\Delta \mathcal{G} f = f - Q^* H_1^* f + D_2 \mathcal{G} f + K^*(\Delta - D_2)\mathcal{G} f. \tag{RT.33}$$

Two pieces of this identity now have the correct source weights. Proposition RT.3 proves the weighted H_1^* estimate. Also, the value bound for $\mathcal{G}$ in CMP99 Theorem 3.13, combined with (3.137), gives

$$\|D_2 \mathcal{G} f\|_{-3,\varkappa';A_0} \le CM\alpha_0 \|f\|_{-3,\varkappa;A_0} \tag{RT.34}$$

for a suitably reserved $0 < \varkappa' < \varkappa$. Indeed, the covariance value estimate takes a current of size h^{-3} to a field of size h^{-1}, while (3.137) multiplies the local field bound by $CM\alpha_0 h^{-2}$. The finite neighboring-block support in (3.137) costs only a fixed endpoint factor. This use of Theorem 3.13 needs no Laplacian estimate.

The complete estimate nevertheless still needs the actual local Q^* bound in the source measures and control of the last term of (RT.33). Neither follows merely by naming K a gauge projection. For example, $K^* D_2 a = DRG' D^* D_2 a$ contains a differentiated scalar propagator on both sides of a potentially rough input. The corresponding scalar estimate in CMP99 (3.44) requires additional input regularity; a uniform supremum bound does not follow from (3.137) alone.

The source gauge identity (3.117) does identify a useful cancellation in part of that last term. At a real background with its actual Wilson current J_U, differentiating the quadratic gauge identity once in a gauge parameter yields

$$D^* \Delta a = \mathcal{B}_{J_U} a, \tag{RT.35}$$

where $\mathcal{B}_{J_U}$ is a local zeroth-order operator. Explicitly, with the source parallel transport R_b, its bilinear form is

$$2\langle \lambda, \mathcal{B}_{J_U} a\rangle = \langle i[\lambda(b_-), a(b)] - i[a(b), R_b\lambda(b_+)], J_U(b)\rangle\,. \tag{RT.36}$$

Hence $|\mathcal{B}_{J_U} a| \le C|J_U||a|$, with only the finite endpoint shifts, and

$$K^*\Delta a = DRG'\mathcal{B}_{J_U} a. \tag{RT.37}$$

This replaces a third derivative by a scalar propagator and a local current vertex. It does not replace J_U by an independent source coordinate. Nor does it control the separate K^*D_2a term or identify the forcing produced by the joined reconstruction. The formula is recorded to make the remaining cancellation problem precise.

The scope restriction is necessary: CMP99 Theorems 3.12 and 3.13 explicitly exclude the covariant-Laplacian bound for G_1 and $\mathcal{G}$. Their value, first-derivative, and Hölder estimates therefore cannot be cited as an unconditional bound for $\Delta\mathcal{G}$. Equations (RT.22)–(RT.37) show what additional structure can replace that missing generic bound and which terms still require an estimate.

Equation (RT.33) is a Wilson quadratic-Hessian identity. Its transfer to the nonlinear current in the actual local Lie-algebra chart, including the current and chart conversion terms, remains part of BM.24.

Finally, to use this cancellation at a fine collar one needs a smooth tangent lift on the reconstructed geometry whose actual determining variation equals that of the original prepared field. A comparison of tangent fields in two separately chosen gauges does not establish this equality. The paired gauge identification, any correction of the averaging mismatch, its physical current norm, and its compatibility with independent J and complex preparation remain part of the actual-model application.

7. Remaining actual theorem

To finish (BM.24), the proof still needs all of the following in one compatible construction:

1. A checked identification of the actual joined determining geometry (RT.1), its data, and the relevant scale conversion, or a direct estimate that does not need this identification.
2. A reconstructed-current estimate preserving the smooth input response through its fine collars. A generic first derivative kernel applied to (RT.10) has the explicit deficit (RT.12).
3. Joint validity for the independently extended source current and the actual complex preparation parameters, with a strictly admitted domain before Cauchy differentiation. Equality on the physical relation $J = J(U)$ cannot justify that joint extension.
4. The marked-path or equivalent local expansion retaining a positive endpoint rate, so the boundary carrier and collar sums can be charged as in the complete sector analysis.

The source's claimed multiscale analytic extension is relevant evidence, but this note does not replace the missing quantitative current-coordinate application by that claim. The present result is therefore a source-checked structural improvement and an explicit obstruction to a shortcut, not a completed C-05 proof.

Primary references and verification scope

- T. Balaban, CMP119 (1988), 243–285: (2.14), (2.34)–(2.39), and the discussion of (3.43) in this and the argument.
- T. Balaban, CMP102 (1985), 277–309: Proposition 9 and (190), with the five different physical output powers retained.
- T. Balaban, CMP99 (1985), 389–434: (3.115)–(3.153), especially the defining quadratic form (3.128), projection (3.147), covariance identity (3.153), kernel estimate (3.133), and the explicit Laplacian exceptions in Theorems 3.12–3.13. Printed pp.420–426 were for the operator signs, adjoints, and constraints.
- T. Balaban, CMP96 (1984), 223–250: Lemma 2.1, (2.60)–(2.61), printed p.234 for the layer-distance and normalized summation bounds.
- T. Balaban, CMP109 (1987), 249–301: article, (1.15)–(1.16) and the explanation immediately following (1.16).
- T. Balaban, CMP122 (1989), 355–392, LFR II: (1.61) and the associated fixed-geometry localization statement, pp.373–375.

Supplement AJ.1: Complete boundary-sector application and a local Wilson difference bound

This part checks the downstream boundary application and proves a useful local Wilson-action estimate. The joined reconstruction estimate (BM.24), including its spatially localized version, is not obtained here. The C-05 statement and its application conditions are unchanged.

1. What the primary source actually localizes See CMP119, pp.251, 254–255, 261–262, 269 and 276–277. The complete boundary ledger of Annex AF, Section 2, is retained. In particular:

- the current Gaussian quadratic form and its retained cross force stay in the exact Gaussian split;
- the classical and nonlinear-constraint terms, Haar and coarea factors, inherited E/R terms assigned to B, inherited B terms, heterogeneous Wilson coefficient and relocation/Jacobian are all present;
- the older cross-component Gaussian terms already assigned to B are covered by the inherited B sector;
- retained-only boundary constants cancel only from the conditional normalized law, and remain in the outer marginal;
- the relocation cutoff is smooth in the old real variable, which remains a parameter. Its analyticity in the small relocation field does not assert analyticity of arbitrary old-selector derivatives.

CMP119 p.276 applies the complete generalized random-walk expansions of CMP99 Theorems 3.7–3.10 to the full determining sequence. It describes the localized terms by current-scale domains and retains only the genuinely regular domains in E/R. Its boundary terms use the separate multiscale domain (3.43). On p.277 the B-origin fluctuation expansion retains the primitive full-tree decay, together with the small response factor. The heterogeneous coefficient and relocation terms are treated immediately afterward.

These are useful primary statements, but their analyticity domain is not a substitute for showing that the actual proposed current-coordinate tube is mapped into it. The proof must retain the full tuple of conditions (2.34)–(2.39), including the reconstructed field and current.

2. A fine-spacing-independent Wilson difference estimate Use any fixed matrix norm on the complexification of the defining representation of SU(N). On a neighborhood of the identity in the invertible matrices define the holomorphic Wilson plaquette function

$$s(P) = N - \frac{1}{2}\operatorname{Tr}(P + P^{-1}). \tag{BC15.1}$$

For a unitary plaquette this is exactly the scalar plaquette observable $N - \operatorname{Re}\operatorname{Tr} P$. No change to the source coefficient dictionary is made.

Lemma BC15.1 — local curvature differences pay the fine-cell count Work in dimension $d = 4$, and let $0 < \eta \le h$. Suppose two plaquette matrices, expressed in one common local trivialization, obey

$$\|P_0 - I\| \le a(\eta/h)^2, \qquad \|P_1 - P_0\| \le u(\eta/h)^2, \qquad a + u \le \tfrac{1}{4}. \tag{BC15.2}$$

Then

$$|s(P_1) - s(P_0)| \le C_N(au + u^2)(\eta/h)^4. \tag{BC15.3}$$

If a proper-scale localization cube has side at most $C_b h$, and its fixed partition function has modulus at most C_ζ, summing the corresponding plaquette differences in that cube gives

$$\left|\sum_p \zeta(p)[s(P_1(p)) - s(P_0(p))]\right| \le C_{N,d,C_b,C_\zeta}(au + u^2). \tag{BC15.4}$$

The constant does not depend on η/h.

Proof. The segment $P_t = P_0 + t(P_1 - P_0)$, $0 \le t \le 1$, lies in the ball of radius 1/4 about the identity. It need not lie in SU(N): the holomorphic extension (BC15.1) is defined there. Differentiating and using cyclicity of trace gives

$$Ds(P)[H] = \frac{1}{2}\,\mathrm{Tr}[(P^{-2} - I)H]. \tag{BC15.5}$$

On that ball, the inverse identity or its convergent geometric series gives $\|P^{-2} - I\| \le C\|P - I\|$. Thus

$$|Ds(P_t)[P_1 - P_0]| \le C_N(a + u)u(\eta/h)^4.$$

Integrating proves (BC15.3). The number of plaquettes in the cube is at most $C_{d,C_b}(h/\eta)^d$. For $d = 4$, this cancels the fourth power in (BC15.3). The bounded partition function contributes only its stated constant. This proves (BC15.4). □

The proof works on a complex admitted contour and is consequently suitable for subsequent Cauchy differentiation. It must use a common chart, or an explicitly bounded gauge comparison. It does not compare unrelated plaquette matrices in different moving gauges.

The primitive output estimate needed in (BC15.2) is a curvature estimate, not merely an undifferentiated link estimate. In the original determining geometry it is supplied by the prepared-field/current estimates (BM.10)–(BM.11) and the exact local current/plaquette calculation in Annex AD. For a reconstructed output on the joined geometry, it remains part of the actual transfer theorem.

3. The heterogeneous coefficient does not need a beta-function sign assumption The source coefficient is the localized recursion CMP119 (2.24). Write $w_l = g_l^{-2}$, $\beta_l = w_{l-1} - w_l$. Then

$$w_k(x) - w_k = \sum_{l=1}^{k} \beta_l(1 - \phi_l(x)). \tag{BC15.6}$$

At a point whose earliest nontrivial cutoff index is n, the uniform cutoff bound and the admitted coupling comparison yield

$$|w_k(x) - w_k| \le C(1 + k - n)^2 g_k^{-2}. \tag{BC15.7}$$

Indeed, bound $|\beta_l|$ by $w_{l-1} + w_l$, use $w_l \le C(1 + k - l)w_k$, and sum. No monotonicity or sign of β_l is used.

The earliest active cutoff layer must still be identified with the appropriate geometrical layer, up to the source's fixed collar convention, when applying a layerwise kernel bound. Interlacing alone is not that check. The construction of Ω_{l+1} in CMP119 (3.2)–(3.5) surrounds the prior bad set before admitting the next region; the fixed cutoff $\phi_l = 1$ on the prescribed interior of Λ_l is used together with that collar. The enlarged bad-complement rectangles do not weaken a regular-to-bad separation: pp.251 and 269 enlarge components of Λ_l^c, so the regular region only shrinks.

Combining (BC15.4), (BC15.7) and a verified original-geometry response

$$u_n \le C g_k P_k\, e^{-cR_k} e^{-\Gamma(k-n-c_0)_+} \tag{BC15.8}$$

gives a local heterogeneous-action value bound proportional to

$$g_k^{-2}(g_k P_k)(1 + k - n)^2 e^{-cR_k} e^{-\Gamma(k-n-c_0)_+}. \tag{BC15.9}$$

Here (BC15.8) has the actual current-to-old-boundary connector: the factor e^{-cR_k} cannot be asserted at an arbitrary point without that boundary anchor. The original-geometry connector in (BM.14) supplies the needed distinction.

This calculation closes the fine-spacing normalization in the heterogeneous sector. A naive Lipschitz estimate on individual link matrices would lose powers of η; the curvature-square structure in (BC15.5) avoids that error. The localized carrier sum still uses the actual prepared-kernel expansion described next.

4. The exact extra interface required of the joined reconstruction An age-only version of (BM.24) proves common-domain admission but does not, by itself, prove an anchored spatial boundary bound. The actual transfer needed by the full C-05 application must retain all of the following:

1. Joint admission. The full reconstructed field/current tuple is holomorphic on one simultaneous tube containing $\|z\|_\infty < 2P_k$, with all required native external and preparation variables. Its normalization is exactly that in (2.34)–(2.39), including the two (p, n) weights recorded in (BM.23).
2. A marked endpoint. A current-to-output response attached at an old output point y retains a positive exponential in the actual joined-geometry path distance from the current input. A scalar age norm with the attachment point forgotten is insufficient.
3. Normalized rows. Its input measure and output normalization yield a uniform kernel row bound, rather than a sum over unnormalized fine bonds.
4. Prepared localization. The same statement holds under the source's complex decoupling parameters, with sufficient unspent walk/tree decay to produce exact current-scale carriers.
5. External directions. The admitted native background/current directions have quantitative radii. A raw field direction is not assigned such a radius merely because the composite activity is analytic on a nonempty set.

The primitive decay (2.42) is a full-tree bound. It therefore combines with a marked minimizing-map path. The path and primitive tree together connect the current anchor to the old boundary. As in (BV.6)–(BV.9), divide the available exponentials into separate shares for the current collar, layer age, row/carrier summation and final spatial localization. One must not spend the same exponential twice.

CMP99 Theorem 3.10 explicitly supplies generalized walk localization and endpoint estimates for admitted determining geometries. The discussion on p.427 also treats the tree-shaped expansions generated by the background-current term. CMP116 (1.7)–(1.11) uses separate walk-localization and endpoint factors: a sufficiently large fixed localization block allows complex decouplings while leaving a positive endpoint rate. Its (1.10) and (1.23) then localize the composite analytic functions; disconnected decoupling derivatives vanish. CMP119 p.276 transfers this procedure to the full multiscale cover.

Those primary facts apply to the joined reconstruction only after its determining geometry, all inputs and output norms have been verified. No same-lattice minimizer-localization identity is used to identify two different reconstructed minima.

5. Quantitative completion once that actual transfer holds This section is an implication with the precise actual-map premise of Section 4. It is not a declaration that the premise has been proved.

Fix the localization parameters first. Reserve enough primitive and prepared decay for the anchored cube count of Annex AH. Choose the effective age rate, after all exponent shares, so that

$$\Gamma > 4\log L + \log 2 + \delta, \qquad \delta > 0. \tag{BC15.10}$$

Increase the fixed admissible separation constant to meet the normalized row entropy condition. Choose one fixed $q > p_0$, one $\kappa_0 \geq 12$, and one geometric cutoff exponent $r > 1$. Only after these choices reduce the common coupling ceiling.

For $\ell_k = \log g_k^{-2}$, put

$$P_k = A_1 \ell_k^{p_0}, \qquad \delta_k = g_k P_k, \qquad \theta_k = \ell_k^{p_0 - q}. \tag{BC15.11}$$

The old B derivative consumes a factor $2^{j-n}/\alpha_{*,n}$. The common-contour map contributes $g_k P_k e^{-\Gamma(k-n)}$, and the running-radius comparison contributes at most $C\sqrt{1+k-n}\,\theta_k$. At fixed $a = k - n$, there are at most $1 + a$ choices of j. Even retaining a conservative L^{4a} old-cell count, the resulting sum is bounded by

$$CB_0\theta_k \sum_{a \geq 0} (1+a)^b e^{-\delta a}, \tag{BC15.12}$$

where one fixed b includes all finite polynomial age costs. The current collar is helpful but is not needed for this row after the common contour has been admitted. It remains indispensable in the heterogeneous row (BC15.9).

The resulting complete conditional boundary fluctuation can be localized as

$$\widetilde{B}_k = \sum_Y b_{k,Y}, \qquad \sup_{\mathcal{C}_k} |b_{k,Y}| \le K_{B,k} e^{-\varkappa_B d_k(Y)}, \tag{BC15.13}$$

with the conservative prefactor

$$\begin{aligned} K_{B,k} \le C_{\mathrm{src}}(1+S_{b,\delta})\big[&\mathfrak{K}_E + C_R g_k^{\varkappa_0 - 11} + B_0\theta_k \\ &+ g_k^{-2}\delta_k^3 + \delta_k + g_k^{-2}\delta_k e^{-cR_k} + g_k^{\nu}\big], \qquad S_{b,\delta} = \sum_{a\ge 0}(1+a)^b e^{-\delta a}. \end{aligned} \tag{BC15.14}$$

The first two terms include E/R-origin carriers meeting the newest boundary, for which no old-boundary collar gain is asserted. The third comes from the inherited B terms. The next two are the cubic classical and measure remainders. The sixth is the heterogeneous term whose fine-spacing normalization is proved in Lemma BC15.1. The final term is the relocation/Jacobian contribution, for any one fixed positive power ν, using the actual exponentially small relocation field and the same complete-action localization. Older cutoff variables stay fixed in this native complex estimate.

The exact current Gaussian cross force is not included in (BC15.14). It remains in the conditional Gaussian mean.

Let $r_z = c_G P_k$ be the scalar-coordinate radius retained by the common contour. The simultaneous-row estimate BD.2 and the actual anchored carrier count give

$$\sup_i \sum_l e^{\gamma\rho_k(i,l)} \|D_i D_l \widetilde{B}_k\|_{\mathrm{op}} \le C_{G,\gamma} K_{B,k} P_k^{-2}. \tag{BC15.15}$$

The value and gradient bounds have respectively no factor P_k^{-1} and one factor P_k^{-1}. For fixed local directions in the admitted slots, define q_s as their norm divided by the actual slot radius. The same carrier sum gives

$$\sum_Y e^{\gamma\,\mathrm{diam}(I_Y)} |D_{h_1}\cdots D_{h_m} b_{k,Y}| \le C N(S_1) K_{B,k} e^m m! \prod_{s=1}^m q_s. \tag{BC15.16}$$

Here S_1 is the support of any chosen first direction, and $N(S_1)$ counts its incident localization cubes. Higher coefficients of an additional nonlinear parameter-to-slot map remain separate chain-rule terms. This estimate is not a dimension-free sum of all absolute order-m tensor coefficients.

All terms on the right side of (BC15.15) tend to zero with the common coupling ceiling. The classical term gives $g_k P_k$; the bounded E term gives $O(P_k^{-2})$; the heterogeneous term gives $g_k^{-1}P_k^{-1}e^{-cR_k}$, which tends to zero because $R_k \ge \ell_k^r$ and $r > 1$. Thus the completed map interface would yield an arbitrarily small uniform boundary Hessian budget, with no hidden volume or distance factor.

6. Application conditions The sector list and downstream numerical/carrier deductions are now explicit. Lemma BC15.1 proves the actual local Wilson normalization needed by the heterogeneous sector. The primitive source localizations have the appropriate form for the final sum.

The joined-reconstruction transfer, in the stronger spatial/prepared form in Section 4, is still required before applying this conclusion to the full boundary action. In particular, the existence of a holomorphic activity on (3.43), the original-geometry bound, or the local two-power averaging result does not establish that transfer.

C-05 remains open in this note.

Primary sources

- T. Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, CMP 119 (1988), 243–285, especially (2.24), (2.34)–(2.42), and pp.269, 276–277.
- T. Balaban, *Propagators for Lattice Gauge Theories in a Background Field*, CMP 99 (1985), 389–434, Theorems 3.7–3.10 and the tree-expansion discussion on p.427.
- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. II. Cluster Expansions*, CMP 116 (1988), 1–22, (1.7)–(1.11), (1.23)–(1.36). DOI: https://doi.org/10.1007/BF01239022.

Annex AK — Full-law reset-jet bounds for moment-controlled marks on an admitted chart

Annex AK. Exact finite conditional identities apply to the specified positive kernel. A uniform Wilson application requires CL on the same retained sigma-field, the relevant complete-potential estimates under CR and the stated spatial and energy budgets. History estimates · Observable transport.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

Scope

Neither H-03 nor H-02 is completed by this note. The new result removes a restriction from the previous native-chart reset theorem: the incoming marks need not be bounded or analytic in the native background. Their full-law moment bounds suffice for the newly generated kernel currents and for the entire frozen-input reset jet. The theorem uses the actual normalized old-component posterior and keeps its nonsmall self action. It does not replace that posterior by the fresh R reference.

The actual joined multiscale coordinate map, the pointwise regulator estimate and the sector/interface part of the M-03 norm remain additional requirements. The exact remaining requirements are listed in Section 6. In particular, closing the boundary reconstruction estimate (BM.24) would not, by itself, close all of H-03.

1. Source input and the full-law disintegration

We retain the fixed eligible component geometry of Annex AG. Write its actual posterior as

$$\pi_x(d\eta) = \frac{e^{H_x(\eta)} M_x(d\eta)}{M_x(e^{H_x})}, \qquad H_x = \sum_{Y\,:\, i\in J(Y)} V_Y. \tag{HP.1}$$

Here x contains the retained native data and the other components' variables, while η contains every variable and old-history label integrated by the component operation $\mathsf{T}'_k(X_i)$. The self-component action from LFR II (1.69), the old characteristic functions, graph sums, reference factors and integration Jacobians are inside M_x. No smallness of that self action is assumed. The law of (x,η) is

$$d\mu(x,\eta) = d\nu(x)\,\pi_x(d\eta). \tag{HP.2}$$

The primary input is LFR II, CMP 122 (1989), pp.379–380, (1.71)–(1.75). On the source's admitted fixed native domain the component integral is its real positive integral times a bounded analytic exponential. The dependence on the native (U,J) variables occurs in the quadratic form and the self-component action; all the original integrated histories remain present. The source controls the variation of that large self action by its localized connections to the current input. Its normalized bound does not estimate the reciprocal of a small real component mass separately.

The exact defining representation of (1.73) permits the same domination argument for integrable marks. One must first verify that a proposed coordinate variation stays within the source domain (1.65). We assume such an admitted chart in Sections 2–4, and do not infer one from the completed regular R activity theorem.

For definiteness, let $|z| < \rho$ parameterize one native direction at each real datum x, with a common fixed support for η. On a sufficiently small concentric disc the full posterior has the representation

$$\pi_{x,z}(d\eta) = r_{x,z}(\eta)\pi_x(d\eta), \qquad r_{x,z}(\eta) = \frac{e^{\theta_x(z,\eta)}}{\pi_x(e^{\theta_x(z,\cdot)})}, \quad \theta_x(0,\eta) = 0. \tag{HP.3}$$

For the actual component operation, θ is the recentered source exponential plus the change of H_x. The quantitative radius of Annex AG, (HL.22), is one sufficient choice. If the parameter radius or the constants depend on x, the conclusions below retain those costs pointwise. A numerical bound uniform in x requires the corresponding actual uniform domain estimate.

2. Domination gives kernel currents without analytic marks

Proposition HP.1 — a dominated complex posterior Suppose throughout $|z| < \rho$ that θ_x is holomorphic in z, measurable in (x, η), and

$$\sup_{x,\eta,|z|<\rho} |\theta_x(z,\eta)| \le \log(3/2). \tag{HP.4}$$

Then the denominator in (HP.3) has absolute value at least $1/2$, and

$$|r_{x,z}(\eta)| \le 3. \tag{HP.5}$$

For every $F(x,\cdot) \in L^1(\pi_x)$, regarded as frozen when z varies, the function $\pi_{x,z}F$ is holomorphic and

$$\left|\partial_z^q \pi_{x,z}F\big|_{z=0}\right| \le 3q!\rho^{-q}\pi_x|F|, \qquad q \ge 0. \tag{HP.6}$$

For the complete finite-difference current on the inner half-disc,

$$\sup_{0<|t|\le\rho/2} \left|\frac{\pi_{x,t}F - \pi_x F}{t}\right| \le \frac{6}{\rho}\pi_x|F|. \tag{HP.7}$$

Proof. The inequalities

$$|\pi_x(e^{\theta_x})| \ge 1 - \pi_x|e^{\theta_x} - 1| \ge 2 - e^{\log(3/2)} = \frac{1}{2}, \qquad |e^{\theta_x}| \le \frac{3}{2}$$

give (HP.5). The integrable majorant $3|F|$ justifies holomorphy of the integral, for example by Morera's theorem and Fubini, followed by continuity from dominated convergence. Cauchy's estimate on any circle of radius less than ρ, then passage to the supremum radius, gives (HP.6). The positive majorant in (HP.6) is independent of the circle angle and of the order of differentiation. Expanding in powers of t gives

$$\left|\frac{\pi_{x,t}F - \pi_x F}{t}\right| \le 3\pi_x|F| \sum_{q\ge1} \rho^{-q}|t|^{q-1} = \frac{3}{\rho - |t|}\pi_x|F|,$$

which proves (HP.7). The normalizer and all its derivatives remain in the analytic density $r_{x,z}$. □

In particular, conditional Jensen gives the actual full-law estimate

$$\left\| \sup_{0<|t|\le\rho/2} \left|\frac{\pi_{x,t}F - \pi_x F}{t}\right| \right\|_{L^p(\nu)} \le \frac{6}{\rho}\|F\|_{L^p(\mu)}, \qquad p \ge 1. \tag{HP.8}$$

This has no inverse real component mass. It is a bound for the new normalized-kernel current with the incoming expression frozen. It is not a bound on a moving hard selector or on a varying coordinate pullback outside the common chart. Direct differences and previously generated current records still follow the exact separate transport formula (M.30).

3. Entire reset jets with only full-law moment inputs

Let I be a set of m distinguished source labels. For every nonempty $J \subseteq I$, let $A_J(x,\eta)$ be the incoming logarithmic-jet coefficient, and suppose

$$\|A_J\|_{L^p(\mu)} \le C^{d+1}(d!)^{p_0} b^d p^{sd}, \quad d = |J|, \quad p \ge 1, \quad C \ge 1,\ p_0 \ge 1,\ s \ge 0. \tag{HP.9}$$

All incoming higher vertices are retained. The source labels are distinguished before repeated field types are identified. Define the frozen-input reset coefficient by the finite formal identity

$$A_I^+(x,z) = \partial_I \log \pi_{x,z} \exp\left(\sum_{\varnothing \neq J \subseteq I} t_J A_J\right)\Big|_{t=0} = \sum_{\mathcal{P}\in\Pi(I)} \kappa_{\pi_{x,z}}(A_J : J \in \mathcal{P}). \tag{HP.10}$$

No actual exponential integrability in the source variables is assumed: (HP.10) is a finite source jet.

Theorem HP.2 — full-law moment and kernel-derivative bounds More generally suppose that (HP.3) is a holomorphic mass-one complex kernel with

$$|r_{x,z}| \le K, \qquad K \ge 1, \tag{HP.11}$$

on a uniform disc of radius ρ. Set

$$B_{C,K} = C(1 + 2KC). \tag{HP.12}$$

Then, for $m \ge 1$, $p \ge 1$ and $q \ge 0$,

$$\boxed{\left\|\partial_z^q A_I^+(\cdot,0)\right\|_{L^p(\nu)} \le q!\rho^{-q} B_{C,K}^m (m!)^{p_0} b^m (mp)^{sm}.} \tag{HP.13}$$

The inner-half-disc finite-difference envelope satisfies the same inequality with $q!\rho^{-q}$ replaced by $2/\rho$. Proposition HP.1 supplies $K = 3$ for the actual old posterior on its admitted shrunken chart.

Proof. We first record the precise conditional-product estimate. Take coefficients $F_1, \dots, F_k$ of degrees $d_1, \dots, d_k$, with $\sum_i d_i = m$, and a partition π of $\{1, \dots, k\}$. Conditional independent replicas of η, one for every block of π, give

$$G_\pi(x) = \prod_{B\in\pi} \pi_x\left(\prod_{i\in B} |F_i|\right).$$

Jensen on this replicated conditional law, followed by Hölder with exponents m/d_i, gives

$$\|G_\pi\|_{L^p(\nu)} \le \prod_{i=1}^k \|F_i\|_{L^{pm/d_i}(\mu)}. \tag{HP.14}$$

Every individual conditional replica has the original full-law marginal μ. There is no product-law assumption on the original histories. By (HP.9), the right side is at most

$$C^{m+k} b^m (mp)^{sm} \prod_{i=1}^k (d_i!)^{p_0}. \tag{HP.15}$$

We dropped only the factors $d_i^{-sd_i} \le 1$.

For the complex kernel, a product of moments occurring in a cumulant is bounded by $K^{|\pi|} G_\pi$. Thus the cumulant coefficient sum costs at most

$$\sum_{\pi\in\Pi_k} (|\pi| - 1)! K^{|\pi|} \le K^k k! 2^{k-1} \le (2K)^k k!. \tag{HP.16}$$

For the outer reset partitions of the source labels, the ordered-composition identity gives

$$k! \sum_{\substack{\mathcal{P}\in\Pi(I)\\ |\mathcal{P}|=k}} \prod_{J\in\mathcal{P}} (|J|!)^{p_0} \le (m!)^{p_0} \binom{m-1}{k-1}. \tag{HP.17}$$

Indeed, after ordering the blocks and their sizes, the left side is $m!\sum_{d_1+\cdots+d_k=m}\prod_i (d_i!)^{p_0-1}$, and each product is at most $(m!)^{p_0-1}$. Combining (HP.15)–(HP.17) bounds the L^p norm of (HP.10), uniformly on the disc, by

$$C^m (m!)^{p_0} b^m (mp)^{sm} \sum_{k=1}^{m} (2KC)^k \binom{m-1}{k-1} \le B_{C,K}^m (m!)^{p_0} b^m (mp)^{sm}. \tag{HP.18}$$

Cauchy's formula and Minkowski's integral inequality prove (HP.13). To prove the supremum finite-difference statement without interchanging a supremum and an L^p norm, retain the positive conditional-product majorant before applying (HP.14). That single majorant controls every circle angle. Its Taylor-coefficient bounds sum to at most $2/\rho$ times the same majorant on the inner half-disc, as in (HP.7). Applying (HP.14) afterward proves the asserted envelope estimate. □

At a physical current-scale input budget $b = h^{-4}$ and $s = 1$, the averaged value part of M-03 has exactly the factor $h^{-4m}(mp)^m$. The new kernel derivative, with its explicit radius cost, also fits the averaged grade $m + 1/2$, because

$$\frac{(mp)^m}{[(m+1/2)p]^{m+1/2}} \le 1 \qquad (m, p \ge 1). \tag{HP.19}$$

This statement assumes the displayed current-scale input budget. If the input is a bare field with $b = a^{-4}$, then the output still carries a^{-4m}; no coarse-scale improvement has been obtained from conditional expectation.

4. Localization errors retain the same moment structure

Let E contain the source dependence of the component base and of the input coefficients. Let

$$R_x(z,\eta) = \sum_{\substack{Y:\, i\in J(Y)\\ Y\not\subseteq E}} V_Y(z,x,\eta), \qquad \sup_{x,\eta,|z|<\rho_0} |R_x(z,\eta)| \le \lambda_E. \tag{HP.20}$$

Interpolate the cross action by $H_t = H^E + tR$, $0 \le t \le 1$. Suppose the common analytic tube constructed in HL.6 is admitted, and denote its radius by ρ. The same real posterior $\pi_x = Q_{1,x,0}$ will be used in every full-law moment. On the real slice,

$$\frac{dQ_{t,x,0}}{d\pi_x} = \frac{e^{-(1-t)R_x(0,\cdot)}}{\pi_x(e^{-(1-t)R_x(0,\cdot)})}, \qquad \left|\frac{dQ_{t,x,0}}{d\pi_x}\right| \le e^{2\lambda_E}. \tag{HP.21}$$

Consequently the complex interpolated kernels are dominated by

$$|Q_{t,x,z}| \le K\pi_x, \qquad K = 3e^{2\lambda_E}. \tag{HP.22}$$

Theorem HP.3 — localized full-jet current errors with moment inputs Under (HP.9), (HP.20) and the common-tube premise, let $A_I^{+,1}$ and $A_I^{+,0}$ denote the full and truncated frozen-input reset jets. There is an explicit constant $D_{C,K}$, depending only on C, K, such that

$$\boxed{\left\|\partial_z^q (A_I^{+,1} - A_I^{+,0})(\cdot,0)\right\|_{L^p(\nu)} \le q!\rho^{-q}\lambda_E D_{C,K}^{m+1} (m!)^{p_0} b^m (mp)^{sm}.} \tag{HP.23}$$

The same inner-half-disc finite-difference envelope has $2/\rho$ in place of $q!\rho^{-q}$. In particular, for all sufficiently large truncation regions for which $\lambda_E \le 1$, one may use the fixed $K = 3e^2$, independent of E.

Proof. The exact derivative of a normalized interpolated integral is

$$\partial_t Q_{t,x,z} F = Q_{t,x,z}(FR_x) - Q_{t,x,z}F\, Q_{t,x,z}R_x. \tag{HP.24}$$

Equation (HP.22) and $\|R_x\|_\infty \le \lambda_E$ give the pointwise majorant

$$|\partial_t Q_{t,x,z} F| \le K(1+K)\lambda_E \pi_x |F|. \tag{HP.25}$$

Differentiate each product of moments in the cumulant formula. For a partition with r blocks, the extra factor is at most

$$rK^{r-1}K(1+K)\lambda_E = rK^r(1+K)\lambda_E.$$

After its cumulant coefficient $(r-1)!$ is included, summing over partitions of k input coefficients costs at most

$$(1+K)\lambda_E K^k k! 2^{k-1}. \tag{HP.26}$$

Apply the conditional-product estimate (HP.14), the source partition count (HP.17), and integrate t from zero to one. For example,

$$D_{C,K} = 2(1+K)C(1+2KC) \tag{HP.27}$$

is more than sufficient. The positive conditional-product majorants remain independent of the circle angle, so Cauchy's formula and the same envelope argument as in Theorem HP.2 prove all the stated derivative and finite-difference bounds. □

Theorem HL.9 supplies an actual source-coordinate bound for eligible components of the form

$$\lambda_E \le C\alpha(1+R_k)^d \exp\{-\omega[r - D_d R_k - 2]_+\} \tag{HP.28}$$

when E is the prescribed radius-r neighborhood. Choose

$$r_0 = D_d R_k + 2 + \omega^{-1} \log^+\!\big(C\alpha(1+R_k)^d\big). \tag{HP.29}$$

Then $\lambda_E \le 1$ for $r \ge r_0$, and the localization error in (HP.23) has an exponential tail in $r - r_0$, with a fixed factorial exponent. The radius ρ remains the actual chart radius from HL.22; (HP.28) is not a proof that its pullback to every retained M-03 coordinate is uniform.

5. A necessary distinction in the spatial carrier summation

The source-coordinate radial estimate (HP.28) is useful, but a particular radial truncation representation is not automatically summable in the strong M-03 carrier weight. That weight is

$$w_h(X) = \exp\{\kappa[|X| + \mathrm{t}_h(X)]\}. \tag{HP.30}$$

If one assigns the n-th radial correction to an entire four-dimensional ball of radius $r_0 + n$, its size can be of order $(r_0+n)^4$. A bound of order $e^{-\omega n}$ then only yields the numerical majorant

$$\sum_{n\ge 0} \exp\{c\kappa(r_0+n)^4 - \omega n\}, \tag{HP.31}$$

which diverges. Thus this route does not prove the desired uniform weighted carrier norm. This calculation says that the upper bounds do not sum; it does not claim that the actual corrections attain those upper bounds or that no other representation works.

A usable M-03 application must retain a suitably thin localized carrier expansion, prove a stronger volume estimate, or establish an equivalent controlled representation. The relative-tree estimate in LFR II (1.68) pays the exterior connections. It does not independently pay the volume of every newly included nonsmall component after its mass has canceled in the conditional posterior. The original factorization and geometry assumptions must remain in that argument.

This is separate from the joined-p reconstruction problem in C-05. A successful proof of the latter supplies domain and response information for the coordinate map; it does not by itself perform the strong marked carrier sum just described.

6. Exact remaining H-03 and H-02 obligations

The original H-03 test is a spatial estimate for the exact reset jet uniform in the remaining history, in the chosen current-scale framework. This note proves its averaged-moment and fixed-chart kernel-current part under explicit chart and source-localization premises. The following actual applications are still open:

1. Prepared multiscale image and retained radii. Compose the actual current/retained map into LFR II (1.65)–(1.75), including every independent-current argument and the nonsmall self action. Check the complete support and one uniform quantitative radius for the directions actually used. The regular integrated R activity is a different object.
2. Pointwise regulator pushforward. M-03 includes a supremum relative to the output regulator, as well as full-law moments. A sufficient pointwise estimate would control the conditional old regulator, its pulled-back input footprints and its path/difference costs in terms of the new regulator. For example, the schematic inequality
$$\pi_x(\widehat{R}^m_{h,X}) \le C^{m+1}(m!)^r \widehat{R}^m_{h',Y} \tag{HP.32}$$
would require its actual scale factor, carrier Y, degree and coordinate-map costs to be specified and proved. Conditional L^p contraction proves the averaged part, not (HP.32). This note does not assume (HP.32).
3. Complete weighted carrier sum. Retain insertion anchors and actual dependency footprints, and sum the value, direct-difference and separate kernel-current records with (HP.30). The radial bound alone does not prove this step.
4. Sector changes and interfaces. For a change that moves a hard selector or changes the common support, use the actual signed-kernel finite difference (M.29). The normalized analytic score on a fixed chart cannot replace that signed measure. If a smoother extension is used off the physical support, record it and prove that it agrees on the entire physical support with a sufficient reserve.
5. Incoming scale budget. The new theorem can use the full input jet and its decorated moment budget. That input must still be the actual transported current-scale jet. Its bare a^{-4m} budget is not relabeled h^{-4m}.

For H-02, the primary source does prove the small factors and unnormalized component estimates in LFR II (1.79)–(1.89), including the recursive reserve for unrepaired regions. The estimate is unnormalized. These inequalities do not bound the normalized probability of an inherited low-margin event after conditioning on retained data. A local ratio/surgery theorem or an actual full-law marked insertion estimate is still needed, with the geometric and physical-scale losses included. None of those missing estimates follows from (HP.13).

7. Application conditions

Item	Result
Actual old-component posterior and nonsmall self action	Retained exactly from (1.71)–(1.75) and Annex AG.
Kernel-current bound with only full-law L^p input	Proved in Proposition HP.1 and Theorem HP.2, on the admitted chart.
All generated higher input vertices	Included in the full reset formula (HP.10).
Localized kernel-current errors with the same moment budget	Proved in Theorem HP.3, with the actual discarded-carrier majorant and radius displayed.
Newly generated factorial loss from reset partitions	None beyond the stated p_0; the moment factor $(mp)^{sm}$ remains explicit.
Strong pointwise/current/geometry carrier norm	Not proved; Sections 5–6 identify its remaining tests.

Primary source: T. Balaban, *Large Field Renormalization. II. Localization, Exponentiation, and Bounds for the R Operation*, CMP 122 (1989), 355–392, pp.379–380 for the actual fixed-support tilt and normalization; pp.380–387 for the

unnormalized large-field reserve. The finite probability, domination, combinatorial and moment arguments above are proved in this note. References to M-03 are to Annex Y.

Annex AL — Exact nonlinear support, joined consistency and bounded gauge conversion

Annex AL. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

This note proves a nonlinear shell-support identity for the source averaging operation, a consistency theorem under an exact constraint-inclusion premise, and an anchored transport-gauge estimate. It also gives an explicit completed splice for which the constraint-inclusion premise holds. Identifying that splice with the incomplete-tuple notation in CMP119 (2.35), and matching the bounded transport gauge to the actual reconstructed representative, remain necessary source checks. The physical family of true minima is not identified with arbitrary independently extended current/background coordinates or with separately prepared kernels. Because of these limitations, the actual-model estimate is not completed here.

1. The lattice convention does not remove the fine collar

Write a for the physical fine spacing, and $h_r = L^r a$ for the physical spacing after r averages. When the j-th coarse spacing is the unit, the fine spacing is $\xi = L^{-j}$. When the p-th coarse spacing is the unit, it is $\eta_p = L^{-p}$. The coordinate change is

$$x^{[p]} = L^{j-p} x^{[j]}, \qquad a = h_j \xi = h_p \eta_p. \tag{JR.1}$$

These are the same physical fine links, expressed in different units. This convention is explicit in the geometric discussion of CMP109, printed p.257, and in its definition of reconstructed currents in (1.15). CMP119 (2.35) and (2.37) retain it. In particular,

$$L^{-2p}(L^n L^{-p})^{-2} = L^{-2n},$$

while multiplying the current bound in (2.37) by h_p^{-3} gives the physical current weight

$$h_p^{-3} L^{n-\min(p,n)} (L^n L^{-p})^{-3} = h_n^{-2} h_{\min(p,n)}^{-1}. \tag{JR.2}$$

Consequently a fine determining collar is not removed by the appearance of L^{-p} in the reconstructed-current formula. Its local determining blocks can still be smaller than $h_{\min(p,n)}$. The deficit identified in RT.12–RT.13 is a real deficit of the generic estimate; the question is whether the actual data enjoy an exact cancellation.

2. What one averaging step uses

Let M_r denote one source averaging step from the lattice of spacing h_r to that of spacing h_{r+1}. Thus

$$M^{r+1} = \mathsf{M}_r \circ M^r. \tag{JR.3}$$

The source average uses the two endpoint blocks of an output bond. In CMP109 (0.12), the relevant paths consist of a path from the first coarse endpoint to a point in its block, the translated straight coarse bond, and a path within the second endpoint block. Its reference path is contained in the same pair of blocks. The symmetric contour averages in (0.11) do not enlarge that support. Hence:

If two input connections agree on the fine bonds in the two endpoint blocks of a coarse bond, including the connecting bonds, their one-step averages agree at that coarse bond.

This is a statement about the actual nonlinear group average. No linearization or Abelian replacement is used. The definition as iterated averaging gives (JR.3); the corresponding derivative composition is also displayed in CMP99 (3.15).

The footprint conclusion must be used carefully. A change inside one endpoint block can change an output bond whose other endpoint lies outside the changed region. Averaging therefore gives star support, not support on bonds with both endpoints in the region.

3. The exact bond-set convention gives a star-support induction

Write $\mathrm{st}_r\, Y$ for the level-r bond set induced by a point set Y: its bonds have at least one endpoint in Y. The source set difference is taken on these bond sets. In particular, it is not generally the star of the difference of the point sets. Let an admitted determining history of depth p be

$$X_0 \supset X_1 \supset \cdots \supset X_p, \qquad X_{p+1} = \varnothing. \tag{JR.4}$$

Each X_r is a union of the appropriate r-blocks. Define the determining bond sets by the actual source convention

$$\Lambda_r = \mathrm{st}_r X_r \setminus \mathrm{st}_r X_{r+1}, \quad 0 \le r \le p, \tag{JR.5}$$

with the exterior also fixed at level zero. The prescribed values are imposed on these bond sets. A bond joining the point shell to the inner region lies in $\mathrm{st}_r X_{r+1}$, so it is not fixed merely because it touches the point shell. This distinction is visible in CMP96 (2.3), whose accompanying text applies the difference separately to the site and bond sets.

Lemma JR.1 — nonlinear shell support induction Suppose U, W agree on the exterior fine bonds and share the complete determining data of (JR.4)–(JR.5). At every step where the averages are defined,

$$\mathrm{supp}(M^r W - M^r U) \subset \mathrm{st}_r X_{r+1}, \qquad 0 \le r \le p. \tag{JR.6}$$

In particular, the p-th averages agree. This statement can be applied on the finite dependency cone of a specified higher average; it does not require defining averages in unrelated large-field regions.

Proof. At level zero, the prescribed Λ_0 values and the fixed exterior imply that every changed fine bond lies in $\mathrm{st}_0 X_1$. Cross-boundary bonds incident to X_1 may still vary; they are retained in this support statement.

Assume (JR.6) at level $r < p$. If both endpoints of an $(r+1)$-bond lie outside X_{r+1}, their endpoint blocks also lie outside X_{r+1}, because that set is a union of these blocks. Every fine bond in the actual two-block averaging footprint has both endpoints in this outside union. It therefore misses $\mathrm{st}_r X_{r+1}$, including the cross-boundary bonds retained in the induction hypothesis. The input fields agree throughout the footprint. Before imposing the next determining data, the changed output bonds consequently belong to $\mathrm{st}_{r+1} X_{r+1}$.

The actual prescribed level-$(r+1)$ datum fixes the set difference $\mathrm{st}_{r+1} X_{r+1} \setminus \mathrm{st}_{r+1} X_{r+2}$. Removing that difference leaves precisely $\mathrm{st}_{r+1} X_{r+2}$, proving the induction step. At $r = p$, the set X_{p+1} is empty and no changed bond remains. □

At no stage is the support asserted to have both endpoints inside the next domain. Cross shell-to-inner bonds are allowed until their actual data are fixed at the appropriate subsequent level. The terminal empty star is sufficient for equality. The larger source separation is needed elsewhere for the admissible variational problem and for patching the action; the support induction itself uses block alignment and the exact set differences.

The conclusion uses the exact group-valued data, not their linearized approximations. It remains an equality under holomorphic continuation wherever the same averaging expressions are defined.

Dependency-cone convention In the actual large-field history, $\Gamma_0 = \Omega_1^c$ can contain fixed data outside the domain on which higher averages are used. The proof does not manufacture a globally defined $M^p U$ there. For an old constrained bond c at level r, consider only the intermediate bonds and paths occurring in the actual iterated expression for $M^r U(c)$. Those intermediate expressions are part of the source's definition of that constraint. On portions of this cone outside the varied collar, the input links coincide. On its remaining portions, Lemma JR.1 is applied using the shell data. Every equality needed for that old constraint is therefore obtained without any new averaging in unrelated bad regions.

4. The source joined tuple and the inclusion of old constraints

The full determining tuple of CMP119 (2.2) is

$$\Gamma_0 = \Omega_1^c, \quad \Gamma_r = \Omega_r^{(r)} \setminus \Omega_{r+1}^{(r)} \ (0 < r < k), \quad \Gamma_k = \Omega_k^{(k)}. \tag{JR.7}$$

Its values are $M^r U$, by (2.11). The reconstruction in (2.35) is written

$$U_{p,X}(M^\bullet U) = U\big(B_p(X) \cup \{\Gamma_i\}_{i<p}, M^\bullet U\big). \tag{JR.8}$$

The two operations involved here have to be kept separate:

1. Truncation at p retains the old determining shells below p, and fills the remaining last region with level-p determining data.
2. Localization to X introduces the source fine collar and the proper-scale boundary splice. The exact splice is (2.14): the old determining set is retained in the interior and the local set supplies the collar. Its generalized version treats each admitted boundary-layer patch separately.

Under this operational reading of (JR.8), the completed localized history has depth at most p. Its interior retains every old shell below p; where a boundary splice replaces an old determining block, it uses that block's scale or finer scales. In particular it refines, rather than coarsens, the old data below p. The source-tuple identification has not been independently established from the displayed notation. The construction below proves a result for the explicitly completed splice, and must not be read as having settled that identification.

4.1 An explicit determining-set dictionary for that identification

Complete the old truncated sequence by

$$\Omega_i^{[p]} = \Omega_i \quad (1 \le i \le p), \qquad \Omega_{p+1}^{[p]} = \varnothing, \qquad \Gamma_i^{[p]} = \mathrm{st}_i\, \Omega_i^{[p]} \setminus \mathrm{st}_i\, \Omega_{i+1}^{[p]}. \tag{JR.11}$$

The level-zero exterior is included as in (JR.7). In particular, $\Gamma_i^{[p]} = \Gamma_i$ for $i < p$; the last set fills the old Ω_p with level-p data.

For one admitted boundary patch lying in old layer n, put $r = \min(n, p)$. Let C_r denote its actual proper-scale collar in the source construction, and I the retained interior. Let $\Lambda_i^{\mathrm{loc},r}$, $0 \le i \le r$, be the determining sets of the source local depth-r problem on that patch. Formula (2.14) gives, component by component,

$$\hat{\Gamma}_i = \big(\Gamma_i^{[p]} \cap I\big) \cup \bigcup_{r \ge i} \big(\Lambda_i^{\mathrm{loc},r} \cap C_r\big). \tag{JR.12}$$

The intersections are taken in the scale of the indicated determining variable. For multiple boundary layers, the source explicitly prescribes the sum of the corresponding right-hand sides. Equation (JR.12) records that prescription; it is not an intersection of two independently invented domain sequences. The fixed exterior supplies the remaining level-zero set.

The completed nested physical domains can be recovered from their determining sites by

$$\hat{\Omega}_i = \bigcup_{s=i}^{p} \bigcup_{y \in \hat{\Gamma}_s^{\mathrm{sites}}} \mathrm{cell}_s(y), \qquad 1 \le i \le p, \qquad \hat{\Omega}_{p+1} = \varnothing. \tag{JR.13}$$

Here $\mathrm{cell}_s(y)$ is the physical block of side h_s; faces are assigned by the common source convention. The determining bond sets are then the differences of the stars of these domains, not stars of site-shell differences. The validity and block compatibility of this admitted local geometry are the source's construction (2.13)–(2.15).

The mesh comparison can now be checked by cases. An interior determining cell is an old truncated cell, so its level is the old level, capped at p. A collar cell in C_r has a level at most r. If that patch lies in old layer $n < p$, then $r = n$, so no collar cell is coarser than the old cell there. If $n \geq p$, then $r = p$, so the same conclusion holds. Across an old layer boundary, the source uses the contributions restricted to each side; no cell from the coarser-side list is assigned to the fine side. Once the point sets have been formed, a bond crossing that boundary is assigned by the star-difference convention, so it is not silently counted as a datum on both sides. This proves the domain inclusion required for the constraint argument:

$$\widehat{\Omega}_i \subset \Omega_i \quad (1 \leq i \leq p). \tag{JR.14}$$

The associated datum at every retained or newly exposed determining bond of level i is the same $M^i U$ evaluated on that bond. There is no averaging of the already assigned heterogeneous tuple, and no substitution of the minimizing field for that datum.

The remaining identification question is narrow: (JR.12) is the source localization of the explicitly completed truncated history, and gives the desired constraint inclusion. The notation (JR.8) must denote that history, with these data, for the result to apply to the displayed B-domain. Its incomplete old-family notation and reversed join order are why this point remains a premise. LFR II uses the same localization join in the reversed order $B(Y) \cup B_k$, so argument order alone cannot distinguish two maps; that observation does not replace the componentwise dictionary above. Although the lower determining site shells uniquely determine the domains $\Omega_1, \dots, \Omega_p$ of a depth-p completion, this uniqueness alone does not specify how that completion is combined with the collar in the notation (JR.8).

The resulting constraint inclusion has a direct verification. Clamp a local competitor to U outside the localized fine collar, and call the patched field $\widetilde{W}$. For old indices $r < p$, Lemma JR.1 and (JR.14) give

$$\mathrm{supp}(M^r \widetilde{W} - M^r U) \subset \mathrm{st}_r\, \widehat{\Omega}_{r+1} \subset \mathrm{st}_r\, \Omega_{r+1}, \tag{JR.15}$$

which is disjoint from the old determining bond set $\Gamma_r = \mathrm{st}_r\, \Omega_r \setminus \mathrm{st}_r\, \Omega_{r+1}$. For old indices $r \geq p$, that lemma first fixes every required p-level intermediate value; iteration (JR.3) then fixes the old r-level datum. Therefore

$$M_B \widetilde{W} = M_B U. \tag{JR.9}$$

There are two boundary checks in this assertion. The local fine collar fixes all links needed to patch the Wilson plaquettes crossing ∂X. Also, an old constrained bond whose endpoint blocks lie wholly outside X cannot change. If its averaging footprint meets X, its datum or its finer replacement is part of the joined tuple. For $X \in D_j$ and $p \leq j$, the source carrier is a union of current j-scale cubes, hence is aligned with the finer block partitions used in this check; the layer-dependent splice retains the additional old-boundary buffers. No estimate of a remote influence is used to replace these exact boundary data.

For $p = 0$, the local tuple consists of fine data and the reconstructed connection equals the supplied connection directly. For $p = k$, (JR.9) is the ordinary localization of the full old history. The same argument treats components and holes of an admitted carrier: all component boundary collars are clamped. A component with no active interior has no variable competitor.

5. Consistency of true constrained minima

Proposition JR.2 — consistency on the actual minimal family Assume the source joined tuple has the constraint inclusion (JR.9), and let $U = U_B(V)$ be a real, regular critical configuration of the original Wilson variational problem. Then its restriction, in the corresponding gauge orbit, is the regular critical configuration for the joined problem with its own determining data $M^\bullet U$.

Proof. The point U belongs to the joined regular class: the new determining mesh is no larger than the old mesh, and the absolute small-field estimates required for the local variational theorem are therefore weaker than the old estimates, after the fixed unit conversion (JR.1). Consider any sufficiently small differentiable variation through the

joined constraint fiber at U. Patch it to U outside the fine collar. Equation (JR.9) makes this a variation through the original constraint fiber. At each finite cutoff, sufficiently small variations stay in the original open regular class. Their Wilson first variations vanish because U is critical there.

The localized and original first variations agree: every potentially affected boundary plaquette is fixed by the fine collar. Thus U is critical in the joined problem. The uniqueness of the regular critical orbit in CMP102 Theorem 1 identifies it with the joined minimum. □

This proof uses local stationarity and uniqueness, rather than claiming that every globally admissible joined competitor satisfies the stronger old regularity bounds. That latter assertion is unnecessary and would not follow from mesh refinement alone.

For scalar gauge-invariant outputs the identity is exact. For covariant current outputs it is equality after the actual change between the two source gauge representatives. The change is not silently discarded: on real fields its adjoint action is isometric, and on a sufficiently small complex chart its parallel-transport factors must be bounded. A global equality between unrelated axial-gauge representatives is not asserted.

6. Complex current variations and the remaining scope distinction

Fix real physical determining data and let $V(z)$ be the actual unprepared current-coordinate insertion into those data. For real z, the field $U_B(V(z))$ is a true Wilson minimizing field. Proposition JR.2 therefore applies throughout the real source chart. Both variational maps are holomorphic in the determining variables on the absolute small-field domain supplied by CMP102 Proposition 9. Once the joined determining data are shown to stay in that absolute domain for $\|z\|_\infty < 2P_k$, the finite-dimensional identity theorem extends the gauge-invariant consistency identity to that whole connected complex current tube.

This admission concerns an absolute map radius, before the smaller running B-domain is tested. The existing original-geometry estimate gives relative determining variations of size at most Cg_kP_k when the joined mesh is no larger than the original mesh. Taking the common coupling ceiling small makes this fit the fixed variational-map radius. The finer source weights do not need to be proved by a generic joined current kernel first: on this family the output equals the original minimizing field.

In physical units, the original current estimate $|\delta J| \le u_n h_n^{-3}$ is stronger than the reconstructed target (JR.2), because

$$h_n^{-3} \le h_{\min(p,n)}^{-1} h_n^{-2}. \tag{JR.10}$$

Thus exact consistency removes the fine-collar amplification on the true-minimizer family. The normalized endpoint and age bounds already proved on the original geometry can be retained there, subject to the bounded gauge representative change. The same reasoning permits admissible variations of the original retained determining data, since they still parameterize true minima on the same fixed history.

It does not prove equality for arbitrary off-minimum backgrounds (U, J), nor for independently altered preparation kernels. In the source's native extension, an independent J is a free analytic coordinate and need not equal the physical Wilson current of U. Equality on $J = J(U)$ is not an identity theorem in that larger space. An original C-05 completion must either prove the required native extension and prepared-domain estimates, or supply a complete actual-boundary derivative argument that does not require this larger intermediate map theorem. Either application requires verification of the stated estimate on its declared domain.

7. A bounded anchored transport gauge, with its matching condition exposed

There is an elementary local gauge construction that resolves the size of a gauge converter once the reconstructed representative is known to obey the corresponding transport condition. It does not require a bound on a generic second derivative of a covariance. The distinction between constructing this gauge and identifying the source's chosen gauge is essential.

Let a complete admitted determining history partition its active sites into aligned blocks $B_s(y)$, whose roots y are determining sites. The block has physical side h_s. Fix in each block a path γ_{yx} from its root to each fine site x, with length at most $c_d h_s$, contained in that block. The constant c_d depends only on dimension; coordinate-ordered shortest paths give such a choice. Put

$$T_U(y,x) = U(\gamma_{yx}), \qquad g_U(x) = T_{U_0}(y,x)^{-1} T_U(y,x), \tag{JR.16}$$

where U_0 is real and $g_U(y) = 1$. For the convention $U^g(x,x') = g(x)U(x,x')g(x')^{-1}$, products telescope and give

$$T_{U^{g_U}}(y,x) = T_{U_0}(y,x). \tag{JR.17}$$

Thus (JR.16) is an explicitly defined, anchored relative transport gauge. All its operations are holomorphic wherever the input fields are holomorphic.

Proposition JR.3 — constraint preservation and a uniform local bound Suppose $U = \exp(iaA)U_0$, with U_0 group valued and

$$\delta_{s,y} = c_d h_s \sup_{b \subset B_s(y)} \|A(b)\| < \delta_*. \tag{JR.18}$$

Then

$$\|g_U(x)\|,\ \|g_U(x)^{-1}\| \le e^{\delta_{s,y}}, \qquad \|g_U(x) - 1\| \le e^{\delta_{s,y}} - 1. \tag{JR.19}$$

The gauge preserves every actual determining datum:

$$M_{\hat{B}}(U^{g_U}) = M_{\hat{B}}(U). \tag{JR.20}$$

Proof. Expand a path product by inserting $\exp(iaA)$ on each edge and move the baseline unitary transports to the outside. The remaining factors are exponentials of unitary conjugates of the edge perturbations. Submultiplicativity bounds their product and its inverse by $\exp(a \sum_{b \in \gamma} \|A(b)\|)$. A telescoping expansion of the difference from the baseline bounds it by the same exponential minus one. The length bound gives (JR.19). Reversed edges obey the same estimate, because the corresponding baseline conjugation is unitary.

For (JR.20), first observe that the two endpoints of every determining bond are determining roots, possibly at different levels. Indeed, a level-r determining bond lies in $\mathrm{st}_r \widehat{\Omega}_r \setminus \mathrm{st}_r \widehat{\Omega}_{r+1}$, so neither endpoint lies in the next inner domain. An endpoint in the level-r site shell is a level-r root. An endpoint in an earlier shell is a root of that finer shell, because the aligned odd-block lattices are nested. The gauge (JR.16) equals one at all these roots. The exact one-step group average is equivariant under endpoint gauge transformations, and composition preserves that property. Every prescribed iterated average therefore has unchanged endpoints and unchanged value. This proves (JR.20). □

The same algebra works with any analytic, endpoint-equivariant averaged transport in place of the single path, provided its local perturbation estimate is supplied. CMP99 (3.55)–(3.59) supplies such local estimates for its hierarchical transport variables. It is not necessary to replace those variables by the single-path choice to use the algebra.

If A has an endpoint-sensitive first variation, direct differentiation of the path product gives

$$\|g_U(x)^{-1} D_i g_U(x)\| \le C e^{C\delta_{s,y}} h_s \sup_{b \subset B_s(y)} \|D_i A(b)\|. \tag{JR.21}$$

Each differentiated edge contributes one factor aD_iA; the sum of the edge lengths is at most $c_d h_s$. No number of fine links remains after this length normalization. If the old geometry bounds the relative field by $u_n h_n^{-1}$ and the joined block has $h_s \le h_n$, (JR.18)–(JR.21) have no factor h_n/h_s. An endpoint exponential remains available when the diameter of the block in the old inverse-mesh metric is bounded, because replacing the distance to an endpoint by the distance to this block changes that distance by a bounded amount. This last statement is a local consequence of the admitted mesh comparison; it does not itself prove the complete carrier and age summations required by C-05.

What would identify this converter with the actual reconstructed field Suppose W is a reconstructed representative in the same anchored gauge orbit as U, and suppose its exact gauge condition is $T_W(y,x) = T_{U_0}(y,x)$. If $W = U^v$ with $v(y) = 1$, covariance of the transport implies

$$T_U(y,x)v(x)^{-1} = T_{U_0}(y,x), \qquad v(x) = g_U(x). \tag{JR.22}$$

Consequently (JR.19)–(JR.21) would be the required bounds for that representative change. However, the actual relative axial condition in CMP102 p.280 is referred to CMP99 pp.75–102, equation (1.19), a different paper from the background-propagator paper CMP99, 389–434. Its complete formula is not used here. CMP109 (0.14)–(0.16) discusses averaged-contour exponential gauge fixing, but this is not a substitute for checking the exact relative axial condition used for its minimizing maps. Therefore the equality of the actual gauge condition with the hypothesis of (JR.22) remains unproved. Real gauge equivalence and analyticity alone do not provide a uniform complex norm bound.

8. Exact current conversion and the remaining nonlinear-chart issue

The physical Wilson current transforms by conjugation. This can be derived without differentiating the gauge in physical space: apply gauge invariance to the first variation of the Wilson action and use invariance of the trace pairing. At a bond whose initial endpoint is x,

$$J(U^g)(b) = g(x)J(U)(b)g(x)^{-1}. \tag{JR.23}$$

The same identity holds for the holomorphic extension of that current and for its fixed changes of lattice units. Differentiating with respect to a parameter i, and setting $K_i = g^{-1}D_i g$, gives the exact formula

$$D_i J(U^g) = g\big(D_i J(U) + [K_i, J(U)]\big)g^{-1}. \tag{JR.24}$$

It follows from (JR.19) and (JR.21) that an admitted converter with $h_s\|D_i A\| \lesssim u_n$ and $\|J(U)\| \lesssim \alpha_n h_n^{-3}$ contributes at most $C\alpha_n u_n h_n^{-3}$ through the commutator. This has the original physical current weight, and hence the weaker reconstructed weight in (JR.2). There is no spatial derivative of g in (JR.23) and no extra fine-collar factor in this calculation.

Equations (JR.23)–(JR.24) concern the actual physical current. If a proof instead uses a different current coordinate or a quadratic-Hessian current, it must still identify that coordinate with the source's nonlinear current and bound the conversion. An independently extended source current J is not made equal to $J(U)$ by this calculation. Thus the transport identity removes a potential nonlinear-current loss after the actual representative and physical graph have been identified; it does not complete that identification or the independent-current extension.

9. Application conditions for C-05

The advances and their exact scopes are:

Result	Status	Limitation
Same physical fine graph under the j- and p-unit conventions, (JR.1)–(JR.2)	Checked against source scale factors	Fine collars remain present.
Nonlinear shell-support induction, Lemma JR.1	Proved for the actual endpoint-block average and admitted complete histories	Uses the actual star-difference bond convention and defined dependency cones.
Constraint inclusion for the explicit completed splice, (JR.11)–(JR.15)	Proved under the source's admitted patch geometry	The notation (JR.8) has not been independently identified with that splice.

Result	Status	Limitation
True-minimum consistency, Proposition JR.2	Proved implication	Needs the exact source-tuple identification, regular admission and fixed boundary patching.
Anchored transport gauge and norm bound, Proposition JR.3	Proved	Its equality with the actual relative axial representative is not checked.
Physical-current conjugation and variation, (JR.23)–(JR.24)	Proved	Does not identify an arbitrary independent current coordinate with the physical current.
Complete actual B pullback on its common complex current contour	Open	All actual tuple, representative, domain and localization premises must hold together.

The remaining work for this route is concrete. First, identify the actual joined determining set in CMP119 (2.35) component by component, including its fine exterior data and every admitted mixed boundary patch. Second, obtain the exact relative gauge condition or an equivalent source theorem that supplies a uniformly bounded analytic converter into the actual representative on the full common contour. Third, prove the resulting native background/current extension and the endpoint-localized application to all B sectors. An alternative extension using an independent displacement from the physical current may be useful, but its agreement with the actual activity and its full source-domain admission require proof. These are additional hypotheses, not consequences of the shell-support identity.

Neither C-05 nor BM.24 is proved here.

10. Sources

- CMP119 (1988), 243–285, (2.2), (2.10)–(2.15), (2.34)–(2.39), and pp.276–277. Printed pp.255–257 and 261 give the determining sets, the tuple and the scale factors.
- CMP102 (1985), 277–309, (1)–(7), Theorem 1, and Proposition 9. The uniqueness assertion is for the regular critical orbit on the source's fixed determining geometry.
- CMP99 (1985), 389–434, (3.15), composition of the actual averaging derivatives; the nonlinear averaging operations referred to there are the iterated operations.
- CMP96 (1984), 223–250, printed p.224, (2.3): the determining sets use the set difference for sites and bonds separately. This is the source of the star-support convention used in the induction.
- CMP109 (1987), 249–301: (0.11)–(0.12), the geometric scaling discussion at p.257, and (1.15)–(1.16). DOI https://doi.org/10.1007/BF01215223.
- The gauge-condition source is Balaban, *Spaces of regular gauge field configurations on a lattice and gauge fixing conditions*, CMP99 (1985), 75–102, DOI 10.1007/BF01466594. Its equation (1.19) is not used here, so the identification of the actual gauge condition remains a hypothesis.

Annex AM — Reference-coordinate current transfer and the local averaging-Hessian estimate

Annex AM. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

Scope

This note proves a local weighted forward averaging row and adjoint-minimizer rows, a weighted regular constrained-covariance estimate, and a physical-minimizer tangent theorem with an explicitly retained matched-lift premise. The last theorem avoids the rough gauge-projection term in the earlier formula (RT.33). It is not a bound for the excluded generic covariant Laplacian of the fully gauge-projected covariance.

The common-gauge lift is constructed in CC.5 for matched source-regularized data. The actual joined geometry, the raw-data matching, and the complete native/carrier application remain necessary to complete C-05. No uniform conclusion is claimed here.

1. Actual norms, geometry and source inputs

Let η be the physical fine mesh. An admitted determining geometry has blocks $\Delta(a)$ of mesh $h_a = L^{r_a}\eta$ and the source distance $d(a,b)$. All constants below depend on the fixed group, dimension $d = 4$, block size and admitted geometric constants. They do not depend on the number of determining layers or the total volume.

The fine inner product uses measure η^4. The determining-variable inner product uses h_b^4. Adjoint symbols refer to these real inner products; their analytic continuations are bilinear transposes, not conjugate-dependent operations.

A positive profile w is admissible at rate β if

$$w_a/w_b \le A_w e^{\beta d(a,b)}. \tag{CC.1}$$

An endpoint factor $e^{-\kappa d(a,A_*)}$ can be included in w_a; it increases the comparison rate by at most κ. Write

$$\|f\|_{F_w} = \sup_a w_a^{-1}\|f\|_{\infty,\Delta(a)}, \quad \|u\|_{E_w} = \sup_a \max\left\{\frac{\|u\|_{\infty,\Delta(a)}}{h_a^2 w_a}, \frac{\|\nabla_U u\|_{\infty,\Delta(a)}}{h_a w_a}, \frac{\|\Delta_U^{\mathrm{cov}} u\|_{\infty,\Delta(a)}}{w_a}\right\}. \tag{CC.2}$$

The last operator is the componentwise covariant Laplacian. It is distinguished from the Wilson quadratic Hessian below.

CMP96 (2.2), Lemma 2.1 and (2.60)–(2.61) supply

$$d(a,b) \ge RM\max\{|r_a - r_b| - 1, 0\}, \quad h_a/h_b \le Le^{c_h d(a,b)}, \quad c_h = \frac{\log L}{RM}, \tag{CC.3}$$

and finite exponential row sums $S(t) = \sup_a \sum_b e^{-td(a,b)}$ at the admitted rates. Choose the fixed geometry so that all rates used below, in particular

$$t_* = \delta - \beta - \kappa - 4c_h > 0, \tag{CC.4}$$

retain the additional reserve required by that source summation lemma. Here δ is a common smaller exponential rate of the source kernels. This is a fixed geometric restriction, not a claim for arbitrary R, M.

We use the original regular Wilson propagator

$$G_0 = (\Delta + DRD^* + Q^* a Q)^{-1}, \quad H_0 = G_0 Q^* (Q G_0 Q^*)^{-1}, \quad P_0 = I - H_0 Q, \quad C_0 = P_0 G_0 = G_0 P_0^*. \tag{CC.5}$$

Δ is the Wilson quadratic Hessian at the real background. G_0 is the operator of CMP99 Theorem 3.3, before the gauge-invariant modification used to define G_1. Theorem 3.3 includes the component-Laplacian estimate. CMP102 (129)–(130) supplies that estimate also for H_0.

In (129)–(130), the printed $H_0 B$ takes dimensionless determining data B. In the convention of (CC.5), its argument is $h^{-1}B$. Thus the source estimate reads

$$\left| D^{(j)}_{\mathrm{out}} (H_0 h^{-1})(x,b) \right| \le C h_a^{-1-j} h_b^{-4} e^{-\delta d(a,b)}, \quad j = 0,1,2, \quad x \in \Delta(a), \tag{CC.6}$$

where $j = 2$ means the component covariant Laplacian, not all arbitrary second partial derivatives. Formula (CC.6) is not an estimate for the fully gauge-projected covariance $\mathcal{G}$.

2. A uniform forward row for the actual averaging map

Proposition CC.1 — forward averaging without a flat-map identification On the admitted real background class, and on its fixed smaller analytic relative-field neighborhood, the source reference-regularized averaging derivative has the local weighted bound

$$\|Qu\|_{\infty,w} \le C_Q A_w e^{\beta c_Q} \|u\|_{\infty,w}. \tag{CC.7}$$

The norms on the two sides use their respective fine and determining blocks and the same physical profile. The fixed constant c_Q allows the actual averaging stencil.

Proof. Fix a proper-scale local gauge cube of size $O(Mh)$, with $h = L^j \eta$. The text immediately following CMP99 (3.78) gives analyticity of the actual relative averages on $\|A\|_\infty < \alpha_1 / h$, and bounds their intermediate logarithms by

$$\|B_r\|_\infty \le C \alpha_1 L^{r-j}, \quad 0 \le r < j. \tag{CC.8}$$

The remaining fixed-L averaging step is analytic and has a fixed derivative bound on the small one-step chart, by (3.79). Consequently the final relative logarithm is bounded by $C_L \alpha_1$ on that disc. The dimensionful relative averaging map from the defining (3.14),

$$\Phi_j(A) = h^{-1} Q_j(U, \eta A) = \frac{1}{ih} \log \overline{(e^{i\eta A})}^{\,j}_{U,\mathrm{rel}}, \qquad D\Phi_j(0) = Q_j(U). \tag{CC.9}$$

Here Q_j is exactly the source nonlinear function in CMP99 (3.13), and the last symbol denotes its reference-corrected relative average, with the background U fixed. It is not the quotient of two uncorrected symmetric averages.

therefore satisfies $\sup \|\Phi_j(A)\| \le C_L \alpha_1 / h$. Cauchy on $A = tu$ gives

$$\|Q_j(U) u\|_\infty \le C_L \|u\|_\infty, \tag{CC.10}$$

uniformly in j. The same argument holds about a smaller complex relative-field center by retaining half the source reserve. There is no constant raised to the number of layers.

The physical support radius of the j-step derivative is at most $c_L h$, since its fixed-range one-step radii sum as a geometric series. Choose the admitted collar with $RM > L c_L$, allowing the endpoint conventions. A point in a determining layer can then meet stencils from only a fixed number of neighboring layers. This follows from the separation in CMP96 (2.2): a layer two or more steps away is separated by a distance greater than the stencil radius. The proper meshes of the remaining layers differ by fixed powers of L, and their source distances are bounded by a fixed c_Q.

Apply (CC.1) on these supports. Real local gauge changes act unitarily. For complex relative fields the real gauge is fixed and the relative field remains in the source neighborhood. This proves (CC.7). □

Operator-family caution. The symmetric contour average in CMP109 (0.12), the reference-transport average used to define CMP99 (3.14), and their gauge corrections must not be interchanged. In particular, the bare symmetric contour derivative at the identity need not be the straight-line operator of CMP95. The preceding proof uses the actual bounded relative average identified in CMP99 (3.14), (3.78) and does not make that flat identification. A stand-alone physical transpose row for the symmetric average has not been proved here and is not needed in Proposition CC.3.

Only those estimates of the CMP98 averaging paper that are explicitly reproduced in CMP99 are used.

3. Adjoint minimizing rows with the full baseline profile retained

Proposition CC.2 — arbitrary slow current profiles Let H be the dimensionful minimizing operator of CMP99 (3.126), or H_1 of (3.129), on their admitted background class. Then

$$\|H^* f\|_{\infty,w} \le C_H A_w L^4 S(\delta - \beta - 4c_h)\|f\|_{\infty,w}. \tag{CC.11}$$

The same formula holds with H_1, and with an endpoint profile included as in (CC.4).

Proof. The value part of source (3.133) is

$$|H(x,b)| \le C h_b^{-4} e^{-\delta d(a,b)}, \qquad x \in \Delta(a).$$

The physical adjoint therefore gives

$$|H^* f(b)| \le C \sum_a (h_a/h_b)^4 e^{-\delta d(a,b)} \|f\|_{\infty,\Delta(a)}. \tag{CC.12}$$

Use (CC.1), then (CC.3), and the admitted exponential row sum. This proves (CC.11). □

The familiar current norm h_a^{-3} reduces the mesh exponent in this argument from four to one; that special case was proved in the argument. Keeping a general profile is useful here because the actual reconstructed baseline has stronger original-layer current restrictions than the generic joined-layer bound.

4. Regular constrained covariance without a rough projection

Proposition CC.3 — current forcing to a second-order field With the stated fixed rate reserves,

$$\|C_0 f\|_{E_w} \le C_0^{\mathrm{bd}} \|f\|_{F_w}. \tag{CC.13}$$

Proof. The localized value, first-derivative and component-Laplacian estimates of CMP99 Theorem 3.3 map a forcing on $\Delta(b)$ to

$$C e^{-\delta d(a,b)} (h_a^2, h_a, 1)\|f\|_{\infty,\Delta(b)}. \tag{CC.14}$$

Summation against (CC.1) bounds $G_0 f$ in E_w. Proposition CC.1 and the fixed neighboring-block mesh comparison then give

$$|(QG_0 f)(b)| \le C h_b^2 w_b \|f\|_{F_w}. \tag{CC.15}$$

Set $B_b = h_b(QG_0 f)(b)$. Thus $|B_b| \le C h_b^3 w_b \|f\|_{F_w}$. Applying the actual source bound (CC.6), with the physical measure h_b^4, gives after normalization by the output factor $h_a^{2-j} w_a$

$$C \sum_b (h_b/h_a)^3 e^{-\delta d(a,b)} w_b/w_a \, \|f\|_{F_w}. \tag{CC.16}$$

The reserved rate in (CC.4) bounds this sum. Both pieces of $C_0 = G_0 - H_0 Q G_0$ are now controlled in E_w, proving (CC.13). □

In particular the two powers gained by G_0 before the average are retained. Applying an output current row to an unrelated bare determining mark would lose that information and reproduce the fine-collar problem (RT.12).

The source local kernels and averaging rows are analytic on their admitted relative-field domains. Their smaller common domain uses the actual strong field/current restrictions and a uniform inverse reserve for QG_0Q^*. One cannot obtain a claim for independently altered, unmatched kernels by applying this proposition to their names alone.

5. Physical stationarity removes the K^*D_2 term

At a physical constrained Wilson minimum W, define

$$T = \Delta - D_2, \qquad \langle A, D_2 A\rangle = 2\langle HC^{(2)}(A), J_W\rangle. \tag{CC.17}$$

The normalization of $C^{(2)}$ is matched to the dimensionful H, as specified after CMP99 (3.127). The source multiplier is $\lambda = H^*J_W$. The exact formula (3.136) retains this multiplier at each determining bond. Its subsequent printed estimate first substitutes $|\lambda_b| \le CM\alpha h_b^{-3}$, and invokes the unavailable averaging-paper inequality (149). That substituted estimate alone would not imply the stronger profile bound needed here.

The needed pointwise derivative information is, however, explicitly displayed in a different published source: CMP102, printed p.289, equation (72). The argument below uses that displayed equation, not the formula (149).

Lemma CC.3a — local multiplier estimate from the reproduced pointwise derivative For the actual reference-regularized averaging and its admitted geometry,

$$|D_2 u(x)| \le Ch_a \sup_{b\in N(a)} |\lambda_b| \sup_{y\in N(a)} |u(y)|, \quad x \in \Delta(a). \tag{CC.18}$$

Here $N(a)$ is a fixed union of neighboring proper-scale blocks. The constants are independent of the fine spacing, number of layers and total volume.

Proof. For a determining bond b of mesh $h_b = L^j\eta$, write $f_b(A) = C_j(h_bA, b)$ in CMP102's notation (44), and let

$$\mathcal{F}_b(A; x) = \frac{\delta f_b}{\delta A}(A; x). \tag{CC.18a}$$

The kernel uses the physical fine measure η^4. CMP102 (72) states the pointwise bound

$$\mathcal{F}_b(A' - HD(A'); x) = h_b \frac{\delta C_j}{\delta A}\big(h_b[A' - HD(A')], b; x\big), \qquad |\mathcal{F}_b(A' - HD(A'); x)| \le 2C_3\epsilon_3 h_b^{1-d}, \quad d = 4. \tag{CC.18b}$$

The paragraph immediately after that equation explains the factor h_b^{-d} as the change to the physical fine-lattice measure. The linearizing correction $D(A')$ is holomorphic, begins at degree two, and is defined on the complex value ball $\|A'\|_{(-1)} < \epsilon_3$, where $\|A'\|_{(-1)} = \sup h|A'|$. The value-ball assertion is explicit in CMP102 p.287; a first-gradient assumption is additional, not a condition needed to define this map.

Fix a test direction V and put $A' = tV$. Since $D(tV) = O(t^2)$, and the remainder C_j has no constant or linear term, the coefficient of t in $\mathcal{F}_b(tV - HD(tV); x)$ is the derivative of the quadratic part $f_b^{(2)}$ at V. Cauchy on $|t| = \epsilon_3/(2\|V\|_{(-1)})$ gives

$$\left|\frac{\delta f_b^{(2)}}{\delta A}(V; x)\right| \le 4C_3 h_b^{1-d}\|V\|_{(-1)}. \tag{CC.18c}$$

The quadratic homogeneity convention is $f_b^{(2)}(A) = h_b^2 C_j^{(2)}(A, b)$, exactly as explained in CMP99 p.421 after (3.127). Therefore

$$\left|\frac{\delta C_j^{(2)}}{\delta A}(V,b;x)\right| \le C h_b^{-1-d}\|V\|_{(-1)}. \tag{CC.18d}$$

This derivative depends on V only in the actual local j-average footprint. Take $V = u$ on that footprint and zero elsewhere. Sharp truncation is allowed because only the value-ball norm is used. The stencil and collar comparison proved in CC.1 makes every proper mesh met by the footprint comparable to h_b, with fixed constants. Thus $\|V\|_{(-1)} \le C h_b \sup_{\mathrm{foot}(b)} |u|$, and

$$\left|\frac{\delta C_j^{(2)}}{\delta A}(u,b;x)\right| \le C h_b^{-d} \sup_{\mathrm{foot}(b)} |u|. \tag{CC.18e}$$

The kernel vanishes when x is outside that footprint. For a fixed fine x, only a fixed number of neighboring determining bonds, at meshes comparable to h_a, have a footprint containing x. Insert (CC.18e) into the exact CMP99 (3.136). Its physical determining factor h_b^{d+1} leaves the factor h_b. The resulting fixed neighboring sum is (CC.18). □

Substituting the generic admitted multiplier bound $|\lambda| \le CM\alpha h^{-3}$ into (CC.18) recovers the printed CMP99 (3.137), and gives

$$\|D_2 u\|_{F_w} \le C_D M\alpha \|u\|_{E_w}. \tag{CC.19}$$

The stronger slow-profile bound (CC.11) can instead be used in (CC.18) when J_W has original-layer current restrictions. The local-majorant step is thus supplied by the reproduced pointwise derivative (72), not inferred from linear dependence on λ or from the already substituted bound (3.137).

Theorem CC.4 — normalized physical current of a matched tangent Suppose $W(t)$ is an admitted differentiable family of physical constrained Wilson minima with its fixed determining geometry. Work in the relative Landau gauge based at $W(0)$. Let u be its physical tangent at zero, and let A be an actual matched lift satisfying

$$QA = Qu, \qquad RD^*A = RD^*u = 0. \tag{CC.20}$$

Assume that A, its Wilson quadratic current ΔA, and the actual baseline multiplier make the following number finite:

$$F_A = \|\Delta A\|_{F_w} + \left\| h_a \sup_{N(a)} |\lambda| \sup_{N(a)} |A| \right\|_{\infty,w}. \tag{CC.21}$$

For $C_0^{\mathrm{bd}} C_D M\alpha \le 1/2$, the difference $e = u - A$ satisfies

$$\|e\|_{E_w} \le 2C_0^{\mathrm{bd}} F_A. \tag{CC.22}$$

It also satisfies a physical Wilson-current bound

$$\|\Delta e\|_{F_w} \le C_{\mathrm{cur}} F_A. \tag{CC.23}$$

Proof. The source's unrestricted constrained variational equation is CMP102 (127), and its derivation in pp.296–297 is essential: admissible variations satisfy $Qv = 0$, without a Landau condition on the variation. Differentiating the constrained stationary equation at a true minimum gives

$$\langle v, Tu\rangle = 0 \quad \text{for every } Qv = 0. \tag{CC.24}$$

This is the actual Lagrange-Hessian equation, with the averaging Hessian term in (CC.17) retained. Since $Qe = 0$ and $RD^*e = 0$, the argument of CMP102 (129)–(133) gives the exact identity

$$e = C_0(D_2 e - TA). \tag{CC.25}$$

There is no $K^* D_2$ term. The covariance in this identity constrains Q, and physical stationarity supplied the unrestricted variation equation. Replacing an arbitrary fully gauge-projected forcing problem by (CC.25) would not be justified.

Proposition CC.3, (CC.18) and (CC.19) yield

$$\|e\|_{E_w} \le C_0^{\mathrm{bd}}(C_D M\alpha \|e\|_{E_w} + F_A).$$

Absorb the first term to obtain (CC.22).

To recover the Wilson quadratic current, use $RD^*e = 0$, hence $DD^*e = DPD^*e$ with $P = I - R$. CMP99 (3.49) bounds DPD^* by a regular exponential kernel of order two. Applying its physical-measure kernel to $|e| \le h^2 w \|e\|_{E_w}$, with the same mesh-rate reserve, gives a bound $Cw\|e\|_{E_w}$. The identity between $D^*D + DD^*$ and the component Laplacian has only the local curvature term displayed in CMP102 (135). The Wilson Hessian differs from D^*D by the bounded local term of CMP99 (3.10), (3.69). Their coefficients are $O(M\alpha h^{-2})$. Combining these facts with (CC.22) proves (CC.23). □

Why the stronger baseline profile matters Let n be the original layer, p the reconstruction scale, $h_* = h_{\min(n,p)}$, and let $h_a \le h_*$ be the joined mesh. The admitted old-B baseline restriction has the form

$$|J_W| \le \alpha_n h_*^{-1} h_n^{-2}. \tag{CC.26}$$

If the actual smooth lift has $|A| \le u_n h_n^{-1}$, Proposition CC.2 and (CC.18), with fixed profile comparison losses, give

$$|D_2 A| \le C\alpha_n u_n \frac{h_a}{h_*} h_n^{-3} \le C\alpha_n u_n h_n^{-3}. \tag{CC.27}$$

This is bounded by the required $u_n h_*^{-1} h_n^{-2}$ profile, since $h_* \le h_n$. Replacing (CC.26) prematurely by the weaker $O(\alpha h_a^{-3})$ would discard this gain.

The nonlinear physical current in a left exponential chart has, at its real center, derivative equal to the quadratic Hessian plus a local chart term bounded by $C\eta|u||J_W|$. This follows by differentiating the local exponential differential, whose first correction is $\frac{1}{2} i\eta \,\mathrm{ad}_u$. The strong baseline bound (CC.26), $\eta \le h_a \le h_*$, and (CC.22) bound that chart term by the same profile. This controls the chart conversion once the actual lift (CC.20) is available.

6. A matched lift for the actual regularized linear constraint

The following strengthens the preceding conditional theorem. It does not identify a raw symmetric-average derivative with the reference-regularized Q. That conversion is kept explicit below.

Lemma CC.5 — the Landau correction needs no second-gradient estimate Use one fixed real physical minimum W, its actual joined operators D, Q, Q', R, G', and its physical current J_W. Suppose the original-layer profiles satisfy (CC.1) with a sufficient rate reserve. Let $h_a \le h_* := h_{\min(n,p)} \le h_n$. Suppose an actual input tangent A obeys

$$|A| \le C_A u_n h_n^{-1}, \qquad |D^*A| \le C_A u_n h_n^{-2}, \qquad |\Delta A| \le C_A u_n h_n^{-3}, \qquad |J_W| \le \alpha_n h_*^{-1} h_n^{-2}. \tag{CC.28}$$

Set $w_2 = u_n h_n^{-2}$, $w_3 = u_n h_n^{-3}$, and define

$$\chi = G' R D^* A, \qquad \widehat{A} = A - D\chi. \tag{CC.29}$$

Then

$$Q\widehat{A} = QA, \qquad RD^*\widehat{A} = 0, \tag{CC.30}$$

and, with constants independent of the fine spacing and the joined collar depth,

$$|\chi| \le CC_A h_a^2 w_2, \qquad |D\chi| \le CC_A h_a w_2, \qquad |\widehat{A}| \le CC_A u_n h_n^{-1}, \qquad |\Delta\widehat{A}| \le CC_A(1+\alpha_n)w_3. \tag{CC.31}$$

Proof. The projection R has range $\Delta_{\mathrm{sc}} \ker Q'$, where $\Delta_{\mathrm{sc}} = D^*D$ is the covariant scalar Laplacian; this is CMP99 (3.21). Thus, if $f \in \operatorname{ran} R$, there exists $\chi \in \ker Q'$ with $\Delta_{\mathrm{sc}}\chi = f$. Since $Q'\chi = 0$, the same χ is $G'f$. In particular

$$Q'G'R = 0, \qquad R\Delta_{\mathrm{sc}}G'R = R. \tag{CC.32}$$

CMP99 (3.115) gives $QD = D^{\mathrm{coarse}}Q'$. Applying these identities to (CC.29) proves (CC.30). These are exactly the projection identities used by the source in (3.119); they are not an assumed right inverse for the raw nonlinear constraint.

For the norms, write $R = I - P$. The physical-measure kernel of P, CMP99 (3.49), is an exponentially decaying operator of order zero. Its row sum against the slow profile w_2 is bounded by the same geometric argument as (CC.10). Thus $|RD^*A| \le CC_A w_2$. The scalar G' value and first-gradient rows in Theorem 3.1, (3.42), then give the first two estimates of (CC.31), with local gains h_a^2 and h_a, respectively. This uses no bound for a general second gradient of G'.

The quadratic Wilson Ward identity is CMP99 (3.117), involving the physical current J_W. Replacing its gauge parameter by $t\chi$, and comparing the coefficient bilinear in t and an arbitrary field υ, gives the exact weak identity

$$2\langle \upsilon, \Delta D\chi\rangle = \langle i[\chi(b_-) + R_b\chi(b_+), \upsilon(b)], J_W(b)\rangle\,. \tag{CC.33}$$

Consequently

$$|(\Delta D\chi)(b)| \le C|J_W(b)|\big(|\chi(b_-)| + |\chi(b_+)|\big). \tag{CC.34}$$

There is no derivative of χ on the right. Its original-layer bound is

$$|J_W|\,|\chi| \le CC_A\alpha_n u_n \frac{h_a^2}{h_* h_n}\, h_n^{-3} \le CC_A\alpha_n w_3. \tag{CC.35}$$

The neighboring-point comparisons are fixed profile constants. Together with $h_a/h_n \le 1$, this proves the remaining assertions of (CC.31). □

Consequence for a physically stationary tangent Suppose the actual relative-Landau tangent u of the reconstructed minimum has the matched regularized datum

$$Qu = QA. \tag{CC.36}$$

Then $\widehat{A}$ satisfies the matching condition (CC.20). Apply Theorem CC.4 with $w = w_3$. Proposition CC.2, (CC.18), and the original value estimate for $\widehat{A}$ give

$$|D_2\widehat{A}| \le CC_A\alpha_n u_n \frac{h_a}{h_*} h_n^{-3} \le CC_A\alpha_n w_3. \tag{CC.37}$$

Thus, for the same fixed smallness reserve as in CC.4,

$$e := u - \widehat{A}, \qquad |e| \le CC_A h_a^2 w_3, \quad |\nabla e| \le CC_A h_a w_3, \quad |\Delta e| \le CC_A w_3, \tag{CC.38}$$

and the physical current derivative of u has the original w_3 bound, including the local exponential-chart correction already described after (CC.27). This is stronger than the reconstructed target $u_n h_*^{-1} h_n^{-2}$.

A general uniform bound for $\nabla D\chi$ has not been asserted. It is unnecessary for the forcing estimate. When the original tangent also has $|\nabla A| \le C_A u_n h_n^{-2}$, restore its infinitesimal gauge after solving:

$$u + D\chi = A + e. \tag{CC.39}$$

The right-hand side has the original value and first-gradient bounds by (CC.38). Gauge covariance changes the physical current derivative by a local adjoint term bounded by $|J_W||\chi|$, which is controlled by (CC.35). Curvature has the corresponding local commutator correction. This proves the regularity in that representative without an unsupported scalar second-gradient estimate.

The actual data conversion that is still required Condition (CC.36) is an equality of the source's regularized data. The raw symmetric averaging M used in the determining constraints, and the reference/gauge-regularized relative average whose differential is Q, are not automatically the same even at a common baseline. In particular, $\chi \in \ker Q'$ generally does not mean that χ vanishes at all determining centers. The projection in (CC.29) therefore preserves Q, but does not by itself prove preservation of the raw M-datum.

A source one-step conversion is visible in CMP99 (3.113), for the averaging operation underlying that reference construction. Write $m(A)$ for its raw relative-average derivative, $q(A) = QA$ for the corrected derivative, and $\xi_y(A)$ for $1/i$ times the derivative at zero of $\overline{R_y U_t'}$, with $U_t' = \exp(i\eta t A)$. Differentiating the printed endpoint factors gives exactly

$$q(A) = m(A) + D^{\mathrm{coarse}}\xi(A), \qquad |\xi_y(A)| \le Ch \sup_{B(y)} |A|. \tag{CC.40}$$

Here $h = L\eta$ for this step, and $D^{\mathrm{coarse}}\xi = (R_c\xi_{c_+} - \xi_{c_-})/h$. The local bound follows by differentiating the transported path logarithms in (3.113); each path has at most Ch/η fine bonds, and each link insertion carries η. At a real baseline all transports are isometries. The corresponding smaller complex chart uses its admitted uniform transport bound. Iterating the same reference construction yields levelwise endpoint corrections $\Xi_j(A)$; the accumulated bound is $C\sum_{r\le j} h_r \sup|Q_{r-1}A| \le Ch_j \sup|A|$, by the forward averaging row and the geometric sum of scales.

The raw average in (CC.40) has not been identified with CMP109's later symmetric M. A complete mixed-history application must check that operator-family conversion, and identify the composed endpoint corrections at the actual determining sites. A level-j bond can meet a site belonging to a finer determining shell, so a collection of levelwise corrections is not automatically one consistent gauge on the mixed determining set. The source's exact relative axialization would provide an alternative verification. For a verified correction, a bounded local gauge parameter is enough for current control by (CC.34); no generic second-gradient projection bound is needed. Neither an operator-family identification nor a mixed-level identification has been replaced by a same-baseline assertion.

7. Independent external parameters and the unresolved application

The independent external source-current variable must not replace J_W in (CC.17) or in the Wilson Ward equation. Here J_W is the physical current of the reconstructed minimum. External variables may enter its determining data through a prepared input map; the minimizing problem remains the physical Wilson problem.

For a genuine common holomorphic family of those data, (CC.24)–(CC.25) continue as identities in the parameters. The bounds then require the corresponding uniform analytic profile and inverse reserves. This does not prove the admission of an arbitrary prepared family, and it is not an estimate for unrelated separately modified propagator pieces.

The remaining application is concrete:

1. Identify the actual joined determining geometry and exact data.
2. Match the raw mixed-history datum to the source regularized constraint. Lemma CC.5 then constructs the common-gauge lift (CC.20) with its original-layer current bound. A same-baseline statement or pointwise comparison in two separately chosen gauges is insufficient.
3. Admit all actual independent external and preparation parameters on one strictly smaller domain.
4. Retain endpoint and marked-path localization through the nonlinear solution and the complete B carrier sums.

The source bond-support convention requires a separate check. In particular a local average generally produces star support; it does not by itself preserve a two-endpoint interior support. No exact reconstruction consistency is assumed in this note.

Primary references

- CMP95 (1984), 17–40: (1.11), (1.18), used to distinguish its straight-line averaging from other source averaging conventions; no unidentified transpose norm is imported.
- CMP96 (1984), 223–250: (2.1)–(2.4), Lemma 2.1, (2.60)–(2.61), admitted geometry, physical support and exponential sums.
- CMP99 (1985), 389–434: Theorems 3.3–3.4; (3.49); (3.78)–(3.81), including its explicit transpose extension; (3.126), (3.133), (3.136)–(3.137), physical minimizing kernels and averaging Hessian.
- CMP102 (1985), 277–309: pp.285–289, (44), (51), (55), (63), (72), including the value-ball analytic linearizing map and the explicitly reproduced pointwise averaging derivative; pp.296–300, especially (123)–(143), unrestricted constrained stationarity and the regular alternative; the explicit H_0 component-Laplacian bound (130).

No technical theorem of the CMP98 averaging paper is used beyond what CMP99 reproduces.

Annex AN — Direct boundary derivatives, all-order summation and explicit application gates

Annex AN. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

1. Target and the alternative considered

The original C-05 statement is to establish the same local derivative bounds for the actual B and boundary-induced activities, including collars, constraints and current-coordinate dependence. It requires constants independent of total volume and hidden carrier distance, with any source formula not verified here retained as a premise. Its catalogue dependency is C-01.

One sufficient way to establish the boundary contour and carrier estimates is a prepared joined reconstruction theorem, with arbitrary complex decoupling parameters. That is stronger than the input required for the derivative argument below. The argument uses the true unlocalized current-to-background map, its common complex contour, and a one-endpoint bound for each primitive boundary activity. It requires no new complex decoupling identity for that map.

This distinction is useful only if the true map is actually checked. In particular, the normalized reconstructed-current restrictions of CMP119 (2.37), the independent-current extension, the fine collars, the fixed coordinate map and the boundary relocation remain part of the application. They cannot be replaced by an estimate for an unrelated scalar analytic function.

2. Source estimates for the direct product-domain shortcut

CMP119, printed pp.261–262, defines the primitive B functions on a multiscale domain containing the original field and independent current, the reconstructed fields and currents, local gauge representatives, and complex relative fields. The source's full primitive carrier bound is (2.42). The component-resummed alternative on p.262 has a weaker carrier weight and is not silently substituted for the full primitive bound.

Printed p.276 is stronger than a statement of primitive analyticity. It explicitly assigns the already localized fluctuation expressions, including those arising from inherited B by the transformations corresponding to CMP109 (3.6)–(3.7), to the product domain (3.43). Its current-variable factor is

$$\{A : \operatorname{supp} A \subset Y \cap \Lambda_{k+1}^{(k)*},\ |A| < C_1 p_1(g_k)\}. \tag{BC16.1}$$

The following paragraph on p.277 describes the use of the decreasing multiscale margins, inherited B bounds, heterogeneous Wilson coefficients and the full relocation action. Thus (3.43), if its radius were matched to the actual current support, would be a genuine direct source input. There would be no reason to re-prove every internal reconstruction solely to differentiate the final functions.

The radius identification is not established. CMP119 p.265 gives the unprimed fluctuation cutoff through

$$\delta_k = (A_1/A_0) g_k p_0(g_k), \qquad P_k = \delta_k / g_k = A_1 (\log g_k^{-2})^{p_0}. \tag{BC16.2}$$

The occurrence of an apparent polynomial/constant typo in the first-step formula on p.247 does not define the symbol in (3.43). In CMP116 the precursor domain (1.34) instead has the absolute current radius ϵ_1/g_k; CMP119 changes

that domain at its multiscale boundary. A later LFR polynomial with exponent $p_1 < p_0$ belongs to a primed preparation threshold and cannot cover (BC16.2) by increasing a fixed coefficient. If that later polynomial were inserted in (BC16.1), the radius-to-support ratio would tend to zero.

Consequently this note does not infer a full-support current contour from (BC16.1), nor enlarge its radius by analytic continuation without estimates. Its alternative is the true-map endpoint theorem below. See CMP119, pp.247, 261, 276 and 277.

3. A finite-volume endpoint theorem

3.1 Hypotheses and notation Let I be the current free-bond coordinate set, with a fixed number D of scalar Lie-algebra coordinates per bond. Distances are periodic lattice distances in current units. All constants below may depend on D, dimension and the fixed localization geometry, but not on the number of coordinates or torus volume.

Let $F_X(z,\eta)$ be the pullback of one primitive activity. The symbol X records its original carrier and old scale. The current field is z; η records the admitted native background and independent-current variables. Old history selectors and old real fluctuation coordinates are fixed input parameters. If the actual source operation changes their arguments, that change is part of F_X, not a reason to suppress its derivative. Consider the simultaneous contour

$$\|z\|_\infty < 2P, \qquad \eta \in \mathcal{D}^+. \tag{BC16.3}$$

Here the sup norm is a scalar-coordinate norm with a fixed conversion from the Lie-algebra norm. The actual support is contained in $\|z\|_\infty \le P$. In η, use a smaller domain $\mathcal{D}$ with fixed positive native radii to the boundary of $\mathcal{D}^+$. All the following estimates must hold on the full larger contour.

Associate to X a nonempty set S_X of current-unit cells covering its physical carrier. Write

$$d_X(i) = \operatorname{dist}(i, S_X), \qquad D_X = \operatorname{diam}(S_X).$$

Assume the one-endpoint gradient estimate

$$|\partial_i F_X(z,\eta)| \le \frac{K_X \vartheta_X}{P}\, e^{-c d_X(i)} \quad \text{on (BC16.3)}, \tag{BC16.4}$$

where $c > 0$ is fixed and $K_X, \vartheta_X \ge 0$. The coordinate i includes its Lie-algebra component; changing to block norms changes fixed constants only. No assumption that F_X depends on only finitely many current coordinates near X is made. Its dependence can propagate through the actual minimizing map.

For m current positions let $\tau(i_1, \dots, i_m)$ denote minimum spanning-tree length in the periodic current metric, with $\tau(i) = 0$.

Proposition BC16.1 — pointwise jets from one endpoint For $\|z\|_\infty \le P$, $\eta \in \mathcal{D}^+$, and every $m \ge 1$,

$$|\partial_{i_1} \cdots \partial_{i_m} F_X| \le K_X \vartheta_X\, (m-1)!(2/P)^m e^{-c \max_s d_X(i_s)}. \tag{BC16.5}$$

Proof. Choose any of the m differentiated coordinates as the first derivative. Estimate that derivative by (BC16.4). A simultaneous scalar polydisc of radius $P/2$ about the real point lies in (BC16.3). Apply the multivariable Cauchy formula to the remaining $m-1$ derivatives. The product of the repeated-coordinate factorials is at most $(m-1)!$. This yields the bound with $e^{-c d_X(i_s)}$ for every chosen first coordinate. Mixed derivatives commute, so take the minimum of these bounds. The harmless factor 2 in (BC16.5) also covers $m = 1$. No power of $|I|$ occurs. □

For $m = 2$, the estimate gives in particular

$$|\partial_i \partial_j F_X| \le \frac{4 K_X \vartheta_X}{P^2} e^{-\frac{c}{2}(d_X(i) + d_X(j))}. \tag{BC16.6}$$

This is the precise use of mixed-derivative symmetry: a one-endpoint estimate is sufficient for a two-endpoint Hessian bound. It does not manufacture a missing endpoint estimate for the actual map.

Proposition BC16.2 — a weighted Hessian row For $0 < \gamma < c/2$, set $t = c/2 - \gamma$. There is a fixed current-lattice counting constant $N_4(t)$ such that

$$\sum_j e^{\gamma\rho(i,j)} |\partial_i \partial_j F_X| \le \frac{4N_4(t) K_X \vartheta_X}{P^2} |S_X| e^{\gamma D_X} e^{-t d_X(i)}. \tag{BC16.7}$$

For example one may take a fixed bond-counting constant times $(1 + 2/(e^t - 1))^4$ for $N_4(t)$.

Proof. Connecting each bond to a nearest point of S_X gives $\rho(i,j) \le d_X(i) + D_X + d_X(j)$, with fixed endpoint constants absorbed in the counting convention. Multiply (BC16.6) by the requested weight. For each j, assign a nearest cell of S_X. The sum of $e^{-t d_X(j)}$ is at most $|S_X| N_4(t)$. The same argument works on a torus, since a periodic lattice shell contains no more points than the corresponding covering-lattice shell up to a fixed bond convention. □

The carrier factor in (BC16.7) is explicit. It must be summed; it is not called a constant independent of carrier size before that summation.

4. Old-scale and carrier sums, including all orders

4.1 Concrete sufficient source majorants Set $a = k - j \ge 0$ for a primitive activity at old scale j. Let $n(X)$ be its equivalent cube-edge tree length, after the fixed comparison with the source's $d_j(X)$. Assume

$$K_X \le K e^{-\kappa n(X)}, \qquad \vartheta_X \le \vartheta_k (1 + a)^b e^{-\Gamma a}. \tag{BC16.8}$$

The primitive carrier encoding gives at most BA^n carriers of length n rooted at any one old localization cell. A root means a fixed chosen incident old cell of the carrier; overcounting all such roots only increases the bound. In current units that old grid has mesh ML^{-a}. Thus, for any fixed $u > 0$,

$$\sup_i \sum_{r \text{ at age } a} e^{-u\rho(i,r)} \le C_{M,u} L^{4a}. \tag{BC16.9}$$

To prove (BC16.9), divide the current lattice into unit cells. The number of old roots in each such cell is at most $C_M L^{4a}$. The current-cell exponential sum is finite uniformly on periodic tori. This proof does not count all old roots in the entire volume without an anchor.

A carrier of tree length n has

$$D_X \le C_M (n + 1), \qquad \#\{i : d_X(i) \le R\} \le C_M (n + R + 1)^4. \tag{BC16.10}$$

The actual old carrier is smaller by L^{-a}; discarding that improvement gives (BC16.10). The estimate follows by covering the old carrier by a ball of radius $C_M(n + 1)$ in current units. It is a counting estimate, not a claim that a ball is an admissible polymer carrier with exponential-volume weight.

The exponent Γ must exceed $4 \log L$, so that

$$S_{b,\Gamma} = \sum_{a \ge 0} (1 + a)^b e^{-(\Gamma - 4 \log L) a} < \infty. \tag{BC16.11}$$

These inequalities are deliberately stated as source application requirements. The primitive carrier count is proved by the usual bounded-degree tree encoding. The missing part is to prove (BC16.4) and the age factor in (BC16.8) for the actual complete pullback; neither follows from the root count.

Theorem BC16.3 — anchored all-order derivative kernels Assume (BC16.3)–(BC16.4) and (BC16.8)–(BC16.10), with $\Gamma > 4 \log L$. Fix the source carrier parameters with $\kappa > \log A + 2$. If necessary decrease the available endpoint rate c, after those parameters are fixed, so that the fixed losses cC_M and the endpoint convention leave a positive carrier reserve. This is always possible because an endpoint estimate with rate c_0 also holds with every smaller positive rate. Fix $0 < \sigma < c/8$. Then there is a fixed C, independent of m, k, volume and history, such that the normalized interaction

$$W(z,\eta)=\sum_X [F_X(z,\eta)-F_X(0,\eta)] \tag{BC16.12}$$

satisfies, for $m\ge 1$,

$$\boxed{\sup_{i_1}\sum_{i_2,\dots,i_m} e^{\sigma\tau(i_1,\dots,i_m)/m}|\partial_{i_1}\cdots\partial_{i_m}W|\le C^{m+1}(m!)^5\,K\vartheta_k S_{b,\Gamma}P^{-m}.} \tag{BC16.13}$$

The finite-volume sum defines (BC16.12) directly. The estimates also justify normal convergence of its anchored differentiated sums and passage to expanding volumes whenever the primitive functions themselves are consistently embedded.

Proof. We give the counting detail because it controls the factorial exponent. For a fixed X, put $R=\max_s d_X(i_s)$. Connecting all positions through nearest points in S_X gives

$$\tau(i_1,\dots,i_m)\le 2mR+mD_X, \tag{BC16.14}$$

after a fixed enlargement of the endpoint convention, if necessary. Hence the product of the weight in (BC16.13) and the exponential in (BC16.5) is bounded by

$$e^{\sigma D_X}e^{-(c-2\sigma)R}.$$

Set $\upsilon=(c-2\sigma)/2>0$. Split this last exponential into $e^{-\upsilon d_X(i_1)}e^{-\upsilon\max_{s\ge 2}d_X(i_s)}$. For $m=1$ the empty maximum has value zero. The exact layer-cake identity

$$e^{-\upsilon\max_{s\ge 2}d_X(i_s)}=\int_{\max_{s\ge 2}d_X(i_s)}^{\infty}\upsilon e^{-\upsilon R'}\,dR'$$

and (BC16.10) give

$$\sum_{i_2,\dots,i_m} e^{-\upsilon\max_{s\ge 2}d_X(i_s)}\le C_M^{m-1}\int_0^{\infty}\upsilon e^{-\upsilon R'}(n+R'+1)^{4(m-1)}dR'. \tag{BC16.15}$$

For a chosen root r of X, $d_X(i_1)\ge\rho(i_1,r)-C_M(n+1)$. Thus root summation is controlled by (BC16.9), at the cost of $e^{\upsilon C_M(n+1)}$. The factor $e^{\sigma D_X}$ has another fixed exponential cost in n. Choose κ leaving a positive rate λ after both costs and the entropy A^n. The old-age sum is (BC16.11). Apart from constants to the power m, the remaining joint size integral is

$$\sum_{n\ge 0}e^{-\lambda n}\int_0^{\infty}\upsilon e^{-\upsilon R'}(n+R'+1)^{4(m-1)}dR'. \tag{BC16.16}$$

Let a_* be half the minimum of $\lambda,\upsilon,1$, and let $r_m=4(m-1)$. Split off $e^{-a_*(n+R')}$ from the exponential. The exponential power series implies

$$(n+R'+1)^{r_m}e^{-a_*(n+R')}\le e^{a_*}r_m!a_*^{-r_m}.$$

The remaining geometric sum and integral are finite fixed constants. Consequently (BC16.16) is at most $C^{m+1}(4(m-1))!$. The multinomial inequality $(4m)!\le 4^{4m}(m!)^4$, together with the Cauchy factor $(m-1)!$ in (BC16.5), gives (BC16.13). The Lie-algebra component multiplicities are at most D^{m-1} and are absorbed into C^m. Absolute convergence permits the differentiated sum. □

In dimension d, the same proof gives the factorial power $d+1$. The value five is not an uncontrolled order-dependent exponent.

For $m = 2$, (BC16.13) gives a weighted Hessian row with weight $e^{\sigma\rho(i,j)/2}$. Alternatively (BC16.7) gives a sharper explicit second-order constant. In either form, a bounded $K\vartheta_k$ and $P_k \to \infty$ make the Hessian budget arbitrarily small. If the primitive value prefactors vary among the sectors, the right side is summed over the finitely many sector classes before choosing the common coupling ceiling.

4.2 Local value densities and native external jets On a finite lattice the exact radial identity is

$$W(z,\eta) = \sum_i b_i(z,\eta), \qquad b_i = z_i \int_0^1 \sum_X \partial_i F_X(tz,\eta)\, dt. \tag{BC16.17}$$

It supplies a spatially anchored value budget

$$\sup_i |b_i| \le CK\vartheta_k S_{b,\Gamma} \quad (\|z\| \le P). \tag{BC16.18}$$

This follows from the same root sum with $m = 1$. The density is quasilocal; it is not asserted to have exact compact carrier support. Its derivatives are controlled by (BC16.13) and differentiation of (BC16.17). The subtraction $F_X(0,\eta)$ cancels only in the conditional current normalization and remains in the outer retained-data marginal.

If a native external slot has a common radius r between $\mathcal{D}$ and $\mathcal{D}^+$, ordinary Cauchy gives external coordinate jets of these anchored quantities with factor $u!r^{-|u|}$. For t unit directions in a norm ball, use parameter radii r/t, giving $t^t r^{-t} \le e^t t! r^{-t}$. Applying this jointly with the current-coordinate argument proves fixed-factorial mixed jets. The radii are those of the actual admitted background and independent-current slots. No arbitrary raw retained-coordinate direction is identified with a native unit direction, and no selector derivative is inferred.

5. How the endpoint premise can follow from an actual map theorem

Let the primitive source function b_X be bounded by K_X on its original multiscale complex domain. Suppose the actual true-map pullback $\Phi_{k,X}(z,\eta)$ lies in a smaller subdomain, leaving a fixed relative margin in each of the restrictions (2.34)–(2.39), simultaneously on (BC16.3). These restrictions include nonlinear reconstructed outputs. Their first derivatives and their value margins do not by themselves constitute a primitive Cauchy radius.

The additional local native-chart certificate used here has the following exact meaning. For every admitted center and every strong native direction v of norm at most one, there is a holomorphic curve $\chi_v(w)$, defined on one fixed disc $|w| < r_0$, with $\chi_v(0)$ equal to that center and $\chi_v'(0) = v$. Every point of the curve belongs to the full primitive B domain, including all reconstructed-field/current restrictions and the specified old-fluctuation arguments. The same fixed $r_0 > 0$ works for every carrier, old scale and admitted center. A verified native-coordinate ball contained in the full domain is a sufficient certificate. A first-order inequality for the derived outputs alone is not.

Assume this certificate and measure a current-induced primitive argument tangent in its strong native norm, with the actual normalization radii included. If the actual map theorem proves

$$\|\partial_i \Phi_{k,X}\|_{\mathrm{margin},X} \le \frac{\vartheta_X}{P_k} e^{-c d_X(i)}, \tag{BC16.19}$$

then Cauchy on the certified curve gives (BC16.4), up to the fixed factor r_0^{-1}. Indeed apply it to $b_X \circ \chi_v$, where v is the normalized tangent. Its derivative at zero is the required primitive derivative in that tangent. The estimate is dimension independent. Uniform chart constants and admission of the whole curve must be checked in the actual geometry.

The need for this distinction has a simple exact counterexample. Put

$$D_N = \{u \in \mathbb{C} : |u| < 1,\ |Nu^2| < 1\}, \qquad f_N(u) = \sqrt{N}\, u. \tag{BC16.19a}$$

Then $|f_N| < 1$ on D_N. At zero the defining outputs $G_N(u) = (u, Nu^2)$ have derivative $G_N'(0) = (1,0)$, and each output threshold is one, yet $f_N'(0) = \sqrt{N}$. A nonlinear output condition can therefore shrink the actual directional disc while

being invisible to its first derivative. This counterexample concerns the proposed Cauchy inference, not Yang–Mills or the actual source map.

The needed age comparison is also explicit. If, in the exact physical normalizations of the source restrictions, an output on layer $n \le j$ has endpoint response bounded by

$$C g_k P_k e^{-\Gamma_0(k-n)} e^{-c d_X(i)}$$

and its old-domain margin is at least $c\alpha_{*,n} 2^{-(j-n)}$, the verified AF comparison and $q > p_0$ give the candidate normalized factor

$$C \ell_k^{p_0-q} \sqrt{1+k-n}\, 2^{j-n} e^{-\Gamma_0(k-n)}. \tag{BC16.20}$$

Writing $a = k - j$ and $r = j - n$, then fixing $\Gamma_0 > 4\log L + \log 2 + 2$, bounds (BC16.20) by

$$C \ell_k^{p_0-q} (1+a)^{1/2} e^{-\Gamma a}$$

for a fixed $\Gamma > 4\log L$, after summing or maximizing over r. Thus it supplies (BC16.8) with a vanishing ϑ_k. This is a numerical deduction from the correctly normalized response. It does not repair a missing two-power factor in the reconstructed current or establish that the true joined map has the stated response.

The important change from the earlier prepared route is precise: (BC16.19) is needed for the actual analytic true-map family and the native external slots. It need not hold for arbitrary new complex decoupling operators. If an exact refinement/consistency identity is valid for the true family and remains valid on its analytic continuation, it can be used here without extending that identity to a different prepared operator family.

5.1 A different auxiliary-current extension that preserves physical jets CMP119, p. 261, permits a useful distinction. In (2.34)–(2.39), the auxiliary current J is restricted by (2.36). The reconstructed current in (2.37) is $J_{p,X}(M^\bullet U)$: it is a function of the determining field U, not of the independently supplied J. Thus changing the auxiliary J while fixing U does not change the reconstructed-field/current requirements.

Write $\mathcal{J}(U)$ for the physical Wilson current in the source normalization. On a complex field neighborhood this notation means its holomorphic extension, using matrix inverses and the complex-linear Lie-algebra projection. In particular the real expression involving $\operatorname{Im} P$ is extended by $(P - P^{-1})/(2i)$, not by complex conjugation of a complex variable. On real group-valued fields this agrees with the physical current.

Let $U(z, b)$ be an actual true-map family with $U_0(b) = U(0, b)$. The parameter b is the specified native physical background or retained-coarse datum. This subsection assumes that this family is holomorphic on the required joint z, b contour and that all U-dependent restrictions of the primitive B domain, including (2.35) and (2.37), have a fixed reserve there. It also assumes a common admitted gauge/chart and the normalized current bound below. The subsection does not prove those true-map facts.

Define the layer-weighted current norm

$$\|\zeta\|_{\mathfrak{J}} = \max_n \frac{h_n^3 \|\zeta\|_{\infty,\Omega_n \setminus \Omega_{n+1}}}{\alpha_{0,n}}, \tag{BC16.21}$$

with the source's innermost-region convention on the final actual layer. For a primitive activity of scale j, the source instead uses the j-norm in its entire innermost region Ω_j. Denote that norm, restricted to the primitive argument set, by $\|\cdot\|_{\mathfrak{J},j}$. On an actual layer $n \ge j$, the ratio of its weight to the weight in (BC16.21) is

$$\frac{h_j^3 \alpha_{0,n}}{h_n^3 \alpha_{0,j}} \le C(1+n-j)^{b_q} L^{-3(n-j)}. \tag{BC16.21a}$$

The admitted AF comparison gives the fixed polynomial bound used here. Hence $\|\zeta\|_{\mathfrak{J},j} \le C_{\mathrm{emb,J}} \|\zeta\|_{\mathfrak{J}}$ for one fixed $C_{\mathrm{emb,J}}$. To use a common contour for every old primitive, define $\|\zeta\|_{\mathfrak{J}}^{\mathrm{com}} = \sup_j \|\zeta\|_{\mathfrak{J},j}$. This is bounded above by $C_{\mathrm{emb,J}}$ times (BC16.21), and includes the final actual-layer norm. In the rest of this subsection the displayed current

inequalities and radii use this common norm. Equivalently every radius in (BC16.21) may be reduced once by $C_{\mathrm{emb,J}}$. This fixed reduction is necessary; the innermost convention is not silently identified across old scales.

Set $s_* = 1 - \beta > 0$, so every coefficient $1 - \beta(1 - 2^{-(j-n)})$ in (2.36) is at least s_*. Suppose the actual joint true-map contour satisfies

$$\|\mathcal{J}(U(z,b))\|_{\mathfrak{F}}^{\mathrm{com}} \le s_*/4. \tag{BC16.22}$$

For an auxiliary complex current ζ define

$$\widehat{F}_X(z,b,\zeta) = b_X\big(U(z,b), \mathcal{J}(U(z,b)) + \zeta, \mathrm{A_{old}}, \mathrm{S}\big), \qquad \|\zeta\|_{\mathfrak{F}}^{\mathrm{com}} < s_*/4. \tag{BC16.23}$$

Proposition BC16.4 — auxiliary-current admission and jet identity Under the stated true-map hypotheses, (BC16.23) is a holomorphic extension of the actual physical primitive pullback, bounded by the same primitive carrier majorant. It retains a current-slot margin at least $s_*/2$ in every primitive normalized current norm. Its auxiliary-current dependence is exactly localized to X. If the true-map endpoint estimate (BC16.19) holds for U and $\mathcal{J}(U)$ and the local native-chart certificate of Section 5 has been proved, it yields the same one-endpoint estimate uniformly on (BC16.23), without differentiating a minimizing operator in an independent off-physical current.

Proof. By (BC16.22) and the radius in (BC16.23), the J argument has normalized size at most $s_*/2$. This is strictly below every source threshold in (2.36), with a margin at least $s_*/2$. Every other source clause is a restriction on U, its reconstructions or relative-field chart, so it is unchanged by adding ζ. The arguments therefore lie in the primitive domain. Holomorphy and the value bound follow from primitive holomorphy and composition.

Only the restriction of ζ to X occurs in the primitive function. Consequently a derivative in $\zeta(x)$, for $x \notin X$, vanishes exactly. In a current coordinate, $\partial_i \zeta = 0$, so the derivative of the output argument is precisely $(\partial_i U, \partial_i \mathcal{J}(U))$. Under the additional native-chart certificate, the one-dimensional Cauchy bound of Section 5 gives (BC16.4), uniformly in ζ. At $\zeta = 0$, the expression is the physical primitive activity evaluated on its actual current graph. □

The point of this proposition is that the original analytic extension of the primitive activity is retained. The new choice concerns how its auxiliary current is pulled back along a true minimizing family. It does not replace the primitive current by an unrelated constant.

To use the original base-current coordinate instead of the centered variable ζ, set

$$J_{\mathrm{base}} = \mathcal{J}(U_0(b)) + \zeta. \tag{BC16.24}$$

The pullback then reads

$$\widehat{F}_X(z,b,J_{\mathrm{base}}) = b_X\big(U(z,b), \mathcal{J}(U(z,b)) + J_{\mathrm{base}} - \mathcal{J}(U_0(b)), \mathrm{A_{old}}, \mathrm{S}\big). \tag{BC16.25}$$

At $z = 0$, this is exactly $b_X(U_0(b), J_{\mathrm{base}}, \dots)$, including its native current dependence. At the physical base-current graph $J_{\mathrm{base}} = \mathcal{J}(U_0(b))$, it equals the actual physical pullback for every admitted z, b.

The change (BC16.24) is triangular and analytic, with derivative equal to the identity in the auxiliary current. Its mixed/background derivatives are precisely the derivatives of $\mathcal{J}(U_0(b))$. If that map is bounded on a larger native b ball, ordinary Cauchy supplies their fixed-factorial bounds. A product contour in b, J_{base} is obtained by taking, for example, an $s_*/16$ current radius about the original center and a b radius small enough that the normalized change of $\mathcal{J}(U_0(b))$ is at most $s_*/16$. Their sum is at most $s_*/8$, leaving half of the auxiliary radius in (BC16.23). The reduction of the b radius is a fixed fraction when the true-map bound and its larger native radius are uniform.

For a holomorphic curve of physical parameters $t \mapsto (z(t), b(t))$, the identity

$$\widehat{F}_X\big(z(t), b(t), \mathcal{J}(U_0(b(t)))\big) = F_X^{\mathrm{physical}}\big(z(t), b(t)\big) \tag{BC16.26}$$

holds throughout its admitted neighborhood. Its derivative of every finite order is therefore identical on both sides, with all chain-rule terms from the base current retained. This proves compatibility of the physical finite jets and the triangular native-coordinate change.

It does not assert that arbitrary off-physical J derivatives of (BC16.25) equal those of a different source-defined off-physical minimizing family. Such numerical equality is unnecessary for a bound obtained using a new explicitly specified analytic extension, but it must not be falsely claimed. If an application demands that particular off-physical family, its matching theorem is still required. No auxiliary variable is integrated or changed in the Wilson measure here; this is an extension used to estimate the same physical functions.

The construction also preserves the relevant gauge rule. In a common fixed chart, $\mathcal{J}(U)$ is gauge covariant. Transforming ζ by the same adjoint action makes (BC16.23) inherit the primitive gauge invariance. A parameter-dependent change between charts must carry its actual analytic adjoint factors and derivative costs; the proposition does not hide those costs in the word covariance. Real compact gauge changes preserve the norm. Bounded complex chart changes modify it by the verified fixed chart constant.

5.2 A uniform numerical slack condition There is no need to spend only the exponentially shrinking difference between two maximal source domains if the actual true-map family lies much farther inside them. Suppose its physical normalized current is bounded on the proposed contour by

$$h_n^3\|\mathcal{J}(U(z,b))\|_{\infty,n} \le C_{\mathrm{bg}} g_n \ell_n^{p_0} + C_{\mathrm{resp}} g_k P_k e^{-\Gamma_0(k-n)} + C_b r_b \alpha_{0,n}. \tag{BC16.27}$$

The last term records a retained/background ball of normalized radius r_b, if it has not already been included in the first two bounds. This is an actual map estimate to be checked, not a new regularity assertion. The AF comparison and $\alpha_{0,n} = C_0 g_n \ell_n^q$, $q > p_0$, imply

$$\|\mathcal{J}(U(z,b))\|_{\mathfrak{J}} \le C_* \ell_*^{p_0-q}\left(1 + \sup_{a\ge 0}\sqrt{1+a}\,e^{-\Gamma_0 a}\right) + C_b r_b, \tag{BC16.28}$$

where ℓ_* is the lower bound on $\log g_n^{-2}$ over the admitted trajectory. Multiply the constants in (BC16.28) by the fixed $C_{\mathrm{emb,J}}$ to bound the common norm. First choose $r_b \le s_*/(16 C_{\mathrm{emb,J}} C_b)$. Then choose one coupling ceiling making the common-norm first term at most $s_*/16$. This proves (BC16.22) with a reserve. The choices do not depend on old-layer depth or later derivative order. The analogous U-dependent bounds, especially the normalized reconstructed-current one, still have to be proved before applying this construction.

This auxiliary-current lemma requires the actual B-domain bounds and true-map physical current control. It does not require an off-physical consistency identity. Exact refinement/minimum consistency is one possible way to prove those true-map bounds; it is not a hypothesis needed merely to add the admitted independent current once the bounds are known.

5.3 Two distinct analytic application routes The local native-chart certificate and a bounded holomorphic actual pullback are different statements. Neither should be substituted for the other without the following additional information.

Primitive-chart route. The actual complete argument map $\Phi_{k,X}$, including any old-field pullbacks, is holomorphic on the joint contour (BC16.3), takes its values in the primitive domain with the specified reserves, and has the normalized endpoint tangent (BC16.19). A uniform local native-chart certificate then bounds the primitive derivative in that tangent. BC16.1–BC16.3 apply. Exact consistency of true minima can simplify the admission check: when the source reconstructed minimum is exactly the original minimum in the required gauge, its reconstructed curvature/current restrictions become the corresponding original-field restrictions, after the explicit scale conversion. The identity must hold for the actual determining tuple and analytic continuation. It does not by itself supply a local directional disc for an arbitrary off-minimum input, nor control a parameter-dependent gauge conversion or an old-field argument.

Direct-pullback route. One may instead establish holomorphy and a bound for the actual scalar function F_X, including all native slots, without parametrizing a local primitive chart. A bound $|F_X| \le K_X$ merely on the common $2P$ current cube gives ordinary Cauchy estimates $m! K_X P^{-m}$. It does not give the endpoint factor in (BC16.4), and summing that bound over all old carriers can introduce a volume factor. A sufficient direct replacement is a uniformly local centered-difference bound. More precisely, on the larger current contour, suppose each actual one-coordinate slice admits a complex disc $|t| < rP$, with one fixed $r > 0$, on which

$$|F_X(z + te_i, \eta) - F_X(z,\eta)| \le K_X \vartheta_X e^{-c d_X(i)}. \tag{BC16.28a}$$

The disc must be admitted for every center used by the subsequent Cauchy estimate; choose one larger fixed cube if needed. Cauchy gives (BC16.4) with the fixed factor r^{-1}. This route requires no separate primitive chart. An actual localized expansion of the scalar pullback, with a common current/native contour and a primitive carrier majorant that pays the root and endpoint sums, is another sufficient direct replacement. In particular the source's already localized (3.43) could serve this role if its radius and its complete parameter/sector scope were identified. Its unresolved p_1 radius remains an obstruction to importing that result here.

These are exact sufficient alternatives, not new estimate definitions. The first route must prove the native-chart/endpoint hypotheses; the second must prove its actual complex centered-difference or localized-carrier bound. A verified bounded physical-current pullback without either local ingredient is progress but does not close C-05. A real-only exact-minimum identity can be analytically continued once both actual families are holomorphic on the same connected neighborhood and agree on its real open family. It cannot establish the existence or size of that neighborhood by the identity theorem.

6. Actual geometry: the age gain is shared by a path and the old carrier

The numerical hypothesis (BC16.8) must not be justified by incorrectly assigning every point of an old B carrier to a layer $n \le j$. Its innermost source region is Ω_j, and that region contains actual determining layers $n > j$. A point close to the current input can have no large kernel age separation. In this case the primitive carrier's connection to the old bad set pays the missing age.

Proposition BC16.5 — the combined age payment Use the actual nested determining geometry and metric of CMP96 (2.2), (2.46)–(2.59), as given in BM.1–BM.5. Write $h_r = L^r h_0$. The physical separation of consecutive regions is at least RMh_r. Let a primitive B carrier of scale j meet its source boundary anchor near $Z_j = \Lambda_j^c$. This anchor is outside Ω_{j+1}, after the fixed endpoint convention in the source collar. Let $y \in X$ belong to actual layer n, and let the current input lie in the admitted deepest layer k, allowing the fixed adjacent-step/stencil offset c_0. Then

$$\begin{aligned} d_\Omega(y,i) &\ge RM(k-n-c_0)_+, \\ C_{\text{tree}} d_j(X) + C_{\text{end}} &\ge R(n-j-2)_+. \end{aligned} \tag{BC16.29}$$

The second inequality is only needed for $n > j$. It is conservative: the physical shell lengths actually grow geometrically relative to h_j.

Proof. The first inequality is the weighted-age path decomposition BM.3, with the fixed current-stencil offset. For the second, connect the primitive boundary anchor to the source cube containing y along a connecting tree. The part of the tree connecting those two points is a path. It starts outside Ω_{j+1}, ends in Ω_n, and therefore crosses at least $(n-j-2)_+$ complete successive collars. A physical crossing of the r-th collar has length at least RMh_r. Since $r \ge j$, dividing by the old carrier unit Mh_j gives at least R per crossing. The fixed source comparison between a tree meeting localization cubes and paths between points in those cubes yields C_{tree} and C_{end}. Neither depends on carrier distance, age or total volume. Repeated crossings only increase length. □

Let the original endpoint bound have rate $\delta > 0$, and retain the primitive factor $e^{-\kappa d_j(X)}$. Allocate one quarter of each exponent to age. Since

$$(k-n-c_0)_+ + (n-j-2)_+ \ge k-j-c_0-2,$$

that allocation supplies

$$e^{-\frac{\delta}{4} d_\Omega(y,i)} e^{-\frac{\kappa}{4} d_j(X)} \le C\, e^{-\Gamma_0(k-j)}, \qquad \Gamma_0 = \min\{\delta RM/4,\ \kappa R/(4C_{\text{tree}})\}. \tag{BC16.30}$$

For $n \le j$, the kernel age alone gives the same conclusion with a changed fixed endpoint constant. The source normalized-row entropy condition is imposed on another reserved endpoint share, as in BM.5. A further share retains spatial decay: every physical path weight is at least h_k^{-1}, up to the fixed adjacent-step conversion, so

$$d_\Omega(y,i) \ge c_L \rho_k(y,i) - C_L. \tag{BC16.31}$$

One quarter of each exponent can still be used for the current-to-old-bad-set collar when needed. The remaining primitive exponent is positive and can be chosen to pay its carrier entropy. More shares, with smaller fixed fractions, can be used if a particular source localization requires them; R is then increased by the corresponding fixed factor before the coupling ceiling is chosen.

Consequently one may first fix the actual primitive decay/localization constants, then fix the admitted geometric separation large enough that Γ_0 exceeds any prescribed finite number, in particular $4\log L + \log 2 + 2$, while meeting the source row condition. The running source collars admit that fixed separation after reducing the common coupling ceiling. This does not replace the weighted multiscale metric by an old-scale Euclidean metric.

6.1 Application to the actual inherited B gradient, with its precise remaining chart premise The original-geometry response BM.6–BM.11, following CMP102 (190), has physical output normalization h_n^s with $s = 1, 2, 3$, and its input integration measure cancels the source kernel's input volume factor. For a marked current coordinate it retains

$$Cg_k e^{-\delta d_\Omega(y,i)} \tag{BC16.32}$$

in those normalized output norms. It is not a sum over unnormalized fine bonds. The local constraint linearization $g_k Cz - hD(g_k Cz)$ has the actual proper-scale stencil and fixed analytic inverse reserve; it changes only c_0 and the fixed constant.

If the true joined consistency statement JR.2 is verified in the actual source gauge, the reconstructed curvature and current on this family are the original physical outputs, with their actual unit conversion. For the reconstructed current, JR.2 gives the target physical normalization $h_n^{-2} h_{\min(p,n)}^{-1}$, which is no stronger than h_n^{-3}. Thus no stand-alone inverse fine-mesh bound is substituted for (BC16.32).

For the inner part of an old j-domain, an actual layer $n > j$ is tested at the coarser source label j. Its field/current norm gains at least $L^{-(n-j)}$, and the current gains three powers before any finer reconstructed normalization. This is sufficient for domain admission but, by itself, insufficient to pay the four-dimensional density of old roots. Proposition BC16.5 supplies the additional age factor at the activity level.

Under the local native-chart certificate of Section 5 and a verified marked gauge conversion, multiply the directional Cauchy estimate by the primitive factor $B_0 e^{-\kappa d_j(X)}$. The output point y may be anywhere in X. Apply (BC16.30)–(BC16.31) before taking its supremum. The AF radius comparison then gives

$$|\partial_i F_X| \le \frac{CB_0\theta_k}{P_k}(1+k-j)^{b_*} e^{-\Gamma(k-j)} e^{-\kappa' n(X)} e^{-c d_X(i)}, \quad \theta_k = \ell_k^{p_0-q}, \tag{BC16.33}$$

with fixed $\Gamma > 4\log L$, $\kappa' > \log A + 2$, and $c > 0$. Polynomial radius/gauge losses which are actually verified are included in the fixed exponent b_*; an unproved loss is not declared polynomial. A residual dyadic domain factor 2^{j-n}, if that smaller margin is used, is paid by the extra $\log 2$ in the fixed rate choice. Strong physical-center slack avoids that factor entirely.

This derives the age/carrier form required by BC16.3 for inherited B from the actual boundary anchor and normalized kernel. The remaining primitive native-chart and marked gauge hypotheses are explicit. They are not consequences of (BC16.32) alone, as the example (BC16.19a) shows.

6.2 The heterogeneous coefficient really begins at the geometrical layer CMP119 (2.24) uses $\phi_l = 1$ on the prescribed one-layer interior of Λ_l. Its construction (3.2)–(3.5) removes several layers of LMR_{l+1}-cubes around Z_l before defining Ω_{l+1}. For a sufficiently small common coupling ceiling, the adjacent-scale bounds (2.6)–(2.8) and the rounding convention (2.5) give $R_l \le LR_{l+1}$. Indeed the ratio of the underlying adjacent logarithmic powers is less than L; rounding to powers of L can then change their exponent by at most one. The new collar contains the old one-layer MR_l-collar. Hence

$$\Omega_{l+1} \subset \Lambda_l^{-1}. \tag{BC16.34}$$

The fixed endpoint conventions are handled by reducing to the prescribed interior collar. Rectangular enlargement of the bad complement shrinks the regular set and does not invalidate this inclusion. Thus for a point in actual layer n, every $l < n$ has $\phi_l(x) = 1$. The exact heterogeneous coefficient difference therefore has the form

$$w_k(x) - w_k = \sum_{l=n}^{k} \beta_l (1 - \phi_l(x)), \qquad |w_k(x) - w_k| \le C(1 + k - n)^2 g_k^{-2}, \tag{BC16.35}$$

with at most the already specified fixed layer offset at a localization-cube boundary. This uses the absolute coupling comparison, not a beta-function sign.

For this sector a primitive B-chart argument is unnecessary: its plaquette function is explicit. On one proper-scale cube, BC15.1 and the normalized curvature derivative (BC16.32) give

$$|\partial_i F_\Delta^{\mathrm{het}}| \le C g_k^{-1} (1 + k - n)^2 e^{-\delta d_\Omega(\Delta, i)}. \tag{BC16.36}$$

The fourth curvature power pays the number of fine plaquettes; there is no inverse fine-spacing remainder. The coefficient support lies in the prescribed old-bad neighborhood, whereas the admitted current support has the current collar. Separate endpoint shares pay this collar, the age, and the spatial anchor. Consequently the sector fits the endpoint theorem with

$$K_{\mathrm{het},k} \le C g_k^{-2} \delta_k e^{-cR_k}, \qquad \delta_k = g_k P_k, \tag{BC16.37}$$

and a fixed polynomial old-age factor. Its weighted current Hessian is bounded by $C g_k^{-1} P_k^{-1} e^{-cR_k}$, which tends to zero for the admitted $R_k \ge \ell_k^r$, $r > 1$. This deduction uses the existing original-geometry kernel scope and common physical current contour; it does not rely on the unresolved joined primitive B Cauchy disc. Native derivatives still use their actual common parameter radii.

6.3 Native projection: exact retraction, with the input norm check retained One possible extension to an off-minimum strong-native background U is

$$\Pi_B(U) = U_B(M_B U), \tag{BC16.38}$$

using the identical original source minimizing function, its determining map, and its fixed gauge slice. Its retraction property is exact on that slice:

$$\Pi_B(U_B(V)) = U_B(M_B U_B(V)) = U_B(V). \tag{BC16.39}$$

Thus this construction need not rotate old fluctuation arguments merely to recover equality at the physical center. If the baseline in C-01 is the next-scale minimum, use that exact baseline function first, followed by the actual current insertion into the preceding-scale determining data. Do not replace either map by an unnamed representative of its gauge orbit.

The source support geometry for estimating M_B is concrete. A determining bond in $\mathrm{st}_j \Omega_j \backslash \mathrm{st}_j \Omega_{j+1}$ has both endpoints outside Ω_{j+1}. Its two-endpoint-block averaging footprint misses that inner domain by block alignment. If the fixed separation obeys $RM > C_{\mathrm{foot}} L$, the footprint cannot reach a layer finer than $j-1$, because it lies within distance $C_{\mathrm{foot}} h_j$ of Ω_j. Only the j and adjacent $j-1$ norms enter. The endpoint-block footprint uses the actual nonlinear averaging operation and its iterated dependency cone; it is not the support of a straight-line substitute for that operation.

Once the actual relative-average analytic norm is verified for this source family, a strong-native input of radius r_b yields normalized determining increments at most $C r_b \alpha_{0,j}$. The source original-geometry map then sums these increments with

$$C \sum_j (1 + |j - n|)^{b_q} e^{-\Gamma_1 |j-n|} < \infty, \tag{BC16.40}$$

where a fixed polynomial bounds $\alpha_{0,j}/\alpha_{0,n}$ by the admitted AF comparison, and the age rate is the same fixed geometric rate as above. This gives a uniform native projection radius after a fixed reduction of r_b. The common contour is chosen first on the absolute variational map domain; the smaller running B thresholds are tested only afterward.

The support calculation and the sum (BC16.40) are proved deductions. They do not identify a relative-average estimate for a different Q or reference average with the actual symmetric M. That source-family identification and the common gauge chart remain checks in this projection route.

Even a proved global projection radius would not, by itself, be the local primitive-chart certificate in Section 5. The projection can transmit an input outside X into X. Using its global boundedness to claim a local primitive directional disc would lose the endpoint estimate. A local matched-lift argument, or an equivalent verified directional disc, is still required for (BC16.33).

7. Complete boundary ledger and current application status

The following source sectors remain in the application:

Sector	Required input to this direct route
Inherited primitive B	Its actual multiscale domain and full primitive carrier bound, followed by (BC16.19) for the true pullback; old retained coordinates are fixed inputs and any change of their arguments is retained.
E and R terms assigned to B	Their source assignment, old-scale subtractions and boundary localizations; regular E/R estimates alone do not remove these terms.
Classical/constraint and measure boundary pieces	The complete constrained action and the local Haar/coarea factors; the bulk estimate does not include unexamined boundary-change determinants.
Heterogeneous Wilson coefficient	The exact coefficient difference, physical plaquette normalization and four-dimensional fine-cell cancellation from BC15.1, combined with a current-to-old-boundary endpoint estimate.
Relocation	The full action change and Jacobian caused by the actual localized change of variables; smooth old selectors are fixed real parameters, not complex analytic variables.
Current Gaussian exterior force	Retained exactly in the conditional Gaussian split. It is linear in current coordinates, has zero current Hessian, and is not discarded from the conditional mean.

The primitive source ledger and the fixed scalar sector budget are unchanged. The direct endpoint theorem can replace the need for a new prepared thin-carrier expansion when proving the current derivative and weighted Hessian output. It does not itself verify all entries of this table.

The old-fluctuation arguments require an explicit distinction. In the C-01 normal form the old coordinates are retained integration data. The current localized gauge/constraint operations cannot be presumed to leave them fixed merely because they are old. CMP102 (4) fixes a variational gauge transformation at determining sites; if an old fluctuation bond has its endpoints at those sites, its adjoint rotation is exactly the identity. The admitted separation between its old bad support and the current inner region must be used to verify those endpoints. Any additional source coordinate change not having this property must retain its old-field pullback and its derivative. The reconstruction gauge converter used only to check a reconstructed-current norm is not automatically applied to the primitive B old-field arguments.

The current relocation in CMP119 p.269 changes the current-scale exterior fluctuation coordinates on S_{k+1}. Its complete induced action and Jacobian remain in the ledger above. Its exponentially small shift gives small coefficients only after the differentiated action terms, including inherited B, are controlled. It is not enough to bound the displacement and omit a derivative of the displaced primitive activity.

The complete local native B-domain chart has not been established. True-minimum consistency and an original-geometry endpoint estimate would resolve the physical reconstructed-output branch, but they do not alone prove the primitive Cauchy disc. The marked source-gauge conversion and every actual old-field/relocation pullback must also be checked. A generic bounded constraint-current coefficient is insufficient for projected derivative terms; the

actual matching issue exposed in Annex AM is retained. The local multiplier-majorant issue is more specific and has been resolved at its proper reference-average scope: the printed CMP99 bound (3.137), which already inserts a global multiplier norm, is not sufficient, but the reproduced CMP102 (72) gives the missing local pointwise derivative. Lemma CC.3a in Annex AM extracts its quadratic coefficient by Cauchy on the value-only ball, then inserts the exact determining measure factor from (3.136). This supplies the local $h_a|\lambda|\sup|A|$ bound without importing the averaging formula (149). It still does not identify that reference-average map with the raw symmetric source map.

C-05 remains open. The new results proved here are the endpoint-to-kernel theorem, the auxiliary-current admission and physical-jet identity, the combined actual path/carrier age payment, and the heterogeneous coefficient/normalization deduction. The sufficient application still needs an exact local native B-domain chart (or equivalent uniformly admitted holomorphic directional discs), the actual source gauge/coordinate conversions and old-field slots, and a checked true-map joint contour. These premises are stronger than first-order output inequalities and are not supplied by assumption.

If those checks are completed, (BC16.33), the complete sector estimates retained from BC15.14, and BC16.3 give a sufficient closure theorem with a fixed prefactor

$$K_{B,k} \le C(1+S_{b,\delta})\left[\aleph_E + C_R g_k^{\kappa_0-11} + B_0 \ell_k^{p_0-q} + g_k^{-2}\delta_k^3 + \delta_k + g_k^{-2}\delta_k e^{-cR_k} + g_k^{\nu}\right]. \tag{BC16.41}$$

The weighted Hessian is then at most $CK_{B,k}/P_k^2$; the direct all-order current kernel version has factorial exponent five and decay length proportional to the order, as in (BC16.13). Choose the source carrier/localization parameters first, the admitted separation R next to pay the combined age and row losses, one fixed $q > p_0$, and then the coupling ceiling to meet all absolute/native radii and the desired Hessian budget. No parameter is changed with derivative order or cutoff.

The native derivative branch is part of this conditional conclusion. It is not removed to declare completion of the current Hessian branch alone. Accordingly C-06 remains a theorem conditional on C-05 while C-05 is open. A proved physical-current Hessian subcase would not establish every native conclusion of C-06.

For clarity, the still unverified application is finite and concrete:

1. Identify the actual true reconstructed minimum with the original one on the mixed determining tuple, in the required analytic source gauges, and establish the joint current/native contour with normalized physical current reserves. An alternative direct source proof with the same full domain is allowed; (3.43) is not yet such a proof because of its radius ambiguity.
2. On that contour prove either the local primitive directional-disc certificate plus endpoint tangent, or a direct centered-difference/localized scalar bound as in Section 5.3. If the former uses a projected matched lift, verify the actual symmetric-average/reference-gauge identification and its source-coordinate matching; the local multiplier bound CC.3a is available at the reference-average scope. First-order reconstructed-output margins do not suffice.
3. Check the actual transformation of every old fluctuation, constraint/measure factor, independent current and relocation term, including their native complex radii and endpoint costs. The auxiliary-current construction above can supply the independent slot once the U-dependent contour is available; it does not supply the remaining U/chart checks.

The age, carrier, derivative and Hessian sums are conditional on these three source applications with the same parameters.

Sources

- T. Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, CMP 119 (1988), 243–285: especially pp.247, 261–262 and 276–277; (2.34)–(2.42), (3.43).
- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. I*, CMP 109 (1987), 249–301: around (3.6)–(3.17). DOI: https://doi.org/10.1007/BF01215223
- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. II. Cluster Expansions*, CMP 116 (1988), 1–22: especially (1.34), Lemmas 1–2 and the localization construction. DOI: https://doi.org/10.1007/BF01239022

Supplement AN.1. Complex-domain assumptions for the localized boundary term

This supplement identifies which complex-domain assumptions enter the boundary estimate.

1. A first-response bound is not a complex-domain certificate The primitive boundary activities are bounded holomorphic functions on a domain defined in part by nonlinear reconstructed-field and reconstructed-current restrictions. Their arguments cannot be treated as independent product coordinates merely because the restrictions have separate displayed bounds.

Proposition DA16.1 — a nonlinear-domain counterexample For an integer $N \geq 1$, define

$$D_N = \{u \in \mathbb{C} : |u| < 1,\ |Nu^2| < 1\}, \qquad f_N(u) = \sqrt{N}\,u, \qquad G_N(u) = (u, Nu^2). \tag{DA16.1}$$

The function f_N is holomorphic on D_N and satisfies $|f_N| < 1$. Both inequalities defining the domain have threshold one. At zero,

$$G_N(0) = 0, \qquad DG_N(0)v = (v, 0), \qquad Df_N(0)v = \sqrt{N}\,v. \tag{DA16.2}$$

Consequently, the first derivative of the constrained output tuple, normalized by its displayed margins, does not bound the first derivative of an arbitrary bounded primitive function on that domain by a constant independent of N.

Proof. The second inequality defining D_N is equivalent to $|u| < N^{-1/2}$, which proves the value bound. The derivative identities follow by direct differentiation. The last derivative diverges for $v = 1$. □

This is a counterexample to an analytic inference. It is not a counterexample to the Yang–Mills model or to the proposed observable route. Its role is to identify the additional certificate needed before applying primitive Cauchy estimates to a boundary function.

2. A sufficient certificate that does permit primitive Cauchy The relevant statement is quantitative and can be checked without demanding an entire source-dependent renormalization construction.

Let E be a finite-dimensional complex normed coordinate space. Let U_* be a baseline point. Suppose each map G_r is holomorphic on the absolute coordinate ball $\|U - U_*\|_E < R_0$. Let the primitive domain be specified by $\|G_r(U)\| < a_r$, together with any independently admitted product slots. Assume

$$\|G_r(U_*)\| \leq (1 - \epsilon)a_r, \qquad \sup_{\|U-U_*\|_E<R_0} \|DG_r(U)v\|/a_r \leq K\|v\|_E \tag{DA16.3}$$

for all the relevant restrictions, with fixed $\epsilon > 0$ and $K < \infty$. The absolute ball must be defined before assuming membership in the smaller primitive domain.

Proposition DA16.2 — uniform native ball and directional derivative Set

$$r_* = \min\{R_0/2, \epsilon/(2K)\}, \tag{DA16.4}$$

with the second term omitted if $K = 0$. Then the ball $\|U - U_*\|_E < r_*$ lies in the primitive domain, leaving at least $\epsilon a_r/2$ reserve in every displayed restriction. If b is holomorphic on that domain and $|b| \leq B$, then

$$|Db(U_*)v| \leq Br_*^{-1}\|v\|_E. \tag{DA16.5}$$

The analogous order-m directional estimates have a fixed factorial bound with cost r_*^{-m}.

Proof. Integrate the derivative of $G_r(U_* + tV)$ over $0 \leq t \leq 1$ to obtain $\|G_r(U_* + V) - G_r(U_*)\| \leq Ka_r\|V\|_E$. For $\|V\|_E < r_*$, this is at most $\epsilon a_r/2$, proving the admitted ball. Apply one-variable Cauchy to $b(U_* + zv)$ on the disc

$|z| < r_*/\|v\|_E$ for (DA16.5). For mixed directions, parameter radii r_*/m give the elementary cost $m^m \le e^m m!$, after normalizing their norms. □

One can replace (DA16.3) by a proved analytic chart with a comparable admitted radius. One can also use a directly proved endpoint derivative of the complete pullback. What is invalid is replacing the uniform native-ball assertion by the derivative at its center alone.

3. Consequence for the proposed C-05 proof The direct endpoint summation theorem of Sections 3–4 (Theorem BC16.3) is useful and has a complete proof. To apply it to the actual primitive B activity, its endpoint hypothesis must be established using a certificate such as Proposition DA16.2, a verified source-native chart, or an independent direct estimate. A bound on the first derivative of the reconstructed outputs along the physical minimizing family does not alone provide this certificate.

Likewise, a holomorphic retraction onto the physical minimizing family can preserve every physical jet, but it must have the required local domain and endpoint estimates. A global projection can introduce dependence outside a primitive carrier; exact locality of the unprojected primitive does not automatically survive that projection. If a local projection is used instead, its fine-collar current and chart estimates must be proved.

The matched-lift calculation in Annex AM supplies a concrete part of this step. In its actual regularized averaging coordinates it controls the projected tangent using a scalar gauge equation and the Wilson Ward identity. It does not identify the derivative of the raw nonlinear determining average with that regularized operator without the corresponding gauge conversion.

4. Representative conventions The CMP102 variational paper explicitly distinguishes its axial representative from the relative Landau field. Its Proposition 9 and formula (190) must be read with that distinction. The axial condition is referred to CMP99, 75–102, equation (1.19), a different paper from the CMP99, 389–434, propagator paper.

That paper is T. Bałaban, *Spaces of regular gauge field configurations on a lattice and gauge fixing conditions*, CMP 99 (1985), 75–102; DOI https://doi.org/10.1007/BF01466594. Its equation (1.19) is not used here, and no secondary source replaces that definition.

The explicit block-transport gauge constructed in Annex AL can be checked algebraically. Calling it the particular source axial gauge requires a checked identification or a separately bounded conversion. Equality of real gauge orbits and unitary invariance of real current norms do not by themselves give that complex representative estimate.

5. Dependence on the boundary estimate The heterogeneous coercivity implication in Annex AI requires the complete boundary derivative estimate on the same law.

Annex AO — Actual joined determining data and the selected source family

Annex AO. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

This annex concerns the actual-model C-05 application. It identifies the determining-set operation declared in the source, proves consistency of true minima on that operation, and gives a bounded auxiliary reconstruction for a smooth true-minimizing family in an explicitly adopted symmetric source gauge. Section 7 identifies the corresponding mixed averaging and normalized inverse-gauge map; Section 9 proves its actual older-fluctuation nodal bounds on the full current support. C-05 is not completed by this annex alone.

1. Why the join in CMP119 (2.35) has a forced completion

Keep the level of each determining variable as part of its label. In CMP119 (2.2), a complete determining set consists of

$$\Gamma_0 = \Omega_1^c, \qquad \Gamma_r = \Omega_r^{(r)} \setminus \Omega_{r+1}^{(r)}, \quad 0 < r < k, \qquad \Gamma_k = \Omega_k^{(k)}. \tag{JI.1}$$

The source explicitly states that this set determines its domain sequence. Its minimal map $U(B, V)$ in (2.12) is defined for such admitted determining histories. The nonlinear data are $M^r U$ on the level-r determining bonds, by (2.11).

The lower family $\Gamma_0, \dots, \Gamma_{p-1}$ determines its depth-p completion uniquely. First Γ_0 fixes Ω_1. Once Ω_r is known, its level-r site set minus Γ_r fixes the site set of Ω_{r+1}. The source block alignment reconstructs the physical domain uniquely from that site set. Termination at depth p forces

$$\Omega_r^{[p]} = \Omega_r \ (1 \le r \le p), \qquad \Omega_{p+1}^{[p]} = \varnothing, \qquad \Gamma_p^{[p]} = \Omega_p^{(p)}. \tag{JI.2}$$

Every datum added in this completion is specified already by the displayed argument $M^{\bullet}U$: its value is $M^p U$. No new boundary value or free source coordinate is introduced.

The symbol combining this family with $B_p(X)$ is the determining-set localization operation declared in (2.14), with its subsequent patchwise generalization. It is not a literal list of mutually overlaid constraints on different scales. Such a literal union need not be an admitted determining set: a local coarse determining cell can overlap an inherited finer determining cell, giving labels incompatible with a single disjoint terminal-block partition. The source has defined $U(B, \cdot)$ on complete determining histories, and (2.14) specifies how to combine their interior with a local collar. No different constraint-union variational problem is introduced in (2.35).

The reversed order of the two entries does not define another operation. LFR II, printed pp.373–374, uses $B(Y) \cup B_k$ for the same local determining history and identifies its interior as the region where the new and old determining sets coincide. Thus the combination in CMP119 (2.35) is obtained by completing the lower family as in (JI.2) and applying the previously declared localization operation to that completion.

This is an unpacking of the declared determining-set notation. It does not assume an unproved equality of minimizing fields, and it does not replace the source join by a pointwise intersection of arbitrary domain sequences.

2. Componentwise dictionary and the mesh bound

For an admitted boundary patch of X in old layer n, set $r = \min(n, p)$. The depth of the completed old history on that patch is r. Let I_r be its retained interior and C_r its collar, with the widths, scale and endpoint conventions of (2.14). Let $\Lambda_i^{r,X}$ be the local determining sets of the source maximal-domain construction on that patch. The declared join has

$$\hat{\Gamma}_i = \big(\Gamma_i^{[p]} \cap I\big) \cup \bigcup_{r \ge i} \big(\Lambda_i^{r,X} \cap C_r\big), \quad 0 \le i \le p, \tag{JI.3}$$

where I records the retained interiors with their proper-scale masks. The fixed level-zero exterior completes the local history. On a boundary meeting several old layers, each right-hand contribution is restricted to its source layer patch and the source instruction after (2.15) sums those restricted contributions. The admitted buffers and compatible partitions are retained; this formula does not authorize an arbitrary overlapping patch family.

There are three exhaustive cases for a terminal cell:

Location	Level in the joined history
Retained old layer $n < p$ in the interior	n
Retained old innermost truncated region Ω_p	p
Proper-scale collar in old layer n	A level at most $\min(n, p)$

Because the global odd-block partitions are nested, every smaller terminal cell is an aligned subcell of the old cell it replaces. At an old layer interface the source restricts the patch contributions before taking their union; an entry from the coarser patch is not assigned to the finer side. The determining bonds are then obtained by differences of bond stars. They are not the stars of the site-shell differences.

If $s(x)$ is the joined terminal level and $n(x)$ the old one, this gives

$$s(x) \le \min\{p, n(x)\}, \qquad h_{s(x)} \le h_{\min(p,n(x))}, \qquad \widehat{\Omega}_i \subset \Omega_i \quad (1 \le i \le p). \tag{JI.4}$$

The lattice spacing L^{-p} in the reconstructed map is a change of units on the same physical fine graph. It does not eliminate the smaller collar cells. In physical units, CMP119 (2.37) asks for a reconstructed current bound proportional to

$$h_n^{-2} h_{\min(p,n)}^{-1}. \tag{JI.5}$$

In particular, an estimate with the original physical-current weight h_n^{-3} is sufficient. A generic inverse-collar current estimate is not substituted for it.

3. Exact preservation of the old constraints

The source bond convention matters. Write $\mathrm{st}_r Y$ for bonds at level r incident to Y. A determining shell is

$$\mathrm{st}_r \widehat{\Omega}_r \setminus \mathrm{st}_r \widehat{\Omega}_{r+1}. \tag{JI.6}$$

If a local competitor W has the same joined data as U, clamp it to U outside the local fixed fine collar. The nonlinear support induction proved as Lemma JR.1 in gives

$$\mathrm{supp}(M^r W - M^r U) \subset \mathrm{st}_r \widehat{\Omega}_{r+1}, \quad 0 \le r \le p. \tag{JI.7}$$

The induction uses the actual nonlinear average: an output bond depends on its two endpoint blocks and their connecting paths. It retains crossing bonds in the star support, then removes only the prescribed star difference. It does not assume support on bonds with both endpoints inside an inner domain.

For $r < p$, (JI.4) makes the support in (JI.7) disjoint from the old level-r determining bond set. At $r = p$, the next joined domain is empty, so every relevant p-th intermediate average agrees. Iteration of the actual one-step averaging operation then fixes every old higher-level constraint. Therefore the patched competitor satisfies

$$M_B W = M_B U. \tag{JI.8}$$

This assertion is restricted to the dependency cones on which the old constrained averages are defined. It does not introduce higher averages in unrelated fixed large-field regions. Near a boundary crossing, the fine collar fixes the links required by every affected Wilson plaquette. At $p = 0$ the reconstruction is the fine-data identity. Components and holes carry their own fixed collars.

Corollary — real consistency of the actual critical orbit Let U be the true regular Wilson minimum for the old determining history. The joined mesh is no larger, so the absolute regularity conditions of the local variational problem are satisfied. Any sufficiently small differentiable variation in the joined constraint fiber extends through the fixed collar to a variation in the old fiber by (JI.8). At finite cutoff, sufficiently small variations remain in the old open regular class. The old Wilson first variation therefore vanishes, and the fixed collar makes it equal to the local first variation. The uniqueness of the regular critical orbit in CMP102 Theorem 1 identifies the joined critical orbit with the restriction of the old one.

No assertion that every large joined competitor satisfies the old regularity conditions is needed. For complex inputs, this identity can be continued only on a common admitted holomorphic chart. Real equality alone does not supply the chart radius or the norm of a representative change.

4. The available source information about the gauge

The following primary statements are separate:

1. CMP119 p.256 initially denotes a minimal configuration as an element of the unique minimal orbit.
2. CMP109 (1.15) specifies that its reconstructed maps are in axial gauges. Thus arbitrary real representative freedom alone does not prove a bound in the complex B domain.
3. CMP109 (0.11) defines the symmetric averaged-contour transport. Its one-step gauge fixing in (0.14)–(0.19) uses these variables. At the real saddle, (2.2) identifies the axial representative by $G(V) = 0$.
4. CMP109 pp.253–254 states that the earlier averaging results remain valid for the symmetric average (0.12). This licenses using the corresponding corrected averaging operators for that average; it does not identify an old corrected operator numerically with a raw nonlinear derivative.
5. CMP99 (3.55) is an explicit hierarchical product of transports at successive averaged levels. Its estimates (3.57)–(3.59) control the relative transport on a terminal block by $Ch_s\|A\|$, uniformly in its depth. The printed paragraph after (3.57) also bounds the coarse correction $v_s(y) - 1$ by that small quantity.

In particular, an unqualified claim that real unitarity and qualitative analyticity imply a uniform complex gauge bound is false. The analytic functions $\exp(iNt)$ already disprove that inference.

5. The adopted symmetric axial slice

Use one explicit convention for all auxiliary reconstructed maps: the one-step axial condition is the zero of the symmetric source gauge-fixing function, and it is iterated only over the active blocks of the admitted determining history. This is the source-compatible symmetric convention defined by CMP109 (0.11), (0.14)–(0.19) and (2.2). It is not asserted to be a verbatim copy of every older relative AX formula.

For real unitary fields, each summand $1 - \operatorname{Re}\operatorname{tr}\mathfrak{t}$ in the gauge-fixing function has the same sign and vanishes precisely when its transport is the identity, using the source normalized trace. Thus the real condition $G(V) = 0$ in (2.2) is equivalent to setting each one-step averaged-contour variable (0.11) to one. The recursion (0.19), followed by the choice of the critical representative in (2.2)–(2.3), imposes this condition successively. On a mixed history it is imposed only in active blocks; determining roots and the fixed exterior are left fixed. The product description below records exactly those iterated conditions. On complex fields it is these individual holomorphic transport equalities that are used, not the scalar equation $G = 0$ by itself.

For a terminal block $B^s(y)$ and a fine point x in it, let $x = x_0, x_1, \dots, x_s = y$ be its nested block centers. Define

$$\mathcal{T}_U(y, x) = \prod_{r=s-1}^{0} \mathfrak{t}_{M^r U}(x_{r+1}, x_r), \tag{JI.9}$$

where $\mathfrak{t}$ is the source one-step averaged-contour transport (0.11), and the product follows the root-to-point order. The group mean itself is not assumed multiplicative. The product in (JI.9) is the explicit hierarchical chain, matching the structure displayed in CMP99 (3.55).

Endpoint equivariance of each factor and gauge covariance of the averaging operation give

$$\mathcal{T}_{U^u}(y,x) = u(y)\mathcal{T}_U(y,x)u(x)^{-1}. \tag{JI.10}$$

The absolute hierarchical slice is

$$\mathcal{T}_U(y,x) = 1 \quad \text{on every terminal block.} \tag{JI.11}$$

Every real orbit of transformations fixed at the determining roots has a unique representative in this slice: take $u(x) = \mathcal{T}_U(y,x)$, with $u(y) = 1$, and use (JI.10). The same algebra gives a holomorphic section on every domain on which the actual averages and group means are defined.

Heredity under aligned refinement Suppose an admitted joined history refines the old one as in (JI.4). Its terminal block $B^t(z)$ is an aligned subblock of an old $B^s(y)$, with $t \le s$. The point $z = x_t$ is an intermediate center in the old chain. Associativity of the displayed product, not multiplicativity of the group mean, gives

$$\mathcal{T}_U(y,x) = \mathcal{T}_U(y,z)\mathcal{T}_U(z,x). \tag{JI.12}$$

The first two factors on the two sides are one when the old slice condition holds. Hence $\mathcal{T}_U(z,x) = 1$. The absolute hierarchical slice is therefore hereditary under the actual aligned refinement.

Consequently, when the input minimum is itself in this slice, real consistency from Section 3 is equality of representatives, not merely equality of gauge orbits. It extends as an equality of holomorphic maps wherever both sides have been admitted.

The convention must be used consistently in the source construction and in the definitions of its auxiliary reconstructed maps. Changing only an auxiliary domain predicate and then importing an estimate for an unrelated legacy extension would be invalid. The assertion here is that the symmetric source construction permits the displayed choice, with its corresponding averaging/reference operators. No numerical equality of arbitrary off-physical extensions is claimed.

Important distinction: the primitive input is kept smooth An absolute axial representative can have fine-edge potential spikes. Even in an Abelian block, moving a smooth coarse variation to an axial representative can put its whole coarse increment on one fine boundary link. Therefore (JI.11) is not used to replace the primitive input U in clause (2.39), nor to estimate its potential or first difference. It fixes only the auxiliary reconstructed representatives whose required outputs in (2.35) and (2.37) are curvature and physical current. Those outputs transform by conjugation, and the following argument bounds that conjugation directly.

6. Bounded auxiliary reconstruction on a smooth true-minimal family

Fix the admitted history and a real true minimum U_0. Let

$$U(z) = e^{iaA(z)}U_0 \tag{JI.13}$$

be an actual smooth holomorphic minimizing family on the common original-geometry contour. For real parameters it is in the real old critical orbit for its own determining data. No assumption that a gauge change fixes those old numerical data is needed: gauge covariance transports the whole old constraint fiber. Assume the existing original-geometry bound on each terminal block is

$$\sup_{b\subset B^s(y)} |A(z,b)| \le u_n h_n^{-1}, \qquad h_s \le h_n, \qquad \sup_n u_n \le u_*. \tag{JI.14}$$

Here u_* is smaller than the fixed absolute map radius. The native field/current endpoint bounds are those already supplied by the original-geometry estimate; they are not conclusions of a bare supremum bound for A.

For this auxiliary joined problem choose the fixed real gauge

$$g_0(x) = \mathcal{T}_{U_0}(y, x) \tag{JI.15}$$

on each joined terminal block. It equals one at every determining root, is group valued, and sends the real reference to the chosen slice. Write $\overline{U}_0 = U_0^{g_0}$. The relative perturbation of $U(z)^{g_0}$ is the adjoint transform of $A(z)$, so its value and covariant norm bounds are unchanged. The fixed real transformation has no parameter derivative cost.

Now put

$$g(z, x) = \mathcal{T}_{U(z)^{g_0}}(y, x), \qquad W(z) = \big(U(z)^{g_0}\big)^{g(z)}. \tag{JI.16}$$

The baseline transport is one. CMP99 (3.55)–(3.59), applied on this terminal block, gives the uniform analytic estimate

$$|g(z, x) - 1| + |g(z, x)^{-1} - 1| \le C h_s \sup_{B^s(y)} |A(z)| \le C u_n \frac{h_s}{h_n}. \tag{JI.17}$$

The use of the symmetric averaging version is the one expressly allowed in CMP109 pp.253–254. The bound is independent of the number of fine bonds and of s. No global path is used. A fixed reduction of u_* ensures that all these transport maps and their inverses lie in the same absolute analytic neighborhood.

Both g_0 and $g(z)$ fix every joined determining root. Every endpoint of a mixed determining bond is such a root, possibly at a finer level, by the nested-grid convention. Hence

$$M_{\hat{B}} W(z) = M_{\hat{B}} U(z), \qquad \mathcal{T}_{W(z)}(y, x) = 1. \tag{JI.18}$$

Here is the endpoint argument in full. If a level-r bond belongs to $\mathrm{st}_r \widehat{\Omega}_r \setminus \mathrm{st}_r \widehat{\Omega}_{r+1}$, neither endpoint belongs to $\widehat{\Omega}_{r+1}$. An endpoint x on the level-r lattice therefore lies in a terminal site shell of some level $t \le r$, unless it is in the fixed fine exterior. The level-r lattice is a subset of the level-t lattice, so x belongs to that shell's determining site set and is its terminal root. Exterior endpoints are fixed directly by the extension of the gauge as the identity there. This proves endpoint fixing also for a coarse bond whose second endpoint lies in an earlier finer shell; it is not an inference merely from proximity to a block center.

For real parameters, Sections 2–3 put $U(z)$ in the joined critical orbit. Equations (JI.18) and uniqueness of the slice identify $W(z)$ with the actual joined minimizing representative in the adopted convention.

Absolute admission precedes running-current estimates The relative raw averages on the joined determining bonds have dimensionless increments bounded by $C h_s \sup |A|$, from the local relative-average bounds used in (3.55)–(3.59), including the bounded coarse correction displayed after (3.57). The supremum here is over the actual two-endpoint-block dependency footprint, not only one bond. At a mixed interface the admitted separation prevents that footprint from reaching an old layer more than one step finer; the resulting fixed power of L and fixed adjacent-layer endpoint loss are included in C. Thus the increments have supremum at most $C u_*$, also at such interfaces. Choose u_* below the fixed absolute Proposition 9 radius before testing the running B-domain inequalities. The real reference data are already regular by the source variational theorem and the finer joined mesh. The source reconstructed branch and (JI.16) are then holomorphic on the common connected parameter chart. Their equality on its real open family extends to the complex chart.

This admission uses the absolute raw-data radius. It does not assume the stronger reconstructed-current conclusion in order to invoke that same conclusion.

Current and curvature bounds Let $F(U) = \partial U - 1$ denote plaquette curvature, with its base point convention, and $J(U)$ the physical Wilson current. Exact gauge covariance yields

$$F(W) = \mathrm{Ad}_{g g_0} F(U), \qquad J(W) = \mathrm{Ad}_{g g_0} J(U). \tag{JI.19}$$

The real adjoint factor of g_0 is isometric and the complex one of g is bounded uniformly by (JI.17). Consequently the old physical weights $a^2 h_n^{-2}$ and h_n^{-3} remain valid, with a fixed factor. Equation (JI.5) is weaker than the latter current weight.

For a marked current coordinate, the analyticity and uniform local bound for the transport map give

$$|g^{-1}\partial_i g| \le C h_s \sup_{B^s(y)} |\partial_i A|. \tag{JI.20}$$

The source original-geometry endpoint estimate bounds the right-hand side by $C g_k (h_s/h_n) e^{-c d_\Omega(y,i)}$. The block diameter in the old inverse-mesh metric is bounded, since $h_s \le h_n$; taking a supremum over that block changes only the fixed endpoint constant. Differentiating (JI.19) gives

$$\partial_i J(W) = \mathrm{Ad}_{g g_0}\left(\partial_i J(U) + [K_i, J(U)]\right), \tag{JI.21}$$

where K_i is $g^{-1}\partial_i g$ conjugated by the fixed real g_0, hence has the same norm. The commutator is bounded by the old current weight times $C\alpha_n g_k e^{-c d_\Omega(y,i)}$, with no inverse-collar factor. The curvature derivative has the identical form, using $[K_i, 1] = 0$. Thus this auxiliary converter retains the original normalized endpoint response and the required reconstructed physical weights.

The primitive arguments remain the smooth $U(z)$, its actual independent-current coordinate, and its specified old-field tuple. Neither (JI.16) nor (JI.19) applies a gauge transformation to those primitive arguments. The argument establishes the reconstructed-output part of their admission on a true-minimizing family; it is not yet a local analytic chart for every off-minimum native direction.

7. The actual symmetric relative-coordinate dictionary

The normalization needed for the Landau-to-axial map is displayed in an available primary formula, CMP99 p.418, (3.113). It need not be inferred from the unavailable formula (106) in the earlier averaging paper. Fix the following single operator family throughout:

Object	Definition in this construction
Raw block average M	CMP109 (0.12)
Endpoint transport $T_U(y, x)$	The same symmetric averaged contour, CMP109 (0.11)
Corrected relative average	The endpoint correction displayed in CMP99 (3.113), using this same M, T
Averaging of a gauge transformation	The anchored logarithmic mean derived below from that correction
Axial slice	The iterated one-step condition on these same transports, Section 5
Linearized Q, Q', remainders, kernels	Those associated with this corrected family, not an unmodified operator from a different path convention

CMP109 pp.253–254 explicitly transfers the earlier averaging results to its symmetric construction. Its Section 2 then actually uses the axial/Landau reduction of CMP102 for that construction. These statements specify which associated operators are meant. They do not assert that the raw derivative of M equals a corrected Q.

7.1 Direct derivation of the one-step normalization For a reference U and a nearby field W, define the relative endpoint correction

$$R_y(W; U) = \exp\left[\sum_{x \in B(y)} L^{-d} \log\big(T_W(y, x) T_U(y, x)^{-1}\big)\right]. \tag{JI.25}$$

All logarithms are in the admitted principal neighborhood. Put $\overline{U} = M(U)$. The corrected relative average at a coarse bond c is the source expression

$$\mathcal{Q}_U(W)(c) = R_{c_-}(W;U)^{-1} M(W)(c) \overline{U}(c)^{-1} \operatorname{Ad}_{\overline{U}(c)} R_{c_+}(W;U). \tag{JI.26}$$

Let $W = U^u$. Raw endpoint covariance gives $M(W)(c) = u(c_-)\overline{U}(c)u(c_+)^{-1}$. The transport in (JI.25) is $u(y)\operatorname{Ad}_{T_U(y,x)} u(x)^{-1}$. Using conjugacy of the principal logarithm gives the exact identity

$$R_y(U^u;U) = u(y)\big(\mathfrak{R}_U u(y)\big)^{-1},$$

where

$$\mathfrak{R}_U u(y) = u(y)\exp\left[\sum_{x\in B(y)} L^{-d}\log\big(u(y)^{-1}\operatorname{Ad}_{T_U(y,x)} u(x)\big)\right]. \tag{JI.27}$$

Substitution in (JI.26) yields

$$\mathcal{Q}_U(U^u)(c) = \mathfrak{R}_U u(c_-)\operatorname{Ad}_{\overline{U}(c)}\big(\mathfrak{R}_U u(c_+)^{-1}\big). \tag{JI.28}$$

This is exactly the calculation printed on p.418, now with the designated symmetric transport. It uses endpoint equivariance and local logarithms; it does not assume that a group mean is multiplicative. Its linearization is the source identity $QD = D^{\text{coarse}}Q'$. The corresponding nonlinear normalization is the iterate of (JI.27), with the successive reference averages, at each terminal determining root. It is not an independently chosen average over an entire large block.

7.2 Uniform bound for the normalized converter Annex AP, Section 9, constructs this exact normalized converter. In a reference frame in the hierarchical slice, put $R_x = \mathcal{T}_L(y,x)$, for $L = e^{iaA}U_0$ on a terminal block of size h_s. The transport bounds give

$$|R_x - 1| \le Ch_sK, \qquad |R_x - R_z| \le Ch_rK \quad (x\in B^r(z)\subset B^s(y)), \quad K = \sup_{B^s(y)}|A|.$$

In this frame (JI.27) is the anchored mean $r_z\exp[L^{-d}\sum_x \log(r_z^{-1}r_x)]$. Its supremum-norm derivative at a constant identity tuple is the normalized arithmetic mean, of norm one. On one fixed neighborhood it is at most two. If E_r is the error between its r-fold iterate and the original value at that root, then

$$E_r \le 2E_{r-1} + Ch_rK, \qquad E_r \le \frac{C}{1-2/L}h_rK. \tag{JI.29}$$

The odd block size satisfies $L \ge 3$. A fixed smallness choice for h_sK therefore keeps every iterate in the same analytic neighborhood, independently of the depth. At the terminal root, the iterated value is $F_s = 1 + O(h_sK)$. The root factor $c_y = F_s^{-1}$ and $u(x) = c_yR_x$ satisfy both the source terminal normalization and the chosen axial condition, with

$$|u-1| + |u^{-1}-1| \le Ch_sK, \qquad |u^{-1}\dot{u}| \le Ch_s\|\dot{A}\|_{\infty,B^s(y)}. \tag{JI.30}$$

The second bound follows by holomorphic differentiation on a smaller fixed absolute ball. It retains an endpoint bound when $\dot{A}$ has one.

For a reference not already in the slice, the exact formula is

$$u(x) = \mathcal{T}_{U_0}(y,x)^{-1}c_y\mathcal{T}_L(y,x). \tag{JI.31}$$

The anchored mean is covariant under a simultaneous frame change of U_0, u; this follows by conjugating each logarithm in (JI.27). Thus one may put the normalization equations in the hierarchical frame and pull the answer back.

For a complex reference near a fixed real one, source transport comparison gives bounded frame factors and the same estimate (JI.30). No propagator or stationary equation is moved into a potentially spiky axial chart. All such operators remain in the admitted smooth reference chart. The frame change is used only in the algebraic normalization equations.

7.3 Raw-data restoration on a mixed history For the zero-data reconstruction around an arbitrary admitted U_0, prescribe $V = M_{\hat{B}} U_0$. The identity relative field has every relative average and endpoint correction equal to one, so its corrected datum is exactly $B = 0$. This conclusion does not require the missing formula for a general nonzero relative datum.

Apply (JI.31) independently on the disjoint actual terminal blocks and set the gauge to one in the fixed fine exterior. This defines a single fine-lattice gauge, with one normalization at each terminal root at its own level. CMP102 Proposition 2 gives the correspondence between its nonlinear corrected fiber and the actual raw-data fiber, including the mixed crossing-bond convention. By uniqueness of that normalized inverse, the explicitly constructed gauge is the gauge in this correspondence. Thus a critical Landau correction on the zero corrected fiber produces the actual raw determining data V. No numerical equality of independently prescribed coarse gauges v_r, v_t is needed at an interface.

This invocation is within the one associated symmetric family in the table above. The local identities (JI.25)–(JI.28), the source's symmetric transfer and the explicit use of the same reduction in CMP109 Section 2 provide its source dictionary. The weighted stationary and current estimates for this corresponding family are proved separately in Annex AP; they are not consequences of the gauge algebra alone. Joint analyticity on the absolute admitted neighborhood extends the real correspondence to its complex chart.

8. Old arguments cannot be discarded by invoking gauge invariance

CMP119 (3.15) evaluates inherited B with the new current insertion in its background argument and the old fluctuation tuple retained as written. A gauge used only to estimate a reconstructed-field/current norm need not be applied to that tuple. This is different from replacing the entire primitive argument by a gauge-transformed argument.

The latter operation must transform every old adjoint slot. Old A_{j-1} lives on a region of the $(j-1)$-lattice inside Ω_j; its base points need not be level-j determining roots. Root fixing alone therefore does not prove its transformation is the identity. LFR II pp.373–374 explicitly retains the splice gauge because two field representations meet at the collar. No gauge-invariance argument here removes that factor.

8.1 A rigorous smooth-chart pullback, when simultaneous covariance is used This subsection is a conditional alternative only. CMP119 (2.34)–(2.42) supplies analyticity in the field/current pair with the older fields fixed; it does not supply the holomorphic old-field domain hypothesized below. The actual application uses the fixed-old-field replacement in Section 9 and AD17.3 instead.

There is nevertheless a valid way to use the covariance of the whole primitive function. State its contour premises first. Suppose the real physical family is written in an axial representative, and an actual analytic gauge $u(z,\eta)$ carries it to a smooth relative-Landau representative $U_L(z,\eta)$. Suppose u is group valued for real parameters and has bounded adjoint factors and marked derivatives on the common complex contour. Let A denote the fixed old integration coordinates. Define directly

$$F_X^{\mathrm{sm}}(z,\eta) = b_X\big(U_L(z,\eta), J_L(z,\eta), \mathrm{Ad}_{u(z,\eta)}\,\mathrm{A}, \mathrm{S}\big), \tag{JI.22}$$

where the current argument is transformed by the same actual gauge, including an independent-current displacement if present. The sign of u is fixed by $U_L = U_{\mathrm{AX}}^{u}$. Require that every displayed argument lies in the source primitive domain with its specified margins.

Then (JI.22) is a holomorphic extension of the physical primitive pullback. For real parameters, the source's simultaneous group-valued covariance gives its equality with $b_X(U_{\mathrm{AX}}, J_{\mathrm{AX}}, \mathrm{A}, \mathrm{S})$. Thus their physical parameter derivatives agree to every finite order. The proof does not invoke complex covariance at an axial point that may fail the relative-field condition (2.39); it defines the holomorphic extension using the admitted smooth-chart arguments. Nor does it change an old integration variable, selector or measure. It retains the explicit current-dependent argument of the primitive function.

8.2 Old sup-norm slots have a uniform relative margin For an old adjoint slot with analytic radius R_j, assume $\|A_j\| \le (1-\sigma)R_j$, with one fixed $\sigma > 0$. If $\|u-1\|, \|u^{-1}-1\| \le \epsilon$ and ϵ is a sufficiently small fixed multiple of σ, then

$$\|\operatorname{Ad}_u A_j\| \le (1+\epsilon)^2(1-\sigma)R_j \le (1-\sigma/2)R_j. \tag{JI.23}$$

Support is preserved exactly. For a marked current coordinate, the old input is fixed, and differentiation gives

$$R_j^{-1}|\partial_i(\operatorname{Ad}_u A_j)| \le C\,|u^{-1}\partial_i u|\,\frac{\|A_j\|}{R_j}. \tag{JI.24}$$

Thus the old logarithmic amplitude cancels against its actual analytic radius. If the source old-slot domain is a product of these balls, using the supremum of their normalized norms produces no factor equal to the number of old scales. An endpoint-local bound for $u^{-1}\partial_i u$ remains an endpoint-local bound for this whole old-slot tangent.

Any additional old-field constraint or gradient clause must be checked in its actual definition. A gauge-covariant norm co-transformed with its background obeys the same adjoint estimate. A norm with a fixed reference background, or a noncovariant constraint chart, does not acquire that property merely from (JI.23). The source old-slot analytic radii and the required marked converter are separate application checks. This lemma therefore does not justify dropping an old argument or a relocation derivative.

9. Actual old-node gauge bounds on the full current support

The converter hypothesis in AD17.3 is supplied by the original-geometry response already proved in Annex AF, together with the normalized inverse gauge above. This section performs that application; it does not assume an independent bound on unknown gauge values.

Write

$$P_k = A_1\ell_k^{p_0}, \qquad \alpha_n = C_\alpha g_n \ell_n^q, \qquad q > p_0, \qquad \vartheta_k = \frac{g_k P_k}{\alpha_k} = \frac{A_1}{C_\alpha}\ell_k^{p_0-q}. \tag{JI.32}$$

The actual nonlinear constraint input is $g_k Cz - hD(g_k Cz)$, not its linear approximation. The full current ball is $\|z\|_\infty < 2P_k$. Proposition E14.1 and the running-radius check E14.24 admit that input in the source's absolute map chart on this entire ball. The original-geometry estimate BM.10 then gives, on output layer $n \le k$,

$$h_n\|A_{\mathrm{L}}(z) - A_{\mathrm{L}}(0)\|_{\infty,\Delta_n} \le C g_k P_k e^{-\gamma(k-n-c_0)_+}. \tag{JI.33}$$

The same estimate holds for the other normalized output rows; BM.11 supplies the physical current row. The proof of BM.10 first retains the endpoint kernel and only then takes its normalized row sum. Consequently a single current coordinate also has

$$h_n|\partial_i A_{\mathrm{L}}(x)| \le C g_k e^{-\delta d_\Omega(i,x)}, \tag{JI.34}$$

after the fixed absolute operator and native-radius reductions, with the current ball still $\|z\|_\infty < 2P_k$. The absolute map has the reserve exhibited in E14.8 before this running current radius is inserted. Constants include the actual constraint stencil and its fixed adjacent-layer enlargement. There is no factor equal to the number of fine bonds or to the current support volume.

The AF comparison used in BM.13 is $\alpha_k/\alpha_n \le C_{\mathrm{AF}}\sqrt{1+k-n}$. Therefore (JI.33) gives

$$\frac{h_n}{\alpha_n}\|A_{\mathrm{L}}(z) - A_{\mathrm{L}}(0)\|_{\infty,\Delta_n} \le C\vartheta_k\sqrt{1+k-n}\,e^{-\gamma(k-n-c_0)_+} \le C'\vartheta_k. \tag{JI.35}$$

Using BM.3, a fixed part of the endpoint exponent in (JI.34) pays this same square-root ratio. A positive endpoint rate remains:

$$\alpha_n^{-1}h_n|\partial_i A_{\mathrm{L}}(x)| \le \frac{C\vartheta_k}{P_k}e^{-\delta' d_\Omega(i,x)}. \tag{JI.36}$$

9.1 From the Landau response to the actual fine gauge Apply (JI.30), or RM.33 and RM.36, on each actual terminal block. Its side is h_n, and its diameter in the source inverse-mesh metric is bounded. Enlarging to the fixed transport dependency neighborhood changes only the constants and the reserved positive endpoint rate. It follows that the actual normalized fine gauge has

$$\alpha_n^{-1}|\log u(x)| \le C\vartheta_k, \qquad \alpha_n^{-1}|\partial_i \log u(x)| \le \frac{C\vartheta_k}{P_k} e^{-\delta'' d_\Omega(i,x)}. \tag{JI.37}$$

The logarithm and inversion use the same fixed small matrix neighborhood. If the convention calls the inverse gauge the axial-to-Landau converter, (JI.37) holds for that inverse as well. The estimate is on the actual fine-lattice values, not merely on their normalized average.

For a varying native reference, retain both inputs in (JI.31). Compare their transports with the fixed real reference U_*, as in RM.35–RM.36. If the relative native reference has value radius $r_b\alpha_n/h_n$, its contribution to the first inequality in (JI.37) is Cr_b. The actual external source chart has that bound by E14.45–E14.48 and the same AF comparison; its fixed coefficient may be reduced after the constants in AD17.3 are known. Current derivatives still differentiate the current input alone, while native derivatives retain the reference term. No reference derivative is dropped. These statements are simultaneous with the independent current variable because the gauge formulas are analytic functions of the same admitted smooth output; the independent current is never identified with its physical value off the physical slice.

There are two distinct identifications here. On real retained determining data the source field is the true minimizing family, and the correspondence is physical. For an arbitrary admitted external native background, use exactly the formal family selected in E14: CMP102 (179)–(180), with its CMP116 (1.17) preparation. CMP102 p.307 defines its axial continuation by applying the normalized inverse gauge to its Landau expression; p.309 extends the operators and analytic estimates to admitted nonminimal external backgrounds. The converter above satisfies the actual same normalization and relative axial equations, so uniqueness of this algebraic gauge construction matches that formal continuation also. It does not assert that the formal continuation is a physical minimum off the minimal reference graph. This argument does not substitute the possibly different formal continuation (174)–(175). The latter is used elsewhere only to read a checked linear kernel. The auxiliary reconstructions required by the primitive B domain, on any such arbitrary smooth input, are the actual zero-data minima treated by NP.5 and RM.3.

For clarity about fixed real gauges, let $t(z,b)$ be the full Landau-to-selected-axial converter, including the rooted reference factor in RM.37, and put $t_* = t(0,0)$. The latter is real and fixed. Rewrite the whole primitive once by its real covariance, replacing its old fields by the fixed real tuple $\mathrm{Ad}_{t_*^{-1}} \mathrm{A}$. The converter from the correspondingly transformed axial input to the smooth field is

$$\gamma(z,b) = t(z,b)^{-1} t_*. \tag{JI.38}$$

It equals one at the center. Transport comparison proves (JI.37) for γ, with the additional Cr_b term. The spatially varying fixed real gauge is not applied to the smooth input chart or to its propagators. Its only purpose here is the exact once-for-all rewriting of the real primitive and its fixed real older arguments. On a reference already in the adopted source slice this factor is simply the identity.

9.2 Actual action on the older fluctuation nodes The earlier field A_j is the logarithm of a quotient of two raw j-scale link configurations with the same endpoint covariance. Hence their quotient transforms by conjugation at the initial point, giving exactly CMP119 (1.17):

$$A_j(b) \longmapsto \mathrm{Ad}_{\gamma(b_-)} A_j(b). \tag{JI.39}$$

Raw averaging restricts the fine gauge to the j-grid. The iterated mean $\Re\gamma$ used to normalize a relative chart is not substituted for this restriction.

By the actual old-slot support and grading proved in AD17.19–AD17.19c, every such basepoint has terminal level $n \in [j, j+2]$. Adjacent running radii and meshes have fixed comparison constants. Equations (JI.37)–(JI.39) therefore imply

$$\begin{aligned} |\log \gamma(x)| &\le C(\vartheta_k + r_b)\alpha_{j+1}, \\ \alpha_{j+1}^{-1}|\partial_i \log \gamma(x)| &\le \frac{C\vartheta_k}{P_k} e^{-\delta''' d_\Omega(i,x)}, \qquad x \in \mathcal{N}_j. \end{aligned} \tag{JI.40}$$

Thus (AD17.20) and its current-endpoint version hold for the actual normalized fine converter. The first coefficient can be made smaller than any fixed required reserve: choose the native coefficient r_b first, then reduce the common coupling ceiling so that ϑ_k is uniformly small. Neither choice depends on age, observable order, fine mesh or total volume.

9.3 Fixed-old-field replacement AD17.3 interpolates these actual nodal values by a smooth fine gauge υ, using disjoint bumps in fixed real local regular frames. It has $\upsilon = \gamma$ at every older fluctuation basepoint, so real simultaneous covariance gives

$$B(U_{\mathrm{AX}}, J_{\mathrm{AX}}, \mathrm{A}) = B(U_{\mathrm{L}}^{\upsilon^{-1}}, \mathrm{Ad}_{\upsilon^{-1}} J_{\mathrm{L}}, \mathrm{A}) \tag{JI.41}$$

after the fixed real convention change just described, if required. The right side has fixed real older fields. It is the admitted holomorphic extension on the contour once NP.5 and the explicit AD17.28 reserve are imposed. No complex old-field analytic radius is invoked. The bump construction retains the endpoint estimate; the curl/current row follows from exact current covariance and the local chart identity, as proved in AD17.25–AD17.27.

This bridge applies to the actual physical pullback and its admitted analytic continuation. It does not assert that a complex decoupled intermediate field is itself a physical minimum. A proof that uses a separately prescribed decoupling family must keep that family's own map identification; the direct endpoint summation BC16 does not require such a replacement. The complete boundary-sector and carrier assembly remains a separate final check.

Primary sources used

- CMP119 (1988), 243–285, (2.2), (2.11)–(2.15), (2.34)–(2.39), (3.15).
- CMP102 (1985), 277–309, Theorem 1, Proposition 2, (124)–(127), (181), Proposition 9.
- CMP99 (1985), 389–434, (3.55)–(3.59) and (3.113); printed pp.401 and 418.
- LFR II, CMP122 (1989), 355–392, pp.373–375, (1.61)–(1.64), including the explicit warning about the retained splice gauge.
- CMP109 (1987), 249–301: (0.11)–(0.19), (1.15), (2.2)–(2.5), and the transfer statement for symmetric averaging on pp.253–254.

The geometry argument uses the source's admitted patch construction. It does not assert a theorem for arbitrary nonaligned overlapping domain families.

Annex AP — Raw-data matching and bounded source gauge conversion

Annex AP. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

Scope

The first route in this note distinguishes the actual symmetric raw average from its associated reference-corrected relative average. It does not equate those maps. The main new route instead takes an arbitrary admitted input configuration as the reference background and prescribes its own exact determining data. The relative datum is then exactly zero. This removes the need to match derivatives of two separately gauged input families.

The estimates below concern the analytic variational problem on one admitted determining geometry. Their application to the actual tuple, complete B domain and selected source representative is retained separately. This annex does not complete C-05.

1. The symmetric operator family must be selected consistently

CMP109, pp.253–254, defines the symmetric averaged transports in (0.11) and the raw block average in (0.12). The same text states that the preceding averaging results apply to this average. This is a primary-source transfer statement. Thus one may use the reference-corrected averaging construction associated with this actual symmetric average and its transports. One may not silently identify the corrected derivative with the uncorrected raw derivative, or combine kernels from different choices.

At one step, write M for the raw source average and $T_U(y,x)$ for its endpoint-equivariant averaged transport. Its associated corrected relative map has the form displayed in CMP99 (3.113): the raw relative average is conjugated at its two endpoints by the averages of the relative transported fields. At a fixed real background its linearization obeys

$$q(A) = m(A) + D^{\mathrm{coarse}}\xi(A), \qquad |\xi(A;y)| \le Ch \sup_{B(y)} |A|. \tag{RM.1}$$

Here m is the derivative of the raw average, q the corrected derivative and h the physical coarse spacing. The bound follows by differentiating the path logarithms and summing their physical lengths. For this associated family, differentiated gauge covariance gives $QD = D^{\mathrm{coarse}}Q'$.

These are levelwise identities. A complete arbitrary-tangent proof still has to match the endpoint corrections on the mixed determining geometry. A coarse determining bond can have an endpoint which belongs to an earlier, finer shell. Independently chosen corrections at every level are not automatically one gauge on that mixed set.

2. Exact zero relative datum

Fix an admitted background U_0, and prescribe the raw determining data

$$V = M_{\mathcal{B}} U_0. \tag{RM.2}$$

In the relative coordinate $U = U'U_0$, the input point is $U' = 1$. Every relative transport, relative average and endpoint correction is the identity there. Consequently the corrected determining datum used in the variational problem is exactly

$$B = 0. \tag{RM.3}$$

This is an equality at the defining relative map, rather than an equality of two different tangent operators. It is valid on a mixed determining set because every component is evaluated at the identical reference field. The assertion does not require an arbitrary off-minimum input to equal its reconstructed minimum.

The constructed correction may use a different analysis coordinate while producing the same physical critical orbit. CMP102 p.287 explicitly permits a different linearizing right inverse h in the map $A = A' - hD(A')$. This freedom is used below to obtain a regular current kernel without an unproved profile extension of a global estimate.

3. A right inverse with checked current kernels

Let H be the dimensionless-data version of the source operator H_1. Thus

$$h_b(Q\mathsf{H}B)(b) = B(b), \qquad RD^*\mathsf{H}B = 0. \tag{RM.4}$$

The source expansion CMP102 (174)–(177) states that its physical relative-Landau field satisfies

$$\mathcal{A}(B) = \mathsf{H}B + O(B^2). \tag{RM.5}$$

The physical field, including the linearizing correction, is used in this expansion. CMP102 (190) gives its value, first-gradient, D^*D and component-Laplacian derivative kernels. Page 309 explicitly extends those bounds to the admitted arbitrary regular external background. Differentiating (RM.5) at zero therefore supplies

$$|\mathcal{L}_r\mathsf{H}(x,b)| \le Ch_a^{-1-r}h_b^{-d}e^{-\delta d(a,b)}, \quad x \in \Delta(a), \quad d = 4, \tag{RM.6}$$

where $r = 0, 1$ means value and covariant first gradient. For $r = 2$, one may use D^*D, the component covariant Laplacian, or the physical quadratic Wilson Hessian Δ. The last follows from the source local bounded curvature correction relating Δ to D^*D. The field $\mathcal{A}(B; U_0)$ used to read this linear kernel need not be interpreted as a physical minimum for an off-minimum external U_0. Only its explicitly stated analytic kernel estimate is used.

Which external-background extension is used The right-inverse and Landau identities in (RM.4) are available before the nonlinear minimizing problem. CMP99 (3.127)–(3.129) defines H_1, for an arbitrary admitted regular external U_0, as the solution of its quadratic variational problem with the restrictions $QA = B$, $RD^*A = 0$. Its formula is $G_1Q^*(QG_1Q^*)^{-1}$, with the dimensionless-data conversion indicated there. These identities do not assume that U_0 is a minimum.

For the current kernel we use specifically the formal analytic extension of the family (174), (175) licensed by CMP102 p.309 and Proposition 9. In that family the remainder V is cubic for every admitted external background: in particular its current-dependent term is $-\langle HD_3, J_0\rangle$, not a linear term. Therefore the fixed-point solution $\mathcal{A}_1(B)$ has $\mathcal{A}_1(0) = 0$ and $\mathcal{A}_1'(0) = 0$, even when $J_0 \neq 0$. The exact expression (174) then gives $\mathcal{A}'(0) = H_1$. The bound (190) is used as the explicitly stated primary bound on this analytic extension.

We do not identify the off-minimum continuation defined by (174), (175) with any continuation obtained from (179), (180) merely because both describe the physical family when the reference is minimal. The selected formal continuation here is used only to read the linear operator kernel; it is not used as an off-minimum physical minimizer.

Its external-background assumptions are the full ones printed in CMP99 (3.35)–(3.38). A real reference has local gauges with value, first difference and d^*d bounds $CM\alpha_0(h^{-1}, h^{-2}, h^{-3})$. A complex reference $e^{i\eta A_0}U_*$, with U_* real, has value, covariant first-gradient and $D^*_{U_*}D_{U_*}$ bounds $\alpha_1(h^{-1}, h^{-2}, h^{-3})$. The source smallness conditions on $M\alpha_0, \alpha_1$ are retained. A native ball used in the application must imply these conditions.

All transposes below use the physical fine measure η^d and determining measure h_b^d. On a complex chart they are analytic bilinear transposes.

4. Profiles and the exact linearizing map

Let h_a be the local joined determining mesh, h_n the old physical mesh at that point, and assume

$$h_a \le h_n, \qquad w_a = \alpha_n h_n^{-3}, \qquad e_a = h_a^3 w_a, \qquad e_* = \sup_a e_a \le \alpha_*. \tag{RM.7}$$

Use the admitted profile comparison and exponential row-sum reserves for w, e, e^2. These follow from the source metric and a fixed sufficiently large layer separation, just as in CC.1–CC.4. No reserve depends on derivative order or total volume.

Set

$$\|A'\|_{X_w} = \sup_a \max\left\{ \frac{\|A'\|_{\infty,\Delta(a)}}{h_a^2 w_a}, \frac{\|\nabla A'\|_{\infty,\Delta(a)}}{h_a w_a} \right\}. \tag{RM.8}$$

Assume $\|A'\|_{X_w} \le \rho$, and that the physical input current satisfies $|J_0| \le \mu w$. The small parameters ρe_* and μe_* are required to fit one fixed absolute source radius.

Write the source averaging remainder as in CMP102 (44),

$$Q_j(A) = h_j Q_j A + C_j(h_j A).$$

Use H in the exact fixed-point equation

$$D(A')_b = C_j(h_b[A' - \mathsf{H}D(A')])_b\,. \tag{RM.9}$$

Then the field

$$A = A' - \mathsf{H}D(A') \tag{RM.10}$$

has corrected nonlinear determining datum hQA'. The identity is algebraic: apply $hQ\mathsf{H} = I$ and (RM.9). In particular $QA' = 0$ gives the desired zero relative datum. Since $RD^*\mathsf{H} = 0$, a Landau field A' produces a Landau field A.

Lemma RM.1 — the quadratic profile is retained For ρe_* sufficiently small, independently of the number of layers, (RM.9) has a unique analytic solution in the fixed-point ball, and

$$|D(A')_b| \le C\rho^2 e_b^2. \tag{RM.11}$$

Consequently

$$\begin{aligned} |\mathsf{H}D(A')|(x) &\le C\rho^2 h_a^{-1} e_a^2, \\ |\nabla\mathsf{H}D(A')|(x) &\le C\rho^2 h_a^{-2} e_a^2, \\ |\Delta\mathsf{H}D(A')|(x) + |\Delta_{\mathrm{cov}}\mathsf{H}D(A')|(x) &\le C\rho^2 h_a^{-3} e_a^2. \end{aligned} \tag{RM.12}$$

The last line is at most $C\rho^2 e_* w_a$.

Proof. The source local remainder bound (44) is quadratic in the actual local dimensionless input. The value kernel in (RM.6), summed against e^2, gives

$$|\mathsf{H}D|(x) \le C h_a^{-1} e_a^2 \|D\|_{e^2}.$$

The local stencil comparison makes the normalized argument of C_j at b at most $C(\rho e_b + \|D\|_{e^2} e_b^2)$. On a ball $\|D\|_{e^2} \le C_D\rho^2$, its square is at most $C\rho^2 e_b^2$, provided ρe_* is small. Choose C_D first and the common smallness ceiling next.

For two candidates, the local analytic derivative of the quadratic remainder is bounded by a constant times the sum of the two local argument norms. The difference of their arguments is bounded by $Ce_b^2\|D_1 - D_2\|_{e^2}$. After division by e_b^2,

the Lipschitz constant is at most $C\rho e_*$. Reduce the ceiling to make it less than 1/2. This proves existence, uniqueness and analyticity in the fixed-point ball. The remaining rows of (RM.6) give (RM.12). All sums use the actual physical determining measure. □

The change of analysis coordinate is locally invertible On one fixed smaller ordinary source ball, $A = A' - \mathsf{H}D(A')$ is an analytic bijection onto a neighborhood of zero. To see this directly, set $A' = A + \mathsf{H}X$ and solve

$$X = D(A + \mathsf{H}X). \tag{RM.12a}$$

The quadratic value bound and the first derivative estimate below make this a contraction on the ordinary dimensionless coarse-field ball, exactly as in CMP102 (59)–(62). Its derivative is $O(\epsilon)$, with ϵ the absolute source size. The first-gradient bounds of H show that the map and inverse preserve the smaller ordinary strong ball.

The identity $\mathcal{Q}(A) = hQA'$ shows that this bijection maps every linear determining fiber onto the corresponding nonlinear corrected fiber. Both directions preserve the Landau condition because $RD^*\mathsf{H} = 0$. Consequently a previously constructed physical critical solution, after its source reduction to this corrected fiber, is carried to a critical solution in these coordinates. This change does not alter its physical orbit, nor any old integration variable.

5. Derivative kernels of the changed linearizing map

The proof of CMP102 (65)–(73) uses the value/gradient kernel of its chosen right inverse and the pointwise derivative (72) of the local averaging remainder. The same differentiation of (RM.9), with (RM.6), gives

$$|D'(A'; b, x)| \le C\epsilon\, h_a^{1-d} e^{-\delta_1 d(a,b)}, \qquad \epsilon = C\rho e_*, \tag{RM.13}$$

where $x \in \Delta(a)$. The constant $\delta_1 > 0$ is a fixed smaller rate.

For completeness, the first term in the differentiated equation is the local kernel $h_b C_j'(h_b[A' - \mathsf{H}D])$; the checked source bound (72) makes it at most $C\epsilon h_b^{1-d}$ on its actual stencil. Composing it with H gives a determining-space kernel bounded by $C\epsilon h_c^{-d} e^{-\delta_1 d(b,c)}$. Its Neumann series converges by the fixed source row-sum reserve. The resulting local-to-global composition is (RM.13). Thus no pointwise transpose estimate for the bare raw averaging operator is inserted.

Let $D_3 = D - C^{(2)}$, with the quadratic normalization used in CMP102 (56). The quadratic coefficient of (RM.9) is still $C^{(2)}$, independently of the selected right inverse. Analytic coefficient extraction from (RM.13), on a fixed larger absolute value ball, gives

$$|D_3'(A'; b, x)| \le C\epsilon^2 h_a^{1-d} e^{-\delta_1 d(a,b)}. \tag{RM.14}$$

Indeed the linear term of $D'(tA')$ is the derivative of $C^{(2)}$. Cauchy on a circle of radius proportional to the fixed absolute analytic radius divided by ϵ bounds the remaining terms by $C\epsilon^2$, with the same smaller rate. This is the same coefficient argument stated after CMP102 (73), now with the selected right inverse.

6. The full unrestricted action, with no discarded gauge term

Use the actual quadratic Wilson Hessian Δ. Define the action in the changed coordinate exactly:

$$\mathcal{F}(A') = S(\exp(i\eta[A' - \mathsf{H}D(A')])U_0)\,. \tag{RM.15}$$

Expand the original Wilson action as $S(\exp(i\eta A)U_0) = S(U_0) + \langle A, J_0\rangle + \frac{1}{2}\langle A, \Delta A\rangle + V_0(A)$. Then

$$\mathcal{F}(A') = S(U_0) + \langle A', J_0\rangle + \frac{1}{2}\langle A', (\Delta - D_{2,\mathsf{H}})A'\rangle + V_{\mathsf{H}}(A'), \tag{RM.16}$$

where

$$\langle A', D_{2,\mathsf{H}}A'\rangle = 2\langle \mathsf{H}C^{(2)}(A'), J_0\rangle$$

and

$$\begin{aligned} V_{\mathsf{H}}(A') = & -\langle \mathsf{H}D_3(A'), J_0\rangle - \langle A', \Delta\mathsf{H}D(A')\rangle \\ & + \tfrac{1}{2}\langle \mathsf{H}D(A'), \Delta\mathsf{H}D(A')\rangle + V_0(A' - \mathsf{H}D(A')). \end{aligned} \tag{RM.17}$$

This identity is defined for unrestricted sufficiently regular A'. It does not replace Δ by a gauge-invariant extension off the Landau slice, and it does not delete the cross term using an orthogonality valid only on a restricted tangent space.

The local Hessian estimate CC.3a applies with the selected right inverse: its proof uses the local averaging derivative and the actual multiplier. The adjoint row in (RM.6) gives

$$|(\mathsf{H}^* J_0)(b)| \le C\mu h_b^{-1} w_b.$$

Equivalently the dimensionful multiplier has size $C\mu w_b$. Thus

$$|D_{2,\mathsf{H}} A'|(x) \le C\mu\rho e_a w_a \le C\mu\rho e_* w_a. \tag{RM.18}$$

Lemma RM.2 — weighted remainder gradient On $\|A'\|_{X_w} \le \rho$, with the above fixed smallness reserves,

$$\|V'_{\mathsf{H}}(A')\|_{F_w} \le C\rho^2 e_*(1+\mu e_*), \qquad \|f\|_{F_w} = \sup_a w_a^{-1}\|f\|_{\infty,\Delta(a)}. \tag{RM.19}$$

On a smaller fixed-fraction ball its derivative has norm

$$\|V''_{\mathsf{H}}(A')\|_{X_w \to F_w} \le C\rho e_*(1+\mu e_*). \tag{RM.20}$$

Proof. Differentiate every term of (RM.17). The exact contributions are

$$\begin{aligned} & -D'_3(A')^* \mathsf{H}^* J_0, \\ & -\Delta\mathsf{H}D(A') - D'(A')^*(\Delta\mathsf{H})^* A', \\ & D'(A')^* \mathsf{H}^* \Delta\mathsf{H} D(A'), \\ & (I - D'(A')^*\mathsf{H}^*)\, V_0'(A' - \mathsf{H}D(A')). \end{aligned} \tag{RM.21}$$

The equality $\mathsf{H}^*\Delta A' = (\Delta\mathsf{H})^* A'$ is an adjoint-kernel identity with the fixed source boundary conditions. It moves the second-order operator onto the checked current kernel, rather than requiring a generic second derivative of A'.

The physical-measure row sums of (RM.6) and (RM.13) give the following mapping rules:

$$\begin{aligned} & \mathsf{H}^* : F_w \longrightarrow h^{-1} w, && (\Delta\mathsf{H})^* : h^2 w \longrightarrow h^{-1} w, \\ & D'(A')^* : h^{-1} w \longrightarrow C\epsilon\, w, && D'_3(A')^* : h^{-1} w \longrightarrow C\epsilon^2 w. \end{aligned} \tag{RM.22}$$

For example the second rule contains the sum

$$h_b^{-1} w_b \sum_a (h_a/h_b)^{d-1} e^{-\delta d(a,b)}\, w_a/w_b,$$

which is bounded by the reserved metric rate. This displays the physical-volume factors; no fine-bond counting factor is hidden.

The first line of (RM.21) is bounded by $C\mu\epsilon^2 w$. By (RM.12), the first term in its second line is at most $C\rho^2 e_* w$; the second is at most $C\rho\epsilon w$. The third line is at most $C\epsilon\rho^2 e_* w$.

For the final line, write $A = A' - \mathsf{H}D(A')$. It has value and first-gradient bounds $C\rho h^2 w, C\rho h w$. The local Wilson remainder estimates of CMP102 (39)–(40), (90)–(96) give

$$|V_0'(A)|(x) \le C\rho^2 h_a^3 w_a^2 = C\rho^2 e_a w_a. \tag{RM.23}$$

The leading cubic terms are bounded by $C|A||\nabla A|$; the remaining analytic plaquette terms have the additional local small-field or η/h_a factors described there. The finite fine-stencil sums preserve the old profile, and $\eta \le h_a$. The prefactor $I - D'^*\mathsf{H}^*$ multiplies this by at most $1 + C\epsilon$, using (RM.22). Combining the four lines proves (RM.19).

All functions are holomorphic on a fixed larger X_w ball because the absolute source radius is fixed and ρe_* is small. One-dimensional Cauchy in a normalized X_w direction, using (RM.19) on that larger ball, proves (RM.20). The constants include only fixed radius fractions. □

7. What this gives to the zero-data bootstrap

For an actual physical critical configuration in the relative Landau slice, the unrestricted constrained first variation of (RM.16) gives

$$\langle v, J_0 + (\Delta - D_{2,\mathsf{H}})A' + V'_{\mathsf{H}}(A')\rangle = 0 \quad \text{for every } Qv = 0. \tag{RM.24}$$

With $QA' = 0$ and $RD^*A' = 0$, the regular constrained covariance yields

$$A' = -C_0 J_0 + C_0 D_{2,\mathsf{H}} A' - C_0 V'_{\mathsf{H}}(A'). \tag{RM.25}$$

Equations (RM.18)–(RM.20), together with CC.3, give a contraction on a ball of radius $C\mu$ in the weighted value/gradient norm, with the component-Laplacian estimate supplied by C_0. The small coefficient is $C\mu e_*$, after fixed constants have been chosen.

Equation (RM.25) by itself does not impose the Landau gauge. The correct identification step is to use the actual physical source solution, transform it by the invertible analytic map (RM.9)–(RM.10), and verify that it satisfies the same equation in the enclosing ordinary source ball. Uniqueness there identifies it with the weighted fixed point. This step, and the holomorphic continuation in the actual input chart, belong to the enclosing bootstrap theorem.

Once that identification is supplied, the physical field correction $A = A' - \mathsf{H}D(A')$ has the old-layer field, gradient and current profiles. In particular the extra linearizing current is bounded by $C\mu^2 e_* w$, with no h_n/h_a loss. The exact current chart conversion and source representative must still be carried into the complete B-domain application.

8. Remaining scope and source analysis

The zero relative datum (RM.3) avoids the arbitrary raw-data tangent matching problem. It does not assert that the associated corrected derivative equals the raw symmetric derivative. The mixed endpoint issue in Section 1 remains relevant to any proof using that stronger tangent identification.

The proof uses the following checked primary statements:

- CMP109 pp.253–254, (0.11)–(0.12), symmetric transport/averaging and the explicit transfer of earlier averaging results.
- CMP99 (3.113)–(3.117), corrected relative averages and physical Ward identities.
- CMP102 pp.285–289, (44), (49), (55), (65)–(73), exact analytic linearization and reproduced pointwise derivative; p.287, freedom to choose the linearizing right inverse.
- CMP102 pp.305–309, (174)–(177), (190), and Proposition 9, the first coefficient H_1 and its regular current kernels, including the stated regular external-background extension.
- CMP102 (39)–(40), (90)–(98), local nonlinear Wilson derivative estimates.
- The checked CC.3/CC.3a covariance and local multiplier bounds.

An independent check is still required that the enclosing bootstrap uses the admitted actual geometry and exact boundary data, that its real critical solution is the physical source solution in the chosen common chart, and that the final B activity argument, including old-field and current slots, remains in its required domain.

9. A blockwise normalized inverse gauge

This section supplies a concrete inverse-gauge bound for the associated symmetric averaging family. The source convention is essential: the designated endpoint transport in the old formula CMP99 (3.113) is replaced throughout by the same symmetric averaged transport $T_U(y,x)$ used in CMP109 (0.11), (0.12) and its symmetric axial gauge. It is not asserted that every individual path holonomy is one in that gauge. The source transfer statement on pp.253–254 is used with this single consistent choice of averaging and reference operations.

The associated-family dictionary can be derived explicitly. For the actual raw average M in CMP109 (0.12), put

$$E_y(U';U) = \exp\left[L^{-d} \sum_{x\in B(y)} \log\big(T_{U'U}(y,x) T_U(y,x)^{-1}\big)\right].$$

Use the relative endpoint correction of CMP99 (3.113), with every designated transport replaced by this same T:

$$\mathcal{U}'_c = E_{c_-}^{-1}\,[M(U'U)]_c[M(U)]_c^{-1}\,\mathrm{Ad}_{[M(U)]_c}\,E_{c_+}.$$

For a pure relative gauge, $U'U = U^u$, transport covariance and the analytic logarithm identity give $E_y = u(y)(\mathfrak{R}_U u)(y)^{-1}$. Substitution into the last display shows that $\mathcal{U}'$ is the coarse pure relative gauge generated by $\mathfrak{R}_U u$ with reference $M(U)$. Repeating the same correction on the active blocks therefore gives the iterated normalization in the source. This derives the nonlinear gauge operator attached to the actual raw average; it does not identify a raw derivative with an older regularized operator.

Write this associated one-step gauge average, as displayed in CMP99 (3.113), as

$$(\mathfrak{R}_U u)(y) = u(y)\exp\left[L^{-d}\sum_{x\in B(y)}\log\big(u(y)^{-1}T_U(y,x)u(x)T_U(y,x)^{-1}\big)\right]. \tag{RM.26}$$

The source normalization is its iterated version at each actual determining root, denoted $\overline{R_0 u}^{\,s}(y) = 1$ in CMP102 Proposition 2. The successive reference backgrounds are the corresponding raw averages of U_0. This is the normalization to be used, not a separately invented average of the whole terminal block.

Fix one terminal block $B^s(y)$. First work in a gauge in which the reference U_0 satisfies the absolute hierarchical transport condition on this block. Every one-step designated transport on every active subblock is then one. For

$$L = e^{i\eta A}U_0$$

put $R_x = \mathcal{T}_L(y,x)$, the actual hierarchical transport from the terminal root. The source estimates (3.55)–(3.59) imply

$$\sup_{x\in B^s(y)}|R_x - 1| \le Ch_sK, \qquad |R_x - R_z| \le Ch_rK \quad (x\in B^r(z)\subset B^s(y)), \quad K = \sup_{B^s(y)}|A|. \tag{RM.27}$$

For the second bound, factor the displayed hierarchical chain at z. The shorter relative transport differs from the corresponding reference transport, which is one, by Ch_rK. The remaining factor has bounded norm. Thus this step uses associativity of the chain, not multiplicativity of the group mean.

In this reference gauge, (RM.26) becomes the anchored logarithmic mean

$$\mathfrak{m}(\{r_x\}; r_z) = r_z\exp\left[L^{-d}\sum_{x\in B(z)}\log(r_z^{-1}r_x)\right]. \tag{RM.28}$$

It is exactly left equivariant. On a fixed sufficiently small matrix ball about the identity its derivative, in the supremum norm of its L^d entries, has norm at most 2. Indeed its derivative at a constant identity tuple is the normalized arithmetic mean, of norm one; analyticity on the finite one-step product gives the fixed neighborhood with derivative norm at most two. The neighborhood depends on d, L, G, not on the depth s.

Let $r^{(0)}(x) = R_x$, and obtain $r^{(r)}$ by applying the actual mean (RM.28) at the successive levels. Define

$$E_r = \sup_z |r^{(r)}(z) - R_z|.$$

The comparison with the tuple of original values at the previous-scale sites, followed by comparison with the constant tuple R_z, gives

$$E_r \le 2E_{r-1} + Ch_rK, \qquad E_0 = 0. \tag{RM.29}$$

Since $h_r = L^r\eta$ and $L \ge 3$,

$$E_r \le CK\sum_{\ell=1}^{r}2^{r-\ell}h_\ell \le \frac{C}{1-2/L}h_rK. \tag{RM.30}$$

Choose the common smallness bound on h_sK after this fixed constant. Induction then keeps every tuple inside the neighborhood used above. In particular

$$F_s = r^{(s)}(y) = 1 + O(h_sK). \tag{RM.31}$$

Set $c_y = F_s^{-1}$ and

$$u(x) = c_yR_x \quad (x\in B^s(y)). \tag{RM.32}$$

Exact left equivariance of (RM.28) at every stage gives

$$\overline{R_0 u}^{s}(y) = c_y F_s = 1.$$

Also the hierarchical transport transforms as $\mathcal{T}_{Lu}(y,x) = u(y)\mathcal{T}_L(y,x)u(x)^{-1} = 1$. Therefore this gauge satisfies both the actual terminal normalization and the relative axial condition. Its bounds are

$$|u(x) - 1| + |u(x)^{-1} - 1| \le C h_s K. \tag{RM.33}$$

The map is holomorphic on the fixed small domain: it is a finite composition of the admitted means and transports, followed by inversion, and the uniform estimates keep the composition in their domains.

For a reference not already in the displayed gauge, the covariant form is

$$u(x) = \mathcal{T}_{U_0}(y,x)^{-1} c_y \mathcal{T}_L(y,x). \tag{RM.34}$$

Here is the full covariance check needed for this formula. If $\widetilde{U} = U^g$ and $\tilde{u}(x) = g(x)u(x)g(x)^{-1}$, endpoint covariance of the same designated transport gives

$$\mathfrak{R}_{\widetilde{U}}\tilde{u}(y) = g(y)(\mathfrak{R}_U u)(y)g(y)^{-1}. \tag{RM.35}$$

Indeed every matrix inside the logarithm in (RM.26) is conjugated by $g(y)$; its exponential and the left root factor transform accordingly. Raw-averaging covariance gives this identity at every successive level. Therefore the terminal normalization is unchanged by a rooted frame change.

Set $g_0(x) = \mathcal{T}_{U_0}(y,x)$ on each terminal block, so $g_0(y) = 1$. Apply this frame change to the normalization equations only. The reference is in the hierarchical transport slice, so each of its designated symmetric endpoint transports is one. The preceding mean proof gives $\tilde{u} = c_y \mathcal{T}_{Lg_0}$; pulling it back by (RM.35) gives exactly (RM.34). Thus the covariant iterates are controlled, rather than being replaced by unrelated means in different frames. No assertion that all individual path holonomies are one is used.

For a real reference g_0 is unitary. For a complex reference near a fixed real U_*, compare its transports with those of U_*. The source estimate gives

$$\mathcal{T}_{U_0}(y,x)\mathcal{T}_{U_*}(y,x)^{-1} = 1 + O(h_s\|E\|), \qquad U_0 = e^{i\eta E}U_*.$$

Thus g_0, g_0^{-1} have bounded matrix norms, uniformly in the depth. Likewise the joint local analytic bound, with the fixed real reference, gives

$$\left|\mathcal{T}_L(z,x)\mathcal{T}_{U_0}(z,x)^{-1} - 1\right| \le C h_r \|A\| \quad (x \in B^r(z)).$$

One obtains this by differentiating the holomorphic transport map in its common absolute ball, or by applying its bound to the local BCH chart relative to U_*. Its derivative has size Ch_r by the same local Cauchy estimate. These comparisons justify (RM.27) in the transformed frame with only fixed norm factors.

This algebraic canonicalization is not an application of a propagator theorem to the axially transformed background. That background may have fine-edge potential spikes. The stationary equation and every propagator remain in the original admitted smooth native chart. The frame change is used only to evaluate the normalization and to bound the auxiliary reconstructed curvature/current.

On a smaller fixed domain, local holomorphic differentiation of (RM.33) gives, for a fixed reference and a perturbation direction $\dot{A}$,

$$\|u^{-1}\dot{u}\|_{\infty, B^s(y)} \le C h_s \|\dot{A}\|_{\infty, B^s(y)}. \tag{RM.36}$$

For a varying reference the corresponding joint derivative estimate follows in the common two-input chart; a derivative of the reference is retained, rather than silently omitted. In the fixed-reference current family, (RM.36) is the normalized endpoint estimate needed at older fluctuation sites.

Apply the construction independently on the disjoint actual terminal blocks and set $u = 1$ on the fixed level-zero exterior. This defines one fine-lattice gauge. Each determining site has its own terminal normalization at its own level. No equality of independently prescribed coarse values at two levels is assumed.

Theorem RM.3 — bounded passage to the actual auxiliary raw-data representative Use the actual symmetric raw average, its associated corrected map just derived, and the corresponding source relative axial condition. Let U_0 be an admitted reference with prescribed data $V = M_{\mathcal{B}}U_0$. Let $L = e^{i\eta A}U_0$ be the actual relative Landau critical configuration on the zero corrected fiber, in the common ordinary radius of CMP102 Proposition 2, and suppose $h_s\|A\|$ is uniformly small on all terminal blocks.

Then the gauge (RM.34), assembled as above, maps L to the relative axial critical configuration with the exact original raw data V. The rooted gauge $g_0(x) = \mathcal{T}_{U_0}(y, x)$ subsequently maps that configuration to the adopted absolute hierarchical axial representative:

$$W = (L^u)^{g_0} = L^{g_0 u}, \qquad M_{\mathcal{B}}W = V, \qquad \mathcal{T}_W(y, x) = 1. \tag{RM.37}$$

On a common admitted complex chart, the maps are holomorphic and

$$\| \mathrm{Ad}_{g_0 u} \| + \| \mathrm{Ad}_{(g_0 u)^{-1}} \| \le C. \tag{RM.38}$$

Consequently all physical curvature and current profile bounds for L pass to W with a fixed factor. For a fixed real reference the parameter-dependent part of this converter also satisfies the local endpoint estimate (RM.36).

Proof. First take real inputs. The constructed u satisfies the actual normalized-average equations and the relative axial equations, exactly, on every terminal block. The uniqueness part of the source axial/Landau reduction identifies it with its normalized inverse gauge. CMP102 Proposition 2 then identifies L^u with the actual raw-data critical configuration for V. This invocation includes the source's special crossing-bond correspondence: no rule replacing a crossing endpoint's actual normalization by a normalization at a different level is inserted.

Every endpoint of an actual mixed determining bond is a determining root at its own level. The nested-grid convention and the terminal partition used here are the ones verified in Annex AO. The gauge g_0 is one at all of these roots, so raw gauge covariance preserves the entire determining datum. Relative axial means $\mathcal{T}_{L^u} = \mathcal{T}_{U_0}$. Thus the final rooted transformation makes its transport one and proves (RM.37).

The real source solution, the weighted Landau solution after physical identification, the associated averaging maps and the displayed converter all lie on the same admitted finite-cutoff analytic chart. Their real identities continue by the holomorphic identity theorem. The bounds for g_0, g_0^{-1} follow by comparison with a fixed real reference, and (RM.33)–(RM.35) bound u, u^{-1}. Their products give (RM.38). Physical curvature and Wilson current transform by conjugation, which proves the stated profile transfer. This transformation is used only to evaluate the auxiliary reconstructed output. It does not replace the actual primitive input or change any old integration variable. □

The associated-family convention must remain fixed in all uses of this result. Using the literal old individual-path gauge-average formula while imposing only the new averaged-transport axial condition would invalidate the reduction to (RM.28); the old individual paths need not be one. The proof above instead derives the gauge average from the actual raw average and its same symmetric endpoint transports, using the explicit primary transfer statement in CMP109 pp.253–254. No numerical equality with an unrelated older operator is claimed.

The theorem remains conditional on actual common admission of the reference, the raw determining data and the physical critical branch. Those hypotheses are the source and native-ball checks in the enclosing projection theorem. It is a scoped intermediate step in that argument, not a completed catalogue estimate.

Annex AQ — A genuine strong native ball for the primitive domain

Annex AQ. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

This note completes the fixed-point and real physical-identification steps for the zero-relative-data construction in Annex AP. It states the strong external-background hypotheses explicitly. Its conclusion is a bound in the source's relative Landau representation. The passage to a specified raw-data representative, and the admission of the complete primitive boundary activity with its retained older fluctuation arguments, are separate application steps. C-05 is not completed by this annex alone.

1. Geometry, norms and the primary inputs actually used

Let an admitted determining history have fine mesh η, terminal meshes h_a, terminal blocks $\Delta(a)$, and source distance $d(a,b)$. Let h_n denote an older physical mesh on the same location. The application considered here has

$$h_a \le h_n, \qquad w_a = \alpha_n h_n^{-3}, \qquad e_a = h_a^3 w_a, \qquad e_* := \sup_a e_a \le \alpha_*. \tag{NP.1}$$

Every profile used in a row sum, including w, e, e^2, must satisfy the comparison inequalities for the actual multiscale metric with a fixed positive rate reserve. This is the geometric condition in CC.1–CC.4 of Annex AM; it is not a comparison with unweighted Euclidean distance. The geometry is fixed before the coupling ceiling. Its constants do not depend on the total volume or on the number of layers.

For the running profiles used below this condition is obtained as follows. The admitted AF comparison bounds α_n/α_m in either direction by $C(1+|n-m|)^b$, with fixed b. Crossing the old layers costs at least $RM(|n-m|-1)_+$ in the old source metric. A refinement makes its path weights no smaller, so the joined metric has the same lower bound. Consequently w_a/w_b is at most a fixed constant times $\exp(\beta d(a,b))$, where β can be chosen proportional to $(3\log L+1)/(RM)$; the polynomial is absorbed with a fixed coefficient. The joined mesh ratios similarly have exponent $c_h = \log L/(RM)$. Thus e_a/e_b and e_a^2/e_b^2 have comparison exponents at most $3c_h+\beta$ and $6c_h+2\beta$, respectively. Choose the admitted geometric separation once so that these exponents and the finitely many physical-volume powers leave all the kernel row-sum reserves used in RM and CC positive. This choice does not depend on the derivative order.

Use the norms

$$\|f\|_{F_w} = \sup_a w_a^{-1}\|f\|_{\infty,\Delta(a)},$$
$$\|A\|_{X_w} = \sup_a \max\left\{(h_a^2 w_a)^{-1}\|A\|_{\infty,\Delta(a)}, (h_a w_a)^{-1}\|\nabla_{U_0}A\|_{\infty,\Delta(a)}\right\}, \tag{NP.2}$$

and let E_w also include the component covariant Laplacian divided by w_a. The component Laplacian is not being identified with every second partial derivative. The ordinary absolute norm is

$$\|A\|_{X_{\mathrm{abs}}} = \sup_a \max\{h_a|A|, h_a^2|\nabla_{U_0}A|\}. \tag{NP.3}$$

It satisfies $\|A\|_{X_{\mathrm{abs}}} \le e_*\|A\|_{X_w}$.

Fix an external background U_0 on this geometry, and prescribe its own actual raw determining data $V = M_{\mathcal{B}}U_0$. This point must be admitted by the source variational construction. For real U_0, its real regularity and the regularity of V are required. For complex U_0, the actual source complex-background conditions are required. In particular the

latter include the second-order relative-field restriction in CMP99 (3.38), in addition to the value and first-gradient restrictions in (3.37). It would be incorrect to invoke Proposition 9 of CMP102 on a value-only background ball.

The source normalization of the relative averages gives zero corrected datum for this choice of V: the relative field at the input point is the identity, so every relative transport, mean and correction is the identity. This is RM.3; it does not assert equality of raw and reference-corrected averaging derivatives on arbitrary tangent vectors.

The following inputs are retained with their stated source scopes:

1. The source critical-orbit existence and uniqueness theorem and the axial/Landau reduction, CMP102 Theorem 1 and Proposition 2. The enlarged-gauge variation argument on pp.296–297 supplies unrestricted variations within the corrected determining fiber at the physical Landau critical point.
2. The regular constrained covariance C_0, with the weighted map $F_w \to E_w$, proved as CC.3 using CMP99 Theorem 3.3 and CMP102 (129)–(130).
3. The chosen dimensionless-data right inverse H, with $hQ\mathsf{H} = I$, $RD^*\mathsf{H} = 0$, and value, gradient and current kernel bounds, as specified in RM.4–RM.6. Here the chosen formal external-background family is the extension of CMP102 (174)–(175). Its first coefficient is H_1, including at an admitted off-minimum external background. The primary extension in Proposition 9 is used for this family; no equality with the alternative off-minimum family (179)–(180) is inferred.
4. The exact changed linearizing map, its local inverse, and the complete unrestricted Wilson-action remainder from RM.9–RM.23. These are the deductions in Annex AP, including every cross term.

All adjoints in analytic formulas are the bilinear transposes for the source physical measures, rather than conjugate-dependent maps. This convention is necessary for holomorphy.

2. The weighted analytic fixed point

Assume

$$\|J_0\|_{F_w} \le \mu, \qquad J_0 = \mathcal{J}(U_0). \tag{NP.4}$$

Write the exact physical relative field as

$$A = A' - \mathsf{H}D(A'), \qquad D(A')_b = C_j(h_b[A' - \mathsf{H}D(A')])_b\,. \tag{NP.5}$$

The corrected determining datum is hQA'. The map is an analytic bijection on a fixed smaller ordinary source ball and preserves the Landau condition in both directions. The action in this coordinate is

$$S(e^{i\eta A}U_0) = S(U_0) + \langle A', J_0\rangle + \tfrac{1}{2}\langle A', (\Delta - D_{2,\mathsf{H}})A'\rangle + V_{\mathsf{H}}(A'). \tag{NP.6}$$

The exact formulas for V_{H} and its derivative are RM.17 and RM.21. In particular the physical Wilson Hessian Δ is used before restricting to Landau; no extension of its quadratic form is silently substituted on unrestricted vectors.

Let K bound $C_0 : F_w \to E_w$. With one fixed c, the estimates of Annex AP give, on $\|A'\|_{X_w} \le \rho$,

$$\begin{gathered}\|D_{2,\mathsf{H}}A'\|_{F_w} \le c\mu e_*\rho,\\ \|V'_{\mathsf{H}}(A')\|_{F_w} \le c\rho^2 e_*(1+\mu e_*), \qquad \|V''_{\mathsf{H}}(A')\|_{X_w\to F_w} \le c\rho e_*(1+\mu e_*).\end{gathered} \tag{NP.7}$$

The derivative bound is taken on a fixed smaller ball within the ball used for the value estimate. Replace c by a larger fixed constant once so that these simultaneous radii are understood throughout.

Proposition NP.1 — a projection estimate driven by physical current Put $R = 2K\mu$. If $\mu = 0$, the fixed point is zero. Otherwise choose the absolute smallness ceiling so that

$$cKe_*\{\mu + 2R(1 + \mu e_*)\} \le \tfrac{1}{4}, \qquad 2Re_* < \varepsilon_{\mathrm{lin}}, \tag{NP.8}$$

where $\varepsilon_{\mathrm{lin}}$ is the fixed radius needed for the changed linearizing map. Then

$$A' = -C_0 J_0 + C_0 D_{2,\mathrm{H}} A' - C_0 V'_{\mathrm{H}}(A') \tag{NP.9}$$

has a unique solution in $\|A'\|_{X_w} \le R$. It satisfies

$$\|A'\|_{X_w} \le 2K\mu, \qquad \|A'\|_{E_w} \le C\mu, \tag{NP.10}$$

and depends holomorphically on all admitted finite-dimensional external parameters on any common domain with these reserves.

Proof. Denote the right side of (NP.9) by $\mathcal{T}(A')$. Its constant term has norm at most $K\mu = R/2$. On the radius-R ball, the other two terms have norm at most

$$Kc\mu e_* R + KcR^2 e_*(1 + \mu e_*),$$

which is at most $R/4$ after (NP.8). The segment integral of the derivative in (NP.7) makes the Lipschitz constant at most 1/4. Thus $\mathcal{T}$ maps the closed ball strictly into itself and is a contraction. Start the Picard iteration at zero. Every iterate satisfies $QA' = 0$, since $QC_0 = 0$. Convergence is uniform in the Banach norm and uniform on smaller external parameter domains. Every iterate is holomorphic there, so the limit is holomorphic. This is a finite-dimensional statement at each cutoff with constants uniform in the cutoff and volume.

Apply the stronger $F_w \to E_w$ covariance bound to the fixed-point equation. The forcing on its right side is bounded by $C\mu w$, which proves the second estimate in (NP.10). No claim that (NP.9) alone imposes the Landau gauge is made in this argument. □

3. Identification with the actual physical solution

The preceding qualification is essential: CMP102 p.300 explicitly distinguishes its equation (143), used for regularity, from the equation that proves existence in the gauge slice. A weighted solution of a projected equation is not automatically a physical critical point.

Proposition NP.2 — uniqueness in an enclosing ordinary ball There is one fixed $\delta > 0$ such that, for admitted external backgrounds with sufficiently small absolute regularity and

$$j_* := \sup_a h_a^3 |J_0| \le \mu e_* \tag{NP.11}$$

sufficiently small, (NP.9) has at most one solution in $\|A'\|_{X_{\mathrm{abs}}} \le \delta$.

Proof. Repeat the actual estimates with the admissible ordinary profile $\bar{w}_a = h_a^{-3}$. Its associated $\bar{e}_a$ is one. The current parameter in those estimates is now j_*, and the field radius is δ. The derivative of the right side of (NP.9) on that ball is at most

$$K_{\mathrm{abs}} c\{j_* + \delta(1 + j_*)\}. \tag{NP.12}$$

Choose δ within the ordinary linearization radius and so that the term independent of j_* is at most 1/8. Then lower the common background and current ceiling until the entire expression is at most 1/4. For any two solutions the segment integral of the derivative gives a contraction estimate for their difference. They must coincide. This argument establishes uniqueness for the changed coordinate equation; it does not assume that the source's uniqueness proposition for a differently written nonlinear equation transfers without proof. □

Theorem NP.3 — physical identification in the relative Landau chart For a real admitted U_0, let the regular critical orbit for $V = M_{\mathcal{B}}U_0$ be the orbit supplied by the source theorem. Assume the ordinary regularity ceiling is chosen so that the source relative Landau representative and its transformed A' lie in the ball of Proposition NP.2. Choose e_* so that also $Re_* < \delta$. Then the weighted fixed point in Proposition NP.1 is precisely that physical representative in the changed linearizing coordinate. Consequently the associated A satisfies

$$Q\text{-corrected datum}(A) = 0, \qquad RD^*A = 0,$$
$$|A| \le C\mu h_a^2 w_a, \quad |\nabla A| \le C\mu h_a w_a, \quad |\Delta_{\mathrm{cov}} A| \le C\mu w_a. \tag{NP.13}$$

On a common holomorphic external chart these identities and bounds continue to complex inputs, in the same relative Landau representation.

Proof. The source reduction of the real critical orbit has zero corrected datum by the exact choice $V = M_{\mathcal{B}}U_0$. Transform it using the local analytic inverse in (NP.5). Since $RD^*\mathsf{H} = 0$, the transformed field remains Landau, and it satisfies $QA' = 0$.

At this physical critical point, the source gauge-orbit argument allows all small variations in the corrected determining fiber, without requiring their variations to be Landau. The changed coordinate map is a local bijection of these fibers. Differentiating the exact action (NP.6) therefore yields

$$\langle v, J_0 + (\Delta - D_{2,\mathsf{H}})A' + V'_{\mathsf{H}}(A')\rangle = 0 \quad (Qv = 0). \tag{NP.14}$$

Since the point itself is Landau, one may insert the regular gauge-fixing quadratic term in this stationarity equation at that point. Apply the regular constrained covariance, as in CMP102 (128)–(143); this gives exactly (NP.9). The actual physical solution and the weighted fixed point both lie in the ordinary radius-δ ball. Proposition NP.2 identifies them. Landau is obtained from this identification, not from an unsupported property of the range of C_0.

The quadratic estimates RM.11–RM.12 applied with $\rho = 2K\mu$ show that the linearizing correction has value, gradient and component-Laplacian bounds at most $C\mu^2 e_*$ times the respective X_w, E_w weights. Combine them with (NP.10) to obtain (NP.13).

For holomorphic external parameters, the operators and fixed point are holomorphic on the common admitted chart. The real identities are identities on a real open parameter set, so they continue by the finite-dimensional holomorphic identity theorem. This continuation requires actual common admission; it does not supply a missing domain radius by itself. □

Physical current in the same representation The Landau condition also gives the physical second-order estimate needed here. The covariant lattice identity used in CMP102 (134)–(140) writes D^*DA as the component covariant Laplacian plus a curvature multiplier, minus DD^*A. Write $I = R + P$ on the scalar gauge space. Since $RD^*A = 0$,

$$DD^*A = DPD^*A. \tag{NP.15}$$

The source smoothing row for DPD^*, CMP99 (3.49), applied with the same profile reserve, maps $h_a^2 w_a$ into w_a. The curvature multiplier is bounded by the admitted absolute regularity times h_a^{-2}; it has the same mapping property. It follows that

$$|D^*DA| + |\Delta A| \le C\mu w_a. \tag{NP.16}$$

Here Δ is the physical Wilson Hessian at U_0. This step uses Landau after physical identification. It is not an L^∞ estimate for arbitrary second derivatives of an arbitrary field.

To compare with the actual current, let $W = e^{i\eta A}U_0$. The derivative of the local exponential chart is an invertible finite-dimensional matrix at every link; its norm and inverse norm are bounded when $\eta|A|$ is within the fixed source radius. The derivative of the physical action in this chart is

$$J_0 + \Delta A + V_0'(A).$$

The local analytic chart factor converts this derivative to $\mathcal{J}(W)$. The Wilson remainder bound RM.23 is $C\mu^2 e_* w_a$. Therefore

$$|\mathcal{J}(W)| \le C\mu w_a. \tag{NP.17}$$

Define the physical curvature norm here by $F(U)(p) = \eta^{-2}(U(\partial p) - 1)$, on the indicated small plaquette chart. This differs by the displayed factor from a convention that calls the unscaled plaquette difference itself F. The corresponding curvature expansion, using the value and first-gradient bounds in (NP.13), gives

$$|F(W) - F(U_0)| \le C\mu h_a w_a \le C\mu\alpha_n h_n^{-2}. \tag{NP.18}$$

In particular (NP.17) has the older current weight $\alpha_n h_n^{-3}$, not the weaker generic joined weight $\alpha_n h_a^{-3}$. This is the point of the zero-data argument.

4. A genuine strong native input ball

Fix a real source background U_* with the local old-layer strong bounds

$$h_n^2|F(U_*)| + h_n^3|\mathcal{J}(U_*)| \le C_{\text{hist}}\varepsilon_n, \qquad \varepsilon_n = A_0 g_n \ell_n^{p_0}. \tag{NP.19}$$

Its local real charts must include the value, first-gradient and second-order bounds in CMP99 (3.35)–(3.36). Let

$$\alpha_n = C_\alpha g_n \ell_n^q, \qquad q > p_0. \tag{NP.20}$$

Use a complex perturbation $U_0(E) = e^{i\eta E} U_*$ with the strong norm

$$\|E\|_{\mathfrak{N}} = \sup_n \max\left\{ \frac{h_n|E|}{\alpha_n}, \frac{h_n^2|\nabla_{U_*}E|}{\alpha_n}, \frac{h_n^3|D^*_{U_*}D_{U_*}E|}{\alpha_n} \right\}. \tag{NP.21}$$

The second-order row required by the primary external-background theorem cannot be omitted. A component-Laplacian row is not imposed on the external perturbation: it is an output supplied by the regular covariance for the correction in NP.10. This distinction matters for a subsequent smooth gauge interpolation, whose curl-current row is controlled by exact current covariance. A ball in (NP.21) is a sufficient external-background domain; no equivalence with a value/gradient-only ball is claimed.

For $\|E\|_{\mathfrak{N}} < r$, the local Wilson expansion and bounded chart factors give

$$|\mathcal{J}(U_0(E))| \le C\left(C_{\text{hist}}\frac{\varepsilon_n}{\alpha_n} + r + r^2\alpha_* \right) w_a. \tag{NP.22}$$

The real-center ratio ε_n/α_n is at most $C\ell_*^{p_0-q}$. The absolute joined-scale norm is smaller than the corresponding old-scale norm because $h_a \le h_n$. Thus a fixed choice of r, followed by a sufficiently small common coupling ceiling, admits the source external-background operators and makes the μ in Proposition NP.1 a fixed small number. The bound is independent of total volume.

Lemma NP.4 — absolute raw-data admission on the whole ball Use the actual joined determining history of JI.1–JI.4 in Annex AO, with its proper-scale collars and star conventions. The determining datum $M_{\hat{B}}U_0(E)$ lies in a common absolute analytic chart throughout a sufficiently small fixed radius r of (NP.21), once the common coupling ceiling is chosen. The constants are uniform over the admitted old histories and carrier sizes.

Proof. At a determining bond of mesh h_s, the relative transport and relative-average estimates in CMP99 (3.55)–(3.59), with the corresponding symmetric operators, bound the raw relative increment by

$$Ch_s \sup_{\text{foot}(b)} |E|. \tag{NP.25}$$

The raw/reference-corrected endpoint factor is retained in this estimate; it is the bounded factor explicitly present after (3.57). No identification of their derivatives is needed. The footprint comprises the actual two endpoint blocks and connecting averaging paths. The source separation permits that footprint to reach at most a fixed neighboring earlier layer. At all other points it is in the same or a coarser old layer. Adjacent meshes and running radii differ by fixed factors. Since the joined mesh is no larger than the old mesh, (NP.25) is at most $Cr\alpha_*$.

The real reference U_* itself witnesses a nonempty regular fiber for its own joined data: the joined meshes are smaller, so its old current and curvature bounds imply the required absolute joined bounds. CMP102 p.278, immediately after (7), explicitly states that a nonempty regular fiber implies the determining-data condition (7), with a fixed multiple of its regularity constant. That statement includes the special crossing plaquettes defined in the preceding paragraph. Thus the real reference data have the absolute bound $CC_{\mathrm{hist}}\varepsilon_*$. No desired running reconstructed-current estimate is used to obtain it. Reduce the common coupling ceiling until that bound and $Cr\alpha_*$ fit fixed fractions of the source real and complex data radii. The resulting chart contains the entire strong ball, not just its real diameter or its first tangent. □

The field space in this lemma is on the actual carrier dependency set, including the fixed collar links that supply its determining data and the stencils used by the strong norm. Those links are external input data and may vary with E. They are not extended by zero across a newly imposed boundary. The unknown correction in NP.5 has the source fixed-boundary conditions relative to each such input. This distinction prevents a spurious fine-mesh derivative from a sharp extension of an input tangent.

Theorem NP.5 — the U-dependent primitive boundary domain has a fixed strong radius Use the normalized converter RM.3 of Annex AP, in the corresponding source symmetric family, followed by its root-fixed conversion to the adopted absolute auxiliary representative. Fix the real source bounds (NP.19), the current/field normalization ratios, and the admitted geometry. There exist fixed $r_0 > 0$ and a common coupling ceiling such that, for every $\|E\|_{\mathfrak{R}} < r_0$, all the U-dependent inequalities CMP119 (2.34), (2.35), (2.37), (2.38), (2.39) hold with a fixed reserve, for every reconstructed level p in that primitive domain. The radius is independent of the fine mesh, layer count and total volume. The independent current slot in (2.36) may be included on the common radius of BC16.21–BC16.23.

Proof. Lemma NP.4 admits the internal source maps first, on their fixed absolute domains. The strong external assumptions in (3.35)–(3.38) hold by (NP.19), (NP.21), the smaller joined meshes, and the same choice of ceiling. Propositions NP.1–NP.3 therefore apply to every reconstruction. The old-layer current and curvature bounds are (NP.17)–(NP.18), with

$$\mu \le C\{C_{\mathrm{hist}}\ell_*^{p_0-q} + r_0 + r_0^2\alpha_*\}. \tag{NP.26}$$

RM.3 gives a holomorphic converter and inverse with a uniform bound on this whole ball. The additional root-fixed gauge used to reach the actual auxiliary axial convention is bounded by comparison to its fixed real reference using the same local transport estimate. Current and plaquette differences transform exactly by conjugation. Thus those bounds acquire only a fixed multiplier. In physical units the required reconstructed-current weight is $\alpha_n h_n^{-2} h_{\min(p,n)}^{-1}$, which is no smaller than $\alpha_n h_n^{-3}$. The reconstructed curvature restriction similarly follows from the old-layer physical curvature bound. The reference field and its real local gauges supply (2.38). The input relative-potential and gradient restrictions are directly (NP.21); an auxiliary axial potential is never used to prove (2.39).

Let $s_* = 1-\beta$. Every numerical coefficient in the listed source restrictions is at least s_*. Let C_{dom} be a common constant in the preceding estimates, including the finitely many normalization ratios and the uniform embedding of actual layers into an older primitive's innermost-region convention. Choose r_0 so that $C_{\mathrm{dom}}(r_0 + r_0^2) \le s_*/8$. Then choose the common coupling ceiling so that $C_{\mathrm{dom}}C_{\mathrm{hist}}\ell_*^{p_0-q} \le s_*/8$, and all absolute smallness requirements in NP.8–NP.12 and RM.3 also hold. Every running output then uses at most a fixed fraction of its allowed radius. The real-reference restrictions and input-field restrictions have the same reserve after one further fixed reduction of r_0, if needed.

For the common older-primitive embedding, the actual layer $n \ge j$ has ratios bounded by a fixed polynomial in $n-j$ times $L^{-r(n-j)}$, with $r = 1, 2, 3$ for value, gradient/curvature and current respectively. The suprema are finite. Include them once in C_{dom}; no factor grows with the number of older activities. The independent current slot does not enter the reconstructed field predicates and is treated as in BC16.4. This proves a genuine ball contained in the complete field/current part of the primitive domain. □

This theorem does not by itself identify a current-dependent smooth representative of the input primitive with the source's original axial pullback when older fluctuation variables are present. That is the distinct fixed-old-variable

gauge interpolation in AD17.3. It also does not supply a joint contour for an unmatched, newly introduced decoupling operator family.

5. Endpoint use and the exact uncompleted application

Suppose the actual primitive input family, in a smooth chart with all older arguments accounted for, has a current-coordinate tangent v_i satisfying

$$\|v_i\|_{\mathfrak{N},X} \le \frac{\vartheta_X}{P_k} e^{-cd_X(i)}. \tag{NP.23}$$

If a fixed radius r_0 of the strong ball above is contained in the complete primitive domain at each such center, then Cauchy on the genuine affine curve in that chart gives

$$|\partial_i b_X(\Phi(z))| \le \frac{K_X}{r_0} \frac{\vartheta_X}{P_k} e^{-cd_X(i)}. \tag{NP.24}$$

The fixed-factorial all-order carrier summation BC16 then applies. The missing assertion cannot be replaced by a bound for first derivatives of the reconstructed outputs. That invalid inference was excluded by the exact counterexample BC16.19a.

At the present scope, the following are still separate checks:

- The chosen native directions and their extensions use the actual local source geometry and remain in its complete strong external-background class.
- The source raw-data gauge converter is uniformly bounded, with the needed endpoint response, on that common ball.
- The actual primitive pullback can be evaluated in a smooth input chart while preserving every old fluctuation, independent current, constraint and relocation argument. Real simultaneous gauge covariance alone does not supply a complex radius for an older variable.
- The complete source sector ledger has the endpoint/carrier prefactors required by BC16 and the earlier boundary value budget.

These are application conditions for C-05, not new estimates. The projection theorem above removes a specific fine-collar loss; it does not remove those conditions.

Primary references

- T. Bałaban, *Propagators for lattice gauge theories in a background field*, Commun. Math. Phys. 99 (1985), 389–434: (3.35)–(3.38), Theorem 3.3, (3.49), (3.127)–(3.133). DOI: https://doi.org/10.1007/BF01240355.
- T. Bałaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, Commun. Math. Phys. 102 (1985), 277–309: Theorem 1, Proposition 2, (123)–(143), (174)–(190), Proposition 9. DOI: https://doi.org/10.1007/BF01229381.
- T. Bałaban, *Convergent renormalization expansions for lattice gauge theories*, Commun. Math. Phys. 119 (1988), 243–285: (2.34)–(2.42), (3.15), (3.43). DOI: https://doi.org/10.1007/BF01217741. These identify the eventual application, rather than supplying its new estimates.

Annex AR — The complete primitive domain and fixed older slots

Annex AR. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

This chapter distinguishes the independent-current extension of the primitive domain from the physical variation of an older raw slot. Applying these estimates to complete Wilson histories requires the composed-response and compatibility hypotheses above.

1. Results and the remaining source question

The direct printed product-domain shortcut has not been justified. CMP119 (3.43) contains a current radius $C_1 p_1(g_k)$, whereas the checked cutoff definitions use $p_0(g_k)$ and the ratio A_1/A_0. The function p_1 in that formula has not been identified by a checked definition in the source chain. The later LFR primed preparation polynomial is a different threshold and is not imported. This note also records a separate printed scaling inconsistency in (3.16); it is not used to shrink the frozen full support.

Two actual argument-map facts can nevertheless be established without this radius identification:

1. In the CMP119 next-step substitution, all earlier fluctuation coordinates A_j, $j < k$, remain fixed inputs. The new current enters the old action through its displayed background-field argument. A gauge conversion used only to bound a reconstructed field need not be applied to those old arguments.
2. The relocation on S_{k+1} changes only exterior current-scale coordinates. Its Jacobian is exactly independent of the inner current variables and cancels from their normalized conditional density. Its full exterior weight remains present. The relocated inherited action may be estimated directly as a composed primitive function; a Taylor estimate in an earlier fluctuation coordinate is not algebraically necessary.

The source also supplies a useful fixed-fraction reserve at the actual real centers. It follows from the physical-current part of the minimizing-field regularity theorem, not from a curvature bound alone. Section 5 states the exact numerical reserve that the native-ball construction of Annex AQ must meet.

2. The p_1 product radius

2.1 Published cutoff and domain formulas The following formulas use the printed notation of CMP119.

Location	Checked content	Consequence for the radius
p.244, introductory large-field estimate	$p_0(g) = A_0(\log g^{-2})^{p_0}$, with a positive integer exponent p_0.	A definition of p_0, not of p_1.
p.247, after (1.8)	The line prints $\delta_0 = g_0 A_1 p_0(g_0) = (A_1/A_0)\varepsilon_0$.	Its first equality is inconsistent with the preceding normalization if read literally; the second equality expresses the intended small-to-large cutoff ratio.
p.250, (1.18)–(1.19)	The normalized first-step fluctuation is restricted by $\lvert A\rvert < g_0^{-1}\delta_0$.	The factor g_0^{-1} is explicit.
p.265, immediately after (3.4)	$\delta_k = g_k(A_1/A_0)p_0(g_k)$.	This is the consistent later unprimed definition.

Location	Checked content	Consequence for the radius
p.267, before (3.15)	The source first defines the unscaled logarithm A'_k, then explicitly rescales $A'_k = g_k A_k$.	The A_k in (3.15) is normalized.
p.268, (3.16)	The displayed cutoff is $\lvert A_k\rvert < \delta_k$, without the g_k^{-1} in the first-step analogue.	This is a separate normalization ambiguity. It does not prove a different full-support convention.
p.276, (3.43)	The new current factor is $\{A : \operatorname{supp} A \subset Y \cap \Lambda_{k+1}^{(k)*},\ \lvert A\rvert < C_1 p_1(g_k)\}$.	The symbol $p_1(g_k)$ is genuinely printed. Its relation to the unprimed support still needs identification.
p.278, after (3.47)	The prefactor is $O(p_0(g_k)^3)$.	It is not an additional use or definition of p_1.

The frozen current support remains

$$P_k = \delta_k/g_k = (A_1/A_0)p_0(g_k) = A_1 \ell_k^{p_0}, \qquad \ell_k = \log g_k^{-2}. \tag{AD17.1}$$

Neither a typographical repair nor a new interpretation of (3.16) is silently made. To use (3.43) on this support one needs the actual inequality

$$C_1 p_1(g_k) \geq (1 + c_*)P_k \quad \text{for one fixed } c_* > 0 \tag{AD17.2}$$

uniformly along the admitted trajectory. No checked definition presently supplies (AD17.2).

2.2 The source chain outside the isolated formula CMP119 carries forward the definitions of its preceding small-field papers. CMP109 Theorem 3 and its fluctuation construction use fixed small constants $\varepsilon_0, \varepsilon_1$. CMP116 (1.20)–(1.22) and Lemma 1 use the absolute current radius $g_k|B| < \varepsilon_1$; its displayed product domain (1.34) has ε_1/g_k, not a defined logarithmic $p_1(g_k)$. Thus those preceding statements provide genuine analytic-map information but do not identify the new notation in CMP119 (3.43).

The earlier reference [16] in that source chain is the three-dimensional stability paper, CMP102 (1985), 255–275, DOI 10.1007/BF01229380. It defines $p(g) = b_0(1 + \log g^{-1})^{p_0}$ near (7), uses enlarged fluctuation cutoffs near (18), (22) and (51), and obtains the smooth minimizing-field representation only modulo a gauge in (52). It does not supply a checked identification of the p_1 in CMP119 (3.43) with the later unprimed cutoff. Several small printed cutoff subscripts or superscripts are ambiguous. No definition is inferred from those ambiguous characters.

CMP119, pp. 247, 250, 261, 267–269, 276 and 278, uses different symbols for the regular radius, the cutoff exponent and the older fluctuation threshold.

The later LFR polynomial bearing the same subscript belongs to the primed preparation threshold and has a lower logarithmic exponent than the unprimed cutoff. Inserting it into (3.43) would make the ratio in (AD17.2) tend to zero. Increasing a fixed coefficient cannot repair that mismatch. This is a reason to reject that proposed identification, not a proof that the published construction fails.

We distinguish three assertions:

- The source's complete construction is intended to evaluate the localized expressions throughout the relevant integration domain.
- Its displayed domain provides a quantitative radius if p_1 is identified.
- The required numerical relation (AD17.2) has been proved.

The first assertion does not, by itself, establish the third. Defining the undefined symbol ourselves to mean P_k would not verify the analytic estimates that the printed domain is supposed to summarize.

2.3 What (3.43) would supply if its radius were matched The source statement is substantive. On p.276 it transfers the completed localization construction to the multiscale boundary, refers to the full generalized random-walk expansion, and places the remaining localized terms on (3.43). The current dependence is localized to $Y \cap \Lambda_{k+1}^{(k)*}$. Inherited B terms are explicitly included by the formulas corresponding to CMP109 (3.6)–(3.7); p.277 retains the shrinking multiscale margins, heterogeneous Wilson coefficient and complete relocation action.

Consequently a verified radius (AD17.2), together with the actual enlarged background/current native margins and the printed complete carrier bound, would allow ordinary Cauchy differentiation of those localized functions. A new theorem about arbitrary prepared minimizing operators would then be unnecessary for this direct application. The already completed localization must be used with its actual carrier majorant and its full variable scope; ordinary boundedness of an unlocalized scalar function does not replace it.

At present the direct source shortcut stops at the exact missing relation (AD17.2). The alternative native-ball proof is a way to establish the required radius independently, without deciding the typographical intention of p_1.

3. Earlier fluctuation coordinates in the actual next-step map

3.1 The source argument list CMP119 (2.20)–(2.23), printed p.258, distinguishes the outer integration operation from the effective action. The operation $T^{(j)}$ retains and integrates A_j on $Z_{j+1} \cap \Omega_{j+1}$, at its own lattice scale. The action is written as

$$A_k(U_k, \mathrm{a}) = -A(g_k^{-2}(\cdot), U_k) + E_k(U_k) + R_k(U_k) + B_k(U_k, \mathrm{a}) - E_k^{\mathrm{vac}}, \tag{AD17.3}$$

where $\mathrm{a} = (A_0, \dots, A_{k-1})$ records the earlier fluctuation inputs. The history sets are suppressed in the source notation but are retained here as fixed parameters.

In (3.15), printed pp.267–268, the new fluctuation A_k enters through

$$U_k\big(\exp i\,[g_k C A_k - hD(g_k C A_k)]\, V^{(k)}\big). \tag{AD17.4}$$

The same inherited action $E_k + R_k + B_k$ is evaluated at that argument and at U_{k+1}. There is no replacement of an earlier A_j in this step. The only coordinate rescaling and constraint elimination described there concern the newly introduced A_k.

Proposition AD17.1 — the old slots are fixed in the direct inherited pullback For the actual current substitution (AD17.4), before the subsequent exterior relocation, a primitive inherited boundary term has the form

$$F_{j,X}(z; U_{k+1}, a, \mathrm{a}, \mathrm{S}) = B^{(j)}(X, U_k(I_k(z, a)), \mathrm{a}, \mathrm{S} \cap X), \tag{AD17.5}$$

where z and a are the inner and exterior pieces of the new A_k, and I_k is the exact source map in (AD17.4). All earlier coordinates a remain fixed. Therefore a current derivative of this direct function differentiates the displayed field map and any source-native independent-current substitution; it does not differentiate an old fluctuation argument.

Proof. The old variables are already integration variables of T_k in (2.18)–(2.23). The construction before (3.15) integrates and changes coordinates only in the new V_k variables, introduces their logarithm A'_k, eliminates their new constraints, and rescales $A'_k = g_k A_k$. The old operation T_k remains outside this new integral. Its earlier fluctuation coordinates are not included in any of those substitutions. Expanding the suppressed argument list of B_k using (2.23) therefore gives exactly (AD17.5). The final split into z and a is a restriction of A_k to disjoint coordinate sets and leaves the old list unchanged. □

This is a statement about the actual fixed-source representative used in (3.15). It does not replace U_k by an arbitrary representative of its gauge orbit. A gauge converter used solely to estimate a reconstructed field/current can be kept inside that norm calculation. A gauge change used to replace the whole primitive B input is different: its action on the earlier fields must be accounted for. Sections 3.3–3.4 make that distinction explicit.

Fixed old coordinates do not imply that the earlier T_k weights, their coefficients or their constraint factors are independent of the native background. Their actual parameter dependence remains in the complete density and conditional

assembly. Proposition AD17.1 removes only old-coordinate chain-rule terms from this direct inherited B pullback; it does not remove those weights or their background derivatives.

3.2 Why fixing determining roots is not a general substitute CMP119 (1.17) gives the adjoint law at the initial endpoint:

$$(R(u)A)(x,x') = u(x)A(x,x')u(x)^{-1}. \tag{AD17.6}$$

CMP102 (4) fixes a variational gauge at the determining sites. Those facts alone do not make every old fluctuation invariant. An earlier A_{j-1} lives at $(j-1)$-lattice sites in $\Omega_j \cap \Lambda_j^c$, whereas the determining data on the surrounding j-layer use j-lattice sites. A noncentral $(j-1)$-site need not be one of the fixed roots. A gauge can equal the identity at the roots and have a nontrivial value at that site, changing (AD17.6).

Thus a proof which actually gauge-transforms a primitive scalar must carry the actual transformed old arguments or prove a stronger identity for that particular gauge. Root fixing by itself is insufficient. The LFR II collar construction makes the same caution concrete: around (1.61), printed pp.373–374, it keeps the splice gauge because two representations meet at the collar. That gauge is not globally removable. Proposition AD17.1 avoids an additional change of old slots in the next current insertion; it does not delete a gauge already present inside the inherited primitive boundary function or inside an actual R-operation.

No analyticity in arbitrary complex old fluctuation coordinates is asserted here. The source primitive B bound is used with its actual old coordinates as fixed parameters, including its stated uniformity over their admitted real values.

3.3 The actual input-gauge issue The first sentence before CMP119 (2.34) specifies analyticity in (U,J). The space in (2.34)–(2.39) contains those variables and their reconstructed fields. Property (ii) after (2.41) is an extension to that space; it does not state a complex product radius for the older fluctuation variables. Property (iii) does state simultaneous real G-gauge covariance of U, J, a.

CMP109, pp.265–267, first converts the relative axial minimizing field into a smooth Landau representative. It uses gauge invariance to expand the Wilson action. The inherited effective action in (2.10) is still written at the original $U_k(\exp iB'V^{(k)})$; subsequent scalar gauge invariance is used in Section 3. In CMP119 the inherited B term also depends on a, so the corresponding replacement rotates those variables. Neither (3.15) nor Proposition AD17.1 proves that its axial input itself satisfies the strong complex relative-field condition (2.39).

This matters even when curvature and physical current are well controlled. Axial representations can put a coarse link increment onto a fine boundary bond. The relative potential on that bond then has size proportional to the increment divided by the fine mesh, whereas (2.39) demands the old physical layer scale. A bound on the gauge matrix alone does not remove this difference.

The literal estimate (3.19) is also narrower than such a claim: it bounds the ratio between $V^{(k)}$ and its localized background average, using (3.18) and the cited gauge estimates. Its gauge compares two background minima. The formula does not by itself bound the different current-dependent converter in (3.15).

For real variables, simultaneous covariance gives the exact identity

$$B(U_{\mathrm{AX}}, J_{\mathrm{AX}}, \mathrm{a}) = B(U_{\mathrm{L}}, J_{\mathrm{L}}, \mathrm{Ad}_u\, \mathrm{a}), \qquad U_{\mathrm{L}} = U_{\mathrm{AX}}^u. \tag{AD17.18}$$

A uniformly bounded complex u does not make the right side an admitted analytic function merely because the real older fields had a norm reserve. A complex old-field domain would have to be proved. The following construction avoids needing one.

3.4 Smooth interpolation of the gauge values on old fluctuation sites The construction uses the actual terminal-layer geometry of the old variables, not a new gauge fixing of the measure. It is performed separately for each primitive $B^{(j)}(X)$, using only its actual older slots. Those slots were introduced by its creation step, so their indices are at most $j-1$; subsequent dependence on newer fields enters its displayed background argument. This termwise choice is an analytic-extension identity for a scalar activity, not a change of integration variables.

Let h_j be the physical mesh of the j-lattice, $h_{j+1} = Lh_j$. Let $\mathcal{N}_j$ consist of the initial sites of oriented bonds carrying an older A_j on which the primitive in question actually depends. By CMP119 (2.20)–(2.21), these bonds are in

$$\Omega_{j+1} \cap Z_{j+1}, \qquad Z_{j+1} = \Lambda_{j+1}^c. \tag{AD17.19}$$

The interlacing (2.1) places the point support inside $\Omega_{j+1} \setminus \Omega_{j+2}$. The bond-star convention can move an initial endpoint by one h_j-step. The source's admitted layer buffers therefore put its actual terminal level $n(x)$ in the fixed range

$$j \le n(x) \le j + 2, \qquad x \in \mathcal{N}_j. \tag{AD17.19a}$$

Here level zero denotes the fine exterior when it occurs. The slightly generous range retains both boundary conventions. No buffer between Λ_{j+1} and Ω_{j+2} is needed for this assertion: a second layer boundary cannot be crossed in one h_j-step because the Ω-history itself is buffered.

This point requires care after R. The sentence on p.256 about domains determined by a renormalization transformation is not a statement that R leaves the geometry unchanged. Indeed p.255 allows $\Lambda_j = \Omega_j$ after R. LFR I, printed p.179, (1.10)–(1.13), explicitly rebuilds the history and states that the resulting Ω'' is admissible; its construction separates the newly introduced successive layers by an M-cube in their corresponding scale. We use this preserved grading, rather than assume that the old domains remain fixed.

For the admitted histories, fix a positive c_{gr} such that two sites of terminal levels differing by at least two satisfy

$$|x - y|_\infty \ge c_{\mathrm{gr}} \max(h_{n(x)}, h_{n(y)}). \tag{AD17.19b}$$

This follows directly from the consecutive-domain separation. For example, if $n(y) \ge n(x) + 2$, then $y \in \Omega_{n(y)}$, whereas $x \notin \Omega_{n(y)-1}$; the separating layer has thickness at least the admitted fixed number of cubes at mesh $h_{n(y)-1}$. The endpoint convention costs at most one step. With the already large block parameter, the conservative choice $c_{\mathrm{gr}} = 1/(4L)$ is available for the one-M-cube version as well as the larger original buffers. Empty layers do not produce old nodes.

Merge repeated nodal sites; their target is the same matrix $u(x)$. For two distinct sites whose terminal levels differ by at most one, the nested odd-block grids and (AD17.19a) give

$$|x - y|_\infty \ge h_{\min(j(x),j(y))} \ge L^{-3} \max(h_{n(x)}, h_{n(y)}).$$

Together with (AD17.19b), this proves uniform separation with

$$c_{\mathrm{sep}} = \min\{c_{\mathrm{gr}}, L^{-3}\}, \qquad r_x = (c_{\mathrm{sep}}/8) h_{n(x)}. \tag{AD17.19c}$$

The open nodal balls of radii r_x are disjoint. Each meets only a fixed adjacent range of terminal layers, so its physical weights and those at its center differ by fixed powers of L and the adjacent AF comparison. This handles the surviving explicit old fluctuation slots on the final admitted history, including the possibility that some slot sets are empty after R.

Fix the real baseline U_*. On each such ball choose one of its real local regular gauges from (2.38), fixed throughout the analytic variation. Choose once a smooth scalar bump ψ, supported in the unit ball, with $\psi(0) = 1$ and fixed first and second derivative bounds. At the very finest scale, a ball smaller than a fine lattice spacing is interpreted by sampling this bump; direct finite-difference bounds give the same conservative constants, and no local smooth gauge outside the fine exterior is needed.

Proposition AD17.3 — fixed-old-field replacement by nodal gauge interpolation. Suppose the actual current/background contour supplies a holomorphic gauge $u(z)$, real G-valued on its real slice, relating the original primitive input to a smooth admitted source family as in (AD17.18). The selected formal family agrees with the minimizing family on its physical real slice; it need not be a minimum for arbitrary independent native parameters. Criticality is not required for the interpolation argument. Normalize the baseline converter to the identity using a fixed real gauge. Suppose its logarithm at the old sites obeys

$$\| \log u(z, x) \| \le r_u \alpha_{j+1}, \qquad x \in \mathcal{N}_j, \tag{AD17.20}$$

uniformly on that contour. The adjacent-layer α ratios are bounded by the admitted AF comparison. Then there is a holomorphic fine-lattice gauge $v(z)$, real G-valued on the real slice, with

$$\upsilon(z,x)=u(z,x)\quad(x\in\mathcal{N}_j,\ \text{all } j),\qquad \|\log\upsilon\|+h_n\|D_{U_*}\log\upsilon\|+h_n^2\|D_{U_*}^2\log\upsilon\|\le C_{\mathrm{node}}r_u\alpha_n \tag{AD17.21}$$

on every actual layer n. The constant depends on the fixed block size and regularity constants, not on the fine mesh or the number of layers. In particular it has no factor h_n/η.

If the smooth family U_{L} lies in a sufficiently small strong-native ball and the actual primitive domain certificate admits that ball, then the function

$$\mathcal{F}_{\mathrm{node}}(z)=B\Big(U_{\mathrm{L}}(z)^{\upsilon(z)^{-1}},\mathrm{Ad}_{\upsilon(z)^{-1}}J_{\mathrm{L}}(z),\mathrm{a}\Big) \tag{AD17.22}$$

is a holomorphic extension of the original real inherited pullback. It has the primitive B value bound on its carrier. Every current or retained-background derivative of the original real function is the corresponding derivative of (AD17.22). No extension of B to arbitrary complex a is required.

Proof. Put $\lambda_x(z)=\log u(z,x)$. Conjugate it by the fixed real local regular gauge at x, multiply by $\psi((y-x)/r_x)$, and conjugate back by that local gauge at y. Let $\lambda(z,y)$ be the sum of these expressions over the distinct old sites and set $\upsilon(z,y)=\exp\lambda(z,y)$. Disjoint supports make this a pointwise local sum with at most one nonzero term. It is holomorphic in z, and unitary on the real slice. Its value at each old site is exactly λ_x, hence $\upsilon=u$ there.

In the local regular gauge, two finite differences of the bump cost at most Cr_x^{-2}, and one costs at most Cr_x^{-1}. The background covariant differences add terms bounded by its regular local potential and derivative. The common smallness ceiling makes these additions harmless. By (AD17.19a)–(AD17.19c), the normalized factors h_n/r_x and h_n^2/r_x^2 are fixed constants depending on L, and not on the number of levels. Equation (AD17.20) and the adjacent-layer comparison therefore give (AD17.21). Sampling the bump on the fine lattice has the same finite-difference bounds, including the case $j=0$. This is the reason a fine axial jump is avoided.

Multiplication by υ^{-1} changes the relative potential by a covariant derivative of λ, plus the usual analytic products with the preexisting relative potential. One additional covariant difference uses $D^2\lambda$. For λ uniformly small, the local logarithm and its first two differentials are bounded on a fixed neighborhood. Thus the value and first-gradient norm of the transformed relative field is at most C times the original norm plus $C_{\mathrm{node}}r_u$.

The sufficient strong-native norm used here has exactly the source-required second-order curl row:

$$\|E\|_{\mathfrak{N}_{\mathrm{curl}}}=\sup_n\alpha_n^{-1}\max\{h_n|E|,\ h_n^2|\nabla_{U_*}E|,h_n^3|D_{U_*}^*D_{U_*}E|\}. \tag{AD17.24}$$

We do not assert a component-Laplacian bound for the interpolated relative potential; that optional row would require an additional argument. The curl row follows from physical-current covariance. Put $W=U_{\mathrm{L}}^{\upsilon^{-1}}=e^{i\eta E}U_*$. The current obeys $\mathcal{J}(W)=\mathrm{Ad}_{\upsilon^{-1}}\mathcal{J}(U_{\mathrm{L}})$. The derivative of the link exponential is an invertible, uniformly bounded local chart matrix. The exact chart identity therefore reads

$$J_*+\Delta_*E+V'_{0,*}(E)=\mathsf{T}(E)\mathcal{J}(W). \tag{AD17.25}$$

Here $\Delta_*=D_{U_*}^*D_{U_*}$ plus the local curvature multiplier, with the source convention, and $\mathsf{T}(E)$ is the bounded chart matrix. The value/gradient Wilson remainder estimate, used in RM.23, bounds the nonlinear term by $Cr^2\alpha_n^2h_n^{-3}$ when the first two rows of (AD17.24) are bounded by r. The real baseline current, the original smooth current, and their difference have their stipulated strong layer bounds. Subtract their values at the center in (AD17.25), and use bounded adjoint matrices. This gives

$$h_n^3|D_{U_*}^*D_{U_*}E|\le C(r_{\mathrm{L}}+C_{\mathrm{node}}r_u)\alpha_n$$

after one common smallness ceiling. The same formula accommodates a nonzero strong real-center current: the chart-factor difference is bounded by the perturbation times that small baseline current and is absorbed in the displayed linear bound. This establishes the curl row without taking a third spatial derivative of λ. The independent-current slot transforms by the same adjoint representation. The reconstructed clauses still use the actual whole-ball primitive domain certificate; they are not inferred from a first-response estimate.

On the real slice, (AD17.18) holds. At every old bond basepoint $\mathrm{Ad}_\upsilon\,\mathrm{a}=\mathrm{Ad}_u\,\mathrm{a}$. Applying the source's simultaneous gauge covariance a second time gives

$$B(U_{\mathrm{L}}, J_{\mathrm{L}}, \mathrm{Ad}_u\, \mathrm{a}) = B(U_{\mathrm{L}}^{\upsilon^{-1}}, \mathrm{Ad}_{\upsilon^{-1}}\, J_{\mathrm{L}}, \mathrm{a}). \tag{AD17.23}$$

The last expression has fixed real older fields and analytic native field/current arguments. Its analyticity and value bound are exactly the primitive source statement once those arguments are admitted. The real equality proves that it is an extension of the actual real pullback and fixes all its jets. One does not first assert that B is defined at the complex axial field, and no integration contour or outer measure has been changed. □

The construction preserves endpoint locality. A marked derivative of a nodal logarithm has the same bound as that of the actual u, up to the uniform logarithm constant. Interpolation uses only sites within the radius in (AD17.19c) of the output. For a primitive born at index j, the relevant node meshes are at most a fixed multiple of h_j. There is no unnecessary later coarse node whose bump reaches the primitive from a distant scale. Thus a source one-endpoint estimate changes by a fixed spatial connector, with the same layer normalization and a fixed decrease of exponential decay. On a primitive carrier this enlarges only its local collar; its cost is absorbed by the available carrier decay. There is no factor proportional to the number of old fields because the interpolation supports have bounded overlap. The number of old nodes per terminal block is bounded by a fixed power of L, by the same separation argument.

More explicitly, suppose the actual smooth family and converter satisfy on the common complex contour

$$\alpha_n^{-1}\big(h_n|\partial_i A_{\mathrm{L}}| + h_n^2|\nabla_{U_*}\partial_i A_{\mathrm{L}}| + h_n^3|\partial_i \mathcal{J}(U_{\mathrm{L}})|\big) \le E_i e^{-cd(i,x)}, \qquad \alpha_n^{-1}|\partial_i \log u(x_j)| \le E_i e^{-cd(i,x_j)} \tag{AD17.26}$$

for the nearby old nodes x_j, with the corresponding independent-current bound. Bump differentiation gives the same bound for $\partial_i\lambda, h_n D\partial_i\lambda, h_n^2 D^2\partial_i\lambda$. Differentiating the analytic relative-potential formula gives the first two rows of $\partial_i E$ with the same majorant. Differentiate the exact covariance of $\mathcal{J}(W)$, then (AD17.25). The new current term is the conjugated original current derivative plus a commutator with $\partial_i\upsilon$; the commutator is bounded by the strong current times the already bounded nodal derivative. The curvature multiplier and the differentiated local remainder have the same normalized estimates. Thus

$$\|\partial_i(E, J_{\mathrm{aux}})\|_{\mathfrak{R}_{\mathrm{curl}},X} \le CE_i e^{-c'd(i,X)}. \tag{AD17.27}$$

This proves the direct one-endpoint tangent estimate in the fixed-old-field input; it is not a tensor-product bound inferred only from a scalar value bound.

If the complete primitive ball has radius r_{nat}, it suffices to take the smooth-family radius and converter reserve so that

$$C(r_{\mathrm{L}} + C_{\mathrm{node}} r_u) < r_{\mathrm{nat}}/4.$$

The converter estimate $r_u \le C_g r_{\mathrm{L}}$, when supplied by the actual normalized inverse-axial gauge, reduces this to the single fixed choice

$$r_{\mathrm{L}} < \frac{r_{\mathrm{nat}}}{8C(1 + C_{\mathrm{node}} C_g)}. \tag{AD17.28}$$

This reserve is independent of age and derivative order. At any resulting point, a genuine affine directional disc of radius at least $r_{\mathrm{nat}}/2$ in the chart remains in the full primitive domain. Cauchy and (AD17.27) give the source-activity first-derivative bound with its unchanged value majorant. Subsequent current derivatives and the old-root/carrier sums are then the already proved BC16 estimates. The complete-domain premise, rather than a bound on the derivative of a domain predicate, is what licenses this last Cauchy step.

Actual inputs identified in Annexes AP and AQ. RM.3 and (RM.26)–(RM.36) in Annex AP construct the normalized inverse-axial gauge in the single chosen symmetric source family. Its unambiguous endpoint row is

$$|u^{-1}\partial_i u|(x) \le Ch_s \sup_{\Delta_s(x)} |\partial_i A_{\mathrm{L}}|.$$

No extra factor α_n is inserted into this row. JI.32–JI.40 in Annex AO apply it to the actual full $2P_k$ current contour, retain the fixed real reference transformation, and compare with the running native radii. This supplies (AD17.20) and (AD17.26) with their actual coupling/age factors. The raw M^j gauge covariance uses the restriction of the fine gauge at j-grid sites; it is not the averaged gauge appearing in corrected relative coordinates. Thus its action on each old A_j is precisely the action used in (AD17.18).

NP.5 in Annex AQ supplies the genuine strong ball for the U-dependent reconstructed clauses, with the independent-current extension specified there. Its input norm is the source-required curl norm in (AD17.24). These results are

subject to their stated source-family and predecessor hypotheses; none asserts an all-source Yang–Mills theorem. Together with AD17.3 they supply the fixed-old-field analytic replacement for the inherited B activity. The complete current/native contour and all boundary-sector prefactors are assembled separately. The absence of a printed complex old-A domain is no longer an additional premise of this replacement.

4. The current-scale exterior relocation

4.1 Its exact support and parameter dependence CMP119 (3.20), p.269, defines

$$\Lambda_{k+1} = \Omega_{k+1}^{\sim -2} \cap (R'_{k+1}{}^{\sim})^c, \qquad S_{k+1} = (\Omega_{k+1} \setminus \Omega_{k+1}^{\sim -1}) \cup R_{k+1}. \tag{AD17.7}$$

Here the enlargement uses the stated LMR_{k+1} cubes. The extra layer between S_{k+1} and Λ_{k+1} also separates their free-bond coordinate sets; it is much wider than the fixed bond endpoint convention. The later rectangular enlargement is of components of Λ_{k+1}^c, so it can only shrink Λ_{k+1} and preserves this separation.

The source defines $H^{(k)}$ from two background averages in (3.22). It depends on U_{k+1} and the fixed geometry, restricted to the stated exterior region $\Lambda_{k+1}^c \cap \Lambda_k$. It is independent of the new inner integration coordinate $z = A_k|_{\Lambda_{k+1}^{(k)*}}$.

For $b \in S_{k+1}^{(k)*}$, the change of variables is (1.22) with $H^{(k)}(b)$ in place of $H_{1,\Lambda_1}(b)$. In current notation it has the form

$$a_b^{\mathrm{old}} = \mathcal{R}_b(a_b; H_b) = \frac{1}{ig_k} \log(\exp(ig_k a_b) \exp[-i\chi_b(a_b) H_b]), \tag{AD17.8}$$

where χ_b is the smooth real cutoff used by the source on the specified subset, and is replaced by (1) on the remaining bonds. It is not a source variable and is not analytically continued as an arbitrary complex function. For each fixed real exterior coordinate a, its value and the derivatives appearing in the Jacobian are fixed coefficients. The source current radius and the logarithm convention ensure that this is the indicated local change of variables.

Proposition AD17.2 — exact cancellation of the relocation Jacobian in the inner conditional law Let $\rho_k(z, a; U_{k+1}, \mathrm{a}, \omega)$ be the full density at this step, with all action terms, nonlinear constraint factors and history information ω retained. The exterior relocation (AD17.8) produces

$$\tilde{\rho}_k(z, a; \cdots) = J_{\mathcal{R}}(a; U_{k+1}, \omega) \rho_k(z, \mathcal{R}(a; H^{(k)}); \cdots). \tag{AD17.9}$$

The Jacobian $J_{\mathcal{R}}$ is independent of z. Wherever the conditional normalization is nonzero,

$$\frac{\tilde{\rho}_k(z, a; \cdots)}{\int \tilde{\rho}_k(z', a; \cdots)\, dz'} = \frac{\rho_k(z, \mathcal{R}(a; H^{(k)}); \cdots)}{\int \rho_k(z', \mathcal{R}(a; H^{(k)}); \cdots)\, dz'}. \tag{AD17.10}$$

In particular the relocation Jacobian's current gradient and current Hessian vanish identically, before any estimate. The cancellation is an identity in the admitted retained/background parameters as well, so every existing mixed current/native jet of this conditional Jacobian contribution is zero. This does not assert that the unnormalized outer Jacobian has zero native derivatives.

Proof. The support statement (AD17.7) shows that the change does not act on an inner free coordinate. Its displacement uses a, U_{k+1} and fixed history information, and none of these is an inner coordinate. Thus the derivative matrix of the full change in ((z,a)) is block triangular with an identity inner block; its nontrivial determinant depends only on the exterior variables. Substitution gives (AD17.9), and the common factor cancels in the quotient. This is an exact algebraic cancellation. It is not an assertion that the Jacobian is one. □

The source's induced $V^{(k)}(S_{k+1})$ includes more than this Jacobian. It also includes the change in every original action term. Those changes do not generally cancel. Nor does the Jacobian disappear from the outer marginal: the integral in the denominator of (AD17.10), multiplied by $J_{\mathcal{R}}$, remains its weight. Native background derivatives of that outer weight are not estimated by the zero inner Hessian.

4.2 Direct treatment of the relocated inherited action Rather than estimate a Taylor expansion of a primitive activity in the displacement, one can retain its exact composition:

$$\widetilde{F}_{j,X}(z;\eta) = B^{(j)}\big(X, U_k(I_k(z, \mathcal{R}(a; H^{(k)}))), \mathrm{a}, \mathrm{S} \cap X\big). \tag{AD17.11}$$

This is the source activity after the actual change of variables. It retains every induced term in that activity. Conditional normalization subtracts its value at a fixed inner reference coordinate only after the whole transformed action has been assembled.

If its actual field/current arguments are admitted to the primitive B domain, the primitive value majorant remains $B_0 e^{-\varkappa d_j(X)}$. No inverse displacement radius is introduced merely by representing the changed activity as a Taylor difference. Its derivative with respect to z is the original exact current-map derivative evaluated at the shifted exterior datum. The derivative of the displacement itself with respect to z is zero. The constraint map I_k still contains its full nonlinear dependence on z and the exterior datum; it is not replaced by its linearization.

This observation removes an unnecessary argument, not the domain requirement. The shifted coarse datum must remain in the common original-map contour, and its entire primitive U-dependent output must pass the native B admission test. The same condition is needed when the native background is complex, because $H^{(k)}$ then has its actual analytic continuation. No holomorphy of the smooth real cutoff in an arbitrary complex exterior coordinate is used.

For the conditional quadratic form, exterior relocation changes the retained linear force and constant term. It does not change its current-current block, because the background precision is fixed in that conditional split. The force must remain in the Gaussian mean. Nonquadratic classical, constraint, measure and E/R terms are likewise retained at the shifted datum; they are not removed by (AD17.10).

Thus the complete source $V^{(k)}$ may either be estimated in its published localized Taylor representation, or handled by the exact fully relocated action plus the explicit exterior Jacobian. Both expressions represent the same density. The second route is sufficient for current derivative estimates if the full shifted-data domain is proved, and does not require first constructing analytic derivatives in an earlier fluctuation variable.

5. Actual real-center reserve for the native B ball

5.1 The physical current is part of the primary regularity theorem CMP102 (2), p.278, defines regular configurations by both a plaquette bound and

$$|D_U^* \partial U| \le \varepsilon_0 \eta^2 h_n^{-3} \quad \text{on an actual layer of proper mesh } h_n. \tag{AD17.12}$$

The physical current used in the later expansion is $J_U = \eta^{-2} \operatorname{Im} D_U^* \partial U$, so (AD17.12) gives $h_n^3 |J_U| \le \varepsilon_0$. Theorem 1, (8), places the actual minimum in this space with $\varepsilon_0 = B_3 \varepsilon_1$. Its (9)–(10) also supplies the local value, first-difference and second-order chart bounds. Thus a strong physical-current bound is a primary result, not an inference from the plaquette bound alone.

CMP119 p.276 states that on a multiscale layer $\Omega_n \setminus \Omega_{n+1}$, the current background has the regularity properties and bounds of an n-scale configuration. At the admitted source-free history scope this gives

$$h_n^2 \|F(U_*)\| + h_n^3 \|J(U_*)\| \le C_{\mathrm{hist}} \varepsilon_n, \qquad \varepsilon_n = A_0 g_n \ell_n^{p_0}, \tag{AD17.13}$$

with the corresponding source local chart bounds. This import retains the actual history geometry and primary smallness conditions. It is not a new proof of the full source-free construction.

5.2 The numerical reserve is fixed, not dyadically shrinking Every coefficient in the source restrictions (2.34)–(2.39) is at least

$$s_* = 1 - \beta > 0. \tag{AD17.14}$$

Choose the already proposed normalization radii

$$\alpha_{0,n} = C_0 g_n \ell_n^q, \qquad \alpha_{1,n} = C_1 g_n \ell_n^q, \qquad q > p_0, \tag{AD17.15}$$

with the fixed ratio required by the source local gauges. Then the real-center ratio in every basic layer norm is bounded by $C\ell_*^{p_0-q}$, where $\ell_* = \inf_n \ell_n$. The current and curvature comparisons between an actual layer $n > j$ and the innermost region of an old j-domain have factors $L^{-3(n-j)}\alpha_{0,n}/\alpha_{0,j}$ and $L^{-2(n-j)}\alpha_{0,n}/\alpha_{0,j}$, respectively. The admitted AF comparison bounds their remaining ratio polynomially in $n-j$; hence both have a fixed finite embedding constant C_{emb}.

The reconstructed-current restriction (2.37), after physical unit conversion, allows the weight $h_n^{-2}h_{\min(p,n)}^{-1}$. The stronger physical profile h_n^{-3} therefore fits it. This is the same unit conversion checked in the argument; it does not prove a reconstructed-current estimate for a new map.

Suppose the actual joined projection proves, on its entire strong-native ball of radius r_b, the output estimate required for those reconstructed profiles,

$$\text{normalized output size} \le C_{\mathrm{map}}\left(C_{\mathrm{hist}}\ell_*^{p_0-q} + r_b\right), \tag{AD17.16}$$

including its actual source gauge and every relevant real/complex relative-field clause. Then the specific numerical choices

$$r_b \le \frac{s_*}{16C_{\mathrm{emb}}C_{\mathrm{map}}}, \qquad C_{\mathrm{emb}}C_{\mathrm{map}}C_{\mathrm{hist}}\ell_*^{p_0-q} \le \frac{s_*}{16} \tag{AD17.17}$$

leave at least a fixed fraction of every threshold in (2.34)–(2.39). The first choice is fixed; the second is achieved by one common coupling ceiling. They do not depend on age, cutoff depth or derivative order. Independent-current additions use the fixed common norm and triangular extension already proved in BC16.4.

The actual mathematical issue is (AD17.16) on a whole ball, not the numerical reserve afterward. A bound at its center or only for its first response does not suffice. The $B = 0$ joined-projection approach is relevant because it seeks this whole-ball bound directly. If proved for the actual determining average and source gauge, it could provide the primitive directional-disc certificate without first establishing an identity between arbitrary prepared maps.

The exterior relocation datum must be included in that same map-admission argument. Its source displacement is $O(\varepsilon_k e^{-R_k})$ before division by g_k, so it is far inside a fixed absolute coarse-data ceiling at real admitted histories. A uniform complex-background version still needs the common analytic map estimate; a real displacement bound alone is not that theorem.

6. Current application conditions

The exact old-slot and relocation reductions above are available for the actual C-05 application. They rule out two unnecessary demands: a proof that every converter is the identity at every earlier fluctuation site, and an inner-current bound on the exterior relocation Jacobian. They do not remove any factor from the full outer law.

The direct source-product-domain route remains blocked at the checked relation (AD17.2). The alternative native-ball route is now supplied at its stated scope by NP.5 and the actual converter identification RM.3/JI.32–JI.40. Proposition AD17.3 then gives the needed fixed-old-field analytic replacement and its endpoint tangent. All inherited B, E/R boundary assignments, classical/constraint/measure terms and the full shifted exterior datum must still be retained in the complete application.

Applying these native-domain estimates to the full boundary perturbation requires the additional hypotheses in Annex AW.

Primary References

- CMP119. T. Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, CMP 119 (1988), 243–285, pp. 247, 250, 261, 267–269, 276 and 278.

- CMP102: T. Balaban, *The Variational Problem and Background Fields in Renormalization Group Method for Lattice Gauge Theories*, CMP 102 (1985), 277–309. (2), (4), Theorem 1/(8)–(10), Proposition 9. The source definitions distinguish physical current regularity from curvature regularity.
- CMP109: T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. I*, CMP 109 (1987), 249–301, Theorem 3, Sections 2–3. DOI: https://doi.org/10.1007/BF01215223
- CMP116: T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. II. Cluster Expansions*, CMP 116 (1988), 1–22, (1.20)–(1.22), (1.34), Lemmas 1–2. DOI: https://doi.org/10.1007/BF01239022
- LFR II: T. Balaban, *Large Field Renormalization. II*, CMP 122 (1989), 355–392. pp.373–374 around (1.61), for the retained collar splice gauge.
- LFR I: T. Balaban, *Large Field Renormalization. I*, CMP 122 (1989), 175–202. p.179, (1.10)–(1.13), for the rebuilt admitted history and its layer grading.
- Earlier three-dimensional stability paper: T. Balaban, *Ultraviolet Stability of Three-Dimensional Lattice Pure Gauge Field Theories*, CMP 102 (1985), 255–275, (7), (18), (22), (51)–(52). DOI: https://doi.org/10.1007/BF01229380. Ambiguous small cutoff symbols are not used as a verified dictionary.
- Related reductions: Annex AN, especially BC16.1–BC16.5 and its explicit native-chart counterexample; Annex AF for physical normalization and original-geometry kernel scope. Their unproved actual-map premises are not treated as completed here.

Annex AS — Ordinary older Gaussian and Schur response

Annex AS. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

1. The current factorization and its response factor

Being supported in the old large-field region does not by itself make an operation independent of a newer current coordinate. CMP119 (1.29), (2.21) contains background-dependent quadratic forms and constrained covariances. LFR II (1.71), pp.379–380, explicitly retains a dependence of the completed operation on its native pair (U, J). If that pair is a function of the newer current, its response must be kept unless an exact determining-data identity makes it constant.

Fix a positive history branch, every old integration coordinate a, and the current free variable z. Suppose its old-operation density is $w(z, a; \upsilon)$, with common support in a on the fixed chart, and choose the reference current $z = 0$. Then, wherever the reference density is positive, the exact additional current exponent is

$$\Theta(z, a; \upsilon) = \log \frac{w(z, a; \upsilon)}{w(0, a; \upsilon)}. \tag{OR17.1}$$

The reference factor $w(0, a; \upsilon)$ remains in the outer measure. Zero sets common to the support do not affect the identity. It is not permissible to replace $w(z, a; \upsilon)$ by $w(0, a; \upsilon)$ without the factor e^{Θ}. If the density is a product of explicit exponentials and fixed selectors, (OR17.1) is the sum of their exact exponent differences and requires no logarithm of a possibly small total integral.

This observation does not establish common support, native analyticity, or a spatial bound for the actual source density. Those are source matching requirements below. A background-dependent delta constraint or selector would need a separate exact coordinate treatment; it cannot be absorbed into an ordinary analytic logarithm.

2. Exact response of a retained Gaussian Schur form

Let $K(\xi)$ be a holomorphic symmetric matrix on fixed free coordinates, positive on the real slice. Split its indices into the already integrated set I and the retained set O. Assume K_{II} is invertible throughout the admitted chart. Define

$$G = K_{II}^{-1}, \qquad S = K_{OO} - K_{OI} G K_{IO}, \qquad \mathcal{E}a = (-GK_{IO}a,\ a). \tag{OR17.2}$$

Here the first component is indexed by I. All transposes below are bilinear transposes in the holomorphic continuation, agreeing with real transposes on the real slice. The exact retained exponent is

$$q(\xi, a) = -\tfrac{1}{2} a^{\mathsf{T}} S(\xi) a = -\tfrac{1}{2} (\mathcal{E}a)^{\mathsf{T}} K(\xi) (\mathcal{E}a). \tag{OR17.3}$$

For any admitted directional variation $\dot{\xi}$,

$$\dot{S} = \mathcal{E}^{\mathsf{T}} \dot{K} \mathcal{E}, \qquad \dot{q} = -\tfrac{1}{2} (\mathcal{E}a)^{\mathsf{T}} \dot{K} (\mathcal{E}a). \tag{OR17.4}$$

Proof. The I-component of $K\mathcal{E}a$ is zero, while the O-component of $\dot{\mathcal{E}}a$ is zero. Thus both terms differentiating $\mathcal{E}$ in (OR17.3) vanish. The remaining term is (OR17.4). Equivalently, differentiate G using $\dot{G} = -G\dot{K}_{II}G$ and collect the four Schur-complement terms. No derivative of the covariance is discarded. The identity holds on the complex chart as an algebraic matrix identity. □

If the source retains only within-component quadratic terms, extend the retained vector by zero outside one component and apply the same identity component by component. Older cross-component terms assigned to primitive B must not be counted again in (OR17.3).

A spatial consequence, with its exact hypotheses Suppose that in old current units, for a local native direction at an anchor x,

$$|\dot{K}_{bc}| \le A\,u(x)e^{-c[\rho(b,x)+\rho(c,x)]}, \quad |\mathcal{E}_{br}| \le Ae^{-c\rho(b,r)}, \quad r \in O. \tag{OR17.5}$$

Let a be supported in $C \subset O$, with $\|a\|_\infty \le P$. Exponential convolution and (OR17.4) give, at reduced fixed rates,

$$|\dot{q}| \le CA^3P^2u(x)\left(\sum_{r\in C} e^{-c'\rho(x,r)}\right)^2 \le C'A^3P^2u(x)\sum_{r\in C} e^{-c''\rho(x,r)}. \tag{OR17.6}$$

The second inequality uses the volume-independent lattice sum $\sup_x \sum_r e^{-c'\rho(x,r)} < \infty$. Lie-algebra coordinate multiplicities enter fixed constants. The same argument works on tori, with their periodic distance, and for principal restrictions. No factor equal to the size of C appears without an exponential anchor.

This is a consequence of a local coefficient-response bound in (OR17.5). Exponential decay of K_{bc} alone is not that bound. A spatially nonlocal dependence of a single entry on a parameter would violate the inference. For the actual application the local response must be obtained from the source operators, their localized expansions or their exact resolvent variations, with their physical measures and old-layer weights retained.

3. What would close the additional old-operation row

For an actual old operation of scale j, the current pullback of its native arguments must be the same admitted family as that operation's source definition, including its local collar reconstructions. The exact C-05 endpoint/current-domain results can be used only after this identification.

A sufficient concrete old-operation row is

$$|\partial_{z_i}\Theta_{j,X}(z,a;\xi)| \le \frac{K_{j,X}}{P_k}\,\theta_k(1+k-j)^b e^{-\Gamma(k-j)-c_{\mathrm{col}}MR_k}e^{-cd_X(i)} \tag{OR17.7}$$

on the full common current/native contour. Here each prefactor and carrier count must be actual, rather than postulated: a Gaussian coefficient can carry P_j^2; a replaced component can have its displayed source polynomial size cost. The available fixed logarithmic and age powers must be stated before summation. Fixed powers of g_j^{-1}, if present, also need an explicit current-collar payment and an AF age comparison.

With a summable carrier decomposition, (OR17.7) supplies the all-order jets by BC16, including the weighted Hessian row. This is a sufficient implication, not an assertion that the source matching is already done.

The Gaussian calculation uses CMP119 (2.21) and CMP99 (3.155)–(3.158), Theorem 3.15. The replaced operation uses LFR II (1.65), (1.71)–(1.75). A normalized-kernel value bound alone does not control its current response.

4. The actual ordinary Gaussian operation

This section applies the response argument to the unreplaced operations of CMP119 (2.21). It does not assert an analogous pointwise application to all hidden variables of a completed R operation. That is a distinct question, and the alternative representation in LFR II (1.103)–(1.104) is treated separately in Annex AT, Section 7.

4.1 Exact source coefficients and their local dependence The source free-coordinate precision is

$$K_j = C_j^* \Delta^{(j)} C_j. \tag{OR17.8}$$

Its definition includes the background-current term in CMP99 (3.156). The source covariance in (3.158) is the inverse principal block in these free coordinates. Thus the within-component quadratic form in CMP119 (2.21) is exactly (OR17.3), with the retained vector supported in that component. This is the same ordinary Gaussian law, not a Laplacian substitution.

The following primary results provide local dependence as well as off-diagonal kernel decay:

- CMP99 p.413, Theorem 3.9: each localized inverse coefficient depends only on the background restricted to the specified fixed enlargement of its carrier.
- CMP99 pp.415–416, Theorem 3.10 and (3.108): the walk representation has the stated walk/tree decay and depends only on the union of the indicated enlarged blocks.
- CMP99 p.422: the inverse coarse averaging operator occurring in (3.156) is handled by the same construction. The current correction and the H_1 factors have the corresponding expansions, including the branches described on p.427; the shortest-tree length replaces the single-line length.
- CMP99 p.432, Theorem 3.15 and (3.187): the actual principal-restriction fluctuation covariance has the same localized expansion and locality consequences.

The analytic coefficient extension is the one in Section B of that paper, pp.399–400, and its stated current extension. The actual native current term is retained. The current external pair in the present application lies in a smaller uniform strong domain by NP.5 and the older-scale embedding used there. A regular complex neighborhood of the principal inverse is obtained either directly from the stated source domain or by the uniform real lower bound and a fixed small complex perturbation. Its radius is fixed before summing over old operations.

The local map C_j introduces neither an extra long-range dependence nor a noncovariant substitution. It solves the actual linear averaging constraint on the fixed exceptional bonds. Under a simultaneous gauge change, that equation and the zero tree conditions transform covariantly. Uniqueness of the exceptional-bond solution therefore gives the endpoint-adjoint transformation rule for C_j. Scalar localization cutoffs commute with this rule. Products, localized inverses and the displayed current correction consequently give endpoint-adjoint covariance for each localized free-coordinate coefficient. This verifies the gauge rule needed below; covariance of a fine Green function alone would not suffice.

Use the actual physical measures in these kernel compositions. The source walk bounds include their normalized row sums; multiplying their kernels is not an unweighted count of fine bonds. The resulting localized expansion of the Schur form has primitive terms

$$q_{j,\mathcal{W},r,s} = -\frac{1}{2} a_r^{\mathsf{T}} S_{j,\mathcal{W}}(r,s;U,J) a_s, \qquad r,s \in C, \tag{OR17.9}$$

where $\mathcal{W}$ denotes the actual finite tree/walk data and its carrier contains the two old endpoints. At every fixed cutoff the sum is exact. Absolute normal convergence and the fixed decay reserves justify its grouped carrier form. With the source parameters chosen as in the existing carrier analysis, its rooted analytic norm is bounded by

$$C P_j^2 e^{-\kappa t_j(\mathcal{W})}. \tag{OR17.10}$$

Here t_j is the source tree/walk length, with the fixed equivalent carrier convention, and the bound includes the summation of internal walk labels at a fixed root. The retained old field is bounded by $C\varepsilon_j/g_j \le C' P_j$, by the actual preliminary small-field restriction; constants include A_0/A_1. No small-field restriction is imposed on a raw large-field link outside that ordinary fluctuation region, since it is not one of the a variables in this quadratic form.

4.2 Gauge and current admission with old fields fixed Each scalar in (OR17.9) has only two displayed old adjoint slots. They are on the ordinary j-grid in $\Omega_{j+1} \cap Z_{j+1}$. Apply AD17.3 to the actual required endpoints, using its proved terminal-layer grading and the converter bound JI.40. Its double real covariance identity gives an admitted holomorphic fixed-old-field expression for (OR17.9). There is no need to rotate an arbitrary hidden raw variable or to assume an unprinted complex old-field product domain.

The strong native ball is simultaneous with the full current contour and with the independent-current slot. The source-local enlarged carrier and the interpolation collar differ by fixed physical-scale factors. Cauchy on that genuine ball, using the original-map one-current endpoint row, gives the analogue of (BC16.4) with primitive prefactor (OR17.10). All derivatives of the actual C_j, constrained Hessian, principal inverse and nonlinear current insertion are included by this composed analytic function. The elementary identity (OR17.4) independently verifies that the covariance response has not been frozen.

Every primitive carrier in (OR17.9) meets an old retained fluctuation site in Z_{j+1}. The actual current-to-boundary path, together with the source tree connector when the native attachment is elsewhere on that carrier, therefore supplies the collar and age factors exactly as in BV.6–BV.12 and BC16.5. Reserve separate fixed portions of the decay for the current collar, the old-age count, and the final spatial weight. This gives (OR17.7), with $K_{j,X}$ carrying the explicit P_j^2 factor.

The distinction between a path to an old boundary and a path to an earlier output layer remains in force. For an output layer earlier than $j+1$, use the actual path to that layer; for a later output layer use the carrier connection to the old endpoint. This is the combined path/carrier argument already established in BC16.5, not an assignment of the same decay to two different costs.

Theorem OR17.1 — ordinary old-operation response budget On the retained source class and the common current/native contour, the sum of the exact ordinary Gaussian density responses has the same all-order anchored current estimate as BC16.13, with prefactor

$$\mathcal{K}^{\mathrm{G}}_{T,k} \le C\,\theta_k P_k^2 e^{-c_{\mathrm{G}} M R_k}, \qquad \theta_k = \ell_k^{p_0-q}. \tag{OR17.11}$$

In particular its weighted current Hessian is at most

$$C\theta_k e^{-c_{\mathrm{G}} M R_k} \longrightarrow 0. \tag{OR17.12}$$

The constants are independent of cutoff depth, volume and admitted history. Fixed native directional jets are obtained on the common smaller strong ball with the Cauchy costs stated in AB17.6.

Proof. The verified native first-response bound, (OR17.10) and the connector give the premises of the BC16 endpoint theorem. The exact AF comparison implies

$$\frac{P_j^2}{P_k^2} = \left(\frac{\ell_j}{\ell_k}\right)^{2p_0} \le C(1+k-j)^{b_0} \tag{OR17.13}$$

for one fixed b_0. Indeed $g_j^{-2} \le C(1+k-j)g_k^{-2}$, so $\ell_j \le \ell_k + \log[C(1+k-j)]$, and ℓ_k has a fixed lower bound. The conservative old-root count contributes $L^{4(k-j)}$, while the current/native radius and finitely many profile embeddings contribute fixed polynomial and dyadic costs. Choose the already fixed layer-separation reserve so that their age sum converges. The walk/tree counting and the all-order current-position count are exactly those in BC16; no derivative-order-dependent geometry parameter is selected. This yields (OR17.11) and a fixed factorial exponent five. Setting the current order to two proves (OR17.12). □

4.3 The other literal factors in an ordinary operation CMP119 (2.21) contains the old raw Haar integration, its exact raw averaging delta, the displayed ζ, χ factors, and the within-component quadratic form. It does not contain an additional determinant to be silently differentiated: CMP119 p.254 after (1.30) assigns the earlier fluctuation normalization to E, and the general step keeps its constants in the corresponding action/vacuum factors. Those assigned terms are already in the appropriate sector of the boundary ledger.

For the fixed raw/delta/selector factors, use their actual local data, not a global-background surrogate. LFR I p.183, after (1.28), states their dependence on field variables localized in the corresponding large-field components. The current substitution, including its one-step constraint stencil, stays beyond the old-component collar and leaves those determining variables unchanged. Thus their supports and values are constant under the current variable in this ordinary-operation chart. This is an exact retained-data statement for these factors. It is not the incorrect inference that the quadratic coefficients are constant merely because their old integration variables have the same support.

The theorem just proved supplies the response for the latter coefficients. The exact exponent (OR17.1) with that Gaussian response gives the correct outer-law factorization without deleting a derivative. Statements about a normalized posterior containing arbitrary marked old observables, or about a completed R operation on its unintegrated hidden variables, remain separate.

Annex AT — Completed operations and their source domains

Annex AT. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

Sections 1–6 preserve the genuine pointwise-kernel issue. Section 7 instead uses the source's fully integrated alternative (1.103)–(1.104), with its changed carrier norm. That alternative removes the hidden old-variable gauge problem for completed eligible components, but requires the separate relative-carrier current estimate and the treatment of unreplaced operations. This note does not claim the full marked-history theorem H-03 or completion of C-05.

1. What the primary formulas do and do not say

CMP119 p.258, (2.19)–(2.21), describes the support and component factorization of the old operation T_k. Its within-component quadratic form contains the background-dependent operators

$$C^*\Delta^{(j)}C, \qquad C^*\Delta^{(j)}C\,C^{(j)}(\Lambda_{j+1})\,C^*\Delta^{(j)}C. \tag{OT.1}$$

An integration supported on an old component need not have coefficients independent of a new current coordinate. The old background is itself a function of the retained and current data.

The issue remains after R. The complete operation $T_j'(X)$ is LFR II (1.71), pp.378–379. The source explicitly states on p.379 that its quadratic form in the R fluctuation and the self-component expression $V(X^\sim)$ depend on the native pair (U, J). It then controls their variation. This is a statement of retained dependence, not exact constancy.

The local representation in (1.61)–(1.65) does not remove that dependence. Printed p.374 explicitly notes that an intermediate localized background still depends nonlocally on the gauge variables V''. Its final extension substitutes the local averages of U into that representation. Keeping the current raw determining data fixed on the old large-field set does not, by itself, keep every new finer collar average in a local determining splice fixed.

Consequently the implication

$$\text{inner current support misses } X \quad\Longrightarrow\quad \partial_z T_j'(X) = 0 \tag{OT.2}$$

has not been proved and is not used here. An exact fixed-data identity for every coefficient, or an actual response estimate, is required.

2. The precise positive source input

Fix an eligible component at the scale j, its admitted outer geometry, and a real native reference. Let η denote every variable and history label still integrated in (1.71), including the old graph sums, cutoffs, field variables and local reference factors. The real reference operation is a finite positive measure $M_0(d\eta)$, of positive but not necessarily uniformly bounded-below mass.

On the source's ordinary native chart, LFR II p.379 writes the analytically deformed operation as that positive reference integral times an exponential:

$$M_\xi(d\eta) = e^{\sigma(\xi,\eta)}M_0(d\eta). \tag{OT.3}$$

After recentering at the chosen real datum, take $\sigma(0,\eta) = 0$. The nonsmall self action is retained in M_0. Only its change occurs in σ.

The primary bound immediately before (1.73) is, up to a fixed source constant,

$$\|\sigma\|_\infty \le C_T B_3^2 A_1^2 C_0 M^d R_j^{d+2} p_1(g_j)^2 q_0(g_j) g_j =: \epsilon_{T,j}. \tag{OT.4}$$

The same bound is stated there for the resummed variation of $V(X^{\sim})$, by its connections to the input set. Constants and fixed enlargements are included in C_T. The bound is uniform in the old variables in the actual integration domain. Recentring or using a fixed smaller chart changes only its fixed multiplier.

All factors in the completed (1.71) remain present in (OT.3). In particular one does not replace the old numerator by the new R reference, remove its self action, or separately estimate the reciprocal of its small real mass. The source's statement that only the quadratic form and $V(X^{\sim})$ depend on (U, J) is made after the denominator and all localization terms have been incorporated into (1.71). A proposed earlier representation must retain its additional normalization responses until that exact rearrangement is made.

The source chart in (1.65) on p.375, has the ordinary complex B space as its input:

$$(U, J) \in \widetilde{\mathcal{U}}_j^c(Y_4, \widetilde{\alpha}_0, \widetilde{\alpha}_1) \longmapsto U_j^0(U, J) \in \widetilde{\mathcal{U}}'^{nc}_j(Y_1, \widetilde{\alpha}_0, \widetilde{\alpha}_1). \tag{OT.5}$$

The complicated modified domain of LFR I (1.64)–(1.69) is the output. It is not another unidentified input ball. The same fixed histories, integrated variables and prepared chain (1.61)–(1.63) are part of this primary implication.

A strong primitive-domain certificate such as NP.5 can be used for the ordinary input only with the corresponding profile families and actual gauge/old-variable representation. If the source uses distinct powers in α_0, α_1, the common profile is bounded by a fixed multiple of their minimum; the minimum power must still exceed p_0.

3. A genuine directional disc gives an ordinary density response

Let $\mathfrak{N}_X$ be a native norm on the dependency set of the component expression. Assume a ball of radius $r_T > 0$ about each point of an inner chart lies in the source domain of (OT.3), with the same old variables and integration support. Assume (OT.4) on that larger ball. Let the actual current/native argument map be $\Phi(z, \xi)$, holomorphic for

$$\|z\|_\infty < 2P_k, \qquad \xi \in \mathcal{D}^+, \tag{OT.6}$$

and suppose its image and every old argument used in (OT.3) are in the preceding inner chart.

For a current coordinate i, suppose

$$\|\partial_i \Phi(z, \xi)\|_{\mathfrak{N}_X} \le \frac{\theta_k}{P_k}\, a_{j,k}\, e^{-c\, d(i,X)}. \tag{OT.7}$$

A fixed enlargement of X and fixed conversion constants may be included in this notation. If the operation uses transformed old fields, their full tangents belong on the left side; they are not silently frozen.

Proposition OT.1 — fixed-old-variable log-density response On the active common support of M_0, the ordinary logarithmic density change

$$F_X(z, \xi; \eta) = \sigma(\Phi(z, \xi), \eta) - \sigma(\Phi(0, \xi), \eta) \tag{OT.8}$$

obeys

$$|\partial_i F_X| \le \frac{C\epsilon_{T,j}}{r_T} \frac{\theta_k}{P_k} a_{j,k} e^{-c\, d(i,X)} \tag{OT.9}$$

uniformly in the old variables. The same statement holds for differences of two numerator or denominator log contributions if each is retained with its actual signed coefficient and admitted disc.

Proof. At the actual argument value, use the affine native line through that point in direction $\partial_i \Phi$. A disc of radius $r_T / \|\partial_i \Phi\|_{\mathfrak{N}_X}$ is admitted. Cauchy applied to the scalar analytic function $\sigma(\,\cdot\,, \eta)$, with bound (OT.4), gives

$$|D\sigma(\Phi, \eta)[\partial_i \Phi]| \le C\epsilon_{T,j} r_T^{-1} \|\partial_i \Phi\|_{\mathfrak{N}_X}.$$

The subtraction in (OT.8) has zero current derivative. Insert (OT.7). No derivative of a moving hard support is taken: the common support is a premise, just as in the source representation. □

On the current support $\|z\|_\infty \le P_k$, all higher current derivatives follow from the full current contour:

$$|\partial_{i_1} \cdots \partial_{i_m} F_X| \le \frac{C\epsilon_{T,j}}{r_T} \theta_k a_{j,k} (m-1)! (2/P_k)^m e^{-c \max_s d(i_s, X)}. \tag{OT.10}$$

This is the finite-dimensional Cauchy argument of BC16.1. It differentiates the full composition, including every higher derivative of the current-coordinate map. Mixed native derivatives use the remaining positive native radii, not an unverified raw-coordinate radius.

4. The component mass has the same endpoint response

The scalar unmarked mass must also retain its denominator. Under the hypotheses above, choose the common ceiling so that, after recentering on the whole contour, $\sup|\sigma| \le \log(3/2)$. Put

$$Z_X(z,\xi) = M_0(e^{\sigma(\Phi(z,\xi),\cdot)}), \qquad \mathcal{F}_X = \log\frac{Z_X(z,\xi)}{Z_X(0,\xi)}. \tag{OT.11}$$

Use the logarithm determined continuously from the real positive reference.

Proposition OT.2 — no inverse-small-mass loss The normalizer is nonzero on the stated chart, and

$$|\partial_i \mathcal{F}_X| \le 3\sup_{\eta}|\partial_i\sigma(\Phi(z,\xi),\eta)|. \tag{OT.12}$$

Thus (OT.9)–(OT.10) hold for $\mathcal{F}_X$ with a further fixed constant.

Proof. Let $m_0 = M_0(1) > 0$ and $\pi_0 = M_0/m_0$. Then

$$|\pi_0(e^\sigma) - 1| \le e^{\|\sigma\|_\infty} - 1 \le \tfrac{1}{2}.$$

Hence $|Z_X| \ge m_0/2$, and a holomorphic logarithm exists on the corresponding ball. Differentiation under the finite positive integral gives

$$\partial_i \log Z_X = \frac{\pi_0(e^\sigma\partial_i\sigma)}{\pi_0(e^\sigma)}.$$

Its modulus is at most $3\sup|\partial_i\sigma|$. The factor m_0 cancels exactly. Apply the current Cauchy estimate afterward. □

This proposition is a response estimate for the scalar mass. It does not permit replacing the conditional law with its mass when an inherited activity or a mark depends on the old integration variables. At fixed old variables, use Proposition OT.1. If a complete old integral is actually performed, retain the dependence of every remaining factor in it.

5. Which ordinary spatial sum would suffice

The fixed-chart propositions require much less than the full M-03 marked carrier norm. For example, if one proves on the actual current map

$$a_{j,k} \le C(1+k-j)^b e^{-\Gamma(k-j)} \tag{OT.13}$$

with enough geometric reserve, and the eligible component footprints have their source bound $\operatorname{diam}_j X \le CMR_j$, their contribution can be summed by the same endpoint and old-root mechanism as BC16.

The old components in a fixed history are not an unrestricted sum over every possible large polymer. At a fixed age their chosen disjoint roots obey the old-grid count $CL^{4(k-j)}$. Their diameter factors must still be paid. If

$$R_j L^{-(k-j)} \le C_R R_k \tag{OT.14}$$

and the actual current guard supplies distance at least $c_R M R_k$, one can choose the fixed spatial exponent small enough that a fixed part of that guard pays $e^{\gamma \operatorname{diam}_k X}$. The remaining guard pays the polynomial diameter factors in the higher-order counting integrals. The estimate

$$\sup_{s\ge 0} s^N e^{-as} \le N! a^{-N} \tag{OT.15}$$

then gives the same fixed factorial type as BC16, rather than a constant depending on the diameter or on derivative order.

The factors $R_j^{d+2} p_1(g_j)^2 q_0(g_j)$ in (OT.4) are fixed polynomials in ℓ_j at the stated source scope. AF comparison bounds their ratios by a fixed polynomial in age and ℓ_k. They can be retained explicitly in the age sum and in a prefactor $Cg_k\ell_k^{N_T}$, with one fixed exponent N_T. The common ceiling makes this prefactor tend to zero.

These are sufficient ordinary component-summation conditions. They must be verified with the actual old component footprints and current-to-component map; this section does not claim their application from a bare scalar bound. In particular an enlarged footprint caused by an unaccounted old gauge transformation must not be discarded.

6. The exact remaining application issue

The source bound (OT.4), the ordinary input of (OT.5), and Propositions OT.1–OT.2 are available. The decisive remaining issue is to match the actual fixed-old-variable current pullback to that native chart and prove (OT.7), without changing its meaning.

For a primitive $B^{(j)}(X)$, AD17.3/JI.39 compensates the actual gauge at the surviving old adjoint-field basepoints. The full $T'_j(X)$ kernel also contains raw V'', graph/delta and gauge-fixing factors and hidden older fields. Primitive-B covariance alone is not pointwise simultaneous covariance of this complete kernel.

There are several logically valid ways to finish the ordinary application, each requiring its own proof:

1. Show that the literal actual axial input is admitted to the ordinary source domain. A bounded axial gauge matrix does not suffice, because a fine-edge potential can still violate the old physical scale.
2. Prove a pointwise fixed-old-variable analytic replacement for the entire variable-dependent part of the kernel. Every old raw endpoint and adjoint slot affected by that replacement must be matched or retained with a quantified analytic radius. If a nodal interpolation is used, its spacing and physical weights must be checked for the actual post-R hidden histories, not borrowed from only the surviving primitive-B slots.
3. Prove an exact joint source-free coordinate change, with its full Jacobian and selector/gauge-fixing transformation, that represents the same conditional density in a common admissible chart. Merely changing variables in $T'_j 1$ does not prove this assertion when the rest of the integrand depends on those old variables.
4. Find an exact primary reassignment of the complete current-dependent coefficient difference into the ordinary B activity and prove its current budget there.

At present this note does not supply one of these matching results. Consequently it does not conclude that $\mathcal{K}_{T,k} = 0$, nor that the ordinary old-operation contribution meets the complete C-05/C-06 current budget.

This ordinary matching issue is narrower than H-03: it requires no renormalized marked input, regulator pushforward, moving-selector finite difference, or strong exponential-volume carrier norm. H-03 remains open even if this ordinary issue is resolved. Conversely deferring all marked reset estimates to H-03 does not erase a current-dependent old factor from a present unmarked conditional density.

7. The fully integrated alternative is a different exact representation

LFR II pp.391–392 supplies an alternative that was not used in Sections 1–6. It should not be conflated with the pointwise operation (1.71). The following records its exact scope and the changes in the localization norm.

7.1 Which components and which factors are integrated At a fixed R step j, let $Y_1, \dots, Y_m$ be the second-class eligible components in LFR II p.378: components of the selected set Z for which the final large-field characteristic function is retained. Write

$$\tau_i = T'_j(Y_i)1, \qquad \Pi_i = \tau_i^{-1} T'_j(Y_i). \tag{OT.16}$$

At real admitted data, $\tau_i > 0$ and Π_i is a probability kernel on the complete old variables in this component. On the primary complex domain, the same analytic normalized operation is defined by the small-tilt argument of Section 4. Its uniformly bounded total variation is precisely the input used on p.391.

Let B_{in} be the sum of the boundary activities whose localization domains meet $\bigcup_i Y_i$. The source identity (1.103) is

$$\Big(\prod_i T'_j(Y_i)\Big) e^{B_{\mathrm{in}}} = \Big(\prod_i \tau_i\Big)\Big(\prod_i \Pi_i\Big) e^{B_{\mathrm{in}}}. \tag{OT.17}$$

The bracket on the right, including every occurrence of the old variables in B_{in}, is then integrated and expanded:

$$\Big(\prod_i \Pi_i\Big) e^{B_{\mathrm{in}}} = \exp\Big(\sum_X \widehat{B}^{(j)}(X)\Big). \tag{OT.18}$$

This is the second use of the convergent expansion (1.98), as prescribed on p.391. It is an exact identity of the source-free density. It is not the replacement $T'_j(Y_i)F \mapsto \tau_i F$.

The final density is (1.104). In that representation the factors associated with Y_i are their retained characteristic functions, their gauge-fixing delta functions, and τ_i, all multiplying the density. There is no remaining integration operation belonging to these Y_i. Printed p.392 states that τ_i depends only on the retained raw field V_j on Y_i, and therefore is not involved in subsequent RG integrations. Thus, for the actual next inner-current operation, these raw variables and these multiplicative factors are fixed.

This fixed-data conclusion belongs to (1.104). It does not assert that the earlier $T'_j(Y_i)$ kernel, with its old variables unintegrated, was independent of the native field or current.

Proposition OT.3 — exact removal of the completed-kernel response Adopt the alternative (1.104) at every completed R step. Then the completed eligible-component sector contributes no inner-current derivative through its multiplicative masses, cutoffs or delta functions. Its entire remaining dependence is in the new boundary activities $\widehat{B}^{(j)}(X)$ produced by (OT.18). The hidden old variables integrated in $\prod_i \Pi_i$ are not arguments of these activities.

Proof. Equations (OT.17)–(OT.18) are the source's exact factorization and cluster expansion. The dependency statement for the factors τ_i is the explicit final statement on p.392. The next RG step integrates raw fields only in the new small-field region; the retained raw fields on Y_i lie outside it. The characteristic and delta factors in (1.104) use those same retained raw arguments. Therefore all their current derivatives are zero at fixed retained data. Every other contribution of the old variables has already been included in (OT.18). □

The operators denoted $T''_j(Z_j)$ in (1.104), for unreplaced components, are still present. Proposition OT.3 makes no assertion of their independence. Their ordinary Gaussian/Schur or other coefficient response must be handled separately.

7.2 Arguments, domain and covariance of the new boundary terms The remaining arguments of $\widehat{B}^{(j)}(X)$ are:

- the native pair (U, J), localized in X;
- the fixed geometric history, component choices and real cutoff support;
- the retained raw data, including $V_j|Y_i$, that were parameters rather than integration variables of (OT.17);
- any old adjoint or raw-field variables outside the integrated Y_i that occur in the incoming activities.

The argument list is obtained by actually applying the kernels to the entire exponential. An integrated hidden field cannot then be treated as a surviving external variable. Conversely a raw retained parameter is not discarded merely because the old histories inside its component have been integrated.

The analytic input is the same ordinary native space

$$\widetilde{\mathcal{U}}^{c}_{j}(X, \widetilde{\alpha}_0, \widetilde{\alpha}_1) \tag{OT.19}$$

used in (1.65), (1.68) and (1.99), with the remaining old arguments fixed in their admitted support. The positive real kernels and their uniformly bounded normalized analytic extensions are used on this space. The convergent majorant is uniform there, so differentiation and integration preserve holomorphy on each smaller native ball. This is a property of the complete normalized integrals, not an inference from a real mass bound alone.

For simultaneous real gauge transformations of all surviving external arguments, the resulting activities have the effective-action covariance required by the source induction. This is the scoped primary import of the final p.392 statement that (1.104) may be used in the inductive description of the effective action and proves Theorem 1. It is also obtained by applying the same real change of integrated variables to the complete expression (OT.17) before its unique convergent localization expansion. No separate covariance of an isolated gauge-fixing delta factor is asserted.

Accordingly NP.5 and the genuine primitive-domain argument can now be applied to these new scalar activities, with their actual retained arguments. If a nodal gauge representation is used, only the surviving raw endpoints and adjoint basepoints must be matched. The infinitely finer hidden coordinates inside an integrated R component do not occur in that test. The supports of the surviving slots must still be checked; an argument for adjoint slots alone is not a license to forget retained raw endpoints.

7.3 What the printed carrier bound actually is The first cluster expansion, (1.97)–(1.99), uses the relative size $d_{j,\mathcal{Y}}(X)$, with $\mathcal{Y} = \bigcup_i Y_i$, and a small coefficient c_1. For activities meeting the retained components the source takes $c_1 = \alpha^{1/3}$. The second cluster expansion in (OT.18) changes this geometry.

The p.391 prescription is explicit:

1. Use ordinary connected components.
2. Omit the first two sums in (1.91)–(1.92).
3. Replace the old first product by $\prod_{Y_i \subset X'} e^3$, arising from the normalized complex kernels.
4. Replace β by $\widehat{\beta} = \beta/4$ and α by $\widehat{\alpha} = Cc_1$.
5. In (1.93), replace the first tree size by $d_j(X' \setminus \mathcal{Y})$, and omit the product over h.
6. Use (1.95) without its first two sums and its first exponential.

Thus the polymer majorant retains the two positive outside reserves

$$\exp\Big[-(1+\widehat{\beta})\kappa\, d_j(X' \setminus \mathcal{Y})\Big], \qquad \exp\left[-\frac{\widehat{\beta}\kappa}{6\cdot 2^d} M^{-d} |X' \setminus \mathcal{Y}|\right], \tag{OT.20}$$

and the small nonempty-polymer factor from $\widehat{\alpha}$. The modified $e^{3N_{\mathcal{Y}}(X')}$ factor occurs in the earlier estimate; the prescribed final majorant has already paid it. Retaining that factor explicitly is a harmless weaker formulation if the corresponding connector reserve is retained as well.

The source then applies the exponentiated cluster expansion and declares (1.98) for (OT.18). It does not print a new equation numbered (1.99) for this second expansion. Its convergence proof yields the analogous outside-carrier bound after a further fixed loss of the positive rates. One can state it without pretending that it is an unchanged full-tree estimate:

$$|\widehat{B}^{(j)}(X; U, J, \eta_{\mathrm{ext}})| \le C_{\mathrm{alt}}\, e^{-c_{\mathrm{alt}} \ell_{\mathcal{Y}}(X)}, \qquad c_{\mathrm{alt}} > 0. \tag{OT.21}$$

Here $\ell_{\mathcal{Y}}$ is a contracted outside-carrier size: every Y_i is a fixed atom, with zero traversal cost inside it; outside M-cube connectivity is charged. It may be chosen smaller than a fixed multiple of the sum of the two sizes in (OT.20). The coefficient C_{alt} is uniform in j, volume and the fixed admitted geometry. It can be made small by the primary common coupling ceiling. The p.391 bookkeeping permits a coefficient $O((Cc_1)^{1/3})$ before fixed norm losses, but no subsequent application needs that particular power.

For clarity, (OT.21) is a weakened consequence of the stated convergent majorant, not an extra printed primary formula. Its weighted form follows by multiplying the connected activities by $e^{s\ell_{\mathcal{Y}}(X)}$ for one small fixed $s > 0$. Contracted tree size is subadditive on an overlapping connected family. A fixed part of the positive reserves in (OT.20) pays this multiplier. The identical absolutely convergent cluster majorant then applies, with the remaining reserves, and gives (OT.21). For a rooted application one must retain the complete outside-volume and counting majorant, rather than infer a uniform sum over all attachments to an atom from (OT.21) alone. A fixed atom may have a polynomial number of boundary attachment sites. The quantitative smallness below pays every fixed such polynomial. No tree length through every interior M-cube of Y_i is restored.

The absence of the Y_i-interior tree factor is intentional. Printed p.391 explains that the domains Y_i are fixed rather than summed over in this expansion. A later spatial current estimate must use this actual norm, not substitute the stronger full-polymer norm from an earlier representation.

Proposition OT.4 — quantitative smallness from the final printed carrier bound Fix the source parameter $\beta_{\mathrm{s}} > 0$. Use a separate fixed analysis value $\beta_{\mathrm{a}} = \beta_{\mathrm{s}}/2$. The coefficient in the final printed nonself estimate (1.68) can be rewritten as

$$|V(Y)| \le \alpha_j^{\mathrm{a}} e^{-(1+2\beta_{\mathrm{a}})\kappa d_{j,Z}(Y)}, \qquad \alpha_j^{\mathrm{a}} \le C_\alpha e^{-c_\alpha R_j}. \tag{OT.24}$$

The source norm with parameter β_{s} is not changed or replaced as an induction hypothesis. Equation (OT.24) is a weaker analysis estimate extracted from it at each step. Thus no rate is repeatedly halved.

Proof and exact source scope. The final (1.68), p.377, already states

$$|V(Y)| \le \alpha e^{-(1+2\beta_{\mathrm{s}})\kappa d_{j,Z}(Y)}$$

on the completed localization domain Y. The assertion that its intersection with $Z^{\sim}$ is a union of whole components is made immediately before that formula. Its nonsmall self case is separately (1.69).

A positive dilation of an eligible component includes its opposite exterior collar cubes. One does not need to identify this dilation with an MR_j-thick layer. Since Z is aligned with the MR_j-grid and $Y \in D_j$ is a union of whole current M-cubes, even a smaller positive dilation makes Y contain adjacent exterior M-cubes on opposite faces. LFR I p.177 gives the eligible component's diameter greater than $20MR_j$ and its rectangular geometry. Distinct original components have a gap at least MR_j, so their first M-collars are free of the other components for $R_j \geq 4$.

The original relative tree (1.67) must connect these opposite exterior cubes. Its total length counts the connecting segment through Z, unlike the later contracted size. Hence

$$d_{j,Z}(Y) \geq c_0 R_j \tag{OT.25}$$

with a fixed $c_0 > 0$. The final source estimate can therefore be split exactly as

$$\begin{aligned} |V(Y)| &\leq \left[\alpha e^{-\beta_s \kappa d_{j,Z}(Y)}\right] e^{-(1+\beta_s)\kappa d_{j,Z}(Y)} \\ &\leq \alpha e^{-c_0 \beta_s \kappa R_j} e^{-(1+2\beta_a)\kappa d_{j,Z}(Y)}. \end{aligned}$$

The primary common ceiling bounds α by a fixed constant. This proves (OT.24).

This proof deliberately uses the final printed (1.68). It does not transfer the earlier p.376 $(1 + 3\beta)$-bound from Y_4 to a larger completed carrier. That route has a genuine carrier-completion gap and is not used.

The self-component expression $V(Y)$ of (1.69) is not assigned the small bound. It stays inside the complete positive operation. Any fixed polynomial factors left by a convention can be absorbed into a further fixed fraction of the extracted exponential. □

The first cluster and the normalized second cluster can be majorized with the one fixed analysis parameter β_a. Their finitely many positive roots of the small coefficient still give

$$\varepsilon_j^R \leq C(1 + \ell_j)^A e^{-cR_j}, \qquad \sup_j (1 + R_j)^s \varepsilon_j^R \longrightarrow 0 \tag{OT.26}$$

for every fixed s, as the common ceiling is lowered. The outside-volume reserve is retained throughout. A per-atom attachment sum is not inferred from a bare contracted pointwise bound.

Repeated use. The primary full-rate norm is retained independently through the actual source induction; the source improvement after T and the prescribed R rate losses are not replaced by this weaker analysis estimate. To retain the more precise tagged norm, the current component's opposite-halo connector must also be controlled in the previous tagged metric. The no-new-large-field waiting depth and/or the new-T guard ensure that old zero-cost tags cannot erase this connector; the detailed argument belongs to the repeated tagged-majorant proof. Promoting the current component itself to a free atom is done only after the nonself estimate. No assertion here bypasses that separate tagged induction.

7.4 Eligible-component geometry and the ordinary current application At its birth scale every Y_i is a whole component of the selected pre-R large-field set. LFR I p.177, condition (i), bounds its containing cube by $100MR_j$. CMP119 p.256 and the actual next-step construction (3.20), p.269, give the common MR_j-cube grid; the small-component rectangularization preserves that grid. LFR II p.386 explicitly treats touching domains as connected. Therefore different selected components do not touch and satisfy

$$\operatorname{diam}_j Y_i \leq C_d 100MR_j, \qquad \operatorname{dist}_j(Y_i, Y_{i'}) \geq c_d MR_j \quad (i \neq i'). \tag{OT.27}$$

The fixed constants allow the Euclidean/supremum norm convention. The corresponding periodic statement is used on a torus.

A contracted tree visiting N different components consequently pays a connector cost at least $cR_j(N-1)$, in M-cube units. An ordinary path through a component costs at most CR_j. Replacing the zero-cost passages along a contracted tree by actual paths therefore gives

$$\operatorname{diam}_j X \leq CM\big(\ell_y(X) + R_j\big), \tag{OT.28}$$

for its connected carrier. Only one component-diameter term remains unpaid. This estimate concerns the diameter and current endpoint tube; it does not pay the volume R_j^d of all interior M-cubes.

In the actual later current step, the growing current-to-old-region guard can pay that one remaining $R_j L^{-(k-j)}$ term. The outside-carrier decay pays the other distances. Together with the actual endpoint map, the fixed geometry and

the usual age reserve, this is the input for the relative-carrier Cauchy sum for the boundary estimate. Completing that sum is separate from the exact factorization proved here.

A marked numerator must still be transported through the normalized old kernels when (OT.17) is used. Positivity gives the usual exact conditional-expectation transport on the positive history space, but it does not give H-03's spatial marked bound. Thus adopting (1.104) is consistent with the frozen source-free observable route, and does not count H-03 as proved.

7.5 Current status of the ordinary old-operation sector The alternative has discharged the pointwise hidden-variable issue for completed eligible R components, by performing the actual integration rather than asserting a false fixed-coefficient identity. Its price is the weaker carrier norm (OT.21).

Two contributions remain to be checked in the full C-05 assembly:

- the quantitative current sum for the new relative-carrier activities, with the actual surviving argument list and the endpoint/current guard;
- the ordinary response of unreplaced operations that remain in (1.104).

The original pointwise route of Sections 3–6 remains a valid alternative if its matching premises are eventually proved. It is not needed for the completed-component sector when (1.104) is adopted.

8. Repeated use: active tags and preparatory failures

The final sentence of LFR II p.392 permits repeated use of (1.104). There is also a relevant explicit primary statement in CMP119 p.262: the boundary induction can be written after resumming each carrier to contain or miss whole large-field components, with $d_j(X)$ replaced by $d_j(X \setminus Z_j)$, and the constructions and proofs remain valid. Thus a relative-carrier induction is a primary option, not an invented interpretation of one scalar identity.

For the present current theorem, however, contracting all of an arbitrarily large unreplaced component would lose too much spatial information. A stronger bookkeeping choice keeps the particular bounded components that have actually been integrated, or are being used as bounded over-carriers. Call these the active tags. The geometric assertions below are proved; preservation of the complete quantitative tagged activity norm must be checked in the assembly and is not inferred from the weaker all-Z_j statement alone.

8.1 All-or-none consumption and uniform mixed-age separation An ordinary next T step has $\Omega_{r+1} \subset \Lambda_r$, so every retained tag stays inside the next pre-R large-field region. A selected current R component is a whole connected component of that region. Thus it contains each older tag completely or is disjoint from it.

When a new eligible component X is processed, old tags inside it may be replaced in the carrier bookkeeping by the single over-carrier X. Its physical diameter is at most $C100MR_rh_r$. Every older tag outside X lies in another current component, separated from X by at least cMR_rh_r. AF comparison gives

$$R_jh_j \le C_RR_rh_r \quad (j \le r), \tag{OT.29}$$

so the older tags have no larger physical diameter than a fixed multiple of this same scale.

Starting with no tags and applying this observation inductively proves that the active tags form an antichain, and that

$$\operatorname{dist}(Y, Y') \ge c_{\text{tag}} \max\{\operatorname{diam} Y, \operatorname{diam} Y'\} \quad (Y \ne Y'). \tag{OT.30}$$

This also holds for tags of different birth ages. The constant may depend on the fixed geometric factors $100, M, L$ and the AF comparison, but not on the age or cutoff.

A tree crossing such separated atoms pays each atom diameter through an adjacent outside connector, apart from one starting atom. The ordinary diameter comparison (OT.28) therefore extends to mixed-age tags, with one remaining largest physical tag radius. This is the geometric input to a tagged ordinary current sum.

8.2 What a failed R preparation actually leaves LFR I p.181, (1.22)–(1.23), pp.182–183, (1.25)–(1.28), and p.188, (1.55), exclude whole originally selected components when a newly introduced large-field alternative occurs. A failed part of an original component is not silently declared a new regular current region.

The density representation nevertheless has changed: preliminary integrations may already have been performed. LFR I p.191 explicitly retains the terms created before stopping as boundary terms. Pages 200–201 state that the operations on the remaining components have several intermediate forms, ranging from the original operation to partly prepared operations. They are not claimed to be the unchanged original T.

Safe carrier bookkeeping is therefore to keep the whole attempted eligible component X as an over-carrier on every failed branch. Its old sub-tags have been absorbed into that same bounded carrier. This preserves (OT.30), never removes an unpaid hole, and is valid even when some inner variables have already been integrated. The final unremoved large-field region in LFR II (1.101)–(1.104) still contains these excluded components.

This geometric statement does not settle their remaining density response. The source's localized selectors, Gaussian forms, prepared backgrounds and any assigned boundary/Jacobian terms still have to be checked in the actual intermediate representation.

Proposition OT.5 — old canonical masses survive every preparatory prefix Suppose the alternative (1.104) has been used at earlier completed steps. Fix a newly selected eligible component at scale r, and put $h = r - N$, as in LFR I p.178. Every older canonical mass in this component belongs to the untouched deep factor of that preparation. It is not partially integrated by a successful or interrupted preparatory prefix.

Proof. The last N steps have no newly created large-field region in the selected component, by eligibility condition (ii). LFR I p.178, (1.2), therefore splits the operation into the last N ordinary one-step operations and $T_h(Z_h)$, with the displayed raw χ_h factor. Canonical masses born at an earlier index $b < h$ are contained in that deep operation. A mass born at $b = h$ is a multiplication factor depending on $V_h|Y_i$, with $Y_i \subset Z_h$.

The actual determining-set formulas (1.15)–(1.17), p.180, integrate the successive shell variables and keep their remaining inner restrictions. Formula (1.11), p.179, reads $Z_s'' = (\Omega_s^{\sim 5})^c \cap Z$ in its stated index range; the index is s, not $s-1$. There is no step integrating a deep $V_h|Z_h$ coordinate into those shells.

The decisive final factorization is printed in (1.76), p.194:

$$\int dV_h|Z_h \, \chi_h(\Omega_h \cap Z_h)\, T_h(Z_h) \quad \text{and a separate integral over the remaining } V''. \tag{OT.31}$$

The source expressly states that these two integration regions are disjoint and that the first integral is left unchanged in the following analysis. This is an exact variable factorization, not an inference from the support of a background-dependent coefficient. In particular the endpoint mass $\tau_h(V_h|Y_i)$ remains in the first factor. The same is true before the full prefix has finished: stopping a sequence of shell integrations cannot have consumed a variable that is explicitly retained throughout (1.15)–(1.17). The failed-branch rule keeps that deep factor.

Eventually the full R component operation integrates it along with all the other old factors. That is precisely the complete T_r' entering (OT.17), not a partial hidden kernel. □

Thus the canonical choice does not resurrect an arbitrary normalized hidden component kernel inside a failed preparation. It leaves fixed raw canonical multipliers, the unchanged deep operation, the actual remaining ordinary/prefix Gaussian factors and fixed selectors, and the boundary activities generated by the completed shell integrations. Their Gaussian/native-domain and selector-relocation responses are still the distinct ordinary-prefix analysis. No independence of those coefficients is inferred from Proposition OT.5.

8.4 The remaining quantitative induction check The tagged norm is preserved by a localization or product graph whenever all its new connecting edges carry the source's physical decay, since contracting a tag only shortens a connecting distance. It is preserved by an overlapping cluster union because contracted tree size is subadditive. A completed normalized old integration can add a zero-cost connection only inside its actual eligible component; that is precisely the new tag just recorded.

A successful removal of an eligible component is different: its interior is no longer a free atom. The source pays this cost explicitly. LFR II p.389, before (1.95), combines the component factor with $2\kappa(100R_j)^d$ and requires the

remaining negative $p_0(g_j)$ reserve. This is the appropriate place to pay a full interior tree/volume if that atom is removed. A failed branch does not have that payment and retains its tag instead.

These observations specify the proof that has to be carried through the actual repeated source expansion. They do not license replacing the original full-tree boundary estimate by an unspecified weaker scalar bound, nor do they prove a marked posterior theorem. The completed C-05 assembly must record the tagged carrier majorant, the successful-removal payment, the prefix/failed-operation response, and the current endpoint sum explicitly.

Primary references

- CMP119 pp.253–254 and 258, especially (2.19)–(2.21): within-component old Gaussian factors and their background dependence.
- LFR II pp.373–375, (1.61)–(1.65): actual localized background dependence and the ordinary input/modified-output domain distinction.
- LFR II pp.378–380, (1.71)–(1.75): completed component integral, the two native-dependent terms, pointwise small exponential deformation and normalized bound.
- LFR II pp.388–392, (1.90)–(1.104): the normalized second expansion, its changed outside-carrier norm, and the explicit fixed-raw-data property of the scalar component masses.
- LFR I pp.177–180 and 194, especially (1.2), (1.11), (1.15)–(1.17), (1.76): eligible components and the exact untouched deep-variable factor during every preparatory prefix.
- CMP119 pp.245, 256, 262, 269: cube conventions, optional relative-carrier induction, common grid and small-component geometry.

Annex AU — Completed R components and the uniform repeated carrier norm

Annex AU. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

Scope of the inference. The endpoint, stopped-prefix and repeated positive carrier estimates on actual all-history data are assumptions in this application. The native estimates cited below apply on their own admitted domains. In particular a derivative of an isolated minimizing map does not establish the derivative of a composed field at fixed older variables.

The scalar carrier argument uses the final majorant of LFR II and the stated birth-index recurrence. Its application to the complete conditional law assumes the relative positive carrier estimates on every repeated history, with the actual support, tag and connector costs.

1. The exact source alternative

LFR II, pp.391–392, applies the normalized component operations to the whole boundary exponential which depends on their variables:

$$\prod_{Y\in\mathcal{Y}_j}(T_j'(Y)1)\left[\prod_{Y\in\mathcal{Y}_j}(T_j'(Y)1)^{-1}T_j'(Y)\,\exp\left(\sum_X B_j(X)\right)\right]. \tag{CR.1}$$

The source constructs a convergent exponentiated expansion of the bracket. Its new scalar activities are denoted here by $\widehat{B}_j(X)$. The remaining multipliers $T_j'(Y)1$ depend only on the raw field V_j restricted to Y; printed p.392 explicitly says that they do not participate in the next renormalization transformation.

This is an exact source-free change of representation. It does not replace $T_j'(F)$ by $T_j'(1)F$. If an observable is present, it must be transported through the same positive normalized operation, with the boundary exponential included in the baseline conditional law. No estimate for such transported marks follows from this algebraic identity. The separate marked-history obligations remain.

The change of representation also changes the convenient carrier estimate. The source retains decay outside the fixed eligible components, with no requirement to cover every M-cube inside one of them by a decay tree. Thus the full primitive estimate (2.42) cannot simply be reused with an unchanged full-tree norm.

The exact printed input to the alternative exponentiation is the modified polymer estimate (1.95): $\widehat{\beta}=\beta/4$, $\widehat{\alpha}=Cc_1$, and the two decays

$$\exp\{-(1+\widehat{\beta})\kappa d_j(X'\setminus\cup Y)\},\qquad \exp\{-c_d\widehat{\beta}\kappa M^{-d}|X'\setminus\cup Y|\}. \tag{CR.2}$$

The normalized-operation factors e^{3N_Y} arise in the modified (1.92); they are not omitted from the subsequent majorant. The paper states that the corresponding modifications give the convergent logarithmic expansion. A precise weighted anchored bound for that expansion must retain the exterior-volume factor and the attachment count. A pointwise contracted-distance bound alone is not a substitute.

2. Actual component geometry

Put $h_j=L^{j-k}$ in the current k-coordinates and $s_j=MR_jh_j$. An eligible component at birth j is a whole component of the pre-R large-field region, not one of its smaller preparatory internal regions.

The following source facts have been checked:

- LFR I p.177 defines eligibility by containment in a cube of side $100MR_j$, and takes a union of whole selected components.

- The pre-R domains produced by the j-th T operation are unions of MR_j-cubes; CMP119 p.256 and the explicit construction (3.20) retain that grid.
- Preparatory exclusions remove whole originally selected components. The internal domains used in the minimizing maps may shrink, but this does not shrink the original component footprint used by the final X_i, Y_i split; LFR II p.378.
- The connectivity convention joins intersecting or touching domains; LFR II p.386. Distinct selected closed grid components consequently have distance at least one grid cell, up to one fixed norm convention.
- The eligible footprints have bounded aspect ratio. Their last N histories are rectangular parallelepipeds, and the ten-layer construction gives every side a lower bound cMR_j, while eligibility gives the upper bound $100MR_j$; LFR I p.177. This lower-side bound is needed when resolving small old tags into current mesh cells.

Therefore, with fixed constants $D, c_{\mathrm{sep}} > 0$,

$$\operatorname{diam} Y \le Ds_j, \qquad \operatorname{dist}(Y, Y') \ge c_{\mathrm{sep}} s_j \quad (Y \ne Y', \text{ same birth } j). \tag{CR.3}$$

The AF calculation (AB17.23b) gives

$$R_j h_j \le B_{\mathrm{AF}} R_k, \qquad B_{\mathrm{AF}} \le 5/4. \tag{CR.4}$$

No factor R_j is replaced by R_k before physical rescaling.

Active tags through further steps Keep a tag for every fixed eligible footprint whose raw multiplier has not yet been integrated by a later complete operation. A later attempted R component is a whole component of the current pre-R large-field region. Hence it contains each older intersected tag in its entirety. It is disjoint from the other tags.

After the nonself estimate for an attempted component has been established, all older tags it contains may be promoted to that whole component as an over-carrier on a retained or failed branch. This is legitimate even on a preparatory failure: the original footprint remains in the large-field region, though some shell integrations or lower-layer geometries have already changed. Promotion enlarges the free set and never strengthens an assumed exterior decay. It does not by itself prove a statement about the changed kernel or selectors. Promoting before the nonself estimate would incorrectly erase a connector needed in Section 7 below.

When a complete successful branch removes such a component, the component's physical size is still bounded by (CR.3). The source's small factor $e^{-p_0(g_j)}$ pays the restoration of every M-cube in it: LFR II p.389 explicitly compares that factor with $\exp\{2\kappa(100R_j)^d\}$ before (1.95)–(1.100). Restoring old free subregions contained in that component costs no more than restoring the whole component.

For two surviving tags of different births, use the grid at the later attempted component. Its separation from the current component containing the older tag is at least cs_{later}. By (CR.4) on those two scales, $s_{\mathrm{older}} \le B_{\mathrm{AF}} s_{\mathrm{later}}$. Thus the active tags form an antichain with

$$\operatorname{dist}(Y, Y') \ge c_* \max(s_Y, s_{Y'}), \qquad \operatorname{diam} Y \le Ds_Y. \tag{CR.5}$$

This is the geometry needed below. It does not collapse an arbitrary, possibly enormous, unreplaced large-field component into a free atom.

3. Diameter control without a full-tree conversion

Let X be a connected carrier which either contains or misses each of its fixed tagged components. Let $\ell(X)$ be the length, in units Mh_j, of a connecting tree after assigning zero traversal cost inside the tags. This is a conservative cost: a source outside-tree length which also charges travel through a tag is no smaller.

Let $s_{\max}$ be the largest physical s_Y among tags contained in X, with $s_{\max} = 0$ if none occurs.

Proposition CR.1 — the diameter estimate For the separated tags (CR.5),

$$\operatorname{diam}_k X \le C\{Mh_j[\ell(X) + 1] + s_{\max}\}. \tag{CR.6}$$

The constant depends only on D, c_* and the fixed lattice conventions.

Proof. Follow a simple path in the contracted connecting tree between two points of X. Consecutive tagged components on this path are separated by at least c_* times the larger of their size parameters. Charge the diameter of every component except the first to the preceding outside connecting segment. A component at an endpoint can be included in the same bound, leaving at most one uncharged diameter. The outside portions have total length at most $Mh_j\ell(X)$. The first component contributes at most $Ds_{\max}$; cube/plaquette endpoints contribute CMh_j. Replace each zero-cost traversal by a path of length at most its component diameter. Taking the supremum over the endpoints proves (CR.6). No traversal tree visiting every cube inside a tag is introduced. □

If the actual source length $d_j(X \setminus \cup Y)$ controls a connected tree visiting all exterior cubes, the sharper bound

$$\operatorname{diam}_k X \le Mh_j d_j(X \setminus \cup Y) + 2Ds_{\max} + CMh_j$$

is available directly. The weaker (CR.6) suffices for the current-position estimate.

The collar is extracted before choosing an interior root All surviving tags are in the old large-field set. The current support is separated from their union by the actual growing collar. A current-to-carrier kernel and the exterior carrier connections can be joined at the first tagged component they reach. No traversal of that component's free interior is needed to reach the old large-field set.

Consequently a fixed share of their combined decay supplies

$$e^{-c_{\mathrm{col}}R_k}. \tag{CR.7}$$

The age connector is handled by the same rule: before the path first reaches an old tagged component, all crossed newer layers are outside that component. The proof of the $n > j$ age payment in BC16 therefore uses the exterior part of the carrier.

Choosing a canonical point deep inside a component first would be incorrect: it would subtract $O(100MR_jh_j)$ from the source guard. The free diameter is paid after (CR.7) has been extracted, with a separate, arbitrarily small final spatial rate.

4. Raw retained variables and the nodal extension

The alternative representation has integrated the fine hidden variables inside the completed component operation. It may still contain the raw V_j restricted to its tagged Y, as a retained argument or in the multiplier. These raw variables are not assigned a new complex ball.

Extend the node list in AD17.3 to include both endpoints of each retained raw bond used by a primitive, as well as the basepoints of its old adjoint fields. At a raw endpoint x of mesh h_b, the surviving support lies outside the subsequent small-field domain. Its actual terminal grade $n(x)$ therefore satisfies

$$b \ge n(x) - c_0 \tag{CR.8}$$

with one fixed star/endpoint allowance c_0. This is the needed one-sided relation: the node grid is no finer than a fixed multiple of the native local mesh. No lower bound on $n(x) - b$ is needed.

Proposition CR.2 — sparse raw endpoints cause no new interpolation loss The AD17.3 nodal construction and its value, first-gradient and curl-current estimates remain valid for this enlarged list, with fixed L^{c_0}-dependent constants.

Proof. Merge repeated nodes; all targets are the same actual fine gauge $u(x)$. At nodes whose native grades differ by at most one, (CR.8) and nesting of the odd grids imply separation at least $L^{-(c_0+2)}$ times the larger native mesh. At larger grade separation, the admitted history buffers give the same fixed separation as in AD17.3. Thus disjoint bumps with radii a fixed fraction of the native mesh can be used. The interpolation is performed in fixed real local regular gauges exactly as in AD17.3; its estimates depend on the native mesh and the target $u(x)$, not on the possibly coarser raw grid.

Raw averaging covariance is the endpoint action

$$V_b(x,x') \longmapsto u(x)V_b(x,x')u(x')^{-1}. \tag{CR.9}$$

Matching $v = u$ at both endpoints therefore cancels the entire raw transformation, while matching the basepoint cancels the adjoint transformation of an old fluctuation. The two real covariance identities give the same fixed-old-variable scalar extension as in AD17.3. The independent-current and physical-current estimates use that proposition's exact current identity. Nodes in the fixed exterior, where the actual converter is the identity, require no bump. □

This proposition concerns the scalar activities after the completed integration. It is not a pointwise covariance theorem for an unintegrated, partially prepared kernel.

5. Which carrier norm the derivative theorem actually needs

The following formulation exposes the precise remaining quantitative input. Let $b_{j,X} \geq 0$ be the absolute cluster majorants of the scalar completed-R activities on their common native contour. They must include the normalized-kernel factors and all sums over the internal polymer expansion. They are not just pointwise values assigned after cancellation.

For some fixed $\lambda > 0$ and integer s, the sufficient anchored input is

$$\sum_{X\,:\,\mathrm{root}=r} b_{j,X}\, e^{\lambda \ell(X)} \leq C B_R (1 + R_j)^s. \tag{CR.10}$$

Here a root is an exterior M-cube or one actual fixed component; the latter is not enumerated as a free polymer. The right side permits its genuine polynomial attachment cost. A constant total-volume factor, an exponential in R_j^4, or an uncounted number of component attachments is not permitted.

The modified source bound (CR.2) is the appropriate starting point for (CR.10). Keeping a slightly smaller positive rate in the original exterior-tree and exterior-volume majorants is essential. An exterior cube can border only a bounded number of the resolved large fixed components, and a whole root component has $O(R_j^d)$ initial attachment sites. Components too small to remove an exterior M-cube do not create additional zero-cost cubes; their geometric slots are fixed rather than independently selected.

These facts remove an interior-domain entropy. They do not, by themselves, prove (CR.10) after logarithmic clustering. In particular many activities can overlap in one fixed component. The nonnegative weighted cluster majorant and quantitative smallness estimate in Section 7 justify this additional passage.

Proposition CR.3 — a polynomial root cost is sufficient for the full current hierarchy Assume (CR.10), the genuine common native contour, and the actual current first-endpoint bound

$$|\partial_i F_{j,X}| \leq \frac{C\theta_k}{P_k} (1 + k - j)^b e^{-\Gamma(k-j)} b_{j,X}\, e^{-cd(i,X)}. \tag{CR.11}$$

Retain in their combined decay the current collar (CR.7), and assume the fixed age rate has the usual $L^{4(k-j)}$ root-count reserve. Then the completed-R boundary family satisfies the same current hierarchy as (AB17.5), with a constant independent of volume and history and fixed factorial power five. Its weighted Hessian contribution is $o(1)$.

Proof. Current Cauchy on the full $2P_k$ contour gives

$$|D_{i_1} \cdots D_{i_m} F_{j,X}| \leq C^m (m-1)! P_k^{-m} \theta_k (1 + k - j)^b e^{-\Gamma(k-j)} b_{j,X} e^{-c \max_\nu d(i_\nu, X)}. \tag{CR.12}$$

This differentiates the complete scalar activity and its nonlinear argument map.

By (CR.6) and (CR.4), the current coordinates at distance at most R from the carrier number at most

$$C\{M h_j [\ell(X) + 1] + CMR_k + R + 1\}^4. \tag{CR.13}$$

A sufficiently small fixed final spatial rate costs at most an exponential in the bracket's first two terms. Spend a fixed share of $\lambda \ell(X)$ on the first and a fixed share of the already extracted collar on MR_k. The rate is chosen independently of the source entropy exponent; no $\exp(+\kappa R_j)$ is introduced.

For the $m - 1$ summed current coordinates, layer-cake counting raises the bracket in (CR.13) to $N = 4(m-1)$. Use the remaining three decays in the single power estimate

$$\sup_{u,v,w \geq 0} (u + v + w + 1)^N e^{-au - av - aw} \leq C_a^{N+1} N!. \tag{CR.14}$$

For example $t^N e^{-at/2} \le N!(2/a)^N$ with $t = u + v + w$; the remaining half-decay is retained for the corresponding sums or integral. Take u proportional to $\ell(X)$, $v = R_k$, and $w = R$. Because $h_j \le 1$, all their coefficients are uniformly bounded. This gives one $N!$, not a product of a carrier factorial and a second collar factorial.

The polynomial root factor in (CR.10) is bounded by a fixed age polynomial times a fixed power of R_k, using AF comparison. A further fixed collar share pays that power. Including it in (CR.14) changes N by a fixed integer and hence changes only C^m.

For each current anchor, the actual root-grid count is at most $CL^{4(k-j)}$ times an anchored exponential sum. The unused age rate sums that factor and every fixed age polynomial. The remaining factorial satisfies

$$(m-1)!\,[4(m-1)+s']! \le C_{s'}^{m+1}(m!)^5$$

for every fixed s'. This proves the claimed hierarchy. At $m = 2$, its prefactor can be bounded by $CB_R\theta_k$ after the collar and age sums, so division by P_k^2 tends uniformly to zero. Mixed native jets use the same smaller genuine native ball as in AB17.6. □

6. Scope of the scalar carrier theorem

The following are now established at their stated scopes:

1. The final source representation removes the hidden completed-R integral while retaining its full boundary exponential and its raw multiplier.
2. Eligible component footprints have the required bounded size and separation. Their active-tag promotion/consumption bookkeeping does not require a new full-tree estimate inside a retained component.
3. The sparse raw endpoint extension of nodal interpolation is valid under the actual support inequality (CR.8).
4. A polynomial anchored cluster norm (CR.10) is sufficient, and its current all-order summation has factorial power five with no extra diameter or volume factor.

Section 7 gives the weighted cluster summation for (CR.10) under its positive primitive attachment, decay and prefix hypotheses. Its use for actual repeated histories requires those uniform input estimates.

The complete application also requires the separately proved partially failed preparation branch estimates. Whole-component promotion controls their carrier geometry. It does not certify their changed density, their hard-selector localization, or their current-coordinate dependence. LFR I pp.200–201 explicitly retains intermediate forms of the operation on excluded components. The supporting note Annex AV supplies their actual matching and the exterior relocations in (1.56)–(1.59); the ordinary coefficient responses are treated in Annex AT and the combined application.

The full boundary derivative estimate additionally requires control of all sectors on the same conditional representation, as stated in Annex AW.

7. Weighted cluster completion

This section uses Proposition OT.4 of Annex AT. Its source input is the final printed (1.68), on the completed domain Y. Charging added halo cubes with the pre-carrier Y_4's p.376 rate is not valid and is not used. The source and analysis rate parameters below are kept separate; neither is repeatedly halved.

7.1 The fresh primitive coefficient is exponentially small in the eligible size After the self terms have been assigned to the positive component operation, the final (1.68) applies to a completed remaining carrier containing a whole positive dilation of an eligible component. Its opposite exterior M-cubes are separated by cMR_j. The relative tree in the final bound must connect them and counts its length through Z. Put $\beta_{\mathrm{a}} = \beta_{\mathrm{s}}/2$, where β_{s} is the fixed source parameter. Then

$$\begin{aligned} |V(Y)| &\le \alpha e^{-(1+2\beta_{\mathrm{s}})\kappa d_{j,Z}(Y)} \\ &\le \left[\alpha e^{-c\beta_{\mathrm{s}}\kappa R_j}\right] e^{-(1+2\beta_{\mathrm{a}})\kappa d_{j,Z}(Y)}. \end{aligned}$$

The primary common ceiling bounds α by a fixed constant. Thus the weaker analysis coefficient obeys

$$\alpha_j \le Ce^{-cR_j}. \tag{CR.15}$$

Self terms are excluded from this estimate. They remain inside T'_j, whose normalized analytic operator norm is bounded by the source (1.75).

The first cluster expansion uses $c_{1,j} = \alpha_j^{1/3}$, and the second replaces its small primitive constant by $Cc_{1,j}$. Every fixed positive root required by the two source resummations can be retained in a single coefficient

$$\varepsilon_j^{\mathrm{R}} \le C(1+\ell_j)^{A} e^{-c_{\mathrm{R}} R_j}, \qquad c_{\mathrm{R}} > 0, \tag{CR.16}$$

with fixed A. This is stronger than a qualitative assertion that a coefficient tends to zero. In particular

$$\sup_j (1+R_j)^s \varepsilon_j^{\mathrm{R}} \longrightarrow 0 \tag{CR.17}$$

for every fixed s, as the common coupling ceiling is decreased.

For repeated tagged input, there are two checks before using (CR.15). First, every active tag was created by a new large-field alternative: either the retained Y_i alternative on LFR II p.378, or an excluded whole component during the preparatory decompositions. A later eligible component has had no such birth during its preceding N steps, by condition (ii) on LFR I p.177. Its contained active tags therefore have ages at least N. For $a \ge N$, the already checked AF estimate gives

$$\frac{s_{\mathrm{old}}}{s_j} \le L^{1-a}(1+\beta_0)(1+a)^{\beta_0}. \tag{CR.18}$$

The source preparation depth $N_0 = N_0(j)$ depends on the coupling: it is defined on LFR I p.179 by $L^{-N_0+1} M R_{j-N_0+1} = M$, and grows with $\log R_j$. The waiting depth $N = N(j)$ is also the actual scale-dependent source choice, with $N > N_0$, not a universal fixed integer. A fixed additional separation in that admitted rule makes the largest old free diameter less than a fixed fraction of the new component's $20MR_j$ span. No constant in this note counts all $N(j)$ levels as a fixed number.

Second, rearranging Proposition CR.1 shows that a tree connecting the two opposite halo points, even with the earlier tags contracted, has charged length at least

$$c\{20MR_j - Ds_{\mathrm{old,max}}\} \ge c' MR_j. \tag{CR.19}$$

The new-T guard gives the same lower bound by stopping at the first old tag. Thus the fixed analysis weight tested on the full-rate source records retains the fresh connector. The attempted current component is made free only after this estimate. The primary full-rate source norm remains the inductive input; the weaker bound extracted here is not substituted back as the next source rate.

7.2 Primitive attachment sums Retain the nonnegative connected-expansion majorant before replacing it by a crude pointwise bound. In particular retain:

- the product of the small primitive coefficients in the modified (1.92);
- the exterior-tree reserve and the exterior-volume reserve in (CR.2);
- the factors e^{3N_Y} from the normalized complex component operations.

The invariant output weight is $e^{\kappa \ell}$, with the same fixed κ as the source induction. The printed $(1+\widehat{\beta})\kappa$ outside-tree rate and the exterior-volume rate leave positive reserves after that weight is inserted. Use those reserves for entropy, the site penalties below, and a separate smaller positive analysis rate $\lambda < \kappa$. The contracted size is no larger than the source tree size before contraction and is subadditive on a connected union. Thus the primitive and both connected sums below are estimated with the full invariant weight, not only with an analysis weight which would be too weak for the next source scale conversion.

For a primitive rooted at an exterior cube, ordinary nearest-neighbor tree encoding gives at most C_d^{n+1} choices for a connected tree of integer length n through its exterior cubes. One rounds the connecting segments to the M-grid and traverses the rounded tree twice; choosing a subset of the visited cubes changes only C_d. Interior cubes of a fixed component are not enumerated.

For a primitive rooted in one fixed component, the first exterior attachment has at most $C(1+R_j)^d$ possibilities. A subsequent distinct component requires an outside connector of length at least cR_j, or the corresponding graded size

in a mixed-age family. A reserved connector exponential pays its polynomial attachment choices. Equivalently, the exterior cube at an entrance meets only a bounded number of resolved tagged components, by (CR.5). Very small tags that remove no whole M-cube introduce no additional exterior-domain choices; their slots are fixed.

These bounds apply to the primitive source tree and its retained product coefficients. They are not a count based only on the contracted length of a final cluster. Multiple branches meeting the same component retain their separate small primitive factors.

Resolve incidence supports at the current birth mesh before counting them. A tag which removes whole current M-cubes is retained as an atom. A finer tag which removes no such cube is assigned to its occupied current M-cells, and overlap inside it is enlarged to ordinary cell overlap. By the actual lower-side and upper-side bounds in Section 2, failure to contain a full current cell forces all its side lengths to be at most a fixed multiple of the current mesh. It therefore meets only boundedly many cells. Diameter and separation alone would not justify this assertion for an arbitrarily thin long region. Many tiny tags in one cell create only that cell's incompatibility, not one new incidence penalty each. The normalized e^{3N_Y} factors apply only to the newly integrated eligible components, whose diameters exceed $20MR_j$; all of them are resolved. Earlier tiny tags already belong to scalar activities and introduce no new e^3 factor at this integration.

Consequently the elementary interaction majorants $\mathfrak{a}_j(Y)$, including their Mayer factor and normalized-operator factors, satisfy

$$\begin{aligned} \sup_{c \text{ exterior}} \sum_{Y \ni c} \mathfrak{a}_j(Y) e^{\kappa \ell(Y) + 4a_0 n(Y)} &\le C\varepsilon_j^{\mathrm{R}}, \\ \sup_{Q \text{ tagged}} \sum_{Y \text{ uses } Q} \mathfrak{a}_j(Y) e^{\kappa \ell(Y) + 4a_0 n(Y)} &\le C\varepsilon_j^{\mathrm{R}} (1 + R_j)^s. \end{aligned} \tag{CR.20}$$

Here $n(Y)$ is the number of exterior cells plus resolved tagged atoms in that incidence support, $a_0 > 0$ is fixed and small, and s is one fixed geometric exponent. It does not count every hidden fine tag. Larger fixed powers from a root convention may be included in s. The exterior-volume term pays $4a_0$ per exterior cell. The connector reserve pays the atom penalties and the e^3 factors for additional atoms; the first atom is included in the constant. No factor $e^{CR_j^d}$ is discarded.

By (CR.17), the common ceiling can make the right sides at most a fixed δ, with $\delta < a_0/16$, uniformly over the actual component family.

7.3 Two connected sums and the rooted logarithmic bound For completeness, the elementary rooted-tree argument behind the use of (CR.20) is given here. Regard an exterior cell and a fixed tagged component as incidence sites. Two objects are incompatible when their incidence supports overlap. If nonnegative activities $w(P)$ satisfy

$$\sum_{Q \sim P} w(Q) e^{a_0 n(Q)} \le a_0 n(P), \tag{CR.21}$$

the rooted-tree iteration

$$T_0(P) = w(P), \qquad T_{r+1}(P) = w(P) \exp\left(\sum_{Q \sim P} T_r(Q) \right)$$

obeys $T_r(P) \le w(P) e^{a_0 n(P)}$, by induction. The exponential accounts for unordered offspring with their $1/q!$ factors. Every rooted connected family has a spanning rooted tree, so its absolute sum is bounded by this iteration. In the logarithmic hard-core polymer expansion, the alternating connected graph coefficient has the usual tree bound: order the vertices, assign each connected graph its ordered exploration tree, and group the graphs with the same tree; optional extra edges give factors $1 - 1$ or 1. Thus its absolute coefficient is at most the number of compatible spanning trees. This is precisely the same majorant.

Summing (CR.20) over incidence sites of P proves (CR.21), with a strict reserve. Apply the iteration first to the connected Mayer families which form the polymer activities in the modified (1.91). It uses at most one of the allocated $a_0 n$ shares and gives a site-rooted polymer norm at most 2δ. Apply it a second time to the logarithmic polymer expansion (1.98). The remaining shares give its site-rooted absolute norm at most 4δ. The contracted weight is preserved at both stages by subadditivity:

$$\ell\left(\bigcup_\nu X_\nu \right) \le \sum_\nu \ell(X_\nu)$$

for an overlapping connected family, where overlap inside a tagged atom is allowed.

This proves the desired anchored norm (CR.10), and in fact the stronger bound with λ replaced by the full source rate κ. Its right side can retain $C\varepsilon_j^{\mathrm{R}}(1+R_j)^s$. That stronger output is retained for the next source induction; the fixed smaller λ-weight is used only in the separate derivative/counting argument. Arbitrarily many branches at one large component are summed by the rooted iteration. They are not estimated by claiming that a bare final pointwise bound has no attachment entropy.

7.4 Preservation through the next source operations The norm just proved is a norm of the absolute convergent majorant, not a cancellation-dependent final coefficient. It is preserved by the actual carrier operations as follows.

An ordinary T step leaves tagged raw multipliers fixed and acts outside their large-field regions. Its Gaussian/localization paths retain their usual decay. Multiplication or connected joining uses subadditivity of the contracted tree size; every new localization path pays its own charged outside length. Rescaling uses the physical tag sizes and the original layer kernel measures. No fine bond count is substituted for a normalized source row sum.

At a completed R step, new retained eligible components absorb every older tag they contain. The union of the free regions can only enlarge on that branch, which weakens the exterior weight. The source's normalized-kernel factors are exactly the factors retained in (CR.20). On a successful removed component, old holes disappear; the lost interior cell coverage costs at most the whole component volume and is paid by the existing $e^{-p_0(g_j)}$ reserve, as explained in Section 2. This does not require restoring a large full-tree weight using the much smaller current collar.

On a preparatory failure, promote the whole attempted component after the available nonself estimate. Already generated boundary terms keep their genuine source majorants; assigning the over-carrier only weakens their geometric weight. The original tag masses inside the deep untouched factor remain positive raw multipliers. LFR I (1.76), p.194, explicitly keeps the V_h, T_h factor separate from the preparatory shell integrals. The source eligibility condition places the earlier tag births in that deep factor. Their geometry is consequently not partly erased by an interrupted shell integration.

These carrier rules are compatible with the primary alternative induction on CMP119 p.262 and LFR II p.392, but retain more information: only actual bounded active tags are free. They do not replace every arbitrary unreplaced large-field component by one atom.

7.5 A uniform birth-index invariant Subadditivity alone would not exclude a factor C^{k-b} after many steps. The invariant uses the source's actual birth labels and its unreduced absolute localization records:

1. Ordinary full-tree B families remain a separate summand, with their source bounds. Only fresh R-connected scalar families enter the normalized cluster bound above.
2. For a family born at b, keep its scalar function, its original birth-scale anchor and carrier dependency set, its fixed assigned absolute majorant, and the slots it actually uses. CMP119 (2.40)–(2.41) writes precisely such a family $B^{(b)}(U_k, \mathrm{A}, \dots)$. An untouched later T step changes its admitted arguments, not that scalar function or its assigned majorant. The embedding uses the original global source map, not a product of one-step norm constants. A promotion to a larger active tag does not regroup these records under the promoted component's anchor.
3. Use the literal add-back convention of the source. An inherited first term reconstructed in LFR II pp.373–375 keeps its old record only when that record's actual slots and boundary anchor survive; the source's domain inclusion gives an admitted image for its arguments. Ordinary full-tree reference terms on the new history stay in the ordinary source family. Differences created by the source identities and the new normalized-R output receive the current birth label. A record whose old variables or supporting tag are actually consumed is retained in that complete integral until the exact output (CR.1) is formed. If its output still uses another surviving tag, including an older tag outside the attempted component, it is a fresh tagged record anchored at such a survivor; its old holes are not restored without payment. The primary classification (1.99)–(1.100), p.390, gives a full-tree e^{-p_0} bound only after all free holes in that output are absent or paid by the actually removed components. It is then a new ordinary B term if it still meets the surviving ordinary large-field set, and a regular R term if it misses the whole new large-field set. "Surviving tag" here includes every active earlier eligible footprint, not only the newly retained Y_i at this R step. The final paragraph on p.392 isolates the earlier-large-field dependence in boundary activities and their raw multipliers; iterating it retains the older surviving tags in that classification. Thus a weak-tag record with a deleted anchor is not carried under the old birth label. No old-B self term is

moved into T'_j merely to simplify a label, and no external factor is silently multiplied into a mass whose raw locality is used. The primary masses in (1.104) are unchanged.

4. A genuinely fresh scale-k output uses a current k-grid incidence root: a resolved active atom, or a current cell meeting an unresolved older tag. It does not inherit that tag's finer birth-grid multiplicity. Tiny tags meeting the same current cell share its incidence convention, as in Section 7.2. This rule is distinct from the original-anchor rule for untouched old records.
5. At every fresh scale, retain the full source rate κ, β_s before extracting the fixed analysis reserve. CMP119 p.262 gives the new T-created B terms the $1+4\beta_s$ improvement. LFR II p.376 supplies the $1+3\beta_s$ scale conversion for older records. This conversion also holds for their retained tagged records: with the original holes physically fixed, the length of the same charged physical path in Mh_b units is L^{k-b} times its length in Mh_k units. Infimizing over the original paths can only increase the old-to-new bound; rounding to current cells has fixed allowances paid by the reserved tree/cell factors. Newly added or promoted holes only weaken the output cost. Thus the old $(1-\beta_s)\kappa$ path share gives the required current $(1+3\beta_s)\kappa$ share, with $L \geq 3$ and the fixed small source β_s, while $\beta_s\kappa$ remains for the old-root sums. This does not identify a contracted old size with the literal uncontracted definition (1.67). The fixed weaker analysis weight is tested against the retained original records; it is never fed back as a smaller source rate.

Here is the numerical invariant. Let $A_{b,k}$ be the anchored sum of the frozen birth majorants, including their fixed root polynomial, for the still-carried records born at b. Its anchors, attachment convention and carrier dependency sets are the original birth-b ones. It is not the norm obtained by aggregating all those records at a subsequently promoted tag. The assigned majorants remain valid because each actual later argument is admitted into the same original analytic domain; they are not replaced by suprema over different evolving subsets.

For a prefix eligible at step k, the contained completed-tag births obey $b \leq h = k - N(k)$. The source keeps the deep T_h, V_h factor and the histories at levels at most h unchanged. The old record's slots and native gauge convention therefore remain available. If a later complete integration consumes those slots, its record belongs to the exact fresh integral/output and is no longer treated as an untouched old record. The scalar first terms which the primary construction explicitly retains use its p.375 domain inclusion.

Removing consumed records cannot increase the nonnegative sum in this fixed birth-anchor convention. Hence

$$A_{b,k} \leq A_{b,k-1} \quad (b < k) \tag{CR.22}$$

for the carried subfamily. This assertion concerns unchanged records, not a re-expansion of their whole function at each step.

The source differences and pre-exponentiation resummations are linear in the incoming action families. The required age gain in their values is not obtained from a future-current derivative estimate. It comes from the explicit value localization factors in (1.66), as follows.

The value connector for an incoming birth-b record Keep the original birth carrier X, its original anchor in Z_b, and its original tree/localization record. A later promoted component is not substituted for that original record's holes when estimating the old residual factor. Such a substitution could erase the very b-to-k collars being counted. Promotion is used only for the incidence convention of the future fresh cluster.

There are three cases in the source reconstruction.

- An unchanged first term is carried under (CR.22); it is not part of the fresh value forcing.
- An exceptional term with empty new localization domains which stays inside one component is a source self term. It remains in the literal primary component operation and is not assigned the nonself estimate.
- Every fresh nonself contribution retains a connector from its original carrier to the active boundary. If X already reaches outside the relevant component, its old tree contains the crossing. If X stays inside it, a nonempty localization is necessary: the first nonempty domain Y'_{i_0} in (1.66) meets the source localization collar and contains its mandated Λ-component, by pp.373–374. Retain both the factor involving the scaled distance from X to that collar and the connecting part of Y'_{i_0}. If the old anchor is outside the attempted component but its carrier meets it, the carrier crosses the corresponding distinct-component gap. Again the old tree supplies that crossing.

In the nonempty case it is not sufficient to call an arbitrary collar point a level-k regular point. The whole source connector, including its Y'_{i_0} path to the mandated current-scale region, is retained. Source (1.66), and the residual

exponential displayed immediately below (1.67), retain fixed positive shares of

$$\beta_{\mathrm{s}}\kappa\, d_b(X), \qquad \delta\, d_\Omega(X,\Sigma_{i_0}), \qquad \sum_{i=2}^{4}(\kappa_1-1)M^{-4}|Y_i'|. \tag{CR.22a}$$

Here $\Sigma_{i_0} = (Y_{i_0}^0)^c \cap Y_{i_0} \cap Y_{i_0}'$, in the source notation. The volume factor pays a connecting path through the corresponding M-cubes. When there is no localization domain, the old tree supplies the full connector instead. These are factors of an absolute value majorant.

For the contained-anchor case, stop the connector at its first exit from the whole selected current component. The source Y_1 contains every current Z-component it meets, and the later Y_i enlarge it. Therefore a retained path from X to the first nonempty localization collar reaches this exterior boundary, up to the fixed $2M_1$ allowance, even if its ultimate collar endpoint lies in another proper-scale region. If X already exits, the old tree provides this portion. This is the boundary crossing used in the age estimate.

Split this connector where it first enters each actual determining layer. If the point at which it leaves the old carrier has grade n, its old tree part crosses at least $(n-b-c_0)_+$ collars, while the retained localization part crosses the remaining $(k-n-c_0)_+$ collars. The metric here is the actual metric of the corresponding original or prepared map. LFR I p.179 gives one M-cell layer in the recent prepared history; it does not give MR_r there. The uniform physical lower width valid for both histories is therefore cMh_r. Measured in the old carrier unit Mh_b, its old-tree cost is at least c for $r \geq b$; measured in the source scaled metric, its kernel part has cost at least cM. The retained localization walk/volume part pays its corresponding current-scale connector with its own fixed $\kappa_1 - 1$ reserve. Original unprepared portions have larger guards, but that improvement is not assigned to prepared portions. For $n \leq b$, the localization part alone pays the age. The fixed c_0 includes the source star, endpoint and collar conventions.

The same path argument applies with the original record's free components: by (CR.5), every noninitial free traversal is charged to an adjacent outside segment with one fixed factor. Start at the first exit from the old anchor set, so no artificial canonical interior-root diameter is subtracted. The new attempted component is not yet contracted. For nested older tags, eligibility and (CR.18) give the same separation, as in (CR.19). Thus, with fixed $c_*, C_* > 0$, a reserved part $\mathcal{L}_{b,k}$ of (CR.22a) obeys

$$\mathcal{L}_{b,k} \geq c_* \min\{\beta_{\mathrm{s}}\kappa, \delta M, \kappa_1 - 1\}(k-b-c_0)_+, \qquad \mathcal{L}_{b,k} \geq c_*\rho_k(r,X) - C_*, \tag{CR.22b}$$

where r is the actual current root/attachment being summed, and the second estimate is used with another fixed share. These inequalities are the source layer-crossing argument of BC16.29–30 applied to a value-localization path. No observable mark or derivative occurs in their proof.

After retaining the full invariant output weight and the primitive entropy shares, choose the admitted source block and localization parameters M, κ, κ_1, in their original order, so that the remaining value-age rate Γ_{v} satisfies

$$\Gamma_{\mathrm{v}} > 4\log L + \lambda + \lambda_{\mathrm{pol}} + 2. \tag{CR.22c}$$

Here $\lambda > 0$ is the chosen age weight, and $\lambda_{\mathrm{pol}} > 0$ is any fixed rate paying the finite AF/prefix polynomials. CMP119 p.260 immediately after (2.31) states the freedom to take the fixed decay parameters arbitrarily large, with the other parameters chosen accordingly. CMP116 (1.7), (1.11) supplies the corresponding block-size/decoupling reserve: choose κ, κ_1 first and then M so that they occupy only the chosen fixed fraction of the available $\delta_1 M$. Thus the minimum in (CR.22b) can exceed the fixed threshold in (CR.22c). This choice is made once, before the coupling ceiling or any derivative order. It uses positive fixed shares of (CR.22a), not a larger derivative rate or a fictitious running prepared collar. The source normalized kernel row sums control the attachment integration. The original root density is $L^{4(k-b)}$. Consequently the contribution of the incoming anchored birth-b family to a fixed new root is bounded by

$$C(1+k-b)^B L^{4(k-b)} e^{-\Gamma_{\mathrm{v}}(k-b)} A_{b,k-1} \leq C e^{-\lambda(k-1-b)} A_{b,k-1}. \tag{CR.22d}$$

Summing this inequality over $b < k$ is the value estimate which supplies the age-weighted input. The inherited ordinary source family has its separate fixed norm; it is not obtained by assigning a new small coefficient to every unchanged old scalar.

Let that ordinary E/R/full-tree B forcing constant be $F_0 \geq 1$. The pre-exponentiation bounds are then at most a fixed multiple of $F_0 + \mathcal{A}_{k-1}$. The source's positive-root weakenings, such as $c_1 = \alpha^{1/3}$, do not preserve literal linearity. This causes no loss: on a fixed bootstrap interval for $\mathcal{A}$, every fixed positive root $0 < r \leq 1$ satisfies $(F_0 + \mathcal{A})^r \leq F_0 + \mathcal{A}$. The corresponding root of the exponentially small coefficient is already included in (CR.16). Once the resulting elementary norm is below the fixed rooted-tree threshold, the two iterations in Section 7.3 change it by at most a fixed multiplier.

Chronological prefix forcing and the recurrence The partially failed-prefix estimates are applied before the current completed-R output is constructed. Eligibility puts its resident tag masses at births at most $h = k - N(k)$, in the untouched deep factor (1.76). Scalar families born outside the attempted component at an intermediate scale can have tails reaching it; those remain earlier-birth inputs $b < k$, with their full carrier connectors. They are not assigned the stronger resident-tag restriction. Both classes are controlled by the induction hypothesis. The fresh prefix, failed-action and failed-Jacobian contributions have the bound proved in PF.22,

$$\varepsilon_k^{\mathrm{PF}} \le C g_k^{-2} (1+\ell_k)^{\mathcal{A}_{\mathrm{PF}}} e^{-c_{\mathrm{PF}} R_k}. \tag{CR.22e}$$

They multiply at most $F_0 + \mathcal{A}_{k-1}$. The coefficient may be weakened once to cover the finitely many positive roots in the ensuing connected sums; it still has the form (CR.22e), with fixed changed constants. It never uses the same-birth completed output as an input. Failure promotions change geometry only after these estimates.

Put

$$\zeta_k = \varepsilon_k^{\mathrm{R}} (1+R_k)^s + \varepsilon_k^{\mathrm{PF}}. \tag{CR.22f}$$

The admissible $R_k \ge \ell_k^r$, $r > 1$, makes $\sup_k \zeta_k \to 0$ when the common ceiling decreases. The fresh tagged and prefix contributions, followed by the successful normalized scalar output, therefore satisfy

$$A_{k,k} \le C_{\mathrm{fresh}}\, \zeta_k (F_0 + \mathcal{A}_{k-1}). \tag{CR.23}$$

The constant C_{fresh} depends on the fixed source parameters and the chosen norm reserves. It does not depend on how many times an untouched record has previously been evaluated. The source's full-scale resummation in (1.49)–(1.68) supplies the linear incoming-action sum; the resolved atom/cell count in (CR.20) and the two rooted sums supply the additional tagged norm. Removed-component coverage is paid by the already present $e^{-p_0(g_k)}$ term, so it adds no multiplicative factor to a carried family.

Take one fixed $\lambda > 0$ from the available age reserve, after the $L^{4(k-b)}$ root count and fixed age polynomials have been paid, and define

$$\mathcal{A}_k = \sum_{b \le k} e^{-\lambda(k-b)} A_{b,k}.$$

Use the total ζ_k in (CR.22f). Equations (CR.22)–(CR.23) give

$$\mathcal{A}_k \le \big(e^{-\lambda} + C_{\mathrm{fresh}} \zeta_k\big) \mathcal{A}_{k-1} + C_{\mathrm{fresh}} F_0 \zeta_k. \tag{CR.24}$$

Choose the common ceiling so that

$$C_{\mathrm{fresh}} \sup_k \zeta_k \le \tfrac{1}{2}(1 - e^{-\lambda}), \qquad \rho := \tfrac{1}{2}(1 + e^{-\lambda}) < 1.$$

Then

$$\mathcal{A}_k \le \rho^k \mathcal{A}_0 + \frac{C_{\mathrm{fresh}} F_0 \sup_b \zeta_b}{1-\rho}. \tag{CR.25}$$

Starting with no completed-R tagged families gives $\mathcal{A}_0 = 0$; choose the ceiling once more to place this bound inside the fixed elementary threshold used in (CR.23). This closes the chronological induction, including the prefix forcing. The forcing tends to zero by (CR.17) and (CR.22e). There is no uncontrolled multiplication by C_{fresh} at each age step, no repeated loss of the source rate, and no dependency on the same-birth completed output.

Old tags outside a new R component introduce no fresh integration factor: their scalar records are carried under (CR.22). Old tags inside it are included once in that component's complete positive operation and the fresh localization under (CR.23). On a failed prefix their original birth records remain in the explicitly untouched deep factor or in its already generated ordinary boundary terms; they are not counted again as newly normalized kernels. Promotion changes the incidence convention for a future fresh cluster; it does not enlarge the old scalar dependency sets or the old nodal gauge list. In the final current application the original root density $L^{4(k-b)}$ is paid explicitly by the age reserve. If one instead collected every old record at a promoted root, monotonicity in (CR.22) would be false; that convention is not used.

Under the repeated-component and prefix majorants, the anchored norm (CR.10) and the derivative estimate of Proposition CR.3 apply to the scalar completed-R family with its actual tags. The complete conditional boundary estimate also requires the ordinary and unreplaced sectors.

Annex AV — Every stopped preparatory prefix and its relocation response

Annex AV. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

Scope of the inference. The endpoint, stopped-prefix and repeated positive carrier estimates on actual all-history data are assumptions in this application. The native estimates cited below apply on their own admitted domains. In particular a derivative of an isolated minimizing map does not establish the derivative of a composed field at fixed older variables.

The stopped preparatory operations retain the actual intermediate variables of LFR I. Their contribution to the complete boundary law requires the relative-carrier norm and the uniform stopped-prefix response bounds.

1. Indices and the exact prefix geometry

Write b for the scale called k in LFR I, $h = b - N$, $t = h + n$, and $b_0 = b - N_0$. The future current scale is $k \geq b$. All meshes in a formula are physical meshes in the same underlying lattice. At birth scale b, put $h_b = 1$; then $h_t = L^{t-b}$. Returning to the future scale multiplies every physical length by L^{b-k}.

Let Z be one actually selected component, with its original eligibility conditions on LFR I p.177. It is contained in a cube of side $100MR_b$. Do not replace Z by a smaller component after a failed test.

The prefixes in (1.15)–(1.16) are the determining histories associated with

$$\widehat{\Omega}_r^{[n]} = \begin{cases} \Omega_r'', & r \leq t, \\ \Omega_r, & r \geq t+1. \end{cases} \tag{PF.1}$$

Outside Z the original history is unchanged. Inside Z, the t-shell is $(\Omega_t'' \setminus \Omega_{t+1})^{(t)}$; the lower shells are the displayed primed shells in (1.16), and the higher shells are the original ones. This is a componentwise reading of the actual formula, with the original site/bond-star convention, not an intersection ansatz for arbitrary patches. The endpoint convention for crossing bonds is unchanged.

The localized prefix uses precisely (1.19)–(1.20): the localization join with $B_b(Z)$, original interior determining data, and the boundary data obtained from $Q_b^s V_b$. Consequently the disagreement between the localized and global determining prescriptions occurs in the source's $2M_1$ boundary collar of Z. The localization join, its actual data, and the older raw variables are retained.

PF.1 — a fixed native core for every prefix A fixed smaller ordinary strong native chart for (PF.1) is contained in the modified source chart (1.64)–(1.69), uniformly in n, b, N, once all its actual real centers are admitted. This statement includes the independent-current and reconstructed p-current clauses.

Verification. Every displayed multiplier L_0^{2a} in (1.64)–(1.69) has $a \geq 0$. Every dyadic coefficient is bounded below by $1 - 3\beta \geq 1/4$, for the common choice $\beta \leq 1/4$. Clause (i) covers the actual lower prefix layers and the outside part of its t-layer. Clause (ii) covers $\Omega_{b_0+1} \setminus \Omega_{t+1}$, the rest of that same layer. Although this set is written as a union of old annuli, its base field, current, relative-field and gradient weights are all the t-weights, as printed in the formulas. Its L_0 factors only enlarge these radii. Clause (iii) is the unchanged higher-layer condition. The same comparison applies to the p-lattice weights and the local real-gauge clauses. For $t \leq b_0$, clause (ii) is empty and the L_0 factors in (i) are absent. Taking a fixed fraction smaller than $1/4$ proves the chart inclusion.

The actual real center can contain the displayed $L_0^{2N_0}\varepsilon$ enhancement. LFR I p.192 gives

$$L_0^{2N_0} \leq C N_0^{\beta_0} R_b, \qquad N_0 = O(\log R_b). \tag{PF.2}$$

Thus one may choose the fixed native logarithmic exponents in CMP119 (2.28) with

$$q_0, q_1 > p_0 + r + 2, \qquad R_b \asymp \ell_b^r, \quad \ell_b = \log g_b^{-2}. \tag{PF.3}$$

This absorbs (PF.2), including its fixed logarithmic factors, into a vanishing fraction of the native radii. These exponents are chosen once, not as functions of derivative order or prefix length. The usual AF comparison includes the fixed adjacent-scale and prefix-age factors. This is a stronger explicit parameter choice than $q > p_0$; it must be included in the combined parameter ledger.

The source itself states the needed successive insertion property on p.191: the insertion from the $(n+1)$-chart lies in the n-chart, including its current, with the proof obtained from (1.44)–(1.52). This is an all-prefix statement. The final $n = N$ theorem LFR II (1.65) is not being used as an unsupported theorem about arbitrary interrupted prefixes. □

2. The relocation is not a generic difference of two averages

The failed branch in question is $1 - \chi'^{(t)}$ in LFR I (1.55). At that point the newly linearized field obeys the earlier bound (1.54), so its being "large" relative to δ_t' does not make it unrestricted: its original small-field coordinate is still bounded by a fixed multiple of δ_t'.

Let $\Sigma_{b,t}$ be the determining-data collar in the actual representation (1.56). Suppressing the fixed mesh inclusion and the source's specified relative gauge, that representation has the form

$$U_b^{(n+1)} = \left(e^{i\eta \mathcal{H}_{b,Z}^{(n+1)}(d_\Sigma)} U_{b,Z}^{(n+1)}\right)^{u^{-1}}, \qquad d_\Sigma = 1_\Sigma \frac{1}{i} \log\left[M^\bullet(U_b^{(n+1)}) M^\bullet(Q_b^s V_b)^{-1}\right]. \tag{PF.4}$$

The indicator in this notation records the exact data identification in (1.19)–(1.20): the actual difference is zero off the collar. It is not an extra cutoff applied to the source map. The source states its width and its real bound directly after (1.56).

For the native extension use this same localized-data formula and the source's selected formal response (179)–(180) of CMP102. The localized background is obtained from its actual mixed raw data; the data difference is still only the collar datum in (PF.4). The independent current enters the analytic response coefficients in the source extension, with its admitted auxiliary-current chart. It is not inserted as a new forcing datum outside Σ. In particular

$$\mathcal{H}(0; U, J) = 0 \tag{PF.5}$$

holds for the selected formal family at admitted off-minimum external backgrounds as well as at physical minima. This is the zero-at-zero identity of (179)–(180); a source-free forcing term from a different external-background equation would invalidate this argument.

At real physical arguments (PF.4) is the actual source identity. The raw/corrected/gauge family is the adopted family in the joined-identification note. The extension used here is the supported formal extension of that physical identity. On an arbitrary off-minimum field it is not asserted to equal the literal difference of its coarse average and the average of its localized minimizer. That literal difference need not be exponentially small.

The equality required for the current theorem is stronger than equality at one point but narrower than an off-minimum identity: for fixed retained physical data, (PF.4)–(1.58) agree with the actual relocation for every real current value in the common current contour. Both current families are holomorphic there, so the equality continues to complex current values and all actual current jets. The native and independent-current variables specify the chosen auxiliary extension. Their mixed jets are the jets of that extension. A later pullback which varies retained physical data must use its actual chain identity, as required in AB17.6; equality on a fixed physical graph is not a substitute for that identity.

Here the localized auxiliary background requires only absolute source regularity. Its admission can be verified without applying the zero-data theorem to hybrid data. Fix its actual real localized prefix minimum as reference. On the native U-ball, each varying mixed raw datum is an average of U; the remaining boundary datum is fixed. If its determining block has mesh h_s and lies in native layer n, the actual join has $h_s \le h_n$, and the normalized raw averaging path bound gives a dimensionless increment at most $Ch_s\alpha_n/h_n \le C\alpha_*$. The actual relative-gauge correction has the same absolute bound. CMP102 p.305, (172), is the unweighted determining-bond supremum condition $|V' - 1| < C_1\varepsilon_1$, equivalently

$|\log V'| < 2C_1\varepsilon_1$ on the whole determining set. There is no additional mesh weight on this input norm. Taking $C\alpha_* < cC_1\varepsilon_1$, with fixed $c < 1$, puts every mixed raw increment on one common smaller absolute data chart. Proposition 9, p.309, gives the output conditions (19)–(21) with the fixed constant $B_5\varepsilon_1$, as well as the endpoint row (190). Thus it admits the actual localized minimizing branch and its absolute regularity uniformly in the number of determining layers. Use the smooth relative Landau representative when invoking the internal kernels. The normalized inverse gauge RM.26–RM.36 recovers the actual raw-data output and is uniformly bounded on the corresponding complex chart. This argument does not claim the older physical-layer current weights for this auxiliary base; those stronger weights are not needed for the coefficient bounds of $\mathcal{H}$. The global action's running weights continue to come from the actual prefix/native chart, not from this auxiliary absolute bound.

The independent current of the response coefficients is extended on a fixed smaller admitted auxiliary-current chart. Its physical value is the localized base current; the independent deviation is introduced by the same triangular physical-current-plus-deviation convention used in BC16. The original strong current increment is also absolutely admissible on the joined mesh because $h_s^3\alpha_n h_n^{-3} \le \alpha_*$. This preserves the physical graph and does not add an unlocalized response datum. The real and complex gauge conversions above are used only for these internal coefficient norms and the actual coarse displacement. No fine axial potential is substituted for the smooth native field.

The native size of the collar datum is bounded by $C\alpha_b$, with the fixed prefix/profile factors included if the enlarged modified chart is used. Indeed its real center has the bound following (1.56); the native increment is controlled by the actual raw averaging derivative on the boundary blocks. Those boundary blocks have the birth-scale physical footprint, and their normalized path lengths are bounded. This is the same raw-data/relative-gauge comparison used in JI.32–JI.40. It gives a datum bound, not a claim that raw M' equals the corrected source Q.

PF.2 — exponential failed-relocation estimate on the native chart Retain the full source separation

$$D_{b,t} := 10M \sum_{s=t+1}^{b-1} R_s + MR_b. \tag{PF.6}$$

On the actual failed-shift bonds, a fixed smaller joint native chart gives

$$|H_Z^{(t)}| \le C\alpha_b \mathcal{P}_{b,t} e^{-\delta D_{b,t}}, \tag{PF.7}$$

where $\mathcal{P}_{b,t}$ is a fixed polynomial in the logarithmic parameters and age. On the ordinary prefix core it is needed only for fixed AF and coordinate comparisons. The rate $\delta > 0$ and the polynomial degree are fixed before the derivative order.

Derivation. On the fixed absolute data chart write the exact segment integral

$$\mathcal{H}(d_\Sigma; U, J) = \int_0^1 D_d \mathcal{H}(sd_\Sigma; U, J)[d_\Sigma]\, ds. \tag{PF.7a}$$

This uses (PF.5). Apply the actual normalized endpoint row (190) of CMP102 Proposition 9, including its chosen complex-background extension, under this integral. Its only data endpoints lie in Σ. The geometric evaluation of the resulting collar-to-output distance is exactly the one in (1.57), namely (PF.6). Complex native perturbations and independent-current coefficients use the remaining fixed absolute radii and the same row reserve; they do not change the support of the data endpoint. Passing from the smooth response to the actual coarse displacement uses (1.58), the analogue of (1.41), and the normalized inverse-gauge bound RM.36. Each gauge increment carries the corresponding block length, so it obeys the same dimensionless bound. This is the complex counterpart of the source's derivation (1.37), (1.57)–(1.58), with the real ε replaced by its admitted native radius. No estimate of a difference of two unrelated reconstructed fields is used. □

The function in (PF.7) is local in Z: this is stated after (1.58), and the localized-data construction preserves it. A native Cauchy derivative in any admitted direction v consequently obeys

$$|DH_Z^{(t)}[v]| \le C\rho^{-1}\alpha_b \mathcal{P}_{b,t} e^{-\delta D_{b,t}} \|v\|_{\mathfrak{N}(Z)}, \tag{PF.8}$$

on a smaller fixed native chart. This is a genuine affine-disc estimate on that chart. It does not infer derivative control from a real value estimate.

Composing (PF.8) with the actual future-current endpoint map gives

$$|\partial_i H_Z^{(t)}| \le C\alpha_b \mathcal{P}_{b,t} e^{-\delta D_{b,t}} \frac{\theta_k}{P_k}(1+k-b)^B e^{-\Gamma(k-b)} e^{-c\, d_k(i,Z)}. \tag{PF.9}$$

Here the age factor is the combined original-kernel/old-carrier estimate, with normalized physical row measures. The bounded old component is the carrier in this application. The same assertion holds after the fixed-old-variable nodal compensation: only the actual surviving graded raw endpoints and adjoint basepoints are involved. It makes no assertion about the dense hidden coordinates of an unintegrated completed-R kernel.

3. The Jacobian is analytic, small, and must be retained

Write $g = g_t$, $\delta' = \delta'_t$, and a for one fixed real rescaled old variable. The actual change (CMP119 (1.22), invoked by LFR I p.188) is

$$\mathcal{R}_H(a) = \frac{1}{ig}\log\!\left(e^{iga} e^{-i\chi(a)H}\right), \tag{PF.10}$$

with the actual smooth real bump χ, or with $\chi = 1$ on the stated remaining bonds. Its derivative scale is

$$\|D\chi\| \le Cg/\delta'. \tag{PF.11}$$

For fixed real a, $\chi(a)$ and its derivatives are fixed coefficients. Thus $H \mapsto D_a\mathcal{R}_H(a)$ is holomorphic on the relevant small complex H-ball even though χ need not be holomorphic as a function of a.

The convergent BCH series gives, uniformly on the actual old support,

$$\|D_a\mathcal{R}_H - I\| \le C|H|/\delta'. \tag{PF.12}$$

The term differentiating the bump gives precisely (PF.11) times H/g; the remaining BCH terms are bounded by $C|H|$, hence by the right side of (PF.12). There is no derivative of H with respect to this old variable: H is determined by the retained next-level background in (1.56), before the variable in (1.53) is integrated or retained.

The AF comparison and (PF.7) imply

$$\epsilon_{b,t} := C\frac{\alpha_b}{\delta'_t}\mathcal{P}_{b,t} e^{-\delta D_{b,t}} \longrightarrow 0 \tag{PF.13}$$

uniformly in the admitted prefix histories as the common ceiling is lowered. The ratio α_b/δ'_t has only a fixed polynomial logarithmic/age cost. Take the ceiling so that $\epsilon_{b,t} < 1/4$.

The logarithm

$$j_{b,t}(a;H) = \log\det D_a\mathcal{R}_H(a) = \mathrm{Tr}\sum_{m\ge 1}\frac{(-1)^{m+1}}{m}(D_a\mathcal{R}_H - I)^m \tag{PF.14}$$

is then defined by its branch at $H = 0$, is holomorphic in the native pair, and obeys

$$|j_{b,t}| \le C\epsilon_{b,t}, \qquad |D_H j_{b,t}[v]| \le C\delta_t'^{-1}|v|. \tag{PF.15}$$

These are local single-bond formulas. The determinant dimension is the fixed Lie algebra dimension, not the total lattice dimension.

PF.3 — the summed failed Jacobian has the required local current estimate There are at most

$$C(MR_b)^4 L^{4(b-t)} \tag{PF.16}$$

shifted t-grid bonds in the selected component. Keep (PF.6) until after this sum. A fixed share of its exponential pays $L^{4(b-t)}$, all fixed age polynomials, and the sum over t. Another share retains e^{-cR_b}. Thus the sum of (PF.14) over every possible stopped prefix in this fixed component has an absolute native bound

$$C\ell_b^{A_J} e^{-c_J R_b} \tag{PF.17}$$

and first future-current endpoint bound

$$\frac{C\theta_k}{P_k}(1+k-b)^{B_J} e^{-\Gamma_J(k-b)} \ell_b^{A_J} e^{-c_J R_b} e^{-c_J d_k(i,Z)}. \tag{PF.18}$$

The constants absorb the actual fixed number of alternatives at a stopping stage; one history stops at its first failed alternative. They do not enumerate exponentially many independent stopping labels for the same component. Subsequent higher current and native jets use the retained common contours and the BC16 Cauchy/counting theorem.

The Jacobian does not cancel from a future conditional current density. Equations (PF.14)–(PF.18) estimate that response explicitly.

4. Selectors, old variables, and the action difference

The selector ledger is exact. LFR I p.183 states that all characteristic functions introduced in (1.22)–(1.28) depend on variables localized in the corresponding old component. After (PF.10), (1.59) removes the last background dependence in the failed branch of (1.55). These are fixed real selectors in a future current calculation. No holomorphic extension of a hard selector is used.

The source does not say that this final failed-shift contribution is one of the successfully integrated $C_b^{(n+1)}$. Immediately after (1.59) it restricts to the small $\chi'^{(t)}$ components before defining (1.60)–(1.63). The failed action difference and (PF.14) must therefore be kept as separate scalar boundary-induced terms.

For the former, take the difference of the complete analytic exponential action before and after (PF.10), on the actual linearized t-current chart. The t-variable is the newly introduced coarse variable of (1.53); the older fluctuation variables remain fixed. The analyticity used here is the original new-current analyticity of that one-step expression, explicitly identified with CMP119 (3.15) in LFR I p.188. It is not an unproved complex old-A radius for a primitive B.

The segment from a to $\mathcal{R}_H(a)$ stays inside a larger fixed t-current contour because

$$g|\mathcal{R}_H(a) - a| \le C|H| \ll \delta_t'. \tag{PF.19}$$

Moreover $6\delta_t'/g_t = O(\ell_t^{p_1}) \ll P_t$, since $p_1 < p_0$. Thus the actual failed real support and this complex perturbation have a fixed margin in the already admitted full t-current contour.

The ordinary action sectors can consequently be estimated before taking the difference:

- The Wilson sector uses the local four-dimensional difference estimate BC15.1, with its fine-cell cancellation and the actual coupling coefficients.
- The E, R, previously generated $C_b^{(s)}$, and inherited scalar B sectors use their source local analytic norms and the fixed-old-field chart. For $C_b^{(s)}$, use its actual modified domain and (1.70), with Section 1's prefix chart inclusion.
- The remaining ordinary Gaussian sector uses OR17.1 with its actual principal Schur covariance. Its background dependence is not frozen.
- Previously completed canonical R masses remain in the deep factor by OT.5; they are not differentiated as a reconstructed hidden kernel. The scalar activities created by their full integration retain their actual relative-carrier majorants.

Here is the intermediate force row, including its coupling normalization. Use the angular new current $\beta = g_t a$. For a scalar action primitive $\mathcal{A}_{s,Y}$ with native value majorant $M_{s,Y}$, the already proved common-disc and endpoint estimates give

$$|\partial_{\beta_u}\mathcal{A}_{s,Y}| \le C\frac{M_{s,Y}}{\alpha_s}\mathsf{K}_{s,t}(Y,u). \tag{PF.19a}$$

The kernel K is the original normalized mixed-history endpoint row, with the two old nodal arguments added for an ordinary quadratic primitive. It includes the physical layer measures, the endpoint decay, and the layer-crossing reserve. For a completed-R scalar primitive its majorant is the anchored absolute majorant, not merely a final pointwise value. For a Wilson term one can use BC15.1 directly; the weaker right side of (PF.19a) also suffices after including its local coupling coefficient in M.

The source value norms in these sectors have at most a fixed inverse-coupling factor g_s^{-2} and fixed logarithmic factors. The ordinary Gaussian sector has only the squared retained-field bound P_s^2, hence is smaller for this purpose. Coarea logarithms and interaction remainders add no worse inverse coupling; pure normalization constants have zero β-derivative. Thus $M_{s,Y}/\alpha_s$ is bounded by $Cg_s^{-3}\ell_s^A$ times its actual local carrier majorant. This displays the native Cauchy cost; it is not hidden in a constant.

For $s \le t$, the kernel age reserve pays the $L^{4(t-s)}$ old-root multiplicity. The remaining sum of $g_s^{-3}\ell_s^A$ relative to $g_b^{-3}\ell_b^A$ has only a fixed age polynomial by AF, so the exponential sums it. Higher old action layers $t < s \le b$ have no finer root-count loss and give at most a fixed polynomial in $1 + b - t$. The scalar relative-carrier norms contribute their fixed polynomial root cost, also paid by the available collar/age reserves. Consequently, per new angular coordinate,

$$\sum_{s,Y}|\partial_{\beta_u}\mathcal{A}_{s,Y}| \le Cg_b^{-3}\ell_b^{A_0}(1+b-t)^{B_0}, \tag{PF.19b}$$

Under the endpoint-localized assumptions above, the derivative estimate applies to the corresponding stopped prefix. The complete-law application requires these bounds uniformly on the same positive histories.

Now $\Delta\beta = g_t(\mathcal{R}_H(a) - a)$ satisfies $|\Delta\beta| \le C|H|$. The segment integral of (PF.19b), followed by (PF.7), gives $Cg_b^{-3}\alpha_b\ell_b^{A_0}\mathcal{P}_{b,t}e^{-\delta D_{b,t}}$. Since $\alpha_b = Cg_b\ell_b^q$, the resulting inverse coupling is precisely g_b^{-2}. No factor g_t^{-1} was discarded: it cancels by using the angular force. Finally sum the genuinely shifted coordinates with (PF.16). The extra prefix multiplicity $L^{4(b-t)}$ is paid separately by (PF.6), as in Section 3.

The resulting bound for the complete failed-shift scalar is of the form

$$|F^{\mathrm{fail}}_{b,t,Z}| \le Cg_b^{-2}\ell_b^{A_F}e^{-c_F D_{b,t}}(MR_b)^4 L^{4(b-t)}, \tag{PF.20}$$

with the current-independent reference value removed when forming a conditional log density. The harmless g_b^{-2} upper bound covers the Wilson coefficient; the other sectors have smaller fixed powers. The inherited scalar-activity majorants in this statement include the separately proved ordinary/relative-carrier value norms. This is the same source input used in the boundary assembly, not a new marked estimate.

The action difference is generally quasi-local. An inherited carrier Y which meets a shifted bond can extend outside Z. Therefore the Cauchy estimate in $\mathfrak{N}(Z)$ used for H alone is not applied to the complete action difference.

For clarity, its segment representation is

$$\Delta\mathcal{A} = \int_0^1 \sum_u \Delta\beta_u\,\partial_{\beta_u}\mathcal{A}(\beta + s\Delta\beta; U, J)\,ds. \tag{PF.20a}$$

A future-current derivative has two types of terms. Differentiating $\Delta\beta$ uses (PF.9). Differentiating the action force uses the whole carrier and kernel record in (PF.19a). A new native endpoint $y \in Y$ is then connected to the shifted bond $u \in Z$ by that record; the future current is connected to y by its original response kernel. The old primitive tree, the kernel path and the future endpoint path are retained together until the BV connector sum is performed. The elementary path inequality then gives a current-to-Z connection. For relative carriers, use CR.1–CR.3 with the actual active tags and retain the current guard to pay the one uncharged physical tag diameter. A fixed share of the original rates remains for all root and age sums. The same argument covers the term differentiating the shifted argument inside the force: it has the additional $\partial_i\Delta\beta$ endpoint, already controlled by (PF.9).

This proves the endpoint-localized version of (PF.20), not a literal compact-support assertion for the scalar difference. After the prefix sum, it is

$$|\partial_i F^{\mathrm{fail}}_{b,Z}| \le \frac{C\theta_k}{P_k}(1+k-b)^{B_F} e^{-\Gamma_F(k-b)} g_b^{-2}\ell_b^{A_F} e^{-c_F R_b} e^{-c_F d_k(i,Z)}. \tag{PF.21}$$

Since $r > 1$, $g_b^{-2}\ell_b^{A_F} e^{-c_F R_b} \to 0$ uniformly as the common ceiling decreases. The actual old-current guard pays the one remaining physical diameter of the bounded component, and the age reserve pays the old-root density. BC16 then gives the fixed factorial-five hierarchy and a vanishing weighted Hessian contribution.

The induction is chronological. PF.19 uses only scalar completed-R families already produced at earlier scales. The tag masses resident inside the newly eligible component have births at most $h = b - N$, by OT.5. A scalar family born outside that component at an intermediate earlier scale can have a tail carrier reaching Z; it is retained with that carrier, rather than incorrectly assigned the stronger resident-tag age restriction. No same-birth completed-R output is used to estimate its own preparatory failure. In the combined invariant the fresh failed contribution enters with

$$\varepsilon_b^{\mathrm{PF}} = C g_b^{-2}\ell_b^{A_F} e^{-c_F R_b}. \tag{PF.22}$$

It is added to the fresh small forcing constant in CR.23–CR.25. It does not multiply an untouched record at every later step. Thus the prefix estimate and the repeated scalar-activity norm close by induction when their shared initial/source value bounds hold; they do not assume one another at the same birth scale.

5. Domain of application

The use of PF.2 requires uniform endpoint-localized response for the actual prepared source family at fixed older variables. The later cluster summation also requires the relative positive carrier bounds on the same histories.

The statements do not establish a marked replacement theorem, operator energy bounds, OS reconstruction, or the Yang–Mills continuum construction. They also do not make the unproved pointwise covariance assertion for dense hidden completed-R coordinates: the source's fully integrated alternative is retained for that sector.

Primary references

- LFR I pp.177–180: eligibility, exact untouched deep factor, prefix determining families and localized data (1.14)–(1.20).
- LFR I pp.181–188: failed alternatives, selector locality, actual linearization, boundary-supported response and relocation (1.22)–(1.59).
- LFR I pp.189–192: successful shell contributions, modified analytic domains, all-prefix insertion statement, and logarithmic L_0 cost.
- LFR I p.194, (1.76): disjoint deep integration left unchanged.
- CMP119 pp.251–252, (1.21)–(1.24), and pp.268–270: the actual bump-dependent change of variables and its Jacobian/action terms.
- CMP102 pp.307–309: the selected formal response, normalized inverse gauge and admitted arbitrary-background extension.

This is a proof and source analysis. It is not an independent verification of the complete construction.

Annex AW — Conditional assembly of the boundary derivative estimate

Annex AW. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

Scope of the inference. The endpoint, stopped-prefix and repeated positive carrier estimates on actual all-history data are assumptions in this application. The native estimates cited below apply on their own admitted domains. In particular a derivative of an isolated minimizing map does not establish the derivative of a composed field at fixed older variables.

The following assembly assumes uniform estimates for the same conditional representation, every composed reconstruction and the complete positive repeated-component and stopped-prefix expansions. These assumptions are required at the points where the corresponding endpoint and carrier estimates are used below.

1. Statement, conventions and exact scope

Use the source-free representation and current conditional split of C-01, with its actual admitted histories, reference measure and consistently normalized coupling trajectory. For completed replacement components choose the exact alternative representation LFR II (1.103)–(1.104), as specified in Section 2. This is an exact marginalization of their hidden variables with the entire incoming boundary exponential. Work on the current k-lattice in its unit physical coordinates. Let I be the independent free bonds integrated in the current regular region, and write $z = (z_i)_{i\in I}$ for their real Lie-algebra coordinates. Surviving earlier fluctuation variables, retained raw component fields, current-scale exterior variables, selectors and history labels are retained.

Put

$$\ell_k = \log g_k^{-2}, \quad P_k = A_1 \ell_k^{p_0}, \quad \delta_k = g_k P_k, \quad \varepsilon_k = A_0 g_k \ell_k^{p_0}, \quad \alpha_k = C_\alpha g_k \ell_k^{q}, \qquad q_0 = q_1 = q > p_0 + r + 2. \tag{AB17.1}$$

Here $r > 1$ is the fixed running-collar exponent, $R_k \asymp \ell_k^r$, with the source rounding convention. Equal native exponents are an allowed fixed choice in CMP119 (2.28); their fixed constants may differ. The stronger inequality includes the real-center enhancement in every preparatory prefix (PF.2–PF.3). The actual real current support is the full unprimed support P_k. Fixed conversions between scalar coordinate norms and the source Lie-algebra norm are included in constants. A current contour denoted $2P_k$ below means twice a fixed coordinate radius containing that full support; it is not a replacement of the support by the smaller LFR primed cutoff.

The native external field chart is the strong source chart

$$\|E\|_{\mathfrak{N}_{\mathrm{curl}}} = \sup_n \alpha_n^{-1} \max\{h_n|E|,\ h_n^2|\nabla_{U_*}E|, h_n^3|D^*_{U_*}D_{U_*}E|\}. \tag{AB17.2}$$

Here h_n is the physical mesh of the actual layer and U_* the fixed real reference. Independent-current arguments use the common weighted radius of BC16.4, including the embedding of actual inner layers into all earlier primitive conventions. A native direction is measured in these norms. It is not assigned a cutoff-independent radius merely because its raw coordinate value is one.

Let $B_k^{\mathrm{cur}}(z,\xi)$ denote the complete current-dependent boundary perturbation in CMP119 (3.25), including every sector listed in Section 5 below. The symbol ξ denotes the admitted native external coordinates; old real variables and selectors remain fixed parameters. The current Gaussian precision, its retained cross force and its Schur-complement scalar remain in their exact C-01 roles. Define

$$\widetilde{B}_k(z,\xi) = B_k^{\mathrm{cur}}(z,\xi) - B_k^{\mathrm{cur}}(0,\xi). \tag{AB17.3}$$

The subtracted value remains in the outer marginal. The subtraction changes no current derivative.

Theorem AB17.1 — boundary budget under the composed-response and representation hypotheses Under the stated C-01/source hypotheses, choose the fixed source parameters and common coupling ceiling in the order specified in Section 7. There are a fixed $\gamma > 0$, constants independent of cutoff depth, total volume and admissible history, and a quantity $\mathcal{K}_{B,k}$ specified in (AB17.24), such that

$$\sup_i \sum_l e^{\gamma\rho(i,l)} \|D_i D_l \widetilde{B}_k\|_{\mathrm{op}} \le C\mathcal{K}_{B,k} P_k^{-2}. \tag{AB17.4}$$

Moreover $\mathcal{K}_{B,k} P_k^{-2} \to 0$ uniformly as the common coupling ceiling tends to zero. Thus the boundary budget can be made smaller than any prescribed fixed share of the source principal coercivity gap.

For $m \ge 1$, a fixed $\sigma > 0$, and the minimum connecting-tree length τ of the current positions,

$$\sup_{i_1} \sum_{i_2,\dots,i_m} e^{\sigma\tau(i_1,\dots,i_m)/m} |D_{i_1} \cdots D_{i_m} \widetilde{B}_k| \le C^{m+1} (m!)^5 \mathcal{K}_{B,k} P_k^{-m}. \tag{AB17.5}$$

Every finite mixed native jet exists on the common smaller external chart. For native directions $h_1, \dots, h_r$, the right side can be multiplied by

$$C^r r! \prod_{s=1}^{r} \frac{\|h_s\|_{\mathrm{native}}}{r_{\mathrm{ext}}}, \tag{AB17.6}$$

where $r_{\mathrm{ext}} > 0$ is the fixed radius actually retained by the joint construction. Further nonlinear pullbacks to arbitrary retained data retain their full higher chain-rule terms; (AB17.6) is not a claim about differentiating selectors or old cutoffs.

The physical current jets are derivatives of the exact reduced real density. The mixed native jets are derivatives of the explicitly specified compatible analytic extensions: each agrees with its physical expression for the entire real current family at fixed admitted retained physical data. An off-minimum formal background is not asserted to be a constrained minimum or to satisfy every identity of a different off-physical extension. A later pullback to moving physical retained data must keep all of its higher chain-rule terms.

The proof uses the direct primitive one-endpoint method of BC16. It does not import the unidentified $p_1(g_k)$ in CMP119 (3.43), nor require a new identity for arbitrary prepared joined minimizers.

2. The outer operator and the current conditional density

CMP119 (2.19)–(2.23) states that T_k is supported in the old large-field region Z_k, factorizes over its components, and contains the retained older integrations, their cutoffs, gauge constraints and within-component old Gaussian pieces. The present free current is supported in Λ_{k+1}. The source constructs Ω_{k+1}, and then Λ_{k+1}, beyond several current-size cube layers surrounding Z_k; see (3.2)–(3.5) and (3.20).

The current substitution is the literal source map

$$I_k(A) = \exp i[g_k CA - hD(g_k CA)]\, V^{(k)}. \tag{AB17.7}$$

The one-step constraint map C, hD has its stated block locality. Its fixed enlargement of the inner support stays inside the buffer and does not change the directly retained coarse data on Z_k. This fact does not prove that every coefficient of T_k, when written through the full background, is independent of the current. A prose factorization cannot supply a missing coefficient identity.

The issue is visible already in CMP119 pp.253–254: the old Gaussian precision and its Schur covariance depend on U_1. The later R-modified operator is written explicitly as $T'_k(X,(U,J))$ in LFR II, pp.379–380. Its old integration support alone does not eliminate this dependence. The nodal identity (AB17.11) changes no integration variables, so it does not itself remove such responses.

The following three source cases exhaust that older response.

2.1 Ordinary unreplaced Gaussian operations On fixed free coordinates put $G = K_{II}^{-1}$, $S = K_{OO} - K_{OI} G K_{IO}$, and $\mathcal{E}a = (-GK_{IO}a, a)$. The exact old retained exponent and its variation are

$$q(U,J,a) = -\tfrac{1}{2} a^{\mathsf{T}} S a, \qquad dq = -\tfrac{1}{2} (\mathcal{E}a)^{\mathsf{T}} (dK)(\mathcal{E}a). \tag{AB17.7a}$$

The latter identity includes the covariance derivative; its cancellation with the variation of the harmonic extension is proved algebraically in OR17.4. The actual $K = C^*\Delta^{(j)}C$ includes the source constraint-current correction. CMP99's localized inverse/walk expansions provide local coefficient dependence as well as off-diagonal decay. Thus the common-disc argument applies to each actual quadratic carrier with value prefactor CP_j^2, not an abstract background-independent precision.

Theorem OR17.1 proves the resulting complete ordinary response bound

$$\mathcal{K}^{\mathrm{G}}_{T,k} \le C\theta_k P_k^2 e^{-c_{\mathrm{G}}MR_k}, \qquad \theta_k = \ell_k^{p_0-q}. \tag{AB17.7b}$$

Old raw Haar, raw averaging delta and selector factors depend on their actual retained local data and are constant under the new inner current. Earlier Gaussian determinant normalizations are already assigned to E/vacuum terms by the source; they are not omitted or counted a second time. The fixed old-field nodal argument applies to the two endpoints of each quadratic scalar carrier.

2.2 Preparatory prefixes and failed branches Use every actual prefix determining history in LFR I (1.15)–(1.16), not only the final completed prefix. PF.1 proves a simultaneous ordinary native core inside the modified domains (1.64)–(1.69). The real-center enhancement $L_0^{2N_0} \le CN_0^{\beta_0}R_b$ is paid by the fixed choice in (AB17.1).

Apply the ordinary response proof to each actual prefix principal constraint/Schur block on that admitted history. Its coefficient is still $C^*\Delta C$, with the same current correction, and its principal inverse is the actual one used at that prefix. The CMP99 localized coefficient, principal-inverse and complex-domain results are uniform on the admitted determining class; they do not require that the history be homogeneous or unprepared. PF.1 supplies precisely that class and the common strong core, including its real-center check. Retained ordinary fluctuation variables obey CP_t; the newly failed variable is bounded by $6\delta_t'/g_t \ll P_t$. Thus OR17.4 and the CP_t^2 primitive majorant apply without freezing a prefix covariance. For the recent one-M-layer prepared collars, retain the fixed δM age reserve as in CR.22b, not an unprepared RM factor. The source block order places the fixed propagator collars inside these admissible blocks before the common ceiling is chosen. PF supplies the remaining selector, domain and relocation changes, while this prefix version of the ordinary response supplies its remaining quadratic coefficients.

The failed-selector displacement uses the exact boundary-supported data in (1.56). Its selected formal extension has zero displacement for zero data. PF.2 admits its actual mixed-data localized background on the primary absolute chart and retains the source collar

$$D_{b,t} = 10M \sum_{s=t+1}^{b-1} R_s + MR_b. \tag{AB17.7c}$$

The displacement and all its current/native responses carry $e^{-cD_{b,t}}$. The supported extension agrees with the actual relocation for every real current at fixed retained physical data. It is not asserted to equal an arbitrary two-average difference on every off-minimum native background.

The hard selectors become functions of retained real component data after (1.59); they are not differentiated. The failed branch retains the whole action difference and its logarithmic coordinate Jacobian. This old failed Jacobian depends on a future current and does not cancel. PF.14–PF.21 bounds it explicitly. The potentially large number of shifted fine coordinates is $C(MR_b)^4L^{4(b-t)}$; (AB17.7c) pays that factor and the prefix sum. In the action difference use the angular coordinate $\beta = g_t a$: its derivative has cost at most $Cg_b^{-3}\ell_b^A$, while $\Delta\beta = O(\alpha_b e^{-cD_{b,t}})$. This yields the stated g_b^{-2} coefficient, including all coupling factors.

The action difference is quasi-local. Its inherited carrier connector and the future endpoint path are kept together, as in PF.20a; it is not replaced by a compactly supported scalar. The resulting absolute fresh coefficient and current endpoint obey

$$\varepsilon_b^{\mathrm{PF}} \le Cg_b^{-2}(1+\ell_b)^{A_F}e^{-c_F R_b}, \qquad |\partial_i F^{\mathrm{PF}}_{b,Z}| \le \frac{C\theta_k}{P_k}(1+k-b)^B e^{-\Gamma(k-b)}\varepsilon_b^{\mathrm{PF}} e^{-cd_k(i,Z)}. \tag{AB17.7d}$$

This includes failures at intermediate stages, their retained older ordinary operations, and the selectors localized by their actual relocation. A whole-component geometric promotion alone would not have established it.

2.3 Completed replacements: exact integration before the next current split For each completed component operation use the literal alternative on LFR II pp.391–392:

$$\prod_Y T'_b(Y)\, e^{B_{\mathrm{in}}} = \prod_Y \tau_Y \left[\prod_Y \Pi_Y\, e^{B_{\mathrm{in}}} \right], \quad \tau_Y = T'_b(Y)1, \quad \Pi_Y = \tau_Y^{-1} T'_b(Y). \tag{AB17.7e}$$

The bracket integrates the entire incoming boundary exponential. Its convergent scalar logarithm is part of the new boundary action. The source proves that each remaining τ_Y depends only on the raw $V_b|_Y$ and explicitly excludes it from the next renormalization step. This assertion is used only for that completed scalar mass; it is false for the earlier pointwise kernel $T'_b(Y,(U,J))$.

The equality is an equality of reduced measures after those hidden variables have been integrated, not pointwise equality at every old hidden coordinate. The unintegrated real variables of ordinary or failed branches remain present. All canonical masses and their normalizations are kept. No old self term is moved into a mass to simplify the argument.

The source output loses full-tree coverage inside retained eligible components. The proof in Annex AU retains the actual outside-tree and outside-volume reserves, the normalized-operation factors, and the polynomial attachment cost of each fixed component. Its two rooted connected sums establish a full-κ absolute carrier norm. The small fresh coefficient follows from the final printed (1.68) on its completed carrier, with a fixed source/analysis-rate split:

$$\varepsilon_b^{\mathrm{R}} \le C(1+\ell_b)^{\mathcal{A}} e^{-c_{\mathrm{R}} R_b}. \tag{AB17.7f}$$

It is not inferred by assigning the newly added completed halo to an earlier pre-carrier estimate.

For later steps keep each untouched activity's original birth anchor, original holes and fixed absolute majorant. Promotion affects future cluster incidence, not old record norms. Source terms whose variables or sole boundary tag are consumed enter the exact fresh output or the primary new-history full-tree reference classification; no nonexistent old anchor is retained. The actual source support classification is distinct from the new quantitative bound.

The uniform repeated bound is chronological. Earlier resident completed masses have births at most $h = b - N(b)$; scalar tails born outside at intermediate scales retain their carrier connectors. They feed PF before the same-birth completed output is constructed. With $F_0 \ge 1$ the fixed ordinary source norm, and with original birth-majorant norms $A_{j,b}$, CR.22–CR.25 proves

$$\begin{aligned} \mathcal{A}_b &= \sum_{j\le b} e^{-\lambda(b-j)} A_{j,b}, \\ \zeta_b &= \varepsilon_b^{\mathrm{R}}(1+R_b)^s + \varepsilon_b^{\mathrm{PF}}, \\ \mathcal{A}_b &\le (e^{-\lambda} + C\zeta_b)\mathcal{A}_{b-1} + CF_0\zeta_b, \\ \mathcal{A}_b &\le \mathcal{A}_* := \frac{CF_0 \sup_j \zeta_j}{1-\rho}, \qquad \rho = \tfrac{1}{2}(1+e^{-\lambda}) < 1, \end{aligned} \tag{AB17.7g}$$

after $C\sup\zeta \le (1-e^{-\lambda})/2$. There are no initial completed families. In particular $\mathcal{A}_* \to 0$ with the common ceiling. The value-age estimate needed in this recurrence uses the old-tree and first-nonempty localization factors of source (1.66), not a future observable derivative. Recent prepared layers have one M-cell collars, so its fixed rate is supplied by $\min(\beta\kappa, \delta M, \kappa_1 - 1)$, not by a fictitious MR_b guard. The growing R_b is used separately in (AB17.7f).

Raw fields surviving this integration are held real and fixed in the native extension. CR.2 extends the nodal construction to both raw bond endpoints, using actual raw endpoint covariance. Fine hidden raw fields inside the completed operation have already been integrated and require no pointwise complex-gauge claim.

2.4 The full older-density budget Combining OR17.1, PF.21–PF.22 and CR.3/CR.25 gives the finite sufficient bound

$$\mathcal{K}_{T,k} \le C\theta_k \left[P_k^2 e^{-c_{\mathrm{G}} M R_k} + \mathcal{A}_* + \varepsilon_*^{\mathrm{PF}} \right], \qquad \varepsilon_*^{\mathrm{PF}} := \sup_b \varepsilon_b^{\mathrm{PF}}. \tag{AB17.7h}$$

Every scalar contribution is assigned once in the actual density; this nonnegative upper bound can conservatively cover a prefix term both directly and in the forcing of a later scalar norm. It does not duplicate a term in the action. The carrier/age sums and the current guard pay each polynomial component attachment cost. There is no total-volume factor. Both $\mathcal{A}_*$ and $\varepsilon_*^{\mathrm{PF}}$ tend to zero because $r > 1$. Hence

$$\mathcal{K}_{T,k} P_k^{-2} \le C\theta_k \left[e^{-c_{\mathrm{G}} M R_k} + P_k^{-2}(\mathcal{A}_* + \varepsilon_*^{\mathrm{PF}}) \right] \longrightarrow 0. \tag{AB17.7i}$$

Older cross-component Gaussian terms which the source already assigned to B remain in the primitive B sector. The present cross term $z^{\mathsf{T}} K_{IO} a$ is different: it stays in the conditional Gaussian mean. Its force need not be small. Current exterior relocation changes that retained force and scalar, but not the current-current precision block.

If marks are present, (AB17.7e) transports their numerators through the same positive conditional operation. The exact quotient identity preserves their integrals. It does not establish local bounds for those transported marks; H-03 and the later observable hierarchy remain separate obligations.

3. A common contour obtained from the actual maps

3.1 Original map and actual gauge family Fix the symmetric raw averaging operation CMP109 (0.11)–(0.12), and use its own associated corrected relative average and inverse-axial gauge. The selected original nonlinear source family is the one used in E14.3, CMP102 (179)–(180) and CMP116 (1.17). Its formal extension at an arbitrary native background is not called a physical minimum. CMP102, pp.307–309, supplies the associated gauge conversion and the stated extension of this analytic family.

BM.10–BM.11 retain the original-geometry value, gradient and current kernel rows, with their actual physical measures and weighted source distance. JI.32–JI.40 and RM.26–RM.36 apply the inverse-axial gauge to this same family. In particular the nodal response uses the actual fine gauge restricted to coarse grid sites, not its iterated gauge mean.

Write

$$\theta_k = \ell_k^{p_0 - q}. \tag{AB17.8}$$

The source kernel rows and AF comparison give, for the current insertion on the full fixed multiple of P_k, a normalized layer response bounded by

$$C\theta_k (1 + k - n)^b e^{-\Gamma_0 (k - n - c_0)_+}, \tag{AB17.9}$$

with fixed b, c_0. Its marked one-coordinate version is the same expression divided by P_k, with a retained spatial endpoint exponential. All exponents below are reduced fixed shares of the original endpoint decay. The source row sums, rather than an unnormalized sum over fine bonds, are used in (AB17.9).

For a native background increment of radius r_{ext}, the same original-family bounds add $C r_{\mathrm{ext}}$ to the local normalized strong size. This parameter is selected small once. Thus the entire smooth source family, including the fixed real exterior current and its shift discussed below, has normalized size

$$C(\theta_k + r_{\mathrm{ext}}) \tag{AB17.10}$$

inside one strong chart. The absolute internal analytic domains are admitted first: $g_k P_k$, α_k, and all their fixed parameter multiples tend to zero. Only afterward is the smaller running primitive radius tested.

3.2 Primitive B domain and all older slots NP.5 proves a genuine strong ball inside all U-dependent restrictions (2.34)–(2.39), including the actual joined reconstructed field/current predicates. Its proof uses the zero-relative-data variational projection on the actual admitted geometry, followed by the bounded actual gauge converter. The second-order input row is (AB17.2); a first derivative of a domain predicate is not used as a certificate.

For each primitive $B^{(j)}(X)$, use the nodal gauge interpolation AD17.3 only on its actual earlier fluctuation slots. For scalar families after completed integration, CR.2 additionally includes both endpoints of every retained raw bond used by that primitive. Ordinary old fluctuation nodes lie within a fixed range of their one-step mesh; surviving raw endpoints satisfy the one-sided grid-to-local-grade relation CR.8. The admitted graded geometry, including the R-rebuilt history, supplies disjoint interpolation balls with radii fixed multiples of those local meshes. Repeated nodes are merged, and each uses the same actual fine-gauge value. Fixed real local regular gauges are used on each ball.

If $U_{\mathrm{L}} = U_{\mathrm{AX}}^{u}$, the interpolating gauge υ_X matches u at the old bond basepoints on which that primitive depends. On real parameters, the two exact source covariance identities give

$$B^{(j)}(X, U_{\mathrm{AX}}, J_{\mathrm{AX}}, \mathrm{A}) = B^{(j)}(X, U_{\mathrm{L}}^{\upsilon_X^{-1}}, \mathrm{Ad}_{\upsilon_X^{-1}} J_{\mathrm{L}}, \mathrm{A}). \tag{AB17.11}$$

The last expression has fixed older fields. The analogous identity for a raw bond uses both endpoint factors, which are canceled by the same nodal matching. Its native field/current arguments lie in the NP.5 ball by (AB17.10) and the fixed reserve (AD17.28). It defines the holomorphic extension of the actual real pullback and therefore all its physical jets. No analytic extension in arbitrary complex older A_j or arbitrary complex retained raw links is required.

AD17.24–AD17.27 show that its one-current tangent has the same normalized endpoint estimate as (AB17.9). The curl row follows from exact current covariance and the local action-chart identity; a third derivative of an interpolated gauge potential is not assumed. The interpolation changes carrier and endpoint constants only by fixed factors. It introduces neither a fine-spacing factor nor a sum over all earlier fields.

3.3 Independent currents and native directions Use the common independent-current profile of BC16.21–BC16.23. Its fixed embedding constant includes every older primitive's innermost-region convention. Under the gauge conversion the auxiliary current is co-transformed with the physical current. The triangular extension

$$J_{\mathrm{aux}} = J_{\mathrm{phys}}(\Phi) + \zeta \tag{AB17.12}$$

has a fixed uniform slack after the common norm comparison. On $\zeta = 0$ it is the original physical expression, so its physical jets agree exactly. Derivatives in ζ are additional admitted native derivatives, not omitted chain-rule responses of J_{phys}.

The field/current argument lists are finite at each cutoff, with uniform sup-type norms. Taking a simultaneous ball in those norms, not one shrinking radius for each old scale, avoids a factor proportional to the number of layers.

For clarity, a raw unit field coordinate δE on a fine bond in layer n has at worst the explicit native cost

$$C\alpha_n^{-1} \max\{h_n,\ h_n^2/\eta,\ h_n^3/\eta^2\}. \tag{AB17.13}$$

The η^{-1}, η^{-2} come from its finite-difference stencil. A raw unit link-angle direction has the additional conversion $E = \eta^{-1}\delta(\text{link angle})$. Neither is called a uniformly bounded native direction. The actual current-coordinate tangent is smooth at the physical layer scale by the source kernel, which is why (AB17.9) has no such fine-mesh loss.

4. The complete exterior relocation on this contour

4.1 Exact composition and Jacobian cancellation AD17.2 proves that the source relocation acts only on the retained current-scale exterior coordinates a, with

$$a' = \mathcal{R}(a; H^{(k)}), \qquad g_k \mathcal{R}_b(a; H) = \frac{1}{i} \log\{\exp(i g_k a_b) \exp[-i\chi_b(a_b) H_b]\}. \tag{AB17.14}$$

The smooth real cutoff $\chi_b(a_b)$, and the derivatives entering its real Jacobian, are fixed coefficients in the native complex continuation. The displacement and Jacobian are independent of z. The Jacobian cancels exactly from the normalized inner law and remains in the outer marginal.

We estimate each sector of the entire relocated action

$$F_X(z, \xi) = b_X\big(\Phi(I_k(z, \mathcal{R}(a; H^{(k)}(\xi))), \xi), \mathrm{A}\big) \tag{AB17.15}$$

directly. There is no Taylor truncation in the exterior displacement. In particular every term in the source's $V^{(k)}$ is retained, either in this exact shifted action or in the outer Jacobian. The nonlinear C, hD dependence in (AB17.7) is retained to all orders.

4.2 Only a coarse native bound on the relocation field is needed For actual real backgrounds the source gives $|H^{(k)}| \le C\varepsilon_k e^{-R_k}$. We do not assume that exponential estimate on an arbitrary off-minimum native ball. The sufficient uniform bound for the present continuation is instead

$$|H^{(k)}(\xi)| \le C\varepsilon_k + C r_{\mathrm{ext}} \alpha_k, \qquad g_k |a'| \le D_k := C(\varepsilon_k + r_{\mathrm{ext}} \alpha_k). \tag{AB17.16}$$

To verify it, use the actual two specified current-scale averages and gauges in CMP119 (3.22), with their own real reference values. The input native field has strong size $r_{\mathrm{ext}}\alpha$. NP.13 bounds the smooth relative Landau correction of each actual localized reconstruction. Cauchy on its genuine NP.5 ball bounds its change from the real reference by $C r_{\mathrm{ext}} \alpha$ in the corresponding local weighted rows. We do not claim a small fine axial potential from a bounded gauge.

Instead apply exact raw averaging covariance. If the source reconstructed representative is $W = L^t$, then

$$M^k(W)(x, x') = t(x)\, M^k(L)(x, x')\, t(x')^{-1}. \tag{AB17.16a}$$

The smooth factor L has the forward averaging estimate of NP.4. The relative change of t from its real value has the RM/JI bound $Ch_s(\|\delta A_{\mathrm{L}}\| + \|\delta E\|)$, not merely a bound on $\|t\|$. The fixed reference gauge factors are unitary. The old physical layers reached by the actual k-averaging footprint are current or fixed adjacent layers, so the combined coarse increment is $Cr_{\mathrm{ext}}\alpha_k$. Joined localization collars may have $h_s \ll h_k$; NP.13 gives $h_s^2 w_{\mathrm{old}}$ there and the factor h_s in the gauge response only improves this estimate. No fine collar is removed from the argument.

At the real reference the ratio of the two source averages has its stated $O(\varepsilon_k e^{-R_k})$ size. Their separate relative changes just estimated keep that ratio in one fixed logarithm chart. This proves the first inequality in (AB17.16), conservatively retaining $C\varepsilon_k$. The local logarithm product in (AB17.14) is bounded by the sum of its small arguments; the original retained real exterior field obeys $g_k|a| \le C\varepsilon_k$ from (3.14). This proves the second inequality.

Since $D_k \le Cg_k\ell_k^q$, the shifted exterior field remains in every fixed absolute internal-map domain after the common ceiling is reduced. It need not stay inside the real current cutoff P_k, because it is a retained parameter on this complex native contour. The integrated current still has the entire $2P_k$ contour.

For the regular E/R arguments, the original absolute-map admission and running-radius comparison are repeated with

$$\frac{g_k P_k + D_k}{\alpha_n} \le C(\theta_k + r_{\mathrm{ext}})(1 + k - n)^b \tag{AB17.17}$$

before the same age kernel is applied. This is a fixed-fraction admission when r_{ext} is chosen first and θ_k made small. Thus the shift is accommodated by their existing native domains, not by pretending that the E14 $2P_k$ current cube already bounded every complex exterior coordinate. The regular primitive value bounds and Ward subtractions remain exactly those used in C-04.

The first current response of (AB17.15) has no derivative of $H^{(k)}$ or a' with respect to z. The differentiated original map at the shifted datum has the same uniform endpoint row on its admitted absolute contour. All higher current responses are obtained by Cauchy from that full composition, so its second and higher nonlinear coordinate derivatives are included.

5. The complete sector majorants

The ledger is the exact exponent C-01 (3.11), followed by the exterior change (3.12), and the source B assignments in CMP119 pp.276–277. No sector is inferred from a determinant calculation alone.

Sector	Function estimated and quantitative input
Inherited B, including old Gaussian terms assigned to B	The primitive full-tree bound CMP119 (2.42), pulled back through (AB17.11), on NP.5's genuine ball.
E/R expressions assigned to the current boundary	The actual layerwise E/R majorants and subtractions of C-04/C-03, restricted to the source's boundary-assigned subfamilies. The shift is admitted by (AB17.17).
Homogeneous nonlinear classical and nonlinear-constraint remainder	The complete source remainder after its actual constrained quadratic part; CMP116 (1.37)–(1.43), transferred to the actual multiscale operators by CMP119 p.276.
Haar and current constraint/coarea factors	The analytic current coordinate density in C-01 (3.6), with its constant factor (3.6a) kept outside the normalized current law.
Ordinary older Gaussian/Schur response	The exact covariance/precision response (AB17.7a), with local normalized walk expansions and prefactor (AB17.7b). Raw/delta/selector factors use their actual retained local data.
Intermediate preparatory and failed branches	Every prefix native chart, the boundary-supported failed relocation, its full action difference and its noncanceling future-current Jacobian; (AB17.7c)–(AB17.7d).

Sector	Function estimated and quantitative input
Completed R operations and surviving raw multipliers	The exact whole-boundary integration (AB17.7e); full-rate relative-carrier majorants, the chronological bound (AB17.7g), and the raw endpoint extension. The canonical raw masses stay fixed.
Heterogeneous Wilson coefficient	The exact difference $A(w_k - g_k^{-2}, \Phi(z)) - A(w_k - g_k^{-2}, \Phi(0))$, with the local recursion (2.24), its transition cutoff collars and the fine-cell estimate BC15.1.
Entire exterior relocation	Every preceding row evaluated at a'; the exterior Jacobian cancels only in the inner conditional quotient by AD17.2.

5.1 Inherited B For a primitive at age $a = k - j$, (2.42) gives

$$K_X = B_0 e^{-\kappa d_j(X)}. \tag{AB17.18}$$

NP.5 and AD17.3 provide a fixed native directional disc at every point of the common current contour. The actual one-endpoint tangent, (AB17.9) and (AD17.27), gives a first-current derivative bounded by the primitive value divided by that fixed native radius, times the tangent norm.

When an attachment is in a layer $n \le j$, the original kernel pays the full age $k - n$. The ratio $g_k P_k/\alpha_n$ has at most a fixed polynomial age cost times θ_k. Summing these earlier layer restrictions costs another fixed polynomial.

When $n > j$, the kernel pays $k - n$. The full primitive tree, which meets Z_j, must connect the attachment to that old boundary. Grading of the actual Ω-history makes its old-j length at least cL^{n-j-c_0}, apart from fixed adjacent-level conventions. A fixed part of its exponential can therefore pay any prescribed finite rate times $n - j$, leaving a positive full-tree reserve. This is the explicit path/carrier split of BC16.5. It is not a claim that the original kernel alone pays age $k - j$ at such an attachment.

The resulting first-current bound has the BC16 form

$$|\partial_i F_X| \le \frac{C B_0 \theta_k}{P_k} (1 + a)^{b_B} e^{-\Gamma a} e^{-\kappa' d_j(X)} e^{-c d_X(i)}. \tag{AB17.19}$$

Here $d_X(i)$ is current-unit distance to the physical carrier, with a fixed collar enlargement from the termwise nodal interpolation. The normalized original kernel rows and the actual carrier tree are both retained. The coupling/volume constants do not hide a carrier-to-current distance.

5.2 Boundary-assigned E and R The C-04 regular contour proof and the C-03 R proof supply the complete layerwise primitive sums

$$\mathfrak{K}_E, \qquad C_R g_k^{\kappa_0 - 11}, \qquad \kappa_0 \ge 12. \tag{AB17.20}$$

Their analytic input is the admitted running native field/current class. Equation (AB17.17) verifies that class after the exact exterior shift, with the same fixed fraction of reserve. Thus their Ward subtractions and old-index summations are not changed.

CMP119 p.276 first performs those estimates on each actual physical layer, then assigns the boundary-crossing localizations to B. Taking the absolute majorant of that subfamily does not increase the already summed regular majorant. In particular carriers meeting the newest boundary use (AB17.20) directly. No old-Z_k collar factor is asserted for them. Remote older-layer pieces retain the original endpoint kernel and are summed with its age reserve as in Section 6.

Scalar gauge invariance of these E/R primitives permits their smooth-family analytic extension without an old-field rotation. Their original exact subtractions, including the E coupling counterterm, remain present. They are not replaced by an unrenormalized vacuum bound.

5.3 Classical, constraint and measure remainder The source constrained Hessian is the Gaussian precision of C-01 (3.8). In particular the quadratic background-current contribution from nonlinear constraint elimination is already in that Hessian. The classical remainder starts at third order in the unscaled current argument; the selected formal extension preserves this source decomposition.

CMP116 (1.39) gives its localized cubic bound, and (1.40)–(1.43) supplies the local coefficient representation. The neighboring argument says the other classical terms have the same bounds. With the actual full unscaled input bounded by $D_k + 2\delta_k$, enlarge D_k by a fixed factor and obtain

$$K_{\mathrm{cl,k}} \le C g_k^{-2} D_k^3. \tag{AB17.21}$$

The localization polynomial in carrier size is paid by a fixed share of the source exponential. The absolute current domain $\epsilon_{\mathrm{abs}}/g_k$ is admitted because $D_k \to 0$. No identification of the p_1 in (3.43) is used.

The Haar and coarea logarithms are analytic in the same small unscaled arguments; after their fixed normalization at zero their conservative localized bound is CD_k. Their higher constraint-map and density derivatives are covered by the same common analytic contour. These functions use the original current-coordinate map, not an ad hoc affine fluctuation.

The exact shifted classical function has induced linear and quadratic terms in z, even though the unshifted remainder starts cubically in the full field. They are included in (AB17.21) by evaluating the full cubic analytic remainder at (z, a'). They are not set to zero by the word "remainder." The current-current Gaussian block is left unchanged, and its separate shifted cross force is retained.

5.4 Heterogeneous coefficient, including transition collars Let $w_l = g_l^{-2}$. The exact recursion gives

$$w_k(x) - w_k = \sum_{l=1}^{k} (w_{l-1} - w_l)(1 - \phi_l(x)). \tag{AB17.22}$$

No sign of $w_{l-1} - w_l$ is assumed. If a point is on actual layer n, the safely sufficient bound is

$$|w_k(x) - w_k| \le C g_k^{-2} \big(1 + k - n + \log \ell_k\big)^{b_\phi}, \tag{AB17.23}$$

with one fixed b_ϕ. The slightly weaker logarithmic factor here avoids silently identifying the earliest active cutoff index with n.

Here is the cutoff-geometry verification. A nonzero $1 - \phi_l$ lies outside the prescribed plateau of Λ_l, hence within at most one fixed localization collar of Z_l. Its physical width is bounded conservatively by $CMR_l h_l$; using the smaller source M-cell convention only improves the bound. For $l < n-1$, interlacing gives $Z_l \subset \Omega_{l+1}^c$. The admitted separation of Ω_n from Ω_{n-1}^c then forces

$$L^{n-l} \le C R_l$$

whenever that collar reaches layer n. The AF comparison gives $\ell_l \le \ell_k + C + \log(1 + k - l)$, and the definition of R_l gives $R_l \le C\ell_l^r$, including its fixed rounding factor. It follows that the number of possible extra older indices, and their ages $k - l$, are bounded by a fixed polynomial in $1 + k - n + \log \ell_k$. For instance absorb $c \log(1 + k - l)$ into half of $k - l$ to obtain $k - l \le C(1 + k - n + \log \ell_k)$. The remaining indices $l \ge n - 1$ already have age at most $1 + k - n$. Bound $|w_{l-1} - w_l| \le w_{l-1} + w_l \le C(1 + k - l) w_k$ and sum. This proves (AB17.23), without any unrecorded equality of cutoff and geometric layers.

The source $k + 1$ collar is formed around Z_k before the current support is chosen. Here the geometric constant can be checked, rather than left as an arbitrary C. The cutoff plateau costs at most one MR_l-cube layer at its own mesh. Choose the small AF comparison parameter $\beta_0 > 0$ so that

$$B_{\mathrm{AF}} := (1 + \beta_0) 2^{\beta_0} \le 5/4. \tag{AB17.23a}$$

Using (2.7) with the fixed power r, then the rounding $R_l < L\ell_l^r$, gives for $a = k - l \ge 1$

$$\frac{R_l h_l}{R_k h_k} \le L^{1-a} (1 + \beta_0)(1 + a)^{\beta_0} \le B_{\mathrm{AF}}. \tag{AB17.23b}$$

The last expression decreases for $a \ge 1$ after, for example, $\beta_0 \le 1/4$; for $a = 0$ the ratio is exactly one. The adjacent version also gives $L R_{k+1} \ge R_k / B_{\mathrm{AF}}$.

Since $Z_l \subset Z_k$, the union of all the older cutoff collars is contained in the $B_{\mathrm{AF}} M R_k h_k$-neighborhood of Z_k. CMP119 p.265 surrounds the compatible cube cover of Z_k by four LMR_{k+1}-cube layers before the first new decomposition; subsequent steps remove further layers before choosing the current free support. Already those four layers give distance at least $4MR_k h_k/B_{\mathrm{AF}}$. Subtracting the entire old cutoff collar leaves

$$(4/B_{\mathrm{AF}} - B_{\mathrm{AF}})\, MR_k h_k \geq (39/20) MR_k h_k. \tag{AB17.23c}$$

Fixed block footprints and plaquette/star conventions subtract only a fixed $C_{\mathrm{foot}} LMh_k$. Taking the common coupling ceiling small enough that $R_k \geq 2C_{\mathrm{foot}} L$ leaves a positive $MR_k h_k$ collar. This also covers the source p.277 shorthand which places the coefficient in Z_k: the proof here has allowed the actual one-layer cutoff transitions rather than dropping them. No unverified equality between a transition index and the geometric layer is used.

The one-current variation of the smooth plaquette field therefore retains $e^{-c_{\mathrm{col}} R_k}$, together with the age and spatial endpoint shares. In dimension four, BC15.1 bounds the plaquette-action difference by the product of two curvature-scale factors. Its fourth power of the fine spacing cancels the number of fine plaquettes in a proper-scale cell. The coefficient (AB17.23) consequently yields the summed prefactor

$$K_{\mathrm{het,k}} \leq C g_k^{-2} \delta_k (1 + \log \ell_k)^{b_\phi} e^{-c_{\mathrm{col}} R_k}.$$

The omitted fixed age polynomial is included in the convergent age sum below. A bound on an individual link without the curvature-square cancellation would not justify this estimate. The running g_k^{-2} is paid by the actual growing collar $R_k \geq \ell_k^r$, not by a fixed number.

5.5 Total prefactor The entire relocation has already been included by shifted evaluation in every row. Its exterior Jacobian has zero current contribution. Thus a sufficient total prefactor is

$$\boxed{\mathcal{K}_{B,k} \leq C_{\mathrm{src}} (1 + S_{b,\Gamma}) \Big[\mathfrak{K}_E + C_R g_k^{\varkappa_0 - 11} + B_0 \theta_k + g_k^{-2} D_k^3 + D_k + g_k^{-2} \delta_k (1 + \log \ell_k)^{b_\phi} e^{-c_{\mathrm{col}} R_k} + \mathcal{K}_{T,k} \Big].} \tag{AB17.24}$$

Here $S_{b,\Gamma}$ is the finite age sum in Section 6. The last summand is the proved explicit older-density budget (AB17.7h); it is not an outstanding premise. Every summand retains the full exactly shifted action. The conservative bound does not demand an exponential current-exterior relocation estimate on an arbitrary native background.

6. Spatial/age summation and all derivative orders

The exponential allocations are made once, before differentiation. For inherited B, combine the current-to-attachment kernel path with a path inside the full primitive carrier to Z_j. The connector inequality is BV.6:

$$d_\Omega(i, \partial_{\mathrm{in}} \Lambda_j) \leq d_\Omega(i, y) + C_M d_j(X) + C_{\mathrm{edge}}.$$

Use separate positive shares for the current collar where required, the age, the normalized kernel/carrier sum, and the final current-distance decay. The passage $n > j$ is paid as in Section 5.1. The actual distance is the weighted source distance, with layer weight h_n^{-1}; it is not replaced by distance divided by the oldest mesh.

CMP96 (2.46)–(2.59), as given in BM.1 and Supplement AF.3, charges one fixed RM separation per complete layer transition. After retaining the source entropy reserve, choose the fixed separation so that the remaining age rate obeys

$$\Gamma > 4 \log L + \log 2 + 2. \tag{AB17.25}$$

The extra $\log 2$ is harmless here because the strong-center proof uses a fixed native reserve rather than the older dyadic reserve; retaining it also covers the conservative prior bookkeeping. At fixed age the number of earlier-layer restrictions and coefficient costs is a fixed polynomial. A conservative old-root density factor is L^{4a}, so

$$S_{b,\Gamma} = \sum_{a \geq 0} (1 + a)^b e^{-(\Gamma - 4 \log L - \log 2) a} < \infty. \tag{AB17.26}$$

This is a sum with an anchored spatial kernel. It is not a count of every old root in the torus.

For the separate prepared value recurrence, recent prefix layers need only have one M-cell separation. The rate used in CR.22b–CR.22d is instead a fixed share of $\min(\beta\kappa, \delta M, \kappa_1 - 1)$, chosen above the same root-entropy threshold. The two estimates refer to their actual respective geometries. A growing R_k is not assigned to every prepared layer.

For the all-history inference assume that each primitive current pullback satisfies BC16.4 on the full $2P_k$ contour with the same uniform sector constants. For composed reconstructions this is a CR hypothesis. For the already localized current-scale classical/E/R pieces it follows directly from their admitted product disc and support; for inherited B and the heterogeneous sector it is the native directional-disc estimate with the retained endpoint path.

At order m, choose each differentiated coordinate in turn as the first derivative, apply Cauchy to the other $m-1$ coordinates on a radius $P_k/2$ polydisc, and take the best estimate. Commutation of analytic derivatives gives

$$|D_{i_1}\cdots D_{i_m}F_X| \le K_X\vartheta_X(m-1)!(2/P_k)^m e^{-c\max_s d_X(i_s)}. \tag{AB17.27}$$

This includes every derivative of the nonlinear current coordinate map and of the nodal representative. It is not obtained by discarding $D^2\Phi$.

BC16.2 sums the $m=2$ row, with a factor $|S_X|e^{\gamma\,\mathrm{diam}\,S_X}$ explicitly paid by the remaining carrier exponential. BC16.3 sums all orders. Its layer-cake argument counts at most $C(n+R+1)^4$ current coordinates within radius R of a length-n carrier. The resulting $4(m-1)$-moment contributes at most $C^m((m-1)!)^4$, and Cauchy contributes the remaining factorial. This gives precisely (AB17.4)–(AB17.5).

For completed-R relative carriers, use CR.1–CR.3 in this last step. Their current diameter is at most $C[Mh_j(\ell(X)+1)+MR_k]$; the one free component diameter is paid by a separate small spatial rate and a share of the already extracted current guard. Keep carrier length, that diameter and the endpoint radius inside the one polynomial moment of degree $4(m-1)$, as in CR.14. This retains factorial power five, instead of multiplying separate factorial losses. Exterior-volume/attachment sums use the rooted absolute norm, not a pointwise contracted decay with its entropy omitted. The exact prefix sum has already been paid in PF.16–PF.21 before this all-order counting.

These estimates are normally convergent in their anchored differentiated sums. They are uniform on a fixed smaller external native chart. Cauchy in any finite family of native directions then gives (AB17.6), with the exact norms (AB17.2) and the independent-current norm. Normal convergence justifies taking those derivatives after the age/carrier sum. No exact thin-polymer decomposition with an exponential volume weight is asserted by this direct endpoint method.

7. Fixed parameter order and quantitative smallness

All choices below are independent of derivative order, cutoff depth, torus volume and the individual admitted history.

1. Fix one admitted odd L, the fixed finite gauge group dimension and the source geometric conventions. Choose one fixed admitted $r>1$, p_0, and integer native exponents $q_0=q_1=q>p_0+r+2$. This explicitly pays every prefix real-center enhancement; the choice does not depend on the number of prefix integrations.
2. Fix the source $\beta>0$, the full invariant carrier rate κ, the localization reserve κ_1, and the block/separation parameters in the primary order. CMP119 after (2.31) permits the large decay choice; CMP116 (1.7), (1.11) allows M large enough to retain all required fractions of $\delta_1 M$. Retain positive shares for full-κ output, primitive entropy, collars, age, normalized anchored summation and final spatial weights. In original histories choose the admitted separation for (AB17.25). In prepared histories require the fixed one-M-layer rate from CR.22b to exceed (CR.22c). Do not spend a rate repeatedly at later births: the source full rate is preserved and a fixed weaker analysis rate is tested against it. This fixes all constants in NP, RM, JI, the nodal interpolation, CR and BC16.
3. Choose the AF comparison parameter

$$0<\beta_0\le\min\left\{\frac{1}{8},\frac{\log(5/4)}{2(1+\log 2)}\right\}. \tag{AB17.28}$$

   The AF theorem permits this fixed choice by reducing the common coupling ceiling. It supplies (AB17.23a) with slack.
4. Let $r_{\mathrm{NP}}>0$ be the genuine whole-ball radius of NP.5, and let $\sigma_{\mathrm{src}}>0$ be the smallest normalized primitive reserve after the auxiliary-current embeddings and old-current comparison. Take one fixed

$$r_{\mathrm{ext}}\le\min\left\{\frac{r_{\mathrm{NP}}}{64C_{\mathrm{map}}},\frac{\sigma_{\mathrm{src}}}{64C_{\mathrm{map}}C_{\mathrm{node}}},\frac{1}{64C_{\mathrm{ER}}}\right\}. \tag{AB17.29}$$

The constants here are the already fixed map, nodal and E/R admission constants. The factors 64 pay the larger contour and a smaller domain for Cauchy; increasing them has no effect on the argument. Set the independent-current radius by the same fixed reserve and the actual BC16 inner-layer embedding constant.

5. Only now reduce the common coupling ceiling so that $C\theta_k$, including the prefix enhancement divided by α_k, occupies at most the same fixed reserve; $D_k + 2\delta_k < \epsilon_{\mathrm{abs}}/8$; all logarithm charts remain in their chosen balls; $R_k \geq 2C_{\mathrm{foot}}L$; and all source small-field hypotheses hold. Also impose the primitive rooted-sum threshold in CR.20 and $C \sup_b \zeta_b \leq (1-e^{-\lambda})/2$, with the resulting $\mathcal{A}_*$ inside that threshold. Since all exponential coefficients tend to zero, these are compatible finitely many choices. The resulting interval applies to every scale of the admitted trajectory.

For all local sectors, the smallness can be read directly from (AB17.24). The fixed external radius does not have to tend to zero with the coupling. Since $D_k \leq C g_k \ell_k^q$,

$$\begin{aligned}
&\mathfrak{K}_E P_k^{-2} && \leq C\ell_k^{-2p_0} \longrightarrow 0,\\
&C_R g_k^{\kappa_0-11} P_k^{-2} && \leq C g_k^{\kappa_0-11}\ell_k^{-2p_0} \longrightarrow 0,\\
&B_0\theta_k P_k^{-2} && \leq C\ell_k^{-q-p_0} \longrightarrow 0,\\
&g_k^{-2} D_k^3 P_k^{-2} && \leq C g_k \ell_k^{3q-2p_0} \longrightarrow 0,\\
&D_k P_k^{-2} && \leq C g_k \ell_k^{q-2p_0} \longrightarrow 0,\\
&g_k^{-2}\delta_k(1+\log \ell_k)^{b_\phi} e^{-c_{\mathrm{col}}R_k} P_k^{-2} && \leq C g_k^{-1}\ell_k^{-p_0}(1+\log \ell_k)^{b_\phi} e^{-c_{\mathrm{col}}\ell_k^r} \longrightarrow 0.
\end{aligned} \tag{AB17.30}$$

Here the R exponent is the admitted $\kappa_0 \geq 12$. The last limit follows from $g_k^{-1} = \exp(\ell_k/2)$ and $r > 1$. All limits are uniform for $0 < g_k \leq \bar{g}$ as $\bar{g} \downarrow 0$, because the right sides have a finite decreasing tail in ℓ_k.

The full older-density term is uniformly vanishing by (AB17.7i). Its two small coefficient norms obey

$$\mathcal{A}_* + \varepsilon_*^{\mathrm{PF}} \leq C \sup_b \left[(1+\ell_b)^A e^{-cR_b} + g_b^{-2}(1+\ell_b)^{A_F} e^{-c_F R_b}\right] \longrightarrow 0, \tag{AB17.31}$$

after enlarging fixed exponents and reducing fixed rates. Here $g_b^{-2} = e^{\ell_b}$ and $R_b \geq \ell_b^r$, $r > 1$. For any prescribed $\varepsilon_B > 0$, choose the ceiling once more so that the finite sum in (AB17.30) and (AB17.7i) is at most $\varepsilon_B/C_{\mathrm{row}}$. Equation (AB17.4) then supplies the specified share of the source principal coercivity gap. No own-sector smallness assumption is left in this final choice.

8. Sector hypotheses and consequences

For the all-history boundary application, the same local derivative bounds are required for the actual complete B and boundary-induced activities, including collars, constraints and current-coordinate dependence. A response of an older reference density is within that statement whenever it actually remains in the current conditional law.

Application component	Result under these hypotheses	Status
Full unprimed current support	The contour contains a fixed multiple of the actual P_k; no p_1 identification is used.	Included in the theorem.
Primitive nonlinear reconstructed field and physical-current domain	NP.5 gives a strong native ball on the specified source chart.	Transfer to a composed Wilson history requires CR and CL.
Symmetric average and source gauge	RM/JI use the specified raw average and corrected chart, with the fine gauge restriction acting on old slots.	The comparison must hold on the same retained data.
Surviving old fluctuation and raw arguments	The proposed native gauge conversion in AD17.3 and CR.2 must control both endpoints of raw links.	Uniform control for the composed reconstruction is a hypothesis.

Application component	Result under these hypotheses	Status
Independent currents and arbitrary admitted native directions	A common radius includes all older embeddings; derivatives use the explicit strong norm (AB17.2).	Supplied at the stated norm, with raw-coordinate costs exposed.
Complete current exterior relocation	The whole shifted action is estimated on an absolute small-field contour; this relocation's Jacobian is independent of the inner current.	Included, with its outer factor retained.
Heterogeneous coefficient and collar	The exact recursion, transition-index bound and four-layer collar calculation retain the running g_k^{-2}.	Included in Section 5.4.
Inherited B, E/R boundary pieces, classical/measure/constraint remainders	All are retained on the shifted common contour with their actual sector majorants.	Included in Sections 3–5.
Ordinary older Gaussian/Schur coefficients	The exact derivative (AB17.7a), local walk dependence and the P_j^2 old-field cost are summed.	OR17.1 supplies the uniform budget.
Every interrupted preparatory branch	Actual modified charts, selector localization, supported failed shift, complete action difference and its noncanceling future-current Jacobian are retained.	PF.1–PF.22 supplies the budget; $q > p_0 + r + 2$.
Completed R components and old reference masses	Exact (1.104) integrates all hidden dependence with the incoming boundary; canonical raw masses remain.	Reduced scalar law is exact; no pointwise hidden-density identity is claimed.
Repeated relative-carrier norm	Actual bounded tags, attachment entropy, full-κ output, original birth anchors and the chronological recurrence are retained.	CR.1–CR.25 supplies the value norm and factorial-five derivative application.
Uniform summed derivatives and weighted Hessian	Assume every actual sector has the common contour and anchored endpoint majorant; the total bound is (AB17.24).	Equations (AB17.4)–(AB17.6), (AB17.7i), (AB17.30)–(AB17.31).

The sector sum supplies the complete derivative estimate provided the stated sector, domain and transport hypotheses hold simultaneously.

Under CL the representation gives the Wilson marginal and the conditional law on the stated retained data. Differentiating that representation requires the current and gauge conventions and the contour used above.

9. Compatibility with the already proved C-06 implication

The exact reduction (AB17.7e) changes which older hidden variables have already been integrated; it does not change the current free-bond coordinates or the source current precision

$$K_{II} = (C^* \Delta^{(k)} C)_{II}. \tag{AB17.32}$$

The background, determining geometry and free-coordinate map in this formula are the actual ones of the reduced source representation. Its surviving retained histories still satisfy C-06's primary admissibility hypotheses. That theorem's principal lower bound is uniform on this entire heterogeneous class. It is not an identity-background bound or an inverse covariance estimate for the interaction.

All current-dependent effects of the older ordinary or failed operations, and the completed scalar logarithms, now belong to the explicit scalar boundary perturbation just bounded. Canonical τ_Y factors are current-independent raw

multipliers; the current interior and its one-step stencil miss their old large-field support. The current exterior relocation is supported in the retained regular exterior region and does not change those raw $V_{b|Y}$. Earlier prefix relocations leave already completed resident masses in the unchanged deep factor (1.76). Their scalar action responses have nevertheless been included where they occur.

The source conditional potential therefore has exactly the form used in HC.15: its unchanged principal quadratic form, its retained exterior linear force, and the complete local/boundary scalar perturbations. If a bulk classical or measure term has already been counted in this note, omit its duplicate from HC's separate local remainder; this only improves the majorant. The exterior force has zero current Hessian and stays in the mean.

Intersect the genuine current/native contours proved here with C-04's and C-06's absolute principal-map reserves, taking a fixed fraction of each radius in its own norm. The strong native rows include the required second-order curl/current row, and the actual current map includes its g_k scaling. The current contour still contains the full real P_k support. At a real exterior datum the source bound is the one used by HC; on the auxiliary complex ball the larger D_k still tends to zero, so it fits the same fixed absolute radius.

With (AB17.4) below the share allotted in (HC.28), the perturbation criterion in Annex AI yields coercivity of this reduced law. The Wilson application requires the complete boundary budget under CR and CL.

Primary references and proof dependencies

- T. Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, CMP 119 (1988), 243–285: pp.253–254 for the first old Gaussian factor; pp.255–259 for admitted histories, AF comparisons, old operators and the localized coupling recursion; pp.261–262 for primitive B domains and covariance; pp.264–270 for the exact current substitution and relocation; pp.276–277 for the complete sector assignments.
- T. Balaban, *Propagators for Lattice Gauge Theories in a Background Field*, CMP 99 (1985), 389–434: corrected relative average, physical-current and constrained-covariance formulas, used with the actual symmetric family specified in RM/JI.
- T. Balaban, *The Variational Problem and Background Fields in Renormalization Group Method for Lattice Gauge Theories*, CMP 102 (1985), 277–309: the original analytic family, reproduced local averaging derivatives, formal extension and inverse gauge on pp.307–309.
- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories I*, CMP 109 (1987), 249–301: (0.11)–(0.12), the actual symmetric raw averaging family.
- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories II*, CMP 116 (1988), 1–22: (1.17), (1.37)–(1.43), the complete localized classical and measure remainder.
- T. Balaban, *Large Field Renormalization I*, CMP 122 (1989), 175–202: pp.179,191–192,201, the changed admissible history and localized replacement.
- T. Balaban, *Large Field Renormalization II*, CMP 122 (1989), 355–392: (1.65)–(1.75), native prepared domains and the background-dependent operation; (1.95)–(1.104), source carrier majorants, classification and the exact integrated alternative.

Annex AX — Dependence of older density coefficients on the current field

Annex AX. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

The coefficient dependence that must be retained

Support of an old integration operation in the old large-field region does not make it independent of the new inner current. The integration variables and its directly retained raw data may be supported there while its quadratic coefficients still depend on the reconstructed background.

CMP119, pp.253–254, explicitly retains the background dependence of the ordinary constrained precision and its Schur complement. LFR II, pp.379–380, likewise retains the native pair (U, J) in the completed component operation. The current factorization on a common active support is

$$w(z, a; \upsilon) = w(0, a; \upsilon) \exp \Theta(z, a; \upsilon), \qquad \Theta(z, a; \upsilon) = \log \frac{w(z, a; \upsilon)}{w(0, a; \upsilon)}.$$

Here a denotes the still retained older integration variables, υ the fixed raw data, and z the current variable. The reference density $w(0, a; \upsilon)$ stays in the positive outer measure. Every surviving current dependence of its coefficients belongs in Θ, and therefore in the complete scalar perturbation that C-05 must estimate. A current-dependent hard selector or constraint is not a smooth factor to be inserted into this logarithm; its actual coordinate reduction and support must first be checked.

For an ordinary old Gaussian operation, its exact Schur response is given and estimated in Annex AS. A currently fixed old integration variable is not replaced by a new independent Gaussian variable. Previously assigned determinant, normalization and cross-component terms remain in their original action or vacuum sectors and are not counted a second time.

The exact completed-R alternative

For completed R components, the chosen source-free representation is the final alternative LFR II (1.103)–(1.104), pp.391–392. It integrates each normalized component operation together with the entire boundary exponential depending on that operation's variables. The resulting scalar activities carry that dependence. The remaining primary masses $T_j'(Y)1$ depend on the retained raw data specified by the source.

This is an equality after the indicated integrations. It is not a pointwise assertion that $T_j'(F) = T_j'(1)F$, and it is not an assertion that the earlier unintegrated kernel has no current response. The actual primary masses are left unchanged. Moving an externally dependent self term into a mass would require its compensating scalar factor to be retained and estimated separately.

With an observable present, the same positive normalized operation transports the observable along with its baseline density. This keeps the marked integral exact. Spatial bounds for that transported observable remain the separate marked-history obligations, including H-03; scalar boundary control alone does not prove them.

Relation to C-01

The C-01 statement and its exact-density objective are unchanged. Any independence of old coefficients from the current field must be read with the qualification above. The complete current exponent contains the old-density responses, the ordinary current action, its explicit retained force, all boundary activities, and every surviving coordinate Jacobian exactly once.

Bounds for those responses are part of the application conditions for C-05 and are not established here. Neither this annex nor the exact completed-R marginalization establishes the Yang–Mills Schwinger hierarchy or the continuum existence claim.

Annex AY — Conditional regular-law moments, native fields and spatial cumulants

Annex AY. The complete Wilson application assumes CL (compatibility of the conditional representation) and CR (uniform composed reconstruction response with the prescribed earlier innovations and raw variables fixed). The ordinary and completed sectors must control the same subtracted primitive, with its actual support, tags, connectors, stopped prefixes and repeated histories. The two-age estimate must retain both the evaluation and forcing ages. These are uniform hypotheses for the all-history application. Boundary induction · Conditional response and the continuum application.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

Scope of the inference. The endpoint, stopped-prefix and repeated positive carrier estimates on actual all-history data are assumptions in this application. The native estimates cited below apply on their own admitted domains. In particular a derivative of an isolated minimizing map does not establish the derivative of a composed field at fixed older variables.

0. Conditional moment problem

The regular-law conclusion is the following.

> Combine the actual quadratic lower bound and complete perturbation budget to prove current-scale mean, all-p field/gradient moments, and spatial cumulant decay for the selected regular conditional law.

The required scope is:

> The law is the one in C-01, with fixed factorial growth and volume-uniform constants; its remaining application to all histories is H-04.

The selected conditional law must satisfy the C-04 perturbation budget, the complete boundary estimates under CL and CR, heterogeneous coercivity, and the convex support/Neumann hypotheses. The Wilson-normalized trajectory and its calibration are additional hypotheses.

The distinction is reflected in the conclusions. The note proves bounds for the actual constrained current field, its covariant current-scale gradient and the native scalar/tensor stencils of that field. It also proves an operator inequality for arbitrary marks in the stated weighted Sobolev spaces. A projected augmented mark can be put into that inequality only with its actual norms; its smoothness is not inferred from the smoothness of the marginal density.

The one-point envelope retains the exterior field. In particular, it is not asserted that every coordinate next to a conditioned exterior has a uniformly bounded mean independent of the logarithmic field radius. Such a statement would fail even for a strictly convex Gaussian law.

1. The exact law after the C-05 reduction

Fix one active retained datum in the admitted source representation. The datum contains the actual determining geometry, coarse fields, surviving older variables, selectors and the current exterior free coordinates. It is held fixed throughout the conditional estimates.

Let `I` be the current free bonds integrated in the regular interior of CMP119 (3.25). Let `O` denote the other free coordinates of the same current fluctuation chart and write `y` for their retained values. No older-layer fluctuation is inserted into `y`: older variables are parameters of the complete perturbation. Distances `rho` are shortest periodic current-lattice distances. Lie-algebra block dimension is fixed.

The exact C-01 law, with the coefficient-response factorization of Annex AX and the completed-component choice, is

$$d\nu_v(z) = Z_v^{-1} 1_D(z) \exp\{-\tfrac{1}{2} z^T K z + b^T z + W(z;v)\}\, dz, \qquad b = -K_{IO} y, \tag{RC18.1}$$

where `K=K_II` is the principal block of the actual constrained precision `C* Delta C`. It includes the quadratic correction caused by the nonlinear averaging constraint. The constraints are solved before the interior/exterior split; they are not solved again after conditioning. The product domain is

$$D = \prod_{i \in I} D_i, \qquad D_i = \{z_i : \|z_i\|_{\text{source}} < r_{k,i}\}. \tag{RC18.2}$$

These are the literal independent-bond norm balls in C-07. They are symmetric and convex and do not vary with retained field values inside the fixed geometry/coupling chart. Write

$$P = P_k = A_1 \ell_k^{p_0}, \quad \ell_k = \log g_k^{-2}, \quad g = g_k. \tag{RC18.3}$$

Assume the literal support in (RC18.2) is contained in the common current contour with the stated margin. This is a hypothesis on the complete conditional representation.

The exact completed-R alternative LFR II (1.103)–(1.104) integrates each normalized completed operation with its entire incoming boundary exponential. This equality moves its scalar effects into `W`. The remaining canonical raw-data masses are independent of this inner current. It is an equality after those integrations, not a pointwise factorization of an unintegrated marked posterior. Ordinary older operations and failed prefixes retain their full current coefficient responses and Jacobians in `W`, as in Annex AW and the C-01 erratum. Constants independent of `z`, including the correct exterior Schur scalar, remain in the outer law. They cancel only from the normalized ratio (RC18.1).

Thus the choice made in C-05 changes no current coordinate, principal precision, retained force or product support in the law now being estimated. It does change which hidden variables have already been integrated, and an observable must pass through those same exact conditional expectations. That observation explains the separate marked application; it does not add a hypothesis to the scalar law (RC18.1).

For the interpolation used below set

$$d\nu_{s,t}(z) = Z_{s,t}^{-1} 1_D(z) \exp\{-\tfrac{1}{2} z^T K z + t b^T z + s W(z)\}\, dz, \quad 0 \le s, t \le 1. \tag{RC18.4}$$

Additional arbitrary real linear tilts are permitted. They change neither the Hessian nor the domain. At finite cutoff all these normalizers are strictly positive and finite on an active fiber.

2. Actual force and complete perturbation envelopes

For a nonnegative scalar matrix define

$$\|A\|_a = \sup_i \sum_j e^{a\rho(i,j)} A_{ij}. \tag{RC18.5}$$

The same notation for a block matrix means the corresponding absolute block norms. On the current four-dimensional lattice,

$$S_a = \sup_i \sum_j e^{-a\rho(i,j)} < \infty \tag{RC18.6}$$

with constants independent of volume and of a restriction to an admitted subset. Counting shortest displacement representatives proves the same bound on periodic tori.

Lemma RC18.1 — the actual derivative and force data After a fixed reduction of the common coupling ceiling, there are fixed constants `C`, `delta>0` and nonnegative, symmetric, coordinate-independent majorants `A_ij` such that throughout the real support and its relevant straight segments

$$\|D_iD_jW(z)\| \le A_{ij}, \qquad \|A\|_\gamma \le CP^{-2}, \qquad |D_iW(0)| \le CP^{-1}. \tag{RC18.7}$$

The ceiling can make `||A||_gamma` smaller than any prescribed positive constant fixed in advance. The exterior force obeys

$$|y_j| \le CP, \qquad |b_i| \le CP\,e^{-\delta d_O(i)}, \qquad d_O(i) = \operatorname{dist}(i, O). \tag{RC18.8}$$

If `O` is empty, set `d_O=+infinity` and `b=0`.

Proof of the exterior bound. CMP119 (3.12)–(3.14) bounds the physical current fluctuation on its complete new active domain before the free-coordinate split. The exact local constraint chart and fixed Lie-algebra norm conversions therefore give `g|y_j|<=C epsilon_k`, hence `|y_j|<=CP`. The fixed constant includes the source cutoff constants; it is not declared small. The actual exterior relocation (3.22) has its admitted small relative-field correction, and the C-05/C-06 compatibility check retains the same bound at the real exterior datum. Surviving variables outside this current chart belong to `W`, not to an unbounded extension of `y`.

C-06 supplies the absolute precision estimate on the full current free-coordinate matrix before restriction,

$$\|(K_{IO})_{ij}\| \le K_0 e^{-\delta_K \rho(i,j)}. \tag{RC18.9}$$

Consequently

$$|b_i| \le CP \sum_{j\in O} e^{-\delta_K\rho(i,j)} \le CPS_{\delta_K/2} e^{-\delta_K d_O(i)/2}. \tag{RC18.10}$$

This proves (RC18.8), including the actual exterior coefficient. No Gaussian recentering or denominator comparison has been used.

Proof of the perturbation bounds. Use the exact ownership of sectors in C-01/C-05/C-06. In a split

$$W = W_{ER} + W_B + W_{\mathrm{cl}} + W_{\mathrm{meas}}, \tag{RC18.11}$$

each scalar term is included once. A local classical or measure term already assigned to the complete C-05 boundary perturbation is omitted from the last two terms. The estimates in the predecessor proofs are sums of absolute primitive majorants, so removing a duplicated subset only improves them.

For `W_ER`, C-04 gives first-derivative and fixed entrywise Hessian majorants with anchored sums

$$d_{ER} \le CK_{ER}P^{-1}, \qquad \|A_{ER}\|_\gamma \le CK_{ER}P^{-2}, \quad K_{ER} = \mathfrak{K}_E + C_R g^{\varkappa_0 - 11}, \quad \varkappa_0 \ge 12. \tag{RC18.12}$$

For `W_B`, Theorem AB17.1 and its primitive endpoint/carrier proof give

$$d_B \le C\mathcal{K}_{B,k}P^{-1}, \qquad \|A_B\|_\gamma \le C\mathcal{K}_{B,k}P^{-2}. \tag{RC18.13}$$

The majorants in these two formulas are formed before summing the carriers and taking the supremum over real current points. This distinction matters: a bound for `sup_z sum_j |D_iD_j W(z)|` would not by itself bound `sum_j sup_z |D_iD_j W(z)|`. In particular the dimension-free row estimate DS.11 does not justify exchanging these operations. Instead use its individual coordinate Cauchy bound and define

$$(A_{ER})_{ij} = \frac{C}{P^2} \sum_{Y:\, i,j\in I_Y} \widehat{K}_{ER} e^{-\widehat{\kappa} n(Y)}. \tag{RC18.13a}$$

The fixed Lie-algebra block dimension is in C. The actual current carrier contains at most C_M(n(Y)+1) free blocks. Its weighted row sum is therefore bounded by the C-04 anchored carrier series with one additional factor C_M(n+1). Reserve a fixed positive part of the source exponential to pay that factor. The resulting constant is larger than DS.10's convenient dimension-free constant, but is still independent of volume, depth and history and has precisely the P^{-2} smallness in (RC18.12). In C-05, (AB17.27) is already a fixed endpoint majorant; its m=2 layer-cake/carrier sum supplies A_B directly. These are the fixed matrices used here. Symmetrizing them costs at most a fixed factor.

The explicit prefactor is uniformly bounded as g decreases. Indeed, with D_k<=C g ell_k^q, its local terms satisfy

$$g^{-2}D_k^3 \le Cg\ell_k^{3q} \longrightarrow 0, \qquad D_k \longrightarrow 0, \qquad \theta_k = \ell_k^{p_0-q} \longrightarrow 0. \tag{RC18.14}$$

The apparently singular heterogeneous term is bounded by a polynomial in ell_k times

$$g^{-1}e^{-c\ell_k^r} = \exp\{\ell_k/2 - c\ell_k^r\} \longrightarrow 0, \qquad r > 1. \tag{RC18.15}$$

The completed and failed older-operation terms have the uniform chronological bounds in (AB17.7h)–(AB17.7i). Thus all terms in (AB17.24), including the old-density responses, are at most one fixed K_*. This is a bound on the actual complete prefactor, not merely on a homogeneous specialization.

For completeness, the remaining local gradient is obtained without assigning a coupling-dependent value to an unverified analytic-domain constant. The degree-three ownership is fixed before splitting the classical localizations into bulk and boundary families. The exact source algebra, as given in Annex AI, is as follows. Before the final constraint substitution write the classical function as F(0)+ell(q')+(1/2)<q',K_raw q'>+R_3(q'), and write D=D_2+D_{>=3}. With q=gC_vt, its remainder after subtraction of the complete constrained quadratic is

$$\begin{aligned} g^{-2}\big\{&R_3(q - hD(q)) - \ell(hD_{\ge 3}(q)) - \langle q, K_{\mathrm{raw}} hD(q)\rangle \\ &+ \tfrac{1}{2}\langle hD(q), K_{\mathrm{raw}} hD(q)\rangle\big\}. \end{aligned} \tag{RC18.15a}$$

Every displayed function starts at degree three (the last at degree four). The removed quadratic includes -ell(hD_2); the stationary first derivative on free coordinates is zero. CMP109 (2.5)–(2.12) gives this algebra; CMP116 (1.37)–(1.39) localizes the representative remainder, and its paragraph before Lemma 2 applies the same localization to the other classical terms. C-06 has checked the actual heterogeneous admission of those localizations.

For clarity one can impose a carrierwise convention which makes the cancellation independent of any shorthand in the localization. Let f_Y(t) be the exact localized functions of the complete classical remainder (RC18.15a), before bulk/boundary assignment. Replace each by its integral third-order Taylor remainder

$$f_Y^{[3]}(t) = \frac{1}{2}\int_0^1 (1-s)^2 D^3 f_Y(st)[t,t,t], ds = f_Y(t) - \sum_{r=0}^{2} \frac{1}{r!} D^r f_Y(0)[t^r]. \tag{RC18.15b}$$

The anchored analytic sums converge at each finite cutoff. The sums of their zero-, first- and second-order coefficients are zero because the constant, stationary linear term and constrained quadratic have already been removed in (RC18.15a). Hence the sum of (RC18.15b) is the same complete classical remainder. The Taylor remainder depends on the same current carrier, and its norm on a fixed smaller absolute ball is bounded by a fixed multiple of the original analytic carrier norm. Its third-derivative bound on that smaller ball gives the cubic norm on every still smaller running ball. Thus this choice preserves the source exponential majorants and, in particular, the classical contribution in AB17.21 up to fixed constants. It is not an assertion that the cancellation persists after deleting an arbitrary noncubic subset.

Now assign these individually cubic carriers to their actual bulk or boundary family. Evaluate the actual exterior relocation in each assigned function. Every induced lower-order current response from that shifted evaluation is retained in W_B, as in AB17.21; it is not discarded as a vanishing remainder. The remaining unrelocated local W_cl(t) is a sum of individually cubic functions. It therefore has vanishing first and second full-current jets at t=0. This fixes the ownership used in (RC18.11) and below.

Fix once the absolute source domain used in HC.19–HC.22. On that fixed domain, the localized classical remainder has an anchored third-derivative majorant

$$\sup_i \sum_{j,l} e^{\gamma\tau(i,j,l)} \sup_t \|D_i D_j D_l W_{\mathrm{cl}}(t)\| \le Cg. \tag{RC18.16}$$

Here `t` is the full current free field, including the later retained exterior. To see the estimate, take three coordinate Cauchy derivatives of the fixed-contour bound HC.20 on a fixed smaller absolute ball. Its value factor is `C g^{-2} epsilon_abs^3` and each derivative costs `C g/epsilon_abs`. The product is `Cg`. The local bulk source expansion is into actual current-layer carriers: the underlying large-field/proper-scale boundary terms belong to `W_B`. At a compact current carrier of tree length `n`, the two summed current-coordinate indices cost at most `C(n+1)^2`, with the fixed `M`-cube block multiplicity included. A fixed share of the source carrier exponential pays this polynomial, the tree weight and the anchored carrier count. If the equivalent endpoint-kernel form of the same localization is used instead, retain its three endpoint exponentials and sum their fixed-order lattice tails; the one-layer endpoint/volume proof of AB17.27–AB17.31 gives a fixed constant at order three. It does not insert the number of microscopic fine bonds. The constants are fixed before `g` varies.

By the carrierwise ownership just specified, the remaining unshifted local classical sum has zero first and second jets in the full current at `t=0`. Taylor's integral formula on the admitted segment from `0` to `t` gives

$$|D_i W_{\mathrm{cl}}(y)| \le CgP^2, \qquad \sup_{z\in D} \|D_I^2 W_{\mathrm{cl}}(z+y)\|_\gamma \le CgP. \tag{RC18.17}$$

The proof of (RC18.16) also supplies fixed entrywise majorants for the second formula. Boundary-classified classical terms, if their relocation produces lower-order responses, are kept in `W_B` and are not subjected to the vanishing-jet assertion.

The Haar and nonlinear exceptional-constraint logarithms have the fixed absolute analytic neighborhood and local trace-loop expansion used in HC.21–HC.22. One and two Cauchy derivatives there give

$$\sup_{z\in D} |D_i W_{\mathrm{meas}}(z+y)| \le Cg, \qquad \|A_{\mathrm{meas}}\|_\gamma \le Cg^2. \tag{RC18.18}$$

The constants absorb the one fixed absolute radius. The background-only linear coarea determinant stays in the outer factor. Neither an extensive determinant nor its volume is used in (RC18.18).

The support and the exterior amplitude imply `g||z+y||_infinity<=CgP`, so the entire segment used above lies in the fixed absolute neighborhood for one small enough ceiling. Combining (RC18.12)–(RC18.18) gives

$$\begin{aligned} d_k &:= \sup_i |D_i W(0)| \le C\{P^{-1} + gP^2 + g\}, \\ \eta_k &:= \|A\|_\gamma \le C\{P^{-2} + gP + g^2\}. \end{aligned} \tag{RC18.19}$$

Since `P=A_1 ell_k^{p_0}`, `gP^3` tends to zero. After fixing a ceiling with `P>=1`, (RC18.19) proves (RC18.7) and its arbitrary smallness. This proof retains the actual first derivative; a Hessian estimate alone would not prove it. ■

3. The full weighted covariance operator on the actual support

C-06 supplies `K>=lambda_0 I` and the weighted principal conjugation estimate. Choose one `gamma>0` so that

$$a_K(\gamma) := \sup_i \sum_j \|K_{ij}\|(e^{\gamma\rho(i,j)} - 1) \le \lambda_0/4. \tag{RC18.20}$$

Reduce it, if necessary, to fit the fixed carrier and chart rates. By Lemma RC18.1 choose the common coupling ceiling so that

$$\eta_k \le \lambda_0/4, \qquad c_0 := \lambda_0 - a_K(\gamma) - \eta_k \ge \lambda_0/2. \tag{RC18.21}$$

The same inequalities hold for every `s,t` in (RC18.4) and every additional real linear tilt. Write $\lambda = \lambda_0 - \eta_k \ge 3\lambda_0/4$ for a fixed unweighted lower bound, decreasing it to `lambda_0/2` when one common constant is preferable.

Lemma RC18.2 — block resolvent and covariance bound For every measure in that family, there is a vector Neumann covariance operator `A_nu^{-1}` with

$$\operatorname{Cov}_\nu(F,G) = \langle \nabla F, \mathcal{A}_\nu^{-1} \nabla G \rangle_{L^2(\nu)}, \tag{RC18.22}$$

and block operator estimates

$$\|\Pi_i \mathcal{A}_\nu^{-1} \Pi_j\|_{L^2 \to L^2} \le c_0^{-1} e^{-\gamma \rho(i,j)}. \tag{RC18.23}$$

Consequently

$$|\operatorname{Cov}_\nu(F,G)| \le \sum_{i,j} G_{ij} \|\nabla_i F\|_{L^2(\nu)} \|\nabla_j G\|_{L^2(\nu)}, \quad G_{ij} = \frac{2}{\lambda_0} e^{-\gamma \rho(i,j)}. \tag{RC18.24}$$

The theorem applies to the indicated weighted Sobolev functions. In particular the stronger bilinear estimate required in CM.9a has now been derived for the actual law.

Proof. Start with smooth convex approximations of each literal bond ball and use their product. For `V= -log` of the smooth density inside the support, let

$$L = -\Delta + \nabla V \cdot \nabla \tag{RC18.25}$$

with scalar Neumann boundary condition. The positive form on vector fields tangent to each boundary face is

$$\begin{aligned} \mathfrak{a}(w,w) = &\int \sum_\alpha |\partial_\alpha w|^2 \, d\nu + \int \langle w, V'' w \rangle \, d\nu \\ &+ \sum_i \int_{\partial D_i \times D_{\neq i}} \mathrm{II}_i(w_{i,\tan}, w_{i,\tan}) \, d\nu_{\partial i}. \end{aligned} \tag{RC18.26}$$

Its boundary term is nonnegative by convexity. The associated operator is the differentiated Neumann realization of `L tensor I+V''`. Solving `Lu=G-nu G` and differentiating the weak equation gives `nabla u=A_nu^{-1}nabla G`. The usual scalar integration-by-parts identity proves (RC18.22).

For a real one-Lipschitz spatial function `f_i`, multiply a whole Lie-algebra block by `M_i=exp(gamma f_i)`. This multiplier is constant in the integration variables. It preserves each face's tangent/normal constraint because it is scalar on that face's block. The scalar differential part commutes with `M`; the boundary form stays nonnegative after the weighted test. The principal conjugation loss is at most `a_K(gamma)` and the complete perturbation costs at most `eta_k`. Thus, for every vector source `h`, the form equation `A_nu w=h` gives

$$c_0 \|Mw\|_2^2 \le |\langle Mw, Mh \rangle|, \qquad \|M \mathcal{A}_\nu^{-1} M^{-1}\| \le c_0^{-1}. \tag{RC18.27}$$

This argument is for arbitrary vector sources, not just gradients. Choosing `f_i=rho(i,j)` proves (RC18.23). Splitting the gradients into their finitely many block components and applying Cauchy–Schwarz in `L^2(nu)` proves (RC18.24).

The smooth approximations can be chosen symmetric and increasing to the literal convex balls. Their product boundary forms have the same sign and the same constants. At fixed cutoff the density and the relevant smooth functions are bounded on the compact product. Dominated convergence proves the covariance inequalities on the actual product support. Approximation in the displayed weighted Sobolev norms gives the general domain. Corners and nonsmooth norm spheres introduce no negative boundary term. This is precisely where C-07's product-support identification is used. ■

4. Current-scale means and all-p field fluctuations

Let a be a fixed positive number less than gamma/4. Fix G as in (RC18.24). The scalar kernel norm is bounded by

$$\|G\|_a \le \frac{2}{\lambda_0} S_{\gamma-a} =: G_a. \tag{RC18.28}$$

Using the arbitrary smallness in (RC18.7), reduce the coupling ceiling once more until

$$\|GA\|_a \le G_a \eta_k \le 1/2. \tag{RC18.29}$$

This is a fixed numerical choice, independent of moment order, cutoff depth, volume and retained history. Define the nonnegative resolvent kernel

$$H = (I - GA)^{-1} G, \qquad \|H\|_a \le 2G_a. \tag{RC18.30}$$

Theorem RC18.3 — actual mean profile and all-p coordinate moments For the actual selected regular law and for the interpolation (RC18.4), there are fixed C,c>0 such that

$$|\nu_{s,t} z_i| \le C\{P^{-1} + Pe^{-cd_O(i)}\} =: M_i, \tag{RC18.31}$$

and, for every p>=1,

$$\|z_i - \nu_{s,t} z_i\|_p \le C\sqrt{p}, \qquad \|z_i\|_p \le M_i + C\sqrt{p}. \tag{RC18.32}$$

More precisely the mean is bounded by the actual force convolution

$$\sup_{s,t} |\nu_{s,t} z_i| \le \sum_j H_{ij} \left(|b_j| + |D_j W(0)| + sigma \sum_l A_{jl} \right), \qquad \sigma = \sqrt{d_{\mathfrak{g}}/\lambda}. \tag{RC18.33}$$

Here d_g is the fixed real block dimension. In particular there is no extensive force norm in the estimate.

Proof. An arbitrary real linear tilt leaves the Hessian and product support unchanged. Applying the unweighted covariance estimate in that tilted law to u dot z_i, where |u|=1, gives

$$\frac{d^2}{dt^2} \log \nu e^{tu\cdot z_i} = \mathrm{Var}_{\nu_t}(u \cdot z_i) \le \lambda^{-1}. \tag{RC18.34}$$

Integrating twice yields a centered sub-Gaussian moment-generating bound. Exponential Markov inequality and integration of its tails prove (RC18.32) componentwise, and the fixed block dimension gives the vector estimate. In particular ||z_i-nu z_i||_2<=sigma.

At s=t=0 the density and support are even, so the coordinate mean is zero. At s=0, interpolate t from zero to one. The normalized derivative is Cov(z_i,b dot z) and (RC18.24) gives the reference bound |nu_0,t z_i|<=sum_j G_ij |b_j|.

For sW, convexity of the support and Taylor's integral formula give

$$|D_j W(z)| \le |D_j W(0)| + \sum_l A_{jl} |z_l|. \tag{RC18.35}$$

Write M_i^*=sup_s,t |nu_s,t z_i|. Differentiate the normalized mean in s, apply (RC18.24), and use (RC18.32) at p=2. The resulting componentwise inequality is

$$M^* \le G|b| + Gd + GA(M^* + \sigma 1). \tag{RC18.36}$$

All vectors are finite at each cutoff. The positive convergent Neumann series in (RC18.30) proves (RC18.33). Lemma RC18.1 bounds the nonforce term by `C/P`. The row norm of `H` then bounds its convolution by `C/P`.

For the exterior term use `H_ij<=2G_a exp(-a rho(i,j))` and (RC18.8). The inequality `d_O(i)<=rho(i,j)+d_O(j)` and a fixed smaller exponential rate imply

$$\sum_j e^{-a\rho(i,j)} e^{-\delta d_O(j)} \le C e^{-c d_O(i)}. \tag{RC18.37}$$

For example take `c<min(a,delta)/2` and retain the remaining `exp[-(a-c)rho]` for the lattice sum. This proves (RC18.31). ■

The term `P exp(-c d_O)` is essential. It permits a conditioned boundary layer with means of order `P` while providing an order-one bound in the bulk at distance of order `log P`. Neither a zero retained mean nor a uniform rare-collar estimate at every boundary bond is asserted.

A local collar consequence is nevertheless available. Suppose the literal bond ball contains a Euclidean ball of radius `c_D P`. If `d_O(i)>=d_0`, where `d_0` is a fixed sufficiently large constant, then (RC18.31) gives `|nu z_i|<=c_D P/4` for a sufficiently small common ceiling. The concentration argument yields

$$\nu\{z_i \notin \tfrac{1}{2} D_i\} \le C e^{-cP^2}. \tag{RC18.38}$$

This last conclusion uses the stated lower radius comparison and therefore is not invoked under a convention in which the literal support radius is smaller than `P`. The mean and moment theorem itself requires only the verified upper support inclusion and applies to either conservative support convention.

5. The actual nonlinear constrained field and its covariant gradient

This section verifies the field maps on which the law acts. The map is the one in CMP119 (3.15), including its nonlinear constraint correction. It is not replaced by the linear chart `exp(i g C z)V`.

Let `V` be the retained reference current configuration, and let the exact local average constraint be

$$\Phi_v(a) = Q_v a + N_v(a), \qquad N_v(0) = D N_v(0) = 0. \tag{RC18.39}$$

With the actual exceptional-bond right inverse `h_v` and kernel injection `C_v`, the physical current chart is

$$q = g C_v t, \qquad a_v(t) = q - h_v D_v(q), \qquad D_v(q) = N_v(q - h_v D_v(q)), \qquad U_e(t) = \exp(i a_{v,e}(t)) V_e. \tag{RC18.40}$$

Here `t=y+iota_I z`. Source normalization factors, including the fixed one-step averaging factor, are in the definition of `Q_v,h_v`. Their local inverses are the ones admitted in C-06, HC.10. The derivative kernels below refer to free `z` coordinates while `v,y` stay fixed.

All the current free blocks share this one-step local chart, even when the retained determining geometry has older layers. The source regular collar admits the local averaging paths and exceptional stencils. The native stencils below are anchored where their whole fixed footprint lies in that admitted current chart. A field whose footprint enters an old nonregular component belongs to the history/matching analysis rather than to the selected regular field family of this theorem.

Lemma RC18.4 — uniform current-chart derivative kernels For fixed positive rates, after one further fixed ceiling reduction,

$$|D_i a_{v,e}(y+z)| \le C g e^{-c\rho(e,i)}, \tag{RC18.41}$$

$$\sup_e \sum_{i,j} e^{c\tau(e,i,j)} \sup_{z\in D} \|D_i D_j a_{v,e}(y+z)\| \le C g^2, \tag{RC18.42}$$

with the analogous anchored row and column sums. These estimates hold uniformly over the admitted actual heterogeneous class.

Proof. The finite averaging paths and the analytic group mean have uniform local derivatives in the fixed small physical neighborhood. The exceptional linear block is uniformly invertible by HC.10. A constraint row uses a fixed number of current coordinates in a fixed number of adjacent unit blocks, and a current coordinate occurs in a fixed number of such rows. Its second derivative uses the same finite dependency footprint. Thus both the row and column matrix sums and each anchored second-derivative tensor sum are bounded uniformly before any inverse is taken. These are current-block counts; no count of microscopic bonds in the reconstructed fine field enters. These facts give fixed weighted matrix and second-derivative tensor bounds for `C_v,h_v,N_v` in every current block; no global gauge choice or old-coordinate inversion is used.

The full current support obeys `g||y+z||_infinity<=CgP`, hence tends to zero. For one fixed spatial exponential norm, `||N'_v(a)h_v||<=C gP` on that support and a fixed surrounding physical neighborhood small enough for the local inverse. Choose the absolute neighborhood so the norm is at most `1/2`. Differentiation of the actual fixed-point equation gives

$$D'_v = (I + N'_v(a)h_v)^{-1}N'_v(a), \tag{RC18.43}$$

$$D''_v = (I + N'_v(a)h_v)^{-1}N''_v(a)[I - h_vD'_v, I - h_vD'_v]. \tag{RC18.44}$$

The inverse is a convergent weighted Neumann series. Finite-range composition and the triangle inequality for the connecting-tree length preserve the corresponding matrix and anchored tensor norms. The substitution `q=gC_vt` gives one factor `g` in (RC18.41) and two factors in (RC18.42). All local constants depend only on the admitted block size, group and source regularity ceiling. The older geometry enters only through the uniformly admitted retained coefficients. This proves the asserted extension of the earlier native chart calculation to the actual current region. ■

Define the normalized constrained fluctuation around its fixed exterior reference by

$$\mathcal{A}_e(z) = g^{-1}[a_{v,e}(y + z) - a_{v,e}(y)], \qquad U_{\mathrm{ref}} = U(y). \tag{RC18.45}$$

The difference of link exponentials is kept as such when forming group observables; (RC18.45) is not an assertion that `exp(i a(y+z)) exp(-i a(y))` equals the exponential of a difference in a noncommutative group.

For neighboring oriented current links, let `nabla_ref` denote any one of the fixed covariant finite-difference stencils with the unitary background `U_ref`. The stencil has uniformly bounded range and coefficients. Physical first differences have the extra factor `h_k^{-1}`; (RC18.46) is in current lattice units.

Theorem RC18.5 — constrained field and gradient moments For `p>=1`,

$$\begin{aligned} \|\mathcal{A}_e - \nu\mathcal{A}_e\|_p + \|\nabla_{\mathrm{ref}}\mathcal{A}_e - \nu\nabla_{\mathrm{ref}}\mathcal{A}_e\|_p &\le C\sqrt{p}, \\ |\nu\mathcal{A}_e| + |\nu\nabla_{\mathrm{ref}}\mathcal{A}_e| &\le C\{P^{-1} + Pe^{-cd_O(e)}\}. \end{aligned} \tag{RC18.46}$$

The uncentered `L^p` bound is the sum of the two displayed right sides. For the full current field `g^{-1}a_v(y+z)`, its fixed reference value `g^{-1}a_v(y)` and its fixed covariant derivative are added explicitly.

Proof. The coordinate mean `m=nu z` belongs to the convex product support. By (RC18.41),

$$|\mathcal{A}_e(z) - \mathcal{A}_e(m)| \le C\sum_i e^{-c\rho(e,i)}|z_i - m_i|. \tag{RC18.47}$$

Minkowski and (RC18.32) give an `L^p` bound `C sqrt p`. Centering costs at most a factor two. The same proof applies to the finite covariant difference stencil.

For the mean, expand about `m`. The linear term has zero expectation. Equation (RC18.42), divided by `g`, and `||z_i-m_i||_2<=sigma` bound the expected quadratic remainder by `Cg`. The value `mathcal A_e(m)` is bounded by

the exponential convolution of |m_i|, by integrating (RC18.41) from 0 to m. Theorem RC18.3 and (RC18.37) bound that convolution by C[P^{-1}+P exp(-c d_O(e))]. Finally g<=C/P at the fixed small ceiling. The same argument applies to the covariant difference. ■

If a physical canonically normalized potential is defined by mathcal A_e/h_k, multiply the potential estimate by h_k^{-1} and its physical gradient estimate by h_k^{-2}. The corresponding unrescaled connection has an additional factor g_k. These dimensional factors have not been absorbed into the constants.

6. Native scalar/tensor fields in the exact selected law

Use the fixed clover family of Annex E, with the scalar trace replaced by the Wilson action density, including its genuine off-diagonal tensor components. Equivalently, take any of its finitely many curvature-bilinear stencils and the Wilson action density. Let P_{alpha,x}(U) denote its dimensionless finite-loop polynomial, with the fixed parallel transports to a common base point retained. Set

$$O_{\alpha,x}(z) = h_k^{-4} g^{-2} P_{\alpha,x}(U(y+z)). \tag{RC18.48}$$

The fixed normalization conversion between the Wilson action and g^{-2} belongs to the declared coupling dictionary and changes only fixed constants. An additional running mixing matrix is not bounded by this theorem.

Let B_x be g^{-1} times a fixed local sum of the reference plaquette distances ||U_ref,p-I|| over the actual stencil and the fixed additional plaquettes needed by its transport paths. By the real source regularity in CMP119 (3.12)–(3.14), the uniform-depth current averaging bound checked below, and |y|<=CP,

$$B_x \le CP. \tag{RC18.49}$$

This conclusion concerns stencils in the admitted current chart. It does not bound an arbitrary old retained field by P. The current reference is the actual V=V^(k)=M^k(U_{k+1}) of CMP119 (3.10); its full averaging depth k must be retained. Here is the precise local application which bounds it.

Take one fixed physical cube, in current units, containing the entire actual k-fold averaging footprints of all bonds in the native stencil and its fixed transport paths. The dependency cone of the actual average has physical diameter at most a fixed multiple of its output spacing, as in the two-endpoint-block convention of BM.2. The native stencil is finite. Thus this cube has side C_b, independent of k. The source current regular region has several MR_k-cube collars inside Omega_{k+1}; for the same small ceiling used in C-05, R_k exceeds the fixed footprint constant. Hence the whole cube and a fixed additional margin lie in the regular current patch on which (3.12) holds. No averaging footprint crosses an old determining interface in this application.

Write eta for the fine spacing in current units. CMP119 (3.12) gives ||U_{k+1}(partial p)-I||<=C epsilon_k eta^2 throughout that cube. Apply one ordered axial gauge on the cube, anchored at a fixed fine vertex. The exact plaquette path-interchange argument in BM.20 gives ||U_{k+1}^u(e)-I||<=C epsilon_k eta and hence a fine potential of magnitude C epsilon_k. Now apply the actual uniform-depth average estimate BM.21 with h_p=h_n=1 and p=k. Its proof uses CMP99 p.406 after (3.78), followed by the fixed final averaging step (E14.39)–(E14.41); it does not iterate a constant greater than one k times. It yields

$$\| \log M^k(U^u_{k+1})_b \| \le C\epsilon_k. \tag{RC18.49a}$$

Raw averaging is gauge covariant with the restriction of the fine gauge to the coarse endpoints. All four coarse plaquette links use the same cube gauge. Multiplying them and using real unitarity shows ||V_p-I||<=C epsilon_k in the original gauge as well. This uses the true M^k, not a substitution by a one-step average or an unrelated corrected operator. CMP102 p.278, immediately after (7), also records the source consequence that regular configurations have regular determining averages, consistently with this local calculation.

Finally telescope the finite current plaquette products from V to U(y). The correction |a_v(y)|<=CgP follows from (RC18.40)–(RC18.44). Unitarity bounds the product differences by the sum of their link differences. Since epsilon_k/g<=CP, this proves (RC18.49). Conjugating a curvature along a finite parallel transporter changes neither its real matrix norm nor this conclusion.

Fix a rate theta>0 smaller than the derivative-kernel and covariance rates. Put

$$Z_x(z)=\sum_i e^{-2\theta\rho(x,i)}|z_i|,\qquad M_x=\sum_i e^{-2\theta\rho(x,i)}M_i,\qquad Q_x=1+B_x+M_x. \tag{RC18.50}$$

The actual derivative kernels prove the deterministic estimates

$$|O_{\alpha,x}(z)|\le Ch_k^{-4}[B_x+Z_x(z)]^2, \tag{RC18.51}$$

$$\left(\sum_i e^{2\theta\rho(x,i)}|D_iO_{\alpha,x}(z)|^2\right)^{1/2}\le Ch_k^{-4}[B_x+Z_x(z)]. \tag{RC18.52}$$

Indeed matrix telescoping and (RC18.41) give one factor g for each differentiated link and an exponentially local first derivative. Each undifferentiated curvature is at most `Cg(B_x+Z_x)`. A parallel-transport derivative multiplies an already small curvature. Its possible contribution after the `g^{-2}` normalization is `C g(B_x+Z_x)^2`, rather than the linear expression from differentiating a curvature factor. On the actual support, however, `B_x+Z_x<=CP` and `gP` is uniformly small. Consequently this term is at most `C(B_x+Z_x)` with the same endpoint decay. Differentiating the two curvature factors and their finite transporters therefore proves (RC18.52); multiplying them proves (RC18.51). For the Wilson scalar use `||U_p-I||_HS^2=2s(U_p)` and its first derivative. The scalar repair is a fixed linear combination and hence satisfies the same estimates. This verifies all components of the fixed native family rather than only its trace.

Theorem RC18.6 — all-p native values and gradients For every native component and every `p>=1`,

$$\|O_{\alpha,x}\|_p\le Ch_k^{-4}(B_x+M_x+\sqrt{p})^2, \tag{RC18.53}$$

$$\|O_{\alpha,x}-\nu O_{\alpha,x}\|_p\le Ch_k^{-4}\{Q_x\sqrt{p}+p\}, \tag{RC18.54}$$

$$\left\|\left(\sum_i e^{2\theta\rho(x,i)}|D_iO_{\alpha,x}|^2\right)^{1/2}\right\|_p\le Ch_k^{-4}(B_x+M_x+\sqrt{p}). \tag{RC18.55}$$

All constants are uniform over the actual selected regular laws. Their displayed background envelopes obey

$$M_x\le C\{P^{-1}+Pe^{-cd_O(x)}\},\qquad Q_x\le CP. \tag{RC18.56}$$

Proof. Theorem RC18.3 and Minkowski's inequality give `||Z_x||_p<=M_x+C sqrt p`. Equations (RC18.51)–(RC18.52) prove (RC18.53) and (RC18.55).

For the centered estimate write `xi=z-m`, where `m=nu z`, and set `X_x=sum_i exp(-theta rho(x,i))|xi_i|`. Then `||X_x||_p<=C sqrt p`. The segment from `m` to `z` lies in the actual convex product domain. The fundamental theorem of calculus, (RC18.52), and Cauchy–Schwarz in the reciprocal spatial weight give

$$|O_{\alpha,x}(z)-O_{\alpha,x}(m)|\le Ch_k^{-4}\{(B_x+M_x)X_x+X_x^2\}. \tag{RC18.57}$$

Taking moments and subtracting its expectation proves (RC18.54). The first formula in (RC18.56) is the convolution inequality (RC18.37); the second uses (RC18.49) and `P>=1`. ■

7. Spatial cumulants, arbitrary Sobolev marks and fixed factorial growth

This section works with the actual complete interacting law (RC18.1), not with its Gaussian reference. For a mark `F_i` anchored at `x_i` and `m>=2`, define

$$B_{i,m} = \|F_i - \nu F_i\|_{2m}, \qquad D_{i,m} = \left\| \left(\sum_b e^{2\theta\rho(b,x_i)} |\nabla_b F_i|^2 \right)^{1/2} \right\|_{2m}. \tag{RC18.58}$$

These are actual conditional norms. No assertion about an unexamined projected mark is hidden in their definition.

Theorem RC18.7 — cut estimate for the exact regular conditional cumulants For any nontrivial label partition `[m]=A disjoint-union B`,

$$|\kappa_\nu(F_1, \dots, F_m)| \le C_0 c_{m-1} \sum_{i\in A, j\in B} e^{-\theta\rho(x_i,x_j)} D_{i,m} D_{j,m} \prod_{l\neq i,j} B_{l,m}, \tag{RC18.59}$$

where `c_n=sum_{pi in Pi_n}(|pi|-1)!<=n!2^{n-1}` and `C_0` is fixed. The theorem is valid whenever the displayed weighted Sobolev moments are finite.

Proof. Center every mark; cumulants of order at least two are unchanged. Lemma RC18.2 gives the covariance resolvent for arbitrary vector sources. Let `W_i` multiply block `b` by `exp(theta rho(b,x_i))`. Weighted conjugation and the triangle inequality give

$$\|W_i^{-1} \mathcal{A}_\nu^{-1} W_j^{-1}\| \le C_0 e^{-\theta\rho(x_i,x_j)}. \tag{RC18.60}$$

For example factor the operator as `(W_i^{-1} A_nu^{-1} W_i)(W_i^{-1}W_j^{-1})`. The first factor is bounded by the same form argument; the second is a block diagonal multiplier of norm at most the displayed exponential.

For disjoint label sets `E,F` with at most `m` labels in their union, expand the gradients of the two products in the covariance identity. Apply (RC18.60) to each differentiated pair. Hölder with the available `L^{2m}` norms gives

$$\left| \mathrm{Cov}_\nu \left(\prod_{i\in E} F_i, \prod_{j\in F} F_j \right) \right| \le C_0 \sum_{i\in E, j\in F} e^{-\theta\rho(x_i,x_j)} D_{i,m} D_{j,m} \prod_{l\in (E\cup F)\setminus\{i,j\}} B_{l,m}. \tag{RC18.61}$$

Write `M_E=nu product_{i in E} F_i` and `M_E^0=M_{E intersect A}M_{E intersect B}`. The latter are the moments of the product of the two marginal laws. Their mixed cumulant is zero. Subtract the two cumulant partition formulas and telescope the product over blocks of each partition. A differing block is exactly the covariance in (RC18.61); all other blocks cost the product of the corresponding `B_l,m`.

For a fixed crossing pair `i,j`, only partitions placing them in one block occur. Merging these two labels identifies that coefficient sum with `c_{m-1}`. There is no additional independent block-count factor. This proves (RC18.59). The coefficient inequality follows, for instance, by encoding ordered partitions as permutations with cuts. Smooth bounded approximation, followed by convergence in the displayed Sobolev norms, proves the asserted domain. ■

Corollary RC18.8 — native all-order decay in the actual law Let `tau(x_1,...,x_m)` be the length of a minimum spanning tree of the current anchors. Then for `m>=2`,

$$\boxed{|\kappa_\nu(O_{\alpha_1,x_1}, \dots, O_{\alpha_m,x_m})| \le C^{m+1} (m!)^2 h_k^{-4m} \prod_{i=1}^m Q_{x_i} \exp\left[-\frac{\theta\tau(x_1,\dots,x_m)}{m-1} \right].} \tag{RC18.62}$$

The full native moments satisfy

$$\left| \nu \prod_{i=1}^m O_{\alpha_i,x_i} \right| \le C^m m! h_k^{-4m} \prod_{i=1}^m Q_{x_i}^2. \tag{RC18.63}$$

The same estimates hold uniformly for the interpolation laws. Constants may depend on the fixed group, block size, source parameters and prescribed rates, but not on m, total volume, cutoff depth or the admitted retained history. The explicit Q_x, P_k and physical factors are not hidden in those constants.

Proof. From Theorem RC18.6, at moment order 2m, one can take

$$B_{i,m} \le C h_k^{-4} Q_{x_i} m, \qquad D_{i,m} \le C h_k^{-4} Q_{x_i} \sqrt{m}. \tag{RC18.64}$$

Delete a longest edge of a minimum spanning tree. Every edge crossing the resulting cut is at least as long as the deleted edge; otherwise replacing it lowers the tree length. That edge has length at least tau/(m-1). The cut sum in (RC18.59) has at most m^2 terms. Its moment factor is m^{m-1} and its partition coefficient is at most (m-1)!2^{m-2}. Thus the total is bounded by C^m(m!)^2, using m^m<=e^m m! and absorbing the remaining fixed polynomial factor in C^m. This proves (RC18.62). For the full moments, use Hölder at order m, (RC18.53), Q_x>=1 and m^m<=e^m m!. ■

The order-dependent range in (RC18.62) has controlled summability. In four current dimensions,

$$\sup_{x_1} \sum_{x_2,\dots,x_m} \exp\left[-\frac{\theta \tau(x_1,\dots,x_m)}{m-1}\right] \le C_\theta^m (m!)^5. \tag{RC18.65}$$

To prove it, dominate the minimum-tree exponential by the sum over all labelled trees. Successively sum each tree's leaves; each edge costs at most C((m-1)/theta)^4. Cayley's count m^{m-2} and the same factorial inequality give (RC18.65), including periodic tori. Combining with (RC18.62), after division by the displayed envelopes, gives an anchored cumulant sum of fixed factorial power seven. No cutoff-independent exponential-source series is required.

8. Application boundary and exact application conditions

Under the common complete-potential, coercivity and support hypotheses, the following estimates apply to the selected regular law.

Requirement	Result
C-01 law with all current responses	Equation (RC18.1), using the factorization of Annex AX and exact completed-R reduction; all scalar effects retained in W.
Actual selected support	Literal independent-bond convex product from C-07; the vector Neumann boundary form is retained with its nonnegative sign.
Retained current force	Source bound (RC18.8), with its spatial boundary profile and its nonzero conditional mean.
Complete first derivative and Hessian	Lemma RC18.1 derives actual fixed majorants, including ordinary/failed/completed old operations, local classical and measure terms.
Current-scale conditional mean	Theorem RC18.3 and its explicit force-resolvent formula.
All-p constrained field and covariant gradient	Theorem RC18.5 uses the exact nonlinear constraint chart.
All native scalar/tensor field and gradient moments	Theorem RC18.6 treats the whole fixed native family, with the actual reference envelope.
Spatial cumulants and fixed factorial growth	Theorem RC18.7 and Corollary RC18.8; volume-uniform constants and summable order-dependent range.
Arbitrary input marks	Theorem RC18.7 applies with their actual Sobolev norms; no native replacement or projection-regularity assertion.

The mean, gradient moment and cumulant estimates use the complete conditional Hessian bound, the retained exterior force, and the regular support hypotheses. Their uniform Wilson application requires CR, CL and the complete sector budgets above.

The remaining conditions include actual projected-mark bounds. That requirement concerns applying the conditional-law theorem in the marked induction; it is not an assertion that those bounds have been proved here. The catalogue dependency graph places M-04, the actual native/transported-mark matching, downstream of C-08, and likewise uses C-08 in the later M-06 induction. Requiring their completed matching as an input to C-08 would reverse that graph. The C-08 statement and its application conditions are met at their law-level scope by the results above, while the precise operator inequality (RC18.59) exposes the Sobolev inputs that those later applications must supply. No criterion is removed and no projected mark is replaced by a native mark.

The conclusion does not finish H-04 or H-05: the conditional background envelope and the resulting retained conditional means still have to be integrated against the complete outer history law. Nor does it finish M-04/M-06: (RC18.48) is the native field in the actual nonlinear current chart, not the unidentified conditional expectation of an arbitrary original microscopic observable. A projected mark with poor Sobolev regularity is not repaired by applying (RC18.59) to a different function. H-03 supplies a different missing estimate for exact history-reset responses. The full continuum Schwinger hierarchy, rotations, ultraviolet identification, energy estimates and phase selection remain outside C-08.

9. Source and proof references

- Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, CMP 119 (1988), 243–285: (3.12)–(3.15), pp.267–268, for the actual physical current bound and nonlinear constraint chart; (3.16)–(3.25), pp.268–270, for selected support, exterior relocation and exact regular integral.
- Balaban, *Propagators for Lattice Gauge Theories in a Background Field*, CMP 99 (1985), 389–434: (3.155)–(3.158), p.428, and Theorem 3.15, p.432, for the actual constrained precision, heterogeneous gap and exponential locality. Used in the already checked C-06 scope.
- Balaban, CMP 99, p.406 after (3.78), and CMP 102, p.278 after (7): the actual uniform-depth relative-average bound and the printed regularity consequence for determining averages. The normalized local application is the already proved BM.20–BM.21 in Annex AF; its exact `M^k` and footprint margin are retained in (RC18.49a).
- Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories II*, CMP 116 (1988), 1–22: the exact fluctuation decomposition, nonlinear constraint terms, and localized classical remainder in Section 1, especially (1.37)–(1.43). The fixed absolute-domain and carrier use is the one as given in C-06.
- Balaban, *Large Field Renormalization II*, CMP 122 (1989), 355–392: (1.103)–(1.104), pp.391–392, for the exact completed-component alternative used in Section 1.

Annex AZ — Exact reset-jet interaction carriers and the relative source theorem

Annex AZ. Exact finite conditional identities apply to the specified positive kernel. A uniform Wilson application requires CL on the same retained sigma-field, the relevant complete-potential estimates under CR and the stated spatial and energy budgets. History estimates · Observable transport.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

Scope

The H-03 estimate is not closed by this annex. It concerns the exact reset in the specified physical value/difference/current framework, uniformly over the remaining history. The source-native result below is a substantive additional part of that estimate, but does not yet control the complete M-03 norm.

The new result is an absolutely convergent thin-carrier expansion of every finite reset jet of the actual normalized old-component posterior. Each activity is carried by the base component, the actual input footprints and the actual interaction carriers. No entire-radius ball is substituted for their union. The factorial exponent of the incoming source hierarchy is preserved. The source bounds make this expansion uniform in a relative norm which charges the exterior of the nonsmall components. The conditional normalizer and the nonsmall self action remain present throughout.

This improves the radial localization theorem HP.3 of Annex AK: it gives the exact individual carrier series and an explicit sufficient condition for its weighted sum. It also identifies precisely why the completed scalar C-05 theorem does not furnish the stronger marked norm. The primary unmarked expansion uses unnormalized small-component factors to pay whole-component volume when those components are summed. Such a factor cannot be reused after normalizing that same component. The alternative normalized expansion keeps the remaining component domains fixed and expressly does not require their own decay factors; it consequently supplies no replacement payment for the stronger marked weight.

The proof is at the declared source-free construction and coupling scope. No new source-free construction or independent asymptotically free trajectory is asserted.

The actual source application in Section 4 is specifically the fixed component operation LFR II (1.71)–(1.72) for which the final primitive bound (1.68) and the eligible-carrier count HL.G3 have been checked. It retains every old variable and internal history still integrated by that operation. It is not asserted for a new, repeatedly normalized carrier geometry with additional surviving older holes merely because the scalar C-05 expansion has a relative norm. The algebraic theorem in Sections 2–3 applies to that later geometry only after its own uniform weighted cross-action sum has been supplied.

1. The actual positive kernel and its analytic reference

Fix an eligible component X_i and a geometry admitted by LFR II (1.71)–(1.72). Let η_i collect all old variables, old history labels, graph choices and local integrations in the positive component operation. Let x collect its retained native data and the other component variables. Write

Here X_i is a generic component label in this note; it is not restricted to the primary source's post-split class denoted X_j, as opposed to its retained large-field class Y_j.

$$M_{i,x}(d\eta_i), \qquad \nu_{i,x} = \frac{M_{i,x}}{M_{i,x}(1)}, \qquad H_{i,x} = \sum_{Y:\, i\in J(Y)} V_Y(x,\eta_i). \tag{HR18.1}$$

The self-component action $V(X_i^{\sim})$, the older selectors, local Jacobians and the real reference factors are in $M_{i,x}$. Its mass is positive on the active sector. No lower bound on that mass is needed. The exact old conditional posterior is

$$Q_x(d\eta_i) = \frac{e^{H_{i,x}(\eta_i)}\nu_{i,x}(d\eta_i)}{\nu_{i,x}(e^{H_{i,x}})}. \tag{HR18.2}$$

The factors from a freshly sampled independent reference field cancel in this old conditional expectation. They do not turn Q_x into the fresh reference law. Also, if a geometry label itself is forgotten, its conditional probability contains its old mass and replacement likelihood; (HR18.2) is not a license to cancel these factors across different geometries.

Use the actual full-law disintegration

$$d\mu(x,\eta_i) = d\bar{\mu}(x)\, Q_x(d\eta_i). \tag{HR18.3}$$

The outer law may be any of the corresponding actual marginal laws for which the input bounds are stated. A moment estimate after conditioning on a rare sector is not inferred from an unconditional moment estimate. All moment hypotheses below are explicitly for the law in (HR18.3).

1.1 Fixed native charts and pointwise domination LFR II pp.379–380, (1.73)–(1.75), represents the native analytic deformation of the old operation by a small pointwise exponential on its original fixed positive support. On a sufficiently small admitted native disc $D_\rho = \{|z| < \rho\}$, after recentering,

$$\nu_{i,x,z}(d\eta_i) = \frac{e^{\sigma_x(z,\eta_i)}\nu_{i,x}(d\eta_i)}{\nu_{i,x}(e^{\sigma_x(z,\cdot)})}, \quad \sigma_x(0,\eta_i) = 0, \quad \sup_{x,z,\eta_i} |\sigma_x| \le \log(3/2). \tag{HR18.4}$$

Consequently the mass-one complex measure satisfies

$$|\nu_{i,x,z}| \le K_0 \nu_{i,x}, \qquad K_0 = 3. \tag{HR18.5}$$

This follows directly from $|\nu_{i,x}(e^\sigma)| \ge 1 - \nu_{i,x}|e^\sigma - 1| \ge 1/2$. The source's self action is included in σ; it is not discarded because its unperturbed value is large. The derivative of the component normalizer is included in (HR18.4).

Assume the actual interaction carriers obey the common-disc bounds

$$\sup_{x,z,\eta_i} |V_Y(x,z,\eta_i)| \le b_Y, \qquad \lambda_0 = \sum_{Y:\ i\in J(Y)} b_Y < \infty. \tag{HR18.6}$$

On the real slice $|H_{i,x}| \le \lambda_0$, so

$$\nu_{i,x} \le e^{2\lambda_0} Q_x, \qquad |\nu_{i,x,z}| \le K Q_x, \quad K = K_0 e^{2\lambda_0}. \tag{HR18.7}$$

This domination uses the true normalized posterior as its positive reference and introduces no inverse component mass.

The assertion in (HR18.4) is about the admitted source-native chart, with the old integration variables and support fixed. It is not automatically a theorem about every elementary raw retained-coordinate variation in M-03. The completed C-05 proof integrates completed components before making some of its scalar complex-gauge extensions. That scalar extension cannot be reused for an arbitrary marked numerator without a pointwise preintegration map or a justified marked change of variables. This scope issue is discussed again in Section 7.

2. Exact expansion of the full source jet

Let I be a set of $m \ge 1$ distinguished source labels. For nonempty $B \subseteq I$, let $A_B(x,\eta_i)$ be the incoming logarithmic-jet coefficient. All such higher vertices are retained. In the square-free finite source algebra, put

$$\Psi(t) = \sum_{\emptyset \ne B \subseteq I} t_B A_B, \qquad t_B = \prod_{a\in B} t_a. \tag{HR18.8}$$

The exact reset coefficient is

$$A_I^+(x,z) = \partial_I \log Q_{x,z}(e^{\Psi(t)})\big|_{t=0}, \quad Q_{x,z}(F) = \frac{\nu_{i,x,z}(Fe^{H_{i,x,z}})}{\nu_{i,x,z}(e^{H_{i,x,z}})}. \tag{HR18.9}$$

No convergent exponential of unbounded source marks is assumed. The source algebra is finite; every coefficient uses finitely many moments. The interaction expansion below is a different, genuinely convergent series in the bounded cross action.

For a partition $\mathcal{P} \in \Pi(I)$, let $k = |\mathcal{P}|$, and let $\mathrm{Y} = (Y_1, \dots, Y_n)$ be an ordered interaction list, allowing repetitions. Define

$$a_{\mathcal{P},\mathrm{Y}}(x,z) = \frac{1}{n!}\,\kappa_{\nu_{i,x,z}}(A_B : B \in \mathcal{P}, V_{Y_1}, \dots, V_{Y_n}). \tag{HR18.10}$$

For $n = 0$, this is the cumulant of the incoming coefficients under the full nonsmall component base. It is not replaced by a product of one-point means.

Proposition HR18.1 — the exact interaction-carrier expansion If $2K\lambda_0 < 1$, with a uniform strict reserve, then

$$A_I^+(x,z) = \sum_{\mathcal{P}\in\Pi(I)} \sum_{n\ge 0} \sum_{Y_1,\dots,Y_n} a_{\mathcal{P},\mathrm{Y}}(x,z) \tag{HR18.11}$$

whenever the required source moments are finite. Under the weighted and moment hypotheses in Theorem HR18.2, the series is absolutely convergent in the displayed weighted L^p norms, uniformly on the common native disc.

Proof. First replace H in (HR18.9) by (uH). For $|u| \le r$,

$$|\nu_{i,x,z}(e^{uH}) - 1| \le K_0(e^{r\lambda_0} - 1). \tag{HR18.12}$$

Thus the denominator is nonzero on a disc containing $u = 1$ if the common ceiling makes λ_0 sufficiently small. For example $2K\lambda_0 \le 1/2$ guarantees a nonzero denominator throughout $|u| \le 3/2$, since $K \ge K_0 \ge 1$ and

$$K_0(e^{(3/2)\lambda_0} - 1) \le K(e^{3/(8K)} - 1) \le e^{3/8} - 1 < 1. \tag{HR18.13}$$

For merely $2K\lambda_0 < 1$, a smaller $r > 1$ works because $K_0(e^{\lambda_0} - 1) < 1$. Integrability of each finite product of input marks and the bounded exponential justify holomorphy of all finite source coefficients. One may equivalently use bounded truncations of the marks and pass to the required finite moments.

Differentiating a normalized exponential tilt gives

$$\frac{d}{du}\kappa_{Q_{u,x,z}}(F_1, \dots, F_k) = \kappa_{Q_{u,x,z}}(F_1, \dots, F_k, H), \tag{HR18.14}$$

for inputs fixed in u. This identity follows by differentiating the finite formal logarithm of the joint moment series; the derivative of its denominator is included. Its n-fold iteration at $u = 0$, followed by the source partition identity, gives (HR18.10) after multilinearly expanding each copy of $H = \sum_Y V_Y$. The Taylor series at $u = 0$ converges to (HR18.9) at $u = 1$. The absolute bounds in Theorem HR18.2 justify all subsequent rearrangements. □

2.1 Actual carriers of the terms Let C_i contain the exact source-coordinate footprint of the component base, and let S_B contain the footprint of the input coefficient A_B. The activity in (HR18.10) is carried by

$$X(\mathcal{P},\mathrm{Y}) = C_i \cup \bigcup_{B\in\mathcal{P}} S_B \cup \bigcup_{j=1}^{n} Y_j. \tag{HR18.15}$$

If a component base or a source coordinate has a larger actual footprint, it must be added to C_i or the corresponding S_B, Y_j. No current-link footprint is inferred from a native storage location.

Every non-self interaction in (HR18.10) meets the reset component. The carriers therefore attach to a common base core. The union in (HR18.15) is used literally; no bounding ball is assigned to it. An input coefficient which is independent of the integrated variables is deterministic for this cumulant. Any cumulant of order at least two containing such an argument vanishes. Its sole order-one, $n = 0$ contribution is its unchanged transport. Thus the genuinely generated reset responses attach only to input vertices which depend on the reset variables.

3. The weighted all-order estimate

Let $w \geq 1$ be a weight on these declared carrier unions. Suppose its geometric union rule is

$$w(X(\mathcal{P}, \mathrm{Y})) \leq W_{\mathcal{P}} \prod_{j=1}^{n} w_i(Y_j), \qquad \Lambda_w = \sum_{Y:\ i \in J(Y)} w_i(Y) b_Y. \tag{HR18.16}$$

Here $W_{\mathcal{P}}$ is the exact base/input carrier cost; it is not set to one unless the chosen relative norm really permits that. A convenient finite-family version has $W_{\mathcal{P}} \leq W_I$ for all partitions of I. An activity-array version follows by summing the input carrier costs before applying the same argument.

Assume the full-law input moment bounds

$$\|A_B\|_{L^p(\mu)} \leq C^{d+1}(d!)^{p_0} b^d p^{sd}, \quad d = |B|, \quad p \geq 1, \quad p_0 \geq 1, \quad s \geq 0. \tag{HR18.17}$$

These are hypotheses on the incoming current-scale jet, not a claim that its microscopic normalization already has that scale. A reset does not change $b = a^{-4}$ to $b = h^{-4}$.

Theorem HR18.2 — thin-carrier reset bound with fixed factorial exponent Suppose (HR18.4)–(HR18.7), (HR18.16)–(HR18.17) hold and

$$2K\Lambda_w \leq \frac{1}{2}. \tag{HR18.18}$$

Put $D = C(1 + 4KC)$. For every $m \geq 1$, $p \geq 1$,

$$\boxed{\sum_{\mathcal{P}, n, \mathrm{Y}} w(X(\mathcal{P}, \mathrm{Y})) \|a_{\mathcal{P}, \mathrm{Y}}(\cdot, z)\|_{L^p(\bar{\mu})} \leq 2W_I D^m (m!)^{p_0} b^m (mp)^{sm}.} \tag{HR18.19}$$

The bound is uniform on the native disc. For $q \geq 0$, the same weighted sum of the q-th native derivatives at its center has the additional factor $q!\rho^{-q}$. On its inner half-disc, the weighted sum of the supremum finite-difference envelopes has the additional factor $2/\rho$. Multiple native directions retain their corresponding Cauchy radii and multi-index factorials.

Proof. For k incoming coefficients of degrees $d_1, \dots, d_k$ summing to m, a product of conditional moments occurring in the cumulant partition formula is bounded by conditional independent replicas and Hölder:

$$\left\| \prod_{D \in \pi} Q_x\left(\prod_{j \in D \cap \{1,\dots,k\}} |A_{B_j}| \right) \right\|_{L^p(\bar{\mu})} \leq \prod_{j=1}^{k} \|A_{B_j}\|_{L^{pm/d_j}(\mu)}. \tag{HR18.20}$$

Blocks with no incoming A contribute one after their bounded interaction factors have been removed. Every replica has the original full-law marginal in (HR18.3). No independence between actual old histories is inserted.

With $r = k + n$, domination (HR18.7), the interaction bounds, and the elementary cumulant coefficient count give

$$\|\kappa_{\nu_{i,x,z}}(A_{B_1}, \dots, A_{B_k}, V_{Y_1}, \dots, V_{Y_n})\|_p \leq (2K)^{k+n}(k+n)! \prod_{j=1}^{n} b_{Y_j} \prod_{j=1}^{k} \|A_{B_j}\|_{L^{pm/d_j}(\mu)}. \tag{HR18.21}$$

Indeed $\sum_{\pi \in \Pi_r} (|\pi| - 1)! \leq r! 2^{r-1}$, obtained by encoding ordered partitions as permutations with cuts, and each moment block costs at most one factor K. The relaxed coefficient $(2K)^r r!$ is sufficient for all $r \geq 1$.

After the interaction carrier sum, the series in its length n is exact:

$$\sum_{n \geq 0} \frac{(k+n)!}{n!} (2K\Lambda_w)^n = \frac{k!}{(1 - 2K\Lambda_w)^{k+1}}. \tag{HR18.22}$$

This is the step that prevents an unbounded factorial loss from arbitrarily many interaction carriers. It is not a truncation of their number.

By (HR18.17), the last product in (HR18.21) is bounded by

$$C^{m+k}b^m(mp)^{sm}\prod_{j=1}^{k}(d_j!)^{p_0}, \tag{HR18.23}$$

where only factors $d_j^{-sd_j}\le 1$ have been dropped. Finally,

$$k!\sum_{\substack{\mathcal{P}\in\Pi(I)\\ |\mathcal{P}|=k}}\prod_{B\in\mathcal{P}}(|B|!)^{p_0}\le (m!)^{p_0}\binom{m-1}{k-1}. \tag{HR18.24}$$

To see this, order the blocks, then sum over their positive sizes; the left side becomes $m!\sum_{d_1+\cdots+d_k=m}\prod_j (d_j!)^{p_0-1}$. Each product is bounded by $(m!)^{p_0-1}$.

Put $\upsilon = 2K\Lambda_w \le 1/2$. Combining (HR18.21)–(HR18.24), the remaining sum is bounded by

$$W_I C^m (m!)^{p_0} b^m (mp)^{sm}\frac{1}{1-\upsilon}\sum_{k=1}^{m}\left(\frac{2KC}{1-\upsilon}\right)^k\binom{m-1}{k-1}\le 2W_I D^m (m!)^{p_0} b^m (mp)^{sm}. \tag{HR18.25}$$

All estimates before taking L^p retain one positive conditional-product majorant independent of the angle of z. Cauchy's formula and Minkowski's inequality therefore apply term by term and to their absolute sum. Summing the Taylor-coefficient majorants on $|t|\le \rho/2$ gives the factor $2/\rho$ for the genuine supremum finite-difference envelope. No interchange of an L^p norm with an uncontrolled supremum is used. □

For $b = h^{-4}$ and $s = 1$, the averaged value grade is exactly the M-03 grade $h^{-4m}(mp)^m$. A native kernel-current costs the stated inverse radius, and fits the averaged difference/current grade (m+1/2) after retaining that cost. This says nothing yet about the pointwise regulator part or the other retained-coordinate directions of M-03.

4. Actual source application in a relative carrier norm

The source geometry here is the fixed eligible geometry of LFR I p.177 and LFR II (1.71)–(1.72), with the coupling and parameter choices retained from the stated source scope. The set Z and the interaction carriers are the literal ones of that operation, with the final bound (1.68). In particular, this paragraph does not replace Z by an arbitrary collection of older holes created by repeated alternative normalizations. Let $R = R_k$. Eligible components are contained in cubes of side $100MR$, so their physical diameters are $O(MRh)$, and their numbers of M-cubes are $O(R^4)$.

The source cross-carrier bound is

$$b_Y \le \alpha_k e^{-\kappa_a d_{k,Z}(Y)}, \qquad \alpha_k \le Ce^{-cR}, \tag{HR18.26}$$

with a fixed positive rate reserve in κ_a. The second inequality is the extraction from the final completed carrier in LFR II (1.68), proved as CR.15 in Annex AU. Self terms are excluded from (HR18.26). They remain in ν_i.

The geometric count HL.G3 gives, for fixed source-admitted geometry,

$$\#\{Y : i\in J(Y),\ n\le d_{k,Z}(Y) < n+1\}\le C(1+R)^4 A_4^{n+1}. \tag{HR18.27}$$

It counts exterior lattice trees and the fixed component attachments. It does not count arbitrary independent shapes inside a component. Its constants are independent of total volume and of the integrated old-history values.

For this theorem, contract each fixed component interior to a site. Also make the base footprint C_i free. Charge the remaining M-cubes and the length of a connecting graph in that contracted graph. Denote these two costs by $N_i(X)$ and $\ell_i(X)$. Use

$$w_i^{\mathrm{rel}}(X) = \exp\{aN_i(X) + \lambda \ell_i(X)\}, \quad a, \lambda > 0. \tag{HR18.28}$$

A fixed small incidence penalty for each included component can be added, if desired. It is paid by the same exterior-tree count. The whole physical interior volume of such a component is not charged by (HR18.28).

All carrier terms attach to the root component. Exterior cube counts are subadditive under union, and their rooted contracted connecting graphs can be joined at that root without an extra interior cost. Hence (HR18.16) holds in this weight with its actual input carrier cost. Rounding the source tree to the M-grid and traversing it twice gives

$$N_i(Y) + \ell_i(Y) \le C_4\{1 + d_{k,Z}(Y)\}. \tag{HR18.29}$$

The source tree meets every exterior cube; a path of length $d_{k,Z}(Y)$ meets at most a dimension-only constant times $1 + d_{k,Z}(Y)$ unit cubes. An included component meets the exterior connected set, and becomes a single contracted site. Thus no factor R^4 appears in (HR18.29).

Choose the fixed weights and source entropy reserve so that

$$\omega = \kappa_a - \log A_4 - C_4(a + \lambda) > 0. \tag{HR18.30}$$

This is a parameter-order requirement: choose the source decay constant sufficiently large, then the fixed analysis weights, then the common coupling ceiling. These constants do not run with k or derivative order. Equations (HR18.26)–(HR18.29) give the actual weighted sum

$$\boxed{\Lambda_{\mathrm{rel},i} = \sum_{Y\,:\,i \in J(Y)} w_i^{\mathrm{rel}}(Y) b_Y \le C_{a,\lambda,\omega}(1+R)^4 e^{-cR} \longrightarrow 0.} \tag{HR18.31}$$

The convergence is uniform in the remaining source-admitted eligible geometry and old history values. Taking the ceiling smaller makes (HR18.18) hold, for example with $K = 6$: the native component density costs at most $K_0 = 3$, and $e^{2\lambda_0} \le 2$ for this choice of ceiling.

Corollary HR18.3 — the actual source-native relative reset theorem For the exact old posterior (HR18.2) of the fixed primary operation just specified, under its admitted fixed native chart and the displayed incoming moment/carrier budgets, every source order has the thin-carrier representation (HR18.11). Its relative weighted sum is bounded by (HR18.19), with fixed source factorial exponent p_0, uniformly in total volume, the other admitted component histories in that operation and its integrated internal old histories. Its native kernel derivatives and finite-difference envelopes have the explicit common-chart radius costs from Theorem HR18.2.

This is a spatial estimate: the core has physical radius $O(MR_k h)$, and the actual exterior connections are exponentially charged. It is stronger than merely bounding an integral or assigning its output to an entire radial ball. It is still a relative estimate; the physical interior footprints of all nonsmall components remain free in this norm.

For a repeated scalar-normalization representation with surviving earlier tagged holes, CR.20 gives a useful scalar absolute norm after its specified incidence resolution and chronological bookkeeping. It cannot simply be substituted for (HR18.27): the carrier family, allowed free sets and pointwise marked kernel have to be identified. An application of Theorem HR18.2 to that representation needs its actual resolved-tag weighted sum Λ_w on the preintegration marked chart. That is a concrete additional transfer check, not an extra claim of Corollary HR18.3.

4.1 No component-mass payment has been reused The proof of (HR18.31) uses the cross-interaction coefficient and geometry, not $M_{i,x}(1)$. That mass canceled in (HR18.2). The $n = 0$ terms retain all base cumulants. For example, two marks inside the same reset component can have order-one covariance throughout that component, and this is compatible with Corollary HR18.3. The source component radius is kept explicitly.

4.2 Families of fixed geometries retained in the outer law One need not condition the whole probability measure on one rare geometry. Let $x = (g, \xi)$, where g is the actual retained geometry label, and take the full disintegration (HR18.3) on the disjoint union of its active component fibers. Suppose the native radius, the density constant K, the weighted cross sum and the base/input cost W_I have the uniform bounds displayed above. Apply the local expansion on each fiber, with that fiber's true carriers and its relative weight. Define its absolute envelope by

$$\mathcal{E}_I(g, \xi) = \sum_{\mathcal{P},n,\mathrm{Y}} w_g(X(\mathcal{P}, \mathrm{Y}))\, |a_{\mathcal{P},\mathrm{Y}}(g, \xi)|. \tag{HR18.31f}$$

Then, under the unconditioned incoming moment hypothesis (HR18.17) for this full law,

$$\|\mathcal{E}_I\|_{L^p(\bar{\mu})} \le 2W_I D^m (m!)^{p_0} b^m (mp)^{sm}. \tag{HR18.31g}$$

No inverse probability of a geometry and no sum of such inverse probabilities occurs. To prove this, use conditional Hölder in each moment block before summing interaction lists. For a fixed input partition $\mathcal{P}$, every product of input moments is bounded by the same positive envelope

$$\mathcal{H}_{\mathcal{P}}(x) = \prod_{B\in\mathcal{P}} \left[Q_x |A_B|^{m/|B|}\right]^{|B|/m}. \tag{HR18.31h}$$

All interactions have already been bounded pointwise by their b_Y. The weighted interaction sums can therefore be made on each fiber, using its uniform Λ_w, before the L^p norm. Finally Hölder in x gives $\|\mathcal{H}_{\mathcal{P}}\|_p \le \prod_{B\in\mathcal{P}} \|A_B\|_{pm/|B|}$. The identical partition count proves (HR18.31g).

This retains the geometry label; it does not average or forget that label. It also does not exchange the norm with a geometry-dependent carrier sum. The global sum of individual carrier L^p norms can be stronger than the L^p norm of (HR18.31f). Proving that stronger global M-03 carrier/current sum still requires the actual common carrier representation and its geometry costs. The fixed-geometry estimate (HR18.19) and the full-law envelope estimate (HR18.31g) have deliberately different left sides.

4.3 A discrete current for changing another fixed component history The thin expansion also localizes an actual class of discrete kernel currents. Keep the outer geometry and native background fixed, and change the old variables/history of one other component $j \neq i$. This keeps the base $\nu_{i,x,z}$ fixed, as required by the source factorization, and changes only those cross terms with $j \in J(Y)$. Freeze the incoming A_B in this comparison, so that only the newly generated kernel current is being estimated. Put

$$\Lambda_{w,ij} = \sum_{Y:\ i,j\in J(Y)} w_i(Y) b_Y. \tag{HR18.31a}$$

Let $\Delta_j a_{\mathcal{P},\mathrm{Y}}$ be the difference of the exact activities in (HR18.10) for these two component histories, on the common chart. Under Theorem HR18.2's hypotheses,

$$\sum_{\mathcal{P},n,\mathrm{Y}} w(X) \|\Delta_j a_{\mathcal{P},\mathrm{Y}}\|_{L^p(\bar{\mu})} \le 16K\Lambda_{w,ij} W_I [C(1 + 8KC)]^m (m!)^{p_0} b^m (mp)^{sm}. \tag{HR18.31b}$$

The expectation on the right still uses the original actual full law in (HR18.3); no moment estimate under the shifted history is inserted. If the incoming coefficients change as well, their direct and previous-current differences are separate terms under (HR18.41).

Proof. Each cumulant in (HR18.10) is multilinear in its interaction arguments. Telescope the n changed arguments. The r-th summand has one difference $\Delta_j V_{Y_r}$, which vanishes unless $i, j \in J(Y_r)$, and otherwise is bounded by $2b_{Y_r}$. The other arguments retain their old or new values, each bounded by its same b_Y. Their weighted ordered-carrier sum is therefore bounded by

$$2n\Lambda_{w,ij}\Lambda_w^{n-1}. \tag{HR18.31c}$$

The positive reference for the cumulant remains Q_x, since the unchanged base obeys (HR18.7) relative to that reference. With $\upsilon = 2K\Lambda_w$, the extra interaction sum is

$$4K\Lambda_{w,ij}\frac{d}{d\upsilon}\frac{k!}{(1-\upsilon)^{k+1}} = 4K\Lambda_{w,ij}\frac{k!(k+1)}{(1-\upsilon)^{k+2}}. \tag{HR18.31d}$$

Use $\upsilon \le 1/2$, $k+1 \le 2^k$, and (HR18.24). The same source-partition sum as before gives (HR18.31b). The common positive majorants also give its native derivatives and inner-disc finite-difference envelopes with their previous radius costs. □

For the relative weight of (HR18.28), this is a genuine remote-component estimate. If two components have separation $d_M(X_i, X_j)$, any carrier using both must have relative size at least $d_M(X_i, X_j) - CR - C$, by the source geometric bound HL.G4. Reserve a fixed half of the rate ω in (HR18.30) for this constraint; the remaining half still sums the source entropy. Then

$$\Lambda_{\mathrm{rel},ij} \le C(1+R)^4 e^{-cR} e^{-c'[\,d_M(X_i,X_j)-CR-C\,]_+}. \tag{HR18.31e}$$

The comparison changes one actual other-component history on fixed geometry, and its right side is uniform over its allowed old values. It does not compare distinct base geometries, erase a geometry label, or bound the sum over every alternative discrete value in the full M-03 current norm. Those operations have different costs and remain explicit in Section 7.

5. The exact stronger carrier test

The original M-03 weight is

$$w_h(X) = e^{\kappa[|X|+\mathrm{t}_h(X)]}, \qquad \kappa > 0 \text{ fixed}. \tag{HR18.32}$$

It charges all physical current blocks in the actual dependency footprint, including the interiors treated as free in (HR18.28). To apply Theorem HR18.2 directly in this norm one needs the actual full-footprint union rule and a sufficiently small sum of the form

$$\Lambda_{\mathrm{full},i} = \sum_{Y\,:\,i\in J(Y)} b_Y\, e^{\kappa\,\mathrm{cost}_{\mathrm{full},i}(Y)}, \tag{HR18.33}$$

together with a controlled base/input-core factor W_I. Here $\mathrm{cost}_{\mathrm{full},i}(Y)$ must count every new component interior and every genuine coordinate footprint in the output carrier. Removing those terms from (HR18.33) changes the norm.

The source estimate (1.68) bounds the relative tree through exterior cubes. The eligibility upper bound yields only a comparison with a running component-size constant; for example the elementary count gives

$$|Y|_M + \mathrm{t}_M(Y) \le C(1+R)^4\{1 + d_{k,Z}(Y)\}. \tag{HR18.34}$$

An included component is adjacent to an exterior cube, there are at most a dimension-only multiple of the number of exterior cubes such components, and each has at most CR^4 cubes. A spanning tree of the connected union has at most its number of cubes minus one edges. This proves the displayed comparison, without asserting that it is sharp.

Using this comparison to sum (HR18.33) gives a series with terms bounded by

$$C(1+R)^4\alpha_k \exp\left\{-\left[\kappa_a - \log A_4 - C\kappa(1+R)^4\right]n + C\kappa(1+R)^4\right\}. \tag{HR18.35}$$

The bracket is not positive uniformly as $R \to \infty$. Choosing a larger but fixed M or a larger but fixed source rate does not fix this running loss. Even improving the elementary comparison using a fixed-width collar leaves a growing volume-to-boundary ratio for four-dimensional boxes. A separate factor e^{-cR} in α_k cannot compensate an exponential full-volume cost per arbitrarily many component attachments. This is a failure of this available majorant, not a lower bound on the actual Yang–Mills activities.

The primary source displays the missing payment explicitly. On LFR II p.389, before (1.95), a factor

$$e^{2\kappa(100R_k)^4} \tag{HR18.36}$$

is paid, in that particular expansion, by the unnormalized suppression factors for the newly summed small components X_j: the product over their indices h in (1.93) is combined with the first product in (1.92), using the bound (1.89). It is not a statement that every retained Y_i has the same suppressed mass. In the normalized alternative on p.391 the Y_i domains are fixed; the corresponding first sums and first suppression product are absent, and the source explicitly explains why its unmarked convergence does not require decay for those fixed domains.

Thus a small-component factor, when available in the unnormalized full law, is potentially useful to a full-law marked-history estimate. Once that component is normalized as in (HR18.2), its mass has canceled. For the fixed Y_i alternative there is also no newly supplied full-interior payment in the normalized expansion. The relatively weighted scalar theorem CR.20–CR.21 deliberately treats such a component as an incidence atom and does not prove (HR18.33).

An explicit operation cost growing like $e^{CR_k^4}$ can be recorded for a bounded one-component core. It does not provide the uniform strong estimate (HL.42), and does not by itself pay the cost of arbitrarily many external component footprints. A valid full-law reserve for those costs belongs to the H-02/H-04 analysis unless a stronger internal marked locality theorem is proved first.

6. Why control of the nonsmall base is a separate mathematical input

The preceding obstruction is not merely an unfortunate bound for the cross action. The $n = 0$ terms in (HR18.11) already contain the normalized self-component law. The following finite probability example isolates the logical role of its external locality.

Let $b_1, \dots, b_N$ be retained signs at distinct sites in one component, and let an old sign σ be stored at an insertion anchor. Fix $a > 0$, and define the positive component measure

$$\nu_b(\sigma) = \frac{e^{a\sigma \prod_{j=1}^N b_j}}{2\cosh a}, \qquad \sigma \in \{-1, 1\}. \tag{HR18.37}$$

Its total mass is one, its cross interaction is zero, and it is independent of any additional current/native variable. Its scalar current Hessian and all scalar native responses are zero. For the bounded local input $F = \sigma$, however,

$$Q_b F = \tanh(a) \prod_{j=1}^N b_j. \tag{HR18.38}$$

Any exact carrier representation of (HR18.38) in which an activity on a proper subset of the b_j is independent of an omitted b_j must include activities whose carrier contains all N sites. Apply the Walsh coefficient

$$\mathcal{W}(f) = 2^{-N} \sum_{b \in \{-1,1\}^N} f(b) \prod_{j=1}^N b_j. \tag{HR18.39}$$

It annihilates every activity missing one site, and $|\mathcal{W}(f)| \le \|f\|_\infty$. Thus the sum of the supremum norms of the full-footprint activities is at least $\tanh(a)$. Their weighted supremum carrier sum is at least

$$e^{\kappa N} \tanh(a). \tag{HR18.40}$$

With the M-03 constant-regulator convention $\widehat{R} = 2$, this gives the same lower bound up to the fixed factor $1/2$ for the pointwise value part of that norm. The input norm can be bounded independently of N: its only active coordinate is the local sign σ, its value and discrete-difference envelopes are bounded, and its regulator can be constant. The example therefore excludes a uniform strong-norm contraction deduced solely from positivity, normalized scalar control, an admitted irrelevant analytic chart and small cross interactions.

This example is not a Yang–Mills model, does not claim that the actual source component has a parity interaction, and does not disprove the intended H-03 theorem. It shows why a marked locality/conditional-regulator argument for the actual nonsmall component cannot be replaced by a scalar boundary estimate.

7. What remains to apply the result in the physical M-03 framework

The exact decorated transport rule remains

$$\delta_\tau(QA) = Q_{\tau x}(B_\tau + J_\tau) + \frac{Q_{\tau x} - Q_x}{t}(A) \tag{HR18.41}$$

for a continuous step $t \neq 0$, with denominator one for a discrete change. It keeps the direct record, previous current and new normalized-kernel current separate. The analytic z-derivative in Theorem HR18.2 estimates the newly generated current only for the admitted fixed-support native variation with the input frozen.

Four actual transfer questions are still unresolved by the sources and arguments in this note.

1. The preintegration marked map. The completed scalar C-05 representation is an exact marginalization. Its scalar gauge extension fixes surviving raw slots. For an arbitrary incoming mark, one still needs a common preintegration chart, or an exact marked coordinate/gauge transformation, for every old variable which the mark uses. The map must stay in LFR II's admitted domain uniformly over its old histories and must keep the direct pullback and its full chain rule. Scalar equality after integration does not identify the marked complex integrand.
2. Raw retained directions and interfaces. Current/native deformations which keep raw retained slots and the outer geometry fixed do not cover every M-03 elementary direction. A retained raw field can change the nonsmall self law and its real reference. A hard selector or a geometry change may change support. The signed measure in (HR18.41), including its normalizer, then replaces a formal analytic score. An improved off-support extension can remove artificial jumps only if it agrees on the entire physical support and has a proved margin. It cannot remove a genuine on-support jump without a corresponding boundary-current estimate.
3. The pointwise regulator. Theorem HR18.2 controls the averaged part of the norm. It does not establish an estimate of the form
$$Q_x\big(\widehat{R}^m_{h,X} \circ \chi\big) \leq C^{m+1}(m!)^r \widehat{R}_{h,Y}(x)^m, \tag{HR18.42}$$
including the actual scale factor, chart path lengths and output footprint. The old self action can contain many old coordinates. Full-law L^p contraction gives no uniform pointwise conditional inequality such as (HR18.42).
4. The complete physical carrier transfer and payment. For the repeatedly normalized representation chosen in C-05, first identify the actual preintegration marked carrier family with its surviving older tags and prove the corresponding weighted cross sum; the original fixed-geometry count (HR18.27) is not a substitute for that check. Then the relative sum has to be upgraded using accurate physical dependency footprints or paid by a rigorously proved full-law marked reserve. The strong carrier series (HR18.33) and the base factor W_I must be bounded. The source's unnormalized mass cannot be used after it has canceled in the conditional law.

The current-scale incoming budget is a hypothesis of an operator estimate such as H-03. Proving that budget at all earlier scales remains M-07 and the matching/renormalization estimates. This note neither assumes that full induction has already closed nor relabels a bare cutoff budget as a current-scale one.

8. Sector hypotheses and consequences

Required item	Result in this note
Exact old posterior and all normalizers	Kept in (HR18.1)–(HR18.7).
Nonsmall self action and old histories	Kept in the positive base; $n = 0$ cumulants are not dropped.
Every higher incoming source vertex	Included through all partitions in (HR18.11).

Required item	Result in this note
Exact thin carrier representation	Proved in (HR18.10)–(HR18.15); no full-radius ball enlargement.
Absolute convergence and fixed source factorial growth	Proved by (HR18.22)–(HR18.25), with convergence radius in the cross-action parameter strictly greater than one.
Actual source-native relative carrier sum	Proved in Corollary HR18.3 for the specified primary fixed component operation and its literal carrier family, with all its internal histories retained.
Repeatedly normalized older-tag geometry	The algebraic theorem applies if its own actual preintegration weighted sum is supplied; no automatic application of the original HL.G3 count is made.
Native kernel derivative and finite-difference records	Proved with their explicit Cauchy-radius cost; input differences remain separate.
Actual preintegration physical marked pullback	Still open.
Raw retained/support-changing currents	Still open.
Pointwise regulator and full-volume weighted norm	Still open; their exact inequalities are displayed.

This annex supplies an additional actual source-level posterior theorem and a precise stronger application test. It does not change any catalogue statement, application criterion or dependency.

Primary sources and local proof dependencies

- T. Balaban, *Large Field Renormalization. I. The Basic Step of the R Operation*, CMP 122 (1989), 175–202: eligible components, p.177; exact replacement, (1.100)–(1.102), p.201.
- T. Balaban, *Large Field Renormalization. II. Localization, Exponentiation, and Bounds for the R Operation*, CMP 122 (1989), 355–392: (1.67)–(1.69), pp.376–377; (1.71)–(1.75), pp.378–380; the component-volume payment on p.389; (1.103)–(1.104), pp.391–392.
- T. Balaban, *Renormalization Group Approach to Lattice Gauge Field Theories. I*, CMP 109 (1987), 249–301: p.257 defines the finite connected M-cube carriers and their tree size. DOI `https://doi.org/10.1007/BF01215223`.
- Annex AG: exact posterior HL.1, source domains HL.6–HL.7, eligible geometry HL.8–HL.9. These scopes are retained.
- Annex AK: the moment-only fixed-chart reset theorem HP.1–HP.3, and the earlier radial-ball summation warning.
- Annex AU: CR.15, actual completed-carrier coefficient extraction; CR.20–CR.21, the deliberately relative scalar cluster norm.
- Annex AW gives the conditional scalar boundary estimate. The marked numerator requires separate control.
- Annex Y: the unchanged strong norm M.11–M.12 and exact decorated conditional transport M.28–M.42.

The finite probability and combinatorial arguments in Sections 2–3 and 6 are proved in this note. The relative-geometry sum in Section 4 is proved from the explicitly identified primary-source geometry and cross-action bounds; it is not an imported marked Yang–Mills theorem.

Annex BA — Marked history geometry, selector currents and regulator limitations

Annex BA. Exact finite conditional identities apply to the specified positive kernel. A uniform Wilson application requires CL on the same retained sigma-field, the relevant complete-potential estimates under CR and the stated spatial and energy budgets. History estimates · Observable transport.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

1. Result and precise scope

The completed scalar C-05 theorem does not by itself complete the marked history reset. Three distinct operations must be separated:

1. The normalized old-component operation in its own fixed native source coordinates. The primary source supplies a pointwise exponential tilt on a common support, including its nonsmall self action. This gives domination for arbitrary marked numerators, not merely for the unmarked scalar integral.
2. Its representation in the actual current coordinates before its hidden variables are integrated. This additionally requires a marked coordinate/gauge identity. The C-05 proof explicitly used exact integration to avoid a pointwise complex-gauge assertion for certain hidden raw fields.
3. Variation of the remaining raw data or the live selector/geometry labels. These changes can move supports and normalizers. Neither a native analytic chart nor a scalar boundary Hessian bound controls them automatically.

The source analysis below identifies an actual selector exception to a blanket gauge-invariance argument. The elementary results proved here give a guarded extension on a product current domain, the exact selector-gluing identity, a quantitative obstruction to bounding the M-03 shifted-history norm from full-law moments, and a necessary payment for a connected response inside a nonsmall component. They are useful checks on an H-03 proof. They are not counterexamples to Yang–Mills.

2. The primary pointwise normalized-kernel statement

Fix a component X and the outer geometry in Balaban's LFR II (1.71)–(1.72). Let η comprise the old component's continuous variables, discrete histories, and graph choices. The real positive unnormalized operation is $M_0(d\eta)$, including the old self-component action. In the native analytic variables used on printed pp.379–380, the source writes

$$M_z(d\eta) = e^{\sigma(z,\eta)} M_0(d\eta), \qquad \sigma(0,\eta) = 0, \qquad \sup_{|z|<\rho,\eta} |\sigma(z,\eta)| \le s. \tag{HG18.1}$$

The support and all the factors outside the exponent are fixed in this particular chart. Printed p.379 says that only the quadratic form in B and the self term $V(X^{\sim})$ depend on (U,J); their first-order variation is small uniformly in the variables still integrated. The estimate of $V(X^{\sim})$ is not an assertion that its value is small. Its value stays in M_0.

The formulas (1.73)–(1.75), printed p.380, apply to an arbitrary function of the integration variables. Thus they justify a marked domination statement in their actual native chart. They do not identify the chart with every possible retained-link variation.

Include the actual cross action $H_z(\eta) = \sum_{Y \ni X} V_Y(z,\eta)$, with a summable source majorant

$$\sup_{|z|<\rho,\eta} |H_z(\eta)| \le \lambda, \qquad \pi_0(d\eta) = \frac{e^{H_0(\eta)} M_0(d\eta)}{M_0(e^{H_0})}. \tag{HG18.2}$$

The exact old posterior is

$$Q_z F = \frac{\pi_0(e^{\Theta_z} F)}{\pi_0(e^{\Theta_z})}, \qquad \Theta_z = \sigma(z,\cdot) + H_z - H_0. \tag{HG18.3}$$

Set $r_* = \log(3/2)$ and

$$\rho_* = \frac{\rho r_*}{s + \lambda + r_*}. \tag{HG18.4}$$

The Cauchy coefficient estimate for the analytic function $\sigma + H$ gives

$$\|\Theta_z\|_\infty \le (s+\lambda)\frac{|z|}{\rho - |z|} \le r_* \quad (|z| \le \rho_*).$$

Consequently

$$\left|\frac{dQ_z}{d\pi_0}\right| \le 3, \quad |\partial_z^q Q_z F|_{z=0} \le 3q!\rho_*^{-q}\pi_0|F|, \tag{HG18.5}$$

and, by summing the Taylor coefficient majorants,

$$\sup_{0<|t|\le\rho_*/2}\left|\frac{Q_t F - Q_0 F}{t}\right| \le \frac{6}{\rho_*}\pi_0|F|. \tag{HG18.6}$$

Indeed $|\pi_0(e^{\Theta_z})| \ge 2 - e^{r_*} = 1/2$, whereas $|e^{\Theta_z}| \le 3/2$. The Cauchy estimate may then be applied to the density relative to π_0, before multiplying by F and integrating. This proves (HG18.5)–(HG18.6) without assuming that the mark is analytic. Under a full incoming law $\mu(dx, d\eta) = \nu(dx)\pi_x(d\eta)$, conditional Jensen gives the corresponding $L^p(\nu)$ bound by $3q!\rho_*^{-q}\|F\|_{L^p(\mu)}$, when the displayed radius is uniform in x.

These statements recover the HP.1 mechanism on its exact source scope. Higher input vertices and the complete finite reset jet are handled by HP.2–HP.3; no new partition loss is introduced here.

What C-05 now supplies, and what it does not identify C-05 supplies the full $2P_k$ current contour, a whole strong native ball, normalized endpoint estimates, and a real gauge identity for every scalar primitive with its surviving old arguments fixed. See AB17.8–AB17.13, AD17.3, and CR.2. A uniform factorization

$$M_{\chi(z)}(d\eta) = e^{\sigma_\chi(z,\eta)} M_{\chi(0)}(d\eta) \tag{HG18.7}$$

on a single padded hidden-variable space, with the source bound in (HG18.1), would apply (HG18.5) to the actual current map χ. The required checks are that the map lands in the source chart, that its change of variables preserves the common hidden support, and that the transformed numerator is the actual transported numerator. The map may be independent of the mark; its full direct response still has to be carried separately.

One must not replace these checks by the scalar result after integrating η. AB17 §2.3 expressly states that fine hidden raw fields inside the completed operation have already been integrated, so no pointwise complex-gauge claim was required there. AB17 §3.2 and CR.2 fix the *surviving* raw endpoints. They do not state (HG18.7) for every hidden endpoint of an arbitrary marked numerator.

3. The exact marked gauge-transfer lemma and its source limitation

Lemma HG18.1 — marked real gauge transfer before complex continuation Let G act measurably on a hidden-variable space. Let M_υ be positive finite measures, and suppose the exact kernel covariance

$$M_{\upsilon g} = g_* M_\upsilon \tag{HG18.8}$$

holds for a specified real gauge g. If F_υ is a covariant marked family,

$$F_{\upsilon g}(\eta^{g}) = F_{\upsilon}(\eta), \tag{HG18.9}$$

then

$$\frac{M_{\upsilon g}(F_{\upsilon g})}{M_{\upsilon g}(1)} = \frac{M_{\upsilon}(F_{\upsilon})}{M_{\upsilon}(1)}. \tag{HG18.10}$$

The same identity holds coefficientwise for the full exponential source jet and its finite logarithm. If a real family $\upsilon(z)$ can be replaced by a source-admitted family $\upsilon(z)^{g(z)}$, where (HG18.8)–(HG18.9) hold for every real z, the source chart can be used to define the holomorphic kernel after this exact real change, provided its stated fixed-support source extension is the one under consideration. Estimates for an arbitrary frozen numerator then follow from (HG18.5). A moving marked family still requires its own coordinate regularity; covariance alone does not make it holomorphic. No change of a compact Haar contour by a complex gauge is used.

Proof. Apply (HG18.8) to the numerator and denominator. The pushforward formula and (HG18.9) identify both integrals. Finite jets are finite algebraic combinations of these identities. □

For an arbitrary frozen function F, rather than the covariant family (HG18.9), the transformed numerator is $F\circ g(z)^{-1}$. Its direct finite difference is present. The lemma does not set that term to zero, and it does not say that separate carrier records obtained by an arbitrary localization are individually gauge invariant.

The source factors that can be checked directly Normalized Haar measure is invariant under real left/right gauge multiplication. Relative Lie-algebra variables transform by the adjoint action, which is orthogonal in the source invariant norm. Their norm cutoffs, zero tree constraints, and covariant quadratic forms therefore transform correctly. Group averaging constraints are covariant under the actual restricted fine gauge; the scalar E/R/B action obeys the source's simultaneous real gauge covariance. The localized R reference numerator and denominator must be transformed together. These identities explain why (HG18.8) is a plausible target for a correctly augmented source construction.

They do not check every archived selector.

The explicit selector exception CMP119, printed p.250, immediately after (1.16), distinguishes two invariance statements. The expressions in (1.14)–(1.16) are invariant under the simultaneous fine gauge transformations (1.17), except for the approximate-fluctuation characteristic function $\chi'_0(Q_1)$. That characteristic function is invariant under a gauge of the new coarse field together with its block-constant extension u_0. The source then introduces the free-field split (1.18) so that the problematic selector can be replaced on the regular cells. The general-step counterpart is the use of (3.16)–(3.19).

This is the precise reason an arbitrary smooth residual gauge from the C-05 converter cannot simply be declared to preserve all bad-history selectors. Archived versions of those factors are part of the earlier operations in (2.21) and of the component operation (1.71). Their fixed-component sums or their gauge disintegrations must be checked at the marked level. Gauge invariance of the original Wilson observable alone is insufficient to prove a statement about a conditional kernel whose selector event is not invariant under that gauge.

There are two possible repairs, neither silently assumed here:

- Prove that summing the actual internal histories of the fixed component restores (HG18.8), retaining the eligibility tests and every normalization; or
- Carry the induced selector/support change as a signed-kernel current and prove its required bounds.

The complete regular marginal has already removed its inner selectors in C-07. That theorem is not a proof that every archived component-conditioned kernel has the same support.

4. A guarded off-support extension that is genuinely local in current coordinates

The M-03 default reference extension can create an artificial jump outside an active chart. There is a useful repair when a full product current margin has actually been proved.

Lemma HG18.2 — product-current clipping Let the real current coordinates be a finite product of normed finite-dimensional spaces. Suppose a positive probability kernel Q_z is defined for

$$\mathcal{D}_{2P} = \{z : \|z_i\| < 2P \text{ for every } i\}, \qquad P \geq 4,$$

and has a bounded analytic continuation on that product domain. The physical support is contained in $\overline{\mathcal{D}_P}$. Define, independently at each coordinate,

$$T_i(z_i) = \begin{cases} z_i, & \|z_i\| \leq P+1, \\ (P+1)z_i/\|z_i\|, & \|z_i\| > P+1. \end{cases} \tag{HG18.11}$$

Then T is the identity on the physical support and on each of its one-coordinate perturbations of size at most one. Each T_i is 2-Lipschitz in its norm, so the extension $\widetilde{Q}_z = Q_{Tz}$ has bounded finite differences whenever Q_z has bounded first derivatives on $\overline{\mathcal{D}_{P+1}}$. The extension introduces no new coordinate dependency: T_i depends only on z_i.

If for a frozen marked numerator one has

$$|D_i Q_z F| \leq L_i(F) \quad (z \in \overline{\mathcal{D}_{P+1}}),$$

then

$$|\widetilde{Q}_{z+te_i}F - \widetilde{Q}_z F| \leq 2|t|L_i(F). \tag{HG18.12}$$

Proof. The radial retraction onto a norm ball has Lipschitz constant at most two: add and subtract the point obtained by using the first radius at the second point, and use the reverse triangle inequality for the norms. Apply the ordinary line-segment derivative bound to the two retracted points. The other conclusions follow coordinatewise. □

This can remove a default extension artifact for an already verified product-current kernel. It is not a proof of (HG18.7). It does not give a uniform extension in arbitrary raw retained variables. In particular, radial retraction of an entire strong native field norm would generally depend on every block and hence could create a global carrier; that operation is not substituted for the coordinatewise construction above.

5. Exact selector gluing and why normalized branches do not glue by themselves

Lemma HG18.3 — unnormalized marked gluing Let $r_y(d\eta)$ be a positive finite measure and let $\chi_h(y,\eta) \geq 0$ be a finite partition of unity. Put

$$r_{h,y} = \chi_h r_y, \quad Z_h(y) = r_{h,y}(1), \quad Z(y) = \sum_h Z_h(y) > 0.$$

On active fibers define $q_{h,y} = r_{h,y}/Z_h(y)$ and $p_h(y) = Z_h(y)/Z(y)$. For every integrable F_y,

$$\sum_h p_h(y) q_{h,y} F_y = \frac{r_y(F_y)}{Z(y)}. \tag{HG18.13}$$

Zero-mass branches contribute zero without assigning their conditional means a canonical value. Every derivative or finite difference must be taken on this exact weighted identity. On a smooth common support it includes both $\sum_h p_h D(q_h F)$ and $\sum_h (Dp_h) q_h F$. Across hard interfaces it is the unnormalized marked numerators that must glue.

If a replacement is geometry dependent, with fresh density $k_h(u' \mid y)$, the actual posterior weights are instead

$$p_h(u', y) = \frac{Z_h(y) k_h(u' \mid y)}{\sum_l Z_l(y) k_l(u' \mid y)}. \tag{HG18.14}$$

The fresh factors cancel only inside a fixed-h old-field conditional expectation. They do not cancel from (HG18.14).

Proof. Multiply each active conditional expectation by its weight and cancel Z_h; then sum the unnormalized measures. Formula (HG18.14) is Bayes' formula for the actual joint positive measure. □

A default extension can fail even when the unsplit law is harmless Let u be uniform on $[0,1]$ and, for $-1/2 < y < 1/2$, select the branch

$$h = 1 \quad \Longleftrightarrow \quad y > 0 \text{ and } u > 1 - y.$$

For $y > 0$, its mass is y and $q_{1,y}u = 1 - y/2$. Extend the inactive kernel for $y \le 0$ by the reference point mass at $u = 0$. The conditional marked function jumps from zero to one at $y = 0$. At $0 < y < 1/2$, the allowed increment $t = -2y$ gives

$$\sup_{0<|t|\le 1} \left| \frac{q_{1,y+t}u - q_{1,y}u}{t} \right| \ge \frac{3}{8y}. \tag{HG18.15}$$

On the actual branch law, whose y-density is proportional to y, the p-th moment of this envelope diverges for every $p \ge 2$. Yet summing the two branches with their exact weights gives the original conditional mean $1/2$. Both facts follow by elementary integration.

This example does not prove that a better extension is impossible. Here extending by the point mass at $u = 1$ on the inactive side fixes the discontinuity for Lipschitz marks. It proves that the reference convention and an interior analytic estimate alone do not supply the M-03 finite-difference bound. An actual matching extension or the full weighted gluing must be exhibited.

There need not always be such a continuous extension. The positive two-state unnormalized weights y_1^2 and y_2^2, with mark $+1$ and -1, give

$$Q_y F = \frac{y_1^2 - y_2^2}{y_1^2 + y_2^2} \quad (y \ne 0). \tag{HG18.16}$$

The numerator and the two weights are entire, but the normalized mean has different limits along the two axes. A common physical margin is therefore substantive information.

6. Full-law moment bounds do not control shifted retained-history envelopes

Proposition HG18.4 — one-block regulator and shift obstruction There is a family of exact positive one-block resets with uniformly bounded incoming M-03 value/direct-difference norm, all unshifted full-law moment bounds of the stipulated degree, and no cross interaction, for which the outgoing M-03 norm is unbounded.

Construction and proof. Take the current scale $h = 1$, one Euclidean coordinate z, and one discrete retained label $y \in \{0,1\}$. Let

$$\Pr(y = 1) = e^{-n}, \qquad z = \sqrt{n}\, y, \qquad n \ge 1. \tag{HG18.17}$$

Use $R_{\text{in}} = 1 + z^2$, the exact M-03 elementary translations $z \mapsto z + t$, $|t| \le 1$, and the discrete replacement of y. Define the degree-one input

$$A(z,y) = z^2, \quad B_{z,t} = 2z + t, \quad B_y = 0, \quad J = 0. \tag{HG18.18}$$

The input value envelope is $(|z|+1)^2$, and the direct-difference envelope is at most $2|z|+1$. The regulator quotients in M.11 are bounded by fixed constants on the entire ambient space. For the averaged value part, the only nonconstant term is controlled by

$$\sup_{p\ge 1} \frac{n e^{-n/p}}{p} = \sup_{0<x\le n} x e^{-x} \le e^{-1}. \tag{HG18.19}$$

The analogous bound for $\sqrt{n} e^{-n/p}/p$ is smaller. The grade-3/2 direct-difference moment is uniformly bounded too, because

$$\sup_{p\ge 1} p^{-3/2}\sqrt{n}\, e^{-n/p} = n^{-1} \sup_{0<x\le n} x^{3/2} e^{-x} \le C. \tag{HG18.20}$$

Thus the incoming single-carrier M-03 norm is bounded independently of n.

Now integrate z conditionally on the retained y. The exact normalized kernel is $q_y = \delta_{\sqrt{n}y}$, so

$$QA(y) = ny. \tag{HG18.21}$$

The output has only the discrete label, hence the specified native output regulator is $R_{\mathrm{out}} = 1$. The allowed discrete replacement of y makes the output sharp value envelope equal to n at both values of y. Its degree-one moment norm is therefore n, already at $p = 1$. The new kernel-current record under the label replacement has magnitude n as well. Hence the output M-03 norm diverges.

On the other hand, the *unshifted* output moments obey

$$\|QA\|_{L^p} = ne^{-n/p} \le p/e, \tag{HG18.22}$$

uniformly. Conditional moment contraction is completely consistent with the failure of the shifted envelope and pointwise regulator. □

The example is an abstract positive reset, not the Wilson law. It proves that the stated H-01 identities, unconditional moment estimates, and harmless scalar boundary action do not imply the missing M-03 transport. It also explains why the actual remaining-history normalization matters. Erasing the label y jointly would give the small constant ne^{-n}; retaining it gives (HG18.21). A proof may choose an exact joint erasure, but it must then estimate the actual posterior (HG18.14), rather than reuse the fixed-label conditional law.

7. A nonsmall component needs a genuine connected-carrier payment

The source's normalized component base may contain arbitrary correlations under the currently imported estimates. To isolate the logical implication, take a connected component with two local hidden spins at sites x and y, at distance r, with law assigning probability $1/2$ to each of $(+1,+1)$ and $(-1,-1)$. Their individual coordinate footprints and incoming marks are local. Put all cross actions equal to zero. Then

$$\kappa(\eta_x, \eta_y) = 1. \tag{HG18.23}$$

Every representation of this connected degree-two response by carriers containing the two prescribed anchors has M-03 geometric weight at least $e^{\kappa(2+r)}$. Indeed $|X| \ge 2$ and $\mathrm{t}_h(X) \ge r$; the triangle inequality prevents a sum of absolute coefficients from costing less than the magnitude one of the response. Thus a fixed-constant strong connected-carrier bound cannot be inferred from the cross-interaction estimates alone when the allowed component diameter grows.

The eligible source components have diameter $O(MR_k)$, so a radius with that explicit running cost is possible. A uniform original M-03 bound needs either an internal marked estimate, a genuine full-law probability payment made before conditional normalization, or a proved alternative carrier representation with the same original requirements. The small masses of the newly summed successful components used in LFR II pp.388–389 have canceled when those same components are normalized conditionally and cannot be spent a second time. Fixed surviving components have a different role: the alternative on p.391 leaves their domains fixed and does not require a tree factor for those domains in that unmarked expansion. The scalar C-05 estimate deliberately uses a weaker exterior-component carrier norm; it does not prove the M-03 exponential volume and connecting-tree bound.

8. Application to the conditional law

The following conclusions are justified:

Item	Scope
Arbitrary marked numerator in the source's fixed native component chart	The pointwise tilt and domination are primary-source statements; (HG18.3)–(HG18.6) retain all normalizers.
Actual current pullback before completed-component integration	Requires the common hidden-variable identity (HG18.7), or the fully checked marked gauge transfer. C-05's integrated scalar identity alone does not identify it.
Product-current off-support convention	Lemma HG18.2 supplies a local clipping repair once the actual full kernel chart exists.
Blanket fine-gauge invariance of archived selectors	Invalid as an unqualified inference; CMP119 p.250 states an explicit exception.
Erasing geometry/selector labels	Lemma HG18.3 gives the exact weighted posterior and the needed gluing identity. Its actual strong bounds remain to be proved.
Pointwise regulator and shifted remaining-history norm	Not supplied by full-law moments; Proposition HG18.4 gives an explicit obstruction to that inference.
Exponential connected-carrier payment for nonsmall components	Not supplied by the scalar exterior-component norm or the small cross action alone.

The remaining source-specific work is therefore concrete: prove the marked common-coordinate identity through the actual archived selectors; control the raw retained/geometry directions, including shifted kernels and regulator transport; and pay the connected-carrier norm with the actual normalized law. The thin interaction expansion of Annex AZ can improve the carrier representation but does not remove these independent requirements.

Sources

- T. Balaban, *Convergent Renormalization Expansions for Lattice Gauge Theories*, CMP 119 (1988), 243–285: printed p.250, (1.16)–(1.18), for the explicit selector covariance exception; (2.21), (2.40)–(2.42), for old operations and scalar primitive covariance; (3.16)–(3.19), for the removal of the old approximate selector on regular cells.
- T. Balaban, *Large Field Renormalization. I*, CMP 122 (1989), 175–202: (1.100)–(1.102), printed p.201, for the localized positive quotient and its cutoff; the relative cutoff is a condition on the actual log coordinate.
- T. Balaban, *Large Field Renormalization. II*, CMP 122 (1989), 355–392: (1.65), printed p.375, for the source prepared-domain map; (1.68)–(1.75), printed pp.377–380, for the self/cross distinction and the positive normalized component tilt; pp.388–389 and (1.103)–(1.104), pp.391–392, for the component-mass and fixed-domain limitations.

All counterexamples in this note test implications between stated abstract estimates. None is asserted to model or disprove the actual Yang–Mills construction.

Annex BB — Exact marked reset maps and selector currents

Annex BB. Exact finite conditional identities apply to the specified positive kernel. A uniform Wilson application requires CL on the same retained sigma-field, the relevant complete-potential estimates under CR and the stated spatial and energy budgets. History estimates · Observable transport.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

0. Scope

This note does not close H-03. It identifies exactly which fresh-coordinate currents vanish in the literal positive R operation, derives the complete selector-crossing current, checks why summing the internal histories of a fixed eligible component does not remove its selection, and gives an explicit positive joint-history reset that preserves the entire source-free output law while changing the marked coupling. It also separates the actual fixed-size continuous integration core from the larger retained-data and history-label footprint.

The last construction is a possible alternative. Its current, regulator and spatial estimates have not been proved. It is not substituted for the fixed-component kernel in the H-03 argument of the earlier annexes.

The source-free construction and admissibility assumptions retain the scope of C-01. Page references below are to the printed pages of Balaban, CMP119 (1988), 243–285, and Large Field Renormalization I and II, CMP122 (1989), 175–202 and 355–392. The marked algebra below is proved directly.

1. What the actual source keeps fixed

LFR I printed p.177 defines an eligible component as a whole component of the existing large-field region. It must fit in a cube of side 100 M R, and no new large-field region has been created inside it during the prescribed preceding N steps. These conditions select histories; they do not declare their union to be the unconstrained history space.

Printed pp.200–201 and (1.99)–(1.100) separate the external component domains from the remaining history sums. The localized replacement reference contains its actual integration region, axial tree, cutoff and background. In particular, the component name X alone is not a proof that two history branches use the same reference kernel. The reference data in (1.100) occur within the remaining history sum. Printed p.177 also explicitly distinguishes this localized reference from the schematic full density in the introduction.

LFR II printed pp.378–379 and (1.71)–(1.72) integrate and sum internal histories while holding the eligible component geometry fixed. The cross action is acted on by the same history sum. It is therefore incorrect to replace the fixed-component sum by an unrestricted partition of unity, or to take the old posterior to be the fresh reference law.

The exact positive quotient operation nevertheless provides a useful stronger statement for a correctly fixed reference descriptor. It is the next theorem.

2. The exact reset when its reference descriptor is fixed

Let gamma collect every discrete datum that selects the replacement kernel: the integration region, the relevant determining geometry, axial/tree convention and any other branch label on which that reference depends. Continuous retained arguments are denoted v. The definition of gamma is deliberately complete; a coarser label is permitted only after proving that it determines the same reference.

Write u for the old integrated variables, including any remaining old labels whose summation does not change this reference, and u’ for the fresh variables. On an active fiber let

$$m_\gamma(v) = \int \rho_\gamma(u, v)\, du > 0, \qquad k_\gamma(u' \mid v) \geq 0, \qquad \int k_\gamma(u' \mid v)\, du' = 1. \tag{MG19.1}$$

Here rho is the entire old positive baseline density at this operation. Any incoming source-free boundary exponential must be included in rho before the posterior is formed. The kernel k is the localized source reference h/H, with the source’s numerator and denominator retained together.

Theorem MG19.1 — all pure fresh-coordinate reset currents vanish The literal replacement and its marked quotient are

$$(P_\gamma f)(u', v) = k_\gamma(u' \mid v) \int f(u, v)\, du, \quad \rho'_\gamma = k_\gamma m_\gamma, \tag{MG19.2}$$

$$Q_\gamma F(u', v) = \frac{P_\gamma(\rho_\gamma F)}{\rho'_\gamma} = \frac{\int \rho_\gamma(u, v) F(u, v)\, du}{m_\gamma(v)}. \tag{MG19.3}$$

Consequently the exact reset output is independent of the fresh u' coordinates. Every finite difference or derivative in a pure fresh direction is zero, wherever the same descriptor and retained datum are used. This holds for every coefficient of the full transported finite source jet and of its formal logarithm.

Proof. Both numerator and denominator contain the same factor k. It cancels on their common nonzero support. The remaining expression has no u' argument. On null output fibers one may use that same right-hand side as an extension; a newly chosen unrelated reference value is unnecessary. Integrating (MG19.2) first in u' proves exact integral preservation. Applying it coefficientwise to a finite source polynomial proves the full-jet statement. Its constant term is one, so the finite logarithm exists and is again independent of u'. All its covariance and higher response vertices are retained by that algebraic operation. No source-series convergence is assumed. □

This is an application to the actual positive quotient in LFR I (1.100)–(1.102), not a claim that the displayed factor T'(X) in a later refactorization is separately independent of every new background argument. If the source divides out a new effective action and moves factors into a boundary exponential, all those factors must be recombined in the actual baseline conditional law before applying (MG19.3).

The conclusion does not make retained v currents vanish. It also does not apply after summing descriptors whose references differ. Coordinate changes that mix fresh variables with v have their own direct pullback records. The theorem identifies the exact zero-current subspace of the reset and prevents a redundant estimate of fresh-reference derivatives that cancel from its true marked quotient.

For an incoming logarithmic jet with coefficients A_B, the output coefficient is the usual full conditional-cumulant partition sum under the old law in (MG19.3). The vanishing above includes every term in that full sum; it is not obtained by retaining only the first input mark.

3. Descriptor changes and the full normalized selector current

3.1 A changing reference cannot be canceled across branches When gamma is itself erased, the actual old posterior at a fixed fresh output is

$$\mathbb{P}(\gamma, du \mid u', v) = \frac{k_\gamma(u' \mid v)\rho_\gamma(u, v)\, du}{\sum_\delta k_\delta(u' \mid v) m_\delta(v)}. \tag{MG19.4}$$

Thus its descriptor probabilities are proportional to m_gamma times k_gamma. A common fixed-component symbol does not remove the second factor. This formula is Bayes' rule for the actual joint positive density and is valid for discrete, continuous or padded sector inputs.

If the references happen to coincide on the erased family, their common factor cancels and the posterior weights reduce to m_gamma divided by their sum. That is a special case to be proved, not a convention for a normalized component.

3.2 Exact finite differences, including moving support Let M_0 and M_t be two positive finite unnormalized kernel measures on a common padded space, with masses Z_0,Z_t>0, and put Q_s=M_s/Z_s. For marks F_0,F_t the exact identity is

$$\frac{Q_t F_t - Q_0 F_0}{t} = Q_t \frac{F_t - F_0}{t} + \frac{1}{tZ_t} \int (F_0 - Q_0 F_0)\, d(M_t - M_0). \tag{MG19.5}$$

The first term is the direct marked response. If F already contains a previous kernel-current record, it is carried inside that direct difference according to its existing compatible decomposition; it is not folded into the newly generated current. The second term is the complete new normalized-kernel current. Equation (MG19.5) remains exact across hard selector interfaces and rank changes represented by the padded sectors.

Proof. Add and subtract Q_t F_0. Then write the difference of the two normalized expectations over the common denominator Z_t, using the equality M_0(F_0-Q_0F_0)=0. □

Suppose on a given comparison chart one can write

$$M_0(d\eta) = \chi_0(\eta) r(d\eta), \qquad M_t(d\eta) = e^{\sigma_t(\eta)} \chi_t(\eta) r(d\eta). \tag{MG19.6}$$

The selector weights may be indicators or positive sums of history weights. The exact new current in (MG19.5) splits into

$$\begin{aligned} J_t(F_0) = {} & \frac{Z_0}{tZ_t} Q_0\big[(e^{\sigma_t} - 1)(F_0 - Q_0 F_0)\big] \\ & + \frac{1}{tZ_t} \int e^{\sigma_t} (\chi_t - \chi_0)(F_0 - Q_0 F_0)\, dr. \end{aligned} \tag{MG19.7}$$

The first line is the smooth density response. The second is the selector/support response. For real positive data, the latter obeys the exact usable upper bound

$$|J_t^{\mathrm{sel}}(F_0)| \le \frac{1}{|t| Z_t} \int e^{\sigma_t} |\chi_t - \chi_0| \big(|F_0| + |Q_0 F_0|\big)\, dr. \tag{MG19.8}$$

This exposes a required crossing-measure or trace estimate. A bounded analytic tilt with fixed support controls only the first line unless chi_t equals chi_0. In particular, a representation M_t=exp(sigma_t) M_0 with uniformly bounded sigma_t requires equivalent supports. It cannot hold where a positive-measure selector branch is newly admitted or removed.

The pointwise native statement LFR II (1.73)–(1.75) is exactly of the fixed-support type. The paper says on p.379 that only its quadratic B form and self V term depend on the native pair (U,J). It supplies no estimate of (MG19.8) for a different physical change that moves an archived selector.

3.3 Why a fixed-component history sum is not the missing gluing For a full positive history partition, write chi_h for its weights and let Gamma(h) be the component/reference event being retained. For a prescribed event E, the sum over the histories compatible with E is

$$w_E = \sum_{h:\, \Gamma(h) \in E} \chi_h. \tag{MG19.9}$$

Only the sum over the whole partition is one. In general w_E is the probability weight or indicator of the retained selection. A fixed eligible component X is such an event. The source's internal sum does not include the histories with no such component: eligibility was imposed before the sum, and the component domains were explicitly separated from the remaining sums on LFR I pp.200–201.

Thus summing those internal histories can simplify selectors, but it does not justify setting the total selected weight to one. A proof that a particular current cancels must exhibit the relevant complete family of branches and its common marked numerator. Moreover, if the branch has a different fresh kernel, (MG19.4) supplies its additional weight.

This is an exact index-set check of the actual source operation. It is not a lower bound on the size of its current and is not a counterexample to Yang–Mills.

4. What gauge covariance is actually available

4.1 The archived approximate-fluctuation selector has a restricted symmetry CMP119 printed p.250, (1.17)–(1.19), explicitly distinguishes arbitrary simultaneous fine gauge covariance of the effective functions from covariance of the approximate-fluctuation characteristic function. The latter is preserved when a gauge of the new coarse field is extended constantly over its fine blocks. The displayed selector contains the product

$$\exp(igA(b))\, U_1(b)\, U_{1,\square}(b)^{-1}. \tag{MG19.10}$$

The local approximant is selected from the coarse field in its prescribed axial convention. Under a coarse gauge and its block-constant extension, the two backgrounds and the relative field transform together, and the invariant matrix norm in the selector is unchanged. Haar measure and the Lie-algebra norm are unchanged as well. This verifies the protected real transformation for that actual selector, not merely for a generic gauge-invariant cutoff.

An arbitrary residual fine gauge need not have this property. For example, a fine gauge may be the identity on coarse sites but nontrivial at an interior fine site. The raw averaged coarse field is then unchanged by endpoint covariance, while the prescribed axial approximant remains the one selected from that coarse field. The separately transformed U_1 factor in (MG19.10) need not transform with that fixed approximant. This explains the source's exception directly. The observation concerns the literal selector; it is not an assertion that every such configuration belongs to every later fixed-history fiber.

The source removes these selectors on the regular cells using (1.18)–(1.19). On the remaining cells, (1.21)–(1.25), printed pp.251–252, makes an exact small relocation so the selectors depend only on the raw variables in the large-field region. It does not prove invariance under every fine gauge. At a later current insertion those raw variables can be held fixed; changing them or using a different residual gauge still requires the corresponding marked pullback or selector current.

4.2 Preliminary R integration has a genuinely gauge-invariant subintegral LFR I p.179 explains why the preliminary integrations are needed: several older cutoff factors are not gauge invariant in the variables one would like to transform. Printed pp.194–196 then identify a specific result of those integrations. The second integral in (1.76), on the stated Bo domain outside the untouched old core, is gauge invariant. The first integral, containing the earlier operation T_h on the old core, is left unchanged. The source subsequently fixes the Bo gauge by the actual tree T_0.

For that gauge-invariant subintegral one may transport a mark through gauge fixing by the exact Haar orbit average. If a marked function is gauge invariant on that subintegral, its value is unchanged. For a general mark it must be replaced by its average over the removed gauge variables. This follows by the finite compact-group change of variables underlying the Faddeev–Popov identity; the Haar average is positive, sends one to one and preserves the marked integral. Its logarithmic source jets retain the generated gauge-orbit covariances.

This supplies a marked version of that specific source gauge-fixing operation. It does not make the untouched core selectors gauge invariant. In particular, the source's statement about the second integral cannot be applied to the first integral or to an arbitrary unintegrated marked kernel.

4.3 The proposed hole-patching shortcut needs an additional domain estimate The source input and output spaces must be kept separate. LFR II p.375, (1.65), has an ordinary new-field input space and a prepared old-field output space. The latter is the multiscale space defined in LFR I (1.64)–(1.69), pp.190–191. Its weaker old-region restrictions do not exempt the corresponding part of the ordinary input from the relative-field condition of CMP119 (2.39).

A proposed converter may agree with the old residual gauge at every hidden raw endpoint and interpolate across a component collar. To keep all old raw variables fixed by a double covariance argument, those endpoint values are prescribed. The sparse-node argument used after scalar integration in C-05 applies because surviving raw nodes have the required terminal mesh grading. Hidden nodes inside T_h do not have that proved grading relative to the newly regular input mesh.

There is an elementary necessary estimate. Let eta be a fine hidden mesh and h the relevant native input mesh. If a gauge interpolant v matches prescribed values at adjacent fine nodes x and x+eta e, then

$$h\,|D_\eta v(x)| = \frac{h}{\eta}\,|v(x+\eta e) - v(x)|. \tag{MG19.11}$$

A value bound on the nodal logarithms alone cannot bound this expression uniformly in h/eta. The analogous second-difference issue enters the native gradient row. A collar-small value exp(−cR) is not by itself a bound for all arbitrarily fine hidden nodes at fixed current R. The current-endpoint estimates in the completed scalar theorem are not spatial finite-difference estimates on those nodes.

This is a limitation of that deduction, not a proof that the actual converter has bad spatial differences. To complete the patching argument, one must prove either the appropriate spatial regularity of the actual nodal values, a controlled-depth reduction of all nodes that really have to be fixed, or a new marked common-domain construction. Fixing only the raw V_h endpoints is less costly, but treating the whole earlier marked T_h operation as unchanged then requires its own exact marked argument. LFR I (1.76) expressly retains that older operation.

4.4 A fixed continuous core is a real improvement, but is not the full kernel carrier LFR I printed p.192, (1.73), states that the prepared domain Lambda is contained in a cube of side 100 M in current units. The eligible original component X only had the larger bound 100 M R. The remaining coordinates of (1.76), printed p.194, comprise the old operation on Z_h and the B_0 variables on the complementary prepared shell. The nested prepared rectangles place these continuous integrations in Lambda. The final versions (1.100) and LFR II (1.71) redistribute an old V_h collar; that collar must be followed explicitly, or included in a fixed current-block enlargement of Lambda. It does not turn the integration volume back into the original 100 M R component. Annex BC, Proposition RN19.1, records that support bookkeeping.

This distinction is useful: for marks which depend on continuous hidden variables only, every input participating in a newly generated nonzero cumulant must meet this bounded physical core. A fixed connected cover can join those input carriers with a constant weight. It is unnecessary to pay the original component volume merely to connect continuous variables which are all located inside that core. The number of fine hidden coordinates inside the bounded region can still grow with the remaining depth, so the spatial nodal regularity issue in Section 4.3 is not resolved by the volume bound.

The source does not identify the full normalized kernel carrier with Lambda. LFR II (1.71) still sums the histories restricted to X, and its self action includes the older boundary terms localized in X. Most decisively, the actual native-variation connection set printed on p.379 is

$$\mathcal{B}_0 \ \cup \ (\Gamma_k'' \cap X^{\sim}), \tag{MG19.17}$$

with subscript k, as checked in the published formula. It is not the old h-scale core. The second set contains the current determining variables inside the component; the printed connection statement therefore does not localize every retained-variable response to Lambda. The associated decay is valuable, but its endpoint set and all full carrier weights must remain in the estimate.

A historical domain label may also describe X or a whole earlier layer even when its remaining continuous coordinates lie in Lambda. By M-03's definition, such a label keeps its full dependency footprint until an exact local reindexing or cancellation is proved. One may retain exterior history labels and reset only core labels, but the resulting theorem must then bound the dependence on those retained exterior labels. A scalar factor independent of all erased coordinates cancels from a normalized posterior. A scalar which varies among the histories being erased supplies their relative weights and cannot be canceled term by term.

The source's fixed-core fact removes a specific overly large connector cost. It does not, by itself, prove the missing preintegration common chart, history-current bound, or output-regulator comparison.

5. An exact joint-history reset preserving the full source-free output law

The previous obstruction concerns a termwise fixed-descriptor coupling. There is an explicit alternative coupling that shows how one could retain component masses before normalization without changing the unmarked output distribution. This construction is finite and exact. Its estimates are a separate question.

Let the complete family of old descriptors have positive joint density rho_gamma(u,v), and put

$$m(v) = \sum_{\gamma} m_\gamma(v), \qquad \rho'_\gamma(u', v) = m_\gamma(v) k_\gamma(u' \mid v). \tag{MG19.12}$$

The sums can be finite or countable with finite positive m on active fibers. The finite cutoff construction has a finite descriptor family after its actual discrete data are fixed. Define a source-independent positive map on the joint old state by

$$(P_{\mathrm{mix}} f)(\gamma', u', v) = \frac{\rho'_{\gamma'}(u', v)}{m(v)} \sum_{\gamma} \int f(\gamma, u, v)\, du. \tag{MG19.13}$$

The zero-mass fibers are null for the baseline and may be assigned the fixed reference convention.

Theorem MG19.2 — exact mixture reset and its full marked jet The map (MG19.13) is positive and integral preserving. On the baseline it has exactly the same full output density, including the retained output descriptor, as the original termwise replacements:

$$P_{\mathrm{mix}} \rho = \rho'. \tag{MG19.14}$$

Its normalized marked transfer is

$$Q_{\mathrm{mix}} F(\gamma', u', v) = \frac{\sum_{\gamma} \int \rho_{\gamma}(u, v) F(\gamma, u, v)\, du}{m(v)}. \tag{MG19.15}$$

It is independent of the fresh output coordinates and output descriptor. It preserves every integrated full source-jet coefficient, hence every original full moment. It is an Lp contraction from the actual old baseline law to the output law, for every 1≤p≤infinity.

Proof. Summing rho'_gamma over gamma and integrating u' gives m. Therefore the prefactor in (MG19.13) is a conditional probability density at fixed v. Integration proves preservation, substitution of f=rho proves (MG19.14), and division gives (MG19.15). One can sample the original old variables conditional on v and independently sample the output pair (gamma',u') with density rho'/m. Formula (MG19.15) is the conditional expectation of the old mark given that output. Jensen proves the Lp contraction. Source jets are finite polynomial expressions, so positivity and linearity apply coefficientwise. Their logarithm retains the full conditional cumulant partition rule. □

The map uses only the source-free masses and kernels. It does not use a nonzero observable source. Thus, if adopted with its actual footprints and estimates checked, it would remain a frozen source-free mark transport. The unmarked output law and every subsequent source-free operation can be left unchanged. The marked coupling to the past, however, is different from the original termwise coupling. Its use must be recorded explicitly in the transport and history ledgers.

What this construction does and does not resolve The old descriptor weights in (MG19.15) are m_gamma/m. Small component masses remain in those weights. The output mark has zero direct dependence on gamma', so the artificial amplification caused by changing that output label is absent. These are exact identities; they need no analyticity or component estimate.

There are nevertheless additional generated correlations. For two marks,

$$\mathrm{Cov}_{Q_{\mathrm{mix}}}(F, G) = \sum_{\gamma} \pi_{\gamma}\, \mathrm{Cov}_{Q_{\gamma}}(F, G) + \mathrm{Cov}_{\pi}(Q_{\gamma} F, Q_{\gamma} G), \quad \pi_{\gamma} = m_{\gamma}/m. \tag{MG19.16}$$

Higher orders obey the full law of total cumulance. The second term in (MG19.16), and its higher counterparts, cannot be discarded. They measure changes of the component means across the erased histories.

Nor does a common conditional density make the map local. If the family of descriptors ranges over arbitrarily large or distant geometry alternatives, the dependence footprint of (MG19.15) can grow accordingly. A useful implementation must specify a local family closed under the required history changes, retain the true normalization, and prove its carrier, shifted-current and pointwise regulator estimates. Full-law moment contraction alone is insufficient, as regulator examples already show.

A no-reset branch can be included in the finite algebra only after representing it with the same retained variables and a common padded integration window: its output kernel is its own normalized old conditional law, not a Dirac identity on variables that the other branches erase. Empty and successful branches may then be mixed by exactly the same construction. Whether that common window and denominator retain the required local bounds is an additional source application. Unrepaired residual components cannot be included merely by renaming them as an empty branch.

The positive-gas insertion/deletion approach examined in Annex BC, Section 6, is relevant to this joint alternative: it can keep unnormalized component masses in an estimate before the final normalized mixture is formed. It does not automatically bound the stronger norm of a fixed selected posterior.

6. Application conditions and remaining tests

The actual fixed-reference reset has the exact zero fresh-current theorem (MG19.1). The full current formula (MG19.5) and selector decomposition (MG19.7) identify the response that a native fixed-support argument omits. The history-index analysis of the source rules out erasing component selection merely by summing its compatible internal histories. The joint-history map (MG19.13) is an explicit alternative positive coupling preserving the original unmarked output, with all marked responses retained.

None of these statements proves the missing physical H-03 estimate. In particular the following tests remain:

1. For the chosen actual marked representation, identify a preintegration common current chart and its full set of fixed and moving hidden variables. The residual-gauge patch must satisfy the ordinary input domain, not just the prepared output bounds.
2. Bound (MG19.8), or prove exact cancellation in a fully specified joint branch sum, for all retained, selector and geometry changes required by M-03.
3. Prove the actual regulator pullback and shifted-envelope estimates, including all direct and previous/new kernel-current records.
4. Establish the full physical carrier sum. If the joint-history alternative is adopted, also bound the descriptor-mean cumulants and its actual enlarged dependency footprint.

The same source assumptions remain visible. No generic counterexample in this annex is presented as a counterexample to the actual Yang–Mills law.

Annex BC — The actual continuous core and two-outer-energy component reserve

Annex BC. Exact finite conditional identities apply to the specified positive kernel. A uniform Wilson application requires CL on the same retained sigma-field, the relevant complete-potential estimates under CR and the stated spatial and energy budgets. History estimates · Observable transport.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

This annex separates three costs that are easily combined too pessimistically: the physical support of the continuously integrated variables after preparation, the retained-data footprint of their normalized law, and the full footprint of an unnormalized successful-component activity. These are different sets and different estimates.

1. The unchanged strong norm

The M-03 norm uses the actual ambient retained variables, the native regulator

$$R_{h,X} = 1 + \sum_{p:\,\mathrm{foot}(p)\subset X} \beta_p s(V_p) + \sum_{i:\,\mathrm{foot}(i)\subset X} |z_i|^2, \qquad w_h(X) = e^{\varkappa(|X|+\mathrm{t}_h(X))}. \tag{RN19.1}$$

Its value envelope takes a supremum over elementary shifted retained states. Its difference/current envelope sums all corresponding finite-difference suprema, including discrete history changes. The pointwise part divides by a power of the shifted output native regulator. The input regulator cannot be retained after its variables have been erased unless a genuine pullback inequality proves the comparison.

The results below do not change these requirements. A pointwise estimate for an unnormalized marked numerator is useful because its small mass can pay an exponential footprint weight. Dividing by that same mass generally loses this payment. Conversely, jointly summing a successful geometry with its mass before normalization can retain the payment; a precise deletion lemma appears in Section 6.

2. The continuously integrated core is smaller than the eligible component

An eligible component X in LFR I printed p.177 has diameter at most 100 M R_k in current units. That is not the size of the final replacement integration domain.

Proposition RN19.1 — the final continuous core After the successful preparatory steps of the literal primary operation, the physical support of the remaining continuously integrated coordinates in that component is contained in a fixed current-block enlargement C of the final domain Lambda, with

$$\mathrm{diam}_h\, C \le C_d\, 100M, \qquad |C|_h \le C_d (100M)^d. \tag{RN19.2}$$

The statement concerns support in physical current blocks. It does not bound the number of old-scale coordinates located in those blocks. It also does not assert that a remaining history label or the retained-data dependence of the kernel has the same footprint.

Proof. LFR I printed p.192 defines Lambda in (1.73) by adding one current M-layer to the prepared domain Z'_k. The text explicitly states that Lambda is a rectangular parallelepiped contained in a cube of side 100M and that it is the domain on which the field variables are integrated. The subsequent integrations must still be checked, rather than identified with the fresh reference denominator.

The first integral in (1.76), printed p.194, is over the older region Z_h and is left unchanged while the second integral is prepared. The nested prepared domains described in (1.10)–(1.12), and displayed again immediately before the gauge construction on p.195, give

$$\Lambda \supset Z'_k \supset \cdots \supset Z'_{h+1} \supset (\Omega''_{h+1})^c \cap X \supset Z_h \cap X. \tag{RN19.3}$$

Here h=k−N is the source's older step index, not the physical spacing used in (RN19.1). The last inclusion uses the ordinary nesting of the next small-field domain inside the current small-field domain. The old operation T_h is localized in Z_h, as in CMP119 (2.19)–(2.21); its integrated coordinates are thus in the last set in (RN19.3).

The other remaining coordinates are the relative fluctuation on

$$B_0 = \mathcal{B}'_k \cap \Lambda \cap \Omega''_{h+1}, \tag{RN19.4}$$

as specified by LFR I (1.81), printed p.195. The exact final operations LFR I (1.100) and LFR II (1.71), pp.378–379, retain these B coordinates, T_h and a V_h integral on the complementary older region with its displayed fixed old-scale collar. In particular, the final V_h domain is written using the enlarged complement of Omega' '_(h+1), not merely the uncollared complement displayed in (RN19.3). LFR I p.195 identifies this older domain with the prepared Z'_h layers; p.199 describes the corresponding boundary layer of thickness 2 M_1 in the h-scale. Its physical thickness is bounded by a fixed number of current blocks because the h-scale is finer than the current scale. Include that collar, the bonds crossing its boundary and the prescribed gauge graphs in C. Then C lies in a fixed enlargement of the 100M cube. No claim that the final collar is literally contained in the unmodified Lambda is needed. Counting the current blocks of this enlarged fixed cube proves (RN19.2). The locality assertion here is for the raw integration coordinates after the stated preparatory operations; reconstructing a fine background from them can have additional retained-data dependence. ■

The same small-domain fact is independently visible in the fresh denominator argument: LFR II printed pp.357–358 gives (1.8) with the fixed constant d(100M)^(d+1). That denominator argument alone would not identify the old posterior. Proposition RN19.1 instead follows every remaining continuous integration in (1.71).

Lemma RN19.2 — joining input carriers through the continuous core Let S_1,...,S_l be actual input carriers, each meeting the fixed continuous core C of Proposition RN19.1, enlarged once to a connected current-block cover if necessary. Its cardinality and tree length retain the fixed bounds in (RN19.2). Then

$$w_h\left(C \cup \bigcup_{j=1}^{l} S_j\right) \le w_h(C) \prod_{j=1}^{l} w_h(S_j), \qquad w_h(C) \le C_{M,d,\kappa}. \tag{RN19.5}$$

Proof. Join a shortest tree of each S_j to the fixed tree of C at a point of intersection. The resulting union is connected, has at most the sum of the cardinalities and tree lengths, and contains the required set. Exponentiate the resulting two subadditivity inequalities. ■

For a fixed geometry and retained datum, a cumulant of order at least two vanishes if one of its arguments is independent of every integrated variable. Hence a genuinely generated cumulant of inputs independent of integrated history labels must have every participating input carrier meet the continuous core. Formula (RN19.5) pays the connector needed for those inputs with a fixed factor. It does not pay additional retained-data dependence of the normalized kernel.

In particular, assigning an arbitrary old core correlation the diameter 100MR_k solely because X has that diameter is unnecessarily pessimistic after the successful preparations. It is still invalid to replace the full kernel carrier by C without proving its retained-data and history-label locality. The actual native-variation connection set printed on LFR II p.379 is B_0 union (Gamma' '_k intersect X-tilde), with subscript k. Its second set contains current determining variables throughout the component and is not an old h-scale core. Annex BB, equation (MG19.17)„ records the underlying size distinction explicitly.

3. An exponential reserve for two actual outer energies

Write T'_k(X;U) for the real positive unnormalized operation in LFR II (1.71). Use exactly its old integration variables, selectors, coarea factors and normalization constants. Among its displayed exponent terms are two nonnegative real functions:

$$\begin{aligned} Q_X &= \tfrac{1}{2}\langle DH''_{1,k,X}B,\, \zeta DH''_{1,k,X}B\rangle, \\ S_X &= A\big((g'_k)^{-2}\zeta_1|_X, U''_{h+2}(\exp(ig_kB), V'')\big), \qquad \mathcal{E}_X = Q_X + S_X. \end{aligned} \tag{RN19.6}$$

The notation is the source's notation in (1.71); S_X is the localized Wilson action with its actual nonnegative coupling weight. Q_X is nonnegative because the real cutoff zeta is nonnegative. These are the two displayed outer energies. They are not defined to be the full M-03 regulator and do not silently include every older free coordinate.

Proposition RN19.3 — fractional-action reserve on the admitted source class Fix theta_0=1/4. The source parameter order and one sufficiently small common coupling ceiling can be chosen so that, uniformly over the literal admitted eligible operations and their real native backgrounds,

$$T'_k(X;U)\big(e^{\theta_0 \mathcal{E}_X}\big) \le e^{-P_0(g_k)}, \qquad P_0(g) = p_0(g) \text{ in LFR II.} \tag{RN19.7}$$

This is an unnormalized estimate. No lower bound for T'_k(X;U)1 is asserted.

Proof and source-parameter check. Multiplication by the exponential in (RN19.7) replaces the coefficients of precisely Q_X and S_X in (1.71) by 3/4. It leaves every cutoff, domain, older operation, field coordinate and normalizing determinant unchanged. In particular, this is an insertion in the original operation; it is not a replacement of the original measure by a newly chosen renormalized trajectory.

There is no hidden change to an older normalized reference operation. CMP119 (2.19)–(2.21), printed p.258, represents T_h as the ordered product of localized positive integrations with their existing characteristic functions, Gaussian forms, renormalization delta constraints and gauge-fixing factors. The source states that an earlier R operation preserves that general form, with its altered characteristic functions and possibly empty fluctuation domain. Those old forms and their normalization factors are left exactly as they occur in the baseline T_h. Any already completed old scalar factor is likewise unchanged. The insertion multiplies only the two displayed outer exponent terms of LFR II (1.71); it neither redefines the old kernels nor invokes a bound for an old normalized kernel with a new reference action. The extraction below is performed on this single unchanged positive integral with the two outer coefficients reduced.

The real extraction proof in LFR II pp.380–383 uses lower bounds for the positive Wilson and quadratic forms to extract the factors associated with old and preparatory large-field selectors. Multiplying either relevant positive form by a number in [3/4,1] reduces the corresponding extraction coefficient by at most 3/4. Nonlinear comparison errors and bounded action terms are treated by the same absolute upper bounds as in that proof. They can be absorbed by reducing the fixed positive lower coefficient once more, for example to gamma_* = gamma_0/2, and enlarging a fixed constant C_* in the positive domain cost.

For clarity, the resulting upper bound has exactly the structure of the printed (1.79):

$$\begin{aligned} \sup_{\text{admitted histories}} \quad & \exp\left\{ C_* \sum_{j \le k} M^d R_j^{d+1} d'_j(Z_j) \right\} \\ & \times \prod_{\text{new bad components at } j} \exp\left\{ -\tfrac{1}{2}\gamma_* A_1^2 P_0(g_j)^2 [d'_j(Z_j^{(i)}) + 1] - 2P_0(g_j) \right\} \\ & \times \prod_{\text{preparatory failures at } j} e^{-P_0(g_j)}. \end{aligned} \tag{RN19.8}$$

The full primary operation sums the admissible histories before this supremum bound is taken. The source's counting and Gaussian volume factors are already included in the first exponential. No additional count of all microscopic bonds of the torus is introduced.

The coefficient reduction is harmless for the preparatory factors as well. For the worst such factor the source proves, on printed p.383, a bound of the form

$$e^{-cR_j^{-(d+5)} p_1(g_j)^2}, \qquad 2p_1 - (d+5)r_0 > p_0. \tag{RN19.9}$$

Replacing c by a fixed positive smaller constant still gives e^{-P_0(g_j)} at one smaller ceiling. The other preparatory factors have stronger bounds. The initial bad-field extraction has a term of order P_0^2, so the fixed store 2P_0 in (RN19.8) also remains available after reducing its quadratic coefficient.

The treatment of integration volumes is particularly important. Let d_G be the Lie-group dimension and let n_B be the number of free B bonds after removing the displayed axial tree. The last line of (1.71) contains exactly

$$\Big[-\tfrac{1}{2}d_G \log g_k^{-2} + \log \sigma_0\Big] n_B, \qquad e^{[-\frac{1}{2}d_G \log g_k^{-2} + \log \sigma_0] n_B} = (\sigma_0 g_k^{d_G})^{n_B}. \tag{RN19.8a}$$

Combined with dB times sigma(g_k B), this is the Haar-coordinate Jacobian under B'=g_k B on the retained small exponential chart. The tree delta removes exactly the tree coordinates, so the factor counts the free coordinates. As stated on printed p.383, the source includes this logarithmic factor in the integration measure, drops the remaining positive B quadratic to zero, and bounds the resulting restricted normalized compact measure by one. The same argument applies to 3Q_X/4. There is no coefficient-dependent Gaussian determinant to cancel for this outer B estimate, and no additional rank loss.

The older Gaussian forms in T_h are unchanged by the insertion. To quantify the maximal cost even if one uses a common fractional-action comparison when extracting old bad-field factors, let r_j be the actual rank of the Gaussian majorant charged to the j-scale volume in LFR II (1.79). These are the older fluctuation forms in CMP119 (2.21), or LFR I (1.25), with the indices relabeled to the volume convention of (1.79); in the original CMP119 convention A_j is integrated on Z_(j+1) intersect Omega_(j+1). Reducing a fixed positive coefficient by 3/4 changes the full Gaussian majorant integral by (4/3)^(r_j/2). This comparison is only an upper bound for the integral; it does not change the actual old kernel or its normalizer. The number of Lie coordinates gives

$$r_j \le C_{d,G,L} |Z_j \cap \Omega_j|_j \le C_{d,G,L} M^d R_j^d d'_j(Z_j), \tag{RN19.8b}$$

where the volume in the first expression is at that old j-scale; the fixed L-dependent constant allows the one-step bond-count conversion in this relabeling. The second inequality is exactly the old-scale volume-to-component bound used on LFR II p.380 for its actual admissible enlarged Z_j and its own d'_j convention. It is not a claim for arbitrary sets with an unrelated diameter convention. Thus any such loss is bounded by $\exp[C \sum_j M^d R_j^d d'_j(Z_j)]$, which is absorbed by the existing positive cost in (RN19.8), since R_j>=1. This argument is valid for arbitrary old depth: it retains the complete scale-by-scale sum, and the chronological induction below pays it. It does not bound the number of old coordinates by the fixed current volume of Proposition RN19.1. The other compact group integrals are bounded by one as in the source. These facts account for every continuous integration class in (1.71).

Finally, the chronological induction (1.80)–(1.88) is an induction for this numerical structure with fixed positive extraction coefficient and fixed positive domain-cost constant. Its proof uses the strict source parameter separations. A sufficient way to state the resulting requirements is

$$\begin{aligned} &P_0(g) \gg C_* M^d R(g)^{d+2}, \\ &\gamma_* A_1^2 P_0(g)^2 \gg P_0(g) + C_* M^d R(g)^{d+2}, \\ &c_* R(g)^{-(d+5)} p_1(g)^2 \gg P_0(g). \end{aligned} \tag{RN19.10}$$

All constants are fixed before the ceiling is chosen. These are the same strict power-order requirements used in the source proof; multiplying their constants by fixed numbers changes only the ceiling. In particular its choice N=R_j in (1.80) is retained. The induction therefore leaves the store 2P_0 of the earliest surviving bad component just as in (1.89), giving exp[-2(1+beta_0)^(-1)P_0(g_k)]. The weaker (RN19.7) follows for the admitted fixed beta_0<1, after taking the common ceiling small enough for the inequalities above. This supplies the claimed source-scoped insertion estimate. ■

The proof depends on the source extraction and admissibility hypotheses exactly as the C-05 argument does. Independent verification of every numerical ceiling and the actual Wilson-normalized trajectory remains part of the base estimates. The useful new assertion is the explicitly identified fractional-action insertion (RN19.6), not an unproved claim that all native regulators are equivalent to it.

Corollary RN19.4 — polynomial moments with fixed factorial growth For every integer m>=0,

$$T'_k(X;U)\big[(1+\mathcal{E}_X)^m\big] \le e^{\theta_0} \theta_0^{-m} m!\, e^{-P_0(g_k)}. \tag{RN19.11}$$

Proof. For s>=0, s^m<=m! theta_0^(-m) exp(theta_0 s). Apply it to s=1+E_X and use (RN19.7). ■

This argument preserves a fixed factorial budget. In contrast, replacing a compact marked field by a bound of order g^(-2m) and then paying it with exp[-c(log g$^{(-2)}$q] produces a bound of order exp(C m^(q/(q−1))) after optimizing in g. That is not a fixed factorial-power estimate. Such a compact-supremum shortcut is not used here.

4. Every finite unnormalized source jet in the energy class

Let I have m labels. Suppose all incoming log-jet coefficients on a component satisfy the actual pointwise bound

$$|A_B(x,\eta)| \le C^{d+1}(d!)^s b^d r(x)^d [1+\mathcal{E}_X(x,\eta)]^d, \qquad d=|B|, \quad s\ge 1, \quad r(x)\ge 1. \tag{RN19.12}$$

The function r is retained data, fixed inside this operation. It is not automatically the output M-03 regulator. For independent scalar variables t_i indexed by I, put $t_B=\prod_{i\in B} t_i$. Define the unnormalized jet by applying T'_k(X) to the finite formal exponential of sum $t_B A_B$. The coefficient selected by $\partial_I=\prod_{i\in I}\partial_{t_i}$ is therefore the usual labeled partition sum; t_B is not an additional independent source coordinate.

Theorem RN19.5 — unnormalized full-jet coefficient bound For every m>=1,

$$\left|\partial_I T_k'(X)\exp\left(\sum_{B\neq\varnothing} t_B A_B\right)\Big|_{t=0}\right| \le e^{-P_0(g_k)} C_1^m (m!)^{s+1} b^m r(x)^m. \tag{RN19.13}$$

The result is compatible under truncation in source order. It requires no convergent exponential-source series.

Proof. The coefficient is the sum over partitions of I of T'_k(X) applied to the product of the corresponding A_B. Its total power of 1+E_X is m. By (RN19.11), that integral costs at most C^m m! exp(-P_0). For a fixed number l of blocks, order them and sum their positive sizes d_1+...+d_l=m. Since s>=1,

$$\sum_{\substack{\pi\in\Pi(I)\\ |\pi|=l}} \prod_{B\in\pi} (|B|!)^s \le \frac{(m!)^s}{l!}\binom{m-1}{l-1}. \tag{RN19.14}$$

The remaining sum over l, including C^l, is at most (1+C)^m. This proves (RN19.13). All operations are in the finite source algebra, so lower-order truncation gives the same coefficient. ■

If a common native chart is used and the complete input coefficient is frozen as a function of the old integration variables at a real chart center, the primary small exponential tilt in LFR II (1.73) bounds the complex unnormalized integral by a fixed multiple of its real absolute integral. Formula (RN19.13) therefore holds on that native disc for this kernel response. Native derivatives of order j cost j! rho^(-j), where j is distinct from the source order m; finite-difference suprema within the inner half-disc cost the corresponding first-derivative radius. Direct changes of the incoming coefficients require their own transported input records. This assertion does not include arbitrary raw retained-coordinate pullbacks or a moving selector support.

There is also a useful quantified polynomial loss rule. Write P_0(g)=ell(g)^(p_0) and R(g)<=C ell(g)^(r_0), with the source's fixed exponents. For any fixed u>=0 and a>0,

$$\sup_g R(g)^{um} e^{-aP_0(g)} \le C_{a,u}^m (m!)^{ur_0/p_0}. \tag{RN19.15}$$

Indeed set v=P_0(g) and maximize v^(ur_0 m/p_0) exp(-a v). Thus a *proved polynomial* regulator comparison in R could be absorbed at a fixed additional factorial power while retaining part of the component mass. A factor exponential in ell times m does not obey this conclusion.

5. Paying the full footprint for unnormalized successful components

For the source-native operation all expressions in (1.71) are localized in X, with the stated fixed collars included. Let X^+ be that actual source-native footprint. Since X lies in a cube of side 100MR_k,

$$|X^+|_h + \mathrm{t}_h(X^+) \le C_M(1+R_k)^d. \tag{RN19.16}$$

The number of possible such current-block sets containing one fixed anchor is at most exp[C_M(1+R_k)^d]. This deliberately crude bound follows by choosing a containing cube, then a subset of its blocks. It bounds the allowed eligible shapes without replacing a more refined source count by an unjustified smaller count.

Because P_0(g_k)/R_k^d tends to infinity under the source parameter order, one can retain a fixed fraction of (RN19.13) and obtain

$$\sup_a \sum_{X^+ \ni a} w_h(X^+) e^{-P_0(g_k)/2} \le C e^{-P_0(g_k)/4}. \tag{RN19.17}$$

This is a full physical footprint bound for these unnormalized component contributions. It is the same type of payment used in the source's (1.92)–(1.97). Printed p.389 explicitly combines the component mass with exp[2 kappa (100R_k)^d], using

$$-2(1+\beta_0)^{-1}P_0(g_k) + 1 + 2\kappa(100R_k)^d \le -\frac{3}{2}P_0(g_k). \tag{RN19.18}$$

The weakened reserve needed in (RN19.17) merely asks for another fixed part of that strict separation. No mass that has already canceled in a conditional ratio is used.

Equations (RN19.13), (RN19.15) and (RN19.17) therefore give a fixed-factorial, fully weighted unnormalized bound for actual successful-component jets in the stated outer-energy class. They do not identify the entire M-03 regulator with that energy, do not bound the normalized fixed-component posterior, and do not control all retained-coordinate currents.

6. A positive normalization lemma for joint successful-geometry erasure

The following precise lemma identifies one way the component payment can survive a normalization. Its geometric hypotheses must be checked in the actual representation before applying it.

Let C range over admissible finite families of compatible components. Suppose the family is closed under deletion. Let M_X be a positive finite measure for the internal variables of X, and write

$$Z = \sum_{C \text{ admissible}} \int e^{H_C(\eta_C)} \prod_{X \in C} M_X(d\eta_X) > 0. \tag{RN19.19}$$

Retained data are fixed here. For a fixed required family S, assume that removing S from any C containing it gives the same remaining measures and coordinates, and that on their common physical support

$$H_{D\cup S}(\eta_D, \eta_S) - H_D(\eta_D) \le \sum_{X \in S} \lambda_X. \tag{RN19.20}$$

Lemma RN19.6 — normalized deletion estimate with marked masses retained For nonnegative component functions F_X,

$$\mathbb{E}\left[1_{\{S \subset C\}} \prod_{X \in S} F_X(\eta_X) \right] \le e^{\sum_{X \in S} \lambda_X} \prod_{X \in S} M_X(F_X). \tag{RN19.21}$$

Proof. In the numerator use (RN19.20) and perform the S-integrals. The remaining sum is over those D for which D union S is admissible. By deletion closure those D form a subfamily of the families summed in Z, with exactly the same positive summands. Their sum is at most Z. Divide by Z. ■

No comparison of partition functions on different tori is required. The proof does not divide by M_X(1). It is therefore compatible with the unnormalized marked reserve of Section 5. If cross interactions have a common geometry-local representation and a uniform absolute incidence sum, that sum supplies the lambda_X in (RN19.20).

For the source's successful X-sector, the factorized form (1.72), exact localization of V(Y) and the mass estimate make this a concrete application to investigate. It is not enough merely to observe a scalar cluster bound: one must verify deletion-closed admissible history families, a common retained field/coordinate representation after deletion, and equality of terms whose full footprints miss the deleted components. The source's localization statements on pp.374, 377 and 379 support that program but do not by themselves specify this entire comparison. The lemma is not imported as an already checked actual deletion theorem.

7. The exact normalization cost for this energy reserve

The unnormalized reserve also gives a sharper diagnostic than multiplying every normalized bound by a reciprocal component mass. It identifies the precise logarithmic mass information that would suffice for polynomial moments of the two outer energies.

Lemma RN19.7 — logarithmic mass cost Let M be any positive finite measure, let $\mathcal{E} \geq 0$, and suppose $M(e^{\theta\mathcal{E}}) \leq A$ with $\theta > 0$. On a positive-mass fiber define

$$m = M(1), \qquad Q = M/m, \qquad L = \log(A/m) \geq 0. \tag{RN19.22}$$

For every real $p \geq 1$,

$$\|1 + \mathcal{E}\|_{L^p(Q)} \leq 1 + \theta^{-1}(L + Cp), \qquad \|(1 + \mathcal{E})^q\|_{L^p(Q)} \leq [1 + \theta^{-1}(L + Cqp)]^q \quad (q \geq 1). \tag{RN19.23}$$

Proof. Markov's inequality gives $Q(\theta\mathcal{E} > L + t) \leq e^{-t}$ for every $t \geq 0$. Thus the nonnegative variable $\theta\mathcal{E}$ is stochastically dominated by $L+E$, where E has the unit exponential distribution. Its p-th norm is $\Gamma(p+1)^{1/p} \leq Cp$. Minkowski gives the first bound; replacing p by qp and raising to q gives the second. ■

In Proposition RN19.3 one may take $A = e^{-P_0(g_k)}$, $\theta = \theta_0$, and the actual $m_X = T_k'(X;U)1$. The resulting quantity is

$$L_X = -P_0(g_k) - \log m_X \geq 0. \tag{RN19.24}$$

No upper bound for L_X has been established here. If a subsequent proof bounded L_X by the actual shifted output native regulator, (RN19.23) would control normalized polynomial outer-energy moments at the correct polynomial grade, with a fixed factorial cost in source order. The source's upper bound for m_X is not such a bound: it supplies no lower bound for the mass and hence no upper bound for L_X. This lemma also says nothing by itself about other native energies, selector currents or the full carrier footprint.

8. What remains for H-03

The work here proves or isolates useful pieces but leaves the estimate open.

1. The full marked chart and labels. Proposition RN19.1 shrinks the continuous integration support. A live history label can still have a larger defining footprint, and the normalized kernel can depend on retained fields outside that support. Those dependencies must be localized, not discarded.
2. The complete native regulator. The explicit outer energy (RN19.6) does not include every term of M-03's retained/native list. A bound such as R_in<=C R_k^u R_out(1+E_X), with the actual pulled-back shifted versions and a fixed u, would be useful; no such full inequality has been proved here. A comparison for only a fresh reference is insufficient.

3. Joint geometry erasure. The successful-sector normalization must be implemented with its actual marked posterior and verified common geometry, as in Section 6. Keeping a fixed component and dividing by its mass instead loses (RN19.17).
4. Residual components and all currents. The fixed Y-components in (1.103)–(1.104) have a different role. Their normalized old posterior still requires the actual retained/selector/geometry currents and an internal or retained-data carrier theorem. The unnormalized reserve is an upper bound; it is not a lower bound for their masses. Raw-coordinate paths and pointwise shifted regulators remain essential.

No actual Yang–Mills counterexample is claimed. In particular, a lower bound on the mean old energy on a selected bad event is not by itself a covariance lower bound, and an auxiliary component-wide reflection label has a component-wide input footprint in M-03. Neither observation can be used to manufacture a false disproof of the original spatial-reset requirement.

Sources

- Balaban, *Large Field Renormalization I*, CMP 122 (1989), printed pp.177, 179, 192–198, especially (1.10)–(1.12), (1.73), (1.76), (1.81), for the prepared domains, the small final continuous core and the exact remaining old integrations.
- Balaban, *Large Field Renormalization II*, CMP 122 (1989), printed pp.357–358, 378–389, especially (1.71), (1.73), (1.77)–(1.89) and (1.92)–(1.97), for the positive outer energies, fractional-action extraction, chronological costs and successful-component volume payment.
- Balaban, *Convergent Renormalization Expansions*, CMP 119 (1988), (2.19)–(2.21), for localization of the old positive operations.
- The existing M-03 construction, Annex Y, and the reset and geometry annexes, are retained with their original scope restrictions.

See LFR II, pp. 378–379 and 383.

Annex BD — Regular conditional and minimizing-reference energy regulators

Annex BD. Exact finite conditional identities apply to the specified positive kernel. A uniform Wilson application requires CL on the same retained sigma-field, the relevant complete-potential estimates under CR and the stated spatial and energy budgets. History estimates · Observable transport.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

0. Selected estimate and exact scope

This annex concerns theorem M-04 of the catalogue, whose exact statement is:

> At one RG step, expand the actual transported scalar insertion in the fixed incoming reference basis of F-01, under an explicit one-step induction budget, into the current scalar basis, identity term and a localized remainder. Include constrained background, E/B/R responses and coarea terms present in that sector. This does not assume the final running Z from F-05.

The required scope is:

> Equality with Q_k F_k and a remainder estimate in M-03, uniform in scale; the left side is not a freshly inserted coarse plaquette.

The following estimates use the regular-law moment bounds of Annex AY, with all its hypotheses. The conditional expectation still requires a separate carrier expansion.

The scalar matching problem requires the specified incoming norm and the conditional moment bounds of Annex AY.

1. The actual regular law used here

Use the reduced regular law of Annexes W, AW and AX subject to CL and the derivative estimates under CR.

$$Q_v(dz) = Z_v^{-1} 1_D(z) \exp\{-\tfrac{1}{2} z^{\mathsf{T}} K z + b^{\mathsf{T}} z + W(z;v)\}\, dz, \qquad b = -K_{IO} y. \tag{NP19.1}$$

Here I is the set of integrated current free coordinates, O the retained exterior coordinates of the same current chart, and $y = (y_j)_{j\in O}$. Older-layer variables remain parameters of W; they are not inserted into y. The literal product support D, constrained precision K, full W, and source assumptions are those in Annex AY, Sections 1–4. In particular no current-independent exterior Schur scalar is lost from the outer law.

C-08 supplies fixed positive rates, a covariance majorant G, a nonnegative entrywise Hessian majorant A, and

$$H = (I - GA)^{-1} G, \qquad \|H\|_a \le C_H, \qquad \|K_{IO}\|_a \le C_K, \tag{NP19.2}$$

for one fixed $a > 0$, uniformly over its admitted retained data. With $d_i = |D_i W(0)|$,

$$\sup_{s,t} |Q_{v;s,t} z_i| \le \sum_l H_{il} \left(|b_l| + d_l + \sigma \sum_n A_{ln} \right), \quad d_l + \sigma \sum_n A_{ln} \le C/P, \tag{NP19.3}$$

and

$$\|z_i - Q_{v;s,t} z_i\|_{L^r(Q_{v;s,t})} \le C\sqrt{r}, \qquad r \ge 1. \tag{NP19.4}$$

The interpolation (s,t) and all additional real linear tilts have the same estimates, with their actual force in (NP19.3). These are RC18.28–RC18.36, not additional assumptions about transported observables.

2. A pointwise conditional regulator with the exterior field retained

The coarse bound $Pe^{-c\,d(i,O)}$ is useful for collars. For a native polynomial regulator one should retain the actual exterior amplitudes.

Theorem NP19.1 — force-resolved local free-coordinate regulator There are fixed $C, c > 0$ and a nonnegative kernel L_{ij}, $i \in I,\ j \in O$, such that

$$L_{ij} \le Ce^{-c\rho(i,j)}, \qquad \sup_i \sum_j L_{ij} \le C, \tag{NP19.5}$$

and, pointwise at every admitted retained datum,

$$|Q_v z_i| \le C/P + \sum_{j\in O} L_{ij}|y_j|, \tag{NP19.6}$$

$$\|z_i\|^2_{L^r(Q_v)} \le C\left[r + \sum_{j\in O} e^{-c\rho(i,j)}|y_j|^2\right], \qquad r \ge 1. \tag{NP19.7}$$

For a finite set X of current free-coordinate blocks put

$$E_X(z) = 1 + \sum_{i\in X\cap I} |z_i|^2, \qquad n_X = 1 + |X \cap I|, \qquad a_{Xj} = \frac{1}{n_X}\sum_{i\in X\cap I} e^{-c\rho(i,j)}. \tag{NP19.8}$$

Set

$$\mathcal{R}_X(y) = n_X\left(1 + \sum_{j\in O} a_{Xj}|y_j|^2\right). \tag{NP19.9}$$

Then, for all real $q \ge 1$ and $p \ge 1$,

$$\|E_X^q\|_{L^p(Q_v)} \le C^q (qp)^q\, \mathcal{R}_X(y)^q. \tag{NP19.10}$$

In particular

$$Q_v(E_X^q) \le C^q q^q \mathcal{R}_X(y)^q. \tag{NP19.11}$$

The constants are independent of the number of integration variables, volume, RG depth, retained history admitted to this regular law, p, and q. The factor n_X^q is explicit. No moment estimate for the outer law of y is assumed.

Proof. Define

$$L = H\,|K_{IO}|, \tag{NP19.12}$$

where absolute values mean block operator norms. Weighted row norms are submultiplicative by the triangle inequality for ρ; hence $\|L\|_a \le C_H C_K$. This proves (NP19.5), after decreasing the rate if needed. Insert the exact inequality $|b_l| \le \sum_j |(K_{IO})_{lj}||y_j|$ in (NP19.3). The nonforce term is at most C/P by the row bound on H, proving (NP19.6).

By weighted Cauchy–Schwarz and the uniform row bound,

$$\left(\sum_j L_{ij}|y_j|\right)^2 \le \left(\sum_j L_{ij}\right)\left(\sum_j L_{ij}|y_j|^2\right) \le C\sum_j e^{-c\rho(i,j)}|y_j|^2. \tag{NP19.13}$$

Combine this with (NP19.4), $P \geq 1$, and the triangle inequality to obtain (NP19.7). Minkowski's inequality, applied to the finite positive sum defining E_X, gives for every $r \geq 1$

$$\begin{aligned} \|E_X\|_{L^r(Q_v)} &\leq 1 + \sum_{i \in X \cap I} \|z_i\|^2_{L^{2r}(Q_v)} \\ &\leq C\left[rn_X + \sum_{i \in X \cap I} \sum_{j \in O} e^{-c\rho(i,j)} |y_j|^2 \right] \\ &\leq Cr\,\mathcal{R}_X(y). \end{aligned} \tag{NP19.14}$$

Take $r = qp$ and use $\|E_X^q\|_p = \|E_X\|_{qp}^q$. This proves (NP19.10) and (NP19.11). □

The earlier logarithmic support bound $|y_j| \leq CP$ would turn the last term in (NP19.7) into CP^2. The new bound avoids that loss by keeping the native retained energies themselves. This matters for the pointwise polynomial part of M-03.

Corollary NP19.2 — admitted unit shifts If one integrated free coordinate is changed by a vector of norm at most one, its local input energy satisfies $E_X(\tau z) \leq 3E_X(z)$, whenever that coordinate is in X. Also, if one exterior free coordinate is changed by a vector of norm at most one,

$$\mathcal{R}_X(\tau y) \leq C\mathcal{R}_X(y). \tag{NP19.15}$$

More precisely,

$$\left\| \sup_{\tau_I} E_X(\tau_I z)^q \right\|_{L^p(Q_v)} \leq C^q (qp)^q \mathcal{R}_X(y)^q, \qquad \sup_{\tau_O \text{ admitted}} \|E_X^q\|_{L^p(Q_{\tau_O v})} \leq C^q (qp)^q \mathcal{R}_X(y)^q. \tag{NP19.15a}$$

In the first expression τ_I changes one integrated free coordinate. In the second it changes one exterior free coordinate and the actual shifted datum must remain in the same admitted regular family. In particular the latter formula at $p = 1$ is a pointwise bound for the supremum of the conditional values. No outer-law moment is inferred from that pointwise statement.

Proof. Use $|u + t|^2 \leq 2|u|^2 + 2$, $E_X \geq 1$, and $\sum_j a_{Xj} \leq C$, which follows from the fixed lattice exponential row sum. Apply Theorem NP19.1 at each shifted admissible datum. The constants are uniform, so the pointwise supremum obeys the same bound. □

This corollary is not an ambient-chart extension theorem. M-03 also takes shifts crossing a chart or selector boundary, with its recorded conventions. Such shifts are not included by asserting that all points of a real interior chart are admissible.

3. The positive regulator majorant has a thin-path summation

The next result concerns the explicit positive majorant (NP19.9), not the still-unknown support of $Q_v F$.

Proposition NP19.3 — exponential carrier payment for the retained energy majorant Suppose X is a connected current block set containing the insertion anchor and the footprints of the listed free coordinates have uniformly bounded current diameter. For every integer $m \geq 1$, expand

$$\mathcal{R}_X(y)^m = n_X^m \sum_{(j_1,\dots,j_m) \in (O \cup \{*\})^m} \left(\prod_{\ell=1}^m a_{Xj_\ell} \right) \prod_{\ell:\, j_\ell \neq *} |y_{j_\ell}|^2, \tag{NP19.16}$$

where $a_{X*} = 1$. For each nonstar index choose one summand in a_{Xj}, with starting point $i \in X$, and connect i to j by a fixed shortest path. Let Z be X, those paths, and the fixed coordinate-footprint enlargements. For sufficiently small fixed $\varkappa > 0$, the sum of the resulting coefficients multiplied by $w_h(Z)/w_h(X)$, with w_h exactly as in M-03, is at most

$$C^m n_X^m. \tag{NP19.17}$$

Each energy monomial in (NP19.16) is bounded by $R_{h,Z}(y)^m$, provided those y_j's are entries in the actual surviving native list. One must include every such entry's full footprint. If a coordinate has a growing or global footprint, its extra cost is not bounded by this proposition.

Proof. Adding a path of length r and a fixed local endpoint footprint increases $|Z| + \mathrm{t}_h(Z)$ by at most $c_1(r+1)$, for one fixed c_1. The ratio of weights is therefore at most the product of $\exp[\varkappa c_1(\rho(i,j)+1)]$ over the chosen paths. Choose $\varkappa c_1 < c/2$. The weighted sum of one endpoint coefficient is bounded by

$$\frac{1}{n_X} \sum_{i \in X \cap I} \sum_{j \in O} e^{-c\rho(i,j)} e^{\varkappa c_1(\rho(i,j)+1)} \le C. \tag{NP19.18}$$

Including the star option changes C by one. Sum the m endpoint choices independently; shared paths and repeated indices only reduce the union's weight. This proves (NP19.17). The regulator inequality follows because each $|y_j|^2$ is one of its nonnegative summands. □

This proves a concrete carrier payment for the majorant. It avoids the volume of balls of growing radius. It does not identify a term of $Q_v F$ that depends only on a particular Z. For that, an actual rooted expansion of the marked conditional integral is still needed. Assigning $Q_v F$ to the small carrier of a pointwise upper bound would be an invalid support claim.

4. Precise source opening for a rooted insertion expansion

CMP119, printed pp.277–278, (3.44)–(3.47), gives a useful exact construction. Its numerator has one marked factor $H'(Z_0)$, and its other activities are the denominator activities. Exponentiating the polymers disjoint from Z_0 cancels those denominator components; the surviving expansion is anchored at the marked root. This is an appropriate starting point for the transported scalar, with every coefficient of its actual projected numerator retained.

The construction there is applied to the specified action-derivative insertions in (3.26), (3.28) and (3.37). Its bounds include the source insertion's own polynomial cutoff factors. It does not state a bounded transport theorem in M-03 for an arbitrary marked input.

CMP116, Section 2, especially (2.14)–(2.26), performs the Gaussian majorant estimates before the final polymer sum. This is the location at which a fixed-degree marked polynomial can in principle be charged to reserved Gaussian energy, rather than bounded by the support radius. The exact current map, physical reference field, and current records must still be carried through that argument. The scalar unmarked activity estimate alone does not do this.

Published sources for these hypotheses:

- T. Bałaban, *Convergent renormalization expansions for lattice gauge theories*, CMP 119 (1988), 243–285, pp.277–278.
- T. Bałaban, *Renormalization group approach to lattice gauge field theories. II. Cluster expansions*, CMP 116 (1988), 1–22, DOI 10.1007/BF01239022, especially pp.12–20.
- T. Bałaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, CMP 102 (1985), 277–309, Proposition 9 and (190).

5. A source-scoped physical-reference regulator on a deep regular patch

Theorem NP19.1 controls the actual integrated free coordinates. A physical incoming plaquette also sees its constrained reference field. In C-08 the current connection has the form

$$U_e(t) = e^{i a_{v,e}(t)} V_e, \qquad a_v(t) = g C_v t - h_v D_v(g C_v t). \tag{NP19.19}$$

The source derivative estimate gives

$$g^{-1}|a_{v,e}(t)| \le C \sum_i e^{-c\rho(e,i)}|t_i|, \tag{NP19.20}$$

on its admitted current segment. For a fixed plaquette, the compact product inequality therefore bounds its normalized energy by the reference energy and a fixed exponential convolution of $|t_i|^2$. Combining that inequality with Theorem NP19.1 controls the fluctuating part.

The reference field here is the actual $V^{(k)} = \mathcal{M}^k(U_{k+1})$. In the source notation U_{k+1} is a minimal fine configuration depending on the retained determining averages. CMP102 p.305 explicitly distinguishes $U_k(V)$ from its determining coordinates V. Thus the fine plaquettes of U_{k+1} cannot simply be inserted into the surviving independent-coordinate regulator.

C-08's RC18.49a proves the absolute estimate $g^{-2}s(V_p^{(k)}) \le CP^2$. That estimate is insufficient by itself for the cutoff-uniform pointwise quotient in M-03. The stronger bridge considered here is

$$g^{-2}s(V_p^{(k)}) \le C\left[1 + \sum_{p' \text{ surviving}} e^{-c\rho(p,p')}\beta_{p'}\, s(V_{p'}^{\text{out}})\right]. \tag{NP19.21}$$

The positive constant in (NP19.21) pays the distant boundary of the large regular patch. Every plaquette on the right belongs to the actual surviving determining configuration, with its stated scale and footprint.

Theorem NP19.4 — the actual local minimizing-field energy bound Work in the current physical units and fix a plaquette p whose complete current averaging stencil is inside a regular cube. The entire cube and the fixed source margins used below must be contained in the deepest original determining region Ω_{k+1}. Suppose the cube has a margin of M current units, where M is an allowed multiple of the fixed source block constant. Use the actual minimizing fine configuration and actual determining averages of CMP119 (3.10), with the source local regularity conditions on that enlarged cube. Write ϵ_0 for their current normalized bound. Assume

$$C_{\text{loc}} M \epsilon_0 \le \epsilon_*, \tag{NP19.22}$$

where the two constants are fixed sufficiently small/large according to the source local minimizing-map domains. The bound ϵ_0 covers both curvature and the source covariant-divergence regularity in CMP102 (2); the curvature inequality CMP119 (3.12) alone is not substituted for that full admitted regularity class. No bound is imposed on a number of microscopic bonds.

Then, with fixed $C, c > 0$,

$$\left\| V^{(k)}(\partial p) - 1 \right\| \le C \sum_{p' \in \mathcal{P}_M} e^{-c\rho(p,p')} \left\| V^{\text{out}}(\partial p') - 1 \right\| + CM\epsilon_0 e^{-cM}. \tag{NP19.23}$$

Here $\mathcal{P}_M$ consists of the actual next-scale determining plaquettes in the chosen cube, not fine plaquettes of the minimizing field. Their current distances are converted to next-scale distances only by the fixed block factor L. Constants depend on the source group and fixed geometric parameters, not M, the fine mesh, depth, volume or the exterior history.

On an actual deep regular patch one can choose $M \asymp R_k \asymp \ell_k^r$, $r > 1$, with $\epsilon_0 \le C g_k \ell_k^q$ from the admitted current regularity class. Then (NP19.22) holds at a fixed sufficiently small coupling ceiling. If the next-step native inverse coupling obeys the source-normalized comparison

$$c_\beta g_k^{-2} \le \beta^{\text{out}} \le C_\beta g_k^{-2}, \tag{NP19.24}$$

the bound (NP19.21) follows, with a uniform constant. The comparison (NP19.24) is part of the same admitted one-step coupling trajectory; this theorem does not prove B-03 or identify a new bare coupling.

Proof. The local construction needed here is already in CMP102, Section F, rather than an unproved interpolation from an arbitrary global field to the identity. Its precise roles are the following.

1. In (144)–(150) the source restricts the actual minimum to a nested family of local cubes and fixes its own auxiliary boundary averages. The source explains immediately after (150) that it remains a minimum for this more restricted local problem. The determining coordinates in the inner deepest grid are the original ones. Artificial finer determining grids occur only in the constructed boundary region. Our deep-patch choice makes the original mixed set $\square_k'^{(k-1)}$ in the proof of (160) empty: the whole inner determining grid is the actual next-scale V_{k+1} grid. The artificial finer boundary data are retained and estimated separately below.

2. Equations (151)–(157) put this local problem, after the stated gauge changes, into the exact nonlinear constraint chart about the identity. No one-step exceptional-bond map is substituted for the fine multiscale constraint. In the source notation,

$$A = A_1 + HB - HD(A_1 + HB), \qquad A_1 + \mathcal{G}\, V'(A_1 + HB) = 0. \tag{NP19.25}$$

These are (158)–(159), with $\mathcal{G}$ denoting the source's constrained covariance operator. The functional V in (157) begins cubically. Its derivative at zero is zero and its derivative Lipschitz constant on the relevant small ball is $C\epsilon$, where $\epsilon = C_{\mathrm{loc}} M \epsilon_0$. The exact constraint correction D begins quadratically. The source explicitly states below (158) that these operators have identity external field. The constructed source coordinates satisfy all the required physical first-, second-, and higher regularity norms with bound $C_{\mathrm{loc}} M \epsilon_0$.

3. The defining gauge is a generalized axial gauge rooted at a point next to the marked plaquette. In (154) the field B is the actual logarithmic determining input for this local problem. Equation (155) bounds it globally by $CM\epsilon_0$. On the inner determining grid, the path argument preceding (160) expresses it in terms of the actual determining plaquettes. Its usual uniform ϵ_1 bound can be sharpened by retaining each plaquette separately, as below.

We first make the weighted estimate in (NP19.25) explicit. Let $d_{\Omega'}$ be the source weighted metric for the nested local cubes, and put

$$w(y) = e^{-\sigma d_{\Omega'}(x_p, y)} \tag{NP19.26}$$

for a fixed sufficiently small $\sigma > 0$. Weight the source physically normalized field norms by w. For example, on a determining block of proper size h_j, the value and first-gradient weights are $h_j|A|$ and $h_j^2|\nabla A|$. The corresponding forcing norm has weight h_j^3. Use those physical factors, not the norm of an unrescaled matrix on the fine lattice.

The triangle inequality gives $w(y)/w(y') \le e^{\sigma d_{\Omega'}(y,y')}$. The pointwise kernels of $H, \mathcal{G}$, and their required output derivatives have a fixed exponential in this metric. Reserving σ of that exponential and applying the same physical weighted row sums used in CMP102 (161)–(162) therefore gives uniform operator bounds on these weighted norms. In particular, the factors h_j^{-1} occurring explicitly in the sum in (161) are retained and summed by that source estimate; they are not replaced by a microscopic counting measure. The geometry of the nested cubes is exactly the source geometry in (144), so these row constants do not grow with M or fine depth.

For the nonlinear functions one uses the same argument as for the weighted contraction in BM.8–BM.9. To specify the factors covered by that argument, let $\mathcal{D} = \partial^{\eta *}\partial^{\eta}$ be the identity-background quadratic operator and set $F(A) = A - HD(A)$. The exact functional in source (157) gives

$$\begin{aligned} V(A) &= \tfrac{1}{2}\langle F(A), \mathcal{D}F(A)\rangle + V_0(F(A)) - \tfrac{1}{2}\langle A, \mathcal{D}A\rangle, \\ V'(A) &= -\mathcal{D}HD(A) - D'(A)^* H^* \mathcal{D}F(A) + (I - D'(A)^* H^*) V_0'(F(A)). \end{aligned} \tag{NP19.26a}$$

All adjoints are in the source physical scalar products. The adjoint, derivative, and propagator compositions in this display are the ones estimated in the source's actual derivative calculation (85)–(97); their differentiated physical kernels and measures are retained. The exponential Hessian estimate (189) is a further explicit source bound for this derivative calculation, with the proper h_j^{-3} forcing factor and the source input norms retained. It supports the same physical weighted contraction; it is not an identification with the derivative of a different composed map. Thus "every occurrence" includes those compositions, not only the undifferentiated H. The local nonlinear averaging equation for D has $N(0) = DN(0) = 0$ and $D = N(A - HD)$; its derivative solves $(I + N'(F)H)D' = N'(F)$. Its weighted Neumann series has the same small fixed ceiling.

Every local analytic factor has bounded proper-scale footprint. In a difference, assign the weight to the differing factor and use the source global small bound for all its other factors. Every physical kernel, including the adjoint/derivative kernels just specified, loses only σ of its exponential. Its row sums, with the source measure weights, remain bounded as above. The cubic V_0, the quadratic D, and their exact products in (NP19.26a) then give one factor of the global small radius in the derivative Lipschitz bounds. This proves, on the actual source ball,

$$\begin{aligned} \|\mathcal{G}[V'(A') - V'(A'')]\|_w &\le C\epsilon \|A' - A''\|_w, \\ \|HD(A')\|_w &\le C\epsilon \|A'\|_w. \end{aligned} \tag{NP19.27}$$

All these are physical source norms. The derivation uses the actual factors in the nonlinear source equations, rather than identifying the derivative of a different composed map with V'. An absolute fine-mesh Schur bound for a matrix with entries of size η^{-2} is neither asserted nor needed.

Choose the fixed ceiling in (NP19.22) so that $C\epsilon \le 1/4$. Since $V'(0) = 0$, (NP19.25) and (NP19.27) imply

$$\|A_1\|_w \le \frac{C\epsilon}{1 - C\epsilon} \|HB\|_w, \qquad \|A\|_w \le C\|HB\|_w \le C\|B\|_w. \tag{NP19.28}$$

The final input norm is explicitly $\|B\|_w = \sup_c w(y_c)|B(c)|$, where y_c is the source determining-block label of c. Thus B has no extra microscopic power. This is the weighted version of the dimensionless input norm in CMP102 (45)–(46). The preceding kernel argument, and the physical row sum in (161)–(162), specify how the scale factors in the operator are paid.

It remains to estimate this input using actual curvature. On this inner deepest grid the source has $B(c) = i^{-1} \log V'(c)$, as in the first case immediately preceding (160). The generalized axial gauge writes the group element $V'(c)$ as a product of conjugates of the actual determining plaquette holonomies in an axial surface Σ_c. All conjugating factors are compact and unitary at the real retained data. The elementary telescoping inequality for products of unitaries yields

$$|B(c)| \le C \sum_{p' \in \Sigma_c} \|V^{\mathrm{out}}(\partial p') - 1\|. \tag{NP19.29}$$

The fixed principal logarithm exists by (155) and (NP19.22), and its norm is bounded by a fixed multiple of the group difference. The surface lies in a coordinate rectangle rooted at x_p; every $p' \in \Sigma_c$ has distance at most $C[\rho(x_p, c) + 1]$ from that root. Its multiplicity is bounded by a fixed polynomial in that distance. Multiplying by $w(c)$, spending half its exponential to pay this polynomial, and then taking the supremum in c gives

$$\|B_{\mathrm{inner}}\|_w \le C \sum_{p' \in \mathcal{P}_M} e^{-c\rho(p,p')} \|V^{\mathrm{out}}(\partial p') - 1\|. \tag{NP19.30}$$

There is no assertion here that the second, adjacent-original-layer case of source (160) is an exact holonomy product of V_{k+1} plaquettes. That case uses the additional finer input and the one-step estimate of the cited gauge-fixing lemma. Applying the present theorem to a cube meeting such an original layer would require retaining its actual finer inputs and scales. The deep-patch hypothesis excludes that extension.

For the auxiliary boundary determining grids use (155), retaining the actual root-to-boundary distance in the first sum of (161). That distance is at least cM; the final uniform estimate in the printed (161) had replaced it by a fixed minimum because it was estimating every point of a large cube simultaneously. A fixed further exponential-rate reserve pays the finer-layer physical factors exactly as in (162). The result is

$$\|HB_{\mathrm{boundary}}\|_w \le CM\epsilon_0 e^{-cM}. \tag{NP19.31}$$

One may use (NP19.31) directly in (NP19.28); no claim that an unweighted fine-node sum is bounded is involved.

At the marked root and a fixed neighborhood the weight in (NP19.26) is bounded below. The bounds on $A, \nabla A$, and the elementary plaquette expansion of $e^{i\eta A}$, give fine curvature bounded by η^2 times the sum of (NP19.30) and (NP19.31). The quadratic terms cost one extra small global norm and are absorbed in C. Finally apply the actual

uniform-depth averaging estimate BM.20–BM.21 on the fixed current stencil. Its entire footprint lies in the marked neighborhood. This gives the actual averaged plaquette bound (NP19.23), not a linearized or a newly inserted average.

For (NP19.21), use $\|U-1\|_{\mathrm{HS}}^2 = 2s(U)$, fixed norm equivalence, the uniform row sum of the exponential plaquette kernel, and weighted Cauchy–Schwarz. The square of the first term in (NP19.23), divided by g_k^2, is at most $C\sum_{p'} e^{-c\rho(p,p')}\beta^{\mathrm{out}}s(V_{p'}^{\mathrm{out}})$ by (NP19.24). The squared boundary error divided by g_k^2 is bounded by $CM^2\ell_k^{2q}e^{-2cM}$ on the chosen patch, hence by a fixed constant. This proves the claimed energy inequality. □

The theorem uses the original source local-minimizer construction and its weighted physical kernels. It does not enlarge a current scalar density theorem to arbitrary raw marked kernels. It supplies a value-regulator estimate on a deep admitted regular patch. The separate derivative/support/history records in M-03 still require their actual constructions.

6. Application conditions

The regulator estimates apply on the stated conditional law and minimizing-reference patch.

For the estimate, the following still require proof for the same actual incoming scalar mark:

1. A root-insertion expansion with termwise native-regulator bounds, complete current records and an M-03 carrier sum; the already-proved positive majorant is not this expansion. A source completed-operation norm that does not charge a fixed nonsmall component's interior cannot simply pay the full M-03 physical carrier weight.
2. The retained-coordinate pullback, common support or compatible extension, and shifted-envelope bounds for all actual directions included in M-03. The stated native inner-current contour does not stand for every raw retained or history direction.
3. The scalar/identity extraction from that exact output in the F-01 reference basis, with the remaining localized terms bounded in the same norm.

Annex BE — Constrained stationarity: an auxiliary conditional theorem

Annex BE. Exact finite conditional identities apply to the specified positive kernel. A uniform Wilson application requires CL on the same retained sigma-field, the relevant complete-potential estimates under CR and the stated spatial and energy budgets. History estimates · Observable transport.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

1. Weighted inverse for an averaged constrained Hessian

Let a finite block graph have distance rho and bounded exponential volume sums. Write the variables as t=(z,y), with free interior blocks I and collar/exterior blocks O. Each coordinate block has a fixed Euclidean norm, or more generally a Hilbert norm with uniform row/column estimates. Let S be a real C^2 function on a neighborhood of the segment joining 0 to t*. The coordinates are assumed to parameterize the exact constraint, not just its tangent space.

Suppose, throughout that segment, the interior Hessian satisfies

$$S''_{II}(st^*) \geq \lambda I, \qquad \sup_{i\in I} \sum_{j\in I} \|S''_{ij}(st^*)\|\big(e^{a\rho(i,j)} - 1\big) \leq \lambda/2, \tag{LS19.1}$$

with the symmetric column estimate. Assume also

$$\sup_{0\leq s\leq 1} \|S''_{ij}(st^*)\| \leq Ke^{-b\rho(i,j)}, \qquad i \in I,\ j \in O, \tag{LS19.2}$$

for fixed a,b,lambda,K>0. Put

$$\overline{K} = \int_0^1 S''(st^*)\, ds. \tag{LS19.3}$$

Both hypotheses pass to this average. For a real one-Lipschitz spatial function f, multiply coordinate block i by M_i=exp(a f_i). The norm of M Kbar_II M^-1-Kbar_II is at most lambda/2, by the row/column Schur estimate in (LS19.1). Hence

$$\|M\overline{K}_{II}^{-1}M^{-1}\| \leq 2/\lambda, \qquad \|(\overline{K}_{II}^{-1})_{ij}\| \leq (2/\lambda)e^{-a\rho(i,j)}. \tag{LS19.4}$$

The first statement follows by a bounded perturbation of the positive matrix Kbar_II, or directly from the real-part coercivity inequality. Choose f_l=rho(l,j) for the second. No diagonal dominance of Kbar_II is assumed.

2. Stationary interior and boundary response

Theorem LS19.1 Suppose t=*(z,y) is stationary in the free interior variables, so D_I S(t)*=0. Let d=D_I S(0). Under (LS19.1)–(LS19.2),

$$z^* = -\overline{K}_{II}^{-1}\big(d + \overline{K}_{IO}y^*\big). \tag{LS19.5}$$

Consequently, for any c<min(a,b)/2,

$$|z_i^*| \leq C\sum_{j\in I} e^{-a\rho(i,j)}|d_j| + C\sum_{j\in O} e^{-c\rho(i,j)}|y_j^*|. \tag{LS19.6}$$

If d is supported in a collar B and bounded by D there, and |y_j*|<=Y, then

$$|z_i^*| \leq C(D+Y)e^{-c'\,\mathrm{dist}(i,O\cup B)} \tag{LS19.7}$$

for any fixed c'<c. Constants are independent of the total volume, provided the stated graph exponential sums and Hessian estimates are uniform.

Proof. The fundamental theorem of calculus gives

$$0 = D_I S(t^*) = d + \overline{K}_{II} z^* + \overline{K}_{IO} y^*.$$

Invert with (LS19.4) to obtain (LS19.5). The matrix product between the inverse and Kbar_IO is bounded by a convolution of two exponential kernels. The triangle inequality and one fixed rate loss give (LS19.6). A further rate loss leaves a summable kernel and proves (LS19.7). The exact constraint has already been built into S, so its first and second derivatives include the corresponding constraint-current terms. Omitting those terms would change the hypotheses and the equation. ■

The theorem concerns a path in an exact free-coordinate chart on a local patch. It does not require a path connecting arbitrary global determining data to a globally flat configuration. This avoids that particular topological assumption, but it does not verify the local chart or its Hessian by itself.

Fine-mesh qualification and weighted-form alternative The absolute matrix hypothesis (LS19.1) is suitable for fixed-spacing coordinate blocks. It must not be claimed automatically uniform on a refining fine lattice. A physical fine Laplacian has neighbor entries of order eta^-2. With distance in physical units, the absolute conjugation sum in (LS19.1) then has a bound of order a/eta. Physical elliptic weighted estimates instead use form cancellations, or the source's uniform Green-function norms.

There is an exact abstract alternative. If the averaged constrained operator has the weighted coercivity property

$$\mathrm{Re}\langle Mw, M\overline{K}_{II} w\rangle \geq c\|Mw\|^2 \tag{LS19.11}$$

for every admissible spatial weight M, on its actual common form domain, then its weighted inverse has norm at most c^-1. Equations (LS19.5)–(LS19.9) follow with that inverse and the corresponding mixed collar kernel bounds. The proof tests the inverse equation with the weighted vector and applies Cauchy–Schwarz. A weighted form estimate of this kind must include constraint/projection commutators and domain preservation; positivity of an unweighted fine Hessian alone does not prove it. Grouping fine variables into coarse blocks likewise does not automatically bound the interface block operators.

In the actual variational source, the global right inverse H and nonlinear correction A'=A-H D(A) are distinct from C-08's local one-step exceptional-bond chart. A multidepth fine-grid application must use the true H, its tails and the source physical norms, or prove a different uniformly controlled exact chart. The one-step h cannot be substituted silently.

3. A local observable and its curvature interpretation

Let Phi_i(t) be a local observable in the same chart with Phi_i(0)=0 and an anchored derivative bound

$$\|D_j \Phi_i(st^*)\| \leq C_\Phi e^{-b_\Phi \rho(i,j)}. \tag{LS19.8}$$

Integrating along the segment and inserting (LS19.6) gives an exponentially localized bound for |Phi_i(t)| *in terms of d and the retained y*. If both are confined to a collar at distance r, then

$$|\Phi_i(t^*)| \leq C C_\Phi (D + Y) e^{-c_\Phi r}. \tag{LS19.9}$$

A plaquette curvature of an exactly flat reference is an example only after its actual constrained chart derivative has been checked. If a current-normalized curvature has an additional factor g^-1, that factor must remain on the right. For a collar R=(log g^{-2})r0 with r0>1, a boundary budget bounded by g^-A times a fixed polynomial in log g^-2 is then suppressed by exp(-cR). This last numerical statement follows from

$$A \log g^{-1} + q \log\log g^{-2} - c(\log g^{-2})^{r_0} \longrightarrow -\infty. \tag{LS19.10}$$

4. Exact application checks still needed for M-04

For the background field in CMP119, the notation U_(k+1) denotes a minimizing fine configuration determined by retained coarse averages. Its fine plaquette energies cannot be inserted into the M-03 output regulator merely by renaming them retained coordinates. The following source-specific checks are necessary to use the theorem above:

1. A local patch whose interior determining averages are flat admits an exact free-coordinate chart for the actual fine minimizer, with the boundary and any crossing constraints confined to a stated collar.
2. The actual minimizer and the radial path in that chart lie in the admitted source domain, uniformly in the fine mesh, including the required curvature/current norms and the nonlinear constraint correction.
3. The fine constrained Hessian and its mixed collar block satisfy the required uniform weighted inverse and collar estimates on that path in the correct physical normalization, through (LS19.1)–(LS19.2) when those absolute bounds hold or the weighted-form alternative (LS19.11) with its domain and commutator checks. A current-lattice C-08 estimate alone is not a fine-grid minimizing-field theorem.
4. The local reference produces d=0 in the interior, and its boundary d and y budgets have the explicit polynomial/coupling costs used in (LS19.10).
5. The curvature derivative estimate (LS19.8) holds for the actual averaging and reconstruction map. Its endpoints, scales and all retained field slots must be identified.

If those are verified, the flat-inner-patch field decays into the interior by (LS19.9). Combining it with a local gauge-covariant response to nonflat determining data can give the desired weighted output-energy bound. None of the five checks is silently imported from the finite-dimensional algebra.

Annex BF — A sharp limitation of logarithmic rarity estimates

Annex BF. Exact finite conditional identities apply to the specified positive kernel. A uniform Wilson application requires CL on the same retained sigma-field, the relevant complete-potential estimates under CR and the stated spatial and energy budgets. History estimates · Observable transport.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

Proposition RG19.1

Fix c>0 and r>1. There is a family of finite positive probability laws with a nonnegative bounded mark X_l, indexed by l>=1, for which

$$0 \le X_l \le e^l, \qquad \mathbb{P}(X_l \neq 0) = e^{-cl^r}, \tag{RG19.1}$$

but no constants C,p independent of the integer m can satisfy

$$\sup_{l\ge 1} \mathbb{E}[X_l^m] \le C^{m+1}(m!)^p \quad \text{for every } m \ge 1. \tag{RG19.2}$$

Proof. Take a two-point law with X_l=e^l on the indicated rare event and X_l=0 otherwise. Then

$$\mathbb{E}[X_l^m] = \exp\{ml - cl^r\}. \tag{RG19.3}$$

For sufficiently large m choose l=(m/(cr))^(1/(r-1)), which is at least one. Its exponent is

$$(r-1)c\left(\frac{m}{cr}\right)^{r/(r-1)}. \tag{RG19.4}$$

This grows faster than (m+1)log C+p log(m!) for every fixed C,p, since r/(r-1)>1 and log(m!)<=m log m. Thus (RG19.2) is impossible. ■

The averaged M-03 grade is also stronger than fixed-order rarity

For the same family and a fixed positive integer m, consider X_l^m with value grade m. Whenever c l^r/m>=1, take the probability moment order q=c l^r/m in (M.8). Direct substitution gives

$$\mathfrak{m}_m(X_l^m) \ge \frac{\exp(ml - m)}{(cl^r)^m}. \tag{RG19.5}$$

It diverges as l tends to infinity. Thus the rare-event bound in (RG19.1) does not yield a uniform averaged M-03 bound even at one fixed source order. The pointwise regulator part requires additional control as well.

Implication for the proof strategy

A compact insertion can have a crude bound proportional to g^-2, with l=log g^-2. Combining that crude supremum with a source small factor exp[-c(log g^{-2})r] may control each fixed derivative order with order-dependent constants, but it does not prove fixed-factorial control of the full hierarchy. The calculation above gives the precise growth obstruction to that deduction.

An actual exponential reserve in a nonnegative coercive energy E supplies the needed different information. If an unnormalized positive measure M satisfies M(exp(theta E))<=A for fixed theta>0, then

$$M(E^m) \le A\, m!\, \theta^{-m}, \tag{RG19.6}$$

because (theta E)^m/m!<=exp(theta E). This bound preserves a fixed factorial exponent. Applying it to the source operation requires identifying which actual energy has the reserve and which incoming marks or regulators it controls. A reserve for an outer quadratic/Wilson energy does not automatically cover every older hidden native coordinate.

Annex BG — A spatial carrier criterion for marked conditional kernels

Annex BG. Exact finite conditional identities apply to the specified positive kernel. A uniform Wilson application requires CL on the same retained sigma-field, the relevant complete-potential estimates under CR and the stated spatial and energy budgets. History estimates · Observable transport.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

For the developments under their corresponding hypotheses, see the history extensions and the scalar matching analysis.

The following is a concrete sufficient estimate for the remaining H-03 problem. It is not proved. It states the unknown in the existing M-03 quantities rather than replacing the estimate by a moment bound. The actual input records are the prepared records entering the R step; their earlier preparation is a separate obligation.

Fix the literal positive source-free R coupling, all retained exterior and residual-history data x, and the actual old conditional probability Q_x obtained after including its self action and all cross activities. Fix a source order m and a partition into incoming vertices F_1,...,F_l, of positive source grades d_1,...,d_l with sum m. Each incoming vertex carries its existing compatible triple (F_i,B_i,J_i) and its true padded dependency footprint. The conditional cumulant is

$$K_d(x) = \kappa_{Q_x}(F_1, \dots, F_l). \tag{KT19.1}$$

For l=1 this denotes Q_x F_1. All higher incoming log-jet vertices are included by allowing every partition and every grade tuple; replacing them by only first-order marks would not meet the test.

Required actual decomposition and estimate

Construct a compatible physical-carrier expansion

$$\mathrm{K}_d = \sum_Z \big(K_Z, B_Z, J_Z^{\mathrm{old}} + J_Z^{\mathrm{new}}\big), \qquad \delta K_Z = B_Z + J_Z^{\mathrm{old}} + J_Z^{\mathrm{new}}, \tag{KT19.2}$$

in the exact current/retained coordinates of this R step. Here B_Z includes the direct coordinate pullback, J_Z^old transports the already present current records, and J_Z^new contains every change of the normalized old law, including selectors, domain/geometry labels and the denominator. It is not permissible to combine B and J before taking their separate norms.

For fixed positive kappa_out less than the available input decay reserve and some fixed C,q, prove

$$\sum_Z e^{\varkappa_{\mathrm{out}}(|Z|+\mathrm{t}_h(Z))} \big[\upsilon_{h,m,Z}(K) + g_{h,m,Z}(B) + j_{h,m,Z}(J^{\mathrm{old}}) + j_{h,m,Z}(J^{\mathrm{new}})\big] \le C^{m+l}(l!)^q \prod_{i=1}^{l} \|\mathrm{F}_i\|_{h,\varkappa_{\mathrm{in}}}. \tag{KT19.3; open}$$

The terms v,g,j and the input norm in this display are exactly M-03 (M.11)–(M.12). In particular their value part includes

$$h^{4m}\mathfrak{m}_m(K_Z^\sharp;\mu_{\mathrm{out}}) + \frac{h^{4m}}{m^m}\sup_x \frac{K_Z^\sharp(x)}{\widehat{R}_{h,Z}(x)^m}, \tag{KT19.4}$$

and their current parts use the grade m+1/2, the summed finite-difference envelope, and the corresponding shifted output native regulator. All ambient shifts in the specified M-03 versions are included; membership in an interior analytic chart is not a substitute for this quantifier. Constants may depend on the fixed source parameters and chosen input-class budget, but not on the cutoff, volume, number of older layers, actual remaining history or the position of the component.

The full jet is the sum of KT19.1 over its labeled partitions. The already recorded finite-jet partition algebra then gives its complete response; it does not remove any part of KT19.3. Subsequent all-scale inductive closure remains with the corresponding marked-induction estimates.

What must enter the new-current record

For one retained direction tau with nonzero continuous displacement t, let M_x and M_(tau x) be the unnormalized kernels on the same padded hidden space, with masses Z_x and Z_(tau x). The mandatory new current at a first moment is

$$J_{\tau}^{\mathrm{new}}(F)(x) = \frac{1}{tZ_{\tau x}} \int (F_x - Q_x F_x)\, d\,(M_{\tau x} - M_x)\,. \tag{KT19.5}$$

For a discrete replacement take t=1. The direct term is Q_(tau x)[(F_(tau x)-F_x)/t], preserving its B and old-J split. If M_x=chi_x r_x and the smooth part changes by exp(sigma), KT19.5 includes the selector measure (chi_(tau x)-chi_x) exp(sigma) r_x. For a cumulant apply these exact rules to every conditional moment in its finite partition formula. This specifies J^new without discarding support changes.

The marked chart problem is to realize KT19.2–KT19.5 for the actual preintegration map, including every old hidden variable. The native fixed-support tilt in LFR II (1.73)–(1.75) bounds one part of KT19.5. It does not identify all the raw/selector/geometry changes in that display with native analytic shifts.

Precisely what is available and what remains

The pure fresh-coordinate directions in the correctly fixed-reference R quotient have J^new=0 exactly, as proved in MG19.1. The continuously integrated support after preparation is inside a fixed current-block core, so its connector cost is bounded. These facts remove two costs from KT19.3.

Still open are the actual retained-direction chart/pullback and selector terms in KT19.5, their pointwise comparison to the shifted output regulator in KT19.4, and their full dependency footprints in KT19.3. The source-native response endpoints include Gamma' '_k intersect X-tilde, so the core volume alone does not pay that full footprint.

An explicitly adopted joint-history coupling can set additional fresh and output-descriptor currents to zero and preserve small component masses. If it is used, Q in KT19.1–KT19.5 must be its actual mixture posterior, including the descriptor-mean cumulants. The same norm estimate must still be proved. Changing the coupling does not, on its own, verify KT19.3.

Annex BH — A scalar matching criterion for the transported physical insertion

Annex BH. Exact finite conditional identities apply to the specified positive kernel. A uniform Wilson application requires CL on the same retained sigma-field, the relevant complete-potential estimates under CR and the stated spatial and energy budgets. History estimates · Observable transport.

The normalized coupling and geometric hypotheses are retained throughout. Whenever the Wilson trajectory is identified through Gaussian calibration, GC includes the error bounded uniformly in depth. Theorem catalogue.

For the developments under their corresponding hypotheses, see the history extensions and the scalar matching analysis.

The following conditions concern exact transport of the physical scalar insertion and its remainder.

The scalar matching problem requires equality with the actual transported scalar insertion $Q_k F_k$, followed by a localized remainder bound in the already specified M-03 norm, uniformly in scale. A new coarse plaquette inserted after the integration is not that left side. The exact estimate statement also retains constrained-background, E/B/R and coarea responses present in the selected regular sector.

Annex BD gives the following partial estimates. They remove a logarithmic support loss from the actual regular conditional value regulator. NP19.1 retains the actual exterior force, NP19.2 covers admitted native unit shifts, and NP19.3 pays thin paths for the resulting positive majorant. NP19.4 controls the actual minimizing reference energy on a patch wholly inside the deepest original Ω_{k+1}, with the declared curvature/current regularity, full averaging stencil, and the same admitted one-step coupling ratio. Its artificial finer boundary inputs remain in the exponentially small boundary term. These estimates do not supply the three requirements below.

1. Exact rooted output with its full physical carrier sum

Specify a genuine one-step incoming budget $\mathfrak{B}_k(F_k)$, in the fixed F-01 incoming basis and the actual augmented variables. It may include stated analytic/local hypotheses that are proved for the input. It must not assume the desired norm of $Q_k F_k$.

Construct the marked numerator using the same denominator and the same complete density as the source-free operation. After cancellation of unmarked components, prove an actual carrier identity

$$Q_k F_k = \sum_X A_{k+1,X}, \tag{M04-R1}$$

where each $A_{k+1,X}$ is a function of the coordinates in its declared full dependency footprint X, with all constrained-background, E/B/R, Haar, coarea and retained-mean terms assigned once. All current and response records are part of the construction.

The terms must satisfy the value part of the actual M-03 carrier bound, with constants controlled by $\mathfrak{B}_k(F_k)$ uniformly in cutoff, volume and the admitted regular sector. Its weight is the existing $w_h(X) = \exp[\varkappa(|X| + \mathfrak{t}_h(X))]$, including full footprints. A source norm that exempts a fixed nonsmall component's interior does not supply this bound. Nor does a small positive majorant prove that the function it bounds has the majorant's support. NP19.3 proves a payment for a majorant only.

The rooted expansion in CMP119 (3.44)–(3.47) is a concrete source starting point. The needed marked estimate has not yet been proved for this input budget and this norm.

2. Exact direct/current records on the full M-03 ambient domain

For every actual M-03 perturbation, retain the exact identity

$$\delta_\tau(Q_\upsilon F_\upsilon) = Q_{\tau\upsilon}\Big(\frac{F_{\tau\upsilon} - F_\upsilon}{t}\Big) + \frac{Q_{\tau\upsilon} - Q_\upsilon}{t}(F_\upsilon), \tag{M04-R2}$$

with denominator $t = 1$ for discrete history changes and the prescribed normalized steps for continuous variables. Pullbacks and extensions in this formula must be the actual chosen versions. The second term includes changes of

the normalized density, support, selectors and reference geometry. It is not discarded because the unmarked normalization is one.

Construct compatible carrier triples $A_{k+1,X}, B_{k+1,X,\tau}, J_{k+1,X,\tau}$ with $\delta_\tau A_{k+1,X} = B_{k+1,X,\tau} + J_{k+1,X,\tau}$. Prove their complete M-03 bound: shifted value envelopes have grade one and direct/current envelopes have grade 3/2 for this one scalar mark. This includes both the specified all-p moments under the actual outer law and the ambient pointwise quotients by the native regulator. The norm counts direct and current records separately.

A common interior native chart proves only its admitted current directions. Raw retained links, archived labels, zero-normalizer extensions and chart boundaries still need their own exact treatment. If improved extensions are used, record and prove their compatibility with the actual on-support kernel. NP19.2 does not provide this gluing result.

3. The scalar and identity readout of that same output

Define and justify the scalar/identity readout in the F-01 reference basis for the output from requirements 1–2, keeping the same incoming mark. Prove

$$Q_k F_k(f) = B_k^{\mathrm{scalar}}(f) + c_k(f)1 + \sum_X R_{k,X}(f), \tag{M04-R3}$$

where B_k^{scalar} is expressed in the current F-01 scalar reference basis, the coefficients and identity subtraction are specified, and the complete remainder triples obey the M-03 bound uniformly in scale under $\mathfrak{B}_k$. The exact selected-law virial readout already recorded in Annex AB is available, but its retained mean, full density gradients, collar expectation and current terms must receive these bounds.

The requested matching does not require assuming the final running normalization from F-05, and these conditions do not add that circular premise. They also do not claim the smaller perturbative mixing size belonging to the later estimate merely from a bounded one-step remainder.

Relation to the marked transport problem

Together the three conditions would give the required scalar matching. The available value estimates address only part of the carrier and current bounds.

Annex Y defines the carrier norm and finite transport rules. Annex AY specifies the conditional regular-law hypotheses.

Annex BI — The complete positive reset posterior and all-order spatial responses — H-03

The purpose of this estimate is to estimate the actual reset law. The native source coordinates used below are not identified with independent physical retained links after a minimizing-field pullback. The resulting distinction is stated explicitly in Section 8.

Every probability statement is on the genuine real compact source-free slice, with the auxiliary analytic current J=0, as in LFR II p.380. The analytic native chart is used only for the already admitted uniform majorants; a general complex or analytically deformed pair (U,J) is not assigned a probability law.

1. The three kinds of history in the literal source operation

Use Balaban, *Large Field Renormalization II*, CMP122 (1989), printed pp.378–392, especially (1.71)–(1.72), before any signed Mayer expansion. The notation in this note is:

- `Zbar`: the unreplaced region omitted from the eligible R operation;
- `Y_1,...,Y_s`: eligible components for which the final fresh small-field test fails; their geometry remains in the output;
- `C={X_1,...,X_n}`: eligible components for which that final test succeeds; their geometry is summed in the braces of (1.72).

The internal state `eta_D` of a completed component D includes all old variables, old restricted domain sequences, surviving older T operations, graph choices and gauge-fixing variables integrated by (1.71). It does not mean only the final B fluctuation. Printed p.378 defines the same T-prime operation for a component of either the X or the Y class.

The exact marked meaning of the successful-component Haar restoration Printed p.378 calls the removal of the fresh axial delta functions an integrated equivalence, not equality of the prior gauge-fixed densities. The following argument specifies the positive marked coupling used here.

At a fixed complete reference descriptor gamma, the literal R quotient in LFR I (1.100) has joint density

$$k_\gamma(u' \mid v_\gamma)\,\rho_\gamma(u, v_\gamma)\,du\,du'\,dv_\gamma, \qquad k_\gamma = h_\gamma/H_\gamma. \tag{HS20.0a}$$

Here u' is the fresh current field on Lambda_gamma, u contains the old integrated field on that domain and every earlier hidden variable, and v_gamma is the retained exterior. A transported old marked numerator multiplies this density by F(u,v_gamma). It is independent of u'. This is the actual fixed-reference identity MG19.1, before replacing it by a refactorization of the effective action.

LFR I p.201 specifies that G_gamma is an axial gauge graph in Lambda_gamma and that the reference denominator integrates precisely the fresh variables on Lambda_gamma. Take real gauge transformations supported on that domain and fixed at its exterior boundary. They leave v_gamma unchanged. The constrained reference minimizing orbit is gauge covariant under these transformations: its determining data transform at their endpoints, the Wilson action is invariant, and the minimizing orbit is the unique orbit supplied by the source variational construction. Thus the localized Wilson factor in h_gamma is invariant along these fresh gauge orbits. No transformation of an old raw field is needed.

On the successful sector, LFR II p.378 proves that the extra fresh cutoff chi(Lambda_gamma) equals one. The exterior cutoff chi_(k,Lambda) is already fixed by v_gamma, and the remaining successful small-field factors are gauge invariant. Therefore the source Faddeev–Popov restoration is the following positive operation: draw the normalized compact gauge-orbit variable and apply it only to u', leaving gamma,u,v_gamma fixed. Haar invariance and the tree gauge identity give the restored reference density. Since every old mark F(u,v_gamma) is fixed during this operation, the same identity holds with any old marked numerator and coefficientwise for every full finite jet. Old archived selectors are untouched.

This is not an assertion of pointwise equality between the old gauge-fixed density and the restored one. It is the explicit positive coupling realizing the source's integrated equivalence. If an analysis coordinate is subsequently changed, its

old mark is pulled back through that exact component coordinate change. It must not be replaced by an unchanged function of a freshly inserted coarse plaquette. The extra compact gauge variable is component-local; conditional on the source component coordinates its old preimage is a local lift. It introduces no cross-component factor into the source action.

The lemma concerns the full transported jet. A prior artificial decomposition into separate fresh-field-dependent records may require recombination or the corresponding orbit averages before this operation; those separate records are not individually declared invariant. This restriction prevents a hidden assumption about the stronger M-03 current norm.

At this stage the new field and the residual output data are common to the successful-component sum. Denote them by v. More precisely, v includes the actual new background U_k, the new current field V_k where it occurs explicitly, the unreplaced Zbar history data, and the Y geometry. Internal eta_Y variables are not included in v for the full posterior below.

The exact common-background statement is LFR II (1.33)–(1.35), printed p.365: the new determining set keeps the old data outside the eligible Z region and uses the current k-grid throughout Z. Its new raw current field there is V'_k V_Lambda. Thus the same complete new V_k and unreplaced exterior define U_k, independently of the internal component histories. The classification on p.378 combines the successful components' final small-field factors into the common new small-field factor. Varying C in (1.72) therefore does not vary v. This is a statement about the exact source representation, not a claim that the old and fresh integration windows were the same before the R operation.

For a successful component X write its real positive unnormalized operation as

$$M_X(v, d\eta_X) = T'_k(X; v)(d\eta_X), \qquad m_X(v) = M_X(v, 1). \tag{HS20.1}$$

All reference ratios, vacuum constants and old normalization factors in (1.71) stay in M_X. Its self activity V(X-tilde) is included there, using exactly the component and fixed-collar convention of (1.71). A carrier touching more than one source component is a cross activity; it is not reclassified as an independent self term by enlarging two collars until they meet. The p.379 localization of every expression in (1.71) is the component-locality statement used below.

The residual Y base needs additional care. The factors written outside T-prime(Y_i) in (1.72), including chi_(k,Lambda_i), chi_i and the fresh axial delta on G_i, can depend on the internal descriptor which determines Lambda_i and G_i. They are not all v-only constants and may not be dropped. Put every such positive component-local factor into the full residual base $\mathcal{B}_i(v, d\eta_{Y_i})$. Only factors genuinely independent of the internal state are canceled as common factors.

The gauge deltas in this assertion are positive coarea/gauge-slice measures, not functions evaluated at a point. A precise common version is obtained componentwise as follows. At a fixed cutoff there are finitely many graph/domain descriptors in Y_i. For each descriptor, first fill the current shell Y_i minus Lambda by normalized Haar measure, and then use its conditional fresh gauge-slice reference on Lambda. LFR I Proposition 1, p.194, (1.77)–(1.78), identifies V_Lambda as a function of V_k restricted to Z intersect Lambda-complement, independent of the hidden continuous old variables; its axial representative is fixed at p.197, (1.83). Thus each descriptor supplies one well-defined positive measure on the full Y_i current variables, even though its axial stratum varies with the shell values. A finite mixture of these measures with strictly positive coefficients is a common dominating reference. This mixes the measures after including their shells; it does not incorrectly mix pointwise slices at one frozen full current field. Use the Radon–Nikodym density of each descriptor's measure with respect to that reference, and multiply it into the corresponding old positive component measure. Include the remaining descriptor-dependent cutoffs there as well. Different Y_i have disjoint source component domains; conditioning on their exterior gives the product of these local references by the p.201 factorization. Local collars are included in the fixed exterior or component domain according to the source convention.

This constructs $\mathcal{B}_i(v, \cdot)$ by positive disintegration for almost every v under the actual output law. Any soft self factor depending on the common U_k is part of that component base. The local reference mixture is the same for every successful family C. Its normalization depends only on v and the fixed Y geometry, so it cancels from the conditional law. No bound on a descriptor count, an inverse Radon–Nikodym density, or a residual component mass is used below.

The remaining interaction is

$$H_C(v,\eta_Y,\eta_C)=\sum_A V_C(A;v,\eta_Y,\eta_C), \tag{HS20.2}$$

where the sum excludes those self activities. The carrier convention is exactly the convention of (1.68)–(1.72).

Normalize only these full fixed Y bases:

$$b_i(v)=\mathcal{B}_i(v,1)>0, \qquad \nu_i=\mathcal{B}_i/b_i. \tag{HS20.3}$$

On active positive-mass fibers, the full conditional reset posterior is

$$\boxed{\pi_v(dC,d\eta_Y,d\eta_C)=\mathcal{Z}_v^{-1}\,1_{\{C\text{ admissible}\}}e^{H_C(v,\eta_Y,\eta_C)}\prod_{i=1}^{s}\nu_i(d\eta_{Y_i})\prod_{X\in C}M_X(v,d\eta_X).} \tag{HS20.4}$$

The products of fixed masses b_i and genuinely outer v-dependent factors cancel from this conditional law. The successful-component masses do not cancel individually. All conditional normalizers in (HS20.4) are retained. Zero-mass candidate X species have no contribution and can be omitted. Positive-mass active fibers suffice; null disintegration fibers have no effect on the marked integrals.

Equation (HS20.4) follows by disintegrating the positive integrand of (1.72). It is not a probability assigned to a signed polymer activity. The source's statement on p.379 that the internal history sums act also on the cross exponential is precisely why H_C remains inside this posterior.

2. The actual component family has a common local coordinate representation

Lemma HS20.1 — deletion and gluing of completed components For fixed v as above, the successful-component family in (1.72) contains C=empty and is closed under deleting any subset of its whole components. The remaining component measures and coordinates are unchanged. The only compatibility condition between separate successful components is their source geometric compatibility with one another and the fixed outside components; it is independent of their internal eta states.

Source-level proof. LFR I p.201, before (1.100), first separates the sum over the outer component domains from the remaining sums. It then explicitly factorizes those remaining sums and integrations in the component domains. Their local indices are the restricted sequences `(Omega_j^c intersect D, Z_j intersect D)` and the local graph choices. CMP119 (2.19)–(2.22), printed p.258, gives the corresponding component factorization of older positive T operations. These are factorization statements about the whole operations, including their cutoffs and delta constraints.

Here is why deletion preserves the actual index family, rather than only a formal product. Every earlier large-field region inside a given final component stays in that component: the nested sequence in CMP119 (2.1)–(2.3) places the older holes in the later large-field regions. The preparatory sequence in LFR I (1.10)–(1.12) has the same property. Delete all old holes and all old large-field labels belonging to one final X. This removes whole old-region components or whole sets of their descendant cubes; it does not cut through a surviving grid cube. Every remaining region is still a union of cubes in its original admitted partition. The nesting relations, buffer inequalities and rectangle conditions for the remaining components are unchanged. Filling the removed holes cannot create a new small gap between two remaining holes.

The forward construction gives the same conclusion. The cover-and-padding operation S in LFR II pp.384–385 distributes over the union of the ancestor cubes, as used explicitly in (1.81). A merger or a rectangular completion is performed within one connected descendant component. Ancestors of a deleted final component do not participate in any other final component's mergers; otherwise those components would already be connected. Removing that genealogy therefore leaves every other component's geometric operations unchanged. The component eligibility tests on LFR I p.177 concern its own size and its own preceding history. They do not introduce a global count or a condition on a disjoint component's state.

Conversely, the component-local histories appearing in the factorized sum can be combined whenever their final domains have the source geometric compatibility. All their earlier regions and local characteristic functions remain in the assigned components. The geometric exclusion is imposed by the outer sum; state-dependent cutoffs remain

within each local T-prime measure. The only remaining coupling in (1.72) is the displayed exponential of V activities. In particular there is no additional hard cutoff involving eta_X and eta_Y outside those measures.

The operation T-prime(D) itself is unchanged when another component is deleted. Its formula (1.71) uses the local history, the local graphs and the common new field v. The residual component factors and dominating reference just specified are local to their fixed Y_i and unchanged as well. The p.379 localization statement and the omission of self activities from the cross exponential fix precisely this convention. The empty family is the displayed n=0 term in (1.72); it has no successful-component factor. Its cross interaction can still depend on the fixed Y states and is not set to zero. This proves the assertions. □

This lemma does not say that all internal histories of a selected component sum to one. That assertion would be false. Deletion compares different terms in the successful-component sum, keeping the exact positive masses.

Lemma HS20.2 — a common carrier family and local differences Index the cross interactions by all finite current localization domains A, using zero for a domain not occurring in the particular component family or history. For two admissible component configurations which agree on every completed component meeting A and on the common v restricted to A, their A activity is identical. Consequently a change of a component D changes only activities whose full source carriers meet D (with the fixed source collar included).

Proof. The localization is constructed before the final resummation. LFR II pp.373–374, (1.61)–(1.64), enlarges each carrier to include every component it intersects and localizes the remaining background dependences. Its final Y_4 expression uses only the new field and the remaining old variables in Y_4. On p.377 the terms are summed by this final localization domain; the self terms are assigned to their own component operation. Printed p.379 adds the explicit history statement: V(A) depends on the old sequences restricted to the completed components contained in A.

Thus both a primitive localization term and its admissibility test are the same when those local component data are the same. A component outside A cannot change which primitive terms are summed into A. Zero-padding the ambient family of A domains therefore gives the asserted common function, rather than comparing unrelated choices of a scalar decomposition. The subsequent subtraction of the reconstructed new action on p.377 uses the same new field v and is local on the same domain, so it preserves this identity.

Deletion can create A activities which run through part of the removed component, because that region is now ordinary exterior. Such A meets that component and is included in the affected set. It is not claimed that the A index families are otherwise literally equal. The common-family statement only identifies terms whose full carriers miss the changed region. □

The full carrier here includes the source collar, graph/data footprint and any old component which the localization completes. It is not the fixed small continuous integration core of the corresponding estimate.

3. Uniform one- and two-component interaction budgets

Use current M-cube units and write R=R_k. Every eligible completed component has diameter at most D_R=C_d R and at most C_d R^d M-cubes, by LFR I p.177. The constants may include the fixed source collars. Choose one anchor z_D in each component and put

$$L_R = 1 + D_R. \tag{HS20.5}$$

The final nonself source bound is

$$|V_C(A)| \le \alpha_k e^{-\kappa_a d_{k,Z_C}(A)}, \qquad \alpha_k \le Ce^{-cR}, \tag{HS20.6}$$

uniformly in the states still integrated. The first part is LFR II (1.68); the fixed decay reserve and the completed-carrier extraction in CR.15/HR18.26 give the second part under the admitted source parameter choices. Here Z_C is the union of the completed eligible components involved in the literal operation, not Zbar or an arbitrary collection of older tags.

For a carrier meeting an eligible component, the source tree count HL.G3 gives at most C(1+R)^d A_d^(n+1) domains with n<=d_(k,Z_C)(A)<n+1. A carrier rooted at a specified exterior cube has the same proof without the component

root-choice factor: its tree already meets that cube. These bounds remain valid for the carriers newly appearing after a deletion. They are rooted in a cube of the deleted component, which has become exterior.

It follows that, for some fixed integer b, all of the following are bounded by

$$\lambda_R = C(1+R)^b \alpha_k \ : \tag{HS20.7}$$

- the cross-action change upon inserting one compatible X;
- the state-dependent part of the conditional interaction of one fixed Y;
- the sum of the absolute affected activities when deleting a component.

For later use, let J_(DE) bound the oscillation in the D-conditional interaction caused by changing E, after discarding terms independent of D. If either change toggles a successful species, take the relevant four-configuration difference. Lemma HS20.2 cancels every term whose carrier misses D or E. Summing the remaining source bounds gives

Here independence of an X coordinate includes its empty/present state. A term constant in the internal present-state variable eta_X, but different from its absent-state value, is not discarded: it changes the presence activity and remains in the full four-configuration difference.

$$J_{DE} \le C(1+R)^b \alpha_k e^{-c_1 |z_D - z_E|/L_R}. \tag{HS20.8}$$

To justify the distance factor, the original source-tree enclosure HL.G4 puts a carrier of relative length n meeting D in the n+C_d R neighborhood of D. A carrier also meeting E must therefore have n at least their separation minus C_d R. Reserve a fixed part of the source exponential for this constraint and sum the remainder with the tree-count entropy. Weakening the resulting rate to c_1/L_R absorbs the additive C_d R with a fixed constant. No component interior is declared to have zero physical diameter.

The same argument supplies spatial row sums. There are at most C(1+r)^d fixed disjoint Y anchors in a radius-r region in M-cube units. The number of candidate successful component shapes per anchor is bounded by

$$N_R \le e^{C(1+R)^d}. \tag{HS20.9}$$

Indeed an eligible component is contained in a cube of side C R; counting all subsets of that cube is an upper bound on the allowed shapes. Internal history labels are already integrated in M_X and are not counted again in N_R.

The source unnormalized estimate (1.89), with the common native-domain reserve in (1.73), gives after reducing the coupling ceiling

$$m_X(v) \le e^{-P_k}, \qquad P_k = p_0(g_k), \qquad P_k/(1+R_k)^d \longrightarrow \infty. \tag{HS20.10}$$

These are uniform in the remaining source-admitted history data. No corresponding lower bound is needed for m_X or m_Y.

4. Coordinate representation of the full posterior

Let the coordinate set contain one coordinate for each fixed Y_i and one coordinate for each candidate successful X. A Y coordinate has state space eta_(Y_i) and base law nu_i. An X coordinate has state space `{0} disjoint-union eta_X`, where 0 means absent. Its unnormalized base measure is

$$\delta_0 + M_X(v, d\eta_X). \tag{HS20.11}$$

The coordinate index set is finite at every cutoff, even though its number of possible X geometries can be large and its component state spaces need not be finite. The allowed occupancy configurations are exactly the source geometric compatible families. The proof uses only admissible outside configurations and the deletion paths below. It does not assume any arbitrary extension of the source potentials to mutually incompatible padded states. No signed activity enters this coordinate law.

For a compatible outside family C not containing X, the exact X conditional is

$$\gamma_X^C(d\zeta) = \frac{\delta_0(d\zeta) + e^{H_{C\cup\{X\}} - H_C} M_X(v, d\eta_X)}{1 + M_X(v, e^{H_{C\cup\{X\}} - H_C})}. \tag{HS20.12}$$

If X is geometrically incompatible with an occupied outside component, its conditional is delta_0. Thus

$$\gamma_X^C(X \text{ present}) \le e^{\lambda_R} m_X(v). \tag{HS20.13}$$

The empty state in (HS20.12) has weight exactly one after dividing by the actual outside density. It is not supplied by an artificial reference denominator. The Y conditional is its fixed base nu_i tilted by the interactions which depend on eta_(Y_i); its state-dependent exponent has oscillation at most a fixed multiple of lambda_R.

Admissible paths for single-coordinate comparison Two allowed outside configurations can be connected by deleting components present only in the first, changing common components' internal states, and adding components present only in the second. Lemma HS20.1 makes all intermediate occupancy families admissible. An internal-state change leaves geometric compatibility unchanged. Thus the usual telescoping comparison of conditional kernels is legitimate on this hard-core family. It does not require passing through a forbidden overlap configuration, and the proof below does not use the arbitrary values of a padded off-support kernel.

5. Weighted influence estimate

For two distinct reset coordinates i,j let c_(ij) be an upper bound on the total-variation change of the i conditional when an admissible outside state is changed only at j. These coefficients are bounded only on the actual admissible pairs; an off-support product-space extension is unnecessary.

The exact conditional formulas and (HS20.7)–(HS20.8) give

$$\begin{aligned} &c_{ij} \le C J_{ij}, && i \text{ a fixed } Y \text{ coordinate},\\ &c_{ij} \le C m_i J_{ij}, && i \text{ an } X \text{ coordinate and both occupancies are allowed},\\ &c_{ij} \le C m_i, && i, j \text{ incompatible } X \text{ geometries}. \end{aligned} \tag{HS20.14}$$

In the last line a change at j switches the i law between a delta at the empty state and a law with presence probability at most (HS20.13). Such incompatible geometries have anchors separated by at most C D_R. Candidate X geometries incompatible with a fixed Y or Zbar geometry have already been excluded from the candidate set.

For the first line, the two positive Y conditionals have the same arbitrary base nu_i and a change of the state-dependent logarithm bounded by J_(ij), after removing its additive constant. Their normalized densities differ in total variation by at most exp(C J_(ij))-1<=C J_(ij). This estimate is independent of b_i. For the second line compare the two numerator measures in (HS20.12). Their absent part is identical, their denominator is at least one, and

$$M_i |e^{D_i} - e^{\widetilde{D}_i}| \le C m_i \|D_i - \widetilde{D}_i\|_\infty.$$

The two normalization errors have the same bound. This proves (HS20.14), including the full conditional normalizers.

Define fixed weights at the chosen v by

$$a_i(v) = \begin{cases} 1, & i = Y_i, \\ \sqrt{m_X(v)}, & i = X. \end{cases} \tag{HS20.15}$$

They are positive after the zero-mass species have been removed. They may depend on v; no derivative of these weights is used. Choose one fixed theta>0 smaller than the spatial reserve in (HS20.8). Then

$$\sup_i \frac{1}{a_i} \sum_j c_{ij} a_j e^{\theta |z_i - z_j|/L_R} \le \varepsilon_R, \qquad \varepsilon_R \le C(1+R)^{b_1} \alpha_k + C(1+R)^{b_1} e^{-P_k/2 + C(1+R)^d}. \tag{HS20.16}$$

Here b_1 is fixed. In particular a single sufficiently small common coupling ceiling makes epsilon_R<=1/4 at every admitted scale.

Proof. For Y-to-Y rows sum (HS20.8) over the disjoint fixed Y anchors; the spatial lattice row sum costs at most C L_R^d. For Y-to-X rows the additional factor sqrt(m_X) pays the candidate-shape multiplicity (HS20.9). For X-to-Y rows division by a_X changes the factor m_X in (HS20.14) to sqrt(m_X). For X-to-X rows the factor is sqrt(m_X)sqrt(m_(X')), including the hard geometric exclusion. Its finite interaction radius costs at most C(1+R)^d; the remaining exponential tail has the same lattice row sum. These are bounded by the two terms on the right. The strict source separation P_k»R_k^d and alpha_k<=C exp(-cR_k) pay every fixed polynomial and shape factor. No count of the old internal history depth enters: those states are inside the already estimated positive component measures. □

The square-root choice is useful even though this source has geometry-only compatibility. It makes both directions of a rare X versus an ordinary Y influence small. No inverse component mass occurs in the final row bound.

6. A spatial cut coupling for the actual posterior

For a set I of reset coordinates define its reset-variable support and its weight by

$$S(I) = \bigcup_{i \in I} D_i, \qquad n_v(I) = \sum_{i \in I} a_i(v), \tag{HS20.17}$$

where D_i is the full source component footprint, with its fixed collar. A coordinate representing an old history or graph has that whole component footprint. It is not assigned only the location of one stored variable. For a candidate X coordinate the absent state and the present state are parts of the same coordinate.

Theorem HS20.3 — true-marginal cut coupling Let I,J be two finite sets of reset coordinates. Under (HS20.16), the actual joint marginal of (eta_I,eta_J) under pi_v and the product of its two actual marginals admit a coupling with disagreement probability bounded by

$$\epsilon_{I,J}(v) \le \min\left\{1,\ C\, n_v(I)\, \exp\left[-\frac{\theta\, \mathrm{dist}(S(I), S(J))}{L_R}\right]\right\}. \tag{HS20.18}$$

The same assertion with a harmless additional constant covers overlapping coordinate sets. Constants are uniform in the remaining admitted history and total volume. The theorem concerns the full posterior (HS20.4), not a conditional law selected on the presence of a rare X.

Proof. We give a finite-coordinate coupling argument. Condition on the J coordinates. For two admissible boundary values beta,beta-prime, use their respective Gibbs conditionals on the complement of J. Couple single-coordinate heat-bath updates by maximal coupling, choosing a coordinate uniformly for each update. Each update preserves the two required conditional marginals.

For two allowed outside configurations the deletion path in Section 4 changes each differing coordinate once and stays admissible. Thus their conditional total-variation difference at i is at most

$$\sum_j c_{ij} 1_{\{\eta_j \neq \eta'_j\}}. \tag{HS20.19}$$

Write F for the complement of J and N for its finite number of coordinates. Start with any coupling of the two actual conditional marginals, for example their independent product. After averaging the boundary choices below, let delta_i(t) be the disagreement probability at time t and let b_j be the fixed boundary disagreement probability. The exact numerical recursion is

$$\delta_i(t+1) \le (1 - N^{-1})\delta_i(t) + N^{-1}\left(\sum_{j \in F} c_{ij}\delta_j(t) + \sum_{j \in J} c_{ij} b_j\right). \tag{HS20.19a}$$

Conjugating the F matrix by the positive diagonal a makes its row norm at most epsilon_R<1. Hence the powers of $T_N = (1 - N^{-1})I + N^{-1}C_{FF}$ decay in that weighted norm. Iterating (HS20.19a) gives

$$\delta_F(t) \le T_N^t \delta_F(0) + N^{-1} \sum_{\ell=0}^{t-1} T_N^\ell C_{FJ} b_J. \tag{HS20.19b}$$

Its limit upper bound is $(I - C_{FF})^{-1} C_{FJ} b_J$. This is a limit of finite numerical inequalities, not a construction of a stationary coupling or an infinite product. At every time the observed first-copy/boundary marginal and the observed second-copy/boundary marginal have the same required distributions, so their fixed total-variation distance is bounded by the error of that time's coupling. Taking the infimum of these bounds proves the limit estimate. Maximal coupling of those two fixed observed laws then realizes it. A very small positive minimum a_X can change the time needed for the first term to decay, but does not appear in the final bound. When F is empty only the boundary estimates are used.

Now draw beta and beta-prime independently from the true pi_v marginal on J and average these inequalities. If j is a Y coordinate, its boundary disagreement probability is at most one. If j is an X coordinate, the two copies agree at their empty state unless at least one is present, so by (HS20.13)

$$\mathbb{P}(\beta_j \ne \beta'_j) \le 2\pi_v(X_j \text{ present}) \le C m_j \le C a_j. \tag{HS20.20}$$

The identical bound supplies an initial a-weighted bound for every free X coordinate in the two conditional marginals. This is the step that prevents a hidden inverse rare-event mass.

Inserting (HS20.20) into the resolvent and using the spatially weighted row bound gives, for i outside J,

$$p_i \le C a_i \frac{\varepsilon_R}{1 - \varepsilon_R} e^{-\theta\, \mathrm{dist}(z_i, \{z_j : j \in J\})/L_R}. \tag{HS20.21}$$

For i in J use (HS20.20) directly. The triangle inequality for anchor distances pays the exponential along every path in the matrix series. Summing the observed disagreement probabilities over i in I proves the displayed bound, because all anchors lie in their declared supports.

Finally, let the first observed copy be sampled conditionally on beta and the second conditionally on beta-prime. After beta-prime is averaged, the second observed copy has the true pi_v I-marginal and is independent of beta. Hence the comparison is between the true joint law and the product of its true marginals. A union bound on the observed coordinates, or maximal coupling of the resulting finite observed marginals, gives (HS20.18). This argument never replaces a rare-sector conditional moment by an unconditional moment. □

If the inputs are specified by physical locations rather than full component-coordinate footprints, their reset-variable supports must first be enlarged to contain every completed component on which they actually depend. The enlargement has radius at most C M R_k h for a mark which depends only on components meeting its initial local set. This is the explicit reset radius; it is not a bound on a nonlocal minimizing-field pullback.

7. All conditional cumulants and the complete finite reset jet

Let F_1,...,F_r be real or complex functions under the actual posterior pi_v. Suppose F_l depends on the reset coordinates in I_l, with v fixed. Write

$$S_l = S(I_l), \qquad \mathcal{S}_l(v) = \sqrt{1 + n_v(I_l)}. \tag{HS20.22}$$

Let T(S_1,...,S_r) be the minimum spanning-tree weight on these r support sets, with edge weight dist(S_i,S_j). This definition is valid even when sets overlap; the edge weights need not themselves define a metric.

Theorem HS20.4 — spatial cumulants of the full reset law For r>=2 and F_l in L^(2r)(pi_v),

$$\boxed{|\kappa_{\pi_v}(F_1, \dots, F_r)| \le C^r r! \prod_{\ell=1}^{r} \left[\mathcal{S}_\ell(v) \|F_\ell\|_{L^{2r}(\pi_v)}\right] \exp\left[-\frac{c\, T(S_1, \dots, S_r)}{(r-1)L_R}\right].} \tag{HS20.23}$$

Every input norm is in the same actual full posterior. The bound is uniform in the remaining admitted history. It includes descriptor-mean fluctuations as well as old fluctuations inside a component.

Proof. Choose a minimum spanning tree on the support sets and delete its longest edge. The resulting nontrivial cut A union B has all cross distances at least the length of that edge; otherwise replacing the edge would reduce the tree weight. Its length is at least T/(r-1). Apply Theorem HS20.3 to the unions of the coordinate sets on the two sides.

Couple the resulting true joint law with the product across the cut. Denote its failure probability by epsilon. A moment of a block K of labels differs between the two laws by at most

$$2\epsilon^{1/2} \prod_{\ell \in K} \|F_\ell\|_{L^{2r}(\pi_v)}. \tag{HS20.24}$$

Indeed the products agree on the coupling success event; on its complement use Hölder with the indicator in L^2 and each of the at most r input factors in L^(2r). The product-across-the-cut law has the same individual input marginals, so the same norms bound both terms. A block contained on one side has no difference.

The mixed cumulant of all r variables under the product law is zero. In the cumulant partition formula, telescope each product of block moments. This costs at most the number of blocks. Consequently its coefficient sum is bounded by

$$2\epsilon^{1/2} \sum_{\pi \in \Pi_r} |\pi|! \prod_{\ell=1}^{r} \|F_\ell\|_{2r} \le 2^r r! \epsilon^{1/2} \prod_{\ell=1}^{r} \|F_\ell\|_{2r}. \tag{HS20.25}$$

The last inequality follows by encoding ordered set partitions by a permutation and cuts. The support factor in the cut bound satisfies 1+sum_(l in A) n_v(I_l)<=product_l(1+n_v(I_l)); using (HS20.18) and taking its square root proves (HS20.23), with a fixed changed rate. □

The decay rate has the explicit r-1 denominator. No stronger tree-product or order-independent rate is claimed. At one component's own scale the theorem permits unrestricted internal dependence; its physical component diameter supplies the reset radius.

Full-law moments without conditioning on a rare event Let mu(dv,deta)=nu(dv) pi_v(deta) be the actual positive source-free joint law, and suppose the support factors in (HS20.22) have their indicated uniform bounds. If supports or candidate species depend on v, choose a deterministic lower bound $T_* \le T(S_1(v), \dots, S_r(v))$ on the active fibers. At one fixed scale L_R is common; otherwise its uniform upper bound is used. Applying (HS20.23) with the larger input exponents 2rp and then conditional Hölder gives, for p>=1,

$$\|\kappa_{\pi_v}(F_1, \dots, F_r)\|_{L^p(\nu)} \le C^r r! \left(\prod_{\ell} \sup_{v} \mathcal{S}_\ell(v) \right) e^{-cT_*/((r-1)L_R)} \prod_{\ell=1}^{r} \|F_\ell\|_{L^{2rp}(\mu)}. \tag{HS20.26}$$

Without such a uniform lower bound, the spatial factor stays inside the Lp(nu) estimate; it is not removed from a v-dependent integrand. For fixed initial physical supports and their uniform component-radius enlargements, the corresponding deterministic separation supplies T_* whenever those enlarged supports are separated.

To see the measure step explicitly, bound each conditional factor by (pi_v |F_l|^(2rp))(1/(2rp)), raise their product to p, apply Hölder over v, and use concavity of the remaining square root. No conditional mass is divided out in this estimateage.

For M-03's averaged grade seminorm, input source grades d_l summing to m give, up to the same explicit support and spatial factors,

$$\mathfrak{m}_m\big(\kappa_{\pi_v}(F_1, \dots, F_r); \nu\big) \le C^m\, m!\, r! \prod_{\ell=1}^{r} \mathfrak{m}_{d_\ell}(F_\ell; \mu). \tag{HS20.27}$$

This follows by substituting the input grade bounds into (HS20.26), using product_l d_l^(d_l)<=^m m and r^m<=^m m<=e^m m!. The h^(4m) factors multiply correctly because this is an R reset at one current scale. This is only the averaged moment part of M-03, not its pointwise regulator, ambient-shift or full carrier-weight assertion.

The complete logarithmic source jet Work in the polynomial source algebra through an arbitrary finite order M. If the incoming logarithmic jet is

$$\Psi(t) = \sum_{\emptyset \neq B} t_B A_B, \qquad t_B = \prod_{i \in B} t_i,$$

define

$$\Psi^+(t) = \log \pi_v(e^{\Psi(t)}). \tag{HS20.28}$$

The output coefficient with label set I is exactly

$$A_I^+ = \sum_{\mathcal{P} \in \Pi(I)} \kappa_{\pi_v}(A_B : B \in \mathcal{P}). \tag{HS20.29}$$

Every higher incoming vertex A_B is retained. The spatial bound (HS20.23) applies to each generated cumulant, using the actual support of each such vertex; its order-one term is the exact conditional mean and need not create a new connection. If the incoming grade moment and support budgets have fixed factorial growth, (HS20.27), the ordered-partition count and the grade multiplication rule give a fixed changed factorial power for the entire finite jet. For example, input bounds of the form

$$\mathcal{S}_B(v)\|A_B\|_{L^p(\pi_v)} \leq K^{d+1}(d!)^{p_0} b^d (dp)^{sd}, \quad d = |B|, \quad p_0 \geq 1, \quad s \geq 0, \tag{HS20.30}$$

give the conservative value bound C^m(m!)(p_0+2s) b^m at total order m, with the partition-specific spatial factor in (HS20.23). The input budget in this example is explicit; it is not inferred from unnormalized component masses. Alternatively (HS20.26) uses the actual full-law grade budgets directly.

These definitions and estimates are compatible under reducing the source truncation order. They require no common convergent exponential-source series. Since (HS20.4) is the full mixture, (HS20.29) retains, in particular,

$$\mathrm{Cov}_{\pi_v}(F, G) = \mathbb{E}_C \,\mathrm{Cov}(F, G \mid C, v) + \mathrm{Cov}_C\big(\mathbb{E}[F \mid C, v], \mathbb{E}[G \mid C, v]\big), \tag{HS20.31}$$

and every higher analogue. Descriptor-mean cumulants have not been dropped.

8. Exact transport, support qualifications and scope

The source's complete R operation already sums the successful geometries and integrates the old internal Y states. Thus (HS20.4) is an explicit positive disintegration of that exact integration, not a new source-dependent action. If an augmented path space is to retain a history coordinate for later use, resample a fresh copy from pi_v independently conditional on v. This preserves the complete augmented source-free law. The corresponding marked quotient is pi_v F, independent of the newly resampled history. It is the normalized history operation required for the reverse-tail construction M-01.

The fresh Haar restoration described in Section 1 is performed with the whole transported jet. Therefore multiplying the actual numerator by its exact old marked expression, followed by (HS20.28), preserves all integrated source coefficients. No fresh coarse plaquette is substituted for an old transported mark.

The physical supports in Theorems HS20.3–HS20.4 are the unions of the actual reset-coordinate component footprints. They give a physical reset radius O(M R_k h) and an exponential spatial response outside it, uniformly in the remaining admitted history, with the stated order loss and support factors. All internal old labels in both X and Y components are included. Unreplaced Zbar labels and the output Y geometry are retained, not silently reset. Their later marked summation is part of H-02/H-04.

The proof does not identify a source background U_k coordinate with an independently variable raw retained link. A projected microscopic insertion can depend on U_k through a nonlocal minimizing map; its actual pullback, localization, old-current records and native regulator must still be supplied. Applying the finite cumulant inequality to pre-existing direct/current records is legitimate if their actual reset-variable support and moment budgets are supplied. The theorem does not bound the new response obtained by differentiating the posterior with respect to every raw retained coordinate or moving across an ambient selector/chart boundary.

Accordingly the following stronger conclusions are not asserted:

- a cutoff-uniform pointwise quotient by the complete shifted output native regulator;

- the full physical exponential volume carrier norm M-03 for an arbitrary transported insertion;
- the complete scalar matching theorem M-04, tensor matching M-05, or one-step response/current algebra M-06;
- the full-history summation H-04 or the all-scale induction M-07.

Theorems HS20.3–HS20.4 and the reset jet (HS20.29) give the spatial result for the mixed posterior under their support and moment hypotheses. KT19.3 is a stronger transport estimate.

Primary source locations used

- CMP119 (1988), pp.254–258, (2.1)–(2.3), (2.19)–(2.22): nested geometries, old component factorization and full local positive operations.
- LFR I (1989), pp.177, 179–180, 192–201, particularly (1.10)–(1.12), (1.73), (1.76), (1.99)–(1.102): eligibility, preparation, complete reference quotient, local history sums and the fresh axial graph.
- LFR II (1989), p.365, (1.33)–(1.35): the common rebuilt determining field on every eligible component.
- LFR II pp.373–380, (1.61)–(1.75): common localized interaction family, positive completed component operation, successful Haar restoration and the exact joint sum.
- LFR II pp.383–392, (1.79)–(1.104): unnormalized component mass, the successful geometry sum, and the fixed residual alternative. In (1.92) the free set is explicitly the union of selected X and Y components; in (1.97) and (1.103) it is the fixed union of eligible Y components. The unreplaced Zbar region is not added to that free set.

The source scopes and coupling/geometry hypotheses are the already declared ones of C-01 and the preceding hypotheses. No unpublished Yang–Mills construction is invoked.

Annex BJ — Old Gaussian energy, rank and native-envelope reserves

This note proves an additional estimate on the actual unnormalized old operation, including all its surviving Gaussian coordinates, and a common positive envelope on its admitted native chart. It also gives an actual-source reason that a curvature-only lower bound for every bad selector is false in the displayed soft-gauge convention. It does not identify the old posterior with the fresh reference law, infer a lower mass from an upper mass, or claim a full M-03 operator theorem. Whether the H-03 spatial-response requirement follows from the separate full-law spatial theorem is not decided here; none of the strong-norm gaps below is removed by it.

1. Exact energies and exact normalizers

Fix a literal eligible component X at step k, with all the admissibility and parameter hypotheses in LFR I–II. Let $M_{X,\xi}$ denote the positive operation $T'_k(X;\xi)$ in LFR II (1.71), on the full remaining old-variable and old-history space. It includes its selectors, coarea factors, old reference normalizers, self action and internal-history sums. Its native retained argument is $\xi = (U, J)$. In positive-measure statements the reference is an actual real compact configuration with $J = 0$, as on LFR II p.380; a general native analytic deformation need not itself be a positive law. Write

$$m_{X,\xi} = M_{X,\xi}(1), \qquad \nu_{X,\xi} = m_{X,\xi}^{-1} M_{X,\xi} \tag{NR20.1}$$

on the active set $m_{X,\xi} > 0$. No lower estimate for this mass is part of the definition.

There are two outer nonnegative energies, exactly as in RN19.6:

$$\begin{aligned} Q_X &= \tfrac{1}{2}\langle DH''_{1,k,X}B, \zeta DH''_{1,k,X}B\rangle, \\ S_X &= A((g'_k)^{-2}\zeta_1|_X, U''_{h+2}(\exp(ig_kB), V'')), \qquad \mathcal{E}_X = Q_X + S_X. \end{aligned} \tag{NR20.2}$$

Here $h = k - N$ is the source's older step index, not the physical spacing in M-03.

Now retain every positive Gaussian exponent still present in T_h. The ordinary factor is CMP119 (2.21), or LFR I (1.25), restricted to its old component. In actual free coordinates it is

$$q_{j,C}(a) = \tfrac{1}{2} a_C^{\mathsf{T}} S_{j,C} a_C, \qquad S_j = K_{OO} - K_{OI}(K_{II})^{-1}K_{IO}, \qquad K_j = C_j^* \Delta^{(j)} C_j. \tag{NR20.3}$$

The actual background-current term, exceptional-bond map and principal covariance are retained. These are the coefficients identified in OR17.8 using CMP99 (3.155)–(3.158) and Theorem 3.15. Cross-component terms assigned to the effective action are not included a second time. On the real admitted source domain these within-component forms have the positivity used on LFR II printed p.383.

Define, branch by branch,

$$\mathcal{G}_X = \sum_{(j,C)\ \text{still integrated}} q_{j,C}, \qquad \mathcal{F}_X = \mathcal{E}_X + \mathcal{G}_X. \tag{NR20.4}$$

Earlier completely integrated operations now represented as scalar factors are not reintroduced as hidden coordinates. A modified old factor contributes its surviving displayed positive Gaussian form on its actual surviving coordinate set; an empty fluctuation set contributes zero. Raw compact variables, soft gauge factors, characteristic functions and already integrated constants remain in the original measure.

$\mathcal{F}_X$ is not defined to be the full M-03 regulator. In particular it does not silently include every native plaquette energy at every surviving old scale. Equation (NR20.3) is a precision energy, not a newly chosen Euclidean-coordinate norm.

2. A reserve for the actual old Gaussian energies

Theorem NR20.1 — fractional reserve with arbitrary old depth There is a fixed $\theta > 0$, which may be chosen sufficiently small with $\theta \le 1/8$, such that the source parameter order and one common small coupling ceiling give

$$\boxed{M_{X,\xi}\big(e^{\theta\mathcal{F}_X}\big) \le e^{-P_0(g_k)}}, \tag{NR20.5}$$

uniformly in the actual admitted real native data, total volume and arbitrary depth of the internal old histories. Here $P_0(g)$ is the function denoted $p_0(g)$ in LFR II. This is an unnormalized estimate.

Proof. The insertion replaces each displayed positive outer and surviving old Gaussian exponent coefficient by $1-\theta$. It changes no chart, constraint, coarea factor, reference density or normalizer. In particular an old normalized reference operation is not replaced by one with a smaller precision. We estimate the original positive integral with the stated insertion.

The source extraction on LFR II pp.380–383 bounds all other exponent terms and normalization terms by a positive domain cost, then uses fixed fractions of the positive Wilson/quadratic energies to extract the old and preparatory bad-field factors. Reducing the positive coefficients preserves that argument with a smaller fixed extraction coefficient. Choose the fractions before the coupling ceiling. The strict source inequalities leave room both for extraction and for the remaining Gaussian majorant. Each energy is reduced once at its own step; the number of fractions is not chosen according to old depth.

The additional integration change beyond RN19.3 is the old Gaussian comparison. On the exact $r_{j,C}$-dimensional free subspace, at fixed remaining data,

$$\int e^{-(1-\theta)q_{j,C}(a)}\,da = (1-\theta)^{-r_{j,C}/2}\int e^{-q_{j,C}(a)}\,da. \tag{NR20.6}$$

This is the full Gaussian majorant after dropping the old characteristic restrictions, exactly where the source invokes Gaussian positivity on p.383. It uses the actual Schur precision and its actual determinant. A coarea factor present before the free-coordinate parametrization is retained on both sides; its remaining-data dependence is treated by the same source domain bound. Equation (NR20.6) is a comparison of majorants, not an equality for the selected integral.

In the relabeled volume convention of LFR II (1.79), the rank satisfies

$$\sum_C r_{j,C} \le C_{d,G,L}|Z_j\cap\Omega_j|_j \le C_{d,G,L}M^d R_j^d d'_j(Z_j). \tag{NR20.7}$$

The first inequality counts the independent Lie coordinates which actually remain. Deleting a tree, imposing the linear constraint or taking a component restriction decreases the count. The fixed L-dependent factor permits the indexing conversion from CMP119, where A_j is on $Z_{j+1}\cap\Omega_{j+1}$. The second is the admissible enlarged-domain estimate on LFR II p.380, with its own d'_j. The total new loss is at most

$$\exp\left[C_\theta\sum_{j\le h}M^d R_j^d d'_j(Z_j)\right]. \tag{NR20.8}$$

Enlarge the existing coefficient of $C\sum_j M^d R_j^{d+1} d'_j(Z_j)$ to absorb this. The entire old-scale sum remains. It is not bounded by current volume or by a bounded count of old variables.

For the outer B integral, the logarithmic factor in the last line of (1.71) is $(\sigma_0 g_k^{d_G})^{n_B}$. Together with $dB\,\sigma(g_kB)$ and its tree delta this is the normalized compact Haar-coordinate measure, as verified in RN19.8a. Drop the remaining positive outer quadratic and bound that restricted compact integral by one, as on p.383. There is no outer Gaussian determinant loss. The other compact integrals and all unchanged reference constants are treated exactly as in the source.

We have therefore obtained source (1.79) with a smaller fixed positive extraction coefficient and a larger fixed domain-cost coefficient. Its worst preparatory factor is still

$$\exp[-c_\theta R_j^{-(d+5)}p_1(g_j)^2], \qquad 2p_1-(d+5)r_0 > p_0. \tag{NR20.9}$$

The chronological induction (1.80)–(1.88), with the source's $N = R_j$, pays the complete scale sum. Its strict requirements include

$$P_0(g) \gg C_\theta M^d R(g)^{d+2}, \qquad c_\theta A_1^2 P_0(g)^2 \gg P_0(g) + C_\theta M^d R(g)^{d+2}. \tag{NR20.10}$$

Changing fixed constants changes only the common ceiling. The final store from (1.89) gives the weaker (NR20.5). This proves arbitrary-depth uniformity with every old reference unchanged. ■

The imported positivity and chronological extraction retain the scope of the established source-free construction. This theorem adds an explicitly identified marked insertion; it is not an independent verification of its coupling trajectory.

The same proof allows an explicit old-coordinate-count store. Let $D_X = \sum_{j,C} r_{j,C}$ be the actual number of surviving independent Gaussian Lie coordinates on a history branch. For each fixed $u \geq 0$, the ceiling may be chosen so that

$$M_{X,\xi}\big[e^{\theta \mathcal{F}_X + uD_X}\big] \leq e^{-P_0(g_k)}. \tag{NR20.9a}$$

Indeed uD_X adds at most $uC \sum_j M^d R_j^q d'_j(Z_j)$ to the same chronological domain cost. No new analytic argument is needed, and u is fixed before source order is varied. Thus polynomial factors counting those old coordinates can also be paid with fixed factorial growth before individual normalization. This is a random rank under the actual history sum, not an assertion that the rank is bounded by the small current core.

Corollary NR20.1a — the ordinary old native free coordinates Let $\mathcal{O}_X$ be the free coordinates still integrated through the ordinary old factors of CMP119 (2.21), in their dimensionless birth-scale convention, and set

$$Z_X = \sum_{(j,i)\in\mathcal{O}_X} |a_{j,i}|^2. \tag{NR20.10a}$$

There is a fixed $\lambda_0 > 0$, independent of old scale and admitted history, such that

$$\mathcal{G}_X \geq \tfrac{1}{2}\lambda_0 Z_X, \qquad M_{X,\xi}\big[e^{\theta_1(\mathcal{E}_X + Z_X)}\big] \leq e^{-P_0(g_k)}, \qquad \theta_1 = \theta \min\{1, \lambda_0/2\}. \tag{NR20.10b}$$

Proof. The paragraph following CMP99 (3.158), printed p.428, states the positive lower bound $K_j \geq \lambda_0 I$ for the actual free-coordinate precision, independent of step and background under the admitted conditions (3.35)–(3.36). This is the primary bound used in C-06, HC.8–HC.10 and HA15.1–HA15.7. The real background admission includes its current and second-order conditions, not only a curvature restriction.

The source's larger current free region is Ω_{j+1} before the inner Λ_{j+1} variables are integrated. That larger region already has the fixed collar from the preceding bad region, by CMP119 (3.2)–(3.5) and the C-06 hypotheses, Section 3. It therefore satisfies the unit-lattice hypothesis of CMP99 Section E. The later restriction to the inner Λ is not used to move an old retained coordinate across a missing collar. Apply the primary bound on this full admitted current free space with its fixed exceptional-bond map.

For the exact Schur complement in (NR20.3),

$$a^{\mathsf{T}} S_j a = \min_b (b, a)^{\mathsf{T}} K_j (b, a) \geq \lambda_0 |a|^2. \tag{NR20.10c}$$

Restricting a to one retained component preserves the lower bound. Sum the resulting inequalities over the actual ordinary old factors. This proves the first inequality in (NR20.10b). The second follows pointwise from $\theta_1(\mathcal{E}_X + Z_X) \leq \theta \mathcal{F}_X$ and Theorem NR20.1. ■

The $a_{j,i}$ here are the actual source dimensionless free variables at their own j-scale, which are the corresponding ordinary slots of M-03's native list. No simultaneous physical rescaling to the current k-scale is made. A mark's physical factor remains separate. The statement includes no raw compact-link plaquette and makes no unverified identification for a hidden coordinate which has undergone a different completed-R chart. A modified factor can be included only when its surviving coordinate and form are literally the same identified ordinary ones. Thus this is a genuine partial native-regulator comparison, not a declaration that the full regulator has been bounded.

More explicitly, the norm in (NR20.10c) is the unweighted sum on the birth-scale unit lattice, in an orthonormal basis for the source Lie-algebra inner product. This is the unit-lattice operator of CMP99 Section E and HC.1/HA15.7; it is not a layer-weighted physical-space norm. A fixed change to M-03's Hilbert–Schmidt Lie basis changes only

λ_0 by a fixed group-normalization factor. A slot is included in the native comparison only when its recorded M-03 dimensionless coordinate coincides with this source coordinate under that fixed conversion. In the convention $\log U = g_j a_j = h_j A_j^{\mathrm{phys}}$, a physical potential instead has $a_j = h_j A_j^{\mathrm{phys}}/g_j$; its square cannot be inserted without that factor. Physical-volume weights, if introduced in another notation, must first be converted back to this unit-lattice quadratic form. None is omitted from a claimed bound on arbitrary raw variables.

Corollary NR20.2 — all finite old-energy powers For real $r \geq 1$,

$$M_{X,\xi}[(1+\mathcal{F}_X)^r] \leq e^{\theta}\left(\frac{r}{e\theta}\right)^r e^{-P_0(g_k)}. \tag{NR20.11}$$

For integers $m \geq 0$, a convenient weaker bound is

$$M_{X,\xi}[(1+\mathcal{F}_X)^m] \leq e^{\theta}\theta^{-m} m! e^{-P_0(g_k)}. \tag{NR20.12}$$

Proof. Apply $x^r \leq (r/(e\theta))^r e^{\theta x}$, or $x^m \leq m!\theta^{-m}e^{\theta x}$, to $x = 1+\mathcal{F}_X$, then use (NR20.5). ■

The finite-jet partition proof in RN19.5 now applies with $\mathcal{F}_X$ in place of $\mathcal{E}_X$. If every incoming degree-d log coefficient has the pointwise bound $C^{d+1}(d!)^s b^d r(\xi)^d (1+\mathcal{F}_X)^d$, $s \geq 1$, the full unnormalized degree-m coefficient is bounded by

$$e^{-P_0(g_k)} C_1^m (m!)^{s+1} b^m r(\xi)^m. \tag{NR20.13}$$

All higher incoming vertices are retained. Membership of a general M-03 coefficient in this energy class is not assumed.

3. A common envelope on the actual native chart

On the fixed-support native chart of LFR II pp.379–380 only Q_X and the self activity $V(X^{\sim})$ depend on the native pair. In those coordinates the outer Wilson term, old operation, old selectors and old Gaussian forms are unchanged. The source bounds the quadratic variation and the resummed variation of V separately. Shrink the common chart so that

$$M_{X,\xi}(d\eta) = e^{\sigma_\xi(\eta)} M_{X,0}(d\eta), \qquad \sup_{\xi,\eta}|\sigma_\xi| \leq \delta_\sigma, \qquad \sup_{\xi,\eta}|Q_{X,\xi} - Q_{X,0}| \leq \delta_Q \tag{NR20.14}$$

with fixed small constants. The first is the actual pointwise comparison (1.73)–(1.75); the second is its separately bounded quadratic contribution, not an inference from cancellation in σ. Define

$$\mathcal{F}_X^{\natural}(\eta) = 1 + \sup_{\xi \text{ admitted native}} |\mathcal{F}_{X,\xi}(\eta)|. \tag{NR20.15}$$

Corollary NR20.3 — native shifted energy envelope For every admitted native ξ, with total variation for its possible complex measure,

$$|M_{X,\xi}|[e^{\theta\mathcal{F}_X^{\natural}}] \leq e^{\delta_\sigma + \theta(1+\delta_Q)} e^{-P_0(g_k)}. \tag{NR20.16}$$

The power bounds (NR20.11) follow for this envelope with the fixed extra factor. This controls one supremum envelope before integrating it.

Proof. Exact native independence of the other energy terms gives $\mathcal{F}_X^{\natural} \leq 1 + \mathcal{F}_{X,0} + \delta_Q$. Use this and $|e^{\sigma_\xi}| \leq e^{\delta_\sigma}$ in the common measure, then apply (NR20.5) at the real center. ■

This is narrower than the full M-03 shifted envelope. It does not cover arbitrary raw retained links, hard selectors, geometry changes or current maps outside the fixed-support chart. Those variations may change the measure or the old forms. Taking a supremum over unavailable charts would not prove their bound.

The normalized cost remains explicit. At the real center set

$$L_X = -P_0(g_k) - \log m_{X,0} \geq 0. \tag{NR20.17}$$

For the positive real slice, RN19.7 and the mass comparison $e^{-\delta_\sigma} m_{X,0} \leq m_{X,\xi} \leq e^{\delta_\sigma} m_{X,0}$ yield

$$\|\mathcal{F}_X^\natural\|_{L^p(\nu_{X,\xi})} \leq C_\theta[1 + L_X + p], \qquad p \geq 1. \tag{NR20.18}$$

This is the enlarged-energy and native-envelope version of the logarithmic estimate. No upper bound for L_X has been proved here. The upper bound $m_X \leq e^{-P_0}$ cannot supply one.

4. Full-footprint payment when the mass is retained

An eligible X, including its fixed dependency collars, fits in a current cube of side $C_M R_k$. Both the cube volume and a spanning-tree length are $O(C_M R_k^d)$. The number of candidate shapes through an anchor, including containing-cube placements, is at most $\exp(C_M R_k^d)$. Therefore, for every fixed footprint rate κ, a smaller common ceiling gives

$$\sup_b \sum_{X \ni b} e^{\kappa(|X^+| + \mathrm{t}(X^+))} |M_{X,\xi}| [(\mathcal{F}_X^\natural)^m] \leq C^{m+1} m! e^{-P_0(g_k)/2}. \tag{NR20.19}$$

Here X^+ is the actual fixed-collar carrier. The sum is over literal eligible geometric species; each measure already contains its internal-history sum. To prove the formula, use the deliberately rough bound $2^{C_M R_k^d}$ on subsets of a containing cube, apply (NR20.12)/(NR20.16), and use $P_0(g_k) \gg R_k^d$. No internal history is counted again.

On the actual positive real slice this is the form needed by a common positive successful-component gas. For example, if its exact deletion identity preserves the retained variables and remaining measures and has

$$H_{D \cup S} - H_D \leq \sum_{X \in S} \lambda_X, \tag{NR20.20}$$

then RN19.6 gives the normalized full-law pinned estimate

$$\mathbb{E}\left[1_{S \subset D} \prod_{X \in S} (\mathcal{F}_X^\natural)^{m_X} \right] \leq \prod_{X \in S} e^{\lambda_X} M_X[(\mathcal{F}_X^\natural)^{m_X}]. \tag{NR20.21}$$

This is an implication until the actual common-gas/deletion identities are verified. There is no component m_X^{-1} on its right. Conversely its small right side does not bound ν_X after conditioning on presence of X; that is a different normalization.

5. An actual selector obstruction to curvature-only shortcuts

This section uses the displayed soft gauge fixing in CMP119 (1.5)–(1.8), pp.246–248. The source also permits an averaged hard tree gauge; the statement is not asserted for that alternative without a separate check.

Proposition NR20.4 — a pure-gauge Q_1-bad configuration For sufficiently small g_0 and the source order $A_0 \gg A_1$, there is a configuration admitted by the soft gauge factor such that every fine plaquette and the first coarse configuration are the identity, $P_0 = P_1 = \varnothing$, but $Q_1 \neq \varnothing$.

Proof. Let ε_0, δ_0 be exactly the source thresholds in (1.1), (1.8), so $\delta_0/\varepsilon_0 = A_1/A_0$ and δ_0/g_0 is a fixed positive multiple of $P_0(g_0)$. Choose an interior noncentral fine site x of one first block and a fixed nonzero Lie generator T. Set $u = 1$ at every site except x, and choose $u(x) = \exp(aT)$ so that $3\delta_0 < \|u(x) - 1\| < 4\delta_0$. Take

$$U(e) = u(e_-) u(e_+)^{-1}. \tag{NR20.22}$$

All fine plaquettes are one. Gauge covariance of the actual block average gives $V_1 = 1$, since u is one at every coarse block center. The axial local minimizing backgrounds $U_{1,\square}(V_1)$ in (1.2) are consequently one, so the P_0, P_1 tests are strictly good.

The soft gauge contours satisfy $U(y,z) = u(y)u(z)^{-1}$. They equal one except at x, where their distance is less than $4\delta_0 < \varepsilon_0$. Every characteristic factor in (1.5) is therefore one and its exponential is positive. An interior bond incident to x is not an exceptional interblock bond $b_0(c)$, and

$$\|U(b)U_{1,\square}(b)^{-1} - 1\| > 2\delta_0. \tag{NR20.23}$$

The selector (1.8) puts a cube whose enlarged tested stencil contains that bond into Q_1. ■

These strict inequalities persist in a neighborhood, by continuity of the admitted local minimizing map. At fixed finite cutoff one can make the Wilson energy of that neighborhood arbitrarily small while keeping the Q_1-bad inequality. Thus a curvature-only implication

$$\text{some actual } P/Q/R \text{ selector is bad} \implies \sum_p \beta_p s(U_p) \geq cP_0(g)^2 \tag{NR20.24}$$

is false for the displayed soft-gauge convention. The soft gauge energy pays this event. This does not contradict the source extraction, which retains gauge factors and positive fluctuation forms, and it is not a counterexample to H-03 or to a theorem for the actual restricted gauge-invariant field family.

Lemma NR20.5 — quantitative local soft-gauge seed weight On the flat physical orbit with coarse center gauges fixed, the normalized gauge coordinates in (1.5) have product density

$$\prod_{z \neq y} \frac{1_{\{\|v_z - 1\| < \varepsilon_0\}} e^{-g_0^{-2} s_0(v_z)}}{z(g_0)}\, dv_z, \qquad s_0 = 1 - \operatorname{Re}\operatorname{tr}. \tag{NR20.25}$$

For a finite local seed region containing D gauge coordinates, put the coordinate at x in a radius-cg_0 ball about the value in Proposition NR20.4 and its other $D-1$ coordinates in such balls about one. This event has conditional probability at least

$$\exp[-CP_0(g_0)^2 - CD]. \tag{NR20.26}$$

Outside coordinates are fixed or retain their product marginal; no torus-volume cost is imposed.

Proof. Compact-group Haar density in a fixed exponential chart is bounded above and below, and $c|v|^2 \leq s_0(\exp v) \leq C|v|^2$. Since $\varepsilon_0/g_0 \to \infty$,

$$cg_0^{d_G} \leq z(g_0) \leq Cg_0^{d_G}. \tag{NR20.27}$$

Every radius-cg_0 numerator ball has Haar volume at least $c' g_0^{d_G}$. Its action is $O(1)$ about one and $O(P_0(g_0)^2)$ at the seed. Divide by the exact normalizers and multiply. The bad margin survives because $\delta_0/g_0 \to \infty$. ■

This is an actual local gauge-selector seed probability, conditional on the stated physical orbit. It is not a lower bound for the later fully integrated $m_X(U)$. It has not proved that every prescribed final eligible X is seed-realizable, that all preparatory coarea fibers have uniformly controlled thickness, or that arbitrary retained data can be compared to the flat case.

6. Exact remaining normalized-regulator questions

Two distinct questions remain.

1. Normalization. A fixed-component posterior estimate needs an upper bound for L_X in (NR20.18), controlled by the actual output native regulator or by an explicitly retained alternative. A seed at birth is insufficient without exact final geometry, chronological restrictions, fine-coordinate volume and retained-data comparisons. The bounded native tilt is local in its admitted chart; it is not a global comparison to the flat datum.
2. The rest of the native-energy dictionary. Corollary NR20.1a now controls the ordinary old dimensionless free slots by their actual uniformly coercive Schur forms. The full shifted M-03 regulator also contains its stated surviving old plaquette list and any differently transformed free coordinates. These still need the correct physical scale factors, actual coordinate maps, footprints and every allowed shift. One reconstructed Wilson action does not automatically dominate an extensive list of old-scale plaquettes.

For successful components in a joint full law, the masses can instead remain available as in (NR20.21), once its actual representation is verified. For residual fixed components, a law-level spatial theorem can use the actual conditional input moments without a mass lower bound. An unconditional moment bound cannot be substituted for a rare-sector conditional one; any global passage must use a true-marginal disintegration or replica construction.

This note thus adds an actual old-Gaussian reserve, a genuine native supremum envelope, full-footprint payment for that energy class and an actual-source selector test. It leaves the full ambient M-03 normalized-regulator theorem unproved.

Annex BK — Exact scalar rooted calculus and a full-carrier summation criterion

Scope: M-04 remains open. This note proves an incoming analytic scalar budget, an exact rooted numerator/denominator expansion with compatible current records, and a quantitative graded carrier criterion for that expansion. It then identifies the source-specific estimate which that criterion still requires. A finite positive model shows why the established regular-law estimates and a relative carrier bound do not alone imply that missing estimate.

The stated M-04 test is equality with the actual transported scalar $Q_k F_k$, followed by a remainder estimate in M-03 uniform in scale. The source-dependent definition of F_k, all responses and the actual normalized law remain fixed. In particular a newly inserted coarse plaquette is not substituted for $Q_k F_k$.

1. A genuine incoming budget which the initial scalar satisfies

Let h be the input spacing, and let S be the fixed plaquette/link stencil of the F-01 scalar. The repaired scalar component is a fixed nonzero multiple of the Wilson density, by F-01 and the identity $\operatorname{Tr} \tilde{q}_h = 2w_h$. A representative centered incoming scalar therefore has the form

$$F_h(U) = h^{-4}\alpha\,\beta \sum_{p\in\mathcal{P}(S)} a_p\, s(U_p) - c, \qquad s(U) = N - \operatorname{Re}\operatorname{Tr} U, \tag{SR20.1}$$

where the number of plaquettes and the coefficients a_p are fixed. The exposed coefficient budget is $b = |\alpha| + h^4|c|$. The actual finite F-01 normalization coefficients may be inserted into this formula; no uniform bound on their running is asserted.

For a real compact configuration, complexify a finite set of links by

$$U_e(\zeta) = \exp\!\left(\frac{\zeta_e T_e}{\sqrt{1+\beta}}\right) U_e, \qquad |\zeta_e| \le \rho, \tag{SR20.2}$$

where the generators have fixed norm, $\rho > 0$ is fixed and the stencil contains a fixed number of links. Negative orientations use the inverse. The holomorphic extension of s is

$$\mathsf{s}(P) = N - \tfrac{1}{2}\operatorname{Tr}(P + P^{-1}). \tag{SR20.3}$$

Lemma SR20.1 — local complex energy control With $E_S(U) = \beta \sum_{p\in\mathcal{P}(S)} s(U_p)$, there is a fixed C_ρ such that

$$\sup_{|\zeta_e|\le\rho} |F_h(U(\zeta))| \le C_\rho b h^{-4}[1 + E_S(U)]. \tag{SR20.4}$$

Consequently every ordered r-fold derivative in the normalized link coordinates at the real center satisfies

$$|D^r F_h(U)| \le C^{r+1} r!\, \rho^{-r} b h^{-4}[1 + E_S(U)]. \tag{SR20.5}$$

The constants are independent of $\beta > 0$, h, volume and the real compact configuration.

Proof. A plaquette has a fixed number of factors. At a real center the trace of each differentiated generator alone is zero. Subtracting the identity from the remaining compact plaquette gives the first variation bound

$$\left|\frac{d}{dt}\mathsf{s}(P(t\zeta))\Big|_{t=0}\right| \le C(1+\beta)^{-1/2}\sqrt{s(P)}\sum_e |\zeta_e|. \tag{SR20.6}$$

Along the complex radial path $t\zeta$, every compact factor has norm one and every added exponential has norm at most $e^{C\rho}$. The second variation is bounded by $C_\rho(1+\beta)^{-1}(\sum_e |\zeta_e|)^2$. Taylor's integral formula, multiplied by β, therefore

bounds the change by

$$C_\rho \left[\sqrt{\beta s(P)} \sum_e |\zeta_e| + \Big(\sum_e |\zeta_e| \Big)^2 \right]. \tag{SR20.7}$$

Here $\beta/(1+\beta) \le 1$. The finite stencil and $2ab \le a^2 + b^2$ prove (SR20.4), including the identity coefficient. For coordinate-chart derivatives apply multivariable Cauchy on a fixed smaller polydisc. For ordered invariant derivatives with noncommuting generators on the same link, introduce r ordered exponential increments with individual radius ρ/r. Their total increment size remains bounded by ρ, so the same energy estimate applies. Cauchy then costs $(r/\rho)^r \le e^r r! \rho^{-r}$. This proves (SR20.5) for both conventions. □

Thus an incoming one-step hypothesis consisting of actual carrier support, a local complex energy bound such as (SR20.4), its exposed coefficient sum, and the already specified direct/current records is nonvacuous. The microscopic scalar satisfies its analytic part before any conditional expectation is taken. The real moment part follows from the same compact Wilson energy reserve used in M-03. This does not prove that every previously transported scalar already satisfies that hypothesis; propagation of the input budget remains an induction task.

The established NP19.1–NP19.4 estimates control the corresponding real regular conditional value majorant, including the actual retained exterior force and, on a wholly deepest original regular patch, the minimizing reference energy. They do not estimate each absolute marked Gaussian/cluster term separately. That distinction is retained below.

2. The exact rooted identity

Fix one geometry and one actual source-free regular integral. After the source's Gaussian and localizations, write the zero-source polymer gas as

$$\Xi(\upsilon) = \sum_{\Gamma \text{ compatible}} \prod_{\gamma \in \Gamma} H_\gamma(\upsilon). \tag{SR20.8}$$

Every activity includes its full field and history dependence. Let υ denote the actual retained variables, rather than only the native analytic parameters used to estimate them. Compatibility is the source relation: two localization domains cannot share a cube or a cube wall. Fixed collar enlargements can be used to express this as intersection of connected covers.

For a localized incoming mark F, carry that mark through the same Gaussian integrations and localizations. Denote the one-root activity by $R_Z(\upsilon)$. In the source numerator this is $H'(Z)$, with exactly one marked integrand. No minimizing field or integration map is changed by the mark. The numerator has the form

$$\mathcal{N}_F(\upsilon) = \sum_Z R_Z(\upsilon)\, \Xi_{\nsim Z}(\upsilon), \tag{SR20.9}$$

where $\Xi_{\nsim Z}$ contains only background polymers compatible with Z. This is the structure of CMP119 (3.44). The mark is linear in this construction. Every finite analytic input for which the localization is performed has this identity at finite cutoff; its uniform marked bound is a separate matter.

Let

$$\log \Xi(\upsilon) = \sum_B E_B(\upsilon) \tag{SR20.10}$$

be the convergent connected cluster representation, with complete activity tags retained. Removing all polymers incompatible with Z deletes precisely those connected clusters containing at least one such polymer. Denote that subfamily by $B \bowtie Z$; the detailed cluster tag is kept if its union alone does not encode the incidence. Then

$$\frac{\Xi_{\nsim Z}(\upsilon)}{\Xi(\upsilon)} = \exp\left[- \sum_{B \bowtie Z} E_B(\upsilon) \right]. \tag{SR20.11}$$

Proposition SR20.2 — actual rooted carrier identity Whenever the corresponding marked series converges absolutely,

$$Q_v F_v = \sum_Z \sum_{n\ge 0} \frac{1}{n!} \sum_{B_1,\dots,B_n:\, B_i \bowtie Z} R_Z(v) \prod_{i=1}^{n} [-E_{B_i}(v)]. \tag{SR20.12}$$

Each displayed term is a function on the actual carrier

$$X = Z \cup B_1 \cup \cdots \cup B_n \tag{SR20.13}$$

with fixed connecting collars included. This assertion requires the factors in that term to have those actual supports. It follows from multiplication of the factors, not from a support property of their numerical bounds.

Proof. Divide (SR20.9) by (SR20.8), use the connected cluster identity (SR20.11), and expand the last exponential. Each B_i is connected internally and meets the root under the stated compatibility relation; hence their covered union is connected. A function depending on the coordinates of each factor depends on their union. Absolute convergence permits the resummation. The same computation is valid first as a formal first-order marked source coefficient; a quantitative sufficient convergence criterion is proved in Section 4. □

The primary construction CMP119 (3.44)–(3.47), pp.277–278, already performs this numerator/denominator cancellation for its specified action-response insertions. CMP116, Section 2, first localizes the Gaussian expressions in their external variables and then bounds the integrals before the polymer resummation. Its unmarked Gaussian majorant is the appropriate place to insert a root with (SR20.4). Neither source states the M-03 marked operator bound proved conditionally in Section 4.

If the regular density has already used LFR II (1.103)–(1.104), its entire boundary exponential and normalized fixed-component operations are inside the baseline of (SR20.8). A root numerator must use that same baseline. Moving one of those factors into an unrelated reference law changes $Q_v F_v$.

3. Compatible direct and normalized-kernel currents

Write

$$R_Z(v) = \mathcal{R}_{Z,v}(F_v) \tag{SR20.14}$$

for the linear marked-root integration/localization operator. On a common padded family of records, let the incoming exact difference be $\delta_\tau F = B_\tau^F + J_\tau^F$. The root records are

$$\begin{aligned} B_{Z,\tau}^R &= \mathcal{R}_{Z,\tau v}(B_\tau^F), \\ J_{Z,\tau}^{R,\mathrm{old}} &= \mathcal{R}_{Z,\tau v}(J_\tau^F), \\ J_{Z,\tau}^{R,\mathrm{new}} &= \frac{\mathcal{R}_{Z,\tau v} - \mathcal{R}_{Z,v}}{t}(F_v). \end{aligned} \tag{SR20.15}$$

For discrete changes take $t = 1$. These formulas include every change of the root operator's density, support, localization, geometry and normalizer. In particular

$$\delta_\tau R_Z = B_{Z,\tau}^R + J_{Z,\tau}^{R,\mathrm{old}} + J_{Z,\tau}^{R,\mathrm{new}}. \tag{SR20.16}$$

Set $e_B = -E_B$. For a tagged term of (SR20.12), suppress its factor $1/n!$ temporarily and write $A = R_Z \prod_i e_{B_i}$. Its compatible records are

$$B_\tau^A = B_{Z,\tau}^R \prod_i e_{B_i}(\tau v), \tag{SR20.17}$$

$$\begin{aligned} J_\tau^A = {} & (J_{Z,\tau}^{R,\mathrm{old}} + J_{Z,\tau}^{R,\mathrm{new}}) \prod_i e_{B_i}(\tau v) \\ & + R_Z(v) \sum_{j=1}^{n} \delta_\tau e_{B_j}(v) \prod_{i<j} e_{B_i}(\tau v) \prod_{i>j} e_{B_i}(v). \end{aligned} \tag{SR20.18}$$

Proposition SR20.3 — exact compatible rooted records For every tagged term,

$$\delta_\tau A = B^A_\tau + J^A_\tau. \tag{SR20.19}$$

The first line of (SR20.18) retains old and new root currents separately; its second line is the denominator-cluster response. Summing them gives the exact normalized-kernel current, including the centering contribution from the denominator.

Proof. Add and subtract $R_Z(\upsilon)\prod_i e_{B_i}(\tau\upsilon)$, and use the finite telescoping product identity on the remaining product. Substitute (SR20.16). This proves (SR20.17)–(SR20.19) for finite differences, without a smooth-score assumption. □

If a history change alters the allowed polymer family or compatibility relation, pad the families and include the corresponding zero/one indicators in the factors before differencing. More explicitly replace e_B by $e_{Z,B}(\upsilon) = -1_{\{B\bowtie_\upsilon Z\}}E_B(\upsilon)$, with the full footprint of that indicator included. Its difference, not only the difference of E_B, occurs in (SR20.18). There can also be indicators inside the root operator in (SR20.15). No bound for these new terms follows merely from the displayed identity. On an admitted fixed geometry all indices are unchanged and this complication is absent.

For these formulas to produce valid M-03 triples on carrier X, the declared support must include the footprints of all its value and current factors and every direction on which the separate records depend. A cancellation $B+J=0$ does not permit deleting nonzero separate records.

4. A full M-03 bound from rooted pre-normalization estimates

This theorem is a quantitative criterion. Its hypotheses concern the actual pre-normalization root activities and complete background cluster activities, not merely the final conditional value.

Use the exact M-03 functions $\upsilon_{h',1,Z}$, $g_{h',1,Z}$, $j_{h',1,Z}$, the actual outer law $\mu_{h'}$, and the full native regulator. Let r_Z be the sum of these three quantities for the compatible root triple, keeping old and new current parts separately in the sum. The root's first source order has value grade one and current grade 3/2.

For a background cluster define

$$a_B = \sup_\upsilon |E_B|^\sharp(\upsilon). \tag{SR20.20}$$

Let $D^\natural_B$ be its summed finite-difference envelope, exactly as in M-03, and put

$$c_B = \mathfrak{m}_{1/2}(D^\natural_B;\mu_{h'}) + 2^{1/2}\sup_\upsilon \frac{D^\natural_B(\upsilon)}{\widehat{R}_{h',B}(\upsilon)^{1/2}}. \tag{SR20.21}$$

The constants and physical h'^4 factor remain in the root norm; no such factor is attached to the zero-source action. All suprema in this section are on the specified ambient versions and include every M-03 elementary shift. Restriction to an admitted chart alone would give only that restricted conclusion.

Use fixed connected covers so that incidence is realized by intersection. Equivalently retain a fixed collar constant C_{inc}. Suppose

$$\begin{aligned}\sup_b \sum_{B\ni b} w_{h',\kappa}(B)a_B &\le \varepsilon_0,\\ \sup_b \sum_{B\ni b} w_{h',\kappa}(B)c_B &\le \varepsilon_1,\end{aligned} \tag{SR20.22}$$

and

$$\mathcal{B}_{\mathrm{root}} := \sum_Z w_{h',\kappa+\delta}(Z)\, r_Z < \infty, \qquad C_{\mathrm{inc}}\varepsilon_0 \le \delta/2. \tag{SR20.23}$$

The constant ε_1 need not be small.

For the geometry-dependent factors just specified, use their actual carrier bounds $a_{Z,B}, c_{Z,B}$ instead. The exact sufficient replacement for (SR20.22) is

$$\sum_B w_\kappa(B)a_{Z,B} \le C_{\mathrm{inc}}\varepsilon_0|Z|, \qquad \sum_B w_\kappa(B)c_{Z,B} \le C_{\mathrm{inc}}\varepsilon_1|Z|, \tag{SR20.22a}$$

for every root, with complete indicator/direction footprints included and each declared factor carrier meeting the root cover. This is a hypothesis to be verified, not a consequence of padding. In the fixed-geometry case (SR20.22) implies (SR20.22a) by counting incident blocks. The proof below uses only the latter two sums, so covers both cases.

Theorem SR20.4 — full graded rooted summation The tagged expansion (SR20.12), with records (SR20.17)–(SR20.18), converges in the actual M-03 norm and obeys

$$\|QF\|_{h',\varkappa} \le C\left(1+\frac{\varepsilon_1}{\delta}\right)\mathcal{B}_{\mathrm{root}}. \tag{SR20.24}$$

The constant is independent of cutoff, volume and the number of integrated coordinates. Any fixed one-step conversion $h' = Lh$ is retained explicitly when estimating $\mathcal{B}_{\mathrm{root}}$ from an incoming norm; it contributes L^4, not an unbounded depth factor.

Proof. For intersecting connected covers,

$$w_\varkappa\left(Z\cup\bigcup_i B_i\right) \le w_\varkappa(Z)\prod_i w_\varkappa(B_i). \tag{SR20.25}$$

The same inequality with fixed incidence constants follows by adding the stated collars. Both cardinality and connecting-tree length are subadditive on such a union.

For averaged grades, Hölder gives the exact inequality

$$\mathfrak{m}_{d+e}(FG) \le \mathfrak{m}_d(F)\mathfrak{m}_e(G). \tag{SR20.26}$$

Indeed choose the two Hölder exponents $(d+e)/d$ and $(d+e)/e$ at each probability moment order p. The two factors become $((d+e)p)^d$ and $((d+e)p)^e$. The pointwise regulator grades satisfy the same product estimate because the union regulator dominates each individual regulator and $d^d e^e \le (d+e)^{d+e}$.

The value envelope of a tagged product is bounded by $R_Z^\sharp\prod_i a_{B_i}$. The direct and root-current envelopes are bounded by their root envelopes times the same product. The remaining term in (SR20.18) is bounded by

$$R_Z^\sharp\sum_j D_{B_j}^\natural\prod_{i\ne j} a_{B_i}. \tag{SR20.27}$$

Formula (SR20.26) with $d=1, e=1/2$, and the corresponding pointwise estimate, give exactly the required current grade 3/2. Thus the norm of the tagged product is at most

$$Cr_Z\left[\prod_i a_{B_i} + \sum_j c_{B_j}\prod_{i\ne j} a_{B_i}\right]. \tag{SR20.28}$$

Every incident cluster touches one of at most $C_{\mathrm{inc}}|Z|$ blocks. Sum the first product over the ordered B_i's and n, including $1/n!$, and use (SR20.22). This costs at most

$$e^{C_{\mathrm{inc}}\varepsilon_0|Z|}. \tag{SR20.29}$$

The second product sum costs in addition at most $C_{\mathrm{inc}}\varepsilon_1|Z|$. Finally,

$$[1+C_{\mathrm{inc}}\varepsilon_1|Z|]\,e^{C_{\mathrm{inc}}\varepsilon_0|Z|} \le C(1+\varepsilon_1/\delta)e^{\delta|Z|}. \tag{SR20.30}$$

The spare δ in (SR20.23) pays this factor. Summing the root carriers proves (SR20.24), with absolute convergence of the separate records. The convergent series therefore represents the same exact conditional value and compatible currents. □

This result solves the support issue when its actual factor hypotheses have been verified. It does not infer those hypotheses from a pointwise bound on QF. In particular (SR20.23) is an intermediate marked Gaussian/localization estimate still to be proved from the genuine incoming budget in Section 1; it is not being renamed an incoming hypothesis to declare M-04 complete.

5. Why the retained fixed-component gap is substantive

The established scalar C-05/C-08 construction controls native current endpoints and source relative carriers. The alternative LFR II (1.103)–(1.104) leaves fixed residual components Y in its scalar action. Printed p.391 explicitly removes their internal tree cost because their domains are not summed in that expansion. That is adequate for the source's scalar logarithm and the previously proved current-position estimates.

For (SR20.22), however, a factor depending on all raw variables in Y has the full physical carrier Y, even if its current endpoint lies outside it. A small coefficient e^{-cR}, where $|Y|$ can be of order R^4, does not by itself pay $e^{\varkappa|Y|}$. The small scalar multiplier $T_k'(Y)1$ remains in the outer probability law; it cannot pay the pointwise part of a normalized conditional mark merely by being small.

The following exact example shows that this is not resolved by the already established regular conditional moment estimates.

Proposition SR20.5 — a sharp full-carrier diagnostic Let Y_R be a connected four-dimensional cube with $N_R \asymp R^4$ sites, at distance comparable to R from the integrated root. Let $\sigma_j \in \{-1, 1\}$, $j \in Y_R$, be retained discrete coordinates. Join Y_R to the root by a fixed thin path. Take $P_R = R + 2$, fix $a > 0$, and put

$$\chi_R(\sigma) = \prod_{j \in Y_R} \sigma_j, \qquad \vartheta_R(\sigma) = \tfrac{1}{2}(1 + \chi_R(\sigma)), \qquad \epsilon_R = \frac{e^{-aR}}{8P_R^2}. \tag{SR20.31}$$

At an anchored integrated site take the positive truncated conditional law

$$q_\sigma(dz) = \frac{1_{[-P_R, P_R]}(z)}{Z_\sigma} \exp\Big[-\tfrac{1}{2}z^2 + \epsilon_R \vartheta_R(\sigma) z^2\Big]\, dz. \tag{SR20.32}$$

Give σ its uniform outer law. This defines a finite-dimensional positive probability model with compact product support for the integrated variable, matching the support structure used in C-07. The root mark is $F(z) = z^2$.

The integrated law has a fixed coercivity gap, zero mean, uniformly Gaussian moment bounds and a perturbation with Hessian at most $2\epsilon_R$. On its entire product support $|W| \le e^{-aR}/8$; on the doubled complex current disc it is at most $e^{-aR}/2$. Its dependence is confined to one current coordinate and one fixed retained set Y_R. If that fixed set's interior has zero carrier cost, the remaining thin path has cost $O(R)$; choosing the fixed a larger than its prescribed relative-weight rate gives a uniformly small relative activity bound. The incoming one-mark M-03 norm is uniformly bounded: the input regulator contains $1 + z^2$, the root carrier is fixed, and the mark is independent of every σ_j.

Nevertheless every actual carrier representation of the conditional output has M-03 value norm at least

$$c\, e^{\varkappa N_R} \epsilon_R, \tag{SR20.33}$$

which is unbounded as $R \to \infty$. The output native regulator is one because no continuous energy coordinate survives.

Proof. Let $m_P(s)$ be the second moment of the normalized density proportional to $e^{-z^2/2 + sz^2}$ on $[-P, P]$. Differentiation on this fixed compact support gives

$$m_P'(s) = \mathrm{Var}_{P,s}(z^2). \tag{SR20.34a}$$

This variance is bounded below by a fixed positive constant for every $P \ge 2$, $0 \le s \le 1/8$. To see this, the two events $|z| \le 1/4$ and $1 \le |z| \le 5/4$ have probabilities bounded below uniformly: their intervals have fixed length, their density numerators have fixed positive lower bounds, and the normalizer is bounded above by the full Gaussian integral with exponent $-3z^2/8$. On two independent copies lying in these two events the difference of their squares is at least $15/16$. The identity $\mathrm{Var}(z^2) = \frac{1}{2}\mathbb{E}[(z^2 - z'^2)^2]$ proves the lower bound.

The exact conditional value consequently has the form

$$Q_\sigma F = a_R + b_R \chi_R, \qquad b_R = \tfrac{1}{2}[m_{P_R}(\epsilon_R) - m_{P_R}(0)] \ge c\epsilon_R, \tag{SR20.34}$$

with $a_R = \frac{1}{2}[m_{P_R}(\epsilon_R) + m_{P_R}(0)]$. For any function independent of at least one coordinate of Y_R, its Walsh coefficient against χ_R under the uniform measure is zero. Hence in any representation $QF = \sum_X A_X$ by actual supported functions, only carriers containing every site of Y_R can contribute the nonzero coefficient b_R. Its magnitude is at

most the sum of the suprema of those A_X's. Each such carrier has weight at least $e^{\varkappa N_R}$, and its shifted output regulator is exactly two. The pointwise value term in M-03 therefore proves (SR20.33). This reasoning applies to any absolutely summable representation, not only to the reference-point decomposition.

For the input, the normalizer is bounded below by its integral over $[-1, 1]$, while its density numerator is bounded above by $e^{-3z^2/8}$. Thus $\|z^2\|_p \leq Cp$ uniformly, including all $p \geq 1$. Its unit-shift value envelope is bounded by $C(1 + z^2)$, on the full ambient Euclidean extension. Its direct finite-difference envelope is at most $2|z| + 1$, with no discrete direction contribution. Both its averaged grade 3/2 and the corresponding pointwise native-regulator quotient are uniformly bounded; its incoming current record is zero.

Finally $e^{\varkappa N_R} \epsilon_R$ diverges because $N_R \asymp R^4$ and $P_R = R + 2$. Subtracting an identity term or any fixed local scalar readout depending on a proper subset of Y_R leaves the highest Walsh coefficient unchanged, so that subtraction cannot remove the lower bound for the remainder. □

This is not a Yang–Mills counterexample. It proves that the regular coercivity/moment estimates, a small native-current perturbation, a small integration core and a relative fixed-component bound do not imply the full physical carrier assertion. Actual localization of the retained dependence, or an actual coupling which retains a small unnormalized mass in each marked term, is additional information.

6. Exact status of M-04

The root expansion (SR20.12) and the separate current formulas are exact and give genuine function supports. The full graded summation theorem shows precisely how those supports and a source-sized absolute root estimate would yield the M-03 bound. The initial scalar has a verifiable incoming analytic energy budget.

The following source applications remain unproved:

1. A termwise marked Gaussian/localization estimate (SR20.23) from that incoming budget, on the same full density as C-08, including every background, E/B/R and coarea contribution. A bound for the whole conditional expectation from NP19 does not bound its absolute localized terms.
2. The full physical version of (SR20.22) for the actual residual fixed-component dependence, together with every ambient raw, selector and history current. The source's relative fixed-Y norm is not that version.
3. The same exact output's scalar/identity extraction in the specified F-01 reference basis, with the resulting remainder and all its records bounded in M-03. This must use the output proved in the first two steps, not a native replacement field.

The sufficient root criterion does not complete M-04, and the diagnostic example does not disprove the actual model. M-04 remains open as stated.

Annex BL — The retained stronger interface and exact remaining M-04 applications

The following estimates specify the relationship between the reset theorem, the stronger KT19 test, and the M-04 requirement.

1. H-03 is a spatial theorem about an exact law

H-03 asks for an estimate of the conditional covariance and higher cumulants generated when a live history label is averaged and resampled, with a carrier radius or decay function uniform in the remaining history. Its requirement is a spatial response estimate for the exact reset jet beyond its integral identity.

The completed result is HS20.1–HS20.4 and the complete finite-jet identity HS20.29. The law is the actual positive mixture of the successful eligible component geometries and the internal old states of both successful and fixed residual eligible components. The remaining data include the new current field, the unreplaced exterior history and the residual component geometry. Every descriptor-dependent residual cutoff and gauge reference remains in its positive base. The successful masses stay unnormalized inside the sum.

For marks supported on actual reset-coordinate footprints, the theorem gives conditional cumulants of every finite order, using true posterior input moments. Its physical radius is a fixed multiple of $MR_k h$, and its decay includes the stated $(r-1)^{-1}$ loss at cumulant order r. The source and parameter hypotheses remain those of the declared Balaban construction. Independent verification of the entire source-free base and Wilson-normalized trajectory is still part of J0.

This is a new full-mixture result, beyond the earlier fixed-geometry estimates. The successful geometry fluctuations and all cumulants of descriptor-dependent means are included. Fixed residual internal states are averaged; their geometry and the unreplaced exterior geometry are retained.

2. The stronger KT19 interface remains unproved

The historical KT19 test asks for substantially more: a bounded operator on the complete M-03 value/direct/current norm, including every elementary raw or discrete shift, a pointwise quotient by the shifted output-native regulator and a sum with an exponential weight on the full physical carrier volume.

HS20 does not assert that theorem. This interpretation of the H-03 statement does not make any KT19 condition disappear. The conditions remain necessary wherever that stronger norm is used in the subsequent induction, particularly M-04, M-05, M-06 and H-04.

The distinction is concrete. A conditional cumulant can decay between its actual reset-variable supports while its dependence on a fixed retained background remains to be analyzed. A smooth source-native parameter also need not be an independently varying raw retained link. Neither statement proves the corresponding marked pullback or its current.

The exact new normalized-kernel current remains

$$J_\tau^{\mathrm{new}}(F)(\upsilon) = \frac{1}{tZ_{\tau\upsilon}} \int (F_\upsilon - Q_\upsilon F_\upsilon)\, d(M_{\tau\upsilon} - M_\upsilon), \qquad Z_\upsilon = M_\upsilon(1). \tag{RA20.1}$$

For a discrete replacement take $t = 1$. The old mark is frozen in this term; its own direct and previous-current differences are transported separately. Moving selectors and changes of support occur in the signed measure difference. HS20's probability theorem does not bound this expression for every ambient M-03 version or shift.

3. M-04 remains open, with three precise applications

The M-04 statement requires equality with $Q_k F_k$ and a remainder estimate in the M-03 norm, uniformly in scale. The left side must be the transported incoming scalar. The five results SR20.1–SR20.5 do not yet meet that whole test.

Application 1: an absolute root estimate in the actual complete density The microscopic scalar has the proved local analytic energy bound in equation (SR20.4), Lemma SR20.1, including the correctly normalized compact-link directions. Once the actual marked-root factorization is supplied, its exact ratio identity is

$$Q_v F_v = \sum_Z R_Z(v) \exp\left[- \sum_{B \bowtie Z} E_B(v) \right]. \tag{RA20.2}$$

Here the root activities must be obtained from the same Gaussian integrations and localizations as the source-free complete law. Providing that factorization with genuine supports for the actual incoming mark is part of this application. Every effective-action, boundary, completed-R, Haar/coarea, exterior-force and retained-mean contribution present in that law must remain included. The algebraic ratio identity is proved in SR20.2; a uniform actual-model application to every admitted incoming mark is not asserted by writing it down.

Let r_Z be the sum of the M-03 value, direct and separate current quantities for the compatible root triple; it includes the physical scalar factor. The missing source estimate is a uniform bound on

$$\mathcal{B}_{\mathrm{root}} = \sum_Z w_{h',\varkappa+\delta}(Z)\, r_Z, \tag{RA20.3}$$

derived from the genuine incoming one-step analytic/energy/carrier budget. It must bound the absolute localized root terms, not only the final conditional expectation. The latter is already better understood through NP19; cancellation in that whole expectation cannot be used to bound an absolute root sum.

Application 2: full retained-component and current localization For the actual background factors, the missing bounds are the full physical versions of SR20.22, including all their direction footprints:

$$\sup_b \sum_{B \ni b} w_{h',\varkappa}(B) a_B \le \varepsilon_0, \qquad \sup_b \sum_{B \ni b} w_{h',\varkappa}(B) c_B \le \varepsilon_1. \tag{RA20.4}$$

The value factor a_B is the actual shifted supremum. The difference factor c_B has the half-grade moment and pointwise regulator bounds specified in SR20.21. The current footprint includes any geometry or incidence indicator. The smallness needed is $C_{\mathrm{inc}}\varepsilon_0 \le \delta/2$; the finite constant ε_1 need not be small. The root-dependent version SR20.22a is available when the allowed family changes with the retained geometry.

Once RA20.3–RA20.4 are proved for these actual factors, the already proved theorem SR20.4 gives

$$\|\mathrm{Q}F\|_{h',\varkappa} \le C(1 + \varepsilon_1/\delta)\mathcal{B}_{\mathrm{root}}. \tag{RA20.5}$$

Thus the remaining issue is an identified estimate on the actual integrands and their dependence. RA20.5 is a proved implication, not a new assertion that its hypotheses hold.

Application 3: scalar/identity extraction from the same output After controlling that exact output, extract its scalar coefficient and identity part in the fixed F-01 reference basis, and estimate the remainder in M-03. The exact selected-law readout already available in Annex AB can be used only with all its retained mean, complete-density gradient and collar terms. A newly inserted native coarse plaquette cannot replace the transported mark in RA20.2.

This final extraction must preserve all the value/direct/current records and the same scale-independent budget. It does not assume a final running normalization matrix from F-05.

4. Why the existing bounds do not yet give Application 2

The primary normalized fixed-component expansion on LFR II p.391 expressly does not charge the interior tree size of a fixed residual component. Printed p.392 gives locality of the final scalar multiplier in the whole component. These are the source's actual statements; neither supplies thin localization of an arbitrary marked numerator in independent raw retained coordinates.

The elementary scale mismatch is

$$|Y| \asymp R^4, \qquad e^{-cR}e^{\kappa|Y|} \longrightarrow \infty \quad (\kappa > 0). \tag{RA20.6}$$

This comparison shows why the presently available majorant cannot pay the required full-volume weight. SR20.5 strengthens that observation with a finite positive truncated Gaussian example: all the listed regular conditional estimates and a small relative activity hold, yet every actual supported decomposition has the full-volume lower bound. That example is not the Wilson model and does not prove that the actual M-04 theorem is false.

The proof must therefore obtain additional actual localization or a different exact full-law payment that really controls the norm being asserted. Treating the small scalar mass of a fixed component as still available after normalizing that very component is invalid.

5. The expanded energy reserve is useful but has a definite boundary

NR20.1 proves an unnormalized exponential reserve for the two outer positive energies and every surviving old Gaussian precision energy. It also permits a fixed exponential store for the actual number of old Gaussian coordinates. NR20.1a identifies a genuine part of the native regulator: the ordinary old dimensionless free coordinates, through their uniformly coercive Schur forms in their own birth-scale units.

NR20.3 proves a common envelope for the admitted native analytic chart using total-variation domination by one real positive reference. NR20.19 pays full eligible-component footprints for this unnormalized energy class. Together with the actual deletion identity now proved in HS20.1, the pinned successful-component implication NR20.21 applies to that exact mixed law.

These statements do not give a lower bound for a fixed residual mass, do not control every raw old plaquette energy, and do not cover every ambient shifted coordinate in M-03. They provide actual additional tools for the remaining application.

Corollary RA20.1 — an actual normalized pinned-energy estimate Fix the actual remaining data υ in HS20 and a finite compatible family $\mathcal{D}$ of successful-component candidates. For nonnegative integers m_X, the exact mixed posterior obeys

$$\mathbb{E}_{\pi_\upsilon}\Big[1_{\{\mathcal{D}\subset C\}}\prod_{X\in\mathcal{D}}(\mathcal{F}_X^\natural)^{m_X}\Big] \le \prod_{X\in\mathcal{D}} e^{\lambda_R}M_X\Big[(\mathcal{F}_X^\natural)^{m_X}\Big] \le \prod_{X\in\mathcal{D}} C^{m_X+1}m_X!\,e^{-P_k+\lambda_R}. \tag{RA20.7}$$

All component measures in this probability assertion are on the actual positive real slice. The native envelope includes the complex-chart supremum only through the common real positive bound proved in NR20.3. Constants are uniform in the remaining admitted history and its old depth.

Proof. In the positive sum defining π_υ, restrict to occupied families containing $\mathcal{D}$ and delete those components. By HS20.1 this produces an admissible family with the same remaining data and component bases. Successively applying HS20.7 bounds the change of the cross action by $|\mathcal{D}|\lambda_R$, uniformly in every old internal state, including residual states. The deleted internal integrations now factor and give the first product in RA20.7. The sum over the remaining admissible families is bounded by the full positive partition function. Dividing by that same partition function proves the first inequality. The energy-power reserve NR20.12/NR20.16 proves the second. No individual inverse component mass occurs. □

In particular the source shape count and $P_k \gg R_k^4$ give, for each fixed component-geometry weight κ,

$$\sup_b \sum_{X\ni b} e^{\kappa(|X^+|+\mathrm{t}(X^+))}\mathbb{E}_{\pi_\upsilon}\Big[1_{\{X\in C\}}(\mathcal{F}_X^\natural)^m\Big] \le C^{m+1}m!e^{-P_k/2}. \tag{RA20.8}$$

This is a normalized estimate for the actual successful-component part of the law. It pays full eligible geometric component weights in that energy class. It does not assert a decomposition of the resulting function of raw retained data into M-03 carriers. It also does not apply to the internal normalized law obtained by conditioning on occupation of a particular rare component, nor does it supply a rare weight for the fixed residual components. These are distinct conditional statements.

6. Obligations retained in the downstream mathematical conditions

Catalogue item	What it must still prove after HS20
M-04	RA20.3–RA20.4 for the actual complete scalar step, followed by exact scalar/identity extraction and a bounded remainder.
M-05	The corresponding exact tensor matching, including its own mixing and current dependence.
M-06	The full arbitrary-order one-step local response sum with every collision, reset and remainder vertex; the complete direct/current and carrier budget with fixed factorial/spatial parameters.
H-04	The full normalized-law one-step sum over regular, repaired, residual and inherited sectors, including actual M-03/M-06 input and output costs. No positive-probability sector or normalizer may be discarded.
M-07	Propagation of the required input budgets through all scales; a conditional one-step theorem does not supply its own inductive hypothesis.

H-02 and H-05 keep their stated normalized history and terminal requirements. None is closed by the one-reset comparison theorem.

7. Continuum assembly remains a separate target

J0 still requires the independent source-free base and coupling verification in the correct Wilson normalization. J1–J2 require the compatible renormalized full hierarchy and its OS growth bound. J3 includes the positivity, translation and rotation requirements. J4 requires quantitative ultraviolet identification and nontriviality in the selected phase. J5 requires the field energy bounds and valid phase transfer. Their stated mathematical conditions are unchanged.

The results above establish H-03 at its stated law-level scope and leave M-04 open. Full four-dimensional Yang–Mills existence is not proved here.

Annex BM — Actual normalized successful-component energy and rank moments

This note combines two estimates for the Wilson law: the complete positive mixed posterior in HS20 and the old energy/rank reserve in NR20. It turns the conditional deletion consequence into a normalized theorem for successful components in the joint law. It does not condition on the presence of a component, control fixed residual-component energies, or claim a full M-03 carrier representation.

1. Fixed actual posterior and a common exponential store

Fix an admitted real retained datum υ. Use exactly the posterior

$$\pi_\upsilon(dC, d\eta_Y, d\eta_C) = Z_\upsilon^{-1} 1_{\{C \text{ admissible}\}} e^{H_C} \prod_i \nu_{Y_i}(d\eta_{Y_i}) \prod_{X \in C} M_X(\upsilon, d\eta_X) \tag{NM20.1}$$

of HS20.4. Each residual base includes all its required old and descriptor-dependent reference factors. Each successful M_X is unnormalized. Its internal histories are summed inside it, and the cross action retains their dependence. There is no signed activity probability and no replacement reference used in place of the old law.

Let $\mathcal{F}_X^\natural$ be NR20.15's common positive native envelope, and D_X the actual surviving Gaussian coordinate rank in NR20.9a. Put

$$B_X = \mathcal{F}_X^\natural + D_X \geq 1. \tag{NM20.2}$$

The ordinary birth-scale native free squares are bounded by a fixed multiple of this energy, by NR20.1a, with the exact coordinate qualification there. No raw old plaquette energy is added to the definition.

Choose the fractional-energy reserve parameter $\theta_0 > 0$ in NR20.1 and a fixed rank parameter $u_0 > 0$ in NR20.9a. Fix once and for all

$$0 < t \leq \tfrac{1}{2} \min\{\theta_0, u_0\}. \tag{NM20.3}$$

The same common source ceiling can then be used for all source orders. Equations NR20.9a and NR20.14–NR20.16 give

$$M_X(\upsilon, e^{tB_X}) \leq C_0 e^{-P_k}, \qquad P_k = P_0(g_k), \tag{NM20.4}$$

uniformly in X, υ and old depth. Indeed the common-envelope bound is $\mathcal{F}_X^\natural \leq 1 + \mathcal{F}_{X,0} + \delta_Q$, and the density comparison contributes at most e^{δ_σ}. The remaining insertion has coefficients $t \leq \theta_0$ and $t \leq u_0$, so the actual reserve applies. C_0 is fixed. This calculation uses a positive real reference; analytic native directions enter only through total variation and the positive envelope.

The source-specific component deletion and interaction bound HS20.1/HS20.7 give

$$H_{C \cup \{X\}} - H_C \leq \lambda_R \tag{NM20.5}$$

whenever insertion is allowed, with λ_R uniformly small at the common ceiling. Reduce that ceiling so that

$$\lambda_R + \log C_0 \leq P_k/2 \tag{NM20.6}$$

at every admitted scale.

2. A normalized exponential theorem

Theorem NM20.1 — pinned successful energy and rank For every compatible finite family $\mathcal{D}$ of candidate successful components,

$$\boxed{\mathbb{E}_{\pi_v}\left[1_{\{\mathcal{D}\subset C\}}\exp\left(t\sum_{X\in\mathcal{D}}B_X\right)\right]\le e^{-P_k|\mathcal{D}|/2}.} \tag{NM20.7}$$

If the prescribed family is incompatible, the left side is zero. The bound is uniform in all remaining source-admitted data and internal old depth. It is a statement about the full normalized posterior (NM20.1).

Proof. In the numerator of (NM20.1), restrict to $C\supset\mathcal{D}$ and write $C=E\cup\mathcal{D}$. HS20.1 says that deletion leaves an admissible E, the same retained datum and exactly the same remaining component measures. In particular it leaves the residual bases unchanged.

Successive deletion, or insertion in the reverse order, gives

$$H_{E\cup\mathcal{D}}-H_E\le|\mathcal{D}|\lambda_R.$$

This inequality holds uniformly in all old states, including the residual states still integrated. Therefore the deleted integrations factor, and the restricted numerator is at most

$$e^{|\mathcal{D}|\lambda_R}\prod_{X\in\mathcal{D}}M_X(e^{tB_X})\sum_{\substack{E\ \text{admissible}\\ E\sim\mathcal{D}}}\int e^{H_E}\prod_i d\nu_{Y_i}\prod_{X\in E}dM_X.$$

Positivity bounds the remaining restricted sum by the same full partition function Z_v. Divide by it and apply (NM20.4)–(NM20.6). No m_X^{-1} is introduced. ■

Corollary NM20.2 — every polynomial grade For nonnegative integers m_X, and also for a single total power $m\ge0$,

$$\mathbb{E}_{\pi_v}\left[1_{\{\mathcal{D}\subset C\}}\prod_{X\in\mathcal{D}}B_X^{m_X}\right]\le e^{-P_k|\mathcal{D}|/2}\prod_{X\in\mathcal{D}}t^{-m_X}m_X!, \tag{NM20.8}$$

$$\mathbb{E}_{\pi_v}\left[1_{\{\mathcal{D}\subset C\}}\left(\sum_{X\in\mathcal{D}}B_X\right)^m\right]\le e^{-P_k|\mathcal{D}|/2}t^{-m}m!. \tag{NM20.9}$$

For $m=0$, interpret the total-power expression as the occupation indicator. For a real power $q\ge1$, replace the integer-power factor $t^{-m}m!$ by $(q/(et))^q$.

Proof. Apply $x^m\le m!t^{-m}e^{tx}$ to each factor or to the total sum and use (NM20.7). ■

The estimates are pointwise in the actual admitted v. Integrating over its true outer marginal preserves them. They are not an estimate for $\pi_v(\cdot\mid\mathcal{D}\subset C)$: that division would remove the occupation probability.

3. Anchored geometry cost and finite marked numerators

For every fixed physical component weight $\varkappa>0$, the source bound on eligible component size and the shape count in HS20.9 imply

$$\sup_b\sum_{X\ni b}e^{\varkappa(|X^+|+\mathrm{t}(X^+))}\mathbb{E}_{\pi_v}\left[1_{\{X\in C\}}B_X^m\right]\le C^{m+1}m!\,e^{-P_k/4} \tag{NM20.10}$$

at one common ceiling. Both the number of shapes through b and the footprint weight are bounded by $\exp[C_\varkappa R_k^d]$, and $P_k\gg R_k^d$; this proves the displayed estimate from (NM20.8). The internal histories have already been summed in M_X and are not counted again.

This is a weighted positive energy majorant in the actual normalized law. It does not state that the resulting function of v has carrier X^+. Its dependence through the full posterior can require additional connecting carriers; constructing and paying those actual carriers remains a separate marked-localization task.

The same reserve controls every finite unnormalized marked numerator inside the normalized law. For a compatible fixed $\mathcal{D}$, suppose the degree-d incoming log-jet coefficients obey

$$|A_B| \le K^{d+1}(d!)^s b^d r(v)^d \left(\sum_{X\in\mathcal{D}} B_X\right)^d, \qquad d = |B|, \quad s \ge 1. \tag{NM20.11}$$

Define the finite formal numerator

$$N_{\mathcal{D}}(t) = \pi_v\left[1_{\{\mathcal{D}\subset C\}} \exp\left(\sum_{B\neq\emptyset} t_B A_B\right)\right]. \tag{NM20.12}$$

The labeled partition identity and (NM20.9) give

$$|\partial_I N_{\mathcal{D}}(0)| \le e^{-P_k|\mathcal{D}|/2} C^m (m!)^{s+1} b^m r(v)^m, \qquad |I| = m. \tag{NM20.13}$$

For completeness, a partition with block sizes $d_1, \dots, d_\ell$ has total energy degree m, so its integral costs $t^{-m}m!$ by (NM20.9). The ordered-size identity bounds

$$\sum_{\pi\in\Pi_m} \prod_{B\in\pi} (|B|!)^s \le C^m (m!)^s, \qquad s \ge 1.$$

Absorb the fixed $K^{m+\ell}$, t^{-m} and the number of compositions into C^m. This proves (NM20.13), with every higher incoming vertex retained. The bounds are compatible under finite-order truncation.

No logarithm of $N_{\mathcal{D}}$ is taken: normalizing this pinned numerator by its constant term would again divide out the rare occupation. The complete unpinned reset logarithmic jet is separately the exact HS20.29 formula.

4. Exact limits of the result

The actual normalized estimate now covers the successful-component energy/rank class. Fixed residual Y components have no absent state in (NM20.1), so this deletion argument supplies no rare mass for their individually normalized energies. It also does not control all raw old plaquette energies, arbitrary raw retained shifts, moving selector boundaries or all separate kernel-current records. Its geometry-weighted positive majorant is not an M-03 carrier expansion of a transported scalar or tensor.

The same deletion argument without the rank variable gives RA20.1. M-04, M-06 and H-04 require stronger marked estimates.

Transverse variation of the blocked covariance

The coupling recurrence is conditional. For the projected sum C_n of the actual constrained Gaussian increments, Gaussian calibration requires identification with the composite constrained Gaussian and $\sup_n |C_n - nd_B| < \infty$. The finite-dimensional projection and the logarithmic shell coefficient do not prove this uniform bound. In the transverse decomposition it requires $\sup_n |\mathcal{B}_n| < \infty$. Level-adapted restriction, real-centered complex reconstruction, the multiscale boundary restriction, flat-vertex constants and a common coupling ceiling are required in the same representation. See Gaussian calibration and complex reconstruction.

1. The constrained Gaussian density and its projection

1.1 Composite identification. Assume parts 1–2 of Theorem S for the constrained Gaussian, which identify the actual blocked map and its scalar projection. Then $\sum_{k<n} d_k^G = C_n := \tilde{\mathcal{P}}_n \log \mathfrak{D}_n$, where $\mathfrak{D}_n(V)$ is the g^0 density (Laplace prefactor) of $\rho_n(V) = \int dU\, \delta(\bar{U}^n(U)V^{-1}) e^{-YA(U)}$ relative to coarse Haar measure, and $\tilde{\mathcal{P}}$ is the second moment CMP109 (1.22) normalised by $\tilde{\mathcal{P}}A = 1$. The finite-dimensional steps use the Jensen bound $\|d'Qa\|^2 \le L^{4-d}\|da\|^2$, the transport identity of Lemma T, and the Laplace expansion at the constrained critical orbit. These calculations do not establish parts 1–2 of Theorem S.

1.2 Notation. Fine lattice $\mathbb{Z}^4$; block centres $L^n\mathbb{Z}^4$ (L odd). $Q_U := D\bar{U}^n(U)$ in right-invariant coordinates $U' = e^{iw}U$; d_U the infinitesimal gauge action, $(d_U\lambda)_b = \lambda(b_-) - \mathrm{Ad}_{U_b}\lambda(b_+)$; E the restriction of site functions to block centres; K the inclusion of $\ker E$; $V_U := d_U K$ (internal gauge directions). Gauge covariance of the average gives, exactly and for every U,

$$Q_U V_U = 0, \qquad Q_U d_U = d'_{\bar{U}^n(U)} E. \tag{1.1}$$

H_U is the Hessian of A at U in the chart $w \mapsto e^{iw}U$. $T'_U := \ker Q_U \cap (\mathrm{Ran}\, V_U)^\perp$. At a constrained minimiser $U = U_n(V)$ with multiplier λ_V, $H^\lambda := \mathrm{Hess}(A - \lambda_V \cdot \bar{U}^n)(U)$ annihilates $\mathrm{Ran}\, V$ and

$$\log \mathfrak{D}_n(V) = \tfrac{1}{2}\log\det(V^*V) - \tfrac{1}{2}\log\det(QQ^*) - \tfrac{1}{2}\log\det\big(H^\lambda|_{T'}\big) \quad \text{(up to a constant)}. \tag{1.2}$$

(This is the Morse–Bott Laplace prefactor; Gaussian transport §3.5, deduction.)

1.3 Reading on fine plane waves.

> Lemma 1.1 conditional. Let F be a gauge-invariant analytic function of fine configurations near 1, invariant under the lattice symmetries that preserve the block structure, such that the Hessian at $V = 1$ of $F \circ U_n - \log \mathfrak{D}_n$ has a symbol $o(|K|^2)$ at small coarse momentum K, and with a second-derivative kernel at $U = 1$ whose Bloch symbol $\hat{F}''(k, k')$ ($k - k' \in 2\pi L^{-n}\mathbb{Z}^4$) is continuous. Then
>
> $$C_n = \lim_{p\to 0} \frac{F''[a_p, a_p]}{A''[a_p, a_p]},$$
>
> where a_p is a transverse plane wave of fine momentum p in a fixed colour direction, and $F''[a_p, a_p]$ is the second variation per unit volume (the diagonal Bloch block $\hat{F}''(p, p)$).

Proof. $F'(1) = 0$ (a colour-invariant linear functional on adjoint fields vanishes), so $\mathrm{Hess}_V(F \circ U_n)(1) = D^{\mathsf{T}}F''D$ with $D = DU_n(1)$. For a coarse transverse wave $b = \beta e^{iK\cdot y}$, $Db = \sum_m \varphi_m e^{ik_m\cdot x}$, $k_m = (K + 2\pi m)/L^n$, and by Steps 2–4 of Lemma T (Gaussian transport §4.2): $\varphi_m = \hat{e}(k_m)s_m + O(K^2)$ for $m \neq 0$ and $\varphi_0 = \beta L^{-n} + \hat{e}(k_0)t + O(K)$. Gauge invariance gives $\hat{F}''(k, k')\hat{e}(k') = 0$, and $\hat{F}''(k_0, \cdot) = O(K)$ because a constant field is a pure gauge. Hence $\langle Db, F''Db\rangle = L^{-2n}\beta^\dagger \hat{F}''(k_0, k_0)\beta + O(K^3)$. The same holds for A (with $\hat{A}''$ diagonal). The ratio of the two coarse second moments is therefore the limit of the ratio of the diagonal blocks. □

So C_n is the coefficient of A in F on slowly varying fine backgrounds, for any admissible extension F. The choice of F matters for what can be proved (§8).

1.4 Three extensions.

$$F^{(\lambda)}: \text{ (1.2) itself, on constrained minimisers;}$$

$$F^{(1)}(U) := \log\det(V^*V) - \tfrac{1}{2}\log\det(QQ^*) - \tfrac{1}{2}\log\det{}_{\ker Q}\big(H_U + V_U V_U^*\big); \quad \text{by (1)}$$

$$\tilde{F}(U) := \tfrac{1}{2}\log\det(V^*V) - \tfrac{1}{2}\log\det(QQ^*) - \tfrac{1}{2}\log\det\big(\tilde{H}_U|_{T'}\big), \qquad \tilde{H}_U := P_U H_U P_U,$$

with P_U the orthogonal projection on $(\operatorname{Ran} d_U)^\perp$. All three are gauge invariant and coincide when U is a critical point of A (then $H_U d_U = 0$).

Lemma 1.2. (i) Under the constrained-gap hypothesis, $F^{(1)}$ satisfies the hypotheses of Lemma 1.1. (ii) Under the additional absolute-convergence hypothesis, $\tilde F$ gives the same limit.

Proof of (i). At $U = 1$ the operator $H + VV^*$ on $\ker Q = \operatorname{Ran} V \oplus T'$ is block diagonal and strictly positive: on T' by the hard gap (CMP96 Lemma 2.4, used here as a spectral input), on $\operatorname{Ran} V$ because $K^*\Delta K$ (lattice Laplacian with Dirichlet condition at the block centres) is positive. (On the Bloch orbits used in , $n = 1$, $L = 3$, the lowest eigenvalue of $H + VV^*$ on $\ker Q$ is 0.30, and that of $\tilde H$ on T' is 1.26.) So the second-variation kernel of $F^{(1)}$ decays exponentially (at a rate that depends on n) and its symbol is analytic. On constrained minimisers $F^{(1)} - F^{(\lambda)}$ is a sum of terms each containing the multiplier λ_V or the current $J = A'(U) = Q^*\lambda_V$. At second order in V: the terms linear in the second-order part of λ vanish by colour invariance; the rest is a bilinear form $B(\lambda_1, b)$ with $\lambda_1 = H_c b = O(K^2)b$. B has an analytic symbol, and $B(\lambda, d'c) = 0$ for all c by gauge covariance (a constant abelian background is a gauge transform of 1, which does not rotate a Cartan-valued λ). An analytic symbol annihilating $\hat e'(K)$ for all K vanishes at $K = 0$. So $B(\lambda_1, b) = O(K^3)$. □

On (ii). $\tilde F - F^{(1)}$ is again proportional to the current, but its kernels contain one massless factor Δ^{-1}, so the symbol of the bilinear form is continuous at 0 (each loop integral converges absolutely at $p = 0$, using $Ed^*G_{T'}dE^* = O(\theta^2)$ at small coarse momentum) but not analytic. Continuity and the same gauge argument give the same limit. This extension requires absolute convergence of the massless kernel integrals uniformly enough to pass to the continuous symbol. That estimate is an additional input for $\tilde F$; the fixed-depth argument above applies directly to $F^{(1)}$.

2. Sector decomposition (abelian background)

Take the background $U_b = e^{ia_b T}$, T in a Cartan subalgebra, $\operatorname{Tr} T^2 = \frac{1}{2}$, $T = \operatorname{diag}(t_1, \dots, t_N)$. Then $A = \sum_p F_p^2/(4N) + O(a^4)$, $F_p = (da)_p$. Fluctuations split into root components $w^{ij}E_{ij} + \overline{w^{ij}}E_{ji}$ (charge $q_{ij} = t_i - t_j$) and Cartan components.

Lemma 2.1. For the plaquette with bonds $1, 2, 3, 4$ (signs $\sigma = (+, +, -, -)$), the quadratic part of $1 - \frac{1}{N}\operatorname{Re}\operatorname{Tr} U_p'$ on the (i, j) component is

$$\frac{1}{N}\Big\{\tfrac{1}{2}|Y|^2\big(\cos F_p t_i + \cos F_p t_j\big) - \big(\sin F_p t_i - \sin F_p t_j\big)\sum_{a<b}\sigma_a\sigma_b \operatorname{Im}(x_a\bar x_b)\Big\},$$

$x_1 = w_1,\ x_2 = e^{iqa_1}w_2,\ x_3 = e^{iq(a_1+a_2-a_3)}w_3,\ x_4 = e^{iqF_p}w_4,\ Y = \sum_a \sigma_a x_a.$

Proof. $U_p' = e^{iX_1}e^{iX_2}e^{-iX_3}e^{-iX_4}U_p$ with X_a the transported fluctuations; the second-order term is $-\frac{1}{2}(\sum\sigma_a X_a)^2 - \frac{1}{2}\sum_{a<b}\sigma_a\sigma_b[X_a, X_b]$; take the trace against e^{iF_pT}. □

To second order in a: $\cos Ft_i + \cos Ft_j = 2 - \frac{1}{2}F^2(t_i^2 + t_j^2)$, $\sin Ft_i - \sin Ft_j = qF$. So, after normalising the form to $\|\operatorname{curl} w\|^2$ at $a = 0$, every component sees (α) a charge-q covariant structure depending on qa only (also through Q_U and d_U), and (β) a relative weight $1 - \varpi F_p^2$ on its plaquette term, $\varpi = \frac{1}{4}(t_i^2 + t_j^2)$ per real root direction and $\frac{1}{2}\langle X, T^2X\rangle$ per Cartan direction. Summing: $\sum_{i<j} q_{ij}^2 = N/2$ and $\sum\varpi = (N^2 - 1)/(4N)$.

Proposition 2.2 deduction, given Lemma 1.1.

$$C_n = 2N^2\kappa_n + \frac{N^2 - 1}{2}E_n, \qquad E_n = \tfrac{1}{2}\big(1 - L^{-4n}\big),$$

where κ_n is defined by: the one-loop log-density of one complex vector field of unit charge, with the weights (β) dropped, is $\kappa_n \sum_p F_p^2$ on slowly varying a.

Proof. The contribution (α) of a pair $\{i,j\}$ is $q_{ij}^2\kappa_n\sum F_p^2$; the Cartan components are not charged and their average is linear, so they contribute to (β) only. A weight change $-\varpi F_p^2|Y_p|^2$ on a real direction shifts the log-density by $+\frac{1}{2}\varpi F_p^2\langle|Y_p|^2\rangle$, the expectation being in the constrained flat Gaussian. Since that Gaussian is $(H_0|_{T'})^{-1}$ and $H_0 = \sum_p \operatorname{curl}_p^*\operatorname{curl}_p$, $\sum_p\langle|Y_p|^2\rangle = \dim T'$: per fine site $4-4L^{-4n}$ (kernel of Q) minus $1-L^{-4n}$ (internal gauge), over 6 plaquettes. So $E_n = \langle|Y_p|^2\rangle = \frac{1}{2}(1-L^{-4n})$ exactly. Total coefficient of $\sum F_p^2$: $\frac{N}{2}\kappa_n + \frac{N^2-1}{8N}E_n$; multiply by $4N$. □ (, `tadpoleE`: $E_1 = 40/81$, $E_2 = 0.4999238 = \frac{1}{2}(1-3^{-8})$ to 15 digits.)

The tadpole term converges geometrically and is harmless for (G-val). From here on only κ_n matters, and the problem is that of a single complex charged vector field: operators d_a, H_a, Q_a depending on a real abelian bond field a, with (complex normalisation)

$$\tilde F_C = \log\det(V^*V) - \log\det(QQ^*) - \log\det(\tilde H|_{T'}), \qquad F_C^{(1)} = 2\log\det(V^*V) - \log\det(QQ^*) - \log\det{}_{\ker Q}(H+VV^*).$$

The charged average. For CMP109 (0.12) with the mean (0.10) (pp. 253–254), at the background e^{iaT}, with θ_Γ the background phase of a contour, $\bar\theta$ the mean over $G(y,x)$, $\delta_\Gamma = \theta_\Gamma - \bar\theta$, ω_Γ the transported sum of w along Γ, $h_-(z) = z/(e^z-1)$, $r(z) = (e^z-1)/z$:

$$\beta_{y,x} = \frac{\sum_\Gamma h_-(i\delta_\Gamma)\,\omega_\Gamma}{\sum_\Gamma h_-(-i\delta_\Gamma)}, \qquad \Omega_x = \beta_{y,x} + e^{i\bar\theta_{y,x}}\omega_{[x,x']} - e^{i(\bar\theta_{y,x}+a_{[x,x']}-\bar\theta_{y',x'})}\beta_{y',x'} - e^{i\varphi_x}\omega_c,$$

$$(Q_a w)(c) = r(i\bar\varphi)\,L^{-4}\sum_{x\in B(y)} h_-(i\varphi_x)\,\Omega_x + e^{i\bar\varphi}\omega_c, \qquad a_c' = \bar\varphi + a(c), \tag{2.1}$$

3. Exact blocked formulas

Write $\Delta = d^*d$, $M = H + dd^*$, $\tilde M = \tilde H + dd^*$, and for coarse objects

$$\mathcal{C} = QM^{-1}Q^*, \quad \tilde{\mathcal{C}} = Q\tilde M^{-1}Q^*, \quad G_E = E\Delta^{-1}E^*, \quad W = Ed^*M^{-1}dE^*, \quad Y = QM^{-1}dE^*.$$

Proposition 3.1. For every fine background (no critical-point assumption), as identities between determinants of finite matrices satisfying (1.1):

$$F_C^{(1)} = 2\log\det\Delta - \log\det M + 2\log\det G_E - \log\det\mathcal{C} - \log\det\bigl(1-W+Y^*\mathcal{C}^{-1}Y\bigr), \tag{3.1}$$

$$\tilde F_C = 2\log\det\Delta - \log\det\tilde M - \log\det\tilde{\mathcal{C}} - \log\det\bigl(d'^*\tilde{\mathcal{C}}^{-1}d'\bigr). \tag{3.2}$$

Proof. (3.1): $\det(V^*V) = \det(K^*\Delta K) = \det\Delta\det G_E$ (Schur, $EE^* = 1$). $H + VV^* = M - (Ed^*)^*(Ed^*)$ because $KK^* = 1 - E^*E$. On $\ker Q$: $\det_{\ker Q}(M - B^*B) = \det_{\ker Q} M\cdot\det(1 - BG_{\ker Q}B^*)$ with $B = Ed^*$, $G_{\ker Q} = M^{-1} - M^{-1}Q^*\mathcal{C}^{-1}QM^{-1}$, and $\det_{\ker Q} M = \det M\det\mathcal{C}/\det(QQ^*)$. The factor $\det(QQ^*)$ cancels. (3.2): $\tilde H d = 0$ exactly, so $\tilde M d = d\Delta$ and $\tilde M$ is block diagonal on $\operatorname{Ran} d \oplus (\operatorname{Ran} d)^\perp$. Then $W = 1$, $Y = d'G_E$ by (1.1), and the chain of (3.1) with $H \to \tilde H$ gives $1 - W + Y^*\tilde{\mathcal{C}}^{-1}Y = G_E d'^*\tilde{\mathcal{C}}^{-1}d'G_E$; it remains to check $\det_{\ker Q}(\tilde H + VV^*) = \det(V^*V)\det(\tilde H|_{T'})$, which holds because $\tilde H V = 0$. □

Interpretation. (3.2) says: one-loop density = (fine one-loop functional) / (coarse one-loop functional), the coarse theory having covariance $\tilde{\mathcal{C}}$ and "ghost" operator $d'^*\tilde{\mathcal{C}}^{-1}d'$. The fine factor $2\log\det\Delta - \log\det\tilde M$ is the Wilson theory alone.

Second variations. With $a = \epsilon(Ae^{ip\cdot x} + \bar A e^{-ip\cdot x})$, an operator O is carried through order ϵ^2 on the three Bloch fibres $\kappa, \kappa\pm p$ ("jets": zeroth-order blocks O_0 on the three fibres, the four first-order blocks between neighbours, the second-order block O_S on the centre). Products, inverses and adjoints are closed at this order. The ϵ^2 coefficient of $\log\det O$ per fibre is

$$\ell_2(O)(\kappa) = \operatorname{tr}\bigl[O_0^{-1}O_S\bigr] - \frac{1}{2}\sum_{\pm}\operatorname{tr}\bigl[O_0(\kappa)^{-1}O_{\mp}(\kappa\pm p)\,O_0(\kappa\pm p)^{-1}O_{\pm}(\kappa)\bigr], \tag{3.3}$$

and (3.1), (3.2) hold for the sums of ℓ_2 over closed orbits $\{\kappa + jp\}$, hence for the integrals over the Brillouin zone. Fine operators are diagonal in the alias index m ($k_m = (\theta + 2\pi m)/L^n$); a coarse operator is an alias average, e.g. $\tilde{\mathcal{C}}(\theta) = L^{-4n}\sum_m Q(k_m)\tilde M(k_m)^{-1}Q(k_m)^*$, computed level by level.

4. The decomposition $\kappa_n = \sum_{k<n} s_2(k) + \mathcal{B}_n$

There are two versions, one for each blocked formula. Let

$$f^{\mathrm{F}}(k;p) := 2\ell_2(\Delta) - \ell_2(M), \qquad \tilde f(k;p) := 2\ell_2(\Delta) - \ell_2(\tilde M)$$

at fine momentum k and fine background momentum p (transverse polarisation): the Wilson one-loop density in the local Feynman form and in the projected form. By (3.1), (3.2), per unit fine volume and with $p_0 = pL^{-n}$,

$$F_C^{(1)\prime\prime} = \int_{[-\pi,\pi]^4} \frac{d^4k}{(2\pi)^4} f^{\mathrm{F}}(k;p_0) + L^{-4n}\int \frac{d^4\theta}{(2\pi)^4}\big[2\ell_2(G_E) - \ell_2(\mathcal{C}) - \ell_2(1 - W + Y^*\mathcal{C}^{-1}Y)\big],$$

$$\tilde F_C'' = \int_{[-\pi,\pi]^4} \frac{d^4k}{(2\pi)^4} \tilde f(k;p_0) - L^{-4n}\int \frac{d^4\theta}{(2\pi)^4}\big[\ell_2(\tilde{\mathcal{C}}) + \ell_2(d'^*\tilde{\mathcal{C}}^{-1}d')\big].$$

Split the fine integral at the cell $c_n = \{|k_i| \le \pi L^{-n}\}$ (the zero alias), and $c_0 \setminus c_n$ into shells $c_k \setminus c_{k+1}$. With $\mathsf{N}(p_0) = 2\sum_{\mu<\nu}|\hat p_\mu A_\nu - \hat p_\nu A_\mu|^2$ the second variation of $\sum F_p^2$ per unit volume ($\hat p_\mu = e^{ip_{0,\mu}} - 1$), define for $f = f^{\mathrm{F}}$ or $\tilde f$

$$\sigma_k(p_0) := \frac{1}{\mathsf{N}(p_0)}\int_{c_k\setminus c_{k+1}} \frac{d^4k}{(2\pi)^4} f(k;p_0), \qquad \beta_n(p) := \frac{1}{\mathsf{N}(p_0)}\Big\{\int_{c_n} \frac{d^4k}{(2\pi)^4} f(k;p_0) + [\text{coarse integral}]\Big\}.$$

For k in a shell and $|p_0|$ small compared with L^{-k-1}, $f(k;\cdot)$ is real-analytic and even, so $\sigma_k(p_0) = s_0(k)|p_0|^{-2} + s_2(k) + O(p_0^2)$. The $|p_0|^{-2}$ terms are the pieces that a sharp momentum cut does not keep gauge invariant; they cancel in the sum because the total is gauge invariant (numerically $a_0 + b_0 = O(10^{-6})$, §7). Set

$$s_2(k) := \big[\sigma_k\big]_{p^0}, \qquad \mathcal{B}_n := \lim_{p\to 0}\Big(\beta_n(p) + |p_0|^{-2}\sum_{k<n} s_0(k)\Big) = \kappa_n - \sum_{k<n} s_2(k), \tag{4.1}$$

with superscripts ^{F} or $\tilde{}$ according to the version. For the Feynman version (4.1) rests on Lemma 1.1 and Lemma 1.2(i) only; for the projected version also on Lemma 1.2(ii). (4.1) is a definition plus the reduction; its content is that $s_2(k)$ involves the Wilson action only, and that $\mathcal{B}_n$ lives at the top scale: in units of the top lattice it is an integral over $\theta \in [-\pi,\pi]^4$ of the zero-alias Wilson density plus the coarse density.

5. The logarithm: $s_2(k) \to \dfrac{11}{24\pi^2}\log L$

Theorem 5.1 (a; step (1) of the proof is argued without explicit constants on compact momentum shells). For both versions, $s_2(k) = \dfrac{11}{24\pi^2}\log L + O(L^{-k})$. In particular $\sum_{k<n} s_2(k) = \dfrac{11}{24\pi^2}\, n\log L + O(1)$.

Proof. (1) *Scaling.* Each of $f^{\mathrm{F}}, \tilde f$ (written f) is a fixed rational function of the jets of d_a and H_a, which are trigonometric polynomials in k, p with denominators $\Delta_0(\kappa) = \sum_\mu 4\sin^2(\kappa_\mu/2)$ at the three fibres. Under $k \to tk,\ p \to tp$ the jets converge, after division by their homogeneity factors, to the jets of the continuum operators

$$d^{\mathrm{c}}\lambda = -D\lambda, \qquad (H^{\mathrm{c}}w)_\nu = -D_\mu(D_\mu w_\nu - D_\nu w_\mu) - iF_{\nu\mu}w_\mu, \qquad D = \partial + ia, \quad [D_\mu, D_\nu] = iF_{\mu\nu},$$

uniformly with error $O(t)$ for (k,p) in compact sets on which the three fibres stay away from 0, together with their p-derivatives (analyticity in a complex neighbourhood and Cauchy). [The sign of the last term of H^{c} is fixed by the Ward identity $H_a d_a = O(\partial F)$: T3 shows $\|H_a d_a\| = O(p^2)$ for the lattice form of Lemma 2.1 and $O(p)$ if the magnetic term is dropped or reversed.] Since f is of second order in a, $f(tk;tp) = t^{-2}[f^{\mathrm{c}}(k;p) + O(t)]$ and for its p^2-coefficient $f_2(tk) = t^{-4}[f_2^{\mathrm{c}}(k) + O(t)]$ with f_2^{c} homogeneous of degree -4. Hence, with $k = L^{-k}\kappa$,

$$s_2(k) = \frac{1}{\mathsf{N}^{(2)}}\int_{c_0\setminus c_1} \frac{d^4\kappa}{(2\pi)^4} f_2^{\mathrm{c}}(\kappa) + O(L^{-k}) = \frac{\log L}{(2\pi)^4\,\mathsf{N}^{(2)}}\int_{S^3} f_2^{\mathrm{c}}\, d\Omega + O(L^{-k}),$$

the last step because the shell between two homothetic star-shaped surfaces of ratio L contributes $\int_{r(\Omega)/L}^{r(\Omega)} dr/r = \log L$ in every direction. $\mathsf{N}^{(2)} = 2p^2|A|^2$ for transverse A.

(2) *Feynman form.* Take $f = f^{\mathrm{F}}$. In the continuum $M^{\mathrm{c}} = -D^2 + \mathsf{E}$, $\mathsf{E}_{\nu\mu} = -2iF_{\nu\mu}$ (Hermitian), so $2\ell_2(\Delta) - \ell_2(M) = -2\ell_2(-D^2) + \mathrm{tr}[G\mathsf{E}_+ G\mathsf{E}_-]$, $G = 1/k^2$.

- *Scalar.* $\ell_2(-D^2) = 2|A|^2/k^2 - 4|k\cdot A|^2/(k^2(k+p)^2)$ (transverse A, symmetrised in $\pm p$). Its p^2-coefficient is $-4|k\cdot A|^2 k^{-4}[-p^2k^{-2} + 4(k\cdot p)^2 k^{-4}]$. With $\langle k_\mu k_\nu\rangle = \frac{k^2}{4}\delta_{\mu\nu}$ and $\langle k_\mu k_\nu k_\rho k_\sigma\rangle = \frac{k^4}{24}(\delta\delta+\delta\delta+\delta\delta)$ on S^3: angular mean $= \frac{1}{3}p^2|A|^2k^{-4} = \frac{1}{6}\mathsf{N}^{(2)}k^{-4}$.
- *Magnetic.* $\mathsf{E}_\pm = -2iF_\pm$, $F_{+\mu\nu} = i(p_\mu A_\nu - p_\nu A_\mu)$, $\mathsf{E}_- = \mathsf{E}_+^\dagger$; $\mathrm{tr}[\mathsf{E}_+\mathsf{E}_-] = 8\sum_{\mu<\nu}|p_\mu A_\nu - p_\nu A_\mu|^2 = 4\mathsf{N}^{(2)}$; the p^0 coefficient of $1/(k^2(k+p)^2)$ is k^{-4}.

So $\langle f_2^{\mathrm{c,Feyn}}\rangle = (4 - \frac{2}{6})\mathsf{N}^{(2)}k^{-4}$, and with $|S^3| = 2\pi^2$: $\frac{2\pi^2}{(2\pi)^4}\cdot\frac{11}{3} = \frac{11}{24\pi^2}$.

(3) *Projected form.* $-\log\det M + \log\det\tilde{M} = -\mathrm{Tr}\log(1+\tilde{M}^{-1}(H-\tilde{H})) = -\mathrm{Tr}[\Delta^{-2}\mathfrak{m}] + O(J^2)$, $\mathfrak{m} := d^*Hd$. In the continuum $H^{\mathrm{c}}D\lambda = -iJ\lambda$ with $J_\nu = \partial_\mu F_{\mu\nu}$, so $\mathfrak{m} = iJ\cdot D$, and $J_+ = -p^2A$ (transverse A) is already of order p^2. The p^2-coefficient of $-\mathrm{Tr}[\Delta^{-2}\mathfrak{m}]$ has integrand $-\{-2\,\mathrm{Re}(J_+\cdot\bar{A})k^{-4} + 8\,\mathrm{Re}[(k\cdot\bar{A})(k\cdot J_+)]k^{-6}\}$, whose angular mean is 0. The $O(J^2)$ terms are $O(p^4)$. So $\tilde{M}$ and M give the same $\langle f_2^{\mathrm{c}}\rangle$. □

The scaling step uses the expansion of the lattice jets on compact momentum shells. It requires convergence of the second momentum derivatives there; it gives no uniform bound on the transverse remainder.

6. The remainder $\mathcal{B}_n$ and hypothesis (U)

(U). $\sup_n|\mathcal{B}_n^{\mathrm{F}}| < \infty$. (Equivalently, given Lemma 1.2(ii): $\sup_n|\tilde{\mathcal{B}}_n| < \infty$, since $\mathcal{B}_n^{\mathrm{F}} - \tilde{\mathcal{B}}_n = \sum_{k<n}(\tilde{s}_2(k) - s_2^{\mathrm{F}}(k))$ converges by Theorem 5.1.)

Theorem 6.1 conditional on Theorem S for the composite constrained Gaussian. For every n

$$C_n - d_B n = 2N^2\Big[\sum_{k<n}\Big(s_2^{\mathrm{F}}(k) - \frac{11\log L}{24\pi^2}\Big) + \mathcal{B}_n^{\mathrm{F}}\Big] + \frac{N^2-1}{4}\big(1 - L^{-4n}\big),$$

the sum converges absolutely, and (G-val) holds if and only if (U) holds, with $B_G \le 2N^2\big(\sum_k|s_2^{\mathrm{F}}(k) - \frac{11\log L}{24\pi^2}| + \sup_n|\mathcal{B}_n^{\mathrm{F}}|\big) + \frac{N^2-1}{4}$. The same formula holds with $\tilde{s}_2, \tilde{\mathcal{B}}_n$ if Lemma 1.2(ii) is granted.

Proof. Lemma 1.1 with Lemma 1.2(i), Proposition 2.2, (3.1), (4.1), Theorem 5.1. □

So the coefficient of n in C_n is d_B, with its sign, if and only if $\mathcal{B}_n^{\mathrm{F}} = o(n)$; and (G-val) is exactly the boundedness of $\mathcal{B}_n^{\mathrm{F}}$. The projected version is described below.

A gauge-invariant form of the coarse factor.

Lemma 6.2. For any ξ (coarse sites → coarse bonds) and any τ (coarse sites → coarse sites), the coarse factor $\log\det\tilde{\mathcal{C}} + \log\det(d'^*\tilde{\mathcal{C}}^{-1}d')$ of (3.2) is unchanged under $\tilde{\mathcal{C}} \to \tilde{\mathcal{C}} + d'\xi^* + \xi d'^* + d'\tau d'^*$. Consequently, with P' the orthogonal projection on $(\mathrm{Ran}\,d')^\perp$,

$$\tilde{F}_C = \big[\log\det\Delta - \log\det{}'(\tilde{H}|_{(\mathrm{Ran}\,d)^\perp})\big] - \big[\log\det(d'^*d') + \log\det{}'(P'\tilde{\mathcal{C}}P')\big].$$

Proof. $\tilde{\mathcal{C}} + d'\xi^* + \xi d'^* + d'\tau d'^* = S\tilde{\mathcal{C}}S^* + d'\tau' d'^*$ with $S = 1 + d'\eta^*$, $\eta = \tilde{\mathcal{C}}^{-1}\xi$. For $S\tilde{\mathcal{C}}S^*$: det picks up $|\det(1+\eta^*d')|^2$ and $d'^*(S\tilde{\mathcal{C}}S^*)^{-1}d' = (1+\eta^*d')^{-*}(d'^*\tilde{\mathcal{C}}^{-1}d')(1+\eta^*d')^{-1}$ loses the same factor. For $\mathcal{C} + d'\tau d'^*$: det picks up $\det(1+\tau\mathcal{G})$, $\mathcal{G} = d'^*\mathcal{C}^{-1}d'$, and by Woodbury $d'^*(\mathcal{C} + d'\tau d'^*)^{-1}d' = \mathcal{G}(1+\tau\mathcal{G})^{-1}$. Choosing ξ, τ to remove the mixed blocks and normalise the longitudinal block gives the second statement; the fine bracket is $2\log\det\Delta - \log\det\tilde{M}$ because $\tilde{M} = \tilde{H} \oplus dd^*$. □

So the one-loop density is exactly (fine orbit volume × fine transverse Gaussian) / (coarse orbit volume × coarse transverse Gaussian), and only the transverse block of the blocked covariance enters. This matters: the full regular

part of $\tilde{\mathcal{C}}$ is *not* uniformly bounded. Numerically, at the flat background, $\|R_n\|$ grows linearly in n (e.g. 0.39, 0.66, 0.89 at the zone corner for $n = 1, 2, 3$), while $\|P' R_n P'\|$ stays bounded and converges (0.0091, 0.0053, 0.0049 at the same point); see. The growth is the contour part of the average: $Q = S + d'\gamma$ with $|\gamma(k)|^2 \sim k^{-2}$ between the block scale and the lattice scale, which gives $\int d^4k/k^4$ in $d'\gamma\tilde{M}^{-1}\gamma^* d'^*$. By Lemma 6.2 it is a pure coarse gauge term and drops out exactly.

Why (U) should hold additional assumption. Write $\tilde{\mathcal{C}} = \mathcal{C}_0 + R$, $\mathcal{C}_0 = L^{-4n} Q_0 \tilde{M}_0^{-1} Q_0^*$ the zero-alias term and R the sum over aliases $m \neq 0$. By (1.1), $Q(k_m) d(k_m) = d'(\theta)$ for every alias. If R is dropped, then $d'^* \mathcal{C}_0^{-1} d' = L^{4n} d_0^* \tilde{M}_0 d_0 = L^{4n} \Delta_0^2$ and

$$\big[2\ell_2(\Delta_0) - \ell_2(\tilde{M}_0)\big] - \ell_2(\mathcal{C}_0) - \ell_2(d'^* \mathcal{C}_0^{-1} d') = -\ell_2(Q_0) - \ell_2(Q_0^*)$$

up to terms that are second differences $g(\theta + p) + g(\theta - p) - 2g(\theta)$, which integrate to boundary terms on the cell. The infrared-singular Wilson density of the zero alias cancels identically against the coarse density; what remains is the ϵ^2 variation of the zero-alias symbol of the average, a regular object. With R restored, the correction is a function of $L^{4n} Q_0^{-1} R Q_0^{-*}$ which tends to 0 at $\theta \to 0$. So $\mathcal{B}_n$ is bounded as soon as, in units of the top lattice, uniformly in n, and modulo the coarse gauge terms of Lemma 6.2,

- (U1) the zero-alias symbol $Q_{n,0}(\theta)$ of the composite average, its inverse, and its first two background jets are bounded on the cell;
- (U2) the regular part $R_n(\theta) = L^{-4n} \sum_{m \neq 0} Q(k_m) \tilde{M}(k_m)^{-1} Q(k_m)^*$ and its first two background jets are bounded.

What (U) is not. It is not a statement about large fields, histories, cutoffs or volumes: it concerns the Gaussian at the flat background, in infinite volume, for one complex scalar-charged vector field. It is the precise form of the bound "uniform $O(1)$", stripped of the logarithm.

Homogeneous curvature projection

Conditional application. Use the stated source, support and fixed-chart hypotheses. Any application of the complete C05 boundary estimates or the C06/C08 law to all original Wilson histories assumes spatial and native-domain control of the iterated reconstruction with the earlier innovations fixed. The actual composed response required for that application remains open.

Assumptions and conclusions

A positive carrier norm with a paid physical diameter suffices for the local curvature projection on a raw-free, gauge-invariant homogeneous or periodic activity family. A full physical interior-volume weight is not needed for that linear readout. The projection costs the expected inverse native radius squared; the old-grid root density is canceled by four physical spacing powers. This is proved below.

This does not yet apply the projection to the complete C05 boundary bank as if that bank were the old-reset difference of the bulk E action. These are different source objects. The actual all-depth bulk E-feedback estimate required in the compensating-counterterm construction, target (6), is a separate input to the projection. The homogeneous feedback estimate is developed in Homogeneous feedback and the causal Wilson recursion under its stated source assumptions. The conditional C05 estimate takes the ordinary E bank as admitted forcing and does not preserve a reset-dependence mark through its homogeneous recursion.

The AF comparisons used below are conditional comparisons on an already admitted signed prefix. They do not prove that the physical Wilson trajectory satisfies those hypotheses.

1. The source quantity being projected

CMP119, printed pp.278–279, constructs the interior $E^{(k+1)}$ terms from (3.37)–(3.47). Terms whose localization domains lie in the regular interior agree with their whole-lattice counterparts. The remaining terms are assigned to $B^{(k+1)}$. The coupling subtraction is then taken from the vacuum-subtracted regular $E^{(k+1)}$, not from an arbitrary sum of boundary scalars on a retained history.

The quadratic curvature coefficient is the gauge-invariant second-order marginal obtained through (3.49)–(3.56), or equivalently by the compatible small-flux readout in the retained coupling analysis. Mature reset contributions may enter the regular E-producing operation. Once they have done so, later ordinary bulk steps propagate them through the same E recursion as all other regular fields.

Thus the relevant reset error is a difference such as

$$\Delta H_{k+1} = H_{k+1}(E^{\mathrm{actual}}_{\le k}, R_{\mathrm{mat}}; Y_{\le k}) - H_{k+1}(E^{0}_{\le k}, 0; Y_{\le k}), \tag{CP.1}$$

with the prescribed comparison and its actual own vacuum/Wilson subtractions specified. It is not just the current mature-R insertion while keeping the incoming E identical. The direct latter difference is already bounded by (D) in the compensating-counterterm construction.

The source's full E bounds and C05's full boundary bounds separately control the two sides of various identities. Such control does not give a small bound for the history-dependent difference (CP.1).

2. A linear projection lemma in the weaker diameter norm

Here is a sufficient readout theorem that separates its genuine hypotheses from the reset-feedback estimate.

Fix a birth scale b with physical mesh h_b, a later reference scale k with $h_k \ge h_b$, and $r_b > 0$, the dimensionless native analytic radius at scale b. Work with the source local dimensionless link coordinates and their fixed strong derivative/current companions. Let

$$F_b(U,J) = \sum_{\Gamma \in \mathfrak{G}_b} F_{b,\Gamma}(U,J) \tag{CP.2}$$

be a raw-free gauge-invariant family with homogeneous or a fixed compatible periodic covariance. Each activity has an actual dependency footprint S_Γ. It may contain bounded declared cores, but its whole physical diameter is retained:

$$D_\Gamma = 1 + \frac{\operatorname{diam} S_\Gamma}{h_b}. \tag{CP.3}$$

The fixed M, field stencil and local gauge collar can be incorporated into D_Γ and fixed constants.

Assume the following.

1. Flat local chart. The activity is holomorphic and bounded by M_Γ on the genuine source native neighborhood of the flat configuration on its dependency footprint and fixed collar. The local affine Cartan curvature curve used for the source readout is admitted there at dimensionless birth flux

$$|f_b| \le \frac{c\, r_b}{D_\Gamma}. \tag{CP.4}$$

   The field and gradient pieces satisfy this by the actual affine Cartan potential; its abelian physical current vanishes in the interior. This condition is a source-chart assertion, not a consequence of boundedness around an arbitrary nonflat retained history.

2. Positive rooted diameter norm. There is a permitted birth-cell anchor for each activity, chosen on an actual exterior cell or a genuine component slot, with

$$\sup_{x \in \mathbb{L}_b} \sum_{\Gamma:\, a(\Gamma)=x} e^{\sigma D_\Gamma} M_\Gamma \le A_b. \tag{CP.5}$$

   Rooting at every actual incident birth cell is stronger and also suffices. A component slot is resolved by its actual birth cell. Arbitrary hidden fine labels are not counted as new anchors.

3. Source local thermodynamic compatibility. The finite-volume coefficients agree with the limiting local coefficients away from the boundary/winding sector, with the same positive bounds. Equivalently, use the infinite-volume local activity family on which the stated coupling projector is defined. This is the ordinary localization/whole-lattice identification used at CMP119 pp.278–279. No projector on one finite torus is silently substituted for it.

Let $\mathcal{P}_k$ be any fixed scalar linear combination of the constant-Cartan-curvature Hessian components per current-scale volume, with the Wilson normalization constants retained. Then

$$\boxed{|\mathcal{P}_k F_b| \le C_\sigma\, r_b^{-2} A_b.} \tag{CP.6}$$

The constant depends on the fixed group, tensor projection, source block parameters and σ, but not on b, k, the number of activities, the torus volume, or the volumes of the declared cores.

Proof

Let φ be the dimensionless curvature at scale k. The same physical curvature has birth plaquette flux

$$f_b = \left(\frac{h_b}{h_k}\right)^2 \varphi. \tag{CP.7}$$

Choose the local Cartan gauge centered in S_Γ. The dimensionless potential on that footprint is bounded by $C|f_b|D_\Gamma$; its first difference is $O(|f_b|)$, and the abelian current is zero. By (CP.4), the activity is holomorphic on the disc

$$|\varphi| \le c r_b \left(\frac{h_k}{h_b}\right)^2 D_\Gamma^{-1}. \tag{CP.8}$$

Second-order Cauchy gives

$$\left|\frac{1}{2} D_\varphi^2 F_{b,\Gamma}(U_\varphi, J_\varphi)\big|_{\varphi=0}\right| \le C r_b^{-2} \left(\frac{h_b}{h_k}\right)^4 D_\Gamma^2 M_\Gamma. \tag{CP.9}$$

The scalar coupling combination changes only a fixed constant.

For a compatible physical torus let V_b, V_k be its birth/current cell counts. They satisfy

$$\frac{V_b}{V_k} = \left(\frac{h_k}{h_b}\right)^4. \tag{CP.10}$$

Since $\sup_{D\geq 1} D^2 e^{-\sigma D} < \infty$, (CP.5) bounds the sum of $D_\Gamma^2 M_\Gamma$ by $C_\sigma A_b V_b$. Divide (CP.9) by V_k and use (CP.10). This proves (CP.6) before the source thermodynamic local-coefficient limit. The same bound and spare diameter exponential give dominated convergence in that limit; winding contributions are removed with the source localization limit, not with an unproved exchange of flux quantization and derivatives.

Gauge invariance and the raw-free argument list are essential to the use of a different centered local gauge for each activity. The estimate has not filled a core interior with a new activity tree. Only its physical diameter appears in the Cauchy radius. □

Periodic coefficients and divided tensor projections

If the coefficient has a compatible birth-grid period, average the local Hessian over one period before taking the scalar contraction. The same proof applies because (CP.5) is uniform in the root and the average has total weight one. There is no extra period-volume factor.

This statement does not itself prove the fractional periodic cancellation of the compensating-counterterm construction §2. That cancellation concerns the marginal tensor left after its correct subtraction inside a localized transfer. It remains necessary where the iteration requires a small irrelevant gain rather than a bounded final readout.

3. Several native grades and creation times

Suppose a sum has source families at grades m, each satisfying the hypotheses of the lemma with radius r_m, and

$$\sum_m r_m^{-2} A_m < \infty. \tag{CP.11}$$

Then the projected series is normally convergent and

$$\left| \mathcal{P}_k \sum_m F_m \right| \leq C \sum_m r_m^{-2} A_m. \tag{CP.12}$$

The cancellation of root density is grade-by-grade. There is no $L^{4(k-m)}$ left in (CP.12).

A new prefix contribution can have a source grade m different from its chronological creation event r. It must retain that grade in this readout. If the actual construction gives $m \in [r - N_r - C, r + C]$, with N_r a fixed logarithmic power, the admitted signed-prefix upper slope implies

$$Y_m \leq Y_r + CN_r + B \leq C' Y_r, \qquad r_m^{-2} \leq C r_r^{-2} (1 + \ell_r)^{C'} \tag{CP.13}$$

at one sufficiently large lower ceiling. Here $Y_j = g_j^{-2}$ and the radius exponents are fixed. One can therefore sum the finitely many true prefix grades at a fixed creation event at the cost of a fixed logarithmic power. This is not permission to replace a fine root by a coarse one before (CP.9)–(CP.10).

The new C05 creation profiles have the form

$$f_r \leq C Y_r^u (1 + \log Y_r)^v \exp[-c(\log Y_r)^s], \qquad s > 1, \tag{CP.14}$$

with fixed exponents, after its finite fractional powers and norm reserves. If a raw-free homogeneous subfamily satisfies (CP.4)–(CP.5) with total grade contribution at event r bounded by f_r, then (CP.12) and (CP.13) give its curvature readout with only fixed inverse-coupling/logarithmic losses.

On an already admitted prefix, the lower AF comparison

$$Y_r \geq c Y_k + a(k - r) \tag{CP.15}$$

makes the readout summable. For every $N > 0$, the function $Y^{u+1}(1 + \log Y)^{v+C'} e^{-c(\log Y)^s}$ is bounded by $C_N Y^{-N-2}$ for large Y. Hence

$$\sum_{r \le k} r_r^{-2}(1 + \ell_r)^{C'} f_r \le C_N \sum_{j \ge 0} (cY_k + aj)^{-N-2} \le C'_N Y_k^{-N-1}. \tag{CP.16}$$

All exponents are fixed before the ceiling. This is a conditional consequence of (CP.15), not a derivation of the physical AF trajectory. It could be used inside a stopped-prefix bootstrap after the actual marked source estimate has been proved.

4. Why the C05 bank cannot simply be substituted into the lemma

There are three separate issues.

4.1 Different argument and symmetry class

A general C05 activity retains older raw group/adjoint slots and a fixed nonhomogeneous history. Its native analyticity holds on that actual admitted source chart. Simultaneous gauge covariance in the native field and the raw/adjoint arguments does not imply gauge invariance in the native field alone with those arguments held fixed.

The source curvature projector acts on the homogeneous regular E family. Its local Cartan gauge change is legitimate because that family has no retained old raw argument and the whole-lattice covariance is known. For a general boundary activity, neither a flat chart with the same fixed raw history nor the independent local gauge change used in (CP.8) follows from C05. The C05 local-native jet is a theorem at its admitted real center; it is not automatically a small-flux theorem at a different flat center.

Thus (CP.6) is a bounded projection on an explicitly identified subspace. It is not a map from every C05 history scalar to the physical coupling.

4.2 The causal reset label has been lost in the ordinary forcing bank

The new C05 history induction keeps completed/failed relative scalar records and their new responses under exact normalizations. It also invokes the source regular E/R families as ordinary inputs with their admitted native majorants. In that use, E_0 is a forcing budget. The proof does not bound the derivative of E with respect to a marker multiplying every earlier mature reset.

A mature reset can change a homogeneous E birth. At later steps that change has no required surviving large-field owner port. It evolves in the ordinary bulk, so the C05 current-to-large-field collar cannot be assigned to it. The term is not restored to the fresh boundary bank merely because a reset caused it historically.

Consequently C05 does not prove that the difference (CP.1) has a small A_m in (CP.11). Assigning its two individually bounded E_0 banks a difference bound gives only $2E_0$; after the readout this costs Cr_m^{-2}, which is insufficient for the B03 remainder.

4.3 A conditional AF tail is not a physical trajectory proof

Even if one identifies a small raw-free family, (CP.16) uses exactly the admitted two-sided/signed-prefix comparisons. B03 has to prove that the actual Wilson output continues to satisfy a closed quantitative recurrence. The C05 use of these comparisons cannot be run backwards to establish them.

A stopped-prefix argument may use them up to the first possible exit and estimate one provisional output. That is the appropriate logical order for the synthetic-marker comparison in the reset-remainder reduction.

5. The remaining source estimate in an actionable form

Let $H_j(z)$ be the single-complex-marker family defined in the compensating-counterterm construction §4: keep the numerical Y_i prescribed, multiply the actual mature reset inputs by z, and use each synthetic $H_i(z)$ with its own

vacuum and Wilson subtraction. The compensating Wilson term keeps the prescribed principal coupling. At $z = 1$ it agrees with the actual bulk construction.

The exact missing positive/analytic input can be stated in either of two forms.

Native-ball form. For one common marker disc $|z| \le A_k$ and all precursor steps, prove the actual raw E output estimate on an invariant genuine native chart,

$$\sup_x \sum_{\Gamma : a(\Gamma)=x} e^{\sigma D_\Gamma} \sup_{\mathcal{D}_j^+, |z|\le A_k} |H_{j,\Gamma}(z)| \le C E_0. \tag{CP.17}$$

The incoming subtractions, the fixed numerical Gaussian/coupling array, the simultaneous source localizer domain and the cutoff are those of the synthetic construction. A signed summed value bound without this source chart is insufficient.

Marked-difference form. Prove directly that the transported reset-dependent homogeneous family has

$$\sum_m r_m^{-2} A_m^{\mathrm{reset}} \le C Y_k^{-1} \tag{CP.18}$$

or the smaller readout error required by the selected coupling coordinate, with actual local flat charts and covariance. This can be obtained from a marker amplification estimate or an actual signed-prefix sensitivity/renewal theorem.

Once (CP.17) is established on $|z| \le A_k$, the projection lemma and a single marker Cauchy estimate give

$$|\mathcal{P} H_{k+1}(1) - \mathcal{P} H_{k+1}(0)| \le \frac{C r_k^{-2} E_0}{A_k - 1}, \tag{CP.19}$$

with any fixed source/logarithmic projection loss included explicitly. There is only one inverse native-radius loss in this final readout; it is not multiplied at every age. This recovers the useful implication (6)⇒(7) of the reset-remainder reduction with the weaker diameter-norm variant proved above.

The remaining work is therefore not a missing elementary curvature projection bound, and not a demand to charge every fixed core's full interior volume. It is the uniform marked homogeneous E transfer on its actual invariant native chart, including the own Wilson subtractions and prescribed-coupling compensation, or an equivalent direct bound for that same feedback difference. No B03 or B05 completion follows from this note alone.

Shooting at common Gaussian record depths

The identification of the cumulative Gaussian marginal with a linear drift up to a uniformly bounded remainder is a separate hypothesis (GC). An arithmetic consequence of that drift does not prove GC or match a prescribed physical endpoint.

Status. The fixed Gaussian calibration selects cofinal cutoff depths at which the literal terminal source inequality holds simultaneously for every admitted bare shot. There are explicit bare brackets of width $O(\log K)$ at these depths. A jump in the terminal readout from below the chosen target to above it remains possible with the presently proved estimates. B-05 remains open. The target below is the preassigned inverse coupling in the source units of B-03.

Common depths before choosing any bare parameter

Write $b = Y_0 = N\beta_W$, $y = Y_{\text{phys}} > F$, where F is a fixed stopping floor above all source ceilings in inverse-coupling units, with the one-step buffer of Appendix F. In the fixed scheme the uncut Gaussian increments have deterministic prefix $G_j = dj + q_j$, $q_0 = 0$, $|q_j| \le Q$, $d = d_B > 0$. Every causal Wilson prefix with inputs $Y_i \ge F$ has the provisional physical update

$$Y_{i+1} = Y_i - (G_{i+1} - G_i) - e_i, \qquad |e_i| \le A/Y_i.$$

The same G for different bare values is essential here. It comes from the fixed uncut stationary Gaussian, including its gauge/coarea and thermodynamic readout. Fix $F > 2Q$ sufficiently large that $\varepsilon = A/F < d$, set $d_\pm = d \pm \varepsilon$, and take $0 < a < d_-$. Keep the source's fixed L, charts and rounding rule.

Let K range over running maxima of $V_j = G_j - (a+\varepsilon)j$, ties allowed. With $c = d - a - \varepsilon > 0$, $V_j = cj + q_j$. If r is the last record through n, then $0 \ge V_n - V_r \ge c(n-r) - 2Q$. In particular a record occurs in every interval of $1 + \lfloor 2Q/c \rfloor$ consecutive nonnegative indices. At a record, the Gaussian suffix has slope at least $a + \varepsilon$. For every admitted shot to this *same* K,

$$a(K-j) \le Y_j - Y_K = G_K - G_j + \sum_{i=j}^{K-1} e_i \le (d_+ + 2Q)(K-j), \qquad 0 \le j \le K. \tag{1}$$

For the upper bound absorb $2Q$ at positive integer age, and use equality at age zero. The lower bound pays the worst possible signed interacting errors over this suffix. Thus the sharp terminal lower envelope no longer requires trimming after choosing the shooting parameter. At a shot with terminal value in a fixed compact interval above F, (1) also gives Appendix H's period/divisor bound uniformly. One can use $a_K = \ell_0 L^{-K}$, the centered compatible periods of PE.19–24, and the original finite-torus replay. Source envelope and geometry, Appendix H §§3–5

A bracket that handles premature stopping

Define a scalar function $T_K(b)$ for $b \ge F$. Follow the actual source until K or the first provisional $Y_t < F$. In the latter case put $e_i = 0$ for $i \ge t$ in the numerical recursion and take its value at K. This completion only defines the scalar bracket. The pre-exit coefficients and Y_t are the original ones. The completed array still has $|e_i| \le \varepsilon$ everywhere. Equation (1) therefore holds even beyond an exit. In particular $T_K(b) \ge F$ at a record implies $Y_j \ge F$ at every $j \le K$, and this shot and all of its replay are genuine.

Let u be the last admitted input index used before K, with $u \le K - 1$. For $i \le u$, the corrected drift gives $Y_i \ge F - 2Q + d_-(u-i)$. With $h = F - 2Q > 0$, put

$$M_K = \frac{A}{h} + \frac{A}{d_-}\log(1 + d_-K/h).$$

Summing the decreasing reciprocal majorant and bounding its tail by an integral gives $|\sum_{i<K} e_i| \le M_K$. Consequently for every sufficiently large record and any fixed $s > 0$, the explicit bare values

$$b_\pm = G_K + y \pm (M_K + s)$$

satisfy $b_- \ge F$, $T_K(b_-) \le y - s$, $T_K(b_+) \ge y + s$. In particular the upper shot is always physical. The bracket is centered at $dK + q_K + y$ and has width $2(M_K + s)$ in native bare units.

One can also obtain genuine lower and upper shots on common cofinal record depths when y has extra weak reserve. Take $Z_* > F + Q + d_+$, $J = \lfloor 2Q/c \rfloor$, and assume

$$Z_* + d_+(1 + J) + Q < y.$$

Run any Appendix H sequence $b_n \to \infty$ to its last corrected index U_n above Z_*. Let K_n be the last Gaussian record through U_n. Then $U_n - K_n \le J$ and

$$F < Z_* - Q \le Y_{K_n} < Z_* + d_+(1 + J) + Q < y.$$

Indeed $Z_{U_n} < Z_* + d_+$ and each preceding corrected step costs at most d_+. These lower shots are actual. Their bare parameters lie below the explicit b_+ at the same K_n by the total-error bound. Choose a strictly increasing subsequence of K_n. This proves a genuine two-sided bracket without assuming every intermediate shot survives. The extended scalar bracket above requires only $y > F$.

The sign of a crossing that shooting actually needs

Bisect the bare bracket at a record depth, retaining a low-value left endpoint and high-value right endpoint. Either a tested bare value hits y, or nested intervals shrink to some b^*. Include constant sequences at b^* when a bracket endpoint stops moving. A sufficient exact test at every possible straddling limit is

$$\liminf_m T_K(x_m^-) \ \ge\ T_K(b^*) \ \ge \limsup_m T_K(x_m^+), \qquad x_m^- \uparrow b^*, \quad x_m^+ \downarrow b^*, \tag{2}$$

where only straddling endpoint sequences need to be tested. Indeed the low values give $T_K(b^*) \le y$ and the high ones give $T_K(b^*) \ge y$. The actual equality label at a rounded threshold occurs in the middle of (2). Downward jumps in increasing bare parameter are allowed. If both inequalities hold up to additive δ_K, the actual value at b^* is within δ_K of y. For $\delta_K < y - F$ the record certificate guarantees physical admission. A cofinal $\delta_K \to 0$ would tune the scalar in the limit. Exact finite-cutoff normalization requires exact hits or a separate exact calibration. Any physical state compatibility required by the frozen normalization would still have to use these same tuned shots.

For a bound available on both sides, take two actual shots to a record K, with physical terminal values at least $w \ge F$. At any common comparison index $j \le K$, telescoping and canceling the common Gaussian sequence give

$$|Y_K^+ - Y_K^-| \le |Y_j^+ - Y_j^-| + 2A \sum_{s=1}^{K-j} \frac{1}{w + as} \le |Y_j^+ - Y_j^-| + \frac{2A}{a} \log(1 + a(K - j)/w). \tag{3}$$

This uses (1) separately on each actual history, including histories with different mature resets. If the first differing incoming grid has index j and the preceding construction has a common limiting Y_j, the input-difference term vanishes in that limit. An outgoing grid change alone, at the very step that installs Y_j, does not alter that step's provisional germ by BP.24 and the subsequent replay argument. The later steps in (3) can use different incoming grids and mature banks. Even a single subsequent step leaves a fixed positive upper bound at fixed w. At large remaining depth the bound grows. It supplies neither the orientation in (2) nor an error tending to zero with cutoff depth. Appendix I passes HS.23–27 on each extracted, labeled branch with the mature reset bank as forcing. Its assertion about one branch supplies no ordering between the two sides in (2). Causal replay, Appendix F pp. 79–80 · Threshold labels and limiting equation, Appendix I §§8–9 · Bałaban, CMP 119, pp. 278–279

The target test (2), or a physical matching estimate that bypasses it, remains a concrete missing input. Cases in which the lower side of a straddle exits before the terminal step are included in the completed bracket and must be covered by a proof of (2). The record argument alone gives no such crossing comparison.

A stopped sequence with a positive weak endpoint

Conditional application. Use the stated source, support and fixed-chart hypotheses. Any application of the complete C05 boundary estimates or the C06/C08 law to all original Wilson histories assumes spatial and native-domain control of the iterated reconstruction with the earlier innovations fixed. The actual composed response required for that application remains open.

Gaussian calibration hypothesis (GC). For the signed Gaussian increments of the actual stationary source section in its thermodynamic normalization, assume $\sum_{k<n} d_k^G = d_B n + \vartheta_n$ with $d_B > 0$ and $\sup_n |\vartheta_n| \le B_G$. The coupling and stopping conclusions use this calibration and the specified source readout and replay. A physical continuum limit also requires the stated uniform normalization and full-law hierarchy estimates.

Stopped-sequence application and the stated B05 test

Assume the actual source-free Wilson recursion has been proved through every admitted stopped prefix, with its infinite-volume coupling readout and finite-torus replay identified. Assume its Gaussian calibration and all-depth error give a bounded-coboundary corrected inverse coupling with a uniformly positive decrement while the physical inverse coupling is at least one fixed large ceiling $Y_{\min}$.

Then one can construct an actual cofinal sequence of bare Wilson coefficients and cubic lattices whose terminal inverse couplings stay in a fixed compact subset of $(Y_{\min}, \infty)$. A subsequence has a strictly positive weak terminal coupling. No continuity across the rounded R_j-grid changes and no shooting at a prescribed integer depth are needed for this conclusion.

The actual-map inputs are now supplied at the retained causal-source and original Gaussian-calibration scope by Homogeneous feedback and the causal Wilson recursion, (BP.17), (BP.20), and (BP.22)–(BP.31), together with The driven homogeneous coupling endpoint, (DE1), and the causal-source compatibility conditions. Thus the stopped sequence below is an application of those results. The hypotheses are repeated in Section 1 so the numerical theorem is standalone; a numerical recurrence without the actual-map/replay identification would not suffice.

The original B05 statement is stronger: it asks for one coherent bare Wilson set satisfying the chosen finite physical normalization at ℓ_0, with compatibility across cutoff depths. A subsequential, emergent scalar endpoint is not automatically that prescribed normalization or a compatible limit of full effective states. Section 8 states the precise remaining distinction.

1. Actual-map assumptions and the corrected coordinate

Use the retained source convention

$$Y_j = g_j^{-2}, \qquad \beta_{W,0} = Y_0/N. \tag{PE.1}$$

Fix the source group, odd block factor L, all static grid/decay/chart constants, and a common admitted lower ceiling $Y_{\min}$ for Y.

The hypothesis is an actual, causal recursion on every prefix for which $Y_j \ge Y_{\min}$, including a provisional next output computed from that admitted input:

$$Y_{j+1} = Y_j - d_j^G - e_j, \qquad \left|\sum_{i=0}^{j-1} (d_i^G - d_B)\right| \le Q, \qquad |e_j| \le A/Y_j, \tag{PE.2}$$

with fixed $d_B > 0, A, Q < \infty$. A different quantified error suffices if it is uniformly smaller than $d_B/2$ on this ceiling. The constants must be independent of total prefix length, bare coupling and finite volume in the stated source readout/replay convention.

At the stated source and calibration scope, (PE.2) is the consequence (BP.23)–(BP.28) of the two stated B03 analytic gates. The source scheme and exponent choices are fixed once before the final weak ceiling. The outgoing scalar is the actual homogeneous infinite-volume projector, and finite cubic lattices use the same scalar prescription in the retained replay convention.

Put

$$q_j = \sum_{i<j}(d_i^G - d_B), \qquad Z_j = Y_j + q_j.$$

Then

$$|q_j| \le Q, \quad q_0 = 0, \qquad Z_{j+1} = Z_j - d_B - e_j. \tag{PE.3}$$

Choose $Y_{\min}$ large enough that $A/Y_{\min} \le d_B/4$, and set, for example,

$$d_- = \frac{3}{4}d_B, \qquad d_+ = \frac{5}{4}d_B.$$

Every admitted step then obeys

$$d_- \le Z_j - Z_{j+1} \le d_+. \tag{PE.4}$$

No sign is asserted for the individual uncorrected decrement $Y_j - Y_{j+1}$. That is why stopping is first done in Z.

Choose a fixed threshold

$$Z_* > Y_{\min} + Q + d_+. \tag{PE.5}$$

Its excess can be increased to meet any finite list of native/source ceiling tolerances. The choice is independent of the bare cutoff.

2. The last admitted corrected stopping point

For an actual trajectory starting at $Y_0 \to \infty$, let K be the last index with $Z_K \ge Z_*$, stopping when the provisional next output satisfies $Z_{K+1} < Z_*$.

This stopping time exists and is finite. Until the crossing,

$$Y_j = Z_j - q_j \ge Z_* - Q > Y_{\min},$$

so every required input is admitted. The next output is computed by the actual provisional recursion; (PE.5) even keeps its physical inverse coupling above $Y_{\min}$, although that extra output need not be used in the final sequence. The positive decrement in (PE.4) forces a crossing in finitely many steps.

At the stop,

$$Z_* \le Z_K < Z_* + d_+, \qquad Z_* - Q \le Y_K < Z_* + d_+ + Q. \tag{PE.6}$$

In particular

$$(Z_* + d_+ + Q)^{-1/2} < g_K \le (Z_* - Q)^{-1/2}. \tag{PE.7}$$

Both endpoints are fixed positive numbers, and the upper one lies below all chosen weak-coupling ceilings.

Moreover

$$\frac{Y_0 - Z_* - d_+}{d_+} < K \le \frac{Y_0 - Z_*}{d_-}. \tag{PE.8}$$

Thus $K \to \infty$ with Y_0.

If the uncorrected coordinate itself has the positive decrement, take $Q = 0$. Then (PE.6) reduces to the usual fixed interval of width d_+. The bounded-coboundary formulation is the one relevant to the currently retained Gaussian calibration.

3. Recovering the literal terminal suffix inequality

The bounded correction gives useful all-pair inequalities immediately:

$$d_-(k-j) - 2Q \le Y_j - Y_k \le d_+(k-j) + 2Q, \qquad j < k \le K. \tag{PE.9}$$

For the exact no-additive-slack terminal inequality printed in CMP109 (0.31), make a bounded final trimming, as in the accepted B02 construction.

Fix $0 < a < d_-$, for example $a = d_-/2$. Let $S_i = Y_0 - Y_i$, and let K' be the last maximizer of

$$S_i - ai, \qquad 0 \le i \le K. \tag{PE.10}$$

Then

$$0 \le K - K' \le J_0 := \left\lceil \frac{2Q}{d_- - a} \right\rceil. \tag{PE.11}$$

Indeed maximality gives $S_K - S_{K'} \le a(K - K')$, while (PE.9) bounds the left side below by $d_-(K - K') - 2Q$.

For every $j \le K'$,

$$a(K' - j) \le Y_j - Y_{K'} \le D(K' - j), \qquad D = d_+ + 2Q, \tag{PE.12}$$

where the upper inequality for positive integer age absorbs $2Q$ into D; at age zero it is exact. The lower inequality is precisely the maximality condition in (PE.10).

Trimming changes the terminal compact interval by only a fixed amount:

$$Y_- := Z_* - Q \le Y_{K'} \le Y_+ := Z_* + d_+(J_0 + 1) + Q. \tag{PE.13}$$

The retained causal trajectory before K' is unchanged. Also $K' \to \infty$.

With $\epsilon = L^{-K'}$, (PE.12) is the printed source terminal envelope, with constants $a/\log L$ and $D/\log L$:

$$Y_{K'} + \frac{a}{\log L} \log[(L^j\epsilon)^{-1}] \le Y_j \le Y_{K'} + \frac{D}{\log L} \log[(L^j\epsilon)^{-1}]. \tag{PE.14}$$

This does not assert positive single-step drift of raw Y_j.

The all-pair bound (PE.9), together with a large fixed $Y_{\min}$, also gives the source pairwise coupling/profile comparisons. For example $Y_j \ge Y_k - 2Q$ makes $g_j/g_k \le 1 + \beta_0$ at a sufficiently large ceiling, while $Y_j/Y_k \le 1 + (d_+ + 2Q)(k - j)/Y_k$ controls the reverse ratio. For any fixed profile exponent p,

$$\left(\frac{\log Y_j}{\log Y_k}\right)^p \le \left(1 + \frac{D(k - j)}{Y_k}\right)^{p/\log Y_k}$$

in the direction $Y_j \ge Y_k$, with the bounded opposite-direction slack absorbed by the chosen ceiling. Taking $Y_{\min}$ large after the finite list of exponents makes the required exponent at most the source tolerance β_0.

For the rounded radii R_j, CMP119 (2.9), p.256, permits the fixed L rounding factor. The preceding logarithmic comparisons give its two inequalities. In particular possible small raw-coupling reversals do not require assuming all R_j are equal.

4. One bare prescription and a cofinal physical mesh

Choose any deterministic unbounded bare sequence, for example

$$Y_{0,n} = Y_{\min} + n. \tag{PE.15}$$

For each n, use the actual same source prescription to generate its trajectory and compute the stops K_n, K'_n. If strictly increasing depths are convenient, retain the deterministic subsequence obtained by taking the next n for which K'_n exceeds its preceding chosen value. Equation (PE.8) ensures this procedure never stops.

Define

$$a_n = \ell_0 L^{-K'_n}, \qquad h_{j,n} = \ell_0 L^{j - K'_n}, \qquad \beta_{W,n} = Y_{0,n}/N. \tag{PE.16}$$

Then $a_n \to 0$, and the final retained scale is exactly ℓ_0. These are genuine bare Wilson coefficients, not independently postulated sequences of effective couplings.

The source lower/upper slopes give the logarithmic bare order

$$c_1 K'_n \le Y_{0,n} \le c_2 K'_n + C,$$

so

$$\beta_{W,n} \asymp \log(\ell_0/a_n). \tag{PE.17}$$

This is an additional useful property absent from the B02 short-window witness.

Under the sharper error in (PE.2), sum the actual recurrence to obtain

$$Y_{0,n} = d_B K'_n + Y_{K'_n} + q_{K'_n} + \sum_{j<K'_n} e_j.$$

Using (PE.12),

$$\sum_{j<K'_n} \frac{1}{Y_j} \le \frac{1}{Y_-} + \frac{1}{a}\log\left(1 + \frac{aK'_n}{Y_-}\right).$$

Consequently

$$\beta_{W,n} = \frac{d_B}{N\log L}\log(\ell_0/a_n) + O(\log\log(\ell_0/a_n)) + O(1), \tag{PE.18}$$

with the fixed terminal interval absorbed in the last constant. This is conditional on the actual all-depth recurrence, not a new proof of its coefficient.

5. Original cubic geometry works at a fixed positive endpoint

Keep exactly the fixed divisors and odd-L centered partitions of the accepted B02 geometry. Write $M = L^u$, $M_2 = L^v$, with the source's once-selected compatible fixed divisors understood. Let

$$R_j = L^{r_j}, \qquad R_j \le C_R[\log Y_j]^r \tag{PE.19}$$

with the source rounding factor included in C_R. The equality convention at a rounding threshold is the actual source convention, not a new one.

At age $s = K'_n - j$, (PE.12)–(PE.13) give

$$Y_j \le Y_+ + Ds.$$

Hence

$$\sup_{n,j\le K'_n} L^{j-K'_n} R_j \le C_R \sup_{s\ge 0} L^{-s}[\log(Y_+ + Ds)]^r =: C_{\text{geom}} < \infty. \tag{PE.20}$$

The supremum is finite because an exponential dominates every power of a logarithm. Thus the physical sizes of all old MR_j and LM_2R_j tiles are bounded by one fixed multiple of ℓ_0, uniformly in the bare depth. They need not be equal.

Choose a fixed integer H_0 such that every such tile exponent

$$j - K'_n + r_j + \max(u, 1+v)$$

is at most $H_0 - q$, where q is fixed with L^q larger than the source's finite halo/component multipliers. Every elementary and earlier compatible grid then divides the cubic period

$$T_n = 2L^{H_n}\ell_0, \qquad H_n \ge H_0. \tag{PE.21}$$

The $100MR_j$ component rectangles and their finitely many fixed enlargements fit as neighborhoods. They do not impose new independent period divisors.

Any choice $H_n \to \infty$ above this fixed bound is cofinal. In particular one may take

$$H_n = \max\{H_0, \lfloor \theta \log_L(2 + K'_n)\rfloor\}, \qquad 0 < \theta < \frac{1}{4}. \tag{PE.22}$$

Then $T_n \to \infty$ and the terminal lattice volume satisfies

$$V_{\text{term},n} = (T_n/\ell_0)^4 \le C(2 + K'_n)^{4\theta} = o(K'_n). \tag{PE.23}$$

This optional slow-volume property is a geometric fact only; it does not imply an interacting terminal estimate.

The common centered origin and odd L give the same site-reflection planes as B02. Explicitly,

$$T_n/(2h_{j,n}) = L^{H_n + K'_n - j} \in \mathbb{N}.$$

The fixed half-ℓ_0 translation used in the accepted construction identifies all center lattices with the corresponding $h_{j,n}\mathbb{Z}^4$. Every compact physical support eventually fits inside the cubic torus.

Finally the compact Wilson reserve obeys

$$\frac{\log(1+\beta_{W,n})}{T_n/a_n} \le \frac{C\log(2+K_n')}{2L^{H_n+K_n'}} \longrightarrow 0. \tag{PE.24}$$

Thus the actual logarithmic bare order is compatible with this geometry and its microscopic compact-moment reserve.

All statements in this section concern original cubic tori. No rectangular extension of the full source-free RG theorem is used. The infinite-volume coupling readout is performed at each finite prefix in the retained B02 convention of The Wilson trajectory on intervals admitting mature resets before choosing the finite torus large enough to replay that prefix.

6. A positive terminal subsequence and fixed physical offsets

By (PE.13), extract a subsequence with

$$Y_{K_n'} \longrightarrow Y_\infty \in [Y_-, Y_+], \qquad g_{K_n'} \longrightarrow g_\infty = Y_\infty^{-1/2} > 0. \tag{PE.25}$$

The upper bound on g_∞ remains below all fixed weak-coupling ceilings.

For every fixed ultraviolet offset $s \ge 0$, the actual couplings at the fixed physical mesh $\ell_0 L^{-s}$ lie in the compact interval

$$Y_{K_n'} + as \le Y_{K_n'-s} \le Y_{K_n'} + Ds.$$

A countable diagonal subsequence therefore yields

$$\widehat{Y}_s = \lim_n Y_{K_n'-s}, \qquad Y_\infty + as \le \widehat{Y}_s \le Y_\infty + Ds. \tag{PE.26}$$

This is a compatible numerical suffix array drawn from the same finite Wilson trajectories. It is not a new independently selected sequence of finite effective trajectories.

For each fixed offset the rounded radius and the finitely many integer source parameters have only finitely many possible values in this compact range. A further diagonal can make those discrete labels eventually constant at every fixed offset. At a rounding threshold, the constant label can be a one-sided limiting label; its equality with the canonical label computed from $\widehat{Y}_s$ requires a closed-graph or chart-matching check. Numeric convergence does not prove that check.

In particular (PE.26) alone does not prove that the limiting suffix array solves the full history-dependent Wilson map. Bounds on an error term pass to subsequences as inequalities. Its exact dependence on the infinite incoming effective state requires convergence and compatibility of that state.

7. What continuity has been avoided, and what has not

The construction uses no continuity of $Y_0 \mapsto Y_K(Y_0)$ across a rounded R_j transition. It never asks a prescribed integer-depth trajectory to land at a specified Y_*. It follows each actual trajectory until a common corrected threshold is crossed and then selects a subsequence.

It also does not compare conditional mean laws from unrelated reset decompositions or replace the full E/R/boundary state by the scalar Y. The native projection may depend on that whole state; its all-depth estimate is part of the hypothesis, and its limiting-state continuity is a separate issue.

A sufficient full-state compatibility statement would be the following. At every fixed physical offset and fixed finite spatial region, the actual effective coefficients, positive history kernels and normalized marginal laws admit a common topology in which:

1. the local coefficient families are compact on their genuine admitted charts;
2. contributions from incoming births beyond m additional ultraviolet levels have a uniform tail tending to zero as $m \to \infty$, in the same norm used to pass the actual projection and normalization;

3. on each retained discrete grid/selector chart, the finite-window source transformations pass to limits, with either closed-graph control at threshold labels or an exact invariance identifying the different decompositions.

Then extract the full states, pass the finite-window identities, and use the uniform tail to obtain an exact limiting RG/history identity. This is a precise sufficient interface. A scalar recurrence and a diagonal array do not establish its three statements. C05/C08 provide several of their uniform scalar-law ingredients, but neither proves arbitrary transported-mark compactness or the full physical field normalization.

8. Comparison with the stated B05 normalization

The following conclusions must be distinguished.

Assertion	Status under the actual-map assumptions (PE.2)
One deterministic cofinal set of genuine bare Wilson coefficients and original cubic tori, with every used source prefix below all weak-coupling ceilings	Proved by (PE.5)–(PE.24), using the fixed B02 source constants and replay convention.
A terminal coupling bounded away from zero and a subsequential positive scalar endpoint at scale ℓ_0	Proved by (PE.13), (PE.25).
A fixed-offset numerical array extracted from those same trajectories, with the source terminal AF envelope	Proved by (PE.26).
A prescribed numerical value g_{phys} at ℓ_0 chosen before the construction	Not proved: g_∞ is an emergent subsequential value in a compact interval.
Compatibility of the complete effective states and actual normalized history maps across cutoffs	Requires the full-state convergence/tail/threshold statement in Section 7, not just scalar diagonalization.
Identification with the program's chosen physical observable normalization, and the renormalized continuum fields	Still requires the corresponding B05/F05/M07 matching statements.

If the physical construction permits choosing some sufficiently small positive renormalization condition after extraction, (PE.25) removes the need to shoot to a preassigned scalar endpoint. That is potentially useful for an existence proof. It must be stated as that choice; it does not meet a requirement for a prescribed normalization.

Changing the physical unit or selecting an integer RG offset does not by itself prove equality with an arbitrary prescribed coupling. Such an identification needs the relevant scale/normalization theorem. Likewise a limiting positive g_∞ is not by itself a proof of nontrivial continuum Yang–Mills fields.

The substantive gain is therefore exact and conditional: once the actual all-depth B03 stopped-prefix estimate is established, cofinal original-lattice geometry and a positive weak subsequential endpoint need no separate shooting continuity argument. The remaining B05 question is the specified physical normalization and full-state compatibility, rather than the elementary existence of an admissible numerical stopping sequence.

Physical matching at cofinal cutoffs

The identification of the cumulative Gaussian marginal with a linear drift up to a uniformly bounded remainder is a separate hypothesis (GC). An arithmetic consequence of that drift does not prove GC or match a prescribed physical endpoint.

B-05 remains open. The normalized F-01 measurements leave a bare-coupling degree of freedom even at a finite cutoff. An independently normalized Wilson observable can be used for a continuous scalar shot if its level set is calibrated to the native marginal projection. The finite-volume Wilson integral gives the requisite continuity for that different observable. A last-step gap example below rules out recovering this calibration or the prescribed native endpoint from the B-03 drift estimate and cofinal selection alone.

Which physical condition is fixed?

the stated criterion asks for one compatible family inside the source ceilings satisfying the chosen finite physical normalization at the common scale ℓ_0. The source-free normalization N2-F fixes the numerical endpoint before the construction. With the corrected dictionary, if the program target is g_*, then

$$Y_K = Y_* := 2N^2 g_*^{-2}, \qquad \beta_{W,K} = Y_{0,K}/N, \qquad a_K = L^{-K}\ell_0.$$

Here $\beta_{W,K}$ is the bare coefficient of $S = \sum_p (N - \Re \operatorname{Tr} U_p)$. If the target is already in native units, it is simply $Y_* = (g_*^B)^{-2}$. At positive depth the prescription is the homogeneous polarization in the chosen decomposition. Its definition does not say it equals the Hessian of an arbitrary normalized finite-torus marginal. B-05's stated criterion does not require all of the stronger differentiability assertions of N2-F. · N2 §§2–3, ICR-N2-1 · Actual projection and replay

At fixed admissible torus and mesh let $b = Y_0/N$, and write $n_h(b)$ for the seven unrenormalized F-01 measurements. All are real analytic in real positive b. Indeed S and the finitely many group stencils are bounded at fixed mesh. The numerator integrals for ordinary moments are entire in b, by the compact Haar integral with weight e^{-bS}; fixed polynomial bare factors, when present, preserve analyticity. The strictly positive partition function is locally nonzero in a complex neighborhood of every real b. Finite products and differences give the centered integrals. Wherever all $n_h > 0$, the seven positive roots and the identity coefficient depend real analytically on b:

$$z_h(b) = \sqrt{\lambda_h / n_h(b)}, \qquad c(b) = z_S(b) m(b).$$

Therefore every b in an open interval around any nonsingular point solves all eight equations after fitting the eight coefficients. The derivative of every resulting normalization residual along that interval is zero. If there is a nonsingular witness for one fixed geometry, the nonnegative analytic functions also give nonsingularity there except at a locally finite set of positive bare values (finite union of isolated zero sets). Neither the exception set nor a lower bound is uniform in the changing lattice. F-01(F.34) is precisely the Jacobian in the eight field coefficients with b fixed. It cannot be inverted to fix the ninth unknown b. The positive comparison on larger ultracold cubes is a witness on those geometries, with a different endpoint. Annex X, F.22–34 · Actual weak-family reflected measurements

A continuous Wilson measurement and the missing equality

For an independently specified bounded physical stencil or bounded multiscale test A_K with a fixed, bare-independent sampling rule, the literal finite Wilson response $M_K(b) = \mathbb{E}_b A_K$ is real analytic, with $M_K'(b) = -\operatorname{Cov}_b(A_K, S)$. The same conclusion holds for ratios with a nonzero denominator on the shooting interval. A fixed random readout conditional on the microscopic links is covered by taking its conditional expectation first. For a bare-dependent history kernel, this proof requires either retaining its derivative/continuity or identifying its expectation with the fixed microscopic test.

Specifically, suppose at cofinal depths there are *admitted* bare intervals on which the native trajectories satisfy the terminal and intermediate geometry tests, and the continuous M_K brackets a predetermined m_*. It suffices to prove, on the same intervals, the scalar implication

$$M_K(b) = m_* \quad \Longrightarrow \quad Y_K(Nb) = Y_*.$$

Here $Y_K(y)$ denotes the actual endpoint starting at native inverse coupling y. An intermediate value gives an exact shot even when Y_K has jumps elsewhere. One sufficient comparison near that level is $|Y_K(Nb) - Y_*| \le C_K|M_K(b) - m_*|$; no uniformity of C_K is required for this finite implication. An additive error tending to zero yields only asymptotic matching. The compact Wilson integral supplies neither sign brackets for a prescribed physical observable nor the implication. The uniform signed prefix controls the principal slope in scale, while the needed equality links two different readouts at a fixed scale. Equality of partition functions or a vacuum subtraction alone cannot fix the split into $YA + E$, since the replacements $Y \mapsto Y + c$, $E \mapsto E - cA$ leave this sum identical. This algebraic observation concerns what follows from equality of densities alone. The source fixes its own split by an additional projection. Native projection, HS.24–27

A last-step gap persists for every cutoff depth. As a test of the numerical implication, take $d > 0$, $c > 0$, M with $c/M < d/2$, $c/M^2 < 1$, and $t > M + 2d$. For admitted $y \ge M$, set

$$F(y) = y - d + \frac{c}{y}\begin{cases} -1, & y < t, \\ +1, & y \ge t. \end{cases}$$

Each branch is strictly increasing. Its decrement is in $[d/2, 3d/2]$ and has remainder bounded by c/y, with zero Gaussian transient. Thus all raw suffix slopes hold, even at one step. Yet the range of *each final step* misses $(t - d - c/t, t - d + c/t)$, so $Y_* = t - d$ cannot even be approached by changing depths, initial values or subsequences. For example, backward iteration on the upper branch from any $y > t$ produces positive endpoints on coherent arbitrarily long admitted trajectories, with a stabilized label and exact compatible scalar recurrence. The example concerns an inference from B-03 and compactness. It gives no evidence that the Wilson projector itself jumps. Appendix H's interval and Appendix I's one-sided label keep precisely the information to which the example applies. PE.13, PE.25–26 and §8 · HS.23–28

A scalar test for compatibility of conditional physical readouts

There is a weaker alternative to total-variation convergence for passing bounded retained probes. On a fixed retained finite geometry, let X_n be the outer raw state in common Polish coordinates, U_n a retained inner output, and $Q_n(X_n, dU)$ its genuine conditional law. Assume the actual joint laws converge weakly to $\mu(dx)Q(x, du)$ on these coordinates. This premise can be obtained on a tight subsequence for ordinary finite-dimensional raw variables. For each real bounded continuous inner probe f, draw two conditionally independent inner continuations from that *same* kernel at the same exterior. Write

$$D_n(f) = \mathbb{E}[f(U_n^{(1)})f(U_n^{(2)})] = \mathbb{E}[(Q_n f)(X_n)^2].$$

A necessary and sufficient condition for approximation of the conditional response by common continuous exterior probes is

$$\lim_n D_n(f) = \int |Qf|^2 \, d\mu \quad \Longleftrightarrow \quad \inf_{\phi \in C_b} \limsup_n \mathbb{E}|(Q_n f)(X_n) - \phi(X_n)|^2 = 0.$$

Indeed for fixed real ϕ the cross moment is $\mathbb{E}[\phi(X_n)f(U_n)]$, which converges by joint weak convergence. Expanding the square gives the limsup defect $\limsup D_n(f) - \|Qf\|^2_{L^2(\mu)} + \|Qf - \phi\|^2_{L^2(\mu)}$. Density of bounded continuous functions gives the equivalence, using the same expansion with liminf and nonnegativity for the universal lower bound. Bounded products of finitely many such responses can then pass by telescoping and Cauchy–Schwarz, clipping each approximating continuous probe to the corresponding fixed sup norm. Countably many fixed probes admit a common diagonal sequence. This tests the response to the chosen probes only. Identifying Q with the specific source kernel still requires its defining conditional identities at the limiting state.

The right side can be tested against actual native integral approximants, without lower bounds on rare component masses if the joint laws keep the selected outer branch mass. A branch of positive but small mass is weighted by that mass once for both continuations. The oscillating example FK.17–18 gives a strictly positive mean-square defect for its raw mass function. A similar defect for bounded conditional means arises by taking Bernoulli continuations whose conditional success parameter equals that mass function. Thus a one-copy weak-law theorem cannot establish this compatibility by itself. Appendix P identifies archive-dependent raw multipliers, the constrained finite-dimensional coordinates, and hard-selector interfaces that still need estimates for the actual model. This criterion gives an observable-specific test for those cases in which nonlinear products of conditional probes are required. It does

not retroactively impose whole-kernel convergence as a necessary clause of the source-free B-05 statement. Appendix P, FK.12–20 · Actual scope of positive-history extraction

The remaining source-free test is either a genuine native shot with compatible actual cutoff data, or an exact calibration and admitted bracket for a prespecified physical observable. A single common subsequence can preserve the HS homogeneous identities once exact shots exist and its uniform extraction hypotheses hold. Where canonical mixed-history operations are asserted, their separate limiting identification still has to be made. Neither selection of the terminal band nor refitting the seven field coefficients resolves the finite physical shot.

Reflected susceptibilities and physical normalization

Conditional application. Use the stated source, support and fixed-chart hypotheses. Any application of the complete C05 boundary estimates or the C06/C08 law to all original Wilson histories assumes spatial and native-domain control of the iterated reconstruction with the earlier innovations fixed. The actual composed response required for that application remains open.

Gaussian calibration hypothesis (GC). For the signed Gaussian increments of the actual stationary source section in its thermodynamic normalization, assume $\sum_{k<n} d_k^G = d_B n + \vartheta_n$ with $d_B > 0$ and $\sup_n |\vartheta_n| \le B_G$. The coupling and stopping conclusions use this calibration and the specified source readout and replay. A physical continuum limit also requires the stated uniform normalization and full-law hierarchy estimates.

Finite-cutoff scope. Nonzero reflected measurements at each cutoff allow finite normalization. Uniform bounds for the seven normalizing factors along one physical sequence and passage of the full normalized hierarchy have separate hypotheses.

Result and exact status

The actual scalar identity coefficient in the stated F01 prescription is controlled by the actual channel multiplier. It is not an additional uncontrolled parameter. On the stopped Wilson sequence of PE,

$$c_N \le \beta_a a^4 m_{a,T} \le C_N, \qquad c_N \frac{z_S}{\beta_a} \le a^4 c_{a,T,\mu} \le C_N \frac{z_S}{\beta_a}. \tag{PN.1}$$

The upper constant may depend on the fixed trajectory and geometric constants; neither constant depends on the ultraviolet depth or total volume. Here m is the exact scalar mean and $c = z_S m$ is precisely the identity coefficient of Annex X, rather than a new subtraction.

Consequently the coefficient budget used in CS and ER satisfies

$$B_a := \frac{\|Z_{a,T,\mu}\|_{\mathrm{op}}}{\beta_a} \le b_a := B_a + a^4 |c_{a,T,\mu}| \le C B_a. \tag{PN.2}$$

The proof below uses the original compact Wilson law, an exact Schwinger–Dyson identity, and the checked Wilson partition-ratio bound. No Gaussian replacement, source-conditioned density, or assumed susceptibility estimate is used.

This closes the identity-coefficient part of the normalization budget once the channel multipliers are controlled. It does not prove the missing logarithmic bounds for those multipliers, the actual F03/F04 mixing estimates, or the prescribed B05 endpoint. F05 as a whole remains open. Sections 6–8 identify its remaining numerical data precisely and prove how the prescribed finite normalization supplies nondegeneracy whenever a common hierarchy limit has been obtained.

The subsequent finite reflected witnesses proves the separate qualitative finite-cutoff positivity needed to use this prescription. Its Corollary FN.5 supplies all seven $n_h > 0$ on one actual PE sequence with the original slow-volume geometry. It therefore removes exact zero-channel and determinant singularities on that sequence; the logarithmic upper and lower bounds remain open.

Statements labeled P are proved here. Statements labeled C are proved implications with an explicitly stated remaining actual-model input. The final table records the unproved inputs.

1. Frozen physical prescription

Use the original four-dimensional periodic Wilson measure on an even cubic n^4 lattice, $n \ge 2$, with fine physical spacing a:

$$d\mu_{\beta,n}(U) = Z_n(\beta)^{-1} e^{-\beta S(U)}\, dU, \qquad S = \sum_p s_p, \qquad s_p = N - \mathrm{ReTr}\, U_p, \qquad G = SU(N). \tag{PN.3}$$

There are $V = n^4$ vertices and $6V$ plaquettes. Haar measure is normalized. The coefficient β is the bare Wilson coefficient, not the terminal effective inverse coupling. In the source units, PE uses $\beta_{W,n} = Y_{0,n}/N$, consistently with Annex X's $\beta = 2Nx_0$ and $Y_0 = 2N^2x_0$.

The tensor space is

$$E = \mathrm{Sym}^2(\Lambda^2\mathbb{R}^4), \qquad \dim E = 21, \qquad e_S = I_6/\sqrt{6}. \tag{PN.4}$$

Retain exactly the clover and Wilson-scalar repair of Annex X (F.4)–(F.7). Only the following exact identities are needed here:

$$\frac{1}{2}\operatorname{Tr}\tilde{q}_a(x) = w_a(x), \qquad w_a(x) = \frac{1}{4a^4}\sum_{\mu<\nu}\sum_{\sigma,\tau=\pm1} s(P_{\sigma e_\mu,\tau e_\nu}(x)), \tag{PN.5}$$

$$\mathbb{E}\tilde{q}_a(x) = m_{a,T}e_S, \qquad m_{a,T} = \frac{2}{\sqrt{6}}\mathbb{E}w_a(0). \tag{PN.6}$$

Translation and hypercubic invariance make all the plaquette means in (PN.5) identical. There are 24 summands, so

$$\mathbb{E}w_a(0) = 6a^{-4}\mathbb{E}s_p, \qquad \boxed{\beta a^4 m_{a,T} = 2\sqrt{6}\,\beta\mathbb{E}s_p.} \tag{PN.7}$$

Let P_h, $h = (S, T_d, W_d, T_o, W_s, P, W_p)$, be the seven fixed orthogonal hypercubic projectors, with dimensions

$$(d_h) = (1, 3, 2, 6, 6, 1, 2). \tag{PN.8}$$

The fixed separated reflected measurements and their positive reference values are those of Annex X (F.19)–(F.28):

$$n_{h,a,T} = \mathcal{N}_{h,a,T}(\tilde{q}_a) \geq 0, \qquad \lambda_h(\mu) > 0. \tag{PN.9}$$

At a cutoff where all seven n_h are positive, the actual prescribed solution is

$$Z = \sum_h z_h P_h, \qquad z_h = \sqrt{\lambda_h/n_h} > 0, \qquad c = z_S m, \qquad \Phi_a = Z\tilde{q}_a - ce_S. \tag{PN.10}$$

Its mean is exactly zero and its seven reflected measurements are exactly λ_h. No change of this prescription is made below.

The stated F05 normalization condition asks for actual mixing products, the inverse matrices, the finite normalization determinant and logarithmic coefficient budgets. Its predecessor F01 is a finite construction; nonnegativity of n_h alone is not quantitative invertibility.

2. An exact lower Wilson-energy bound

Theorem PN.1 (P): uniform lower energy from one-link integration

Equip $\mathfrak{su}(N)$ with the metric $\langle X, Y\rangle = -\operatorname{Tr}(XY)$, and put

$$d = N^2 - 1, \qquad c_F = d/N. \tag{PN.11}$$

For every $\beta > 0$, every finite cubic torus as above, and every plaquette p,

$$\boxed{\beta\,\mathbb{E}_{\beta,n}s_p \geq \frac{c_F N\beta}{12\beta + c_F}.} \tag{PN.12}$$

In particular, for $\beta \geq 1$, the right side is at least $c_F N/(12 + c_F) > 0$, uniformly in volume.

Proof. Let $X_1, \ldots, X_d$ be an orthonormal basis of the Lie algebra, and let $\Delta = \sum_\alpha D_{X_\alpha}^2$ be the nonpositive group Laplacian. Adjoint invariance makes $\sum_\alpha X_\alpha^2$ a scalar matrix. Taking its trace gives

$$\sum_\alpha X_\alpha^2 = -c_F I. \tag{PN.13}$$

For $s(U) = N - \mathrm{ReTr}\, U$,

$$\Delta s = c_F(N - s). \tag{PN.14}$$

Also $\mathrm{ReTr}\, X_\alpha = 0$. Bessel's inequality in the real Hilbert–Schmidt space, together with $\|U - I\|_{HS}^2 = 2s(U)$, gives

$$|\nabla s(U)|^2 = \sum_\alpha \big(\mathrm{ReTr}[X_\alpha(U - I)]\big)^2 \le 2s(U). \tag{PN.15}$$

Left and right translations and inversion preserve these statements for a plaquette viewed as a function of any one of its links.

Fix a positive link b and all the other raw Wilson links. Exactly six plaquettes contain b. Their sum

$$A_b(U_b) = \sum_{p \ni b} s_p(U_b) \tag{PN.16}$$

is the entire U_b-dependent action. Its true conditional law is proportional to $e^{-\beta A_b(U_b)}\, dU_b$. Equations (PN.14)–(PN.15) give, pointwise,

$$\Delta_b A_b = c_F(6N - A_b), \qquad |\nabla_b A_b|^2 \le 6 \sum_{p \ni b} |\nabla_b s_p|^2 \le 12 A_b. \tag{PN.17}$$

There is no boundary term on the compact group. Integration by parts under this normalized conditional law yields

$$c_F\big(6N - \mathbb{E}[A_b \mid U_{b^c}]\big) = \beta\, \mathbb{E}[|\nabla_b A_b|^2 \mid U_{b^c}] \le 12\beta\, \mathbb{E}[A_b \mid U_{b^c}]. \tag{PN.18}$$

Thus

$$\mathbb{E}[A_b \mid U_{b^c}] \ge \frac{6c_F N}{12\beta + c_F}. \tag{PN.19}$$

Average the actual exterior marginal. Translation and hypercubic symmetry give $\mathbb{E}A_b = 6\mathbb{E}s_p$, proving (PN.12). □

The conditional assertion (PN.19) is valid for the literal one-link Wilson conditional law. An additional selector depending on U_b would change that law and is not included. No such selector is present when deriving the full-law mean used in (PN.7).

3. The matching upper energy bound on the actual sequence

Theorem PN.2 (P): first-moment upper bound with its finite-volume term

For each fixed $0 < s < 1$, the checked partition-ratio estimate gives

$$\beta \mathbb{E}_{\beta,n} s_p \le C_{N,s} + \frac{d}{3sV} \log(1 + \beta). \tag{PN.20}$$

In particular this is uniformly bounded on the actual stopped sequence of PE.

Proof. The exact compact bounds CA.3, CA.6 and CA.9a in the Wilson energy normalization give

$$c_N(1 + \beta)^{-d/2} \le z(\beta) \le C_N(1 + \beta)^{-d/2}, \tag{PN.21}$$

$$Z_n(\beta) \ge e^{-C_N V}(1 + \beta)^{-d(3V+1)/2}, \qquad Z_n(\beta) \le z(\beta)^{3V-3}. \tag{PN.22}$$

The lower bound comes from an exact spanning-tree gauge and a product identity neighborhood in the $3V + 1$ remaining group variables. The upper bound integrates the recursive open plaquette forest in CA.9a–CA.9b; it is not a product assumption for the full action. Comparing the two at β and $(1 - s)\beta$ gives

$$\log \frac{Z_n((1 - s)\beta)}{Z_n(\beta)} \le C_{N,s} V + 2d \log(1 + \beta). \tag{PN.23}$$

The normalization remains in this ratio. Since

$$\mathbb{E}_{\beta,n} e^{s\beta S} = \frac{Z_n((1 - s)\beta)}{Z_n(\beta)}, \qquad \mathbb{E}S = 6V \mathbb{E}s_p, \tag{PN.24}$$

Jensen's inequality proves (PN.20).

For the actual PE sequence,

$$\beta_{W,n} \le C(1 + K'_n), \qquad T_n/a_n = 2L^{H_n+K'_n}, \qquad V_n = (T_n/a_n)^4. \tag{PN.25}$$

Therefore $\log(1+\beta_{W,n})/V_n$ is bounded, and tends to zero as the ultraviolet depth grows. The finitely many initial members can be absorbed into the same constant. This uses PE.17 and PE.21, not an unidentified physical normalization of the coupling. □

The proof of this first-moment bound does not require the more substantial local chessboard estimate. That estimate, CA.16–CA.18, is used only for the higher moments in Section 5.

4. The identity coefficient is now controlled

Theorem PN.3 (P): the actual F01 scalar counterterm budget

On the stopped sequence, use the precise finite F01 prescription where it is nonsingular. There are fixed $0 < c_m \le C_m < \infty$ such that

$$c_m \le e_a := \beta_a a^4 m_{a,T} \le C_m. \tag{PN.26}$$

Consequently

$$a^4 c_{a,T,\mu} = e_a \frac{z_{S,a,T,\mu}}{\beta_a} > 0, \tag{PN.27}$$

and (PN.1)–(PN.2) follow.

Proof. Combine the exact scalar identity (PN.7) with Theorems PN.1–PN.2. Orthogonality of the seven projectors and positivity of their multipliers give $\|Z\|_{\mathrm{op}} = \max_h z_h$. Hence

$$B_a \le b_a = B_a + e_a z_S/\beta_a \le (1 + C_m)B_a. \tag{PN.28}$$

All these quantities refer to the same Wilson law and the same finite physical prescription. □

In particular, a proved channel budget

$$\frac{\|Z_a\|_{\mathrm{op}}}{\beta_a} \le C[1 + \log(\ell_0/a)]^\gamma \tag{PN.29}$$

would imply the same logarithmic power for b_a, and

$$|c_a| \le Ca^{-4}[1 + \log(\ell_0/a)]^\gamma. \tag{PN.30}$$

There is no further logarithmic loss in the identity term. Channel-specific upper and lower powers for z_S/β_a transfer directly to $a^4 c_a$ through (PN.27).

The bare identity coefficient is naturally of order a^{-4} times its scalar channel budget. A bounded dimensionless $a^4 c_a$ is not a bounded unscaled c_a.

The same estimates hold for the explicit totalization F.35 wherever that bookkeeping definition is used, because it still sets $c = z_S m$. At a zero measurement, however, the totalization does not solve that channel's physical normalization equation. This observation supplies no missing nondegeneracy assertion.

5. Consequences for the actual transported source

Corollary PN.4 (P): one coefficient budget suffices in CS and ER

For the actual normalized cutoff fields of (PN.10), the compact Wilson local-energy reserve gives

$$\|\Phi_{\alpha,a}(x)\|_{L^p(\mu_a)} \le CB_a\, a^{-4} p, \qquad p \ge 1. \tag{PN.31}$$

Thus b_a may be replaced by a fixed constant times B_a in the source collar and ancient-history estimates CS.17, CS.19, CS.22 and ER.4–ER.6.

Proof. The finite clover identities give the pointwise bound

$$|\tilde{q}_{\alpha,a}(x)| \le C_N a^{-4} \sum_{p \in \mathcal{P}_x} s_p, \tag{PN.32}$$

where $\mathcal{P}_x$ is the fixed finite stencil. It follows from $\|U_p - I\|_{HS}^2 = 2s_p$, the finite clover sum, and Hilbert–Schmidt Cauchy–Schwarz; the scalar repair adds only the same finite energy sum.

CA.16–CA.18 on the PE geometry imply

$$\|\beta_a \tilde{q}_{\alpha,a}(x)\|_p \le C_N a^{-4} p. \tag{PN.33}$$

Those estimates apply because their proof only needs the compatible fixed reflection blocks and the bounded partition-ratio density. Both are supplied by PE.21 and (PN.23)–(PN.25). They do not require the unproved physical endpoint matching.

Now

$$\Phi_a = Z_a(\tilde{q}_a - m_a e_S) \tag{PN.34}$$

exactly. Minkowski's inequality and the finite number of components give (PN.31). This is the same original microscopic field in every source-free conditional kernel. Conditional expectation remains an L^p contraction, so the CS arguments apply without another identity budget. □

The full one-point Schwinger function of these fields is identically zero at every cutoff. Their conditional first moments at retained cuts can be nonzero. Equation (PN.34) does not discard those outer means or the higher source vertices of CS.10–CS.11.

6. What the remaining finite normalization problem actually is

Introduce the dimensionless physical data

$$D_{h,a} := \beta_a^2 n_{h,a,T}, \qquad t_{h,a} := z_{h,a,T,\mu}/\beta_a, \qquad \hat{c}_a := a^4 c_a. \tag{PN.35}$$

The exact finite system is now

$$t_h^2 D_h = \lambda_h, \qquad \hat{c} = e_a t_S, \qquad c_m \le e_a \le C_m. \tag{PN.36}$$

Proposition PN.5 (P): budget and determinant equivalences

Whenever $D_h > 0$ for all seven channels,

$$t_h = \sqrt{\lambda_h / D_h}, \qquad B_a = \max_h \sqrt{\lambda_h / D_h}. \tag{PN.37}$$

For any positive prescribed scale function L_a, the following are equivalent up to fixed constants:

$$b_a \le C L_a \iff B_a \le C' L_a \iff D_{h,a} \ge c\, \lambda_h L_a^{-2} \quad \text{for every } h. \tag{PN.38}$$

Also

$$\|\beta_a Z_a^{-1}\|_{\mathrm{op}} = \max_h \sqrt{D_{h,a}/\lambda_h}. \tag{PN.39}$$

Thus logarithmic upper and lower bounds on the seven D_h give the corresponding logarithmic bounds on the scaled normalization and its inverse. A polynomial version of the lower bound, with $L_a = a^{-\gamma}$, is precisely the coefficient budget required for ER.2, with no separate scalar-identity input.

The Jacobian of (PN.36) in $(t_h, \hat{c})$, with the seven measurement equations first, has determinant

$$\det J = -2^7 \prod_h t_h D_h = -2^7 \prod_h \sqrt{\lambda_h D_h}. \tag{PN.40}$$

Its inverse has the explicit action

$$\delta t_h = \frac{t_h}{2\lambda_h}\, \delta F_h, \qquad \delta \hat{c} = e_a \delta t_S - \delta F_0, \tag{PN.41}$$

where δF_h and δF_0 are changes in the equation values with the data held fixed. In particular its norm is at most $C(1 + \max_h t_h)$. The scalar mean introduces no additional ill-conditioning.

Proof. Equations (PN.37)–(PN.39) follow from the exact orthogonal diagonalization and (PN.28). The last row of the Jacobian has e_a in the scalar-channel column and -1 in its last column. The first seven rows are diagonal with entries $2t_h D_h = 2\lambda_h/t_h$. This proves (PN.40)–(PN.41). □

This proposition does not supply the lower bounds in (PN.38). The existing raw compact estimate gives only

$$D_{h,a} \le C_{\mu,N} a^{-8}, \tag{PN.42}$$

by Cauchy–Schwarz and (PN.33) on the fixed normalization lobe. That upper bound is useful for finite-cutoff control but is weaker than the logarithmic upper bound required for the inverse budget. In particular, an upper moment bound on a microscopic field cannot be substituted for a lower separated reflected norm.

7. An additional microscopic nondegeneracy estimate

Proposition PN.6 (P): conditional small-ball bound

For $\beta \ge 1$, any plaquette p, any one of its links b, and the true conditional law with all other raw links fixed, there are constants $C_N, u_N > 0$, independent of the exterior, such that

$$\mathbb{P}_{\beta,n}\{\beta s_p \le u \mid U_{b^c}\} \le C_N u^{d/2}, \qquad 0 < u \le u_N. \tag{PN.43}$$

Consequently, for every $0 < q < d/2$,

$$\mathbb{E}_{\beta,n}[(\beta s_p)^{-q} \mid U_{b^c}] \le C_{N,q}. \tag{PN.44}$$

Proof. The conditional potential can be written $A_b(V) = 6N - \mathrm{ReTr}(VM)$, where M is the sum of six unitary staples, after the orientation conventions are absorbed. Thus $\|M\|_{\mathrm{op}} \le 6$. Let V_0 minimize A_b. Its first derivative vanishes, and for a unit Lie-algebra direction X,

$$\left|\frac{d^2}{dt^2} A_b(V_0 e^{tX})\right| \le 6. \tag{PN.45}$$

On a fixed identity chart this implies $A_b(V_0 e^X) \le A_b(V_0) + 3\|X\|^2$. The Haar Jacobian is bounded below there. Integrating a ball of radius $c\beta^{-1/2}$ gives the conditional partition bound

$$\int_G e^{-\beta A_b(V)}\, dV \ge c_N \beta^{-d/2} e^{-\beta A_b(V_0)}. \tag{PN.46}$$

The conditional Haar density is therefore at most $C_N \beta^{d/2}$, uniformly in the staples.

For fixed exterior, the plaquette holonomy as a function of V is a left/right translate of V or V^{-1}, so its Haar distribution is Haar. The exact identity $s(U) = \|U - I\|_{HS}^2/2$, the nondegenerate identity chart and compactness give

$$\mathrm{Haar}\{U : s(U) \le v\} \le C_N v^{d/2} \quad (0 < v \le v_N). \tag{PN.47}$$

Multiply this by the conditional density bound, with $v = u/\beta$, to obtain (PN.43). Choose $u_N \le v_N$.

Finally split the tail integral for $(\beta s_p)^{-q}$ at a fixed small u_N. The small-u part is bounded by a constant times $\int_0^{u_N} u^{d/2-q-1}\, du$, which is finite precisely for $q < d/2$; the remainder is bounded by u_N^{-q}. This proves (PN.44). □

This is microscopic energy nondegeneracy, not the separated reflected norm of a centered composite. The latter is the square of a conditional mean on the reflection-plane links. Centered local fluctuations can have positive variance and still have zero such conditional mean. Therefore neither (PN.12) nor (PN.43) proves the missing lower D_h bound.

8. What exact physical normalization gives in a common limit

Theorem PN.7 (C): the prescribed nondegeneracy survives hierarchy convergence

Suppose that along one cofinal sequence:

1. all seven $n_{h,a,T}$ are positive and the exact F01 prescription (PN.10) is used;
2. the mixed full Schwinger functions of these same fields converge on compact coincidence-flat joint tests, on one common subsequence.

Hypothesis 1 is now supplied on the slow-volume stopped sequence of Corollary FN.5 in finite reflected witnesses. Hypothesis 2 remains the physical hierarchy input; bounded-probe kernel compactness alone does not establish it.

Then the limiting one-point functions vanish and the limiting separated reflected channel measurements satisfy

$$\mathcal{N}_h(S_2) = \lambda_h(\mu) > 0. \tag{PN.48}$$

In particular no hypercubic channel disappears from the limiting field hierarchy. More precisely, after averaging the reflected Gram matrices over the finite signed hypercubic symmetry group, the Gram matrix on E is

$$\overline{G} = \sum_h \lambda_h(\mu) P_h, \qquad \langle u, \overline{G}u\rangle \geq \lambda_{\min}(\mu)\|u\|^2. \tag{PN.49}$$

Thus every nonzero tensor direction has a nonzero separated field measurement in the finite symmetry orbit of the fixed physical normalization probe.

Proof. The normalization lobe and its reflection stay a fixed positive physical distance apart. Each tensor contraction in F.22 is therefore a compact smooth joint test vanishing in an entire neighborhood of the coincidence diagonal. It belongs to the common test space in hypothesis 2. Its finite-cutoff measurement equals λ_h exactly by (PN.10), so its limit has that value. The one-point assertion follows from the exact cutoff centering.

Let G be the 21×21 reflected Gram matrix at the fixed lobe and plane. It is positive semidefinite by the original Wilson reflection positivity and passage of this finite family of joint tests. Its channel traces obey $\mathrm{Tr}(P_h G) = d_h \lambda_h$. Average $D(g)^* G D(g)$ over the finite signed hypercubic group. This positive selfadjoint average commutes with every $D(g)$. The seven sectors are the fixed inequivalent real irreducible sectors of F01. The eigenspaces of a commuting selfadjoint operator are invariant, so it is scalar on each sector, and its off-sector blocks vanish. The preserved channel traces identify the scalar as λ_h, proving (PN.49).

Equivalently, this average is taken over the physically rotated and reflected probe geometries. Their covariance follows from the exact finite hypercubic tensor law, which passes to the same limiting hierarchy. If a nonzero tensor direction vanished as a field distribution, all its symmetry-related measurements would vanish, contradicting (PN.49). This argument requires no full continuous-rotation restoration. □

The theorem proves passage of the specified nondegeneracy, not hypothesis 1 or hypothesis 2. A free curvature-bilinear hierarchy can also satisfy these positive measurements. They do not establish the interacting ultraviolet coefficient, a nonzero physical coupling in a specified continuum scheme, or the complete Yang–Mills identification.

9. The narrow remaining reflected-susceptibility input

The exact site-reflection square formula identifies

$$n_{h,a,T} = \frac{1}{d_h} \sum_{i=1}^{d_h} \left\| \mathbb{E}\big[\langle e_{h,i}, B_a(\tilde{q}_a)\rangle \mid U_\partial\big] \right\|_2^2, \tag{PN.50}$$

where U_∂ are the shared links on the two site-reflection planes. All conditional and exterior measures in this formula are the actual normalized Wilson measures. It follows from the same complete source and its actual centering, as in the reflected normalization convention.

For a vector of centered plane-measurable witnesses W, the Hilbert-space projection inequality gives a useful sufficient lower bound. If

$$V_W = \mathbb{E}[WW^*] > 0, \qquad C_{hW} = \mathbb{E}[\beta_a P_h B_a(\tilde{q}_a)\, W^*], \tag{PN.51}$$

then

$$D_{h,a} \geq \frac{1}{d_h} \mathrm{Tr}(C_{hW} V_W^{-1} C_{hW}^*). \tag{PN.52}$$

This follows by orthogonally projecting the conditional mean in (PN.50) onto the span of W; its squared norm is the displayed matrix expression. It is valid with the Moore–Penrose inverse after restriction to the actual positive covariance range.

Equation (PN.52) is a sufficient test, not a computed Wilson lower bound. Using a Gaussian covariance in place of the actual C_{hW}, or a current-scale stencil in place of the transported microscopic $B_a(\tilde{q}_a)$, would change the problem. The matrix must be estimated through the exact full source transport, including its outer conditional means.

Remaining conditions

Obligation	Status after this note
Actual scalar mean in the F01 identity direction	P: two-sided bound (PN.26), uniformly on the PE sequence.
Independent identity-coefficient budget	P: reduced exactly to the scalar channel multiplier, with no extra logarithmic exponent; (PN.27)–(PN.30).
Separate identity assumption in CS/ER's coefficient budget	P: eliminated through (PN.2) and PN.4.
Microscopic Wilson energy small-ball nondegeneracy	P: conditional (PN.43)–(PN.44); its distinct scope is explicit.
Finite normalization and inverse-Jacobian algebra	P: exact formulas (PN.36)–(PN.41); quantitative data bounds remain required.
Strict positivity of all seven actual measurements on one slow-volume PE sequence	P in subsequent Corollary FN.5: the actual finite normalization and determinant are nonsingular on that sequence.
Nondegeneracy of a common physically normalized hierarchy limit	C: proved under the two explicit hypotheses of PN.7.
Uniform logarithmic upper/lower bounds for the seven positive reflected data	O: these remain the quantitative actual-model normalization estimates.
Actual seven one-step mixing coefficients and summable matrix remainder	O: F03/F04 are not established by the source-free C05 estimate.
Prescribed physical B05 endpoint and full state compatibility	O: the stopped emergent positive endpoint alone does not establish them.
ER.7 for the recent separated full microscopic hierarchy	O: the source estimate remains necessary; only its identity-budget input is simplified here.

Retained sigma-fields and conditional expectations

These estimates concern current-rooted derivatives on a specified conditional law. Use the retained coordinates and Haar factors, including clause (2″), throughout. The purely native derivatives, retained-data derivatives, and current-independent outer density are separate. For collared box estimates the two-layer constraint is required. A statement about a single holed record does not control overlapping records at a T step; the collective crossing estimate is open. See external branches.

Conditional coordinates. Clause (2″) in Retained coordinates specifies all variables integrated, including the completed preparation stages and auxiliary gauge variables. On a relocated bond the log Jacobian is combined with the Haar density ratio.

3. The σ-fields, stated precisely

Setting. Fix a finite cutoff and an admitted geometry. Let $(\Omega, \mathfrak{F}, \mathbb{P})$ be the positive path space of Annex W §4.1, built on the main-line representation (F1–F6). It is standard Borel, and every variable ever introduced (fine field, labels, gauge variables, fresh reference variables) is a coordinate. Order the elementary operations chronologically as sub-steps $s = 1, 2, \ldots$: each T step, each R step, each label exposure and each fresh sampling. Let $\mathfrak{F}_s$ be the σ-field of all variables created up to and including sub-step s. After sub-step s, let:

- C_s: the current field and the exposed labels, i.e. the data (h, V) of Annex W (4.2);
- P_s: every variable introduced by sub-step s that the main-line representation has not yet integrated. This includes S-a, S-c and S-d variables, the parts of stopped preparations that are not yet integrated, and fresh variables still present;
- $T_s \subset P_s$: the hidden variables of second-class components (S-c and S-d).

Define

$$\mathcal{G}_s^{\star} = \sigma(C_s, P_s), \qquad \mathcal{G}_s^{B} = \sigma(C_s, P_s \setminus T_s), \qquad \mathcal{G}_s^{\alpha} = \sigma(C_s),$$

so that $\mathcal{G}_s^{\alpha} \subset \mathcal{G}_s^{B} \subset \mathcal{G}_s^{\star}$. [Boundary estimate e]'s Fork A is $\mathcal{G}_s^{A} = \sigma(C_s, P_s \setminus T_s^{\mathrm{new}})$, where T_s^{new} are the hidden variables of tags born at the R step immediately before the T step s. So $\mathcal{G}_s^{B} \subset \mathcal{G}_s^{A} \subset \mathcal{G}_s^{\star}$ at that T step, and $\mathcal{G}_s^{A} = \mathcal{G}_s^{\star}$ at all other steps.

The one-step law at a T step s is the regular conditional law of the current free coordinates z_s given the chosen σ-field. Under $\star$ it is the law of z_s given $\mathcal{G}^{\star}$, where $\mathcal{G}^{\star}$ contains everything except z_s. The transport of a mark is $F_s = \mathbb{E}[F_{s-1} \mid \mathcal{G}_s], F_0 = \Phi$.

Premise ($\mathrm{Mem}^{\mathcal{G}}$). For every s: (i) $\mathcal{G}_{s+1} \subset \mathcal{G}_s \vee \sigma(\xi_{s+1})$, where ξ_{s+1} is the tuple of new auxiliary variables (labels, fresh fields, gauge variables) introduced at $s+1$; and (ii) the conditional law of ξ_{s+1} given $\mathfrak{F}_s$ is a function of the $\mathcal{G}_s$-data. In words: every partition of unity, reference kernel and gauge average used at the next sub-step is evaluated on retained data only.

- For $\mathcal{G}^{\star}$, (i) holds by construction. Blocking is a function of the current field. New postponed variables are restrictions of the current field or fresh variables. Consumed variables leave the state. (ii) is the statement that the source applies every operation to an integrand that is a function of not-yet-integrated variables only ([III] (2.18), F1, F6). That is [RB]'s open item (M), restricted to what the source integrates. source input; additional assumption, to be checked on LFR II pp.365–379.
- For $\mathcal{G}^{B}$, (ii) additionally requires that no later kernel uses tag hidden variables. This is plausible. The reference kernel uses the constrained filling V_{Λ}, which LFR I Prop. 1 makes a function of raw data ([BI] l.294, assumed input). Preparations act on levels $> h$ and leave the deep factor untouched ([C05U] l.3189, assumed input). additional assumption
- For $\mathcal{G}^{\alpha}$, (ii) fails in the actual construction. The coarse-field selector P_{j+1} tests "the local minimizing background determined by the coarse field" ([MS] l.13880). That background is $U_{\mathrm{B}}(V)$, which depends on postponed belts (F1). The preparatory R tests are tests on postponed old variables ([MS] l.13884; LFR I p.178 "we try to do a part of the integrations in the operations $\mathrm{T}^{(j)}(Z_j)$"). source for the inputs, additional assumption that the dependence is genuine.

4. Measure-theoretic relations (proved)

Throughout, $\mathcal{G} \subset \mathcal{G}'$ are sub-σ-fields of a standard Borel probability space, and Z is a random variable in $\mathbb{R}^I$.

Lemma 4.1 (coarse law = mixture of fine laws). Let $\nu'(\cdot, dz)$ and $\nu(\cdot, dz)$ be regular conditional laws of Z given $\mathcal{G}'$ and $\mathcal{G}$. Then $\nu(\cdot, A) = \mathbb{E}[\nu'(\cdot, A) \mid \mathcal{G}]$ a.s. for every Borel A. If $\mathcal{G}' = \mathcal{G} \vee \sigma(\eta)$ and $\nu'(dz) = p'(z;\eta)\, dz$, then $\nu(dz) = p(z)\, dz$ with $p(z) = \int p'(z;\eta)\, \kappa(d\eta)$, where $\kappa = \mathbb{P}(\eta \in \cdot \mid \mathcal{G})$.

Proof. For $B \in \mathcal{G}$, $\mathbb{E}[1_B 1_{Z\in A}] = \mathbb{E}[1_B\, \mathbb{E}(1_{Z\in A} \mid \mathcal{G}')] = \mathbb{E}[1_B \nu'(A)]$. Conditioning ν' on $\mathcal{G}$ disintegrates over κ. □

Proposition 4.2 (exact transport needs (Mem)). Let $\mathcal{G}_s$ satisfy (Mem). Then $F_s := \mathbb{E}[F_{s-1} \mid \mathcal{G}_s]$, $F_0 = \Phi \in L^1(\mathfrak{F}_0)$, satisfies $F_s = \mathbb{E}[\Phi \mid \mathcal{G}_s]$ for all s. Without (Mem) one still has $\mathbb{E}F_s = \mathbb{E}\Phi$, but in general $F_s \neq \mathbb{E}[\Phi \mid \mathcal{G}_s]$.

Proof. Induction. Since ξ_{s+1} is conditionally independent of $\mathfrak{F}_s$ given $\mathcal{G}_s$ by (Mem)(ii), and $\Phi \in \mathfrak{F}_s$, we have $\mathbb{E}[\Phi \mid \mathcal{G}_s \vee \sigma(\xi_{s+1})] = \mathbb{E}[\Phi \mid \mathcal{G}_s]$. Then (Mem)(i) and the tower property give $\mathbb{E}[\Phi \mid \mathcal{G}_{s+1}] = \mathbb{E}\big[\mathbb{E}[\Phi \mid \mathcal{G}_s] \mid \mathcal{G}_{s+1}\big]$. The expectation identity is the law of total expectation. □

Example 4.3 (lift α is not a transport filtration). Let $X \in \{0,1\}$ be fair, postponed, and the only variable. Let $\mathcal{G}_s^\alpha$ be trivial. Let the label at $s+1$ be $\xi = X$, exposed by the partition of unity $1 = 1_{X=1} + 1_{X=0}$, and $\mathcal{G}_{s+1}^\alpha = \sigma(\xi)$. Take $\Phi = X$. Then $\mathbb{E}[\Phi \mid \mathcal{G}_s^\alpha] = \frac{1}{2}$ and $\mathbb{E}[\frac{1}{2} \mid \mathcal{G}_{s+1}^\alpha] = \frac{1}{2}$, but $\mathbb{E}[\Phi \mid \mathcal{G}_{s+1}^\alpha] = X$. With $\mathcal{G}_s^\star = \sigma(X)$, (Mem) holds and the chain is exact.

Consequence (deduction; given §3). α-marks $\mathbb{E}[\Phi \mid C_s]$ are well-defined, as projections $\mathbb{E}[F_s^\star \mid \mathcal{G}_s^\alpha]$. They cannot be propagated by α-steps alone. Annex I makes an α-type chain exact by resampling every postponed variable at every step (Theorem 2.1, Cor. 2.2). The price is a reset cumulant response at every step, which Annex I says "can be nonlocal" and which no proposition estimates.

Proposition 4.4 (retain vs average-and-resample). Let the future after sub-step s be generated from (W, η) and new auxiliary variables by fixed kernels, where W is the $\mathcal{G}^B$-type data at s and η are tag variables. Draw $\eta' \sim \mathbb{P}(\eta \in \cdot \mid W)$ independently, and generate the future from (W, η') by the same kernels. Then the joint law of $(W, \eta', \text{future})$ equals that of (W, η, future). In particular, every subsequent conditional law given $\sigma(W, \eta', \dots)$ has the same form as the one given $\sigma(W, \eta, \dots)$.

Proof. $(W, \eta') \overset{d}{=} (W, \eta)$ by construction, and the future is the same measurable function of them and of independent noise. □

Application (conditional; given (Mem*) and [BI] §8). At an R step, H-03's π_υ conditions on $\upsilon \supseteq (U_k, V_k, \text{Zbar data, Y geometry})$. Everything later depends on (υ, η_Y). So "average η_Y at birth and resample" (A1) and "retain η_Y from birth" (A2) give the same joint law and identical T-step laws. They differ only in the transported mark: A1 averages over η_Y at birth; A2 keeps the dependence until consumption.

Proposition 4.5 (what transfers from fine $\mathcal{G}'$ to coarse $\mathcal{G}$). In the notation of Lemma 4.1:

1. Event and label probabilities. $\mathbb{P}(E)$ does not depend on the conditioning, and $\mathbb{P}(E \mid \mathcal{G}) = \mathbb{E}[\mathbb{P}(E \mid \mathcal{G}') \mid \mathcal{G}]$. Joint bounds of the H-02 form $\mathbb{P}(\bigcap_i E_i) \le \prod a_i$ are statements about $\mathbb{P}$ and are identical in every lift. Annex W (4.2) gives $\mathbb{P}(H_k = h) = Z_{k,h}/\mathcal{Z}$ in every lift.
2. Values and moments. $\mathbb{E}_\nu f = \mathbb{E}[\mathbb{E}_{\nu'} f \mid \mathcal{G}]$. Any bound of the form "$\mathbb{E}_{\nu'} f \in K$ a.s." with K closed and convex (sup bounds, L^p moment bounds, regulator-relative bounds with a $\mathcal{G}$-measurable regulator) passes to ν.
3. First derivatives at real points. If $p'(\cdot;\eta) > 0$ is C^1, then $\partial_i \log p(z) = \int \partial_i \log p'(z;\eta)\, \kappa_z(d\eta)$ with $\kappa_z \propto p'(z;\cdot)\kappa$. Hence $|\partial_i \log p(z)| \le \sup_\eta |\partial_i \log p'(z;\eta)|$. This is the only sense in which "derivative bounds transfer by sup".
4. Second derivatives. $\partial_i\partial_l \log p = \mathbb{E}_{\kappa_z}[\partial_i\partial_l \log p'] + \mathrm{Cov}_{\kappa_z}(\partial_i \log p', \partial_l \log p')$. The covariance matrix is positive semidefinite, so (i) $-D^2 \log p \preceq \mathbb{E}_{\kappa_z}[-D^2 \log p']$: coercivity can only be lost; and (ii) a weighted Hessian-row bound for p needs a row bound on the covariance. That is the two-attachment term of [Rep21] (R29) and [Boundary estimate d] §1.4, and it is not implied by row bounds for p'. Counterexample to transfer of coercivity: $p'(z;\eta) = \phi(z-\eta)$, $\eta = \pm a$ with probability 1/2 each, gives $-(\log p)''(0) = 1 - a^2 < 0$ for $a > 1$.
5. Complex domains. Zero-freeness of $p'(\cdot;\eta)$ on a polydisc does not give zero-freeness of p. [Boundary estimate e] Example N2 (checked there) exhibits a zero for kernels of uniformly small variation. Cauchy-type derivative bounds for $\log p$ on complex polydiscs therefore do not transfer.
6. Cumulants. By the law of total cumulance, $\kappa_\nu(f_1, \dots, f_r) = \sum_\pi \kappa_{\mathcal{G}}\big(\kappa_{\nu'}(f_B)_{B\in\pi}\big)$. Spatial decay of the ν'-cumulants gives nothing for the $\pi \neq \{\text{one block}\}$ terms, i.e. for the outer cumulants of conditional means.

7. Retained-coordinate differences (Annex Y (M.4)). For a $\mathcal{G}$-coordinate y, $\delta_y \mathbb{E}[F \mid \mathcal{G}] = \mathbb{E}[\delta_y F \mid \mathcal{G}] +$ (score term from the change of κ in y), cf. [BI] BL (J^{new}). Difference bounds do not transfer by sup unless κ does not depend on y.
8. Coarse → fine. Nothing transfers in general: ν' can be singular with respect to ν.

Proof. 1–2: Lemma 4.1 and Jensen. 3–4: differentiate $\log \int p' \kappa$ under the integral. 5: the cited example. 6: the standard partition formula for conditional cumulants. 7: product rule for the quotient $\int F\, p'\, d\kappa / \int p'\, d\kappa$. 8: take $\eta = Z$. □

Proposition 4.6 (pinning preserves a Dobrushin condition) (a abstract; b for H-03). Let a finite specification on coordinates I have total-variation influence coefficients c_{ij} on admissible pairs, with $\sup_i a_i^{-1} \sum_j c_{ij} a_j e^{\theta d(i,j)} \le \varepsilon < 1$. For $J \subset I$ and an admissible pinned value on J, the conditional law on $I \setminus J$ has the same single-site conditionals. Its influence matrix is $c|_{(I\setminus J)^2}$, whose weighted row sums are at most ε.

Proof. Single-site conditionals given all other coordinates do not change when some of those coordinates are additionally fixed. Restricting a nonnegative matrix reduces row sums. □

Application conditional. The proofs of HS20.3–HS20.4 use (HS20.13), (HS20.16) and (HS20.20). (HS20.13) holds for every admissible outside configuration, including fixed Y states. The proof of HS20.3 already "Condition[s] on the J coordinates" ([BI] l.546). So HS20.3–HS20.4 hold for $\pi_v(\cdot \mid \eta_Y)$, the law needed in variant A2, with the same constants. The posterior disintegration and the corresponding closure-bound properties are required as inputs.

Ordinary stopped-density oscillation on mixed scales

Composed-response hypothesis (CR). Each complete boundary conclusion below assumes the displayed native-chart and spatial response estimates for the actual iterated reconstruction, with the prescribed earlier innovations and raw variables fixed, uniformly over the admitted histories. A sufficient estimate for the latent contribution is (A.8) in Conditional-law estimates and the continuum application, refined for the anchored exterior derivative in PG.18–PG.20. The two endpoint bounds concern the identical Wilson-subtracted primitive, its actual cutoff and its independently extended current. A direct minimizing-field derivative alone does not establish this hypothesis. The application to every required original Wilson history remains open. Complete boundary estimates and their C06/C08 applications in this chapter are conditional on CR and the other displayed source, AF, domain and support assumptions. Exact density identities and abstract inequalities retain their individually stated hypotheses.

In the projective density formulation, a sufficient comparison is

$$\sup_{\zeta,x,a}\left|\log\frac{W_{\zeta}(x,a)W_0(x,a_*)}{W_0(x,a)W_{\zeta}(x,a_*)}\right|\le\delta_r<\pi/2,$$

on the stipulated chart with the required uniform rates and actual fixed support. The composed ordinary response is the unresolved input in its derivation. The exact partial-gauge transport and the surviving fixed-exterior derivative are given in the chapter Gauge covariance of partial integrals and exterior scores, equations PG.2–PG.6 and the subsequent anchored-score criterion.

1. Evaluation age and forcing age

The boundary-induction argument's OR.14–OR.16 did not justify OR.19. An ordinary term has a birth index b, an actual evaluation layer n, and an integration stage j. On an interrupted history, $b\le n<j$ is possible. CMP119 (3.56)–(3.67) gives an irrelevant remainder measured by $n-b$. It does not replace that exponent by $j-b$.

Write

$$a=n-b,\qquad c=j-n,\qquad \lambda=5-\vartheta>4. \tag{SR.1}$$

The missing calculation cannot be supplied by performing the convergent sum OR.15 again. Nor does the physical-input/native-radius conversion RL.25 alone supply spatial decay. The decay comes from the actual determining-data response and the retained carrier joining its attachment to the term's root.

This repair uses a weaker sufficient estimate. It obtains two bounds on the identical finite difference and interpolates their minimum. The first uses the Ward cancellation at the two endpoints. The second uses a direct, deliberately crude response bound, for which the E term and its own Wilson term may be estimated separately. There is no differentiation of a Ward identity asserted only on the physical-current graph.

The boundary-induction argument's final-mass restoration, moving-normalizer calculation, and local quadratic cumulant estimate are not changed by this argument. Their application still requires the ordinary bound proved here.

2. Keep the source object and its cutoff exactly

At fixed birth b, the inherited source expression is

$$\mathcal{E}_b(U)=E^{(b)}(\Lambda_b,U)-E^{(b)}(\Lambda_b,1)-\beta_bA(\phi_b,U). \tag{SR.2}$$

The coefficient β_b, cutoff ϕ_b, domain Λ_b, and primitive bank are fixed during a preparatory prefix. The prefix creates the additional C bank; it does not replace the ordinary E bank by a newly projected one. This is the content of LFR I (1.60)–(1.63).

CMP119 (2.26)–(2.29) and (3.65) give the exact partition

$$\begin{aligned}
\mathcal{E}_b(U) &= \sum_{z\in\Lambda_b^0} \mathcal{R}_{b,z}(U) + \mathcal{B}_b^{\mathrm{cut}}(U),\\
\mathcal{R}_{b,z}(U) &= \sum_{X\ni z,\ X\subset\Lambda_b} E^{(b)}(X,U,z) - E^{(b)}(\Lambda_b,1,z) - \beta_b A(h_z,U),\\
\mathcal{B}_b^{\mathrm{cut}}(U) &= -\beta_b \sum_{z\notin\Lambda_b^0} A(h_z\phi_b,U).
\end{aligned} \tag{SR.3}$$

Here $h_z\phi_b = h_z$ for $z \in \Lambda_b^0$. The last line must be retained. In particular, an interior root estimate must not silently delete the remaining cutoff terms.

The marginal cancellation is a statement about the complete root aggregate $\mathcal{R}_{b,z}$. The Euclidean tensor reduction uses the summed primitive catalog. It is not a cancellation available for each absolute X-summand. CMP119 (3.61)–(3.66) identifies its coefficient with the same β_b in (SR.2).

Both estimates below are applied to

$$\Delta_j\mathcal{R}_{b,z} = \mathcal{R}_{b,z}(U^1) - \mathcal{R}_{b,z}(U^0), \tag{SR.4}$$

with the actual two stopped configurations and the earlier raw variables fixed. No subtraction, reference field, or cutoff is changed between the bounds.

3. An elementary response interpolation theorem

Let S_j be the actual forcing cells in a stage-j strip. Fix $0 < \eta < 1$ and put $u = 1 - \eta$. Suppose the same difference (SR.4) satisfies

$$\begin{aligned}
|\Delta_j\mathcal{R}_{b,z}| &\le C_A E_0 \ell_r^{D_A}(1+a)^{s_A}L^{-\lambda a} =: A_{b,n},\\
|\Delta_j\mathcal{R}_{b,z}| &\le C_B E_0 \ell_r^{D_B}(1+a+c)^{s_B}e^{-\Gamma c}\sum_{s\in S_j} w_j(z,s)^{1/u} =: B_{b,n,j,z}.
\end{aligned} \tag{SR.5}$$

Normalize the spatial weights by a fixed constant so that $0 \le w_j \le 1$, and choose their decay with sufficient reserve that

$$\sum_{z \text{ at birth mesh } b \text{ and layer } n} w_j(z,s) \le C(1+a+c)^{s_0}L^{4(a+c)}. \tag{SR.6}$$

The four-dimensional density is explicit. It is not hidden inside a source row that has already included it.

Since $\min(A,B) \le A^\eta B^u$ and $(\sum_s v_s)^u \le \sum_s v_s^u$, (SR.5) implies

$$|\Delta_j\mathcal{R}_{b,z}| \le CE_0\ell_r^D(1+a+c)^s L^{-\eta\lambda a}e^{-u\Gamma c}\sum_{s\in S_j} w_j(z,s). \tag{SR.7}$$

Take $D \ge \max(D_A, D_B)$, increasing a fixed polynomial exponent if necessary. Sum first over the literal roots using (SR.6), and then over the two ages:

$$\begin{aligned}
\sum_{b\le n\le j}\sum_z |\Delta_j\mathcal{R}_{b,z}| &\le CE_0\ell_r^D|S_j|\,\mathcal{S},\\
\mathcal{S} &= \sum_{a,c\ge 0}(1+a+c)^s L^{-(\eta\lambda-4)a}e^{-[u\Gamma-4\log L]c}.
\end{aligned} \tag{SR.8}$$

This sum is finite if

$$\eta\lambda > 4, \qquad (1-\eta)\Gamma > 4\log L. \tag{SR.9}$$

Such an η exists precisely when

$$\Gamma > \frac{4\lambda}{\lambda-4}\log L. \tag{SR.10}$$

For example, choose

$$\vartheta = \tfrac{1}{2}, \quad \lambda = \tfrac{9}{2}, \quad \eta = \tfrac{11}{12}, \quad \Gamma \geq 60\log L. \tag{SR.11}$$

The summand in (SR.8) is then bounded by $(1+a+c)^s L^{-a/8}L^{-c}$. Factor $1+a+c \leq (1+a)(1+c)$ to see convergence directly. All constants are independent of the total history depth and volume.

This proof does not assign a Ward bound to each marked forcing derivative. The Ward bound applies once to the complete two-endpoint difference. Only afterwards does subadditivity distribute its majorant over the forcing cells. It also does not retain the original spatial rate for free: the response bound must reserve $w^{1/u}$, since interpolation leaves w.

4. Source domains: a fixed reserve before any estimate

The interpolation theorem is unconditional mathematics. Its Wilson application requires actual endpoint and ordinary-response admission. The following reserve records the intended source application explicitly.

Use a fixed external/map profile and a larger ordinary primitive profile,

$$\gamma_n = g_n \ell_n^{q+Q}, \qquad \alpha_n = g_n \ell_n^{q+Q+D_{\mathrm{prep}}+2}. \tag{SR.12}$$

Here D_{prep} is fixed before Q; it bounds the finite preparation and top-band losses, including $L_0^{2N_0}$. It does not depend on the number of derivatives or the history depth. The source exponents, map/localizer parameters, and common coupling ceiling are then selected once. This reserve concerns the ordinary source family, not an enlargement of arbitrary completed relative inputs.

PF.1–PF.3 in Annex AV of the retained theorem group prove that a fixed strong native core is contained in every modified prefix domain, including the independent-current and reconstructed-current clauses. They use the actual mixed determining sets in LFR I (1.14)–(1.17). The source's forward insertion statement on p.191 supplies the additional step from a later prefix chart to the earlier one. Mere inclusion of the input core would not by itself prove a bound on every possible modified output.

For the actual outputs, the displayed modified norms give a bound $C\ell_r^{D_{\mathrm{prep}}}\gamma_n$. The stipulated AF comparison yields $\ell_n \geq \ell_r - C$ for $n \leq r$. Consequently

$$\frac{C\ell_r^{D_{\mathrm{prep}}}\gamma_n}{\alpha_n} \leq \frac{C'}{\ell_n^2}, \tag{SR.13}$$

after reducing the common coupling ceiling. Thus the actual endpoint jets lie strictly within the larger ordinary native profile. The C02/E14 admission calculations, including E14.26–E14.32 and E14.40–E14.48, then use this reserve for the genuine minimizing map, local gauges, and independent-current substitution.

For an ordinary interpolation between the two allowed input tuples, retain the exact constraint correction $g_j CB_j - hD(g_j CB_j)$. The normalized input is $g_j P_j$, not P_j. With $c = j-n$, the full smooth determining-data row gives a bound of the form

$$C\ell_r^{D_{\mathrm{prep}}}\frac{g_j P_j}{\alpha_n}(1+c)^{\upsilon}e^{-\gamma c} \leq \frac{\beta}{8}\,2^{-c}, \tag{SR.14}$$

by the fixed profile reserve, a sufficiently large fixed geometric rate, and the common weak-coupling ceiling. Summing these contributions preserves a fixed fraction of the dyadic margins in LFR I (1.64)–(1.69). This is ordinary primitive analyticity; it does not assert Ward cancellation at every point of that interpolation.

The independent current is carried by the actual source extension and its complete current-output row. It is not replaced by a physical current off the physical slice. Likewise, no bare fine-link tangent is substituted for the admitted collection of field, gradient, projection, Laplacian, and reconstructed-current norms.

For the problematic stratum $n < j$, the determining sets below j are unchanged by the stage-j integration: LFR I (1.15)–(1.17) retain their already-prepared lower shells. Thus the two endpoints have a common lower evaluation layer. The changing layer $n = j$ has no missing age. Coarser outputs $n > j$ keep their true mesh and spatial cover and are treated separately in Section 8.1; they are not relabeled as grade j. A stopped component is removed as a whole from subsequent integrations; it is not silently assigned the grading of a still-active component.

5. The endpoint bound keeps the cancellation and cutoff error

At either actual endpoint, apply the source's nearby/far split at its true layer n. The far-carrier estimate is CMP119 (3.48). For the nearby part, CMP109 (3.28)–(3.36), Lemma 4, and CMP119 (3.49)–(3.67) give the Ward expansion with the same Wilson subtraction.

The chosen analytic extension matters. CMP109 carries the independent current through its analytic gauges, then permits the regular complex insertion A of (3.31). The C02/E14 calculations track this prescribed pullback and its running radii. They do not appeal to uniqueness of an extension from a physical-current graph.

The coefficient-order decomposition retains an order p denominator α_b^{-p} together with field degree at least p, for $2 \le p \le 5$. The full-lattice marginal quadratic coefficient cancels with its own β_b. The ordinary irrelevant remainder has geometric factor $L^{-\lambda(n-b)}$, but a restricted-bank coefficient error must also be retained if a prepared local patch crosses the old cutoff. The AF profile comparison gives

$$\left(\frac{\alpha_n}{\alpha_b}\right)^p \le C_Q(1+n-b)^{p/2}. \tag{SR.15}$$

Fixed preparation powers can be included in $\ell_r^{D_A}$, with D_A independent of Q. No claim of $j-b$ Ward decay is made.

5.1 Ancient births: the true layer gain

For $b < h$, the preserved inclusion $\Omega'_{b+1} = \Omega_{b+1} \subset \Lambda_b$ implies that a root at prepared layer n is separated from Λ_b^c by a fixed multiple of h_n, except for harmless adjacent-layer cases. The same graded admissibility used in (SR.17) proves this separation. A primitive carrier omitted in extending the coefficient sum to the full lattice therefore has $d_b(X) \ge cL^{n-b}$. Its reserved full-tree decay is bounded by any fixed irrelevant power. Thus both endpoints satisfy the first line of (SR.5), with their exact restricted bank and own Wilson subtraction.

5.2 Recent births: retain the truncated-tensor error

For $h \le b \le n \le j$, the preparation can expand a determining domain through the old Λ_b cutoff. It would be incorrect to assert that the preceding separation always holds. Instead, stop the source coefficient calculation at (3.56), and keep the error in extending the restricted X-sum to the full-lattice tensor in (3.57).

Since $z \in \Lambda_b^0$, any omitted carrier has to cross the original MR_b interior buffer. Reserving a share of its literal full-tree decay gives $e^{-\kappa_{\text{tail}}R_b}$. The tensor is calculated at the flat reference $(1,0)$, where these primitive coefficients are defined. No evaluation of a missing primitive at a field outside its admitted domain is needed. The retained coefficient-order/field-degree decomposition bounds its field factors by fixed age polynomials through (SR.15). Consequently the actual two-endpoint bound is

$$|\Delta_j \mathcal{R}_{b,z}| \le C_Q E_0 \ell_r^{D_A}(1+a)^{s_A}\left[L^{-\lambda a} + e^{-\kappa_{\text{tail}}R_b}\right]. \tag{SR.15a}$$

The counterterm remains the original β_b. The second term measures the part of the finite-domain tensor not canceled by that coefficient; it is not a newly selected Wilson subtraction.

The response bound in Section 6 is for the entire original difference. Thus it also handles (SR.15a) without a separate response theorem for a manufactured error term. For positive numbers,

$$\min(A_1 + A_2, B) \le (A_1 + A_2)^{\eta} B^{1-\eta} \le (A_1^{\eta} + A_2^{\eta}) B^{1-\eta}. \tag{SR.15b}$$

The first summand gives (SR.8). For the second, $a \le N_r$, and the recent-window AF comparison gives $R_b \ge c_R R_r$, with one fixed $c_R > 0$. LFR II p.361, immediately after (1.20), requires $N_r \le R_r$; retaining a fixed endpoint/rounding factor, write $N_r \le C_N R_r$. The retained waiting-depth choice on p.384 is compatible with this relation. There is no substitution of a shorter preparation window.

After the full root count, the extra contribution is bounded by

$$\begin{aligned} CE_0 \ell_r^{D} |S_j| (1 + N_r)^{s+1} \exp[4N_r \log L - \eta \kappa_{\mathrm{tail}} c_R R_r] \\ \times \sum_{c \ge 0} (1 + c)^{s} e^{-[(1-\eta)\Gamma - 4\log L]c}. \end{aligned} \tag{SR.15c}$$

Choose the remaining carrier-tail rate, after allocating the spatial and entropy shares, so that

$$\eta \kappa_{\mathrm{tail}} c_R > 4 C_N \log L + \epsilon, \qquad \epsilon > 0. \tag{SR.15d}$$

The source permits the ordinary carrier exponent to be fixed sufficiently large, with the larger block/localization parameters and the weak-coupling ceiling chosen afterwards. This is an explicit strengthened fixed rate inequality. It changes neither the real cutoffs nor the preparation depth nor the class of histories at those parameters. The expression (SR.15c) is bounded by a fixed polynomial times $e^{-\epsilon R_r}$.

Thus the recent restricted-bank error is summable. A finite number of births alone would not have paid $L^{4(n-b)}$; the retained cutoff buffer and (SR.15d) are essential.

6. The crude bound keeps the response

For this second bound, work with the original $E^{(b)}(X, U, z)$, before any nearby minimizing-map replacement. Ordinary native Cauchy gives its differential at cost $CE_0 e^{-\kappa d_b(X)}/\alpha_b$. Compose with the actual response row and keep the physical forcing $g_j P_j$.

For an attachment at native grade $m \ge b$, the source conversion for an output of physical order $p \ge 1$ contains $(h_b/h_m)^p$. In particular there is no positive power of L^{m-b}. The ratio

$$\frac{g_j P_j}{\alpha_b} \le C_Q \ell_r^{D_1} (1 + j - b)^{\upsilon_1} \tag{SR.16}$$

has no uncanceled inverse coupling. The fixed logarithmic degree here is independent of the chosen enlargement exponent Q.

6.1 The response must be rooted at the same z

The source row initially reaches an attachment y of grade m, which may differ from the root's grade n. If $m \le n + 1$, the row pays $j - n$, up to a fixed endpoint factor. If $m \ge n + 2$, the literal full carrier contains a path from the root z to y. Graded admissibility gives

$$\mathrm{dist}(z, y) \ge c M_1 h_{m-1}, \qquad d_b(X) \ge c' L^{m-n}, \tag{SR.17}$$

where fixed block ratios and endpoint conventions are absorbed in c'. Indeed $z \notin \Omega_{n+1}$ and $y \in \Omega_m$, so the interlayer separation may be applied at layer $m - 1$. The same argument uses the actual prepared determining domains.

For every fixed required $G > 0$,

$$e^{-\kappa' d_b(X)} \le C_G e^{-G(m-n)} \tag{SR.18}$$

using one reserved share of the full carrier. The remaining row pays $(j-m)_+$. Together they pay $j-n$; remaining positive shares are kept for spatial summation and carrier entropy. This argument uses the ordinary full E carrier, not a contracted quotient tree with an artificial root.

Choose the source geometric constants so that the remaining age rate is at least Γ in (SR.11), and keep enough spatial decay for the power $w^{1/u}$ in (SR.5). CMP119 fixes M_1 and $M_2 = L^2 M_1$ before choosing M large; LFR I (1.47) supplies the rate proportional to M/M_1. The parameter choice is fixed before the weak-coupling ceiling and never depends on a history's depth.

The original source measure factors in the normalized reconstruction row must be paired before counting roots. Summing the primitive catalog with the remaining full-tree reserve yields the E part of the second line of (SR.5).

6.2 The Wilson term has two curvature factors

The birth coefficient satisfies

$$|\beta_b| \le C E_0 \alpha_b^{-2}. \tag{SR.19}$$

The Wilson plaquette action has a quadratic zero at the identity. Its difference contains one background-curvature factor and one response-curvature factor. On a birth cell whose actual evaluation layer is n, the fine-plaquette count cancels their physical mesh factors, giving

$$|\beta_b \Delta_j A(h_z \phi_b, U)| \le C E_0 \frac{\alpha_n}{\alpha_b} \frac{g_j P_j}{\alpha_b} L^{-4(n-b)} e^{-\Gamma(j-n)} \sum_{s \in S_j} w_j(z,s)^{1/u}, \tag{SR.20}$$

up to the already retained fixed profile and age polynomials. The two radii in the denominator are canceled by the two physical field factors. Replacing β_b by a purported uniform constant would hide the calculation.

For an interior root, discard the favorable $L^{-4(n-b)}$ in this crude estimate and combine it with the E response. This proves the second line of (SR.5) for the same complete difference (SR.4). The Ward estimate is not used in this calculation.

7. The remaining cutoff terms

Formula (SR.3) still contains the literal cutoff remainder. It is handled by (SR.20), retaining its four curvature powers. For a cutoff label outside Λ_b^0, choose an actual support anchor in $\operatorname{supp}(h_z \phi_b) \subset \Lambda_b$; the label itself may lie just outside Λ_b. This changes the root count by a bounded multiplicity and ensures $n \ge b$. The local Wilson stencil has comparable native grades by the same interlayer separation.

Put $h = r - N_r$, the lower end of the current preparation window.

7.1 Ancient births $b < h$

LFR I (1.11) and (1.15)–(1.17) leave the determining layers through h unchanged. A cutoff support anchor lies within $CMR_b h_b$ of a point outside the literal Λ_b. By $\Omega_{b+1} \subset \Lambda_b$, that point belongs to an unchanged lower-grade region. CMP102 p.277, condition (1), applied to the prepared admissible sequence therefore gives

$$L^{n-b} \le C R_b. \tag{SR.21}$$

This is a logarithmic allowance, not an assertion that $n = b$. If an anchor has $n \ge b+2$, compare it to that outside point using the separation between $(\Omega'_{n-1})^c$ and Ω'_n; the cases $n = b, b+1$ are absorbed in the constant.

The factor L^{-4a} in (SR.20) cancels the birth-to-layer portion L^{4a} of the root density. Use

$$1_{\{L^a \le CR_b\}} \le CR_b L^{-a}, \qquad R_b \le C\ell_r^{\rho}(1+a+c)^{\rho}. \tag{SR.22}$$

The second inequality follows from the stipulated AF comparison and the fixed logarithmic cutoff profile. The resulting old-birth sum is bounded by

$$CE_0\ell_r^D|S_j| \sum_{a,c\ge 0} (1+a+c)^s L^{-a} e^{-[\Gamma - 4\log L]c} < \infty. \tag{SR.23}$$

No Ward estimate for an artificial boundary root is used.

7.2 Recent births $b \ge h$

There are at most $N_r + 1$ such births in the current bank. For $b \le n \le j$, retain the L^{-4a} in (SR.20); it cancels the a-density exactly. The remaining c-sum converges for $\Gamma > 4\log L$, and the number of possible a values costs at most a fixed polynomial in N_r.

If $b > j$, keep the entire coarse birth atom. An actual stage-j cell intersects only a bounded number of coarser birth cells; a fictitious factor $L^{4(b-j)}$ is not inserted. The same-profile Wilson bound is CE_0, and the ordinary primitive bound is CE_0, with the actual response and its spatial row retained. Again only the finite window is counted.

At the start of a new attempt these are the current canonical cutoffs, not every historical cutoff ever consumed. LFR II p.377 reconstructs the new ordinary action and subtracts the same added terms into the remainder; p.390–391 returns the full source induction hypothesis. The ancient-layer argument is then applied within the new attempt's literal geometry.

8. Replacement for OR.19

8.1 Coarser outputs $b \le j < n$

This case is different from a recent birth $b > j$. Retain the actual output cover at grade n. A forcing cell at grade j belongs to only a bounded number of compatible coarser n-cells. The remaining native spatial row, at the post-interpolation rate, has a bounded sum over those output cells. The birth-root count in one such cell is $L^{4(n-b)}$, not $L^{4(j-b)}$.

Put $a = n - b$. The endpoint estimate still uses $L^{-\lambda a}$, while the crude response has no required missing-age factor. The identical min calculation, now with the spatial kernel normalized on the actual n-cover, leaves

$$L^{4a}L^{-\eta\lambda a} = L^{-(\eta\lambda - 4)a}. \tag{SR.23a}$$

This is summable in the old birth. There are at most N_r output grades $j < n \le r$, so that enumeration costs a fixed polynomial in the preparation window. For the recent truncated-tensor error, $h \le b \le j < n \le r$, and the worst root density is $L^{4(r-b)} \le L^{4N_r}$. The same fixed inequality (SR.15d) pays it. For ancient births the cutoff distance again supplies the actual irrelevant-age gain.

For the literal Wilson cutoff residual, its $L^{-4(n-b)}$ cancels the actual n-cell birth density. Recent births and output grades are finite in the window. Ancient cutoff anchors retain (SR.21) at their true layer n; the indicator estimate (SR.22) supplies a summable $L^{-(n-b)}$. The remaining spatial row is used with its original native measures. Since $0 \le w \le 1$, a stronger raw factor $w^{1/u}$ may be weakened to w for this Wilson estimate.

No fine subdivision of a coarse forcing/output atom, or implicit factor $L^{4(n-j)}$ beyond the displayed actual birth-root count, is introduced. The separate $b > j$ case remains the whole-coarse-atom calculation in Section 7.2.

8.2 The scalar bound

Combining the ancient/interior estimate (SR.8), the recent truncated-tensor estimate (SR.15c), the literal cutoff estimate (SR.23), the recent-birth count, and the existing actual strip count gives

$$\sup_{\mathcal{D}^{++}} |D^E_{\mathrm{ord}}| \le C_Q E_0 \, \ell_r^{D_*} N_r^{s_*} L_0^{c_* N_0} \sum_{j=h}^{r-1} D_j(P) \le C_Q \ell_r^{D_E}. \tag{SR.24}$$

The degree D_E is fixed before Q. The constants and common weak-coupling ceiling may depend on the selected source profiles. The comparison keeps all earlier real variables fixed and retains both stopped endpoint configurations.

This is the form of OR.19 that the ordinary phase argument needs. It is obtained without claiming the stronger per-term factor $L^{-\lambda(j-b)}$. The rates actually paid after summation can be the explicit $L^{-a/8} L^{-c}$ of (SR.11).

9. Conditions for the two-age implication

The following conditions enter the application:

1. The two endpoint fields use the prescribed analytic C02/E14 extension, including the independent current, and meet the native norms after the fixed reserve (SR.12).
2. The response bound is for the original primitive bank on the actual fixed-raw family, including all required smooth/current rows; the physical g_j factor survives constraint elimination and relocation.
3. The root-to-attachment carrier and the source row refer to the same primitive record, and retain the spatial reserve lost in interpolation.
4. The literal Wilson cutoff remainder is retained and uses the ancient/recent split above.
5. The recent truncated E-bank error is kept as in (SR.15a), and its full density cost is paid with the actual $N_r \le C_N R_r$ relation and the explicit remaining-rate inequality (SR.15d).
6. The range $b \le j < n$ keeps its actual output mesh and birth-root density, as in Section 8.1. It is separate from $b > j$. The spatial weights are normalized to lie between zero and one.

Equation (SR.24) therefore supplies the missing ordinary E input to the boundary-induction argument's already repaired normalization and positive-root arguments. Together with the retained fixed-fiber normalization, local cumulant estimate, once-paid linear transport and running-tail induction, under CR this gives the C05 estimates on every history admitted by that fixed source construction and the stipulated AF trajectory. The original stronger per-term $j - b$ Ward assertion is not restored or required.

These implications assume CR. The AF trajectory is not constructed here. C06/C08 retain their own direct derivative/readout requirements, and the remaining Yang–Mills existence obligations keep their separate status.

Conditional law on admissible heterogeneous backgrounds

Composed-response hypothesis (CR). Each complete boundary conclusion below assumes the displayed native-chart and spatial response estimates for the actual iterated reconstruction, with the prescribed earlier innovations and raw variables fixed, uniformly over the admitted histories. A sufficient estimate for the latent contribution is (A.8) in Conditional-law estimates and the continuum application, refined for the anchored exterior derivative in PG.18–PG.20. The two endpoint bounds concern the identical Wilson-subtracted primitive, its actual cutoff and its independently extended current. A direct minimizing-field derivative alone does not establish this hypothesis. The application to every required original Wilson history remains open. Complete boundary estimates and their C06/C08 applications in this chapter are conditional on CR and the other displayed source, AF, domain and support assumptions. Exact density identities and abstract inequalities retain their individually stated hypotheses.

In the projective density formulation, a sufficient comparison is

$$\sup_{\zeta,x,a}\left|\log\frac{W_{\zeta}(x,a)W_0(x,a_*)}{W_0(x,a)W_{\zeta}(x,a_*)}\right|\le\delta_r<\pi/2,$$

on the stipulated chart with the required uniform rates and actual fixed support. The composed ordinary response is the unresolved input in its derivation. The exact partial-gauge transport and the surviving fixed-exterior derivative are given in the chapter Gauge covariance of partial integrals and exterior scores, equations PG.2–PG.6 and the subsequent anchored-score criterion.

Scope: the eight conditional-law estimates C-01 through C-08, on the retained C01/J0 source class and stipulated two-sided AF trajectory. A source-hypothesis theorem is not a construction of that trajectory or of continuum Yang–Mills fields.

1. Conditional finite-law theorem

Fix the gauge group, source blocking and localization parameters, cutoff profiles and the retained AF comparison constants. Use the original compact Wilson construction, with its actual constrained free-coordinate maps, canonical operations, selector-supported real histories and normalized conditional laws. Keep the corrected C01 allocation of action, Gaussian, Schur, Haar/coarea, boundary and normalization factors. An inherited raw variable remains real and fixed whenever its declared interface makes it a parameter.

Let an admitted history mean exactly this retained source class. Let a conditional fiber have positive mass, and let its current coordinates be the independent free-bond coordinates of C01. All estimates below are uniform in finite volume, ultraviolet cutoff depth, admitted history and positive fiber. Native derivatives use the declared strong source chart; current derivatives use the full real current support and a common complex increment margin around it.

Theorem CS.1 (C-series assembly). Assume CR for the actual composed response. With one fixed selection of source rates, native radii and a common sufficiently weak coupling ceiling, the retained C01–C04 and C07 results, the conditional two-age C05 estimate, and the applications below establish the eight conditional-law conclusions at this scope. In particular, the exact regular conditional law has a uniformly positive complete Hessian, an exponentially decaying pointwise inverse Hessian, an exponentially localized interacting covariance inequality on its literal product support, retained-force mean bounds, all-order current field and gradient moments, and fixed-factorial spatial cumulants of the stated native scalar/tensor family.

This theorem assembles the existing C-series predicates; it does not add the desired C06 Hessian or C08 covariance bound as a hypothesis. The required boundary score and Hessian estimates are obtained from the repaired C05 construction. The parameter dependencies are specified below.

2. Hypotheses and conclusions

Theorem group	Exact result used or established	Role in the assembly
C-01	Exact conditional density and factor/coordinate/domain decomposition	Identifies the measure being estimated; its Gaussian and normalizers are kept in their actual positions.
C-02	Actual regular E contour and analytic-domain transfer	Admits the regular E composition on its stated current/native chart.
C-03	Actual normalized regular R contour and marginal transfer	Transfers the regular R family on its declared chart.
C-04	Combined regular E/R derivative-carrier sum	Controls the complete regular family; B-assigned outputs remain in C05.
C-05	Complete boundary current and unsubtracted native estimates	Supplies the full boundary score and weighted Hessian, including stopped histories.
C-06	Complete heterogeneous derivative/coercivity theorem	Gives the positive Hessian and pointwise inverse decay on the actual chart.
C-07	Exact product-support and nonnegative Neumann boundary form	Justifies scalar covariance estimates on the actual conditional domain.
C-08	Complete regular conditional means, moments and spatial cumulants	Applies the preceding estimates to the original normalized conditional law.

Use C01–C04 and C07 as source hypotheses. Under CR the two-age estimate supplies the ordinary C05 input. The stated radius choices then give the C06 precision estimate and the C08 regular-law estimates.

Averaging the regular-law estimate over the full history distribution requires H04. Conversion to the stronger M03 norm also requires control of selector-support flux.

3. One radius dictionary and one order of choices

The two different uses of the symbol q in earlier notes must be distinguished. Use

$$q_{\mathrm{val}} = q_{\mathrm{tar}} + Q_1, \qquad q_{\mathrm{phase}} = q_{\mathrm{val}} + Q, \qquad q_{\mathrm{prim}} = q_{\mathrm{phase}} + D_{\mathrm{prep}} + 2. \tag{CS.1}$$

Here q_{tar} is the target native exponent, q_{val} is the fixed working activity-value exponent, q_{phase} is the larger ordinary external exponent, and q_{prim} is the original primitive profile used to leave room for the actual composed outputs. The corresponding radii are g_n times the appropriate power of ℓ_n. In particular the q in two-age SR.12 is read as q_{val} for the ordinary phase comparison. Equivalently, if that q is read as q_{tar}, its displayed enlargement must also pay Q_1. We use the first convention throughout this theorem group. A conservative local-output bound may use $q_{\mathrm{loc}} = q_{\mathrm{val}} + D_{\mathrm{loc}}$, where $D_{\mathrm{loc}} \geq 0$ is fixed before Q; $D_{\mathrm{loc}} = 0$ under the direct CE.2 bound. This is a bound on the output, not a redefinition or enlargement of the working input chart.

Fix q_{tar} above the source/current threshold and Q_1 above the native preparation/counting losses. Fix the geometric and primitive decay rates needed by the two-age age and cutoff estimates. The ordinary polynomial degree D_E and the preparation degree D_{prep} are fixed before Q. Then choose Q greater than D_E and the finitely many external-composition losses, with a strict margin. Constants may depend on these once-selected exponents. No choice depends on the number of scales or on the order of a derivative.

The two-age calculation uses the two-age inequalities

$$\eta\lambda > 4, \qquad (1-\eta)\Gamma > 4\log L, \qquad \eta\kappa_{\mathrm{tail}} c_R > 4C_N \log L + \epsilon. \tag{CS.2}$$

The first two pay the physical birth-root density after interpolation; the third pays the recent finite-domain tensor tail with the original $N_r \le C_N R_r$ preparation depth. For example $\lambda = 9/2$, $\eta = 11/12$ and $\Gamma \ge 60 \log L$ satisfy the age conditions. The primitive/full-carrier and reconstruction rates retain separate positive portions for spatial sums and their fractional-power loss.

Choose the final displayed physical rates small enough to satisfy the C05 creation/diameter reserves and the C06 weighted conjugation bound. This is compatible with CS.2: those conditions demand large source decay rates, while the final physical weights only need to be positive and sufficiently small. Finally lower one common g_* to satisfy all source admissions, projective-density estimates, chronological capacity bounds, local classical/measure bounds and coercivity margins.

This resolves a notation ambiguity rather than imposing an additional history restriction. The original real cutoffs and preparation depth are unchanged. See the dependency analysis for the primary-source parameter order.

4. Why the repaired OR.19 gives the required boundary input

The polynomial ordinary estimate and the final current prefactor are different quantities. Two-age SR.24 bounds the original ordinary E difference on the larger external chart by $C_Q \ell_r^{D_E}$. If the fixed external composition costs a power D_{map}, the actual radial disk has radius $c\ell_r^{Q-D_{\mathrm{map}}}$; radial Cauchy comparison contributes $\ell_r^{D_{\mathrm{map}}-Q}$. Keep this conservative loss and take $D_2 \ge D_E + D_{\mathrm{map}}$. Together with the independently controlled quadratic and rare terms, this gives the projective density estimate

$$\delta_{\mathrm{ord},r} \le C g_r \ell_r^{D_1} + C_Q \ell_r^{D_2-Q} + C g_r^{-u} \ell_r^{D_3} e^{-cR_r}. \tag{CS.3}$$

Here Q is measured from q_{val} as in CS.1. The old variables are fixed, and the comparison is for the same two stopped endpoints and same Wilson subtraction. The finite-domain tensor and literal Wilson cutoff residuals remain in SR.24. This is the corrected input to FP.8–FP.12, not a direct estimate of a final current Hessian.

At each chronological step, the incoming relative catalog uses the previously proved capacity A_{r-1}. The full fixed-fiber density is compared to one common positive law, with all masses left in their actual locations. Its bounded complex norm transfers the positive catalog on common Cauchy contours. The one-time linear transport accounting and local quadratic cumulant estimate give FP.20. Its running-tail induction proves

$$A_r \le 2F_r \le Ce^{-c_1 \ell_r^p}, \qquad p > 1, \tag{CS.4}$$

and the incoming anchored relative contribution tends uniformly to zero. A bounded operation norm is not multiplied once per earlier scale. The order is: admit the step's density using A_{r-1}, estimate its rooted output, and obtain A_r.

The complete current prefactor is then the separate positive sector budget FP.26/CE.19:

$$\begin{aligned} K_{B,k} \le C[&\mathfrak{K}_E + C_R g_k^{\kappa_0-11} + (B_0 + C_0)\theta_k + g_k^{-2} D_k^3 + D_k \\ &+ g_k^{-2}\delta_k (1 + \log \ell_k)^{A_\phi} e^{-c_\phi R_k} \\ &+ \theta_k P_k^2 (1 + \ell_k)^{A_G} e^{-c_G R_k} + \theta_k (A_k + \mathcal{F}_*)], \end{aligned} \tag{CS.5}$$

where $\theta_k = \ell_k^{p_0 - q_{\mathrm{tar}}}$, $D_k \le C g_k \ell_k^{q_{\mathrm{loc}}}$, $\delta_k = C g_k P_k$, $\kappa_0 \ge 12$, and $\mathcal{F}_*$ is the retained convergent failed-preparation budget. The separate fixed output exponent q_{loc} is defined in Section 3.

To check uniform boundedness directly, write $g_k = e^{-\ell_k/2}$. The classical factor $g_k^{-2} D_k^3$ is at most $C g_k \ell_k^{3q_{\mathrm{loc}}}$; the measure term is $C g_k \ell_k^{q_{\mathrm{loc}}}$; the heterogeneous term is a fixed polynomial times $\exp(\ell_k/2 - c\ell_k^p)$; the old Gaussian factor keeps its P_k^2 and is a fixed polynomial times $\exp(-c\ell_k^p)$. These tend to zero. The B/C and relative factors have θ_k with $q_{\mathrm{tar}} > p_0$. The E prefactor is uniformly finite. Thus $K_{B,k} \le K_B^* < \infty$ at one ceiling.

The complete current contour is the one already admitted in C05: a fixed cP_k increment polydisc around every point of the actual full current support. The first endpoint estimate and the rooted spatial catalog yield

$$\sup_i |D_i W_B| \le C K_B^*/P_k, \qquad \sup_z \sup_i \sum_j e^{\gamma_B \rho(i,j)} \|D_i D_j W_B(z)\| \le C K_B^*/P_k^2. \tag{CS.6}$$

The first bound is the $m = 1$ case of FP.3; it is not inferred from the Hessian. The second is FP.2. The physical endpoint calculation CE.4–CE.22 retains actual grades, complete atoms, Gaussian P_j^2 factors and every carrier diameter. The stronger DB.59 value-localization formulation is a sufficient alternative, not a new premise inserted here.

For a fixed matrix of entrywise Hessian suprema, take one entry in CS.6 and then sum at a smaller rate:

$$\begin{aligned} A_{ij}^B &:= \sup_z \|D_i D_j W_B(z)\| \le C P_k^{-2} e^{-\gamma_B \rho(i,j)}, \\ \sup_i \sum_j e^{\gamma_B \rho(i,j)/2} A_{ij}^B &\le C P_k^{-2} \sum_{n \ge 0} (n+1)^3 e^{-\gamma_B n/2}. \end{aligned} \tag{CS.7}$$

This is a fixed loss of spatial rate. It does not exchange a supremum and an endpoint sum.

5. C06 on the complete conditional potential

The exact law has the form

$$d\nu_\upsilon(z) = Z_\upsilon^{-1} 1_{D_\upsilon}(z) \exp\{-\tfrac{1}{2} z^T K z + b^T z + W(z;\upsilon)\}\, dz, \tag{CS.8}$$

with the actual principal constrained precision K, exterior force $b = -K_{IO} y$, and disjoint $W = W_{ER} + W_B + W_{\mathrm{loc}}$. Older fields are retained parameters, not appended to the current vector y. The current exterior force is allowed to be large near the boundary.

The primary principal-operator estimate gives $K \ge \lambda_0 I$ and exponentially decaying entries. C03/C04, CS.6, and the actual constrained classical/Haar/coarea estimates give a small complete Hessian perturbation. With the parameter choices in the C06 companion,

$$H(z) := D_z^2[-\log \nu_\upsilon(z)] = K - W''(z) \ge \tfrac{3}{4}\lambda_0 I \tag{CS.9}$$

on the interior of the full literal support. This expression refers to the smooth potential, not derivatives of its indicator.

For a 1-Lipschitz physical function f, conjugate by $D_f = \mathrm{diag}(e^{\gamma f_i})$. The off-diagonal precision change has norm at most γA_K, while both weighted Schur sums of the conjugated perturbation are small. One fixed γ gives accretivity at least $\lambda_0/2$. Choosing $f_i = \rho(i,j)$ proves

$$\|[H(z)^{-1}]_{ij}\| \le (2/\lambda_0) e^{-\gamma \rho(i,j)}. \tag{CS.10}$$

The complete native derivative clauses are retained on the common strong domain, including current-independent native scalars. The detailed proof, matrix identifications, local carrier estimates and contour checks are in Heterogeneous constrained precision and coercivity. The pointwise matrix argument does not need C07's Neumann conclusion, so there is no C06/C07 cycle.

6. C08: interacting covariance, means and moments

C07 identifies D_υ as the actual product of symmetric convex free-bond balls after the complete source operation. On smooth inner approximations of these balls, the scalar Neumann form has a differentiated vector form whose face curvature term is nonnegative. Blockwise constant spatial conjugations preserve the tangency conditions. Using CS.9 and the same weighted perturbation estimate gives the scalar inequality

$$|\operatorname{Cov}_{\nu_\upsilon}(F,G)| \le \sum_{i,j} \mathsf{G}_{ij} \|\nabla_i F\|_{L^2(\nu_\upsilon)} \|\nabla_j G\|_{L^2(\nu_\upsilon)}, \qquad \mathsf{G}_{ij} = (2/\lambda_0) e^{-\gamma \rho(i,j)}. \tag{CS.11}$$

Pass this scalar inequality to the literal norm balls by convergence of the densities and Sobolev approximation. No identity equates interacting covariance with the matrix $H(z)^{-1}$. No convergence assertion for an unspecified arbitrary-vector boundary realization is needed.

The fixed complete score and Hessian majorant satisfy

$$d_i = |D_i W(0)| \le C/P_k, \qquad \sup_i \sum_j e^{a\rho(i,j)} A_{ij} \le C/P_k^2, \qquad A_{ij} \ge \sup_z \|D_i D_j W(z)\|. \tag{CS.12}$$

The actual exterior has $|y| \le CP_k$. The precision row gives $|b_i| \le CP_k e^{-\delta d_O(i)}$. In the source law with interpolated force and interaction, the even zero-force Gaussian on the same support has mean zero. Differentiating normalized means retains the covariance with the complete score. If M_i is the supremum of the means over this interpolation, CS.11 gives

$$M \le \mathsf{G}(|b| + d) + \mathsf{G}A(M + \sigma 1), \qquad \sigma = (d_{\mathfrak{g}}/\lambda)^{1/2}. \tag{CS.13}$$

Choose the common ceiling so the weighted norm of $\mathsf{G}A$ is at most $1/2$. The positive numerical resolvent $(I - \mathsf{G}A)^{-1}\mathsf{G}$ and the distance inequality then imply

$$|\nu_v z_i| \le C(P_k^{-1} + P_k e^{-c d_O(i)}), \qquad \|z_i - \nu_v z_i\|_{L^p(\nu_v)} \le C\sqrt{p}. \tag{CS.14}$$

The second estimate follows from the same unweighted covariance bound under every real linear tilt. The constants are uniform in p except for the displayed $\sqrt{p}$, and uniform in the actual fiber. The full nonlinear constrained chart transfers these estimates to the stated current field and finite covariant-gradient stencils, with the reference field and physical mesh powers kept explicit.

For the prescribed native scalar/tensor fields $O_{\alpha,x}$, put $Q_x = 1 + B_x + M_x$, retaining the actual reference-curvature and conditional-mean envelopes. The companion proves

$$\|O_{\alpha,x} - \nu_v O_{\alpha,x}\|_p \le C h_k^{-4}(Q_x\sqrt{p} + p), \tag{CS.15}$$

and, for $m \ge 2$,

$$|\kappa_{\nu_v}(O_{\alpha_1,x_1}, \dots, O_{\alpha_m,x_m})| \le C^{m+1}(m!)^2 h_k^{-4m} \prod_i Q_{x_i} e^{-\theta\tau(x_1,\dots,x_m)/(m-1)}. \tag{CS.16}$$

A longest-edge cut of a minimum spanning tree reduces the cumulant to a scalar covariance of products. The partition coefficient is bounded by a fixed exponential times $(m-1)!$, and the native L^{2m} norms supply one further factorial. Summing the divided-tree kernel over the other anchors in four dimensions costs at most $C^m(m!)^5$. The anchored cumulant bound therefore has fixed factorial power seven. The detailed partition identity and all approximation qualifications are in Regular conditional covariance, moments and cumulants and its retained C8A proof.

7. Additional results and exact limits of their use

The C08 proof also supplies finite-support quadratic regulator moments for the actual centered conditional current. For any real symmetric positive semidefinite matrix T and $0 \le t < \lambda/(2\|T\|)$, joint sub-Gaussian linear tilts and Gaussian auxiliary integration give

$$\nu_v \exp\{t(z-m)^T T(z-m)\} \le \det(I - 2tT/\lambda)^{-1/2}, \qquad m = \nu_v z. \tag{CS.17}$$

The matrix may be restricted to any finite set of actual free-coordinate components. Its trace/volume and any nonzero center cost remain explicit. This is not a volume-free bound for every strong history regulator.

For the exterior part y of the same newly introduced constrained-current field, the full current chart gives a mixed inner/exterior derivative row. The normalized response is

$$D_{y_a}\nu_y F = \nu_y D_{y_a} F + \mathrm{Cov}_{\nu_y}(F, S_a), \qquad S_a = D_{y_a} \log p_y. \tag{CS.18}$$

The centered score is bounded in L^p by $C\sqrt{p}$, and its kernel response is exponentially localized, with the $O(1)$ precision contribution retained. The statement includes the source's real first-derivative relocation at fixed geometry. It does not differentiate arbitrary older raw arguments, varying selector surfaces, or a changing history chart.

The following remain outside the fixed regular-law C-series conclusion: the AF trajectory construction; quantitative physical normalization and microscopic-field matching; all-history integration of conditional Q_x envelopes; the strong M03/M07 regulator conversion and moving-selector flux SS; the full marked H04/M04/M06 induction; continuum hierarchy, rotation restoration and Yang–Mills identification. These are not used as premises of CS.9–CS.18.

8. Proof and analysis map

Exact conditional densities and deterministic Hessian envelopes identifies all eight fixed targets, their accepted predecessor locations, parameter compatibility and absence of a C06/C07 or C05/C08 dependency cycle. Heterogeneous constrained precision and coercivity supplies the corrected C05-to-C06 application and common-domain proof. Regular conditional covariance, moments and cumulants supplies the actual-law covariance, means, moment and cumulant application, the regulator corollary and the same-current response analysis.

The Wilson subtraction is taken for the same term at both stopped endpoints. The native-radius choices are those in CS.1.

The fixed radial-map loss enters the chart comparison. The covariance and regulator statements retain their scalar Neumann, determinant, trace and mean hypotheses.

Gauge covariance of partial integrals and exterior scores

Conclusion and source comparison

Proved identity. A change of variables in a stopped integral may transport the variables actually integrated. A simultaneous gauge transformation of the earlier fixed tuple instead relates two different conditional fibers. The resulting partial integral is covariant along that exterior orbit; it need not be constant at one fixed earlier tuple.

Useful repair criterion. The exterior response need not vanish. For the original C05 projective density argument, its part that varies with the still-integrated tuple must satisfy a bound, while a scalar common to the whole tuple can remain in the mass decomposition. This is weaker than requiring the entire exterior derivative to be small, but stronger than a completed-integral Ward identity.

This appendix does not embed the homogeneous MB.12 example into a literal C05 stopped history. The actual graded preparation and the same Wilson-subtracted source object still have to be identified. It therefore supplies an exact comparison and a sufficient criterion, not a counterexample to C05.

The objects being compared are:

- Stopped densities and projective comparison, SD.1 and SD.5: the actual conditional density at fixed earlier raw tuple x.
- SD.6d–SD.6e: the actual pullbacks Φ_j, their complete factors J_j, and the final mass. Unrolling cancels only identical intermediate masses.
- Wilson-subtracted primitive response and interpolation, ST.2–ST.3: the same rootwise bank

$$\mathcal{R}^{H}_{b,z}(U) = E^{(b)}(\Lambda_b, U, z) - E^{(b)}(\Lambda_b, 1, z) - \beta_b A(h_z \phi_b, U). \tag{PG.1}$$

  Its birth coefficient, cutoff, root and localization sum stay fixed.
- Moving ancestor bases, fixed innovations, and exact latent-gauge current transport, MB.9–MB.11 and MB.18–MB.25: exact fixed-target covariance and normalized latent transport.

1. Exact finite partial-integral identity

Write a for all variables actually integrated at the stop and x for every earlier raw coordinate that is fixed. Include all smooth positive multipliers, Jacobians and archived masses in a full coordinate density $\rho_t(x, a)$. Moving a native-independent multiplier out of SD.1's reference measure and back into ρ_t is only notation; it is not dropped.

Suppose an actual real gauge path has a fiber-preserving coordinate map

$$G_t(x, a) = (E_t x, T_t(x, a)), \qquad J^x_t(x) = |\det DE_t(x)|, \quad J^a_t(x, a) = |\det D_a T_t(x, a)|. \tag{PG.2}$$

Here the new exterior tuple $E_t x$ is independent of a. Require the complete source covariance identities

$$\begin{aligned} \rho_t(E_t x, T_t(x, a)) J^x_t(x) J^a_t(x, a) &= \rho_0(x, a), \\ S_t(E_t x) &= T_t(x, S_0(x)), \\ f_t(E_t x, T_t(x, a)) &= f_0(x, a). \end{aligned} \tag{PG.3}$$

Every saved target, Q reference, selected factor and normalizer occurring in the source must participate. If covariance is established only after summing branches, PG.3 cannot be asserted separately on each fixed branch.

Define the actual partial mass and normalized expectation

$$Z_t(x) = \int_{S_t(x)} \rho_t(x, a)\, da, \qquad H_t(x) = Z_t(x)^{-1} \int_{S_t(x)} \rho_t(x, a) f_t(x, a)\, da. \tag{PG.4}$$

Then

$$\boxed{Z_t(E_t x) J^x_t(x) = Z_0(x), \qquad H_t(E_t x) = H_0(x).} \tag{PG.5}$$

For the conditional coordinate density $p_t = \rho_t / Z_t$,

$$p_t(T_t(x, a) \mid E_t x) J^a_t(x, a) = p_0(a \mid x). \tag{PG.6}$$

Proof. Substitute $a' = T_t(x,a)$ in both integrals. The same factor $J_t^x(x)^{-1}$ multiplies numerator and denominator, and cancels only in their ratio. This proves every identity without differentiating a sharp cutoff. □

Real blockwise adjoint maps have determinant one. This removes the displayed Jacobian factors in that case, but does not remove $E_t x$. At a prescribed fixed exterior point x, PG.5 instead says

$$H_t(x) = H_0(E_t^{-1}x). \tag{PG.7}$$

Thus transporting the integrated tuple is a legitimate integration change of variables; replacing x by $E_t^{-1}x$ changes the conditional fiber being evaluated.

Let $v_t = \dot{E}_t \circ E_t^{-1}$. At differentiability points,

$$(\partial_t + v_t \cdot D_x)H_t = 0, \qquad \partial_t Z_t + \operatorname{div}_x(v_t Z_t) = 0. \tag{PG.8}$$

The same formulas hold weakly when the finite transport identities are the available formulation. A scalar normalized density ratio relative to another source-covariant reference transforms like H_t, because their common exterior Jacobian cancels.

2. What remains after the integrated gauge currents cancel

Let $w_t(x,a)$ be the integrated-coordinate velocity induced by G_t. On an active smooth stratum the complete density has

$$\partial_t \rho_t + \operatorname{div}_x(v_t\rho_t) + \operatorname{div}_a(w_t\rho_t) = 0. \tag{PG.9}$$

The corresponding equation for the conditional density, extended with its actual support, is

$$\partial_t p_t + v_t \cdot D_x p_t + \operatorname{div}_a(w_t p_t) = 0. \tag{PG.10}$$

This is an equation for a density, not a pointwise assertion that its gauge derivative vanishes.

At fixed x, the original normalized derivative contains insertion, score covariance, and boundary terms, exactly as in SD.15 and MB.19. Apply MB.22 only to the integrated tuple a. Its insertion transport, density divergence and Q-boundary terms combine, leaving precisely

$$\partial_t H_t(x) = -D_x H_t(x)[v_t(x)]. \tag{PG.11}$$

This equality is the differentiated finite identity PG.7, so it retains the same numerator, final mass and normalizer. An integration by parts in a cannot eliminate its x-derivative.

There are genuine restricted cancellation cases. If the gauge belongs to the stabilizer of the entire fixed tuple, $v_t(x) = 0$, then PG.11 vanishes. It also vanishes for a particular insertion if the resulting H_t is independently proved constant along that exterior orbit. A general admitted nonzero earlier innovation need satisfy neither condition.

If the proposed gauge sends the exterior tuple to a value depending on a, it does not preserve the conditioning fibers in PG.2. One then gets a continuity equation with exterior flux $\int \rho_t w_t^x \, da$, rather than PG.8 with a deterministic exterior velocity. Its normalized insertion equation includes the corresponding conditional covariance flux. Such a transformation supplies no simpler argument for dropping the exterior response.

3. A criterion that really gives the original projective density bound

The preceding real covariance does not by itself imply SD.10's domination at the same fixed (x,a). In particular, PG.6 compares a density with its pullback at a different exterior point and a different integrated coordinate.

Here is a direct sufficient criterion that preserves the original statement. On the fixed support S_x, let $W_t(x,a)$ be the same nonvanishing analytic interpolation of the complete SD.1 weight, with $W_0 > 0$, and set

$$q_t(x,a) = \partial_t \log W_t(x,a).$$

Choose one actual-support anchor $a_*(x)$. Suppose

$$\int_0^1 \sup_{a \in S_x} |q_t(x,a) - q_t(x,a_*)| \, dt \le \delta < \pi/2. \tag{PG.12}$$

This must hold uniformly on the required native complex family when used for C05's analytic bound. Then

$$W_1(x,a) = W_0(x,a)e^{c(x)+\sigma(x,a)}, \qquad |\sigma(x,a)| \le \delta, \tag{PG.13}$$

where $c(x) = \int_0^1 q_t(x,a_*)\,dt$. Consequently the exact SD.1 conditional density obeys

$$|p_1(a \mid x)| \le e^{2\delta} \sec\delta\; p_0(a \mid x). \tag{PG.14}$$

The mass retains $c(x)$.

Proof. Integrate the difference of the two logarithmic derivatives. This gives PG.13 on the same fixed fiber, and SD.8–SD.10's sector argument gives PG.14. □

For a genuine pure gauge path, on smooth positive strata the complete score from PG.9 is

$$q_t = -D_x \log\rho_t[v_t] - D_a \log\rho_t[w_t] - \operatorname{div}_x v_t - \operatorname{div}_a w_t. \tag{PG.15}$$

The exterior divergence is independent of a and disappears from the anchored difference. For the literal adjoint action the diagonal divergences vanish. The two derivative terms still need their anchored estimates. In a completed owned integral the integrated-coordinate term may cancel by the full transport identity. In the pointwise density criterion PG.12, it is not zero and must not be deleted.

Thus one useful sufficient repair is to bound the exterior anchored score

$$\sup_{a\in S_x} \left| D_x \log\rho_t(x,a)[v_t(x)] - D_x \log\rho_t(x,a_*)[v_t(x)] \right|, \tag{PG.16}$$

together with the actual integrated-coordinate and transverse contributions needed in PG.12. No bound is needed on the component of the exterior score common to every a; it is exactly an allowed mass scalar. This is a sharper target than controlling the entire exterior derivative.

If an argument uses only complete coefficient integrals, it can instead use MB.22 before taking absolute values, retaining the exterior derivative PG.11 and all original labels. To recover C05's particular positive catalog estimate SD.14 that way, it must prove the resulting common domination or equivalent estimate with the same analytic supremum, weights and origin assignments. A cancellation between separately prescribed absolute labels is insufficient.

4. Applying the criterion to the same Wilson-subtracted bank

Set

$$F_t(x,a) = \mathcal{R}^H_{b,z}(U_t(x,a)) \tag{PG.17}$$

with exactly PG.1's bank. A simultaneous gauge change of all its native arguments leaves this scalar unchanged when the source covariance hypotheses hold. This licenses expressing both values in a common output gauge; it does not license omitting the motion of the earlier raw parameters used to describe those values.

For example, after pulling back the integrated tuple,

$$\widehat{F}_t(x,a) := F_t\big(x, T_t(E_t^{-1}x, a)\big) = F_0(E_t^{-1}x, a). \tag{PG.18}$$

Writing $x_t = E_t^{-1}x$, the paired difference has exact derivative

$$\frac{d}{dt}\{\widehat{F}_t(x,a) - \widehat{F}_t(x,a_*)\} = \{D_xF_0(x_t,a) - D_xF_0(x_t,a_*)\}[\dot{x}_t]. \tag{PG.19}$$

The old tuple is identical at the two entries of this difference, so the same Wilson subtraction is retained. But that tuple varies along the estimating path. PG.19 is the term a claimed response estimate must bound.

A constructive sufficient exterior criterion for the disputed $b \le n < j$ range is therefore an estimate of the form

$$|\{D_xF_0(x_t,a) - D_xF_0(x_t,a_*)\}[\dot{x}_t]| \le B^x_{b,z;j,s}(t), \tag{PG.20}$$

where, after integration in t, the positive majorant has the age and physical spatial row required of the crude response in ST.13–ST.14. It may then be combined with the endpoint Ward estimate for this same bank. Contributions from

integrated-coordinate pullbacks must also be restored when the original target keeps those coordinates fixed: PG.18 is not an identity identifying their fixed-coordinate derivatives. All current, background and cutoff pieces of the same primitive remain included.

PG.20 is potentially weaker than a pointwise age bound on the entire reconstruction: it only asks for the anchored scalar response after the actual Wilson subtraction. Its feasibility is a substantive model-specific question. Neither MB.11 nor gauge invariance proves it. Conversely, MB.12's coarse plaquette detector alone does not disprove it, because that detector has not been identified with PG.1 on the literal stopped preparation geometry, and the subtraction could remove its leading response.

Finally, MB.9–MB.25 concern a real covariant path. C05's contour supremum additionally needs a common admitted native analytic family and its quantitative majorant. A real gauge identity, or a parameter-dependent choice of real gauges, does not establish that complex-family statement.

Status. The partial-integral identities and the anchored-score implication PG.12–PG.14 are proved. The all-history bound PG.20, the other anchored score pieces required by PG.12, and their application on the actual C05 graded preparation remain to be verified. No original all-history C05 closure is inferred here.

Inner normalizer covariance and outer conditional laws

Result and scope

(P, as an implication from the explicit C06/C08 input inequalities.) For any fixed-history conditional law satisfying (OH.3)–(OH.6), the covariance term generated by differentiating its inner normalizer is uniformly exponentially localized when the stated spatial majorants hold. More sharply, its quadratic form is bounded above by the exact principal Gaussian Schur contribution plus a quantity of order `P_k^{-2}`. The proof uses the complete interacting conditional law on its literal convex product cutoff. It makes no Gaussian replacement and retains the normalizer derivative. This note does not independently verify the open C05 actual-history estimate that is used upstream to supply those inequalities uniformly on every required Wilson history; the Schur implication remains rigorous independently of that application question.

(C.) This gives a concrete sufficient test for coercivity of the same-current outer marginal: after extracting the principal exterior quadratic, one must control an explicitly identified outer curvature defect and the actual outer support. On a fixed convex outer domain satisfying that test, the outer covariance application follows.

(O.) The full retained-history law has not passed that test. Retained-only selectors, geometry changes, the relocation Jacobian and history factors remain in its outer density; they did not disappear when the inner law was normalized. C08 consequently does not establish H04/H05, the amended all-history M04/M07 estimate, or the ER.7 physically normalized hierarchy bound.

For M04, the original full-footprint norm has a first-step obstruction from the geometry of the integration support. A local-mask amendment preserves the physical law, but its arbitrary-depth common-family and current bounds remain unproved. This is an interface correction, not a counterexample to Yang–Mills existence.

1. The exact current variables and outer factor

Use the notation of Response to exterior current coordinates, RC.1–RC.4 and RC.23–RC.24. At one fixed admitted history, let `z` be the integrated free current coordinates `I` and let `y` be the retained free coordinates `O` of the same current chart. In the source convention

$$O = (\Omega_{k+1} \cap \Lambda_{k+1}^{c})^{(k)*}. \tag{OH.1}$$

The coarse datum, earlier raw fields, coupling and geometry are fixed. These variations are not arbitrary older retained-coordinate or history variations. The full constrained principal matrix was formed before conditioning:

$$\mathcal{K} = \begin{pmatrix} A & B \\ B^T & D_0 \end{pmatrix}, \qquad A = K_{II}, \quad B = K_{IO}, \quad D_0 = K_{OO}. \tag{OH.2}$$

The source identities `QC=0`, `Qh=I` and `D(q)=N(q-hD(q))` imply that changing `y` within this chart keeps the prescribed coarse field and `mathcal K` fixed. The exact conditional law is

$$\nu_y(dz) = Z(y)^{-1} 1_{D_I}(z) e^{H(z,y)}\, dz, \qquad H(z,y) = -\tfrac{1}{2} z^T A z - z^T B y + W(z,y), \tag{OH.3}$$

where `D_I` is C07's literal fixed product of symmetric convex free-bond balls. All terms with nonzero `z`-gradient occur in `W` in their actual source allocation. Write the continuous part of the actual outer density, on a fixed active stratum, as

$$\pi(dy) \propto 1_{D_O}(y)\, q(y) e^{-\frac{1}{2} y^T D_0 y} Z(y)\, dy. \tag{OH.4}$$

Here `q` means every remaining outer factor after the displayed principal exterior quadratic has been extracted. The indicator notation does not assert that `D_O` is convex, a product, or even described by one smooth boundary. Factors depending only on `y` include the applicable retained selectors, exterior coarea/measure factors, relocation Jacobian and inherited normalization/history factors. They canceled from (OH.3), but they remain in (OH.4). Other continuous or discrete retained data may still lie outside this chart altogether.

The distinction is already explicit in RC.3–RC.4: the mixed current estimate does not assert a full exterior-law Hessian. The present note makes the normalizer part of that exterior Hessian quantitative.

2. The complete inner and mixed inequalities used as inputs

The input inequalities appearing in the conditional-law C06/C08 conditional theorem are, at a fixed smaller spatial rate,

$$A \succeq \lambda_0 I, \qquad \|W_{zz}(z,y)\|_{2\to 2} \le \varepsilon_k, \qquad \|W_{zy}(z,y)\|_{2\to 2} \le \varepsilon_k, \qquad \varepsilon_k = C_* P_k^{-2}. \tag{OH.5}$$

Increase `C_*` to cover both sectors. For the implication proved here, require these bounds uniformly in the chosen real inner and same-current exterior chart, with fixed entrywise matrices having bounded weighted rows and columns. They must not be obtained by interchanging a supremum and a row sum. Their application to all original histories is not a new conclusion of this note. For sufficiently weak coupling, `epsilon_k < lambda_0/2`. Set

$$H_I(z,y) = A - W_{zz}(z,y) \succeq (\lambda_0 - \varepsilon_k) I. \tag{OH.6}$$

The exact source precision has uniform exponentially summable rows, so `kappa = ||B||_{2->2}` is bounded independently of volume and history. Block dimensions are fixed by the gauge group and free-coordinate convention.

The same-current score is

$$S(z,y) = \nabla_y H(z,y) = -B^T z + \nabla_y W(z,y), \qquad \hat{S} = S - \nu_y S, \tag{OH.7}$$

and

$$D_{y_a} \nu_y = \hat{S}_a \nu_y. \tag{OH.8}$$

The denominator derivative in (OH.8) is essential. RC.7–RC.9 and C8Y.3–C8Y.5 already prove the corresponding first normalized response and its uniform centered `L^p` bound.

3. A matrix Brascamp–Lieb consequence on the literal cutoff

Lemma OH.1 — the actual inner variance form

For every real scalar `f` in the indicated Sobolev domain,

$$\mathrm{Var}_{\nu_y}(f) \le \nu_y\big[(\nabla_z f)^T H_I^{-1} (\nabla_z f)\big]. \tag{OH.9}$$

Proof. First replace each literal norm ball by an increasing smooth convex inner approximation, as in C8A.12–C8A.14. On its product, the differentiated scalar Neumann form for a vector field `u` is the sum of the nonnegative kinetic term, the multiplication term `u^TH_Iu`, and the nonnegative boundary second-fundamental-form term. Let its operator be `mathscr A`. The scalar covariance representation gives

$$\mathrm{Var}_{\nu_y}(f) = \langle \nabla f, \mathcal{A}^{-1} \nabla f \rangle.$$

Use the variational expression for the inverse quadratic form. Since the form of `mathscr A` is at least the multiplication form of `H_I`,

$$\begin{aligned} \langle v, \mathcal{A}^{-1} v \rangle &= \sup_{u \in \mathcal{D}(\mathfrak{a})} \{2\langle v, u\rangle - \mathfrak{a}(u,u)\} \\ &\le \sup_{u \in L^2} \{2\langle v, u\rangle - \nu_y[u^T H_I u]\} = \nu_y[v^T H_I^{-1} v]. \end{aligned}$$

The last equality is pointwise completion of the square. Taking `v=gradient f` proves (OH.9) on each approximation. The coefficients and the positive lower bound are continuous on the fixed admitted compact chart. Scalar density convergence and Sobolev approximation therefore pass the inequality to the literal product, exactly as in the existing C08 scalar argument. This proves the asserted scalar inequality without asserting convergence of an arbitrary-vector boundary realization. ■

Theorem OH.2 — a sharp one-sided normalizer covariance bound

Let

$$C_\nu(y) = \mathrm{Cov}_{\nu_y}(S, S).$$

Then, for `0 <= epsilon_k < lambda_0`,

$$0 \preceq C_\nu(y) \preceq B^T A^{-1} B + \delta_k I, \tag{OH.10}$$

where the explicit uniform error is

$$\delta_k = \frac{\kappa^2 \varepsilon_k}{\lambda_0(\lambda_0 - \varepsilon_k)} + \frac{2\kappa\varepsilon_k + \varepsilon_k^2}{\lambda_0 - \varepsilon_k} = O(P_k^{-2}). \tag{OH.11}$$

Proof. For a real exterior vector `a`, apply (OH.9) to `f=a^TS`. Its exact inner derivative is

$$\nabla_z(a^T S) = (-B + W_{zy})a.$$

Consequently

$$a^T C_\nu a \le \nu_y\big[a^T(-B + W_{zy})^T H_I^{-1}(-B + W_{zy})a\big]. \tag{OH.12}$$

Write `E=W_zy`. Resolvent comparison and (OH.5) give

$$\|H_I^{-1}\| \le (\lambda_0 - \varepsilon_k)^{-1}, \qquad \|H_I^{-1} - A^{-1}\| \le \frac{\varepsilon_k}{\lambda_0(\lambda_0 - \varepsilon_k)}. \tag{OH.13}$$

Expand the expression in (OH.12) around `B^TA^{-1}B`. Its first difference is `B^T(H_I^{-1}-A^{-1})B`. The two cross terms are bounded in absolute quadratic form by `2 kappa epsilon_k/(lambda_0-epsilon_k)`, and the final `E^TH_I^{-1}E` term by `epsilon_k^2/(lambda_0-epsilon_k)`. Their sum is (OH.11), pointwise in `z`. Averaging preserves the inequality. Nonnegativity is the covariance identity itself. ■

This is a one-sided Loewner bound. It is not the assertion `||C_nu-B^TA^{-1}B||=O(P_k^{-2})`. A convex cutoff can reduce a Gaussian covariance by order one. No lower Gaussian comparison is used or needed.

At `W_zz=W_zy=0`, the error vanishes and (OH.10) retains the precise Gaussian Schur contribution even on the literal truncated domain, at every retained exterior force. The mean need not be small. This feature avoids paying a spurious `P_k` factor in the normalizer covariance.

4. Its absolute spatial majorant is also uniform

Let `A^IO_{ia}` be the fixed mixed derivative envelope in C8Y.1/RC.4 and put

$$C_{ia} = \|B_{ia}\| + A^{IO}_{ia}, \qquad G_{ij} = \frac{2}{\lambda_0} e^{-\gamma\rho(i,j)}. \tag{OH.14}$$

For unit scalar directions in exterior blocks `a,b`, C08's covariance inequality applied to the two scores gives

$$\|(C_\nu)_{ab}\|_{\mathrm{op}} \le (C^T G C)_{ab}. \tag{OH.15}$$

Indeed `||D_{z_i}S_a|| <= C_{ia}` pointwise; insert those deterministic bounds in C8A.12 and take the supremum over the two unit directions. For any fixed rate `beta` below the available rates, the triangle inequality yields

$$\|C^T G C\|_\beta \le \|C^T\|_\beta \|G\|_\beta \|C\|_\beta < \infty. \tag{OH.16}$$

Both row and column sums are retained. Thus the covariance part of the induced outer Hessian is exponentially localized, with constants independent of volume, cutoff depth and history wherever the displayed C08 and mixed-score input bounds hold uniformly. This is a proved consequence of those inequalities, not an independent certification of their all-history application. Unlike the error in the Loewner upper bound, its full absolute norm is generally order one, because it contains the principal cross precision `B`.

5. The exact remaining outer curvature and support test

On a smooth positive stratum of (OH.4), differentiation of the same normalizer gives

$$D_y^2 \log Z(y) = \nu_y W_{yy} + C_\nu(y). \tag{OH.17}$$

No derivative of `B` or `A` occurs because this is the same-current variation at fixed coarse datum. For older native/physical directions those derivatives must be restored, as in RC.19–RC.22.

The outer smooth potential is

$$\overline{V}(y) = \tfrac{1}{2} y^T D_0 y - \log q(y) - \log Z(y).$$

Define the remaining defect

$$\mathcal{R}(y) = D_y^2 \log q(y) + \nu_y W_{yy}(z, y). \tag{OH.18}$$

Then the exact identity and proved bound are

$$\begin{aligned} \overline{V}''(y) &= D_0 - \mathcal{R}(y) - C_\nu(y), \\ \overline{V}''(y) &\ge \underbrace{D_0 - B^T A^{-1} B}_{\Sigma} - \mathcal{R}(y) - \delta_k I. \end{aligned} \tag{OH.19}$$

The defect is invariant under shifting a scalar `c(y)` from the outer factor into the inner exponent: `W -> W+c`, `q -> q exp(-c)` leaves (OH.18) unchanged. Thus the missing estimate cannot be removed by reallocating a current-independent-in-z scalar.

If the full principal matrix in (OH.2) satisfies the source gap `mathcal K >= lambda_0 I`, then

$$\Sigma \succeq \lambda_0 I. \tag{OH.20}$$

To see this, minimize `(x,a)^T mathcal K (x,a)` over `x`. Its minimum is `a^T Sigma a`, and the full lower bound gives at least `lambda_0 |a|^2`. This keeps the actual principal inverse and all cross coefficients together.

Corollary OH.3 — sufficient fixed-current outer application

Suppose, additionally, that the actual outer law on this fixed chart has:

1. a fixed convex product support (or a separately proved replacement support theorem whose nonnegative boundary form also admits the same spatial weighted-conjugation estimate);
2. the smooth defect bound `mathcal R <= lambda_0 I/4`;

3. an exponentially summable deterministic entrywise majorant for `mathcal R`, uniformly on that support.

After lowering the common coupling ceiling so that `delta_k <= lambda_0/4`,

$$\overline{V}'' \geq \tfrac{1}{2}\lambda_0 I. \tag{OH.21}$$

The weighted-conjugation and scalar Neumann arguments of C06/C08 then give a fixed exponentially localized covariance kernel for this outer law. The entries of `D_0`, of `C_nu` by (OH.15), and of the stipulated defect are exponentially summable; conjugation at a sufficiently small fixed rate preserves a positive form gap. A product support keeps its boundary form local to the individual coordinate blocks. Nonnegative curvature alone on a general convex nonproduct support would prove a variance bound, but would not justify this spatial conclusion: a global convex constraint can create nonlocal boundary couplings. This proves the stated conditional implication with its stated support hypothesis.

The corollary is not an application to the full history law. C05 controls its identified analytic activity families in their admitted current/native domains, and the conditional-law derivation preserves current-independent native scalars. Those facts do not prove Items 1–3 for all factors and strata in (OH.4). In particular:

- retained current `R`-bad tests can restrict `y` to complements of cutoff balls and more general selected sets;
- changing a mask or a determining component changes the domain/coordinate problem and is not a derivative in the fixed `(z,y)` chart;
- the exterior relocation Jacobian cancels only from the inner normalized ratio, not from `q`;
- an old history mass kept as a fixed parameter by C05 becomes a live factor when the corresponding history is subsequently averaged.

The native derivative bound for an assigned scalar does not supply the missing global selected-support theorem. On open strata an indicator can be constant, while its boundary flux remains essential to covariance estimates.

6. Why a positive smooth Hessian is insufficient on the outer bad sectors

Proposition OH.4 — an elementary selected-domain obstruction

Let `Y` be a standard Gaussian vector in `R^d`, conditioned on `|Y| >= P`. The smooth potential has Hessian exactly `I`, and the conditional mean is zero. Nevertheless

$$\operatorname{Var}(Y_1 \mid |Y| \geq P) = \frac{1}{d}\mathbb{E}[|Y|^2 \mid |Y| \geq P] \geq \frac{P^2}{d}. \tag{OH.22}$$

Proof. Reflection symmetry gives zero mean. Rotational symmetry gives equal second moments whose sum is the radial second moment. The event gives the lower bound. ■

Thus no cutoff-independent Poincare/covariance constant follows from the local Hessian `I` on that nonconvex selected support. This example is not a counterexample to the Wilson theorem; it demonstrates precisely why the `R`-bad support cannot be ignored when trying to extend the inner C07 argument to the outer variables. Its small probability can be useful in the full law, but that probability and the actual insertion costs must be estimated together. This is the type of H02/H04 sector payment the conditional theorem does not perform.

Proposition OH.5 — even perfectly local conditional response does not control the outer mixture

Let a finite lattice have variables `z_i` in a fixed symmetric interval `[-P,P]`, with `P>2`, and define

$$\nu_y(dz) = \prod_i \frac{1_{[-P,P]}(z_i) e^{-z_i^2/2 + y_i z_i}\, dz_i}{\int_{-P}^{P} e^{-u^2/2 + y_i u}\, du}. \tag{OH.23}$$

Its conditional Hessian is `I`, its support is a convex product, its off-diagonal conditional covariance is zero, and its response to `y_i` is exactly local. Take the outer law to put probability one half on each configuration `y_i=+1` for every `i` and `y_i=-1` for every `i`. If `m(P)>0` denotes the one-site mean at `y_i=1`, then for distinct sites

$$\mathrm{Cov}_{\pi\nu}(z_i, z_j) = m(P)^2, \tag{OH.24}$$

independent of their separation.

Proof. Odd symmetry gives the means `+m(P)` and `-m(P)`. Positivity follows because the numerator of the mean is the integral over `u>0` of `u e^{-u^2/2}(e^u-e^{-u})`, which is positive. The conditional off-diagonal covariance is zero, so total covariance is the covariance of the conditional means. ■

Again this is a logical counterexample to an inference from conditional estimates alone, not a Wilson counterexample. It shows why an estimate for the actual outer law is necessary in the composition kernel `G + T G_out T^T` of RC.26.

7. Three M04 distinctions required in the existence route

The M04 analysis contains three relevant distinctions.

First, First ordinary step: an active geometry pair and a strong-carrier lower bound gives two active admitted first-step histories with the same nearby numerical data and a changed integration domain controlled by distant selector data. Under the original defining-footprint rule, the transported scalar has a nonzero local gap but every distinguishing carrier pays the distant footprint. The resulting lower bound grows like

$$\frac{e^{c\kappa M R_1}}{1+P^2}, \tag{OH.25}$$

and exceeds the microscopic incoming budget. This is an obstruction to the original full-footprint interface, not as an unproved estimate expected to become true by another C08 application.

Second, An explicit local-mask interface for the actual positive RG law introduces independent ambient current masks and a full padded retained regulator. On supported states the mask is still determined by the actual source geometry. The supported laws, marked integrals and finite source jets agree exactly with the old construction, LM.5–LM.9. Passive selection while retaining the old raw state has zero new mark current, LM.10–LM.12. After elimination, however, the actual mask current is the complete normalized difference LM.14–LM.15. A local mask representation does not itself bound that difference or make old global component data local.

Third, Moving ancestor bases, fixed innovations, and exact latent-gauge current transport separates a coherent critical-chain derivative from a chronological fixed-innovation derivative. The exact source identity is

$$D_W \Xi[\mathfrak{d}_W \xi] = \mathfrak{d}_\Xi \tilde{\xi} - D_x \Xi[A_{\tilde{\xi}} x]. \tag{OH.26}$$

Its second term is a physical innovation response and cannot be dropped by passing to a quotient gauge. MB.12 gives a concrete admitted fixed-innovation response without the desired ancestor-distance suppression. A latent change of variables can cancel the complete gauge response, but only with the insertion, density/divergence and Q-boundary terms all retained:

$$\begin{aligned}\dot{\mathbb{E}}_t f_t = {} & \mathbb{E}_t[(\partial_t + w_t \cdot \nabla) f_t] + \mathrm{Cov}_t(f_t, s_t + \delta_{\rho_t} w_t) \\ & + Z_t^{-1}(\mathcal{B}_t - \mathcal{F}_{w,t})[\rho_t(f_t - \mathbb{E}_t f_t)].\end{aligned} \tag{OH.27}$$

The current C08 theorem controls suitable fixed-support same-current covariance terms once their actual Sobolev budgets are given. It does not prove the common arbitrary-depth pullback family, the Q-boundary term, the originwise absolute carrier sum, or joint energy–velocity bounds in the amended M04 norm. Three regular layers and their first source jets are meaningful finite-depth progress, but they do not establish the uniform-depth induction.

The resolvent target MB.30–MB.33 illustrates the exact remaining depth issue: on one actual common family, one needs a small transverse error and estimates for the retained innovation forcing and labelled divergence/boundary terms. C08 cannot justify deleting the forcing term in (OH.26).

8. Remaining applications

Claim	Status	What is supplied here
Fixed-history normalized score	Retained C08 result	Exact variables and source allocation recovered
Spatial locality of the induced normalizer covariance	Proved from C08	OH.14–OH.16
Gaussian Schur upper bound plus `O(P_k^{-2})`	Proved from C08/C06	OH.9–OH.13, with literal convex cutoffs and no small-mean assumption
Fixed-current outer coercivity	Conditional	OH.18–OH.21; missing outer curvature/support test isolated
Full H04/H05 estimate	Open	Bad-sector weights, moving support and older-history contributions remain
Original fixed full-footprint M04 bound	Obstructed	Must use and prove an explicit amended interface
Amended all-history transport and ER.7	Open	Uniform depth, physical normalization and marked outer law are still required
Yang–Mills existence	Not proved	This note supplies one quantitative normalization bridge only

Sources

The following links refer to the accompanying mathematical estimates.

1. Heterogeneous constrained precision and coercivity, C6.6–C6.10 and C6.35–C6.49.
2. Regular conditional covariance, moments and cumulants, C8A.5–C8A.14 and C8Y.1–C8Y.7.
3. Response to exterior current coordinates, RC.1–RC.9, RC.19–RC.28; in particular its actual outer factorization and explicit application boundary.
4. Complete boundary activities and normalized component induction, FP.1–FP.6, read with its two-age correction; and Current endpoint estimates for boundary activities, CE.23–CE.30 for the full same-current support and native continuation.
5. Sufficient conditions for a continuum hierarchy, ER.1–ER.8 and the post-C05 scope.
6. Scalar transport: An explicit local-mask interface for the actual positive RG law, LM.1–LM.19 and its amendment scope.
7. Scalar transport: Moving ancestor bases, fixed innovations, and exact latent-gauge current transport, MB.9–MB.33, including the fixed-innovation obstruction and full normalized transport identity.
8. Scalar transport: A conditional composition interface for origin-preserving path currents, PC.3–PC.13, with its primitive admission and joint supported reserve hypotheses; for the third step it is replaced only where the finite three-layer proofs explicitly apply.

Independent check of the companion continuum calculation

The companion Physical normalization, ultraviolet bounds and source transport, EC.6–EC.20, was checked independently at the level of its stated C08 and physical-profile premises. Its new estimate has the correct physical and factorial accounting: the `h^{4m}` quadrature cancels the `h^{-4m}` native field factor; sixth-order diagonal flatness supplies `h^6`; the sole absolute physical root sum costs `h^{-4}`; and the anchored divided-tree sum adds factorial power five to the pointwise power two. Fixed source-block offsets are harmless after replacing `tau^6` by `(1+tau)^6`. The remaining fixed polynomial in `m` is absorbed in `C^m`.

Its fixed logarithmic profile `P(h) <= C(1+log(1/h))^b` costs exactly an additional factorial power `b` after absorption into `h^{2-delta}`, for any fixed `0<delta<2`. That profile requires the stated common physical-scale control, including a bounded terminal inverse-coupling interval where the cited AF comparison is used; it is not inferred solely from a weak-coupling upper ceiling. The geometric ultraviolet sum is then valid. A joint Schwartz seminorm with six derivatives and an integrable root weight controls the Taylor step and is dominated by the displayed linear-order seminorm.

The outer-average conclusion is correct for an outer probability supported on the admitted class, but it remains an average of conditional cumulants. The squarefree-source proof of the complete total-cumulance formula in EC.19 has multiplicity one for each set partition. Its other partition terms, as well as actual microscopic transport, remain outside that conditional shell estimate. This section does not promote EC.15 to ER.7.

An explicit local-mask interface for the actual positive RG law

This is an amended interface, not a proof of the M04 statement. It changes the ambient support convention for the present integration mask and specifies a corresponding off-support conditional version. On the actual supported state it preserves the original positive law, every marked integral, and every finite source jet.

The selection/lift answer is exact: adding history and mask labels while retaining all old coordinates costs nothing in the value/direct/current norm of a history-independent incoming raw mark, provided the old regulator is retained at that intermediate stage. Coordinate pullback and actual elimination remain separate operations and need their own bounds. Merely appending redundant mask bits, without changing which ambient data the conditional formula uses, does not remove the previous obstruction.

1. The amended coordinates

Fix a finite cutoff and one ordinary regular step. Let J be the finite universe of possible present independent free-coordinate slots after the actual source exceptional-bond elimination. Its slot labels include the native scale, coupling, storage block, Lie-algebra convention, and physical bond footprint. These labels are external and do not change when a mask value changes.

For each i∈J introduce two independent ambient coordinates:

$$r_i \in E_i, \qquad m_i \in \{0,1\}. \tag{LM.1}$$

Here m_i=1 means that this slot is integrated in the present regular step, while m_i=0 means that its value r_i is retained. The mask is a primitive coordinate of the ambient product, with footprint equal to the fixed physical slot footprint. One may bundle the finitely many such slots in one current block into a finite mask vector. The number per block at this one ordinary scale is bounded in terms of fixed L and the gauge group. No bounded count is asserted for an arbitrary collection of all historical modes.

Keep the original retained data θ, including all live source histories, older coordinates, coarse fields and required gauge conventions. Their old dependency footprints are unchanged. In particular a global old history flag or a whole-component response has not been declared local. The amended ambient output is

$$\widetilde{\mathsf{Y}} = \mathsf{Y}_{\mathrm{other}} \times \prod_{i\in J} (E_i \times \{0,1\}). \tag{LM.2}$$

The original source determines a mask M(θ) from its live geometry, including the literal bond-incidence convention. The actual law is supported on

$$m = M(\theta), \qquad r_i = 0 \quad \text{when } m_i = 1, \tag{LM.3}$$

as well as all original source admissibility conditions. The value zero is the old reference padding. There is no assertion that the masks are stochastically independent of θ. Ambient coordinate independence and probabilistic independence are different statements; only the former is used in (LM.2).

This note promotes only the present integration/retention mask. It does not automatically make the old determining geometry Ω_j, component identity, hull, eligibility history, tree choice, or reference field a local function. If a later argument wishes to promote additional typed occupancy fields, it must specify the corresponding primitive slots and repeat the conditional equivalence check. Encoding a whole component descriptor in one locally stored value is not the construction in (LM.1).

1.1 Full padded native regulator

At the output use the actual native compact-coordinate list, padded by identity where inactive, and the full padded retained free list:

$$R^{\mathrm{mask}}_{h,X} = 1 + \sum_{p\colon \mathrm{foot}(p)\subset X} \beta_p s(V_p) + \sum_{i\colon \mathrm{foot}(i)\subset X} |r_i|^2. \tag{LM.4}$$

The second sum has no factor 1−m_i. On the actual law (LM.3), it is exactly the native retained free-coordinate energy. Off support it also counts arbitrary numerical values in slots whose mask is one. This is an explicit part of the amendment to the ambient regulator.

That convention matters for the summed mask-current envelope. If a state has m_i=1 at many slots and arbitrary padded values r_i, separate one-mask changes can expose each of their $|r_i|^2$ contributions. A regulator $\Sigma_i(1-m_i)|r_i|^2$ followed by a supremum over one edit would see at most the largest newly exposed value, rather than their sum. The full padded list in (LM.4) does not have this defect.

No eliminated microscopic energy is added to the actual output law. On support the newly counted inactive entries are zero. During a passive lift before elimination, the regulator is instead the unchanged incoming native regulator, as explained in Section 3.

2. Exact measurable equivalence on the supported state

The source finite padded state spaces are standard Borel. A finite history decomposition is a positive kernel, and its actual coordinate, gauge/coarea and normalized conditional operations are the fixed ones from B-01/H-01 and C-01/C-07.

First suppose m=M(y) is a function of the geometry already retained in the original output y. Define

$$\iota(y) = (y, M(y)), \qquad \pi(y,m) = y, \qquad \widetilde{\mu} = \iota_* \mu. \tag{LM.5}$$

Redundant original geometric coordinates can be retained in y. They still carry their original footprints. The graph $\Gamma=\iota(Y)$ is Borel; $\pi\circ\iota$ is identity, and $\iota\circ\pi$ is identity on Γ. Thus the original supported probability space and the amended supported probability space are measurably isomorphic. Nothing was conditioned away or resampled.

Let P(dx,dy) be the original joint law before/after a regular step, and q_y(dx) its actual backward conditional. The lifted joint law is

$$\widetilde{P}(dx, d\tilde{y}) = (\mathrm{id}, \iota)_* P. \tag{LM.6}$$

For any positive probability kernel extension satisfying

$$\tilde{q}_{\iota(y)} = q_y \quad \text{on the actual admitted support,} \tag{LM.7}$$

one has

$$(\widetilde{Q}F)(\iota(y)) = (QF)(y), \qquad \int G\,\widetilde{Q}F\,d\widetilde{\mu}_{\mathrm{out}} = \int G(\iota(y))F(x)\,P(dx,dy). \tag{LM.8}$$

This is just the defining disintegration identity, so it also covers singular coarea measures and changing ranks. If the incoming stage is lifted as well, push q_y forward by its incoming embedding. Repeating this operation along the finite path gives conjugate Markov operators on the supported graphs. Hence all tower identities, joint marked moments, and finite formal source jets agree exactly. In particular

$$\int e^{\sum_a t_a F_a}\,d\widetilde{\mu} = \int e^{\sum_a t_a F_a}\,d\mu \tag{LM.9}$$

whenever the displayed finite-cutoff integrals are defined; the same identity coefficientwise needs only the corresponding finite moments.

Condition (LM.7) does not bound an ambient norm. There are many off-graph probability extensions, and some have arbitrarily bad or infinite currents. Sections 4–5 specify how the masks have to be used for a substantive amendment.

2.1 Do not reveal already averaged histories

The preceding isomorphism requires that M be a function of the state which the original operation already retains. It does not permit recovering a hidden successful component descriptor after H03/C08 has averaged it. Revealing such a descriptor can change the conditional mark. An independent posterior resampling can preserve an already transported mark, but needs its own exact source identity and compatibility with the next operation. That different construction is not assumed here.

3. Passive selection and lift have exactly zero new mark current

Let x be the complete incoming raw state, still retained. Let ℓ_x(dh) be any normalized positive source history-selection kernel; it can be discontinuous and nonlocal. Include any actual normalized gauge-choice auxiliaries in h. Let m=M(x,h) be its generated mask. The lifted law is

$$\mu^{+}(dx, dh, dm) = \mu(dx)\, \ell_x(dh)\, \delta_{M(x,h)}(dm). \tag{LM.10}$$

For every old mark F(x), its lift is

$$(LF)(x, h, m) = F(x). \tag{LM.11}$$

The exact reverse kernel is the point mass at the retained current old state x. Choose that same version on every ambient label:

$$q^{\mathrm{lift}}_{x,h,m}(dx') = \delta_x(dx'). \tag{LM.12}$$

This is a probability kernel even on an inconsistent or zero-selection- weight label. It is not the specified inactive extension δ_{x_*}. The choice (LM.12) is recorded as part of the amended interface.

For the direct/current filing, use its equivalent common-coordinate presentation with a single dummy input u=, *reconstruction* χ(,x,h,m)=x, and the constant kernel δ_*. The old B/J records then belong to the direct coordinate pullback exactly as before, and no new kernel current is created. If instead one keeps the physical point masses δ_x in a pushforward-only filing, an old x variation is represented by a point-mass kernel difference and its total-variation derivative can be infinite. That different majorization is not used in the norm theorem.

For a new history or mask perturbation, the old state x does not move, the kernel in (LM.12) does not move, and F does not move. Both direct and kernel-current records are therefore exactly zero. For an old coordinate perturbation, the old direct and current records are simply carried along. No derivative of ℓ or of M is needed.

The same cancellation can be seen without a reverse kernel. A signed finite difference of the normalized selection measure has total mass zero. Pairing it with the mark F(x), constant in its selection variables, gives F(x) times zero. This applies to hard selector jumps as well as smooth positive selection probabilities. It must be done before any absolute-value majorization of the separate unnormalized branches.

Theorem LM.1 — exact norm preservation at passive lift

At the passive stage keep the full incoming native coordinate and energy list, its external scale/coupling labels and its carrier footprints. Give (LF)_X the same carrier as F_X. Lift each old compatible record and put every newly introduced history/mask record equal to zero. Then the M.11–M.12 value/direct/current norm, using the amended coordinate inventory, is exactly the incoming norm.

Proof. The lift ignores all the new coordinates. Its value envelope under old or new single-coordinate perturbations is exactly the old value envelope, and the extra difference directions contribute zero. The regulator is the unchanged incoming regulator and likewise ignores the new coordinates, so its shifted envelope is unchanged. The x marginal of (LM.10) is μ, giving equality of every averaged moment. Carriers, weights and the physical unit h are unchanged. This proves each summand of the norm identity. The product rule shows the same statement coefficientwise for every finite jet. ■

This theorem in particular covers the microscopic F01 Wilson functions, which have their global compact-link formula independently of the auxiliary history labels. It also covers an old mark already depending on retained old history, with that dependence's existing cost kept.

It would be wrong to remove the soon-to-be-integrated energies from the regulator at this passive stage, while the corresponding raw coordinates are still present in F. It would also be wrong to cite LM.1 after discarding x and conditioning only on the new masks. Those are actual integration operations; their conditional marks need not be constant in the mask.

Keeping x here is an intermediate exact disintegration, not permission to keep eliminated microscopic fields or their energies in the final coarse native list.

4. Rank padding and the complete conditional current algebra

Write the old current field in a common noise space. For each slot take a fixed normalized reference probability π_i on its source free cutoff domain. If a coordinate is integrated use its noise z_i; if it is retained use r_i:

$$a_i(z,r,m)=\begin{cases} z_i, & m_i=1,\\ r_i, & m_i=0. \end{cases} \tag{LM.13}$$

Unused z_i are dummy internal noises with probability π_i and are ignored by this assembly. They are not new retained state coordinates and do not enter the output native regulator. This normalized padding is optional: one may instead use point masses on inactive noise slots and the finite signed-measure formulas below still hold.

Let χ_m(z,r,θ) be the actual complete source reconstruction, including the exceptional solve, any coordinate relocation and every gauge convention which has not already been undone by its exact identity. Let ν_{θ,r,m} be its complete normalized conditional law, represented on the common noise space with its actual density, constraints, coarea factors, determinants, selectors and normalizer. On the graph (LM.3) require the exact match (LM.7).

For a discrete one-mask change τ write ν'=ν_{τy} and ν=ν_y. Then

$$\begin{aligned} \widetilde{Q}F(\tau y)-\widetilde{Q}F(y)=\nu'\big[F\circ\chi_{\tau y}-F\circ\chi_y\big]\\ +(\nu'-\nu)\big[F\circ\chi_y\big]. \end{aligned} \tag{LM.14}$$

The first term is a direct reconstruction/rank response. The second is the normalized-kernel response. Since (ν'−ν)1=0, the second can be centered by subtracting ν(F∘χ_y). No denominator other than one is used for a discrete mask replacement.

This direct/kernel assignment is for the explicitly pulled-back common-noise filing of (LM.13). In the physical push-forward filing used by the Gaussian local-mask theorem and the primitive physical quadratic-mask estimate, both the embedding and density changes belong to J. The two exact filings are alternatives; their direct and current records are not silently identified term by term.

More generally, if the pulled-back mark has compatible records δA=B+J, its transported records are exactly

$$\begin{aligned} A'&=\nu_y A,\\ B'_\tau&=\nu_{\tau y}B_\tau,\\ J'_\tau&=\nu_{\tau y}J_\tau+(\nu_{\tau y}-\nu_y)A, \end{aligned} \tag{LM.15}$$

with division by the native step for continuous directions. These are M.30's identities on the new product coordinates. They neither discard old currents nor assume a constant physical rank.

One may push ν_y through χ_y and work directly with physical old-state probabilities. A bare physical mark then has no explicit retained-data dependence and the new response is entirely a kernel current. In that filing, changing masks may produce mutually singular physical kernels; the finite signed-measure difference is still exact. A continuous total-variation derivative may be infinite, so that filing does not replace a needed chart-path estimate for a general incoming mark.

The product rules, finite jet logarithms/exponentials and Hölder grade identities of M.23–M.25 remain unchanged. The regulator (LM.4) is monotone under carrier union, and the same physical weight exp{κ(|X|+t_h(X))} is used. Consequently the support/current algebra is preserved. No uniform operator estimate follows merely from this algebra.

4.1 Coordinatewise localization avoids unrelated storage closure

A global old flag may be stored in the same block as a local mask. Blindly taking the full storage closure of that block can charge the unused global flag again. Use instead one reset projector E_c for each primitive coordinate c, including the mask coordinates. They commute. For a finite coordinate set S define

$$L_S = \prod_{c\in S}(1-E_c)\prod_{c\notin S}E_c, \qquad f=\sum_S L_S f. \tag{LM.16}$$

Its declared carrier contains the insertion anchors and the full footprints of the coordinates in S. The footprints of the original non-mask coordinates have not been shortened. If f is independent of a global flag c, every term with c∈S is zero. Thus unrelated metadata stored at the same block cannot create an artificial carrier cost.

For a direction in coordinate c, set its localized records to zero when c∉S. When c∈S omit the factor 1−E_c from the record-localizing operator. The identity

$$\delta_c L_S f = \Big[\prod_{d\in S\setminus\{c\}}(1-E_d)\prod_{d\notin S}E_d\Big]\delta_c f \tag{LM.17}$$

follows because δ_cE_c=0 and δ_c commutes with every other reset. It proves compatibility of the separately localized direct and kernel records. This is the coordinatewise version of M.36a–c. It is a supported representation, not a claim that every term has a small carrier or that the resulting absolute sum converges uniformly.

5. A concrete positive all-mask extension at the first flat step

The first-step geometry obstruction has a particularly transparent amended version. Fix Ω_1 full, V_1=1, and the literal soft-gauge operation. Undo the exterior relocation if necessary. At these flat data it is already exactly the identity. The complete primitive free-coordinate density is a strictly positive smooth function

$$W_g(a)=\exp\{-\tfrac{1}{2}a^*Ka+\mathcal{W}_g(a)\} \tag{LM.18}$$

on a fixed small physical chart. Here W_g includes Wilson, soft gauge, Haar, coarea and exceptional-coordinate factors, each once. The source R-good product cutoffs are imposed separately. All current R-bad and remaining approximate-fluctuation factors are retained-only on an actual source geometry; the normalized conditional cancels them. Initial redundant P/gauge selectors are one there. This is the source factorization used in C7 and in the GO analysis.

Choose P=δ_0/g_0 and a fixed buffered radius bP containing every actual retained first-chart value in this sector. Let s:E_i→E_i be a bounded coordinatewise smooth saturation, identity on that admitted retained domain, with image in a slightly larger b'P ball. For sufficiently small g_0, g_0b'P lies inside the same fixed constraint chart. Its Lipschitz constant is fixed. Use s(r_i) instead of r_i off that domain. This is an explicit ambient continuation; it is not an estimate of the original field at arbitrarily large ghost values.

Let π_i be normalized Lebesgue measure on the literal norm ball C_P. For every ambient mask, define

$$\begin{aligned} a_i^s(z,r,m) &= m_i z_i + (1-m_i)s(r_i),\\ \nu_{r,m}(dz) &= \frac{W_g(a^s(z,r,m))}{\int W_g(a^s(w,r,m))\prod_i \pi_i(dw_i)}\prod_i \pi_i(dz_i). \end{aligned} \tag{LM.19}$$

The denominator is positive and finite at every fixed finite cutoff. Unused π_i integrate to one. On a consistent actual retained datum, s(r)=r and (LM.19) is precisely the actual source conditional over the coordinates I_m, with its product cutoff. Constants such as the Lebesgue volume of C_P, Gaussian normalizations and source scale powers cancel within this fixed-fiber ratio. They remain in the unchanged actual outer law μ_out; no history-dependent factor is removed from its branch probabilities.

Current R-history labels do not appear in (LM.19). Their exact selection probabilities and their consistency with m are retained in the outer law, not inserted as a zero-normalizer gate in this ambient conditional. At fixed r,m, changing such an otherwise redundant label therefore has zero output current in this first sector. A change of m has the genuine direct/kernel responses (LM.14).

The physical reconstruction in this first chart uses the local C and exceptional solve at the same fixed Ω_1,V_1. A one-mask assembly change alters only one independent coordinate before those fixed-L maps. Thus its direct physical reconstruction footprint is a fixed-L neighborhood. The normalized conditional law can still respond throughout the system. Bounding that response, its absolute supported expansion, shifted envelopes and all source orders is an analytic problem; it is not asserted by (LM.19).

This explicit construction shows that the proposed repair is more than renaming the old global flag. At the two active GO data, the local mask near the root differs. A nearby carrier can now distinguish them. Changing a remote R label while holding the independent local mask fixed leaves (LM.19) unchanged. The old lower-bound premise no longer holds for this amended conditional version.

5.1 What would make the repair only notational

There are two ineffective alternatives:

1. Append m but continue to evaluate Q using the derived set Λ(H), ignoring the new mask. Then the lifted output is the same old function of H, with the same nonlocal dependency. Nothing was repaired.
2. Set the ambient conditional equal to the source ratio only when m=M(H), and use δ_reference otherwise. The consistency indicator itself then remains in the output formula. A remote H change at fixed m can create an order-one response, and the full footprint cost can return. Positivity and on-law equivalence alone do not bound that version.

The useful extension (LM.19) instead makes the independent mask the actual rank argument off support and removes the canceled retained-only selection gates from the normalized conditional. The outer supported law is still exactly the original law.

6. The source-coordinate costs which remain

The following distinctions are essential beyond the flat first sector.

Actual coordinate pullback. A raw incoming F(x) becomes F(χ_m(z,r,θ)). Even if F originally ignores history, χ_m can depend on the geometry, old reference fields and coarea conventions. The direct record in (LM.14) and the M.28a weighted chart path must be estimated. Passive lift theorem LM.1 does not make χ local or give it unit cost.

Exterior relocation. CMP119 (1.22), and its later analogue, change retained exterior coordinates using a geometry-dependent reference. At the GO centers they are identity. In general one can undo the exact change, including its Jacobian, and use a pre-relocation retained chart whose ordinary background is attached to Ω rather than the new Λ. That is an equality of measures. Its pullback footprint and normalized coordinate-path bounds are additional obligations. Treating the two retained coordinate arrays as identical without that map would be false.

Inherited complete density. At a later step, old Gaussian coefficients and completed E/B/R responses can depend on the live old history and on the current reconstructed field. Merely replacing the current integration mask does not remove those dependencies. All source factors and normalizations must remain. A claim that the conditional factors through only local masks requires an actual factorization identity for that density, not a relabeling of its geometrical arguments.

Arbitrary nonflat ambient data. Formula (LM.19) has a complete positive extension on the fixed first flat chart. For a general native background, source minima and coordinate charts have admission conditions. A piecewise reference

extension exists measurably, but can have infinite continuous currents. A useful global extension needs a buffered chart/patch construction and actual estimates. None is assumed for an unproved complete-history family.

History erasure and future operations. Consistency remains in the outer law. If a later operation averages masks or primitive histories, its posterior depends on that actual law and can create nonlocal normalized currents. A full revised induction must bound them. The passive-selection cancellation does not apply after the physical variables on which F depends have been discarded.

Rank and moment budgets. The final regulator contains only retained native fields on support. Conditional moments of the eliminated variables, changed-rank normalizers and direct path lengths are still to be paid. The full padded off-support extension in (LM.4) handles ghost retained values, not the actual eliminated microscopic energy.

7. The amended norm is not equivalent to the specified norm

The repair is deliberately substantive. For one mask slot b, the function

$$f_b(r, m, \theta) = m_b \tag{LM.20}$$

has a bounded local carrier in the amended interface. Its values and one-mask differences are at most one, its other coordinate differences are zero, and its moment norms are bounded for every actual supported law. With a local anchor its M.11–M.12 norm is bounded by a fixed constant.

Its restriction to the original graph is the derived membership function M_b(H). In the actual pair of source histories constructed in M04 First Step Geometry Obstruction, this bit near the extended hole face changes when a remote R seed is added. All nearby primitive data are unchanged and the native regulators are only polynomial in P. The supported-carrier lemma GO.19 therefore forces an old-interface cost at least c $\exp(c\kappa MR_1)/(1+P^2)$.

Thus restriction along the consistency embedding is not a uniformly bounded map from the amended norm to the specified M03 norm. There is no contradiction with the exact law isomorphism in Section 2: the probability laws agree on their graphs, while the pointwise ambient support/current norms are different. Nor is there a contradiction with LM.1: its isometry concerns marks independent of the newly added coordinates, whereas (LM.20) depends on a new mask.

A physical perturbation which changes a primitive R test and then updates all masks to restore consistency is a composed path in the new coordinates. It can change many nearby mask bits at once. The amended elementary current at fixed independent masks is not a bound for that composed path. Any downstream requirement for such a consistency-preserving response must include its true path/current cost. The previous obstruction cannot be hidden in a claimed bounded change of norm.

8. Exact changes from the original interface

Item	Original construction	Amendment in this note
Present Λ/activation data	Derived from primitive P/Q/R geometry, with full construction footprint	Independent ambient per-slot mask; its actual law remains constrained to that same derived value
Other histories and descriptors	Original full footprints	Unchanged; no whole-component payload is made local
Passive-lift null labels	Default reference-input extension can create an unnecessary version jump	Reverse kernel δ_x at the retained old state for every new label
Present conditional off support	Original generated version or consistency-gated ratio	Independent mask supplies the rank argument; canceled retained-only gates stay only in the outer law; (LM.19) is an explicit first-chart example

Item	Original construction	Amendment in this note
Padded retained energy off support	History-sensitive native-list reading can omit inactive ghost values	Full padded Σ
Carrier construction	Blockwise reset/closure can overcharge unrelated stored flags	Coordinatewise supported reset representation (LM.16)–(LM.17)
Physical law and marked integrals	Original exact positive law	Unchanged on support, by (LM.5)–(LM.10)
Downstream completion	Original M03/M04 obligations	Requires an explicitly revised induction and norm-conversion/dependency analysis; no proof claimed

The outcome is an exact, explicit alternative interface. Its passive selection step introduces no hidden mark cost. Its genuinely difficult steps remain the source reconstruction, complete conditional response, physical carrier sums, and later averaging of the constrained history law. Those estimates must be proved in the amended interface before it can support a proof of M04.

Sources and companion results

- Annex W §2.2: exact positive primitive history inventory and derived geometric domains.
- Annex Y (M.1)–(M.12), (M.23)–(M.39): native regulators, padded sectors, finite current identities, and compatible localization.
- C7.1 and CMP119 (1.18)–(1.26): actual fixed free-product regular conditional and retained-only selector cancellation.
- M04 First Step Geometry Obstruction and its companion analyses: actual active pair, source normalization, and the old-norm lower bound which this amendment does not purport to prove away.

Three actual regular integrations: density, scalar and four path origins

This note assembles the actual third homogeneous regular integration from the companion analytic/Gaussian construction TC, nested real-core construction NC, and three-innovation marked calculation TR. It proves the density and one-microscopic-scalar conclusions in the explicitly R-averaged, native, diagonal-path interface stated below. Constants and the weak-coupling ceiling may depend on these three fixed levels.

1. The operation and the two inherited defect generations

Use three homogeneous ordinary integrations with the full lattice as each P/Q-regular region. At each stage perform the explicitly specified physical averaging operation of BA.5a and sum its R partition with the actual masses. The operation acts on the preceding averaged density. This does not assert that the original history-dependent future kernel is unchanged after erasing its history labels.

Set $h_j = L^j h_0$, $j = 0, 1, 2, 3$, and use the supplied zero-microscopic-source couplings g_j. Adjacent coupling ratios are bounded at these three fixed levels. They, the localization grids, cutoffs, gauges and normalization conventions stay fixed during a current or source derivative. No coupling-flow theorem is assumed for a new auxiliary scheme.

Let P_0 denote the first comparison radius, $P_1 = P_{\mathrm{aux},1}$ the broad second comparison radius of AD, and $P_2 = P_{\mathrm{aux},2}$ the third broad comparison introduced after the third R sum. Their associated physical Q radii remain distinct. Each auxiliary radius is a fixed sufficiently small fraction of its own Q radius; it is not the shrinking later R-good radius printed in CMP119 (3.16).

The two-step density identity is

$$\rho_{QQ,2}(V_2) = \rho_{aa,2}(V_2) \exp\{E_2^{(0)}(V_2) + E_2^{(1)}(V_2)\}, \tag{TS.1}$$

where $\rho_{aa,2} = \rho_{\mathrm{aux},2}$ is the complete first-P/second-auxiliary density of AD,

$$E_2^{(0)} = \Delta_2, \qquad E_2^{(1)} = \delta E_{\mathrm{new},2}. \tag{TS.2}$$

Each tag of $E_2^{(0)}$ retains a distinguished first-generation Q defect; each tag of $E_2^{(1)}$ retains a second-generation Q defect. A first-generation tag can also contain additional later cutoff factors. Its distinguished colour is a bookkeeping label for the retained first reserve, not a denial of those factors.

Use their full latent integral representations. They include all ordinary connector supports, Gaussian copies, exact first/second normalizers, local references and hard domains. The bound on their integrated values does not authorize a holomorphic substitution into a moving hard domain.

2. Exact third density and analytic bookkeeping

TC supplies the actual third constrained reconstruction

$$V_2 = \Psi_{3,W}(c)$$

and the complete smooth third normal form

$$D_2(c; W) = \exp\{-\tfrac{1}{2} c^{\mathsf{T}} K_2(W) c + \mathcal{W}_2(c; W)\}. \tag{TS.3}$$

It is obtained from the complete density $\rho_{aa,2}$, including its determinant and vacuum terms. The precision retains the source constraint-force corrections and the nonlinear averaged soft-gauge coefficient. The perturbation retains the inherited analytic action difference, classical/gauge remainders, Haar factors and exceptional-bond coarea. Any difference between the auxiliary comparison's projected action coefficient and the supplied running-coupling convention remains in that actual analytic action; it is not silently set to zero.

Keep all retained-only factors in the same $C_2(W)$. Write $Q_2(c, W)$ for the literal third Q indicator and set

$$Z(\lambda_0, \lambda_1; W) = \int Q_2(c, W) D_2(c; W) \exp\Big\{\lambda_0 E_2^{(0)}(\Psi_{3,W}(c)) + \lambda_1 E_2^{(1)}(\Psi_{3,W}(c))\Big\}\, dc. \tag{TS.4}$$

The three physical densities used here are

$$\rho_{QQQ,3} = \chi_3 C_2 Z(1,1), \qquad \rho_{aaQ,3} = \chi_3 C_2 Z(0,0), \qquad \rho_{aaa,3} = \chi_3 C_2 Z_{a,3}, \tag{TS.5}$$

where $Z_{a,3}$ is the same smooth third integral with the fixed auxiliary product domain in place of Q_2.

Define

$$\begin{aligned} E_3^{(0)} &= \log Z(1,1) - \log Z(0,1), \\ E_3^{(1)} &= \log Z(0,1) - \log Z(0,0), \\ E_3^{(2)} &= \log Z(0,0) - \log Z_{a,3}. \end{aligned} \tag{TS.6}$$

The intermediate $Z(0,1)$ is an auxiliary positive integral. No identification with an original unaveraged history is required. The exact density identity is

$$\rho_{QQQ,3} = \rho_{aaa,3} \exp\{E_3^{(0)} + E_3^{(1)} + E_3^{(2)}\}. \tag{TS.7}$$

All logarithms in TS.6 are after the complete third integration. The full old exponential occurs once in the actual $Z(1,1)$ denominator.

3. Two-colour normalization against one actual gas

3.1 What must be retained before the next integration

For each old payload occurrence keep its entire Gaussian integration tree. A copied two-step logarithmic factor has its own middle latent variables and its own old latent copies; equal physical supports do not identify those variables. Each tree has depth at most two. The third integration variable is shared.

NC freezes the complete earlier nonanalytic input core: the promoted old core already includes its middle reference boxes, whole middle-Q factors, fixed-target data and frame-registry inputs. A genuine second-generation Q core is added separately. Ordinary analytic connectors receive fixed collars only.

TC's conditional certificate integrates all oldest variables first, then the middle variables, at fixed real third variable. For a primitive forest it has the schematic bound

$$A_{\mathcal{J}} \exp\left\{\eta \sum_{\text{old copies}} r_i\right\} \exp\{-c(n_0 P_0^2 + n_1 P_1^2)\}. \tag{TS.8}$$

Every actual Gaussian rank is charged to its own primitive before connected summation. There is no remaining third-variable quadratic source tilt per unmarked copied payload. The one common third Gaussian, its frame and its complete interaction are then integrated once.

The outer core cost is additive in the actual defect occurrences:

$$C_L n_0 (R_1/L^2 + R_2/L + R_3 + m)^4 + C_L n_1 (R_2/L + R_3 + m)^4. \tag{TS.9}$$

At three fixed levels the appropriate birth tails pay these costs and the polynomial incidence factors. This is a finite allocation of the existing strict reserves.

3.2 Exact component identity

Expand the exponential in TS.4 with its exact factors $1/(m_0! m_1!)$, retaining separate latent copies for every occurrence. Use TC/NC's common coefficient, constrained-frame, covariance and determinant registry. A component containing no old payload is exactly its unmarked third-Q component, including the literal domains and complete analytic density. Its equality is the same root-deletion identity required by CN; approximate agreement would be insufficient.

Thus, after one identical retained prefactor, TS.4 is a gas with base species z_γ and species containing one or both old colours. The actual denominator for a normalized observable at $(1,1)$ is this full gas. A disjoint vacuum component cancels against that same denominator. The excluded-vacuum factor on a rooted carrier is CN's e^{-T_Y}, with all connector supports included.

Lemma TS.1 — retaining each old colour

Suppose the actual primitive bounds just described give old anchored input budgets $\varepsilon_0, \varepsilon_1$, with a fixed strict component and decorated connected-cluster margin. Then the connected logarithm terms containing a specified old colour i have anchored value/once-current bound $C\varepsilon_i$. The subseries containing both distinguished old colours has bound $C\varepsilon_0\varepsilon_1$.

Proof. Choose a fixed small intensity a_* within the ordinary component and connected-cluster margins. In the positive majorant multiply each occurrence of old colour i by s_i. Evaluate the majorant at $s_i^* = a_*/(2\varepsilon_i)$, after decreasing the coupling so both values are at least two. The inflated total old input is at most a_*, so the same majorant is bounded there by a fixed constant, including the reserved first-current decorations.

Every term containing colour i has degree at least one in s_i, so returning to $s_i = 1$ gains $1/s_i^*$. Terms containing both colours gain the product. Coefficients of this majorant are nonnegative, so no cancellation is used in this inference. The factorials of the actual exponential are retained in the component enumeration. No Gaussian action is changed by these counting variables; they multiply already bounded primitive old vertices. This proves the assertions. ■

Applying the lemma to the exact differences TS.6 identifies the surviving colours. TC/NC supply $\varepsilon_i \le C_L e^{-cP_i^2}$, $i = 0, 1$. The new third-Q comparison is the same integrated annulus/side-chain construction on its actual inherited precision, with radius P_2.

Theorem TS.2 — third native regular density

For the actual third-step family constructed in TC and NC, the three terms in TS.6 have connected raw-supported decompositions satisfying

$$\|E_3^{(i)}\|_{\kappa,\mathrm{anc},1} \le C_L e^{-cP_i^2}, \qquad i = 0, 1, 2, \tag{TS.10}$$

and hence

$$\|E_3^{(0)} + E_3^{(1)} + E_3^{(2)}\|_{\kappa,\mathrm{anc},1} \le C_L \sum_{i=0}^{2} e^{-cP_i^2}. \tag{TS.11}$$

The norm is TD.5 at the third scale, with the native buffered paths and the original continuous direction normalization.

Proof. TS.8–9 give the actual positive old-colour inputs, uniformly over arbitrary copied multiplicity. The same component expansion gives the strict full gas used by Lemma TS.1. Taking its connected logarithm and subtracting as in TS.6 keeps colour zero in the first difference and colour one in the second. The new-Q difference contains a third-generation annulus or boundary-chain defect. Its inherited covariance, determinant, measure and interaction families are exactly those in TC. Their native shell and chain bounds give the third term.

A current differentiates one primitive factor, coordinate map or actual domain. Old and new domain derivatives are the real boundary currents in NC. The old conditional certificates are uniform in the outer real variables, so a later boundary current retains an earlier defect's tail. Analytic density currents retain both determinant and inverse responses and all inherited measure terms. Their polynomial costs are allocated from the already reserved rate and tail margins before applying the coloured majorant. This gives TS.10 for values and currents. The common prefactor restores TS.7 without alteration. ■

4. One microscopic mark and four physical origins

Keep the original microscopic field

$$F(U) = \alpha\beta_0\sigma_{h_0}(x; U) - c_0, \qquad b_F = |\alpha| + h_0^4|c_0|. \tag{TS.12}$$

Use b_F here to distinguish the incoming scalar budget from a Gaussian birth index.

For a selected chain η, let $q_{0,\eta}$, $q_{1,\eta}$, $q_{2,\eta}$ be its actual zero-source normalized conditionals, including the inherited prefix density in each next kernel once. Let z_0, z_1, z_2 be their original independent coordinates. Their physical reconstructed microscopic readout is $f_\eta(z_0, z_1, z_2, W)$, and

$$\mathcal{F}_\eta(W) = q_{2,\eta}q_{1,\eta}q_{0,\eta}[f_\eta]. \tag{TS.13}$$

The chains of primary interest are the full QQQ chain and the first-P/second-auxiliary/third-Q baseline aaQ.

Define the microscopic origin by differentiating f at fixed original coordinates. Define origin i by differentiating the actual normalized kernel q_i, holding its displayed conditional test fixed. All factors inside a given kernel density or its denominator belong to that kernel's origin, including inherited masses that arose from earlier integrations. Equal-valued occurrences elsewhere keep their own labels.

An exact occurrence-level source expression is

$$\mathcal{G}_\eta(\tau, W) = \frac{1}{Z_{2,\eta}(W)} \int n_{2,\eta}(z_2, W) \frac{1}{Z_{1,\eta}(V_2)} \int n_{1,\eta}(z_1, V_2) \frac{1}{Z_{0,\eta}(V_1)} \int n_{0,\eta}(z_0, V_1) e^{\tau f_\eta}\, dz_0\, dz_1\, dz_2. \tag{TS.14}$$

Characteristic domains are part of the n_i. Form values and origin-labelled first jets of every occurrence before canceling a prefix mass against a conditional normalizer. Each inverse uses the same first-jet inverse rule as DP.22b. In the common component family,

$$q_{i,s}(1) = 1, \qquad \dot{q}^{\,i}_{i,s}(1) = 0, \qquad i = 0, 1, 2. \tag{TS.15}$$

The source-free microscopic jet is zero as well. Thus each physical origin separately annihilates a constant microscopic mark.

NC's three-level triangular map and its full Lie/divergence identities turn the sum of these four physical jets into the path derivative of the same continued integral. They are constructed together with the value at every outside parameter. Coefficient extraction and the normally convergent connected expansion therefore preserve

$$\frac{d}{d\theta} \mathcal{F}_{\eta,Z}(W(\theta)) = \sum_{r=M,0,1,2} j^{r}_{\eta,Z,d}(W(\theta)) \quad \text{a.e.} \tag{TS.16}$$

This applies to the actual analytic lifted paths using the retained primitive domains. It does not evaluate stored elementary current numbers as the components of a differential on a tie stratum.

5. The rare microscopic correction

Set $\mathcal{E}_F = \mathcal{F}_{QQQ} - \mathcal{F}_{aaQ}$ and subtract the two physical responses within each origin before any absolute estimate.

The source versions of the two-step density ratios give a useful value-level factorization. Let $B_2(\tau)$ be the logarithm of the normalized aa two-step source expectation. Let $E_2^{(1)}(\tau)$ be the log ratio of the unnormalized aQ and aa source densities, and $E_2^{(0)}(\tau)$ the log ratio of QQ and aQ source densities. Then $B_2(0) = 0$, the latter two reduce to TS.2 at zero source, and the full old source density is exactly

$$\rho_{aa,2} \exp\{B_2(\tau) + E_2^{(0)}(\tau) + E_2^{(1)}(\tau)\}. \tag{TS.17}$$

This factorization does not reassign the origin jets of TS.14. Those retain their original occurrence labels through every cancellation.

TR's bounded native source certificate permits one fixed source circle

$$|\tau| \le c/M_F, \qquad M_F = C_L b_F h_0^{-4} P_{\max}^2, \qquad P_{\max} = 1 + \max(P_{Q,0}, P_{Q,1}, P_{Q,2}). \tag{TS.18}$$

It is fixed over the entire real path, every admitted component/source tube and all relevant contours. At three fixed levels these cutoff radii have bounded ratios. No source derivative changes the saddle, running coupling or geometric partition.

5.1 Colour survives separately in each physical origin

Here is an algebraic check of the per-origin assertion. For one physical path direction use the finite-dimensional first-jet algebra

$$A = \mathbb{C}[\epsilon_M, \epsilon_0, \epsilon_1, \epsilon_2]/(\epsilon_r \epsilon_s \,:\, r, s \in \{M, 0, 1, 2\}). \tag{TS.18a}$$

An occurrence belonging to origin r is replaced by its value plus ϵ_r times its own physical derivative. Products retain the Leibniz rule, and the inverse rule is exactly $(a + \epsilon_r b)^{-1} = a^{-1} - \epsilon_r b a^{-2}$. Apply these rules to the uncanceled

nested expression TS.14. A mass inside a newer kernel and an equal-valued mass inside an older normalizer are distinct occurrences in this algebra. In particular the old mass inside the third kernel carries ϵ_2.

Introduce primitive birth-counting variables ξ_0, ξ_1, assigning one factor to each actual first or second Q-defect occurrence in the common integral expansions. These are not the density-toggle variables λ_i in TS.4. Every numerator and its own normalizer use the same birth-counted family and the same owned jet. The positive component majorants converge with both birth intensities inflated, as in Lemma TS.1; integration, inversion, source differentiation and coefficient extraction therefore act on normally convergent A-valued series.

At $\xi_0 = \xi_1 = 0$, the first Q/P difference disappears, the second Q domain becomes its auxiliary P domain, both inherited correction masses become one, and the entire third density becomes its analytic-base Q density. This statement includes every occurrence's first jet: it is an evaluation of the common primitive family, not a new continuation chosen after identifying its values. Thus, writing G_ξ for TS.14 in this algebra,

$$\mathrm{G}_\xi(0) = 1, \qquad \mathrm{G}_{(0,0)}(\tau) = \mathrm{G}_{aaQ}(\tau), \qquad \mathrm{G}_{(1,1)}(\tau) = \mathrm{G}_{QQQ}(\tau). \tag{TS.18b}$$

The first equality holds in the full algebra because every normalized conditional integrates 1 to 1 at every outside parameter. The microscopic jet of $e^{\tau f}$ also vanishes at $\tau = 0$.

Expand $\log \mathrm{G}_\xi - \log \mathrm{G}_{(0,0)}$ in the same connected rooted family. Each surviving tag has positive degree in an old birth variable and positive source degree. Differentiating in τ and taking any one ϵ_r coefficient preserves that positive old degree. Hence the difference of the two origin-r responses is itself a sum of source-containing, old-defect-containing tags before any absolute estimate. Source-free jets and vacuum components disjoint from the full root cancel by TS.18b and the identical component restrictions, rather than by a bound on an extensive partition-function score.

One may check the same algebra directly. If two normalized kernels have unnormalized numerators N_Q, N_A, masses Z_Q, Z_A, and $\delta N = N_Q - N_A$, then

$$q_Q - q_A = Z_Q^{-1}\{\delta N - q_A\, \delta N[1]\}. \tag{TS.18c}$$

It is an identity of test-function measures and of the owned first-jet algebra. Every term in its value or jet difference has a changed numerator or its signed derivative. Telescope the three conditional compositions with this identity; every term has a changed kernel, its changed jet, or the changed source-containing lower test. The connected construction just described keeps their normalizer responses combined until disjoint components have canceled. It never takes separate absolute bounds on the extensive terms that appear if TS.18c is differentiated and expanded prematurely.

5.2 The actual source and current majorant

Add the bounded source vertices and their four origin jets to the same full coloured primitive family before its normalized connected expansion. TC and NC give its component identity and its same-denominator normalizations; TR gives the source-containing primitive bounds. Section 5.1 identifies the old colour in each physical-origin difference.

TC/NC give the unmarked copied-tree certificate. TR supplies the one distinguished source/current root and its actual supported velocity. Cauchy costs M_F once; a fixed fraction of the appropriate old tail pays $P_{\max}^2$, the surface powers and the defect-core costs. Ordinary rank/incidence costs use the strict ordinary spatial margin. No growing halo is assigned to a source-only analytic connector.

Define

$$J^{r,\mathrm{path}}_{\Delta,Z,d,t} = \frac{1}{t}\int_0^t j^r_{\Delta,Z,d}(W(\theta))\, d\theta. \tag{TS.19}$$

The common deterministic envelopes and TS.16 give the exact coefficientwise finite identity and bound

$$\Delta^{\mathrm{norm}}_{d,t}\mathcal{E}_{F,Z} = \sum_{r=M,0,1,2} J^{r,\mathrm{path}}_{\Delta,Z,d,t}, \tag{TS.20}$$

$$h_3^4 \sum_{Z\ni x^{(3)}} e^{\varkappa n(Z)} \left[\sup_W |\mathcal{E}_{F,Z}| + \sup_W \sum_d \sup_{0<|t|\le 1} \sum_{r=M,0,1,2} |J^{r,\mathrm{path}}_{\Delta,Z,d,t}|\right] \le C_L L^{12} b_F \left(e^{-cP_0^2} + e^{-cP_1^2}\right). \tag{TS.21}$$

The factor $h_3^4/h_0^4 = L^{12}$ is retained.

6. Absolute scalar and fixed-basis extraction

TR supplies the absolute aaQ value and its own four diagonal derivative records in a common energy-graded first-jet family. Add the rare family from Section 5 with a common reindexing. This defines the QQQ value tags and their currents together; no unrelated specified value representation receives new currents afterward.

For the final regulator $R_Z(W) = 1 + \sum_{p\subset Z} \beta_3 s(W_p)$ and its actual shifted version, the resulting bound is

$$h_3^4 \sum_{Z\ni x^{(3)}} e^{\varkappa n(Z)} \left[\sup_W \frac{|A_Z^{QQQ}(W)|}{R_Z(W)} + \sup_W \sum_d \sup_{0<|t|\le 1} \frac{\sum_{r=M,0,1,2} |J^{r,\mathrm{path}}_{Z,d,t}(W)|}{\widehat{R}_Z(W)^{3/2}} \right] \le C_L L^{12} b_F. \tag{TS.22}$$

The unweighted rare estimate is absorbed because the regulators are at least one.

Apply the same bounded symmetry-averaged finite-probe projector SE to this value family at scale h_3:

$$T_{QQQ}F = a_{\mathrm{out}}\beta_3\sigma_{h_3} + c_{\mathrm{out}} + \mathcal{R}. \tag{TS.23}$$

The fixed native probes have bounded final energy and a uniform nonzero scalar denominator. Thus the scalar, identity and exact remainder have the bound implied by TS.22. The scalar subtraction has its explicit extraction-current record; the four microscopic origins remain recorded. Conversion to the unchanged incoming F01 normalization retains β_3/zS_{in}. No new incoming budget, continuum nondegeneracy or outer-law moment estimate is inferred.

7. Scope and what the third step does not establish

The completed statement is a native theorem at three fixed homogeneous regular levels, for the explicitly R-averaged density operation and one microscopic scalar source derivative. Its finite currents use the amended diagonal path convention. It does not identify them with the specified canonical endpoint rectangles.

The finite number of coefficient-rate losses, real/complex admission collars, rank tilts, conditional trace stores and geometric covering constants are chosen for these three levels. This proof does not claim that they are uniform when the number of levels grows. The new real AP/SG estimates remain useful separate inputs but do not, by themselves, replace the complex interacting and moving-domain certificates.

The original support-interface obstruction, ambient and averaged M03 requirements, complementary P/Q sectors, arbitrary histories/resets, complete outer connected hierarchy and continuum Yang–Mills existence remain outside this fixed-three statement.

8. Concrete inputs and scope boundaries

The following are the actual companion constructions used in this proof. They are not additional hypotheses to be supplied by an arbitrary previously integrated field.

Required input	Construction and location
Complete analytic input, actual third precision, inherited action and determinant	TC.3–10, including the literal classical factor $(g_2/g_1)^2$ and the separately normalized soft gauge
Positive primitive reserve before old logarithms are aggregated	TC.11–11a; the additional Gaussian/rank and vertex stores are inserted in the majorants, not the physical action
Simultaneous real admission of the entire old determining cores	NC.2–12, retaining the full earlier frame domain and making separate real completions for separate latent copies
One current-scale Gaussian for arbitrarily many old copies	TC.17–22, with actual conditional covariances, inverse/determinant pairs and per-copy ranks
Same connected gas and complete normalizer	TC.23–29b and NC Section 6; TS.18a–c retains each origin through the uncanceled nested normalization

Required input	Construction and location
Actual lifted paths, velocity ratios, fiber Jacobians and Q tie rule	NC.17–21 and Section 8; TR.19–24
Three physical innovation leaves and one deterministic joint root store	TR.6–26, including force and prescribed-average mismatch leaves
Three exact marked normalizations and a positive rate after all costs	TR.27–34, especially the tunable rate ledger TR.33a
One native source circle, boundary currents and old colour in every origin	TR.35–38g and TS Sections 5.1–5.2

TC is M04 Third Step Analytic Base and Joint Certificate; NC is M04 Third Step Nested Core Admission; TR is M04 Three Layer Adaptive Marked Root. Their primary-source equations and earlier companion dependencies are listed in those files.

This verifies PC's needed primitive composition inputs for the particular third microscopic construction above. It does not prove PC.10 for every abstract field satisfying only the old integrated norm. Nor does it give a fourth-step closure from the finite numbers $J^{r,\mathrm{path}}$ alone: the full latent integral, domain, covariance and first-jet representation must continue to be retained.

A support-preserving non-Abelian completion of the full local Hessian

This note constructs an explicit linear projector from an invariant local analytic coefficient to a globally defined gauge-invariant function with the same quadratic Taylor coefficient. It uses only loops made from the coefficient's own input edges. It neither fills those loops with new input plaquettes nor assigns an enclosing box as their support. Its nonlinear complement has a small native weak-curvature value and composite first-current norm for the actual fixed-depth smooth source, using the bounded-source envelopes in PJ and the scalar equivariantization in EC.

There are three distinct qualifications. The finite-torus estimate has a separate winding term. The current estimate is for the derivative of the completed scalar function, not a separate estimate of every microscopic and kernel origin. The actual application concerns the smooth three-layer comparison family. It does not establish the specified M04 claim or an induction through arbitrary sharp-Q histories.

1. Supported input and conventions

Work with SU(N), N at least 2. Positive oriented bonds are the independent variables; negative orientations mean inverse matrices. Write physical link coordinates as

$$U_e = \exp(iB_e), \qquad B_e = B_e^*, \quad \operatorname{Tr} B_e = 0. \tag{GC.1}$$

The complexification permits complex traceless B. PJ uses an anti-Hermitian convention; multiplication by i identifies the two physical charts without changing a bound. A normalized field is B=ga, where g is the retained-scale coupling. Fix a real orthonormal basis of traceless Hermitian matrices for the trace inner product. All constants can depend on N and on the fixed block and localization choices.

Let f_X be a scalar coefficient with actual finite raw edge-input graph G_X. No graph edges outside this determining input list are needed below. G_X may be disconnected; a forest will handle that case. Set m_X=1+|E(G_X)|. The complete connected carrier X also records the source anchor, normalization connectors, and any needed determining stencils. The native size n(X) satisfies

$$m_X \le Cn(X), \qquad \operatorname{diam} G_X \le Cn(X), \qquad n(X) \ge 1. \tag{GC.2}$$

The input f_X is invariant under all real gauge transformations at the endpoints of its input edges. It is holomorphic on a fixed physical polydisc about the identity. EC.1 supplies such termwise invariance for a complete scalar coefficient family, provided its already proved real-orbit tube is used in the Haar projection. Projection of arbitrary overlapping uncontracted factors is not allowed.

For the quantitative result, let M_X be deterministic nonnegative envelopes. On a physical polydisc of radius 2r_0 about the flat field, assume

$$|f_X(e^{iB})| \le g^{-2} M_X. \tag{GC.3}$$

At every admitted native real center V, assume the same envelope bounds the coefficient on a relative physical polydisc of a fixed radius r_0:

$$|f_X(e^{iB}V)| \le Cg^{-2} M_X, \qquad \|B\|_\infty < r_0. \tag{GC.4}$$

These are coefficientwise hypotheses, stronger than a bound on the total normalized expectation. All complete raw input slots are varied in these charts. The bounds are uniform on real gauge orbits. Constants can be absorbed into M_X. Suppose, for some positive rate nu,

$$\sup_n \sum_{X \ni n} e^{\nu n(X)} M_{n,X} \le A. \tag{GC.5}$$

An unanchored coefficient family is treated with the corresponding fixed-point incidence norm. When more than one envelope is available, their sum gives a common envelope at the smaller rate.

For the actual smooth three-layer mark, PJ.1 and PJ.1b give (GC.3)–(GC.5), after the support-preserving operation EC.1, with

$$g = g_3, \qquad A \le C_L B_3, \qquad B_3 = L^{12}|\alpha|(g_3/g_0)^2. \tag{GC.6}$$

The identity insertion is removed exactly as in PJ.2 before applying the construction. It is restored separately. The three original conditional normalizers, source normalization, force correction, Haar/coarea terms and inherited action remain in the coefficients furnishing these envelopes.

2. The exact completion

Choose a spanning tree in each connected component of G_X and one root in each tree. These choices are fixed combinatorial data, independent of U and g. For each nontree edge e, take the based fundamental loop gamma_e: the tree path from its root to e_-, then e, then the reverse tree path from e_+ to the root. Its length is at most 2|E(G_X)|+1. Its holonomy is denoted Omega_e(U).

Tree gauge makes the original coefficient a function of these holonomies. More precisely, put the tree edges equal to I and the nontree edges equal to W_e, and define

$$\varphi_X((W_e)_{e\notin T}) = f_X(U_T = I, U_e = W_e\ (e \notin T)). \tag{GC.7}$$

For real U, a gauge at vertex x equal to the ordered product along its root-to-x tree path sends every tree edge to I and every nontree edge to Omega_e(U). Hence, wherever both expressions belong to the admitted family,

$$f_X(U) = \varphi_X((\Omega_e(U))_{e\notin T}). \tag{GC.8}$$

The remaining gauge at a tree root conjugates all its based holonomies simultaneously. Distinct connected components have independent remaining conjugations. Since the adjoint representation of su(N) has no invariant vector,

$$D\varphi_X(0) = 0. \tag{GC.9}$$

Here and below derivatives of varphi are in W_e=exp(iC_e) at C=0. Uniqueness of an invariant bilinear form on the complexified simple Lie algebra gives

$$D^2\varphi_X(0)[C,C] = \sum_{e,f\notin T} h_{ef}\,\mathrm{Tr}(C_e C_f), \qquad h_{ef} = h_{fe}. \tag{GC.10}$$

Mixed coefficients between different connected components vanish, by their independent conjugations. This also covers a forest without any nontree edges: its invariant scalar coefficient is constant near the flat field.

Introduce the globally defined matrix function

$$\Xi(W) = \frac{W - W^{-1}}{2i} - \frac{\mathrm{Tr}(W - W^{-1})}{2iN} I. \tag{GC.11}$$

It is holomorphic on SL(N,C), equivariant under conjugation, real Hermitian and traceless on SU(N), and uniformly bounded there. For traceless Hermitian C,

$$\Xi(e^{iC}) = C + O(\|C\|^3). \tag{GC.12}$$

Define the completion and the nonlinear complement by

$$(\mathcal{C}_X f_X)(U) = \frac{1}{2}\sum_{e,f\notin T} h_{ef}\,\mathrm{Tr}\big[\Xi(\Omega_e(U))\Xi(\Omega_f(U))\big], \tag{GC.13}$$

$$\mathcal{N}_X(U) = f_X(U) - f_X(I) - (\mathcal{C}_X f_X)(U). \tag{GC.14}$$

Theorem GC.1 – exact support-preserving quadratic projector

The map f_X to C_X f_X is linear and depends only on the Hessian of f_X at the identity. Its output is a globally defined real gauge-invariant function of the original edge inputs. If f_X is real on real fields, its completion is real there. It has zero constant and linear Taylor coefficients and exactly the same Hessian as f_X. In particular,

$$D^j \mathcal{N}_X(I) = 0 \quad (j = 0, 1, 2), \qquad \mathcal{C}_X^2 = \mathcal{C}_X. \tag{GC.15}$$

No logarithm or branch choice is used in the globally defined output (GC.13). No filled surface, bounding box, or additional raw edge becomes part of its support.

Proof. Each based holonomy in a connected component transforms by conjugation at that component's root. Its trace pairing in (GC.13) is invariant. Cross-component pairings are zero. Every loop uses only original input edges, which proves the support assertion. Unitarity proves boundedness and reality on real fields; inverses are holomorphic on SL(N,C).

For a raw tangent B, write the first-order expansion of a fundamental holonomy as

$$\Omega_e(e^{itB}) = I + it\, L_e(B) + O(t^2), \tag{GC.16}$$

where L_e is the oriented tree-cycle sum of the raw angles. Equations (GC.8)–(GC.10) show that the quadratic coefficient of f_X along this tangent is one half of the sum h_ef Tr(L_e(B)L_f(B)). Equations (GC.11)–(GC.13) give exactly the same coefficient for C_X f_X. Polarization proves equality of the full Hessians, not only of their diagonal evaluations. Both linear coefficients vanish by (GC.9).

If the tree edges are set to I, the based fundamental holonomies in (GC.13) are exactly the nontree variables. Thus applying (GC.10) to the completion recovers exactly h_ef. Its second application in (GC.13) is unchanged. This proves idempotence. All operations are linear in the original Hessian. □

The operator is a completion of the whole quadratic kernel. It is not the scalar Wilson projector of WQ or NW. Those projectors first select the marginal quadratic part. The same construction can be applied to an already transverse remainder kernel, using its supported invariant coefficient representation. A fixed finite lattice-symmetry average can be imposed on a correspondingly closed carrier, with the explicit finite carrier/rate cost recorded in NW; it is not a free preservation of every spatial rate.

Lemma GC.2 – bounded completion and local derivatives

For fixed q, on real raw-link fields the completion satisfies

$$|\mathcal{C}_X f_X| \le C g^{-2} m_X^2 M_X, \qquad \|D_B^q(\mathcal{C}_X f_X)\|_{\infty,\dots,\infty} \le C_q g^{-2} m_X^{q+2} M_X. \tag{GC.17}$$

The derivative formula is valid at real points in physical angle charts of fixed bounded size around any real field. A sum over raw first-coordinate derivatives adds at most one further factor C m_X. Multiplication by g converts such a derivative to its normalized raw-link derivative.

For any strict rate kappa below nu, there is a smaller fixed positive physical radius r_1 for which the completed family has a holomorphic carrier norm at most C A/g^2 on that radius, at rate kappa.

Proof. The restriction (GC.7) lies in the same raw physical polydisc as (GC.3). Cauchy estimates give |h_ef| at most C r_0^{-2}M_X/g2; the fixed colour dimension is absorbed in C. There are at most m_X^2 pairs. A based loop has at most C m_X factors, all unitary at a real field. Differentiating its product q times in physical angle directions costs at most C_q m_X^q times the product of their sup norms. Derivatives of the matrix exponential at Hermitian arguments are bounded by its unitary integral formula. The inverse loop satisfies the same estimates. Product differentiation of the two Xi factors proves (GC.17).

On a complex physical chart of radius r_1 about a real field, a loop product or its inverse has norm at most exp(C m_X r_1). The same coefficient bound therefore gives at most C g^{-2}m_X2 M_X exp(C m_X r_1) for the completion. Choose r_1 so that the last exponent fits a fixed part of nu-kappa; the remaining margin pays m_X^2. This gives the holomorphic family bound. No claim that complex long-loop norms are uniformly unitary was used. □

3. A real estimating gauge without changing support

Assume the native real field obeys

$$\|I - U(p)\| \le \varepsilon = C_0 g P \quad \text{for every plaquette in the required native region,} \qquad P \ge 1. \tag{GC.18}$$

On the infinite lattice, place a carrier in a coordinate box of side at most C n(X). For a torus, first restrict this construction to carriers with a coordinate box that lifts injectively to the lattice. Fix a box root and coordinate-ordered paths from it to every vertex. Use their parallel transports as the estimating gauge.

Lemma GC.3 – short-carrier small links

In this gauge the edges read by f_X satisfy

$$\|I - U_e^{\mathrm{est}}\| \le C n(X)^2 \varepsilon. \tag{GC.19}$$

Whenever the right side is sufficiently small, the Hermitian logarithms of these edges satisfy

$$\|B^{\mathrm{est}}\|_\infty \le C n(X)^2 g P. \tag{GC.20}$$

Proof. Compare the coordinate-ordered path to the far endpoint of an edge with the coordinate-ordered path to its near endpoint followed by the edge. Adjacent interchanges of coordinate steps, and cancellations of inverse steps, reduce the resulting loop to at most C n(X)^2 elementary plaquettes. This is an exact non-Abelian product identity, with conjugated plaquette holonomies. All conjugations are unitary at a real field. For unitary A_j, the telescoping inequality |I-product A_j| at most sum |I-A_j| proves (GC.19). The ordinary local logarithm estimate near I proves (GC.20). □

This gauge may read links elsewhere in the estimating box. Equality of the original gauge-invariant function with its value in this gauge is pointwise. By EC.2 and GC.1, those extra reads do not become arguments of f_X or of its completion. The box is used only to prove a bound on their values. Differentiation below holds the chosen real gauge fixed at the base field; it never differentiates this estimating rule.

4. The native cubic remainder

The following explicit small-coupling condition is sufficient:

$$0 < g < g_*, \qquad P \ge 1, \qquad g P^2 \le 1. \tag{GC.21}$$

To conclude that the value bound tends to zero also require gP^3 to tend to zero. The actual logarithmic radii satisfy both conditions. The weaker condition gP to zero alone would not justify the logarithmic carrier split used here.

For a normalized raw coordinate derivative write

$$D^{(g)}_{e,a} F(U) = \frac{d}{dt} F(\dots, e^{itgT_a} U_e, \dots)\Big|_{t=0}. \tag{GC.22}$$

Theorem GC.4 – infinite-lattice or injectively localized estimate

Fix 0<kappa<nu. Under (GC.2)–(GC.5) and (GC.18)–(GC.21), for sufficiently small g,

$$\sup_n \sum_{X \ni n} e^{\kappa n(X)} \sup_{U \text{ native}} |\mathcal{N}_{n,X}(U)| \le C A g P^3, \tag{GC.23}$$

$$\sup_n \sum_{X \ni n} e^{\kappa n(X)} \sup_{U \text{ native}} \sum_{e,a} |D^{(g)}_{e,a} \mathcal{N}_{n,X}(U)| \le C A g P^2. \tag{GC.24}$$

These formulas hold on the infinite lattice, or for an explicitly restricted family all of whose small carriers have the injective boxes in Lemma GC.3. The finite-torus statement without that restriction is GC.5 below. Fixed spatial moments may be inserted after reserving another strict rate margin. A coefficientwise common deterministic envelope gives the same bound for admitted normalized coordinate path increments of length at most one, with P replaced by P+C.

Proof. Set delta=nu-kappa. Choose D so large that delta D/4 exceeds 4, and split the carrier family at

$$n_*(g) = D \log(1/g). \tag{GC.25}$$

By (GC.21), gP is at most sqrt(g). Therefore n_^*2 gP tends to zero. Every carrier with n(X) at most n_* admits the fixed flat chart after applying Lemma GC.3, for sufficiently small g. Its gauged logarithm B has norm at most C n(X)^2 gP.

By GC.15 the remainder in these raw logarithm coordinates has zero jets through order two at 0. On the real segment from 0 to B, the third derivative of f_X is bounded by C M_X/g^2 in the multilinear sup-direction norm, by polarized Cauchy estimates on the fixed physical polydisc in (GC.3). This directional estimate does not sum coordinate indices; polarization uses only a fixed number of scalar complex directions and has no factor depending on the number of edges. The third derivative of its completion is bounded by C m_X^5 M_X/g^2, by (GC.17). Taylor's integral formula, (GC.2), and (GC.20) imply

$$|\mathcal{N}_X(U)| \le C(1 + n(X))^{12} M_X g P^3. \tag{GC.26}$$

The power 12 is a harmless common upper bound; it is not claimed optimal.

For a raw derivative at U, hold the estimating real gauge fixed. The normalized left link variation then becomes a normalized left variation conjugated by a fixed unitary matrix. On the small logarithm chart its angle velocity has norm at most Cg. Differentiating the Taylor remainder, and using its zero first and second jets, bounds one derivative by C g^{-2}M_X times a polynomial in n(X) times |B|^2 g. Summing all raw edge/colour directions adds at most C m_X. Thus

$$\sum_{e,a} |D^{(g)}_{e,a} \mathcal{N}_X(U)| \le C(1 + n(X))^{12} M_X g P^2. \tag{GC.27}$$

A fixed gauge at the start of a coordinate path works throughout that path: its additional physical link angle is at most Cg. Alternatively the same uniform coefficientwise envelopes can be integrated along the path. No derivative of a field-dependent gauge is hidden in (GC.27).

For n(X)>n_*, no small-link gauge is required. Equations (GC.3)–(GC.4) and (GC.17) give

$$|\mathcal{N}_X(U)| \le C(1 + n(X))^2 M_X g^{-2}, \tag{GC.28}$$

$$\sum_{e,a} |D^{(g)}_{e,a} \mathcal{N}_X(U)| \le C(1 + n(X))^4 M_X g^{-1}. \tag{GC.29}$$

In (GC.29), the coefficient f_X uses Cauchy on the native-center physical tube (GC.4), rather than an unjustified flat chart. The derivatives of f_X(I) are zero. Spare exponential rate pays all displayed polynomials. The remaining tail factor is at most exp(-delta n_*/2), up to adjustment of a fixed constant. The choice of D makes its product with g^-2 at most Cg and its product with g^-1 at most Cg. Since P is at least 1, these are bounded by the right sides of (GC.23)–(GC.24). Summation of (GC.26)–(GC.27) uses the same spare rate and (GC.5). This proves both bounds. □

The supremum of the sum in (GC.24) is obtained here from a common deterministic coefficientwise envelope. It is not an interchange of different coordinate-path suprema. Dividing a coefficientwise bound by a native energy regulator at least one can only decrease it. This observation does not supply any missing support or origin-composition property of the specified M03 interface.

Theorem GC.5 – explicit finite-torus winding term

Let ell be the shortest compatible torus period in retained lattice units. Suppose a carrier of size less than c_0 ell has an injective estimating box, as follows from (GC.2) after fixing c_0 sufficiently small. For the complete finite-torus family,

$$\|\mathcal{N}\|_{\kappa,0} \le CA\left(gP^3 + g^{-2}e^{-c\ell}\right), \tag{GC.30}$$

$$\|D^{(g)}\mathcal{N}\|_{\kappa,1} \le CA\left(gP^2 + g^{-1}e^{-c\ell}\right). \tag{GC.31}$$

The notation denotes exactly the value and summed-first-derivative norms in (GC.23)–(GC.24). The constants are independent of ell and g within the stated fixed-depth family. The exponential terms include all carriers for which an injective small box is unavailable; they need not be assigned a topologically winding loop individually.

Proof. Apply the preceding small-carrier argument only when both n(X) is at most n_*(g) and n(X)<c_0 ell. The tail n(X)>n_* is treated as before. The additional tail n(X) at least c_0 ell is estimated using (GC.28)–(GC.29) and its retained exponential carrier weight. This gives (GC.30)–(GC.31). □

If ell is at least a sufficiently large constant times log(1/g), these extra terms fit the respective small remainders. Without such a relation they must stay in the theorem. A flat real field on a short torus can have nontrivial winding holonomy even when every plaquette is I; weak curvature alone does not make all its links small in a periodic gauge. GC.30 therefore does not assert a volume-uniform cubic bound without a winding remainder. This agrees with the distinction already made in SW and FH.

5. Actual smooth-source consequence and remaining interfaces

Apply EC.1 to the complete supported coefficient family of the actual local smooth mark Phi_n in PJ. The same full determining carriers and common component restrictions are retained. PJ.1 gives the flat physical tube, PJ.1b gives the native-center bounded-source envelope needed in the long-carrier estimate, and EC.3 retains the normalized coupling powers and common derivative envelopes. Thus GC.1–GC.5 apply to that actual family, with (GC.6).

The resulting identity is coefficientwise and exact:

$$\Phi_{n,X}(U) = \Phi_{n,X}(I) + (\mathcal{C}_X\Phi_{n,X})(U) + \mathcal{N}_{n,X}(U). \tag{GC.32}$$

Every term is fully gauge invariant. The completion has the exact raw Hessian, including the full nonlinear loop completion selected by (GC.13). On nonwinding native fields the nonlinear complement has actual bounds

$$\|\mathcal{N}\|_{\kappa,0} \le C_L B_3 g_3 P^3, \qquad \|D^{(g_3)}\mathcal{N}\|_{\kappa,1} \le C_L B_3 g_3 P^2, \tag{GC.33}$$

with the terms in GC.30–GC.31 on a general finite torus. The chosen P must bound the actual retained native plaquette radius and satisfy GC.21; the source logarithmic choices do so for sufficiently weak coupling. The microscopic identity source is transported exactly and is not part of this nonlinear estimate.

This is stronger than a raw Taylor subtraction on a global small-link slice: both pieces in GC.32 are genuine gauge-invariant functions on their real domains, and the estimate uses local estimating gauges for the original supported coefficients. The statement remains limited in the following concrete ways.

1. A coefficient estimate on the admitted native weak-curvature class is not, by itself, a compact-extension theorem for every raw field in a specified global interface.
2. Completion of a quadratic kernel does not prove that its nonlinear conditional transfer has the Gaussian contraction in FH. The full interacting comparison must be supplied separately, with its exact inherited precision and normalized density.
3. A bound on the derivative of GC.32 does not split that derivative into microscopic insertion and each fixed-test kernel origin. Existing physical current labels and their moving-reference velocities remain part of the induction state.

4. No hard-Q indicator has been analytically continued or differentiated beyond its established real first-current representation. The real boundary/core problem persists.
5. All constants inherited from PJ and EC concern the specified finite three-layer smooth family. No depth-uniform native orbit tube, old-core registry, or repeated contraction constant is inferred here.

These qualifications prevent using a valid finite-depth non-Abelian completion as an unproved claim of M04 closure or Yang–Mills existence.

6. Dependencies and checks

The algebraic theorem GC.1 and geometric estimate GC.3 are proved in this note. The analytic and large-carrier estimates explicitly use both kinds of coefficient envelope in GC.3–GC.4. Their model-specific discharge is PJ.1/PJ.1b, not a probabilistic bound on an extensive whole-volume expectation. EC supplies support-preserving scalar invariance, uniform real-orbit control, and the first-jet change-of-basis estimate. WQ and NW identify the marginal Wilson part separately. FH concerns the actual inherited classical Gaussian lift, with its stated scope.

Primary source facts used by PJ/EC are CMP109 (1.7)–(1.19), (2.6)–(2.18), CMP116 (1.13)–(1.21), Lemma 2 and (2.5)–(2.26), and the final gauge-orbit paragraph on CMP116 printed pp.21–22. The full force/gap and raw pullback estimates are those of BA, BR/FC and TC. None of these citations is a claim that the original papers state GC.13 or the present nonlinear-complement estimate; those constructions and deductions are given above.

Global Gaussian source families and conditional contraction of the first-order remainder

The Gaussian result applies at all orders to the global energy budget. The interacting order-one conclusion below requires separate local, geometric, regulator and covariance hypotheses. The fixed source radius is taken sufficiently small for convergence of the local polymer expansion.

1.1 Setting

Let $\varphi \in \mathbb{R}^N$ be a centred Gaussian vector with covariance $C_0 > 0$ (the free lattice link field at cutoff a on the torus; N is the number of link components). Let the "block fields" be $\psi_k = B_k\varphi$, $B_k : \mathbb{R}^N \to \mathbb{R}^{N_k}$ surjective and nested ($B_{k+1} = b_k B_k$), $k = 0, \dots, K$, $B_0 = I$. Conditionally on ψ_k, φ is Gaussian with mean $T_k\psi_k$, $T_k = C_0 B_k^T (B_k C_0 B_k^T)^{-1}$, and covariance

$$C_k := C_0 - T_k B_k C_0, \qquad 0 = C_0^{\mathrm{cond}} \le C_k \le C_{k+1} \le C_K \le C_0. \tag{1.1}$$

Marks: n centred quadratic forms $\Phi_l = \varphi^T A_l \varphi - \mathrm{Tr}(C_0 A_l)$, A_l real symmetric (for $\mathrm{Tr}\, F^2$ smeared with f_l, $A_l = \sum_x f_l(x)\, \mathsf{s}_x$ with s_x the curl–curl stencil at x). Write $A(z) = \sum_l z_l A_l$, $A_l = A_l^+ - A_l^-$ with $A_l^\pm = \sum_x f_l^\pm(x)\mathsf{s}_x \ge 0$ for the curvature stencil; any positive decomposition with the stated bound suffices, and the insertion size

$$\mathfrak{a}_l := \|(A_l^+)^{1/2} C_0 (A_l^+)^{1/2}\| + \|(A_l^-)^{1/2} C_0 (A_l^-)^{1/2}\|, \qquad \mathfrak{a}(z) := \sum_l |z_l|\mathfrak{a}_l, \qquad \lambda := \Big(4\sum_l \mathfrak{a}_l\Big)^{-1}. \tag{1.2}$$

For the curvature field $\mathfrak{a}_l \le C g_0^2 \|f_l\|_\infty$ uniformly in a, because $\partial(-\Delta)^{-1}\partial$ is bounded on ℓ^2; this is the only place the dimension-four nature of the insertion enters. Let $E_k(\psi) := \psi^T (B_k C_0 B_k^T)^{-1}\psi = (T_k\psi)^T C_0^{-1}(T_k\psi)$ be the coarse Gaussian energy (Annex Y's regulator (M.5) in the free case).

Theorem 1.1 (exact analytic-family form of the free transported jet)

For $z \in \mathcal{D}_\lambda := \{|z_l| \le \lambda\}$ and every $k \le K$,

$$G_k(z;\psi) := \mathbb{E}\Big[e^{\sum_l z_l \Phi_l} \,\Big|\, \psi_k = \psi\Big] = \exp\big(\mathfrak{L}_k(z) + \psi^T \mathfrak{B}_k(z)\psi\big), \tag{1.3}$$

$$\mathfrak{L}_k(z) = -\sum_l z_l\, \mathrm{Tr}(C_0 A_l) - \tfrac{1}{2}\,\mathrm{Tr}\log\big(I - 2C_k^{1/2} A(z) C_k^{1/2}\big), \qquad \mathfrak{B}_k(z) = T_k^T A(z)\big(I - 2C_k A(z)\big)^{-1} T_k, \tag{1.4}$$

both analytic on $\mathcal{D}_\lambda$, with the uniform field-exponent budget

$$\boxed{\sup_{z\in\mathcal{D}_\lambda} |\psi^T \mathfrak{B}_k(z)\psi| \le \frac{\mathfrak{a}(z)}{1 - 2\mathfrak{a}(z)} E_k(\psi) \le \tfrac{1}{2} E_k(\psi) \quad \text{for all } k, K, N.} \tag{1.5}$$

The one-step map is the exact Möbius recursion

$$\mathfrak{B}_{k+1} = t_k^T \mathfrak{B}_k \big(I - 2c_k \mathfrak{B}_k\big)^{-1} t_k, \qquad \mathfrak{L}_{k+1} = \mathfrak{L}_k - \tfrac{1}{2}\,\mathrm{Tr}\log\big(I - 2c_k^{1/2}\mathfrak{B}_k c_k^{1/2}\big), \tag{1.6}$$

where $t_k\psi_{k+1}$ and c_k are the conditional mean map and covariance of ψ_k given ψ_{k+1}. Consequently the connected coefficients of order m in z of the field-dependent exponent obey, by Cauchy,

$$\Big|\partial_z^I \big(\psi^T \mathfrak{B}_k(z)\psi\big)\big|_{z=0}\Big| \le m!\, \lambda^{-m}\, \tfrac{1}{2} E_k(\psi), \qquad |I| = m, \tag{1.7}$$

a fixed-factorial budget with exponent 1, uniform in k, K, N, relative to the regulator E_k: exactly the shape of the M-03 sup-part (M.11) at grade m, and exactly what [I, Prop 3.8(c)] showed the per-step coefficient iteration cannot reproduce.

Proof. For a Gaussian vector with mean μ and covariance C and complex symmetric A with $\|C^{1/2} A C^{1/2}\| < \tfrac{1}{2}$, completing the square gives $\mathbb{E} e^{\varphi^T A \varphi} = \det(I - 2C^{1/2} A C^{1/2})^{-1/2} \exp(\mu^T A (I - 2CA)^{-1}\mu)$, the branch of the square

root being the one analytic in A and equal to 1 at $A = 0$. Apply this to the conditional law (1.1) with $\mu = T_k\psi$, $C = C_k$, $A = A(z)$; the centring produces the linear term of $\mathfrak{L}_k$. For the domain: $C_k \le C_0$ and $A_l^{\pm} \ge 0$ give $\|C_k^{1/2}A_l^{\pm}C_k^{1/2}\| = \|(A_l^{\pm})^{1/2}C_k(A_l^{\pm})^{1/2}\| \le \|(A_l^{\pm})^{1/2}C_0(A_l^{\pm})^{1/2}\|$, hence $\|C_k^{1/2}A(z)C_k^{1/2}\| \le \mathfrak{a}(z) \le \frac{1}{4}$ on $\mathcal{D}_\lambda$, uniformly in k. For (1.5) put $w = T_k\psi$, $u = C_0^{-1/2}w$ (so $|u|^2 = E_k(\psi)$), $\tilde{A} = C_0^{1/2}A(z)C_0^{1/2}$, $\tilde{C}_k = C_0^{-1/2}C_kC_0^{-1/2} \in [0, I]$; then $\psi^T\mathfrak{B}_k\psi = \sum_{j\ge 0} u^T\tilde{A}(2\tilde{C}_k\tilde{A})^j u$ and $\|\tilde{A}\| \le \mathfrak{a}(z)$ by the same splitting, so the series is bounded by $\mathfrak{a}\sum_j(2\mathfrak{a})^j|u|^2$. (1.6) is the same Gaussian formula applied to one step (tower property), and composing two Möbius maps of the form (1.6) reproduces the closed form with c_k added to the covariance, which is (1.1). (1.7) is Cauchy on $\mathcal{D}_\lambda$. □

The covariance bands can satisfy an operator-order bound on their sum while the sum of their individual operator norms increases proportionally to depth. The Gaussian result uses the exact composition (1.6).

2. The analytic-family induction hypothesis and its closure

2.1 Objects

Fix finitely many smeared sources $f_1, \dots, f_n$ (cell-supported C^5 profiles as in Transport of insertions on a reference torus §1.4 and the preamble, or any fixed finite test family) and the complex source vector $z \in \mathbb{C}^n$. The bare insertion is $\sum_l z_l\Phi_a(f_l)$, $\Phi_a(f) = \sum_x a^4 f(x)\Phi_a(x)$ in the F-01 basis (M.20). The transported jet at scale k is $G_k(z; x) = Q_{0\to k}\exp(\sum_l z_l\Phi_a(f_l))(x)$, x the augmented state of Annex Y §1.

Definition 2.1 (analytic-family state). The state at scale k consists of: 1. the running local coefficients $t_{\rho,k} > 0$, v_k of [I, Def. 1.1]; 2. the renormalized source $\zeta_l := t_{\rho(l),k}\, z_l$ (channel of the component carried by f_l) and the fixed polydisc $\mathcal{D}_\lambda = \{|\zeta_l| \le \lambda\}$; 3. the c-number counterterm $\mathfrak{L}_k(z)$, a polynomial in z of degree $\le m_{\max}$ with explicit coefficients (the identity kernels $C_{I,1}^{(k)}$ of [I, (1.5)–(1.6)]); 4. the local field kernels $\mathfrak{K}_k(z; x) = \sum_l z_l t_{\rho,k} e_k(f_l^{(k)})(x) + \sum_{2\le|I|\le m_{\max}} z^I \sum_{P\neq 1} C_{I,P}^{(k)}P(x)$, with $f_l^{(k)}$ the block-averaged profile; 5. the irrelevant polymer remainder $\Psi_k(z; x) = \sum_X \Psi_k(X; z; x|_X)$, a sum over connected carriers X of the scale-k block graph of activities depending on the state only inside X, each analytic in $\zeta \in \mathcal{D}_\lambda$ and in the field variables on the common contour (RM.12)/(E14.49) of Theorems E14.3/R14.3, with $\Pi_k^{(m)}\Psi_k = 0$ coefficientwise (no local polynomial part);

such that, exactly,

$$G_k(z; x) = \exp\big(\mathfrak{L}_k(z) + \mathfrak{K}_k(z; x) + \Psi_k(z; x)\big) \quad \text{as finite jets of order } \le m_{\max}. \tag{2.1}$$

The norm on the remainder is the Kotecký–Preiss type weighted supremum

$$\|\Psi_k\|_{\kappa,\lambda} := \sup_b \sum_{X\ni b} e^{\kappa|X|} \sup_{\zeta\in\mathcal{D}_\lambda} \sup_{x\in\mathcal{C}_k} \frac{|\Psi_k(X; z; x)|}{\widehat{R}_{h_k,X}(x)}, \tag{2.2}$$

with $\widehat{R}$ the Annex Y regulator (M.6) and $\mathcal{C}_k$ the contour. The analytic-family budget is

$$(\mathsf{AF}_k) \qquad \|\Psi_k\|_{\kappa,\lambda} \le \psi_* \quad (\psi_* \text{ below the KP threshold } \psi_0(\kappa)). \tag{2.3}$$

At $k = 0$: $\Psi_0 = 0$, $\mathfrak{L}_0 = 0$, $\mathfrak{K}_0 = \sum_l z_l\Phi_a(f_l)$ with $t_{\rho,0} = z_{\rho,a,T,\mu}$. The bare a^{-4} size of Φ_a sits entirely in the local term $\mathfrak{K}_0$, whose one-step image is computed by exact extraction, never by a sup bound of its exponential. This is why no "radius conversion" is needed for the remainder: its radius is λ from the start.

2.2 The one-step hypothesis (to be supplied by M-04/M-05/M-06 and H-04; §3)

(OS_k). There are $\theta \in (0, 1)$, $\beta_*, C_\Pi < \infty$ and a KP threshold ψ_0, independent of k, K, T, history and $m_{\max}$, such that for every admitted $k < K$ and every state satisfying (AF_k), the full-law one-step operation Q_k (all sectors, Annex W (4.2)–(4.9)) produces a state of the form (2.1) at scale $k + 1$ with

$$t_{\rho,k+1} = M_{\rho,k}t_{\rho,k}, \quad \|M_k - I - u_k\Gamma\| \le C_4u_k^2; \qquad \mathfrak{L}_{k+1}, \mathfrak{K}_{k+1} \text{ given by the exact extraction } \Pi_{k+1}^{(\le m_{\max})}\log Q_k e^{\mathfrak{K}_k+\Psi_k}; \tag{2.4}$$

$$\boxed{\|\Psi_{k+1}\|_{\kappa,\lambda} \le \theta\,\|\Psi_k\|_{\kappa,\lambda} + \beta_*.} \tag{2.5}$$

Here the logarithm exists because (AF_k) with $\psi_* < \psi_0$ puts e^{Ψ_k} in the KP convergence region on the polydisc (the local part $e^{\mathfrak{K}_k}$ is handled by the extraction, not by a smallness condition), and the births in β_* are the irrelevant parts of the collisions of local terms and of the residual history/selector currents.

Theorem 2.3 (closure at all orders; conditional on (OS_k) for all $k < K$)

Let $\psi_* := \beta_*/(1-\theta)$ and suppose $\psi_* < \psi_0$. Then (AF_k) holds for every $k \le K$ with the same ψ_*, uniformly in K, the torus and the history. Consequently, for every $|I| = m \le m_{\max}$, the irrelevant connected coefficient satisfies

$$\sup_b \sum_{X \ni b} e^{\varkappa|X|} \sup_{x \in \mathcal{C}_k} \frac{|\partial_z^I \Psi_k(X;0;x)|}{\widehat{R}_{h_k,X}(x)} \le m!\, \lambda^{-m} \prod_{i \in I} t_{\rho(i),k}\, \psi_*, \tag{2.6}$$

i.e. the M-03 budget [I, (B_k)] relative to the explicit normalization [I, (1.6)] with factorial exponent $p = 1$ and no logarithmic accumulation; the local kernels $\mathfrak{L}_k, \mathfrak{K}_k$ are explicit; and the terminal remainder is [I, Prop 3.7].

Proof. Induction on k: $\Psi_0 = 0 \le \psi_*$; if $\|\Psi_k\| \le \psi_*$ then (2.5) gives $\|\Psi_{k+1}\| \le \theta\psi_* + \beta_* = \psi_*$. No partition sum, no product of lower-order budgets and no per-order constant occur: the births are already summed inside β_* as a sup over the polydisc. (2.6) is Cauchy in ζ on $\mathcal{D}_\lambda$ followed by $\partial_{z_l} = t_{\rho(l),k}\partial_{\zeta_l}$. Uniformity in K, volume and history is inherited from the constants of (OS_k). □

Remark 2.5 (comparison with [I, Thm 3.6]). The coefficient-norm closure needed (G3-strong) with $\sum_k \varepsilon_k < \infty$, false in the M-03 norm (Cor. 1.2). The analytic-family closure needs only (2.5). The price is that (OS_k) must be proved for the exponent on a complex polydisc and on the field contour, i.e. in Bałaban's inductive space (CMP 119 p. 262, second remark: "RT transforms the space with the index k into the space with the index $k+1$"), which is precisely the form in which E14.3/R14.3 are already proved for the source-free E/R activities.

Lemma 9.1 (the extraction F-06a at orders $m = 1, 2$; exact recursions)

Define $\Pi_{k+1}^{(\le)}$ on an activity $\Phi(X')(V';z)$ analytic in the coarse variables F_P, $P \in X'$, as the removal of the Taylor monomials at x_* of (a) field degree $r = 0$ (pure source polynomials: the identity kernels, any j), and (b) field degree $r = 2$ with $j = 0$ (block-averaged sources times a quadratic form in the F_P, $P \in X'$); the quadratic form is further split into its diagonal/clover part, which is the span of the 21 fields $e_{k+1,\beta}$ at the anchor, and its separated part $\sum_{P \ne P'} Q_{PP'} F_P F_{P'}$, which is kept as a (nonlocal, dimension-four) remainder. Then $\mathcal{L}_{k+1}^{(m)}$ of Definition 2.1 is span$\{1 \cdot$ (source monomials), $\bar f^{\,m} e_{k+1,\beta}\}$, and the local kernels obey, exactly,

$$\mathfrak{L}_{k+1}(z) = \mathfrak{L}_k(z) + \Delta\mathfrak{L}_k(z), \qquad \Delta\mathfrak{L}_k(z) = \big[\Pi_{k+1}^{(\le)} \log Q_k e^{\mathfrak{K}_k + \Psi_k}\big]_{r=0}, \tag{9.1}$$

$$m = 1: \quad t_{\rho,k+1} = M_{\rho,k} t_{\rho,k}, \qquad m = 2: \quad c_{2,\beta}^{(k+1)} = M_{\rho(\beta),k}\, c_{2,\beta}^{(k)} + b_{2,\beta}^{(k)}, \tag{9.2}$$

where $c_{2,\beta}^{(k)}$ is the coefficient of $\bar f^{\,2} e_{k,\beta}$ in $\mathfrak{K}_k$, $M_{\rho,k}$ is the one-field ratio of [I, (1.3)] (the same number for every monomial carrying e_β, by linearity of Q_k and equivariance), and $b_{2,\beta}^{(k)} := \big[\Pi_{k+1}^{(\le)} \kappa_{Q_k}(\text{two linear insertions})\big]_{\bar f^2 e_\beta}$ is the extracted degree-two, order-two collision. Equivariance makes (9.2) channel-diagonal.

Proof. The Taylor expansion of an analytic function of finitely many F_P at x_* is unique, so $\Pi^{(\le)}$ is a well-defined linear projection commuting with the H_4 action and with the block-average structure of the sources. Degree-0 monomials are V'-independent, hence Q_k-invariant up to the extracted collisions, giving (9.1). For $m = 1$, the coefficient of z_l in $\mathfrak{K}_k$ is $t_{\rho,k}\bar f_l e_{k,\cdot}$; its image under Q_k has $r = 2$, $j = 0$ part $M_{\rho,k} t_{\rho,k} \bar f_l e_{k+1,\cdot}$ by [I, Lemma 1.1] (the $r \ge 3$ and $j \ge 1$ parts are not extracted). For $m = 2$, the coefficient of $z_l z_{l'}$ in $\log Q_k e^{\mathfrak{K}_k + \Psi_k}$ is $Q_k[c_{2,\beta}^{(k)} \bar f \bar f' e_{k,\beta}] + \kappa_{Q_k}(t\bar f e, t\bar f' e) +$ (terms with Ψ_k); the first has $r = 2$, $j = 0$ part $M_{\rho,k} c_{2,\beta}^{(k)} \bar f \bar f' e_{k+1,\beta}$ by the same lemma; the second's extracted part is $b_{2,\beta}^{(k)}$; terms with Ψ_k have no $r \le 2$, $j = 0$ content at scale k by the induction hypothesis, but their transported images can acquire an $r = 2$, $j = 0$ part at scale $k+1$ only through the separated-to-local merging of Theorem 9.2(c) below, which is included in $b_{2,\beta}^{(k)}$. □

Remark 9.1a (free-field check of (9.2)). In the free case $b_{2,\beta}^{(k)}$ is the zeroth Taylor moment of $F_x\, C_k^F(x-y)\, F_y$ in y around x, with C^F the curvature covariance. In the continuum $C^F_{\mu\nu,\rho\sigma}(y) = (\partial_\mu\partial_\rho\delta_{\nu\sigma} - \dots)G(y)$, G harmonic away from 0, so every angular average $\fint \partial_\mu\partial_\rho G = \frac{1}{4}\delta_{\mu\rho} \fint \Delta G = 0$: the free composite kernel $\bar f^2 e_\beta$ receives no logarithm, only an $O(1)$ lattice-artifact contribution from $|x-y| \lesssim a$ and nothing from scales $j \ge 1$. This is consistent with

Theorem 1.1 (the free exponent is bounded on a fixed polydisc) and shows that the running of the $m \geq 2$ composite kernels is an interaction effect, $b^{(k)}_{2,\beta} = O(u_k)$ in the interacting theory; (9.2) then runs like Annex P (R.18)–(R.20) with an additive $O(u_k)$ source, i.e. a further explicit logarithmic power. These kernels are explicit outputs; they are not bounded uniformly and are not claimed to be.

Order-one contraction at fixed block parameters

Fix admissible block parameters L and M. Assume the age-n real-centre and relative second-order curvature transfer with exponential tails, (I2), (I3), a common hull-regular contour on the small carriers, transverse complex regulator control, the sector hypotheses and the Kotecký–Preiss smallness region. The separated-carrier improvement additionally requires translation covariance of the extracted coefficient and the stated physical first-difference estimate. All constants in the following estimate may depend on L and M.

For the n-step composition with extraction at its terminal scale, one has

$$\|\Psi^{(1)}_{k+n}\|_{\varkappa,\lambda} \leq \theta_n \|\Psi^{(1)}_k\|_{\varkappa,\lambda} + nC'(L,M)\beta_*.$$

The separated-within-block bilinear row has L^{4n} (block count) $\times$ $(L^{-2n})^2$ (two curvature factors) $\times$ L^{-n} (one fine difference, (12.2)) $= L^{-n}$, which gives the following estimate:

$$\theta_n = C_R(L,M)\, n\, L^{4n}\Big[\frac{(2C'_\Psi L^{-2n})^3}{1-2C'_\Psi L^{-2n}} + C_{\mathrm{sep}} L^{-5n}\Big] + L^{4n} C_R C'^2_\Psi\, B_d A_d^{C_{L^n}} e^{-(\varkappa-\log A_d)(1-C'/L^n)C_{L^n}/2} \lesssim C(L,M)\big(nL^{-2n} + nL^{-n}\big) + e^{-c\varkappa L^n}, \tag{13.1}$$

so $\theta_n \to 0$ and $n_0(L,M)$ exists; for the displayed entropy bound one chooses $\varkappa$ with a strict margin above $3\log A_d$ and then n large enough that $(\varkappa - \log A_d)(1 - C'/L^n) > 2\log A_d$. This absorbs the explicitly displayed $A_d^{C_{L^n}}$. The source decoupling radius must leave room for this choice and the intermediate-step window below.

13.2 Carrier weights over the n intermediate steps: symbols and the window

Symbols: $\varkappa$ is M-07's carrier weight in the norm (2.2) (also denoted κ); κ_1 is Bałaban's decoupling radius (E14.5); κ_* the activity decay rate of E14.3/R14.3; κ_{KP} the Kotecký–Preiss loss per cluster step (a universal constant, $\kappa_{\mathrm{KP}} \leq 1$ for the weight $e^{\varkappa|X|}$ with $a(X) = |X|$).

Per intermediate step $i < n$. The activities after step i live on scale-$(k+i+1)$ carriers. Two facts: (i) reblocking never loses weight: every fine block lies in exactly one coarse block, so $|X^{(i+1)}| \leq |X^{(i)}|$ and $e^{\varkappa|X^{(i+1)}|} \leq e^{\varkappa|X^{(i)}|}$ for every carrier, small or large (the Bałaban bound $|X'| \leq C_d + C'|X|/L$ of (11.3) is used only to obtain a *gain* for $|X| \geq C_L$; the factor $e^{\varkappa C_d}$ in (11.3) is therefore not a loss and is replaced by the trivial bound for small carriers); (ii) the decoupling tails of step i (rate κ_1) are summable against the weight $e^{\varkappa|\cdot|}$ iff $\varkappa < \kappa_1 - \log A_d$, with no loss of rate; the cluster/KP step loses κ_{KP}. Hence the weight sequence is

$$\varkappa_0 = \varkappa, \qquad \varkappa_{i+1} = \varkappa_i - \kappa_{\mathrm{KP}} \quad (i < n), \qquad \varkappa_n = \varkappa - n\kappa_{\mathrm{KP}}, \tag{13.2}$$

and the extraction at scale $k+n$, together with the reblocking gain (11.3) for $|X| \geq C_{L^n}$, restores $\varkappa$ provided the entropy sums converge at the end, i.e. $\varkappa_n > \log A_d$. The window is therefore

$$\log A_d + n_0\kappa_{\mathrm{KP}} \;<\; \varkappa \;<\; \kappa_1 - \log A_d. \tag{13.3}$$

Non-circularity. $n_0 = n_0(L,M)$ is determined by (13.1), whose constants $C_R, C'_\Psi, C_{\mathrm{sep}}$ come from (I-L/n) and (I3) at $s \equiv 1$ (the physical minimizer and the one-scale fluctuation covariance), hence do not depend on κ_1; the decoupling bound $K = B_0 e^{16\kappa_1}$ enters only the births β_1 of Theorem 8.1 and the KP constant, not θ_n. So the order is: fix (L,M); fix $n = n_0(L,M)$; fix $\varkappa$ with $\varkappa > \log A_d + n_0\kappa_{\mathrm{KP}}$; fix $\kappa_1 > \varkappa + \log A_d$ (compatible with Bałaban's "κ_1 large" and the parameter order of B-03 in §19.3 of homogeneous_projection.md, which also needs $\kappa_1 \geq 16\varkappa + 1$); then K is fixed; then λ small enough that $C(L,M)^n\, n\, C'\beta_*(\lambda,K)/(1-\theta_n) < \psi_0(\varkappa)$; then the coupling ceiling. (13.3) is non-empty and there is no cycle. If, contrary to the reading in §12.2(b), the constants in (13.1) did depend on κ_1 (through the localized C_H), the window would require $n_0(\kappa_1)\kappa_{\mathrm{KP}} < \kappa_1 - 2\log A_d$, i.e. n_0 sublinear in κ_1: with $C'_\Psi \propto e^{c\kappa_1}$ one gets $n_0 \asymp c\kappa_1/\log L$ and the window is non-empty iff $L > e^{c\kappa_{\mathrm{KP}}}$, a condition on the fixed L — still not circular, but it would make L_1 depend on the (unknown) c. This is why (I-L/n) is stated at $s \equiv 1$.

13.3 Translation covariance behind the extra L^{-1} in Lemma 9.3(b) / Theorem 11.4(b)

The extraction of $\sum_{p\subset P} F_p^2$ as the local clover $e_{k+n,\beta}(P)$ plus terms of dimension ≥ 6 uses the cancellation of the linear term $(x_p - c_P)\cdot \nabla F$ when summed over the positions $p \subset P$ (symmetric block, L^n odd), and the identification of the resulting coefficient with the same number for every block. The first is geometric; the second requires

(TC) translation covariance. On the clean sector, $\Psi_{k\leftarrow k+n}$ commutes with translations by scale-$(k+n)$ lattice vectors and with the block reflections; equivalently the extraction coefficient $[\Pi^{(1)}_{k+n} Q^{(n)}_k e_k]_{e_{k+n,\beta}}$ is independent of the anchor block. *Source:* the Euclidean covariance of the inductive assumption, CMP 119 (2.29), restated for the layer functions on p. 275 ("when defined on the whole lattice, it is Euclidean covariant"); it holds on the homogeneous clean sector and fails on sectors with large-field geometry, where the extraction coefficient becomes sector-dependent — exactly the sector-dependent $t_{\rho,k,h}$ of §10.1. (TC) is added to the hypotheses of Theorem 11.4.

Choose a fixed n with contraction factor below one, followed by the carrier weight and decoupling radius in the stated window. Taking the source radius sufficiently small puts the uniform block-end bound $nC'\beta^*/(1-\theta_n)$ and the finitely many intermediate bounds inside the convergence region. This gives the uniform first-order result conditional on the required physical inputs. At order two and above the extraction recursion for composite sources must be shown compatible with their renormalization. The full-law applications also require selector flux and the marked reset quotient.

A conditional rotation theorem

Conditional rotation theorem

Set the fixed physical unit $\ell_0 = 1$, take $h_l = L^{-l}$ and cutoff $a_K = h_K$, $L > 1$. Increasing l means moving toward the ultraviolet; the physical terminal level is $l = 0$. Throughout, all fields, counterterms and histories are those of the specified Wilson hierarchy. No averaging of rotated theories and no replacement normalization are permitted.

For m insertions let $V = \mathrm{Sym}^2(\Lambda^2\mathbb{R}^4)$ and take $V^{\otimes m}$-valued coincidence-flat Schwartz tests. The seminorm is exactly the joint-test convention, with a maximum over orthonormal tensor components:

$$\nu_s(F) = \max_{|\alpha|\le s} \sup_X (1 + |X|^2)^{s/2} |\partial^\alpha F(X)|.$$

Let $T_{m,K}$ and $S_{m,K}$ denote the actual connected and full distributions. A unit plane rotation generator acting simultaneously on coordinates and labels is denoted D_e, for one of the six planes e.

The hypotheses below produce sequences $q_l \ge 0$, $Q = \sum_{l\ge 0} q_l < \infty$, and numbers $C_0, \delta > 0$. Define

$$b_d = C_0(1 + d)L^{-\delta d}, \tag{1}$$

$$\eta_K = b_K + \sum_{l=0}^{K} q_l b_{K-l} + a_K^2. \tag{2}$$

Every constant in these formulas is independent of source order, volume, history and cutoff depth. For $d \ge 0$, one can choose a fixed B with $b_d \le BL^{-\delta d/2}$. Consequently

$$\eta_K \le BL^{-\delta K/2} + BQL^{-\delta K/4} + B \sum_{l>\lfloor K/2 \rfloor} q_l + a_K^2 \longrightarrow 0. \tag{3}$$

This is an explicit rate in terms of the supplied summable majorant; no polynomial tail for that majorant is assumed.

Theorem 1 (Conditional R-02–R-04). *Under hypotheses P0–P4 in Section 2, the action and observable/contact contributions to every connected angular Ward identity separately obey*

$$|G^A_{m,K,e}(F)| + |G^O_{m,K,e}(F)| + |C_{m,K,e}(F)| \le \eta_K C^{m+1}(m!)^p \nu_{qm}(F),$$

for fixed q, p, C and every $m \ge 1$, plane e and cutoff K. Their corresponding marked full-moment contributions obey the same form of estimate. In particular,

$$|S_{m,K}(D_e F)| \le \eta_K C^{m+1}(m!)^p \nu_{qm}(F). \tag{4}$$

For every $R \in SO(4)$, uniformly over all endpoints and the intermediate rotations used to reach them,

$$|S_{m,K}(U_R F - F)| \le 6\pi\, \eta_K C_*^{m+1}(m!)^p \nu_{qm}(F). \tag{5}$$

Thus the action-defect conclusion of R-02, the all-label and all-order observable/contact conclusion of R-03, and the common path-budget conclusion of R-04 all follow. The proof is completed below. Verification of P0–P4 for the actual model is not claimed.

External premises, stated at the primitive level

These premises must hold for the *same* physical matching, cofinal sequence and normalized full Wilson law. Constants may depend on the fixed group, blocking factor and physical units. Uniformity in arbitrary source order, volume and admitted history is part of each bound. They cannot be supplied by estimates only inside one chosen regular history.

P0: exact marked representation. The connected angular identity has the exact decomposition

$$T_{m,K}(D_e F) = G^A_{m,K,e}(F) + G^O_{m,K,e}(F) + C_{m,K,e}(F). \tag{6}$$

Here A is the action-score insertion, O is the field/matching insertion, and C consists of point or shrinking-stencil contacts. For $\sigma = A, O$, exact marked transport gives

$$G^\sigma_{m,K,e} = \mathcal{T}^\sigma_{m,K,e}(u^\sigma_{K,0,e}) + \sum_{l=0}^{K} \mathcal{L}^\sigma_{m,K,l,e}(u^\sigma_{K,l,e}). \tag{7}$$

The two kinds of operators are respectively terminal and shell response maps. They act linearly on explicit local defect coefficients in a fixed normed space; they are not defined by dividing an unknown entire correlation by a small number. For a source-jet space, the norm includes its fixed factorial weights. All changes in conditional normalizers, outer means, retained histories, and collisions are included by the exact marked transport. The decomposition must be an identity of joint distributions, not merely an identity on separated tensor-product tests.

For the action, the extracted local marginal terms are vacuum constants and the reflection-even hypercubic-invariant gauge-polynomial terms of engineering dimension at most four. Their scalar part has zero angular variation; it is removed algebraically before forming u^A. For observables, u^O includes the *compensated* matching commutator in the prescribed physical basis, all 21 channels and all higher source jets. It is not the commutator of a raw bare-to-renormalized matrix alone. Sections 3 and 4 justify these algebraic distinctions.

P1: coefficient contraction and controlled forcing. The two coefficient families in

$$eq : marked$$

satisfy, in their fixed norms,

$$\left\| u^\sigma_{K,K,e} \right\| \le C_u, \tag{8}$$

$$u^\sigma_{K,l,e} = B^\sigma_{K,l,e} u^\sigma_{K,l+1,e} + f^\sigma_{K,l,e}, \qquad 0 \le l < K, \tag{9}$$

$$\left\| B^\sigma_{K,l,e} \right\| \le L^{-\delta}, \qquad \left\| f^\sigma_{K,l,e} \right\| \le C_f L^{-\delta(K-l)}. \tag{10}$$

The physical angular marginal defect must be absent or included in a compensated coefficient family satisfying this recursion. A nonzero uncontracted angular marginal term is not allowed to disappear by terminology. The same fixed $\delta > 0$ works for both families and all six planes. The observable coefficient norm dominates each compensated matrix commutator and the corresponding source-jet commutators with their fixed factorial weights.

P1 is a new, unverified actual-model requirement. General bounds on an RG remainder, ordinary hypercubic symmetry, and the original dependency lists do not themselves imply it. It is strictly a coefficient-level estimate; it assumes neither small angular correlations nor rotational invariance of the limiting hierarchy.

P2: response-kernel and terminal majorants. For $m \ge 2$ the shell response to u has a representation

$$\mathcal{L}(u)(F) = \sum_{|\alpha| \le r_* m} \int K_\alpha(X)\, \partial^\alpha F(X)\, d\lambda_K^{\otimes m}(X),$$

where $r_* \ge 0$ is fixed. The measure λ_K is either Lebesgue measure or normalized counting measure $a_K^4 \sum_{x \in a_K \mathbb{Z}^4} \delta_x$, restricted to the admitted finite-volume embedding when appropriate. Thus this premise permits the derivative distributions of an actual lattice hierarchy. For a fixed $s_* \ge 0$, the sum of the component kernel magnitudes satisfies

$$\sum_\alpha |K_\alpha(X)| \le \|u\|\, A^{m+1} (m!)^{p_0} h_l^{-4m} w_l (1 + |x_1|^2)^{s_* m/2} \exp\left[-\frac{c\,\tau(X)}{(m + d_0) h_l} \right]. \tag{11}$$

Here τ is the minimum Euclidean spanning-tree length, $d_0 \ge 0$, $c > 0$, and $p_0 \ge 0$ are fixed. Every carrier sum, differentiated preparation, marked position, and history contribution assigned to that shell has already been included in this nonnegative bound. The common weights satisfy

$$q_l = h_l w_l, \qquad Q = \sum_{l \ge 0} q_l < \infty.$$

The terminal response satisfies

$$|\mathcal{T}^{\sigma}_{m,K,e}(u)(F)| \le \|u\| A^{m+1}(m!)^{p_T} \nu_{q_T m}(F). \tag{12}$$

Each one-point marked contribution is required to satisfy the terminal bound with coefficient $u_{K,0,e}$, or to vanish by a separately verified identity. Centering the unmarked field alone does not imply that its covariance with an action-score insertion vanishes. No one-point tree gain is inferred from coincidence flatness. These are full-law marked-insertion versions of the kernel and terminal inputs, not estimates merely for unmarked correlations.

P3: point contacts or quantitative shrinking contacts. There are no coincidence contacts at source order one; any one-point local matching term is included in O and must satisfy the one-point clause of P2. At each fixed source order, a contact is either supported exactly on a partial diagonal, with finite distribution order, or has a representation

$$C_{m,K,e} = \sum_{|\alpha| \le r_0 m} \partial^{\alpha} \mu_{\alpha,m,K,e} \tag{13}$$

for $m \ge 2$. Each measure in this finite sum is supported within distance $c_0 a_K$ of some partial diagonal. The constants $r_0, b \ge 0, c_0 > 0$ are fixed, and for a fixed integer $q_C \ge r_0 + \lceil b \rceil + 4$,

$$\sum_{\alpha} \int (1 + |X|^2)^{-q_C m/2} d|\mu_{\alpha,m,K,e}|(X) \le A_C^{m+1} (m!)^{p_C} a_K^{-bm}. \tag{14}$$

For tensor contacts the component sum is included or absorbed into A_C^{m+1}. Angularly differentiated contacts must admit this representation with these constants; this includes derivatives of contact coefficients and finite stencils. Compatibility under truncation in m is retained. Exact point contacts do not need a coefficient bound.

P4: the base hierarchy and joint-test partition theorem. The actual unmarked connected distributions obey

$$|T_{m,K}(F)| \le A_B^{m+1} (m!)^{p_B} \nu_{q_B m}(F),$$

and the completed S-01 joint-distribution tensor-product, extension-independence and stability hypotheses hold. We use its theorem with $p \ge 1$. This is not an assumption that every arbitrary distributional family with a scalar seminorm bound has a well-defined unrestricted product. S-01 supplies the required joint-test construction.

Dependencies. P0 requires marked and history transport of the kind assigned to M-07, H-04/H-05 and S-04. P2 requires marked versions of C-08, H-04, S-03 and S-05. P3 is a quantitative realization of F-06/S-06. P4 uses the actual common hierarchy bound and the already completed S-01 bridge. These correspondences describe required interfaces, not claims that the named catalogue items already prove the premises. In particular P1 and the exact marginal/compensated extraction in P0 remain additional primitive verification obligations. No R-02, R-03, R-04 or R-05 conclusion is used as a premise.

The exact Wilson starting identity

On the finite link manifold $SU(N)^E$, let L_b^A be left-invariant derivatives and $\mathcal{V} = \sum_{b,A} v_b^A L_b^A$ a smooth link vector field. For the normalized Wilson law $d\mu = Z^{-1} e^{-S} dU$, set

$$W = \mathcal{V}S - \sum_{b,A} L_b^A v_b^A.$$

Haar integration by parts gives

$$\mathbb{E}W = 0, \qquad \mathbb{E}(\mathcal{V}O) = \mathrm{Cov}(O, W).$$

For a microscopic product observable $O_m(F)$ this yields the exact split

$$S_m(D_e F) = \mathrm{Cov}(O_m(F), W) + \mathbb{E}\{O_m(D_e F) - \mathcal{V} O_m(F)\}. \tag{15}$$

One can choose the local curvature vector field

$$v_{x,\mu} = a^{-1} \sum_{\nu \ne \mu} \xi_{\nu}(x) \Big[\tfrac{1}{2}(U_{\nu\mu} - U_{\nu\mu}^{\dagger}) - \tfrac{1}{2N} \mathrm{Tr}(U_{\nu\mu} - U_{\nu\mu}^{\dagger}) I \Big],$$

with a smooth compactly supported angular vector ξ. Then $W = \sum_b w_b$, where $w_b = \sum_A (v_b^A L_b^A S - L_b^A v_b^A)$ has a fixed local link carrier. This does not establish P1 or P2, but it specifies the microscopic insertions to which P0 must apply, including the Haar-divergence term. Passing from full to connected source coefficients is an exact formal operation at every finite order.

Exact conditional transport must preserve

$$\mathrm{Cov}(O, W) = \mathrm{Cov}(\mathbb{E}[O \mid \mathcal{G}], \mathbb{E}[W \mid \mathcal{G}]) + \mathbb{E}\,\mathrm{Cov}(O, W \mid \mathcal{G}).$$

Thus P0/P2 explicitly retain the second term and the outer covariance, along with their higher-order versions. Boundary cutoffs used to localize ξ must be included in these inputs and their summable majorants. Their removal is not inferred from microscopic locality alone.

Why scalar marginal isotropy is algebraic

Lemma 2 (Reflection-even curvature quadratic). *Let $SU(N)$ be fixed. A local scalar quadratic in curvature that is gauge invariant and invariant under signed coordinate permutations is a multiple of*

$$\sum_{\mu<\nu} \langle F_{\mu\nu}, F_{\mu\nu} \rangle.$$

Proof. Gauge invariance contracts the two adjoint color indices by a multiple of the invariant quadratic form of the simple Lie algebra. A product of two distinct two-form components changes sign under a reflection of an index occurring in exactly one pair. Distinct pairs always have such an index, including complementary pairs. Hence every cross coefficient vanishes. Coordinate permutations act transitively on the six pairs, making the diagonal coefficients equal. The resulting quadratic is rotationally invariant. □

This applies to the extracted quadratic-curvature marginal sector in P0. Vacuum constants have zero angular variation and cancel from a normalized law. A reflection-odd pseudoscalar is excluded by the stated symmetry. This lemma does not classify a nonlocal effective action as local or prove that all remaining angular terms satisfy P1; those are precisely the retained primitive obligations. It also says nothing about the seven-channel normalization of the observables.

For the latter, one has, in the fixed Frobenius metric,

$$\sum_e \|[J_e, Z]\|_{\mathrm{HS}}^2 = 48\big[(z_{T_d} - z_{T_o})^2 + (z_{W_d} - z_{W_s})^2 + (z_{W_p} - z_{W_s})^2\big]. \tag{16}$$

The included exact rational certificate reconstructs the six 21×21 generators and seven projectors and verifies every block coefficient. The same formula may be used for a compensated matching matrix that has this seven-channel form. For a general matching matrix its full commutator is included in u^O. Assuming commutation of a raw matrix in place of compensated matching would change P0 and is not allowed.

Contraction gives the primitive cutoff factor

Lemma 3 (Uniform solution of P1). *P1 implies $\left\| u^{\sigma}_{K,l,e} \right\| \le b_{K-l}$ with $C_0 = C_u + C_f$ (enlarged if necessary). In particular the terminal coefficient tends to zero.*

Proof. Iterating $d = K - l$ steps, the initial term has norm at most $C_u L^{-\delta d}$. A forcing created at layer $r \in \{l, \dots, K-1\}$ is multiplied by at most $L^{-\delta(r-l)}$, and its norm is at most $C_f L^{-\delta(K-r)}$. Each of the d forcing contributions is therefore at most $C_f L^{-\delta d}$. Add them. Finally $(1+d)L^{-\delta d/2}$ is bounded on the nonnegative integers, proving the exponential envelope used in

$$eq : rateexplicit$$

. □

For a compensated matrix $M_{K,l}$ whose commutators are included in u^O, the same coefficient bound also gives an operator-level finite-angle estimate. On orthogonal source and target representations, integrate the commutator along the six-plane path to get

$$\left\| D(R) M_{K,l} - M_{K,l} D(R) \right\| \le 6\pi b_{K-l}.$$

For a multilinear source jet, use its induced source and target representations and the weighted jet norm in P1. Thus the same statement holds with its fixed factorial factor. At every fixed physical layer l, these defects tend to zero as $K \to \infty$; the near-cutoff layers are instead handled by the shell sum below.

There is no derivative-order dependent choice of δ here. Any amplification from inverse running matrices or preparations must already be in the actual norms and maps of P1; it cannot be assigned norm one without proof.

The full shell integral, including approaching diagonals

Lemma 4 (A flat-test shell gain). *For $m \geq 2$, $0 < a \leq h \leq 1$, and $F \in \mathcal{S}_0$, a response kernel as in P2 at scale h, including its test derivatives and root weight, satisfies*

$$|\mathcal{L}(u)(F)| \leq \|u\|\, hw\, C^{m+1}(m!)^{p_0+5}\nu_{q_* m}(F), \qquad q_* = \lceil r_* \rceil + \lceil s_* \rceil + 6. \tag{17}$$

No positive separation of the test support from a partial diagonal is required. The same constants work for integration and normalized lattice summation.

Proof. Hold x_1 and all variables other than x_2 fixed, and Taylor expand each $\partial^\alpha F$ in x_2 about $x_2 = x_1$ to order four. All those Taylor coefficients vanish by flatness. Along the Taylor segment the first variable stays x_1, so the weighted derivative bound gives

$$|\partial^\alpha F(X)| \leq C\nu_{q_* m}(F)(1 + |x_1|^2)^{-(q_* m)/2}|x_2 - x_1|^5.$$

The directional derivative costs only the fixed dimension four, and $|\alpha| + 5 \leq q_* m$. Multiplication by the root factor in P2 leaves a weight bounded by $(1 + |x_1|^2)^{-3}$. Its integral and its normalized lattice sum for $a \leq 1$ are uniformly finite.

Let $\mathfrak{T}_m$ be the labeled spanning trees and $\ell_T(X)$ their lengths. Since $\tau = \min_T \ell_T$, for $s = (m + d_0)h$,

$$|x_2 - x_1|^5 e^{-c\tau/s} \leq \sum_{T \in \mathfrak{T}_m} \ell_T^5 e^{-c\ell_T/s} \leq Cs^5 \sum_T e^{-c\ell_T/(2s)}.$$

The first inequality follows by selecting a minimizing tree, whose path from 1 to 2 has length at least $|x_2 - x_1|$; adding the remaining nonnegative tree terms only enlarges it. Root each tree at 1 and use its edge differences as integration variables. The Jacobian is one and each of its $m-1$ edge integrals costs Cs^4. For a lattice sum the edge transformation is a bijection of lattice variables. Since $s = (m + d_0)h \geq a$, comparison with lattice cubes gives

$$a^4 \sum_{y \in a\mathbb{Z}^4} e^{-c|y|/(2s)} \leq Cs^4.$$

The distance to the cube center is at most $2a$, costing only a fixed exponential. Restricting a finite-volume sum only reduces the absolute majorant when the Euclidean embedding in P2 holds. There are m^{m-2} labeled trees. The bound is therefore

$$C^{m+1}(m!)^{p_0}\, \|u\|\, \nu_{q_* m}(F) w h^{-4m} m^{m-2}[(m + d_0)h]^{4m+1}.$$

The power of h is exactly one. For fixed d_0, the remaining factor $m^{m-2}(m + d_0)^{4m+1}$ is at most $C^{m+1}(m!)^5$, by $m! \geq (m/e)^m$ and absorption of a fixed polynomial in m into C^m. The sum over derivative indices is already included in P2. Summing tensor components costs at most 21^m, absorbed into C^{m+1}. □

Combining Lemmas 3 and 4 gives the normally convergent bound

$$\left|\sum_{l=0}^{K} \mathcal{L}^{\sigma}_{m,K,l,e}(u^{\sigma}_{K,l,e})(F)\right| \leq \left(\sum_{l=0}^{K} q_l b_{K-l}\right) C^{m+1}(m!)^{p_0+5}\nu_{q_* m}(F).$$

The terminal term adds $b_K A^{m+1}(m!)^{p_T}\nu_{q_T m}(F)$. This proves the action and matching shell conclusions with all old scales and all spatial carriers accounted for by P2.

To verify

$$eq : rateexplicit$$

, split the shell sum at $l = \lfloor K/2 \rfloor$. On the first part $K - l \geq K/2$, so $b_{K-l} \leq BL^{-\delta K/4}$; on the second part use $b_{K-l} \leq B$. Summability gives the stated tail tending to zero. This argument remains valid when the ultraviolet weights have no known power-law tail.

Contacts: exact annihilation or a proved small error

Exact finite-order distributions supported on partial diagonals annihilate F, D_eF and U_RF, since simultaneous rotations preserve those diagonals and the angular vector fields are tangent to them. This is valid at each finite source order regardless of the size of the contact coefficients.

Lemma 5 (Shrinking-contact budget). *Under the shrinking-contact alternative of P3,*

$$|C_{m,K,e}(F)| \le a_K^2 C^{m+1}(m!)^{p_C}\nu_{q_C m}(F).$$

Proof. Set $n = \lceil b\rceil m + 2$. For each X in a measure support choose a pair of arguments whose distance is at most a fixed multiple of a_K, and move one to the other, leaving the other coordinates fixed. Taylor's formula for $\partial^\alpha F$ has zero coefficients at the coincidence point, to every order. The nth remainder requires at most $|\alpha| + n \le q_C m$ derivatives.

The displacement is $O(a_K)$ in a fixed number of coordinates. Along this segment the ratio of $1 + |X|^2$ to the corresponding shifted weight is bounded by a fixed constant. Raising it to $q_C m/2$ costs at most a constant to the power m. Taylor's formula and the fixed-dimensional directional derivative bound therefore give

$$|\partial^\alpha F(X)| \le C^{m+1}\nu_{q_C m}(F) a_K^{\lceil b\rceil m+2}(1 + |X|^2)^{-q_C m/2}.$$

There is no need to partition the measure measurably by the selected pair: this pointwise upper bound is identical for every choice. Integrate against each total variation measure and use

$$eq : contactmeasure$$

, which already sums the derivative indices. The resulting exponent is $(\lceil b\rceil - b)m + 2 \ge 2$, proving the assertion for $a_K \le 1$. □

A fixed physical-size contact carrier does not meet P3. Neither does a coefficient growth faster than the stated linear-in-m cutoff power. Such terms must be estimated elsewhere in P0–P2; they are not declared zero by flatness.

Connected-to-full passage without losing source orders

Take

$$p = \max\{1, p_0 + 5, p_T, p_C, p_B\}, \qquad q = \max\{q_*, q_T, q_C, q_B\},$$

with integer q enlarged once if the established S-01 interface requires it. The preceding lemmas and P4 bound the base connected hierarchy by $C^{m+1}(m!)^p\nu_{qm}$, and each marked connected family by η_K times that bound.

Here is the marked use of the completed S-01 theorem. For a marked family G and $\eta_K > 0$, form the formal connected family $T + zG/\eta_K$. On $|z| \le 1$ its connected bound has the same q, p with a fixed larger constant. S-01 constructs its full partition family on joint flat tests with the same q, p, uniformly on that disk. At source order m this is a polynomial in z. Cauchy's estimate at zero bounds its derivative by that common bound. Multiplying back by η_K gives

$$\left|\sum_{\pi}\sum_{B\in\pi} G_{|B|}\bigotimes_{B'\in\pi\setminus\{B\}} T_{|B'|}(F)\right| \le \eta_K C^{m+1}(m!)^p\nu_{qm}(F). \tag{18}$$

The products in this formula are the S-01 joint-distribution products, not products defined only on simple tensor tests. If $\eta_K = 0$, the marked families vanish and the assertion is immediate.

Apply

$$eq : markedpartition$$

separately to G^A, G^O and C. Differentiation of the finite partition identity distributes the angular generator over its blocks, exactly producing their sum. Thus

$$eq : genfinal$$

follows with the same normalization, all labels and every finite source order. Compatibility in m follows from P0/P3 and the partition construction, not from a truncation of the list of generated vertices.

Uniform rotation paths and completion of the proof

For every orthogonal R and tensor test,

$$\nu_s(U_R F) \le 21^{m/2} 2^s \nu_s(F). \tag{19}$$

Each coordinate derivative is mixed by a row of a 4×4 orthogonal matrix with absolute row sum at most 2. The tensor product action has dimension 21^m, giving row sum at most $21^{m/2}$. The radial weight is invariant. Flatness is preserved. These facts prove

$$eq : testrotate$$

with no dependence on the rotation or its path.

Givens elimination writes any $R \in SO(4)$ as at most six coordinate-plane rotations, each with angle in $[-\pi, \pi]$. Choosing positive pivots in the first three columns leaves last entry $+1$ because the determinant is one. Interpolate these factors by left multiplication, obtaining a piecewise smooth path of total angular length at most 6π. Along each segment,

$$\frac{d}{dt} U_{R(t)} F = \dot{\theta}(t) D_e(U_{R(t)} F).$$

Apply

$$eq : genfinal$$

to the test $U_{R(t)}F$, then

$$eq : testrotate$$

with $s = qm$, and integrate the path length. The factor $21^{m/2} 2^{qm}$ is absorbed into C_*^{m+1}. This proves

$$eq : rotfinal$$

uniformly at intermediate points and endpoints. Every estimate used the same η_K, which tends to zero by

$$eq : rateexplicit$$

.

This completes the proof of Theorem 1: the action, observable/contact, and common-path conclusions have all been derived from P0–P4, with fixed source-order exponents and no additional uniformity assumption. A convergent subsequence of this same hierarchy would therefore be rotation covariant, but its existence, positivity and remaining reconstruction conditions are not asserted here.

A finite certificate for the Wilson thermal increment

1 The same transfer operator at every auxiliary temporal length

Fix one spatial torus with lattice spacing a, physical side T, and an even number $M = T/a \geq 4$ of sites per direction. This excludes no eventual cofinal members. Its actual Wilson coupling is the one selected by the matched construction; it does not change when the temporal length below changes. Write

$$s(U) = N - ReTrU, \qquad \beta = Y_0/N = 2Nx_0 > 0, \qquad d = N^2 - 1.$$

(TS20.1)

The trace is the ordinary fundamental matrix trace; Haar measure has total mass one. The numbers of spatial links and spatial plaquettes are both $L_s = P_s = 3M^3$. Use the retained exact transfer construction of Annex M (1.1)–(1.2):

$$\begin{gathered} Z_1(\beta) = \int_{SU(N)} e^{-\beta s(U)} dU, \\ \left(C_\beta f\right)(u) = Z_1(\beta)^{-1} \int e^{-\beta s(uv^{-1})} f(v) dv, \\ K = \bigotimes_{l=1}^{L_s} C_\beta, \qquad V_s = \beta \sum_{p\ spatial} s\left(U_p\right), \\ A = e^{-V_s/2}, \qquad \mathsf{T}_0 = AKA. \end{gathered}$$

(TS20.2)

Let T be the physical gauge-invariant restriction, let $\lambda_0 > 0$ be its largest eigenvalue, and set $P = \mathsf{T}/\lambda_0 = e^{-aH}$, $H \geq 0$. The retained positivity, compactness and character-coefficient facts make these positive trace-class operators at the finite cutoff, with a simple physical ground state. All traces below are physical traces. Equivalently the physical operator may be extended by zero on the gauge-null complement.

For $u \geq a$, define

$$F(u) = \log Tr\mathsf{T}^{u/a}, \qquad f(u) = \log Tre^{-uH} = F(u) - \frac{u}{a} \log \lambda_0.$$

(TS20.3)

Thus f is nonnegative and nonincreasing. The desired increment is

$$\Gamma(T) = f(T/2) - f(T).$$

(TS20.4)

The powers used below have integer exponent when M is even. These are exact powers of one spatial operator, not an asserted extension of a source-free RG theorem from cubes to rectangles. Spatial volume and coupling remain fixed within each dyadic sum; they change with the cofinal index.

2 An explicit entropy reserve for the interacting Wilson transfer

The next bound is extensive and cannot by itself close E-05. Its role is to give a uniform explicit error for truncating a positive dyadic sum.

Lemma TS20 1 normalized Haar ball

Give $SU(N)$ the bi-invariant metric $ds^2 = -Tr\left[\left(U^{-1}dU\right)^2\right]$. Set

$$v_d = \frac{\pi^{d/2}}{\Gamma(1 + d/2)}, \qquad V_N = \frac{\sqrt{N}(2\pi)^{N(N+1)/2-1}}{\prod_{j=1}^{N-1} j!},$$

$$c_N = \min\left\{1, \frac{2^{-N(N-1)}v_d}{V_N}\right\}, \qquad b_N = 4^{-d}c_N.$$

(TS20.5)

For $0 < r \le 1$, normalized Haar measure of

$$B_r = \{e^{iX} : X = X^*,\ TrX = 0,\ TrX^2 \le r^2\}$$

is at least $c_N r^d$.

Proof. The exponential map is injective here: all eigenvalues of X lie in $[-1,1]$, within the principal Hermitian logarithm interval. If they are $x_1, \dots, x_N$, the exponential-coordinate volume Jacobian is

$$J(X) = \prod_{i<j}\left[\frac{\sin\left(\left(x_i - x_j\right)/2\right)}{\left(x_i - x_j\right)/2}\right]^2.$$

(TS20.6)

To see this, left translate the differential of the exponential to the identity; it is the integral of $Y \mapsto e^{-itX}Ye^{itX}$ over $0 \le t \le 1$. It is the identity on diagonal directions, and its determinant on each off-diagonal real two-plane is the corresponding square in (TS20.6). Since $\left|x_i - x_j\right|/2 \le 1/\sqrt{2} < 1$ and $\sin z/z \ge 1/2$ on that interval, $J(X) \ge 2^{-N(N-1)}$. Integrating over the Euclidean d-ball and dividing by V_N proves the claim.

The metric volume V_N can be checked explicitly. In $SU(k-1) \to SU(k) \to S^{2k-1}$, the quotient metric at the identity has factor 2 in the $2k-2$ off-diagonal real directions and factor $k/(k-1)$ in the last phase direction relative to the unit round sphere. Its volume is

$$2^{k-1}\sqrt{\frac{k}{k-1}}\frac{2\pi^k}{(k-1)!} = \sqrt{\frac{k}{k-1}}\frac{(2\pi)^k}{(k-1)!}.$$

Multiplication from $k = 2$ to N gives (TS20.5). The trace-metric conventions also agree with G. Moore's notes, (12.10)–(12.13), cited below. ■

Theorem TS20 2 explicit actual Wilson trace reserve

With (TS20.1)–(TS20.5), put

$$\boxed{B_{N,\beta,M} = 3M^3\left[\frac{17}{32} + 3\log\left(b_N^{-1}\right) + \frac{3d}{2}\log(1+\beta)\right].}$$

(TS20.7)

Then

$$0 \le f(u) \le f(a) \le B_{N,\beta,M} \qquad (u \ge a).$$

(TS20.8)

Proof. Set $r_\beta = \left[4\sqrt{1+\beta}\right]^{-1}$. On B_{r_β},

$$s\left(e^{iX}\right) = \frac{1}{2}\parallel I - e^{iX} \parallel_{HS}^2 \le \frac{1}{2}TrX^2 \le \left[32(1+\beta)\right]^{-1}.$$

The Haar-ball estimate therefore gives

$$Z_1(\beta) \ge e^{-1/32}b_N(1+\beta)^{-d/2}.$$

(TS20.9)

On the event that every spatial link belongs to B_{r_β}, unitary telescoping along a plaquette yields $\| I - U_p \|_{HS} \le 4 r_\beta$, hence $\beta s(U_p) \le 1/2$. Consequently $A \ge e^{-P_s/4}$ on that event and

$$\int A \, dU \ge e^{-P_s/4} b_N^{L_s} (1+\beta)^{-dL_s/2}.$$

(TS20.10)

The one-link convolution has eigenvalue one on constants and nonnegative eigenvalues on their orthogonal complement. Thus $K \ge \Pi_1$. Both the constant function and A are gauge invariant. The physical Rayleigh principle gives

$$\begin{aligned} \lambda_0 &\ge \langle 1, AKA1 \rangle = \langle A, KA \rangle \ge (\int A \, dU)^2 \\ &\ge e^{-P_s/2} b_N^{2L_s} (1+\beta)^{-dL_s}. \end{aligned}$$

(TS20.11)

The diagonal of the full-link kernel of K equals $Z_1(\beta)^{-L_s}$. Since $0 < A \le 1$ and the physical space is reducing for the positive operator,

$$Tr_{phys} \mathsf{T} \le Tr \mathsf{T}_0 = Z_1(\beta)^{-L_s} \int A^2 \, dU \le Z_1(\beta)^{-L_s}.$$

(TS20.12)

Combining (TS20.9)–(TS20.12),

$$\begin{aligned} f(a) &= \log Tr_{phys} \mathsf{T} - \log \lambda_0 \\ &\le P_s/2 + L_s/32 + 3L_s \log\left(b_N^{-1}\right) + (3dL_s/2) \log(1+\beta). \end{aligned}$$

This is (TS20.7), since $P_s = L_s = 3M^3$. A positive contraction with eigenvalue one has $f(u) \ge 0$ and nonincreasing in u, completing the proof. ■

This proof uses the full interacting transfer, including gauge and harmonic sectors. No Gaussian replacement, removal of physical flat holonomies, weak-coupling expansion, or spatially uniform gap is assumed. The bound is nevertheless too large to prove the desired rate: $B_{N,\beta,M}/T$ is of order $T^2 a^{-3} [1 + \log(1+\beta)]$.

3 Positive temporal dilation defects cancel the ground energy

For $t > a$, let $\rho_t = e^{-tH}/Tre^{-tH}$. Trace class at a controls every $H^k e^{-tH}$ at strictly later times by a scalar exponential majorant, so the derivatives below exist.

Theorem TS20 3 ground independent dyadic identity

Define

$$\begin{aligned} D(t) &= F(t) - \tfrac{3}{2} F(2t) + \tfrac{1}{2} F(4t) \\ &= \tfrac{1}{2} \log \frac{\left[Tr \mathsf{T}^{t/a}\right]^2 Tr \mathsf{T}^{4t/a}}{\left[Tr \mathsf{T}^{2t/a}\right]^3}. \end{aligned}$$

(TS20.13)

Then $D(t) \ge 0$. For $t_j = 2^j T/2$ and each integer $J \ge 0$,

$$\boxed{\Gamma(T) = \sum_{j=0}^{J-1} 2^{-j} D\left(t_j\right) + 2^{-J}\left[f\left(t_J\right) - f\left(2t_J\right)\right].}$$

(TS20.14)

In particular,

$$\boxed{0 \le \Gamma(T) - \sum_{j=0}^{J-1} 2^{-j} D\left(t_j\right) \le 2^{-J} B_{N,\beta,M}.}$$

(TS20.15)

Thus the increasing positive partial sums determine the increment with an explicit error requiring neither the largest eigenvalue, ground vector, nor a finite-volume gap.

Proof. The linear term in (TS20.3) cancels because $t - 3t + 2t = 0$, so D is unchanged if F is replaced by f. Set $\mathcal{E}(u) = Tr\left(H\rho_u\right)$. The spectral theorem gives

$$f'(u) = -\mathcal{E}(u), \qquad \mathcal{E}'(u) = -Var_{\rho_u}(H) \le 0.$$

Therefore

$$\begin{aligned} D(t) &= [f(t) - f(2t)] - \tfrac{1}{2}[f(2t) - f(4t)] \\ &= \int_t^{2t} [\mathcal{E}(u) - \mathcal{E}(2u)]\, du \\ &= \int_t^{2t} \int_u^{2u} Var_{\rho_v}(H)\, dv\, du \ge 0. \end{aligned}$$

(TS20.16)

Writing $\Gamma_j = f\left(t_j\right) - f\left(2t_j\right)$ gives $D\left(t_j\right) = \Gamma_j - \Gamma_{j+1}/2$. Multiplication by 2^{-j} and telescoping proves (TS20.14). The remainder is bounded because $0 \le \Gamma_J \le f\left(t_J\right) \le B_{N,\beta,M}$. This also proves convergence of the series with its finite error; no large-time convergence theorem is needed to discard the tail. ■

Replacing T by $c\mathsf{T}$ changes $F(u)$ by $(u/a)\log c$ and leaves D unchanged. An exactly constant contribution to F also cancels, since $1 - 3/2 + 1/2 = 0$. Thus the defect does not charge a raw multiplicity entropy when that entropy has negligible effect on the thermal increment.

Thermal purity and the checked factor

The normalized thermal state is unchanged by using T instead of P. Its Rényi entropy of order two is

$$S_2(t) = -\log Tr\rho_t^2 = 2F(t) - F(2t) = 2f(t) - f(2t).$$

(TS20.17)

Consequently

$$D(t) = \frac{1}{2}\left[S_2(t) - S_2(2t)\right], \qquad \Gamma(2t) = \frac{1}{2}\left[S_2(t) - f(2t)\right] \le \frac{1}{2} S_2(t).$$

(TS20.18)

Bounding the whole entropy is only sufficient. The positive dyadic series is exact and retains cancellations that such a bound can lose.

4 A finite certificate on the changing cubic family

Restore the cofinal index n. Put $B_n = B_{N,\beta_n,T_n/a_n}$ and $t_{n,j} = 2^j T_n/2$. For arbitrary $\varepsilon_n \downarrow 0$ and fixed reference physical length $\ell_0 > 0$, choose

$$J_n = \max\left\{0, \left\lceil \log_2 \frac{B_n \ell_0}{\varepsilon_n T_n} \right\rceil\right\}, \qquad C_n = \sum_{j=0}^{J_n-1} 2^{-j} D_n\left(t_{n,j}\right).$$

(TS20.19)

The reference length keeps the logarithm dimensionless. Then

$$\boxed{0 \le \frac{\ell_0 \Gamma_n(T_n)}{T_n} - \frac{\ell_0 C_n}{T_n} \le \varepsilon_n.}$$

(TS20.20)

The original vanishing trace-rate condition is therefore equivalent to $C_n/T_n \to 0$ with this explicit finite prescription. The bounded-rate variation in the homogeneous comparison 1 is equivalent to boundedness of C_n/T_n. For $J_n \ge 1$, adjacent defects reuse trace values: only $J_n + 2$ raw traces are needed, at lengths $T_n/2, T_n, 2T_n, \dots, 2^{J_n} T_n$ on the same spatial torus. If $J_n = 0$, the tail bound alone applies. For example, setting $\varepsilon_n = \ell_0/T_n$ makes the number of trace values $O(\log(T_n/a_n) + \log(1 + \log(1 + \beta_n)))$.

This argument gives an explicit finite error at each member of the changing family. It does not take a fixed-volume limit and silently substitute a different Hamiltonian. It still leaves the actual positive sum C_n to be bounded.

A connected thermal energy criterion

Define the dimensionless total heat capacity $\mathcal{C}_n(u) = u^2 Var_{\rho_{n,u}}(H_n)$. If, for fixed $p \ge 0$ and numbers $A_n \ge 0$,

$$\mathcal{C}_n(u) \le A_n (T_n/u)^p \qquad (u \ge T_n/2),$$

(TS20.21)

then

$$\Gamma_n(T_n) \le \begin{cases} A_n \log 2, & p = 0, \\ A_n (2^p - 1)/[p(p+1)], & p > 0. \end{cases}$$

(TS20.22)

Proof. Spectral dominated convergence gives $\mathcal{E}_n(u) \to 0$ as $u \to \infty$ at the fixed finite operator. Hence

$$\mathcal{E}_n(u) = \int_u^\infty Var_{\rho_{n,v}}(H_n)\, dv \le \frac{A_n T_n^p}{p+1} u^{-p-1}.$$

Integrate from $T_n/2$ to T_n, using Annex M (4.21). ■

Thus $A_n = o(T_n/\ell_0)$ suffices for a vanishing rate; $A_n = O(T_n/\ell_0)$ suffices for a bounded rate. No such bound is proved here. The energy is the exact logarithm of the transfer. Its variance is not automatically the variance of a sum of bare scalar plaquettes or of the prescribed renormalized scalar field. An anisotropic or logarithmic-transfer identification requires its actual coefficients, counterterms and identity contribution. The finite difference (TS20.13) avoids that unproved identification.

5 What temporal polymer cancellation would have to establish

The three-length defect cancels every exactly time-linear contribution, including genuine bulk vacuum normalizations when the same contribution has been identified at all three lengths. A possible route is a linked expansion of the log partition function separating its time-linear part from clusters sensitive to the temporal period.

This is stronger than a local polymer decomposition of the effective action. After integrating the remaining coarse links and taking a logarithm, correlations of arbitrarily many short supported functions can form long connected chains. Locality of action terms alone does not show that only small time-wrapping contributions survive.

Conditional cancellation estimate with a power law alternative

Suppose the actual raw trace at a fixed spatial torus has a compatible decomposition, for every $u \geq T/2$,

$$F(u) = ub + R(u), \qquad |R(u)| \leq A\,(T/\ell_0)^3\,(\ell_0/u)^p\,, \qquad p > 0.$$

(TS20.23)

Then

$$\boxed{\Gamma(T) \leq \frac{2^p\left[1 + \frac{3}{2}\,2^{-p} + \frac{1}{2}\,4^{-p}\right]}{1 - 2^{-(p+1)}}\,A\,(T/\ell_0)^{3-p}\,.}$$

(TS20.24)

Proof. The linear term cancels. The three remainder bounds give

$$D\left(t_j\right) \leq A\,(T/\ell_0)^3\left(\ell_0/t_j\right)^p\left[1 + \frac{3}{2}\,2^{-p} + \frac{1}{2}\,4^{-p}\right].$$

Multiply by 2^{-j} and sum the exact series of Theorem TS20.3. ■

For fixed A, $p > 2$ suffices for the vanishing trace rate; $p = 2$ suffices for a bounded rate. A proved $p = 3$ remainder gives a uniform increment. Thus exponential clustering is more than this route requires.

If a full linked expansion instead yields

$$|R(u)| \leq A\,(T/\ell_0)^3\,(u/\ell_0)\,e^{-cu/\ell_0}, \qquad c > 0,$$

(TS20.25)

then

$$\Gamma(T) \leq 3A\,(T/\ell_0)^4\,\frac{e^{-cT/(2\ell_0)}}{1 - e^{-cT/(2\ell_0)}}.$$

(TS20.26)

Indeed, $D(t) \leq 6A\,(T/\ell_0)^3\,(t/\ell_0)\,e^{-ct/\ell_0}$, and $2^j \geq j + 1$ bounds the resulting series. For varying A_n, this implies a vanishing rate when $\log^+ A_n = o\,(T_n/\ell_0)$. These are conditional implications, not established expansions for the matched Wilson family.

The actual stopping scale obstruction

The completed coupling work matches one weak finite coupling at the physical scale ℓ_0. It does not integrate all scales between ℓ_0 and the growing size T_n. Even if all ultraviolet bulk factors first cancel compatibly in (TS20.13), a terminal compact gauge integral remains at each auxiliary temporal length. Its logarithm is an infrared contribution to the same three-length defect.

The source-free stability bounds control this integral extensively. A separate bound of size $C\,(T_n/\ell_0)^3\,(u/\ell_0)$ at each length proves neither (TS20.23) nor (TS20.25). Subtracting three numbers known only within such errors does not give a subextensive error. The terminal Wilson action at fixed positive coupling, physical flat holonomies, and connected correlations of remaining links cannot be discarded. Fixed physical auxiliary supports and exact coupling shooting ensure compatible definitions, but do not provide these connected infrared estimates.

A valid wrapping argument would require compatible period-independent cluster weights; absolute summability of the connected log-partition expansion including terminal integration; and a bound on clusters whose time extent makes them sensitive to the temporal identification. “Wrapping” here must include long clusters that lose a unique short lift, not just loops assigned nonzero winding. These properties have not been verified for the residual terminal measure. A cubic source-free theorem also cannot be applied to all auxiliary rectangles without a separate geometry justification.

6 Renormalized marks and the physical scale

The thermal estimate is one part of actual field passage. Annex M (4.5) also requires

$$\frac{\Gamma_n + 2\log^+ \| X_{F,n} \|}{T_n} \to 0$$

for every fixed polynomial F in the chosen renormalized hierarchy. If the actual mixing, identity and product-subtraction coefficients obey the finite polynomial cutoff budgets of Annex M (4.17)–(4.20), then $\| X_{F,n} \| \le C_F a_n^{-d_F}$, with degree-dependent d_F. A compatible choice $\log(1/a_n)/T_n \to 0$ pays that budget once the thermal certificate has actually been controlled. Replacing the actual coefficients by bare compact ones does not verify the premise.

For slower geometry, the homogeneous comparison 1’s separated-time argument provides another conditional route using renormalized one-field slabs and off-diagonal limiting moment regularity. Its bounded-rate version allows one finite universal factor in the reduced OS inequality. Those premises are not supplied by the finite trace results here. The order of the continuum, smearing and time-shortening limits remains unchanged.

Proved here: an explicit trace reserve for the full interacting finite Wilson operator; a nonnegative three-length defect with exact scalar and ground-energy cancellation; an exact finite dyadic certificate with a uniform explicit error; and the stated conditional heat-capacity and period-sensitive remainder implications.

Still unproved: a vanishing or bounded cofinal rate for the actual positive certificate, or another original E-05 verification route with the actual hierarchy. The cofinal E-05 application remains open.

Conditional-law estimates and the continuum application

Conclusion: the two-age minimum-of-two-bounds argument repairs the numerical two-age summation if its fixed-raw composed-response premise is established. A new comparison with the M04 moving-base analysis exposes an additional source-interface check: the direct mixed-minimizer response has not yet been identified with every unrolled fixed-innovation response required by the stopped-density proof. This analysis therefore does not certify all-history C05 closure. The C06/C08 implications remain available with their explicit low-order boundary tests. The separate B03/stopped-sequence result retains its source/calibration-scoped emergent positive weak scalar endpoint; it does not establish full-state convergence or continuum Yang–Mills fields.

The question is whether the conditional estimates apply to the original Wilson law on every required history.

1. What the conditional-law theorem actually quantifies over

Conditional law on admissible heterogeneous backgrounds, Theorem CS.1, fixes one source-parameter tuple, one sufficiently weak ceiling, the C01/J0 admitted-history class, and the stipulated two-sided AF comparison. Its current variables are the actual independent free-bond coordinates; older raw variables remain fixed parameters at interfaces declaring them as such. Its conclusions are uniform in finite volume, ultraviolet depth, admitted history and positive conditional fiber.

The theorem includes the complete boundary score and Hessian, coercivity, pointwise inverse-Hessian decay, an interacting covariance inequality on the literal product support, current means and moments, and cumulant bounds for its specified native scalar/tensor marks. It does not assert those mark estimates for every microscopic observable transported through the earlier kernels. In fact C8A.21 expressly identifies native fields, while C8A.24 requires the actual Sobolev and moment norms of more general marks as separate input.

The source and predecessor interfaces are explicit in Exact conditional densities and deterministic Hessian envelopes §§2–4:

Interface	What is used	What it does not establish by itself
C01/J0	Exact coordinate density, selectors, factors and actual real centers	An independent construction of all source hypotheses
C02/PF	Strong native/current charts and actual prepared-prefix map bounds	An interacting covariance estimate, or arbitrary off-graph analytic extensions
C03	Completed R primitives, with their true normalizers	The law of all archived histories
C05	Chronological stopped-density control and boundary derivatives	Physical normalization or compactness of transported microscopic marks
C06–C08	Uniform regular-fiber analytic and probabilistic estimates	A complete ultraviolet continuum hierarchy after averaging every history

The label “conditional” therefore describes a precise theorem scope. It is not a finding that the C05 conclusion was reinserted as its own hypothesis. Conversely, a scope declaration is not proof that a proposed original-Wilson sequence satisfies every premise. That application must be checked against the separate trajectory construction and source realization.

2. The old OR.19 objection and the checked two-age repair

2.1 The three scales are retained

For a primitive born at b, evaluated at actual layer n, and changed by an integration at j, put

$$a = n - b, \qquad c = j - n, \qquad b \le n < j.$$

The source Ward calculation supplies decay in a, not automatically in $a+c$. This is the real boundary-induction issue; replacing $n - b$ by $j - b$ inside that source estimate is not justified.

The two-age calculation proposes to control the same scalar difference

$$\Delta_j \mathcal{R}_{b,z} = \mathcal{R}_{b,z}(U^1) - \mathcal{R}_{b,z}(U^0),$$

with the same old coefficient, Wilson subtraction, fixed cutoff, restricted primitive bank and earlier raw tuple. Its interpolation requires the following two estimates for that one difference:

1. An endpoint Ward estimate with the source evaluation-age gain $L^{-\lambda a}$, where $\lambda > 4$.
2. An ordinary analytic response estimate with the extra determining-data gain $e^{-\Gamma c}$, polynomial age/profile factors, and retained spatial/carrier decay.

The path for the second estimate need not satisfy a new Ward identity. Its role is to stay in the admitted ordinary primitive chart at fixed raw variables. The minimum inequality

$$\min(A, B) \le A^{\eta} B^{1-\eta}, \qquad 0 < \eta < 1,$$

is applied to the complete same-term difference before distributing its response majorant over forcing leaves. This avoids assigning an endpoint cancellation to a different differentiated term.

After the four-dimensional root density $L^{4(j-b)}$, the essential age sum is

$$\sum_{a,c \ge 0} (1 + a + c)^s L^{-(\eta\lambda - 4)a} e^{-[(1-\eta)\Gamma - 4\log L]c}. \tag{A.1}$$

It converges when

$$\eta\lambda > 4, \qquad (1 - \eta)\Gamma > 4 \log L. \tag{A.2}$$

The displayed choice $\lambda = 9/2$, $\eta = 11/12$, $\Gamma \ge 60 \log L$ gives $L^{-a/8} L^{-c}$, apart from the summable polynomial. Thus the arithmetic actually pays both ages; it does not silently relabel one age as the other.

2.2 The source direction and parameter reserve were checked

The relevant primary source is the actual Ward/polarization calculation in Bałaban, *Convergent expansions in lattice gauge theories*, CMP 119 (1988), equations (3.56)–(3.67), printed pp.281–283. It supports the evaluation-layer gain and matching own curvature subtraction. It does not directly supply the disputed $j - b$ gain.

The primitive-response mechanism uses the determining-data estimates in CMP 102 (1985), equation (190), and the prepared-layer separation in *Large field renormalization I*, CMP 122 (1989), equations (1.45)–(1.47), printed p.186. The latter explicitly converts scaled spatial separation into an integration-age exponential by a fixed geometric choice and states the scale independence of those choices. CMP 119, printed p.260 following (2.31), also records the available large source-decay parameter choice. These source passages support the direction of the response/rate allocation used in the repair.

For a root of actual grade n and an attachment of grade $m > n + 1$, the two-age analysis also keeps the entire birth carrier. Source admissibility gives the geometric lower bound

$$d_b(X) \ge c L^{m-n} - C.$$

A retained fraction of its exponential can pay any prescribed finite exponential rate in $m - n$. Combining this with the actual map row in $j - m$ pays $j - n$, including attachments above j. This is the root-to-attachment argument of

Ward estimates, cutoff residuals and polarization tensors, Addendum B.3. It uses actual grades, not a fictitious grade equal to the birth label. The remaining spatial envelope must still be retained with enough rate after the fractional power in the minimum inequality.

2.3 Two cutoff terms must remain distinct

The repaired argument is not just (A.1). The analysis retains two cutoff effects:

- The literal Wilson-cutoff residual keeps the physical-order-two curvature response. Its birth-root plaquette count leaves $L^{-4(n-b)}e^{-\Gamma(j-n)}$. The first factor cancels the corresponding old-root density. Recent births cost only the finite preparation window; ancient cutoff collars have $n-b \le C+\log_L R_b$, leaving a polynomial times a summable response-age exponential. See Addendum A, SA.A7–SA.A13.
- The restricted polarization-tensor tail may only have $e^{-\kappa_{\mathrm{tail}}R_b}$, rather than a pure Ward-age gain. It is retained separately and pays the worst recent-window count using

$$\eta\kappa_{\mathrm{tail}}c_R > 4C_N \log L + \epsilon, \qquad N_r \le C_N R_r.$$

  This is Addendum C and CS.2. Dropping this term would revive an identified endpoint gap.

The enlarged ordinary chart also has an explicit fixed reserve. CS.1 separates $q_{\mathrm{tar}}, q_{\mathrm{val}}, q_{\mathrm{phase}}, q_{\mathrm{prim}}$; Addendum B.1 places the actual prepared outputs inside the larger primitive chart. It does not reverse the inclusion between a strong native core and the modified prefix chart.

Revised conclusion on this step: the minimum-of-two-bounds construction is a valid repair of the summation, conditional on the response estimate for the actual composed family. The direct-source calculations and explicit cutoff bookkeeping checked here do not by themselves prove that identification. Section 6 identifies the obstruction: the unrolled fixed-innovation family has a possible latent-gauge term absent from a direct fixed-determining-data derivative. No literal stopped-prefix counterexample has been established, but the current proof cannot be certified merely by citing the direct row. The ordinary primitive induction, admitted source charts and fixed AF comparisons remain additional retained inputs.

3. Does the later B03 work actually supply a trajectory?

Yes, at its stated source/calibration scope the later record asserts and derives more than an abstract endpoint assumption. The distinction matters.

3.1 The reset-feedback argument is not simply C05 assuming its conclusion

Homogeneous feedback and the causal Wilson recursion constructs a coefficient-replacement family on a prescribed numerical coupling prefix. Geometry, source maps and cutoffs are fixed; each substituted homogeneous E coefficient receives its own vacuum and curvature subtractions. An outgoing compensating Wilson term is installed only after the raw coefficient birth. At marker $z = 1$ it vanishes for the actual recursion, identifying the raw coefficients with the original ones.

The prescribed prefix class includes weak coupling and signed two-sided age comparisons. BP.17 regenerates one full native E ball on a common marker disk. Taking mature reset exponent $\nu + D$ and

$$A_n = (C_g g_n)^{-D}$$

permits amplification on that same disk through the entire preceding prefix. Cauchy extraction then gives

$$|\mathcal{P}H_{n+1}(1) - \mathcal{P}H_{n+1}(0)| \le CE_0 r_n^{-2}(A_n - 1)^{-1} \le C g_n^{D-2}\ell_n^{-2q}. \tag{A.3}$$

For $D > 4$, this is $O(Y_n^{-1})$, allowing fixed logarithmic readout losses. The common disk and regenerated ball matter: a fixed norm loss is not multiplied through arbitrarily many births.

The endpoint $z = 0$ is addressed separately in The driven homogeneous coupling endpoint. It regenerates fixed absolute physical charts for the pure-E recursion, freezes the actual symmetric cutoff, and interpolates its physical fluctuation amplitude τ. It verifies its own flat Hessian gap and localized amplitude/background score estimates, then uses a covariance-of-products method to bound

$$|\partial_\tau^2 D_{u_x} D_{u_y} \log Z_n(\tau, 0)| \le C e^{-c\,\mathrm{dist}(x,y)}.$$

The exact symmetry $Z_n(-\tau, u) = Z_n(\tau, u)$ removes the first amplitude derivative. The Taylor remainder gives

$$|\mathcal{P} H_{n+1}(0) - d_n^G| \le C g_n^2. \tag{A.4}$$

This pure-E argument does not merely assume the full heterogeneous C08 theorem on an unconstructed AF trajectory. It verifies the inputs needed for its narrower application. I therefore did not identify a simple C05-to-AF-to-C05 circularity from these dependencies.

3.2 The remaining calibration input and the actual bootstrap

BP §10 retains the signed Gaussian prefix calibration

$$\sum_{k<n} d_k^G = d_B n + \vartheta_n, \qquad |\vartheta_n| \le B_G, \qquad d_B > 0. \tag{A.5}$$

The finite-flux calculation derives the coefficient through the actual constrained Gaussian/coarea determinant and a finite-flux proper-time calculation, retaining native analytic, covariance and heat-kernel inputs. This analysis inspected that interface; it has not independently re-proved the entire heat-kernel calculation. “Retained Gaussian calibration” is an accurate description; silently treating it as a newly established theorem of the conditional-law derivation is not.

Combining (A.3)–(A.5), the actual scalar map has

$$Y_{k+1} = Y_k - d_k^G - e_k, \qquad |e_k| \le C/Y_k.$$

For the corrected coordinate $\widetilde{Y}_k = Y_k + \vartheta_k$, every sufficiently weak step has positive drift bounded above and below. BP.25–BP.34 use this in a slack-class first-exit bootstrap, rather than assuming future AF comparisons. The calculation orders trial raw birth, coefficient readout, outgoing numerical coupling, source-admission checks and actual finite-volume replay. Its homogeneous interior projector does not depend on the newly chosen boundary partition.

Thus the C theorem alone stipulates an AF class, while a separate B03 argument is intended to produce actual weak prefixes belonging to that numerical class. Those statements are compatible.

3.3 What the stopped endpoint does and does not prove

A stopped sequence with a positive weak endpoint, PE.5–PE.26, constructs a cofinal sequence of genuine bare Wilson coefficients and original compatible cubic tori, with all used prefixes below the source ceilings. After corrected-threshold stopping and a bounded last-maximizer trim,

$$Y_- \le Y_{K'_n} \le Y_+, \qquad as \le Y_{K'_n - s} - Y_{K'_n} \le Ds.$$

A subsequence has $Y_{K'_n} \to Y_\infty \in [Y_-, Y_+]$, hence a positive weak scalar endpoint $g_\infty = Y_\infty^{-1/2}$. The finite numerical suffix arrays come from those same actual trajectories. No continuity shooting through rounded-grid transitions is needed for this emergent endpoint.

PE §§6–8 explicitly withhold stronger assertions:

1. The terminal coupling is not an arbitrarily prescribed physical value fixed before the construction.

2. Convergence of the scalar suffix array does not imply convergence of the complete E/R/boundary state or its exact nonlinear history map.
3. At a rounding threshold, a retained one-sided chart label need not equal the canonical label of the limiting scalar without a closed-graph or chart-identification argument.
4. A positive running scalar endpoint does not identify nontrivial renormalized continuum Yang–Mills fields.

PE §7 states a concrete sufficient full-state interface: local compactness on genuine charts; a uniform tail for births beyond a growing ultraviolet window in the norm used by the actual maps; and finite-window limit compatibility, including threshold labels. This is an explicit remaining problem, not something obtained from scalar diagonalization.

Resolved scope tension: retain the actual scalar-trajectory progress under its source/calibration hypotheses. Do not promote it to complete physical normalization or full-state convergence. The metadata flag that source/AF construction was not independently proved in the conditional-law file is not evidence that the later B03/PE result is absent.

4. A rigorous obstruction to an automatic inference from C08 to UV compactness

This section proves a logical nonimplication. It is not a counterexample to the Wilson construction, its C01 identities, or the Yang–Mills existence conjecture.

C8A.23 and C8A.26 retain the scaling factors

$$\begin{aligned} \|O_{h,x} - \nu O_{h,x}\|_p &\le C h^{-4}(Q_x\sqrt{p} + p), \\ |\kappa_m(O_{h,x_1}, \dots, O_{h,x_m})| &\le C^{m+1}(m!)^2 h^{-4m} \prod_i Q_{x_i} e^{-\theta\tau/(m-1)}. \end{aligned} \tag{A.6}$$

Their constants are uniform in cutoff depth, but the quantities on the right are not cutoff independent. Even setting every $Q_x = 1$ and removing all outer histories does not make (A.6) a uniform bound for physically smeared fields.

Proposition A.1

There is a family of product convex-support laws with Hessian equal to the identity, uniform moments, and estimates (A.6) for which the variance of a fixed physical smearing diverges like h^{-4}.

Proof. On the four-dimensional mesh of a fixed box, let the coordinates be independent with density

$$d\mu_P(z) = Z_P^{-1} e^{-z^2/2} 1_{[-P,P]}(z)\, dz, \qquad P \ge 2.$$

The potential Hessian is 1; the support is a product of symmetric convex intervals. The centered linear-coordinate moments are bounded by $C\sqrt{p}$, uniformly in P. Put

$$X_x = z_x^2 - \mathbb{E} z_x^2, \qquad O_{h,x} = h^{-4} X_x.$$

Uniform Gaussian domination gives $\|X_x\|_p \le C p$ and the corresponding gradient norm $\|2z_x\|_p \le C\sqrt{p}$. The moment generating function is uniformly bounded in a fixed complex neighborhood of zero and uniformly close to 1 on a smaller disk. Cauchy estimates for its logarithm give

$$|\kappa_m(X_x)| \le C^m m!.$$

Every mixed-site cumulant vanishes by independence. Therefore (A.6), including its spatial exponential, holds with $Q_x = 1$. The variance $v_P = \mathrm{Var}(z^2)$ is uniformly bounded below for $P \ge 2$: the events $|z| < 1/2$ and $3/2 < |z| < 2$ have uniformly positive probabilities and separated values of z^2.

For a nonzero smooth compactly supported f, the ordinary physical smearing is

$$F_h(f) = h^4 \sum_x f(x) O_{h,x} = \sum_x f(x) X_x.$$

Consequently

$$\operatorname{Var} F_h(f) = \upsilon_P \sum_x f(x)^2 \asymp h^{-4} \int f(x)^2 \, dx \longrightarrow \infty.$$

All constants used above can be uniform when $P = P_h \geq 2$ grows with the cutoff. This proves the proposition. □

The variance uses the joint test $f(x)f(y)$, which is generally not coincidence-flat. Thus this proposition does not disprove a restricted conditional C08 cumulant bound on coincidence-flat tests, including the separate shell estimate proved in Physical normalization, ultraviolet bounds and source transport. It isolates why a conditional microscopic/native estimate needs additional physical matching, renormalization and summability before it becomes a full smeared-field estimate. It also respects the qualification about coincidence-flat tests: on tests vanishing on all coincidence sets, the centered independent example has zero off-diagonal moments. Thus such a restricted hierarchy can be trivial even while the usual smeared variance diverges. The example does not refute the proposed ER.7 estimate (see physical normalization); it shows that neither a nontrivial continuum field nor the full physical distributional hierarchy follows from the displayed C08 bounds alone.

5. The usable conclusion for the existence program

The C05 interpolation can remove the numerical two-age obstruction once the fixed-raw composed-response interface in Section 6 is proved. Until that check is supplied, the assembled C06/C08 source application must remain conditional on the complete boundary tests it requires. Their matrix and covariance arguments remain usable, including as a route through directly established first and second boundary derivatives. The separate B03/PE construction provides its stated path to an actual common original-Wilson sequence with a positive weak scalar endpoint under retained inputs; the source-application issue identified here does not, by itself, refute that independent pure-E/reset-feedback construction.

There are now two distinct levels of unfinished work: the composed-map response needed to certify the all-history C05 application, and the later continuum transition. The latter includes the explicitly open physical coefficient bound ER.2 and actual same-sequence hierarchy bound ER.7 in Sufficient conditions for a continuum hierarchy; their transported microscopic marks and physical normalization are not the native marks of C8A.21. Full-state compatibility in PE §7 remains distinct from the scalar stopping argument.

Accordingly, this analysis supplies no complete proof of continuum Yang–Mills existence and leaves the complete Wilson-history C05 application conditional on CR. It preserves the verified conditional algebra, the scoped scalar-trajectory progress, and the useful low-order C06/C08 route. A defensible next C05 target is the exact fixed-innovation response or an integrated replacement that preserves the stopped-fiber and positive-catalog requirements. The later existence target remains a same-sequence physically normalized transported-mark estimate strong enough for ER.7, with ER.2 and exact source matching retained.

6. Composed response: direct minimizers versus unrolled innovations

Governing update. This section qualifies the earlier source-application assessment. It does not invalidate the minimum interpolation in (A.1), the restored final stopped mass, or the local cumulant algebra.

6.1 The two map families are different

The primary LFR I equations (1.14)–(1.18), printed pp.179–180, define a background as one constrained minimum on a mixed determining set:

$$U^{(n)} = U(\mathbb{B}_k^{(n)}, V).$$

A derivative changing one component of its determining data while the other determining data stay fixed is the natural scope of CMP102 (190). LFR I (1.45)–(1.47) adds spatial separation across the actual prepared annuli. Neither statement is a universal derivative bound for an arbitrary chronology with all lower innovations held fixed.

The boundary-induction stopped-density proof uses the actual one-step identities

$$W_{i+1}(y) = \int J_i(a_i, y)\, W_i(\Phi_i(a_i, y))\, da_i$$

and then cancels intermediate masses. SD.6c–SD.6e identify the result as the incoming density pulled back through the composed coordinate maps, with the actual Jacobians. The tuple includes the prefix innovations A_i, B_i. SD.4–SD.5 also permits selected earlier entries to be fixed after that functional cancellation.

This unrolling is algebraically valid. It does not assert that the pulled-back argument is a single minimum with every lower determining value fixed. Such an identification, or a direct bound on the differentiated composition, is an additional analytic assertion.

The two-age same-term note ST.7–ST.8 says to compose the literal source maps, retain the old raw tuple, and use the actual normalized row. ST.7b asserts the resulting age decay. The missing check is the chain rule at that interface, including field-dependent changes of the gauge in which the retained innovations are specified.

6.2 The exact latent term exposed by the M04 calculation

For the actual one-step fixed-target solve $\Xi(W, x)$, gauge covariance gives

$$\Xi(W^{\upsilon}, T_{\widetilde{\upsilon}} x) = \Xi(W, x)^{\widetilde{\upsilon}}.$$

Differentiation at fixed innovation yields MB.11:

$$D_W \Xi[\mathfrak{d}_W \xi] = \mathfrak{d}_{\Xi(W,x)} \widetilde{\xi} - D_x \Xi[A_{\widetilde{\xi}} x]. \tag{A.7}$$

The first term is an output gauge direction. The second is an innovation response, and can be physically visible even to a gauge-invariant scalar. Therefore gauge invariance of the ordinary E bank removes the first term, not automatically the second.

The M04 note constructs a finite homogeneous chain and a local gauge-invariant detector for which the second term has no terminal-age factor. It is enough to refute a generic identification of all fixed-innovation responses with direct critical-chain quotient responses.

It is not yet a counterexample to the precise C05 stopped-prefix estimate. In C05 the root grade n is its actual prepared geometric grade; it cannot simply be set equal to the M04 innovation birth ℓ. The source prefix also has disjoint proper strips and protected deep data, unlike an arbitrary homogeneous chronology. Its forcing is the actual constrained free increment $g_j C B_j - h\widetilde{D}(g_j C B_j)$, while the M04 example varies an arbitrary terminal pure-gauge target. Exclusion of that direction by the literal source gauge/constraint convention is another possible distinction that has to be checked, rather than assumed. An actual C05 counterexample would have to satisfy those geometries, charts and selector margins, use the same Wilson-subtracted bank, and violate the required summed estimate. No such construction has been completed here.

6.3 The predecessor support argument does not close the bridge as written

Density and root compatibility §2 invokes the actual-history argument of Conditioning, deletion and mass factors for the retained history (Section 13.10, H32–H36). That argument explicitly says that recent old A_j in the integrated preparation collars have already been integrated out. Its surviving deep nodes have bounded old-grid/prepared-grade mismatch; the actual normalized inverse-gauge and nodal interpolation then transfer the direct response to those fixed old slots.

The stopped-density proof subsequently reopens the integrated prefix to obtain a joint pointwise comparison. The preceding statement about which nodes survive does not automatically hold for the newly reintroduced innovations. Reusing H32–H36 for that joint composition therefore requires a new demonstration. It cannot be inferred merely from the validity of the same source theorem before unrolling.

Two possible estimates are:

- Finite-window payment. Every new innovation is within the current preparation window, but the full length is $N_r \le C_N R_r$, with R_r a fixed power of the source logarithm. A loss L^{cN_r} is not a harmless logarithmic polynomial. Only the overlapping top band $N_0 = O(\log R_r)$ has that property. The recent tensor-tail repair pays its exponential count with an explicit $e^{-\kappa_{\mathrm{tail}} R_b}$; no corresponding rare factor has been displayed for every latent-innovation contribution.
- Spatial support payment. LFR I pp.179–192 gives disjoint lower proper strips and a short overlapping top band. This suggests a possible repair: prove that the actual lifted gauge generator vanishes, is constant in a harmless simultaneous rotation, or has the required annular response on every earlier active innovation support. However, disjoint integration regions alone do not identify the support of the moving-reference inverse gauge. The quantitative bound for those latent terms, and its preservation under the full finite composition, is not in the source formulas examined here.

The second route remains plausible. It must use the literal block lifts and source gauges, rather than replacing them by a convenient representative that changes the fixed raw tuple.

6.4 A concrete sufficient estimate

Let $\mathcal{F}_{H,a}(t)$ be the literal unrolled source field along a stage-j interpolation with all other raw innovations fixed. Subtract from its differential the direct mixed-data contribution already controlled by the source row. Denote the residual gauge/innovation chain-rule contribution, after applying the same complete Wilson-subtracted root bank, by $\mathcal{L}_{b,z;j}(t,a)$.

It would suffice to prove uniformly on the actual admitted joint support

$$\int_0^1 |\mathcal{L}_{b,z;j}(t,a)|\,dt \le C E_0 \ell_r^D (1+n-b+j-n)^s e^{-\Gamma(j-n)} w_j(z,S_j)^{1/(1-\eta)}, \tag{A.8}$$

with the spatial reserve and two-age inequalities already used in (A.1). A direct estimate of the complete root-summed difference can replace (A.8), provided it supplies the same projective-density and positive-catalog bounds without an unrecorded fine-root factor.

For fully integrated gauge directions, the MB.22–MB.24 transport identity cancels insertion, centered density score and domain current before absolute values. That can be a repair mechanism. At a stopped fiber with earlier innovations fixed, however, transporting those retained innovations changes the conditioning unless its residual retained-variable derivative is explicitly included. Replacing a pointwise fixed-fiber estimate by cancellation only after their integration does not prove SD.7 or SD.14. The full kernel, current denominator and sharp support response must remain in any integrated alternative.

The separate Gauge covariance of partial integrals and exterior scores, PG.18–PG.20, refines the retained-variable part of this target to the anchored derivative $\{D_x F_0(x_t,a) - D_x F_0(x_t,a_*)\}[\dot{x}_t]$. This keeps the same Wilson-subtracted bank and permits cancellation of terms independent of the integrated tuple. The integrated-coordinate score and boundary contributions are still required for a pointwise stopped-density comparison.

Remaining estimate. (A.8), an equivalent geometric latent-term bound, or a sufficient integrated replacement has not been established in this analysis. Consequently the two-age all-history source application and the conditional-law application through it are not independently certified here. This is a precise unresolved bridge, not a proof that C05 or Yang–Mills existence is false.

6.5 The lower-order C06/C08 route remains available

The matrix/covariance implications need less than every higher native C05 jet. In particular, on the actual common current chart it suffices to establish directly

$$\sup_i \|D_i W_B\| \le C P_k^{-1}, \qquad \sup_i \sum_j e^{\beta\rho(i,j)} \sup_{\text{admitted real data}} \|D_i D_j W_B\| \le C P_k^{-2}, \tag{A.9}$$

with the actual source support, regular E/R and local-factor bounds, and retained-force profile.

These are the first/second derivative tests of C6.29–C6.30, strengthened to the fixed-entry matrix form used in C6.36–C6.38. They are obtained through the C05 capacity/endpoint argument; they have not been independently established after bypassing the new composed-map question. Nonetheless they are logically sufficient hypotheses. C6.41–C6.47 then prove coercivity and inverse decay. Together with C07 and the specified native mark-gradient bounds, the C08 covariance, means, moments and cumulant argument proceeds without higher derivatives of the potential at every order. The original complete C06 native-jet clause remains a separate stronger requirement.

For the outer same-current response, one additionally needs the corresponding mixed current bound on W_{zy}. The outer-law theorem examined next is an implication from those explicit bounds; it must not be presented as independently proving them.

7. Independent conditions of the outer-law Schur estimate

This section concerns Inner normalizer covariance and outer conditional laws, OH.9–OH.21, on its explicitly fixed same-current chart.

7.1 Convex-support Brascamp–Lieb inequality

The inequality

$$\operatorname{Var}_\nu f \le \nu\big[(\nabla f)^T H_I^{-1} \nabla f\big], \qquad H_I = V_{zz} \succ 0,$$

is valid on the literal convex product cutoff. An independent proof uses the scalar Neumann solution $Lu = f - \nu f$. The integrated Bochner identity gives

$$\nu[(Lu)^2] \ge \nu[(\nabla u)^T H_I \nabla u],$$

because both the Hessian-square term and the convex-boundary second-fundamental-form term are nonnegative. Integration by parts and weighted Cauchy give

$$(\operatorname{Var}_\nu f)^2 = \big(\nu[\nabla f \cdot \nabla u]\big)^2 \le \nu[(\nabla f)^T H_I^{-1} \nabla f]\, \nu[(\nabla u)^T H_I \nabla u],$$

which proves OH.9. Smooth convex approximation and scalar Sobolev convergence handle the literal support. For this unweighted inequality one may approximate the whole convex product, avoiding any ambiguity about corners of products of individually smooth factors.

7.2 The error and normalization signs

For $S = -B^T z + W_y$ and $E = W_{zy}$,

$$\nabla_z (a^T S) = (-B + E)a.$$

The inverse comparison and expansion around $B^T A^{-1} B$ give exactly

$$\delta = \frac{\|B\|^2 \varepsilon}{\lambda_0(\lambda_0 - \varepsilon)} + \frac{2\|B\|\varepsilon + \varepsilon^2}{\lambda_0 - \varepsilon}.$$

Thus OH.10 is the correct one-sided Loewner bound. There is no asserted two-sided norm approximation, and no missing factor from a possibly large conditional mean.

The normalizer identity has the correct sign:

$$(\log Z)_{yy} = \nu W_{yy} + \operatorname{Cov}_\nu(S, S).$$

Consequently

$$\overline{V}'' = D_0 - (\log q)'' - \nu W_{yy} - \operatorname{Cov}_\nu(S, S).$$

The principal Schur contribution cancels in the lower bound exactly as in OH.19. A full principal gap $\mathcal{K} \succeq \lambda_0 I$ implies $\Sigma = D_0 - B^T A^{-1} B \succeq \lambda_0 I$ by minimizing over the inner vector. The stated defect and δ budgets therefore give OH.21. The invariance of the defect under $W \mapsto W + c(y)$, $q \mapsto q e^{-c(y)}$ is also exact.

7.3 Support and outer-only factors are indispensable

OH.17 differentiates an inner normalizer only while its inner support is independent of the same-current y chart. A moving support would add boundary terms. The outer-only factor q remains in the outer density even though it cancels from the inner conditional quotient. These qualifications are mathematically necessary.

For an exponentially localized outer covariance theorem, a replacement for product support must supply the weighted boundary-domain compatibility used by C08, not only a nonnegative unweighted boundary form. General convex global constraints can induce long-range correlations despite Hessian I. For example, an independent standard Gaussian vector conditioned on $\sum_i y_i \le 0$ has off-diagonal covariance $-2/(\pi N)$ on N sites, inconsistent with a fixed exponential bound at arbitrarily large separation. The unweighted convex-domain Brascamp–Lieb inequality still holds. Literal local product factors admit the required block-scalar spatial weights; the revised outer note now states this qualification.

Outer-law implication. OH.9–OH.21 is a valid conditional theorem with the explicit inner/mixed derivative and weighted-support hypotheses. It does not establish the actual full retained-history defect or support theorem. Its selected-bad-domain and mixture examples correctly illustrate those missing inferences.

Physical normalization, ultraviolet bounds and source transport

Scope: existence without a mass-gap assumption; the original compact Wilson law, its prescribed 21 curvature-bilinear fields and seven physical channel normalizations.
Result: an explicit new ultraviolet summation consequence of the C08 conditional cumulant theorem, together with the exact remaining full-law and normalization requirements. No Yang–Mills existence theorem is claimed.

1. Source record and the target that has to be met

1. Regular conditional covariance, moments and cumulants, especially C8A.23–C8A.28 and Sections 8–11.
2. Sufficient conditions for a continuum hierarchy, ER.1–ER.8.
3. Reflected susceptibilities and physical normalization, PN.26–PN.42 and PN.48–PN.52.
4. Separated marked estimates for successful local resets, RS.29–RS.37 and its final scope statement.
5. A stopped sequence with a positive weak endpoint, PE.18–PE.23 and its distinction between an emergent scalar endpoint and prescribed physical matching.

The retained C08 conclusion is a theorem about an actual selected *regular conditional law*. It supplies native current fields and their gradients, the full score, a fixed Hessian majorant, covariance, conditional means, and native scalar/tensor cumulants. It does not identify every transported microscopic insertion with a native stencil and does not control the entire outer distribution of conditional means. Those restrictions matter in every application below.

The continuum estimates below assume the displayed C08 inequalities. Their application to every actual Wilson history requires the composed response with earlier innovations fixed.

For a joint test on $(\mathbb{R}^4)^m$, use the theorem group's dominating Schwartz seminorm

$$\nu_r(F) = \max_{|\alpha|\le r} \sup_X (1+|X|^2)^{r/2} |\partial^\alpha F(X)|, \tag{EC.0}$$

with a maximum over its finitely many tensor coefficients when present. Coincidence-flat means that every derivative vanishes whenever any pair of arguments coincides.

The fixed existence target in the theorem group decomposition requires one original Wilson sequence and one common continuum hierarchy, with physical normalization, corrected OS regularity, full Euclidean covariance, positivity, the requested energy/phase properties and the stipulated interacting ultraviolet coefficients. No mass gap is needed here. Not every sufficient technical norm in the fixed decomposition is logically necessary for every possible proof: ER.7 is already a weaker route to the hierarchy than proving every original intermediate M03 norm.

2. The two sharp numerical inputs for the existing continuum reduction

Write the actual physical field budget as in ER.1:

$$b_a = \frac{\|Z_a\|}{\beta_a} + a^4 |c_a|. \tag{EC.1}$$

The physical-normalization note proves, on its admitted stopped sequence,

$$b_a \asymp B_a := \frac{\|Z_a\|}{\beta_a} = \max_h \sqrt{\frac{\lambda_h}{D_{h,a}}}, \qquad D_{h,a} = \beta_a^2 n_{h,a}. \tag{EC.2}$$

Here h labels the seven hypercubic channels; it is not a lattice mesh in this equation. The fixed targets $\lambda_h > 0$ are the F01 reflected normalization values. Formula EC.2 uses PN.28 and PN.37–PN.38. The identity coefficient adds no independent growth budget.

Proposition EC.1: the weak normalization requirement is seven lower bounds

For a fixed finite $\gamma \geq 0$, the coefficient estimate

$$b_a \leq C a^{-\gamma} \tag{EC.3}$$

is equivalent, up to fixed channel constants, to

$$D_{h,a} \geq c\lambda_h a^{2\gamma} \quad \text{in every channel, on the same sequence.} \tag{EC.4}$$

Proof. Square the maximum in EC.2 and use that the number of channels and all λ_h are fixed. □

Strict positivity of every $D_{h,a}$ at every finite cutoff does not imply EC.4. For example a positive numerical sequence $D_a = e^{-1/a}$ has inverse square root larger than every fixed power of a^{-1}. This is an elementary warning about the inference, not a counterexample for the Wilson data. The finite positivity theorem FN and its countable avoidance construction supply nonsingularity but no rate of avoidance of zero; they must not be substituted for EC.4.

The second input is ER.7. With its original recent-history event, microscopic position collar, same cofinal sequence, and same physical fields,

$$R_{m,n}(F) = a_n^{4m} \sum_{\alpha,X} (1 - \chi_n(X)) F_\alpha(X) \mathbb{E}_n \left[1_{E_n^c} \prod_{i=1}^{m} \Phi_{\alpha_i,n}(x_i) \right]$$

must obey

$$|R_{m,n}(F)| \leq C^{m+1} (m!)^p \nu_{qm}(F) \quad \text{with fixed } C, p, q. \tag{EC.5}$$

The expectation is the actual full Wilson/history expectation, unnormalized on E_n^c. Replacing it by a selected regular conditional law changes EC.5. It includes recent and intermediate histories, successful, residual and unreplaced sectors, outer means, all higher source partners and the actual normalization.

For the exact F01 fields, the one-point case is already reduced by their full-law centering. Since $\mathbb{E}_n \Phi_{\alpha,n}(x) = 0$ and ER sets $\chi_n = 0$ when $m = 1$,

$$R_{1,n}(F) = -a_n^4 \sum_{\alpha,x} F_\alpha(x) \mathbb{E}_n[1_{E_n} \Phi_{\alpha,n}(x)]. \tag{EC.5a}$$

Under EC.3, ER.6 and its once-chosen H2S age threshold make EC.5a tend to zero in its displayed fixed Schwartz seminorm (one may take ν_8 there). It also gives a uniform one-point bound after increasing the constant to cover finitely many initial cutoffs. Thus the one-point recent full moment is not an additional unproved requirement once the exact centering and the normalization budget are retained. This cancellation does not say that conditional means vanish, and it does not delete their outer cumulants at higher orders.

The already written ER.1–ER.2 reduction proves that EC.3 and EC.5 remove the old-history and position-collar errors and yield one common coincidence-flat mixed hierarchy. It is unnecessary to reprove that diagonal extraction. The work still required is to establish the two actual estimates, and then the additional Euclidean, reconstruction and identification inputs of Section 6 below.

3. A new conditional ultraviolet bound obtained from C08

This section gives a concrete strengthening of the continuum use of C08. It controls native conditional cumulants on flat tests with a fixed factorial exponent, not only at fixed separated points.

Rescale the fixed reference length to one. Let $0 < h \leq 1$ be the current physical mesh, and let $\nu_{h,v}$ be an admitted selected regular conditional law. Its actual source anchors form a subset of a four-dimensional lattice of spacing h with fixed bounded local multiplicity. Distances ρ_h are in current lattice units. Fixed blocking ratios are absorbed

in constants. On the central-half-torus convention of ER, physical distances of supported anchors agree with their nonwrapped distances, and

$$|x - y| \le Ch(1 + \rho_h(x, y)). \tag{EC.6}$$

Let τ_h be the minimum spanning-tree length in ρ_h. For the C08 native scalar/tensor family, C8A.26 gives, for $m \ge 2$,

$$|\kappa_{\nu_{h,v}}(O_{\alpha_1,x_1}, \dots, O_{\alpha_m,x_m})| \le C^{m+1}(m!)^2 h^{-4m} P(h)^m e^{-\theta\tau_h/(m-1)}. \tag{EC.7}$$

Here we used the actual bound $Q_x \le CP(h)$ stated after C8A.23. The constants are uniform in the admitted retained datum v. C8A.28 supplies

$$\sup_{x_1} \sum_{x_2,\dots,x_m} e^{-\theta\tau_h/(m-1)} \le C_\theta^m (m!)^5. \tag{EC.8}$$

Define the native conditional cumulant functional

$$\mathcal{K}_{m,h,v}(F) = h^{4m} \sum_{\alpha,X} F_\alpha(X) \kappa_{\nu_{h,v}}(O_{\alpha_1,x_1}, \dots, O_{\alpha_m,x_m}). \tag{EC.9}$$

This definition is not an assertion that $\mathcal{K}$ is an exact connected shell of the original microscopic source. Identifying such shells remains an actual transport problem.

Theorem EC.2: flat-test conditional cumulant bound

Assume EC.6–EC.8. For every smooth joint test flat on the coincidence diagonals, supported in the ER central region when the lattice is periodic,

$$|\mathcal{K}_{m,h,v}(F)| \le C^{m+1}(m!)^7 h^2 P(h)^m \nu_{6m}(F), \qquad m \ge 2. \tag{EC.10}$$

No coupling ceiling depends on m, and no probabilistic independence is used.

Proof. Fix the pair of arguments 1 and 2. Flatness gives $D_2^\alpha F(x_1, x_1, x_3, \dots, x_m) = 0$ for every multi-index α. Taylor's theorem along the segment from x_1 to x_2, to order six, therefore gives

$$|F(X)| \le C|x_2 - x_1|^6 (1 + |x_1|^2)^{-3} \nu_6(F). \tag{EC.11}$$

The derivative is evaluated at a point whose first coordinate is still x_1, so the displayed spatial weight is valid throughout the segment. The directional sixth derivative costs only a fixed four-dimensional combinatorial constant. The same estimate holds for every tensor coefficient.

The path in any spanning tree connects x_1 to x_2; hence $\rho_h(x_1, x_2) \le \tau_h$. Combining EC.6 and EC.11 gives the factor $Ch^6(1 + \tau_h)^6$. The additive one permits fixed source-block/stencil offsets without a false zero-distance identification. For $m \ge 2$,

$$(1 + \tau_h)^6 e^{-\theta\tau_h/(m-1)} \le C_\theta (m-1)^6 e^{-\theta\tau_h/[2(m-1)]}. \tag{EC.12}$$

Use EC.8 at $\theta/2$, and the four-dimensional lattice root estimate

$$\sum_{x_1} (1 + |x_1|^2)^{-3} \le Ch^{-4}. \tag{EC.13}$$

The estimate is uniform on subsets and on the central part of compatible growing tori; fixed local multiplicity changes only C. Inserting these bounds in EC.9 yields

$$|\mathcal{K}_{m,h,v}(F)| \le C^{m+1}(m!)^2 P(h)^m h^6 \nu_6(F)\, h^{-4}(m-1)^6 C^m (m!)^5$$
$$\le C_1^{m+1}(m!)^7 h^2 P(h)^m \nu_{6m}(F).$$

The factors h^{4m} and h^{-4m} cancel before the root count. The fixed power $(m-1)^6$ is absorbed into C_1^m, not into a new m-dependent ceiling. The finite number of tensor labels is absorbed in the same way. Finally $\nu_6 \le \nu_{6m}$. This proves EC.10. □

The power h^2 is the difference between the sixth-order vanishing used on the test and the remaining four-dimensional physical root count. It is not an extra dynamical irrelevant factor claimed for the bare operator.

Corollary EC.3: uniform logarithmic absorption and scale summation

Suppose, in addition, that the *same* physical-scale profile has

$$P(h) \le C(1 + \log(1/h))^b \tag{EC.14}$$

for one fixed finite b. Then for every fixed $0 < \delta < 2$,

$$|\mathcal{K}_{m,h,v}(F)| \le C_\delta^{m+1}(m!)^{7+b} h^\delta \nu_{6m}(F). \tag{EC.15}$$

For the geometric meshes $h = L^{-j}$,

$$\sum_{h \le \varepsilon} |\mathcal{K}_{m,h,v_h}(F)| \le C_\delta^{m+1}(m!)^{7+b} \frac{\varepsilon^\delta}{1 - L^{-\delta}} \nu_{6m}(F). \tag{EC.16}$$

The retained data v_h can vary arbitrarily within the admitted class. The same estimates hold after an arbitrary positive outer probability average supported on that admitted class, applied to each of these conditional cumulant functions.

Proof. Put $t = \log(1/h)$. Maximizing $e^{-(2-\delta)t}(1+t)^{bm}$ for $t \ge 0$, followed by $m! \ge (m/e)^m$, gives

$$e^{-(2-\delta)t}(1+t)^{bm} \le C_{b,\delta}^{m+1}(m!)^b. \tag{EC.17}$$

This proves EC.15 from EC.10. The geometric series proves EC.16. A uniform pointwise-in-v absolute bound passes through any probability average. That last operation averages the conditional cumulant; it does not identify the result with the full-law cumulant. □

The postulated fixed-physical-scale AF envelope implies EC.14 when its terminal inverse coupling is bounded uniformly: $g(h)^{-2} \le C(T + \log_L(1/h))$ with fixed T, and $P(h) = A_1[\log g(h)^{-2}]^{p_0}$, give a bound of the form EC.14. PE.20 records precisely the bounded-terminal-interval age comparison in its retained source/calibration scope. If the terminal inverse coupling diverges with the cutoff, EC.14 with cutoff-independent constants must be checked separately. This corollary is not a new construction of the trajectory.

Corollary EC.4: fixed separated supports have a superalgebraic fine-scale tail

At fixed m, suppose the support of F lies in a fixed compact physical set and its anchor spanning-tree length is at least $d > 0$. Under EC.7 and EC.14,

$$|\mathcal{K}_{m,h,v}(F)| \le C_{m,F} h^{-4m}(1 + \log(1/h))^{bm} e^{-cd/[(m-1)h]}. \tag{EC.18}$$

Consequently, for every N, the sum over $h \le \varepsilon$ is at most $C_{m,F,d,N}\varepsilon^N$, uniformly in v.

Proof. The physical lattice quadrature of $|F|$ is bounded on its fixed compact support, while $\tau_h \ge cd/h - C_m$, with the fixed-order offset absorbed in $C_{m,F}$. Apply EC.7. An exponential $e^{-c'/h}$ absorbs any prescribed fixed power of h^{-1} and logarithm. Leave one additional positive power to sum the geometric meshes. □

EC.15 is stronger for the all-order flat-test application because its seminorm order and factorial exponent are fixed. EC.18 makes the physical separation mechanism transparent at each fixed order.

Corollary EC.4a: fixed inverse-coupling powers in spatially localized reset jets

There is a useful restricted improvement over a rare-value-only optimization. Suppose the literal successful-local coefficients in RS.30, at current physical mesh h, satisfy

$$|h^{4|B|}A_{B,X}| \le C_0^{|B|}(|B|!)^s \left[(g(h)^{-2})^\alpha(1+\log g(h)^{-2})^b\right]^{|B|} B_X^{|B|}, \quad \alpha, b \ge 0, \tag{EC.22}$$

with fixed s, α, b, the exact local support of RS.30, and the same admitted conditional law as RS.32. This is an explicit conditional input, not a claimed microscopic-to-current-scale matching theorem. Put $T(h) = (g(h)^{-2})^\alpha(1+\log g(h)^{-2})^b$. In every source partition, $\sum_{B\in\pi}|B| = m$. Hence its coefficient factor is precisely $T(h)^m$, and the proof of RS.32 gives the complete finite-jet bound

$$|\Gamma_{[m]}(v;X)| \le C^{m+1}(m!)^{s+3}h^{-4m}T(h)^m e^{-cP(h)}e^{-\gamma\tau_h(X)/m}. \tag{EC.23}$$

For $m \ge 2$, apply the same order-six flat-test Taylor argument as EC.11–EC.13. The anchored divided-tree sum costs $C^m(m!)^5$, so

$$\left|h^{4m}\sum_X F(X)\Gamma_{[m]}(v;X)\right| \le C^{m+1}(m!)^{s+8}h^2T(h)^m e^{-cP(h)}\nu_{6m}(F). \tag{EC.24}$$

If, on this same family, $g(h)^{-2} \le C(T_0 + \log(1/h))$ with fixed T_0, then $T(h) \le C(1+\log(1/h))^{\alpha+b}$. The optimization EC.17 therefore proves, for each fixed $0 < \delta < 2$,

$$\left|h^{4m}\sum_X F(X)\Gamma_{[m]}(v;X)\right| \le C_\delta^{m+1}(m!)^{s+8+\alpha+b}h^\delta e^{-cP(h)}\nu_{6m}(F). \tag{EC.25}$$

The geometric fine-scale sum obeys the analogue of EC.16. An input $g(h)^{-\sigma|B|}$ has $\alpha = \sigma/2$. All ceilings and exponents are chosen once, independently of source order.

This conclusion is consistent with the warning after RS.36a: the rare factor alone cannot absorb a fixed inverse-coupling power per source grade at fixed factorial cost. Here the extra h^2 comes from physical flat-test summation and supplies that cost, using the actual physical AF comparison. The argument requires a conditional spatially localized jet and current-scale normalization. It does not turn a bare a^{-4m} microscopic insertion into an h^{-4m} insertion, and it does not control an outer value term for which no spatial decay has been established.

4. Why these estimates do not yet prove ER.7

There are two distinct missing identifications.

First, EC.7 concerns the native C08 scalar/tensor family. The actual input in ER.7 is the *transported microscopic physical source*, including every higher generated source partner. The fixed-depth M04 results do not identify its arbitrary-depth output with that family plus a remainder satisfying EC.15. In particular, a proof for three native regular layers does not supply one uniform estimate after an unbounded number of moving-Q operations, residual sectors and history changes. The new scale-sum theorem does remove a possible downstream counting obstacle once that identification is supplied.

Second, an average of conditional cumulants is only one part of the full cumulant. The exact identity is useful at every order.

Proposition EC.5: complete law of total cumulance

Let $\mu(dv, dz) = \pi(dv)\nu_v(dz)$, with all joint moments of the finite family $X_1, \dots, X_m$ available. For a nonempty block $B \subseteq [m]$, put

$$k_B(v) = \kappa_{\nu_v}(X_i : i \in B).$$

Whenever the displayed products are integrable,

$$\kappa_\mu(X_1,\dots,X_m)=\sum_{\mathcal{P}\in\Pi_m}\kappa_\pi(k_B(\upsilon):B\in\mathcal{P}). \tag{EC.19}$$

Proof. Work in the finite squarefree source algebra, so no moment-generating-function radius is needed. The conditional logarithmic source is $\sum_{\varnothing\neq B\subseteq[m]} t_B k_B(\upsilon)$. The tower identity gives

$$\log \mathbb{E}_\mu e^{\sum_i t_i X_i}=\log \mathbb{E}_\pi \exp\left(\sum_{\varnothing\neq B} t_B k_B(\upsilon)\right).$$

Taking the coefficient of $t_1\cdots t_m$ produces every set partition of the labels once. The outer cumulant of its block coefficients is exactly the corresponding term in EC.19. All operations are finite algebraic identities. □

For m=2 this is total covariance. At order three it reads

$$\begin{aligned}\kappa_\mu(X_1,X_2,X_3)=\mathbb{E}_\pi k_{123}+\kappa_\pi(k_1,k_2,k_3)\\ +\operatorname{Cov}_\pi(k_1,k_{23})+\operatorname{Cov}_\pi(k_2,k_{13})+\operatorname{Cov}_\pi(k_3,k_{12}).\end{aligned} \tag{EC.20}$$

The new conditional ultraviolet estimate controls the one-block term $\mathbb{E}_\pi k_{[m]}$ for its native marks. It does not control the outer cumulants in all the other partitions. Their arguments include conditional means and higher conditional vertices depending on common retained data. Dropping them changes the original Wilson law.

The successful-reset note already retains the analogous full finite-jet partition identity RS.35c. It obtains a complete *value* bound for its specified successful-local class but expressly no spatial decay for an arbitrary outer marginal. This is genuine useful progress and does not constitute the missing general ER.7 estimate. A residual component whose mass has canceled in its normalized base cannot receive the successful-presence rare factor.

5. A concrete sufficient replacement for the remaining transport application

The point of EC.15 is that the next estimate can be stated more weakly than the strongest original M03 carrier/current norm, if one pursues ER rather than the entire fixed transport program.

A sufficient next input would be an exact connected decomposition of the actual recent physical source into:

1. native conditional scale functionals controlled by EC.15, with actual channel coefficients having fixed per-order logarithmic budgets;
2. the successful-local finite jets controlled by RS.32–RS.36, after their microscopic matching is proved;
3. residual/unreplaced and outer-partition terms with joint-test bound $C^{m+1}(m!)^p\nu_{qm}$, uniformly in cutoff and total depth;
4. the actual terminal and one-point pieces, and compatible contact/approximation errors on the same test space.

Every item must refer to the same physical normalization and the original full law. Merely defining the residual as the difference between the desired source and the controlled classes supplies no estimate for it. A stronger route may instead prove M04–M07/H04–H05 in their fixed norms. The direct decomposition above is a sufficient alternative, not a theorem that those remaining terms already obey the displayed bounds.

The scalar endpoint progress should also be preserved accurately. The BP/DE/PE chain supplies, at its retained source and calibration scope, an emergent positive scalar endpoint on one cofinal family. It does not identify an arbitrarily prescribed physical coupling, prove the full effective-state/history-map compatibility, or provide the quantitative physical normalization data in EC.4. The C-to-existence analysis addresses its exact dependencies; this note does not erase that progress or turn it into a completed physical matching theorem.

Existence at some suitable positive endpoint does not logically require solving the fixed B05 problem of matching a preassigned endpoint. If the PE sequence satisfies the source hypotheses and the needed physical identification is established there, it is an eligible sequence for an existence proof. Similarly, a direct proof of EC.5 for original Wilson expectations can bypass a theorem asserting convergence of every effective state or history map. The direct ER route needs its exact input estimates, not every stronger sufficient interface of an earlier plan.

6. A nonduplicative completion checklist for this route

The following are proof interfaces for the stated program, not a claim of mathematical necessity for every conceivable construction.

Interface	Existing usable result	Actual missing input
One physically identified Wilson sequence	Exact positive source representation, stopped compatible geometry and emergent positive scalar endpoint at the retained source/calibration scope	Verify the actual source/physical dictionary and all hypotheses on the chosen sequence. Prescribed B05 endpoint matching and convergence of every effective state/history map are stronger route-specific tasks, not prerequisites for a direct original-law ER.7 proof at an eligible emergent endpoint.
Quantitative field normalization	Explicit F01 system, seven-channel finite positivity, PN identity-budget reduction, exact determinant formulas	Seven lower bounds EC.4 (or stronger logarithmic bounds), actual channel mixing and summable normalized remainders where the chosen route uses them
Full microscopic hierarchy	C08 native conditional estimates; successful-local reset jets; exact source algebra; the new EC.15–EC.16 scale sum; bounded mixed-kernel composition	ER.7/EC.5 for the actual physically normalized transported microscopic fields, all sectors and outer partitions included
Full Euclidean covariance	Exact discrete symmetries, finite hypercubic tensor decomposition and conditional angular reduction	Vanishing action, observable, running-matrix and contact angular defects on this same interacting hierarchy, uniformly along the finite rotation paths
Reconstruction and a physical state	ER conditional common extraction, exact finite Wilson reflection positivity, corrected reconstruction and phase-selection bridge theorems	Verification of corrected OS hypotheses, actual field/energy domains and periodic-state passage, then the common phase set and required disconnected symmetries
Yang–Mills identification	Bare Wilson action and group conventions, native/comparator UV calculations	Physical two-point running, a specified nonzero interacting connected coefficient with uniform remainder and correct order of limits, and the continuum local operator/identity correspondence required by the target

The shortest currently written route therefore starts with the admitted PE family, proves EC.3 and EC.5 for its original physically normalized Wilson correlations, and applies ER's common flat-test compactness implication. It need not wait for every fixed M03 norm, an arbitrary prescribed B05 endpoint, or a separate limit theorem for every history map. EC.3 and EC.5 do not alone provide the last three rows.

The normalization and nondegeneracy tasks should not be counted twice. PN.7 already proves that, if the exact F01-normalized full hierarchy converges on one common sequence, then all seven separated reflected measurements remain exactly $\lambda_h > 0$. A finite symmetry average gives a positive Gram form on the complete 21-dimensional field family. Thus no separate post-limit nonzero-field axiom is needed once that actual convergence is proved. On the other hand, free curvature bilinears can also have such positive norms. Those norms do not identify the continuum dynamics as interacting Yang–Mills.

Likewise, the existence of a connected three-point function somewhere is not a general interaction certificate for composite fields: free-field bilinears can have nonzero higher connected moments. The fixed U02 target specifies

the physical channel and coefficient, while U03 requires its interacting remainder and the order of limits. The work record explicitly distinguishes a nonzero stress-channel comparator from the separated scalar Gaussian coefficient, which vanishes in the stated comparator. Those distinctions must survive the final identification argument.

The rotation requirement can be presented as the concrete sufficient bound

$$|S_{m,a}(F_R) - S_{m,a}(F)| \le C^{m+1}(m!)^p \epsilon_a \nu_{qm}(F), \qquad \epsilon_a \longrightarrow 0, \tag{EC.21}$$

with the proper tensor action and constants uniform along the actual finite rotation paths. Hypercubic invariance alone does not imply EC.21, and an estimate on a zero-terminal-coupling comparison family does not establish it on the physically matched family. R02–R05 separate the action, observable/contact and path-uniformity requirements.

Finally, the A02 reconstruction explicitly reconstructs the initial mixed hierarchy without assuming clustering. The later E/V applications supply the requested energy and phase properties. No mass gap is used in EC.1–EC.21, and no claim about a spectral gap is needed to formulate these remaining tasks.

7. Conclusion of this calculation

The new proved calculation is the ultraviolet estimate EC.10–EC.18, conditional only on the precisely displayed C08 native cumulant bounds and, for its scale-summed physical form, the fixed physical AF envelope. Its factorial growth and test-seminorm order are fixed. It works under arbitrary positive averaging of those conditional cumulant functions.

The same calculation identifies why it is not yet the desired existence theorem: the actual microscopic matching, the non-one-block terms of EC.19, and the quantitative reflected data EC.4 remain unproved here. The existing ER compactness theorem becomes applicable only after those actual-model estimates have supplied EC.3 and EC.5. Full rotations, corrected reconstruction/physical-state applications and Yang–Mills ultraviolet identification still have their own explicit requirements.

This note supplies a usable regular-sector continuum estimate and a precise next target. It does not change an open original-law requirement into an assumption and then report Yang–Mills existence as proved.

Independent calculation conditions

8. A compensated normalized-source route around a pointwise innovation estimate

The M04 analysis gives an exact reason to combine the response terms before taking absolute values. In Moving ancestor bases, fixed innovations, and exact latent-gauge current transport, MB.9–MB.11 imply

$$D_W \Xi[\mathfrak{d}_W \xi] = \mathfrak{d}_\Xi \tilde{\xi} - D_x \Xi[A_{\tilde{\xi}} x]. \tag{EC.26}$$

At a fixed raw innovation x, removal of the output gauge leaves the second term. MB.12–MB.17 gives a finite gauge-invariant plaquette detector of that term, without the proposed ancestor-distance suppression. This does not establish a counterexample to the literal C05 stopped history class: its prepared grade is a geometric datum, and a fixed innovation level is not automatically that grade. It does show why a direct critical-minimum row cannot be transferred to a fixed-innovation composition without checking the extra term.

The issue is concrete in the C-05 argument. SD.6d–SD.6e restores the original density pulled back through the actual composed source maps, while ST.7–ST.8 applies a normalized response row to that composition. The actual-history argument H32–H36 of Conditioning, deletion and mass factors for the retained history used the integration of recent preparation-collar innovations. Such a statement is not, by itself, a pointwise estimate after those innovations have been reintroduced by unrolling. Nor is an exponential loss across the full prefix automatically harmless: the full depth satisfies $N_r \le C_N R_r$, while only the distinct top band $N_0 = O(\log R_r)$ permits L^{CN_0} to be absorbed as a fixed profile polynomial. These are source-application questions requiring a literal variable/geometry identification or a compensated estimate.

The following algebraic identity supplies an alternative interface. It generalizes MB.22 to every finite source grade and retains the source-dependent denominator exactly.

Proposition EC.6: normalized finite-source material-current identity

Let D_t be a moving integration domain, $\rho_t > 0$ its complete unnormalized density, and $\Psi_t(u,x)$ a finite squarefree source polynomial with zero constant source coefficient. All earlier source partners, actual Jacobians, Gaussian factors, archived masses and retained-data dependence are included in these objects. Put

$$\mathfrak{Z}_t = \int_{D_t} \rho_t, \qquad N_t(u) = \int_{D_t} \rho_t e^{\Psi_t(u)}, \qquad G_t(u) = \log \frac{N_t(u)}{\mathfrak{Z}_t}. \tag{EC.27}$$

Assume the ordinary differentiation/transport formulas hold, including their boundary terms. A finite disjoint history decomposition can be included by summing every displayed integral and boundary functional over its literal branches. Let $s_t = \partial_t \log \rho_t$, choose a latent coordinate velocity w_t, and write

$$D_w = \partial_t + w_t \cdot \nabla, \qquad A_t = s_t + \rho_t^{-1} \operatorname{div}(\rho_t w_t). \tag{EC.28}$$

If $\mathfrak{B}_t(f)$ denotes the actual moving-domain boundary functional and $\mathfrak{F}_{w,t}(f) = \int_{\partial D_t} f\, w_t \cdot n$, put $\mathfrak{C}_t^w = \mathfrak{B}_t - \mathfrak{F}_{w,t}$. Define $\mathbb{E}_t^\Psi f = N_t(u)^{-1} \int \rho_t e^{\Psi_t} f$ formally in the finite source algebra, and let $\mathbb{E}_t$ be the positive zero-source law. Then

$$\begin{aligned} \partial_t G_t(u) = {} & \mathbb{E}_t^\Psi[D_w \Psi_t] + \mathbb{E}_t^\Psi[A_t] - \mathbb{E}_t[A_t] \\ & + \frac{\mathfrak{C}_t^w(\rho_t e^{\Psi_t})}{N_t(u)} - \frac{\mathfrak{C}_t^w(\rho_t)}{\mathfrak{Z}_t}. \end{aligned} \tag{EC.29}$$

The two denominators are distinct and neither is omitted. Because $N_t(0) = \mathfrak{Z}_t > 0$, their formal inverses and the logarithm exist at every finite grade without an analytic source radius.

Proof. Differentiate the numerator, then integrate the latent divergence once:

$$\partial_t N_t = \int_{D_t} \rho_t e^{\Psi_t}(D_w \Psi_t + A_t) + \mathfrak{C}_t^w(\rho_t e^{\Psi_t}). \tag{EC.30}$$

Setting the source to zero gives $\partial_t \mathfrak{Z}_t = \int \rho_t A_t + \mathfrak{C}_t^w(\rho_t)$. Subtract $\partial_t \log \mathfrak{Z}_t$ from $\partial_t \log N_t$. This is EC.29. Each source coefficient is a finite algebraic identity. □

For a pure gauge motion, use the actual induced latent velocity, together with the pushed-forward density, domain and source labels. Where this is an exact symmetry, the material source, centered material score and residual flux vanish as a completed expression. This requires transforming the complete retained exterior as well, or restricting the motion to its stabilizer. A partial conditional integral with an earlier raw exterior x fixed need not be invariant: when its covariance identity is $H_t(T_t x) = H_0(x)$, differentiation gives

$$\partial_t H_t(x) = -D_x H_t(x)[v_t(x)], \qquad v_t = \partial_t T_t \circ T_t^{-1}. \tag{EC.29a}$$

That exterior material derivative must be retained or canceled by its later exact outer integration. The complete symmetry cancellation can hold even when the fixed-innovation integrand derivative in EC.26 is nonzero. This is precisely why the completed current, including its boundary and normalization terms, should be estimated before its origins are separated.

Moment form retaining the recent-history event

For any finite source-dependent f_t, the same proof gives

$$\partial_t \mathbb{E}_t f_t = \mathbb{E}_t[D_w f_t] + \operatorname{Cov}_t(f_t, A_t) + \frac{\mathfrak{C}_t^w\big(\rho_t (f_t - \mathbb{E}_t f_t)\big)}{\mathfrak{Z}_t}. \tag{EC.31}$$

Take $f_t = 1_{E_n^c} e^{\Psi_t}$ on the actual fixed-branch decomposition. Count selector motion exactly once: either as the weak derivative of the indicator on the unpartitioned domain, or as the selected-branch boundary current with its indicator

constant on each branch. EC.31 then retains the *unnormalized* recent-history moments required by ER.7; no division by $\mathbb{P}(E_n^c)$ has been introduced. Its denominator remains the complete original $\mathfrak{Z}_t$, not a selected-event mass. The physical F01 normalization belongs to the original source coefficients and must be retained through the identity. If outer variables are still present, apply their exact positive expectation, including their source-dependent weights when using logarithmic source coordinates. The whole-source moment formulation automatically retains the outer terms; an average of conditional log coefficients alone would not.

A precise sufficient estimate, still open for the full Wilson construction

Suppose an exact chronological interpolation of the same microscopic recent source has been constructed, with an endpoint bound and completed increments given by EC.31. After the actual outer expectation and physical source-position sum, let $\mathcal{J}_{m,h,t}(F)$ denote the complete m-source coefficient of its right side. The following are sufficient:

$$\begin{aligned} |R_{m,n}^{\text{endpoint}}(F)| &\le C^{m+1}(m!)^{p_0}\nu_{q_0 m}(F), \\ |\mathcal{J}_{m,h,t}(F)| &\le C^{m+1}(m!)^{p}h^{\delta}\nu_{qm}(F), \qquad 0 \le t \le 1, \end{aligned} \tag{EC.32}$$

with fixed $\delta > 0$, geometric physical meshes h, and an exact telescoping identity for the original source. Integrating in t and summing the geometric h-powers proves ER.7, with fixed factorial and seminorm exponents. If a coarser endpoint contains finitely many scales, they are included in the first bound.

EC.32 estimates one normalized, integrated, physically tested response. It does not require an ancestor-decay bound for every pointwise fixed-innovation derivative. EC.24–EC.25 show how a spatially localized conditional part can meet the required ultraviolet summation, while EC.29–EC.31 specify the additional compensated density and boundary pieces that must be retained. The actual all-history estimate EC.32, its exact source matching and the endpoint bound have not been proved here. This is a weaker sufficient interface for ER.7, not a completion claim for C05, M04 or Yang–Mills existence.

Conditional reconstruction from Wilson source derivatives

Fix $SU(N)$, $N \geq 2$, and count each unoriented plaquette once. Write $s_p = N - \mathrm{ReTr}\, U_p$ and take the Wilson weight $\exp(-\beta_a \sum_p s_p)$, with $\beta_a = 2Nx_{0,a}$. The inverse coupling multiplying $\sum_p s_p/N$ is $N\beta_a$. The fields take values in $E = \mathrm{Sym}^2(\Lambda^2\mathbb{R}^4)$, of dimension 21. The following implication concerns a single cofinal family of these Wilson measures. Its analytic hypotheses include all large-field histories and the same physical normalization for all source orders.

Finite source derivatives and local products

For a source j in a finite-dimensional test family, write $Z_n[j]$ for its cutoff Wilson integral. Let $C_n[0] = 0$ and put

$$\mathcal{M}_n[j] = e^{-C_n[j]} Z_n[j]/Z_n[0], \qquad W_n[j] = \log \mathcal{M}_n[j].$$

At any fixed order these identities also make sense as identities of Taylor polynomials, using the truncated exponential and logarithm. Let $S_{m,n}$ and $S^c_{m,n}$ be the degree-m derivatives at zero of $\mathcal{M}_n$ and W_n, respectively. For elementary tests,

$$S_{m,n}(f_1, \dots, f_m) = \sum_{\pi \in \mathcal{P}_m} \prod_{B \in \pi} S^c_{|B|,n}(f_i : i \in B). \tag{1}$$

This follows by differentiating the exponential. Reflection positivity involves the full moments on the left. Applying (1) to joint Schwartz tests requires a continuous distribution associated with each partition term and estimates for their sum.

Fix a physical scale μ. The one-field prescription can include a matrix $Z_{\alpha\beta}(a, \mu)$ and identity terms $c_{\alpha,a} \int f$. Products can also have local subtractions on partial diagonals. A finite-order distribution supported on a partial diagonal vanishes on a Schwartz test which is flat there. A cutoff approximation with shrinking support needs a quantitative distribution bound to justify that conclusion in the limit. Dependence on source values throughout a fixed physical cell is a different support condition, as the example below shows.

A sufficient set of hypotheses

Let $a_n \downarrow 0$ and $T_n \uparrow \infty$ be the spacings and periods of one admitted family. For example $a_n = L^{-K_n}\ell_0$ and $T_n = 2L^{M_n}\ell_0$ give cofinal sizes for an admitted L when $K_n, M_n \to \infty$. Admissibility includes the actual geometric, small-coupling and trajectory conditions of the construction. Write $\mathcal{S}_0((\mathbb{R}^4)^m)$ for Schwartz functions flat with all derivatives on every coincidence $x_i = x_j$. Give tensor-valued tests the fixed component norm and set

$$\nu_s(F) = \max_{|\alpha| \leq s} \sup_X (1 + |X|^2)^{s/2} |\partial^\alpha F(X)|. \tag{2}$$

J0. Wilson law and trajectory. The source-free integrations, changes of coordinates, reference measures and positive history decomposition represent the normalized Wilson law exactly on this sequence. The physical-scale trajectory is identified in the same coupling convention, including any uniform Gaussian calibration needed to obtain it.

J1. Compatible renormalized fields. At every finite order the full source derivatives on this Wilson law, including the averages over its histories, give distributions $S_{m,n}$ for a common one-field and product prescription. The subtractions have coincidence support in the limit. The fields obey the specified tensor rules and the finite normalization conditions at μ.

J2. Uniform joint-test estimate. There are fixed constants C, p, q, independent of n, m, for which

$$|S_{m,n}(F)| \leq C^{m+1}(m!)^p \, \nu_{qm}(F) \qquad (F \in \mathcal{S}_0((\mathbb{R}^4)^m)). \tag{3}$$

The cutoff distributions here act on a common test space, for example through a periodization compatible with the stated bounds. A derivation starting with separate test factors must justify the passage to this joint-test norm.

J3. Euclidean symmetries and the positive-time form. Reality, permutation symmetry and reflection positivity, including the tensor signs, hold in the limit on the Osterwalder–Schrader test algebra. More explicitly, for each finite positive-time vector the cutoff reflected quadratic form has nonnegative limit inferior. Translation defects

and, for each fixed $R \in O(4)$, the defects $S_{m,n}(R \cdot F) - S_{m,n}(F)$ tend to zero. The tensor action on each field is $A \mapsto (\Lambda^2 R) A (\Lambda^2 R)^t$.

J4. Ultraviolet identification. In the normalization of J0–J1 the cutoff and increasing-volume limits precede the short-distance limit. Uniform remainders identify the stipulated scalar two-point coefficient and the first nonzero interacting connected three-point coefficient on separated tests. The selected phase must retain these same coefficients. The canonical comparison for $Q = F^2/4$ has $c_2 = 3(N^2-1)/\pi^4$ and $c_3 = 11(N^2-1)/(16\pi^8)$ at order $g^2 = N g_{\mathrm{YM}}^2$, with the finite matching factors of the physical scalar applied before the comparison.

J5. Energy bounds and vacuum phase. The reconstructed fields satisfy the required polynomial energy estimates on a common invariant domain. The hypotheses for vacuum-phase decomposition, its symmetries including the disconnected operations, uniqueness of the vacuum and clustering in the specified phase hold. There is one phase in which the energy estimates and the ultraviolet identification apply together.

Conditional assembly theorem. Under J0–J3 a subsequence defines an infinite-volume continuum hierarchy for the stated fields with bound (3), full Euclidean covariance, permutation symmetry and reflection positivity. Corrected Osterwalder–Schrader reconstruction on the indicated test domain gives a relativistic theory, possibly with a degenerate vacuum. Under J4–J5 this theory has a phase with the energy bounds, symmetries and nonzero ultraviolet coefficient specified in the continuum theorem.

Proof. For each order and each label tuple take a countable dense subset of the coincidence-flat Schwartz space. This space is a closed subspace of a separable Schwartz space. Estimate (3) bounds each numerical evaluation. Simultaneous diagonal extraction gives convergence on all the dense sets. Approximation in ν_{qm} extends convergence to each test, on the same subsequence, and preserves (3). For a finite positive-time test vector the Osterwalder–Schrader quadratic form is a finite sum of evaluations of full distributions on the corresponding reflected, time-ordered tests. Convergence and the nonnegative lower bound pass to the sum. Permutation, rotation and translation invariance follow from convergence and J3. Exact fine-lattice translations can also be approximated by translations of a fixed test, using continuity in the seminorm in (3). Apply corrected reconstruction with its order-growth hypothesis to this limiting hierarchy. J4 identifies the short-distance laws, and J5 supplies the common field domain and phase to which that identification applies. The phase and operator requirements are used at this final step. Osterwalder and Schrader, *Axioms for Euclidean Green's Functions II*, §IV.1.

Exact transport of source derivatives

Let v be retained variables and u, u' old and replacement variables. Let $h_0(u', v) \geq 0$ be a source-independent kernel with $0 < H_0(v) = \int h_0(u', v)\, du' < \infty$ on the active data. For a density ρ define

$$(P_h \rho)(u', v) = \frac{h_0(u', v)}{H_0(v)} \int \rho(u, v)\, du, \qquad QF = \frac{P_h(\rho F)}{P_h \rho} \tag{4}$$

on the nonzero support. Sample (u, v) from the normalized old law μ and sample u' conditionally from h_0/H_0. The output law is μ', and QF is conditional expectation given (u', v). Thus $Q1 = 1$, $\int QF\, d\mu' = \int F\, d\mu$, and $\|QF\|_{L^r(\mu')} \leq \|F\|_{L^r(\mu)}$ for $1 \leq r \leq \infty$. On a positive history lift this applies sector by sector to the positive density, before any signed polymer expansion. The selected coordinates, selectors and underlying reference kernel are fixed throughout the source differentiation.

For a normalized source polynomial $G = \exp \Psi$ with $G(0) = 1$, put $G' = QG$, $\Psi' = \log G'$, where the prime labels the new variables. The derivative rules are

$$A_i' = QA_i, \qquad A_{ij}' = QA_{ij} + \mathrm{Cov}_Q(A_i, A_j), \qquad A_i = \partial_i \Psi(0), \quad A_{ij} = \partial_i \partial_j \Psi(0). \tag{5}$$

The conditional cumulants appearing at higher order satisfy, for $A_i \in L^m(\mu)$,

$$\int |\kappa_Q(A_1, \dots, A_m)|\, d\mu' \leq m! 2^{m-1} \prod_i \|A_i\|_{L^m(\mu)}. \tag{6}$$

Indeed, for a partition block B, Jensen followed by Hölder bounds the $L^{m/|B|}$ norm of $Q(\prod_{i \in B} A_i)$ by $\prod_{i \in B} \|A_i\|_m$. Hölder across blocks gives the same bound for every partition term. The sum of their unsigned coefficients is at most the number of ordered partitions, bounded by $m! 2^{m-1}$ by writing a permutation and inserting cuts. This estimate retains the size of the original insertions on its right-hand side. Obtaining (3) still requires spatial estimates and uniform bounds on the physically normalized derivatives.

Bounded pressures and separated correlations

A counterterm on one fixed cell. Let $A_{a,c}$ be a linear average of the lattice source in a cell c of fixed side ℓ_0. Choose reflection-symmetric normalized cubature with $A_{a,c}(f) \to \ell_0^{-4} \int_c f$ for smooth f. In a representation that permits a cell-local pressure term L_a and a compensating remainder R_a, consider

$$F_{a,c}(j) = 1 - \cos(n(a)A_{a,c}(j)), \qquad L'_{a,c} = L_{a,c} + F_{a,c}, \qquad R'_a = R_a - \sum_c F_{a,c}, \quad n(a) \longrightarrow \infty. \tag{7}$$

All real-source pressure changes in the separate terms are bounded by 2 per cell and their sum is unchanged. The modification is analytic at every finite cutoff and respects the cell reflections. Take smooth f, g with disjoint supports in the interior of the same cell and nonzero integrals. Then

$$\partial_s \partial_t F_{a,c}(sf + tg)\big|_0 = n(a)^2 A_{a,c}(f) A_{a,c}(g). \tag{8}$$

Using L' as the counterterm therefore changes the connected two-point function by the negative of (8), whose absolute value diverges at fixed positive separation. A bounded real-source witness on each cell leaves this freedom. Coincidence support or quantitative control of the counterterm derivatives restricts it.

Two reflected cells. Give signs $\sigma, \tau \in \{-1, 1\}$ the joint probability $(1 + c\sigma\tau)/4$, $0 < c < 1$. Reflection interchanges the cells. For $f(\sigma) = a + b\sigma$ the reflected quadratic form is $|a|^2 + c|b|^2$. Insert $X_n = n\sigma$, $Y_n = n\tau$, and use the cell counterterm $L_n(s) = \log\cosh(ns)$. Direct summation gives

$$\mathcal{M}_n(s,t) = e^{-L_n(s)-L_n(t)} \mathrm{E} e^{sX_n + tY_n} = 1 + c \tanh(ns)\tanh(nt). \tag{9}$$

On real sources $\mathcal{M}_n \in [1 - c, 1 + c]$. Its reflection kernel is positive semidefinite, being the sum of the constant rank-one kernel and the rank-one kernel with feature $\sqrt{c}\,\tanh(ns)$. Nevertheless,

$$\partial_s \partial_t \log \mathcal{M}_n(0,0) = cn^2. \tag{10}$$

Since the insertions lie in different cells, adding any sum of single-cell counterterms leaves this mixed derivative unchanged. These two finite examples identify the additional information in the separated and joint derivative bounds of J1–J2.

Reflection positivity after subtractions

Let $K_{ij} = \mathcal{M}_0(\Theta \overline{f}_i + f_j)$ be positive semidefinite for positive-time sources. Suppose a counterterm satisfies $C(\Theta \overline{f}_i + f_j) = \overline{C(f_i)} + C(f_j)$. For $D_{jj} = e^{-C(f_j)}$ the subtracted kernel is $D^* K D$, hence positive semidefinite. Its finite statement for polynomial vectors follows either by finite differences near zero or by the induced triangular transformation of the positive-time polynomial algebra. Local counterterms with the appropriate reality and reflection rules have this splitting for sources supported strictly away from the reflection plane. Shrinking stencils have the same separation for each fixed positive-time margin at sufficiently fine cutoff. A local source redefinition also preserves this argument when it intertwines reflection and splits additively across the separated half-spaces. These are sufficient hypotheses on the actual field and product prescription in J3.

Vacuum moments and energy bounds

Let A be one standard normal random variable and set $\phi(x) = A$ at all spacetime points. The moments are invariant under Euclidean motions and permutations. The reflected form is $\mathrm{E}\,|P(A)|^2$ on polynomial vectors. The full hierarchy acts by $S_m(F) = \mathrm{E}(A^m) \int F(X)\, dX$. For $q \geq 6$, estimate (2) bounds the absolute value of the integral by $C^m \nu_{qm}(F)$, since

$$(1 + |X|^2)^{-3m} \leq \prod_{i=1}^{m} (1 + |x_i|^2)^{-3}.$$

The one-variable weight is integrable in four dimensions, and the Gaussian moments have factorial growth. The hierarchy thus satisfies a bound of the form (3).

Reconstruction gives $L^2(\mathbb{R}, d\gamma)$ for the standard normal law $d\gamma$, because the polynomials are dense. Time translations act identically, so $H = 0$. For a test with $\int f \neq 0$, $\phi(f)$ is unbounded multiplication by $A \int f$, and

$$\|\phi(f)(1 + H)^{-r}\| = \infty \qquad (r \geq 0). \tag{11}$$

Thus the energy assertion in J5 calls for estimates on general polynomial vectors. The example satisfies the vacuum moment and reflection hypotheses used above.

A related distinction occurs in phase selection. Suppose a translation-invariant theory has been decomposed into translation-invariant phases γ with adequate moment integrability. For scalar tests f_i with $\int f_i = 0$, every phase one-point function vanishes. The total connected three-point function is then the average of the phase connected three-point functions. A nonzero value implies a nonzero value in at least one phase. Retaining the stipulated short-distance coefficient requires control of the short-distance limits in the chosen phase, as specified in J4–J5.

Theorem catalogue and open problems

The statements below refer to one fixed group $SU(N)$ and to the same sequence of four-dimensional Wilson lattices. Each entry states the required theorem, its present scope, and the hypotheses used to formulate or apply it. A dependency can supply a definition or a conditional implication; it need not be an established estimate.

Paper · Mathematical scope · Contents

Common hypotheses

GC · Uniform Gaussian calibration

Identify the signed sum of projected Gaussian decrements with the composite constrained Gaussian in the same coordinates and prove a bounded remainder from its linear drift, uniformly in depth on the admitted family. The logarithmic coefficient and the finite Schur algebra alone do not bound the transverse remainder. Status: open for the full application. Derivation.

CL · Compatibility of the conditional law

Identify the conditioning variables, selectors, marginal and conditional normalizers, moving support, and derivative coordinates for each estimate used together. Equality of the final Wilson integral under two factorizations is not equality of their conditional laws or their derivatives. Status: open for the full application. Derivation.

Wilson action and its coupling

Supporting derivations.

B-01 · Expose the exact positive source-free representation

For the chosen SU(N), identify the actual localized RG/R integral, its reference kernels, selectors, coordinate Jacobians and positive history lift at each scale. Give the equality with the Wilson measure, using $\beta = 2Nx_0$ and $X_B = N\beta$ throughout.

Scope. Source identity, under the published domain and admissibility assumptions. A signed polymer activity is not a history probability.

Required conclusion. The published hypotheses and exact integral identity, with every unavailable premise identified; signed polymer activities are never used as history probabilities.

Dependencies: definitions and stated hypotheses.

B-02 · Choose one admissible cofinal geometry

Choose one admitted odd L and integers K_n,M_n giving $a_n = L^{-K_n}\ell_0$ and $T_n = 2L^{M_n}\ell_0$. Verify every minimum size, coarse-grid, reflection-plane and coupling ceiling in the actual source-free statements.

Scope. Geometric construction. Permitting an expanding sequence of cubic tori does not determine the running coupling or the physical endpoint.

Required conclusion. An explicit sequence satisfying all displayed inequalities and permitting every compact support to embed; no unproved rectangular extension.

Dependencies: B-01.

B-03 · Extract the actual one-step coupling remainder

In the source-free Wilson recursion, identify a marginal coupling coordinate u and prove its ultraviolet recurrence $u_{j+1} = u_j - bu_j^2 + r_j$ with $b > 0$ and $|r_j| \le Au_j^3$, or a comparably quantified trajectory theorem retaining dependence on earlier couplings.

Scope. Conditional on the homogeneous charts and GC. The universal coefficient and the recurrence for a calibrated coordinate have different hypotheses.

Required conclusion. The coefficient dictionary, sign, history dependence and uniform remainder are proved for the actual map; a numerical iteration or a native frozen derivative is insufficient.

Dependencies: B-01. Additional condition: GC.

B-04 · Prove stability and asymptotics of the perturbed AF recurrence

Given b>0 and the recurrence in B-03, prove an invariant positive small-coupling interval, reciprocal linear bounds, $u_j = [b(j+\kappa)]^{-1} + O(\log(1+j/\kappa)/(j+\kappa)^2)$, and square-summable tails, uniformly over all admissible remainders.

Scope. Arithmetic theorem for a recurrence satisfying the stated uniform error. Its application to the Wilson map depends on B-03.

Required conclusion. A complete elementary proof with explicit smallness conditions and constants; actual Wilson identification remains B-03.

Dependencies: definitions and stated hypotheses.

B-05 · Match the trajectory to the finite-cutoff construction

Construct one coherent set of bare Wilson coefficients whose finite-depth RG trajectories stay inside the admitted source-free domains and satisfy the chosen finite physical normalization at ell0.

Scope. Open for a prescribed physical endpoint. A positive emergent weak endpoint, conditional on calibration, and positivity of seven finite measurements do not impose that prescription.

Required conclusion. Compatibility across cutoff depths and all coupling ceilings, including any shooting or matching argument; no independently chosen incompatible finite trajectories.

Dependencies: B-02, B-03, B-04.

Physical field normalization and products

Supporting derivations.

F-01 · Fix a separating finite field-normalization prescription

Specify the 21 curvature-bilinear representatives, Wick/reflection conventions, seven hypercubic projectors, identity direction, and explicit finite normalization functionals at mu. State a checkable nondegeneracy condition for the finite normalization system.

Scope. Finite prescription under its nondegeneracy condition; the uniform inverse problem is F-05.

Required conclusion. One concrete prescription whose coefficients and tensor laws can be checked, rather than an unspecified choice of Z; actual invertibility is checked in F-05.

Dependencies: B-02.

F-02 · Absorb logarithmic running into a fixed factorial budget

For a family bounded by $a^{\delta} C^{m+1} (m!)^{p} [1 + \log(\ell_0/a)]^{\gamma m+\eta} \nu_{qm}$, prove that every $0 < \theta < \delta$ gives $a^{\theta} C_{\theta}^{m+1} (m!)^{p+\gamma} \nu_{qm}$. Include finite label mixing and logarithmically many errors.

Scope. Factorial-budget inequality; no identification with the Wilson fields is required for this implication.

Required conclusion. The factorial exponent is fixed, constants are independent of a,m, and convergence is quantitative; no logarithmic factor is hidden in a cutoff-dependent constant.

Dependencies: definitions and stated hypotheses.

F-03 · Compute the seven actual first mixing coefficients

Project the exact regular transported one-field outputs onto the seven fixed hypercubic channels and identify Gamma in $M_j = I + u_j \Gamma + R_j$, with the prescribed field and coupling normalization.

Scope. Native Gaussian and regular-channel calculations are available. Open for the exactly transported physical fields.

Required conclusion. Seven explicit coefficient formulas or finite invariant integrals with their evaluated normalization; Gaussian/native coefficients must be connected to the exact transport.

Dependencies: F-01, M-04, M-05.

F-04 · Bound the one-step mixing matrix error

Prove $\|M_j - I - u_j \Gamma\| \le C u_j^2$ (or a stated summable substitute) for the actual channel matrices, uniformly in cutoff, carriers, retained backgrounds and histories admitted to this regular calculation.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. A uniform matrix norm with the exact source of every derivative and remainder; symmetry alone does not supply its size.

Dependencies: F-03, C-08.

F-05 · Construct the actual running Z and identity coefficients

Apply the proved commuting-product theorem to the actual matrices, solve the finite normalization system at mu, and construct identity coefficients matching that same prescription. Establish the precise logarithmic coefficient budgets.

Scope. Open along a common sequence. Positivity of the seven finite reflected measurements does not bound their reciprocals uniformly.

Required conclusion. The inverse matrices, finite normalization determinant and scalar identity coefficients are controlled on the chosen cofinal sequence.

Dependencies: B-05, F-01, F-04.

F-06 · Construct compatible local product subtractions

Specify all source-jet contact coefficients generated by the exact induction using shrinking stencils or point jets; prove compatibility under truncation in m and a linear-in-m spatial distribution order.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. One compatible hierarchy of contacts with explicit bounds and reality/reflection/tensor laws; fixed physical-cell locality is excluded.

Dependencies: F-01, M-06.

F-07 · Close the weighted additive-remainder budget

For the exact matching expansion, bound the sum of inverse-running-product-weighted additive vertices, including identity and collision responses, in the joint observable norm required by S-06.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. The actual series $\sum_j \|T_{j+1}^{-1} R_j\|$ is summable with fixed source-order growth; unweighted summability does not count.

Dependencies: F-02, F-05, F-06, M-07.

Transport of microscopic fields

Supporting derivations.

M-01 · Realize arbitrary frozen kernels as reverse-tail innovations

For a source-free positive Markov RG/R chain on standard Borel history spaces, realize Q_{0:k}F on one path space as a reverse-tail conditional expectation, prove orthogonal innovations and the infinite-depth covariance/tail identity for L2 marks.

Scope. Positive-chain identity. The terminal tail is retained.

Required conclusion. Replacement noise is included in the dilation; the terminal tail covariance is retained unless a separate triviality/orthogonality hypothesis proves it vanishes. The old finite telescope is a separate identity.

Dependencies: definitions and stated hypotheses.

M-02 · Extend exact microscopic matching to shrinking mesoscopic blocks

Integrate compact Wilson variables by their true conditional laws on independent factor scopes, retaining inherited marginal factors. Prove a scope-radius recurrence and an all-order full-product contact estimate when the final physical radius tends to zero; include blocks of side O(sqrt(a)).

Scope. Finite-scale identity and shrinking-block contact estimate. No all-depth transport assertion.

Required conclusion. A direct full-Wilson proof for every SU(N), fixed q,p and a vanishing error, with no restart of the original one-link action and no claim that a^-4 becomes the coarse-scale field budget.

Dependencies: definitions and stated hypotheses.

M-03 · Define a current-scale norm for exact transported marks

Choose one explicit localized value/gradient/current norm on the augmented history variables. Show its initial finite-cutoff field values, its carrier geometry and its compatibility with conditional expectation and the source-jet algebra.

Scope. Norm and initial bounds specified. For moving support, uniform derivatives require support and normalizer responses; see M-04.

Required conclusion. A fully specified norm and initial bounds, including retained-variable dependence; renaming a bare Lp norm a current-scale norm does not meet the test.

Dependencies: B-01, F-01.

M-04 · Match the exact scalar mark through one regular step

At one RG step, expand the actual transported scalar insertion in the fixed incoming reference basis of F-01, under an explicit one-step induction budget, into the current scalar basis, identity term and a localized remainder. Include

constrained background, E/B/R responses and coarea terms present in that sector. This does not assume the final running Z from F-05.

Scope. Open in the required norm: the original ambient derivative can jump with the integration support. The local-mask construction reaches three regular integrations with separate current origins. Uniform transport of the completed quadratic and moving cores remains open.

Required conclusion. Equality with Q_k F_k and a remainder estimate in M-03, uniform in scale; the left side is not a freshly inserted coarse plaquette.

Dependencies: M-03, C-08.

M-05 · Match the remaining tensor channels through one regular step

Perform the same exact one-step matching for all tensor channels, with their hypercubic transformations and every generated retained-field response, in the same norm and normalization.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. All 21 components, including Weyl-type directions, are represented with uniform coefficients; scalar trace matching alone is insufficient.

Dependencies: M-04, F-01.

M-06 · Close the full finite-jet response algebra at one step

For arbitrary source order m, give an exact local expansion for the full jet transported through one step, retaining every collision, reset and remainder vertex; establish a fixed factorial budget for the expansion coefficients.

Scope. Open for the uniform iterated application. Bounds on individual coefficients can lose a fixed factor at each step. The analytic-family formulation needs the stated contact and regulator estimates.

Required conclusion. Compatibility as m varies and a carrier-bound sum with fixed q,p; a finite truncation of the number of one-field labels is not a truncation of the generated vertices.

Dependencies: M-05, C-08.

M-07 · Prove the all-scale exact matching induction

Iterate the exact one-step expansion over the actual source-free histories with a current-scale field budget and summable irrelevant gains; give the resulting local fields, response kernels and terminal remainder.

Scope. Open at arbitrary depth. The clean first-order transfer is conditional; higher-order composite subtractions and marked large-field estimates do not follow from it.

Required conclusion. Constants are uniform in total RG depth and volume, with the outer retained-data responses explicitly present; finite-depth mesoscopic elimination does not itself prove this induction.

Dependencies: M-06, H-04, F-04.

The regular conditional measure

Supporting derivations.

C-01 · Identify every term of the actual conditional density

Write the constrained quadratic form, complete classical action, inherited E/R/B activities, coarea, selectors and boundary terms once each in common current coordinates.

Scope. Density identity on a specified representation. CL is required to transfer any derivative or support estimate from a different representation.

Required conclusion. A term-by-term identity on the stated domain, not a schematic density; any additional premises are stated.

Dependencies: B-01. Additional condition when changing the conditioning law: CL.

C-02 · Prove a simultaneous current-coordinate E contour

For every localized E carrier, prove a quantitative complex margin for the joint current/background variables used by the marked derivatives, including the independently extended current and preparation factors.

Scope. Regular analytic contour on the stated common chart; application requires the same representation as C-01.

Required conclusion. An explicit radius and prefactor whose Cauchy derivative bound can be summed over old scales; membership in an open analytic set alone is insufficient.

Dependencies: C-01. Additional condition when changing the conditioning law: CL.

C-03 · Prove the corresponding R contour and marginal transfer

Prove a simultaneous contour for each actual localized R pullback and its marginal/coordinate transfer, retaining the current-dependent extension and denominators appearing in the true construction.

Scope. Regular localized contour and marginal transfer on the specified charts. A change to the local R test requires CL.

Required conclusion. A uniform holomorphic norm and carrier decay with the verified preparation losses; the source-free R value bound is not differentiated without this domain.

Dependencies: C-01. Additional condition when changing the conditioning law: CL.

C-04 · Sum the actual E/R derivative carriers

Sum the current-coordinate value, gradient and Hessian majorants over old scales and carriers with the actual running radii, coupling sequence and preparation losses.

Scope. Carrier-summation theorem under the given radii, scale and preparation bounds. It does not include the whole boundary response.

Required conclusion. A displayed small weighted-Hessian budget plus local source-derivative budgets, not just convergence of an unmarked value sum.

Dependencies: C-02, C-03. Additional condition when changing the conditioning law: CL.

C-05 · Control the complete boundary activities

Establish the same local derivative bounds for the actual B and boundary-induced activities, including collars, constraints and current-coordinate dependence.

Scope. Open for all required Wilson histories. Two-age summation is valid when both estimates apply to the same subtracted quantity. The anchored response of the stopped reconstruction at fixed earlier innovations, after retaining the common mass, is still needed.

Required conclusion. Boundary constants do not grow with total volume or hidden carrier distance; source formulas not verified here remain premises until checked.

Dependencies: C-01.

C-06 · Control heterogeneous blocks and nonconstant backgrounds

Extend the admitted regular derivative/coercivity estimates to the heterogeneous retained backgrounds and layer geometry actually encountered, with explicit weighted norms and domain margins.

Scope. Conditional heterogeneous coercivity. Application requires the complete C-05 Hessian estimate on the same domain.

Required conclusion. One uniform theorem on the specified admissible class; homogeneous identity-background bounds are not extrapolated.

Dependencies: C-04, C-05.

C-07 · Treat selectors and the integration support

Justify the covariance/IBP argument on the true selected domain, by an admitted convex-core representation with controlled boundary corrections or another complete support theorem.

Scope. Support/IBP identity on the stated regular product support; this does not independently establish coercivity on all heterogeneous histories.

Required conclusion. Every selector derivative and boundary flux is accounted for; a Hessian inequality on an unrelated open set is insufficient.

Dependencies: C-06.

C-08 · Obtain the complete regular conditional moment and decay estimate

Combine the actual quadratic lower bound and complete perturbation budget to prove current-scale mean, all-p field/gradient moments, and spatial cumulant decay for the selected regular conditional law.

Scope. Conditional regular moment and cumulant theorem. Applications require the complete score, Hessian and true selected support; it does not estimate outer history cumulants.

Required conclusion. The law is the one in C-01, with fixed factorial growth and volume-uniform constants; its remaining application to all histories is H-04.

Dependencies: C-04, C-05, C-06, C-07.

Retained histories and their response

Supporting derivations.

H-01 · Expose normalized history probabilities

Express the law of every retained history and reset under the normalized Wilson measure, with all conditional normalizers and the resampling kernels used in the exact transport.

Scope. Normalized path and reset identity; the law includes each conditional mass.

Required conclusion. An exact normalized identity suitable for marked expectation; an absolute signed activity is not called a probability.

Dependencies: B-01, C-01.

H-02 · Bound inherited low-margin and unreplaced events

Prove current-scale joint probability or directly marked bounds for inherited low-margin and unreplaced sectors, including overlap and retained-data dependence.

Scope. Open. Arbitrary retained large fields obstruct a uniform pointwise bound. An integrated substitute uses native-scale joint events, their shapes, and marked flat-test estimates; it must pay the actual losses in H-04 and S-05.

Required conclusion. The exponent and geometric cost compensate the actual insertion/carrier losses in the stated norm; isolated microscopic rarity is not substituted for a coarse event.

Dependencies: H-01, M-03.

H-03 · Localize the exact history-reset responses

Estimate the conditional covariance and higher cumulants generated when a live history label is averaged and re-sampled, with a carrier radius or decay function uniform in the remaining history.

Scope. Spatial cumulants for the mixed reset posterior under its specified law. Its marked use in arbitrary bad histories remains part of H-04.

Required conclusion. A spatial response estimate for the exact reset jet, beyond the stated integral identity.

Dependencies: H-01, M-01, M-03. Additional condition when changing the conditioning law: CL.

H-04 · Sum marked bad-history contributions

At one arbitrary scale, sum regular, repaired, residual and inherited bad sectors in the full normalized law for input jets satisfying the M-03/M-06 induction budget, keeping all exact output responses and a fixed factorial growth budget, uniformly over admissible scales and histories.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. One full-law one-step estimate that can feed M-07, with no discarded positive-probability sector and no hidden denominator/volume loss. The all-scale input bounds to be proved in M-07 are not assumed.

Dependencies: H-02, H-03, C-08, M-06.

H-05 · Control the outer retained-mean covariance and cumulants

Bound the variations of the retained conditional means and their higher joint cumulants at each scale and at the terminal physical scale.

Scope. Open. A one-sided inner Schur comparison controls a specific normalizer contribution. It does not control the retained marginal's outside curvature or moving support.

Required conclusion. The law-of-total-covariance outer term and its higher-order analogues satisfy the required spatial/terminal budget; a within-sector covariance estimate alone does not pass.

Dependencies: H-04, M-07.

Connected kernels and joint distributions

Supporting derivations.

S-01 · Pass between connected and full joint distributions

From connected functionals on coincidence-flat joint Schwartz spaces with a common bound $A^{m+1}(m!)^{p}\nu_{qm}$, p>=1, construct partition full moments on joint tests, independently of contact extensions, with the same q,p; include inverse Mobius and stability budgets.

Scope. Joint-test distribution theorem for connected versus full moments under the stated seminorm bounds.

Required conclusion. An explicit tensor-product/distribution argument; bounds only on simple tensor tests do not count. Preserve extension independence on flat tests.

Dependencies: definitions and stated hypotheses.

S-02 · Assemble exact connected shells and terminal distributions

Given an exact connected-shell identity, a uniform physical tree-kernel bound and summable shell weights, combine the flat-shell estimate, terminal and one-point budgets into a full joint-test OS hierarchy, with a uniform ultraviolet tail and cofinal convergence.

Scope. Assembly theorem conditional on the kernel identity, weights and convergence. It does not supply the Wilson-law bounds.

Required conclusion. A quantified theorem with fixed q,p and a correct convergence hypothesis; actual shell identity and kernel bounds remain S-04/S-05.

Dependencies: S-01.

S-03 · Prove terminal and one-point distribution budgets

At fixed ell0, prove the normalized identity/one-point prescription and all-order joint-test terminal distribution bounds uniformly as the torus expands.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. Fixed q,p,C and the same fields as F-05, with no infrared mass-gap assumption used unless independently established.

Dependencies: F-05, M-07, H-05.

S-04 · Identify the actual connected physical shell kernels

Represent exact connected Wilson correlations by the physical shell kernels and terminal term required by S-02, including outer means, collision vertices, history resets and normalization.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. An exact equality on joint tests with precisely located kernel/remainder terms; an algebraic label for the whole unknown correlation is not a localized representation.

Dependencies: M-07, H-05, F-06.

S-05 · Prove the all-order full-law tree-kernel majorant

Prove a uniform bound $A^{m+1}(m!)^{p_0} h_j^{-4m} w_j \exp[-c\tau/((m+d_0)h_j)]$ for the actual kernels, with one common nonnegative weight majorant satisfying $\sum_j h_j w_j < \infty$, or a quantitatively sufficient substitute with a proved fixed-factorial summation theorem.

Scope. Open for the complete law. An all-order ultraviolet estimate on flat native tests does not control the transported and retained terms of the physically normalized fields.

Required conclusion. Constants, factorial exponent, range parameters and the summable weight majorant are uniform in cutoff, volume, m and history. The normalized complete law is used.

Dependencies: S-04, C-08, H-04.

S-06 · Realize the complete contact and approximation budget

Bound all finite-cutoff stencil, normalization and contact approximation errors in a single vanishing or summable joint-test norm, compatible with the exact hierarchy and positive-time algebra. Specify whether the error is connected or full; use S-01 inverse stability before inserting a full-moment error into a connected-shell identity.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. A fixed q_E,p_E and a cofinal rate, including partial diagonals. Absorb a connected error into the terminal bound or use S-01 stability. The final OS exponents include q_E,p_E; the illustrative (6,7) is not automatic.

Dependencies: F-07, M-02, S-04.

Positivity and translations

Supporting derivations.

P-01 · Verify the actual normalization lies in the positive architecture

Check that the constructed field/source substitutions are real, local, reflection-compatible, and split across separated time half-spaces, including all required tensor Wick phases.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. Apply the existing Gram-congruence theorem to the actual coefficients; arbitrary symmetry of a counterterm does not replace the splitting identity.

Dependencies: F-05, F-06, S-06.

P-02 · Verify translation and permutation compatibility of the realization

For the chosen lattice representatives and any auxiliary block skeleton, prove exact fine-lattice translation symmetry or a vanishing translation defect, plus permutation symmetry of the full hierarchy.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. The common seminorm controls approximation by fine translations; a fixed physical cell-grid symmetry is insufficient.

Dependencies: P-01, S-05, S-06.

Rotations of the action and fields

Supporting derivations.

R-01 · Identify all finite hypercubic rotational splits

Write the three compatibility relations that join the seven hypercubic channels into the four rotational tensor types for the actual field normalization.

Scope. Finite-dimensional compatibility relations. Their validity for the interacting normalization is a separate estimate.

Required conclusion. Explicit channel coefficients and finite normalization conditions; the representation classification is an algebraic input.

Dependencies: F-03.

R-02 · Bound the action’s angular defect

Express the angular variation of the normalized effective action as local defect carriers plus controlled remainders and show their contribution vanishes on fixed physical tests.

Scope. Conditional on coefficient contraction and full-law localization on the chosen Wilson sequence.

Required conclusion. A bound for each rotation generator on the actual full law, including nonlocal tails and inherited terms; hypercubic invariance alone is insufficient.

Dependencies: C-08, H-04.

R-03 · Bound the observable and contact angular defects

Prove the matching fields, running matrices and product contacts transform with a vanishing defect under each continuum angular generator.

Scope. Conditional on exact marked transport, contact control and coefficient matching.

Required conclusion. All 21 labels and all source orders with the same normalization as S0; scalar action isotropy does not determine tensor normalization.

Dependencies: R-01, M-07, F-06.

R-04 · Obtain a common rotation-defect budget

Combine action and observable defects into a uniform joint-test estimate stable under the finite parameter paths used to integrate each fixed rotation.

Scope. Conditional joint-test theorem integrating generator bounds uniformly along a rotation path.

Required conclusion. A rate tending to zero with the cutoff for each fixed rotation and joint test, with no unproved uniformity over intermediate rotations.

Dependencies: R-02, R-03, S-05.

R-05 · Apply the angular reconstruction and reflection checks

Integrate the stated generator reduction and use the exact discrete reflection to establish full O(4) covariance of the limit, including tensor labels.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. Full covariance for the same limiting hierarchy, without averaging rotated theories or replacing the original hierarchy.

Dependencies: R-04, P-01, P-02, A-01.

Short-distance coefficients

Supporting derivations.

U-01 · Identify the normalized two-point leading coefficient

Compute the prescribed scalar/tensor two-point coefficient in the exact matched normalization and coupling dictionary, including logarithmic running.

Scope. Comparison coefficients available; open for the physically normalized interacting hierarchy.

Required conclusion. A specified distribution and coefficient; the computed native/Wick coefficient is used only after its matching is proved.

Dependencies: F-05, M-07.

U-02 · Identify a specified nonzero connected three-point coefficient

Compute the target connected three-point coefficient on an explicit admissible profile and prove it is nonzero in the same physical normalization.

Scope. Separated nonzero comparison terms available; connected identification and uniform interacting errors remain open.

Required conclusion. A definite coefficient with a checkable nonvanishing witness and correct distinction between connected and full moments.

Dependencies: F-05, M-07.

U-03 · Bound the interacting two- and three-point UV remainders

Prove the coefficient formulas with remainders uniform in cutoff and expanding volume before the short-distance limit, retaining all exact matching and history contributions.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. The order of limits is justified and the nonzero coefficient survives; a finite-order formal expansion is insufficient.

Dependencies: U-01, U-02, S-05.

U-04 · Prove the reflected four- and six-field block estimates

For the two positive-time UV block observables used in Annex L, prove the interacting vacuum norm remainder after subtracting the prescribed scalar coefficient.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. A summable logarithmic-scale error on a common test-space norm, with the correct reflected two-block geometry.

Dependencies: S-05, U-03.

U-05 · Control scale derivatives and all compact UV profiles

Establish the same interacting block norm budget for the logarithmic scale derivative and the quantitative profile continuity required for the continuous-scale phase-transfer theorem.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. The error is integrable/summable in the specified scale norm and uniform on the required bounded test sets, not only a countable sequence of numerical profiles.

Dependencies: U-04.

Energy bounds and periodic states

Supporting derivations.

E-01 · Prove a local-cylinder alternative for periodic state passage

Derive the exact local periodic/vacuum path-law density and a sufficient local total-variation or entropy comparison, with explicit cutoff observable budgets, that transfers vacuum slab estimates to periodic OS forms without the global thermal trace condition.

Scope. Local-cylinder transfer criterion; the required cofinal comparison remains E-05.

Required conclusion. A quantitative bridge covering the physical a^-4m losses and fixed polynomial vectors; actual convergence/rate is E-05.

Dependencies: definitions and stated hypotheses.

E-02 · Prove the renormalized scalar arbitrary-vector bound

Bound the actual scalar logarithmic transfer response or periodic OS insertion after physical renormalization, including its prescribed identity coefficient, uniformly in the cutoff.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. An arbitrary-vector heat or graph estimate with the required common domain; pressure derivatives and vacuum moments are insufficient.

Dependencies: F-05, M-07.

E-03 · Prove the renormalized pseudoscalar seed bound

Establish the corresponding operator estimate for the correctly normalized pseudoscalar field, including any electric/magnetic mixed realization and domain.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. Uniform physical energy weight and Schwartz seminorm for the actual field; a scalar or bare plaquette estimate is not substituted.

Dependencies: F-05, M-07.

E-04 · Prove the renormalized off-diagonal magnetic seed bound

Upgrade the compact magnetic comparison to the actual continuum-normalized B1B2 seed, controlling physical powers, running mixing and temporal admixtures.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. A cutoff-uniform physical estimate; the existing dimensionless compact bound with a^-4 loss remains an input, not the conclusion.

Dependencies: F-05, M-07.

E-05 · Verify the chosen periodic-state transfer criterion

On one admissible cubic cofinal sequence prove either the local state-comparison rate in E-01, the retained thermal trace-increment criterion, or a direct periodic OS insertion bound.

Scope. Open on cofinal cubic tori. Fixed-volume trace bounds and a finite-torus comparison criterion do not give the required uniform local state comparison or a gap.

Required conclusion. The criterion applies to the actual renormalized hierarchy and all fixed polynomial degrees; fixed-volume large-time convergence alone does not count.

Dependencies: E-01, B-02.

E-06 · Apply the field-domain and three-seed theorems

Pass the established seed bounds to the reconstructed fields on the common graph core and then use the finite Lorentz-orbit theorem to obtain all 21 field estimates.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. The field identified by reconstruction is the bounded extension under discussion, and all domain/covariance hypotheses are checked.

Dependencies: E-02, E-03, E-04, E-05, R-05, A-02.

Decomposition into vacuum phases

Supporting derivations.

V-01 · Characterize disconnected symmetry in selected phases

For a valid translation-ergodic decomposition on a specified separable bounded algebra, prove a criterion for almost-everywhere finite/disconnected symmetry invariance using central defects P0(alpha_g(A)-A)Omega and macroscopic translation averages.

Scope. Central-defect criterion conditional on a valid phase decomposition; a symmetric mixed state need not have symmetric individual phases.

Required conclusion. A simultaneous common-null-set theorem and exact defect norm formula; invariance of the mixed state or a common UV coefficient alone is not used to infer phase invariance.

Dependencies: definitions and stated hypotheses.

V-02 · Verify the actual decomposition theorem hypotheses

Apply a precisely scoped phase decomposition theorem to the actual finite field family, with its generalized energy or proved intrinsic-locality hypotheses, measurable fields and symmetry representation.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. A valid field decomposition with measurable closed fields, proper-orthochronous covariance, unique translation-invariant vacuum and clustering in the phases, and the required phasewise field energy domains. Disconnected symmetry is separately V-04. Vacuum OS moments alone do not verify the decomposition hypotheses.

Dependencies: A-02, E-06.

V-03 · Transfer the specified UV coefficients to one common phase set

Apply the existing block-variance/continuous-scale theorem simultaneously to both required coefficients, all specified profiles and normalizations.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. One common full-measure set retaining the prescribed coefficients, not merely some phase with a nonzero cumulant.

Dependencies: V-02, U-04, U-05.

V-04 · Verify the required disconnected phase symmetries

Prove the actual central symmetry defects vanish for the finite disconnected symmetries required by P1, or give another valid selection theorem supplying them.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. A phasewise symmetry statement for the selected theory; symmetric mixtures of symmetry-breaking phases are not accepted.

Dependencies: V-01, V-02.

V-05 · Select a phase with all required properties

Intersect the verified full-measure sets and select a phase with the required finite field family, common energy domains, proper-orthochronous covariance, the required disconnected symmetries, unique vacuum, clustering and the fixed UV coefficients.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. The selected phase is nonempty and each requested property is established for the same field normalization and phase.

Dependencies: V-03, V-04, E-06.

Continuum existence and identification

Supporting derivations.

A-01 · Extract the compatible continuum hierarchy

Apply the proved cofinal compactness theorem to every order and field tuple on a countable dense flat-test family, then extend by the common seminorm bound.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. One subsequence for the entire hierarchy, preserving normalization, positivity and symmetry; no uniqueness assertion.

Dependencies: S-02, S-03, S-05, S-06, P-01, P-02.

A-02 · Apply corrected OS reconstruction to this hierarchy

Verify the corrected OS II regularity and covariance hypotheses for the constructed full moments and reconstruct the relativistic fields without imposing clustering at the initial mixed stage.

Scope. Open for the stated Wilson hierarchy.

Required conclusion. A source-scoped reconstruction application with the actual fields and common domains; an abstract reconstruction theorem is not counted as construction of its inputs.

Dependencies: A-01, R-05.

A-03 · Identify the limiting fields with the stated existence target

Identify each conclusion of the existence theorem with its corresponding estimate, field normalization and phase, and obtain an identification with no unsupported cross-reference or scope extension.

Scope. Open identification theorem for the stated fixed-group existence target. No positive spectral gap is derived.

Required conclusion. The unconditional conclusion requires every Wilson-law hypothesis to be proved. A remaining Wilson-law estimate prevents a claim of Yang–Mills existence.

Dependencies: A-02, V-05, U-03.